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VISUAL C#

SIXTH EDITION

TONY GADDIS

STARTING OUT WITH

Visual C#®

Sixth Edition

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Haywood Community College

Pearson

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Preface

Welcome to *Starting Out with Visual C#*, Sixth Edition. This book is intended for an introductory programming course and is ideal for students with no prior experience. Students who are new to programming will appreciate the clear, down-to-earth explanations and the detailed walk-throughs that are provided by the hands-on tutorials. More experienced students will appreciate the depth of detail as they learn about .NET, databases, Language-Integrated Query, and other topics.

As with all the books in the *Starting Out With* series, the hallmark of this text is its clear, friendly, and easy-to-understand writing. In addition, it is rich in example programs that are concise and practical. The programs in this book include short examples that highlight specific programming topics, as well as more involved examples that focus on problem solving. Each chapter provides numerous hands-on tutorials that guide the student through each step of the development of an application. In addition to detailed, step-by-step instructions, the tutorials also provide the application's completed code and screen captures of the completed forms.

New to This Edition

1. All the tutorials have been updated for Visual Studio 2022 and the latest versions of C# and .NET.

2. All the VideoNotes have been updated and revised to reflect the diverse learners we serve.

3. A new chapter, *Chapter 11: More about Classes*, has been added. This new chapter covers intermediate to advanced topics, including:

- Returning objects from methods
- The `ToString` methods
- Working with `null`
- The `this` reference
- Copy constructors
- Aggregation and composition
- Anonymous objects
- Mutable and immutable classes
- Class collaboration

4. In *Chapter 3, Processing Data*, the material on integer division has been expanded to include a discussion on using cast operators to prevent integer division.

5. In the previous edition, the discussion on ListBoxes started in Chapter 4 and then continued in Chapter 5. In this edition, all the ListBox material has been consolidated in Chapter 4.

6. Changes in *Chapter 7, Arrays and Lists*:

- A discussion on getting the length of a two-dimensional array and its dimensions has been added.
- A discussion on passing two-dimensional arrays to methods has been added.

7. Changes in *Chapter 8, Text Processing*:
   - Interpolated strings are introduced and used throughout the text, from this point forward.
   - The section on Working with Strings has been split into two sections: *Working with Subtrings*, and *Manipulating Strings*.
   - A new section on the `StringBuilder` class has been added.
   - A detailed discussion of method chaining and how it works has been added.
8. In *Chapter 9, Structured Data Types*:
   - A discussion on returning a structure from a method has been added.
   - The introductory material on dictionaries has been expanded and improved.
   - A new section on Tuples has been added.
9. In *Chapter 10, Introduction to Classes*:
   - New discussions on the scope of fields, shadowing, uninitialized fields, and initializing fields at declaration have been added.
   - The discussion of Passing an Object to a Method has been expanded into a full section.
   - A discussion of the `readonly` keyword and how it can be used in field declarations has been added.
   - Private `set` accessors are discussed.
   - The discussion of Auto-Properties has been expanded into a full section.
   - The section on Storing Class Type Objects in Arrays and Lists now has a detailed discussion on how arrays and Lists hold references to objects.
10. In Chapter 12, Inheritance, Polymorphism, and Interfaces, (this was Chapter 11 in the previous edition):
   - A discussion on `protected` access has been added.
   - The material on interface properties has been expanded.

A GUI-Based Approach

Beginning students are more motivated to learn programming when their applications have some sort of graphical element, such as a graphical user interface (GUI). Students using this book will learn to create GUI-based, event-driven, Visual C# applications. The Visual Studio environment is used to create forms that are rich with user interface controls and graphical images.

Learn to Use Objects Early, Learn to Write Classes Later

This book explains what an object is very early and shows the student how to create objects from classes that are provided by .NET. It then introduces the student to the fundamentals of input and output, control structures, methods, arrays and lists, and file operations. Then, the student learns to write their own classes and explores the topics of inheritance and polymorphism.

Brief Overview of Each Chapter

Chapter 1: Introduction to Computers and Programming. This chapter begins by giving a very concrete and easy-to-understand explanation of how computers work, how data is stored and manipulated, and why we write programs in high-level languages. In this chapter, the student learns what an object is and sees several examples by studying the objects that make up a program's GUI. The chapter discusses steps in the programming development cycle. It also gives an introduction to the Visual Studio environment.

Chapter 2: Introduction to Visual C#. In this chapter, the student learns to create forms with labels, buttons, and picture boxes and learns to modify control properties. The student is introduced to C# code and learns the organizational structure of namespaces, classes, and methods. The student learns to write simple event-driven applications that respond to button clicks or provide interaction through clickable images. The importance of commenting code is also discussed.

Chapter 3: Processing Data. This chapter introduces variables and data types. It discusses the use of local variables and variables declared as fields within a form class. The student learns to create applications that read input from TextBox controls, perform mathematical operations, and produce formatted output. The student learns about the exceptions that can occur when the user enters invalid data into a TextBox and learns to write simple exception-handling code to deal with those problems. Named constants are introduced as a way of representing unchanging values and creating self-documenting, maintainable code. The student also learns more intricacies of creating graphical user interfaces. The chapter concludes with a discussion and tutorial on using the Visual Studio debugger to locate logic errors by single-stepping through an application's code.

Chapter 4: Making Decisions. In this chapter, the student learns about relational operators and Boolean expressions and is shown how to control the flow of a program with decision structures. The `if`, `if-else`, and `if-else-if` statements are covered. Nested decision structures, logical operators, and the `switch` statement are also discussed. The student learns to use the `TryParse` family of methods to validate input and prevent exceptions. Radio buttons, check boxes, and list boxes are introduced as ways to let the user select items in a GUI.

Chapter 5: Loops, Files, and Random Numbers. This chapter shows the student how to use loops to create repetition structures. The `while` loop, the `for` loop, and the `do-while` loop are presented. Counters, accumulators, and running totals are also discussed. This chapter also introduces sequential file input and output and using text files. The student learns various programming techniques for writing data to text files and reading the contents of test files. The chapter concludes with a discussion of pseudorandom numbers, their applications, and how to generate them.

Chapter 6: Modularizing Your Code with Methods. In this chapter, the student first learns how to write and call `void` methods as well as value-returning methods. The chapter shows the benefits of using methods to modularize programs and discusses the top-down design approach. Then, the student learns to pass arguments to methods. Passing by value, by reference, and output parameters are discussed. The chapter concludes with a discussion and tutorial on debugging methods with the Visual Studio step-into, step-over, and step-out, commands.

Chapter 7: Arrays and Lists. Arrays and lists are reference-type objects in C# so this chapter begins by discussing the difference between value type and reference type objects in the C# language. Then, the student learns to create and work with single-dimensional and two-dimensional arrays. The student learns to pass arrays as arguments to methods, transfer data between arrays and files, work with partially filled arrays, and create jagged arrays. Many examples of array processing are provided, including examples of finding the sum, average, highest, and lowest values in an array. Finally, the student learns to create `List` objects and store data in them, and use the ImageList control, a data structure for storing and retrieving images.

Chapter 8: Text Processing. This chapter discusses various techniques for working with text. The topics include interpolated strings, working with characters, working with substrings, tokenizing strings, and formatting strings. The `StringBuilder` class is also discussed.

Chapter 9: Structured Data Types. In this chapter, the student learns to use structures to encapsulate several variables into a single item. The student next learns to create and use enumerated types. Then the student learns to use the `Dictionary` collection to store and work with data as key–value pairs. Last, the student learns how to work with tuples.

Chapter 10: Introduction to Classes. Up to this point, the student has extensively used objects that are instances of .NET classes. In this chapter, students learn to write classes to create their own objects with fields, methods, and constructors. The student learns how to implement various types of properties within a class, including auto-properties and read-only auto-properties. Creating arrays of objects and storing objects in a `List` are also discussed. A primer on finding the classes in a problem as well as their responsibilities is provided. The chapter shows the student how to create multiple form classes in a project, instantiate those classes, and display them. A tutorial is given where the student creates a multiform application in which the code in one form accesses controls on another form. The chapter concludes by discussing static members and static classes.

Chapter 11: More about Classes. In this chapter, the student continues to learn about classes and objects. Returning objects from methods, the `ToString` method, working with `null`, the `this` reference, and copy constructors are covered. Aggregation and the "has a" relationship is introduced, and the difference between aggregation and composition is discussed. The student also learns about anonymous objects, mutable and immutable classes, and class collaboration.

Chapter 12: Inheritance, Polymorphism, and Interfaces. The study of classes continues in this chapter with the subjects of inheritance and polymorphism. The topics covered include base classes, derived classes, how constructors functions work in inheritance, method overriding, and polymorphism. Abstract classes and abstract methods are also discussed. The chapter discusses extension methods, and concludes with a section on interfaces.

Chapter 13: Databases. This chapter introduces the student to basic database concepts. The student first learns about tables, rows, and columns and how to create an SQL Server database in Visual Studio. The student then learns how to connect a database to a Visual C# application and display a table in a DataGridView control, a Details view, and other data-bound controls. Finally, the student learns how to write SQL Select statements to retrieve data from a table.

Chapter 14: Delegates and Lambda Expressions. Lambda expressions have become commonplace in C# programming. Before the student can really understand lambda expressions, they must first understand delegates and anonymous methods. The goal of this chapter is to build a foundation of knowledge about how delegates and anonymous methods work, and then build an understanding of lambda expressions.

Chapter 15: Language-Integrated Query (LINQ). This chapter begins by introducing the student to LINQ as a tool for querying the data in common data structures such as arrays and Lists, using LINQ to Objects. It discusses both query syntax and method syntax. Several useful LINQ extension methods are also discussed. The chapter concludes with a section on LINQ to SQL and discusses how to use LINQ to query a database.

Appendix A: C# Primitive Data Types. This appendix gives an overview of the primitive data types available in C#.

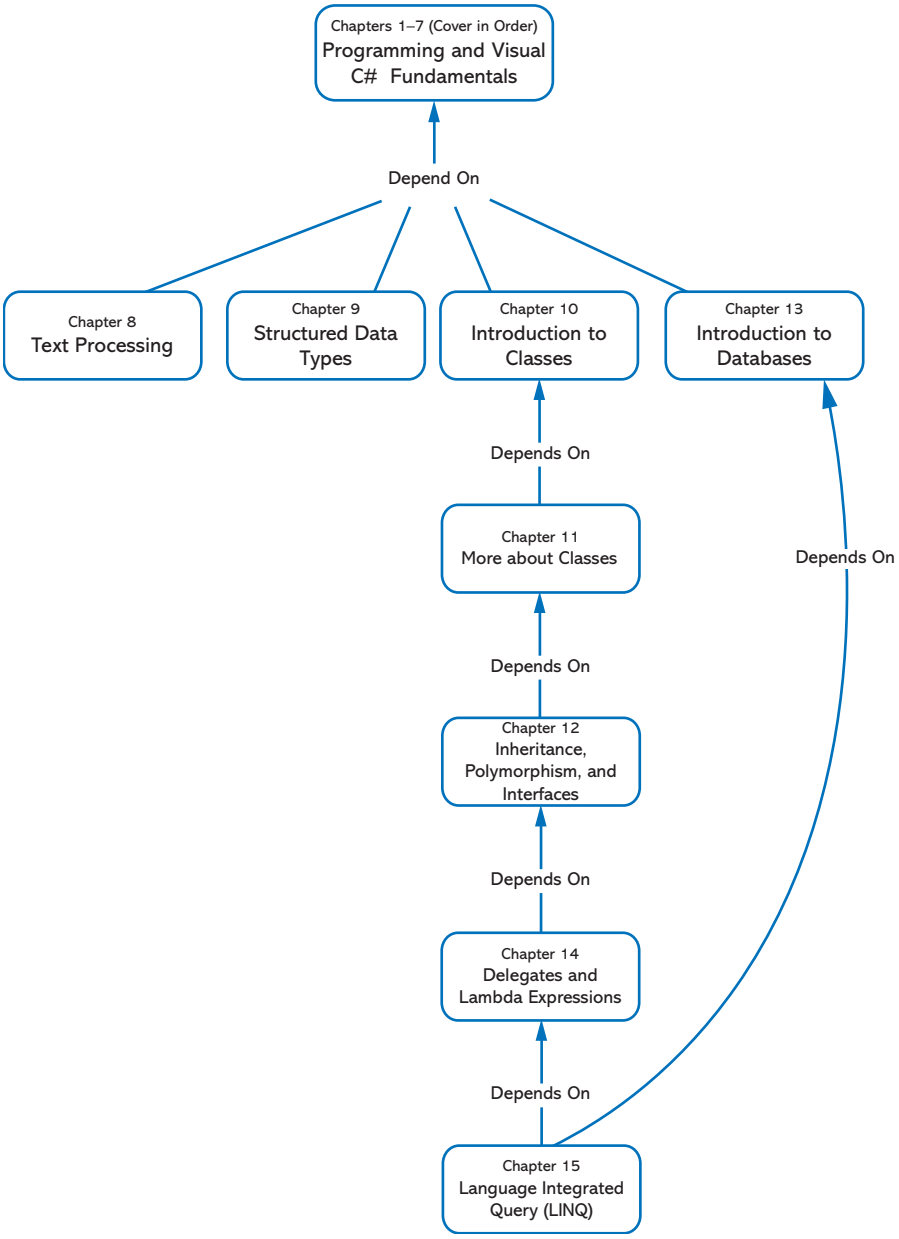
Appendix B: Additional User Interface Controls. This appendix shows how to create a variety of controls such as ToolTips, combo boxes, scroll bars, TabControls, WebBrowser controls, ErrorProvider components, and menu systems.

Appendix C: ASCII/Unicode Characters. This appendix lists the ASCII (American Standard Code for Information Interchange) character set, which is also the Latin Subset of Unicode.

Appendix D: Answers to Checkpoint Questions. This appendix provides the answers to the Checkpoint questions that appear throughout each chapter in the book.

Appendix E: Installing LINQ to SQL. This appendix shows how to use the Visual Studio Installer to download and install LINQ to SQL.

Figure P-1 Chapter dependencies



Organization of the Text

The text teaches Visual C# step by step. Each chapter covers a major set of programming topics, introduces controls and GUI elements, and builds knowledge as the student progresses through the book. Although the chapters can be easily taught in their existing sequence, there is some flexibility. Figure P-1 shows the chapter dependencies. As shown in the figure, Chapters 1–7 present the fundamentals of Visual C# programming and should be covered in sequence. Then, you can move directly to Chapter 8, 9, 10, or 13. Chapter 11 should be covered after Chapter 10. Then Chapters 12, 14, and 15. can be covered in order.

Features of the Text

Concept Statements. Each major section of the text starts with a concept statement. This statement concisely summarizes the main point of the section.

Tutorials. Each chapter has several hands-on tutorials that guide the student through the development of an application. Each tutorial provides detailed, step-by-step instructions, as well as the application's completed code and screen captures of the completed forms.

Example Programs. Each chapter has an abundant number of code examples designed to highlight the current topic.

Notes. Notes appear at several places throughout the text. They are short explanations of interesting or often misunderstood points relevant to the topic at hand.

Tips. Tips advise the student on the best techniques for approaching different programming or animation problems.

Warnings. Warnings caution students about programming techniques or practices that can lead to malfunctioning programs or lost data.

Checkpoints. Checkpoints are questions placed at intervals throughout each chapter. They are designed to query the student's knowledge quickly after learning a new topic. The answers to the Checkpoint questions can be found in Appendix D.

Review Questions. Each chapter presents a thorough and diverse set of Review Questions. They include Multiple Choice, True/False, Algorithm Workbench, and Short Answer.

Programming Problems. Each chapter offers a pool of Programming Problems designed to solidify the student's knowledge of the topics currently being studied.

VideoNotes. Each tutorial in the book has an accompanying online VideoNote that can be accessed on the book's companion Web site www.pearson.com/cs-resources. Students can follow along with the author as they work through each tutorial in the videos. Also, one programming problem at the end of each chapter has an accompanying VideoNote that shows the student how to create the solution.

Supplements

Student The following supplementary material is available at www.pearson.com/cs-resources:

- Source code and files required for the chapter tutorials.
- Access to the VideoNote tutorials that complete set of online VideoNotes that take the student through each tutorial in the book. Also, one programming problem from each chapter has an accompanying VideoNote that helps the student create a solution. You may access the VideoNotes by going to www.pearson.com/cs-resources.

Instructor The following supplements are available to qualified instructors on the Pearson Education Instructor Resource Center:

- Answers to all Review Questions in the text
- Solutions for all Programming Problems in the text
- Completed versions of all tutorials
- PowerPoint presentation slides for every chapter
- Test bank

For information on how to access these supplements, visit the Pearson Education Instructor Resource Center at www.pearsonhighered.com/irc.

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Tony Gaddis is the principal author of the *Starting Out With* series of textbooks. Tony has 20 years experience teaching computer science courses at Haywood Community College in North Carolina. He is a highly acclaimed instructor who was previously selected as the North Carolina Community College Teacher of the Year and has received the Teaching Excellence Award from the National Institute for Staff and Organizational Development.

The *Starting Out With* series includes introductory books using the C++ programming language, the Java™ programming language, Microsoft® Visual Basic®, Microsoft® C#®, Python, Programming Logic and Design, MIT App Inventor, and Alice, all published by Pearson Education.

Attention Students

Installing Visual Studio

To complete the tutorials and programming problems in this book, you need to install

Visual Studio 2022, or a later version, on your computer. We recommend that you download Visual Studio Community 2022 from the following website, and install it on your system:

www.visualstudio.com

Visual Studio Community 2022 is a free, full-featured development environment, and is a perfect companion for this textbook.

NOTE: If you are working in your school's computer lab, there is a good chance that **Microsoft Visual Studio** has already been installed. If this is the case, your instructor will show you how to start Visual Studio.

Installing the Student Sample Program Files

The Student Sample Program files that accompany this book are available for download from the book's companion Web site at:

www.pearson.com/cs-resources

These files are required for many of the book's tutorials. Simply download the Student Sample Program files to a location on your hard drive where you can easily access them.

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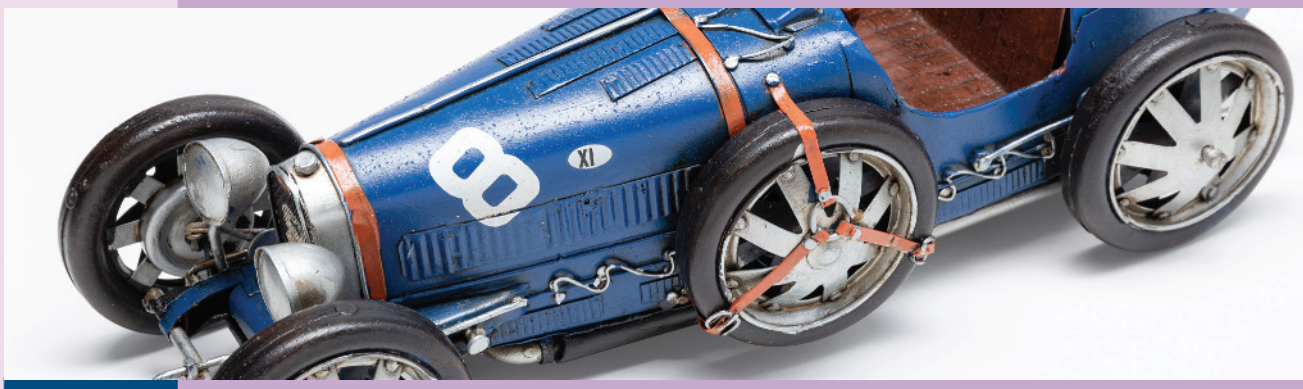


CHAPTER 1

Introduction to Computers and Programming

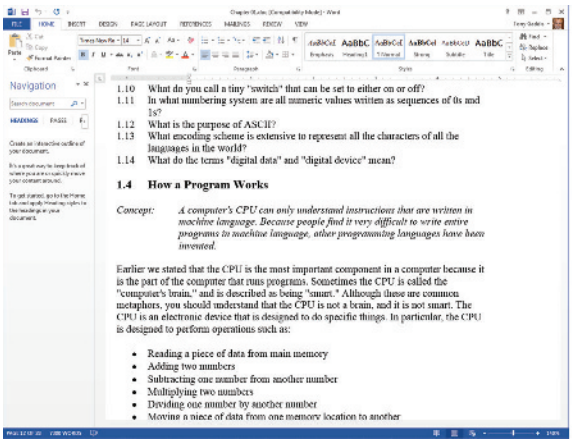
TOPICS

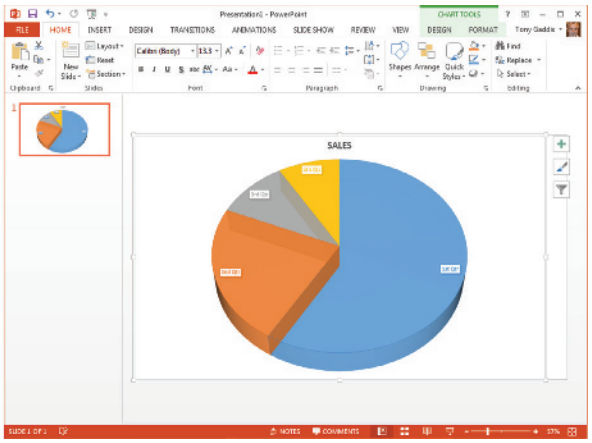
1.1 Introduction
1.2 Hardware and Software
1.3 How Computers Store Data
1.4 How a Program Works
1.5 Graphical User Interfaces
1.6 Objects
1.7 The Program Development Process
1.8 Getting Started with Visual Studio

1.1 Introduction

Think about some of the different ways that people use computers. In school, students use computers for tasks such as writing papers, searching for articles, sending e-mail, and participating in online classes. At work, people use computers to analyze data, make presentations, conduct business transactions, communicate with customers and coworkers, control machines in manufacturing facilities, and do many other things. At home, people use computers for tasks such as paying bills, shopping online, staying connected with friends and family, and playing computer games. And don't forget that smart phones, tablets, car navigation systems, and many other devices are computers as well. The uses of computers are almost limitless in our everyday lives.

Figure 1-1 A word processing program and a presentation program





Computers can do such a wide variety of things because they can be programmed, which means that computers are designed not to do just one job, but to do any job that their programs tell them to do. A **program** is a set of instructions that a computer follows to perform a task. For example, Figure 1-1 shows screens from two commonly used Microsoft programs: Word and PowerPoint. Word is a word processing program that allows you to create, edit, and print documents. PowerPoint allows you to create graphical slides and use them as part of a presentation.

Programs are commonly referred to as **software**. Software is essential to a computer because without software, a computer can do nothing. All the software that makes our computers useful is created by individuals known as programmers, or software developers. A **programmer**, or **software developer**, is a person with the training and skills necessary to design, create, and test computer programs. Computer programming is an exciting and rewarding career. Today, programmers work in business, medicine, government, law enforcement, agriculture, academics, entertainment, and almost every other field.

This book introduces you to the fundamental concepts of computer programming using the C# programming language. Before we begin exploring those concepts, you need to understand a few basic things about computers and how they work. This chapter provides a solid foundation of knowledge that you will continually rely on as you study computer science. First, we discuss the physical components that computers are commonly made of. Then, we look at how computers store data and execute programs. Next, we introduce you to two fundamental elements of modern software design: graphical user interfaces and objects. Finally, we give a quick introduction to the software used to write C# programs.

1.2 Hardware and Software

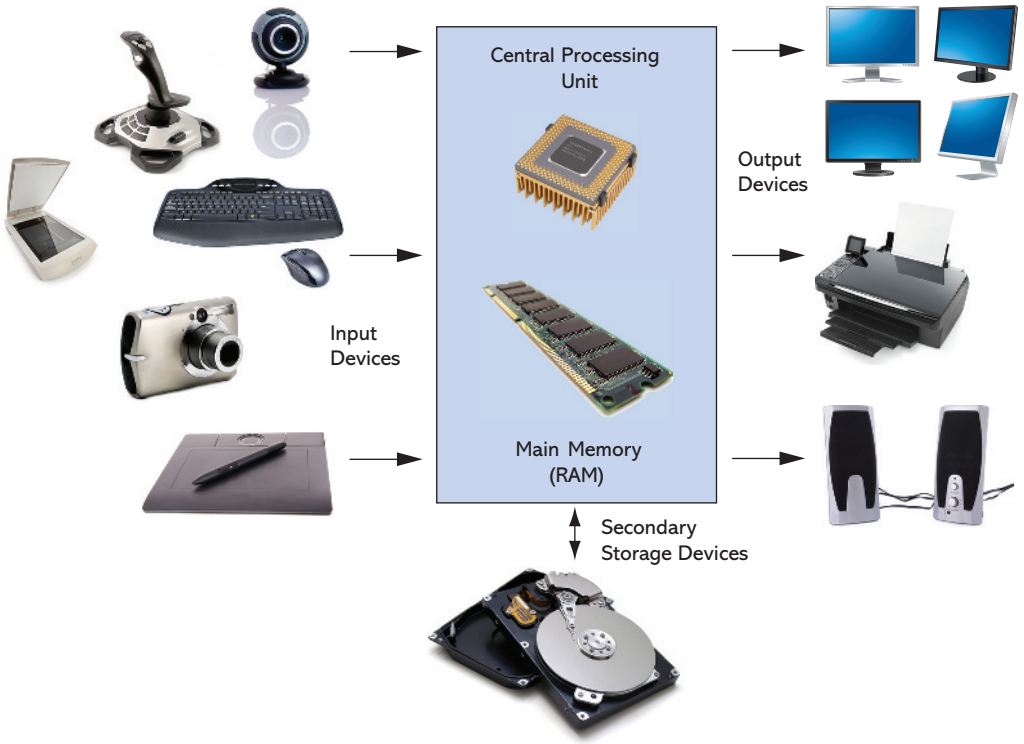
CONCEPT: **The physical devices that a computer is made of are referred to as the computer's hardware. The programs that run on a computer are referred to as software.**

Hardware

Hardware refers to all the physical devices, or components, of which a computer is made. A computer is not one single device but is a system of devices that all work together. Like the different instruments in a symphony orchestra, each device in a computer plays its own part.

If you have ever shopped for a computer, you have probably seen sales literature listing components such as microprocessors, memory, disk drives, video displays, graphics cards, and so forth. Unless you already know a lot about computers or at least have a friend who

Figure 1-2 Typical components of a computer system



(*Photo credits*: **Webcam** Iko/Shutterstock, **Joystick** Nikita Rogul/Shutterstock, **Scanner** Feng Yu/Shutterstock, **Keyboard** Chiyacat/Shutterstock, **Camera** Elkostas/Shutterstock, **Tablet** Tkemot/Shutterstock, **Hard disk** Vitaly Korovin/Shutterstock, **Speakers** StockPhotosArt/Shutterstock, **Printer** Jocic/Shutterstock, **Monitors** Art gallery/Shutterstock, **RAM** Peter Guess/Shutterstock, **Chip** Aquila/Shutterstock).

does, understanding what these different components do can be confusing. As shown in Figure 1-2, a typical computer system consists of the following major components:

- Central processing unit (CPU)
- Main memory
- Secondary storage devices
- Input devices
- Output devices

Let's take a closer look at each of these components.

The CPU

When a computer is performing the tasks that a program tells it to do, we say that the computer is **running** or **executing** the program. The **central processing unit**, or **CPU**, is the part of a computer that actually runs programs. The CPU is the most important component in a computer because without it, the computer could not run software.

In the earliest computers, CPUs were huge devices made of electrical and mechanical components such as vacuum tubes and switches. Figure 1-3 shows such a device. The two women in the photo are working with the historic ENIAC computer. The **ENIAC**, considered by many to be the world's first programmable electronic computer, was built in 1945 to calculate artillery ballistic tables for the U.S. Army. This machine, which was primarily one big CPU, was 8 feet tall and 100 feet long and weighed 30 tons.

Today, CPUs are small chips known as **microprocessors**. Figure 1-4 shows a photo of a lab technician holding a modern-day microprocessor. In addition to being much smaller than the old electromechanical CPUs in early computers, microprocessors are also much more powerful.

Figure 1-3 The ENIAC computer (courtesy of US Army Center of Military History)

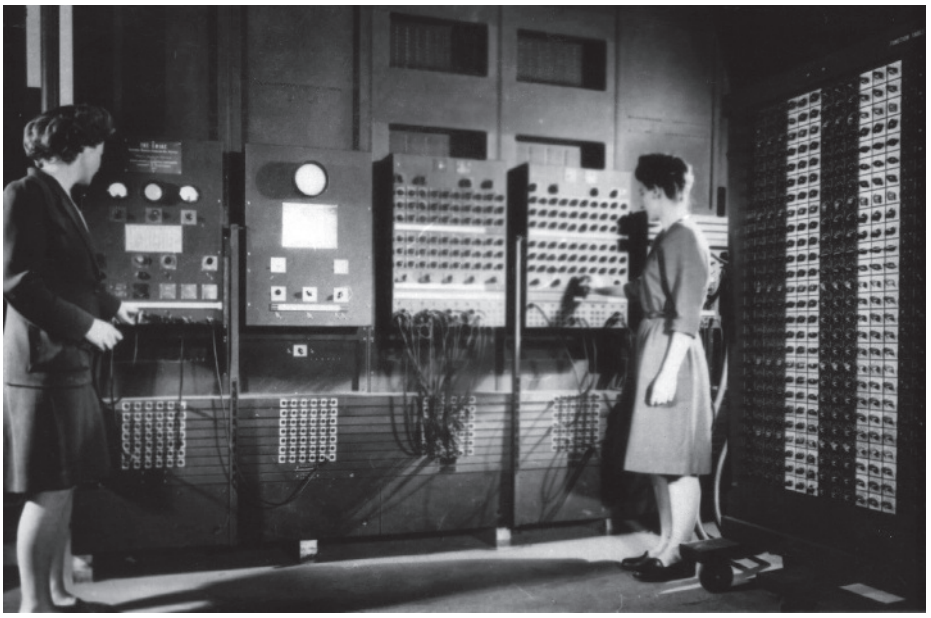
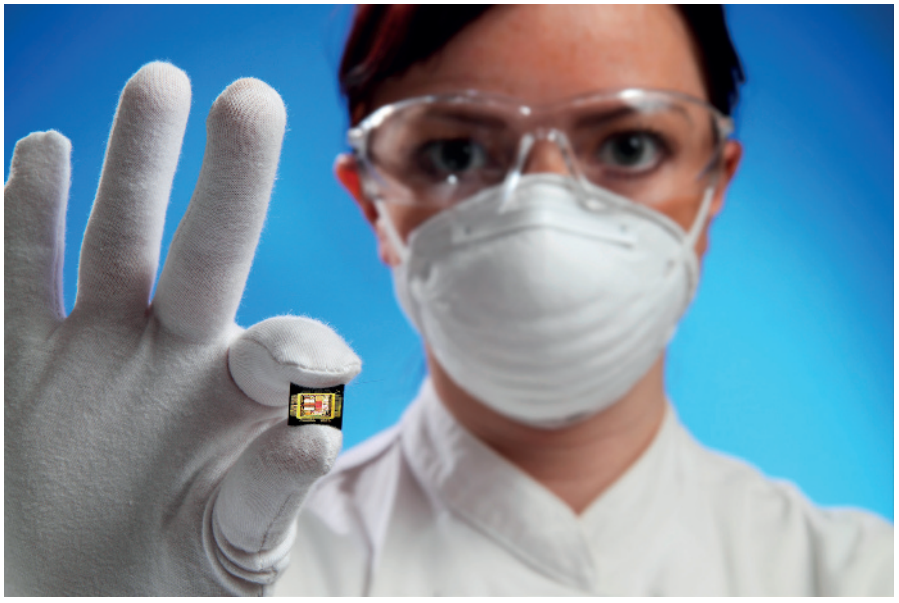


Figure 1-4 A lab technician holds a modern microprocessor (courtesy of Vadim Kolobanov/Shutterstock)



Main Memory

You can think of **main memory** as the computer's work area. This is where the computer stores a program while the program is running, as well as the data that the program is working with. For example, suppose you are using a word processing program to write an essay for one of your classes. While you do this, both the word processing program and the essay are stored in main memory.

Main memory is commonly known as **random access memory**, or **RAM**. It is called this because the CPU is able to quickly access data stored at any random location in RAM. RAM is usually a **volatile** type of memory that is used only for temporary storage while

a program is running. When the computer is turned off, the contents of RAM are erased. Inside your computer, RAM is stored in small microchips.

Secondary Storage Devices

Secondary storage is a type of memory that can hold data for long periods of time, even when there is no power to the computer. Programs are normally stored in secondary memory and loaded into main memory as needed. Important data, such as word processing documents, payroll data, and inventory records, is saved to secondary storage as well.

The most common type of secondary storage device is the **disk drive**. A traditional disk drive stores data by magnetically encoding it onto a spinning circular disk. **Solid-state drives**, which store data in solid-state memory, are increasingly becoming popular. A solid-state drive has no moving parts and operates faster than a traditional disk drive.

Most computers have some sort of secondary storage device, either a traditional disk drive or a solid-state drive, mounted inside their case. External storage devices are also available, which connect to one of the computer's communication ports, or plug into a memory slot. External storage devices can be used to create backup copies of important data or to move data to another computer. For example, **USB (Universal Serial Bus) drives** and **SD (Secure Digital) memory cards** are small devices that appear in the system as disk drives. They are inexpensive, reliable, and small enough to be carried in your pocket.

Input Devices

Input is any data the computer collects from people and from other devices. The component that collects the data and sends it to the computer is called an **input device**. Common input devices are the keyboard, mouse, touchscreen, scanner, microphone, and digital camera. Disk drives and optical drives can also be considered input devices because programs and data are retrieved from them and loaded into the computer's memory.

Output Devices

Output is any data the computer produces for people or for other devices. It might be a sales report, a list of names, or a graphic image. The data is sent to an **output device**, which formats and presents it. Common output devices are screens and printers. Storage devices can also be considered output devices because the system sends data to them in order to be saved.

Software

If a computer is to function, software is not optional. Everything that a computer does, from the time you turn the power switch on until you shut the system down, is under the control of software. There are two general categories of software: system software and application software. Most computer programs clearly fit into one of these two categories. Let's take a closer look at each.

System Software

The programs that control and manage the basic operations of a computer are generally referred to as **system software**. System software typically includes the following types of programs:

Operating Systems

An **operating system** is the most fundamental set of programs on a computer. The operating system controls the internal operations of the computer's hardware, manages all the devices connected to the computer, allows data to be saved to and retrieved from storage devices, and allows other programs to run on the computer. Windows, Linux, macOS, iOS, and Android are examples of operating systems.

Utility Programs

A **utility program** performs a specialized task that enhances the computer's operation or safeguards data. Examples of utility programs are virus scanners, file-compression programs, and data-backup programs.

Software Development Tools

The software tools that programmers use to create, modify, and test software are referred to as **software development tools**. Assemblers, compilers, and interpreters, which are discussed later in this chapter, are examples of programs that fall into this category.

Application Software

Programs that make a computer useful for everyday tasks are known as **application software**. These are the programs that people normally spend most of their time running on their computers. Figure 1-1, at the beginning of this chapter, shows screens from two commonly used applications—Microsoft Word, a word processing program, and Microsoft PowerPoint, a presentation program. Some other examples of application software are spreadsheet programs, e-mail programs, Web browsers, and game programs.

Checkpoint

1.1 What is a program?

1.2 What is hardware?

1.3 List the five major components of a computer system.

1.4 What part of the computer actually runs programs?

1.5 What part of the computer serves as a work area to store a program and its data while the program is running?

1.6 What part of the computer holds data for long periods of time, even when there is no power to the computer?

1.7 What part of the computer collects data from people and from other devices?

1.8 What part of the computer formats and presents data for people or other devices?

1.9 What fundamental set of programs control the internal operations of the computer's hardware?

1.10 What do you call a program that performs a specialized task, such as a virus scanner, a file-compression program, or a data-backup program?

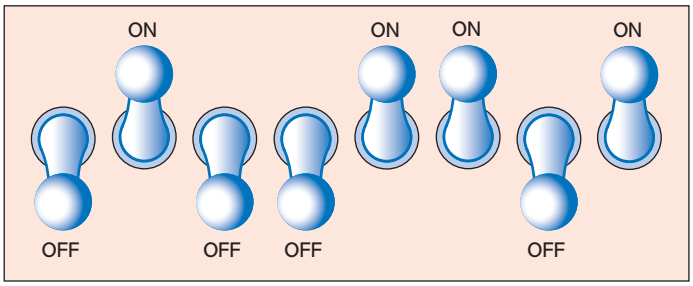
1.11 Word processing programs, spreadsheet programs, e-mail programs, Web browsers, and game programs belong to what category of software?

1.3 How Computers Store Data

CONCEPT: All data stored in a computer is converted to sequences of 0s and 1s.

A computer's memory is divided into tiny storage locations known as bytes. One **byte** is enough memory to store only a letter of the alphabet or a small number. In order to do anything meaningful, a computer has to have lots of bytes. Most computers today have millions, or even billions, of bytes of memory.

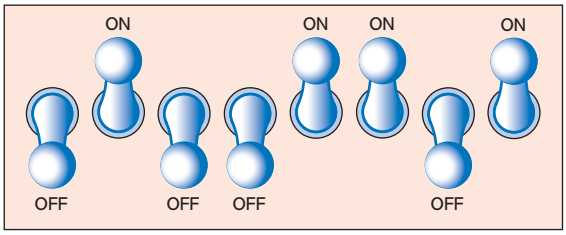
Figure 1-5 A byte thought of as eight switches



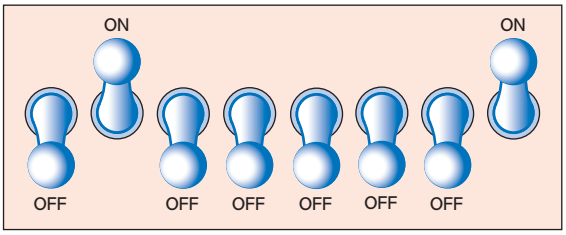
Each byte is divided into eight smaller storage locations known as bits. The term **bit** stands for **binary digit**. Computer scientists usually think of bits as tiny switches that can be either on or off. Bits aren't actual "switches," however, at least not in the conventional sense. In most computer systems, bits are tiny electrical components that can hold either a positive or a negative charge. Computer scientists think of a positive charge as a switch in the *on* position and a negative charge as a switch in the *off* position. Figure 1-5 shows the way that a computer scientist might think of a byte of memory: as a collection of switches that are each flipped to either the on or the off position.

When a piece of data is stored in a byte, the computer sets the eight bits to an on/off pattern that represents the data. For example, the pattern shown on the left in Figure 1-6 shows how the number 77 would be stored in a byte, and the pattern on the right shows how the letter A would be stored in a byte. In a moment, you will see how these patterns are determined.

Figure 1-6 Bit patterns for the number 77 and the letter A



The number 77 stored in a byte.



The letter A stored in a byte.

Storing Numbers

A bit can be used in a very limited way to represent numbers. Depending on whether the bit is turned on or off, it can represent one of two different values. In computer systems, a bit that is turned off represents the number 0 and a bit that is turned on represents the number 1. This corresponds perfectly to the **binary numbering system**. In the binary numbering system (or **binary**, as it is usually called), all numeric values are written as sequences of 0s and 1s. Here is an example of a number that is written in binary:

```
10011101
```

The position of each digit in a binary number has a value assigned to it. Starting with the rightmost digit and moving left, the position values are 2^0, 2^1, 2^2, 2^3, and so forth, as shown in Figure 1-7. Figure 1-8 shows the same diagram with the position values calculated. Starting with the rightmost digit and moving left, the position values are 1, 2, 4, 8, and so forth.

To determine the value of a binary number, you simply add up the position values of all the 1s. For example, in the binary number 10011101, the position values of the 1s are 1, 4, 8, 16, and 128. This is shown in Figure 1-9. The sum of all these position values is 157. So, the value of the binary number 10011101 is 157.

Figure 1-7 The values of binary digits as powers of 2

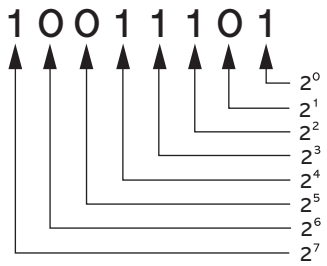


Figure 1-8 The values of binary digits

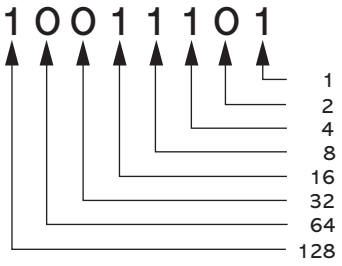


Figure 1-9 Determining the value of 10011101

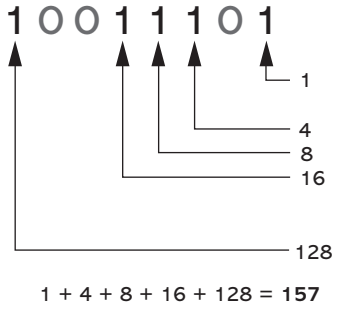


Figure 1-10 shows how you can picture the number 157 stored in a byte of memory. Each 1 is represented by a bit in the on position, and each 0 is represented by a bit in the off position.

When all the bits in a byte are set to 0 (turned off), then the value of the byte is 0. When all the bits in a byte are set to 1 (turned on), then the byte holds the largest value that can be stored in it. The largest value that can be stored in a byte is $1 + 2 + 4 + 8 + 16 + 32 + 64 + 128 = 255$. This limit exists because there are only eight bits in a byte.

What if you need to store a number larger than 255? The answer is simple: use more than 1 byte. For example, suppose we put 2 bytes together. That gives us 16 bits. The position values of those 16 bits would be 2^0, 2^1, 2^2, 2^3, and so forth, up through 2^{15}. As shown

Figure 1-10 The bit pattern for 157

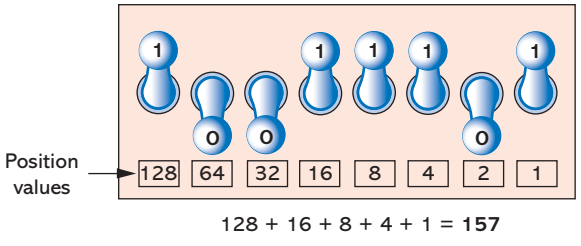
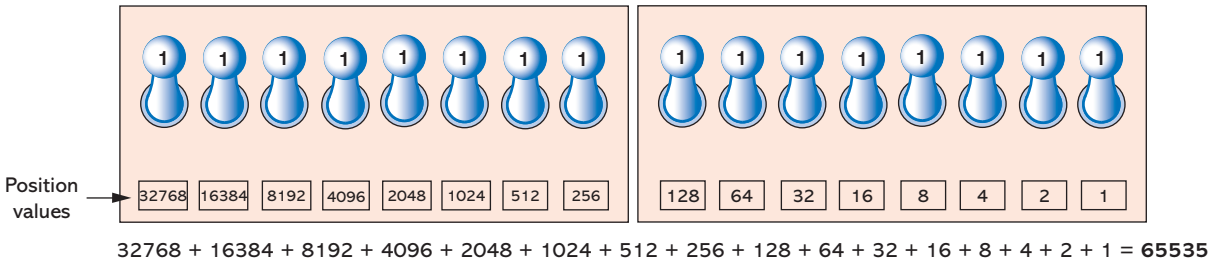


Figure 1-11 Two bytes used for a large number



in Figure 1-11, the maximum value that can be stored in 2 bytes is 65,535. If you need to store a number larger than this, then more bytes are necessary.

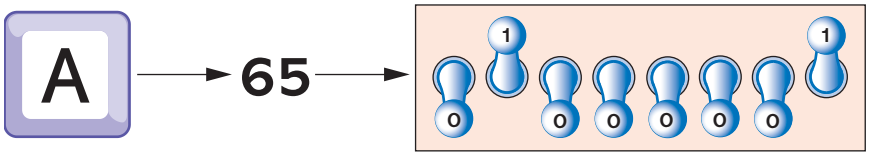
TIP: In case you're feeling overwhelmed by all this, relax! You will not have to actually convert numbers to binary while programming. Knowing that this process is taking place inside the computer will help you as you learn, and in the long term this knowledge will make you a better programmer.

Storing Characters

Any piece of data that is stored in a computer's memory must be stored as a binary number. That includes characters such as letters and punctuation marks. When a character is stored in memory, it is first converted to a numeric code. The numeric code is then stored in memory as a binary number.

Over the years, different coding schemes have been developed to represent characters in computer memory. Historically, the most important of these coding schemes is **ASCII**, which stands for the **American Standard Code for Information Interchange**. ASCII is a set of 128 numeric codes that represent the English letters, various punctuation marks, and other characters. For example, the ASCII code for the uppercase letter A is 65. When you type an uppercase A on your computer keyboard, the number 65 is stored in memory (as a binary number, of course). This is shown in Figure 1-12.

Figure 1-12 The letter A stored in memory as the number 65



TIP: The acronym ASCII is pronounced "askee."

In case you are curious, the ASCII code for uppercase B is 66, for uppercase C is 67, and so forth. Appendix C shows all the ASCII codes and the characters they represent.

The ASCII character set was developed in the early 1960s and was eventually adopted by almost all computer manufacturers. ASCII is limited, however, because it defines codes for only 128 characters. To remedy this, the Unicode character set was developed in the early 1990s. **Unicode** is an extensive encoding scheme that is compatible with ASCII and can also represent the characters of many of the world's languages. Today, Unicode is quickly becoming the standard character set used in the computer industry.

Advanced Number Storage

Earlier you saw how numbers are stored in memory. Perhaps it occurred to you then that the binary numbering system can be used to represent only integer numbers, beginning with 0. Negative numbers and real numbers (such as 3.14159) cannot be represented using the simple binary numbering technique we discussed.

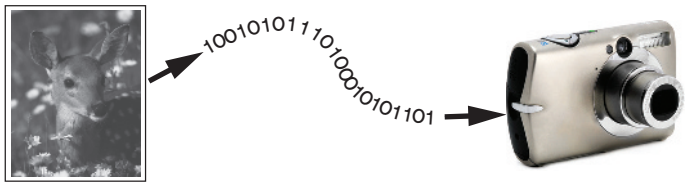
Computers are able to store negative numbers and real numbers in memory, but to do so they use encoding schemes along with the binary numbering system. Negative numbers are encoded using a technique known as **two's complement**, and real numbers are encoded in **floating-point notation**. You don't need to know how these encoding schemes work, only that they are used to convert negative numbers and real numbers to binary format.

Other Types of Data

Computers are often referred to as digital devices. The term **digital** can be used to describe anything that uses binary numbers. **Digital data** is data that is stored in binary, and a **digital device** is any device that works with binary data. In this section, we have discussed how numbers and characters are stored in binary, but computers also work with many other types of digital data.

For example, consider the pictures that you take with your digital camera. These images are composed of tiny dots of color known as **pixels**. (The term pixel stands for **picture element**.) As shown in Figure 1-13, each pixel in an image is converted to a numeric code that represents the pixel's color. The numeric code is stored in memory as a binary number.

Figure 1-13 A digital image stored in binary format (photo on right courtesy by Elkostas/Shutterstock)



The music that you play on your CD player, iPod, or MP3 player is also digital. A digital song is broken into small pieces known as **samples**. Each sample is converted to a binary number, which can be stored in memory. The more samples that a song is divided into, the more it sounds like the original music when it is played back. A CD-quality song is divided into more than 44,000 samples per second!

Checkpoint

1.12 What amount of memory is enough to store a letter of the alphabet or a small number?

1.13 What do you call a tiny "switch" that can be set to either on or off?

1.14 In what numbering system are all numeric values written as sequences of 0s and 1s?

1.15 What is the purpose of ASCII?

1.16 What encoding scheme is extensive enough to represent all the characters of many of the languages in the world?

1.17 What do the terms *digital data* and *digital device* mean?

1.4 How a Program Works

CONCEPT: **A computer's CPU can understand only instructions written in machine language. Because people find it very difficult to write entire programs in machine language, other programming languages have been invented.**

Earlier, we stated that the CPU is the most important component in a computer because it is the part of the computer that runs programs. Sometimes the CPU is called the "computer's brain" and is described as being "smart." Although these are common metaphors, you should understand that the CPU is not a brain, and it is not smart. The CPU is an electronic device that is designed to do specific things. In particular, the CPU is designed to perform operations such as the following:

- Reading a piece of data from main memory
- Adding two numbers
- Subtracting one number from another number
- Multiplying two numbers
- Dividing one number by another number
- Moving a piece of data from one memory location to another
- Determining whether one value is equal to another value

As you can see from this list, the CPU performs simple operations on pieces of data. The CPU does nothing on its own, however. It has to be told what to do, which is the purpose of a program. A program is nothing more than a list of instructions that cause the CPU to perform operations.

Each instruction in a program is a command that tells the CPU to perform a specific operation. Here's an example of an instruction that might appear in a program:

```
10110000
```

To you and me, this is only a series of 0s and 1s. To a CPU, however, this is an instruction to perform an operation.[1] It is written in 0s and 1s because CPUs understand only instructions that are written in **machine language**, and machine language instructions are always written in binary.

A machine language instruction exists for each operation that a CPU is capable of performing. For example, there is an instruction for adding numbers; there is an instruction for subtracting one number from another; and so forth. The entire set of instructions that a CPU can execute is known as the CPU's **instruction set**.

The machine language instruction that was previously shown is an example of only one instruction. It takes a lot more than one instruction, however, for the computer to do

NOTE: There are several microprocessor companies today that manufacture CPUs. Some of the more well-known microprocessor companies are Intel, AMD, and Motorola. If you look carefully at your computer, you might find a tag showing a logo for its microprocessor.

Each brand of microprocessor has its own unique instruction set, which is typically understood only by microprocessors of the same brand. For example, Intel microprocessors understand the same instructions, but they do not understand instructions for Motorola microprocessors.

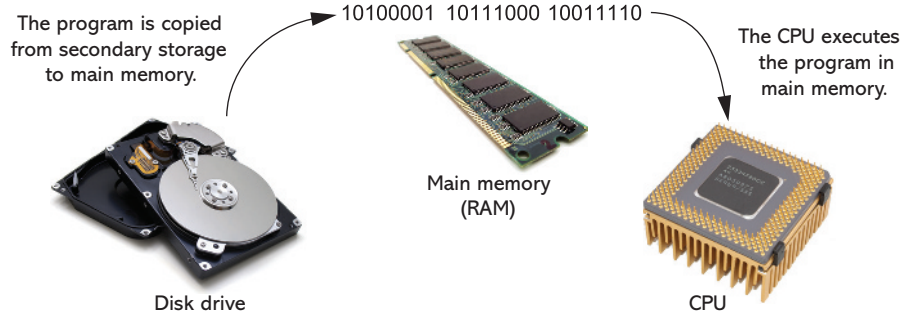
[1]The example shown is an actual instruction for an Intel microprocessor. It tells the microprocessor to move a value into the CPU.

anything meaningful. Because the operations that a CPU knows how to perform are so basic in nature, a meaningful task can be accomplished only if the CPU performs many operations. For example, if you want your computer to calculate the amount of interest that you will earn from your savings account this year, the CPU will have to perform a large number of instructions, carried out in the proper sequence. It is not unusual for a program to contain thousands or even a million or more machine language instructions.

Programs are usually stored on a secondary storage device such as a disk drive. When you install a program on your computer, the program is typically copied to your computer's disk drive from a CD-ROM or perhaps downloaded from a Web site.

Although a program can be stored on a secondary storage device such as a disk drive, it has to be copied into main memory, or RAM, each time the CPU executes it. For example, suppose you have a word processing program on your computer's disk. To execute the program, you use the mouse to double-click the program's icon. This causes the program to be copied from the disk into main memory. Then, the computer's CPU executes the copy of the program that is in main memory. This process is illustrated in Figure 1-14.

Figure 1-14 A program being copied into main memory and then executed (courtesy of Lefteris Papaulakis/Shutterstock, Garsya/Shutterstock and marpan/Shutterstock)



When a CPU executes the instructions in a program, it is engaged in a process that is known as the **fetch-decode-execute cycle**. This cycle, which consists of three steps, is repeated for each instruction in the program. The steps are as follows:

1. **Fetch** A program is a long sequence of machine language instructions. The first step of the cycle is to fetch, or read, the next instruction from memory into the CPU.
2. **Decode** A machine language instruction is a binary number that represents a command that tells the CPU to perform an operation. In this step the CPU decodes the instruction that was just fetched from memory, to determine which operation it should perform.
3. **Execute** The last step in the cycle is to execute, or perform, the operation.

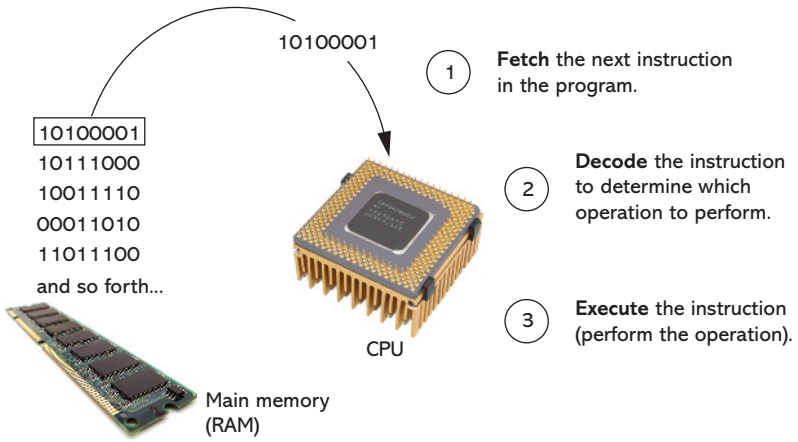
Figure 1-15 illustrates these steps.

From Machine Language to Assembly Language

Computers can execute only programs that are written in machine language. As previously mentioned, a program can have thousands or even a million or more binary instructions, and writing such a program would be very tedious and time consuming. Programming in machine language would also be very difficult because putting a 0 or a 1 in the wrong place would cause an error.

Although a computer's CPU understands only machine language, it is impractical for people to write programs in machine language. For this reason, **assembly language** was

Figure 1-15 The fetch-decode-execute cycle (courtesy of Garsya/Shutterstock, marpan/Shutterstock)

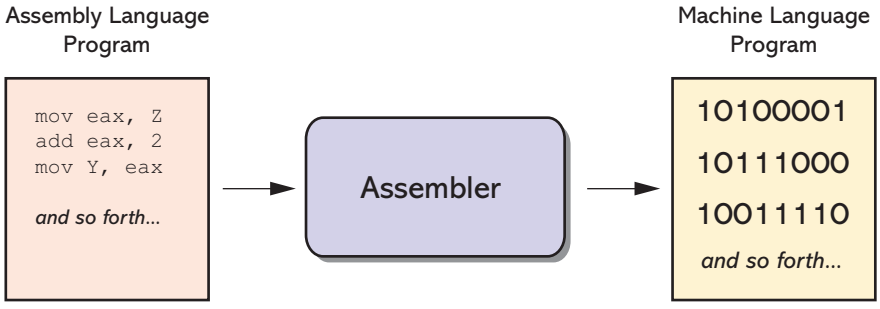


created in the early days of computing[2] as an alternative to machine language. Instead of using binary numbers for instructions, assembly language uses short words that are known as **mnemonics**. For example, in assembly language, the mnemonic `add` typically means to add numbers, `mul` typically means to multiply numbers, and `mov` typically means to move a value to a location in memory. When a programmer uses assembly language to write a program, he or she can write short mnemonics instead of binary numbers.

NOTE: There are many different versions of assembly language. It was mentioned earlier that each brand of CPU has its own machine language instruction set. Each brand of CPU typically has its own assembly language as well.

Assembly language programs cannot be executed by the CPU, however. The CPU understands only machine language, so a special program known as an **assembler** is used to translate an assembly language program to a machine language program. This process is shown in Figure 1-16. The CPU can then execute the machine language program that the assembler creates.

Figure 1-16 An assembler translating an assembly language program to a machine language program



High-Level Languages

Although assembly language makes it unnecessary to write binary machine language instructions, it is not without difficulties. Assembly language is primarily a direct substitute for machine language, and like machine language, it requires that you know a lot

[2]The first assembly language was most likely developed in the 1940s at Cambridge University for use with a historical computer known as the EDSAC.

about the CPU. Assembly language also requires that you write a large number of instructions for even the simplest program. Because assembly language is so close in nature to machine language, it is referred to as a **low-level language**.

In the 1950s, a new generation of programming languages known as high-level languages began to appear. A **high-level language** allows you to create powerful and complex programs without knowing how the CPU works and without writing large numbers of low-level instructions. In addition, most high-level languages use words that are easy to understand. For example, if a programmer were using COBOL (which was one of the early high-level languages created in the 1950s), he or she would write the following instruction to display the message *Hello world* on the computer screen:

```
DISPLAY "Hello world"
```

Doing the same thing in assembly language would require several instructions and an intimate knowledge of how the CPU interacts with the computer's video circuitry. As you can see from this example, high-level languages allow programmers to concentrate on the tasks they want to perform with their programs rather than the details of how the CPU will execute those programs.

Since the 1950s, thousands of high-level languages have been created. Table 1-1 lists several of the more well-known languages.

Table 1-1 Programming languages

Language	Description
Ada	Ada was created in the 1970s, primarily for applications used by the U.S. Department of Defense. The language is named in honor of Ada Lovelace, a 19th-century mathematician who published an algorithm that is considered by many to be the first computer program.
BASIC	**B**eginners **A**ll-purpose **S**ymbolic **I**nstruction **C**ode is a general-purpose language that was originally designed in the early 1960s to be simple enough for beginners to learn. Today, there are many different versions of BASIC.
C and C++	C and C++ (pronounced "c plus plus") are powerful, general-purpose languages developed at Bell Laboratories. They are widely used for systems-level programming. The C language was created in 1972, and the C++ language was created in 1983.
C#	Pronounced "C Sharp," this language was created by Microsoft around the year 2000 for developing applications based on the Microsoft .NET platform.
COBOL	**Co**mmon **B**usiness-**O**riented **L**anguage was created in the 1950s and was designed for business applications.
FORTRAN	**FOR**mula **TRAN**slator was the first high-level programming language. It was designed in the 1950s for performing complex mathematical calculations.
Go	Go was created at Google and is primarily used to develop systems software and large server-based systems.
Java	Java is an object-oriented language invented at Sun Microsystems and is now owned by Oracle. It may be used to develop stand-alone applications that operate on a single computer, mobile apps, or applications that run over the Internet from a Web server.
JavaScript	JavaScript, created in the 1990s, can be used in Web pages. Despite its name, JavaScript is not related to Java.

Language	Description
Kotlin	Although Kotlin is a general-purpose programming language, it is widely used to develop Android mobile apps.
Pascal	Pascal was created in 1970 and was originally designed for teaching programming. The language was named in honor of the mathematician, physicist, and philosopher Blaise Pascal.
PHP	A programming language used primarily for developing Web server applications and dynamic Web pages.
Python	Python is a general-purpose language created in the early 1990s. It is popular for business and academic purposes and is widely used for data science applications.
R	The R programming language is used for statistical computation and working with large sets of data.
Ruby	Ruby is a general-purpose language that was created in the 1990s. It is increasingly becoming a popular language for programs that run on Web servers.
Rust	The Rust programming language is designed for high performance, memory safety, and concurrent execution. It was announced in 2010 by Mozilla Research.
Swift	Swift was created at Apple, primarily for writing apps that run on iOS and macOS devices.
Visual Basic	Visual Basic (commonly known as VB) is a Microsoft programming language and software development environment that allows programmers to create Windows-based applications quickly. VB was originally created in the early 1990s.

Keywords, Operators, and Syntax: An Overview

Each high-level language has its own set of predefined words that the programmer must use to write a program. The words that make up a high-level programming language are known as **keywords** or **reserved words**. Each keyword has a specific meaning and cannot be used for any other purpose. Table 1-2 shows the keywords in the C# programming language.

In addition to keywords, programming languages have **operators** that perform various operations on data. For example, all programming languages have math operators that perform arithmetic. In C#, as well as most other languages, the + sign is an operator that adds two numbers. The following adds 12 and 75:

```
12 + 75
```

There are numerous other operators in the C# language, many of which you will learn about as you progress through this text.

In addition to keywords and operators, each language also has its own **syntax**, which is a set of rules that must be strictly followed when writing a program. The syntax rules dictate how keywords, operators, and various punctuation characters must be used in a program. When you are learning a programming language, you must learn the syntax rules for that particular language.

The individual instructions that you use to write a program in a high-level programming language are called **statements**. A programming statement can consist of keywords, operators, punctuation, and other allowable programming elements, arranged in the proper sequence to perform an operation.

Table 1-2 The C# keywords

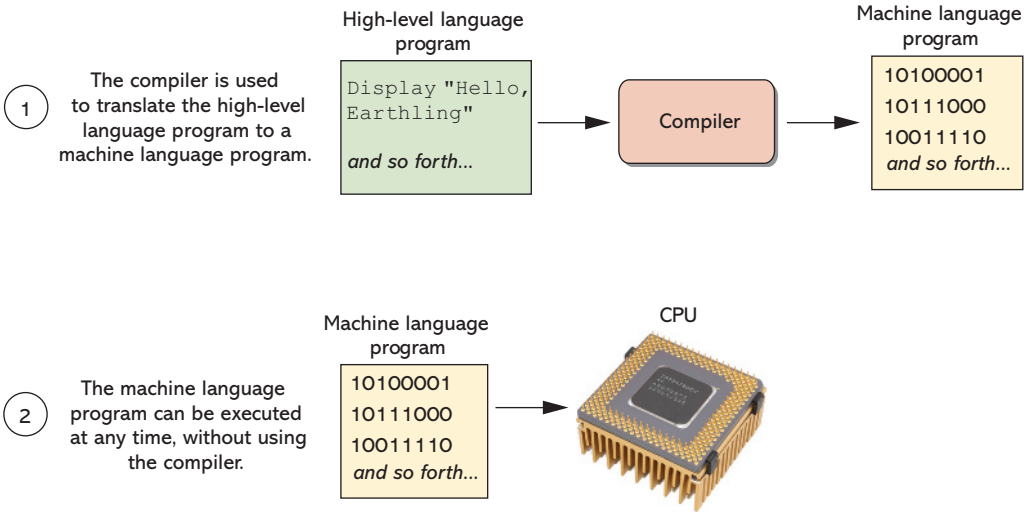
abstract	as	base	bool
break	byte	case	catch
char	checked	class	const
continue	decimal	default	delegate
do	double	else	enum
event	explicit	extern	false
finally	fixed	float	for
foreach	goto	if	implicit
in	int	interface	internal
is	lock	long	namespace
new	null	object	operator
out	override	params	private
protected	public	readonly	ref
return	sbyte	sealed	short
sizeof	stackalloc	static	string
struct	switch	this	throw
true	try	typeof	uint
ulong	unchecked	unsafe	ushort
using	virtual	void	volatile
while			

Compilers and Interpreters

Because the CPU understands only machine language instructions, programs that are written in a high-level language must be translated into machine language. Depending on the language in which a program has been written, the programmer will use either a compiler or an interpreter to make the translation.

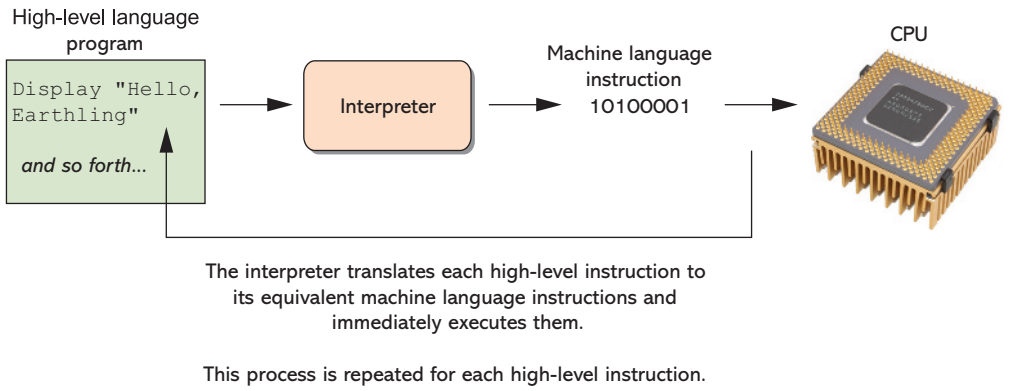
A **compiler** is a program that translates a high-level language program into a separate machine language program. The machine language program can then be executed any time it is needed. This is shown in Figure 1-17. As shown in the figure, compiling and executing are two different processes.

Figure 1-17 Compiling a high-level program and executing it (courtesy of marpan/Shutterstock)



Some programming languages use an **interpreter**, which is a program that both translates and executes the instructions in a high-level language program. As the interpreter reads each individual instruction in the program, it converts it to a machine language instruction and then immediately executes it. This process repeats for every instruction in the program. This process is illustrated in Figure 1-18. Because interpreters combine translation and execution, they typically do not create separate machine language programs.

Figure 1-18 Executing a high-level program with an interpreter (courtesy of marpan/Shutterstock)



The statements that a programmer writes in a high-level language are called **source code**, or simply **code**. Typically, the programmer types a program's code into a text editor and then saves the code in a file on the computer's disk. Next, the programmer uses a compiler to translate the code into a machine language program or an interpreter to translate and execute the code. If the code contains a syntax error, however, it cannot be translated. A **syntax error** is a mistake such as a misspelled keyword, a missing punctuation character, or the incorrect use of an operator. When this happens, the compiler or interpreter displays an error message, indicating that the program contains a syntax error. The programmer corrects the error and then attempts once again to translate the program.

NOTE: Human languages also have syntax rules. Do you remember when you took your first English class and you learned all those rules about commas, apostrophes, capitalization, and so forth? You were learning the syntax of the English language.

Although people commonly violate the syntax rules of their native language when speaking and writing, other people usually understand what they mean. Unfortunately, compilers and interpreters do not have this ability. If even a single syntax error appears in a program, the program cannot be compiled or executed.

Checkpoint

1.18 A CPU understands instructions that are written only in what language?

1.19 A program has to be copied into what type of memory each time the CPU executes it?

1.20 When a CPU executes the instructions in a program, it is engaged in what process?

1.21 What is assembly language?

1.22 What type of programming language allows you to create powerful and complex programs without knowing how the CPU works?

1.23 Each language has a set of rules that must be strictly followed when writing a program. What is this set of rules called?

1.24 What do you call a program that translates a high-level language program into a separate machine language program?

1.25 What do you call a program that both translates and executes the instructions in a high-level language program?

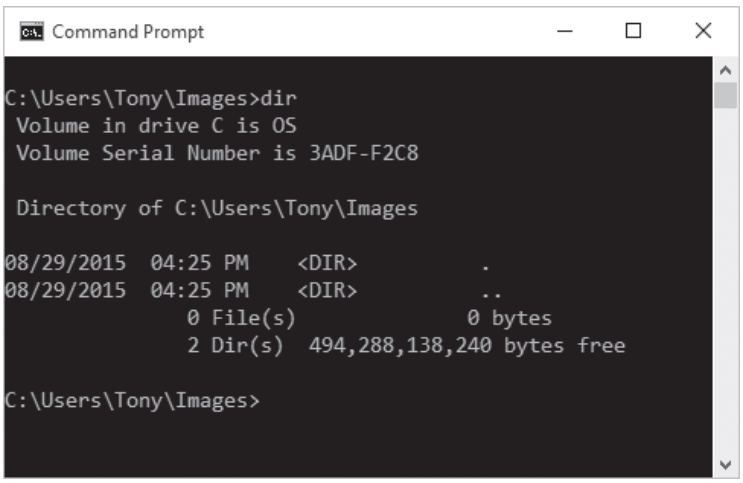
1.26 What type of mistake is usually caused by a misspelled keyword, a missing punctuation character, or the incorrect use of an operator?

1.5 Graphical User Interfaces

CONCEPT: **A graphical user interface allows the user to interact with a program using graphical elements such as icons, buttons, and dialog boxes.**

Programmers commonly use the term **user** to describe any hypothetical person that might be using a computer and its programs. A computer's **user interface** is the part of the computer with which the user interacts. One part of the user interface consists of hardware devices, such as the keyboard and the video display. Another part of the user interface involves the way that the computer's operating system and application software accepts commands from the user. For many years, the only way that the user could interact with a computer was through a command line interface. A **command line interface**, which is also known as a **console interface**, requires the user to type commands. If a command is typed correctly, it is executed and the results are displayed. If a command is not typed correctly, an error message is displayed. Figure 1-19 shows the Windows command prompt window, which is an example of a command line interface.

Figure 1-19 A command line interface



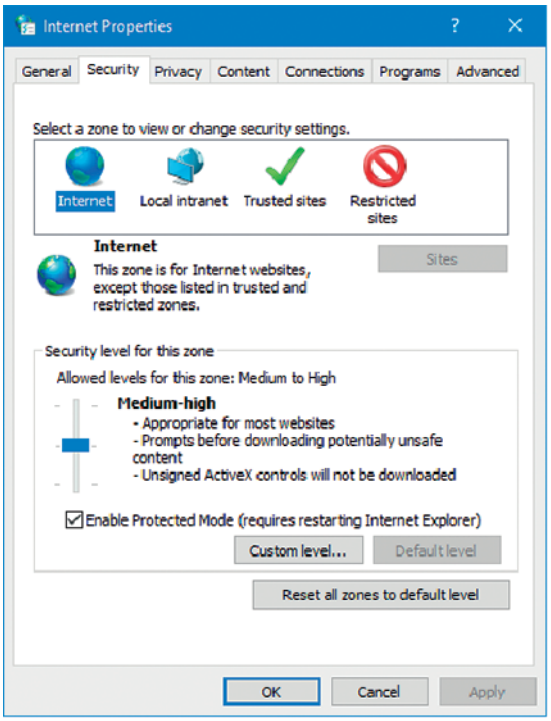
Many computer users, especially beginners, find command line interfaces difficult to use. This is because there are many commands to be learned, and each command has its own syntax, much like a programming statement. If a command isn't entered correctly, it will not work.

In the 1980s, a new type of interface known as a graphical user interface came into use in commercial operating systems. A **graphical user interface**, or **GUI** (pronounced "gooey"),

allows the user to interact with the operating system and application programs through graphical elements on the screen. GUIs also popularized the use of the mouse as an input device. Instead of requiring the user to type commands on the keyboard, GUIs allow the user to point at graphical elements and click the mouse button to activate them.

Much of the interaction with a GUI is done through windows that display information and allow the user to perform actions. Figure 1-20 shows an example of a window that allows the user to change the system's Internet settings. Instead of typing cryptic commands, the user interacts with graphical elements such as icons, buttons, and slider bars.

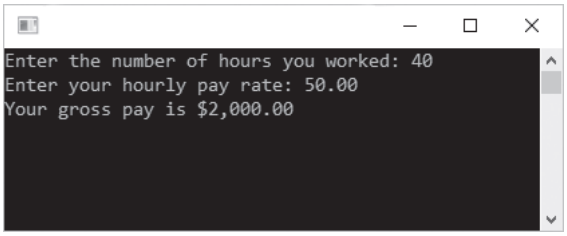
Figure 1-20 A window in a graphical user interface



Event-Driven GUI Programs

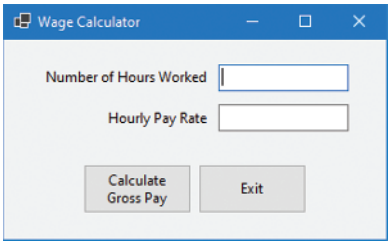
In a text-based environment, such as a command line interface, programs determine the order in which things happen. For example, Figure 1-21 shows the interaction that has taken place in a text environment with a program that calculates an employee's gross pay. First, the program told the user to enter the number of hours worked. In the figure, the user entered 40 and pressed the Enter key. Next, the program told the user to enter his or her hourly pay rate. In the figure, the user entered 50.00, and pressed the Enter key. Then, the program displayed the user's gross pay. As the program was running, the user had no choice but to enter the data in the order requested.

Figure 1-21 Interaction with a program in a text environment



In a GUI environment, however, the user determines the order in which things happen. For example, Figure 1-22 shows a GUI program that calculates an employee's gross pay. Notice that there are boxes in which the user enters the number of hours worked and the hourly pay rate. The user can enter the hours and the pay rate in any order he or she wishes. If the user makes a mistake, the user can erase the data that was entered and retype it. When the user is ready to calculate the area, he or she uses the mouse to click the *Calculate Gross Pay* button and the program performs the calculation.

Figure 1-22 A GUI program



Because GUI programs must respond to the actions of the user, they are said to be **event driven**. The user causes events, such as the clicking of a button, and the program responds to those events.

This book focuses exclusively on the development of GUI applications using the C# programming language. As you work through this book, you will learn to create applications that interact with the user through windows containing graphical objects. You will also learn how to program your applications to respond to the events that take place as the user interacts with them.

Checkpoint

1.27 What is a user interface?

1.28 How does a command line interface work?

1.29 When the user runs a program in a text-based environment, such as the command line, what determines the order in which things happen?

1.30 What is an event-driven program?

1.6 Objects

CONCEPT: **An object is a program component that contains data and performs operations. Programs use objects to perform specific tasks.**

Have you ever driven a car? If so, you know that a car is made of a lot of components. A car has a steering wheel, an accelerator pedal, a brake pedal, a gear shifter, a speedometer, and numerous other devices with which the driver interacts. There are also a lot of components under the hood, such as the engine, the battery, the radiator, and so forth. A car is not just one single object, but rather a collection of objects that work together.

This same notion also applies to computer programming. Most programming languages that are used today are **object oriented**. When you use an object-oriented language, you create programs by putting together a collection of objects. In programming, an object

is not a physical device, however, like a steering wheel or a brake pedal. Instead, it is a software component that exists in the computer's memory. In software, an object has two general capabilities:

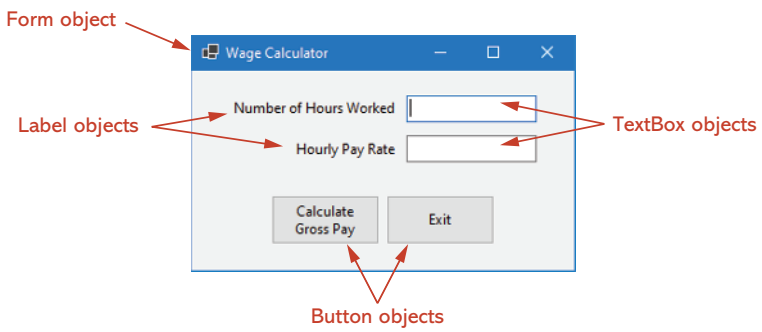
- An object can store data. The data stored in an object are commonly called **fields**, or **properties**.
- An object can perform operations. The operations that an object can perform are called **methods**.

When you write a program using an object-oriented language, you use objects to accomplish specific tasks. Some objects have a visual part that can be seen on the screen. For example, Figure 1-23 shows the wage-calculator program that we discussed in the previous section. The graphical user interface is made of the following objects:

Form object	A window that is displayed on the screen is called a **Form object**. Figure 1-23 shows a Form object that contains several other graphical objects.
Label objects	A **Label** object displays text on a form. The form shown in Figure 1-23 contains two Label objects. One of the Label objects displays the text *Number of Hours Worked* and the other Label object displays the text *Hourly Pay Rate*.
TextBox objects	A **TextBox** object appears as a rectangular region that can accept keyboard input from the user. The form shown in Figure 1-23 has two TextBox objects: one in which the user enters the number of hours worked and another in which the user enters the hourly pay rate.
Button objects	A **Button** object appears on a form as a button with a caption written across its face. When the user clicks a Button object with the mouse, an action takes place. The form in Figure 1-23 has two Button objects. One shows the caption *Calculate Gross Pay*. When the user clicks this button, the program calculates and displays the gross pay. The other button shows the caption *Exit*. When the user clicks this button, the program ends.

Forms, Labels, TextBoxes, and Buttons are just a few of the objects that you will learn to use in C#. As you study this book, you will create applications that incorporate many different types of objects.

Figure 1-23 Objects used in a GUI



Visible versus Invisible Objects

Objects that are visible in a program's graphical user interface are commonly referred to as **controls**. We could say that the form shown in Figure 1-23 contains two Label controls, two TextBox controls, and two Button controls. When an object is referred to as a control, it simply means that the object plays a role in a program's graphical user interface.

Not all objects can be seen on the screen, however. Some objects exist only in memory for the purpose of helping your program perform some task. For example, there are objects that read data from files, objects that generate random numbers, objects that store and sort large collections of data, and so forth. These types of objects help your program perform tasks, but they do not directly display anything on the screen. When you are writing a program, you will use objects that can help your program perform its tasks. Some of the objects that you use will be controls (visible in the program's GUI), and other objects will be invisible.

Classes: Where Objects Come From

Objects are very useful, but they don't just magically appear in your program. Before a specific type of object can be used, that object has to be created in memory. And, before an object can be created in memory, you must have a class for the object.

A **class** is code that describes a particular type of object. It specifies the data that an object can hold (the object's fields and properties), and the actions that an object can perform (the object's methods). You will learn much more about classes as you progress through this book, but for now, just think of a class as a code "blueprint" that can be used to create a particular type of object.

.NET

C# is a very popular programming language, but there are a lot of things it cannot do by itself. For example, you cannot use C# alone to create a graphical user interface, read data from files, work with databases, or many of the other things that programs commonly need to do. C# provides only the basic keywords and operators that you need to construct a program.

To perform operations beyond the most fundamental, C# relies on a software platform known as **.NET** (pronounced "dot net"). .NET provides an extensive collection of classes and tools that can be used in programs that are written in C#, as well as many other languages. The classes and tools provided by .NET allow you to write fully functional desktop applications and Web applications. As you work through this book you will not only learn C#, but you will also learn about many of the classes and other features provided by .NET.

Writing Your Own Classes

.NET provides many prewritten classes ready for use in your programs. There will be times, however, that you will wish you had an object to perform a specific task, and no such class will exist in .NET. This is not a problem because in C# you can write your own classes that have the specific fields, properties, and methods that you need for any situation. In Chapter 10, you will learn to create classes for the specific objects that you need in your programs.

Checkpoint

1.31 What is an object?

1.32 What type of language is used to create programs by putting together a collection of objects?

1.33 What two general capabilities does an object have?

1.34 What term is commonly used to refer to objects such as TextBoxes, Labels, and Buttons that are visible in a program's graphical user interface?

1.35 What is the purpose of an object that cannot be seen on the screen and exists only in memory?

1.36 What is a class?

1.37 What is .NET?

1.38 Why might you need to write your own classes?

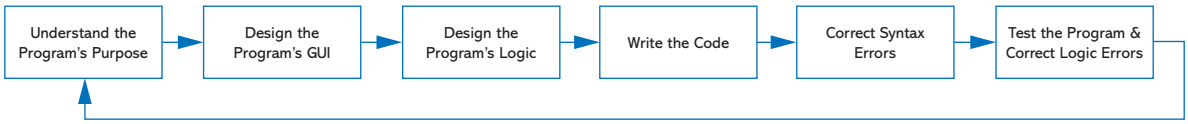
1.7 The Program Development Process

CONCEPT: **Creating a program requires several steps, which include designing the program's logic, creating the user interface, writing code, testing, and debugging.**

The Program Development Cycle

Previously in this chapter, you learned that programmers typically use high-level languages such as C# to create programs. There is much more to creating a program than writing code, however. The process of creating a program that works correctly typically requires the six phases shown in Figure 1-24. The entire process is known as the **program development cycle**.

Figure 1-24 The program development cycle



Let's take a closer look at each stage in the cycle.

1. **Understand the Program's Purpose**

   When beginning a new programming project, it is essential that you understand what the program is supposed to do. Most programs perform the following three-step process:

   Step 1. Input is received.
   Step 2. Some process is performed on the input.
   Step 3. Output is produced.

   Input is any data that the program receives while it is running. Once input is received, some process, such as a mathematical calculation, is usually performed on it. The results of the process are then sent out of the program as output. If you can identify these three elements of a program (input, process, and output), then you are on your way to understanding what the program is supposed to do.

   For example, suppose you have been asked to write a program to calculate and display the gross pay for an hourly paid employee. Here is a summary of the program's input, process, and output:

   Input:

   - Input the number of hours that the employee worked.
   - Input the employee's hourly pay rate.

Process:

- Multiply the number of hours worked by the hourly pay rate. The result is the employee's gross pay.

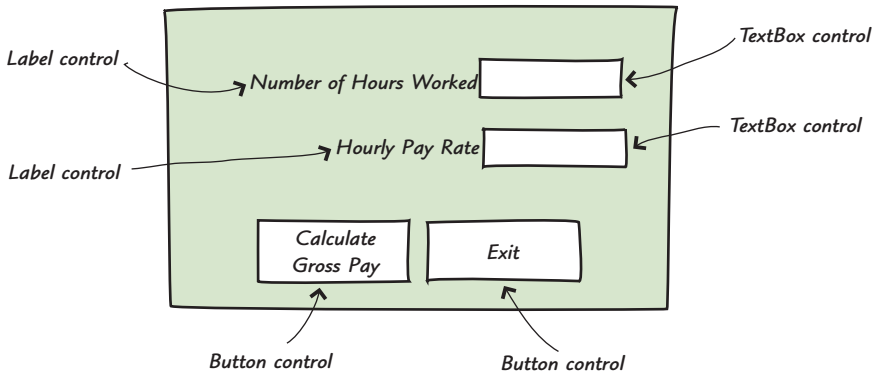
Output:

- Display the employee's gross pay on the screen.

2. **Design the Graphical User Interface**

   Once you clearly understand what the program is supposed to do, you can begin designing its graphical user interface. Often, you will find it helpful to draw a sketch of each form that the program displays. For example, if you are designing a program that calculates gross pay, Figure 1-25 shows how you might sketch the program's form.

Figure 1-25 Form sketch



Notice that the sketch identifies each type of control (GUI object) that will appear on the form. The TextBox controls will allow the user to enter input. The user will type the number of hours worked into one of the TextBoxes and the employee's hourly pay rate into the other TextBox. Notice that Label controls are placed on the form to tell the user what data to enter. When the user clicks the Button control that reads *Calculate Gross Pay*, the program will display the employee's gross pay on the screen in a pop-up window. When the user clicks the Button control that reads *Exit*, the program will end.

Once you are satisfied with the sketches that you have created for the program's forms, you can begin creating the actual forms on the computer. As a Visual C# programmer, you have a powerful environment known as Visual Studio at your disposal.

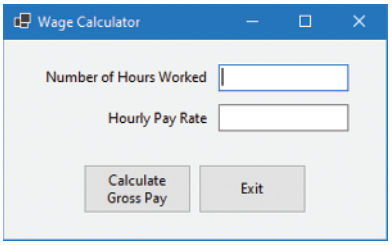
Visual Studio is a professional **integrated development environment** (**IDE**), which means that it provides all the necessary tools for creating, testing, and debugging software. It can be used to create applications not only with Visual C#, but also with other languages such as Visual Basic and Visual C++.

Visual Studio gives you a "what you see is what you get" editor that allows you to visually design a program's forms. You can use Visual Studio to create the program's forms, place all the necessary controls on the forms, and set each control's properties so it has the desired appearance. For example, Figure 1-26 shows the actual form that you might create for the wage-calculator program, which calculates gross pay.

3. **Design the Program's Logic**

   In this phase you break down each task that the program must perform into a series of logical steps. For example, if you look back at Figure 1-26, notice that the pay-calculating program's form has a Button control that reads *Calculate Gross Pay*.

Figure 1-26 Form for the wage-calculator program



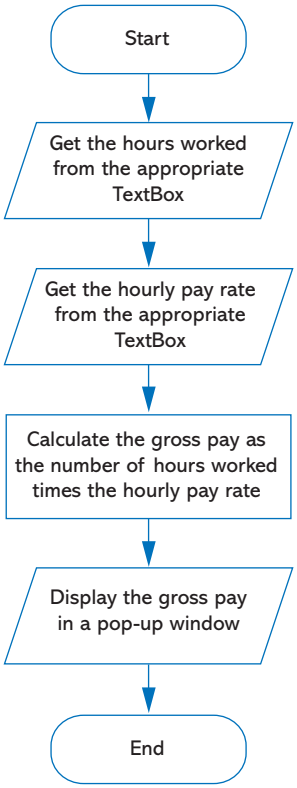
When the user clicks this button, you want the program to display the employee's gross pay. Here are the steps that the program should take to perform that task:

Step 1. Get the number of hours worked from the appropriate TextBox.
Step 2. Get the hourly pay rate from the appropriate TextBox.
Step 3. Calculate the gross pay as the number of hours worked times the hourly pay rate.
Step 4. Display the gross pay in a pop-up window.

This is an example of an **algorithm**, which is a set of well-defined, logical steps that must be taken to perform a task. An algorithm that is written out in this manner, in plain English statements, is called **pseudocode**. (The word *pseudo* means fake, so pseudocode is fake code.) The process of informally writing out the steps of an algorithm in pseudocode before attempting to write any actual code is very helpful when you are designing a program. Because you do not have to worry about breaking any syntax rules, you can focus on the logical steps that the program must perform.

Flowcharting is another tool that programmers use to design programs. A **flowchart** is a diagram that graphically depicts the steps of an algorithm. Figure 1-27 shows how you might create a flowchart for the wage-calculator algorithm. Notice that

Figure 1-27 Flowchart for the wage-calculator program



there are three types of symbols in the flowchart: ovals, parallelograms, and a rectangle. Each of these symbols represents a step in the algorithm, as described here:

- The ovals, which appear at the top and bottom of the flowchart, are called **terminal symbols**. The **Start terminal** symbol marks the program's starting point and the **End terminal** symbol marks the program's ending point.
- Parallelograms are used as **input symbols** and **output symbols**. They represent steps in which the program reads input or displays output.
- Rectangles are used as **processing symbols**. They represent steps in which the program performs some process on data, such as a mathematical calculation.

The symbols are connected by arrows that represent the "flow" of the program. To step through the symbols in the proper order, you begin at the *Start* terminal and follow the arrows until you reach the *End* terminal.

4. **Write the Code**

   Once you have created a program's GUI and designed algorithms for the program's tasks, you are ready to start writing code. During this process, you will refer to the pseudocode or flowcharts that you created in Step 3 and use Visual Studio to write C# code.

5. **Correct Syntax Errors**

   You previously learned in this chapter that a programming language such as C# has rules, known as syntax, that must be followed when writing a program. A language's syntax rules dictate things such as how keywords, operators, and punctuation characters can be used. A syntax error occurs if the programmer violates any of these rules. If the program contains a syntax error or even a simple mistake such as a misspelled keyword, the program cannot be compiled or executed.

   Virtually all code contains syntax errors when it is first written, so the programmer will typically spend some time correcting these. Once all the syntax errors and simple typing mistakes have been corrected, the program can be compiled and translated into an executable program.

6. **Test the Program and Correct Logic Errors**

   Once the code is in an executable form, you must then test it to determine whether any logic errors exist. A **logic error** is a mistake that does not prevent the program from running but causes it to produce incorrect results. (Mathematical mistakes are common causes of logic errors.) If the program produces incorrect results, the programmer must **debug** the code. This means that the programmer finds and corrects logic errors in the program. Sometimes, during this process, the programmer discovers that the program's original design must be changed. In this event, the program development cycle starts over and continues until no errors can be found.

Checkpoint

1.39 List the six steps in the program development cycle.

1.40 What is an algorithm?

1.41 What is pseudocode?

1.42 What is a flowchart?

1.43 What do each of the following symbols mean in a flowchart?
- Oval
- Parallelogram
- Rectangle

1.8 Getting Started with Visual Studio

CONCEPT: **Visual Studio provides a collection of tools that you use to build Visual C# applications. The first step in using Visual C# is learning about these tools.**

Installing Visual Studio

To complete the tutorials and programming problems in this book, you need to install Visual Studio 2022, or a later version, on your computer. You can download Visual Studio 2022 Community Edition from the following website and install it on your system:

```
www.visualstudio.com
```

Visual Studio 2022 Community Edition is a free, full-featured development environment, and is a perfect companion for this textbook.

NOTE: If you are working in your school's computer lab, there is a good chance that Microsoft Visual Studio has already been installed. If this is the case, your instructor will show you how to start Visual Studio.

Installing the Student Sample Program Files

The Student Sample Program files that accompany this book are available for download from the book's companion Web site at

```
www.pearson.com/cs-resources
```

These files are required for many of the book's tutorials. Simply download the Student Sample Program files to a location on your hard drive where you can easily access them.

Creating a New Project

Once you have installed Visual Studio on your system, you can use it to create Visual C# applications. Each Visual C# application that you create is called a **project**. When you are ready to create a new application, you start a new project. Tutorial 1-1 leads you through the steps of starting Visual Studio and creating a new Visual C# project.

Tutorial 1-1:
Starting Visual Studio and Creating a Visual C# Project

VideoNote

Tutorial 1-1: Starting Visual Studio and Creating a Visual C# Project

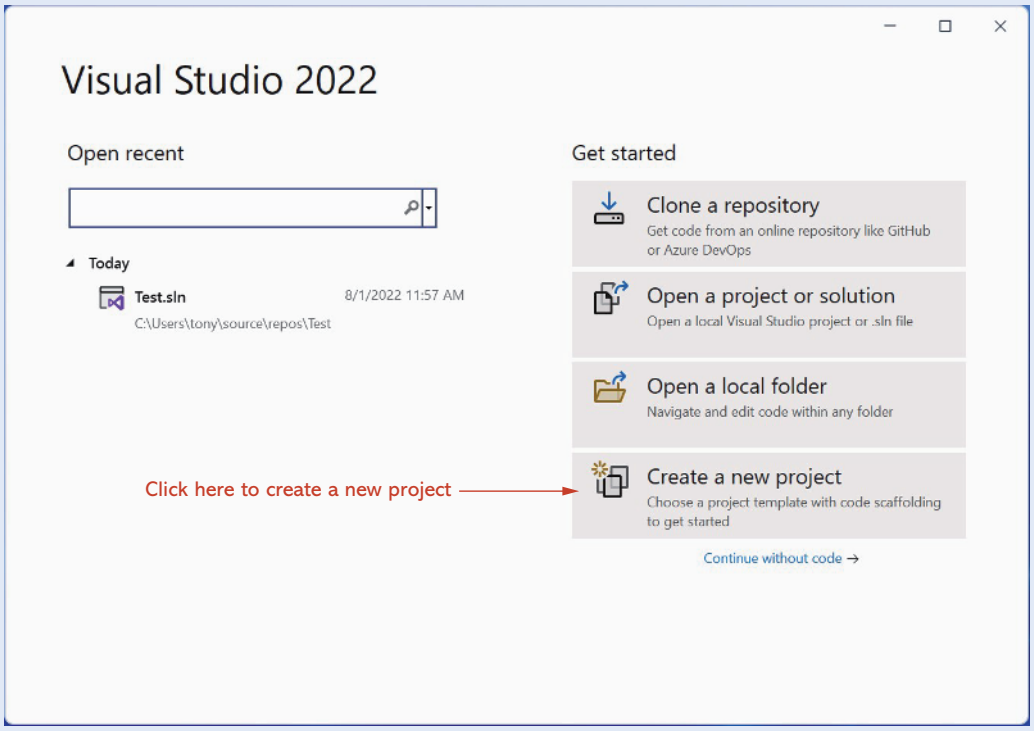
Step 1: Depending on the version of Windows you are running, use one of the following procedures to start Visual Studio:

Windows 11: Click the search icon on the Taskbar. This will take you to the Search screen. In the Search box at the top of the screen, start typing *Visual Studio*. When you see *Visual Studio 2022 App* (or a more recent version) appear in the search results, click it.

Windows 10: In the Windows search bar, start typing *Visual Studio*. When you see *Visual Studio 2022 App* (or a more recent version) appear in the search results, click it.

Step 2: The screen shown in Figure 1-28 will appear. This is known as the *Start* window. Click *Create a new project* in the lower-right area of the window.

Figure 1-28 The Visual Studio *Start* Window



Step 3: The *Create a new project* window, shown in Figure 1-29, will appear next. In this window, you will specify the type of project that you want to create. Notice in the upper-right half of the screen, the three drop-down lists that read *All languages*, *All platforms*, and *All project types*. As shown in Figure 1-30, use these drop-down lists to select *C#, Windows*, and *Desktop*.

Step 4: As shown in Figure 1-31, select the project template that is titled *Windows Forms App*, and then click *Next*.

NOTE: Notice in Figure 1-31, there are two project templates for Windows Forms Apps. The first one is titled *Windows Forms App*, and the second one is titled *Windows Forms App (.NET Framework)*. In this book, we will primarily use the one that is titled *Windows Forms App*. Except in Chapters 13 and 15, we will not use the template that is titled *Windows Forms App (.NET Framework)* because it uses an older version of .NET.

Figure 1-29 The *Create a New Project* window

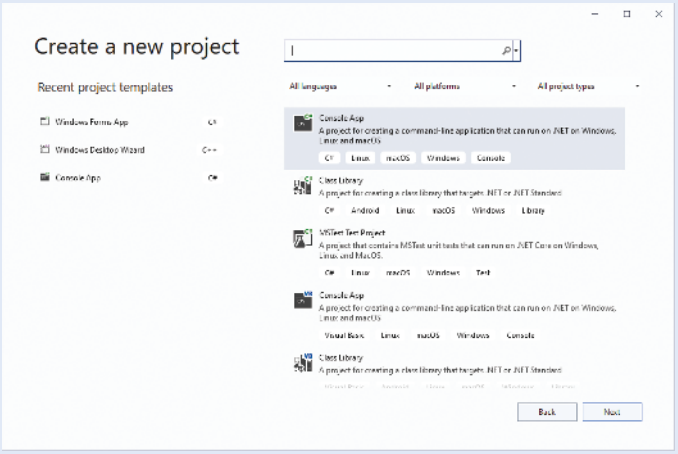


Figure 1-30 The *Create a New Project* window

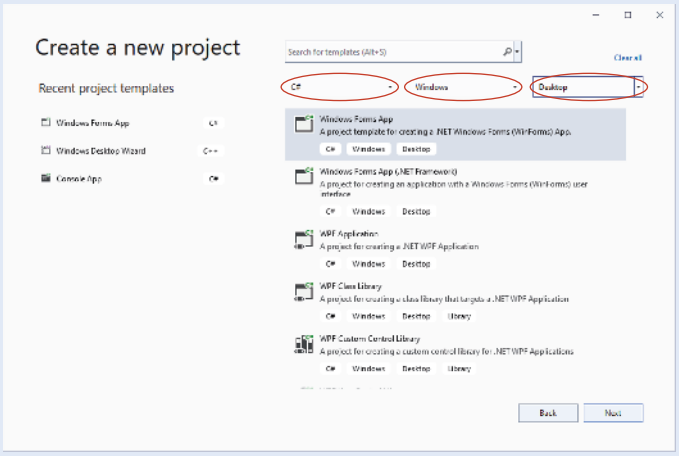
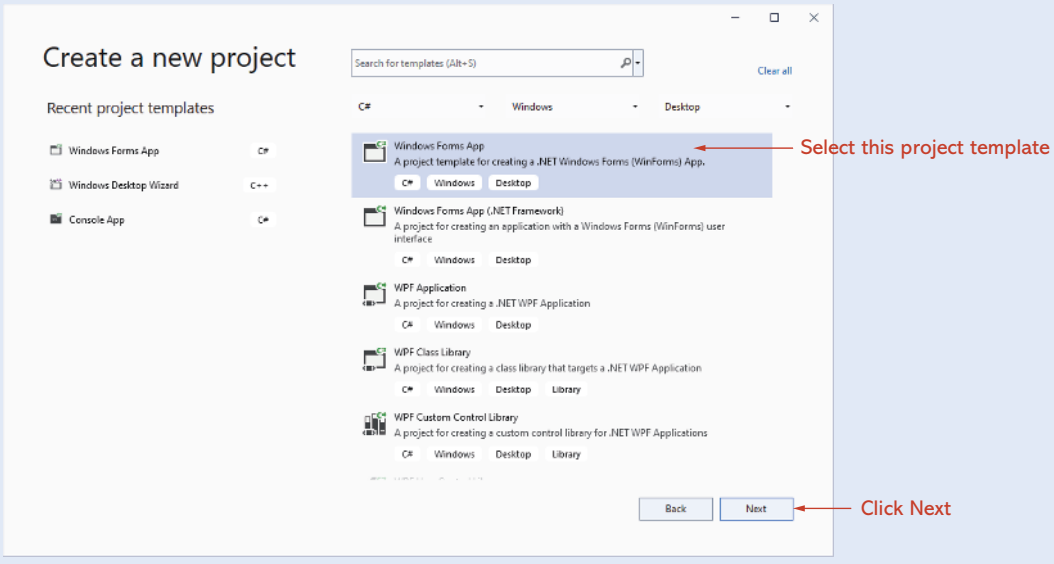


Figure 1-31 Select the type of project you want to create



Step 5: The *Configure your project* window appears next, as shown in Figure 1-32.

At the top of the window, you see a *Project Name* text box. This is where you enter the name of your project. The *Project Name* text box will be automatically filled in with a default name. In Figure 1-32, the default name is *WinFormsApp1*. Change the project name to *MyFirstProject*, as shown in Figure 1-33.

Just below the *Name* text box, you will see a *Location* text box and a Solution name text box.

Figure 1-32 The *Configure Your Project* window

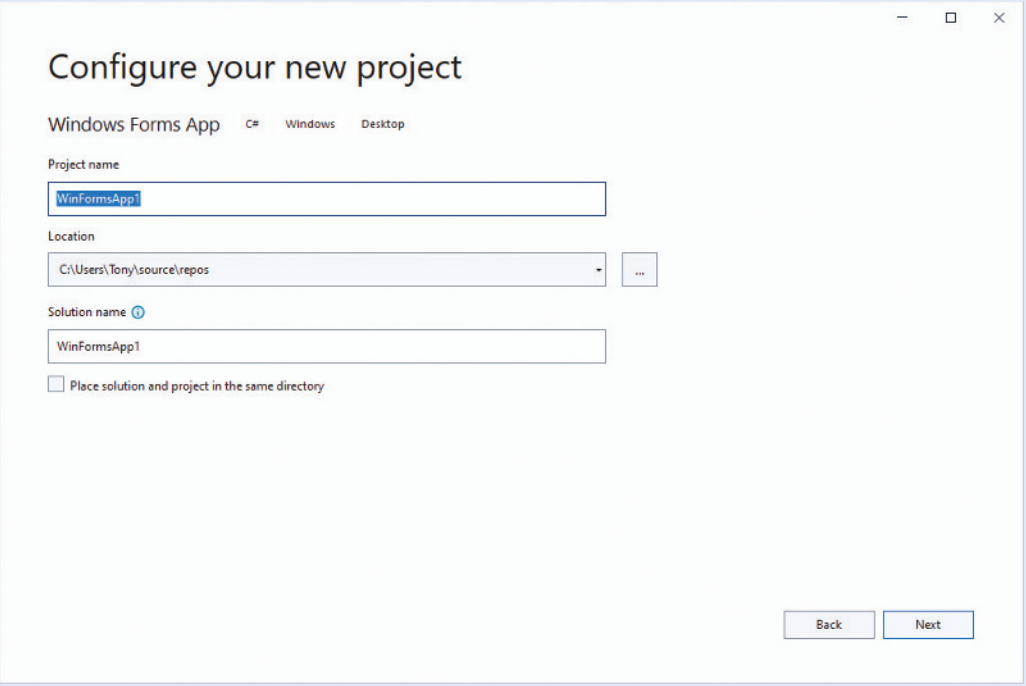
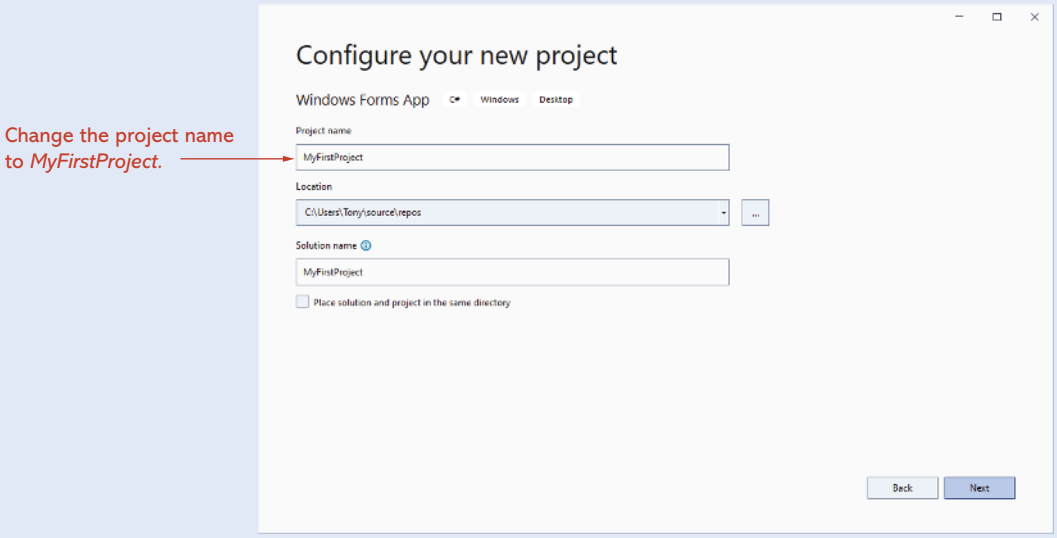


Figure 1-33 Change the project name



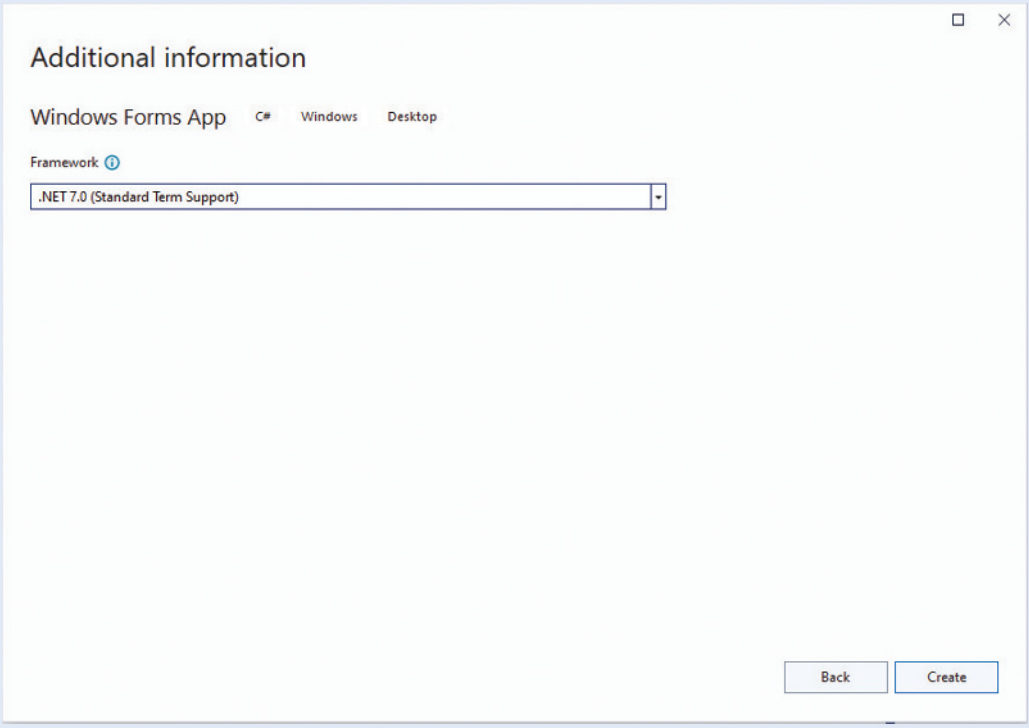
- The *Location* text box shows where a folder will be created to hold the project. If you wish to change the location, click the Browse button and select the desired location.
- A solution is a container that holds one or more projects, and the *Solution name* text box shows the name of the solution that will hold this project. By default, the solution name is the same as the project name. For all the projects that you create in this book, you should keep the solution name the same as the project name.

Step 6: The *Additional information* window appears next, as shown in Figure 1-34. This window allows you to select the .NET version you wish to use with the project. You can click the drop-down list to see the versions that are installed on your computer. Be sure to select the highest version number that you see in the list. Click the *Create* button to create the project.

It might take a moment for the project to be created. Once it is, the Visual Studio environment should appear, similar to Figure 1-35. Notice that the name of the project, *MyFirstProject*, is displayed in the title bar at the top of the Visual Studio window.

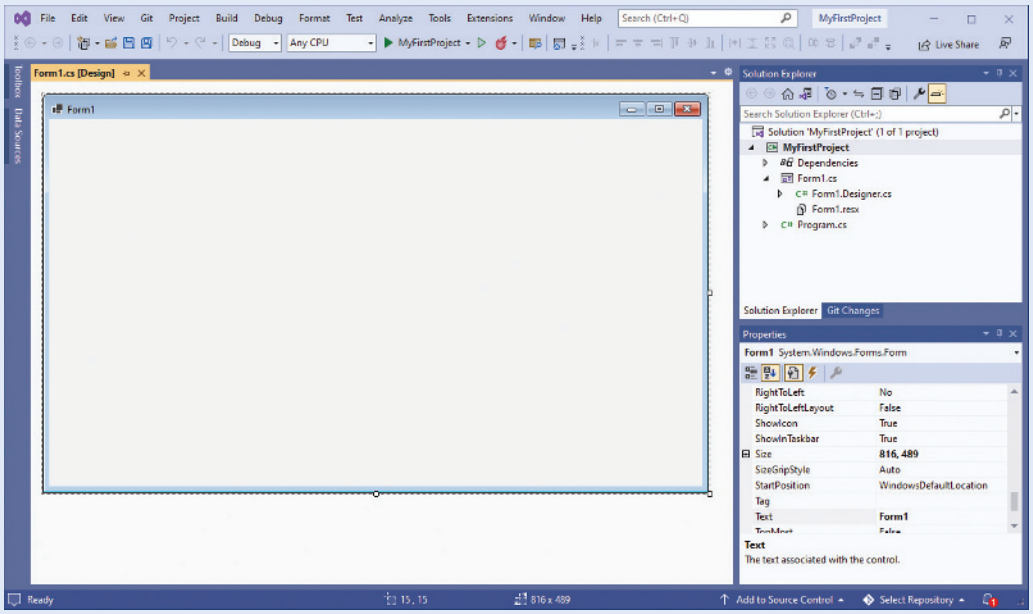
Leave Visual Studio running and complete the next tutorial.

Figure 1-34 The *Additional Information* window



NOTE: As you work through this book, you will create a lot of Visual C# projects. As you do, you will find that default names such as WinFormsApp1 do not help you remember what each project does; therefore, you should always change the name of a new project to something that describes the project's purpose.

Figure 1-35 The Visual Studio environment with a new project open



In Tutorial 1-2 you will practice saving and closing a project.

Tutorial 1-2:
Saving and Closing a Project

VideoNote
Tutorial 1-2: Saving and Closing a Project

As you work on a project, you should get into the habit of saving it often. In this tutorial you will save the *MyFirstProject* application and then close it.

Step 1: Visual Studio should still be running from the previous tutorial. To save the project that is currently open, click *File* on the menu bar and then select *Save All*.

Step 2: To close the project, click *File* on the menu bar and then click *Close Solution*. The solution will be closed and the Start window will be displayed.

The Visual Studio Environment

The Visual Studio environment consists of a number of windows that you will use on a regular basis. Figure 1-36 shows the locations of the following windows that appear within the Visual Studio environment: the ***Designer* window**, the ***Solution Explorer* window**, and the ***Properties* window**. Here is a brief summary of each window's purpose:

- **The *Designer* Window**

  You use the *Designer* window to create an application's graphical user interface. The *Designer* window shows the application's form and allows you to visually design its appearance by placing the desired controls that will appear on the form when the application executes.

- **The *Solution Explorer* Window**

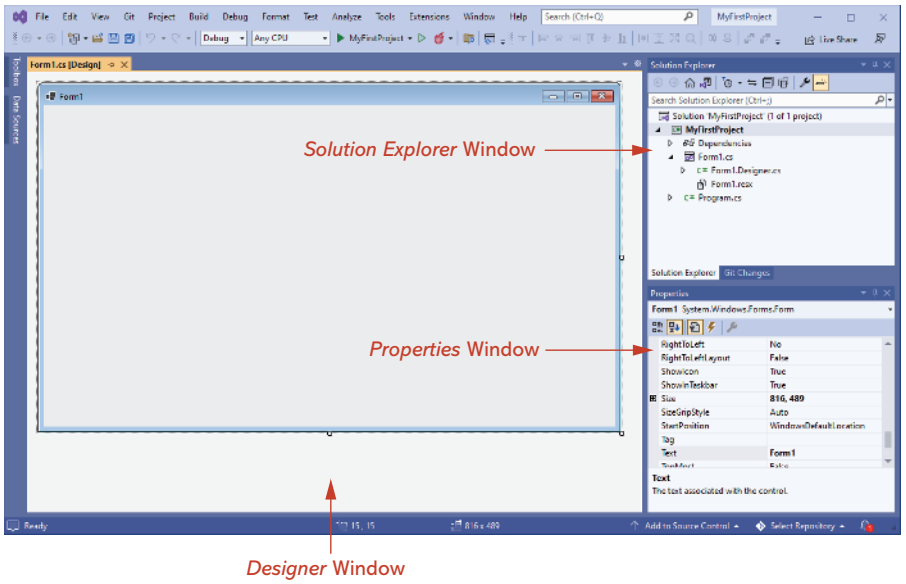
  A solution is a container for holding Visual C# projects. (We discuss solutions in greater detail in a moment.) When you create a new C# project, a new solution is automatically created to contain it. The *Solution Explorer* window allows you to navigate among the files in a Visual C# project.

- **The *Properties* Window**

  A control's appearance and other characteristics are determined by the control's properties. When you are creating a Visual C# application, you use the *Properties* window to examine and change a control's properties.

  Remember that Visual Studio is a customizable environment. You can move the various windows around, so they may not appear in the exact locations shown in Figure 1-36 on your system.

Figure 1-36 The *Designer* window, *Solution Explorer* window, and *Properties* window



Displaying the *Solution Explorer* and *Properties* Windows

If you do not see the *Solution Explorer* or the *Properties* window, you can follow these steps to make them visible:

- If you do not see the *Solution Explorer* window, click *View* on the menu bar. On the *View* menu, click *Solution Explorer*.
- If you do not see the *Properties* window, click *View* on the menu bar. On the *View* menu, click *Properties* window.

Using Auto Hide

Many windows in Visual Studio have a feature known as **Auto Hide**. When you see the push-pin icon in a window's title bar, as shown in Figure 1-37, you know that the window has Auto Hide capability. You click the pushpin icon to turn Auto Hide on or off for a window.

When Auto Hide is turned on, the window is displayed only as a tab along one of the edges of the Visual Studio environment. This feature gives you more room to view your application's forms and code. Figure 1-38 shows how the *Solution Explorer* and *Properties* windows appear when their Auto Hide feature is turned on. Notice the tabs that read

Figure 1-37 *Auto Hide* pushpin icon

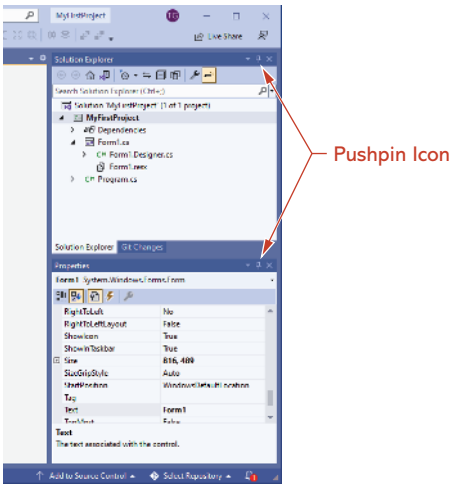
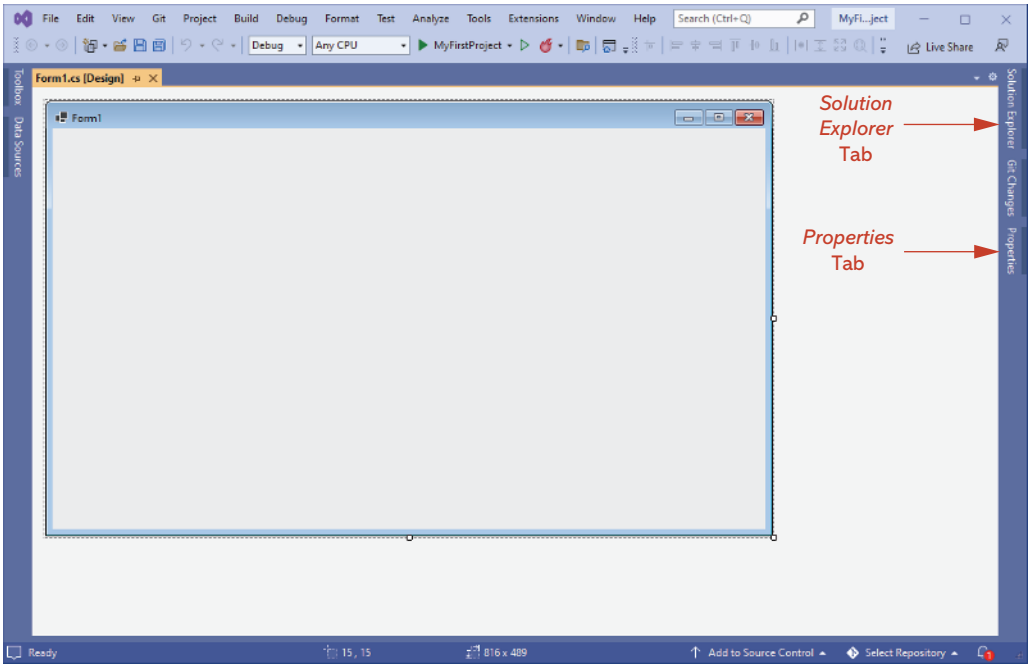


Figure 1-38 The *Solution Explorer* and *Properties* windows hidden



Solution Explorer and *Properties* along the right edge of the screen. (Figure 1-38 also shows a *Team Explorer* tab. We do not discuss the *Team Explorer* in this book.)

The Menu Bar and the Standard Toolbar

You've already used the Visual Studio menu bar several times. This is the bar at the top of the Visual Studio window that provides menus such as *File*, *Edit*, *View*, *Project*, and so forth. As you progress through this book, you will become familiar with many of the menus.

Below the menu bar is the standard toolbar. The **standard toolbar** contains buttons that execute frequently used commands. All commands that are displayed on the toolbar may also be executed from a menu, but the standard toolbar gives you quicker access to them. Figure 1-39 identifies the standard toolbar buttons that you will use most often, and Table 1-3 gives a brief description of each.

Figure 1-39 Visual Studio toolbar buttons

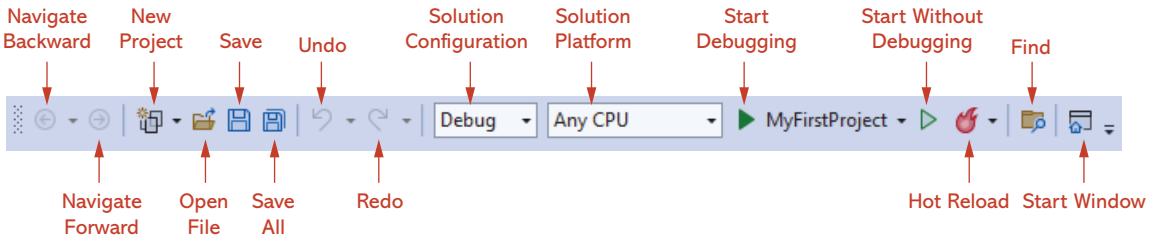


Table 1-3 Visual Studio toolbar buttons

Toolbar Button	Description
Navigate Backward	Moves to the previously active tab in the *Designer* window
Navigate Forward	Moves to the next active tab in the *Designer* window
New Project	Starts a new project
Open File	Opens an existing file
Save	Saves the file named by *filename*
Save All	Saves all the files in the current project
Undo	Undoes the most recent operation
Redo	Redoes the most recently undone operation
Solution Configurations	Configures your project's executable code
Solution Platform	Lets you select the platform on which the application will run
Start Debugging	Executes your program with debugging capability
Start Without Debugging	Executes your program without the ability to debug
Hot Reload	Gives you the ability to edit a running program and reload it
Find	Searches for text in your application code
Start Window	Displays the Start Window

The *Toolbox*

The ***Toolbox*** is a window that allows you to select the controls that you want to use in an application's user interface. When you want to place a Button, Label, TextBox, or other control on an application's form, you select it in the *Toolbox*. You will use the *Toolbox* extensively as you develop Visual C# applications.

The *Toolbox* typically appears on the left side of the Visual Studio environment. If the *Toolbox* is in Auto Hide mode, its tab will appear as shown in Figure 1-40. Figure 1-41 shows the *Toolbox* opened, with Auto Hide turned off.

NOTE: If you do not see the *Toolbox* or its tab along the side of the Visual Studio environment, click *View* on the menu bar and then click *Toolbox*.

Figure 1-40 The *Toolbox* tab (Auto Hide turned on)

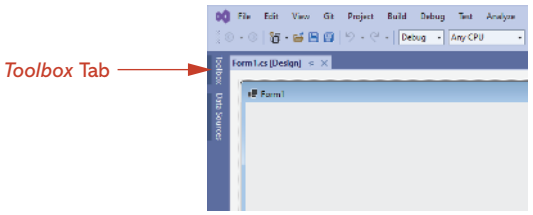
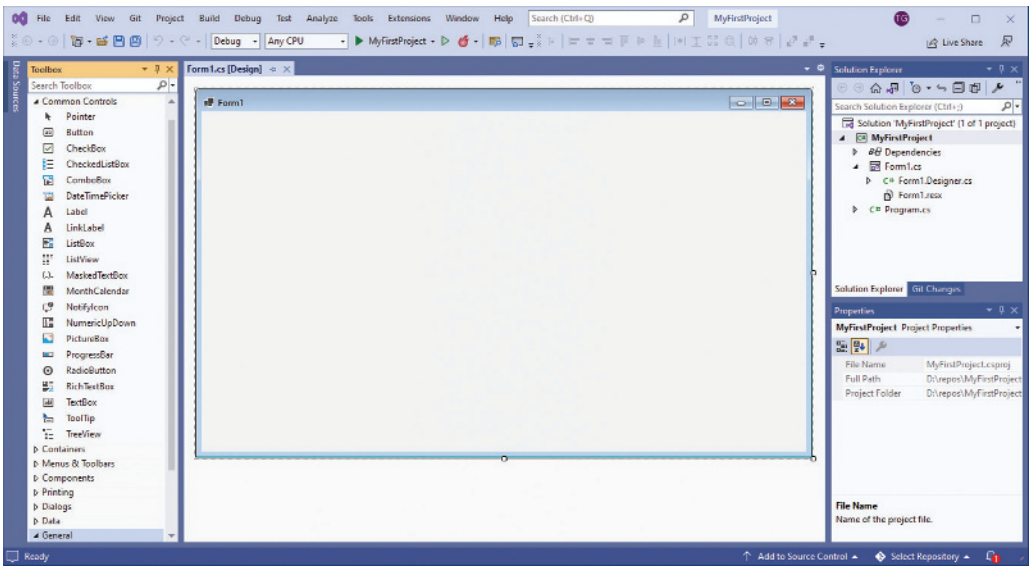


Figure 1-41 The *Toolbox* opened (Auto Hide turned off)



The *Toolbox* is divided into sections, and each section has a name. In Figure 1-41 you can see the *All Windows Forms* and *Common Controls* sections. If you scroll the *Toolbox*, you will see many other sections. Each section can be opened or closed.

If you want to open a section of the *Toolbox*, you simply click on its name tab. To close the section, click on its name tab again. In Figure 1-41, the *Common Controls* section is open. You use the *Common Controls* section to access controls that you frequently need, such as Buttons, Labels, and TextBoxes. You can move any section to the top of the list by dragging its name with the mouse.

Using ToolTips

A **ToolTip** is a small rectangular box that pops up when you hover the mouse pointer over a button on the toolbar or in the *Toolbox* for a few seconds. The ToolTip box contains a short description of the button's purpose. Figure 1-42 shows the ToolTip that appears when the cursor is left sitting on the *Save All* button. Use a ToolTip whenever you cannot remember a particular button's function.

Docked and Floating Windows

Figure 1-41 shows the *Toolbox*, *Solution Explorer,* and *Properties* windows when they are **docked**, which means they are attached to one of the edges of the Visual Studio window. Alternatively, the windows can be **floating**. You can control whether a window is docked or floating as follows:

- To change a window from docked to floating, right-click its title bar and select *Float*.
- To change a window from floating to docked, right-click its title bar and select *Dock*.

Figure 1-42 *Save All* ToolTip

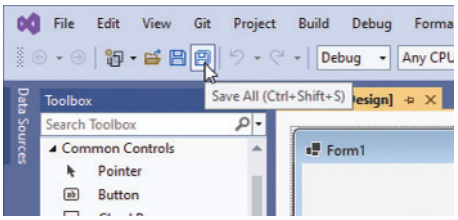


Figure 1-43 *Toolbox*, *Solution Explorer,* and *Properties* windows floating

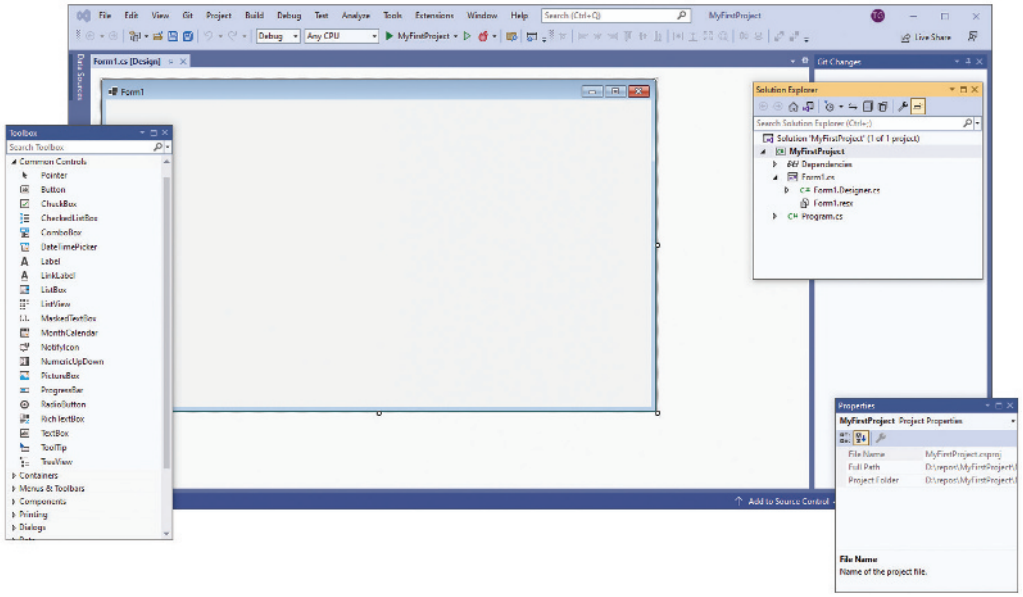


Figure 1-43 shows Visual Studio with the *Toolbox*, *Solution Explorer,* and *Properties* windows floating. When a window is floating, you can click and drag it by its title bar around the screen. You may use whichever style you prefer—docked or floating. When windows are floating, they behave as normal windows. You may move or resize them to suit your preference.

NOTE: A window cannot float if its Auto Hide feature is turned on.

TIP: You can always reset the default window layout by clicking *Window* on the menu bar and then selecting *Reset Window Layout*. If you accidentally close the *Designer* window, the *Solution Explorer* window, or the *Properties* window, you can use the *View* menu to redisplay them.

Projects and Solutions

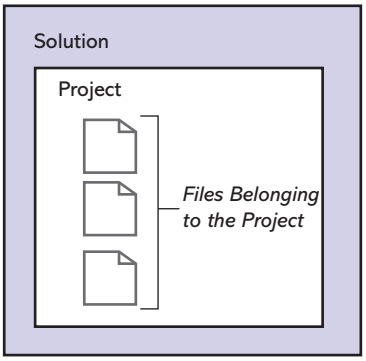
As you learn to program in Visual C#, you will see the terms *project* and *solution* used often. These terms do not mean the same thing, but they are sometimes used interchangeably. Let's briefly discuss the difference between a project and a solution.

Each Visual C# application that you create is called a project. A Visual C# project consists of several files. You can think of a project as a collection of files that belong to a single application.

A **solution** is a container that holds one or more Visual C# projects. If you are developing applications for a large organization, you might find it convenient to store several related projects together in the same solution.

Although it is possible for a solution to hold more than one project, each project that you will create in this book will be saved in its own solution. Each time you create a new project, you will also create a new solution to hold it. Figure 1-44 illustrates this concept. Typically, the solution will be given the same name as the project.

Figure 1-44 Solution and project organization



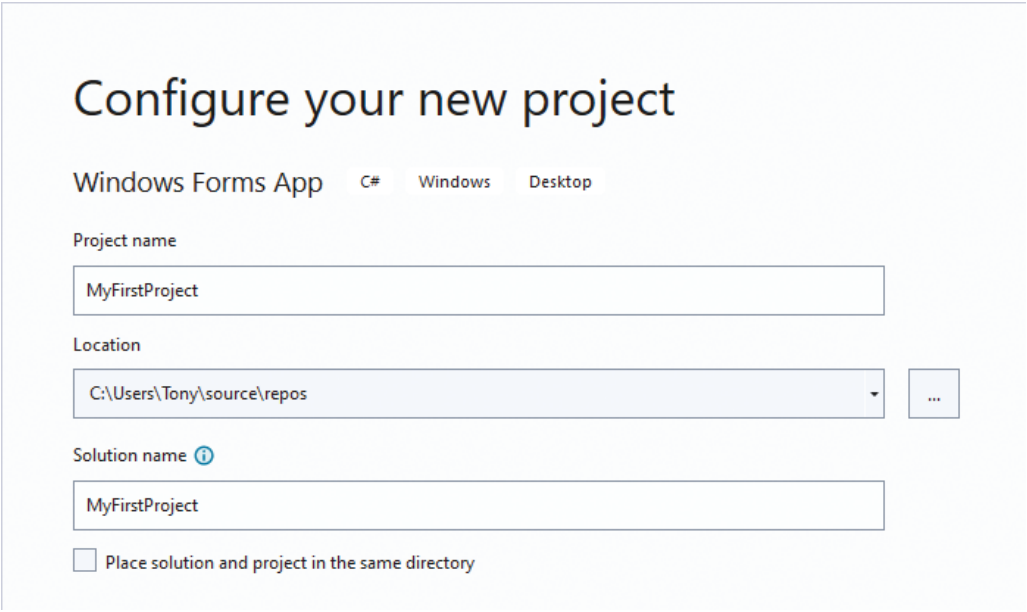
Typical Organization of Solutions and Projects on the Disk

When you create a new project, you specify the project's name, the solution's name, and a location on the disk where the solution should be stored. You specify this information on the *Configure your new project* window, as shown in Figure 1-45.

Let's use Figure 1-45 to see an example of how the files for the *My First Project* solution and project will be organized on the disk. Notice that in Figure 1-45, the following location is shown for the solution:

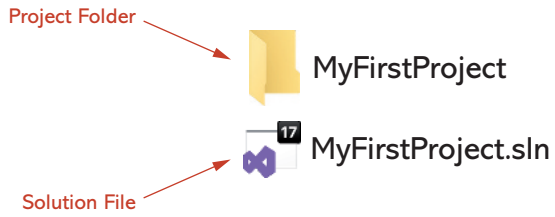
C:\Users\Tony\source\repos

Figure 1-45 Specifying the project name, solution name, and location



On your system, the location will not be exactly the same as this, but it will be something similar. At this location, a **solution folder** named *MyFirstProject* will be created. If we use Windows to look inside that folder, we will see the two items shown in Figure 1-46. Notice that one of the items is another folder named *MyFirstProject*. That is the **project folder**, which contains various files related to the project. The other item is the **solution file**. In Windows, you can double-click the solution file to open the project in Visual Studio.

Figure 1-46 Contents of the *MyFirstProject* solution folder



Opening an Existing Project

If Visual Studio is already running, you can perform the following steps to open an existing project:

- Click *File* on the Visual Studio menu bar, then select *Open*, and then select *Project/Solution*....
- The *Open Project/Solution* window will appear. Navigate to the desired solution folder, select the solution file, and click *Open*.

Tutorial 1-3: Opening an Existing Project

VideoNote
Tutorial 1-3: Opening an Existing Project

In this tutorial you will reopen the *MyFirstProject* application that you created in Tutorial 1-1.

Step 1: Visual Studio should still be running from the previous tutorial.

Click *File* on the menu bar; select *Open* and then select *Project/Solution*.... The *Open Project* window will appear. Navigate to the *MyFirstProject* solution, select the solution file, and click the *Open* button.

MyFirstProject should be opened. If you plan to complete the next tutorial, leave Visual Studio running with *MyFirstProject* opened.

Displaying the *Designer* (When It Does Not Automatically Appear)

Sometimes when you open an existing project, the project's form will not be automatically displayed in the *Designer*. Figure 1-47 shows an example of the Visual Studio environment with an opened project but no form displayed in the *Designer*. When this happens, perform the following steps to display the project's form in the *Designer*:

- As shown in Figure 1-48, right-click Form1.cs in the *Solution Explorer*.
- Click *View Designer* in the pop-up menu.

Figure 1-47 A project opened with no form displayed in the *Designer*

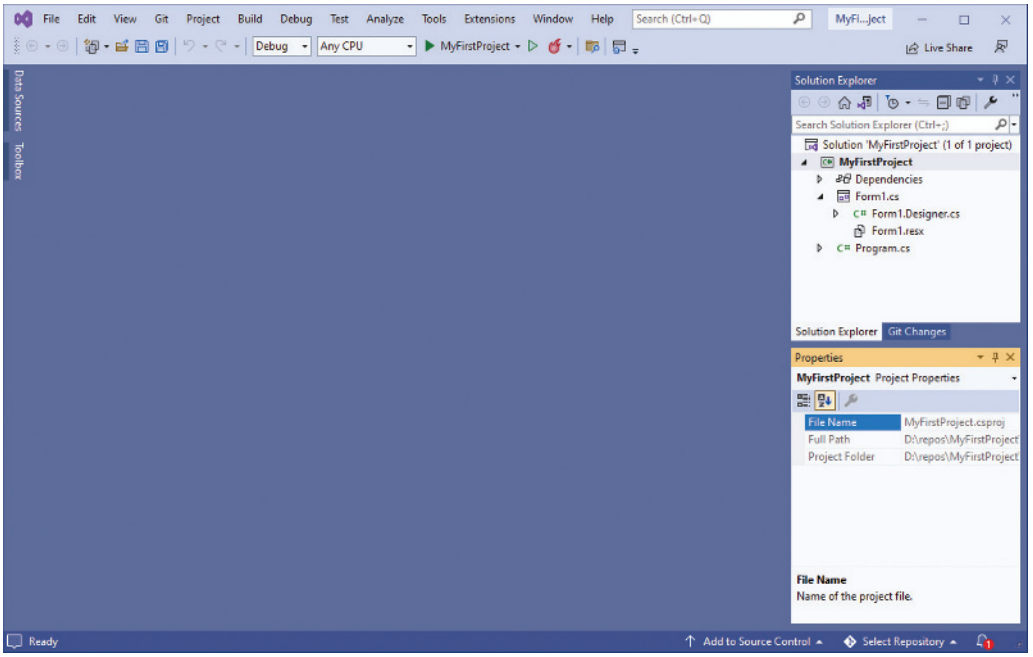
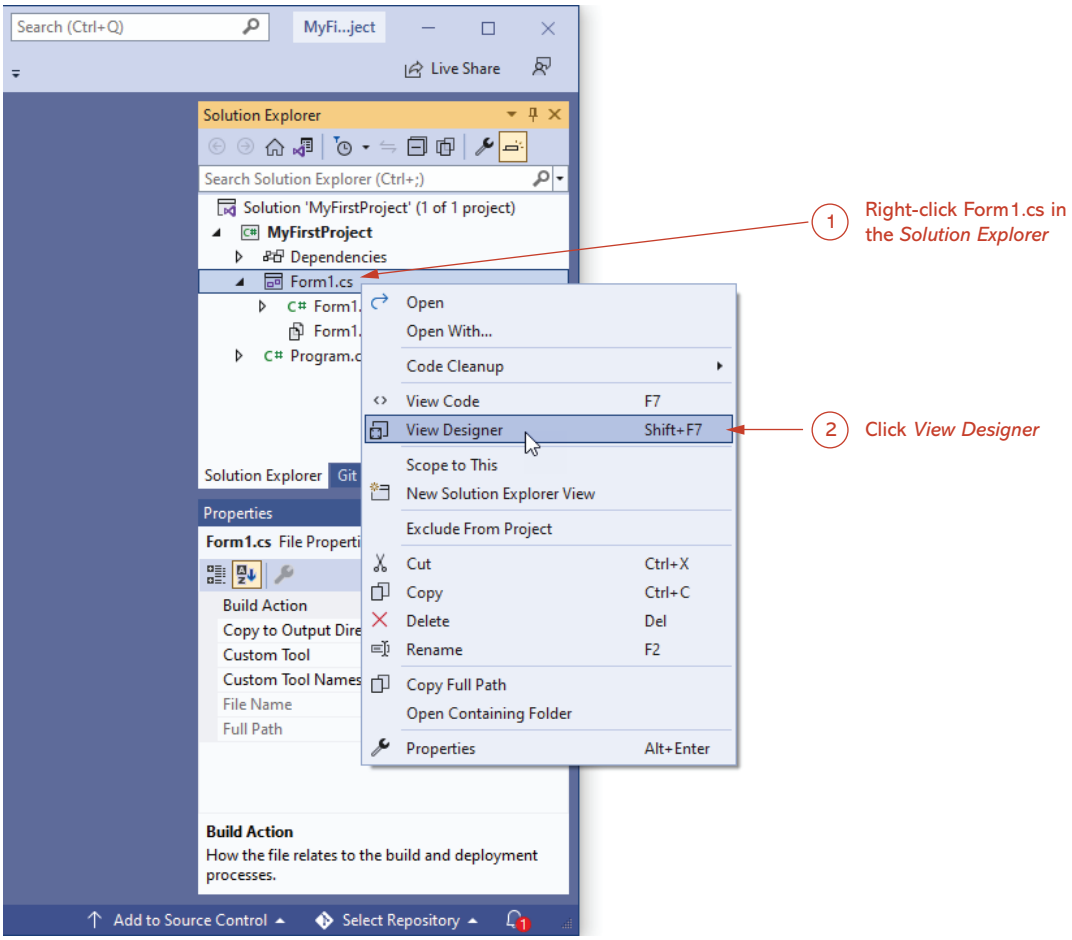


Figure 1-48 Using the *Solution Explorer* to open a form in the *Designer*



Accessing the Visual Studio Documentation

You can access the documentation for Visual Studio by Clicking *Help* on the menu bar, and then selecting *View Help*. (Or, you can press Control+F1, and then press V on the keyboard.) This launches your Web browser and opens the Visual Studio Documentation website. This website provides complete documentation for Visual C# as well as the other programming languages included in Visual Studio. You will also find code samples, tutorials, articles, and access to tutorial videos.

In Tutorial 1-4 you will gain additional practice using the Solution Explorer, Properties window and the Toolbox.

Tutorial 1-4:
Getting Familiar with the Visual Studio Environment

VideoNote

Tutorial 1-4: Getting Familiar with the Visual Studio Environment

This exercise will give you practice interacting with the *Solution Explorer* window, the *Properties* window, and the *Toolbox*.

Step 1: If Visual Studio is still running on your computer from the previous tutorial, continue to Step 2. If Visual Studio is not running on your computer, repeat the steps in Tutorial 1-3 to open *MyFirstProject*.

Step 2: Practice turning the Auto Hide feature on and off for the *Solution Explorer* window, the *Properties* window, and the *Toolbox*. Recall from our previous discussion that clicking the pushpin icon in each window's title bar turns Auto Hide on and off. When you are finished practicing, make sure Auto Hide is turned off for each of these windows.

Step 3: Practice floating and docking the *Solution Explorer* window, the *Properties* window, and the *Toolbox*. Recall from our previous discussion that you can make any of these windows float by right-clicking its title bar and selecting *Float*. You dock a floating window by right-clicking its title bar and selecting *Dock*.

Step 4: The *Toolbox*, *Solution Explorer*, and *Properties* windows each have a *Close* button ( ) in their upper-right corner. Close each of these windows by clicking its *Close* button.

Step 5: Do you remember which buttons on the toolbar restore the *Solution Explorer*, *Properties* window, and *Toolbox*? If not, move your mouse cursor over any button on the toolbar and leave it there until the ToolTip appears. Repeat this procedure on different buttons until you find the ones whose ToolTips read *Solution Explorer*, *Properties Window*, and *Toolbox*. (Refer to Figure 1-39 and Table 1-3 for further assistance.)

Step 6: Click the appropriate buttons on the toolbar to restore the *Solution Explorer*, the *Properties* window, and the *Toolbox*.

Step 7: Exit Visual Studio by clicking *File* on the menu bar and then clicking *Exit*. You may see a dialog box asking whether you wish to save changes to a number of items. Click *Yes*.

Checkpoint

1.44 Briefly describe the purpose of the *Solution Explorer* window.

1.45 Briefly describe the purpose of the *Properties* window.

1.46 Briefly describe the purpose of the standard toolbar.

1.47 What is the difference between the toolbar and the *Toolbox*?

1.48 What is a ToolTip?

1.49 What is a project?

1.50 What is a solution?

Key Terms

algorithm
American Standard Code for Information Interchange (ASCII)
application software
assembler
assembly language
Auto Hide
binary
binary digit
binary numbering system
bit
Button
byte
central processing unit (CPU)
class
code
command line interface
compiler
console interface
controls
CPU
debug
Designer window
digital
digital data
digital device
disk drive
docked (window)
End terminal
ENIAC
event driven
executing
fetch-decode-execute cycle
fields
floating (window)
floating-point notation
flowchart
Form
graphical user interface (GUI)
hardware
high-level languages
input
input device
input symbols
instruction set
integrated development environment (IDE)
interpreter
keywords
Label
logic error
low-level language
machine language
main memory
methods
microprocessors
mnemonics
.NET
object oriented
operating system
operators
output
output device
output symbols
picture element
pixel
processing symbols
program
program development cycle
programmer

project
project folder
properties
Properties window
pseudocode
random access memory (RAM)
reserved words
running
samples
secondary storage
secure digital (SD) memory card
software
software developer
software development tools
Solid-state drives
solution
Solution Explorer window
solution file
solution folder
source code
standard toolbar
start terminal
statements
syntax
syntax error
system software
terminal symbol
TextBox
Toolbox
ToolTip
two's complement
Unicode
universal serial bus (USB) drives
user
user interface
utility program
volatile
Visual Studio

Review Questions

Multiple Choice

1. A(n) _____ is a set of instructions that a computer follows to perform a task.
   a. compiler
   b. program
   c. interpreter
   d. programming language

2. The physical devices that a computer is made of are referred to as _____.
   a. hardware
   b. software
   c. the operating system
   d. tools

3. The part of a computer that runs programs is called _____.
   a. RAM
   b. secondary storage
   c. main memory
   d. the CPU

4. Today, CPUs are small chips known as _____.
   a. ENIACs
   b. microprocessors
   c. memory chips
   d. operating systems

5. The computer stores a program while the program is running, as well as the data that the program is working with, in _____.
   a. secondary storage
   b. the CPU
   c. main memory
   d. the microprocessor

6. _____ is a volatile type of memory that is used only for temporary storage while a program is running.
   a. RAM
   b. Secondary storage
   c. The disk drive
   d. The USB drive

7. A type of memory that can hold data for long periods of time—even when there is no power to the computer—is called _____.
   a. RAM
   b. main memory
   c. secondary storage
   d. CPU storage

8. A component that collects data from people or other devices and sends it to the computer is called _____.
   a. an output device
   b. an input device
   c. a secondary storage device
   d. main memory

9. A video display is a(n) _____ device.
   a. output
   b. input
   c. secondary storage
   d. main memory

10. A _____ is enough memory to store a letter of the alphabet or a small number.
    a. byte
    b. bit
    c. switch
    d. transistor

11. A byte is made up of eight _____.
    a. CPUs
    b. instructions
    c. variables
    d. bits

12. In the _____ numbering system, all numeric values are written as sequences of 0s and 1s.
    a. hexadecimal
    b. binary
    c. octal
    d. decimal

13. A bit that is turned off represents the following value: _____.
    a. 1
    b. -1
    c. 0
    d. “no”

14. A set of 128 numeric codes that represent the English letters, various punctuation marks, and other characters is _____.
    a. binary numbering
    b. ASCII
    c. Unicode
    d. ENIAC

15. An extensive encoding scheme that can represent the characters of many of the languages in the world is _____.

a. binary numbering
b. ASCII
c. Unicode
d. ENIAC

16. Negative numbers are encoded using the _____ technique.

a. two's complement
b. floating point
c. ASCII
d. Unicode

17. Real numbers are encoded using the _____ technique.

a. two's complement
b. floating point
c. ASCII
d. Unicode

18. The tiny dots of color that digital images are composed of are called _____.

a. bits
b. bytes
c. color packets
d. pixels

19. If you were to look at a machine language program, you would see _____.

a. C# code
b. a stream of binary numbers
c. English words
d. circuits

20. In the _____ part of the fetch-decode-execute cycle, the CPU determines which operation it should perform.

a. fetch
b. decode
c. execute
d. immediately after the instruction is executed

21. Computers can execute only programs that are written in _____.

a. C#
b. assembly language
c. machine language
d. Java

22. The _____ translates an assembly language program to a machine language program.

a. assembler
b. compiler
c. translator
d. interpreter

23. The words that make up a high-level programming language are called _____.

a. binary instructions
b. mnemonics
c. commands
d. keywords

24. The rules that must be followed when writing a program are called _____.

a. syntax
b. punctuation
c. keywords
d. operators

25. A(n) _____ is a program that translates a high-level language program into a separate machine language program.

a. assembler
b. compiler
c. translator
d. utility

26. A _____ is any hypothetical person using a program and providing input for it.

a. designer
b. user
c. guinea pig
d. test subject

27. A _____ error does not prevent the program from running but causes it to produce incorrect results.

a. syntax
b. hardware
c. logic
d. fatal

28. A(n) _____ is a set of well-defined logical steps that must be taken to perform a task.

a. logarithm
b. plan of action
c. logic schedule
d. algorithm

29. An informal language that has no syntax rules and is not meant to be compiled or executed is called _____.

a. faux code
b. pseudocode
c. C#
d. a flowchart

30. A _____ is a diagram that graphically depicts the steps that take place in a program.

a. flowchart
b. step chart
c. code graph
d. program graph

31. Objects that are visible in a program's graphical user interface are commonly referred to as _____.

a. buttons
b. controls
c. forms
d. windows

32. A _____ is code that describes a particular type of object.
    a. namespace
    b. blueprint
    c. schema
    d. class

33. _____ is a collection of classes and other code that can be used, along with a programming language such as C#, to create programs for the Windows operating system.
    a. .NET
    b. The Standard Template Library
    c. The GUI framework
    d. The MSDN Library

34. The _____ is the part of a computer with which the user interacts.
    a. central processing unit
    b. user interface
    c. control system
    d. interactivity system

35. Before GUIs became popular, the _____ interface was the most commonly used.
    a. command line
    b. remote terminal
    c. sensory
    d. event-driven

36. _____ programs are usually event driven.
    a. Command line
    b. Text-based
    c. GUI
    d. Procedural

True or False

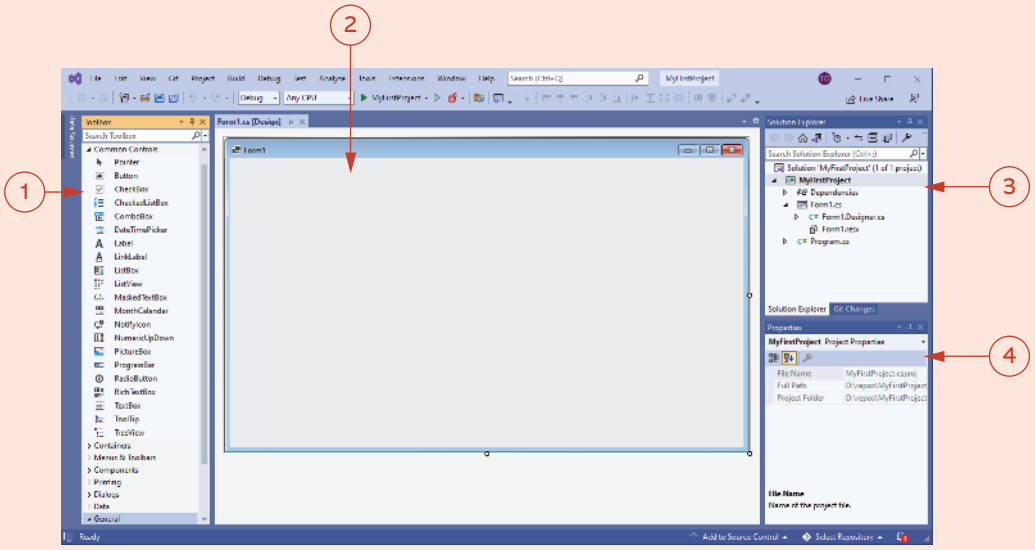
1. Today, CPUs are huge devices made of electrical and mechanical components such as vacuum tubes and switches.
2. Main memory is also known as RAM.
3. Any piece of data that is stored in a computer's memory must be stored as a binary number.
4. Images, such as the ones you make with your digital camera, cannot be stored as binary numbers.
5. Machine language is the only language that a CPU understands.
6. Assembly language is considered a high-level language.
7. An interpreter is a program that both translates and executes the instructions in a high-level language program.
8. A syntax error does not prevent a program from being compiled and executed.
9. Windows, Linux, macOS, iOS, and Android are all examples of application software.
10. Word processing programs, spreadsheet programs, e-mail programs, Web browsers, and games are all examples of utility programs.

11. Programmers must be careful not to make syntax errors when writing pseudocode programs.
12. C# provides only the basic keywords and operators that you need to construct a program.

Short Answer

1. Why is the CPU the most important component in a computer?
2. What number does a bit that is turned on represent? What number does a bit that is turned off represent?
3. What would you call a device that works with binary data?
4. What are the words that make up a high-level programming language called?
5. What are the short words that are used in assembly language called?
6. What is the difference between a compiler and an interpreter?
7. What type of software controls the internal operations of the computer's hardware?
8. What is pseudocode? What is a flowchart?
9. When a program runs in a text-based environment, such as a command line interface, what determines the order in which things happen?
10. What does a class specify about an object?
11. Can you use C# alone to perform advanced operations such as creating GUIs, reading data from a file, or working with databases? Why or why not?
12. Figure 1-49 shows the Visual Studio IDE. What are the names of the four areas that are indicated in the figure?
13. What is the purpose of the *Toolbox* in the Visual Studio environment?
14. How can you access the documentation for Visual Studio?
15. What steps must you take to open an existing project?
16. How can you view the project's form if it is not automatically displayed in the *Designer*?

Figure 1-49 The Visual Studio IDE



Programming Problems

1. Use what **you've** learned about the binary numbering system in this chapter to convert the following decimal numbers to binary:
   11
   65
   100
   255

2. Use what you've learned about the binary numbering system in this chapter to convert the following binary numbers to decimal:
   1101
   1000
   101011

3. Look at the ASCII chart in Appendix C and determine the codes for each letter of your first name.

4. Suppose your instructor gives three exams during the semester and you want to write a program that calculates your average exam score. Answer the following:
   a. What items of input must the user enter?
   b. Once the input has been entered, how will the program determine the average?
   c. What output will the program display?

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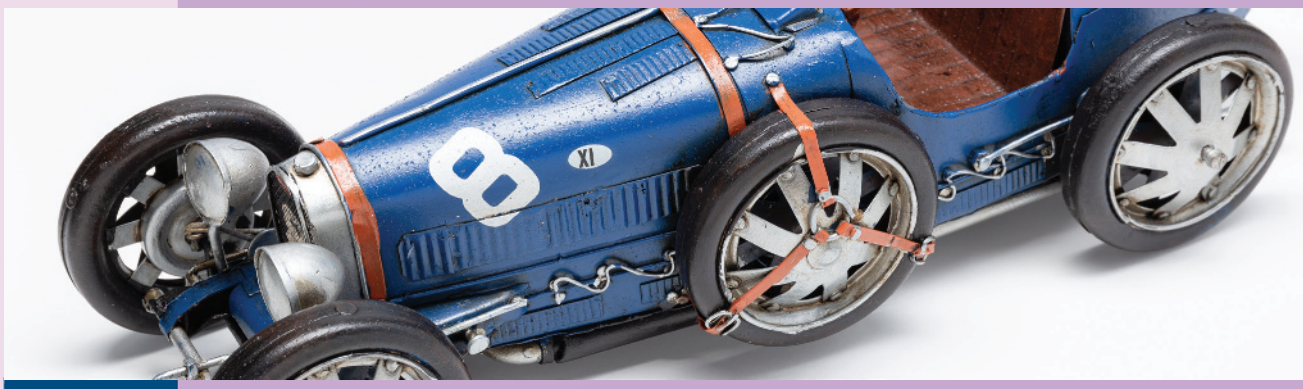


CHAPTER 2

Introduction to Visual C#

TOPICS

2.1 Getting Started with Forms and Controls
2.2 Creating the GUI for Your First Visual C# Application: The *Hello World* Application
2.3 Introduction to C# Code
2.4 Writing Code for the *Hello World* Application
2.5 Label Controls
2.6 Making Sense of IntelliSense
2.7 PictureBox Controls
2.8 Comments, Blank Lines, and Indentation
2.9 Writing the Code to Close an Application's Form
2.10 Dealing with Syntax Errors

2.1 Getting Started with Forms and Controls

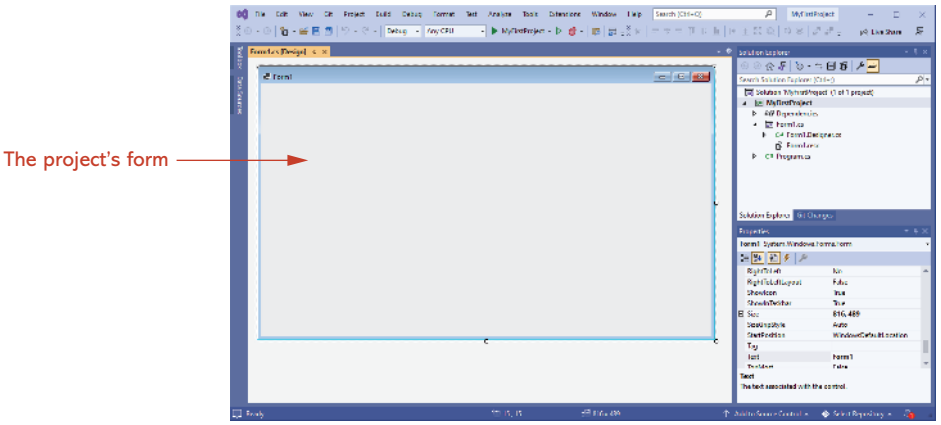
CONCEPT: The first step in creating a Visual C# application is creating the application's GUI. You use the Visual Studio *Designer*, *Toolbox*, and *Properties* window to build the application's form with the desired controls and set each control's properties.

In this chapter, you will create your first Visual C# application. Before you start, however, you need to learn some fundamental concepts about creating a GUI in Visual Studio. This section shows the basics of editing forms and creating controls.

The Application's Form

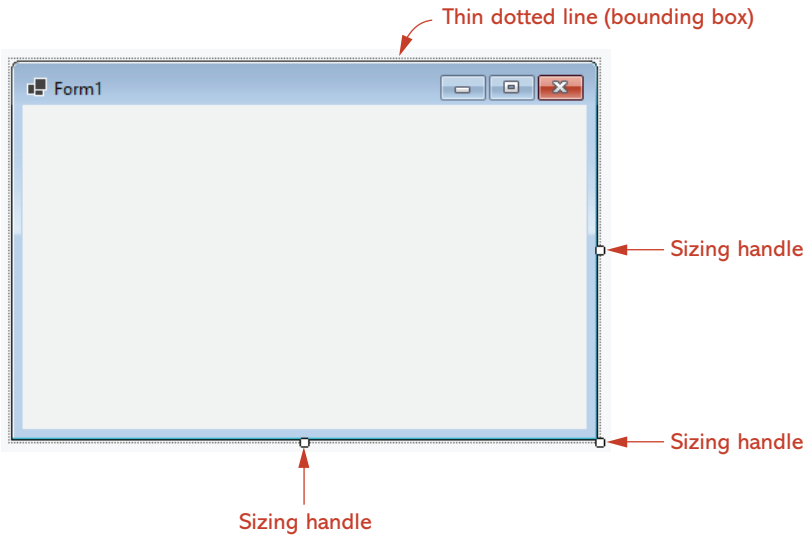
When you start a new Visual C# project, Visual Studio automatically creates an empty form and displays it in the *Designer*. Figure 2-1 shows an example. Think of the empty form as a blank canvas that can be used to create the application's user interface. You can add controls to the form, change the form's size, and modify many of its characteristics. When the application runs, the form will be displayed on the screen.

Figure 2-1 A new project with a blank form displayed in the *Designer*



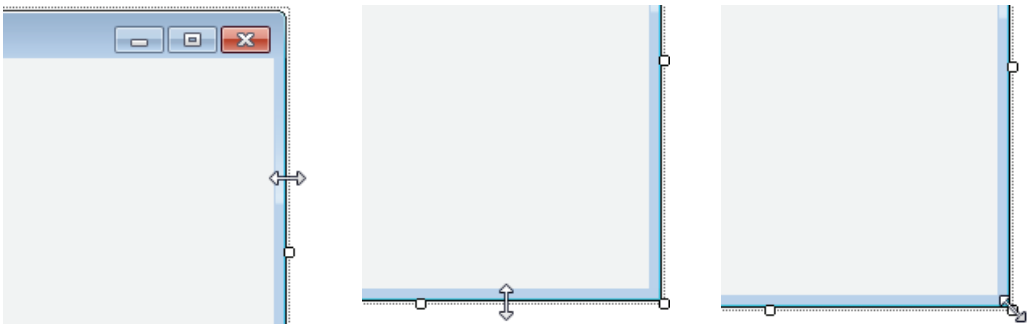
If you take a closer look at the form, you will notice that it is enclosed by a thin dotted line, known as a **bounding box**. As shown in Figure 2-2, the bounding box has small **sizing handles**, which appear on the form's right edge, bottom edge, and lower-right corner. When a bounding box appears around an object in the *Designer*, it indicates that the object is selected and is ready for editing.

Figure 2-2 The form's bounding box and sizing handles



You can easily resize the form with the mouse. When you position the mouse cursor over any edge or corner that has a sizing handle, the cursor changes to a two-headed arrow (⇔). Figure 2-3 shows examples. When the mouse cursor becomes a two-headed arrow, you can click and drag the mouse to resize form.

Figure 2-3 Using the mouse to resize the form



Identifying Forms and Controls by Their Names

An application's GUI is made of forms and various controls. Each form and control in an application's GUI must have a name that identifies it. The blank form that Visual Studio initially creates in a new project is named Form1.

NOTE: Later in this book, you will learn how to change a form's name, but for now, you will keep the default name, Form1.

The *Properties* Window

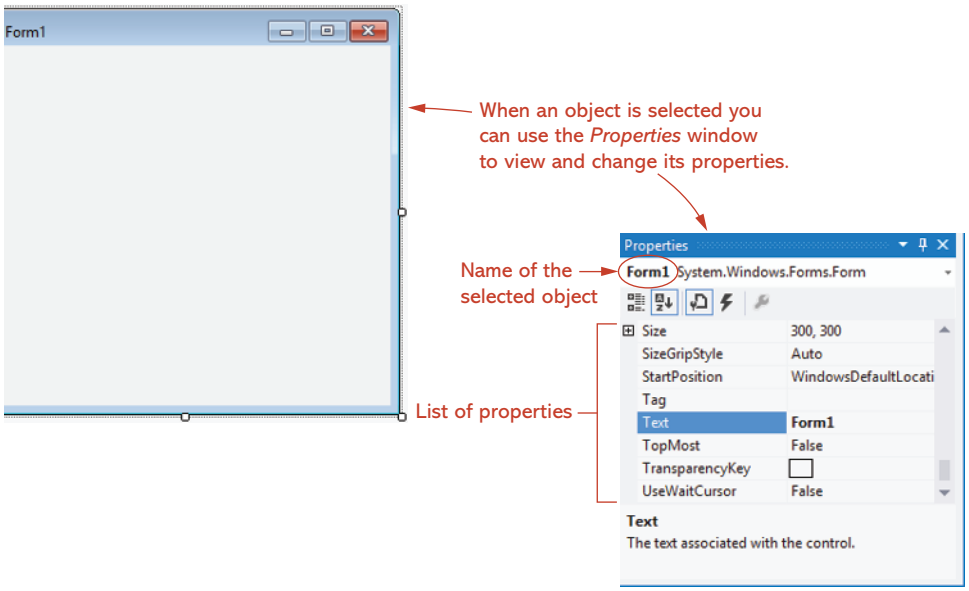
The appearance and other characteristics of a GUI object are determined by the object's properties. When you select an object in the *Designer*, that object's properties are displayed in the *Properties* window. For example, when the Form1 form is selected, its properties are displayed in the *Properties* window, as shown in Figure 2-4.

TIP: Recall from Chapter 1 that if the *Properties* window is in Auto Hide mode, you can click its tab to open it. If you do not see the *Properties* window, click *View* on the menu bar. On the *View* menu, click *Properties Window*.

The area at the top of the *Properties* window shows the name of the object that is currently selected. You can see in Figure 2-4 that the name of the selected object is *Form1*. Below that is a scrollable list of properties. The list of properties has two columns: the left column shows each property's name, and the right column shows each property's value. For example, look at the form's Size property in Figure 2-4. Its value is *300*, *300*. This means that the form's size is 300 pixels wide by 300 pixels high. Next, look at the form's Text property. The Text property determines the text that is displayed in the form's title bar (the bar that appears at the top of the form). Its current value is Form1, so the text *Form1* is displayed in the form's title bar.

When a form is created, its Text property is initially set to the same value as the form's name. When you start a new project, the blank form that appears in the *Designer* will always be named Form1, so the text *Form1* will always appear in the form's title bar. In most cases, you want to change the value of the form's Text property to something more

Figure 2-4 The *Properties* window, showing the selected object's properties



meaningful. For example, assume the Form1 form is currently selected. You can perform the following steps to change its Text property to *My First Program*.

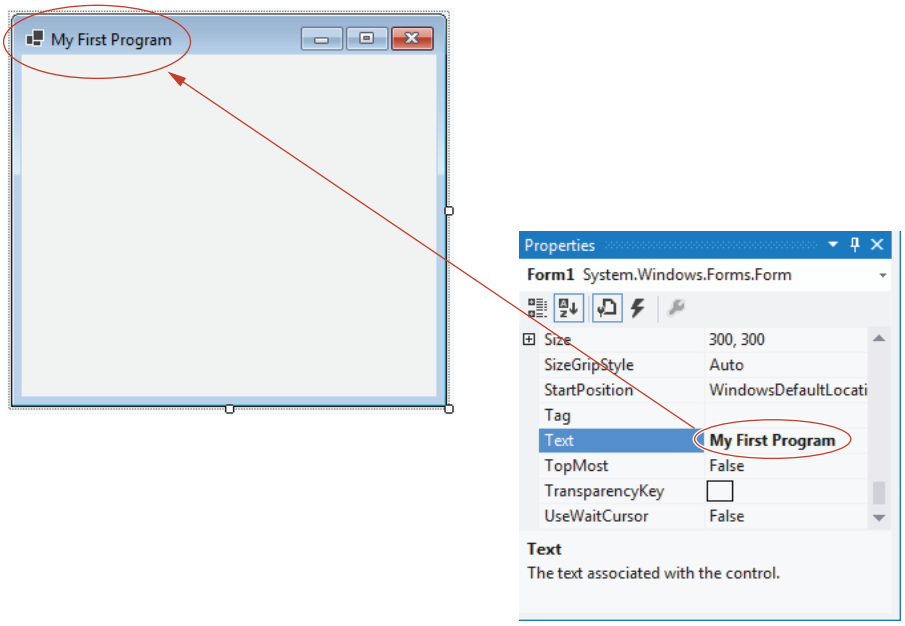
Step 1: In the *Properties* window, locate the Text property.

Step 2: Double-click the word *Form1* that currently appears as the Text property's value, and then use the Delete key to delete it.

Step 3: Type *My First Program* in its place and press the Enter key. The text *My First Program* will now appear in the form's title bar, as shown in Figure 2-5.

NOTE: Changing an object's Text property does not change the object's name. For example, if you change the Form1 form's Text property to *My First Program*, the form's name is still Form1. You have changed only the text that is displayed in the form's title bar.

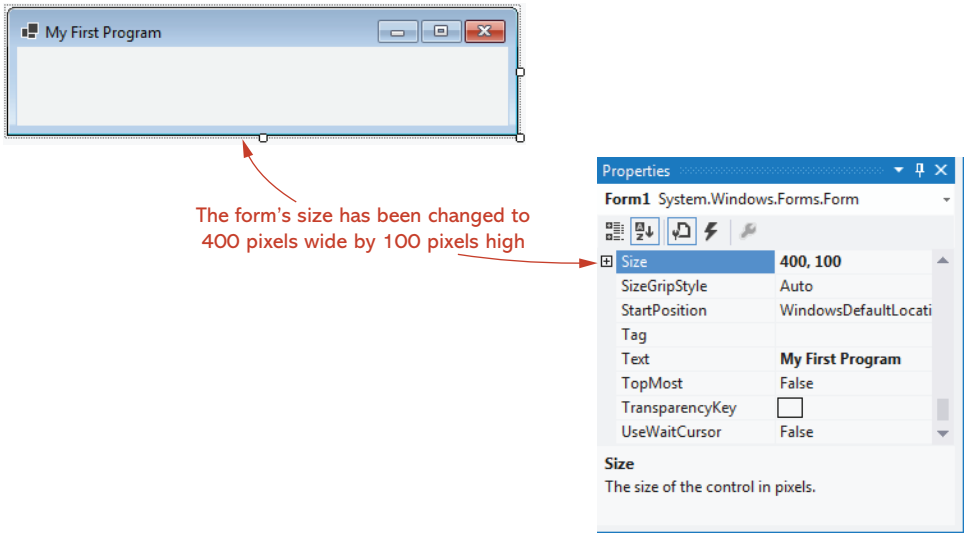
Figure 2-5 The form's Text property value displayed in the form's title bar



Earlier, we discussed how to use the mouse to resize a form in the *Designer*. An alternative method is to change the form's Size property in the *Properties* window. For example, assume the Form1 form is currently selected. You can perform the following steps to change its size to 400 pixels wide by 100 pixels high.

Step 1: In the *Properties* window, locate the Size property.
Step 2: Click inside the area that holds the Size property's value, and then delete the current value.
Step 3: Type *400, 100* in its place and press the Enter key. The form will be resized as shown in Figure 2-6.

Figure 2-6 The form's size changed to 400 by 100



NOTE: Notice in Figure 2-6 that the **Alphabetical button** ( ) is selected near the top of the *Properties* window. This causes the properties to be displayed in alphabetical order. Alternatively, the **Categorized button** ( ) can be selected, which causes the properties to be displayed in groups. The alphabetical listing is the default selection, and most of the time, it makes it easier to locate specific properties.

Adding Controls to a Form

When you are ready to create controls on the application's form, you use the *Toolbox*. Recall from Chapter 1 that the *Toolbox* usually appears on the left side of the Visual Studio environment. If the *Toolbox* is in Auto Hide mode, you can click its tab to open it. Figure 2-7 shows an example of how the *Toolbox* typically appears when it is open.

TIP: Recall from Chapter 1 that if you do not see the *Toolbox* or its tab, click *View* on the menu bar and then click *Toolbox*.

The *Toolbox* shows a scrollable list of controls that you can add to a form. To add a control to a form, you simply find it in the *Toolbox* and then double-click it. The control will be created on the form. For example, suppose you want to create a Button control on the form. You find it in the *Toolbox*, as shown in Figure 2-8, double-click it, and a Button control will appear on the form.

Figure 2-7 The *Toolbox*

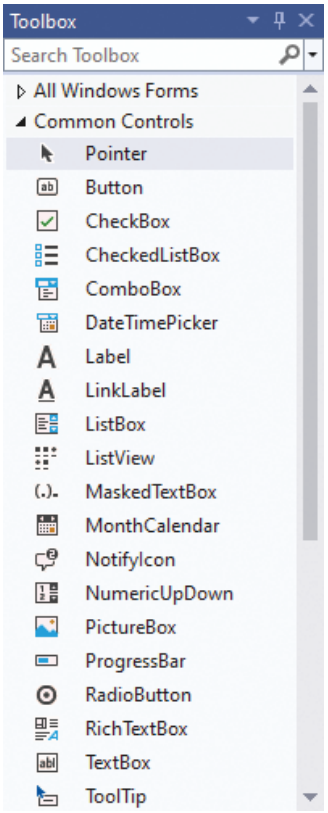
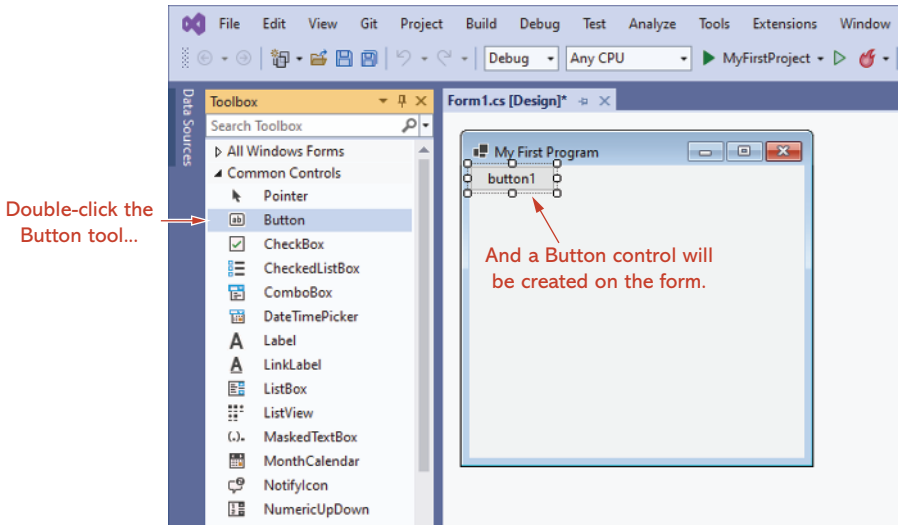


Figure 2-8 Creating a Button control



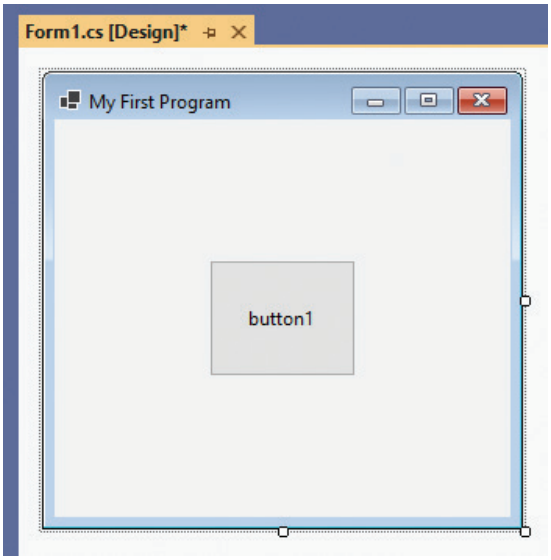
TIP: You can also click and drag controls from the *Toolbox* onto the form.

Resizing and Moving Controls

Take a closer look at the Button control that is shown on the form in Figure 2-8. Notice that it is enclosed in a bounding box with sizing handles. This indicates that the control is currently selected. When a control is selected, you can use the mouse to resize it in the same way

that you learned to resize a form earlier. You can also use the mouse to move a control to a new location on the form. Position the mouse cursor inside the control, and when the mouse cursor becomes a four-headed arrow (), you can click and drag the control to a new location. Figure 2-9 shows a form with a Button control that has been enlarged and moved.

Figure 2-9 A Button control resized and moved



Deleting a Control

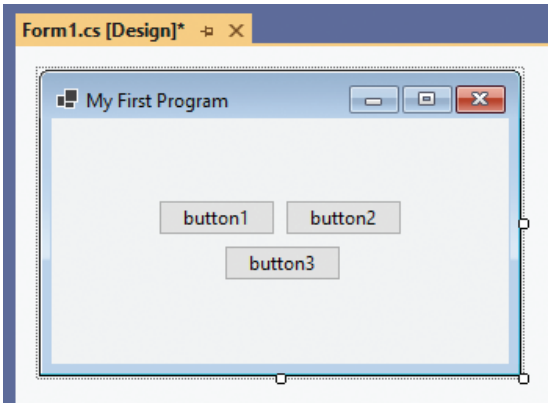
Deleting a control is simple: you select it and then press the Delete key on the keyboard.

More about Button Controls

You learned earlier that each form and each control in an application's GUI must have a name that identifies it. When you create Button controls, they are automatically given default names such as `button1`, `button2`, and so forth.

Button controls have a Text property, which holds the text that is displayed on the face of the button. When a Button control is created, its Text property is initially set to the same value as the Button control's name. As a result, when you create a Button control, its name will be displayed on the face of the button. For example, the form in Figure 2-10 contains three Button controls named `button1`, `button2`, and `button3`.

Figure 2-10 A form with three Button controls



After you create a Button control, you should always change its Text property. The text that is displayed on a button should indicate what the button will do when it is clicked. For example, a button that calculates an average might have the text *Calculate Average* displayed on it, and a button that prints a report might have the text *Print Report* displayed on it. Here are the steps you perform to change a Button control's Text property:

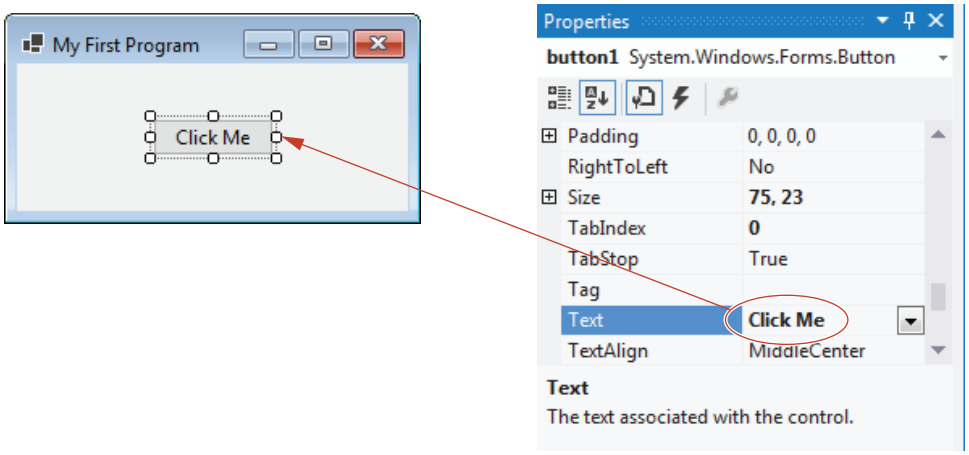
Step 1: Make sure the Button control is selected. (If you don't see the bounding box and sizing handles around the control, just click the control to select it.)

Step 2: In the *Properties* window, locate the Text property.

Step 3: Click inside the area that holds the Text property's value, and then delete the current value. Then, type the new text in its place and press the Enter key. The new text will be displayed on the button.

Figure 2-11 shows an example of how changing a Button control's Text property changes the text displayed on the face of the button.

Figure 2-11 A Button control's Text property changed



Changing a Control's Name

A control's name identifies the control in the application's code and in the Visual Studio environment. When you create a control on an application's form, you should always change the control's name to something that is more meaningful than the default name that Visual Studio gives it. A control's name should reflect the purpose of the control.

For example, suppose you've created a Button control to calculate an amount of tax. A default name such as `button1` does not convey the button's purpose. A name such as `calculateTaxButton` would be much better. When you are working with the application's code and you see the name `calculateTaxButton`, you will know precisely which button the code is referring to.

You can change a control's name by changing its Name property. Here are the steps:

Step 1: Make sure the control is selected. (If you do not see the bounding box and sizing handles around the control, just click the control to select it.)

Step 2: In the *Properties* window, scroll up to the top of the list of properties. You should see the Name property, as shown in Figure 2-12. (The Name property is enclosed in parentheses to make it appear near the top of the alphabetical list of properties. This makes it easier to find.)

Step 3: Click inside the area that holds the Name property's value and then delete the current name. Then, type the new name in its place and press the Enter key. You have successfully changed the name of the control.

Figure 2-12 The Name property

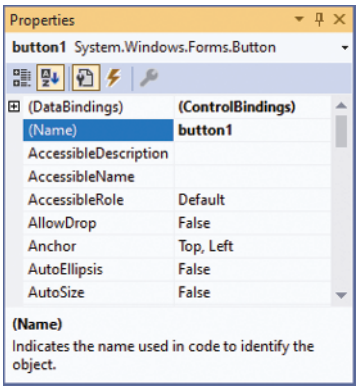
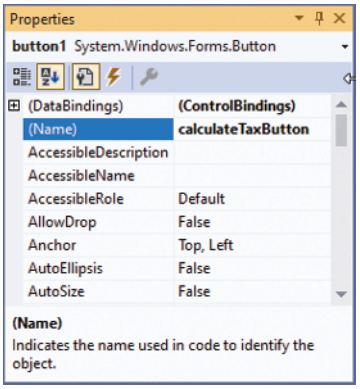


Figure 2-13 shows the *Properties* window after a Button control's name has been changed to `calculateTaxButton`.

Figure 2-13 The Name property changed to `calculateTaxButton`



Rules for Naming Controls

Control names are also known as **identifiers**. When naming a control, you must follow these rules for C# identifiers:

- The first character must be one of the letters *a* through *z* or *A* through *Z* or an underscore character (_).
- After the first character, you may use the letters *a* through *z* or *A* through *Z*, the digits 0 through 9, or underscores.
- The name cannot contain spaces.

Table 2-1 lists some identifiers that might be used for Button control names and indicates whether each is a legal or illegal identifier in C#.

Table 2-1 Legal and illegal identifiers

Identifier	Legal or Illegal?
`showDayOfWeekButton`	Legal
`3rdQuarterButton`	Illegal because identifiers cannot begin with a digit.
`change*color*Button`	Illegal because the * character is not allowed.
`displayTotalButton`	Legal
`calculate Tax Button`	Illegal because identifiers cannot contain spaces.

Because a control's name should reflect the control's purpose, programmers often find themselves creating names that are made of multiple words. For example, consider the following Button control names:

```
calculatetaxbutton
printreportbutton
displayanimationbutton
```

Unfortunately, these names are not easily read by the human eye because the words are not separated. Because we cannot have spaces in control names, we need to find another way to separate the words in a multiword control name to make it more readable to the human eye.

Most C# programmers address this problem by using the **camelCase** naming convention for controls. camelCase names are written in the following manner:

- You begin writing the name with lowercase letters.
- The first character of the second and subsequent words is written in uppercase.

For example, the following control names are written in camelCase:

```
calculateTaxButton
printReportButton
displayAnimationButton
```

NOTE: This style of naming is called camelCase because the uppercase characters that appear in a name are sometimes reminiscent of a camel's humps.

Checkpoint

2.1 When you start a new Visual C# project, what is automatically created and displayed in the *Designer*?

2.2 How can you tell that an object is selected and ready for editing in the *Designer*?

2.3 What is the purpose of an object's sizing handles?

2.4 What must each form and control in an application's GUI have to identify it?

2.5 What is the purpose of the *Properties* window?

2.6 What does the Alphabetical button do when it is selected in the *Properties* window?

2.7 What does the Categorized button do when it is selected in the *Properties* window?

2.8 What does a form's Text property determine?

2.9 What does a form's Size property determine?

2.10 What is shown in the *Toolbox*?

2.11 How do you add a control to a form?

2.12 What should the text that is displayed on a button indicate?

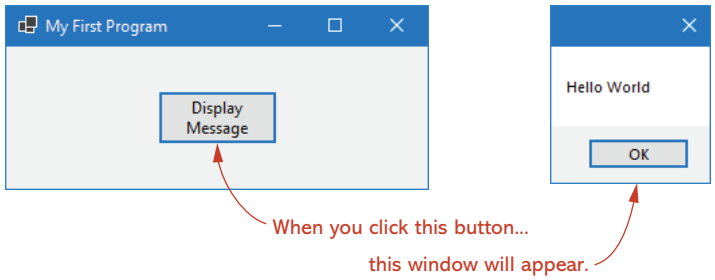
2.13 What are the rules for naming controls?

2.14 What naming convention do most C# programmers use to separate words in a multiword identifier?

2.2 Creating the GUI for Your First Visual C# Application: The *Hello World* Application

When a student is learning computer programming, it is traditional to start by learning to write a *Hello World* program. A *Hello World* program is a simple program that merely displays the words *"Hello World"* on the screen. In this chapter you will create your first Visual C# application, which will be an event-driven *Hello World* program. When the finished application runs, it will display the form shown on the left in Figure 2-14. Notice that the form contains a button that reads *Display Message*. When you click the button, the window shown on the right in the figure will appear.

Figure 2-14 Screens displayed by the completed *Hello World* program



The process of creating this application is divided into two parts. First, you will create the application's GUI, and second, you will write the code that causes the *Hello World* message to appear when the user clicks the *Display Message* button. Tutorial 2-1 leads you through the process of creating the GUI.

Tutorial 2-1:
Creating the GUI for the *Hello World* Application

VideoNote
Tutorial 2-1: Creating the GUI for the *Hello World* Application

Step 1: Start Visual Studio. When the Start window appears, click *Create a new project*.

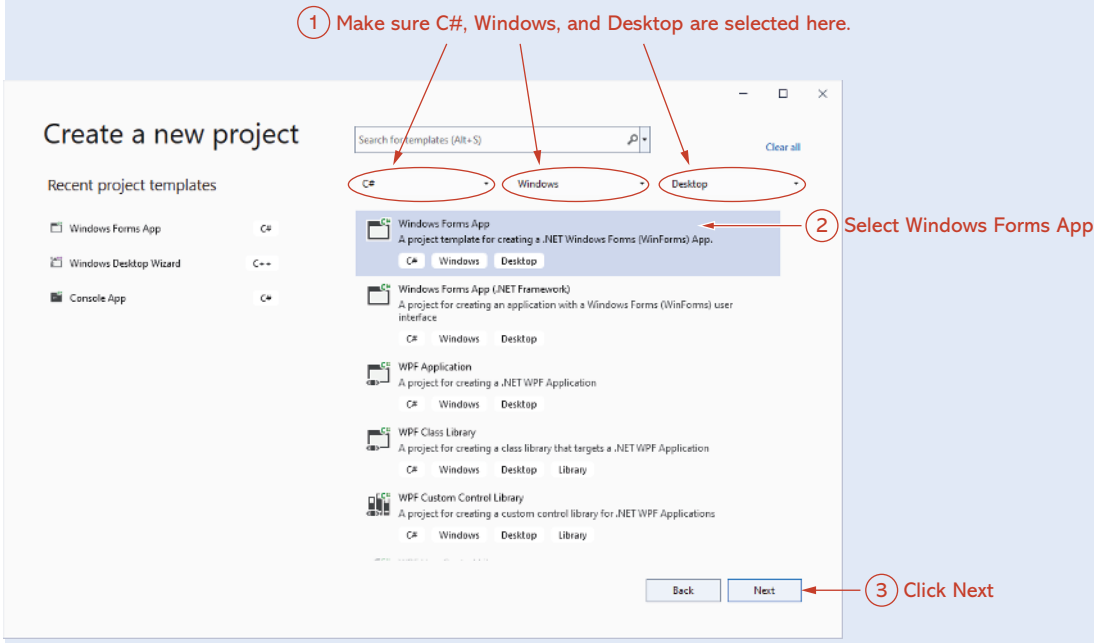
TIP: If Visual Studio is already running, you can start a new project by clicking *File* on the menu bar and then clicking *New*, and then clicking *Project*....

Step 2: The *Create a new project* window will appear. As shown in Figure 2-15, use the drop-down lists to select C#, *Windows*, and *Desktop*. Then, select the *Windows Forms App* template and click *Next*.

Step 3: The *Configure your project* window appears next. In the *Project Name* text box, change the name of the project to *HelloWorld* and then click the *Next* button.

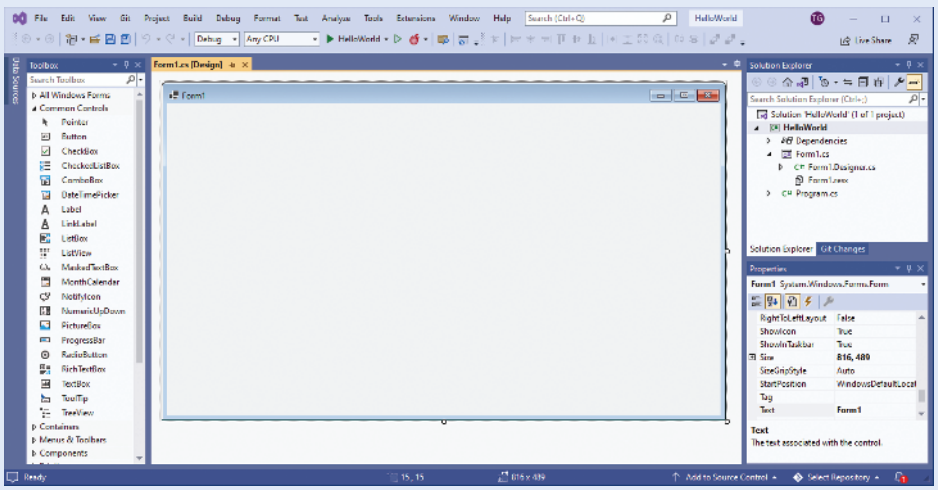
Step 4: The *Additional information* window will appear next. This window allows you to select the .NET version you wish to use with the project. Make sure the highest version number is selected in the drop-down list. Click the *Create* button to create the project.

Figure 2-15 The *Create a new project* Window



Step 5: The Visual Studio environment will appear next. Make sure the *Toolbox*, the *Solution Explorer*, and the *Properties* window are visible and that Auto Hide is turned off for each of these windows. The Visual Studio environment should appear as shown in Figure 2-16.

Figure 2-16 The Visual Studio environment



Step 6: Change the Form1 form's Text property to *My First Program*, as shown in Figure 2-17.

Step 7: The form's default size is too large for this application, so you need to make it smaller. Use the technique discussed in the previous section to adjust the form's size with the mouse. The form should appear similar to that shown in Figure 2-18. (Don't worry about the form's exact size. Just make it appear similar to Figure 2-18.)

Figure 2-17 The form's Text property changed to *My First Program*

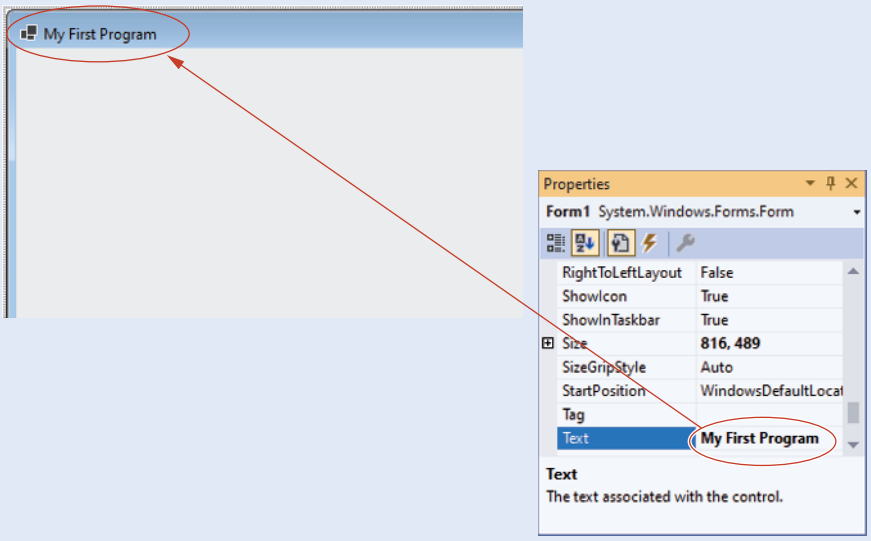
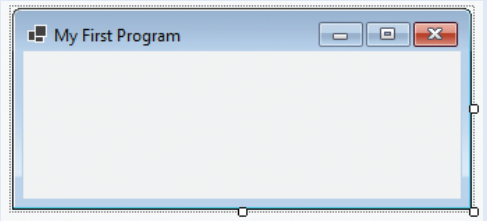


Figure 2-18 The form resized



Step 8: Now you are ready to add a Button control to the form. Locate the Button tool in the *Toolbox* and double-click it. A Button control should appear on the form, as shown in Figure 2-19. Move the Button control so it appears approximately in the center of the form, as shown in Figure 2-20.

Figure 2-19 A Button control created on the form

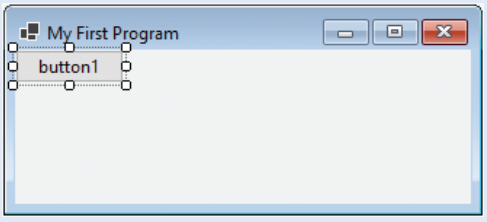
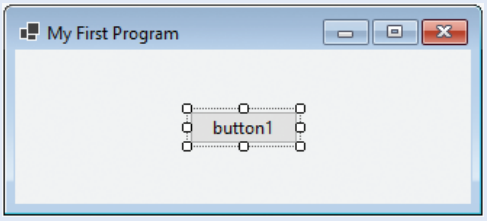
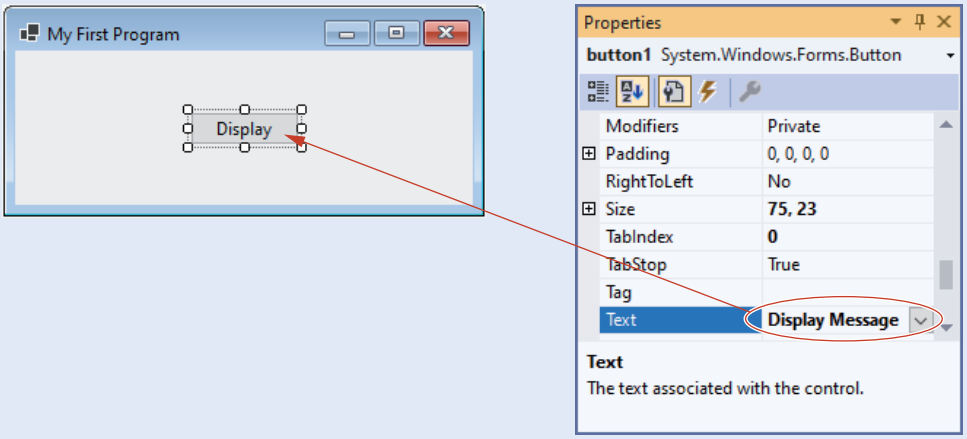


Figure 2-20 The Button control moved



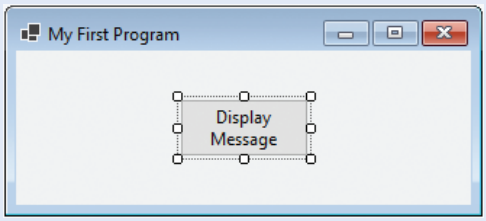
Step 9: Change the value of the Button control's Text property to *Display Message*. After doing this, notice that the text displayed on the button has changed, as shown in Figure 2-21.

Figure 2-21 The Button control's Text property changed



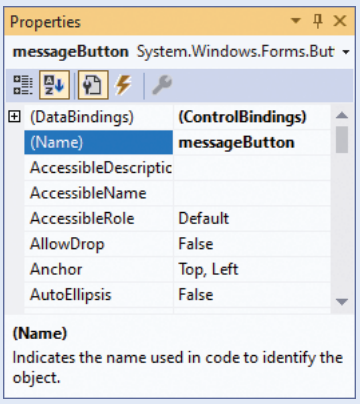
Step 10: The Button control isn't quite large enough to accommodate all of the text that you typed into its Text property, so enlarge the Button control, as shown in Figure 2-22.

Figure 2-22 The Button control enlarged



Step 11: As discussed in the previous section, a control's name should reflect the purpose of the control. The Button control that you created in this application will cause a message to be displayed when it is clicked. The name `button1` does not convey that purpose, however. Change the Button control's Name property to `messageButton`. The *Properties* window should appear as shown in Figure 2-23.

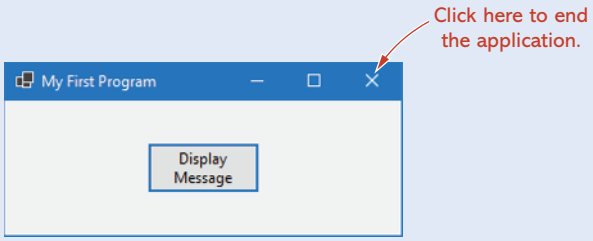
Figure 2-23 The Button control's Name property changed to `messageButton`



Step 12: Click *File* on the Visual Studio menu bar and then click *Save All* to save the project.

Step 13: You're only partially finished with the application, but you can run it now to see how the GUI looks on the screen. To run the application, press the F5 key on the keyboard or click the *Start Debugging* button (▶) on the toolbar. This causes the application to be compiled and executed. You will notice the appearance of the Visual Studio environment change somewhat, and you will see the application's form appear on the screen as shown in Figure 2-24.

Figure 2-24 The application running



Although the application is running, it is not capable of doing anything other than displaying the form. If you click the *Display Message* button, nothing will happen. That is because you have not yet written the code that executes when the button is clicked. You will do that in the next tutorial. To end the application, click the standard Windows close button (✕) in the form's upper-right corner.

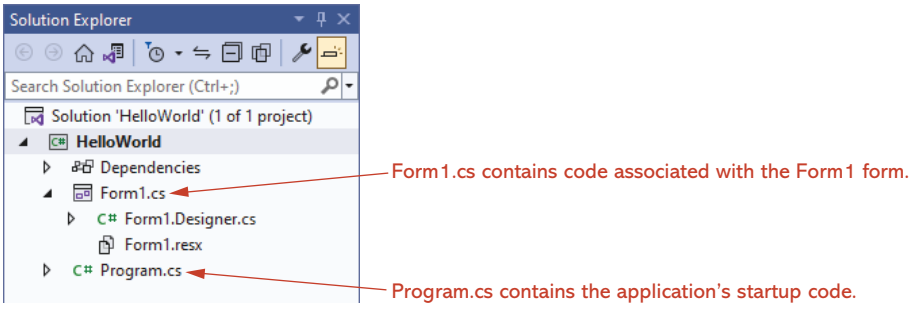
2.3 Introduction to C# Code

CONCEPT: **You use the Visual Studio code editor to write an application's code. Much of the code that you will write in an application will be event handlers. Event handlers respond to specific events that take place while an application is running.**

In the previous sections of this chapter, you learned the basics of creating an application's GUI. An application is more than a user interface, however. If you want your application to perform any meaningful actions, you have to write code. This section introduces you to Visual C# code and shows how to program an application to respond to button clicks.

A file that contains program code is called a **source code file**. When you start a C# Windows Forms Application project, Visual Studio automatically creates several source code files and adds them to the project. If you look at the *Solution Explorer*, as shown in Figure 2-25, you will see the names of two source code files: Form1.cs and Program.cs. (C# source code files always end with the .cs extension.)

Figure 2-25 Source code files shown in the *Solution Explorer*



Here is a brief description of the two files:

- The Program.cs file contains the application's start-up code, which executes when the application runs. The code in this file performs behind-the-scenes initialization tasks that are necessary to get the application up and running. It is important that you do not modify the contents of this file because doing so could prevent the application from executing.
- The Form1.cs file contains code that is associated with the Form1 form. When you write code that defines some action related to Form1 (such as responding to a button click), you will write the code in this file.

NOTE: You might see additional source code files in the *Solution Explorer*, other than those shown in Figure 2-25.

The Form1.cs file already contains code that was generated by Visual Studio when the project was created. You can think of this auto-generated code as an outline to which you can add your own code as you develop the application.

Let's take a look at the code. If you still have the *HelloWorld* project open from the previous tutorial, right-click Form1.cs in the *Solution Explorer*. A pop-up menu will appear, as shown in Figure 2-26. On the pop-up menu, click *View Code*. The file's contents will be displayed in the Visual Studio code editor, as shown in Figure 2-27.

Figure 2-26 Opening Form1.cs in the code editor

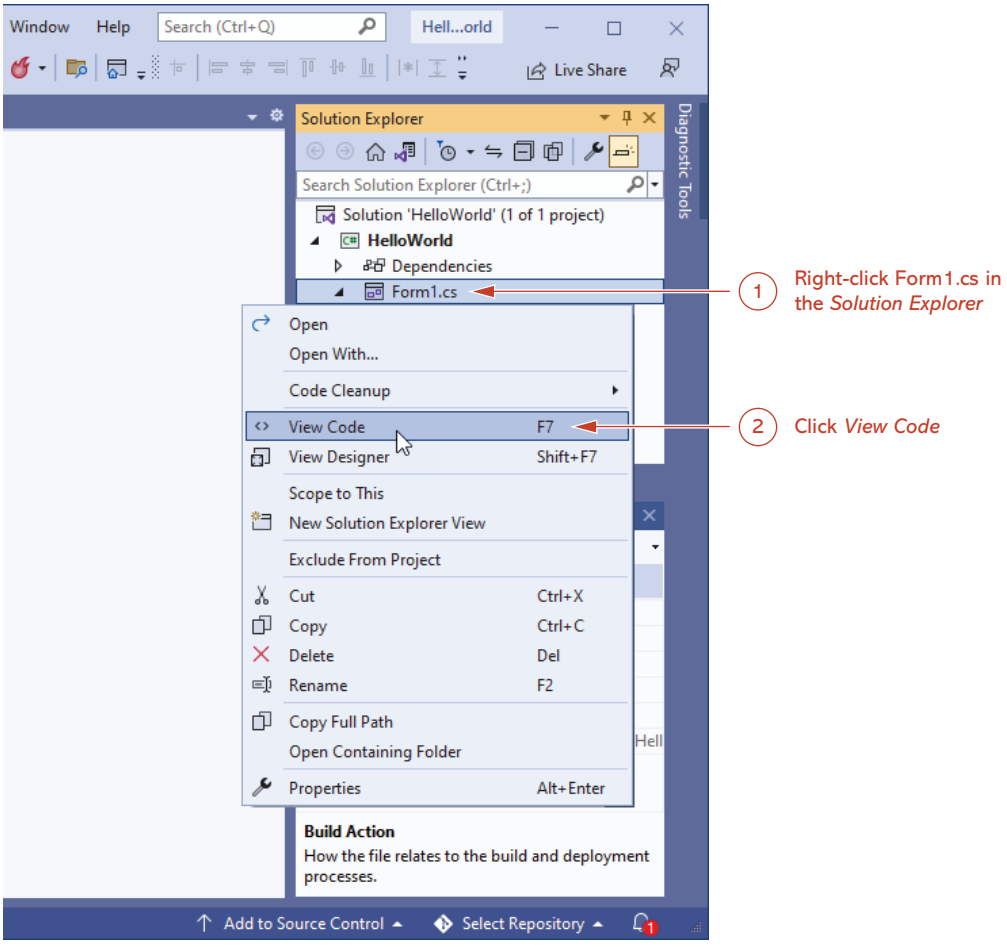
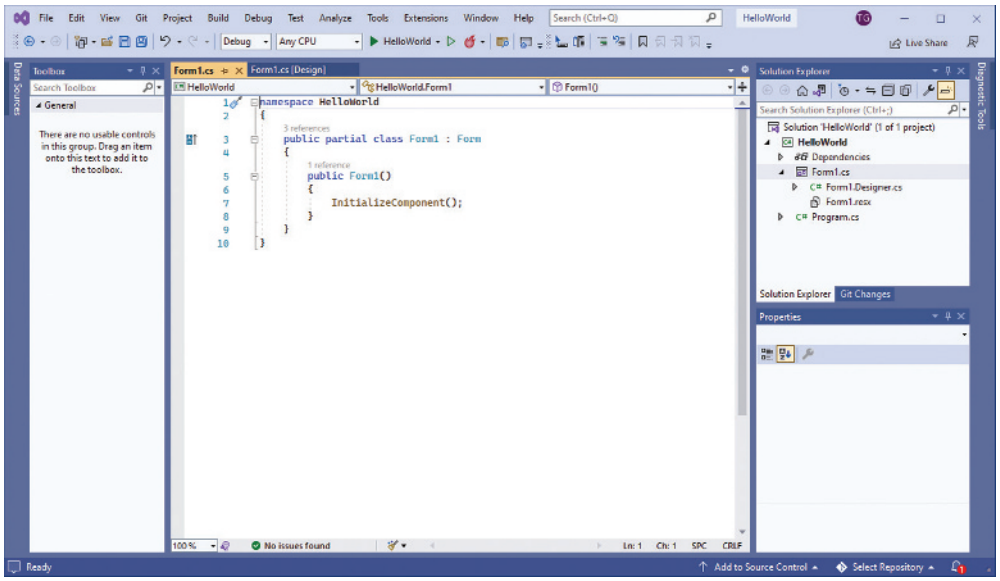


Figure 2-27 Form1.cs code displayed in the Visual Studio code editor

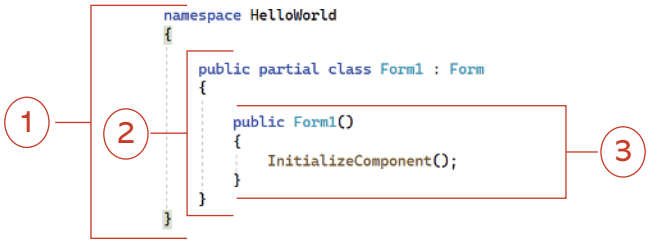


At this point, it's not necessary for you to understand the meaning of the statements that you see in this code. It will be helpful for you to know how this code is organized, however, because later you will add your own code to this file. C# code is primarily organized in three ways: namespaces, classes, and methods. Here's a summary:

- A **namespace** is a container that holds classes.
- A class is a container that holds methods (among other things).
- A method is a group of one or more programming statements that performs some operation.

So, C# code is organized as methods, which are contained inside classes, which are contained inside namespaces. With this organizational structure in mind, look at Figure 2-28.

Figure 2-28 Organization of the Form1.cs code



The figure shows four different sections of the code, marked with the numbers 1, 2, and 3. Let's discuss each section of code.

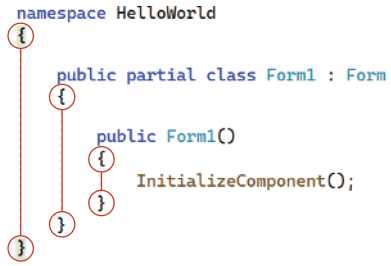
1. This section of code creates a namespace for the project. You can think of a namespace as a container that organizes code. The line that reads `namespace HelloWorld` marks the beginning of a namespace named `HelloWorld`. Notice that the next line contains an opening brace (`{`) and that the last line in the file contains a corresponding closing brace (`}`). All the code that appears between these braces is inside the `HelloWorld` namespace.

(2) This section of code is a class declaration. The line that reads `public partial class`, and so forth, marks the beginning of the class. The next line contains an opening brace (`{`), and the last line in this section of code contains a corresponding closing brace (`}`). All the code that appears between these braces is inside the class.

(3) This section of code is a method. The line that reads `public Form1()` marks the beginning of the method. The next line contains an opening brace (`{`), and the last line in this section of code contains a corresponding closing brace (`}`). The code that appears between these braces is inside the method.

It's important to point out that code containers, such as namespaces, classes, and methods, use **braces** (`{}`) to enclose code. Each opening brace (`{`) must have a corresponding closing brace (`}`) at some later point in the program. Figure 2-29 shows how the braces in Form1.cs are paired.

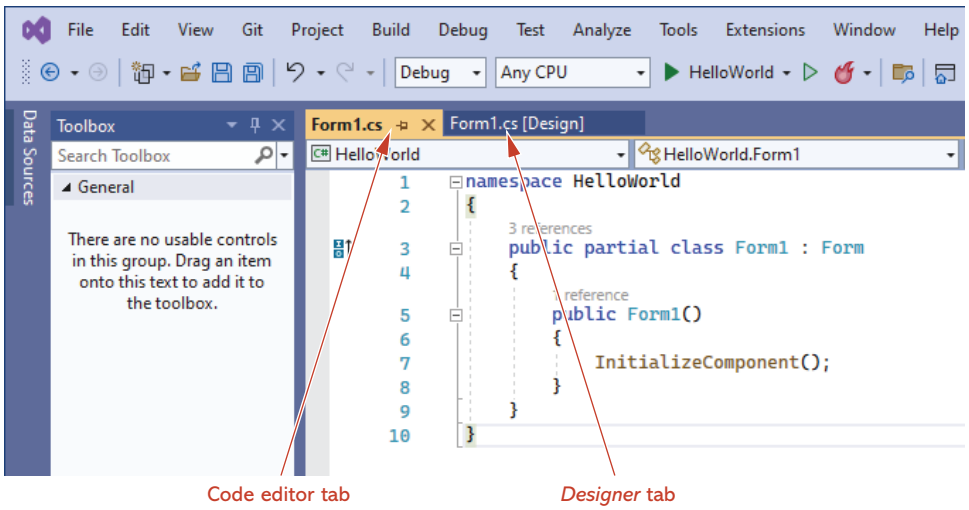
Figure 2-29 Corresponding braces



Switching between the Code Editor and the *Designer*

When you open the code editor, it appears in the same part of the screen as the *Designer*. While developing a Visual C# application, you will often find yourself needing to switch back and forth between the *Designer* and the code editor. One way to quickly switch between the two windows is to use the tabs shown in Figure 2-30. In the figure, notice that the leftmost tab reads *Form1.cs*. That is the tab for the code editor. The rightmost tab reads *Form1.cs* [*Design*]. That is the tab for the *Designer*. (The tabs may not always appear in this order.) To switch between the *Designer* and the code editor, you simply click the tab for the desired window.

Figure 2-30 Code editor and *Designer* tabs



You can also detach the code editor and move it to another part of the screen. This allows you to see the code editor and the *Designer* at the same time. As shown in Figure 2-31, click the code editor tab and drag it to the desired location on the screen. (If you have multiple monitors connected to your computer, you can even drag the code editor to a different monitor.) To return the code editor to its position within the IDE, right-click the tab for the source code file in the code editor window and select *Move to Main Document Group*. This is shown in Figure 2-32.

Figure 2-31 Detaching the code editor by clicking and dragging

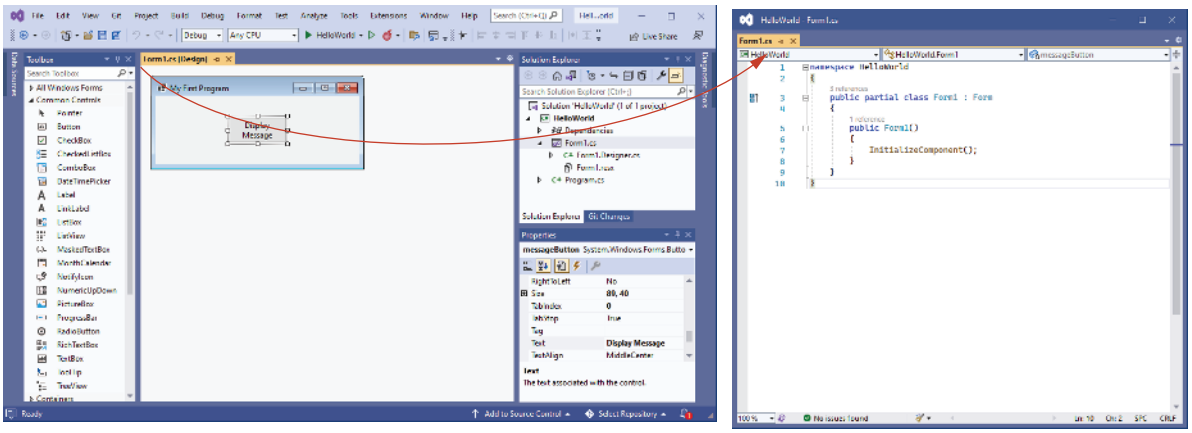
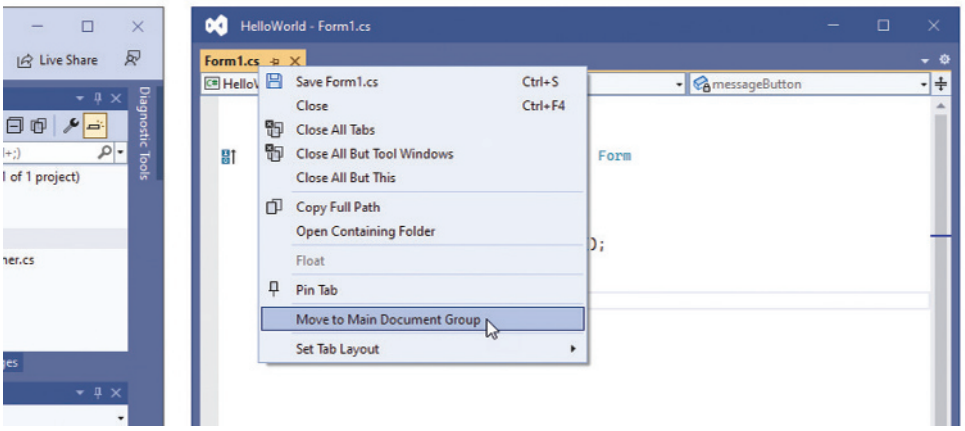


Figure 2-32 Returning the code editor to its docked position



Adding Your Own Code to a Project

Now you are ready to learn how to add your own code to a project. Suppose you have created a project named *CodeDemo* and set up the project's form with a Button control, as shown in Figure 2-33. The Button control's name is `myButton`, and its Text property is set to *Click Me!*.

Suppose you want the application to display the message *Thanks for clicking the button!* when the user clicks the button. To accomplish that, you need to write a special type of method known as an event handler. An **event handler** is a method that executes when a specific event takes place while an application is running. In this project you need to

write an event handler that will execute when the user clicks the `myButton` control. To create the event handler, you double-click the `myButton` control in the *Designer*. This opens the Form1.cs file in the code editor, as shown in Figure 2-34, with some new code added to it.

Figure 2-33 A form with a Button control

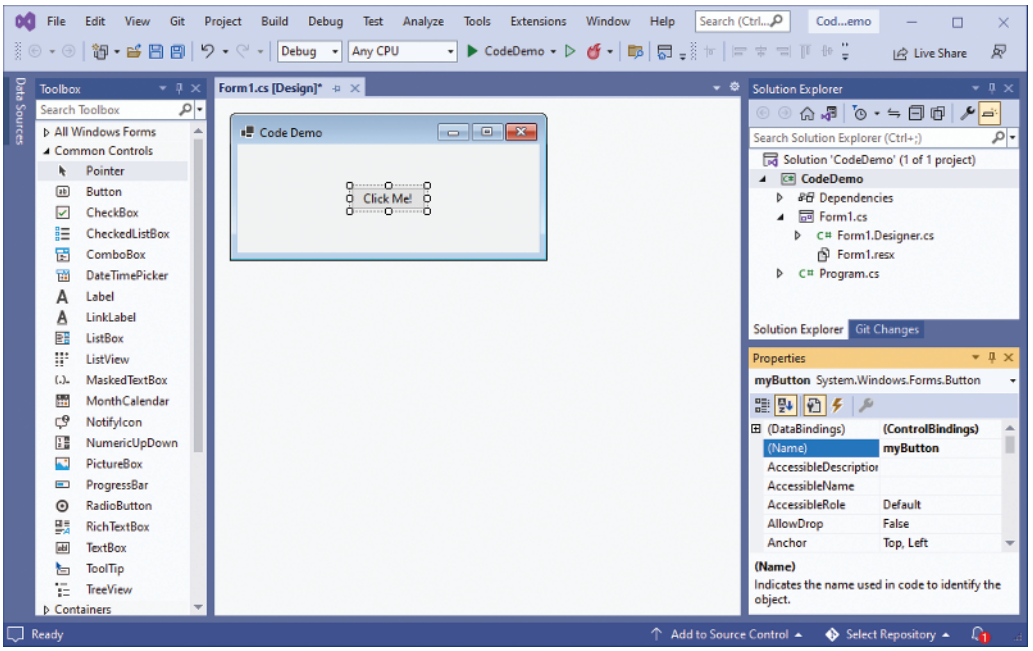
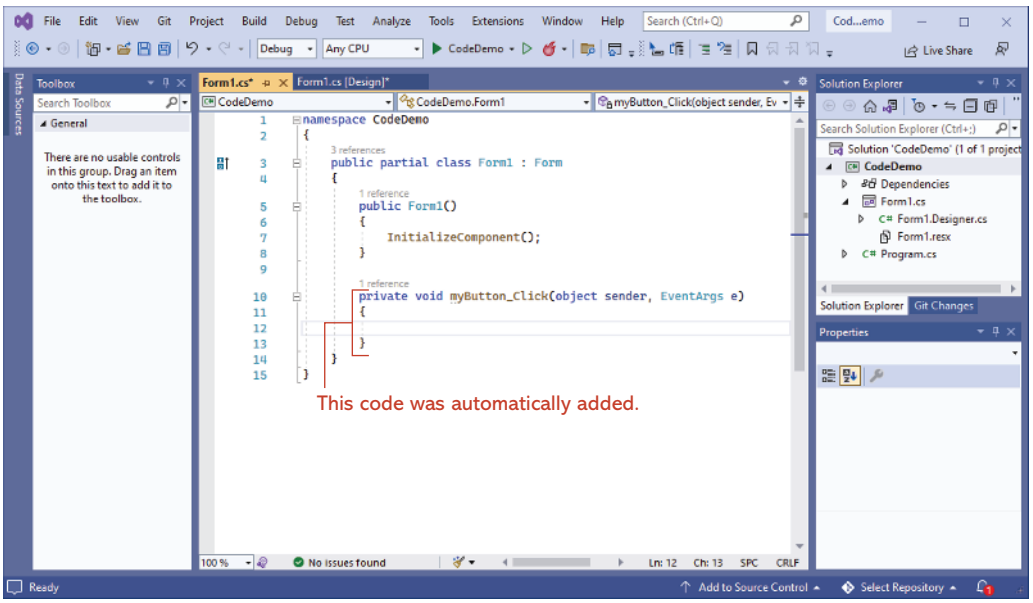


Figure 2-34 The code window opened with event handler code generated

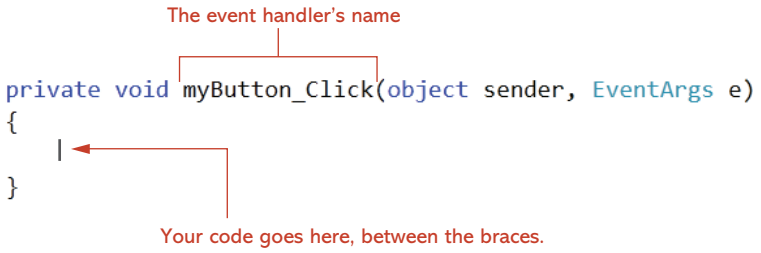


When an application is running and the user clicks a control, we say that a **Click event** has occurred on the control. The code that has been added to the Form1.cs file (shown in Figure 2-34) is an event handler that will execute when a Click event occurs on the

myButton control. For now you do not need to understand all parts of the event handler code. At this point, you need to understand only the following concepts:

- As shown in Figure 2-35, the event handler's name is myButton_Click. The myButton portion of the name indicates that the event handler is associated with the myButton control, and the Click portion of the name indicates that the event handler responds to Click events. This is the typical naming convention that Visual Studio uses when it generates event handler code. When you see the name myButton_Click, you understand that it is an event handler that executes when a Click event occurs on the myButton control.

Figure 2-35 A closer look at the event handler code



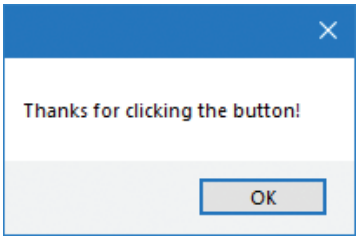
- The event handler that Visual Studio generates doesn't actually do anything. You can think of it as an empty container to which you can add your own code. Notice that the second line of the event handler is an opening brace ({) and the last line is a closing brace (}). Any code that you want executed when the user clicks the myButton control must be written between these braces.

Now you know how to create an empty Click event handler for a Button control. But what code do you write inside the event handler? In this example, we write code that displays the message *Thanks for clicking the button!* in a message box, which is a small pop-up window.

Message Boxes

A **message box** is a small window, sometimes referred to as a **dialog box**, that displays a message. Figure 2-36 shows an example of a message box displaying the message *Thanks for clicking the button!* Notice that the message box also has an *OK* button. When the user clicks the *OK* button, the message box closes.

Figure 2-36 A message box



.NET provides a method named MessageBox.Show that you can use in Visual C# to display a message box. If you want to execute the MessageBox.Show method, you write a statement known as a **method call**. (Programmers refer to the act of executing a method

as *calling* the method.) The following statement shows an example of how you would call the `MessageBox.Show` method to display the message box shown in Figure 2-36:

```
MessageBox.Show("Thanks for clicking the button!");
```

When you call the `MessageBox.Show` method, you write a string of characters inside the parentheses. (In programming, we use the term **string** to mean string of characters.) The string that is written inside the parentheses will be displayed in the message box. In this example, the string `"Thanks for clicking the button!"` is written inside the parentheses.

Notice that the string is enclosed in double quotation marks in the code. When the message is displayed (as shown in Figure 2-36), however, the double quotation marks do not appear. The double quotation marks are required in the code to indicate the beginning and the end of the string.

Also notice that a semicolon appears at the end of the statement. This is required by C# syntax. Just as a period marks the end of a sentence, a **semicolon** marks the end of a programming statement in C#.

Getting back to our *Code Demo* example project, Figure 2-37 shows how you can call the `MessageBox.Show` method from the `myButton_Click` event handler. After typing the statement as shown in the figure, you can press the F5 key on the keyboard or click the *Start Debugging* button (▶) on the toolbar to compile and run the application. When the application runs, it will display the form shown on the left in Figure 2-38. When you click the button, the message box shown on the right in the figure will appear. You can click the *OK* button on the message box to close it.

Figure 2-37 Event handler code for displaying a message box

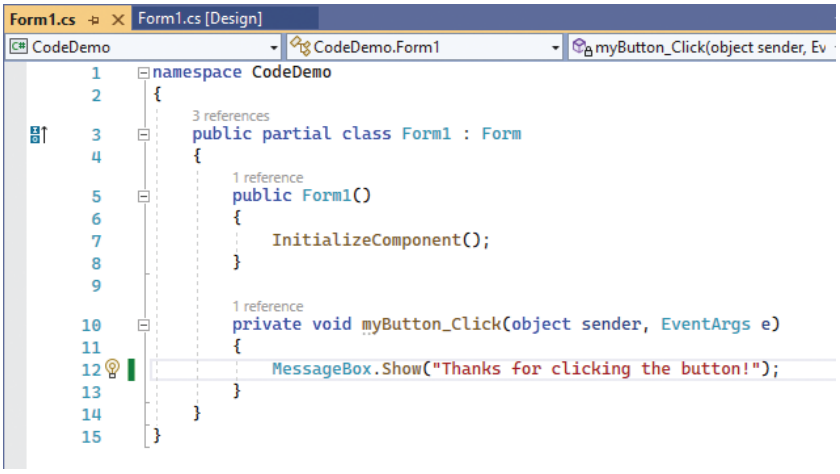
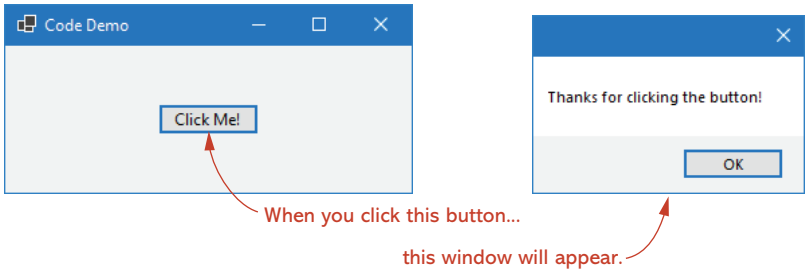


Figure 2-38 The *Code Demo* project running



NOTE: When writing a Click event handler for a Button control, you might be wondering if it is necessary to first double-click the Button control in the *Designer*, creating the empty event handler code. After all, couldn't you just skip this step and instead open the code editor and write all the event handler code yourself? The answer is no, you cannot skip this step. When you double-click a control in the *Designer*, Visual Studio not only creates an empty event handler, but it also writes some code that you don't see, elsewhere in the project. This other code is necessary for the event handler to properly function.

String Literals

Programs almost always work with data of some type. For example, the code shown in Figure 2-37 uses the following string when it calls the `MessageBox.Show` method:

```
"Thanks for clicking the button!"
```

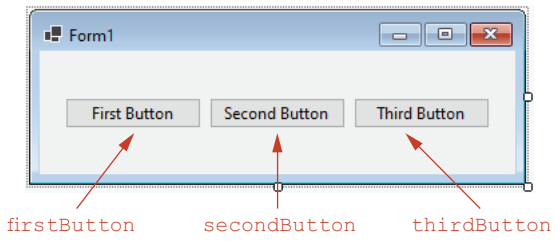
This string is the data that is displayed by the program. When a piece of data is written into a program's code, it's called a **literal** (because the data is *literally* written into the program). When a string is written into a program's code, it's called a *string literal*. In C#, string literals must be enclosed in double quotation marks.

NOTE: Programmers sometimes say that literals are values that are *hard coded* into a program because the value of a literal cannot change while the program is running.

Multiple Buttons with Event Handlers

The *CodeDemo* project previously shown has only one button with a Click event handler. Many of the applications that you will develop will have multiple buttons, each with its own Click event handler. For example, the form shown in Figure 2-39 has three Button controls. As shown in the figure, the controls are named `firstButton`, `secondButton`, and `thirdButton`.

Figure 2-39 A form with multiple Button controls



To create Click event handlers for the buttons, you simply double-click each Button control in the *Designer* and an empty event handler will be created in the form's source code file. The names of the Click event handlers will be `firstButton_Click`, `secondButton_Click`, and `thirdButton_Click`. Figure 2-40 shows an example of the form's source code after the three event handlers have been created and a `MessageBox.Show` statement has been added to each one.

Figure 2-40 Source code with three Click event handlers

```
namespace MultipleButtons
{
    3 references
    public partial class Form1 : Form
    {
        1 reference
        public Form1()
        {
            InitializeComponent();
        }

        1 reference
        private void firstButton_Click(object sender, EventArgs e)
        {
            MessageBox.Show("You clicked the first button.");
        }

        1 reference
        private void secondButton_Click(object sender, EventArgs e)
        {
            MessageBox.Show("You clicked the second button.");
        }

        1 reference
        private void thirdButton_Click(object sender, EventArgs e)
        {
            MessageBox.Show("You clicked the third button.");
        }
    }
}
```

Click event handler for `firstButton`

Click event handler for `secondButton`

Click event handler for `thirdButton`

Design Time and Run Time

When you have a project open in Visual Studio, the time during which you build the GUI and write the application's code is referred to as **design time**. During design time, you can use the *Designer* and the *Toolbox* to place controls on the form, use the *Properties* window to set property values, use the code editor to write code, and so forth. This is the phase during which you create or modify the application.

When you are ready to run a project that you have open in Visual Studio, you press the F5 key on the keyboard or click the *Start Debugging* button (▶) on the toolbar. The project will be compiled, and if there were no errors, it will be executed. The time during which an application is executing is referred to as **run time**. During run time, you can interact with the running application, but you cannot use the *Designer*, the *Toolbox*, the *Properties* window, the code editor, or parts of Visual Studio to make changes to it.

NOTE: In computing literature and on the Web, you will see the term *run time* also spelled as *runtime* or *run-time*. All these variations typically mean the same thing.

Checkpoint

2.15 A file that contains program code is known as what type of file?

2.16 What must you do if you want your application to perform any meaningful actions?

2.17 What does the Program.cs file contain?

2.18 What does the Form1.cs file contain?

2.19 How is C# code organized?

2.20 What is a namespace?

2.21 What characters do code containers, such as namespaces, classes, and methods, use to enclose code?

2.22 How do you switch between the *Designer* and the code editor?

2.23 How do you create an event handler for a button?

2.24 What is a Click event?

2.25 What method do you use in Visual C# to display a message box?

2.26 What is a literal?

2.27 What are string literals enclosed in?

2.28 How do you run a project that you have open in Visual Studio?

2.4 Writing Code for the *Hello World* Application

Now you know everything necessary to complete the *Hello World* application. In Tutorial 2-2, you will open the project and add a Click event handler for the `messageButton` control. The event handler will call the `MessageBox.Show` method to display a message box with the message *Hello World*.

Tutorial 2-2:
Writing Code for the *Hello World* Application

VideoNote
Tutorial 2-2: Writing Code for the *Hello World* Application

Step 1: If Visual Studio is not already running, start it. Open the *HelloWorld* project that you started in Tutorial 2-1.

Step 2: Make sure the Form1 form is visible in the *Designer,* as shown in Figure 2-41. If it is not, right-click Form1.cs in the *Solution Explorer* and then select *View Designer* from the pop-up menu.

Step 3: In the *Designer*, double-click the `messageButton` control. This should cause the code editor to appear as shown in Figure 2-42. Notice that an empty event handler named `messageButton_Click` has been created.

Figure 2-41 The *HelloWorld* project loaded with Form1 shown in the *Designer*

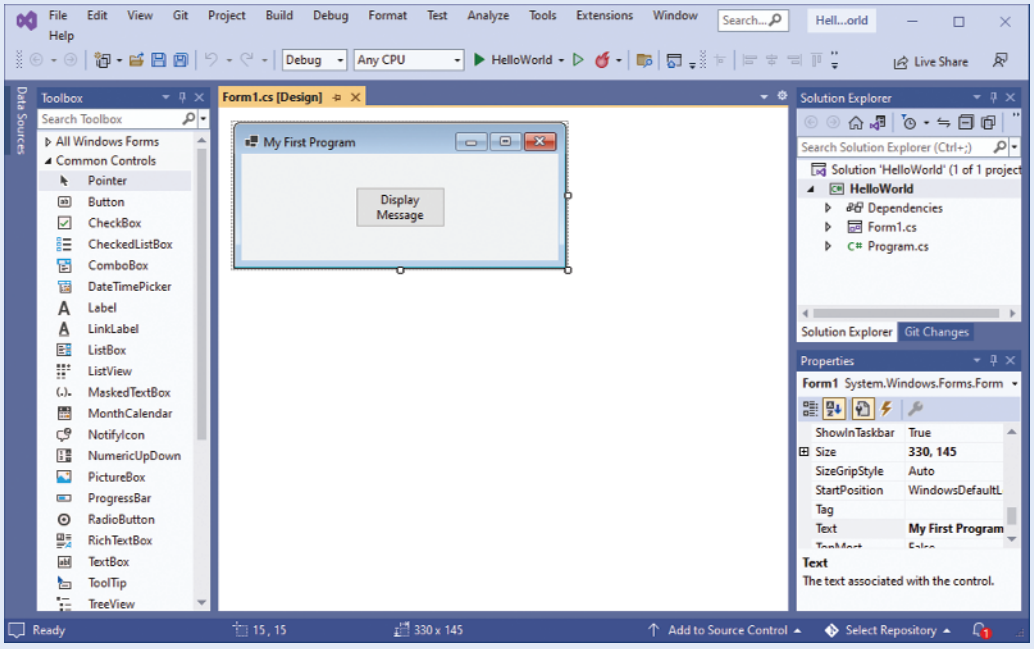
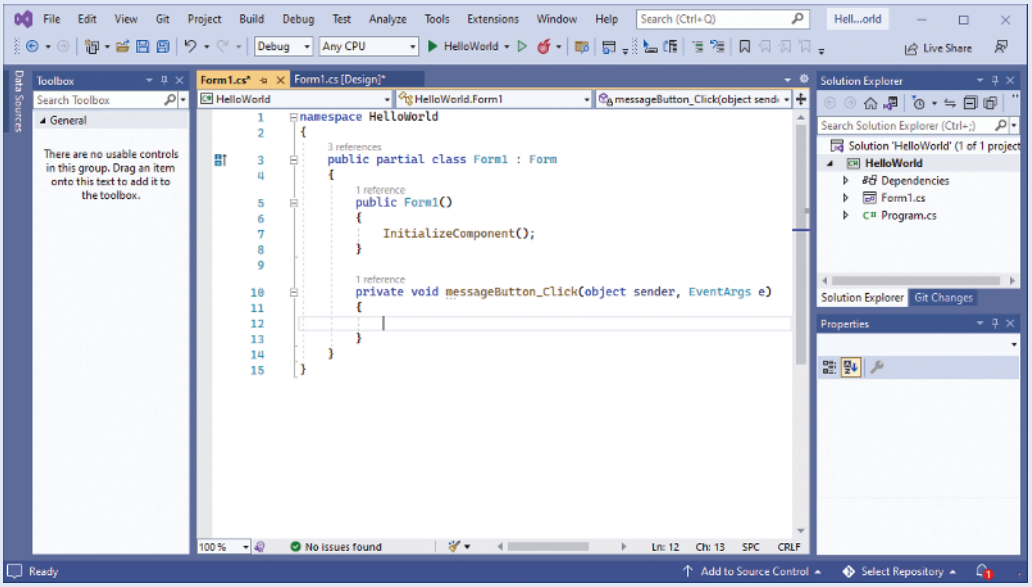


Figure 2-42 Code editor with an empty event handler

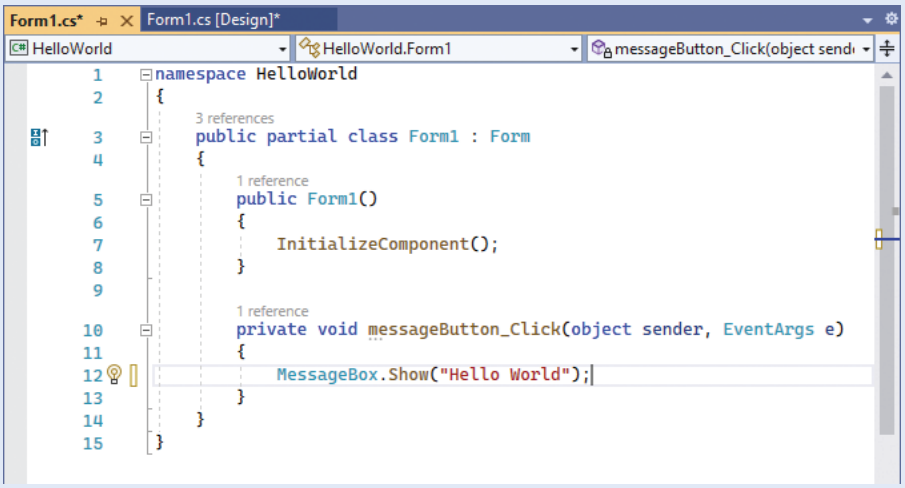


Step 4: Inside the `messageButton_Click` event handler, type the following statement exactly as it is shown:

```
MessageBox.Show("Hello World");
```

Don't forget to type the semicolon at the end of the statement! When you have finished, the code window should look like Figure 2-43.

Figure 2-43 Statement written inside the event handler



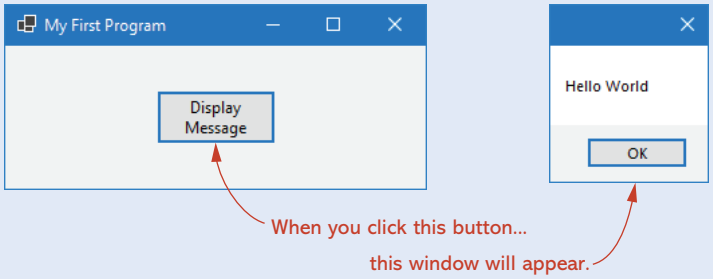
Step 5: Save the project.

Step 6: Press the F5 key on the keyboard or click the *Start Debugging* button ( ) on the toolbar to compile and run the application.

NOTE: If you typed the statement correctly inside the `messageButton_Click` event handler (in Step 4), the application should run. If you did not type the statement correctly, however, a window will appear reporting build errors. If that happens, click the *No* button in the window and then correct the statement so it appears exactly as shown in Figure 2-43.

When the application runs, it will display the form shown on the left in Figure 2-44. When you click the *Display Message* button, the message box shown on the right in the figure will appear. You can click the *OK* button on the message box to close it.

Figure 2-44 The *Hello World* application running

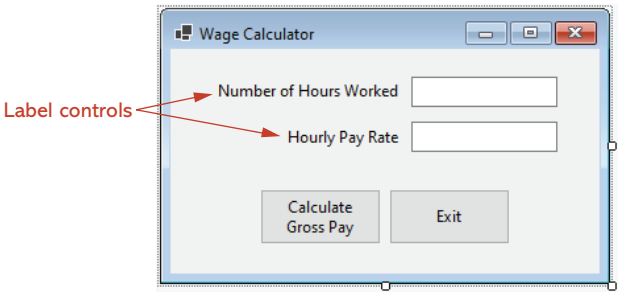


2.5 Label Controls

CONCEPT: **A Label control displays text on a form. Label controls have various properties that affect the control's appearance. Label controls can be used to display unchanging text or program output.**

When you want to display text on a form, you use a **Label control**. Figure 2-45 shows an example of a form with two Label controls. Once you have placed a Label control on a form, you set its Text property to the text that you want to display. For example, in Figure 2-45, the upper Label control's Text property is set to *Number of Hours Worked*, and the lower Label control's Text property is set to *Hourly Pay Rate*.

Figure 2-45 A form with Label controls



You'll find the Label control tool in the *Common Controls* group of the *Toolbox*, as shown in Figure 2-46. To create a Label control on a form, you double-click the Label control tool in the *Toolbox*. As shown in Figure 2-46, a Label control will be created on the form. (Alternatively, you can click and drag the Label control tool from the *Toolbox* onto the form.) Notice that a bounding box appears around the Label control in the figure. This indicates that the control is currently selected.

When you create Label controls, they are automatically given default names such as `label1`, `label2`, and so forth. A Label control's Text property is initially set to the same value as the Label control's name. So, a Label control will display its own name when it is created, as shown by the example in Figure 2-46. When a Label control is selected in the *Designer*, you can use the *Properties* window to change its Text property. Figure 2-47 shows a Label control after its Text property has been changed to *Programming in Visual C# is fun!*

Figure 2-46 Creating a Label control

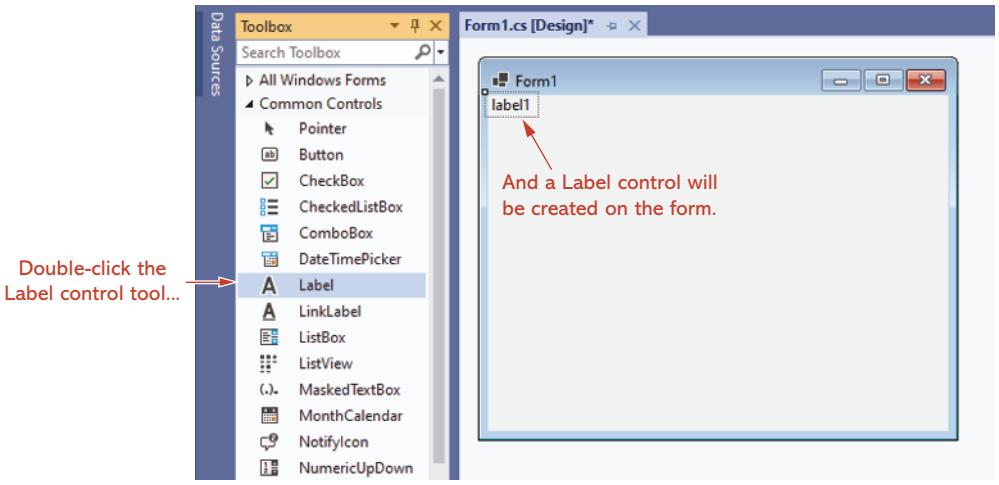
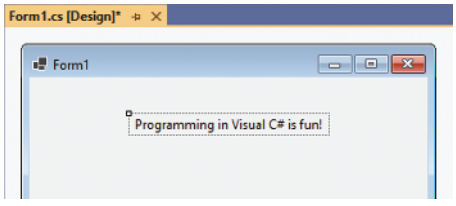


Figure 2-47 A Label control displaying a message

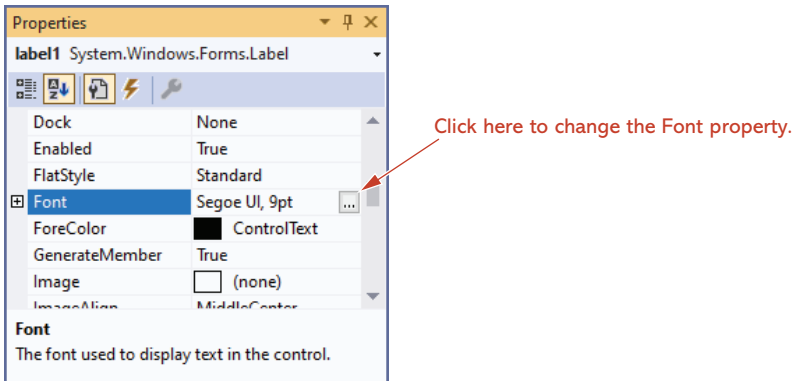


You can also use the *Properties* window to change a Label control's name. It's always a good idea to change a control's name to something that is more meaningful than the default name that Visual Studio gives it.

The Font Property

If you want to change the appearance of a Label control's text, you can change the control's Font property. The Font property allows you to set the font, font style, and size of the control's text. When you select the Font property in the *Properties* window, you will notice that an ellipses button (...) appears next to the property's value, as shown in Figure 2-48.

Figure 2-48 The Font property



When you click the ellipses button, the *Font* dialog box appears, as shown in Figure 2-49. Select a font, font style, and size, and click *OK*. The text displayed by the control will be updated with the selected attributes. For example, Figure 2-50 shows a Label control with the following Font property attributes:

Font: Lucida Handwriting

Font Style: Italic

Size: 10 point

Figure 2-49 The *Font* dialog box

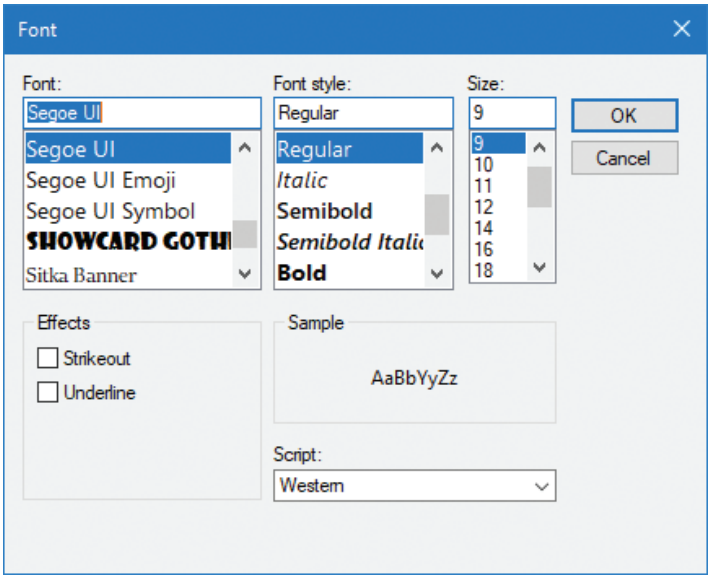


Figure 2-50 A label's appearance with altered font attributes

The BorderStyle Property

Label controls have a BorderStyle property that allows you to display a border around the control's text. The BorderStyle property may have one of three values: None, FixedSingle, or Fixed3D. The property is set to None by default, which means that no border will appear around the control's text. If the BorderStyle property is set to FixedSingle, the control's text will be outlined with a thin border. If the BorderStyle property is set to Fixed3D, the control's text will have a recessed 3D appearance. Figure 2-51 shows an example of Label controls with each BorderStyle setting. The image on the left in the figure shows the form in the Designer, and the image on the right shows the form at runtime.

To change the BorderStyle property, select it in the *Properties* window and then click the down-arrow button ( ) that appears next to the property's value. As shown in Figure 2-52, a drop-down list will appear containing the three possible values for this property. Select the desired value and the control's text will be updated.

Figure 2-51 BorderStyle examples

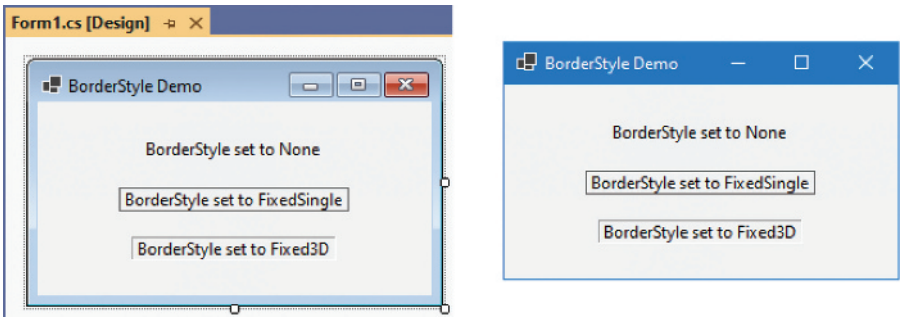
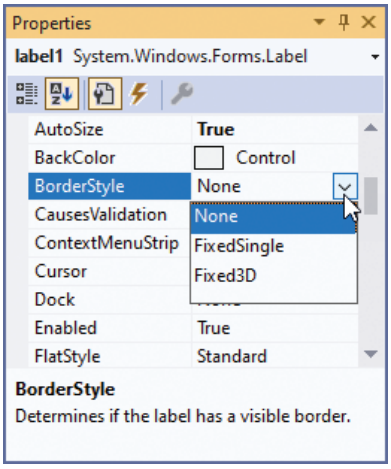


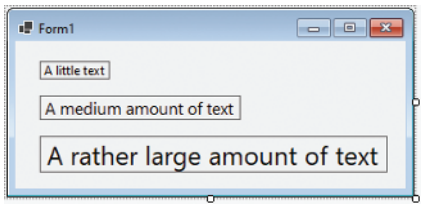
Figure 2-52 BorderStyle selections



The AutoSize Property

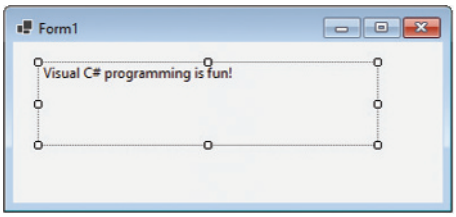
Label controls have an AutoSize property that controls the way they can be resized. The AutoSize property is a **Boolean property**, which means that it can be set to one of two possible values: True or False. By default, a Label control's AutoSize property is set to True, which means that the control automatically resizes itself to accommodate the size of the text it displays. For example, look at the three Label controls in Figure 2-53. Each of the controls displays different amounts of text at different font sizes. Because each control's BorderStyle property is set to FixedSingle, you can see that each control is just large enough to accommodate its text.

Figure 2-53 Label controls with AutoSize set to True



When a Label control's AutoSize property is set to True, you cannot manually change the size of the control by clicking and dragging its bounding box. If you want to manually change the size of a Label control, you have to set its AutoSize property to False. When AutoSize is set to False, sizing handles will appear around the control, allowing you to click and drag the bounding box to resize the control. Figure 2-54 shows an example. In the figure, the Label control has been resized so it is much larger than the text it displays.

Figure 2-54 Label control with AutoSize set to False



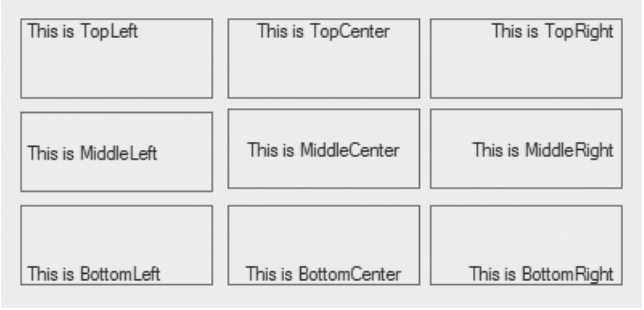
NOTE: When a Label control's AutoSize property is set to True, the label's text will always appear on one line. When the AutoSize property is set to False, the label's text will wrap across multiple lines if it is too long to fit on one line.

The TextAlign Property

When you set a Label control's AutoSize property to False and then manually resize the control, it sometimes becomes necessary to change the way the label's text is aligned. By default, a label's text is aligned with the top and left edges of the label's bounding box. For example, look at the label shown in Figure 2-54. Notice how the text is positioned in the label's upper-left corner.

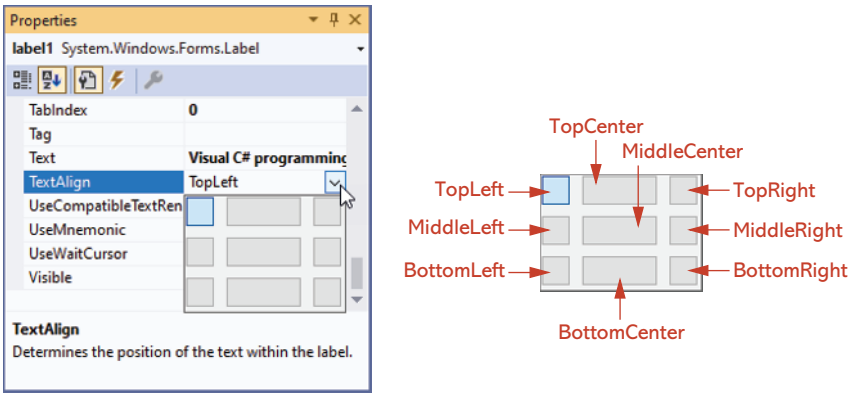
What if we want the text to be aligned differently within the label? For example, what if we want the text to be centered in the label or positioned in the lower-right corner? We can change the text's alignment in the label with the TextAlign property. The TextAlign property may be set to any of the following values: TopLeft, TopCenter, TopRight, MiddleLeft, MiddleCenter, MiddleRight, BottomLeft, BottomCenter, or BottomRight. Figure 2-55 shows nine Label controls, each with a different TextAlign value.

Figure 2-55 Text alignments



To change the TextAlign property, select it in the *Properties* window and then click the down-arrow button ( ) that appears next to its value. This causes a dialog box with nine buttons, as shown in the left image in Figure 2-56, to appear. As shown in the right image in the figure, the nine buttons represent the valid settings of the TextAlign property.

Figure 2-56 Setting the TextAlign property



Using Code to Display Output in a Label Control

In addition to displaying unchanging text on a form, Label controls are also useful for displaying output while an application is running. For example, suppose you are creating an application that performs a calculation and you want to display the result of the calculation at a specific location on the form. Using a Label control to display the output would be an ideal solution. Here are the general steps that you would follow:

Step 1: While creating the application's GUI, you place a Label control on the form at the location where you want the result to be displayed. Then, in the *Properties* window, you erase the contents of the Label control's Text property. Because the control's Text property is empty, the control will not initially display anything when the application runs.

Step 2: In the application's code, you write the necessary statements to perform the calculation and then you store the result of the calculation in the Label control's Text property. This causes the result to be displayed on the form in the Label control.

NOTE: We do not discuss calculations until Chapter 3, so in this chapter we look at examples that display nonmathematical data as output in Label controls.

In code, you use an **assignment statement** to store a value in a control's property. For example, suppose you have created a Label control and named it `outputLabel`. The following assignment statement stores the string `"Thank you very much"` in the control's Text property.

```
outputLabel.Text = "Thank you very much";
```

The equal sign (=) is known as the **assignment operator**. It assigns the value that appears on its right side to the item that appears on its left side. In this example, the item on the left side of the assignment operator is the expression `outputLabel.Text`. This is simply the `outputLabel` control's Text property. The value on the right side of the assignment operator is the string `"Thank you very much"`. When this statement

executes, the string "Thank you very much" is assigned to the outputLabel control's Text property. When this statement executes, the text *Thank you very much* is displayed in the Label control.

WARNING! When writing assignment statements, remember that the item receiving the value must be on the left side of the = operator. The following statement, for example, is wrong and will cause an error when you compile the program:

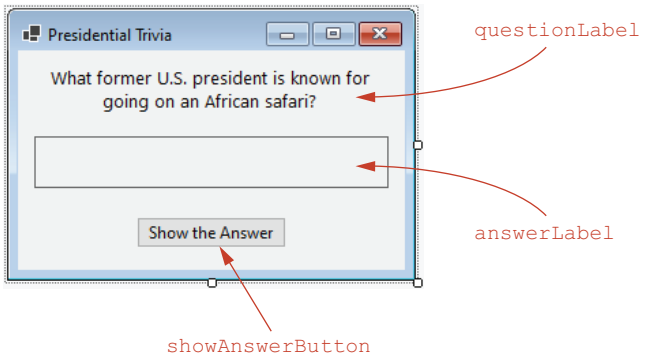
```
"Thank you very much" = outputLabel; ← ERROR!
```

NOTE: The standard notation for referring to a control's property in code is:

ControlName.PropertyName

Let's look at an example application that uses a Label control to display output. Make sure you have downloaded the student sample programs from the book's companion Web site (at www.pearsonhighered.com/gaddis). In the *Chap02* folder, you will find a project named *Presidential Trivia*. The purpose of the application is to display a trivia question about a former U.S. president. When the user clicks a button, the answer to the trivia question is displayed on the form. The project's form appears as shown in Figure 2-57.

Figure 2-57 Presidential Trivia form



As shown in the figure, the form has the three controls:

- A Label control named questionLabel. This label displays the trivia question.
- A Label control named answerLabel. This label initially appears empty, but will be used to display the answer to the trivia question.
- A Button control named showAnswerButton. When the user clicks this button, the answer to the trivia question is displayed.

Table 2-2 lists the property settings for each control of which you should take note.

If we open the Form1.cs file in the code editor, we see the code shown in Figure 2-58. (To open the file in the code window, right-click Form1.cs in the *Solution Explorer* and then select *View Code*.) Notice the method named showAnswerButton_Click in lines 10 through 13. This is the Click event handler for the showAnswerButton control. It contains the following statement in line 12:

```
answerLabel.Text = "Theodore Roosevelt";
```

When this statement executes, it assigns the string "Theodore Roosevelt" to the answerLabel control's Text property. As a result, Theodore Roosevelt is displayed in the label control.

Table 2-2 Control property settings

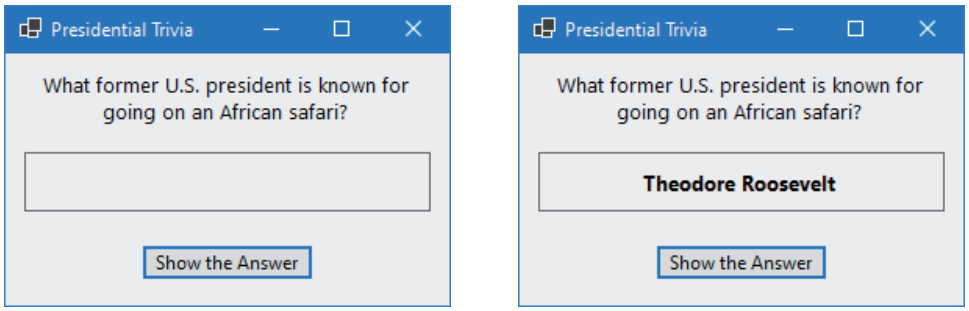
Control Name	Control Type	Property Settings
`questionLabel`	Label	**AutoSize:** False **BorderStyle:** None **Font:** Segoe UI (Style: Regular, Size: 10 point) **Text:** *What former U.S. president is known for going on an African safari?* **TextAlign:** MiddleCenter
`answerLabel`	Label	**AutoSize:** False **BorderStyle:** FixedSingle **Font:** Segoe UI (Style: Bold, Size: 10 point) **Text:** (The contents of the Text property have been erased.) **TextAlign:** MiddleCenter
`showAnswerButton`	Button	**Size:** 110, 23 **Text:** *Show the Answer*

Figure 2-58 Form1.cs code

```
1   namespace Presidential_Trivia
2   {
        3 references
3       public partial class Form1 : Form
4       {
            1 reference
5           public Form1()
6           {
7               InitializeComponent();
8           }
9
            1 reference
10          private void showAnswerButton_Click(object sender, EventArgs e)
11          {
12              answerLabel.Text = "Theodore Roosevelt";
13          }
14      }
15  }
```

When you run the application, the form appears as shown on the left in Figure 2-59. Click the *Show the Answer* button and the answer to the trivia question appears as shown on the right in the figure.

Figure 2-59 The *Presidential Trivia* application running



The Text Property Accepts Strings Only

It is important to point out that the Label control's Text property can accept strings only. You cannot assign a number to the Text property. For example, let's assume that an application has a Label control named `resultLabel`. The following statement will cause an error because it is attempting to store the number 5 in the `resultLabel` control's Text property:

```
resultLabel.Text = 5; ← ERROR!
```

This does not mean that you cannot display a number in a label, however. If you put quotation marks around the number, it becomes a string. The following statement will work:

```
resultLabel.Text = "5";
```

Clearing a Label

In code, if you want to clear the text that is displayed in a Label control, simply assign an empty string (`""`) to the control's Text property, as shown here:

```
answerLabel.Text = "";
```

In Tutorial 2-3, you will work with some of the Label control properties that we have discussed in this section.

Tutorial 2-3:
Creating the *Translator* Application

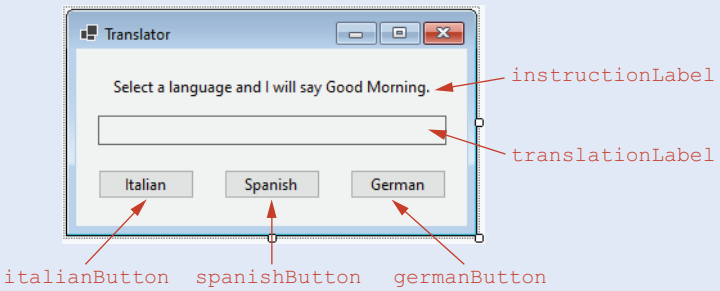
VideoNote
Tutorial 2-3: Creating the *Translator* Application

In this tutorial, you will create an application that displays the phrase "Good Morning" in different languages. The form will have three buttons: one for Italian, one for Spanish, and one for German. When the user clicks any of these buttons, the translated phrase will appear in a Label control.

Step 1: Start Visual Studio and begin a new Windows Forms App project named *Translator*.

Step 2: Set up the application's form as shown in Figure 2-60. Notice that the form's Text property is set to *Translator*. The form has two Label controls and three Button controls. The names of the controls are shown in the figure. As you place each of the controls on the form, refer to Table 2-3 for the relevant property settings.

Figure 2-60 The Translator form



Step 3: Once you have the form and its controls set up, you can create the Click event handlers for the Button controls. In the *Designer*, double-click the `italianButton` control. This will open the code editor, and you will see an empty event handler named `italianButton_Click`. Write the following statement inside the event handler:

```
translationLabel.Text = "Buongiorno";
```

Table 2-3 Control property settings

Control Name	Control Type	Property Settings
`instructionLabel`	Label	**Text:** *Select a language and I will say Good Morning.*
`translationLabel`	Label	**AutoSize:** False **BorderStyle:** FixedSingle **Font:** Segoe UI (Style: Bold, Size: 10 point) **Text:** (The contents of the Text property have been erased.) **TextAlign:** MiddleCenter
`italianButton`	Button	**Text:** *Italian*
`spanishButton`	Button	**Text:** *Spanish*
`germanButton`	Button	**Text:** *German*

Step 4: Switch your view back to the *Designer* and double-click the `spanishButton` control. In the code editor you will see an empty event handler named `spanishButton_Click`. Write the following statement inside the event handler:

```
translationLabel.Text = "Buenos Dias";
```

Step 5: Switch your view back to the *Designer* and double-click the `germanButton` control. In the code editor you will see an empty event handler named `germanButton_Click`. Write the following statement inside the event handler:

```
translationLabel.Text = "Guten Morgen";
```

Step 6: The form's code should now appear as shown in Program 2-1. Note that the line numbers are not part of the code. The line numbers are shown so that you and your instructor can more easily refer to different parts of the program. The lines that appear in boldface are the ones that you typed. Make sure the code you typed appears exactly as shown here. (Don't forget the semicolons!)

Program 2-1 Completed Form1 code for the *Translator* project

```
 1 namespace Translator
 2 {
 3   public partial class Form1 : Form
 4   {
 5     public Form1()
 6     {
 7       InitializeComponent();
 8     }
 9
10     private void italianButton_Click(object sender, Even-
       tArgs e)
11     {
12       translationLabel.Text = "Buongiorno";
13     }
14
```

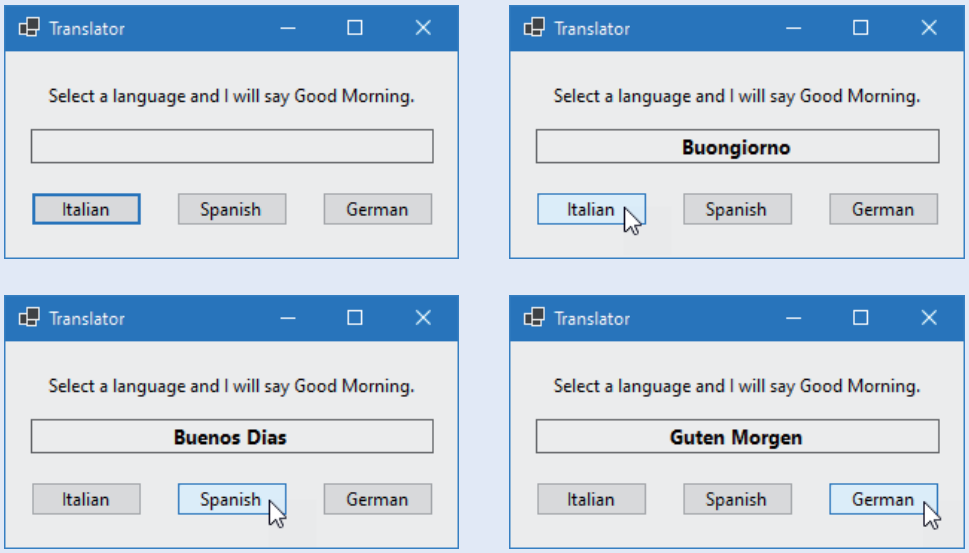
```
15      private void spanishButton_Click(object sender,
        EventArgs e)
16      {
17        translationLabel.Text = "Buenos Dias";
18      }
19
20      private void germanButton_Click(object sender,
        EventArgs e)
21      {
22        translationLabel.Text = "Guten Morgen";
23      }
24    }
25 }
```

Step 7: Save the project. Then, press the F5 key on the keyboard or click the *Start Debugging* button (▶) on the toolbar to compile and run the application.

NOTE: If you typed the statements correctly inside the event handlers, the application should run. If you did not type the statements correctly inside the event handlers, a window will appear reporting build errors. If that happens, click the *No* button in the window and then correct the code so it appears exactly as previously shown.

Figure 2-61 shows the application's form when it starts running and after you have clicked each of the Button controls. After you have tested each button, close the application's form.

Figure 2-61 The *Translator* application running



Checkpoint

2.29 In which group of the *Toolbox* can you find the Label control tool?

2.30 Once you have placed a Label control on a form, which property do you use to set the text that you want to display?

2.31 What property can you use to change the appearance of a Label control's text?

2.32 What is the default value of a label's BorderStyle property?

2.33 How do you change the BorderStyle property of a control in the *Properties* window?

2.34 What property determines whether a label can be resized?

2.35 What property determines the way text is aligned in a Label control?

2.36 How can you use a Label control to display output while a program is running?

2.37 What happens if you assign an empty string to a control's Text property in code?

2.6 Making Sense of IntelliSense

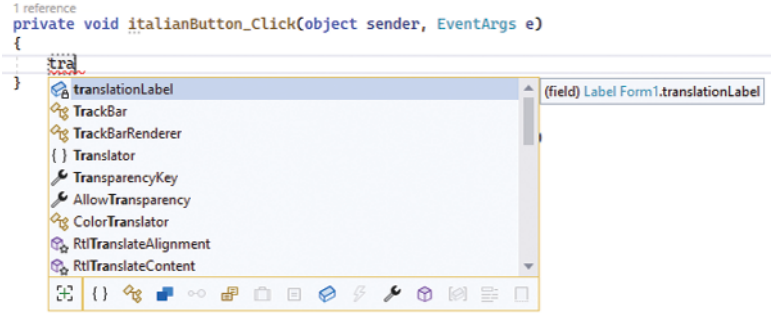
CONCEPT: **As you type code in the Visual Studio code editor, IntelliSense boxes pop up to assist you. You can use the IntelliSense boxes to automatically complete some programming statements after typing only the first few characters.**

IntelliSense is a feature of Visual Studio that provides automatic code completion as you write programming statements. Once you learn how to use IntelliSense, it helps you write code faster. If you've worked through the previous tutorials in this chapter, you've already encountered IntelliSense. For example, in Step 3 of Tutorial 2-3, you were instructed to write the following statement in the `italianButton_Click` event handler:

```
translationLabel.Text = "Buongiorno";
```

Did you notice that as soon as you started typing the statement, a box popped up on the screen? This is known as an IntelliSense list box. The contents of the list box changes as you type. Figure 2-62 shows the IntelliSense list box after you have typed the characters `tra`.

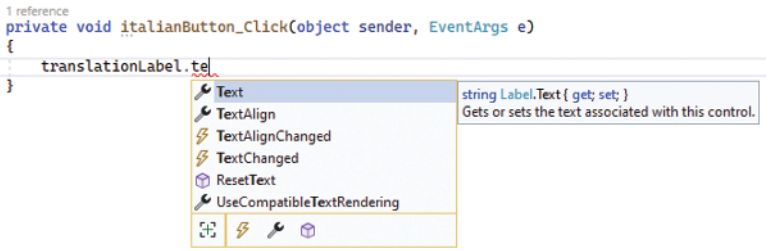
Figure 2-62 IntelliSense list box displayed



The IntelliSense system is anticipating what you are about to type, and as you type characters, the content of the list box is reduced. The list box shown in Figure 2-62 shows all the names starting with `tra` that might be a candidate for the statement you are typing. Notice that `translationLabel` is selected in the list box. With that item selected, you can press the Tab key on the keyboard, and the `tra` that you previously typed becomes `translationLabel`.

Next, when you type a period, an IntelliSense list pops up showing every property and method belonging to the `translationLabel` control. Type `te` and the Text property becomes selected, as shown in Figure 2-63. When you press the Tab key to select the Text property, your statement automatically becomes `translationLabel.Text`. At this point, you can continue typing until you have completed the statement.

Figure 2-63 IntelliSense list box after typing ".te"



Now that you have an idea of how IntelliSense works, you are encouraged to experiment with it as you write code in future projects. With a little practice, it will become intuitive.

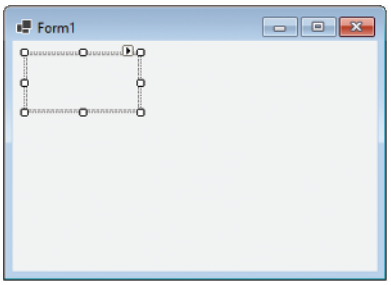
2.7 PictureBox Controls

CONCEPT: **A PictureBox control displays a graphic image on a form. PictureBox controls have properties for controlling the way the image is displayed. A PictureBox control can have a Click event handler that responds when the user clicks the control at run time.**

You can use a **PictureBox control** to display a graphic image on a form. A PictureBox control can display images that have been saved in the bitmap, GIF, JPEG, metafile, or icon graphics formats.

In the *Toolbox*, the PictureBox tool is located in the *Common Controls* group. When you double-click the tool, an empty PictureBox control is created on the form, as shown in Figure 2-64. Although the control does not yet display an image, it has a bounding box that shows its size and location, as well as sizing handles. When you create PictureBox controls, they are automatically given default names such as `pictureBox1`, `pictureBox2`, and so forth. You should always change the default name to something more meaningful.

Figure 2-64 An empty PictureBox control

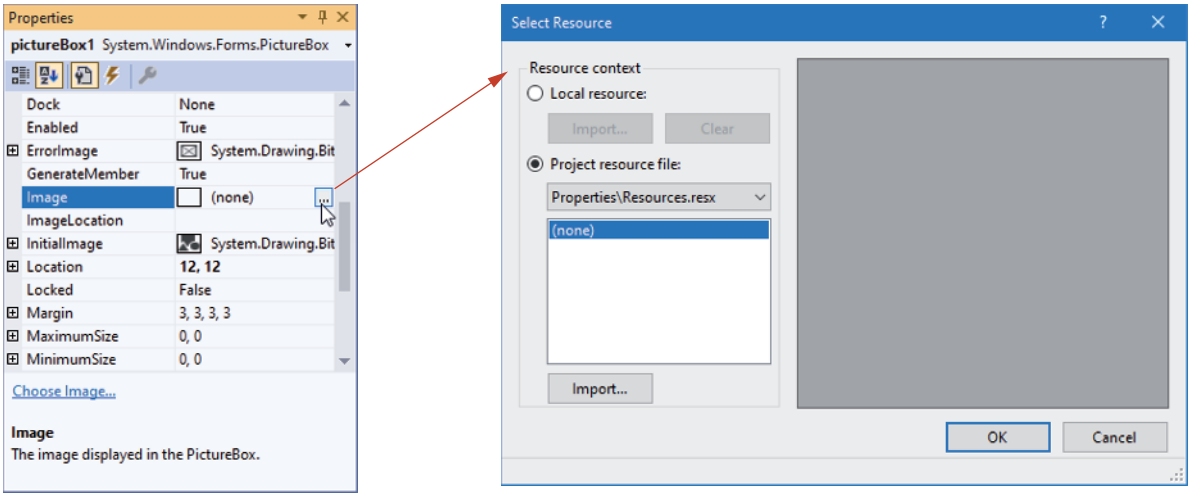


Once you have created a PictureBox control, you use its Image property to specify the image that it will display. Follow these steps:

Step 1: Click the Image property in the *Properties* window. An ellipses button (...) will appear, as shown on the left in Figure 2-65.

Step 2: Click the ellipses button and the *Select Resource* window, shown on the right in Figure 2-65, will appear.

Figure 2-65 The Image property's *Select Resource* window



Step 3: In the *Select Resource* window, click the *Import* button. An *Open* dialog box will appear. Use the dialog box to locate and select the image file that you want to display.

Step 4: After you select an image file, you will see its contents displayed in the *Select Resource* window. This indicates that the image has been imported into the project. Figure 2-66 shows an example of the *Select Resource* window after we have selected and imported an image.

Step 5: Click the *OK* button in the *Select Resource* window, and the selected image will appear in the PictureBox control. Figure 2-67 shows an example. Depending on the size of the image, you might see only part of it displayed. This is the case in Figure 2-67 because the image is larger than the PictureBox control. Your next step is to set the SizeMode property and adjust the size of the control.

Figure 2-66 An image selected and imported

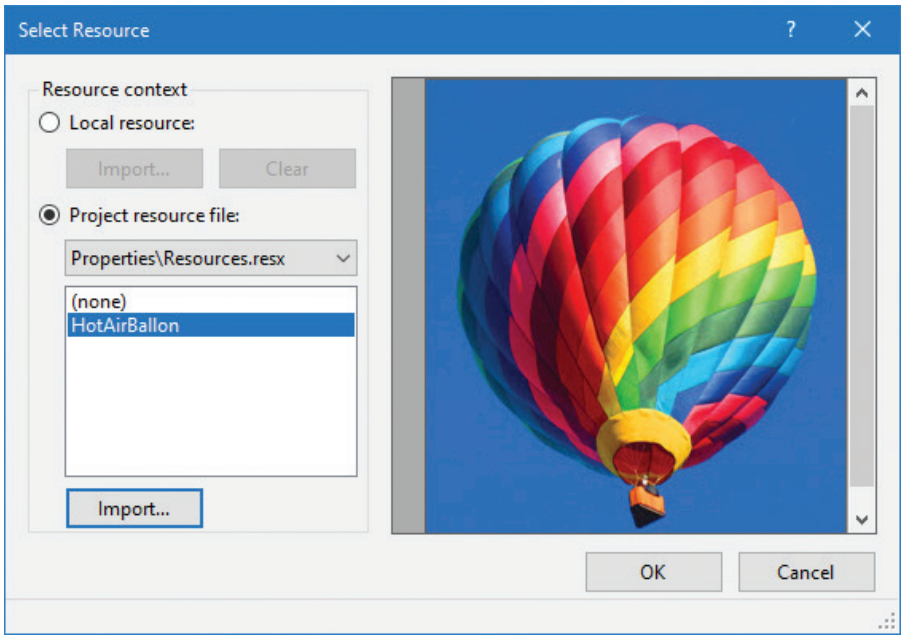
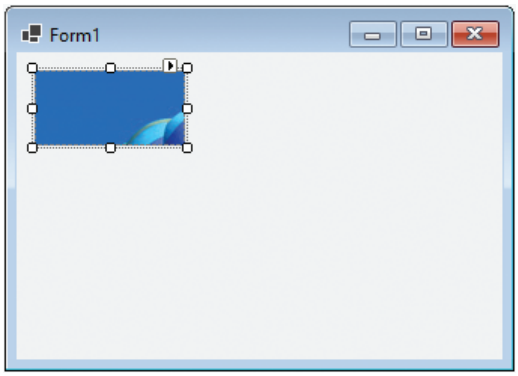


Figure 2-67 The image partially displayed in the PictureBox control



The SizeMode Property

The PictureBox control's SizeMode property specifies how the control's image is to be displayed. It can be set to one of the following values:

- **Normal**

  Normal is the default value. The image will be positioned in the upper-left corner of the PictureBox control. If the image is too big to fit in the PictureBox control, it will be clipped.

- **StretchImage**

  StretchImage resizes the image both horizontally and vertically to fit in the PictureBox control. If the image is resized more in one direction than the other, it will appear stretched.

- **AutoSize**

  With AutoSize, the PictureBox control is automatically resized to fit the size of the image.

- **CenterImage**

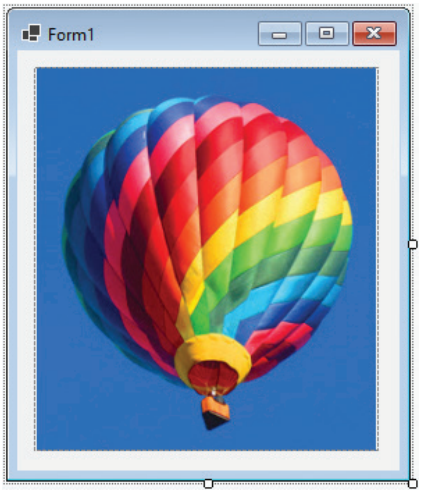
  CenterImage centers the image in the PictureBox control without resizing it.

- **Zoom**

  Zoom uniformly resizes the image to fit in the PictureBox without losing its original aspect ratio. (**Aspect ratio** is the image's width to height ratio.) This causes the image to be resized without appearing stretched.

Figure 2-68 shows an example of an image displayed in a PictureBox control. The control's SizeMode is set to Zoom, so it can be resized without appearing stretched.

Figure 2-68 An image resized with SizeMode set to Zoom



NOTE: PictureBox controls also have a BorderStyle property that works just like a Label control's BorderStyle property.

Creating Clickable Images

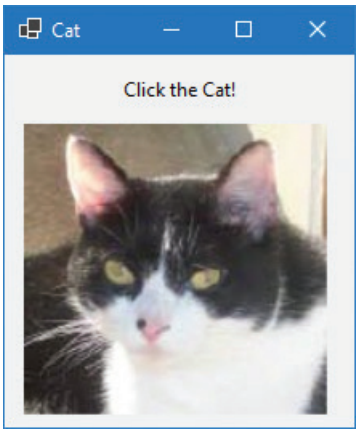
Buttons aren't the only controls that can respond to Click events. PictureBox controls can, too. That means an application can display an image and perform some action when the user clicks the image.

To make an image clickable, you simply have to create a Click event handler for the PictureBox control that displays the image. You create a Click event handler for a PictureBox control in the same way that you create a Click event handler for a Button control:

- You double-click the PictureBox control in the *Designer*. This creates an empty Click event handler in the form's source code file.
- In the code editor you write statements inside the event handler that you want to execute when the image is clicked.

As an example, look at the *Cat* project that is in the *Chap02* folder of the Student Sample Programs that accompany this textbook. Figure 2-69 shows the application's form. The PictureBox control's name is `catPictureBox`. Its image is the Cat.jpg file, which is also found in the *Chap02* folder of the Student Sample Programs. The SizeMode property is set to Zoom, and the BorderStyle property is set to FixedSingle.

Figure 2-69 The Cat form



Open the Form1.cs file in the code editor and you will see that we have already created a Click event handler for the `catPictureBox` control, as shown in Figure 2-70. If you run the application and click the PictureBox, a message box will appear displaying the string *Meow*.

Figure 2-70 Code for the *Cat* project's Form1.cs file

```
1   namespace Cat
2   {
        3 references
3       public partial class Form1 : Form
4       {
            1 reference
5           public Form1()
6           {
7               InitializeComponent();
8           }
9
            1 reference
10          private void catPictureBox_Click(object sender, EventArgs e)
11          {
12              MessageBox.Show("Meow");
13          }
14      }
15  }
```

Click event handler for the `catPictureBox` control

Tutorial 2-4 gives you a chance to practice using PictureBox controls. In the tutorial, you will create an application with three clickable PictureBox controls, displaying images that are provided in the Student Sample Program files that accompany this book.

Tutorial 2-4:
Creating the *Flags* Application

VideoNote
Tutorial 2-4: Creating the *Flags* Application

In this tutorial, you will create an application that displays the flags of Finland, France, and Germany in PictureBox controls. When the user clicks any of these PictureBoxes, the name of that flag's country will appear in a Label control.

Step 1: Start Visual Studio and begin a new Windows Forms App project named *Flags*.

Step 2: Set up the application's form as shown in Figure 2-71. Notice that the form's Text property is set to *Flags*. The names of the controls are shown in the figure. Refer to Table 2-4 for each control's relevant property settings.

Figure 2-71 The Flags form

Table 2-4 Control property settings

Control Name	Control Type	Property Settings
`instructionLabel`	Label	**Text:** *Click a flag to see the name of the country.*
`finlandPictureBox`	PictureBox	**Image:** Select and import the Finland.bmp file from the *Chap02* folder of the Student Sample Programs. **BorderStyle:** FixedSingle **SizeMode:** AutoSize
`francePictureBox`	PictureBox	**Image:** Select and import the France.bmp file from the *Chap02* folder of the Student Sample Programs. **BorderStyle:** FixedSingle **SizeMode:** AutoSize
`germanyPictureBox`	PictureBox	**Image:** Select and import the Germany.bmp file from the *Chap02* folder of the Student Sample Programs. **BorderStyle:** FixedSingle **SizeMode:** AutoSize
`countryLabel`	Label	**AutoSize:** False **BorderStyle:** FixedSingle **Font:** Segoe UI (Style: Bold, Size: 10 point) **Text:** (The contents of the Text property have been erased.) **TextAlign:** MiddleCenter

Step 3: Once you have the form and its controls set up, you can create the Click event handlers for the PictureBox controls. In the *Designer*, double-click the `finlandPictureBox` control. This will open the code editor, and you will

see an empty event handler named `finlandPictureBox_Click`. Write the following statement inside the event handler:

```
countryLabel.Text = "Finland";
```

Step 4: Switch your view back to the *Designer* and double-click the `francePictureBox` control. This will open the code editor, and you will see an empty event handler named `francePictureBox_Click`. Write the following statement inside the event handler:

```
countryLabel.Text = "France";
```

Step 5: Switch your view back to the *Designer* and double-click the `germanyPictureBox` control. This will open the code editor, and you will see an empty event handler named `germanyPictureBox_Click`. Write the following statement inside the event handler:

```
countryLabel.Text = "Germany";
```

Step 6: The form's code should now appear as shown in Program 2-2. As was mentioned in the previous tutorial, the line numbers are shown for reference only, and are not part of the code. The lines that appear in boldface are the ones that you typed. Make sure the code you typed appears exactly as shown here. (Don't forget the semicolons!)

Program 2-2 Completed Form1 code for the *Flags* project

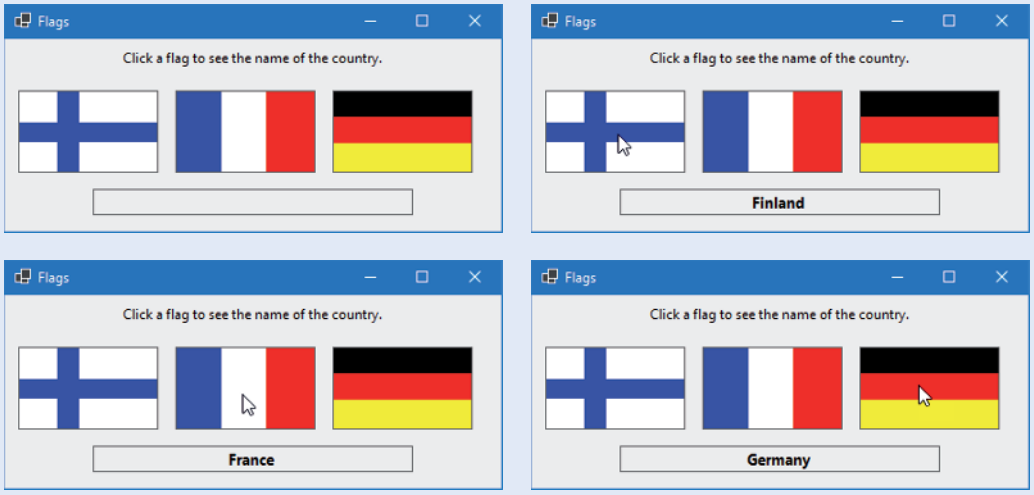
```
 1 namespace Flags
 2 {
 3   public partial class Form1 : Form
 4   {
 5     public Form1()
 6     {
 7       InitializeComponent();
 8     }
 9
10     private void finlandPictureBox_Click(object sender,
       EventArgs e)
11     {
12       countryLabel.Text = "Finland";
13     }
14
15     private void francePictureBox_Click(object sender, Even-
       tArgs e)
16     {
17       countryLabel.Text = "France";
18     }
19
20     private void germanyPictureBox_Click(object sender,
       EventArgs e)
21     {
22       countryLabel.Text = "Germany";
23     }
24   }
25 }
```

Step 7: Save the project. Then, press the F5 key on the keyboard or click the *Start Debugging* button ( ) on the toolbar to compile and run the application.

NOTE: If you typed the statements correctly inside the event handlers, the application should run. If you did not type the statements inside the event handlers correctly, a window will appear reporting build errors. If that happens, click the *No* button in the window, then correct the code so it appears exactly as previously shown.

Figure 2-72 shows the application's form when it starts running and then after you have clicked each of the PictureBox controls. After you have clicked each flag to make sure the application works correctly, close the form.

Figure 2-72 The *Flags* application running



The Visible Property

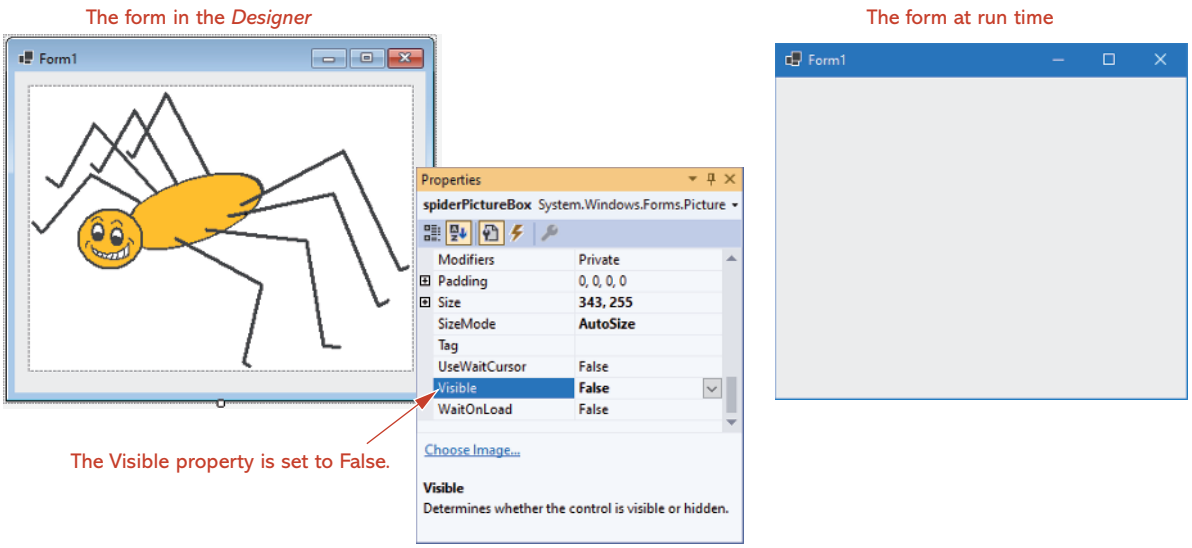
Most controls have a Visible property that determines whether the control is visible on the form at run time. The Visible property is a Boolean property, which means it can be set only to the values True or False. If a control's Visible property is set to True, the control will be visible on the form at run time. If a control's Visible property is set to False, however, the control will not be visible at run time. By default, the Visible property is set to True.

When you use the *Properties* window to change a control's Visible property at design time, the control will still be visible in the *Designer*. When you run the application, however, the control will not be visible on the form. For example, the image on the left in Figure 2-73 shows a form in the *Designer*. The PictureBox control's Visible property is set to False, but the control can still be seen in the *Designer*. The image on the right shows the form while the application is running. At run time, the control is not visible.

A control's Visible property can also be modified in code by an assignment statement, which makes it possible to hide or display a control while the application is running. For example, the PictureBox control shown in Figure 2-73 is named `spiderPictureBox`. The following statement sets the control's Visible property to `true`:

```
spiderPictureBox.Visible = true;
```

Figure 2-73 A PictureBox control with its Visible property set to False



When this statement executes, the `spiderPictureBox` control will become visible. Likewise, the following statement sets the control's Visible property to `false`:

```
spiderPictureBox.Visible = false;
```

When this statement executes, the `spiderPictureBox` control will become invisible.

NOTE: When you write the values `true` and `false` in code, as shown in the previous assignment statement, they must be written in all lowercase letters. The words `true` and `false` are C# keywords, and an error will occur if you don't write them in lowercase. However, when you use the *Properties* window to set a Boolean property, such as Visible, the values True and False will be shown with an initial capital. Try not to let this inconsistency confuse you!

In Tutorial 2-5, you will create an application that uses the Visible property of two PictureBox controls to simulate a card being flipped over.

Tutorial 2-5:
Creating the *Card Flip* Application

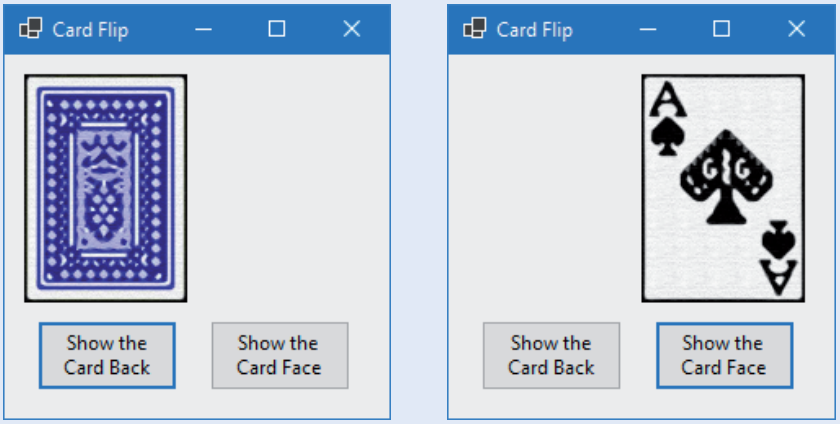
VideoNote
Tutorial 2-5: Creating the *Card Flip* Application

In this tutorial, you will create an application that simulates a card being flipped over. When the application runs, it will display the form shown on the left in Figure 2-74. The form initially displays the back of a poker card. When the user clicks the *Show the Card Face* button, the card will be flipped over to show its face, as shown in the form on the right. When the user clicks the *Show the Card Back* button, the card is flipped back over to show its back.

The simulation of the card being flipped will be accomplished using the following logic:

- When the user clicks the *Show the Card Face* button, the PictureBox showing the card's back will be made invisible and the PictureBox showing the card's face will be made visible.
- When the user clicks the *Show the Card Back* button, the PictureBox showing the card's face will be made invisible and the PictureBox showing the card's back will be made visible.

Figure 2-74 The *Card Flip* application



Step 1: Start Visual Studio and begin a new Windows Forms App project named *CardFlip*.

Step 2: Set up the application's form as shown in Figure 2-75. Notice that the form's Text property is set to *Card Flip*. The names of the controls are shown in the figure. Use the *Properties* window to make the property settings shown in Table 2-5. (In particular, note that the `cardBackPictureBox` control's Visible property is set to True, and the `cardFacePictureBox` control's Visible property is set to False.)

Figure 2-75 The *CardFlip* project's form

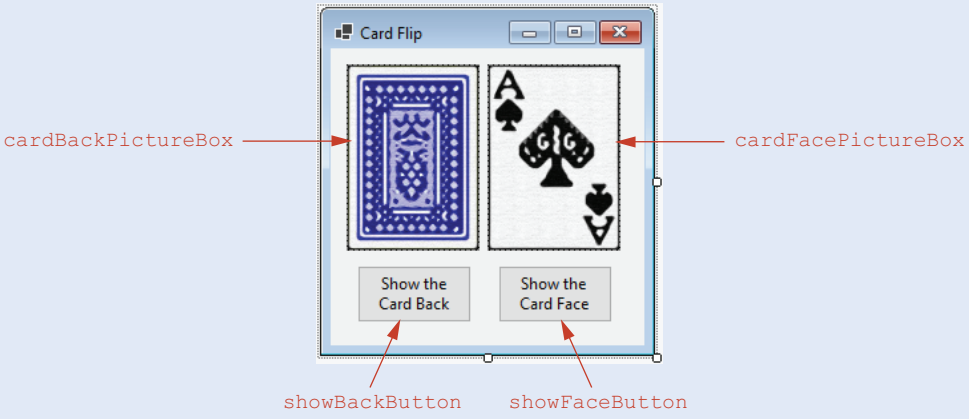


Table 2-5 Control property settings

Control Name	Control Type	Property Settings
`cardBackPictureBox`	PictureBox	**Image:** Select and import the Backface_Blue.jpg file from the *Chap02* folder of the Student Sample Programs. **Size:** 100, 140 **SizeMode:** Zoom **Visible:** True
`cardFacePictureBox`	PictureBox	**Image:** Select and import the Ace_Spades.jpg file from the *Chap02* folder of the Student Sample Programs. **Size:** 100, 140 **SizeMode:** Zoom **Visible:** False
`showBackButton`	Button	**Text:** *Show the Card Back* (Manually resize the button to accommodate the text, as shown in Figure 2-75.)
`showFaceButton`	Button	**Text:** *Show the Card Face* (Manually resize the button to accommodate the text, as shown in Figure 2-75.)

Step 3: Once you have the form and its controls set up, you can create the Click event handlers for the Button controls. In the *Designer*, double-click the `showBackButton` control. This will open the code editor, and you will see an empty event handler named `showBackButton_Click`. Write the following statements inside the event handler:

```
cardBackPictureBox.Visible = true;
cardFacePictureBox.Visible = false;
```

Step 4: Switch your view back to the *Designer* and double-click the `showFaceButton` control. This will open the code editor, and you will see an empty event handler named `showFaceButton_Click`. Write the following statements inside the event handler:

```
cardBackPictureBox.Visible = false;
cardFacePictureBox.Visible = true;
```

Step 5: The form's code should now appear as shown in Program 2-3. Remember, the line numbers are shown for reference only and are not part of the code. The lines that appear in boldface are the ones that you typed. Make sure the code you typed appears exactly as shown here. (Don't forget the semicolons!)

Program 2-3 Completed Form1 code for the *CardFlip* project

```
1 namespace CardFlip
2 {
3   public partial class Form1 : Form
4   {
```

```
 5      public Form1()
 6      {
 7        InitializeComponent();
 8      }
 9
10      private void showBackButton_Click(object sender,
        EventArgs e)
11      {
12        cardBackPictureBox.Visible = true;
13        cardFacePictureBox.Visible = false;
14      }
15
16      private void showFaceButton_Click(object sender,
        EventArgs e)
17      {
18        cardBackPictureBox.Visible = false;
19        cardFacePictureBox.Visible = true;
20      }
21    }
22  }
```

Step 6: Save the project. Then, press the F5 key on the keyboard or click the *Start Debugging* button (▶) on the toolbar to compile and run the application.

Test the application by clicking the buttons. When you click the *Show the Card Face* button you should see the card's face (and the back of the card should be invisible). When you click the *Show the Card Back* button you should see the card's back (and the card's face should be invisible). When you are finished, close the application.

NOTE: If you typed the statements correctly inside the event handlers, the application should run. If you did not type the statements inside the event handlers correctly, a window will appear reporting build errors. If that happens, click the *No* button in the window and then correct the code so it appears exactly as previously shown.

NOTE: In addition to PictureBoxes, many other types of controls have a Visible property. For example, you can make a Label control visible or invisible by setting the value of its Visible property.

Sequential Execution of Statements

In Tutorial 2-5, the event handlers that you created each contained more than one statement. For example, here is the `showBackButton_Click` method:

```
private void showBackButton_Click(object sender, EventArgs e)
{
  cardBackPictureBox.Visible = true;
  cardFacePictureBox.Visible = false;
}
```

This method has two assignment statements. When the method executes, the statements in the method execute in the order that they appear, from the beginning of the method to the end of the method. This statement executes first:

```
cardBackPictureBox.Visible = true;
```

Then this statement executes:

```
cardFacePictureBox.Visible = false;
```

When the application is running, however, you can't really tell that the statements are executing in this order simply by watching the action take place on the screen. When you click the `showBackButton` control, the Click event handler executes so quickly that it appears as though both statements execute simultaneously. It's important for you to understand, however, that the statements execute one at a time, in the order that they appear in the method.

In this particular method, it doesn't really matter which assignment statement is written first. If we reverse the order of the statements, we will not be able to see the difference on the screen because the application executes so quickly. In most applications, however, the order in which you write the statements in the event handlers is critically important. In Chapter 3 you will start writing event handlers that perform several steps, and in most cases, the steps must be performed in a specific order. Otherwise, the program will not produce the correct results.

Checkpoint

2.38 What is a PictureBox control used for?

2.39 Where is the PictureBox tool located in the *Toolbox*?

2.40 How do you display an image in the PictureBox?

2.41 What is the default value of the PictureBox control's SizeMode property?

2.42 How does setting the SizeMode property to Zoom affect the image that is to be displayed in the PictureBox control?

2.43 How do you create a clickable image?

2.44 Does the value of a control's Visible property change how the image appears at run time and design time?

2.8 Comments, Blank Lines, and Indentation

CONCEPT: **Comments are brief notes that are placed in a program's source code to explain how parts of the program work. Programmers commonly use blank lines and indentation in program code to give the code visual organization and make it easier to read.**

Comments

Comments are short notes that are placed in different parts of a program, explaining how those parts of the program work. Comments are not intended for the compiler. They are intended for any person who is reading the code and trying to understand what it does.

In C#, there are three types of comments: line comments, block comments, and documentation comments. A **line comment** appears on one line in a program. You begin a line

comment with two forward slashes (//). Everything written after the slashes, to the end of the line, is ignored by the compiler. The following code sample shows how we might use line comments in the showBackButton_Click event handler from Tutorial 2-5. Each line comment explains what the very next line of code does.

```
private void showBackButton_Click(object sender, EventArgs e)
{
  // Make the image of the back of the card visible.
  cardBackPictureBox.Visible = true;

  // Make the image of the face of the card invisible.
  cardFacePictureBox.Visible = false;
}
```

A line comment does not have to occupy an entire line. Anything appearing after the // symbol, to the end of the line, is ignored. So, a comment can appear after an executable statement. The following code sample shows an example.

```
private void showBackButton_Click(object sender, EventArgs e)
{
  cardBackPictureBox.Visible = true;  // Show the card back.
  cardFacePictureBox.Visible = false; // Hide the card face.
}
```

A **block comment** can occupy multiple consecutive lines in a program. A block comment starts with /* (a forward slash followed by an asterisk) and ends with */ (an asterisk followed by a forward slash). Everything between these markers is ignored. The following code sample shows how block comments may be used.

```
/*  Click event handler for the showBackButton control.
    This method makes the image of the back of the card
    visible and makes the image of the card's face
    invisible.
*/
private void showBackButton_Click(object sender, EventArgs e)
{
  cardBackPictureBox.Visible = true;  // Show the card back.
  cardFacePictureBox.Visible = false; // Hide the card face.
}
```

The first five lines in this code sample are a block comment that explains what the showBackButton_Click method does. Block comments make it easier to write long explanations because you do not have to mark every line with a comment symbol.

Remember the following advice when using block comments:

- Be careful not to reverse the beginning symbol (/*) with the ending symbol (*/).
- Do not forget the ending symbol.

Each of these mistakes can be difficult to track down and will prevent the program from compiling correctly.

The third type of comment is known as a documentation comment. **Documentation comments** are used by professional programmers to embed extensive documentation in a program's source code. Visual Studio can extract information from the documentation comments and generate external documentation files. Single-line documentation comments begin with three forward slashes (///). Block documentation comments begin with /** and end with */. Although documentation comments are useful for professional programmers, we do not use them in this book.

As a beginning programmer, you might resist the idea of writing a lot of comments in your programs. After all, it's a lot more interesting to write code that actually does something; however, it's crucial that you take the extra time to write comments. They will almost

certainly save you time in the future when you have to modify or debug the program. Even large and complex programs can be made easy to read and understand if they have proper comments.

Using Blank Lines and Indentation to Make Your Code Easier to Read

Programmers commonly use blank lines and indentations in their code to create a sense of visual organization. This is similar to the way that authors visually arrange the text on the pages of a book. Instead of writing each chapter as one long series of sentences, they break the text into paragraphs that are visually separated on the page. This does not change the information in the book, but it makes it easier to read.

For example, in the following code sample, we have inserted a blank line inside the method to visually separate the code into two sets of statements. The blank line is not required, but it makes the code easier for humans to read.

```
private void showBackButton_Click(object sender, EventArgs e)
{
  // Make the image of the back of the card visible.
  cardBackPictureBox.Visible = true;

  // Make the image of the face of the card invisible.
  cardFacePictureBox.Visible = false;
}
```

Programmers also use indentations to visually organize code. You may have noticed that in the code editor, all the statements that appear inside a set of braces (`{}`) are indented. For example, all the statements inside a namespace are indented, all the statements inside a class are indented, and all the statements inside a method are indented. In fact, Visual Studio is normally set up to automatically indent the code that you write in this fashion.

Although the indentation is not required, it makes your code much easier to read. By indenting the statements inside a method, you visually set them apart. As a result, you can tell at a glance which statements are inside the method. The same is true for classes and namespaces. This practice of indentation is a convention that virtually all programmers follow.

Checkpoint

2.45 What purpose do comments serve?

2.46 How are line comments and block comments different?

2.47 What should you be careful to remember about the beginning and ending symbols of block comments?

2.48 Why do programmers insert blank lines and indentations in their code?

2.9 Writing the Code to Close an Application's Form

CONCEPT: **To close an application's form in code, you use the statement this.** `Close();`

All the applications that you created in this chapter's tutorials required the user to click the standard Windows close button ( ) to close the application. The standard Windows close button appears in the upper-right corner of almost every window. In many applications,

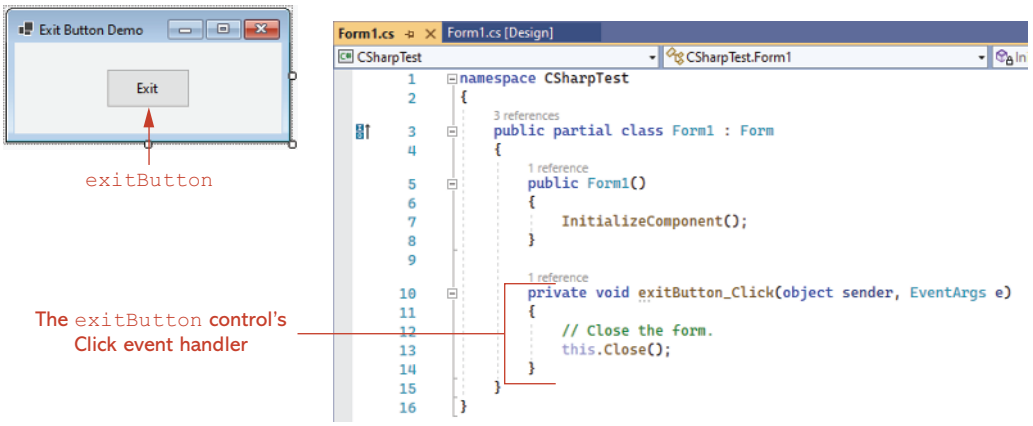
however, you will want to give the user an alternative way to close the application. For example, you might want to create an *Exit* button that closes the application when it is clicked.

To close an application's form, you execute the following statement:

```
this.Close();
```

Let's look at an example of how this statement can be used. Figure 2-76 shows the form and code from a project named *Exit Button Demo*. The Button control that you see on the form is named `exitButton`. In the form's code you can see that we have created a Click event handler for the button. When the user clicks the button, it closes the form, thus closing the application.

Figure 2-76 A form with an *Exit* button



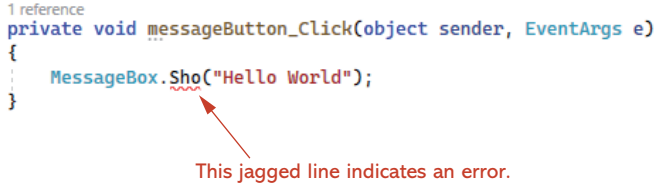
2.10 Dealing with Syntax Errors

CONCEPT: **The Visual Studio code editor examines each statement as you type it and reports any syntax errors that are found. This allows you to quickly correct syntax errors.**

Writing code requires a lot of precision. Even small errors, such as using an uppercase letter where you are supposed to use a lowercase letter or forgetting to end a statement with a semicolon, will prevent an application's code from compiling and executing. Recall from Chapter 1 that these types of mistakes are known as syntax errors.

The Visual Studio code editor does a good job of reporting syntax errors soon after you type them. When you enter a statement into the editor, Visual Studio analyzes it, and if a syntax error is found, it is underlined with a jagged line. Figure 2-77 shows an example. If you hold the mouse cursor over the jagged a line, a description of the error will pop up in a ToolTip window. The description usually gives you enough information to determine the cause of the error and how to fix it.

Figure 2-77 Error underlined



If a syntax error exists in a project's code and you attempt to compile and execute it (by pressing the F5 key on the keyboard, or clicking the *Start Debugging* button ( ) on the toolbar), you will see the window shown in Figure 2-78, reporting build errors. Click the *No* button to close the window, and you will see the *Error List* shown in Figure 2-79.

Figure 2-78 Window reporting build errors

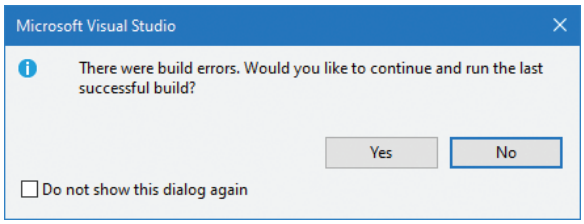
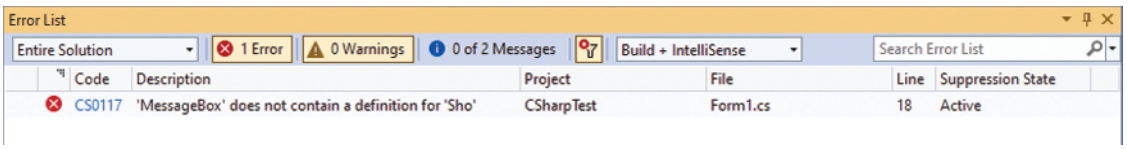


Figure 2-79 *Error List* window



Notice that the *Error List* window shows a description of the error, the source code file that contains the error, the line number and column number of the error, and the name of the project. If you double-click the error message that is displayed in the *Error List* window, the code editor will highlight the code that caused the error.

Checkpoint

2.49 What statement do you use to close an application's form in code?

2.50 How can you tell that Visual Studio has found a syntax error?

2.51 What happens if you hold the mouse cursor over a jagged line in the code editor?

2.52 What happens if you attempt to compile and execute a program that contains syntax errors?

Key Terms

Alphabetical button
aspect ratio
assignment operator
assignment statement
block comment
Boolean property
bounding box
braces
camelCase
Categorized button
Click event
design time
dialog box
documentation comments
event handler
identifiers
IntelliSense
Label control
line comment
message box
method call
namespace
PictureBox control
run time
semicolon
sizing handles
source code file
string

Review Questions

Multiple Choice

1. A(n) _____ is the thin dotted line that encloses an object in the *Designer*.
   a. selection marker
   b. control binder
   c. bounding box
   d. object container

2. The small squares that appear on the right edge, bottom edge, and lower-right corner of a form's bounding box are called _____ .
   a. sizing hooks
   b. form edges
   c. bounding tags
   d. sizing handles

3. _____ is the name of the blank form that Visual Studio initially creates in a new project.
   a. Form1
   b. Main
   c. New1
   d. Blank

4. The _____ property holds the text that is displayed on the face of the button.
   a. Name
   b. Text
   c. Tag
   d. Face

5. A file that contains program code is called a(n) _____ .
   a. destination code file
   b. executable file
   c. machine language file
   d. source code file

6. A namespace is a container that holds _____ .

a. methods
b. names
c. spaces
d. classes

7. A(n) _____ is a method that executes when a specific event takes place while an application is running.

a. action process
b. event handler
c. runtime procedure
d. event method

8. The statement `MessageBox.Show("Hello World");` is an example of a(n) _____ .

a. method call
b. namespace
c. Click event
d. event handler

9. In programming we use the term *string* to mean _____ .

a. many lines of code
b. parallel memory locations
c. string of characters
d. virtually anything

10. A(n) _____ marks the end of a programming statement in C#.

a. semicolon
b. period
c. hyphen
d. underscore

11. A piece of data that is written into a program's code is a(n) _____.

a. identifier
b. specifier
c. keyword
d. literal

12. The time during which you build the GUI and write the application's code is referred to as _____.

a. run time
b. design time
c. code time
d. planning

13. The time during which an application is executing is referred to as _____.

a. go time
b. design time
c. execution
d. run time

14. When you want to display text on a form, you use a _____ control.

a. Button
b. PictureBox
c. Label
d. TextBox

15. The _____ property allows you to set the font, font style, and size of the control's text.

a. Style
b. AutoSize
c. Text
d. Font

16. A _____ property can be set to one of two possible values: True or False.

a. Boolean
b. Logical
c. Binary
d. Dual

17. Label controls have a(n) _____ property that controls the way they can be resized.

a. Stretch
b. AutoSize
c. Dimension
d. Fixed

18. The _____ property can be used to change the text's alignment in the label.

a. TextPosition
b. AutoAlign
c. TextCenter
d. TextAlign

19. In code, you use a(n) _____ to store a value in a control's property.

a. Click event
b. method call
c. assignment statement
d. Boolean value

20. The equal sign (=) is known as the _____.

a. equality symbol
b. assignment operator
c. equality operator
d. property position

21. The standard notation for referring to a control's property in code is _____ .

a. *ControlName.PropertyName*
b. *ControlName=PropertyName*
c. *PropertyName.ControlName*
d. *PropertyName=ControlName*

22. _____ is a feature of Visual Studio that provides automatic code completion as you write programming statements.

a. AutoCode
b. AutoComplete
c. IntelliSense
d. IntelliCode

23. You can use a(n) _____ control to display a graphic image on a form.

a. Graphics
b. PictureBox
c. Drawing
d. ImageBox

24. Once you have created a PictureBox control, you use its _____ property to specify the image that it will display.

a. Image
b. Source
c. DrawSource
d. ImageList

25. The PictureBox control's _____ property specifies how the control's image is to be displayed.

a. RenderMode
b. DrawMode
c. SizeMode
d. ImageMode

26. _____ is the image's width to height ratio.

a. Aspect ratio
b. Size ratio
c. Projection ratio
d. Area ratio

27. Most controls have a _____ property that determines whether the control is visible on the form at run time.

a. Render
b. Viewable
c. Visible
d. Draw

28. A(an) _____ appears on one line in a program.

a. inline comment
b. line comment
c. forward comment
d. block comment

29. A _____ can occupy multiple consecutive lines in a program.

a. block comment
b. square comment
c. multiline comment
d. machine comment

30. Programmers commonly use blank lines and indentations in their code to create a sense of _____.

a. logic
b. visual organization
c. documentation
d. program flow

31. To close an application's form in code, you use the statement _____.

a. `Close();`
b. `Close.This();`
c. `Close()`
d. `this.Close();`

True or False

1. Changing an object's Text property also changes the object's name.
2. When a form is created, its Text property is initially set to the same value as the form's name.
3. The form's title is displayed in the bar along the top of a form.
4. C# source code files always end with the .cs extension.
5. You add your own code to the Progam.cs file as you develop an application.
6. C# code is organized as methods, which are contained inside classes, which are contained inside namespaces.
7. In C# code, each opening brace must have a corresponding closing brace at some point later in the program.
8. When you double-click a control in the *Designer*, Visual Studio not only creates an empty event handler, but it also writes some code that you don't see, elsewhere in the project that is necessary for the event handler to properly function.
9. A Label control's Text property is initially set to the same value as the Label control's name.
10. When a Label control's AutoSize property is set to True, you cannot manually change the size of the control by clicking and dragging its bounding box.
11. By default, a label's text is aligned with the bottom and right edges of the label's bounding box.
12. Label controls are useful for displaying output while an application is running.
13. The assignment operator assigns the value that appears on its left side to the item that a ppears on its right side.
14. PictureBox controls also have a BorderStyle property that works just like a Label control's BorderStyle property.
15. Buttons are the only controls that can respond to Click events.
16. The Visible property is a Binary property, which means it can be set only to the values 1 and 0.
17. When you write the values `true` or `false` in code, they must be written in all lowercase letters.
18. In C#, there are three types of comments: line comments, block comments, and documentation comments.
19. To close an application's form in code, you use the statement `Close.This();`
20. The Visual Studio code editor examines each statement as you type it, and reports any syntax errors that are found.

Short Answer

1. What does a bounding box indicate about an object in the *Designer*?
2. What happens when you position the mouse cursor over an edge or corner of a bounding box that has sizing handles?
3. What determines an object's appearance and other characteristics?
4. What is shown by each column in the *Properties* window?

5. What steps must you perform to change a form's Text property?
6. What steps must you perform to change a form's Size property in the *Properties* window?
7. How do you move a control to a new location on the form using the mouse?
8. What steps do you perform to change a Button control's Text property?
9. Briefly describe the contents of the Form1.cs file.
10. In code, what characters do you enclose a string literal in?
11. When creating an event handler for a button, is it possible to skip a step by opening the code editor and writing all the event handler code yourself? Why or why not?
12. Briefly describe the difference between design time and run time.
13. Describe the appearance of a Label control that's BorderStyle property is set to Fixed3D.
14. What does it mean when a Label control's AutoSize property is set to True?
15. What are the values that the TextAlign property may be set to?
16. How do you clear the text that is displayed in a Label control in code?
17. What are the different image formats that a PictureBox control can display?
18. List the values that the SizeMode property of a PictureBox control can be set to.
19. What are the three types of comments you can use in Visual C#?
20. How does Visual Studio help you to quickly correct syntax errors?

Algorithm Workbench

1. What statement would you write to display *Good Afternoon* in a message box?
2. What statement would you write to display your name in a message box?
3. Suppose an application's GUI has a Label control named `dogLabel`. Write a statement that causes *Fido* to be displayed in the `dogLabel` control.
4. Suppose an application's GUI has a Label control named `outputLabel`. Write a statement that clears any text that happens to be displayed by the control.
5. Suppose an application's GUI has a PictureBox control named `myPicture`. Write a statement that makes the control invisible.

Programming Problems

1. **Latin Translator**

   Look at the following list of Latin words and their meanings:

Latin	English
autem	up
descendit	down
medium	center

   Create an application that translates the Latin words to English. The form should have three buttons, one for each Latin word. When the user clicks a button, the application should display the English translation in a Label control.

VideoNote
Solving the Clickable Number Images Problem

2. **Clickable Number Images**

In the *Chap02* folder, in the Student Sample Program files, you will find the image files shown in Figure 2-80. Create an application that displays these images in PictureBox controls. The application should perform the following actions:

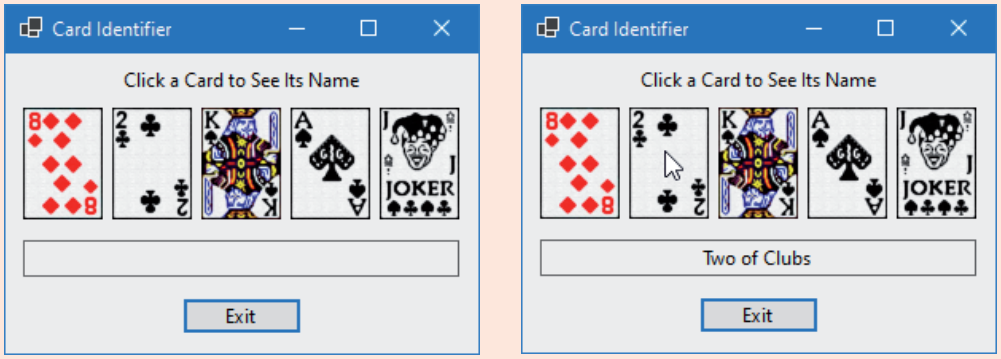
- When the user clicks the 1 image, the application should display the word *One* in a message box.
- When the user clicks the 2 image, the application should display the word *Two* in a message box.
- When the user clicks the 3 image, the application should display the word *Three* in a message box.
- When the user clicks the 4 image, the application should display the word *Four* in a message box.
- When the user clicks the 5 image, the application should display the word *Five* in a message box.

Figure 2-80 Image files

3. **Card Identifier**

In the Student Sample Programs that accompany this book, you will find a folder named *Images\Cards\Poker Large*. In that folder you will find JPEG image files for a complete deck of poker cards. Create an application with five PictureBox controls. Each PictureBox should display a different card from the set of images. When the user clicks any of the PictureBox controls, the name of the card should be displayed in a Label control. Figure 2-81 shows an example of the application running. The image on the left shows the application's form when it starts running. The image on the right shows the form after the user has clicked the two of clubs card.

Figure 2-81 *Card Identifier* application



4. **Joke and Punch Line**

A joke typically has two parts: a setup and a punch line. For example, this might be the setup for a joke:

How many programmers does it take to change a lightbulb?

And this is the punch line:

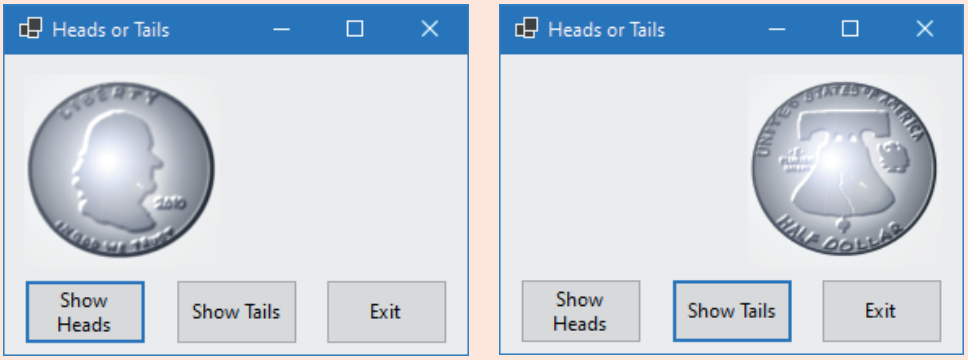
None. That's a hardware problem.

Think of your favorite joke and identify its setup and punch line. Then, create an application that has a Label and two buttons on a form. One of the buttons should read "Setup" and the other button should read "Punch line." When the *Setup* button is clicked, display the joke's setup in the Label control. When the *Punch line* button is clicked, display the joke's punch line in the Label control.

5. **Heads or Tails**

In the Student Sample Programs that accompany this book you will find a folder named *Images\Coins* that contains images showing the heads and tails sides of a coin. Create an application with a *Show Heads* button and a *Show Tails* button. When the user clicks the *Show Heads* button, an image of the heads side of a coin should appear. When the user clicks the *Show Tails* button, an image of the tails side of a coin should appear. Figure 2-82 shows examples of how the application's form might appear.

Figure 2-82 The *Heads or Tails* application

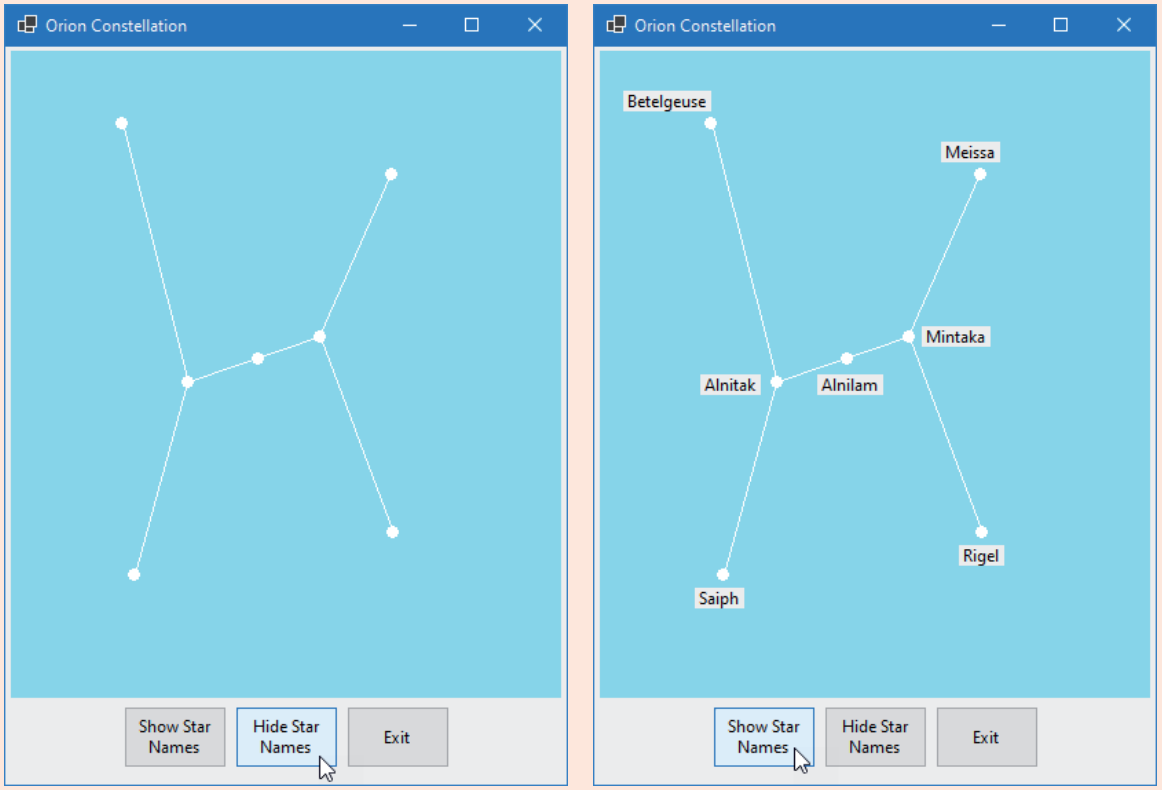


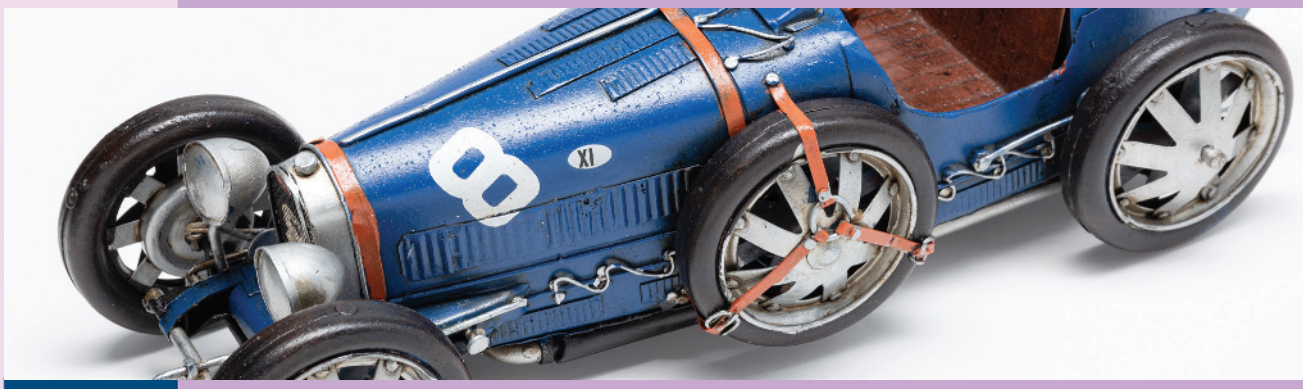
6. **Orion Constellation**

Orion is one of the most famous constellations in the night sky. In the *Chap02* folder of the Student Sample Programs that accompany this book, you will find an image file named Orion.bmp, which contains a diagram of the Orion constellation. Create an application that displays the Orion image in a PictureBox control, as shown on the left in Figure 2-83. The application should have a button that, when clicked, displays the names of each of the stars, as shown on the right in Figure 2-83. The application should have another button that, when clicked, hides the star names. The names of the stars are *Betelgeuse*, *Meissa*, *Alnitak*, *Alnilam*, *Mintaka*, *Saiph*, and *Rigel*.

Hint: Place the PictureBox control with the Orion image on the form. Then, place Label controls containing the star names on top of the PictureBox. Use the *Properties* window to set each of the Label control's Visible property to False. That will cause the labels to be invisible when the application runs. The *Show Star Names* button will set each of the Label control's Visible property to `true`, and the *Hide Star Names* button will set each of the Label control's Visible property to `false`.

Figure 2-83 The *Orion Constellation* application





CHAPTER 3

Processing Data

TOPICS

3.1 Reading Input with TextBox Controls
3.2 A First Look at Variables
3.3 Numeric Data Types and Variables
3.4 Performing Calculations
3.5 Inputting and Outputting Numeric Values
3.6 Formatting Numbers with the `ToString` Method
3.7 Simple Exception Handling
3.8 Using Named Constants
3.9 Declaring Variables as Fields
3.10 Using the Math Class
3.11 More GUI Details
3.12 Using the Debugger to Locate Logic Errors

3.1 Reading Input with TextBox Controls

CONCEPT: **The TextBox control is a rectangular area that can accept keyboard input from the user.**

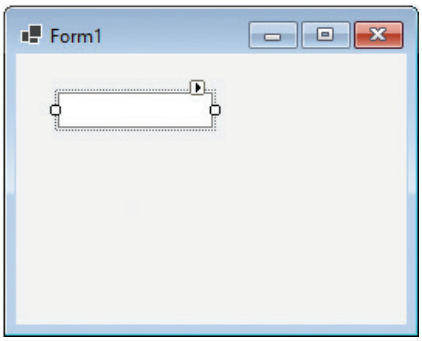
Many of the programs that you will write from this point forward will require the user to enter data. The data entered by the user will then be used in some sort of operation. One of the primary controls that you will use to get data from the user is the TextBox control.

A **TextBox control** appears as a rectangular area on a form. When the application is running, the user can type text into a TextBox control. The program can then retrieve the text that the user entered and use that text in any necessary operations.

In the *Toolbox*, the TextBox tool is located in the *Common Controls* group. When you double-click the tool, a TextBox control is created on the form, as shown in Figure 3-1. When you create TextBox controls, they are automatically given default names such as `textBox1`, `textBox2`, and so forth. As you learned in Chapter 2, you should always change a control's default name to something more meaningful.

When the user types into a TextBox control, the text is stored in the control's Text property. In code, if you want to retrieve the data that has been typed into a TextBox, you simply retrieve the contents of the control's Text property.

Figure 3-1 A TextBox control



NOTE: When you retrieve the contents of the Text property, you *always* get a string. Any operation that can be performed on a string can be performed on a control's Text property.

Let's look at an example. Make sure you have downloaded the student sample programs from the book's companion Web site (at www.pearsonhighered.com/gaddis). In the *Chap03* folder, you will find a project named *TextBoxDemo*. Figure 3-2 shows the form, with most of the control names specified, and Figure 3-3 shows the form's code.

Notice in Figure 3-3 that the `readInputButton` control's Click event handler performs the following assignment statement:

```
outputLabel.Text = nameTextBox.Text;
```

Figure 3-2 The *TextBoxDemo* application

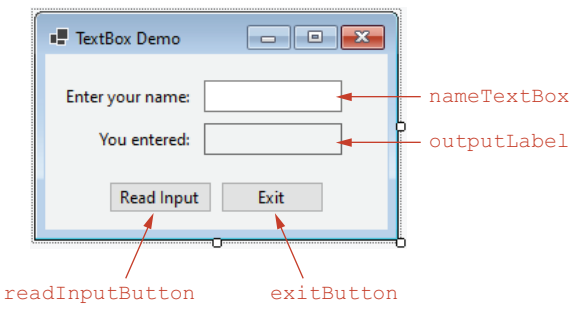
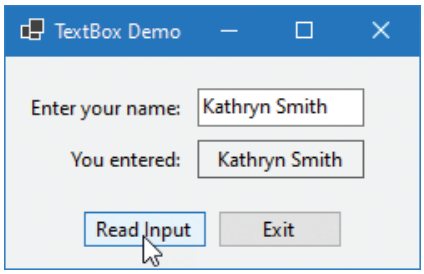


Figure 3-3 The form's code

```
Form1.cs
TextBoxDemo                    TextBoxDemo.Form1                readInputB
 1  namespace TextBoxDemo
 2  {
        3 references
 3      public partial class Form1 : Form
 4      {
            1 reference
 5          public Form1()
 6          {
 7              InitializeComponent();
 8          }
 9
            1 reference
10          private void readInputButton_Click(object sender, EventArgs e)
11          {
12              // Assign the name entered by the user to the
13              // outputLabel control's Text property.
14              outputLabel.Text = nameTextBox.Text;
15          }
16
            1 reference
17          private void exitButton_Click(object sender, EventArgs e)
18          {
19              // Close the form.
20              this.Close();
21          }
22      }
23  }
```

This statement assigns the value of the `nameTextBox` control's Text property to the `outputLabel` control's Text property. In other words, it gets any text that has been entered by the user into the `nameTextBox` control and displays it in the `outputLabel` control. If you run the application, Figure 3-4 shows an example of how the form appears after you have entered *Kathryn Smith* and clicked the `readInputButton` control.

Figure 3-4 The user's name displayed in the label



Clearing the Contents of a TextBox Control

You can clear the contents of a TextBox control in the same way that you clear the contents of a Label control: you assign an empty string (`""`) to the control's Text property. For example, the following statement clears the contents of the `nameTextBox` control:

```
nameTextBox.Text = "";
```

When this statement executes, the `nameTextBox` control will appear empty on the application's form.

Checkpoint

3.1 What control can be used to gather text input from the user?

3.2 In code, how do you retrieve data that has been typed into a TextBox control?

3.3 What type of data does a control's Text property always contain?

3.4 How do you clear the contents of a TextBox control?

3.2 A First Look at Variables

CONCEPT: A variable is a storage location in memory that is represented by a name.

Most programs store data in the computer's memory and perform operations on that data. For example, consider the typical online shopping experience: you browse a Web site and add the items that you want to purchase to the shopping cart. As you add items to the shopping cart, data about those items is stored in memory. Then, when you click the *checkout* button, a program running on the Web site's computer calculates the cost of all the items you have in your shopping cart, applicable sales taxes, shipping costs, and the total of all these charges. When the program performs these calculations, it stores the results in the computer's memory.

Programs use variables to store data in memory. A **variable** is a storage location in memory that is represented by a name. For example, a program that manages a company's customer mailing list might use a variable named `lastName` to hold a customer's last name, a variable named `firstName` to hold the customer's first name, a variable named `address` to hold the customer's mailing address, and so forth.

In C#, you must declare a variable in a program before you can use it to store data. You do this with a **variable declaration**, which specifies two things about the variable:

1. The variable's data type, which is the type of data the variable will hold
2. The variable's name

A variable declaration statement is written in this general format:

```
DataType VariableName;
```

Let's take a closer look at each of these.

Data Type

A variable's **data type** indicates the type of data that the variable will hold. Before you declare a variable, you need to think about the type of value that will be stored in the variable. For example, will the variable hold a string or a number? If it will hold a number, what kind of number will it be, an integer or a real number? When you have determined the kind of data that the variable will hold, you select one of the data types that C# provides for a variable.

The C# language provides many data types for storing fundamental types of data, such as strings, integers, and real numbers. These data types are known as **primitive data types**. We will look at several of them in this chapter.

Variable Name

A **variable name** identifies a variable in the program code. When naming a variable, you should always choose a meaningful name that indicates what the variable is used for. For example, a variable that holds the temperature might be named `temperature`, and a variable that holds a car's speed might be named `speed`. You may be tempted to give variables

short, nondescript names such as `x` or `b2`, but names such as these give no clue as to the purpose of the variable.

In addition, there are some specific rules that you must follow when naming a variable. The same rules for identifiers that apply to control names also apply to variable names. We discussed these rules in Chapter 2, but we review them now:

- The first character must be one of the letters *a* through *z* or *A* through *Z* or an underscore character (_).
- After the first character, you may use the letters *a* through *z* or *A* through *Z*, the digits 0 through 9, or underscores.
- The name cannot contain spaces.

When naming variables, we use the same camelCase naming convention that we introduced in Chapter 2 for control names. For example, if we are declaring a variable to hold an employee's gross pay, we might name it `grossPay`. Or, if are declaring a variable to a customer number, we might name it `customerNumber`.

`string` Variables

The first primitive data type we consider is the `string` data type. A variable of the `string` data type can hold any string of characters, such as a person's name, address, password, and so forth. Here is an example of a statement that declares a `string` variable named `productDescription`:

```
string productDescription;
```

After the variable has been declared, you can use the assignment operator (=) to store a value in the variable. Here is an example:

```
productDescription = "Italian Espresso Machine";
```

When this statement executes, the string literal `"Italian Espresso Machine"` is assigned to the `productDescription` variable. When writing an assignment statement, remember that the assignment operator assigns the value that appears on its right side to the variable that appears on its left side.

Once you have assigned a value to a variable, you can use the variable in other operations. For example, assume `productLabel` is the name of a Label control. The following statement assigns the `productDescription` string to the `productLabel` control's Text property:

```
productLabel.Text = productDescription;
```

After this statement executes, the string that is stored in the `productDescription` variable is displayed in the `productLabel` control. The following statement shows another example:

```
MessageBox.Show(productDescription);
```

When this statement executes, the string that is stored in the `productDescription` variable is displayed in a message box.

String Concatenation

A common operation that performed on strings is **concatenation**, or appending one string to the end of another string. In C# you use the `+` operator to concatenate strings. The `+` operator produces a string that is the combination of the two strings used as its operands. The following code shows an example:

```
string message;
message = "Hello " + "world";
MessageBox.Show(message);
```

The first statement declares a `string` variable named `message`. The second statement combines the strings `"Hello "` and `"world"` to produce the string `"Hello world"`.

The string "Hello world" is then assigned to the message variable. The third statement displays the contents of the message variable in a message box. When the message box is displayed, it shows the string *Hello world*.

Let's look at an application that further demonstrates string concatenation. In the *Chap03* folder of this book's student sample programs (available for download at www.pearsonhighered.com/gaddis), you will find a project named *StringVariableDemo*. Figure 3-5 shows the form, with most of the control names specified, and Figure 3-6 shows the form's code. (In Figure 3-6, to conserve space on the page, we have scrolled past the using directives that appear at the top of the code file.)

Figure 3-5 The *StringVariableDemo* project's form

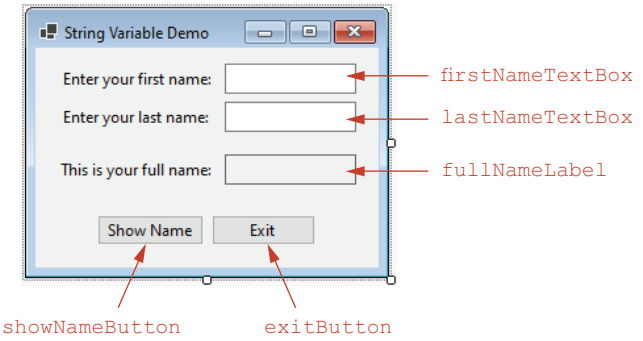


Figure 3-6 The form's code

```
namespace StringVariableDemo
{
    3 references
    public partial class Form1 : Form
    {
        1 reference
        public Form1()
        {
            InitializeComponent();
        }

        1 reference
        private void showNameButton_Click(object sender, EventArgs e)
        {
            // Declare a string variable to hold the full name.
(1)         string fullname;

            // Combine the names with a space between them. Assign the
            // result to the fullname varible.
(2)         fullname = firstNameTextBox.Text + " " + lastNameTextBox.Text;

            // Display the fullname variable in the fullNameLabel control.
(3)         fullNameLabel.Text = fullname;
        }

        1 reference
        private void exitButton_Click(object sender, EventArgs e)
        {
            // Close the form.
            this.Close();
        }
    }
}
```

In Figure 3-6, three statements in the showNameButton_Click event handler are pointed out:

(1) This statement is a variable declaration. It declares a string variable named fullName.

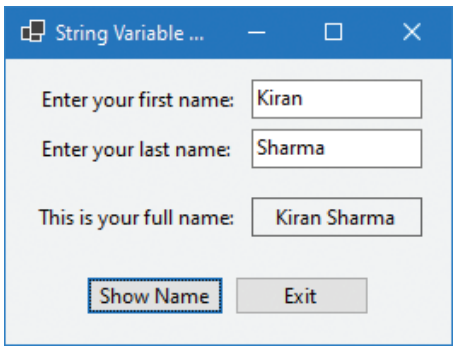
(2) This statement assigns the result of a string concatenation to the fullName variable. The string that is assigned to the variable begins with the value of the firstNameTextBox control's Text property, followed by a space (" "), followed by the value of the lastNameTextBox control's Text property. For example, if the user has entered

Joe into the `firstNameTextBox` control and *Smith* into the `lastNameTextBox` control, this statement will assign the string `"Joe Smith"` to the `fullName` variable.

(3) This statement assigns the `fullName` variable to the `fullNameLabel` control's Text property. As a result, the string that is stored in the `fullName` variable is displayed in the `fullNameLabel` control.

If you run the application, Figure 3-7 shows an example of how the form appears after you have entered *Kiran* for the first name and *Sharma* for the last name and clicked the `showNameButton` control.

Figure 3-7 The user's full name displayed in the label



Declaring Variables before Using Them

The purpose of a variable declaration statement is to tell the compiler that you plan to use a variable of a specified name to store a particular type of data in the program. A variable declaration statement causes the variable to be created in memory. For this reason, a variable's declaration statement must appear *before* any other statements in the method that use the variable. This makes perfect sense because you cannot store a value in a variable if the variable has not been created in memory.

Local Variables

Notice that the `fullName` variable in Figure 3-6 is declared inside the event handler method. Variables that are declared inside a method are known as local variables. A **local variable** belongs to the method in which it is declared, and only statements inside that method can access the variable. (The term *local* is meant to indicate that the variable can be used only locally, within the method in which it is declared.)

An error will occur if a statement in one method tries to access a local variable that belongs to another method. For example, let's go over the sample code shown in Figure 3-8:

(1) This statement declares a `string` variable named `myName`. The variable is declared inside the `firstButton_Click` event handler, so it is local to that method.

Figure 3-8 One method trying to access a variable that is local to another method

```
private void firstButton_Click(object sender, EventArgs e)
{
    // Declare a string variable.
①   string myName;

    // Assign a value to the myName variable.
②   myName = nameTextBox.Text;
}

private void secondButton_Click(object sender, EventArgs e)
{
    // Display the value of the myName variable in a Label.
③   outputLabel.Text = myName;   <-- ERROR!
}
```

(2) This statement, which is also in the `firstButton_Click` event handler, assigns the `nameTextBox` control's Text property to the `myName` variable.

(3) This statement, which is in the `secondButton_Click` event handler, *attempts* to assign the `myName` variable to the `outputLabel` control's Text property. This statement will not work, however, because the `myName` variable is local to the `firstButton_Click` event handler, and statements in the `secondButton_Click` event handler cannot access it.

Scope of a Variable

Programmers use the term **scope** to describe the part of a program in which a variable may be accessed. A variable is visible only to statements inside the variable's scope.

A local variable's scope begins at the variable's declaration and ends at the end of the method in which the variable is declared. As you saw in the previous example, a local variable cannot be accessed by statements that are outside the method. In addition, a local variable cannot be accessed by code that is inside the method but before the variable's declaration.

Lifetime of a Variable

A variable's **lifetime** is the time period during which the variable exists in memory while the program is executing. A local variable is created in memory when the method in which it is declared starts executing. When the method ends, all the method's local variables are destroyed. So, a local variable's lifetime is the time during which the method in which it is declared is executing.

Duplicate Variable Names

You cannot declare two variables with the same name in the same scope. For example, if you declare a variable named `productDescription` in an event handler, you cannot declare another variable with that name in the same event handler. You can, however, have variables of the same name declared in different methods.

Assignment Compatibility

You can assign a value to a variable only if the value is compatible with the variable's data type. Only strings are compatible with the `string` data type, so the assignment statement in the following code snippet works:

```
1  // Declare a string variable.
2  string productDescription;
3
4  // Assign a value from a TextBox to the variable.
5  productDescription = userInputTextBox.Text;
```

Assume the application has a TextBox control named `userInputTextBox`. In line 5 we assign the value of the TextBox control's Text property to the `string` variable `productDescription`. This works because the value in a control's Text property is always a string.

The following code will not work, however, because it attempts to assign a nonstring value to a `string` variable:

```
1 // Declare a string variable.
2 string employeeID;
```

```
3
4 // Assign a value to the variable. Will this work?
5 employeeID = 125; ← ERROR!
```

In line 5, we are attempting to assign the number 125 to a `string` variable. Numbers are not assignment compatible with `string` variables, so this statement will cause an error when the code is compiled.

NOTE: Although you cannot store the number 125 in a `string` variable, you can store the string literal `"125"` in a `string` variable.

A Variable Holds One Value at a Time

Variables can hold different values while a program is running, but they can hold only one value at a time. When you assign a value to a variable, that value will remain in the variable until you assign a different value to the variable. For example, look at the following code sample:

```
1 // Declare a string variable.
2 string productDescription;
3
4 // Assign a value to the variable.
5 productDescription = "Large Medium-Roast Coffee";
6
7 // Display the variable's value.
8 MessageBox.Show(productDescription);
9
10 // Assign a different value to the variable.
11 productDescription = "Chocolate Truffle";
12
13 // Display the variable's value.
14 MessageBox.Show(productDescription);
```

The following comments explain what we did:

- Line 2 declares a `string` variable named `productDescription`.
- Line 5 assigns the string `"Large Medium-Roast Coffee"` to the `productDescription` variable.
- Line 8 displays the value of the `productDescription` variable in a message box. (The message box will display *Large Medium-Roast Coffee*.)
- Line 11 assigns a different value to the `productDescription` variable. After this statement executes, the `productDescription` variable will hold the string `"Chocolate Truffle"`.
- Line 14 displays the value of the `productDescription` variable in a message box. (The message box will display *Chocolate Truffle*.)

This code sample illustrates two important characteristics of variables:

- A variable holds only one value at a time.
- When you store a value in a variable, that value replaces the previous value that was in the variable.

Tutorial 3-1 gives you some practice using variables. You will create an application that uses TextBox controls to get input values, stores those values in variables, and uses the variables in operations.

Tutorial 3-1:
The *Birth Date String* Application

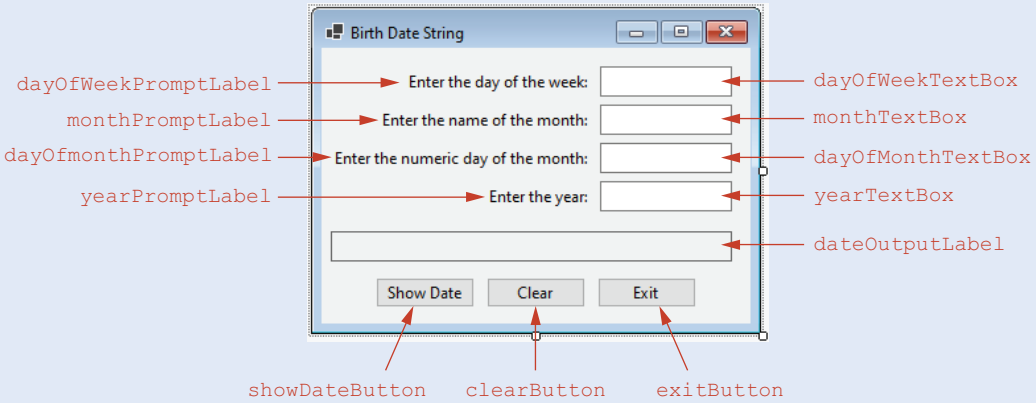
VideoNote
Tutorial 3-1: The *Birth Date String* Application

In this tutorial, you create an application that lets the user enter the following information about their birthdate:

- The day of the week (Monday, Tuesday, etc.)
- The name of the month (January, February, etc.)
- The numeric day of the month
- The year

Figure 3-9 shows the application's form, along with the names of all the controls. When the application runs, the user enters each piece of data into a separate TextBox. When the user clicks the *Show Date* button, the application concatenates the contents of the TextBoxes into a string such as *Friday, June 1, 1990*. The string is displayed in the `dateOutputLabel` control. When the user clicks the *Clear* button, the contents of the TextBoxes and the `dateOutputLabel` control are cleared. The *Exit* button closes the application's form.

Figure 3-9 The *Birth Date String* form



Step 1: Start Visual Studio and begin a new Windows Forms App project named *BirthDateString*.

Step 2: Set up the application's form as shown in Figure 3-9. Notice that the form's Text property is set to *Birth Date String*. The names of the controls are shown in the figure. As you place each control on the form, refer to Table 3-1 for the relevant property settings.

Step 3: Once you have set up the form with its controls, you can create the Click event handlers for the Button controls. At the end of this tutorial, Program 3-1 shows the completed code for the form. You will be instructed to refer to Program 3-1 as you write the event handlers. (Remember, the line numbers that are shown in Program 3-1 are not part of the program. They are shown for reference only.)

In the *Designer*, double-click the `showDateButton` control. This will open the code editor, and you will see an empty event handler named `showDateButton_Click`. Complete the `showDateButton_Click` event handler by typing the code shown in lines 12–22 in Program 3-1.

Let's take a closer look at the code:

Line 13: This statement declares a `string` variable named `output`.

Table 3-1 Control property settings

Control Name	Control Type	Property Settings
`dayOfWeekPromptLabel`	Label	**Text:** *Enter the day of the week*
`monthPromptLabel`	Label	**Text:** *Enter the name of the month*
`dayOfMonthPromptLabel`	Label	**Text:** *Enter the numeric day of the month*
`yearPromptLabel`	Label	**Text:** *Enter the year*
`dayOfWeekTextBox`	TextBox	No properties changed
`monthTextBox`	TextBox	No properties changed
`dayOfMonthTextBox`	TextBox	No properties changed
`yearTextBox`	TextBox	No properties changed
`dateOutputLabel`	Label	**AutoSize:** False **BorderStyle:** FixedSingle **Text:** (The contents of the Text property have been erased.) **TextAlign:** MiddleCenter
`showDateButton`	Button	**Text:** *Show Date*
`clearButton`	Button	**Text:** *Clear*
`exitButton`	Button	**Text:** *Exit*

Lines 16–19: These lines are actually one long statement, broken up into multiple lines. The statement concatenates the Text properties of the TextBox controls, along with appropriately placed commas and spaces, to create the date string. The resulting string is assigned to the `output` variable.

For example, suppose the user has entered the following input:

- *Friday* in the `dayOfWeekTextBox` control.
- *June* in the `monthTextBox` control.
- *1* in the `dayOfMonthTextBox` control.
- *1990* in the `yearTextBox` control.

The concatenation in this statement produces the string "June 1, 1990"; it is assigned to the `output` variable.

Line 22: This statement assigns the `output` variable to the `dateOutputLabel` control's Text property. When this statement executes, the contents of the `output` variable are displayed in the `dateOutputLabel` control.

Step 4: Switch your view back to the *Designer* and double-click the `clearButton` control. In the code editor, you will see an empty event handler named `clearButton_Click`. Complete the `clearButton_Click` event handler by typing the code shown in lines 27–34 in Program 3-1.

Let's take a closer look at the code:

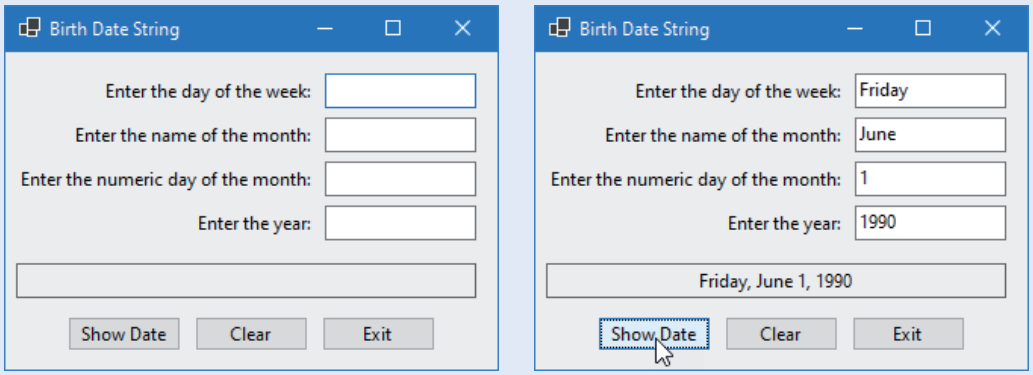
Lines 28–31: Each statement assigns an empty string ("") to the Text property of one of the TextBox controls. When these statements have finished executing, the TextBox controls will appear empty.

Line 34: This statement assigns an empty string ("") to the `dateOutputLabel` control's Text property. After the statement has executed, the label appears empty.

Step 5: Switch your view back to the *Designer* and double-click the `exitButton` control. In the code editor, you will see an empty event handler named `exitButton_Click`. Complete the `exitButton_Click` event handler by typing the code shown in lines 39–40 in Program 3-1.

Step 6: Save the project. Then, press the F5 key on the keyboard, or click the *Start Debugging* button ( ) on the toolbar to compile and run the application. The form will appear as shown in the image on the left in Figure 3-10. Test the application by entering values into the TextBoxes and clicking the *Show Date* button. The date should be displayed, similar to the image shown on the right in the figure. Click the *Clear* button, and the contents of the TextBoxes and the Label control should clear. Click the *Exit* button and the form should close.

Figure 3-10 The *Birth Date String* application



Program 3-1 Completed Form1 code for the *BirthDateString* project

```
1  namespace BirthDateString
2  {
3      public partial class Form1 : Form
4      {
5          public Form1()
6          {
7              InitializeComponent();
8          }
9
10         private void showDateButton_Click(object sender, EventArgs e)
11         {
12             // Declare a string variable.
13             string output;
14
15             // Concatenate the input and build the output string.
16             output = dayOfWeekTextBox.Text + ", " +
```

```
17                  monthTextBox.Text + " " +
18                  dayOfMonthTextBox.Text + ", " +
19                  yearTextBox.Text;
20
21              // Display the output string in the dateOutputLabel
                control.
22              dateOutputLabel.Text = output;
23          }
24
25          private void clearButton_Click(object sender, EventArgs e)
26          {
27              // Clear the TextBoxes.
28              dayOfWeekTextBox.Text = "";
29              monthTextBox.Text = "";
30              dayOfMonthTextBox.Text = "";
31              yearTextBox.Text = "";
32
33              // Clear the dateOutputLabel control.
34              dateOutputLabel.Text = "";
35          }
36
37          private void exitButton_Click(object sender, EventArgs e)
38          {
39              // Close the form.
40              this.Close();
41          }
42      }
43  }
```

NOTE: In Tutorial 3-1, the statement in lines 16–19 shows an example of how you can break up a statement into multiple lines. Quite often, you will find yourself writing statements that are too long to fit entirely inside the *Code* window. Your code will be hard to read if you have to horizontally scroll the *Code* window to view long statements. In addition, if you or your instructor chooses to print your code, the statements that are too long to fit on one line of the page will wrap around to the next line and make your code look unorganized. For these reasons, it is usually best to break a long statement into multiple lines.

When typing most statements, you can simply press the Enter key when you reach an appropriate point to continue the statement on the next line. Remember, however, that you cannot break up a keyword, a quoted string, or an identifier (such as a variable name or a control name).

Initializing Variables

In C#, a local variable must be assigned a value before it can be used. For example, look at this code:

```
string productDescription;
MessageBox.Show(productDescription);
```

This code declares a `string` variable named `productDescription` and then tries to display the variable's value in a message box. The only problem is that we have not assigned

a value to the variable. When we compile the application containing this code, we will get an error message such as *Use of unassigned local variable 'productDescription'*. The C# compiler will not compile code that tries to use an unassigned local variable.

One way to make sure that a variable has been assigned a value is to **initialize** the variable with a value when you declare it. For example, the following statement declares a `string` variable named `productDescription` and immediately assigns the string literal `"Chocolate Truffle"` to it:

```
string productDescription = "Chocolate Truffle";
```

We say that this statement *initializes* the `productDescription` variable with the string `"Chocolate Truffle"`. Here is another example:

```
string lastName = lastNameTextBox.Text;
```

Assume that this statement belongs to an application that has a TextBox named `lastNameTextBox`. The statement declares a `string` variable named `lastName` and initializes it with the value of the `lastNameTextBox` control's Text property.

Here is another example. The following statement declares a `string` variable named `productID` and initializes it with an empty string:

```
string productID = "";
```

Declaring Multiple Variables with One Statement

You can declare multiple variables of the same data type with one declaration statement. Here is an example:

```
string lastName, firstName, middleName;
```

This statement declares three `string` variables named `lastName`, `firstName`, and `middleName`. Notice that commas separate the variable names. Here is an example of how we can declare and initialize the variables with one statement:

```
string lastName = "Jones", firstName = "Jill", middleName = "Rebecca";
```

Remember, you can break up a long statement so it spreads across two or more lines. Sometimes you will see long variable declarations written across multiple lines, like this:

```
string lastName = "Jones",
       firstName = "Jill",
       middleName = "Rebecca";
```

Checkpoint

3.5 What is the purpose of a variable?

3.6 Give an example of a variable declaration that will store the name of your favorite food.

3.7 For each of the following items, determine whether the data type should be an integer, string, or real number.

a. pet name

b. sales tax

c. mailing address

d. video game score

3.8 Indicate whether each of the following is a legal variable name. If it is not, explain why.

a. `pay_Rate`

b. `speed of sound`

c. `totalCost`

d. `2ndPlaceName`

3.9 What will be stored in the `message` variable after the following statement is executed?

```
string message = "He" + "ll" + "o!";
```

3.10 What is the lifetime of a variable that is declared inside of a Click event handler?

3.11 Assuming the local variable `greeting` has not been assigned a value, what will be the result of the following statement?

```
MessageBox.Show(greeting);
```

3.12 Will the following statement cause an error? Why or why not?

```
string luckyNumber = 7;
```

3.13 Write a single declaration statement for the variables `name`, `city`, and `state`.

3.3 Numeric Data Types and Variables

CONCEPT: **If you need to store a number in a variable and use that number in a mathematical operation, the variable must be of a numeric data type. You select a numeric data type that is appropriate for the type of number that you need to store.**

In the previous section, you read about `string` variables. Variables of the `string` data type can be used to store text, but they cannot store numeric data for the purpose of performing mathematical operations. If you need to store numbers and perform mathematical operations on them, you have to use a numeric data type.

The C# language provides several primitive data types. You can read about all the C# primitive data types in Appendix A. Many of the data types provided by C# are for specialized purposes beyond the scope of this book. When it comes to numeric data, most of the time you will use the three numeric primitive data types described in Table 3-2.

Here are examples of declaring variables of each data type:

```
int speed;
double distance;
decimal grossPay;
```

The first statement declares an `int` variable named `speed`. The second example declares a `double` variable named `distance`. The third statement declares a `decimal` variable named `grossPay`.

Numeric Literals

You learned in Chapter 2 that a literal is a piece of data written into a program's code. When you know, at the time that you are writing a program's code, that you want to store a specific value in a variable, you can assign that value as a literal to the variable.

Table 3-2 The primitive numeric data types that you will use most often

Data Type	Description
`int`	A variable of the `int` data type can hold whole numbers only. For example, an `int` variable can hold values such as 42, 0, and −99. An `int` variable cannot hold numbers with a fractional part, such as 22.1 or −4.9. The `int` data type is the primary data type for storing integers. We use it in this book any time we need to store and work with integers. An `int` variable uses 32 bits of memory and can hold an integer number in the range of −2,147,483,648 through 2,147,483,647.
`double`	A variable of the `double` data type can hold real numbers, such as 3.5, −87.95, or 3.0. A number that is stored in a `double` variable is rounded to 15 digits of precision. We use variables of the `double` data type to store any number that might have a fractional part. The double data type is especially useful for storing extremely great or extremely small numbers. In memory a `double` variable uses 64 bits of memory. It is stored in a format that programmers call *double precision floating-point notation*. Variables of the `double` data type can hold numbers in the range of $\pm 5.0 \times 10^{-324}$ to $\pm 1.7 \times 10^{308}$.
`decimal`	A variable of the `decimal` data type can hold real numbers with greater precision than the `double` data type. A number that is stored in a `decimal` variable is rounded to 28 digits of precision. Because `decimal` variables store real numbers with a great deal of precision, they are most commonly used in financial applications. In this book, we typically use the `decimal` data type when storing amounts of money. In memory, a `decimal` variable uses 128 bits of memory. It is stored in a format that programmers call *decimal notation*. Variables of the `decimal` data type can hold numbers in the range of $\pm 1.0 \times 10^{-28}$ to $\pm 7.9 \times 10^{28}$.

A **numeric literal** is a number that is written into a program's code. For example, the following statement declares an `int` variable named `hoursWorked` and initializes it with the value 40:

```
int hoursWorked = 40;
```

In this statement, the number 40 is a numeric literal. The following shows another example:

```
double temperature = 87.6;
```

This statement declares a `double` variable named `temperature` and initializes it with the value 87.6. The number 87.6 is a numeric literal.

When you write a numeric literal in a program's code, the numeric literal is assigned a data type. In C#, if a numeric literal is an integer (not written with a decimal point) and it fits within the range of an `int` (see Table 3-2 for the minimum and maximum values), then the numeric literal is treated as an `int`. A numeric literal that is treated as an `int` is called an **integer literal**. For example, each of the following statements initializes a variable with an integer literal:

```
int hoursWorked = 40;
int unitsSold = 650;
int score = -23;
```

If a numeric literal is written with a decimal point and it fits within the range of a `double` (see Table 3-2 for the minimum and maximum values), then the numeric literal is treated as a `double`. A numeric literal that is treated as a `double` is called a **`double`** literal. For example, each of the following statements initializes a variable with a `double` literal:

```
double distance = 28.75;
double speed = 87.3;
double temperature = -10.0;
```

When you append the letter `M` or `m` to a numeric literal, it is treated as a `decimal` and is referred to as a **`decimal`** literal. Here are some examples:

```
decimal payRate = 28.75m;
decimal price = 8.95M;
decimal profit = -50m;
```

TIP: Because `decimal` is the preferred data type for storing monetary amounts, remembering that "m" stands for "money" might help you to remember that `decimal` literals must end with the letter `M` or `m`.

Assignment Compatibility for `int` Variables

You can assign `int` values to `int` variables, but you cannot assign `double` or `decimal` values to `int` variables. For example, look at the following declarations:

```
int hoursWorked = 40; ← This works
int unitsSold = 650m; ← ERROR!
int score = -25.5;    ← ERROR!
```

The first declaration works because we are initializing an `int` variable with an `int` value. The second declaration causes an error, however, because you cannot assign a `decimal` value to an `int` variable. The third declaration also causes an error because you cannot assign a `double` value to an `int` variable.

You cannot assign a `double` or a `decimal` value to an `int` variable because such an assignment could result in a loss of data. Here are the reasons:

- The `double` and `decimal` values may be fractional, but `int` variables can hold only integers. If you were allowed to store a fractional value in an `int` variable, the fractional part of the value would have to be discarded.
- The `double` and `decimal` values may be much larger or much smaller than allowed by the range of an `int` variable. A `double` or a `decimal` number can potentially be so large or so small that it will not fit in an `int` variable.

Assignment Compatibility for `double` Variables

You can assign either `double` or `int` values to `double` variables, but you cannot assign `decimal` values to `double` variables. For example, look at the following declarations:

```
double distance = 28.75; ← This works
double speed = 75;       ← This works
double sales = 6500.0m;  ← ERROR!
```

The first declaration works because we are initializing a `double` variable with a `double` value. The second declaration works because we are initializing a `double` variable with an `int` value. The third declaration causes an error, however, because you cannot assign a `decimal` value to a `double` variable.

It makes sense that you are allowed to assign an `int` value to a `double` variable because any number that can be stored as an `int` can be converted to a `double` with no loss of data. When you assign an `int` value to a `double` variable, the `int` value is implicitly converted to a `double`.

You cannot assign a `decimal` value to a `double` variable because the `decimal` data type allows for much greater precision than the `double` data type. A `decimal` value can have up to 28 digits of precision, whereas a `double` can provide only 15 digits of precision. Storing a `decimal` value in a `double` variable could potentially result in a loss of data.

Assignment Compatibility for `decimal` Variables

You can assign either `decimal` or `int` values to `decimal` variables, but you cannot assign `double` values to `decimal` variables. For example, look at the following declarations:

```
decimal balance = 9280.73m; ← This works
decimal price = 50;         ← This works
decimal sales = 6500.0;     ← ERROR!
```

The first declaration works because we are initializing a `decimal` variable with a `decimal` value. The second declaration works because we are initializing a `decimal` variable with an `int` value. When you assign an `int` value to a `decimal` variable, the `int` value is implicitly converted to a `decimal` with no loss of data. The third declaration causes an error, however, because you cannot assign a `double` value to a `decimal` variable. A `double` value can potentially be much larger or much smaller than allowed by the range of a `decimal`.

Explicitly Converting Values with Cast Operators

Let's consider a hypothetical situation. Suppose you've written an application that uses a `double` variable, and for some reason, you need to assign the contents of the `double` variable to an `int` variable. In this particular situation, you know that the `double` variable's value is something that can be safely converted to an `int` without any loss of data (such as 3.0, or 98.0). However, the C# compiler will not allow you to make the assignment because `double` values are not assignment compatible with `int` variables. Isn't there a way to override the C# rules in this particular situation and make the assignment anyway?

The answer is yes, there is a way. You can use a **cast operator** to explicitly convert a value from one numeric data type to another, even if the conversion might result in a loss of data. A cast operator is the name of the desired data type, written inside parentheses and placed to the left of the value that you want to convert. The following code sample demonstrates:

```
1 // Declare an int variable.
2 int wholeNumber;
3
4 // Declare a double variable.
5 double realNumber = 3.0;
6
7 // Assign the double to the int.
8 wholeNumber = (int)realNumber;
```

The following points describe the code:

- Line 2 declares an `int` variable named `wholeNumber`.
- Line 5 declares a `double` variable named `realNumber`, initialized with the value 3.0.
- Line 8 uses a cast operator to convert the value of `realNumber` to an `int` and assigns the converted value to `wholeNumber`. After this statement executes, the `wholeNumber` variable is assigned the value 3.

Table 3-3 shows other code examples involving different types of cast operators.

Table 3-3 Examples of uses of cast operators

Code Example	Description

`int wholeNumber;`
`decimal moneyNumber = 4500m;`
`wholeNumber = (int)moneyNumber;`
	The `(int)` cast operator converts the value of the `moneyNumber` variable to an `int`. The converted value is assigned to the `wholeNumber` variable.

`double realNumber;`
`decimal moneyNumber = 625.70m;`
`realNumber = (double)moneyNumber;`
	The `(double)` cast operator converts the value of the `moneyNumber` variable to a `double`. The converted value is assigned to the `realNumber` variable.

`decimal moneyNumber;`
`double realNumber = 98.9;`
`moneyNumber = (decimal)realNumber;`
	The `(decimal)` cast operator converts the value of the `realNumber` variable to a `decimal`. The converted value is assigned to the `moneyNumber` variable.

When you use a cast operator, you are essentially telling the compiler that you know what you are doing and you are willing to accept the consequences of the conversion. It is still possible that a loss of data can occur. For example, look at the following code sample:

```
int wholeNumber;
double realNumber = 8.9;
wholeNumber = (int)realNumber;
```

In this example, the `double` variable contains a fractional number. When the cast operator converts the fractional number to an `int`, the part of the number that appears after the decimal point is dropped. The process of dropping a number's fractional part is called **truncation**. After this code executes, the `wholeNumber` variable contains the value 8.

It's important to realize that when a cast operator is applied to a variable, it does not change the contents of the variable. The cast operator merely returns the value that is stored in the variable, converted to the specified data type. In the previous code sample, when the `(int)` cast operator is applied to the `realNumber` variable, the cast operator returns the value 8. The `realNumber` variable remains unchanged, however, still containing the value 8.9.

Declaring Local Variables with the `var` Keyword

C# provides an alternative way to declare local variables, using the **`var` keyword** and an initialization value. Here is an example:

```
var amount = 100;
```

Notice that this statement uses the word `var` instead of a data type. The `var` keyword tells the compiler to determine the variable's data type from the initialization value. In this example, the initialization value 100 is an `int`, so `amount` will be an `int` variable. Here are other examples:

```
var interestRate = 12.0;
var stockCode = "D465U";
var accountBalance = 1000.0m;
```

In this code, the `interestRate` variable will be a `double` because the initialization value 12.0 is a `double`. The `stockCode` variable will be a `string` because the initialization value `"D465U"` is a `string`. The `accountBalance` variable will be a `decimal` because the initialization value `1000.0m` is a `decimal`.

You cannot declare multiple variables in one statement with the `var` keyword. For example, the following statement will cause an error:

```
var x, y, z = 99;    // ERROR!
```

You must provide an initialization value when you declare a variable with the `var` keyword. The following statement will cause an error:

```
var myvalue;
```

These examples show how to use the `var` keyword, but they don't really show its usefulness. The `var` keyword is intended to simplify the syntax of declarations that are more complex than the ones shown here. Later in the book, you will see examples of how the `var` keyword can improve the readability of complex definition statements.

Checkpoint

3.14 Specify the appropriate primitive numeric data type to use for each of the following values:

a. 24 dollars

b. 12 bananas

c. 14.5 inches

d. 83 cents

e. 2 concert tickets

3.15 Which of the following variable declarations will cause an error? Why?

a. `decimal payRate = 24m;`

b. `int playerScore = 1340.5;`

c. `double boxWidth = 205.25;`

d. `string lastName = "Holm";`

3.16 Write a programming statement that will convert the following `decimal` variable to an `int` and store the result in an `int` variable named `dollars`:

```
decimal deposit = 976.54m;
```

3.17 What value will the `wholePieces` variable contain after the following code executes?

```
double totalPieces = 6.5;
int wholePieces = (int)totalPieces;
```

3.18 Given the following declaration statements, what is the data type of each variable?

a. `var idNumber1 = "007";`

b. `var idNumber2 = 0.007;`

c. `var idNumber3 = 7;`

d. `var idNumber1 = 7m;`

3.4 Performing Calculations

CONCEPT: **You can use math operators to perform simple calculations. Math expressions can be written using the math operators and parentheses as grouping symbols. The result of a math expression can be assigned to a variable.**

Most programs require calculations of some sort to be performed. A programmer's tools for performing calculations are **math operators**. C# provides the math operators shown in Table 3-4.

Table 3-4 Math operators

Operator	Name of the Operator	Description
+	Addition	Adds two numbers
–	Subtraction	Subtracts one number from another
*	Multiplication	Multiplies one number by another
/	Division	Divides one number by another and gives the quotient
%	Modulus	Divides one integer by another and gives the remainder

Programmers use the operators shown in Table 3-4 to create math expressions. A **math expression** performs a calculation and gives a value. The following is an example of a simple math expression:

```
12 * 2
```

The values on the right and left of the * operator are called **operands**. These are values that the * operator multiplies together. The value that is given by this expression is 24.

Variables may also be used in a math expression. For example, suppose we have two variables named `hoursWorked` and `payRate`. The following math expression uses the * operator to multiply the value in the `hoursWorked` variable by the value in the `payRate` variable:

```
hoursWorked * payRate
```

When we use a math expression to calculate a value, we have to do something with the value. Normally we want to save the value in memory so we can use it again in the program. We do this with an assignment statement. For example, suppose we have another variable named `grossPay`. The following statement assigns the value `hoursWorked` times `payRate` to the `grossPay` variable:

```
grossPay = hoursWorked * payRate
```

Here are some other examples of statements that assign the result of a math expression to a variable:

```
total = price + tax;
sale = price - discount;
commission = sales * percent;
half = number / 2;
```

The modulus operator (`%`) performs division between two integers, but instead of returning the quotient, it returns the remainder. The following statement assigns 2 to `leftOver`:

```
leftOver = 17 % 3;
```

This statement assigns 2 to `leftover` because 17 divided by 3 is 5 with a remainder of 2. You will not use the modulus operator frequently, but it is useful in some situations. It is commonly used in calculations that detect odd or even numbers, determine the day of the week, or measure the passage of time and in other specialized operations.

The Order of Operations

You can write mathematical expressions with several operators. The following statement assigns the sum of 17, the variable `x`, 21, and the variable `y` to the variable `answer`.

```
answer = 17 + x + 21 + y;
```

Some expressions are not that straightforward, however. Consider the following statement:

```
outcome = 12 + 6 / 3;
```

What value will be stored in `outcome`? The number 6 is used as an operand for both the addition and division operators. The `outcome` variable could be assigned either 6 or 14, depending on when the division takes place. The answer is 14 because the **order of operations** dictates that the division operator works before the addition operator does.

The order of operations can be summarized as follows:

1. Perform any operations that are enclosed in parentheses.
2. Perform any multiplications, divisions, or modulus operations as they appear from left to right.
3. Perform any additions or subtractions as they appear from left to right.

Mathematical expressions are evaluated from left to right. Multiplication and division are always performed before addition and subtraction, so the statement

```
outcome = 12 + 6 / 3;
```

works like this:

1. 6 is divided by 3, yielding a result of 2.
2. 12 is added to 2, yielding a result of 14.

It could be diagrammed as shown in Figure 3-11.

Table 3-5 shows some other sample expressions with their values.

Figure 3-11 The order of operations at work

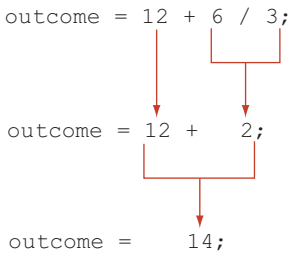


Table 3-5 Some math expressions and their values

Expression	Value
`5 + 2 * 4`	13
`10 / 2 - 3`	2
`8 + 12 * 2 - 4`	28
`6 - 3 * 2 + 7 - 1`	6

Grouping with Parentheses

Parts of a mathematical expression may be grouped with parentheses to force some operations to be performed before others. In the following statement, the variables `a` and `b` are added together, and their sum is divided by 4:

```
result = (a + b) / 4;
```

But what if we left the parentheses out, as shown here?

```
result = a + b / 4;
```

We would get a different result. Without the parentheses, `b` would be divided by 4 and the result added to `a`. Table 3-6 shows some math expressions that use parentheses and their values.

Table 3-6 More expressions and their values

Expression	Value
`(5 + 2) * 4`	28
`10 / (5 – 3)`	5
`8 + 12 * (6 – 2)`	56
`(6 – 3) * (2 + 7) / 3`	9

Mixing Data Types in a Math Expression

When you perform a math operation on two operands, the data type of the result will depend on the data type of the operands. If the operands are of the same data type, the result will also be of that data type. For example:

- When an operation is performed on two `int` values, the result will be an `int`.
- When an operation is performed on two `double` values, the result will be a `double`.
- When an operation is performed on two `decimal` values, the result will be a `decimal`.

It's not uncommon, however, for a math expression to have operands of different data types. C# handles operations involving `int`, `double`, and `decimal` operands in the following ways:

- When a math expression involves an `int` and a `double`, the `int` is temporarily converted to a `double`, and the result is a `double`.
- When a math expression involves an `int` and a `decimal`, the `int` is temporarily converted to a `decimal`, and the result is a `decimal`.
- Math expressions involving a `double` and a `decimal` are not allowed unless a cast operator is used to convert one of the operands.

For example, suppose a pay-calculating program has the following variable declarations:

```
int hoursWorked;  // To hold the number of hours worked
decimal payRate;  // To hold the hourly pay rate
decimal grossPay; // To hold the gross pay
```

Then, later in the program this statement appears:

```
grossPay = hoursWorked * payRate;
```

The math expression on the right side of the = operator multiplies an `int` by a `decimal`. When the statement executes, the value of the `hoursWorked` variable is temporarily converted to a `decimal` and then multiplied by the `payRate` variable. The result is a `decimal` and is assigned to the `grossPay` variable.

When possible, you should avoid math operations that use a mixture of `double` and `decimal` operands. C# does not allow operations involving these two types unless you use a cast operator to explicitly convert one of the operands. For example, suppose a program that calculates the cost of a product has the following variable declarations:

```
double weight;         // The product weight
decimal pricePerPound; // The price per pound
decimal total;         // The total cost
```

Later in the program you need to calculate the total cost, like this:

```
total = weight * pricePerPound; ← ERROR!
```

The compiler will not allow this statement because `weight` is a `double` and `pricePerPound` is a `decimal`. To fix the statement, you can insert a cast operator, as shown here:

```
total = (decimal)weight * pricePerPound;
```

The cast operator converts the value of the `weight` variable to a `decimal`, and the converted value is multiplied by `pricePerPound`. The result of the expression is a `decimal` and is assigned to `total`.

Integer Division

When you divide an integer by an integer in C#, the result is always given as an integer. If the result has a fractional part, it is truncated. For example, look at the following code:

```
int length;           // Declare length as an int
double half;          // Declare half as a double
length = 75;          // Assign 75 to length
half = length / 2;    // Calculate half the length
```

The last statement divides the value of `length` by 2 and assigns the result to half. Mathematically, the result of 75 divided by 2 is 37.5. However, that is not the result that we get from the math expression. The `length` variable is an `int`, and it is being divided by the numeric literal 2, which is also treated as an `int`. The result of the division is truncated, which gives the value 37. This is the value that is assigned to the `half` variable. It does not matter that the `half` variable is declared as a `double`. The fractional part of the result is truncated before the assignment takes place.

When you are dividing an integer by another integer and you want to keep the fractional part of the result, simply use a cast operator to convert one of the operands to a floating-point type. Here is an example:

```
int pies = 10;
int people = 4;
double piesPerPerson;
piesPerPerson = (double)pies / people;
```

In this code, both `pies` and `people` are `int` variables. The last statement divides `pies` by `people`; however, the cast operator converts the value of `pies` to a `double` before the division takes place. This means that one of the division operator's operands is a `double`, so the result of the division will be a `double`.

In the last statement of the previous code snippet, we can use the cast operator to convert either `pies` or `people` to a `double`, and the result of the division will be a `double`. For example, the statement could have been written as follows:

```
piesPerPerson = pies / (double)people;
```

In this statement, the cast operator converts the value of `people` to a `double` before the division takes place. Because one of the division operator's operands is a `double`, the result of the division will be a `double`.

WARNING! The cast operator can be applied to an entire expression enclosed in parentheses. For example, look at the following statement:

```
piesPerPerson = (double)(pies / people);
```

This statement does not convert the value in `pies` or `people` to a `double`, but converts the result of the expression `pies / people`. If this statement were used, an integer division operation would still have been performed. Here's why: The result of the expression `pies / people` is 2 (because integer division takes place). The value 2 converted to a `double` is 2.0. To prevent the integer division from taking place, the cast operator should be applied to one of the operands.

Combined Assignment Operators

Sometimes you want to increase a variable's value by a certain amount. For example, suppose you have a variable named `number` and you want to increase its value by 1. You can accomplish that with the following statement:

```
number = number + 1;
```

The expression on the right side of the assignment operator calculates the value of `number` plus 1. The result is then assigned to `number`, replacing the value that was previously stored there. This statement effectively adds 1 to `number`. For example, if `number` is equal to 6 before this statement executes, it is equal to 7 after the statement executes.

Similarly, the following statement subtracts 5 from `number`:

```
number = number - 5;
```

If `number` is equal to 15 before this statement executes, it is equal to 10 after the statement executes. Here's another example. The following statement doubles the value of the `number` variable:

```
number = number * 2;
```

If `number` is equal to 4 before this statement executes, it is equal to 8 after the statement executes.

These types of operations are very common in programming. For convenience, C# offers a special set of operators known as **combined assignment operators** that are designed specifically for these jobs. Table 3-7 shows the combined assignment operators.

Table 3-7 Combined assignment operators

Operator	Example Usage	Equivalence
`+=`	`x += 5;`	`x = x + 5;`
`-=`	`y -= 2;`	`y = y - 2;`
`*=`	`z *= 10;`	`z = z * 10;`
`/=`	`a /= b;`	`a = a / b;`
`%=`	`c %= 3;`	`c = c % 3;`

As you can see, the combined assignment operators do not require the programmer to type the variable name twice. Also, they give a clear indication of what is happening in the statement.

Checkpoint

3.19 List the operands for the following math expression:

```
length * width
```

3.20 Summarize the mathematical order of operations.

3.21 Rewrite the following code segment so that it does not cause an error.

```
decimal pricePerFoot = 2.99m;
double boardLength = 10.5;
decimal totalCost = boardLength * pricePerFoot;
```

3.22 Assume `result` is a `double` variable. When the following statement executes, what value will be stored in `result`?

```
result = 4 + 10 / 2;
```

3.23 Assume `result` is an `int` variable. When the following statement executes, what value will be stored in `result`?

```
result = (2 + 5) * 10;
```

3.24 Assume `result` is a `double` variable. When the following statement executes, what value will be stored in `result`?

```
result = 5 / 2;
```

3.25 Rewrite the following statements using combined assignment operators:

a. `count = count + 1;`
b. `amount = amount - 5;`
c. `radius = radius * 10;`
d. `length = length / 2;`

3.5 Inputting and Outputting Numeric Values

CONCEPT: **If the user has entered a number into a TextBox, the number will be stored as a string in the TextBox's Text property. If you want to store that number in a numeric variable, you have to convert it to the appropriate numeric data type. When you want to display the value of a numeric variable in a Label control or a message box, you have to convert it to a string.**

Getting a Number from a TextBox

Graphical User Interface (GUI) applications typically use TextBox controls to read keyboard input. Any data that the user enters into a TextBox control is stored in the control's Text property as a string, even if it is a number. For example, if the user enters the number 72 into a TextBox control, the input is stored as the string `"72"` in the control's Text property.

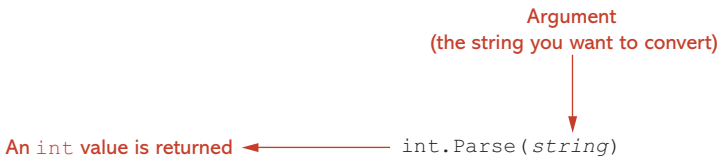
If the user has entered a numeric value into a TextBox control and you want to assign that value to a numeric variable, you have to convert the control's Text property to the desired numeric data type. Unfortunately, you cannot use a cast operator to convert a string to a numeric type.

To convert a string to any of the numeric data types, we use a family of .NET methods known as the **`Parse` methods**. In computer science, the term **parse** typically means to analyze a string of characters for some purpose. The `Parse` methods are used to convert a string to a specific data type. There are several `Parse` methods in .NET, but because we are primarily using the `int`, `double`, and `decimal` numeric data types, we need only three of them:

- We use the `int.Parse` method to convert a string to an `int`.
- We use the `double.Parse` method to convert a string to a `double`.
- We use the `decimal.Parse` method to convert a string to a `decimal`.

When you call one of the `Parse` methods, you pass a piece of data known as an **argument** into the method, and the method returns a piece of data back to you. The argument that you pass to the method is the string that you want to convert, and the piece of data that the method returns back to you is the converted value. Figure 3-12 illustrates this concept using the `int.Parse` method as an example.

Figure 3-12 The `int.Parse` method

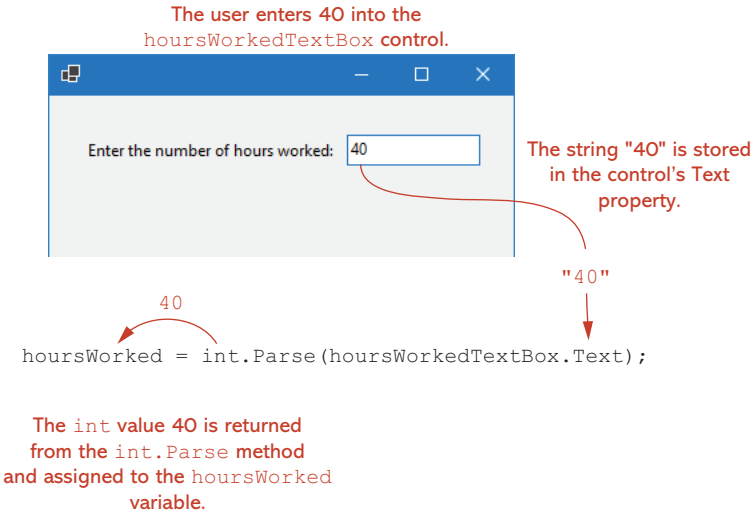


The following code sample shows how to use the `int.Parse` method to convert a control's Text property to an `int`. Assume that `hoursWorkedTextBox` is the name of a TextBox control.

```
1 // Declare an int variable to hold the hours worked.
2 int hoursWorked;
3
4 // Get the hours worked from the TextBox.
5 hoursWorked = int.Parse(hoursWorkedTextBox.Text);
```

Let's assume that the user has entered the value 40 into the `hoursWorkedTextBox` control. In line 5 of the code sample, on the right side of the = operator is the expression `int.Parse(hoursWorkedTextBox.Text)`. This expression calls the `int.Parse` method, passing the value of `hoursWorkedTextBox.Text` as an argument. Because the user has entered 40 into the TextBox, the string `"40"` is the value that is passed as the argument. The method converts the string `"40"` to the `int` value 40. The `int` value 40 is returned from the method and the = operator assigns it to the `hoursWorked` variable. Figure 3-13 illustrates this process.

Figure 3-13 Converting TextBox input to an `int`



The following code sample demonstrates the `double.Parse` method. Assume that `temperatureTextBox` is the name of a TextBox control.

```
1 // Declare a double variable to hold the temperature.
2 double temperature;
3
4 // Get the temperature from the TextBox.
5 temperature = double.Parse(temperatureTextBox.Text);
```

Line 5 passes `temperatureTextBox.Text` as an argument to the `double.Parse` method. That value is converted to a `double`, returned from the `double.parse` method, and assigned to the `temperature` variable.

The following code sample demonstrates the `decimal.Parse` method. Assume that `moneyTextBox` is the name of a TextBox control.

```
1 // Declare a decimal variable to hold an amount of money.
2 decimal money;
3
4 // Get an amount from the TextBox.
5 money = decimal.Parse(moneyTextBox.Text);
```

Line 5 passes `moneyTextBox.Text` as an argument to the `decimal.Parse` method. That value is converted to a `decimal`, returned from the `decimal.parse` method, and assigned to the `money` variable.

NOTE: If you look at the top of a form's source code in the code editor, you should see a directive that reads `using System;`. That directive is required for any program that uses the `Parse` methods.

Invalid Conversions

The `Parse` methods work only if the string that is being converted contains a valid numeric value. If the string contains invalid characters or contains a number that cannot be converted to the specified data type, an error known as an exception occurs. An **exception** is an unexpected error that occurs while a program is running, causing the program to halt if the error is not properly dealt with.

For example, assume that `hoursWorked` is an `int` variable and `hoursWorkedTextBox` is a TextBox control. Suppose the user has entered *xyz* into the TextBox and the following statement executes:

```
hoursWorked = int.Parse(hoursWorkedTextBox.Text);
```

Obviously, the string `"xyz"` cannot be converted to an `int`, so an exception occurs. (When an exception occurs, programmers say an exception is "thrown.") Depending on how you executed the application, you will see one of the error messages displayed in Figure 3-14 or Figure 3-15.

- If you see the message in Figure 3-14, you can stop the application by clicking the *Stop Debugging* button (), or by pressing Shift + F5, or by clicking *Debug* and then *Stop Debugging*.
- When you see the window shown in Figure 3-15, in most situations you should click the *Quit* button to stop the application.

Figure 3-14 Exception reported

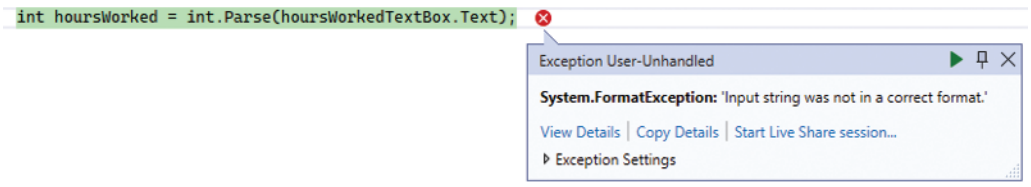
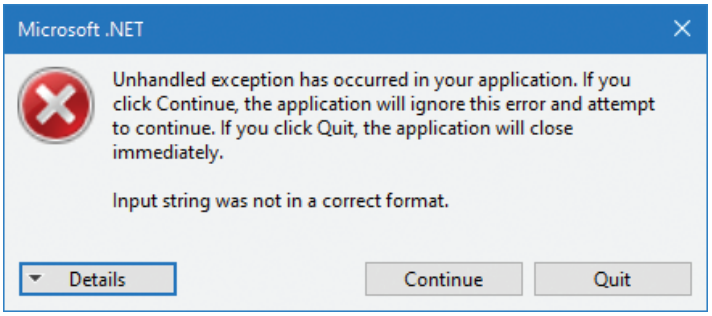


Figure 3-15 Exception reported



Later in this chapter, you will learn how to catch errors like this and prevent the program from halting.

Displaying Numeric Values

Suppose an application has a `decimal` variable named `grossPay` and a Label control named `grossPayLabel`. You want to display the variable's value in the Label control. To accomplish this, you must somehow get the value of the `grossPay` variable into the `grossPayLabel` control's Text property. The following assignment statement will not work, however:

```
grossPayLabel.Text = grossPay; ← ERROR!
```

You cannot assign a numeric value to a control's Text property because only strings can be assigned to the Text property. If you want to display the value of a numeric variable in a Label control, you have to convert the variable's value to a string.

Luckily, all variables have a `ToString` method that you can call to convert the variable's value to a string. You call the `ToString` method using the following general format:

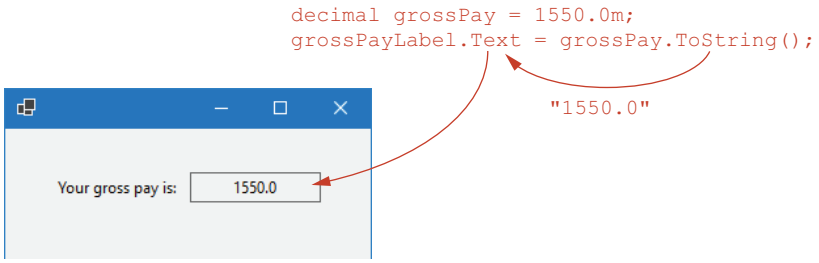
```
variableName.ToString()
```

In the general format, `variableName` is the name of any variable. The expression returns the variable's value as a string. Here is a code sample that demonstrates:

```
decimal grossPay = 1550.0m;
grossPayLabel.Text = grossPay.ToString();
```

The first statement declares a `decimal` variable named `grossPay` initialized with the value 1,550.0. In the second statement, the expression on the right side of the = operator calls the `grossPay` variable's `ToString` method. The method returns the string `"1550.0"`. The = operator then assigns the string `"1550.0"` to the `grossPayLabel` control's Text property. As a result, the value *1550.0* is displayed in the `grossPayLabel` control. This process is illustrated in Figure 3-16.

Figure 3-16 Displaying numeric data in a Label control



You must also convert a numeric variable to a `string` before passing it to the `MessageBox.Show` method. The following example shows how an `int` variable's value can be converted to a string and displayed in a message box:

```
int myNumber = 123;
MessageBox.Show(myNumber.ToString());
```

The first statement declares an `int` variable named `myNumber`, initialized with the value 123. In the second statement the following takes place:

- The `myNumber` variable's `ToString` method is called. The method returns the string `"123"`.
- The string `"123"` is passed to the `MessageBox.Show` method. As a result, the value *123* is displayed in a message box.

Implicit String Conversion with the + Operator

In this chapter, you've learned that the + operator has two uses: string concatenation and numeric addition. If you write an expression using the + operator and both operands are strings, the + operator concatenates the strings. If both operands are numbers of compatible types, then the + operator adds the two numbers. But what happens if one operand is a string and the other operand is a number? The number will be implicitly converted to a string, and both operands will be concatenated. Here's an example:

```
int idNumber = 1044;
string output = "Your ID number is " + idNumber;
```

In the second statement, the `string` variable `output` is initialized with the string `"Your ID number is 1044"`. Here is another example:

```
double testScore = 88.5;
MessageBox.Show("Your test score is " + testScore);
```

The second statement displays a message box showing the string `"Your test score is 88.5"`.

In Tutorial 3-2, you will use some of the techniques discussed in this section. You will create an application that reads numeric input from TextBox controls, and displays numeric output in a Label control.

Tutorial 3-2:
Calculating Fuel Economy

VideoNote
Tutorial 3-2: Calculating Fuel Economy

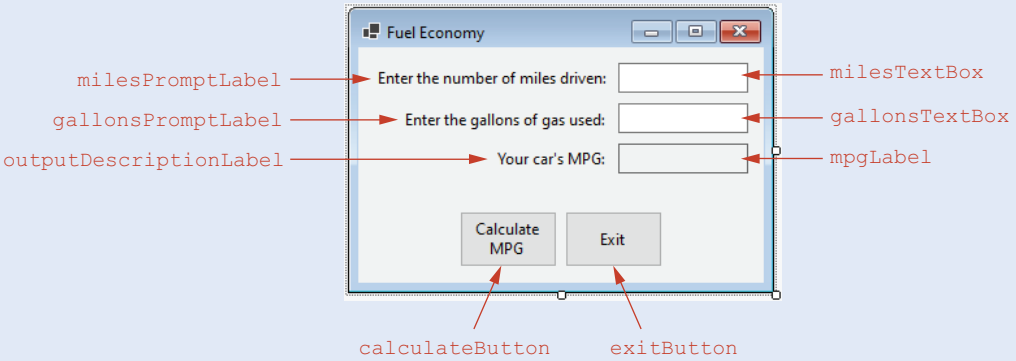
In the United States, a car's fuel economy is measured in miles per gallon, or MPG. You use the following formula to calculate a car's MPG:

$$MPG = Miles\ driven \div Gallons\ of\ gas\ used$$

In this tutorial, you will create an application that lets the user enter the number of miles driven and the gallons of gas used. The application will calculate and display the car's MPG.

Figure 3-17 shows the application's form, with the names of all the controls. When the application runs, the user enter the number of miles driven into the `milesTextBox` control and the gallons of gas used into the `gallonsTextBox` control. When the user clicks the `calculateButton` control, the application calculates the car's MPG and displays the result in the `mpgLabel` control. The `exitButton` control closes the application's form.

Figure 3-17 The *Fuel Economy* form



Step 1: Start Visual Studio and begin a new Windows Forms App project named *FuelEconomy*.

Step 2: Set up the application's form as shown in Figure 3-17. Notice that the form's Text property is set to *Fuel Economy*. The names of the controls are shown in the figure. As you place each of the controls on the form, refer to Table 3-8 for the relevant property settings.

Table 3-8 Control property settings

Control Name	Control Type	Property Settings
`milesPromptLabel`	Label	**Text:** *Enter the number of miles driven:*
`gasPromptLabel`	Label	**Text:** *Enter the gallons of gas used:*
`outputDescriptionLabel`	Label	**Text:** *Your car's MPG:*
`milesTextBox`	TextBox	No properties changed
`gallonsTextBox`	TextBox	No properties changed
`mpgLabel`	Label	**AutoSize:** False
		BorderStyle: FixedSingle
		Text: (The contents of the Text property have been erased.)
		TextAlign: MiddleCenter
`calculateButton`	Button	**Text:** *Calculate MPG*
`exitButton`	Button	**Text:** *Exit*

Step 3: Once you have set up the form with its controls, you can create the Click event handlers for the Button controls. At the end of this tutorial, Program 3-2 shows the completed code for the form. You will be instructed to refer to Program 3-2 as you write the event handlers. (Remember, the line numbers that are shown in Program 3-2 are not part of the program. They are shown for reference only.)

In the *Designer*, double-click the `calculateButton` control. This opens the code editor, and you will see an empty event handler named `calculateButton_Click`. Complete the `calculateButton_Click` event handler by typing the code shown in lines 12–28 in Program 3-2.

Let's take a closer look at the code:

Line 12: This statement declares a `double` variable named `miles`. This variable is used to hold the number of miles driven.

Line 13: This statement declares a `double` variable named `gallons`. This variable is used to hold the number of gallons used.

Line 14: This statement declares a `double` variable named `mpg`. This variable is used to hold the MPG, which will be calculated.

Line 18: This statement converts the `milesTextBox` control's Text property to a `double` and assigns the result to the `miles` variable.

Line 22: This statement converts the `gallonsTextBox` control's Text property to a `double` and assigns the result to the `gallons` variable.

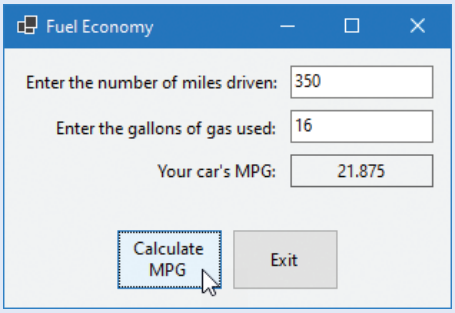
Line 25: This statement calculates MPG. It divides the `miles` variable by the `gallons` variable and assigns the result to the `mpg` variable.

Line 28: This statement converts the `mpg` variable to a string and assigns the result to the `mpgLabel` control's Text property. This causes the value of the `mpg` variable to be displayed in the `mpgLabel` control.

Step 4: Switch your view back to the *Designer* and double-click the `exitButton` control. In the code editor you will see an empty event handler named `exitButton_Click`. Complete the `exitButton_Click` event handler by typing the code shown in lines 33–34 in Program 3-2.

Step 5: Save the project. Then, press the F5 key on the keyboard or click the *Start Debugging* button (▶) on the toolbar to compile and run the application. Test the application by entering values into the TextBoxes and clicking the *Calculate MPG* button. The MPG should be displayed, similar to Figure 3-18. Click the *Exit* button and the form should close.

Figure 3-18 The *Fuel Economy* application



Program 3-2 Completed Form1 code for the *FuelEconomy* project

```
 1 namespace FuelEconomy
 2 {
 3     public partial class Form1 : Form
 4     {
 5         public Form1()
 6         {
 7             InitializeComponent();
 8         }
 9
10         private void calculateButton_Click(object sender,
           EventArgs e)
11         {
12             double miles;     // To hold miles driven
13             double gallons;   // To hold gallons used
14             double mpg;       // To hold MPG
15
16             // Get the miles driven and assign it to
17             // the miles variable.
18             miles = double.Parse(milesTextBox.Text);
19
20             // Get the gallons used and assign it to
21             // the gallons variable.
22             gallons = double.Parse(gallonsTextBox.Text);
23
24             // Calculate MPG.
25             mpg = miles / gallons;
26
27             // Display the MPG in the mpgLabel control.
28             mpgLabel.Text = mpg.ToString();
29         }
30
```

```
31          private void exitButton_Click(object sender, EventArgs e)
32          {
33              // Close the form.
34              this.Close();
35          }
36      }
37  }
```

Checkpoint

3.26 What method converts the string literal `"40"` to a value of the `int` data type?

3.27 Write a statement that converts each of the following `string` values to the `decimal` data type using the `decimal.Parse` method.

a. `"9.05"`

b. `grandTotal`

c. `"50"`

d. `priceTextBox.Text`

3.28 Suppose an application has a `decimal` variable named `total` and a Label control named `totalLabel`. What will be the result when the following assignment statement is executed?

```
totalLabel.Text = total;
```

3.29 Write a statement that displays each of the following numeric variables in a message box.

a. `grandTotal`

b. `highScore`

c. `sum`

d. `width`

3.30 Write a statement that will store the value of an `int` variable named `result` in the Text property of a Label control named `resultLabel`.

3.6 Formatting Numbers with the `ToString` Method

CONCEPT: **The `ToString` method can optionally format a number to appear in a specific way.**

When you display large numbers, you usually want to format them with commas so they are easy to read. For example, 487,634,789.0 is easier to read than 487634789.0. Also, when you display amounts of money, you usually want to round them to two decimal places and display a currency symbol, such as a dollar sign ($).

When you call the `ToString` method, you can optionally pass a formatting string as an argument to the method. The **formatting string** indicates that you want the number to appear formatted in a specific way when it is returned as a string from the method. For example, when you pass the formatting string `"c"` to the `ToString` method, the number is returned formatted as currency. Assuming that you are in the United States, numbers formatted as currency are preceded by a dollar sign ($), are rounded to two decimal

places, and have comma separators inserted as necessary. The following code sample demonstrates:

```
decimal amount = 123456789.45678m;
MessageBox.Show(amount.ToString("c"));
```

Notice in the second statement that the `"c"` formatting string is passed to the `amount` variable's `ToString` method. The message box that the statement displays appears as shown in Figure 3-19.

There are several other format strings that you can use with the `ToString` method, and each produces a different type of formatting. Table 3-9 shows a few of them.

Figure 3-19 A number formatted as currency

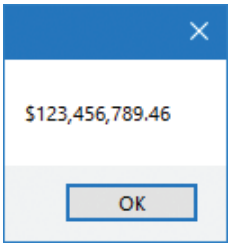


Table 3-9 A few of the formatting strings

Format String	Description
`"N"` or `"n"`	Number format
`"F"` or `"f"`	Fixed-point scientific format
`"E"` or `"e"`	Exponential scientific format
`"C"` or `"c"`	Currency format
`"P"` or `"p"`	Percent format

Number Format

Number format (`"n"` or `"N"`) displays numeric values with comma separators and a decimal point. By default, two digits display to the right of the decimal point. Negative values are displayed with a leading minus sign. An example is −2,345.67.

Fixed-Point Format

Fixed-point format (`"f"` or `"F"`) displays numeric values with no thousands separator and a decimal point. By default, two digits display to the right of the decimal point. Negative values are displayed with a leading minus (-) sign. An example is −2345.67.

Exponential Format

Exponential format (`"e"` or `"E"`) displays numeric values in scientific notation. The number is displayed with a single digit to the left of the decimal point. The letter e appears in front of the exponent, and the exponent has a leading + or - sign. By default, six digits display to the right of the decimal point, and a leading minus sign is used if the number is negative.

An example is −2.345670e+003.

Currency Format

Currency format ("c" or "C") displays a leading currency symbol (such as $), digits, comma separators, and a decimal point. By default, two digits display to the right of the decimal point. Negative values are surrounded by parentheses. An example of a negative value is ($2,345.67).

Using Percent Format

Percent format ("p" or "P") causes the number to be multiplied by 100 and displayed with a trailing space and % sign. By default, two digits display to the right of the decimal point. Negative values are displayed with a leading minus sign. For example, the number 0.125 would be formatted as 12.5 % and the number −0.2345 would be formatted as -23.45 %.

Specifying the Precision

Each numeric format string can optionally be followed by an integer that indicates how many digits to display after the decimal point. For example, the format "n3" displays three digits after the decimal point. Table 3-10 shows a variety of numeric formatting examples, based on the North American locale.

Table 3-10 Numeric formatting examples (North American locale)

Number	Format String	ToString( ) Return Value
12.3	"n3"	12.300
12.348	"n2"	12.35
1234567.1	"N"	1,234,567.10
123456.0	"f2"	123456.00
123456.0	"e3"	1.235e+005
.234	"P"	23.40 %
-1234567.8	"C"	($1,234,567.80)

Rounding

Rounding can occur when the number of digits you have specified after the decimal point in the format string is smaller than the precision of the numeric value. Suppose, for example, that the value 1.235 were displayed with a format string of "n2". Then the displayed value would be 1.24. If the next digit after the last displayed digit is 5 or higher, the last displayed digit is rounded *away from zero*. Table 3-11 shows examples of rounding using a format string of "n2".

Using Leading Zeros with Integer Values

You can use the "d" or "D" formatting strings with integers to specify the minimum width for displaying the number. Leading zeros are inserted if necessary. Table 3-12 shows examples.

Table 3-11 Rounding examples using the `"n2"` display format string

Number	Formatted As
1.234	1.23
1.235	1.24
1.238	1.24
-1.234	-1.23
-1.235	-1.24
-1.238	-1.24

Table 3-12 Formatting integers using the `"d"` or `"D"` formatting strings

Integer Value	Format String	Formatted As
23	"d"	23
23	"d4"	0023
1	"d2"	01

In Tutorial 3-3, you will create an application that uses currency formatting to display a dollar amount.

Tutorial 3-3:
Creating the *Sale Price Calculator* Application with Currency Formatting

VideoNote
Tutorial 3-3: Creating the *Sale Price Calculator* Application with Currency Formatting

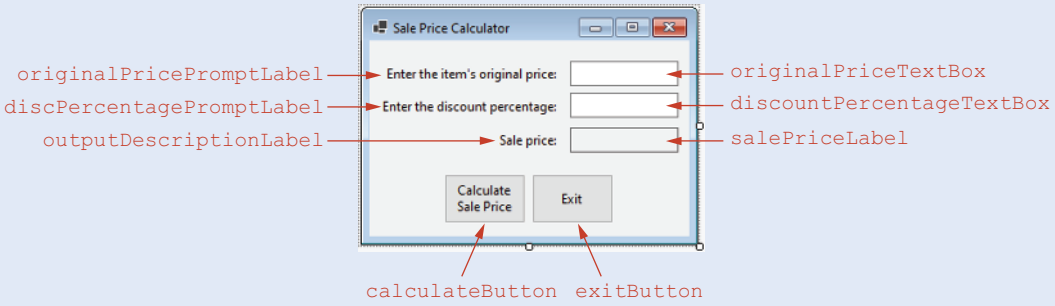
If you are writing a program that works with a percentage, you have to make sure that the percentage's decimal point is in the correct location before doing any math with the percentage. This is especially true when the user enters a percentage as input. Most users will enter the number 50 to mean 50 percent, 20 to mean 20 percent, and so forth. Before you perform any calculations with such a percentage, you have to divide it by 100 to move its decimal point to the left two places.

Suppose a retail business is planning to have a storewide sale in which the prices of all items will be reduced by a specified percentage. In this tutorial, you will create an application to calculate the sale price of an item after the discount is subtracted. Here is the algorithm, expressed as pseudocode:

1. *Get the original price of the item.*
2. *Get the discount percentage. (For example, 20 is entered for 20 percent.)*
3. *Divide the percentage amount by 100 to move the decimal point to the correct location.*
4. *Multiply the percentage by the original price. This is the amount of the discount.*
5. *Subtract the discount from the original price. This is the sale price.*
6. *Display the sale price.*

Figure 3-20 shows the application's form, with the names of all the controls. When the application runs, the user enters an item's original price into the `originalPriceTextBox`

Figure 3-20 The *Sale Price Calculator* form



control and the discount percentage into the `discountPercentageTextBox` control. When the user clicks the `calculateButton` control, the application calculates the item's sale price and displays the result in the `salePriceLabel` control. The `exitButton` control closes the application's form.

Step 1: Start Visual Studio and begin a new Windows Forms App project named *SalePriceCalculator*.

Step 2: Set up the application's form, as shown in Figure 3-20. Notice that the form's Text property is set to *Sale Price Calculator*. The names of the controls are shown in the figure. As you place each of the controls on the form, refer to Table 3-13 for the relevant property settings.

Table 3-13 Control property settings

Control Name	Control Type	Property Settings
`originalPricePromptLabel`	Label	**Text:** *Enter the item's original price:*
`discPercentagePromptLabel`	Label	**Text:** *Enter the discount percentage:*
`outputDescriptionLabel`	Label	**Text:** *Sale price:*
`originalPriceTextBox`	TextBox	No properties changed
`discountPercentageTextBox`	TextBox	No properties changed
`salePriceLabel`	Label	**AutoSize:** False **BorderStyle:** FixedSingle **Text:** (The contents of the Text property have been erased.) **TextAlign:** MiddleCenter
`calculateButton`	Button	**Text:** *Calculate Sale Price*
`exitButton`	Button	**Text:** *Exit*

Step 3: Once you have set up the form with its controls, you can create the Click event handlers for the Button controls. At the end of this tutorial, Program 3-3 shows the completed code for the form. You will be instructed to refer to Program 3-3 as you write the event handlers. (Remember, the line numbers that are shown in Program 3-3 are not part of the program. They are shown for reference only.)

In the *Designer*, double-click the `calculateButton` control. This will open the code editor, and you will see an empty event handler named `calculateButton_Click`. Complete the `calculateButton_Click` event handler by typing the code shown in lines 12–33 in Program 3-3.

Let's take a closer look at the code:

Line 12: This statement declares a `decimal` variable named `originalPrice`. This variable will hold the item's original price.

Line 13: This statement declares a `decimal` variable named `discountPercentage`. This variable will hold the discount percentage.

Line 14: This statement declares a `decimal` variable named `discountAmount`. This variable will hold the amount of discount that will be taken from the item's original price. This amount will be calculated.

Line 15: This statement declares a `decimal` variable named `salePrice`. This variable will hold the item's sale price. This amount will be calculated.

Line 18: This statement converts the `originalPriceTextBox` control's Text property to a `decimal` and assigns the result to the `originalPrice` variable.

Line 21: This statement converts the `discountPercentageTextBox` control's Text property to a `decimal` and assigns the result to the `discountPercentage` variable.

Line 24: This statement divides `discountPercentage` by 100 and stores the result back in `discountPercentage`. This moves the decimal point in the `discountPercentage` variable to the left two places.

Line 27: This statement calculates the amount of the discount. It multiplies `originalPrice` by `discountPercentage` and assigns the result to `discountAmount`.

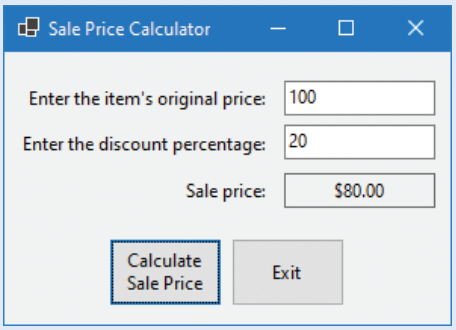
Line 30: This statement calculates the item's sale price. It subtracts the `discountAmount` variable from the `originalPrice` variable and assigns the result to the `salePrice` variable.

Line 33: This statement displays the item's sale price as a currency amount. It converts the `salePrice` variable to a string and assigns the result to the `salePriceLabel` control's Text property. Notice that the format string `"c"` is passed to the `salePrice` variable's `ToString` method.

Step 4: Switch your view back to the *Designer* and double-click the `exitButton` control. In the code editor you will see an empty event handler named `exitButton_Click`. Complete the `exitButton_Click` event handler by typing the code shown in lines 38–39 in Program 3-3.

Step 5: Save the project. Then, press the F5 key on the keyboard or click the *Start Debugging* button (▶) on the toolbar to compile and run the application. Test the application by entering values into the TextBoxes and clicking the *Calculate Sale Price* button. The sale price is displayed, similar to Figure 3-21. Click the *Exit* button and the form closes.

Figure 3-21 The *Sale Price Calculator* application



Program 3-3 Completed Form1 code for the *SalePriceCalculator* project

```
1  namespace SalePriceCalculator
2  {
3      public partial class Form1 : Form
4      {
5          public Form1()
6          {
7              InitializeComponent();
8          }
9
10         private void calculateButton_Click(object sender,
           EventArgs e)
11         {
12             decimal originalPrice;       // The item's
               original price
13             decimal discountPercentage;  // The discount
               percentage
14             decimal discountAmount;      // The amount of
               discount
15             decimal salePrice;           // The item's sale
               price
16
17             // Get the item's original price.
18             originalPrice = decimal.Parse(originalPriceTextBox.Text);
19
20             // Get the discount percentage.
21             discountPercentage = decimal.
               Parse(discountPercentageTextBox.Text);
22
23             // Move the percentage's decimal point left two spaces.
24             discountPercentage = discountPercentage / 100;
25
26             // Calculate the amount of the discount.
27             discountAmount = originalPrice * discountPercentage;
28
29             // Calculate the sale price.
30             salePrice = originalPrice - discountAmount;
31
32             // Display the sale price.
33             salePriceLabel.Text = salePrice.ToString("c");
```

```
34          }
35
36          private void exitButton_Click(object sender, EventArgs e)
37          {
38              // Close the form.
39              this.Close();
40          }
41      }
42  }
```

Checkpoint

3.31 Write a programming statement that displays the string value of a variable named `salary` in a message box using currency format.

3.32 The following variable names give an indication of the data each stores. For each variable, specify the format string that you think is most appropriate.

a. `discountPercentage`
b. `atomicWeight`
c. `retailPrice`
d. `quantityPurchased`
e. `degreesKelvin`

3.33 What value will be displayed in the message box when the following code segment is executed?

```
double apples = 12.0;
MessageBox.Show(apples.ToString("n0"));
```

3.34 Examine the following integer variables and specify the number of leading zeros to use with the `d` or `D` format strings so that all the numbers are equal in width.

```
int valueA = 234, valueB = 56, valueC = 7, valueD = 89123;
```

3.35 Write a programming statement that uses the `ToString` method of a variable named `millimeters` so that it displays a precision of four digits after the decimal point in fixed-point scientific format.

3.7 Simple Exception Handling

CONCEPT: **An exception is an unexpected error that happens while a program is running. If an exception is not handled by the program, the program will abruptly halt.**

An exception is an unexpected error that occurs while a program is running, causing the program to halt if the error is not properly dealt with. Exceptions are usually caused by circumstances that are beyond the programmer's control. For example, suppose the user has entered a value into a TextBox, and that value is expected to be a number. The program uses one of the `Parse` methods to convert the control's Text property to a numeric data type, but the string contains invalid characters and it cannot be converted. The `Parse` method cannot continue, so an exception occurs. (To use the proper terminology, we say that a method throws an exception when an unexpected error occurs and it cannot continue operating.)

Let's look at an example. If you have completed the *Fuel Economy* project from Tutorial 3-2, open it in Visual Studio and then either click the *Start Debugging* button ( ) or press F5 to run the application. On the application's form, enter *300* for the number of miles driven and then enter a nonnumeric sequence of characters for the gallons of gas used. Figure 3-22 shows an example where the user has entered *wxyz*. Then, click the *Calculate MPG* button. Because the invalid string that you entered for the gallons of gas cannot be converted to a `double`, an exception is thrown. The application stops running and Visual Studio goes into a special mode known as **break mode**. The exception window shown in Figure 3-23 is displayed, and the line of code that was executing when the exception was thrown is highlighted.

Figure 3-22 Invalid data entered for gallons

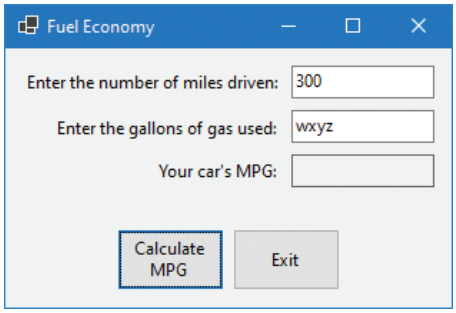
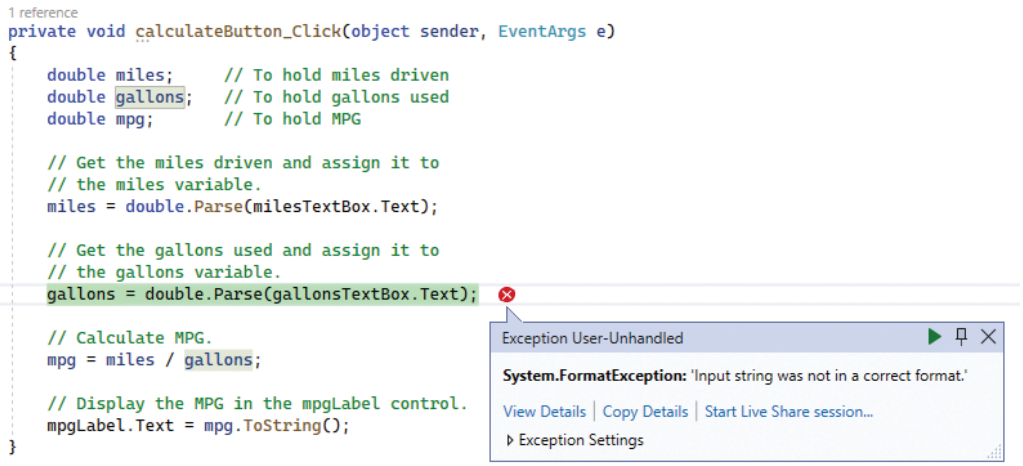


Figure 3-23 Exception reported



If you look at the line just below the exception window's title bar, you see the message *Input string was not in a correct format*. That is a description of what happened to cause the exception. To get out of break mode, click the *Stop Debugging* button ( ), or press Shift + F5. Visual Studio then returns to its normal mode.

Handling Exceptions

C#, like most modern programming languages, allows you to write code that responds to exceptions when they are thrown and prevents the program from abruptly crashing. Such code is called an **exception handler**, and is written with the **`try-catch` statement**. There

are several ways to write a `try-catch` statement, but the following general format is the simplest variation:

```
try
{
  statement;
  statement;
  etc.
}
catch
{
  statement;
  statement;
  etc.
}
```

First the keyword `try` appears, followed by a group of one or more statements that appears inside a set of braces. This group of statements is known as a **try block**. One or more of the statements inside the try block can potentially throw an exception.

After the try block, a **`catch` clause** appears. The `catch` clause is followed by a group of one or more statements enclosed inside a set of braces. This group of statements is known as a **catch block**.

When a `try-catch` statement executes, the statements in the try block are executed in the order that they appear. If a statement in the try block throws an exception, the program immediately jumps to the `catch` clause and executes the statements in the catch block. If all the statements in the try block execute with no exception, the catch block is skipped.

Let's see how a `try-catch` statement can be used in the *Fuel Economy* application. Here is a modified version of the application's `calculateButton_Click` event handler:

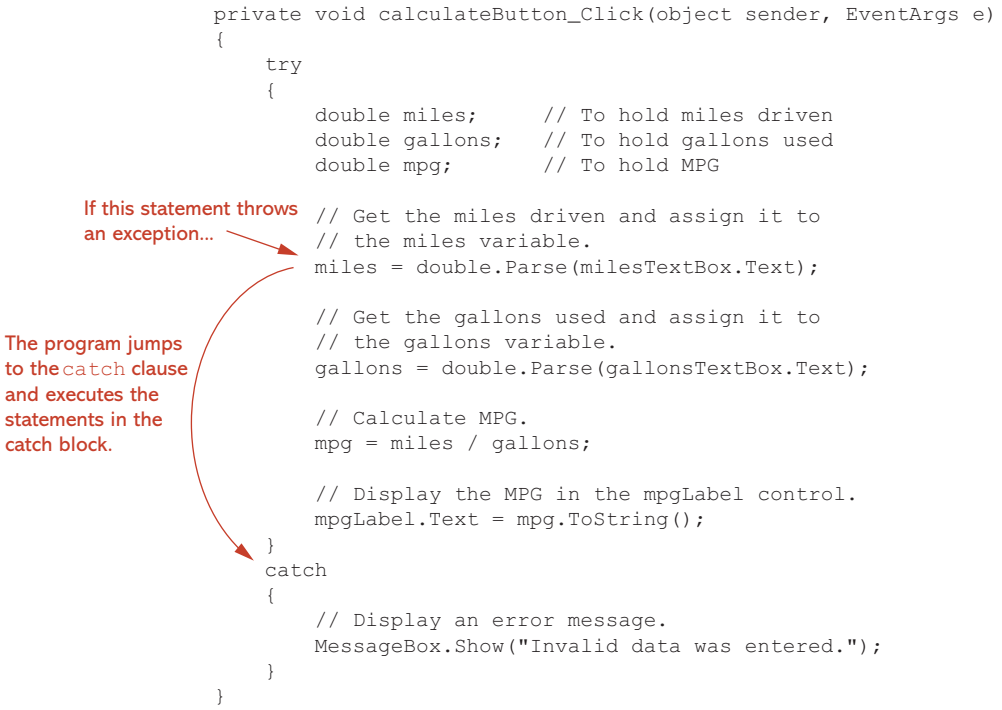
```
 1 private void calculateButton_Click(object sender, EventArgs e)
 2 {
 3     try
 4     {
 5       double miles;   // To hold miles driven
 6       double gallons; // To hold gallons used
 7       double mpg;     // To hold MPG
 8
 9       // Get the miles driven and assign it to
10       // the miles variable.
11       miles = double.Parse(milesTextBox.Text);
12
13       // Get the gallons used and assign it to
14       // the gallons variable.
15       gallons = double.Parse(gallonsTextBox.Text);
16
17       // Calculate MPG.
18       mpg = miles / gallons;
19
20       // Display the MPG in the mpgLabel control.
21       mpgLabel.Text = mpg.ToString();
22     }
23     catch
24     {
25       // Display an error message.
26       MessageBox.Show("Invalid data was entered.");
27     }
28 }
```

When you write a `try-catch` statement, you put all the code that might throw an exception inside the try block. In this version of the event handler, the try block appears in lines 5–21. (In this example, we have put all the statements that previously appeared in the event handler inside the try block.) If any statement inside the try block throws an exception, the program will immediately jump to the `catch` clause in line 23. Then, the statements in the catch block (lines 25–26) will execute.

Let's say that the application is running and the user enters invalid input into the `milesTextBox` control. When the event handler executes, the statement in line 11 throws an exception because the `double.Parse` method is not able to convert the control's Text property to a `double`. The program will immediately jump to the `catch` clause in line 23 and then execute the statements inside the catch block. Line 26 displays a message box with an error message. Figure 3-24 illustrates this process.

On the other hand, if all the statements inside the try block execute and no exceptions are thrown, the catch block will be skipped.

Figure 3-24 Handling an exception




```
private void calculateButton_Click(object sender, EventArgs e)
{
    try
    {
        double miles;     // To hold miles driven
        double gallons;   // To hold gallons used
        double mpg;       // To hold MPG

        // Get the miles driven and assign it to
        // the miles variable.
        miles = double.Parse(milesTextBox.Text);

        // Get the gallons used and assign it to
        // the gallons variable.
        gallons = double.Parse(gallonsTextBox.Text);

        // Calculate MPG.
        mpg = miles / gallons;

        // Display the MPG in the mpgLabel control.
        mpgLabel.Text = mpg.ToString();
    }
    catch
    {
        // Display an error message.
        MessageBox.Show("Invalid data was entered.");
    }
}
```

Displaying an Exception's Default Error Message

When an exception is thrown, an object known as an **exception object** is created in memory. The exception object has various properties that contain data about the exception. When you write a `catch` clause, you can optionally assign a name to the exception object, as shown here:

```
catch (Exception ex)
```

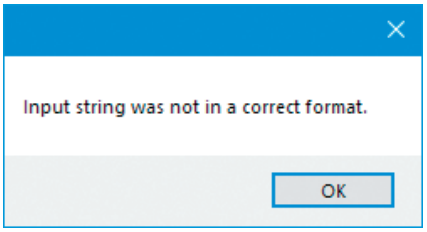
The expression that appears inside the parentheses specifies that we are assigning the name `ex` to the exception object. (There is nothing special about the name `ex`. That is simply the name that we've chosen for the examples. You can use any name that you wish.) Inside the catch block, we can use the name `ex` to access the exception object's properties. One of these is the Message property, which contains the exception's default error

message. The following code shows how this can be done. This is another modification of the *Fuel Economy* project's `calculateButton_Click` event handler.

```
1 private void calculateButton_Click(object sender, EventArgs e)
2 {
3    try
4    {
5       double miles;   // To hold miles driven
6       double gallons; // To hold gallons used
7       double mpg;     // To hold MPG
8
9       // Get the miles driven and assign it to
10      // the miles variable.
11      miles = double.Parse(milesTextBox.Text);
12
13      // Get the gallons used and assign it to
14      // the gallons variable.
15      gallons = double.Parse(gallonsTextBox.Text);
16
17      // Calculate MPG.
18      mpg = miles / gallons;
19
20      // Display the MPG in the mpgLabel control.
21      mpgLabel.Text = mpg.ToString();
22   }
23   catch (Exception ex)
24   {
25      // Display the default error message.
26      MessageBox.Show(ex.Message);
27   }
28 }
```

The statement in line 26 simply passes the exception object's Message property to the `MessageBox.Show` method. This causes the default error message to be displayed in a message box. Figure 3-25 shows an example of the message that is displayed as a result of the user entering invalid input for either the `milesTextBox` or the `gallonsTextBox` controls.

Figure 3-25 A message box showing an exception's default error message



In Tutorial 3-4, you create an application that uses a `try-catch` statement to handle exceptions that are thrown when the user enters invalid data into a TextBox control.

Tutorial 3-4:
Creating the *Test Average* Application with Exception Handling

VideoNote
Tutorial 3-4: Creating the *Test Average* Application with Exception Handling

Determining the average of a group of values is a simple calculation: You add all the values and then divide the sum by the number of values. Although this is a straightforward calculation, it is easy to make a mistake when writing a program that calculates an average. For example, let's assume that the variables `a`, `b`, and `c` each hold a value and we want to calculate the average of those values. If we are careless, we might write a statement such as the following to perform the calculation:

```
average = a + b + c / 3.0;
```

Can you see the error in this statement? When it executes, the division will take place first. The value in `c` will be divided by 3, and then the result will be added to `a + b`. That is not the correct way to calculate an average. To correct this error we need to put parentheses around `a + b + c`, as shown here:

```
average = (a + b + c) / 3.0;
```

In this tutorial, you will create an application that calculates the average of three test scores. Figure 3-26 shows the application's form, with the names of all the controls. When the application runs, the user will enter the test scores into the TextBox controls. When the user clicks the `calculateButton` control, the application will calculate the average test score and display the result in the `averageLabel` control. The `clearButton` control will clear the contents of the TextBoxes and the `averageLabel` control. The `exitButton` control closes the application's form.

Step 1: Start Visual Studio and begin a new Windows Forms App project named *TestAverage*.

Step 2: Set up the application's form as shown in Figure 3-26. Notice that the form's Text property is set to *Test Average*. The names of the controls are shown in the figure. As you place each of the controls on the form, refer to Table 3-14 for the relevant property settings.

Step 3: Once you have set up the form with its controls, you can create the Click event handlers for the Button controls. At the end of this tutorial, Program 3-4 shows the completed code for the form. You will be instructed to refer to

Figure 3-26 The *Test Average* form

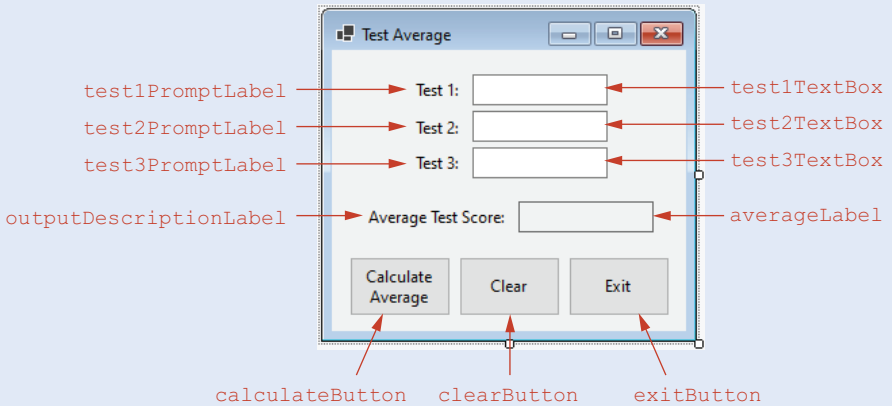


Table 3-14 Control property settings

Control Name	Control Type	Property Settings
`test1PromptLabel`	Label	**Text:** *Test 1:*
`test2PromptLabel`	Label	**Text:** *Test 2:*
`test3PromptLabel`	Label	**Text:** *Test 3:*
`outputDescriptionLabel`	Label	**Text:** *Average Test Score:*
`test1TextBox`	TextBox	No properties changed
`test2TextBox`	TextBox	No properties changed
`test3TextBox`	TextBox	No properties changed
`averageLabel`	Label	**AutoSize:** False **BorderStyle:** FixedSingle **Text:** (The contents of the Text property have been erased.) **TextAlign:** MiddleCenter
`calculateButton`	Button	**Text:** *Calculate Average*
`clearButton`	Button	**Text:** *Clear*
`exitButton`	Button	**Text:** *Exit*

Program 3-4 as you write the event handlers. (Remember, the line numbers that are shown in Program 3-4 are not part of the program. They are shown for reference only.)

In the *Designer*, double-click the `calculateButton` control. This will open the code editor, and you will see an empty event handler named `calculateButton_Click`. Complete the `calculateButton_Click` event handler by typing the code shown in lines 12–35 in Program 3-4. Let's take a closer look at the code:

Line 12: This is the beginning of a try-catch statement. The try block appears in lines 14–29, and the catch block appears in lines 33–34.

Lines 14–17: These statements declare the following `double` variables: `test1`, `test2`, `test3`, and `average`. The variables will hold the three test scores and the average test score.

Line 20: This statement converts the `test1TextBox` control's Text property to a `double` and assigns the result to the `test1` variable.

Line 21: This statement converts the `test2TextBox` control's Text property to a `double` and assigns the result to the `test2` variable.

Line 22: This statement converts the `test3TextBox` control's Text property to a `double` and assigns the result to the `test3` variable.

Line 25: This statement calculates the average of the `test1`, `test2`, and `test3` variables and assigns the result to the `average` variable.

Line 29: This statement converts the `average` variable to a string and assigns the result to the `averageLabel` control's Text property. Notice that the `"n1"` format string is passed as an argument to the `ToString` method. This causes the number to be rounded to one decimal point.

Step 4: Switch your view back to the *Designer* and double-click the `clearButton` control. In the code editor you will see an empty event handler named `clearButton_Click`. Complete the `clearButton_Click` event handler by typing the code shown in lines 40–44 in Program 3-4.

Lines 41–43: Each of these statements assigns an empty string (`""`) to the Text property of one of the TextBox controls. When these statements have finished executing, the TextBox controls appear empty.

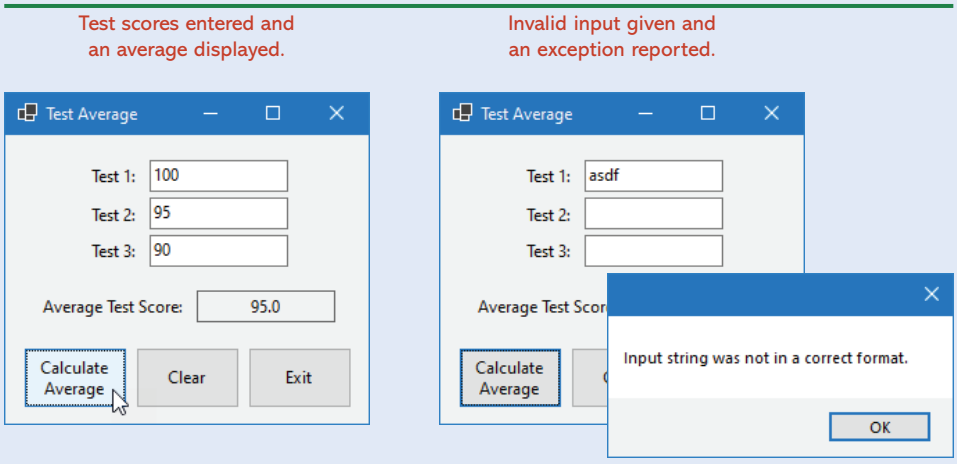
Line 44: This statement assigns an empty string (`""`) to the `averageLabel` control's Text property. After the statement has executed, the label appears empty.

Step 5: Switch your view back to the *Designer* and double-click the `exitButton` control. In the code editor you will see an empty event handler named `exitButton_Click`. Complete the `exitButton_Click` event handler by typing the code shown in lines 49–50 in Program 3-4.

Step 6: Save the project. Then, press the F5 key on the keyboard or click the *Start Debugging* button (▶) on the toolbar to compile and run the application.

First, test the application by entering valid numeric values into the TextBoxes and clicking the *Calculate Average* button. A test average should be displayed, similar to the image shown on the left in Figure 3-27.

Figure 3-27 The *Test Average* application



Next, click the *Clear* button to clear the TextBoxes and the average. Then enter a nonnumeric value for test 1, and click the *Calculate Average* button. An exception will be thrown, and you should see the message box shown in the image on the right in Figure 3-27.

Continue to test the application as you wish. When you are finished, click the *Exit* button and the form should close.

Program 3-4 Completed Form1 code for the *TestAverage* project

```
 1 namespace TestAverage
 2 {
 3     public partial class Form1: Form
 4     {
 5         public Form1()
 6         {
 7             InitializeComponent();
 8         }
 9
10         private void calculateButton_Click(object sender, Even-
           tArgs e)
11         {
12             try
13             {
14                 double test1;    // To hold test score #1
15                 double test2;    // To hold test score #2
16                 double test3;    // To hold test score #3
17                 double average;  // To hold the average test
                   score
18
19                 // Get the three test scores.
20                 test1 = double.Parse(test1TextBox.Text);
21                 test2 = double.Parse(test2TextBox.Text);
22                 test3 = double.Parse(test3TextBox.Text);
23
24                 // Calculate the average test score.
25                 average = (test1 + test2 + test3) / 3.0;
26
27                 // Display the average test score, with
28                 // the output rounded to 1 decimal point.
29                 averageLabel.Text = average.ToString("n1");
30             }
31             catch (Exception ex)
32             {
33                 // Display the default error message.
34                 MessageBox.Show(ex.Message);
35             }
36         }
37
38         private void clearButton_Click(object sender, EventArgs e)
39         {
40             // Clear the input and output controls.
41             test1TextBox.Text = "";
42             test2TextBox.Text = "";
43             test3TextBox.Text = "";
44             averageLabel.Text = "";
45         }
46
47         private void exitButton_Click(object sender, EventArgs e)
48         {
49             // Close the form.
50             this.Close();
51         }
52     }
53 }
```

Checkpoint

3.36 What can cause an application to throw an exception?

3.37 How do you get out of break mode when an exception is thrown?

3.38 What kind of code does the try block of a `try-catch` statement contain?

3.39 What causes the program to jump to the `catch` clause and execute the catch block of a `try-catch` statement?

3.40 How can you display the default error message when an exception is thrown?

3.41 Write a `try-catch` statement for an application that calculates the sum of two whole numbers and displays the result. The application uses two TextBox controls named `value1TextBox` and `value2TextBox` to gather the input and a Label control named `sumLabel` to display the result.

3.8 Using Named Constants

CONCEPT: A named constant is a name that represents a value that cannot be changed during the program's execution.

Assume that the following statement appears in a banking program that calculates data pertaining to loans:

```
amount = balance * 0.069;
```

In such a program, two potential problems arise. First, it is not clear to anyone other than the original programmer what 0.069 is. It appears to be an interest rate, but in some situations, there are fees associated with loan payments. How can the purpose of this statement be determined without painstakingly checking the rest of the program?

The second problem occurs if this number is used in other calculations throughout the program and must be changed periodically. Assuming the number is an interest rate, what if the rate changes from 6.9 percent to 7.2 percent? The programmer would have to search through the source code for every occurrence of the number.

Both these problems can be addressed by using named constants. A **named constant** is a name that represents a value that cannot be changed during the program's execution. The following is an example of how you can declare a named constant in C#:

```
const double INTEREST_RATE = 0.129;
```

This statement declares a named constant named `INTEREST_RATE` initialized with the value 0.129. It looks like a regular variable declaration, except that the word `const` appears before the data type name and the name of the variable is written in uppercase characters. The keyword `const` is a qualifier that tells the compiler to make the variable read only. If a statement attempts to change the constant's value, an error will occur when the program is being compiled. When you declare a named constant, an initialization value is required.

It is not required that the constant name be written in uppercase letters, but many programmers prefer to write them this way so they are easily distinguishable from regular variable names. When you are reading a program's code and you see an uppercase identifier, you know instantly that it is a constant.

NOTE: Writing the names of constants in uppercase letters is traditional in many programming languages, and that practice is followed in this book. Within the C# community, some programmers adhere to this practice and some do not. In the classroom, you should use the naming convention that your instructor prefers.

An advantage of using named constants is that they make programs more self-explanatory. The statement

```
amount = balance * 0.069;
```

can be changed to read

```
amount = balance * INTEREST_RATE;
```

A new programmer can read the second statement and know what is happening. It is evident that `balance` is being multiplied by the interest rate. Another advantage to this approach is that widespread changes can easily be made to the program. Let's say the interest rate appears in a dozen different statements throughout the program. When the rate changes, the initialization value in the declaration of the named constant is the only value that needs to be modified. If the rate increases to 7.2 percent, the declaration can be changed to the following:

```
const double INTEREST_RATE = 0.072;
```

The new value of 0.072 will then be used in each statement that uses the `INTEREST_RATE` constant. In Tutorial 3-5, you will create an application that uses named constants.

3.9 Declaring Variables as Fields

CONCEPT: **A field is a variable that is declared at the class level. A field's scope is the entire class.**

So far in this chapter, all the variables with which we have worked have been local variables. A local variable is declared inside a method and is visible only to statements in that method. Another type of variable is a **field**, which is a variable that is declared inside a class but not inside of any method. A field's scope is the entire class, so when you declare a field, all the methods in the class can access the variable.

Typically, fields are declared at the top of a class declaration, before any methods. Figure 3-28 shows where you would write field declarations inside a form class. When you are about to write a field declaration, you can insert some blank lines after the class's opening brace ({) and write the field declaration in that space.

Figure 3-28 Where to insert field declarations

```
namespace ExampleProject
{
    3 references
    public partial class Form1 : Form
    {
                                    Insert field declarations here.
        1 reference
        public Form1()
        {
            InitializeComponent();
        }
    }
}
```

Let's look at an example of a field declaration. Assume that the following statement appears inside a class declaration but not inside any methods. This statement declares an `int` field named `number`, initialized with the value 0:

```
private int number = 0;
```

Field declarations are written like any other variable declaration, except that an access modifier usually appears before the data type. In this example, the keyword `private` is the access modifier. An **access modifier** specifies how a class member can be accessed by code outside the class. When you use the `private` access modifier in a field declaration, the field cannot be accessed by code outside the class. It can be accessed only by the methods that are inside the class.

It is a good programming practice to make fields `private` because `private` fields are hidden from code outside the class. That prevents code outside the class from changing the values of a class's fields and helps prevent bugs from creeping into your program. You will learn more about this in Chapter 10. Until then, if you declare fields in a class, you should get in the habit of making them `private`.

NOTE: There are other access modifiers, as you will learn later in this book. If you don't write an access modifier in a field declaration, C# will automatically make the field `private`. It is still a good idea to write the `private` access modifier because it makes it evident to anyone reading the code that the field is indeed `private`.

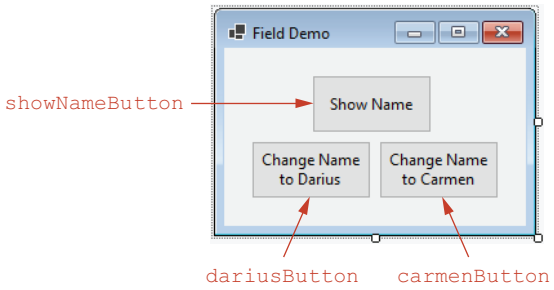
In the previous field-declaration example, the `number` field is initialized with the value 0. If a field is a variable of a numeric data type (such as `int`, `double`, or `decimal`), it will be initialized to 0 by default if you do not explicitly initialize it with a value. It is always a good idea to explicitly initialize a field, however, even if you want it to begin with the value 0. This clearly indicates the field's starting value for anyone reading the code.

WARNING! If you do not initialize a `string` field, it begins with a special value known as `null`. An error will occur if you attempt to use a `string` that is set to `null`.

NOTE: You cannot use the `var` keyword to declare a field. The `var` keyword can be used only to declare local variables.

In a form, fields are useful for storing pieces of data that must be shared among the form's event handlers. For example, in the *Chap03* folder of this book's student sample programs (available for download at www.pearsonhighered.com/gaddis), you will find a project named *FieldDemo*. Figure 3-29 shows the application's form, along with the names of the Button controls.

Figure 3-29 The *Field Demo* form



Program 3-5 shows the Form1 code. Notice that in line 16 a `string` variable named `name` is declared as a field and initialized with the value `"Charles"`. Next, look at the button event handlers:

- In the `showNameButton_Click` event handler, line 15 displays a message box showing the value of the `name` variable.
- In the `dariusButton_Click` event handler, line 20 changes the value of the `name` variable to `"Darius"`.
- In the `carmenButton_Click` event handler, line 25 changes the value of the `name` variable to `"Carmen"`.

As you can see, all of the event handlers in the Form1 class have access to the `name` variable. If you run the application and click the *Show Name* button, a message box will appear displaying *Charles*, which is the `Name` field's initial value. If you click the *Change Name to Darius* button and then click the *Show Name* button, a message box will appear displaying *Darius*. If you click the *Change Name to Carmen* button and then click the *Show Name* button, a message box will appear displaying *Carmen*.

Program 3-5 Form1 code for the *FieldDemo* project

```
1  namespace FieldDemo
2  {
3      public partial class Form1: Form
4      {
5          // Declare a private field to hold a name.
6          private string name = "Charles";
7
8          public Form1()
9          {
10             InitializeComponent();
11         }
12
13         private void showNameButton_Click(object sender,
           EventArgs e)
14         {
15             MessageBox.Show(name);
16         }
17
18         private void dariusButton_Click(object sender, EventArgs e)
19         {
20             name = "Darius";
21         }
22
23         private void carmenButton_Click(object sender, EventArgs e)
```

```
24          {
25              name = "Carmen";
26          }
27      }
28  }
```

The Lifetime of a Field in a Form Class

When you declare a field in a form class, the field's lifetime is the time during which the form exists. This means that the field will exist in memory as long as the form exists. This is different than the lifetime of a local variable, which exists only while the method in which it is declared is executing. Local variables come and go in memory, but a form's fields exist as long as the form exists.

You can see this in the *FieldDemo* project. The `name` field is created in memory when the Form1 form is created, and it continues to exist as long as Form1 exists. When one of the event handlers stores a value in the field, that value remains in the field until it is changed again, perhaps by a different event handler. So, fields give you a way of storing values that must not disappear when a particular method ends.

Precautions

Although fields make it easy to share data among the methods in a class, you should be careful about using them. The overuse of fields can make debugging a class's code difficult, especially if the class has many methods. If an incorrect value is being stored in a field, you will have to track down every statement in the class that accesses the field to determine where the incorrect value is coming from. In most cases, fields should be used only for data that must be shared among multiple methods and must continue to exist in memory when the methods are not executing.

Constant Fields

A **constant field** is a field that, once given an initial value, cannot be changed. An error will occur if the compiler finds a statement that tries to change the value of a constant field. A constant field is declared with the `const` keyword and initialized with a value. Here is an example:

```
private const decimal INTEREST_RATE = 0.075m;
```

This statement declares a constant `decimal` field named `INTEREST_RATE`, initialized with the value 0.075. Constant fields are typically used to represent unchanging values that are needed by many of a class's methods. For example, suppose a banking program uses a constant field to represent an interest rate. If the interest rate is used in several methods, it is easier to create a constant field, rather than a local named constant in each method. This also simplifies maintenance of the code. If the interest rate changes, only the declaration of the constant field has to be changed, instead of several local declarations.

NOTE: Because a constant field's value cannot be changed by other statements in the class, you do not have to worry about many of the potential debugging problems that are associated with the overuse of regular, nonconstant fields.

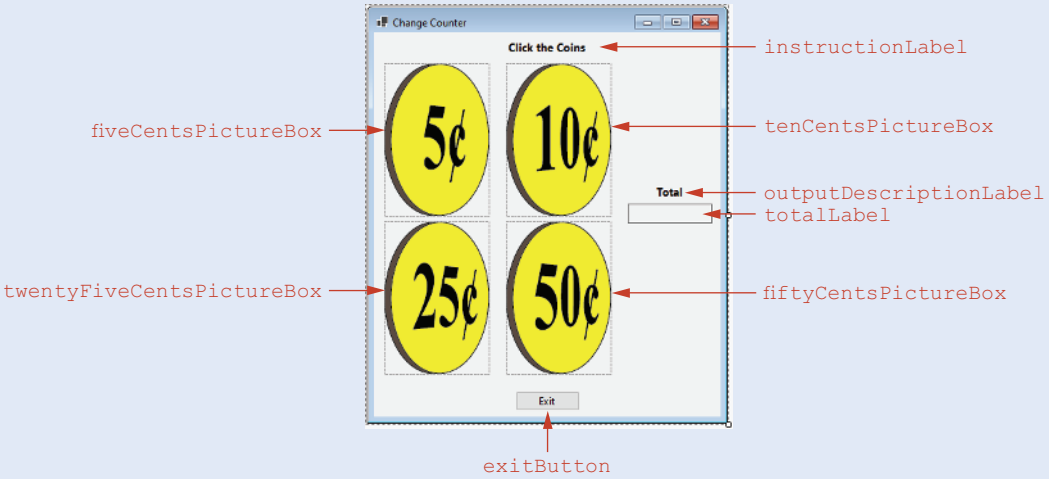
In Tutorial 3-5, you will create an application that uses a field in a form class to hold data, as well as constant fields to represent nonchanging values.

Tutorial 3-5:
Creating the *Change Counter* Application

VideoNote
Tutorial 3-5: Creating the *Change Counter* Application

In this tutorial, you will create the *Change Counter* application. The application displays images of four coins, having the values 5 cents, 10 cents, 25 cents, and 50 cents. Each time the user clicks on a coin image, the value of that coin is added to a total, and the total is displayed. Figure 3-30 shows the application's form, with the names of all the controls.

Figure 3-30 The *Change Counter* form



Step 1: Start Visual Studio and begin a new Windows Forms App project named *ChangeCounter*.

Step 2: Set up the application's form as shown in Figure 3-30. Notice that the form's Text property is set to *Change Counter*. The names of the controls are shown in the figure. As you place each of the controls on the form, refer to Table 3-15 for the relevant property settings.

Step 3: Once you have set up the form with its controls, you can begin writing code. At the end of this tutorial, Program 3-6 shows the completed code for the form. You will be instructed to refer to Program 3-6 as you write the form's code. (Remember, the line numbers that are shown in Program 3-6 are not part of the program. They are shown for reference only.)

First, you write the declarations for the fields. Switch your view to the code editor (right-click *Form1.cs* in the *Solution Explorer* and select *View Code* from the pop-up menu). Write the declarations shown in lines 6–13 in Program 3-6. Let's take a closer look at the code:

Line 6: This statement declares a constant `decimal` field named `FIVE_CENTS_VALUE`, initialized with the value 0.05. This constant represents the value of the 5-cent coin.

Line 7: This statement declares a constant `decimal` field named `TEN_CENTS_VALUE`, initialized with the value 0.10. This constant represents the value of the 10-cent coin.

Table 3-15 Control property settings

Control Name	Control Type	Property Settings
`instructionLabel`	Label	**Font**: Segoe UI (Style: Bold, Size: 10 point) **Text**: *Click the Coins*
`fiveCentsPictureBox`	PictureBox	**Image**: Select and import the 5cents.png file from the *Chap03* folder of the Student Sample Programs. **SizeMode**: AutoSize
`tenCentsPictureBox`	PictureBox	**Image**: Select and import the 10cents.png file from the *Chap03* folder of the Student Sample Programs. **SizeMode**: AutoSize
`twentyFiveCentsPictureBox`	PictureBox	**Image**: Select and import the 25cents.png file from the *Chap03* folder of the Student Sample Programs. **SizeMode**: AutoSize
`fiftyCentsPictureBox`	PictureBox	**Image**: Select and import the 50cents.png file from the *Chap03* folder of the Student Sample Programs. **SizeMode**: AutoSize
`outputDescriptionLabel`	Label	**Font**: Segoe UI (Style: Bold, Size: 10 point) **Text**: *Total*
`totalLabel`	Label	**AutoSize**: False **BorderStyle**: FixedSingle **Text**: (The contents of the Text property have been erased.) **TextAlign**: MiddleCenter
`exitButton`	Button	**Text**: *Exit*

Line 8: This statement declares a constant `decimal` field named `TWENTY_FIVE_CENTS_VALUE`, initialized with the value 0.25. This constant represents the value of the 25-cent coin.

Line 9: This statement declares a constant `decimal` field named `FIFTY_CENTS_VALUE`, initialized with the value 0.50. This constant represents the value of the 50-cent coin.

Line 13: This statement declares a `decimal` field named `total`, initialized with the value 0. This field is used to keep the total value of the coins that the user clicks.

Step 4: Now, you can create the Click event handlers for the PictureBox controls. Switch your view back to the *Designer* and double-click the `fiveCentsPictureBox` control. This opens the code editor, and you will see an empty event handler named `fiveCentsPictureBox_Click`. Complete the `fiveCentsPictureBox_Click` event handler by typing the code shown in lines 22–26 in Program 3-6. Let's take a closer look at the code:

Line 23: This statement adds the value of the `FIVE_CENTS_VALUE` constant to the `total` field.

Line 26: This statement converts the `total` variable to a string and assigns the result to the `totalLabel` control's Text property. The `"c"` format string causes the number to be formatted as currency.

Step 5: Switch your view back to the *Designer* and double-click the `tenCentsPictureBox` control. This opens the code editor, and you will see an empty event handler named `tenCentsPictureBox_Click`. Complete the `tenCentsPictureBox_Click` event handler by typing the code shown in lines 31–35 in Program 3-6. Let's take a closer look at the code:

Line 32: This statement adds the value of the `TEN_CENTS_VALUE` constant to the `total` field.

Line 35: This statement converts the `total` variable to a string and assigns the result to the `totalLabel` control's Text property. The `"c"` format string causes the number to be formatted as currency.

Step 6: Switch your view back to the *Designer* and double-click the `twentyFiveCentsPictureBox` control. This opens the code editor, and you will see an empty event handler named `twentyFiveCentsPictureBox_Click`. Complete the `twentyFiveCentsPictureBox_Click` event handler by typing the code shown in lines 40–44 in Program 3-6. Let's take a closer look at the code:

Line 41: This statement adds the value of the `TWENTY_FIVE_CENTS_VALUE` constant to the `total` field.

Line 44: This statement converts the `total` variable to a string and assigns the result to the `totalLabel` control's Text property. The `"c"` format string causes the number to be formatted as currency.

Step 7: Switch your view back to the *Designer* and double-click the `fiftyCentsPictureBox` control. This opens the code editor, and you will see an empty event handler named `fiftyCentsPictureBox_Click`. Complete the `fiftyCentsPictureBox_Click` event handler by typing the code shown in lines 49–53 in Program 3-6. Let's take a closer look at the code:

Line 50: This statement adds the value of the `FIFTY_CENTS_VALUE` constant to the `total` field.

Line 53: This statement converts the `total` variable to a string and assigns the result to the `totalLabel` control's Text property. The `"c"` format string causes the number to be formatted as currency.

Step 8: Now you write the event handler for the *Exit* button. Switch your view back to the *Designer* and double-click the `exitButton` control. This opens the code editor, and you will see an empty event handler named `exitButton_Click`. Complete the `exitButton_Click` event handler by typing the code shown in lines 58–59 in Program 3-6.

Step 9: Save the project. Then, press the F5 key on the keyboard or click the *Start Debugging* button (▶) on the toolbar to compile and run the application. Test the application by clicking the coin images in any order you wish. The total shown on the form should update by the correct amount each time you click a coin. When you are finished, click the *Exit* button and the form should close.

Program 3-6 Completed Form1 code for the *ChangeCounter* project

```
 1 namespace ChangeCounter
 2 {
 3     public partial class Form1: Form
 4     {
 5         // Constant fields
 6         const decimal FIVE_CENTS_VALUE = 0.05m;
 7         const decimal TEN_CENTS_VALUE = 0.10m;
 8         const decimal TWENTY_FIVE_CENTS_VALUE = 0.25m;
 9         const decimal FIFTY_CENTS_VALUE = 0.50m;
10
11         // Field variable to hold the total,
12         // initialized with 0.
13         private decimal total = 0m;
14
15         public Form1()
16         {
17             InitializeComponent();
18         }
19
20         private void fiveCentsPictureBox_Click(object sender,
           EventArgs e)
21         {
22             // Add the value of 5 cents to the total.
23             total += FIVE_CENTS_VALUE;
24
25             // Display the total, formatted as currency.
26             totalLabel.Text = total.ToString("c");
27         }
28
29         private void tenCentsPictureBox_Click(object sender,
           EventArgs e)
30         {
31             // Add the value of 10 cents to the total.
32             total += TEN_CENTS_VALUE;
33
34             // Display the total, formatted as currency.
35             totalLabel.Text = total.ToString("c");
36         }
37
38         private void twentyFiveCentsPictureBox_Click(object
           sender, EventArgs e)
39         {
40             // Add the value of 25 cents to the total.
41             total += TWENTY_FIVE_CENTS_VALUE;
42
43             // Display the total, formatted as currency.
44             totalLabel.Text = total.ToString("c");
45         }
```

```
46
47          private void fiftyCentsPictureBox_Click(object
            sender, EventArgs e)
48          {
49              // Add the value of 50 cents to the total.
50              total += FIFTY_CENTS_VALUE;
51
52              // Display the total, formatted as currency.
53              totalLabel.Text = total.ToString("c");
54          }
55
56          private void exitButton_Click(object sender, Even-
            tArgs e)
57          {
58              // Close the form.
59              this.Close();
60          }
61      }
62  }
```

Checkpoint

3.42 What are two advantages of using named constants?

3.43 Write a programming statement that declares a named constant for a 10 percent discount.

3.44 Where should you place field declarations in a program?

3.45 What access modifier should you use when declaring a field? Why?

3.46 How is the lifetime of a field different from the lifetime of a local variable?

3.47 Write a programming statement that declares a constant field for a 5.9 percent interest rate.

3.10 Using the `Math` Class

CONCEPT: **.NET's `Math` class provides several methods for performing complex mathematical calculations.**

.NET provides a class named `Math`, which contains numerous methods that are useful for performing advanced mathematical operations. Table 3-16 gives a summary of several of the `Math` class methods. (For a comprehensive list of all the methods provided by the `Math` class and more details on any of the methods, just search for *Math class* in the online .NET documentation at docs.microsoft.com.)

These methods typically accept one or more values as arguments, perform a mathematical operation using the arguments, and return the result. For example, the `Math.Pow` method raises a number to a power. Here is an example of how the method is called:

```
double result;
result = Math.Sqrt(4.0, 2.0);
```

Table 3-16 Many of the Math class methods

Math Class Method	Description
`Math.Abs(x)`	Returns the absolute value of `x`.
`Math.Acos(x)`	Returns the arccosine of `x`, in radians. The argument `x` is a `double`, and the value that is returned is a `double`.
`Math.Asin(x)`	Returns the arcsine of `x`, in radians. The argument `x` is a `double`, and the value that is returned is a `double`.
`Math.Atan(x)`	Returns the arctangent of `x`, in radians. The argument `x` is a `double`, and the value that is returned is a `double`.
`Math.Ceiling(x)`	Returns the least integer that is greater than or equal to `x` (a `decimal` or a `double`).
`Math.Cos(x)`	Returns the cosine of `x` in radians. The argument `x` is a `double`, and the value that is returned is a `double`.
`Math.Exp(x)`	Returns e^x. The argument `x` is a `double`, and the value that is returned is a `double`.
`Math.Floor(x)`	Returns the greatest integer that is less than or equal to `x` (a `decimal` or a `double`).
`Math.Log(x)`	Returns the natural logarithm of `x`. The argument `x` is a `double`, and the value that is returned is a `double`.
`Math.Log10(x)`	Returns the base-10 logarithm of `x`. The argument `x` is a `double`, and the value that is returned is a `double`.
`Math.Max(x, y)`	Returns the greater of the two values `x` and `y`.
`Math.Min(x, y)`	Returns the lesser of the two values `x` and `y`.
`Math.Pow(x, y)`	Returns the value of `x` (a `double`) raised to the power of `y` (also a `double`). The value that is returned is a `double`.
`Math.Round(x)`	Returns the value of `x` (a `double` or a `decimal`) rounded to the nearest integer.
`Math.Sin(x)`	Returns the sine of `x` in radians.
`Math.Sqrt(x)`	Returns the square root of `x` (a `double`). The value that is returned is a `double`.
`Math.Tan(x)`	Returns the tangent of `x` in radians. The argument `x` is a `double`, and the value that is returned is a `double`.
`Math.Truncate(x)`	Returns the integer part of `x` (a `double` or a `decimal`).

The method takes two `double` arguments. It raises the first argument to the power of the second argument and returns the result as a `double`. In this example, 4.0 is raised to the power of 2.0. This statement is equivalent to the following algebraic statement:

$$result = 4^2$$

The following code sample shows another example of a statement using the `Math.Pow` method. It assigns 3 times 6^3 to `x`:

```
double x;
x = 3 * Math.pow(6.0, 3.0);
```

The `Math.Sqrt` method accepts an argument and returns the square root of the argument. Here is an example of how it is used:

```
double result;
result = Math.Sqrt(16.0);
```

The statement that calls the `Math.Sqrt` method passes 16.0 as an argument. The method returns the square root of 16.0 (as a `double`), which is then assigned to the `result` variable.

The `Math.PI` and `Math.E` Named Constants

The `Math` class also provides two predefined named constants, `Math.PI` and `Math.E`, which are assigned mathematical values for *pi* and *e*. You can use these variables in equations that require their values. For example, the following statement, which calculates the area of a circle, uses `Math.PI`.

```
area = Math.PI * radius * radius;
```

NOTE: If you look at the top of a form's source code in the code editor, you should see a directive that reads `using System;`. That directive is required for any program that uses the `Math` class.

Checkpoint

3.48 Write a programming statement that uses the `Math.Pow` method to square the number 12 and store the result in a `double` variable named `product`.

3.49 What method of the `Math` class can be used to determine the larger of two values?

3.50 What method of the `Math` class can be used to determine the smaller of two values?

3.11 More GUI Details

In Chapter 2, you learned the basics of creating a GUI by placing controls on a form and setting various properties. In this section, you learn to fine-tune many aspects of an application's GUI.

Controlling a Form's Tab Order

When an application is running and a form is displayed, one of the form's controls always has the focus. The control having the **focus** is the one that receives the user's keyboard input. For example, when a TextBox control has the focus, it receives the characters that the user enters on the keyboard. When a button has the focus, pressing the Enter key executes the button's Click event handler.

NOTE: Only controls capable of receiving some sort of input, such as text boxes and buttons, may have the focus.

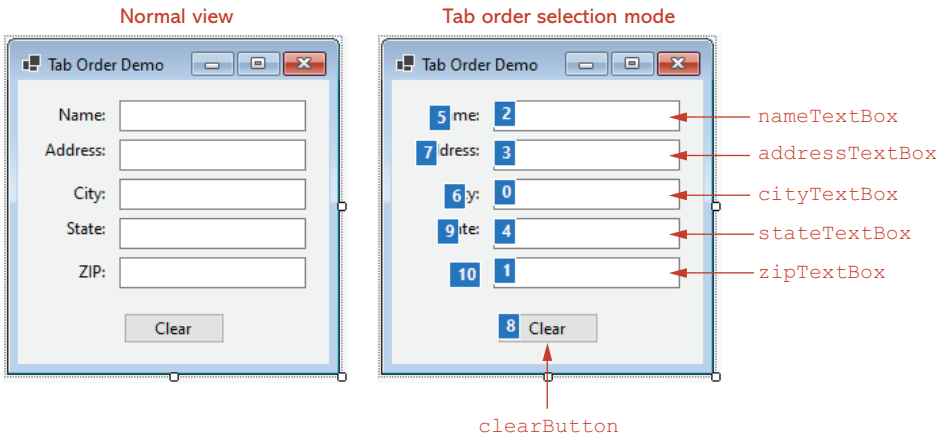
You can tell which control has the focus by looking at the form at run time. When a TextBox control has the focus, a blinking text cursor appears inside it, or the text inside the TextBox control might appear highlighted. When a button has the focus, a thin dotted line usually appears around the control.

When an application is running, pressing the Tab key changes the focus from one control to another. The order in which controls receive the focus is called the **tab order**. When you place controls on a form in Visual C#, the tab order is in the same sequence in which you created the controls. In many cases this is the tab order you want, but sometimes you rearrange controls on a form, delete controls, and add new ones. These modifications often lead to a disorganized tab order, which can confuse and irritate the users of your application.

Users want to tab smoothly from one control to the next, in a logical sequence. You can modify the tab order by changing a control's TabIndex property. The TabIndex property contains a numeric value, which indicates the control's position in the tab order. When you create a control, Visual C# automatically assigns a value to its TabIndex property. The first control you create on a form has a TabIndex of 0, the second has a TabIndex of 1, and so forth. The control with a TabIndex of 0 is the first control in the tab order. The next control in the tab order is the one with a TabIndex of 1. The tab order continues in this sequence.

You may change the tab order of a form's controls by selecting them, one by one, and changing their TabIndex property in the *Properties* window. An easier method, however, is to click *View* on the Visual Studio menu bar and then click *Tab Order*. This causes the form to be displayed in **tab order selection mode**. The image on the left in Figure 3-31 shows a form in the normal view, and the image on the right shows the form in tab order selection mode. We have also inserted the names of the TextBox and Button controls in the image on the right for reference purposes.

Figure 3-31 A form displayed in tab order selection mode



In tab order selection mode, each control's existing TabIndex value is displayed in a small box in the control's upper-left corner. Notice the following in the image on the right in Figure 3-31:

- The `nameTextBox` control's TabIndex is 2.
- The `addressTextBox` control's TabIndex is 3.
- The `cityTextBox` control's TabIndex is 0.
- The `stateTextBox` control's TabIndex is 4.
- The `zipTextBox` control's TabIndex is 1.
- The `clearButton` control's TabIndex is 8.

NOTE: Although the Label controls have TabIndex values, those values are irrelevant in this example because Label controls cannot receive the focus.

As you look at Figure 3-31, think about the order in which the controls will receive the focus when the application runs.

- The `cityTextBox` control has the lowest TabIndex value (0), so it will receive the focus first.
- If you press the Tab key, the focus will jump to the `zipTextBox` control because it has the next lowest TabIndex value (1).
- Press the Tab key again and the focus will jump to the `nameTextBox` control (TabIndex is set to 2).
- Press the Tab key again and the focus will jump to the `addressTextBox` control (TabIndex is set to 3).
- Press the Tab key again and the focus will jump to the `stateTextBox` control (TabIndex is set to 4).
- Press the Tab key again and the focus will jump to the `clearButton` control (TabIndex is set to 8).

This is a very confusing tab order and should be rearranged. When a form is displayed in tab order selection mode, you establish a new tab order by simply clicking the controls with the mouse in the order you want. To fix the disorganized tab order shown in Figure 3-31, we perform the following:

- First, click the `nameTextBox` control. The control's TabIndex value changes to 0.
- Next, click the `addressTextBox` control. The control's TabIndex value changes to 1.
- Then, click the `cityTextBox` control. The control's TabIndex value changes to 2.
- Next, click the `stateTextBox` control. The control's TabIndex value changes to 3.
- Then, click the `zipTextBox` control. The control's TabIndex value changes to 4.
- Finally, click the `clearButton` control. The control's TabIndex value changes to 5.

When you are finished, exit tab order selection mode by pressing the Esc key. Now when the application runs, the focus will shift smoothly in an order that makes sense to the user.

Changing the Focus with the `Focus` Method

Often, you want to make sure a particular control has the focus. For example, look at the form shown in Figure 3-31. The purpose of the *Clear* button is to clear any input that the user has entered and reset the form so it is ready to accept a new set of input. When the *Clear* button is clicked, the TextBox controls should be cleared and the focus should return to the `nameTextBox` control. This would make it unnecessary for the user to click the TextBox control in order to start entering another set of information.

In code, you move the focus to a control by calling the Focus method. The method's general syntax is:

```
ControlName.Focus();
```

where ControlName is the name of the control. For instance, you move the focus to the `nameTextBox` control with this statement:

```
nameTextBox.Focus();
```

After the statement executes, the `nameTextBox` control will have the focus. Here is an example of how the `clearButton` control's Click event handler could be written:

```
1 private void clearButton_Click(object sender, EventArgs e)
2 {
3     // Clear the TextBox controls.
4     nameTextBox.Text = "";
5     addressTextBox.Text = "";
6     cityTextBox.Text = "";
7     stateTextBox.Text = "";
8     zipTextBox.Text = "";
9
10    // Set the focus to nameTextBox.
11    nameTextBox.Focus();
12 }
```

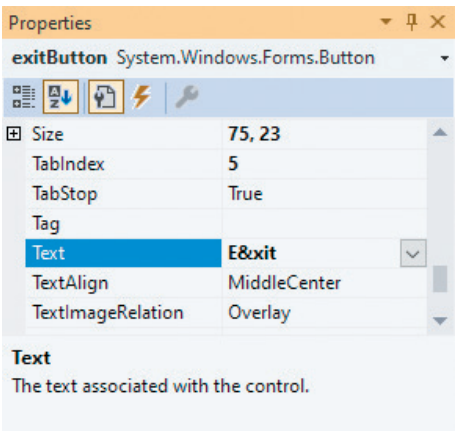
The statements in lines 4–8 clear the contents of the TextBox controls. Then, the statement in line 11 sets the focus to the `nameTextBox` control.

Assigning Keyboard Access Keys to Buttons

An **access key**, also known as a **mnemonic**, is a key that is pressed in combination with the Alt key to access quickly a control such as a button. When you assign an access key to a button, the user can trigger a Click event either by clicking the button with the mouse or by using the access key. Users who are quick with the keyboard prefer to use access keys instead of the mouse.

You assign an access key to a button through its Text property. For example, assume an application has a button whose Text property is set to *Exit*. You wish to assign the access key Alt + X to the button so the user may trigger the button's Click event by pressing Alt + X on the keyboard. To make the assignment, place an ampersand (&) before the letter *x* in the button's Text property: `E&xit`. Figure 3-32 shows how the Text property appears in the *Properties* window.

Figure 3-32 Text property `E&xit`



Although the ampersand is part of the Button control's Text property, it is not displayed on the button. With the ampersand in front of the letter *x*, the letter will appear underlined as shown in Figure 3-33. This indicates that the button may be clicked by pressing Alt + X on the keyboard. (You will see the underlining at design time. At run time, however, the underlining may not appear until the user presses the Alt key.)

Figure 3-33 Button control with `E&xit` Text property

NOTE: Access keys do not distinguish between uppercase and lowercase characters. There is no difference between Alt + X and Alt + X.

Suppose we store the value `&Exit` in the button's Text property. The ampersand is in front of the letter *E*, so Alt + E becomes the access key. The button will appear as shown in Figure 3-34.

Figure 3-34 Button control with `&Exit` Text property

Assigning the Same Access Key to Multiple Buttons

Be careful not to assign the same access key to two or more buttons on the same form. If two or more buttons share the same access key, a Click event is triggered for the first button created when the user presses the access key.

Displaying the & Character on a Button

If you want to display an ampersand character on a button, use two ampersands (`&&`) in the Text property. Using two ampersands causes a single ampersand to display and does not define an access key. For example, if a button's Text property is set to `Save && Exit`, the button will appear as shown in Figure 3-35.

Figure 3-35 Button control with `Save && Exit` Text property

Accept Buttons and Cancel Buttons

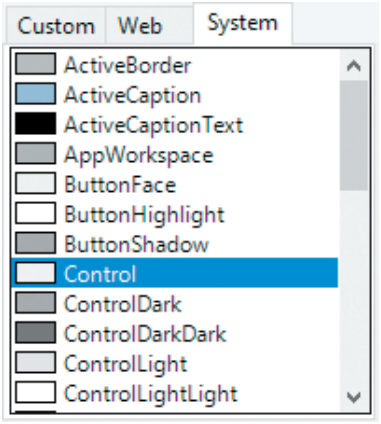
An **accept button** is a button on a form that is automatically clicked when the user presses the Enter key. A **cancel button** is a button on a form that is automatically clicked when the user presses the Esc key. Forms have two properties, AcceptButton and CancelButton, which allow you to designate an accept button and a cancel button. When you select these properties in the *Properties* window, a down-arrow button ( ) appears, which displays a drop-down list when clicked. The list contains the names of all the buttons on the form. You select the button that you want to designate as the accept button or cancel button.

Any button that is frequently clicked should probably be selected as the accept button. This will allow keyboard users to access the button quickly and easily. *Exit* or *Cancel* buttons are likely candidates to become cancel buttons.

The BackColor Property

Forms and most controls have a BackColor property that allows you to change the object's background color. When you select an object's BackColor property in the *Properties* window, a down-arrow button ( ) appears, which displays a drop-down list of available colors when clicked, as shown in Figure 3-36.

Figure 3-36 Drop-down list of colors



The drop-down list has three tabs: *Custom*, *Web*, and *System*. The *System* tab lists colors defined in the current Windows configuration. The *Web* tab lists colors displayed with consistency in Web browsers. The *Custom* tab displays a color palette. Select a color from one of the tabs and the object's background color will be set to that color.

The ForeColor Property

Controls that display text have a ForeColor property that allows you to change the color of the text. When you select a control's ForeColor property in the *Properties* window, a down-arrow button ( ) appears, which displays the drop-down list of available colors shown in Figure 3-36 when clicked. Select a color from one of the tabs and the text that is displayed by the control will be set to that color.

Setting Colors in Code

In addition to using the *Properties* window, you can also set the values of the BackColor and ForeColor properties with code. .NET provides numerous values that represent colors and can be assigned to the ForeColor and BackColor properties in code. The following are a few of the values:

```
Color.Black
Color.Blue
Color.Cyan
Color.Green
Color.Magenta
Color.Red
Color.White
Color.Yellow
```

For example, assume an application has a Label control named `messageLabel`. The following code sets the label's background color to black and foreground color to yellow:

```
messageLabel.BackColor = Color.Black;
messageLabel.ForeColor = Color.Yellow;
```

.NET also provides values that represent default colors on your system. For example, the value `SystemColors.Control` represents the default control background color and `SystemColors.ControlText` represents the default control text color. The following statements set the `messageLabel` control's background and foreground to the default colors:

```
messageLabel.BackColor = SystemColors.Control
messageLabel.ForeColor = SystemColors.ControlText
```

NOTE: If you have an event handler in a form's source code file and you want the event handler to change the form's BackColor property, use the `this` keyword to refer to the form. For example, the following statement changes the color of the form to blue:

```
this.BackColor = Color.Blue;
```

Background Images for Forms

In Chapter 2, you learned about displaying images with PictureBox controls. An image can also be displayed as the background for a form. Forms have a property named BackgroundImage that allows you to import and display an image on the form. If you know how to use the PictureBox control's Image property, then you already know how to use a form's BackgroundImage property. They both work the same way:

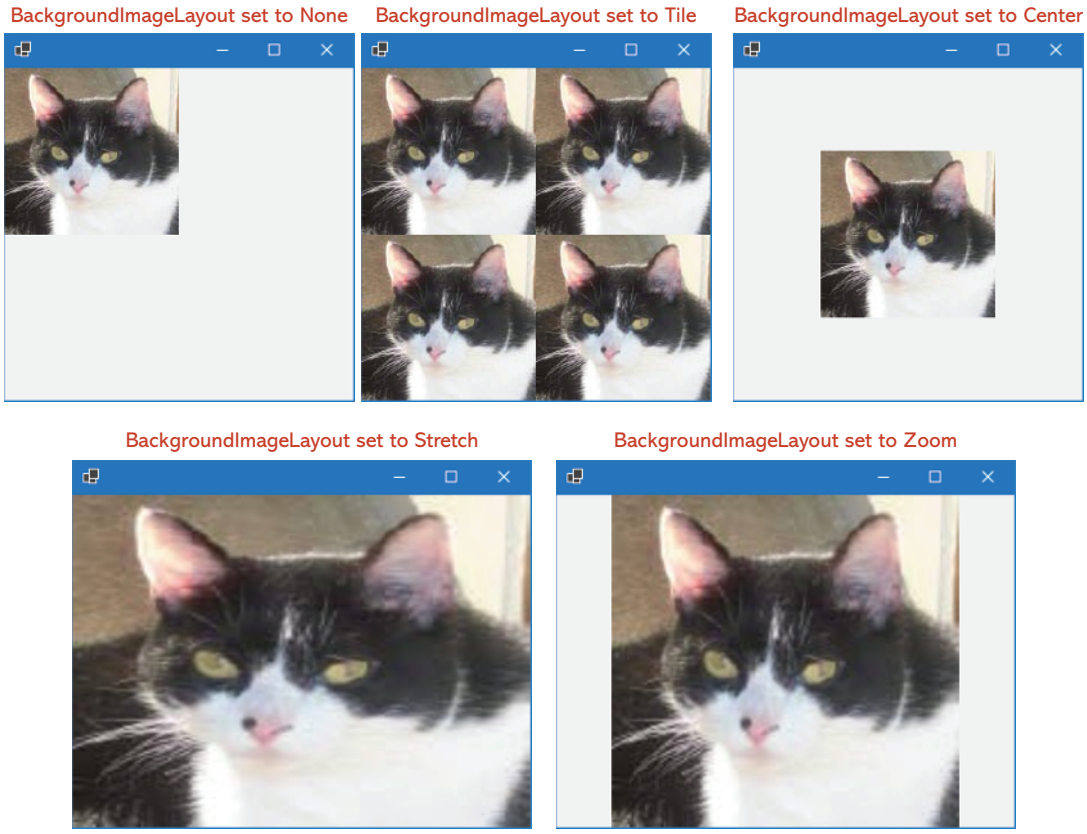
- Click the BackgroundImage property in the *Properties* window. An ellipses button (...) will appear.
- Click the ellipses button and the *Select Resource* window will appear.
- In the *Select Resource* window, click the *Import* button. An *Open* dialog box will appear. Use the dialog box to locate and select the image file that you want to display.
- Click the *OK* button in the *Select Resource* window, and the selected image will appear as the form's background.

A form's BackgroundImageLayout property is similar to the PictureBox control's SizeMode property. It specifies how the background image is to be displayed. It can be set to one of the following values:

- **None**
  The image is positioned in the upper-left corner of the form. If the image is too big to fit in the form, it is clipped.
- **Tile**
  This is the default value. The image is tiled (repeatedly displayed) across the form.
- **Center**
  The image is centered in the form without being resized.
- **Stretch**
  The image is resized both horizontally and vertically to fit in the form. If the image is resized more in one direction than the other, it appears stretched.
- **Zoom**
  The image is uniformly resized to fit in the form without losing its original aspect ratio. This causes the image to be resized without appearing stretched.

Figure 3-37 shows examples of each of these settings.

Figure 3-37 Different settings for the BackgroundImageLayout property



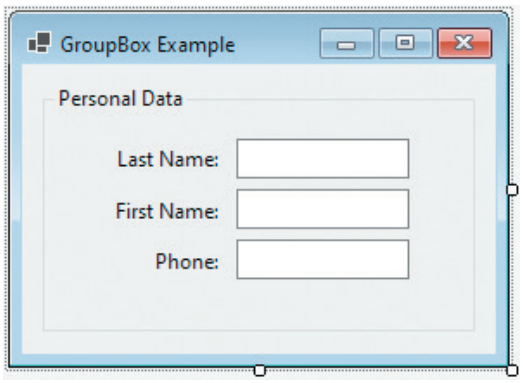
Organizing Controls with GroupBoxes and Panels

A **GroupBox control** is a rectangular control that appears with a thin border and an optional title in its upper-left corner. It is a container that can hold other controls. You can use GroupBoxes to create a sense of visual organization on a form.

The GroupBox control is found in the *Toolbox*, in the *Containers* section. When you create a GroupBox control, you can set its Title property to the text that you want displayed in the GroupBox's upper-left corner. If you don't want a title displayed on the GroupBox, you can clear the contents of its Text property.

Figure 3-38 shows a form with a GroupBox control. The control's Text property is set to *Personal Data*, and several other controls are inside the GroupBox.

Figure 3-38 A GroupBox containing other controls



Creating a GroupBox and Adding Controls to It

Suppose you've just created a GroupBox control. To add another control to the GroupBox, select the GroupBox control and then double-click the desired tool in the *Toolbox* to place another control inside the group box.

Moving an Existing Control to a GroupBox

If an existing control is not inside a GroupBox but you want to move it to the GroupBox, follow these steps:

1. Select the control you wish to add to the GroupBox.
2. Cut the control to the clipboard.
3. Select the GroupBox.
4. Paste the control.

Moving and Resizing a GroupBox

If a GroupBox is selected in the *Designer*, a four-headed arrow ( ) will appear in the GroupBox's upper-left corner. Click and drag the four-headed arrow to move the GroupBox. Any controls inside the GroupBox move with it.

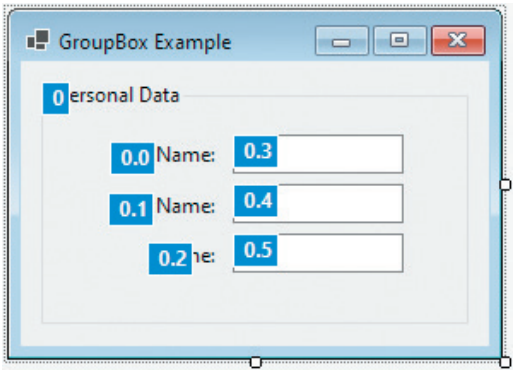
Deleting a GroupBox

To delete a GroupBox, simply select it in the *Designer* and then press the Delete key. Any controls inside the GroupBox are deleted as well.

Group Box Tab Order

The value of a control's TabIndex property is handled differently when the control is placed inside a GroupBox control. GroupBox controls have their own TabIndex property, and the TabIndex value of the controls inside the group box are relative to the GroupBox control's TabIndex property. For example, Figure 3-39 shows a GroupBox control displayed in tab order selection mode. As you can see, the GroupBox control's TabIndex is set to 0. The TabIndex of the controls inside the group box is displayed as 0.0, 0.1, 0.2, and so forth.

Figure 3-39 GroupBox TabIndex values



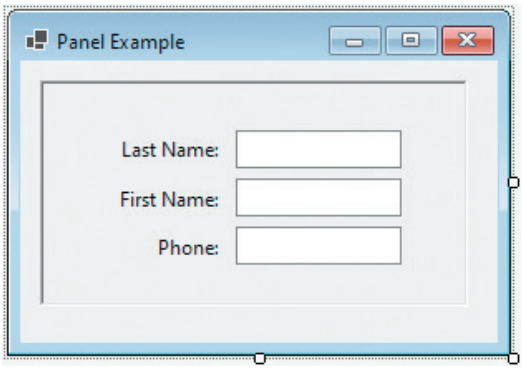
NOTE: The TabIndex properties of the controls inside the group box will not appear this way in the *Properties* window. They will appear as 0, 1, 2, and so forth.

A **Panel control** is a rectangular container for other controls, like a GroupBox. There are several primary differences between a Panel and GroupBox:

- A Panel cannot display a title and does not have a Text property.
- A Panel's border can be specified by its BorderStyle property. The available settings are None, FixedSingle, and Fixed3D. The property is set to None by default, which means that no border will appear. If the BorderStyle property is set to FixedSingle, the control will be outlined with a thin border. If the BorderStyle property is set to Fixed3D, the control will have a recessed 3D appearance.

Figure 3-40 shows an example of a form with a Panel. The Panel's BorderStyle property is set to Fixed3D.

Figure 3-40 A Panel containing other controls



3.12 Using the Debugger to Locate Logic Errors

CONCEPT: **The Visual Studio debugger allows you to pause a program and then execute statements one at a time. After each statement executes, you may examine variable contents and property values.**

A **logic error** is a mistake that does not prevent an application from running, but causes the application to produce incorrect results. Mathematical mistakes, assigning a value to the wrong variable, or assigning the wrong value to a variable are examples of logic errors. Logic errors can be difficult to find. Finding and fixing a logic error usually requires a bit of detective work on the part of the programmer. Fortunately, Visual Studio provides you with debugging tools that make locating logic errors easier.

Visual Studio allows you to set breakpoints in your program code. A **breakpoint** is a line you select in your source code. When the application is running and it reaches a breakpoint, the application pauses and enters **break mode**. While the application is paused, you may examine variable contents and the values stored in certain control properties.

Visual Studio allows you to **single-step** through an application's code once its execution has been paused by a breakpoint. This means that the application's statements execute one at a time, under your control. After each statement executes, you can examine variable and property values. This process allows you to identify the line or lines of code causing the error. In Tutorial 3-6, you single-step through an application's code.

Tutorial 3-6:
Single-stepping through an Application's Code at Runtime

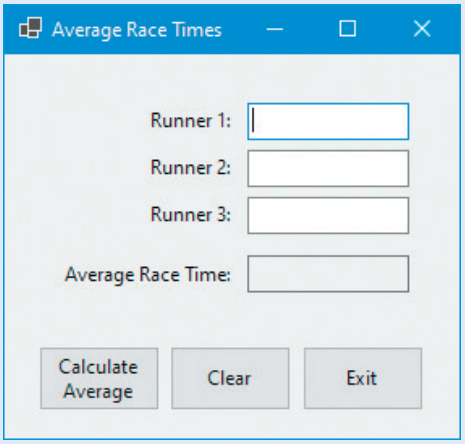
VideoNote
Tutorial 3-6: Single-stepping through an Application's Code at Runtime

In this tutorial, you will open the *Average Race Times* application, and test it for logic errors. The application is simple: It lets the user enter the finishing times for three runners, and then click a button to calculate their average time. To determine whether the application correctly calculates an average, you will perform a simple test. You will enter the value 25 for each runner, and then click the button to calculate the average. The result should also be 25. If the application produces any other value, then you know that a logic error exists somewhere in the application's code.

Step 1: Start Visual Studio. Open the project named *AverageRaceTimes* in the *Chap03* folder of the Student Sample Programs.

Step 2: Run the application. The application's form appears, as shown in Figure 3-41.

Figure 3-41 The *Average Race Times* application running



Step 3: This application allows you to enter the finishing times of three runners in a race, and then see their average time. Enter 25 as the time for all three runners.

Step 4: Click the *Calculate Average* button. The application displays the incorrect value 58.3 as the average time. (The correct value should be 25.)

Step 5: Click the *Exit* button to stop the application.

Step 6: Open the code editor (click *View* on the menu bar, and then select *Code*) and locate the following line of code, which appears in the `calculateButton_Click` event handler:

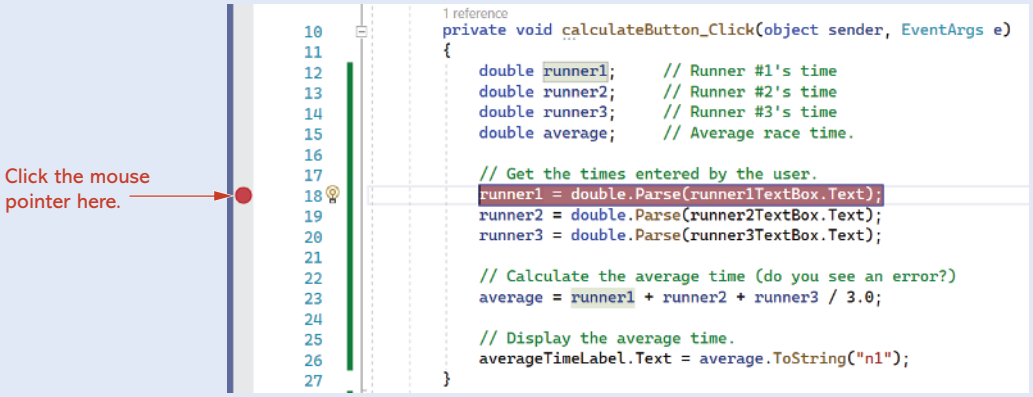
```
runner1 = double.Parse(runner1TextBox.Text);
```

This line of code is where we want to pause the execution of the application. We must make this line a breakpoint.

Step 7: Click the mouse in the left margin of the code editor, next to the line of code, as shown in Figure 3-42. Notice that a red dot appears next to the line in the left margin, and the line of code becomes highlighted. The dot indicates that a breakpoint has been set on this line. Another way to set a breakpoint is to move the text cursor to the line you wish to set as a breakpoint, and then press F9.

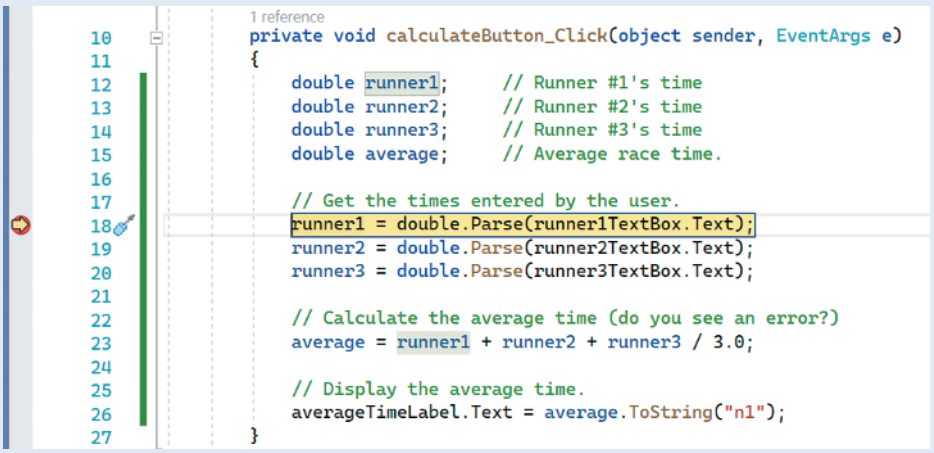
Step 8: Now that you have set the breakpoint, run the application. When the form appears, enter 25 as the time for each runner.

Figure 3-42 Setting a breakpoint



Step 9: Click the *Calculate Average* button. When program execution reaches the breakpoint, it goes into break mode and the code editor reappears. The breakpoint line is shown with yellow highlighting and a small yellow arrow appears in the left margin, as shown in Figure 3-43. The yellow highlighting and small arrow indicate the application's current execution point. The **execution point** is the next line of code that will execute. (The line has not yet executed.)

Figure 3-43 Breakpoint reached



NOTE: If the highlighting and arrow appear in a color other than yellow, the color options on your system may have been changed.

Step 10: To examine the contents of a variable or control property, hover the cursor over the variable or the property's name in the code editor. A small box will appear showing the variable or property's contents. For example, Figure 3-44 shows the result of hovering the mouse pointer over the expression `runner1TextBox`.

Figure 3-44 `runner1TextBox.Text` property contents revealed

```
// Get the times entered by the user.
runner1 = double.Parse(runner1TextBox.Text);
runner2 = double.Parse(runner2TextBox.Tex
runner3 = double.Parse(runner3TextBox.Text);
```

runner1TextBox.Text View "25"

`Text` in the highlighted line. The box indicates that the property is currently set to "25."

Step 11: Now, hover the mouse pointer over the variable name `runner1`. A box appears indicating that the variable is set to 0. Because the highlighted statement has not yet executed, no value has been assigned to this variable.

Step 12: You may also examine the contents of variables with the *Locals*, and *Watch* windows. (The *Immediate* window is beyond the scope of this chapter.)

A description of each window follows:

- The ***Immediate* window** allows you to type debugging commands using the keyboard. This window is generally used by advanced programmers.
- The ***Locals* window** displays a list of all the variables within the scope of the currently executing method. The current value and the data type of each variable are also displayed.
- The ***Watch* window** allows you to add the names of variables you want to watch. This window displays only the variables you have added. Visual Studio lets you open multiple *Watch* windows.

You can open any of these windows by clicking *Debug* on the menu bar, then selecting *Windows,* and then selecting the window that you want to open. Use this technique to open the *Locals* window now.

The *Locals* window should appear, similar to Figure 3-45.

Figure 3-45 *Locals* window displayed

Locals

Search (Ctrl+E) Search Depth: 3

Name	Value	Type
this	{AverageRaceTimes.Form1, Text: Average Race Times}	AverageRaceTimes.Form1
sender	{Text = "Calculate Average"}	object {System.Windows.Forms.Button}
e	{X = 54 Y = 35 Button = Left}	System.EventArgs {System.Windows.Forms.MouseEventArgs}
runner1	0	double
runner2	0	double
runner3	0	double
average	0	double

Locals Watch 1

Step 13: Now you are ready to single-step through each statement in the event handler. To do this, use the ***Step Into*** command. (The *Step Over* command, which is similar to *Step Into*, is covered in Chapter 6.) You activate the *Step Into* command by one of the following methods:

- Press the F11 key.
- Click *Debug* on the menu bar, and then select *Step Into* from the *Debug* menu.

When you activate the *Step Into* command, the highlighted statement is executed. Press the F11 key now. Look at the *Watch* window and notice that the `runner1` variable is now set to 25. Also notice that the next line of code is now highlighted.

Step 14: Press the F11 key two more times. The variables `runner1`, `runner2`, and `runner3` should display values of 25 in the *Locals* window.

Step 15: The following statement, which is supposed to calculate the average of the three scores, is now highlighted:

```
average = runner1 + runner2 + runner3 / 3.0;
```

After this statement executes, the average of the three numbers should display next to `average`. Press F11 to execute the statement.

Step 16: Notice that the *Locals* window now reports that `average` holds the value 58.333333333333336. This is not the correct value, so there must be a problem with the math statement that just executed. Can you find it? The math statement does not calculate the correct value because the division operation takes place before any of the addition operations. You must correct the statement by inserting a set of parentheses.

From the menu, select *Debug*, and then click *Stop Debugging* to halt the application. In the *Code* window, insert a set of parentheses into the math statement so it appears as follows:

```
average = (runner1 + runner2 + runner3) / 3.0;
```

Step 17: Next, you will clear the breakpoint so the application will not pause again when it reaches that line of code. To clear the breakpoint, use one of the following methods:

- Click the mouse on the breakpoint dot in the left margin of the code editor.
- Press Control + Shift + F9.
- Select *Debug* from the menu bar, and then select *Delete All Breakpoints* from the *Debug* menu.

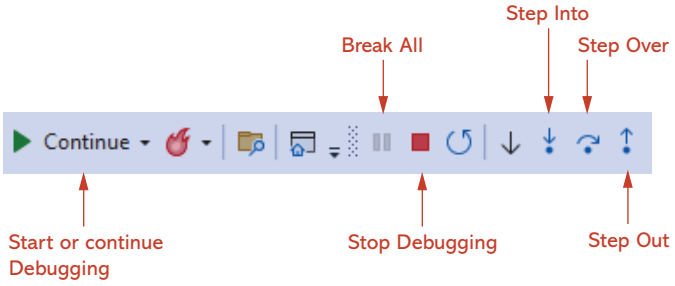
Step 18: Run the application again. Enter 25 as each runner's time, and then click the *Calculate Average* button. This time the correct average, 25.0, is displayed.

Step 19: Click the *Exit* button to stop the application.

Debugging Commands in the Toolbar

Visual Studio provides a toolbar for debugging commands, shown in Figure 3-46.

Figure 3-46 Debug toolbar commands



Checkpoint

3.51 What happens if you press the Enter key while a Button control has the focus?

3.52 How do you display a form in tab order selection mode? How do you exit tab order selection mode?

3.53 Write a programming statement that gives the focus to a TextBox control named `numberTextBox`.

3.54 How do you assign an access key to a Button control?

3.55 How do you display an ampersand (&) character on a Button control?

3.56 Write the code that will change the BackColor property of a Label control named `resultLabel` to the color white and the ForeColor property to the color red.

3.57 List the different values of a form's BackgroundImageLayout property.

3.58 When a GroupBox control is deleted, what happens to the controls that are inside?

3.59 How are the TabIndex properties of the controls inside the group box organized?

3.60 How is a Panel control different from a GroupBox control?

Key Terms

accept button
access key
access modifier
argument
break mode
breakpoint
cancel button
cast operator
catch block
`catch` clause
combined assignment operators
concatenation
constant field
data type
`decimal` literal
`double` literal
exception
exception handler
exception object
execution point
field
focus
formatting string
GroupBox Control
Immediate window
initialize
integer literal
lifetime
local variable
Locals window
logic error
math expression
math operators
mnemonic
named constant
numeric literal
operands
order of operations
Panel control
parse
`Parse` methods
primitive data types
scope
single-step
tab order
tab order selection mode
TextBox control
truncation
try block
`try-catch` statement
`var` keyword
variable
variable declaration
variable name
Watch window

Review Questions

Multiple Choice

1. When the user types into a TextBox control, the text is stored in the control's _____ property.
   a. Input
   b. Text
   c. String
   d. Data

2. A _____ is a storage location in memory that is represented by a name.
   a. mnemonic
   b. data type
   c. namespace
   d. variable

3. In C#, you must _____ a variable before you can use it to store data.
   a. cite
   b. associate
   c. declare
   d. instance

4. A variable's _____ indicates the type of data that the variable will hold.
   a. name
   b. data type
   c. scope
   d. value

5. Fundamental types of data, such as strings, integers, and real numbers, are known as _____.
   a. primitive data types
   b. fundamental variables
   c. logical digits
   d. literal data types

6. A _____ identifies a variable in the program code.
   a. binary number
   b. variable name
   c. unique global identifier
   d. hexadecimal value

7. A common operation performed on strings is _____, or appending one string to the end of another string.
   a. addition
   b. merging
   c. concatenation
   d. tying

8. A _____ belongs to the method in which it is declared, and only statements inside that method can access the variable.
   a. method variable
   b. primitive variable
   c. temporary variable
   d. local variable

9. Programmers use the term _____ to describe the part of a program in which a variable may be accessed.
   a. range
   b. scope
   c. focus
   d. field

10. A variable's _____ is the time period during which the variable exists in memory while the program is executing.
    a. lifetime
    b. run time
    c. time to live
    d. half life

11. One way to make sure that a variable has been assigned a value is to _____ the variable with a value when you declare it.
    a. concatenate
    b. initialize
    c. delimit
    d. restrict

12. You can use a _____ to explicitly convert a value from one numeric data type to another, even if the conversion might result in a loss of data.

a. transpose statement
b. cast operator
c. conversion operator
d. literal conversion

13. The process of dropping a number's fractional part is called _____.

a. shifting
b. twos complement
c. numeric rounding
d. truncation

14. A programmer's tools for performing calculations are _____.

a. math operators
b. numeric literals
c. local variables
d. parsed literals

15. A _____ performs a calculation and gives a value.

a. numeric literal
b. math expression
c. machine instruction
d. programming statement

16. C# offers a special set of operators known as _____ that are designed specifically for changing the value of a variable without having to type the variable name twice.

a. combined assignment operators
b. advanced math operators
c. variable modifiers
d. assignment sequencers

17. In computer science, the term _____ typically means to analyze a string of characters for some purpose.

a. compile
b. compute
c. debug
d. parse

18. A(n) _____ is a piece of data that is passed into a method.

a. variable
b. argument
c. string
d. literal

19. A(n) _____ is an unexpected error that occurs while a program is running, causing the program to halt if the error is not properly dealt with.

a. breakpoint
b. bug
c. syntax error
d. exception

20. The _____ indicates that you want the number to appear formatted in a specific way when it is returned as a string from the `ToString` method.

a. formatting string
b. `insert` method
c. data type
d. variable name

21. You have started an application by clicking the start *Debugging* button (▶) or by pressing F5 on the keyboard. If an exception is thrown, the application stops running and Visual Studio goes into a special mode known as _____.

a. exception mode
b. break mode
c. debug mode
d. crash mode

22. Code that responds to exceptions when they are thrown and prevents the program from abruptly crashing is called a(n) _____.

a. exit strategy
b. fail safe
c. event handler
d. exception handler

23. A _____ is a name that represents a value that cannot be changed during the program's execution.

a. named literal
b. named constant
c. variable signature
d. key term

24. A _____ is a variable that is declared inside a class but not inside any method.

a. term
b. class variable
c. field
d. mnemonic

25. A(n) _____ specifies how a class member can be accessed by code outside the class.

a. namespace
b. access modifier
c. scope delimiter
d. class directive

26. A _____ is a field that cannot be changed by any statement in the class.

a. static field
b. class name
c. key field
d. constant field

27. .NET provides a class named _____, which contains numerous methods that are useful for performing advanced mathematical operations.

a. `Math`
b. `Calc`
c. `Trig`
d. `Linq`

28. When a control has the _____, it receives the user's keyboard input.

a. text
b. tab order
c. focus
d. input allocator

29. The order in which controls receive the focus is called the _____.
    a. order of operations
    b. program flow
    c. execution sequence
    d. tab order

30. The _____ contains a numeric value, which indicates the control's position in the tab order.
    a. IndexOf property
    b. TabIndex property
    c. ControlOrder property
    d. TabOrder property

True or False

1. You can clear the contents of a TextBox control in the same way that you clear the contents of a Label control.
2. In C#, you must declare a variable in a program before you can use it to store data.
3. If you declare a variable using the `var` keyword, you must provide an initialization value.
4. When you append the letter `D` or `d` to a numeric literal, it is treated as a `decimal` and is referred to as a `decimal` literal.
5. The order of operations dictates that the division operator works before the addition operator does.
6. All variables have a `ToString` method that you can call to convert the variable's value to a string.
7. When you pass the formatting string `"C"` or `"c"` to the `ToString` method, the number is returned formatted as currency.
8. When you declare a named constant, an initialization value is required.
9. An error will occur if the compiler finds a statement that tries to change the value of a constant field.
10. Forms and most controls have a Preferences property that allows you to change the object's background color.

Short Answer

1. In the *Toolbox*, in which group is the TextBox tool located?
2. What two things does a variable declaration specify about a variable?
3. Give an example of a programming statement that uses string concatenation.
4. What is the term used for a number that is written into a program's code?
5. Write a programming statement that assigns an integer literal to a variable.
6. What are the values on the right and left of an operator called?
7. Name the family of .NET methods that can be used to convert a string to any of the numeric data types.
8. What object is created in memory when an exception is thrown and has various properties that contain data about the exception?
9. What is the purpose of a `try-catch` statement?

10. Which .NET class provides predefined named constants that are assigned the mathematical values for *pi* and *e*?
11. In code, what function do you call to move the focus to a control?
12. What property allows you to change the color of a control's text?

Programming Problems

1. **Name Formatter**

   Create an application that lets the user enter the following pieces of data:

   - The user's first name
   - The user's middle name
   - The user's last name
   - The user's preferred title (Mr., Mrs., Ms., Dr., etc.)

   Assume the user has entered the following data:

   - First name: *Kelly*
   - Middle name: *Jane*
   - Last name: *Smith*
   - Title: *Ms.*

   The application should have buttons that display the user's name formatted in the following ways:

   - *Ms. Kelly Jane Smith*
   - *Kelly Jane Smith*
   - *Kelly Smith*
   - *Smith, Kelly Jane, Ms.*
   - *Smith, Kelly Jane*
   - *Smith, Kelly*

VideoNote
Solving the Tip, Tax, and Total Problem

2. **Tip, Tax, and Total**

   Create an application that lets the user enter the food charge for a meal at a restaurant. When a button is clicked, the application should calculate and display the amount of a 15 percent tip, 7 percent sales tax, and the total of all three amounts.

3. **Distance Traveled**

   Assuming there are no accidents or delays, the distance that a car travels down an interstate highway can be calculated with the following formula:

   $$\text{Distance} = \text{Speed} \times \text{Time}$$

   Create an application that allows the user to enter a car's speed in miles per hour.

   The application should have buttons that display the following:

   - The distance the car will travel in 5 hours
   - The distance the car will travel in 8 hours
   - The distance the car will travel in 12 hours

4. **Sales Tax and Total**

   Create an application that allows the user to enter the amount of a purchase. The program should then calculate the state and county sales tax. Assume the state sales tax is 4 percent and the county sales tax is 2 percent. The program should display the amount of the purchase, the state sales tax, the county sales tax, the total sales tax, and the total of the sale (which is the sum of the amount of purchase plus the total sales tax).

5. **Celsius and Fahrenheit Temperature Converter**

   Assuming that C is a Celsius temperature, the following formula converts the temperature to Fahrenheit:

   $$F = \frac{9}{5} C + 32$$

   Assuming that F is a Fahrenheit temperature, the following formula converts the temperature to Celsius:

   $$C = \frac{5}{9}(F - 32)$$

   Create an application that allows the user to enter a temperature. The application should have Button controls described as follows:

   - A button that reads *Convert to Fahrenheit*. If the user clicks this button, the application should treat the temperature that is entered as a Celsius temperature and convert it to Fahrenheit.
   - A button that reads *Convert to Celsius*. If the user clicks this button, the application should treat the temperature that is entered as a Fahrenheit temperature, and convert it to Celsius.

6. **Ingredient Adjuster**

   A cookie recipe calls for the following ingredients:

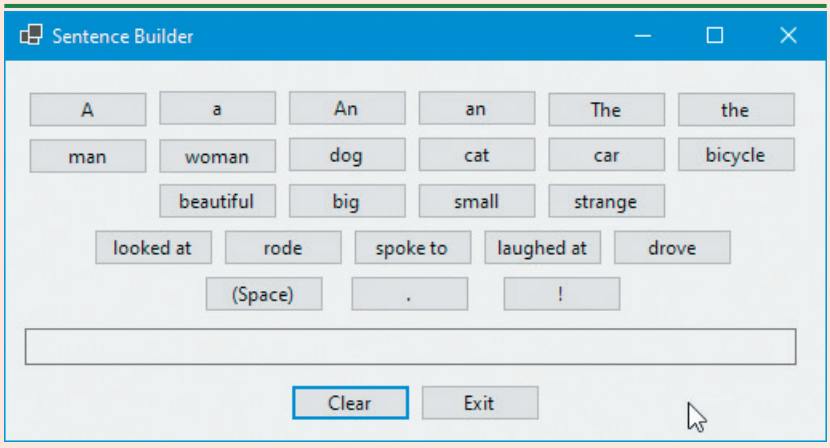
   - 1.5 cups of sugar
   - 1 cup of butter
   - 2.75 cups of flour

   The recipe produces 48 cookies with this amount of the ingredients. Create an application that lets the user enter the number cookies needed, then displays the number of cups of each ingredient needed for the specified number of cookies.

7. **Sentence Builder**

   The form in Figure 3-47 contains buttons showing various words, phrases, and punctuation. Create an application with a form similar to this one. When the application runs, the user clicks the buttons to build a sentence, which is shown in a Label control. You can use the same buttons as shown in the figure or make up your own. The *Clear* button should clear the sentence so the user can start over.

Figure 3-47 The *Sentence Builder* form



8. **How Much Insurance?**

   Many financial experts advise that property owners should insure their homes or buildings for at least 80 percent of the amount it would cost to replace the structure. Create an application that lets the user enter the replacement cost of a

building and then displays the minimum amount of insurance they should buy for the property.

9. **Lion and Tiger Percentages**

   Create an application that lets the user enter the number of lions and the number of tigers in the Big Cat exhibit at the local zoo. The application should display the percentages of lions and tigers in the exhibit.

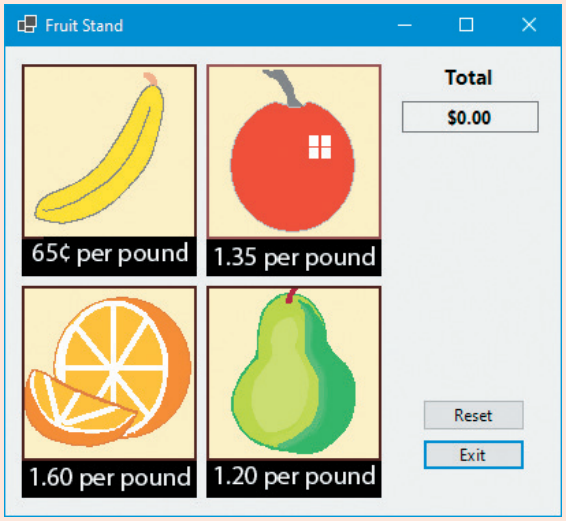
   For example, suppose there are 8 lions and 12 tigers in the exhibit, for a total of 20 cats. The percentage of lions can be calculated as $8 \div 20 = 0.4$, or 40 percent. The percentage of tigers can be calculated as $12 \div 20 = 0.6$, or 60 percent.

10. **Fruit Stand**

    Create an application with a form that resembles Figure 3-48. The PictureBox controls display the images of four fruits (a banana, an apple, an orange, and a pear) and each fruit's price-per-pound. You can find these images in the *Chap03* folder of the Student Sample Programs.

    The application keeps a total of the fruit that is purchased. When the application starts, the total displayed should be zero. Each time the user clicks one of the PictureBoxes, the price for a pound of that fruit should be added to the total, and the total should be displayed. When the user clicks the Reset button, the total should be reset to zero.

Figure 3-48 *Fruit Stand* form



11. **Automobile Costs**

    Create an application that lets the user enter the monthly costs for the following expenses incurred from operating their automobile: loan payment, insurance, gas, oil, tires, and maintenance. The program should then display the total monthly cost of these expenses and the total annual cost of these expenses.

12. **Paint Job Estimator**

    A painting company has determined that for every 115 square feet of wall space, 1 gallon of paint and 8 hours of labor will be required. The company charges $20.00 per hour for labor. Create an application that allows the user to enter the square feet of wall space to be painted and the price of the paint per gallon. The program should display the following data:

    - The number of gallons of paint required
    - The hours of labor required

- The cost of the paint
- The labor charges
- The total cost of the paint job

13. Property Tax

If you own real estate in a particular county, the property tax that you owe each year is calculated as 64 cents per $100 of the property's value. For example, if the property's value is $10,000, then the property tax is calculated as follows:

$$Tax = \$10{,}000 \div 100 \times 0.64$$

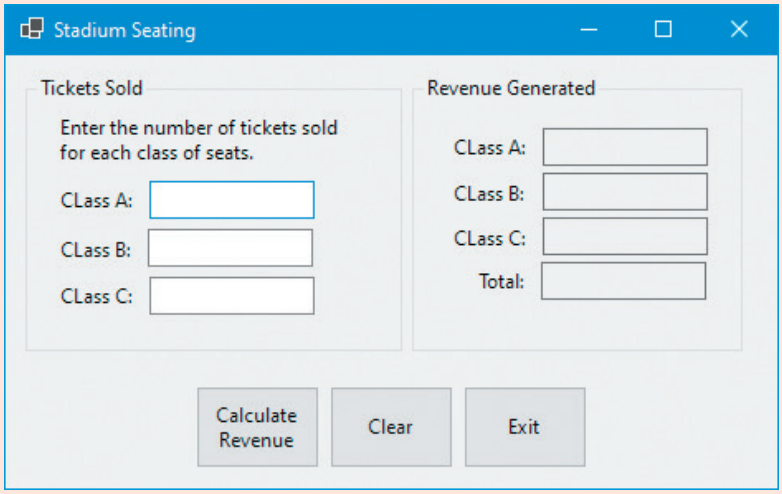
Create an application that allows the user to enter a property's value and displays the property tax on that property.

14. Stadium Seating

There are three seating categories at an athletic stadium. For a baseball game, Class A seats cost $15 each, Class B seats cost $12 each, and Class C seats cost $9 each. Create an application that allows the user to enter the number of tickets sold for each class. The application should be able to display the amount of income generated from each class of ticket sales and the total revenue generated. The application's form should resemble the one shown in Figure 3-49.

Use the following sets of test data to determine if the application is calculating properly:

Figure 3-49 *Stadium Seating* form



Ticket Sales	Revenue
Class A: 320	Class A: $4,800.00
Class B: 570	Class B: $6,840.00
Class C: 890	Class C: $8,010.00
	Total Revenue: $19,650.00
Class A: 500	Class A: $7,500.00
Class B: 750	Class B: $9,000.00
Class C: 1,200	Class C: $10,800.00
	Total Revenue: $27,300.00
Class A: 100	Class A: $1,500.00
Class B: 300	Class B: $3,600.00
Class C: 500	Class C: $4,500.00
	Total Revenue: $9,600.00

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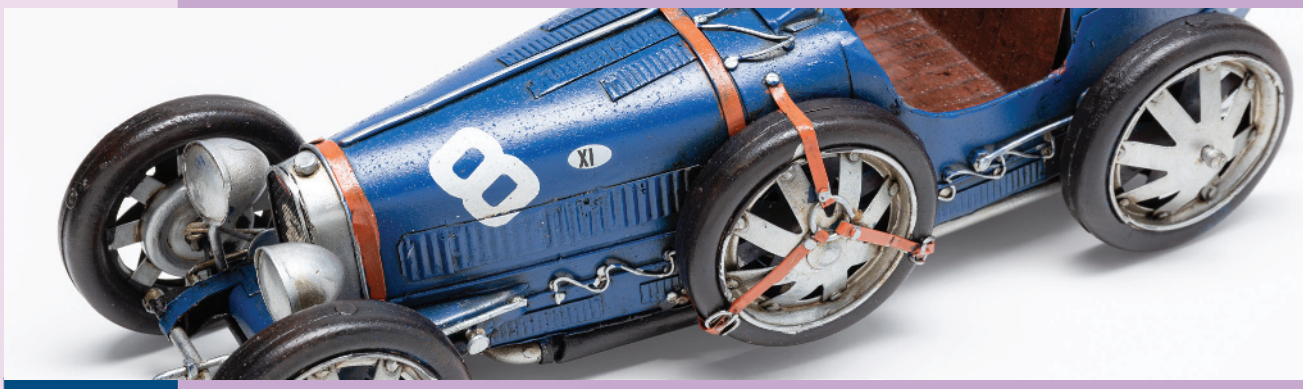


CHAPTER 4

Making Decisions

TOPICS

4.1 Decision Structures and the `if` Statement
4.2 The `if-else` Statement
4.3 Nested Decision Structures
4.4 Logical Operators
4.5 `bool` Variables and Flags
4.6 Comparing Strings
4.7 Preventing Data Conversion Exceptions with the `TryParse` Methods
4.8 Input Validation
4.9 Radio Buttons and Check Boxes
4.10 The `switch` Statement
4.11 Introduction to List Boxes

4.1 Decision Structures and the `if` Statement

CONCEPT: **A decision structure allows a program to perform actions only under certain conditions. In code, you can use the `if` statement to write a simple decision structure.**

A **control structure** is a logical design that controls the order in which a set of statements execute. So far in this book, we have used only the simplest type of control structure: the

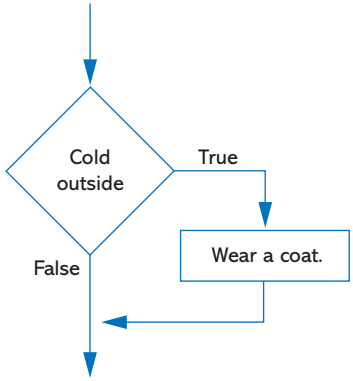
sequence structure. A **sequence structure** is a set of statements that execute in the order that they appear. For example, the following code sample is a sequence structure because the statements execute from top to bottom.

```
int ageInYears, ageInDays;
ageInYears = int.Parse(ageTextBox.Text);
ageInDays = ageInYears * 365;
daysLabel = ageInDays.ToString();
```

Although the sequence structure is heavily used in programming, it cannot handle every type of task. Some problems simply cannot be solved by performing a set of ordered steps, one after the other. For example, consider a pay-calculating program that determines whether an employee has worked overtime. If the employee has worked more than 40 hours, he or she gets paid extra for all hours over 40. Otherwise, the overtime calculation should be skipped. Programs like this require a different type of control structure: one that can execute a set of statements only under certain circumstances. This can be accomplished with a **decision structure**. (Decision structures are also known as **selection structures**.)

In a decision structure's simplest form, a specific action is performed only if a certain condition exists. If the condition does not exist, the action is not performed. The flowchart shown in Figure 4-1 shows how the logic of an everyday decision can be diagrammed as a decision structure. The diamond symbol represents a true-false condition. If the condition is true, we follow one path, which leads to an action being performed. If the condition is false, we follow another path, which skips the action.

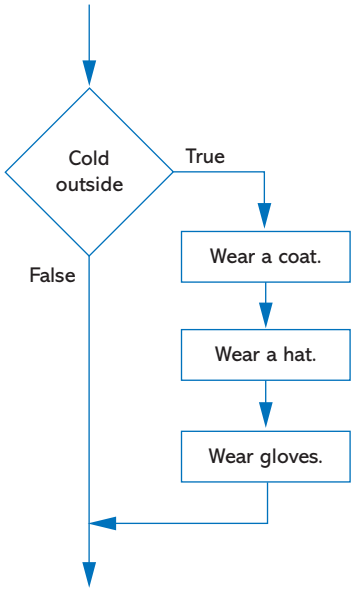
Figure 4-1 A simple decision structure



In the flowchart, the diamond symbol indicates some condition that must be tested. In this case, we are determining whether the condition *Cold outside* is true or false. If this condition is true, the action *Wear a coat* is performed. If the condition is false, the action is skipped. The action is **conditionally executed** because it is performed only when a certain condition is true.

Programmers call the type of decision structure shown in Figure 4-1 a **single-alternative decision structure** because it provides only one alternative path of execution. If the condition in the diamond symbol is true, we take the alternative path. Otherwise, we exit the structure. Figure 4-2 shows a more elaborate example, where three actions are taken only when it is cold outside.

Figure 4-2 A decision structure that performs three actions if it is cold outside



In C#, you use the `if` statement to write a single-alternative decision structure. Here is the general format of the `if` statement:

```
if (expression)
{
    statement;
    statement;
    etc.
}
```

The statement begins with the word `if`, followed by an *`expression`* enclosed in a set of parentheses. Beginning on the next line is a set of statements enclosed in curly braces.

The `expression` that appears inside the parentheses is a Boolean expression. A **Boolean expression** is one that can be evaluated as either true or false. When the `if` statement executes, the Boolean expression is tested. If it is true, the statements that appear inside the curly braces are executed. If the Boolean expression is false, however, the statements inside the curly braces are skipped. We say that the statements inside the curly braces are *conditionally executed* because they are executed only if the Boolean expression is true.

If you are writing an `if` statement that has only one conditionally executed statement, you do not have to enclose the conditionally executed statement inside curly braces. Such an `if` statement can be written in the following general format:

```
if (expression)
    statement;
```

When an `if` statement written in this format executes, the Boolean expression is tested. If it is true, the one statement that appears on the next line is executed. If the Boolean expression is false, however, that one statement is skipped.

Although the curly braces are not required when there is only one conditionally executed statement, it is still a good idea to use them, as shown in the following general format:

```
if (expression)
{
    statement;
}
```

This is a good style for writing `if` statements because it minimizes errors. Remember, if you have more than one conditionally executed statement, those statements *must* be enclosed in curly braces. If you get into the habit of always enclosing the conditionally executed statements in a set of curly braces, it's less likely that you will forget them.

Boolean Expressions and Relational Operators

Boolean expressions are named in honor of the English mathematician George Boole. In the 1800s, Boole invented a system of mathematics in which the abstract concepts of true and false can be used in computations.

Typically, the Boolean expression that is tested by an `if` statement is formed with a relational operator. A **relational operator** determines whether a specific comparative relationship exists between two values. For example, the greater than operator (`>`) determines whether one value is greater than another. The equal to operator (`==`) determines whether two values are equal. Table 4-1 lists the relational operators that are available in C#.

Table 4-1 Relational operators

Operator	Meaning
>	Greater than
<	Less than
>=	Greater than or equal to
<=	Less than or equal to
==	Equal to
!=	Not equal to

The following is an example of an expression that uses the greater than (`>`) operator to compare two variables, `length` and `width`:

```
length > width
```

This expression determines whether the value of the `length` variable is greater than the value of the `width` variable. If `length` is greater than `width`, the value of the expression is true. Otherwise, the value of the expression is false. The following expression uses the less than operator (`<`) to determine whether `length` is less than `width`:

```
length < width
```

Table 4-2 shows examples of several Boolean expressions that compare the variables `x` and `y`.

Table 4-2 Boolean expressions using relational operators

Expression	Meaning
`x > y`	Is `x` greater than `y`?
`x < y`	Is `x` less than `y`?
`x >= y`	Is `x` greater than or equal to `y`?
`x <= y`	Is `x` less than or equal to `y`?
`x == y`	Is `x` equal to `y`?
`x != y`	Is `x` not equal to `y`?

The >= and <= Operators

Two of the operators, >= and <=, test for more than one relationship. The >= operator determines whether the operand on its left is greater than *or* equal to the operand on its right. The <= operator determines whether the operand on its left is less than *or* equal to the operand on its right.

For example, assume the variable a is assigned 4. All the following expressions are true:

```
a >= 4
a >= 2
8 >= a
a <= 4
a <= 9
4 <= a
```

The == Operator

The == operator determines whether the operand on its left is equal to the operand on its right. If the values of both operands are the same, the expression is true. Assuming that a is 4, the expression `a == 4` is true and the expression `a == 2` is false.

NOTE: The equality operator is two = symbols together. Don't confuse this operator with the assignment operator, which is one = symbol.

The != Operator

The != operator is the not equal to operator. It determines whether the operand on its left is not equal to the operand on its right, which is the opposite of the == operator. As before, assuming a is 4, b is 6, and c is 4, both `a != b` and `b != c` are true because a is not equal to b and b is not equal to c; however, `a != c` is false because a is equal to c.

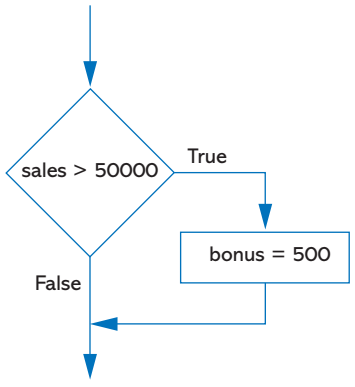
Putting It All Together

Let's look at the following example of the if statement:

```
if (sales > 50000)
{
    bonus = 500;
}
```

This statement uses the > operator to determine whether sales is greater than 50,000. If the expression `sales > 50000` is true, the variable bonus is assigned 500. If the expression is false, however, the assignment statement is skipped. Figure 4-3 shows a flowchart for this section of code.

Figure 4-3 Example decision structure

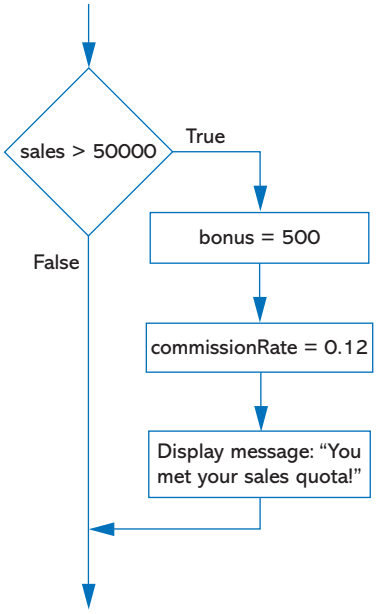


The following code sample conditionally executes three statements. Figure 4-4 shows a flowchart for this section of code.

```
if (sales > 50000)
{
    bonus = 500;
    commissionRate = 0.12;
    MessageBox.Show("You met your sales quota!");
}
```

When you write an `if` statement, Visual Studio automatically indents the conditionally executed statements, as shown in the previous examples. The indentation is not required, but it makes the code easier to read and debug. By indenting the conditionally executed statements, you visually set them apart from the surrounding code. This allows you to tell at a glance what part of the program is controlled by the `if` statement. Most programmers use this style of indentation when writing `if` statements.

Figure 4-4 Example decision structure



In Tutorial 4-1, you will write code that uses an `if` statement to display a message only when a Boolean expression is true.

Tutorial 4-1:
Completing the *Test Score Average* Application

VideoNote
Tutorial 4-1: Completing the *Test Score Average* application

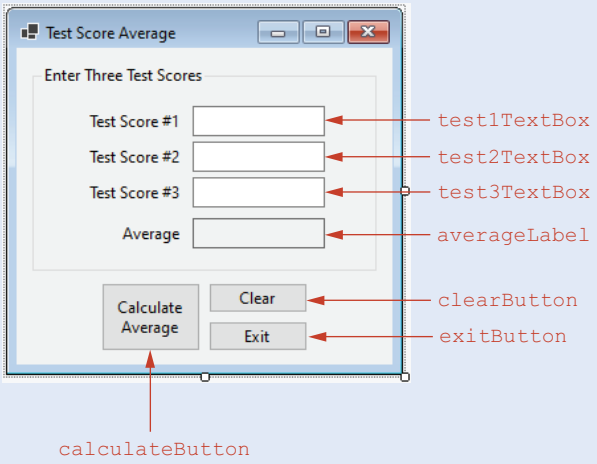
In this tutorial, you will complete an application that allows the user to enter three test scores and calculates the average of the test scores. If the average is greater than 95, the application also displays a message congratulating the user.

To save time, the project has already been started for you, and the application's form has already been created. To complete the project, follow the steps in this tutorial.

Step 1: Start Visual Studio. Open the project named *TestScoreAverage* in the *Chap04* folder of this book's Student Sample Programs, available for download at www.pearsonhighered.com/gaddis.

Step 2: Open the Form1 form in the *Designer*. The form is shown, along with the names of the important controls, in Figure 4-5.

Figure 4-5 The *TestScoreAverage* project's form



Step 3: Now, you will create the Click event handlers for the Button controls. At the end of this tutorial, Program 4-1 shows the completed code for the form. You will be instructed to refer to Program 4-1 as you write the event handlers.

In the *Designer*, double-click the `calculateButton` control. This will open the code editor, and you see an empty event handler named `calculateButton_Click`. Complete the `calculateButton_Click` event handler by typing the code shown in lines 12–39 in Program 4-1.

Let's take a closer look at the code:

Line 12: This is the beginning of a `try-catch` statement. The try block appears in lines 14–33, and the catch block appears in lines 37–38. The purpose of this `try-catch` statement is to gracefully respond if the user enters invalid input for any of the test scores. If any of the statements in lines 18, 19, and 20 throw an exception, the program does not crash. Instead, it jumps to the catch block, and line 38 displays an error message.

Line 14: This statement declares a constant `double` named `HIGH_SCORE`, set to the value 95.0. We use this constant to determine whether the average is high. If the average is greater than this constant, the program displays a message congratulating the user.

Line 15: This statement declares the following `double` variables: `test1`, `test2`, `test3`, and `average`. The variables hold the three test scores and the average test score.

Line 18: This statement converts the `test1TextBox` control's Text property to a `double` and assigns the result to the `test1` variable.

Line 19: This statement converts the `test2TextBox` control's Text property to a `double` and assigns the result to the `test2` variable.

Line 20: This statement converts the `test3TextBox` control's Text property to a `double` and assigns the result to the `test3` variable.

Line 23: This statement calculates the average of the `test1`, `test2`, and `test3` variables and assigns the result to the `average` variable.

Line 26: This statement converts the `average` variable to a string (rounded to 1 decimal place) and assigns the result to the `averageLabel` control's Text property.

Line 30: This `if` statement determines whether `average` is greater than `HIGH_SCORE`. If it is, the statement in line 32 is executed, displaying a message box with a congratulatory message. If `average` is not greater than `HIGH_SCORE`, the statement in line 32 is skipped.

Step 4: Switch your view back to the *Designer* and double-click the `clearButton` control. In the code editor, you will see an empty event handler named `clearButton_Click`. Complete the `clearButton_Click` event handler by typing the code shown in lines 44–51 in Program 4-1.

Lines 45–47: Each of these statements assigns an empty string (`""`) to the Text property of one of the TextBox controls. When these statements have finished executing, the TextBox controls appear empty.

Line 48: This statement assigns an empty string (`""`) to the `averageLabel` control's Text property. After the statement has executed, the label appears empty.

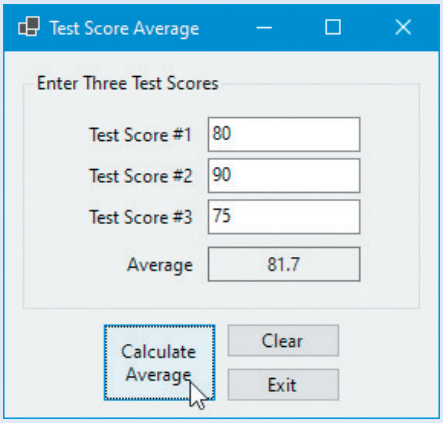
Line 51: This statement sets the focus to the `test1TextBox` control. This makes it more convenient for the user to start entering a new set of test scores.

Step 5: Switch your view back to the *Designer* and double-click the `exitButton` control. In the code editor you will see an empty event handler named `exitButton_Click`. Complete the `exitButton_Click` event handler by typing the code shown in lines 56–57 in Program 4-1.

Step 6: Save the project. Then, press F5 on the keyboard or click the *Start Debugging* button ( ) on the toolbar to compile and run the application.

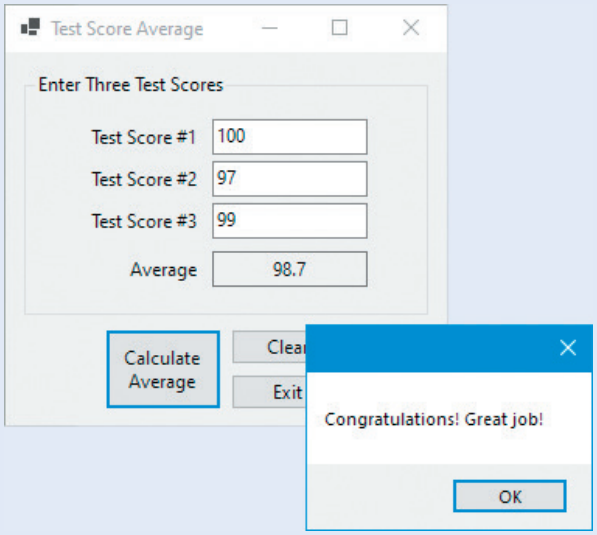
First, enter the following test scores in the TextBoxes: *80*, *90*, and *75*. Click the *Calculate Average* button and the average should appear as shown in Figure 4-6.

Figure 4-6 Average displayed



Next, click the *Clear* button to clear the TextBoxes and the average. Now, enter the following test scores in the TextBoxes: *100*, *97*, and *99*. Click the *Calculate Average* button. This time, in addition to displaying the average, the application displays the message box shown in Figure 4-7.

Figure 4-7 Average and message displayed



Continue to test the application as you wish. When you are finished, click the *Exit* button, and the form should close.

Program 4-1 Completed Form1 code for the *TestScoreAverage* project

```
 1 namespace TestScoreAverage
 2 {
 3     public partial class Form1 : Form
 4     {
 5         public Form1()
 6         {
 7             InitializeComponent();
 8         }
 9
10         private void calculateButton_Click(object sender,
           EventArgs e)
11         {
12             try
13             {
14                const double HIGH_SCORE = 95.0;  // High Score value
15                double test1, test2, test3, average; // Variables
16
17                // Get the test scores from the TextBoxes.
18                test1 = double.Parse(test1TextBox.Text);
19                test2 = double.Parse(test2TextBox.Text);
20                test3 = double.Parse(test3TextBox.Text);
21
22                // Calculate the average test score.
23                average = (test1 + test2 + test3) / 3.0;
24
25                // Display the average, rounded to 2 decimal places.
26                averageLabel.Text = average.ToString("n1");
27
28                // If the average is a high score, congratulate
29                // the user with a message box.
30                if (average > HIGH_SCORE)
31                {
```

```
32                      MessageBox.Show("Congratulations! Great job!");
33                  }
34              }
35              catch (Exception ex)
36              {
37                  // Display the default error message.
38                  MessageBox.Show(ex.Message);
39              }
40          }
41
42          private void clearButton_Click(object sender, EventArgs e)
43          {
44              // Clear the TextBoxes and the averageLabel control.
45              test1TextBox.Text = "";
46              test2TextBox.Text = "";
47              test3TextBox.Text = "";
48              averageLabel.Text = "";
49
50              // Reset the focus to test1.
51              test1TextBox.Focus();
52          }
53
54          private void exitButton_Click(object sender, EventArgs e)
55          {
56              // Close the form.
57              this.Close();
58          }
59      }
60  }
```

Checkpoint

4.1 What is a control structure?

4.2 What is a decision structure?

4.3 What is a single-alternative decision structure?

4.4 What is a Boolean expression?

4.5 What types of relationships between numeric values can you test with relational operators?

4.6 Write an `if` statement that determines whether the variable `y` is equal to 20. If it is, assign 0 to the variable `x`.

4.7 Write an `if` statement that determines whether the variable `sales` is greater than or equal to 10,000. If it is, assign 0.2 to the variable `commissionRate`.

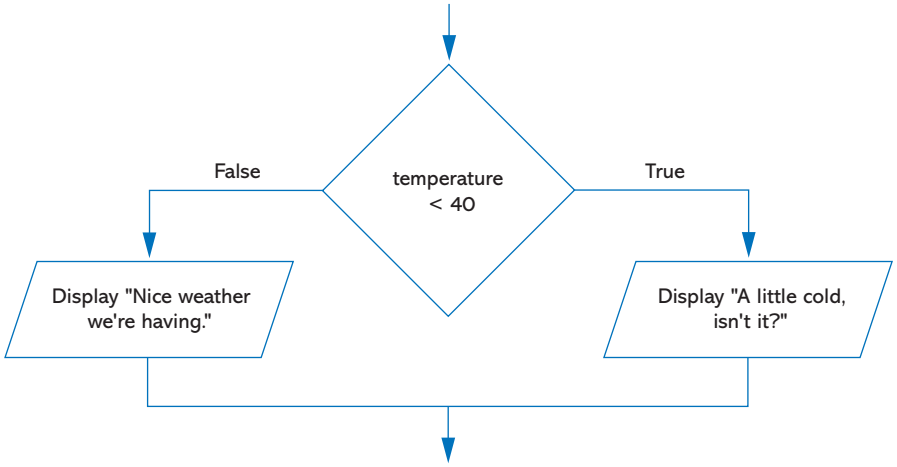
4.2 The `if-else` Statement

CONCEPT: **An `if-else` statement will execute one block of statements if its Boolean expression is true or another block if its Boolean expression is false.**

The previous section introduced the single-alternative decision structure (the `if` statement), which has one alternative path of execution. Now, we will look at the **dual-alternative decision structure**, which has two possible paths of execution—one path is taken if the

Boolean expression is true, and the other path is taken if the Boolean expression is false. Figure 4-8 shows an example flowchart for a dual-alternative decision structure.

Figure 4-8 A dual-alternative decision structure



The decision structure in the flowchart tests the expression `temperature < 40`. If this expression is true, the message "A little cold, isn't it?" is displayed. If the expression is false, the message "Nice weather we're having." is displayed.

In code, we write a dual-alternative decision structure as an **`if-else` statement**. Here is the general format of the `if-else` statement:

```
if (expression)
{
    statement;
    statement;
    etc.
}
else
{
    statement;
    statement;
    etc.
}
```

If the Boolean expression is true, this set of statements is executed.

If the Boolean expression is false, this set of statements is executed.

An `if-else` statement has two parts: an `if` clause and an `else` clause. Just like a regular `if` statement, the `if-else` statement tests a Boolean expression. If the Boolean expression is true, the set of statements following the `if` clause is executed. If the Boolean expression is false, the set of statements following the `else` clause is executed.

The `if-else` statement has two sets of conditionally executed statements. One set is executed only under the condition that the Boolean expression is true, and the other set is executed only under the condition that the Boolean expression is false. Under no circumstances are both sets of conditionally executed statements executed.

If either set of conditionally executed statements contains only one statement, the curly braces are not required. For example, the following general format shows only one statement following the `if` clause and only one statement following the `else` clause:

```
if (expression)
    statement;
else
    statement;
```

Although the curly braces are not required when there is only one conditionally executed statement, it is still a good idea to use them, as shown in the following general format:

```
if (expression)
{
    statement;
}
else
{
    statement;
}
```

When we discussed the regular `if` statement, we mentioned that this is a good style of programming because it cuts down on errors. If there is more than one conditionally executed statement following either the `if` clause or the `else` clause, those statements *must* be enclosed in curly braces. If you get into the habit of always enclosing the conditionally executed statements in a set of curly braces, it's less likely that you will forget them.

In Tutorial 4-2, you will complete an application that uses an `if-else` statement.

Tutorial 4-2:
Completing the *Payroll with Overtime* Application

VideoNote
Tutorial 4-2: Completing the *Payroll with Overtime* application

At a particular business, if an employee works more than 40 hours in a week, it is said that the employee has worked *overtime*. For example, an employee that has worked 45 hours in a week has worked 5 overtime hours. Employees that work overtime get paid their regular hourly pay rate for the first 40 hours plus 1.5 times their regular hourly pay rate for all hours over 40. In this tutorial, you will complete a payroll application that calculates an employee's gross pay, including overtime pay.

The application allows the user to enter the number of hours worked and the hourly pay rate into TextBoxes. When the user clicks a button, the gross pay is calculated in the following manner:

If the hours worked is greater than 40:

base pay = hourly pay rate × 40
overtime hours = hours worked − 40
overtime pay = overtime hours × hourly pay rate × 1.5
gross pay = base pay + overtime pay

Else:

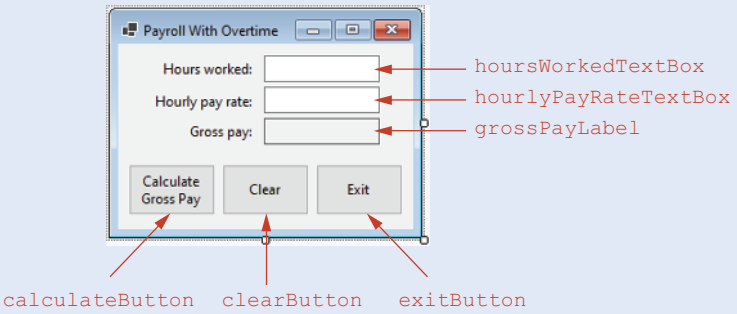
gross pay = hours worked × hourly pay rate

To save time, the project has already been started for you, and the application's form has already been created. To complete the project, follow the steps in this tutorial.

Step 1: Start Visual Studio. Open the project named *PayrollWithOvertime* in the *Chap04* folder of this book's Student Sample Programs.

Step 2: Open the Form1 form in the *Designer*. The form is shown, along with the names of the important controls, in Figure 4-9.

Figure 4-9 The *PayrollWithOvertime* project's form



Step 3: Now, you will create the Click event handlers for the Button controls. At the end of this tutorial, Program 4-2 shows the completed code for the form. You will be instructed to refer to Program 4-2 as you write the event handlers.

In the *Designer*, double-click the `calculateButton` control. This opens the code editor, and you see an empty event handler named `calculateButton_Click`. Complete the `calculateButton_Click` event handler by typing the code shown in lines 12–58 in Program 4-2.

Let's take a closer look at the code:

Line 12: This is the beginning of a `try-catch` statement. The try block appears in lines 14–52, and the catch block appears in lines 56–57. The purpose of this `try-catch` statement is to gracefully respond if the user enters invalid input. If an exception is thrown by any statement inside the try block, the program jumps to the catch block, and line 57 displays an error message.

Lines 15–16: These statements declare the following named constants:

- `BASE_HOURS`, a constant `decimal` set to the value 40. This is the number of hours an employee can work in a week without getting overtime pay.
- `OT_MULTIPLIER`, a constant `decimal` set to the value 1.5. This is the pay rate multiplier for overtime hours.

Lines 19–24: These statements declare the following variables:

- `hoursWorked`, a `decimal` variable to hold the number of hours worked
- `hourlyPayRate`, a `decimal` variable to hold the hourly pay rate
- `basePay`, a `decimal` variable to hold the pay for 40 or less hours
- `overtimeHours`, a `decimal` variable to hold the number of overtime hours worked
- `overtimePay`, a `decimal` variable to hold the amount of overtime pay
- `grossPay`, a `decimal` variable to hold the gross pay

Line 27: This statement converts the `hoursWorkedTextBox` control's Text property to a `decimal` and assigns the result to the `hoursWorked` variable.

Line 28: This statement converts the `hourlyPayRateTextBox` control's Text property to a `decimal` and assigns the result to the `hourlyPayRate` variable.

Line 31: This `if` statement determines whether `hoursWorked` is greater than `BASE_HOURS` (40). If so, the statements in lines 33–43 are executed. Otherwise, the statements in lines 47–48 are executed.

Lines 33–43: These statements, which are executed only if the hours worked are greater than 40, make all the necessary calculations to determine gross pay with overtime:

- Line 34 calculates the base pay, which is the amount of pay for the first 40 hours.
- Line 37 calculates the number of overtime hours, which is the number of hours over 40.
- Line 40 calculates the amount of overtime pay, which is the pay for the hours over 40.
- Line 43 calculates the gross pay, which is the amount of base pay plus the amount of overtime pay. The result is assigned to the `grossPay` variable.

Line 48: This statement, which is executed only if the hours worked are 40 or less, calculates the gross pay and assigns the result to the `grossPay` variable.

Line 52: This statement converts the value of the `grossPay` variable to a string, formatted as currency, and assigns the result to the `grossPayLabel` control's Text property.

Step 4: Switch your view back to the *Designer* and double-click the `clearButton` control.

In the code editor, you see an empty event handler named `clearButton_Click`. Complete the `clearButton_Click` event handler by typing the code shown in lines 63–69 in Program 4-2. These statements clear the TextBoxes and the `grossPayLabel` control and sets the focus to the `hoursWorkedTextBox` control.

Step 5: Switch your view back to the *Designer* and double-click the `exitButton` control. In the code editor you see an empty event handler named `exitButton_Click`. Complete the `exitButton_Click` event handler by typing the code shown in lines 74–75 in Program 4-2.

Step 6: Save the project and run the application. First, enter 40 for the number of hours worked and 20 for the hourly pay rate. Click the *Calculate Gross* Pay button, and the application should display $800.00 as the gross pay. No overtime hours were worked, so the gross pay is simply calculated as hours worked × hourly pay rate.

Click the *Clear* button. Enter 50 for the number of hours worked and 20 for the hourly pay rate. Click the *Calculate Gross* Pay button, and the application should display $1,100.00 as the gross pay. This time, more than 40 hours were worked, so the application calculated the gross pay to include overtime pay.

Continue to test the application as you wish. When you are finished, click the Exit button, and the form should close.

Program 4-2 Completed Form1 code for the *PayrollWithOvertime* project

```
1 namespace PayrollWithOvertime
2 {
3   public partial class Form1 : Form
4   {
5     public Form1()
6     {
7       InitializeComponent();
8     }
9
```

```
10    private void calculateButton_Click(object sender, EventArgs e)
11    {
12      try
13      {
14        // Named constants
15        const decimal BASE_HOURS = 40m;
16        const decimal OT_MULTIPLIER = 1.5m;
17
18        // Local variables
19        decimal hoursWorked;    // Number of hours worked
20        decimal hourlyPayRate;  // Hourly pay rate
21        decimal basePay;        //  Pay not including overtime
22        decimal overtimeHours;  // overtime hours worked
23        decimal overtimePay;    // overtime pay
24        decimal grossPay;       // total gross pay
25
26        // Get the hours worked and hourly pay rate.
27        hoursWorked = decimal.Parse(hoursWorkedTextBox.Text);
28       hourlyPayRate = decimal.Parse(hourlyPayRateText Box.Text);
29
30        // Determine the gross pay.
31        if (hoursWorked > BASE_HOURS)
32        {
33          // Calculate the base pay (without overtime).
34          basePay = hourlyPayRate * BASE_HOURS;
35
36          // Calculate the number of overtime hours.
37          overtimeHours = hoursWorked - BASE_HOURS;
38
39          // Calculate the overtime pay.
40          overtimePay = overtimeHours * hourlyPayRate *
              OT_MULTIPLIER;
41
42          // Calculate the gross pay.
43          grossPay = basePay + overtimePay;
44        }
45        else
46        {
47          // Calculate the gross pay.
48          grossPay = hoursWorked * hourlyPayRate;
49        }
50
51        // Display the gross pay.
52        grossPayLabel.Text = grossPay.ToString("c");
53      }
54      catch (Exception ex)
55      {
56        // Display an error message.
57        MessageBox.Show(ex.Message);
58      }
59    }
60
61    private void clearButton_Click(object sender, EventArgs e)
62    {
63      // Clear the TextBoxes and gross pay label.
64      hoursWorkedTextBox.Text = "";
65      hourlyPayRateTextBox.Text = "";
66      grossPayLabel.Text = "";
67
```

```
68        // Reset the focus.
69        hoursWorkedTextBox.Focus();
70      }
71
72      private void exitButton_Click(object sender, EventArgs e)
73      {
74        // Close the form.
75        this.Close();
76      }
77    }
78 }
```

Checkpoint

4.8 Describe how a dual-alternative decision structure works.

4.9 In an `if-else` statement, under what circumstances do the statements that appear after the `else` clause execute?

4.10 Write an `if-else` statement that works like this: If the `sales` variable is greater-than or equal-to 50,000, the `commissionRate` variable should be assigned the value 0.2. Otherwise, the `commissionRate` variable should be assigned the value 0.1.

4.3 Nested Decision Structures

CONCEPT: **To test more than one condition, a decision structure can be nested inside another decision structure.**

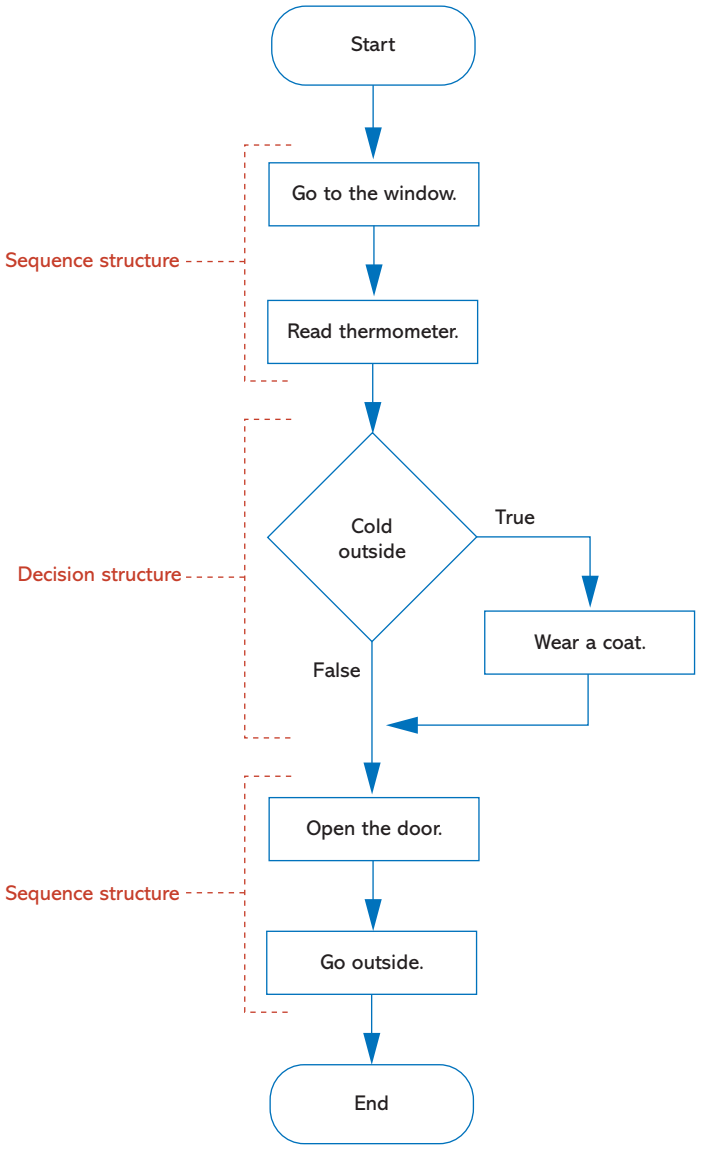
In Section 4.1, we mentioned that a control structure determines the order in which a set of statements execute. Programs are usually designed as combinations of different control structures. For example, Figure 4-10 shows a flowchart that combines a decision structure with two sequence structures.

The flowchart in Figure 4-10 starts with a sequence structure. Assuming you have an outdoor thermometer in your window, the first step is *Go to the window*, and the next step is *Read thermometer*. A decision structure appears next, testing the condition *Cold outside*. If this is true, the action *Wear a coat* is performed. Another sequence structure appears next. The step *Open the door* is performed, followed by *Go outside*.

Quite often, structures must be nested inside other structures. For example, look at the partial flowchart in Figure 4-11. It shows a decision structure with a sequence structure nested inside. The decision structure tests the condition *Cold outside*. If that condition is true, the steps in the sequence structure are executed.

You can also have **nested decision structures**, which are decision structures that appear inside other decision structures. This is commonly done in programs that need to test more than one condition. For example, consider a program that determines whether a

Figure 4-10 Combining sequence structures with a decision structure



bank customer qualifies for a loan. To qualify, two conditions must exist: (1) the customer must earn at least $40,000 per year, and (2) the customer must have been employed at his or her current job for at least 2 years. Figure 4-12 shows a flowchart for an algorithm that could be used in such a program. Assume that the `salary` variable contains the customer's annual salary, and the `yearsOnJob` variable contains the number of years that the customer has worked on his or her current job.

If we follow the flow of execution, we see that the Boolean expression `salary >= 40000` is tested. If this expression is false, there is no need to perform further tests; we know that the customer does not qualify for the loan. If the expression is true, however, we need to test the second condition. This is done with a nested decision structure that tests the Boolean expression `yearsOnJob >= 2`. If this expression is true, then the customer qualifies for the loan. If this expression is false, then the customer does not qualify. In Tutorial 4-3, you create an application that performs this algorithm.

Figure 4-11 A sequence structure nested inside a decision structure

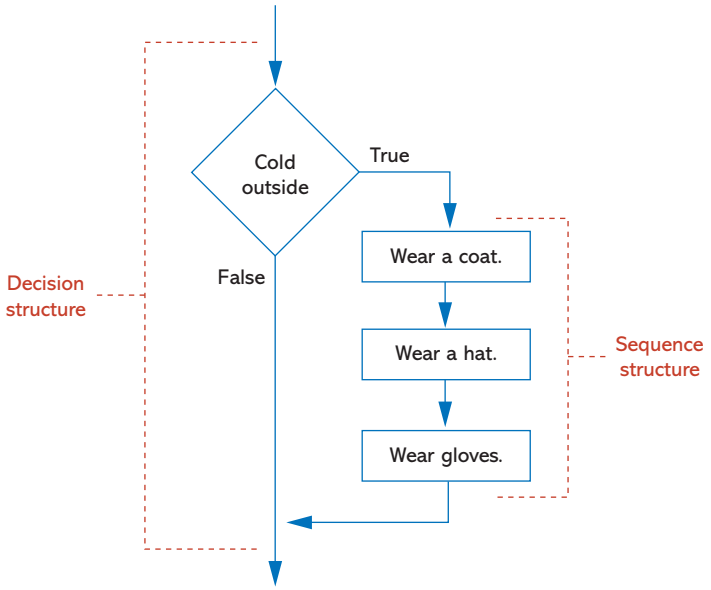
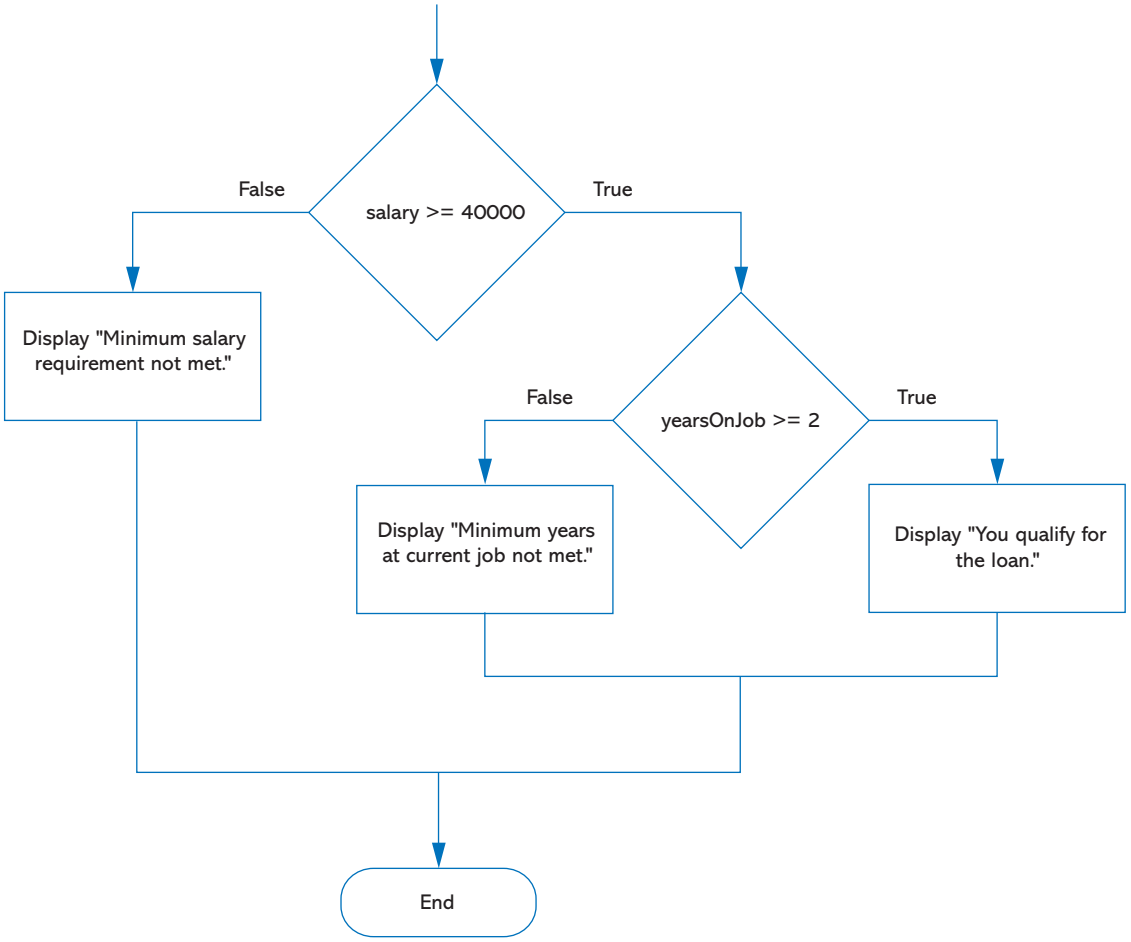


Figure 4-12 A nested decision structure



Tutorial 4-3:
Completing the *Loan Qualifier* Application

VideoNote
Tutorial 4-3: Completing the *Loan Qualifier* application

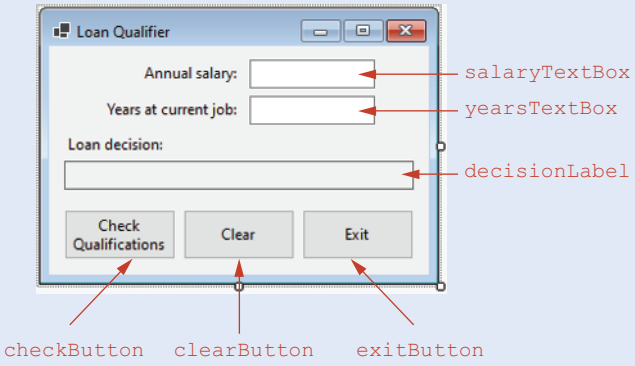
In this tutorial, you complete an application that determines whether a person qualifies for a loan. To qualify for the loan, the person must earn a salary of at least $40,000 and must have been employed at his or her current job for at least 2 years.

To save time, the project has already been started for you, and the application's form has already been created. To complete the project, follow the steps in this tutorial.

Step 1: Start Visual Studio. Open the project named *LoanQualifier* in the *Chap04* folder of this book's Student Sample Programs.

Step 2: Open the Form1 form in the *Designer*. The form is shown, along with the names of the important controls, in Figure 4-13.

Figure 4-13 The *LoanQualifier* project's form



Step 3: Now, you will create the Click event handlers for the Button controls. At the end of this tutorial, Program 4-3 shows the completed code for the form. You will be instructed to refer to Program 4-3 as you write the event handlers.

In the *Designer*, double-click the `checkButton` control. This opens the code editor, and you see an empty event handler named `checkButton_Click`. Complete the `checkButton_Click` event handler by typing the code shown in lines 12–52 in Program 4-3. Let's take a closer look at the code:

Line 12: This is the beginning of a `try-catch` statement. The try block appears in lines 14–46, and the catch block appears in lines 50–51. The purpose of this `try-catch` statement is to gracefully respond if the user enters invalid input. If an exception is thrown by any statement inside the try block, the program jumps to the catch block, and line 51 displays an error message.

Lines 15–16: These statements declare the following named constants:

- `MINIMUM_SALARY`, a constant `decimal` set to the value 40,000, which is the minimum salary a person must earn to qualify for the loan
- `MINIMUM_YEARS_ON_JOB`, a constant `int` set to the value 2, which is the minimum number of years a person must have been at his or her current job to qualify for the loan

Lines 19–20: These statements declare the following variables:

- `salary`, a `decimal` variable to hold the salary
- `yearsOnJob`, an `int` variable to hold the number of years at the current job

Lines 23–24: These statements get the salary and years at the current job from the TextBox controls and assign those values to the `salary` and `yearsOnJob` variables.

Line 27: This `if` statement determines whether `salary` is greater than or equal to `MINIMUM_SALARY`. If so, the program continues at line 29. Otherwise, the program jumps to the `else` clause in line 41, and in lines 44–45 the string "Minimum salary requirement not met." is assigned to the `decisionLabel` control's Text property.

Line 29: This `if` statement determines whether `yearsOnJob` is greater than or equal to `MINIMUM_YEARS_ON_JOB`. If so, the program continues at line 32, where the string "You qualify for the loan." is assigned to the `decisionLabel` control's Text property. Otherwise, the program jumps to the `else` clause in line 34, and in lines 37–38 the string "Minimum years at current job not met." is assigned to the `decisionLabel` control's Text property.

Step 4: Switch your view back to the *Designer* and double-click the `clearButton` control. In the code editor, you see an empty event handler named `clearButton_Click`. Complete the `clearButton_Click` event handler by typing the code shown in lines 57–63 in Program 4-3.

Step 5: Switch your view back to the *Designer* and double-click the `exitButton` control. In the code editor, you see an empty event handler named `exitButton_Click`. Complete the `exitButton_Click` event handler by typing the code shown in lines 68–69 in Program 4-3.

Step 6: Save the project and run the application. First, enter 45000 for the salary and 1 for the years at current job. Click the *Check Qualifications* button, and the application should display the message "Minimum years at current job not met."

Click the *Clear* button. Enter 35000 for the salary and 5 for the years at current job. Click the *Check Qualifications* button, and the application should display the message "Minimum salary requirement not met."

Click the *Clear* button. Enter 45000 for the salary and 5 for the years at current job. Click the *Check Qualifications* button, and the application should display the message "You qualify for the loan."

Continue to test the application as you wish. When you are finished, click the *Exit* button, and the form should close.

Program 4-3 Completed Form1 code for the *LoanQualifier* project

```
 1 namespace LoanQualifier
 2 {
 3   public partial class Form1 : Form
 4   {
 5     public Form1()
 6     {
 7       InitializeComponent();
 8     }
 9
10     private void checkButton_Click(object sender, EventArgs e)
11     {
12       try
13       {
14         // Names constants
15         const decimal MINIMUM_SALARY = 40000m;
16         const int MINIMUM_YEARS_ON_JOB = 2;
```

```
17
18          // Local variables
19          decimal salary;
20          int yearsOnJob;
21
22          // Get the salary and years on the job.
23          salary = decimal.Parse(salaryTextBox.Text);
24          yearsOnJob = int.Parse(yearsTextBox.Text);
25
26          // Determine whether the user qualifies.
27          if (salary >= MINIMUM_SALARY)
28          {
29            if (yearsOnJob >= MINIMUM_YEARS_ON_JOB)
30            {
31              // The user qualifies.
32              decisionLabel.Text = "You qualify for the loan.";
33            }
34            else
35            {
36              // The user does not qualify.
37              decisionLabel.Text = "Minimum years at current " +
38                "job not met.";
39            }
40          }
41          else
42          {
43            // The user does not qualify.
44            decisionLabel.Text = "Minimum salary
                requirement " +
45              "not met.";
46          }
47        }
48        catch (Exception ex)
49        {
50          // Display an error message.
51          MessageBox.Show(ex.Message);
52        }
53      }
54
55      private void clearButton_Click(object sender, EventArgs e)
56      {
57        // Clear the TextBoxes and the decisionLabel.
58        salaryTextBox.Text = "";
59        yearsTextBox.Text = "";
60        decisionLabel.Text = "";
61
62        // Reset the focus.
63        salaryTextBox.Focus();
64      }
65
66      private void exitButton_Click(object sender, EventArgs e)
67      {
68        // Close the form.
69        this.Close();
70      }
71    }
72  }
```

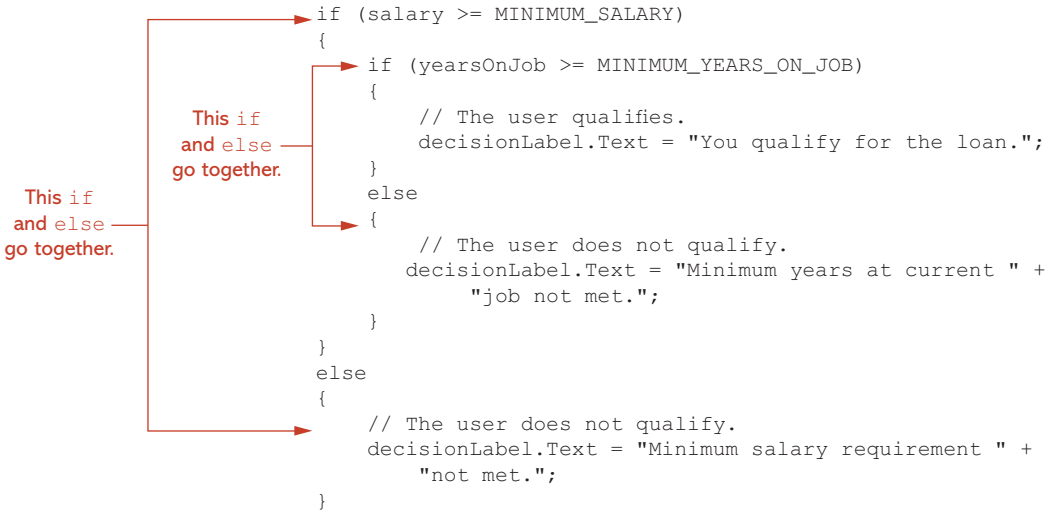
Indentation and Alignment in Nested Decision Structures

For debugging purposes, it's important to use proper alignment and indentation in a nested `if` statement. This makes it easier to see which actions are performed by each part of the structure. For example, the following code is functionally equivalent to lines 27–46 in Program 4-3. Although this code is logically correct, it would be very difficult to debug because it is not properly indented.

```
if (salary >= MINIMUM_SALARY)
{
if (yearsOnJob >= MINIMUM_YEARS_ON_JOB)
{
// The user qualifies.
decisionLabel.Text = "You qualify for the loan.";
}
else
{
// The user does not qualify.
decisionLabel.Text = "Minimum years at current " +
   "job not met.";
}
}
else
{
// The user does not qualify.
decisionLabel.Text = "Minimum salary requirement " +
   "not met.";
}
```

Fortunately, Visual Studio automatically indents and aligns the statements in a decision structure. Proper indentation and alignment make it easier to see which `if` and `else` clauses belong together, as shown in Figure 4-14.

Figure 4-14 Alignment of `if` and `else` clauses



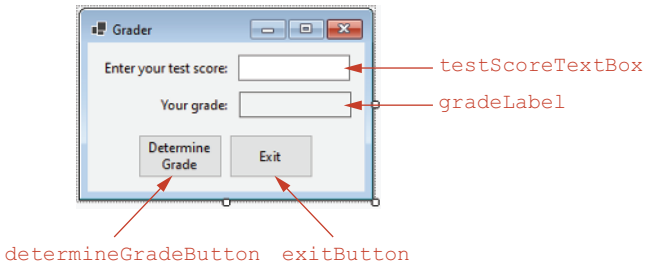
Testing a Series of Conditions

In Tutorial 4-3, you saw how a program can use nested decision structures to test more than one Boolean expression. It is not uncommon for a program to have a series of Boolean expressions to test and then perform an action, depending on which expression

is true. One way to accomplish this it to have a decision structure with numerous other decision structures nested inside it. For example, look at the *Grader* application in the *Chap04* folder of this book's Student Sample Programs.

Figure 4-15 shows the application's form, with the names of several controls. When you run the application, you enter a numeric test score into the `testScoreTexBox` control and click the `determineGradeButton` control; a grade is then displayed in the `gradeLabel` control.

Figure 4-15 The *Grader* application's form



The following 10-point grading scale is used to determine the grade:

Test Score	Grade
90 and above	A
80–89	B
70–79	C
60–69	D
Below 60	F

The logic of determining the grade can be expressed like this:

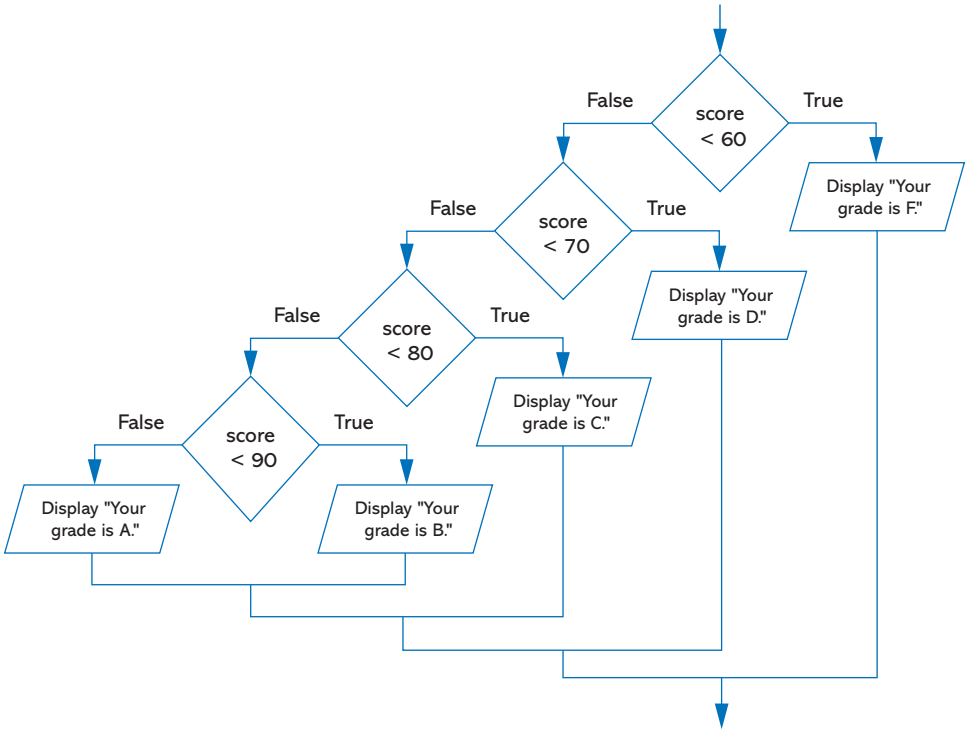
If the test score is less than 60, then the grade is "F."
Otherwise, if the test score is less than 70, then the grade is "D."
Otherwise, if the test score is less than 80, then the grade is "C."
Otherwise, if the test score is less than 90, then the grade is "B."
Otherwise, the grade is "A."

This logic requires several nested decision structures, as shown in the flowchart in Figure 4-16.

Open the code editor and look at the `determineGradeButton_Click` event handler, shown in the following code sample. The nested decision structure appears in lines 12–41.

```
 1 private void determineGradeButton_Click(object sender, EventArgs e)
 2 {
 3    try
 4    {
 5         // Variable to hold the test score.
 6         double testScore;
 7
 8         // Get the test score.
 9         testScore = double.Parse(testScoreTextBox.Text);
10
11         // Determine the grade.
12         if (testScore < 60)
13         {
14             gradeLabel.Text = "F";
15         }
```

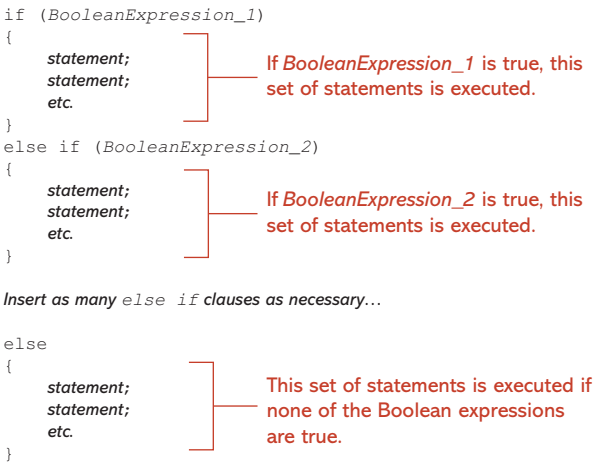
Figure 4-16 Nested decision structure to determine a grade




```
16          else
17          {
18              if (testScore < 70)
19              {
20                  gradeLabel.Text = "D";
21              }
22              else
23              {
24                  if (testScore < 80)
25                  {
26                      gradeLabel.Text = "C";
27                  }
28                  else
29                  {
30                      if (testScore < 90)
31                      {
32                          gradeLabel.Text = "B";
33                      }
34                      else
35                      {
36                          gradeLabel.Text = "A";
37                      }
38                  }
39              }
40          }
41      }
42      catch (Exception ex)
43      {
44          // Display an error message.
45          MessageBox.Show(ex.Message);
46      }
47  }
```

The `if-else-if` Statement

Even though the *Grader* application previously shown is a simple example, the logic of the nested decision structure is fairly complex. C# provides a special version of the decision structure known as the **if-else-if statement**, which makes this type of logic simpler to write. You write the `if-else-if` statement using the following general format:



When the statement executes, *BooleanExpression_1* is tested. If *BooleanExpression_1* is true, the set of statements that immediately follows is executed, and the rest of the structure is skipped. If *BooleanExpression_1* is false, however, the program jumps to the very next `else if` clause and tests *BooleanExpression_2*. If it is true, the set of statements that immediately follows is executed, and the rest of the structure is then skipped. This process continues until a Boolean expression is found to be true, or no more `else if` clauses are left. If none of the Boolean expressions are true, the set of statements following the final `else` clause is executed.

For example, look at the *Grader2* application in the *Chap04* folder of this book's Student Sample Programs. This application works just like the *Grader* application that was previously discussed. The user enters a numeric test score, and the application displays a grade. Its form is identical to the form shown in Figure 4-15. The *Grader2* application, however, uses an `if-else-if` statement to determine the grade instead of nested `if-else` statements. The *Grader2* application's `determineGradeButton_Click` event handler is shown here:

```
 1 private void determineGradeButton_Click(object sender, EventArgs e)
 2 {
 3     try
 4     {
 5         // Variable to hold the test score.
 6         double testScore;
 7
 8         // Get the test score.
 9         testScore = double.Parse(testScoreTextBox.Text);
10
11         // Determine the grade.
12         if (testScore < 60)
13         {
14             gradeLabel.Text = "F";
15         }
16         else if (testScore < 70)
17         {
```

```
18              gradeLabel.Text = "D";
19          }
20          else if (testScore < 80)
21          {
22              gradeLabel.Text = "C";
23          }
24          else if (testScore < 90)
25          {
26              gradeLabel.Text = "B";
27          }
28          else
29          {
30              gradeLabel.Text = "A";
31          }
32      }
33      catch (Exception ex)
34      {
35          // Display an error message.
36          MessageBox.Show(ex.Message);
37      }
38  }
```

Notice the alignment and indentation that is used with the `if-else-if` statement: The `if`, `else if`, and `else` clauses are all aligned, and the conditionally executed statements are indented.

You never have to use the `if-else-if` statement because its logic can be coded with nested `if-else` statements; however, a long series of nested `if-else` statements has two particular disadvantages when you are debugging code:

- The code can grow complex and become difficult to understand.
- Because indenting is important in nested statements, a long series of nested `if-else` statements can become too long to be displayed on the computer screen without horizontal scrolling. Also, long statements tend to wrap around when printed on paper, making the code even more difficult to read.

The logic of an `if-else-if` statement is usually easier to follow than a long series of nested `if-else` statements. And, because all the clauses are aligned in an `if-else-if` statement, the lengths of the lines in the statement tend to be shorter.

Checkpoint

4.11 Convert the following set of nested `if-else` statements to an `if-else if` statement:

```
if (number == 1)
{
   MessageBox.Show("One");
}
else
{
   if (number == 2)
   {
      MessageBox.Show("Two");
   }
 else
 {
      if (number == 3)
      {
         MessageBox.Show("Three");
      }
```

```
            else
            {
               MessageBox.Show("Unknown");
            }
         }
      }
```

4.4 Logical Operators

CONCEPT: **The logical AND operator (`&&`) and the logical OR operator (`||`) allow you to connect multiple Boolean expressions to create a compound expression. The logical NOT operator (`!`) reverses the truth of a Boolean expression.**

The C# language provides a set of operators known as **logical operators**, which you can use to create complex Boolean expressions. Table 4-3 describes these operators.

Table 4-3 Logical operators

Operator	Meaning
`&&`	This is the logical AND operator. It connects two Boolean expressions into one compound expression. Both subexpressions must be true for the compound expression to be true.
`||`	This is the logical OR operator. It connects two Boolean expressions into one compound expression. One or both subexpressions must be true for the compound expression to be true. It is necessary for only one of the subexpressions to be true, and it does not matter which.
`!`	This is the logical NOT operator. It is a unary operator, meaning it works with only one operand. The operand must be a Boolean expression. The not operator reverses the truth of its operand. If it is applied to an expression that is true, the operator returns false. If it is applied to an expression that is false, the operator returns true.

Table 4-4 shows examples of several compound Boolean expressions that use logical operators.

Table 4-4 Compound Boolean expressions using logical operators

Expression	Meaning
`x > y && a < b`	Is `x` greater than `y` *AND* is `a` less than `b`?
`x == y || x == z`	Is `x` equal to `y` *OR* is `x` equal to `z`?
`! (x > y)`	Is the expression `x > y` *NOT* true?

The && Operator

The **&& operator** is the **logical AND operator**. It takes two Boolean expressions as operands and creates a compound Boolean expression that is true only when both subexpressions are true. The following is an example of an `if` statement that uses the `&&` operator:

```
if (temperature < 20 && minutes > 12)
{
  MessageBox.Show("The temperature is in the danger zone.");
}
```

In this statement, the two Boolean expressions `temperature < 20` and `minutes > 12` are combined into a compound expression. The `MessageBox.Show` statement is executed only if `temperature` is less than 20 *and* `minutes` is greater than 12. If either of the Boolean subexpressions is false, the compound expression is false and the message is not displayed.

Table 4-5 shows a truth table for the `&&` operator. The truth table lists expressions showing all the possible combinations of true and false connected with the `&&` operator. The resulting values of the expressions are also shown.

Table 4-5 Truth table for the `AND` operator

Expression	Value of the Expression
`true && false`	`false`
`false && true`	`false`
`false && false`	`false`
`true && true`	`true`

As the table shows, both sides of the `&&` operator must be true for the operator to return a true value.

The || Operator

The **|| operator** is the **logical OR operator**. It takes two Boolean expressions as operands and creates a compound Boolean expression that is true when either of the subexpressions is true. The following is an example of an `if` statement that uses the `||` operator:

```
if (temperature < 20 || temperature > 100)
{
  MessageBox.Show("The temperature is in the danger zone.");
}
```

The `MessageBox.Show` statement executes only if `temperature` is less than 20 *or* `temperature` is greater than 100. If either subexpression is true, the compound expression is true. Table 4-6 shows a truth table for the `||` operator.

Table 4-6 Truth table for the || operator

Expression	Value of the Expression
`true || false`	`true`
`false || true`	`true`
`false || false`	`false`
`true || true`	`true`

All it takes for an || expression to be true is for one side of the || operator to be true. It doesn't matter if the other side is false or true.

Short-Circuit Evaluation

Both the && and || operators perform **short-circuit evaluation**. Here is how it works with the && operator: if the expression on the left side of the && operator is false, the expression on the right side is not checked. Because the compound expression is false if only one of the subexpressions is false, it would waste CPU time to check the remaining expression. So, when the && operator finds that the expression on its left is false, it short-circuits and does not evaluate the expression on its right.

Here's how short-circuit evaluation works with the || operator: if the expression on the left side of the || operator is true, the expression on the right side is not checked. Because it is necessary for only one of the expressions to be true, it would waste CPU time to check the remaining expression.

The ! Operator

The ! **operator** is the **logical NOT operator**. It is a unary operator that takes a Boolean expression as its operand and reverses its logical value. In other words, if the expression is true, the ! operator returns false, and if the expression is false, the ! operator returns true. The following is an if statement using the NOT operator:

```
if ( !(temperature > 100) )
{
    MessageBox.Show("This is below the maximum temperature.");
}
```

First, the expression (temperature > 100) is tested and a value of either true or false is the result. Then the ! operator is applied to that value. If the expression (temperature > 100) is true, the ! operator returns false. If the expression (temperature > 100) is false, the ! operator returns true. The previous code is equivalent to asking "Is the temperature not greater than 100?"

Notice that in this example, we have put parentheses around the expression temperature > 100. This is necessary because the ! operator has higher precedence than the relational operators. If we do not put the parentheses around the expression temperature > 100, the ! operator will be applied just to the temperature variable.

Table 4-7 shows a truth table for the ! operator.

Table 4-7 Truth table for the ! operator

Expression	Value of the Expression
! true	false
! false	true

Precedence of the Logical Operators

We mentioned earlier that the ! operator has higher precedence than the relational operators. The && and || logical operators have lower precedence than the relational operators. For example, look at the following expression:

```
creditScore > 700 || accountBalance > 9000
```

When this expression is evaluated, the > operators work first, and then the || operator works. The expression is the same as the following:

```
(creditScore > 700) || (accountBalance > 9000)
```

Many programmers choose to enclose the expressions that are to the left and the right of a logical operator in parentheses, as shown here. Even though the parentheses are not required in many situations, using them makes the compound expression easier to understand.

Checking Numeric Ranges with Logical Operators

Sometimes you need to write code that determines whether a numeric value is within a specific range of values or outside a specific range of values. When determining whether a number is inside a range, it is best to use the && operator. For example, the following if statement checks the value in x to determine whether it is in the range of 20 through 40:

```
if (x > 20 && x < 40)
{
    MessageBox.Show("The value is in the acceptable range.");
}
```

The compound Boolean expression being tested by this statement is true only when x is greater than 20 *and* less than 40. The value in x must be between the values of 20 and 40 for this compound expression to be true.

When determining whether a number is outside a range, it is best to use the || operator. The following statement determines whether x is outside the range of 20 through 40:

```
if (x < 20 || x > 40)
{
    MessageBox.Show("The value is outside the acceptable range.");
}
```

It is important not to get the logic of the logical operators confused when testing for a range of numbers. For example, the compound Boolean expression in the following code would never test true:

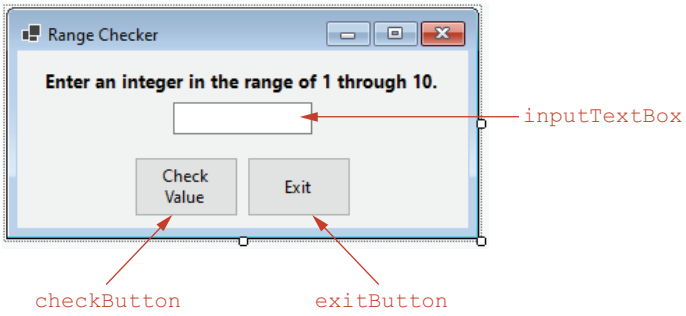
```
// This is an error!
if (x < 20 && x > 40)
{
    MessageBox.Show("The value is outside the acceptable range.");
}
```

Obviously, x cannot be less than 20 and at the same time be greater than 40.

Let's look at an example application that checks the range of a value entered by the user. Open the *RangeChecker* project in the *Chap04* folder of this book's Student Sample Programs. Figure 4-17 shows the application's form, along with the names of some of the controls. When you run the application, you enter an integer into the inputTexBox control and click the checkButton control. If you enter a number in the range of 1 through 10, a message box appears letting you know that the number is acceptable. Otherwise, a message box appears letting you know that the number is *not* acceptable.

The following code sample shows the checkButton_Click event handler. Line 7 declares an int variable named number, initialized with the value that has been entered into the inputTextBox control. The if statement that begins in line 10 determines whether number is greater than or equal to 1 *AND* number is less than or equal to 10. If the Boolean expression is true, the statement in line 12 executes. Otherwise, the statement in line 16 executes.

Figure 4-17 The *RangeChecker* project's form




```
1  private void checkButton_Click(object sender, EventArgs e)
2  {
3      try
4      {
5          // Declare a variable and initialize it with
6          // the user's input.
7          int number = int.Parse(inputTextBox.Text);
8
9          // Check the number's range.
10         if (number >= 1 && number <= 10)
11         {
12             MessageBox.Show("That number is acceptable.");
13         }
14         else
15         {
16             MessageBox.Show("That number is NOT acceptable.");
17         }
18     }
19     catch (Exception ex)
20     {
21         // Display an error message.
22         MessageBox.Show(ex.Message);
23     }
24 }
```

Checkpoint

4.12 What is a compound Boolean expression?

4.13 The following truth table shows various combinations of the values `true` and `false` connected by a logical operator. Complete the table by circling T or F to indicate whether the result of such a combination is `true` or `false`.

Logical Expression	Result (circle T or F)	
`true && false`	T	F
`true && true`	T	F
`false && true`	T	F
`false && false`	T	F
`true || false`	T	F
`true || true`	T	F
`false || true`	T	F
`false || false`	T	F
`! true`	T	F
`! false`	T	F

4.14 Assume the variables `a` = 2, `b` = 4, and `c` = 6. Circle T or F for each of the following conditions to indicate if it is true or false.

```
a == 4 || b > 2          T    F
6 <= c && a > 3          T    F
1 != b && c != 3         T    F
a >= -1 || a <= b        T    F
!(a > 2)                 T    F
```

4.15 Explain how short-circuit evaluation works with the `&&` and `||` operators.

4.16 Write an `if` statement that displays the message "The number is valid" in a message box if the variable `speed` is within the range 0 through 200.

4.17 Write an `if` statement that displays the message "The number is not valid" in a message box if the variable `speed` is outside the range 0 through 200.

4.5 `bool` Variables and Flags

CONCEPT: You can store the values `true` and `false` in `bool` variables, which are commonly used as flags.

The C# language provides the **`bool` data type** that you can use to create variables that hold `true` or `false` values. Here is an example of the declaration of a `bool` variable:

```
bool grandMaster;
```

This declares a `bool` variable named `grandMaster`. In the program, we can assign the values `true` or `false` to the variable, as shown here:

```
if (points > 5000)
{
    grandMaster = true;
}
else
{
    grandMaster = false;
}
```

Variables of the `bool` data type are commonly used as flags. A **flag** is a variable that signals when some condition exists in the program. When the flag variable is set to `false`, it indicates that the condition does not yet exist. When the flag variable is set to `true`, it means the condition does exist. For example, the previous code might be used in a game to determine whether the user is a "grand master." If he or she has earned more than 5,000 points, we set the `grandMaster` variable to `true`. Otherwise, we set the variable to `false`. Later in the program we can test the `grandMaster` variable, like this:

```
if (grandMaster)
{
    powerLevel += 500;
}
```

This code performs the following: if `grandMaster` is `true`, add 500 to `powerLevel`. Here is another example:

```
if (!grandMaster)
{
    powerLevel = 100;
}
```

This code performs the following: if `grandMaster` is not `true`, set `powerLevel` to 100.

Checkpoint

4.18 What special values can you store in a `bool` variable?

4.19 What is a flag variable?

4.6 Comparing Strings

CONCEPT: You can use certain relational operators and methods to compare strings.

You can use the == operator to compare two strings. For example, look at the following code sample:

```
string name1 = "Mary";
string name2 = "Mark";
if (name1 == name2)
{
    MessageBox.Show("The names are the same.");
}
else
{
    MessageBox.Show("The names are NOT the same.");
}
```

The == operator compares `name1` and `name2` to determine whether they are equal. Because the strings "Mary" and "Mark" are not equal, the `else` clause displays the message "The names are NOT the same."

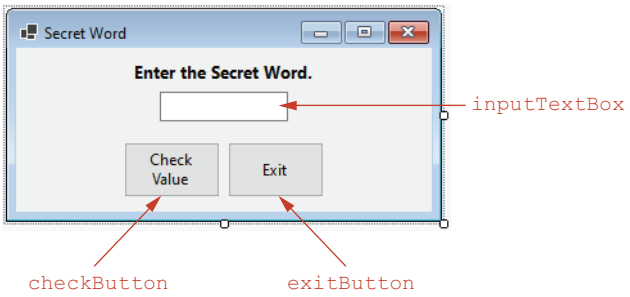
You can compare `string` variables with string literals as well. Assume `month` is a `string` variable. The following code sample uses the `!=` operator to determine whether `month` is not equal to `"October"`.

```
if (month != "October")
{
    statement;
}
```

Look at the *SecretWord* project in the *Chap04* folder of this book's Student Sample Programs. Figure 4-18 shows the application's form, with the names of some of the controls. The form prompts you to enter the secret word into the `inputTexBox` control. When you click the `checkButton` control, the application compares the string that you entered to `"Ariel."`

The following code sample shows the `checkButton_Click` event handler. Line 5 declares a `string` variable named `secretWord`, initialized with the value that has been entered into the `inputTextBox` control. The `if` statement that begins in line 8 compares the `secretWord` variable to the string literal `"Ariel"`. If the two are equal, the statement in line 10 executes. Otherwise, the statement in line 14 executes.

Figure 4-18 The *SecretWord* project's form




```
1 private void checkButton_Click(object sender, EventArgs e)
2 {
3     // Declare a string variable and initialize it with
4     // the user's input.
5     string secretWord = inputTextBox.Text;
6
7     // Did the user enter the correct secret word?
8     if (secretWord == "Ariel")
9     {
10        MessageBox.Show("That is the correct secret word.");
11    }
12    else
13    {
14        MessageBox.Show("Sorry, that is NOT the secret word.");
15    }
16 }
```

Other String Comparisons

In addition to determining whether strings are equal or not equal, you can use the `String.Compare` method to determine whether one string is greater than or less than another string. This is a useful capability because sometimes you need to sort strings in some order. Before we look at how the method works, we should review how characters are stored in memory.

Recall from Chapter 1 that computers do not actually store characters, such as *A*, *B*, *C*, and so forth, in memory. Instead, they store numeric codes that represent the characters. We mentioned in Chapter 1 that C# uses Unicode to represent characters. Here are some facts about the Unicode system:

- The uppercase characters *A* through *Z* are represented by the numbers 65 through 90.
- The lowercase characters *a* through *z* are represented by the numbers 97 through 122.
- When the digits 0 through 9 are stored in memory as characters, they are represented by the numeric codes 48 through 57. (For example, the string "abc123" is stored in memory as the codes 97, 98, 99, 49, 50, and 51.)
- A blank space is represented by the number 32.

In addition to establishing a set of numeric codes to represent characters in memory, Unicode also establishes an order for characters. The character *A* comes before the character *B*, which comes before the character *C*, and so forth.

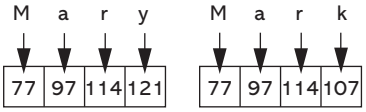
When a program compares characters, it actually compares the codes for the characters. The character *A* would be considered less than the character *B* because the character *A*'s numeric code is less than the character *B*'s numeric code.

Let's look at how strings containing more than one character are compared. Suppose we have the strings "Mary" and "Mark" stored in memory, as follows:

```
string name1 = "Mary";
string name2 = "Mark";
```

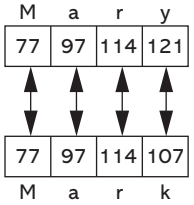
Figure 4-19 shows how the strings "Mary" and "Mark" are stored in memory using character codes.

Figure 4-19 Character codes for the strings "Mary" and "Mark"



When you compare these strings in a program, they are compared character by character, beginning with the first, or leftmost, characters. This is shown in Figure 4-20.

Figure 4-20 Comparing each character in a string



Here is how the comparison takes place:

1. The *M* in "Mary" is compared with the *M* in "Mark." These are the same, so the next characters are compared.
2. The *a* in "Mary" is compared with the *a* in "Mark." Because these are the same, the next characters are compared.
3. The *r* in "Mary" is compared with the *r* in "Mark." These are the same, so the next characters are compared.
4. The *y* in "Mary" is compared with the *k* in "Mark." Because these are not the same, the two strings are not equal. The character *y* has a higher character code (121) than *k* (107), so it is determined that the string "Mary" is greater than the string "Mark."

If one of the strings in a comparison is shorter than the other, only the corresponding characters are compared. If the corresponding characters are identical, then the shorter string is considered less than the longer string. For example, suppose the strings "High" and "Hi" are compared. The string "Hi" is considered less than "High" because it is shorter.

In C# you cannot use relational operators to determine whether one string is greater than or less than another string. Instead, you use the `String.Compare` method. You use the following general format to call the method:

```
String.Compare(string1, string2)
```

In the general format, *`string1`* and *`string2`* are the strings that are being compared. The method returns an integer value indicating the result of the comparison. The integer value will be one of the following:

- Greater than zero if *`string1`* is greater than *`string2`*.
- Zero if *`string1`* is equal to *`string2`*.
- Less than zero if *`string1`* is less than *`string2`*.

Here is a code sample that uses the method to display two names in alphabetical order:

```
1 string str1 = "Joe";
2 string str2 = "Kerry";
3
4 if (String.Compare(str1, str2) < 0)
5 {
6     MessageBox.Show(str1 + " " + str2);
7 }
8 else
9 {
10    MessageBox.Show(str2 + " " + str1);
11 }
```

The `if` statement in line 4 calls the `String.Compare` method, passing `str1` and `str2` as arguments. If we execute this code, the method will return a value that is less than 0 because the string "Joe" is less than the string "Kerry". As a result, the statement in line 6 will display *Joe Kerry*.

The `String.Compare` method performs a case sensitive comparison, which means that uppercase characters are not considered the same as their lowercase counterparts. For example, the strings "Joe" and "joe" are not equal because the case of the first character is different in each. You can pass the Boolean value `true` as an optional third argument to the `String.Compare` method if you want it to perform a case insensitive comparison. Here is an example:

```
1 string str1 = "JOE";
2 string str2 = "joe";
3
4 if (String.Compare(str1, str2, true) == 0)
5 {
6     MessageBox.Show(str1 + " and " + str2 + " are equal.");
7 }
8 else
9 {
10    MessageBox.Show(str1 + " and " + str2 + " are NOT equal.");
11 }
```

Notice that the `if` statement in line 4 passes true as the third argument to the `String.Compare` method. This specifies that we want a case insensitive comparison. As a result, the method will return 0 and the statement in line 6 will display *JOE and joe are equal*.

Checkpoint

4.20 If the following code were part of a complete program, what would it display?

```
if (String.Compare("z", "a") < 0)
{
    MessageBox.Show("z is less than a.");
}
else
{
    MessageBox.Show("z is not less than a.");
}
```

4.21 If the following code were part of a complete program, what would it display?

```
string s1 = "New York";
string s2 = "Boston";
if (String.Compare(s1, s2) > 0)
{
    MessageBox.Show(s2);
```

```
        MessageBox.Show(s1);
    }
    else
    {
        MessageBox.Show(s1);
        MessageBox.Show(s2);
    }
```

4.7 Preventing Data Conversion Exceptions with the TryParse Methods

CONCEPT: **Exceptions should be prevented when possible. You can use the `TryParse` methods to prevent exceptions as a result of the user entering invalid data.**

In Chapter 3, you learned that the `Parse` methods throw an exception when you try to use them to convert nonnumeric data to a numeric data type. If you use one of the `Parse` methods to convert a TextBox control's Text property to a number, there is always the possibility of an exception being thrown. After all, the user is free to enter anything he or she wants into a TextBox control. To handle the exceptions that are caused by the `Parse` methods, we have been using the `try-catch` statement.

Although many exceptions happen for reasons that the programmer cannot anticipate (such as a system malfunction), some exceptions are predictable. For example, you know that using a `Parse` method to convert nonnumeric input to a numeric data type will throw an exception. In situations like that, where an exception is predictable, you should write your code to prevent the exception. It is a better programming practice to prevent an exception instead of allowing it to happen and then letting a `try-catch` statement react to it. You should use `try-catch` statements primarily for those exceptions that are beyond your control.

NOTE: Until now, we've simply been allowing exceptions to happen and letting a `try-catch` statement respond to them. After reading the previous paragraph, you might be wondering why we haven't been preventing exceptions all along. The reason is that you need to know how to write `if` statements to perform the techniques that we discuss in this section. Now that you know how to write `if` statements, you can add more sophistication to your code.

Now that you know how to write `if` statements, you can use a family of .NET methods known as the `TryParse` methods. With the **`TryParse` methods**, you can determine whether a string (such as a control's Text property) contains a value that can be converted to a specific data type *before* it is converted to that data type. The `TryParse` methods do not throw an exception, so you can use them without a `try-catch` statement.

There are several `TryParse` methods in .NET. For now, we are using the `int`, `double`, and `decimal` numeric data types, so we will discuss three of them:

- We use the `int.TryParse` method to convert a string to an `int`.
- We use the `double.TryParse` method to convert a string to a `double`.
- We use the `decimal.TryParse` method to convert a string to a `decimal`.

When you call one of the `TryParse` methods, you pass two arguments: (1) the string that you want to convert, and (2) the name of the variable in which you want to store the

converted value. First, let's look at the `int.TryParse` method. Here is the general format of how the `int.TryParse` method is called:

```
int.TryParse(string, out targetVariable)
```

In the general format, *`string`* is the string that you want to convert, and *`targetVariable`* is the name of an `int` variable. The method tries to convert the *`string`* argument to an `int`. If the conversion is successful, the converted value is stored in the *`targetVariable`*, and the method returns the Boolean value `true` to indicate that the conversion was successful. If the conversion is not successful, the method does not throw an exception. Instead, it stores 0 in the *`targetVariable`* and returns the Boolean value `false` to indicate that the *`string`* could not be converted.

Look carefully at the general format and notice that the word out appears before the *`targetVariable`*. The **`out` keyword** is required, and it specifies that the *`targetVariable`* is an output variable. An **output variable** is a variable that is passed as an argument to a method, and when the method is finished, a value is stored in the variable.

Because the `TryParse` methods return either `true` or `false`, they are commonly called as the Boolean expression in an `if` statement. The following code shows an example using the `int.TryParse` method. In the example, assume that `inputTextBox` is the name of a TextBox control.

```
 1 int number;
 2
 3 if (int.TryParse(inputTextBox.Text, out number))
 4 {
 5     MessageBox.Show("Success!");
 6 }
 7 else
 8 {
 9     MessageBox.Show("Enter a valid integer.");
10 }
```

The purpose of this code sample is to convert the value of the `inputTextBox` control's Text property to an `int` and assign that value to the `number` variable. In line 3, the `if` statement calls the `int.TryParse` method, passing `inputTextBox.Text` as argument 1 and `number` as argument 2. Here's what happens:

- If `inputTextBox.Text` is successfully converted to an `int`, the resulting value is assigned to the `number` variable, and the method returns `true`. That causes the statement in line 5 to execute.
- If `inputTextBox.Text` cannot be converted to an `int`, the value 0 is assigned to the `number` variable, and the method returns `false`. That causes the statement in line 9 (after the `else` clause) to execute.

The other `TryParse` methods work in a similar manner. Here is the general format of how the `double.TryParse` method is called:

```
double.TryParse(string, out targetVariable)
```

In the general format, `string` is the string that you want to convert, and *`targetVariable`* is the name of a `double` variable. If the *`string`* can be converted to a `double`, its value is stored in the *`targetVariable`*, and the method returns the Boolean value `true` to indicate that the conversion was successful. If the conversion was not successful, the method stores 0 in the *`targetVariable`* and returns the Boolean value `false` to indicate that the *`string`* could not be converted.

The following code shows an example using the `double.TryParse` method. In the example, assume that `inputTextBox` is the name of a TextBox control.

```
double number;

if (double.TryParse(inputTextBox.Text, out number))
{
    MessageBox.Show("Success!");
}
else
{
    MessageBox.Show("Enter a valid double.");
}
```

Here is the general format of how the `decimal.TryParse` method is called:

```
decimal.TryParse(string, out targetVariable)
```

In the general format, *`string`* is the string that you want to convert, and *`targetVariable`* is the name of a `decimal` variable. If the *`string`* can be converted to a `decimal`, its value is stored in the *`targetVariable`*, and the method returns the Boolean value `true` to indicate that the conversion was successful. If the conversion was not successful, the method stores 0 in the *`targetVariable`* and returns the Boolean value `false` to indicate that the *`string`* could not be converted.

The following code shows an example using the `decimal.TryParse` method. In the example, assume that `inputTextBox` is the name of a TextBox control.

```
decimal number;

if (decimal.TryParse(inputTextBox.Text, out number))
{
    MessageBox.Show("Success!");
}
else
{
    MessageBox.Show("Enter a valid decimal.");
}
```

Validating the Data in Multiple TextBoxes

If a form has multiple TextBoxes, then the user has multiple opportunities to enter an invalid piece of data. A well-designed program should validate the contents of each TextBox individually. When a piece of invalid data is found, the program should display an error message that tells the user specifically which TextBox contains the bad input.

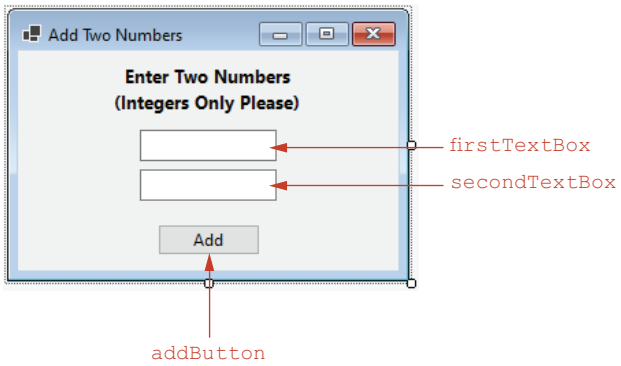
This technique requires a set of nested `if` statements. For example, suppose a form has two TextBoxes. The following pseudocode shows the logic for validating each TextBox. (In the pseudocode, a set of dotted lines connects each If statement with its corresponding Else clause and its ending.)

```
If the data in the first TextBox is good, then
    If the data in the second TextBox is good, then
        Process the data in both TextBoxes
    Else
        Display an error message about the second TextBox
    End if
Else
    Display an error message about the first TextBox
End if
```

Let's see how that logic looks in actual C# code. In the *Chap04* folder of this book's Student Sample Programs, you will find a project named *AddTwoNumbers*. The application's form is shown in Figure 4-21. When you run the application, enter an integer into

each of the TextBox controls and then click the *Add* button. A message box will appear showing the sum of the two numbers. If you enter anything other than an integer into either TextBox, an error message will appear telling you which TextBox contains the invalid data.

Figure 4-21 The *AddTwoNumbers* project's form



Here is the code for the `addButton_Click` event handler:

```
1  private void addButton_Click(object sender, EventArgs e)
2  {
3      // Local variables
4      int first, second, sum;
5
6      if (int.TryParse(firstTextBox.Text, out first))
7      {
8          if (int.TryParse(secondTextBox.Text, out second))
9          {
10             // Add the two numbers and display the sum.
11             sum = first + second;
12             MessageBox.Show(sum.ToString());
13         }
14         else
15         {
16             // Display an error message about the second TextBox.
17             MessageBox.Show("The second TextBox contains invalid
               data.");
18         }
19     }
20     else
21     {
22         // Display an error message about the first TextBox.
23         MessageBox.Show("The first TextBox contains invalid data.");
24     }
25 }
```

Let's take a closer look:

- Line 4 declares three `int` variables: `first`, `second`, and `sum`.
- The `if` statement in line 6 tries to convert `firstTextBox.Text` to an `int`. If the conversion is successful, the result is stored in the `first` variable, and the program continues executing at line 8. If the conversion is not successful, the program jumps to the `else` clause in line 20, and line 23 displays an error message regarding the first TextBox control.
- The `if` statement in line 8 tries to convert `secondTextBox.Text` to an `int`. If the conversion is successful, the result is stored in the `second` variable, and the program continues executing at line 10. If the conversion is not successful, the program

jumps to the `else` clause in line 14, and line 17 displays an error message regarding the second TextBox control.

- The statements in lines 11 and 12 execute only if both TextBox controls contain valid integer values. These statements add the `first` and `second` variables and display their sum.

If you need to validate three TextBox controls, you will write a set of three nested `if` statements. Here's the pseudocode:

```
If the data in the first TextBox is good, then
    If the data in the second TextBox is good, then
        If the data in the third TextBox is good, then
            Process the data in all three TextBoxes
        Else
            Display an error message about the third TextBox
        End if
    Else
        Display an error message about the second TextBox
    End if
Else
    Display an error message about the first TextBox
End if
```

In Tutorial 4-4, you will complete an application that uses the `TryParse` methods to validate data entered into two TextBox controls.

Tutorial 4-4:
Calculating Fuel Economy

VideoNote
Tutorial 4-4: Calculating Fuel Economy

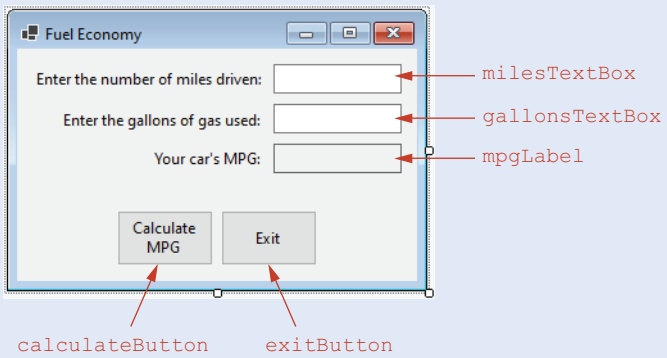
In Tutorial 3-2, you created an application that calculates a car's fuel economy in miles per gallon (MPG). Recall that the application lets the user enter the number of miles he or she has driven and the gallons of gas used. The application calculates and displays the car's MPG. In this tutorial, you will create a new version of the application that validates the data entered by the user.

To save time, the project has already been started for you, and the application's form has already been created. To complete the project, follow the steps in this tutorial.

Step 1: Start Visual Studio. Open the project named *FuelEconomyWithTryParse* in the *Chap04* folder of this book's Student Sample Programs.

Step 2: Open the Form1 form in the *Designer*. The form is shown, along with the names of the important controls, in Figure 4-22.

Figure 4-22 The *FuelEconomyWithTryParse* project's form



Step 3: Now, you will create the Click event handlers for the Button controls. At the end of this tutorial, Program 4-4 shows the completed code for the form. You will be instructed to refer to Program 4-4 as you write the event handlers.

In the *Designer*, double-click the `calculateButton` control. This will open the code editor, and you see an empty event handler named `calculateButton_Click`. Complete the `calculateButton_Click` event handler by typing the code shown in lines 12–38 in Program 4-4.

Let's take a closer look at the code:

Lines 12–14: These lines declare the `double` variables `miles`, `gallons`, and `mpg`. The variables hold the miles driven, the gallons of gas used, and the MPG, respectively.

Line 17: This `if` statement tries to convert `milesTextBox.Text` to a `double`. If the conversion is successful, the result is stored in the `miles` variable, and the program continues executing at line 19. If the conversion is not successful, the program jumps to the `else` clause in line 34, and line 37 displays the error message "Invalid input for miles."

Line 20: This `if` statement tries to convert `gallonsTextBox.Text` to a `double`. If the conversion is successful, the result is stored in the `gallons` variable, and the program continues executing at line 22. If the conversion is not successful, the program jumps to the `else` clause in line 28, and line 31 displays the error message "Invalid input for gallons."

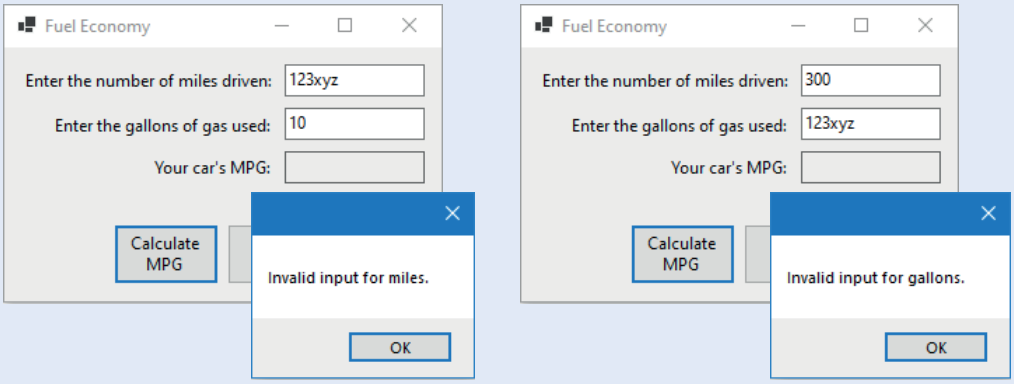
Lines 22–26: These lines are executed only if both the `milesTextBox` and `gallonsTextBox` contain valid data. Line 23 calculates MPG and assigns the result to the `mpg` variable, and line 26 displays the value of the `mpg` variable in the `mpgLabel` control.

Step 4: Switch your view back to the *Designer* and double-click the `exitButton` control. In the code editor, you see an empty event handler named `exitButton_Click`. Complete the `exitButton_Click` event handler by typing the code shown in lines 43–44 in Program 4-4.

Step 5: Save the project and run the application. First, enter 300 for the miles and 10 for the gallons. Click the *Calculate MPG* button, and the application should display 30.0 as the MPG.

Now, change the miles to an invalid entry, such as *123xyz*, and click the *Calculate MPG* button. The message "Invalid input for miles." should appear in a message box, as shown on the left in Figure 4-23.

Figure 4-23 Invalid input entered and the resulting error messages



Now change the miles back to 300, change the gallons to an invalid entry, such as *123xyz*, and click the *Calculate MPG* button. The message *"Invalid input for gallons."* should appear in a message box, as shown on the right in Figure 4-23.

Continue to test the application as you wish. When you are finished, click the *Exit* button, and the form should close.

Program 4-4 Completed Form1 code for the *FuelEconomyWithTryParse* project

```
 1 namespace FuelEconomyWithTryParse
 2 {
 3   public partial class Form1 : Form
 4   {
 5     public Form1()
 6     {
 7       InitializeComponent();
 8     }
 9
10     private void calculateButton_Click(object sender, EventArgs e)
11     {
12       double miles;   // To hold miles driven
13       double gallons; // To hold gallons used
14       double mpg;     // To hold MPG
15
16       // Validate the milesTextBox control.
17       if (double.TryParse(milesTextBox.Text, out miles))
18       {
19         // Validate the gallonsTextBox control.
20         if (double.TryParse(gallonsTextBox.Text, out gallons))
21         {
22           // Calculate MPG.
23           mpg = miles / gallons;
24
25           // Display the MPG in the mpgLabel control.
26           mpgLabel.Text = mpg.ToString("n1");
27         }
28         else
29         {
30           // Display an error message for gallonsTextBox.
31           MessageBox.Show("Invalid input for gallons.");
32         }
33       }
34       else
35       {
36         // Display an error message for milesTextBox.
37         MessageBox.Show("Invalid input for miles.");
38       }
39     }
40
41     private void exitButton_Click(object sender, EventArgs e)
42     {
43       // Close the form.
44       this.Close();
45     }
46   }
47 }
```

Checkpoint

4.22 What value does a `TryParse` method return if the string argument is successfully converted? What value does it return if the string is not converted?

4.23 If a `TryParse` method successfully converts the string argument, where is the result stored?

4.24 If a `TryParse` method cannot convert the string argument, what is stored in the second argument?

4.25 What does the keyword `out` mean when it is written before an argument to a method call?

4.8 Input Validation

CONCEPT: Input validation is the process of inspecting data that has been input to a program to make sure it is valid before it is used in a computation.

In the previous section, you learned about using the `TryParse` methods to validate the type of data entered by the user. You should also validate the accuracy of the data that is entered by the user. One of the most famous sayings among computer programmers is "garbage in, garbage out." This saying, sometimes abbreviated as *GIGO*, refers to the fact that computers cannot tell the difference between good input and bad input. If a user provides bad data as input to a program, the program will process that bad data and, as a result, will produce bad data as output.

For example, consider a payroll program that accepts the number of hours that an employee has worked in a given week as input. If the payroll clerk accidentally enters 400 hours instead of 40 hours, an unusually large check will be written because there are less than 400 hours in a week! The computer, however, is unaware of this fact, and unless the program is written to catch such errors, it will process the bad data just as if it were good data.

Sometimes stories are reported in the news about computer errors that mistakenly cause people to be charged thousands of dollars for small purchases or to receive large tax refunds to which they were not entitled. These "computer errors" are rarely caused by a computer; however, they are more commonly caused by software bugs or bad data that was read into a program as input.

The integrity of a program's output is only as good as the integrity of its input. For this reason, you should write your programs in such a way that bad input is never accepted. When input is given to a program, it should be inspected before it is processed. If the input is invalid, the program should discard it and prompt the user to enter the correct data. This process is known as **input validation**.

For example, in a payroll program we might validate the number of hours worked like this:

```
1 if (int.TryParse(hoursWorkedTextBox.Text, out hours))
2 {
3     if (hours > 0 && hours <= 168)
4     {
5         // Continue to process the input.
6     }
7     else
```

```
 8      {
 9          MessageBox.Show("Invalid number of hours entered.");
10      }
11  }
12  else
13  {
14      MessageBox.Show("The hours worked must be an integer.");
15  }
```

Let's assume the application uses a TextBox named `hoursWorkedTextBox` to get the hours worked. Also assume that the variable `hours` has already been declared as an `int`. The outer `if` statement (line 1) uses the `int.TryParse` method to make sure the user has entered an integer. If so, the value is stored in the `hours` variable and the program continues to the inner `if` statement (line 3). The inner `if` statement ensures that we process the input only if `hours` is greater than 0 *and* `hours` is less than or equal to 168. This is because we cannot write a paycheck for 0 hours worked, and 168 is the maximum number of hours in a week.

Let's look at another example. The following code comes from an application that gets a test score as input. A valid test score is an integer in the range of 0 through 100.

```
 1  if (int.TryParse(testScoreTextBox.Text, out testScore))
 2  {
 3      if (testScore >= 0 && testScore <= 100)
 4      {
 5          // Continue to process the input.
 6      }
 7      else
 8      {
 9          MessageBox.Show("Test score must be in the range 0 - 100.");
10      }
11  }
12  else
13  {
14      MessageBox.Show("The test score must be an integer.");
15  }
```

Let's assume the application uses a TextBox named `testScoreTextBox` to get the test score. Also assume that the variable `testScore` has already been declared as an `int`. The outer `if` statement (line 1) uses the `int.TryParse` method to make sure the user has entered an integer. If so, the value is stored in the `testScore` variable and the program continues to the inner `if` statement (line 3). The inner `if` statement ensures that we process the input only if `testScore` is greater than or equal to 0 *and* `testScore` is less than or equal to 100.

4.9 Radio Buttons and Check Boxes

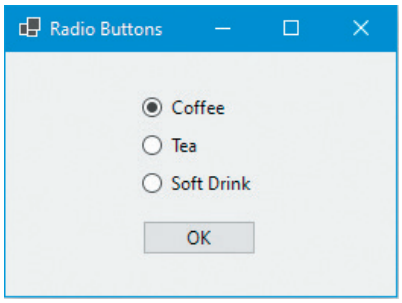
CONCEPT: GUIs commonly use radio buttons and check boxes to let the user select items.

Radio Buttons

Radio buttons are useful when you want the user to select one choice from several possible choices. Figure 4-24 shows a form, at run time, with a group of three radio buttons. The radio buttons in the figure allow the user to select *Coffee*, *Tea*, or *Soft Drink*.

A radio button may be either selected or deselected. Each radio button has a small circle that appears filled in when the radio button is selected, and appears empty when the radio button is deselected. In Figure 4-24, the *Coffee* radio button is selected and the other radio buttons are deselected.

Figure 4-24 Radio buttons



At run time, only one radio button in a group may be selected at a time. Clicking on a radio button selects it, and automatically deselects any other radio button in the same group. We call this **mutually exclusive selection**.

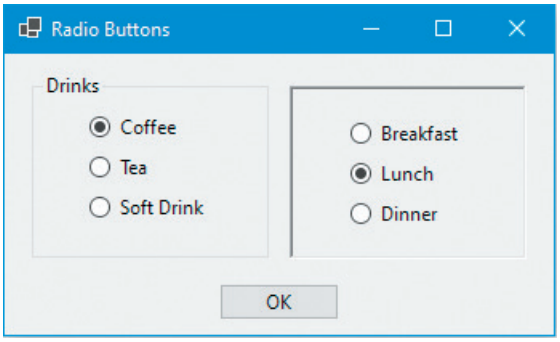
NOTE: The name *radio button* refers to the old car radios that had push buttons for selecting stations. Only one button could be pushed in at a time. When you pushed a button, it automatically popped out the currently selected button.

When you want to create a group of radio buttons on a form, you use the **RadioButton control**, which is found in the *Common Controls* section of the *Toolbox*. RadioButton controls are normally grouped in one of the following ways:

- You place them inside a GroupBox control. All RadioButton controls that are inside a GroupBox are members of the same group.
- You place them inside a Panel control. All RadioButton controls that are inside a Panel are members of the same group.
- You place them on a form but not inside a GroupBox or a Panel. All RadioButton controls that are on a form but not inside a GroupBox or Panel are members of the same group.

Figure 4-25 shows a form, at run time, with two groups of RadioButton controls. The group on the left is inside a GroupBox control, and the group on the right is inside a Panel control. When the application runs, the user will be able to select only one RadioButton from each group. In the figure, *Coffee* is selected in the left group and *Lunch* is selected in the right group.

Figure 4-25 A form with two groups of RadioButton controls



The RadioButton Control's Text Property

RadioButton controls have a Text property, which holds the text that is displayed next to the radio button's circle. For example, the radio buttons shown in Figure 4-24 have their Text properties set to *Coffee*, *Tea*, and *Soft Drink*.

The RadioButton Control's Checked Property

RadioButton controls have a Checked property that determines whether the control is selected or deselected. The Checked property is a Boolean property, which means that it may be set to either True or False. When the Checked property is set to True, the RadioButton is selected, and when the Checked property is set to False, the RadioButton is deselected. By default, the Checked property is set to False.

You can use the *Properties* window to set the initial value of a RadioButton control's Checked property. Keep in mind that the Checked property of only one RadioButton in a group can be set to True at a given time. When you set a RadioButton control's Checked property to True in the *Properties* window, the Checked properties of all the other RadioButtons in the same group automatically are set to False.

TIP: When you create a group of RadioButton controls, you should always set one of the control's Checked property to True in the *Properties* window. If all the RadioButtons in a group have their Checked property set to False, then the Checked property of the RadioButton with the lowest TabIndex value will automatically be set to True when you run the application.

Working with Radio Buttons in Code

In code, you can determine whether a RadioButton control is selected by testing its Checked property. For example, suppose a form has a RadioButton control named `choice1RadioButton`. The following `if` statement determines whether it is selected:

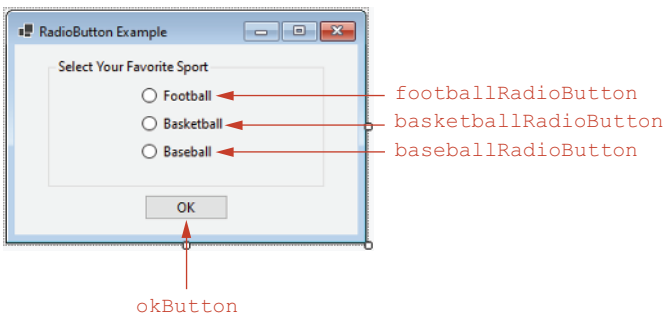
```
if (choice1RadioButton.Checked)
{
    MessageBox.Show("You selected Choice 1");
}
```

Notice that we did not have to use the `==` operator to explicitly compare the Checked property to the value `true`. This code is equivalent to the following:

```
if (choice1RadioButton.Checked == true)
{
    MessageBox.Show("You selected Choice 1.");
}
```

Let's look at an example using multiple RadioButton controls. Open the *RadioButton* project in the *Chap04* folder of this book's Student Sample Programs. The application's form is shown in Figure 4-26. When you run the application, select one of the radio buttons and then click the *OK* button. A message box appears showing the sport that you selected.

Figure 4-26 The *RadioButton Example* form



Here is the code for the `okButton_Click` event handler:

```
 1 private void okButton_Click(object sender, EventArgs e)
 2 {
 3     if (footballRadioButton.Checked)
 4     {
 5         MessageBox.Show("You selected Football.");
 6     }
 7     else if (basketballRadioButton.Checked)
 8     {
 9         MessageBox.Show("You selected Basketball.");
10     }
11     else if (baseballRadioButton.Checked)
12     {
13         MessageBox.Show("You selected Baseball.");
14     }
15 }
```

When the event handler executes, the `if` statement in line 3 determines whether the `footballRadioButton` control's Checked property is true. If it is, the message *You selected Football.* is displayed in line 5. Otherwise, line 7 determines whether the `basketballRadioButton` control's Checked property is true. If it is, the message *You selected Basketball.* is displayed in line 9. Otherwise, line 11 determines whether the `baseballRadioButton` control's Checked property is true. If it is, the message *You selected Baseball.* is displayed in line 13.

Check Boxes

A **check box** appears as a small box with some accompanying text. Figure 4-27 shows an example. They are called check boxes because clicking on an empty check box causes a check mark to appear in it. If a check mark already appears in a check box, clicking it removes the check mark.

Figure 4-27 A check box

Check boxes are similar to radio buttons, except that check boxes are not mutually exclusive. You can have one or more check boxes in a group, and any number of them can be selected at any given time. When you want to create a check box on a form, you use the **CheckBox control**, which is found in the *Common Controls* section of the *Toolbox*.

The CheckBox Control's Text Property

CheckBox controls have a Text property, which holds the text that is displayed next to the check box. For example, the CheckBox control shown in Figure 4-27 has its Text property set to *Pepperoni*.

The CheckBox Control's Checked Property

Like radio buttons, CheckBox controls have a Checked property. When a CheckBox control is selected, or checked, its Checked property is set to True. When a CheckBox control is deselected, or unchecked, its Checked property is set to False.

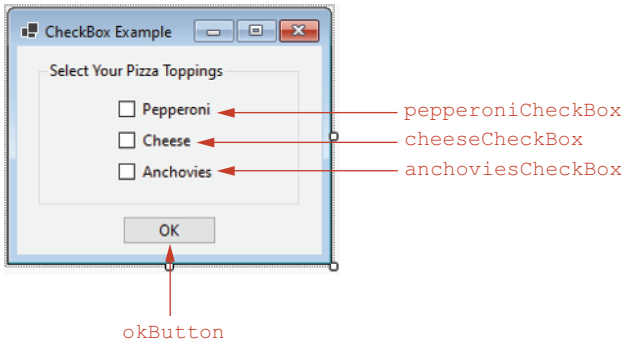
Working with CheckBox Controls in Code

In code, you can determine whether a CheckBox control is selected by testing its Checked property. For example, suppose a form has a CheckBox control named `option1CheckBox`. The following `if` statement determines whether it is selected:

```
if (option1CheckBox.Checked)
{
   MessageBox.Show("You selected Option 1.");
}
```

Let's look at an example program. Open the *CheckBox* project in the *Chap04* folder of this book's Student Sample Programs. The application's form is shown in Figure 4-28. When you run the application, select any of the check boxes and then click the *OK* button. One or more message boxes will appear, showing you the items that you selected.

Figure 4-28 The *CheckBox* project's form



Here is the code for the `okButton_Click` event handler:

```
 1 private void okButton_Click(object sender, EventArgs e)
 2 {
 3     if (pepperoniCheckBox.Checked)
 4     {
 5         MessageBox.Show("You selected Pepperoni.");
 6     }
 7
 8     if (cheeseCheckBox.Checked)
 9     {
10         MessageBox.Show("You selected Cheese.");
11     }
12
13     if (anchoviesCheckBox.Checked)
14     {
15         MessageBox.Show("You selected Anchovies.");
16     }
17 }
```

Notice that we have three separate `if` statements. The `if` statement in line 3 determines whether the `pepperoniCheckBox` control is selected. If so, line 5 displays the message *You selected Pepperoni*. The `if` statement in line 8 determines whether the `cheeseCheckBox` control is selected. If so, line 10 displays the message *You selected Cheese*. The `if` statement in line 13 determines whether the `anchoviesCheckBox` control is selected. If so, line 15 displays the message *You selected Anchovies*.

The CheckedChanged Event

Any time a RadioButton or a CheckBox control's Checked property changes, a **Checked Changed event** happens for that control. If you want some action to immediately take place when the user selects (or deselects) a RadioButton or CheckBox control, you can create a CheckedChanged event handler for the control and write the desired code in that event handler.

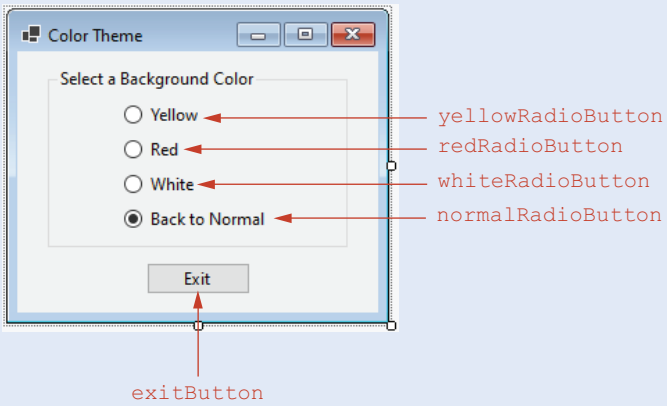
To create a CheckedChanged event handler for a RadioButton or a CheckBox, simply double-click the control in the *Designer*. An empty CheckedChanged event handler is created in the code editor. You can then write code inside the event handler. Tutorial 4-5 leads you through the process.

Tutorial 4-5:
Creating the *Color Theme* Application

VideoNote
Tutorial 4-5 : Creating the *Color Theme* application

In this tutorial, you create an application that allows the user to select a color using RadioButton controls. When the user selects a color, the form's background color is changed to that color immediately. Figure 4-29 shows the application's form, with the names of all the controls.

Figure 4-29 The *ColorTheme* project's form



Step 1: Start Visual Studio and begin a new Windows Forms App project named *ColorTheme*.

Step 2: Set up the application's form as shown in Figure 4-29. Notice that the form's Text property is set to *Color Theme*. The names of the controls are shown in the figure. As you place each of the controls on the form, refer to Table 4-8 for the relevant property settings.

Step 3: Once you have set up the form with its controls, you can create the Checked-Changed event handlers for the RadioButton controls. At the end of this tutorial, Program 4-5 shows the completed code for the form. You will be instructed to refer to Program 4-5 as you write the event handlers.

In the *Designer*, double-click the `yellowRadioButton` control. This opens the code editor, and you see an empty event handler named `yellowRadioButton_CheckedChanged`. Complete the `yellowRadioButton_CheckedChanged` event handler by typing the code shown in lines 12–15 in Program 4-5.

Table 4-8 Control property settings

Control Name	Control Type	Property Settings
`colorGroupBox`	GroupBox	**Text:** *Select a Background Color*
`yellowRadioButton`	RadioButton	**Text:** *Yellow* **Checked:** False
`redRadioButton`	RadioButton	**Text:** *Red* **Checked:** False
`whiteRadioButton`	RadioButton	**Text:** *White* **Checked:** False
`normalRadioButton`	RadioButton	**Text:** *Back to Normal* **Checked:** True
`exitButton`	Button	**Text:** *Exit*

The event handler is easy to understand. The `if` statement in line 12 determines whether the `yellowRadioButton` control is checked. If so, line 14 sets the form's background to yellow.

Step 4: Switch your view back to the *Designer* and double-click the `redRadioButton` control. This opens the code editor, and you see an empty event handler named `redRadioButton_CheckedChanged`. Complete the `redRadioButton_CheckedChanged` event handler by typing the code shown in lines 20–23 in Program 4-5.

Step 5: Switch your view back to the *Designer* and double-click the `whiteRadioButton` control. This opens the code editor, and you see an empty event handler named `whiteRadioButton_CheckedChanged`. Complete the `whiteRadioButton_CheckedChanged` event handler by typing the code shown in lines 28–31 in Program 4-5.

Step 6: Switch your view back to the *Designer* and double-click the `normalRadioButton` control. This opens the code editor, and you see an empty event handler named `normalRadioButton_CheckedChanged`. Complete the `normalRadioButton_CheckedChanged` event handler by typing the code shown in lines 36–39 in Program 4-5.

Step 7: Switch your view back to the *Designer* and double-click the `exitButton` control. In the code editor, you see an empty event handler named `exitButton_Click`. Complete the `exitButton_Click` event handler by typing the code shown in lines 44–45 in Program 4-5.

Step 8: Save the project and run the application. Notice that the *Back to Normal* radio button is initially selected. That's because you set its Checked property to True in the Properties window. Click the other *Radio* buttons and notice that the form's background color changes immediately. When you are finished testing the application, click the *Exit* button to close it.

Program 4-5 Completed Form1 code for the *ColorTheme* project

```
1 namespace ColorTheme
2 {
3   public partial class Form1 : Form
4   {
5     public Form1()
6     {
```

```
7        InitializeComponent();
8      }
9
10     private void yellowRadioButton_CheckedChanged(object
       sender, EventArgs e)
11     {
12       if (yellowRadioButton.Checked)
13       {
14         this.BackColor = Color.Yellow;
15       }
16     }
17
18     private void redRadioButton_CheckedChanged(object sender,
       EventArgs e)
19     {
20       if (redRadioButton.Checked)
21       {
22         this.BackColor = Color.Red;
23       }
24     }
25
26     private void whiteRadioButton_CheckedChanged(object sender,
       EventArgs e)
27     {
28       if (whiteRadioButton.Checked)
29       {
30         this.BackColor = Color.White;
31       }
32     }
33
34     private void normalRadioButton_CheckedChanged(object
       sender, EventArgs e)
35     {
36       if (normalRadioButton.Checked)
37       {
38         this.BackColor = SystemColors.Control;
39       }
40     }
41
42     private void exitButton_Click(object sender, EventArgs e)
43     {
44       // Close the form.
45       this.Close();
46     }
47   }
48 }
```

Checkpoint

4.26 If several RadioButton controls have been created in the same GroupBox, how many of them may be selected at one time?

4.27 If several CheckBox controls have been created in the same GroupBox, how many of them may be selected at one time?

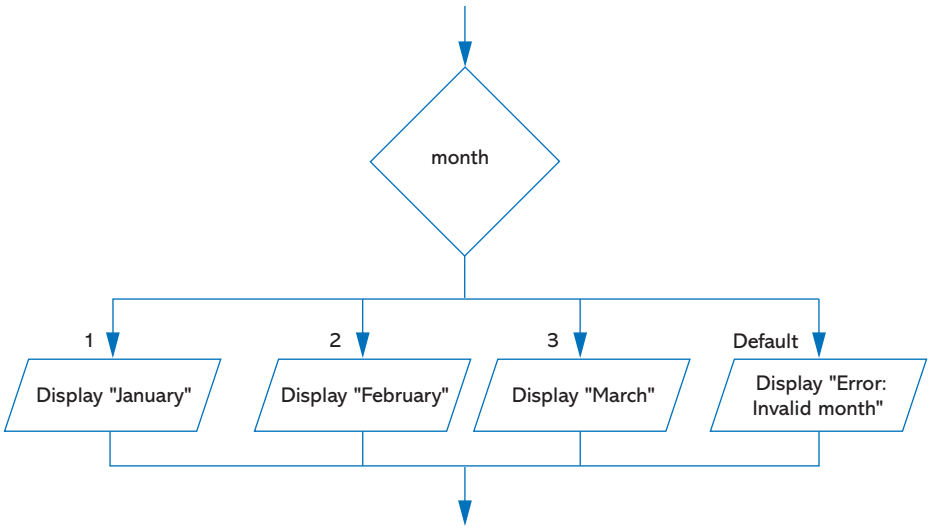
4.28 In code, how do you determine whether a RadioButton or a CheckBox control has been selected?

4.10 The switch Statement

CONCEPT: **The switch statement lets the value of a variable or an expression determine which path of execution the program will take.**

The **switch statement** is a **multiple-alternative decision structure.** It allows you to test the value of a variable or an expression and then use that value to determine which statement or set of statements to execute. Figure 4-30 shows an example of how a multiple-alternative decision structure looks in a flowchart.

Figure 4-30 A multiple-alternative decision structure



In the flowchart, the diamond symbol shows `month`, which is the name of a variable. If the `month` variable contains the value 1, the program displays "January." If the `month` variable contains the value 2, the program displays "February." If the `month` variable contains the value 3, the program displays "March." If the `month` variable contains none of these values, the action that is labeled `Default` is executed. In this case, the program displays "Error: Invalid month."

Here is the general format of a `switch` statement in C# code:

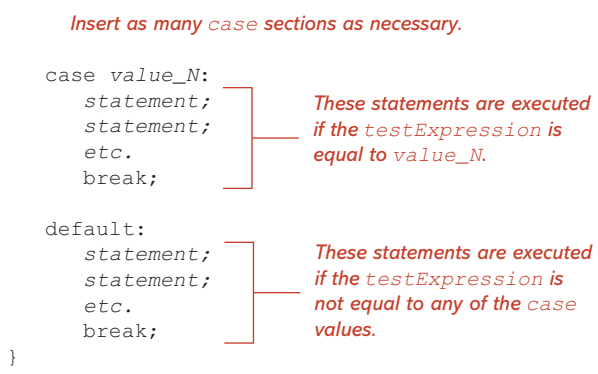
```
switch (testExpression)
{
   case value_1:
      statement;
      statement;
      etc.
      break;

   case value_2:
      statement;
      statement;
      etc.
      break;
```

The *testExpression* is a variable or expression.

These statements are executed if the testExpression is equal to value_1.

These statements are executed if the testExpression is equal to value_2.



The first line of the statement starts with the word `switch`, followed by a *`testExpression`*, which is enclosed in parentheses. The *`testExpression`* is a variable or an expression that gives an integer, string, or `bool` value. (Several other data types that we have not discussed yet are also permissible. The important thing to remember is that the *`testExpression`* cannot be a floating-point or `decimal` value.)

Beginning at the next line is a block of code enclosed in curly braces. Inside this block of code are one or more `case` sections. A `case` section begins with the word `case`, followed by a value, followed by a colon. Each `case` section contains one or more statements, followed by a `break` statement. Each `case` section must end with a `break` statement. At the end is an optional `default` section. The `default` section must also end with a `break` statement.

When the `switch` statement executes, it compares the value of the *`testExpression`* with the values that follow each of the `case` statements (from top to bottom). When it finds a `case` value that matches the *`testExpression`*'s value, the program branches to the `case` statement. The statements that follow the `case` statement are executed, until a `break` statement is encountered. At that point the program jumps out of the `switch` statement. If the *`testExpression`* does not match any of the `case` values, the program branches to the `default` statement and executes the statements that immediately follow it.

For example, the following code performs the same operation as the flowchart in Figure 4-30:

```
switch (month)
{
   case 1:
      MessageBox.Show("January");
      break;

   case 2:
      MessageBox.Show("February");
      break;

   case 3:
      MessageBox.Show("March");
      break;

   default:
      MessageBox.Show("Error: Invalid month");
      break;
}
```

In this example the *`testExpression`* is the `month` variable. If the value in the `month` variable is 1, the program branches to the `case 1:` section and executes the `MessageBox.Show ("January")` statement that immediately follows it. If the value in the `month` variable is 2, the program branches to the `case 2:` section and executes the `MessageBox.Show("February")` statement that immediately follows it. If the value in the `month` variable is 3, the program branches to the `case 3:` section and executes the `MessageBox.`

`Show("March")` statement that immediately follows it. If the value in the `month` variable is not 1, 2, or 3, the program branches to the `default:` section and executes the `MessageBox.Show("Error: Invalid month")` statement that immediately follows it.

The `switch` statement can be used as an alternative to an `if-else-if` statement that tests the same variable or expression for several different values. For example, the previously shown `switch` statement works like this `if-else-if` statement:

```
if (month == 1)
{
    MessageBox.Show("January");
}
else if (month == 2)
{
    MessageBox.Show("February");
}
else if (month == 3)
{
    MessageBox.Show("March");
}
else
{
    MessageBox.Show("Error: Invalid month");
}
```

To see an application that uses a `switch` statement, look at the *Switch Example* project in the *Chap04* folder of this book's Student Sample Programs.

Checkpoint

4.29 Convert the following `if-else-if` code to a `switch` statement:

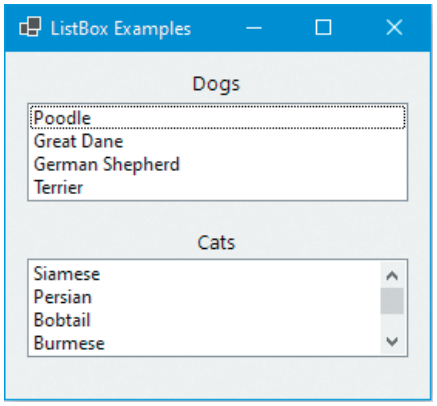
```
if (choice == 1)
{
    MessageBox.Show("You chose 1.");
}
else if (choice == 2)
{
    MessageBox.Show("You chose 2.");
}
else if (choice == 3)
{
    MessageBox.Show("You chose 3.");
}
else
{
    MessageBox.Show("Make another choice.");
}
```

4.11 Introduction to List Boxes

CONCEPT: **List boxes display a list of items and allow the user to select an item from the list.**

A list box displays a list of items and allows the user to select one or more items from the list. In Visual C#, you use the **ListBox control** to create a list box on an application's form. Figure 4-31 shows a form, at run time, with two ListBox controls. At run time, the user may select one of the items, causing the item to appear selected.

Figure 4-31 ListBox examples



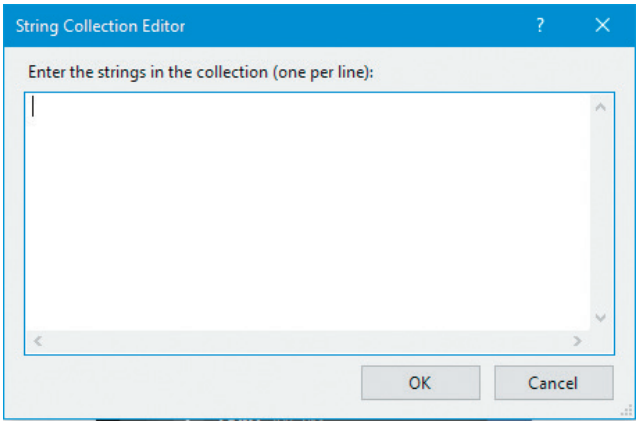
The topmost ListBox in Figure 4-31 does not have a scroll bar, but the bottom one does. A scroll bar appears when a ListBox contains more items than can be displayed in the space provided. In the figure, the top ListBox has four items (Poodle, Great Dane, German Shepherd, and Terrier), and all items are displayed. The bottom ListBox shows four items (Siamese, Persian, Bobtail, and Burmese), but because it has a scroll bar, we know there are more items in the ListBox than those four.

You will find the ListBox control in the *Common Controls* section of the *Toolbox*. Once you create a ListBox control, you add items to its Items property. The items that you add to a ListBox's Items property are displayed in the ListBox.

To store values in the Items property at design time, follow these steps:

1. Select the ListBox control in the *Designer* window.
2. In the *Properties* window, the setting for the Items property is displayed as *(Collection)*. When you select the Items property, an ellipses button (...) appears.
3. Click the ellipses button. The *String Collection Editor* dialog box appears, as shown in Figure 4-32.
4. Type the values that are to appear in the ListBox into the *String Collection Editor* dialog box. Type each value on a separate line by pressing the Enter key after each entry.
5. When you have entered all the values, click the *OK* button.

Figure 4-32 The *String Collection Editor* dialog box



NOTE: Once you acquire the necessary skills, you will be able to fill the Items collection of list boxes from external data sources (such as databases).

The SelectedItem Property

When the user selects an item in a ListBox, the item is stored in the ListBox's SelectedItem property. For example, suppose an application has a ListBox control named `fruitListBox` and a `string` variable named `selectedFruit`. The `fruitListBox` control contains the items *Apples*, *Pears*, and *Bananas*. If the user has selected *Pears*, the following statement assigns the string `"Pears"` to the variable `selectedFruit`:

```
selectedFruit = fruitListBox.SelectedItem.ToString();
```

Notice that you have to call the SelectedItem property's `ToString` method to retrieve the value as a string.

Determining Whether an Item Is Selected

An exception will occur if you try to get the value of a ListBox's SelectedItem property when no item is selected in the ListBox. For that reason, you should always make sure that an item is selected before reading the SelectedItem property. You do this with the SelectedIndex property.

The items that are stored in a ListBox each have an index. The **index** is simply a number that identifies the item's position in the ListBox. The first item has the index 0, the next has the index 1, and so forth. The last index value is $n - 1$, where n is the number of items in the ListBox. When the user selects an item in a ListBox, the item's index is stored in the ListBox's SelectedIndex property. If no item is selected in the ListBox, the SelectedIndex property is set to -1.

You can use the SelectedIndex property to make sure that an item is selected in a ListBox before you try to get the value of the SelectedItem property. You simply make sure the SelectedIndex property is not set to –1 before trying to read the SelectedItem property. Here is an example:

```
if (fruitListBox.SelectedIndex != -1)
{
    selectedFruit = fruitListBox.SelectedItem.ToString();
}
```

In Tutorial 4-6, you will create an application that lets the user select an item from a ListBox control.

Tutorial 4-6:
Creating the *Time Zone* Application

VideoNote

Tutorial 4-6: Creating the *Time Zone* application

In this tutorial, you create an application that allows the user to select a city from a ListBox control. When the user clicks a button, the application displays the name of the city's time zone. Figure 4-33 shows the application's form, with the names of all the controls.

Step 1: Start Visual Studio and begin a new Windows Forms App project named *TimeZone*.

Step 2: Set up the application's form, as shown in Figure 4-33. Notice that the form's Text property is set to *Time Zone*. The names of the controls are shown in the figure. As you place each of the controls on the form, refer to Table 4-9 for the relevant property settings.

Figure 4-33 The *TimeZone* project's form

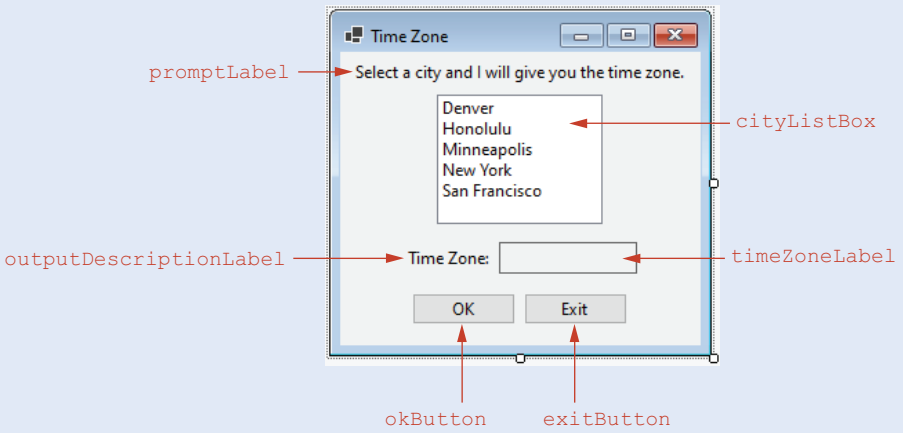


Table 4-9 Control property settings

Control Name	Control Type	Property Settings
`promptLabel`	Label	**Text:** *Select a city and I will give you the time zone.*
`cityListBox`	ListBox	**Items:** *Denver* *Honolulu* *Minneapolis* *New York* *San Francisco*
`outputDescriptionLabel`	Label	**Text:** *Time Zone:*
`timeZoneLabel`	Label	**AutoSize:** False **BorderStyle:** FixedSingle **Text:** (The contents of the Text property have been erased.) **TextAlign:** MiddleCenter
`okButton`	Button	**Text:** *OK*
`exitButton`	Button	**Text:** *Exit*

Step 3: Once you have set up the form with its controls, you can create the Click event handlers for the Button controls. At the end of this tutorial, Program 4-6 shows the completed code for the form. You will be instructed to refer to Program 4-6 as you write the event handlers.

In the *Designer*, double-click the `okButton` control. This opens the code editor, and you see an empty event handler named `okButton_Click`. Complete the `okButton_Click` event handler by typing the code shown in lines 12–43 in Program 4-6. Let's take a closer look at the code:

Line 12: This line declares a `string` variable named `city`. It is used to hold the name of the city that the user selects from the ListBox.

Line 14: This `if` statement determines whether the user has selected an item in the `cityListBox` control. If an item is selected, the control's SelectedIndex property is set to the item's index (a value of 0 or greater), and the program continues to line 16. If no item is selected, however, the control's SelectedIndex property is set to -1, and the program jumps to the `else` clause in line 39.

Line 17: This statement gets the selected item from the ListBox and assigns it to the `city` variable.

Line 20: This `switch` statement tests the city variable and branches to one of its `case` statements, depending on the variable's value:

- If the `city` variable equals `"Honolulu"`, the program jumps to the `case` statement in line 22.
- If the `city` variable equals `"San Francisco"`, the program jumps to the `case` statement in line 25.
- If the `city` variable equals `"Denver"`, the program jumps to the `case` statement in line 28.
- If the `city` variable equals `"Minneapolis"`, the program jumps to the `case` statement in line 31.
- If the `city` variable equals `"New York"`, the program jumps to the `case` statement in line 34.

Step 4: Switch your view back to the *Designer* and double-click the `exitButton` control. In the code editor, you see an empty event handler named `exitButton_Click`. Complete the `exitButton_Click` event handler by typing the code shown in lines 48–49 in Program 4-6.

Step 5: Save the project and run the application. Select a city in the ListBox control and click the *OK* button to see its time zone. Test each city, and when you are finished, click the *Exit* button, and the form should close.

Program 4-6 Completed Form1 code for the *TimeZone* project

```
1 namespace TimeZone
2 {
3   public partial class Form1 : Form
4   {
5     public Form1()
6     {
7       InitializeComponent();
8     }
9
10    private void okButton_Click(object sender, EventArgs e)
11    {
12       string city;  // To hold the name of a city
13
14       if (cityListBox.SelectedIndex != -1)
15       {
16         // Get the selected item.
17         city = cityListBox.SelectedItem.ToString();
18
19         // Determine the time zone.
20         switch (city)
21         {
22           case "Honolulu":
23             timeZoneLabel.Text = "Hawaii-Aleutian";
24             break;
25           case "San Francisco":
26             timeZoneLabel.Text = "Pacific";
27             break;
28           case "Denver":
29             timeZoneLabel.Text = "Mountain";
30             break;
```

```
31            case "Minneapolis":
32              timeZoneLabel.Text = "Central";
33              break;
34            case "New York":
35              timeZoneLabel.Text = "Eastern";
36              break;
37          }
38        }
39        else
40        {
41          // No city was selected.
42          MessageBox.Show("Select a city.");
43        }
44      }
45
46      private void exitButton_Click(object sender, EventArgs e)
47      {
48        // Close the form.
49        this.Close();
50      }
51    }
52 }
```

Working with ListBox Controls in Code

Items can be added to, or removed from, a ListBox control **programmatically**. This means you can write code in your application that adds items to a ListBox, removes items from a ListBox, or clears the contents of a ListBox.

Adding an Item to a ListBox

To add an item to an existing ListBox control, you use the Items property's `Add` method. You call the method using the following general format:

```
ListBox.Items.Add(NewItem);
```

In the general format, *`ListBox`* is the name of the ListBox control and *`NewItem`* is the item you are adding to the ListBox.

For example, suppose an application has an empty ListBox named `daysListBox`. The following code adds three items to the ListBox:

```
daysListBox.Items.Add("Monday");
daysListBox.Items.Add("Tuesday");
daysListBox.Items.Add("Wednesday");
```

After this code executes, the `daysListBox` control will contain the items "Monday", "Tuesday", and "Wednesday". These items are stored in the order that they are added, so "Monday" will be stored at index 0, "Tuesday" will be stored at index 1, and "Wednesday" will be stored at index 2.

The `Add` method returns the index of the added item, so you can optionally assign that index to a variable when you call the method. The following code shows an example. In the code, assume `monIndex`, `tueIndex`, and `wedIndex` are `int` variables that have already been declared:

```
monIndex = daysListBox.Items.Add("Monday");
tueIndex = daysListBox.Items.Add("Tuesday");
wedIndex = daysListBox.Items.Add("Wednesday");
```

This code adds the items "Monday", "Tuesday", and "Wednesday" to the `daysListBox` control. After the code executes, the `monIndex` variable will contain the index of "Monday", the `tueIndex` variable will contain the index of "Tuesday", and the `wedIndex` variable will contain the index of "Wednesday".

Getting the Number of Items in a ListBox

The Items property has an integer **subproperty** named Count that contains the number of items in the ListBox. Here is an example of how we would get the number of items that are stored in the `daysListBox` control and assign that number to the `numItems` variable:

```
numItems = daysListBox.Items.Count;
```

Removing an Item From a ListBox

To remove an item from a ListBox control, you use the Items property's `Remove` or `RemoveAt` methods. The `Remove` method removes a specified item from the ListBox. Here is the general format:

```
ListBox.Items.Remove(ItemToRemove);
```

In the general format, *`ListBox`* is the name of the ListBox control and *`ItemToRemove`* is the item you want to remove from the ListBox. For example, the following statement removes "Tuesday" from the `daysListBox` control:

```
daysListBox.Items.Remove("Tuesday");
```

If the specified item does not exist in the ListBox, nothing will be removed and no error will occur.

The `RemoveAt` method removes the item at a specified index from the ListBox. Here is the general format:

```
ListBox.Items.RemoveAt(Index);
```

In the general format, *`ListBox`* is the name of the ListBox control and *`Index`* is the index of the item you want to remove. For example, the following statement removes the item at index 2 from the `daysListBox` control:

```
daysListBox.Items.RemoveAt(2);
```

When using the `RemoveAt` method, the specified index must be a valid index. It cannot be less than 0, or greater than or equal to the number of items in the ListBox. If the index is invalid, the `RemoveAt` method will throw an exception. The following code shows a simple way to avoid the exception. In the code, assume that `index` is an `int` variable that has already been assigned a value.

```
if (index < 0 || index >= daysListBox.Items.Count)
{
    daysListBox.Items.RemoveAt(index);
}
else
{
    MessageBox.Show("Invalid index.");
}
```

NOTE: When an item is removed from a ListBox, the indexes will be adjusted for all items appearing after the removed item.

Clearing a ListBox

You can remove all the items that are stored in a ListBox control with the Item's property's `Clear` method. You call the method using the following general format:

```
ListBox.Items.Clear();
```

In the general format, `ListBox` is the name of the ListBox control. Here is an example:

```
daysListBox.Items.Clear();
```

After this statement executes, the `daysListBox` control will be empty.

Checkpoint

4.30 How do you add items to a ListBox control using the *Properties* window?

4.31 How do you get the item that is selected in a ListBox?

4.32 How can you determine whether an item has been selected in a ListBox?

4.33 How do you add a new item to a ListBox control in code?

4.34 In code, how do you get the number of items that are stored in a ListBox control?

4.35 Suppose `namesListBox` is the name of a ListBox control. Write a statement that removes "Jamal" from the ListBox.

4.36 Suppose `namesListBox` is the name of a ListBox control. Write a statement that removes the item at index 3 from the ListBox.

4.37 Suppose `namesListBox` is the name of a ListBox control. Write a statement that removes all items from the ListBox.

Key Terms

`!` operator
`&&` operator
`||` operator
`bool` data type
Boolean expression
check box
CheckBox control
CheckedChanged event
conditionally executed
control structure
decision structure
dual-alternative decision structure
flag
`if-else` statement
`if-else-if` statement
index
input validation
ListBox control
logical AND operator
logical NOT operator
logical operator
logical OR operator
multiple-alternative decision structure
mutually exclusive selection
nested decision structure
`out` keyword
output variable
programmatically
radio buttons
RadioButton control
relational operator
selection structure
sequence structure
short-circuit evaluation
single-alternative decision structure
subproperty
`switch` statement
`TryParse` method

Review Questions

Multiple Choice

1. A _____ structure executes a set of statements only under certain circumstances.

a. sequence
b. circumstantial
c. decision
d. Boolean

2. A _____ structure provides one alternative path of execution.

a. sequence
b. single-alternative decision
c. one-path alternative
d. single-execution decision

3. A(n) _____ expression has a value of either true or false.

a. binary
b. decision
c. unconditional
d. Boolean

4. The symbols >, <, and == are all _____ operators.

a. relational
b. logical
c. conditional
d. ternary

5. A _____ structure tests a condition and then takes one path if the condition is true or another path if the condition is false.
   a. multibranch statement
   b. single-alternative decision
   c. dual-alternative decision
   d. sequence

6. You use a(n) _____ statement to write a single-alternative decision structure.
   a. `test-jump`
   b. `if`
   c. `if-else`
   d. `if-call`

7. You use a(n) _____ statement to write a dual-alternative decision structure.
   a. `test-jump`
   b. `if`
   c. `if-else`
   d. `if-call`

8. A _____ decision structure is written inside another decision structure.
   a. nested
   b. tiered
   c. dislodged
   d. hierarchical

9. `&&`, `||`, and `!` are _____ operators.
   a. relational
   b. logical
   c. conditional
   d. ternary

10. A compound Boolean expression created with the _____ operator is true only if both of its subexpressions are true.
    a. `&&`
    b. `||`
    c. `!`
    d. `both`

11. A compound Boolean expression created with the _____ operator is true if either of its subexpressions is true.
    a. `&&`
    b. `||`
    c. `!`
    d. `either`

12. The _____ operator takes a Boolean expression as its operand and reverses its logical value.
    a. `&&`
    b. `||`
    c. `!`
    d. `either`

13. A _____ is a Boolean variable that signals when some condition exists in the program.
    a. flag
    b. signal

c. sentinel
d. siren

14. The _____ family of methods can be used to convert a string to a specific data type without throwing an exception.

a. `TryConvert`
b. `Parse`
c. `TryParse`
d. `SafeConvert`

15. If several _____ controls exist in a GroupBox, only one of them may be selected at a time.

a. CheckBox
b. RadioButton
c. ListBox
d. SelectionButton

16. You use the _____ statement to create a multiple-alternative decision structure.

a. `menu`
b. `branch`
c. `select`
d. `switch`

17. The _____ section of a `switch` statement is branched to if none of the `case` values match the test expression.

a. `else`
b. `default`
c. `case`
d. `otherwise`

18. A ListBox's index numbering starts at _____.

a. 0
b. 1
c. −1
d. any value you specify

19. You can use the _____ property to determine whether an item is selected in a ListBox.

a. Index
b. SelectedItem
c. SelectedIndex
d. Items.SelectedIndex

20. The _____ property holds the item that is selected in a ListBox control.

a. Index
b. SelectedItem
c. SelectedIndex
d. Items.SelectedIndex

True or False

1. You can write any program using only sequence structures.
2. A single-alternative decision structure tests a condition and then takes one path if the condition is true or another path if the condition is false.
3. The `if-else` statement is a dual-alternative decision structure.
4. A decision structure can be nested inside another decision structure.

5. A compound Boolean expression created with the `&&` operator is true only when both subexpressions are true.
6. The `TryParse` methods throw an exception if the string argument cannot be converted.
7. Multiple CheckBox controls in the same GroupBox can be selected at the same time.
8. The test expression in a `switch` statement can be a `double` or a `decimal` value.
9. If an item is not selected in a ListBox, the control's SelectedIndex property will be set to 0.
10. To store items in a ListBox, you add them to the control's Text property.

Short Answer

1. What is meant by the term *conditionally executed*?
2. You need to test a condition and then execute one set of statements if the condition is true. If the condition is false, you need to execute a different set of statements. What structure will you use?
3. Briefly describe how the `&&` operator works.
4. Briefly describe how the `||` operator works.
5. When determining whether a number is inside a range, which logical operator is it best to use?
6. What is a flag and how does it work?
7. What are the two arguments that you pass to a `TryParse` method?
8. How do you determine in code whether a RadioButton control or a CheckBox control is selected?
9. How do you add items to a ListBox using the *Properties* window?
10. How can you read the selected item from a ListBox while preventing an exception from occurring if no item is selected?

Algorithm Workbench

1. Write an `if` statement that assigns 20 to the variable `y` and assigns 40 to the variable `z` if the variable `x` is greater than 100.
2. Write an `if` statement that assigns 0 to the variable `b` and assigns 1 to the variable `c` if the variable `a` is less than 10.
3. Write an `if-else` statement that assigns 0 to the variable `b` if the variable `a` is less than 10. Otherwise, it should assign 99 to the variable `b`.
4. Write nested decision structures that perform the following: if `amount1` is greater than 10 and `amount2` is less than 100, display the greater of `amount1` and `amount2`.
5. Write an `if-else` statement that displays "Speed is normal" if the value of the `speed` variable is at least 24 but no more than 56. If the `speed` variable's value is outside this range, display "Speed is abnormal."
6. Write an `if-else` statement that determines whether the value of the `points` variable is less than 9 or greater than 51. If this is true, display "Invalid points." Otherwise, display "Valid points."
7. Assume `pointsTextBox` is the name of a TextBox control and `points` is the name of an `int` variable. Write an `if-else` statement that uses one of the `TryParse`

methods to convert the `pointsTextBox` control's Text property to an `int` and stores the result in the `points` variable. If the conversion is not successful, display an error message in a message box.

8. Rewrite the following `if-else-if` statement as a `switch` statement.

```
if (selection == 1)
{
   MessageBox.Show("You selected 1.");
}
else if (selection == 2)
{
   MessageBox.Show("You selected 2.");
}
else if (selection == 3)
{
   MessageBox.Show("You selected 3.");
}
else if (selection == 4)
{
   MessageBox.Show("You selected 4.");
}
else
{
   MessageBox.Show("Not good with numbers, eh?");
}
```

9. Assume `nameListBox` is a ListBox control. Write code that reads the selected item from the ListBox. Be sure to prevent an exception from occurring in case no item has been selected.

10. Assume `monthsListBox` is an empty ListBox control. Write code that adds "January", "February", and "March" to the control.

11. Assume `instrumentsListBox` is a ListBox control that contains some items. Write code that removes the item "guitar" from the control.

12. Assume `productsListBox` is a ListBox control that contains some items. Write code that removes the item at index 22 from the control. Make sure the code does not cause an exception in case the index is invalid.

13. Assume `productsListBox` is a ListBox control that contains some items. Write code that removes all the items from the control.

Programming Problems

1. **Roman Numeral Converter**

   Create an application that allows the user to enter an integer between 1 and 10 into a TextBox control. The program should display the Roman numeral version of that number. If the number is outside the range of 1 through 10, the program should display an error message.

   The following table lists the Roman numerals for the numbers 1 through 10.

Number	Roman Numeral
1	I
2	II
3	III
4	IV
5	V

Number	Roman Numeral
6	VI
7	VII
8	VIII
9	IX
10	X

VideoNote
Solving the Mass and Weight Problem

2. **Mass and Weight**

Scientists measure an object's mass in kilograms and its weight in Newtons. If you know the amount of mass of an object, you can calculate its weight, in Newtons, with the following formula:

$$Weight = Mass \times 9.8$$

Create an application that lets the user enter an object's mass and then calculates its weight. If the object weighs more than 1,000 Newtons, display a message indicating that it is too heavy. If the object weighs less than 10 Newtons, display a message indicating that it is too light.

3. **Magic Dates**

The date June 10, 1960, is special because when it is written in the following format, the month times the day equals the year:

6/10/60

Create an application that lets the user enter a month (in numeric form), a day, and a two-digit year. The program should then determine whether the month times the day equals the year. If so, it should display a message saying the date is magic. Otherwise, it should display a message saying the date is not magic.

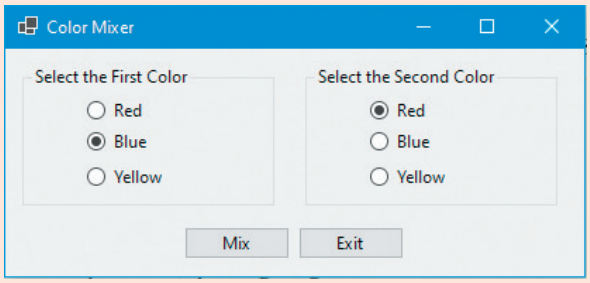
4. **Color Mixer**

The colors red, blue, and yellow are known as the primary colors because they cannot be made by mixing other colors. When you mix two primary colors, you get a secondary color, as shown here:

- When you mix red and blue, you get purple.
- When you mix red and yellow, you get orange.
- When you mix blue and yellow, you get green.

Create an application that lets the user select two primary colors from two different sets of *Radio* buttons. The form should also have a *Mix* button. When the user clicks the *Mix* button, the form's background should change to the color that you get when you mix the two selected primary colors. Figure 4-34 shows an example of how the form should appear.

Figure 4-34 The *Color Mixer* form



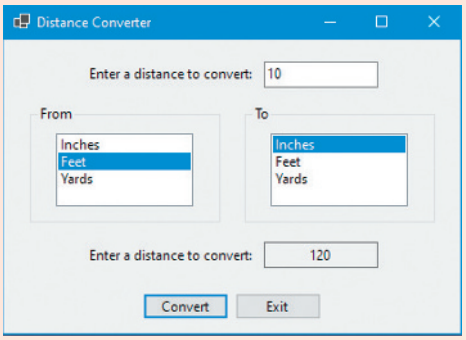
Note: If the user picks the same color from both sets of *Radio* buttons, set the form's background to that color.

5. **Distance Converter**

   In the English measurement system, 1 yard equals 3 feet and 1 foot equals 12 inches. Use this information to create an application that lets the user convert distances to and from inches, feet, and yards.

   Figure 4-35 shows an example of how the application's form might appear. In the example, the user enters the distance to be converted into a TextBox. A ListBox allows the user to select the units being converted from, and another ListBox allows the user to select the units being converted to.

Figure 4-35 The *Distance Converter* form



   Note: Be sure to handle the situation where the user picks the same units from both list boxes. The converted value will be the same as the value entered.

6. **Book Club Points**

   Serendipity Booksellers has a book club that awards points to its customers based on the number of books purchased each month. The points are awarded as follows:

   - If a customer purchases 0 books, they earn 0 points.
   - If a customer purchases 1 book, they earn 5 points.
   - If a customer purchases 2 books, they earn 15 points.
   - If a customer purchases 3 books, they earn 30 points.
   - If a customer purchases 4 or more books, they earn 60 points.

   Create an application that lets the user enter the number of books they have purchased this month and displays the number of points awarded.

7. **Software Sales**

   A software company sells a package that retails for $99. Quantity discounts are given according to the following table:

Quantity	Discount
10–19	20%
20–49	30%
50–99	40%
100 or more	50%

   Create an application that lets the user enter the number of packages purchased. The program should then display the amount of the discount (if any) and the total amount of the purchase after the discount.

8. **Change for a Dollar Game**

   Create a change-counting game that gets the user to enter the number of coins required to make exactly one dollar. The program should let the user enter the number of pennies, nickels, dimes, and quarters. If the total value of the coins entered is

equal to one dollar, the program should congratulate the user for winning the game. Otherwise, the program should display a message indicating whether the amount entered was more than or less than one dollar.

9. Time Calculator

Create an application that lets the user enter a number of seconds and works as follows:

- There are 60 seconds in a minute. If the number of seconds entered by the user is greater than or equal to 60, the program should display the number of minutes in that many seconds.
- There are 3,600 seconds in an hour. If the number of seconds entered by the user is greater than or equal to 3,600, the program should display the number of hours in that many seconds.
- There are 86,400 seconds in a day. If the number of seconds entered by the user is greater than or equal to 86,400, the program should display the number of days in that many seconds.

10. Restaurant Selector

You have a group of friends coming to visit for your high school reunion, and you want to take them out to eat at a local restaurant. You aren't sure if any of them have dietary restrictions, but your restaurant choices are as follows:

Joe's Gourmet Burgers	*Main Street Pizza Company*	*Corner Café*
Vegetarian: No	Vegetarian: Yes	Vegetarian: Yes
Vegan: No	Vegan: No	Vegan: Yes
Gluten-Free: No	Gluten-Free: Yes	Gluten-Free: Yes

Mama's Fine Italian	*The Chef's Kitchen*
Vegetarian: Yes	Vegetarian: Yes
Vegan: No	Vegan: Yes
Gluten-Free: No	Gluten-Free: Yes

Create an application that lets you specify whether any members of your party are vegetarian, vegan, or gluten-free, and then displays only the restaurants to which you may take the group.

Figure 4-36 shows one example of the application's user interface. The user selects dietary restrictions using the CheckBox controls, and when the *Show Restaurants* button is clicked, the names of the applicable restaurants are displayed in the ListBox control.

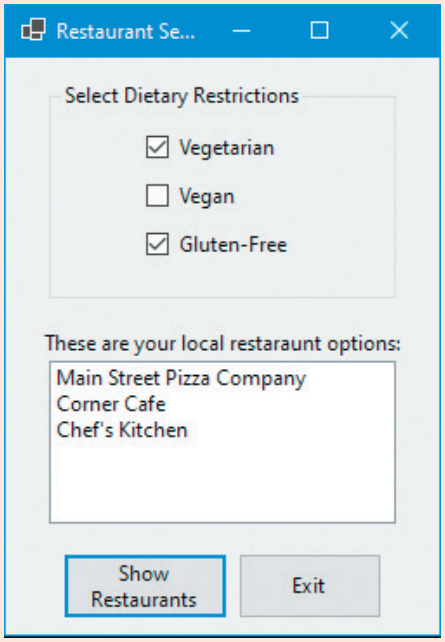
11. Workshop Selector

The following table shows a training company's workshops, the number of days of each, and their registration fees.

Workshop	Number of Days	Registration Fee
Handling Stress	3	$1,000
Time Management	3	$800
Supervision Skills	3	$1,500
Negotiation	5	$1,300
How to Interview	1	$500

The training company conducts its workshops in the six locations shown in the following table. The table also shows the lodging fees per day at each location.

Figure 4-36 Example Form for the Restaurant Selector Application



Location	Lodging Fees per Day
Austin	$150
Chicago	$225
Dallas	$175
Orlando	$300
Phoenix	$175
Raleigh	$150

When a customer registers for a workshop, they must pay the registration fee plus the lodging fees for the selected location. For example, here are the charges to attend the Supervision Skills workshop in Orlando:

Registration: $1,500

Lodging: $300 × 3 days = $900

Total: $2,400

Create an application that lets the user select a workshop from one ListBox and a location from another ListBox. When the user clicks a button, the application should calculate and display the registration cost, the lodging cost, and the total cost.

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CHAPTER 5

Loops, Files, and Random Numbers

TOPICS

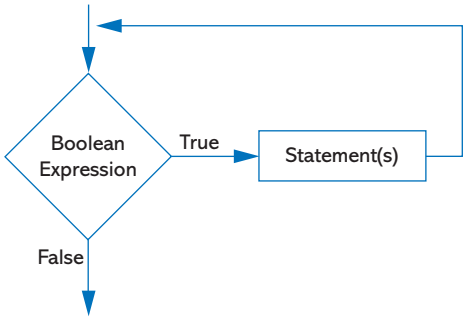
5.1 The `while` Loop
5.2 The `++` and `--` operators
5.3 The `for` Loop
5.4 The `do-while` Loop
5.5 Using Files for Data Storage
5.6 The OpenFileDialog and SaveFileDialog Controls
5.7 Random Numbers
5.8 The Load Event

5.1 The `while` Loop

CONCEPT: **The `while` loop causes a statement or set of statements to repeat as long as a Boolean expression is true.**

The `while` loop gets its name from the way it works: *While a Boolean expression is true, do some task*. The loop has two parts: (1) a Boolean expression that is tested for a true or false value and (2) a statement or set of statements that is repeated as long as the Boolean expression is true. Figure 5-1 shows the logic of a `while` loop.

Figure 5-1 The logic of a `while` loop



The diamond symbol represents the Boolean expression that is tested. Notice what happens if the expression is true: one or more statements are executed and the program's execution flows back to the point just above the diamond symbol. The Boolean expression is tested again, and if it is true, the process repeats. If the Boolean expression is false, the program exits the loop. Each time the loop executes its statement or statements, we say the loop is iterating, or performing an **iteration**.

Here is the general format of the `while` loop:

```
while (BooleanExpression)
{
     statement;
     statement;
     etc.
}
```

This set of statements is repeated while the Boolean expression is true.

We refer to the first line as the **`while` clause**. The `while` clause begins with the word `while`, followed by a Boolean expression that is enclosed in parentheses. Beginning on the next line is a set of statements enclosed in curly braces. This block of statements is known as the **body** of the loop.

When the `while` loop executes, the Boolean expression is tested. If the Boolean expression is true, the statements that appear in the body of the loop are executed, and then the loop starts over. If the Boolean expression is false, the loop ends and the program resumes execution at the statement immediately following the loop.

We say that the statements in the body of the loop are conditionally executed because they are executed only under the condition that the Boolean expression is true. If you are writing a `while` loop that has only one statement in its body, you do not have to enclose the statement inside curly braces. Such a loop can be written in the following general format:

```
while (BooleanExpression)
    statement;
```

When a `while` loop written in this format executes, the Boolean expression is tested. If it is true, the one statement that appears on the next line is executed, and then the loop starts over. If the Boolean expression is false, however, the loop ends.

Although the curly braces are not required when there is only one statement in the loop's body, it is still a good idea to use them, as shown in the following general format:

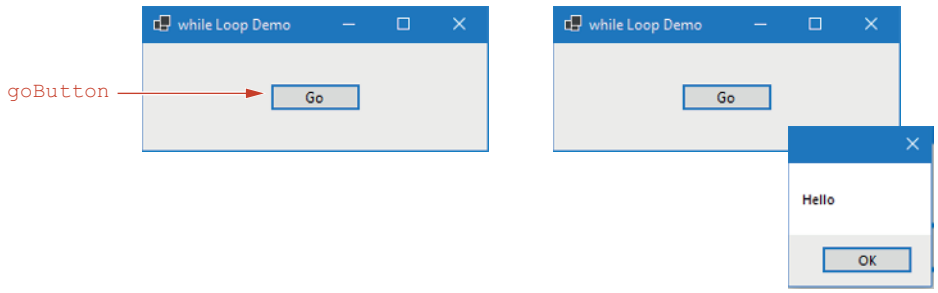
```
while (BooleanExpression)
{
    statement;
}
```

When we discussed the various `if` statements in Chapter 4, we mentioned that this is a good style of programming because it cuts down on errors. If you have more than one statement in the body of a loop, those statements *must* be enclosed in curly braces. If you get into the habit of always enclosing the conditionally executed statements in a set of curly braces, it's less likely that you will forget them.

You should also notice that the statements in the body of the loop are indented. As with `if` statements, this indentation makes the code easier to read and debug. By indenting the statements in the body of the loop, you visually set them apart from the surrounding code.

Let's look at an example. In the *Chap05* folder of this book's Student Sample Programs, you will find a project named *WhileLoopDemo*. Figure 5-2 shows the application's form at run time. As shown in the image on the left, the Button control's name is `goButton`. At run time, when you click the `goButton` control, the message box shown in the image on the right is displayed. When you click the *OK* button to close the message box, another identical message box is displayed. The message box is displayed a total of five times.

Figure 5-2 The *while Loop Demo* application



Here is the code for the `goButton_Click` event handler:

```
 1 private void goButton_Click(object sender, EventArgs e)
 2 {
 3     // Declare a variable to count the loop iterations.
 4     int count = 1;
 5
 6     // Display "Hello" in a message box five times.
 7     while (count <= 5)
 8     {
 9         // Display the message box.
10         MessageBox.Show("Hello");
11
12         // Add one to count.
13         count = count + 1;
14     }
15 }
```

Let's take a closer look at this code. In line 4 an `int` variable named `count` is declared and initialized with the value 1. A `while` loop begins in line 7. Notice that the `while` loop tests the expression `count <= 5`. The statements in the body of the `while` loop repeat as long as the `count` variable is less than or equal to 5. Inside the body of the loop, line 10 displays "Hello" in a message box, and then line 13 adds one to the `count` variable. This is the last statement in the body of the loop, so after it executes, the loop starts over. It tests the Boolean expression again, and if it is true, the statements in the body of the loop are executed. This cycle repeats until the Boolean expression `count <= 5` is false, as illustrated in Figure 5-3. A flowchart for the loop is shown in Figure 5-4.

Figure 5-3 The `while` Loop

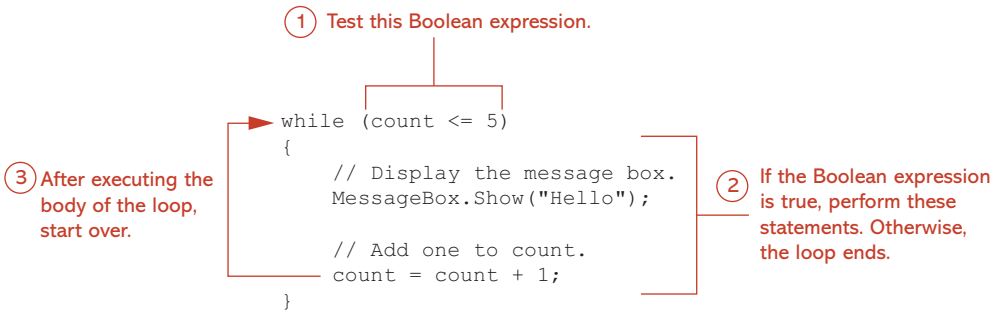
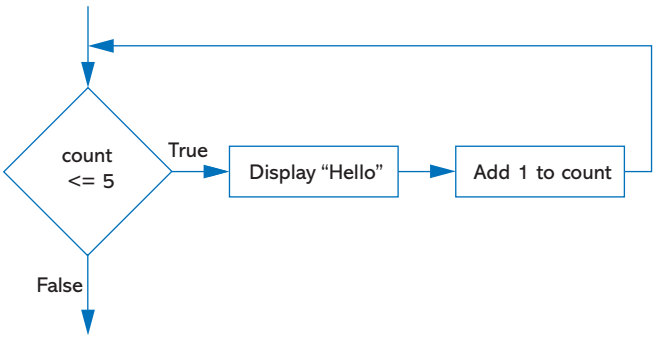


Figure 5-4 Flowchart for the `while` Loop



The `while` Loop Is a Pretest Loop

The `while` loop is known as a **pretest loop**, which means it tests its condition *before* performing an iteration. Because the test is done at the beginning of the loop, you usually have to perform some steps prior to the loop to make sure that the loop executes at least once. Notice the declaration of the `count` variable in the *WhileLoopDemo* program:

```
int count = 1;
```

The `count` variable is initialized with the value 1. If `count` had been initialized with a value that is greater than 5, as shown in the following program sample, the loop would never execute:

```
 1 private void goButton_Click(object sender, EventArgs e)
 2 {
 3     // Declare a variable to count the loop iterations.
 4     int count = 6;
 5
 6     // This loop will never iterate!
 7     while (count <= 5)
 8     {
 9         // Display the message box.
10         MessageBox.Show("Hello");
11
12         // Add one to count.
13         count = count + 1;
14     }
15 }
```

An important characteristic of the `while` loop is that the loop will never iterate if the Boolean expression is false to start with. If you want to be sure that a `while` loop executes the first time, you must initialize the relevant data in such a way that the Boolean expression starts out as true.

Counter Variables

In the *WhileLoopDemo* project the variable `count` is initialized with the value 1, and then 1 is added to the variable `count` during each loop iteration. The loop executes as long as `count` is less than or equal to 5. The variable `count` is used as a **counter variable**, which means it is regularly incremented in each loop iteration. In essence, the `count` variable keeps count of the number of iterations the loop has performed. Counter variables are commonly used to control the number of times that a loop iterates.

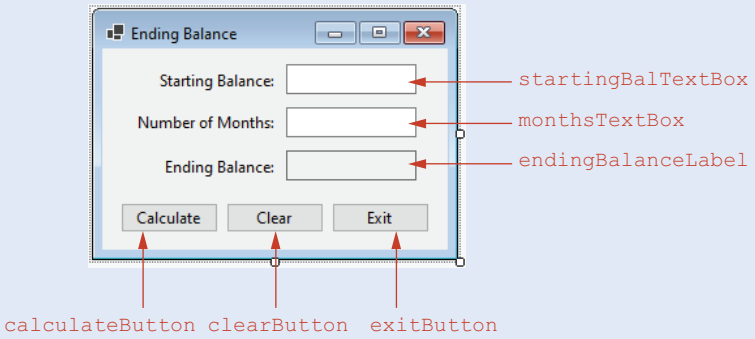
Tutorials 5-1 and 5-2 will give you some practice writing a loop and using a counter variable. In these tutorials, you will write a `while` loop that calculates the amount of interest earned by a bank account each month for a number of months.

Tutorial 5-1:
Using a Loop to Calculate an Account Balance

VideoNote
Tutorial 5-1: Using a Loop to Calculate an Account Balance

In this tutorial, you complete the *EndingBalance* project. The project has already been started for you and is located in the *Chap05* folder of this book's Student Sample Programs. The application's form is shown in Figure 5-5.

Figure 5-5 The *Ending Balance* form



When you complete the application, it will allow the user to enter an account's starting balance into the `startingBalTextBox` control and the number of months that the account will be left to earn interest into the `monthsTextBox` control. When the user clicks the `calculateButton` control, the application calculates the account's balance at the end of the time period. The account's monthly interest rate is 0.005, and the interest is compounded monthly.

Step 1: Start Visual Studio. Open the project named *EndingBalance* in the *Chap05* folder of this book's Student Sample Programs.

Step 2: Open the Form1 form in the *Designer*. The form is shown, along with the names of the important controls, in Figure 5-5.

Step 3: Now you will create the Click event handlers for the Button controls. At the end of this tutorial, Program 5-1 shows the completed code for the form. You will be instructed to refer to Program 5-1 as you write the event handlers.

In the *Designer*, double-click the `calculateButton` control. This opens the code editor, and you will see an empty event handler named `calculateButton_Click`. Complete the `calculateButton_Click` event handler by typing the code shown in lines 12–49 in Program 5-1. Let's take a closer look at the code:

Line 13: This statement declares a constant `decimal` named `INTEREST_RATE`, set to the value 0.005. This is the monthly interest rate.

Lines 16–18: These statements declare the following variables:

- `balance`, a `decimal` variable to hold the account balance.
- `months`, an `int` variable to hold the number of months that the account will be left to earn interest.
- `count`, an `int` that is used to count the months as a loop iterates. Notice that the `count` variable is initialized with the value 1.

Line 21: This `if` statement tries to convert `startingBalTextBox.Text` to a `decimal`. If the conversion is successful, the result is stored in the `balance` variable, and the program continues executing at line 23. If the conversion is not successful, the program jumps to the `else` clause in line 45, and line 48 displays the error message *Invalid value for starting balance*.

Line 24: This `if` statement tries to convert `monthsTextBox.Text` to an `int`. If the conversion is successful, the result is stored in the `months` variable, and the program continues executing at line 26. If the conversion is not successful, the program jumps to the `else` clause in line 39, and line 42 displays the error message *Invalid value for months*.

Line 27: This is the beginning of a `while` loop. The loop executes as long as the expression `count <= months` is true.

Lines 29–33: These statements are the body of the loop. Line 30 calculates the monthly interest and adds it to the `balance` variable. Line 33 adds 1 to the `count` variable.

Line 37: This statement executes after the loop has finished all of its iterations. It converts the value of the `balance` variable to a string (formatted as currency) and assigns the resulting string to the `endingBalanceLabel` control's Text property.

Step 4: Switch your view back to the *Designer* and double-click the `clearButton` control. In the code editor you will see an empty event handler named `clearButton_Click`. Complete the `clearButton_Click` event handler by typing the code shown in lines 54–60 in Program 5-1.

Step 5: Switch your view back to the *Designer* and double-click the `exitButton` control. In the code editor you will see an empty event handler named `exitButton_Click`. Complete the `exitButton_Click` event handler by typing the code shown in lines 65–66 in Program 5-1.

Step 6: Save the project. Then, press F5 on the keyboard, or click the *Start Debugging* button ( ) on the toolbar to compile and run the application.

First, enter *1000* as the starting balance and *48* as the number of months. Click the *Calculate Average* button and $1,270.49 should appear as the ending balance. Think about the value that you entered for the number of months. How many times did the `while` loop in line 36 iterate? (Answer: 48 times.)

Next, click the *Clear* button to clear the TextBoxes and the ending balance. Now, enter *100* as the starting balance and *1* as the number of months. Click the *Calculate Average* button and $100.50 should appear as the ending balance. How many times did the `while` loop iterate this time? (Answer: 1 time.)

Continue to test the application as you wish. When you are finished, click the *Exit* button and the form should close. (If you plan to continue to the next tutorial, leave this project open in Visual Studio.)

Program 5-1 Completed Form1 code for the *EndingBalance* project

```
 1 namespace EndingBalance
 2 {
 3     public partial class Form1: Form
 4     {
 5         public Form1()
 6         {
 7             InitializeComponent();
 8         }
 9
10         private void calculateButton_Click(object sender,
           EventArgs e)
11         {
12             // Constant for the monthly interest rate.
13             const decimal INTEREST_RATE = 0.005m;
14
15             // Local variables
16             decimal balance;           // The account balance
17             int months;                // The number of months
18             int count = 1;             // Loop counter, initialized
                                          with 1
19
20             // Get the starting balance.
21             if (decimal.TryParse(startingBalTextBox.Text,
               out balance))
22             {
23                 // Get the number of months.
24                 if (int.TryParse(monthsTextBox.Text, out months))
25                 {
26                     // The following loop calculates the ending
                       balance.
27                     while (count <= months)
28                     {
29                         // Add this month's interest to the
                           balance.
30                         balance = balance + (INTEREST_RATE *
                           balance);
31
32                         // Add one to the loop counter.
33                         count = count + 1;
34                     }
35
36                     // Display the ending balance.
37                     endingBalanceLabel.Text = balance.ToString("c");
38                 }
39                 else
40                 {
41                     // Invalid number of months was entered.
42                     MessageBox.Show("Invalid value for months.");
43                 }
44             }
45             else
```

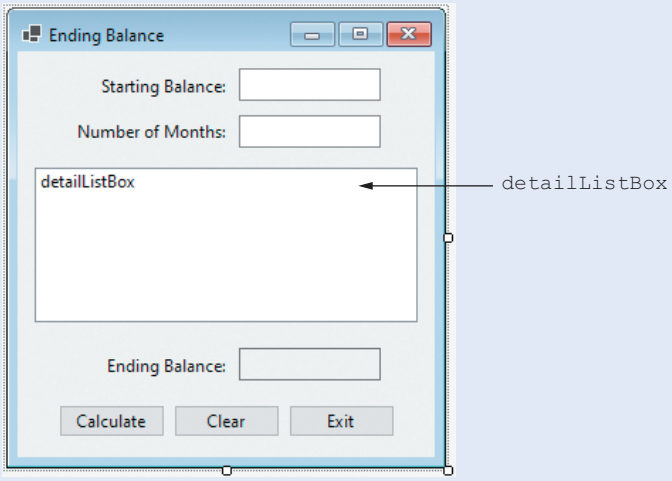
```
            {
                // Invalid starting balance was entered.
                MessageBox.Show("Invalid value for starting
                balance.");
            }
        }

        private void clearButton_Click(object sender, EventArgs e)
        {
            // Clear the TextBoxes and the endingBalanceLabel
            control.
            startingBalTextBox.Text = "";
            monthsTextBox.Text = "";
            endingBalanceLabel.Text = "";

            // Reset the focus.
            startingBalTextBox.Focus();
        }

        private void exitButton_Click(object sender, EventArgs e)
        {
            // Close the form.
            this.Close();
        }
    }
}
```

Tutorial 5-2:
Enhancing the *Ending Balance* Application

VideoNote
Tutorial 5-2: Enhancing the *Ending Balance* Application

In this tutorial, you enhance the *EndingBalance* project that you created in Tutorial 5-1. First, add a ListBox control to the application's form, as shown in Figure 5-6. Then modify the `calculateButton_Click` event handler so it displays each month's ending balance in the ListBox. Figure 5-7 shows an example of how the form will appear at run time, when the user has entered 1000 for the starting balance and 8 for the months.

Figure 5-6 The modified *Ending Balance* form

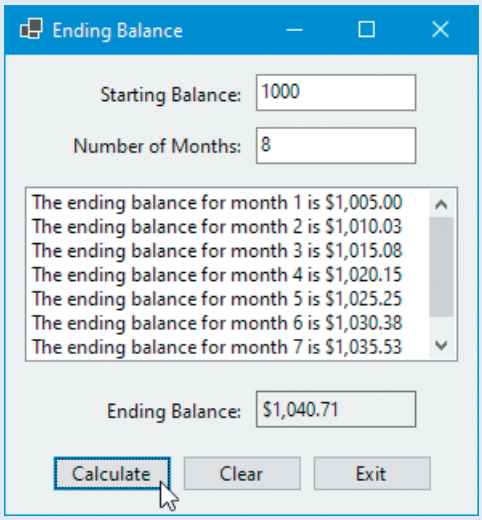


Step 1: Start Visual Studio and open the *EndingBalance* project that you completed in Tutorial 5-1.

Step 2: Enlarge the form so it is roughly the size shown in Figure 5-6. (310 pixels wide by 325 pixels high should be sufficient.)

Step 3: Create a ListBox control named `detailListBox`. Resize the ListBox as shown in Figure 5-6.

Figure 5-7 Example output



Step 4: Switch to the code editor and insert the code shown in lines 32–35 in Program 5-2. (The new lines of code are shown in bold.) The statement in lines 33–35 adds a string to the `detailListBox` control. If you examine the statement carefully, you will see that it uses concatenation to create a string in the following format:

The ending balance for month *count* is *balance*.

In the actual string that is created, *count* will be the value of the `count` variable and *balance* will be the value of the `balance` variable, formatted as currency.

Step 5: Find the `clearButton_Click` event handler in the code editor. Update the comment as shown in lines 59–60, and insert the line of code shown in line 64. (The lines are shown in bold.) The statement in line 64 clears the contents of the `detailListBox` control.

Step 6: Save the project. Then, press F5 on the keyboard or click the *Start Debugging* button (▶) on the toolbar to compile and run the application.

As shown in Figure 5-7, enter *1000* as the starting balance and *8* as the number of months. Click the *Calculate Average* button. Your output should look like that shown in Figure 5-7. Click the *Clear* button and enter any other values you wish to test the application further. When you are finished, click the *Exit* button and the form should close.

Program 5-2 Completed Form1 code for the *EndingBalance* project

```
 1 namespace EndingBalance
 2 {
 3     public partial class Form1: Form
 4     {
 5         public Form1()
 6         {
 7             InitializeComponent();
 8         }
 9
10         private void calculateButton_Click(object sender,
           EventArgs e)
11         {
12             // Constant for the monthly interest rate.
13             const decimal INTEREST_RATE = 0.005m;
14
15             // Local variables
16             decimal balance;           // The account balance
17             int months;                // The number of months
18             int count = 1;             // Loop counter,
                                             initialized with 1
19
20             // Get the starting balance.
21             if (decimal.TryParse(startingBalTextBox.Text, out
               balance))
22             {
23                 // Get the number of months.
24                 if (int.TryParse(monthsTextBox.Text, out
                   months))
25                 {
26                     // The following loop calculates the ending
                       balance.
27                     while (count <= months)
28                     {
29                         // Add this month's interest to the
                           balance.
30                         balance = balance + (INTEREST_RATE *
                           balance);
31
32                         // Display this month's ending balance.
33                         detailListBox.Items.Add("The ending
                           balance " +
34                             "for month " + count + " is " +
35                             balance.ToString("c"));
36
37                         // Add one to the loop counter.
38                         count = count + 1;
39                     }
40
41                     // Display the ending balance.
42                     endingBalanceLabel.Text = balance.ToString("c");
43                 }
44                 else
45                 {
46                     // Invalid number of months was entered.
47                     MessageBox.Show("Invalid value for months.");
48                 }
49             }
```

```
50              else
51              {
52                  // Invalid starting balance was entered.
53                  MessageBox.Show("Invalid value for starting
                    balance.");
54              }
55          }
56
57          private void clearButton_Click(object sender, EventArgs e)
58          {
59              // Clear the TextBoxes and the endingBalanceLabel
                control,
60              // and the ListBox.
61              startingBalTextBox.Text = "";
62              monthsTextBox.Text = "";
63              endingBalanceLabel.Text = "";
64              detailListBox.Items.Clear();
65
66              // Reset the focus.
67              startingBalTextBox.Focus();
68          }
69
70          private void exitButton_Click(object sender, EventArgs e)
71          {
72              // Close the form.
73              this.Close();
74          }
75      }
76 }
```

Infinite Loops

In all but rare cases, loops must contain a way to terminate within themselves. This means that something inside the loop must eventually make the loop's Boolean expression false. The loop in Program 5-2 stops when the expression `count <= months` is false. If a loop does not have a way of stopping, it is called an infinite loop. An *infinite loop* continues to repeat until the program is interrupted. Infinite loops usually occur when the programmer forgets to write code inside the loop that makes the test condition false. In most circumstances, you should avoid writing infinite loops.

The following code sample demonstrates an infinite loop. In line 1, the `count` variable is declared and initialized with the value 1. The `while` loop that begins in line 5 executes as long as `count` is less than or equal to 5. There is no code inside the loop to change the `count` variable's value, so the expression `count <= 5` in line 5 is always true. As a consequence, the loop has no way of stopping.

```
1 // Declare a variable to count the loop iterations.
2 int count = 1;
3
4 // How many times will this loop iterate?
5 while (count <= 5)
6 {
7     // Display the message box.
8     MessageBox.Show("Hello");
9 }
```

Checkpoint

5.1 What is a loop iteration?

5.2 What is a counter variable?

5.3 What is a pretest loop?

5.4 Does the `while` loop test its condition before or after it performs an iteration?

5.5 What is an infinite loop?

5.2 The ++ and -- operators

CONCEPT: **To increment a variable means to increase its value, and to decrement a variable means to decrease its value. C# provides special operators to increment and decrement variables.**

To **increment** a variable means to increase its value and to **decrement** a variable means to decrease its value. Both of the following statements increment the variable `num` by 1:

```
num = num + 1;
num += 1;
```

And `num` is decremented by 1 in both the following statements:

```
num = num - 1;
num -= 1;
```

Incrementing and decrementing is so commonly done in programs that C# provides a set of simple unary operators designed just for incrementing and decrementing variables. The increment operator is `++`, and the decrement operator is `--`. The following statement uses the `++` operator to add 1 to `num`:

```
num++;
```

After this statement executes, the value of `num` is increased by 1. The following statement uses the `--` operator to subtract 1 from `num`:

```
num--;
```

NOTE: The `++` operator is pronounced "plus plus," and the `--` operator is pronounced "minus minus." The expression `num++` is pronounced "num plus plus," and the expression `num--` is pronounced "num minus minus."

In these examples, we have written the `++` and `--` operators after their operands (or, on the right side of their operands). This is called **postfix mode**. The operators can also be written before (or, on the left side) of their operands, which is called **prefix mode**. Here are examples:

```
++num;
--num;
```

When you write a simple statement to increment or decrement a variable, such as the ones shown here, it doesn't matter if you use prefix mode or postfix mode. The operators do the same thing in either mode; however, if you write statements that mix these operators

with other operators or with other operations, there is a difference in the way the two modes work. Such complex code can be difficult to understand and debug. When we use the increment and decrement operators, we will do so only in ways that are straightforward and easy to understand, such as the statements previously shown.

We introduce these operators at this point because they are commonly used in loops. The following code segment shows an example. In the code, the `count` variable is initialized with the value 1. The `while` loop that begins in line 5 iterates as long as `count` is less than or equal to 5. The statement in line 11 increments `count`. The loop will iterate 5 times.

```
1 // Declare a variable to count the loop iterations.
2 int count = 1;
3
4 // Display "Hello" in a message box five times.
5 while (count <= 5)
6 {
7     // Display the message box.
8     MessageBox.Show("Hello");
9
10    // Increment count.
11    count++;
12 }
```

In the next section, which discusses the `for` loop, you will see these operators used often.

Checkpoint

5.6 What messages will the following code sample display?

```
int number = 5;
number++;
MessageBox.Show(number.ToString());
number--;
MessageBox.Show(number.ToString());
```

5.7 How many times will the following loop iterate?

```
int count = 0;
while (count < 4)
{
   MessageBox.Show(count.ToString());
   count++;
}
```

5.3 The `for` Loop

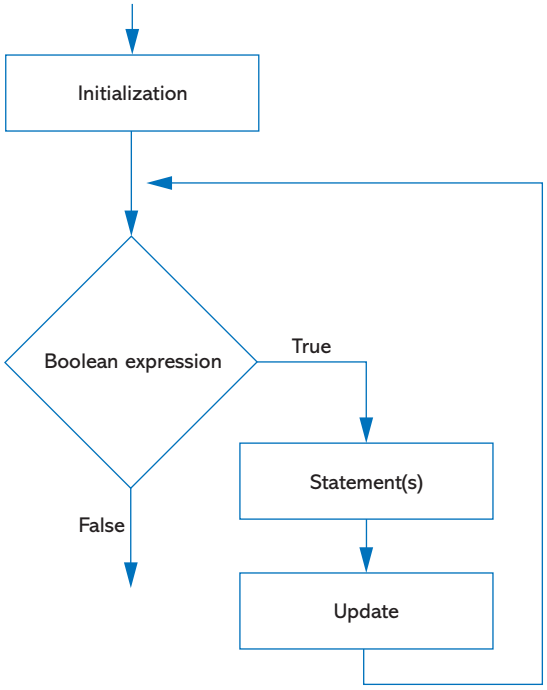
CONCEPT: The `for` loop is ideal for performing a known number of iterations.

The `for` loop is specifically designed for situations requiring a counter variable to control the number of times that a loop iterates. When you write a `for` loop, you specify three actions:

- Initialization: This action takes place when the loop begins. It happens only once.
- Test: A Boolean expression is tested. If the expression is true, the loop iterates. Otherwise, the loop stops.
- Update: This action takes place at the end of each loop iteration.

Figure 5-8 shows how these three actions are used in the logic of a `for` loop.

Figure 5-8 Logic of a `for` loop



Here is the general format of the `for` loop:

```
for (InitializationExpression; TestExpression; UpdateExpression)
{
    statement;
    statement;
    etc.
}
```

The statements that appear inside the curly braces are the body of the loop. These are the statements that are executed each time the loop iterates. As with other control structures, the curly braces are optional if the body of the loop contains only one statement, as shown in the following general format:

```
for (InitializationExpression; TestExpression; UpdateExpression)
   statement;
```

The first line of the `for` loop is the *loop header*. After the key word `for`, there are three expressions inside the parentheses, separated by semicolons. (Notice that there is not a semicolon after the third expression.)

The first expression is the **initialization expression**. It is normally used to initialize a counter variable to its starting value. This is the first action performed by the loop, and it is only done once. The second expression is the **test expression**. This is a Boolean expression that controls the execution of the loop. As long as this expression is true, the body of the `for` loop will repeat. The `for` loop is a pretest loop, so it evaluates the test expression before each iteration. The third expression is the **update expression**. It executes at the end of each iteration. Typically, this is a statement that increments the loop's counter variable.

Let's assume that `count` is an `int` variable that has already been declared. Here is an example of a simple `for` loop that displays `"Hello"` in a message box 5 times:

```
for (count = 1; count <= 5; count++)
{
   MessageBox.Show("Hello");
}
```

In this loop, the initialization expression is `count = 1`, the test expression is `count <= 5`, and the increment expression is `count++`. The body of the loop has one statement, which is the call to `MessageBox.Show` method. This is a summary of what happens when this loop executes:

1. The initialization expression `count = 1` is executed. This assigns 1 to the `count` variable.
2. The expression `count <= 5` is tested. If the expression is true, continue with Step 3. Otherwise, the loop is finished.
3. The statement `MessageBox.Show("Hello");` is executed.
4. The update expression `count++` is executed. This adds 1 to the `count` variable.
5. Go back to Step 2.

Figure 5-9 illustrates this sequence of events. Notice that Steps 2–4 are repeated as long as the test expression is true. Figure 5-10 shows the logic of the loop as a flowchart.

Figure 5-9 Sequence of events in the `for` loop

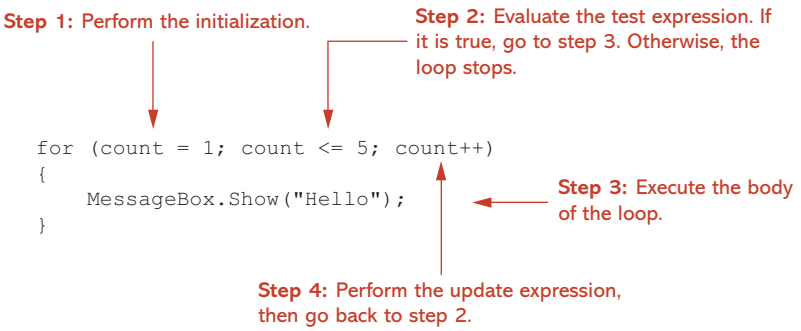
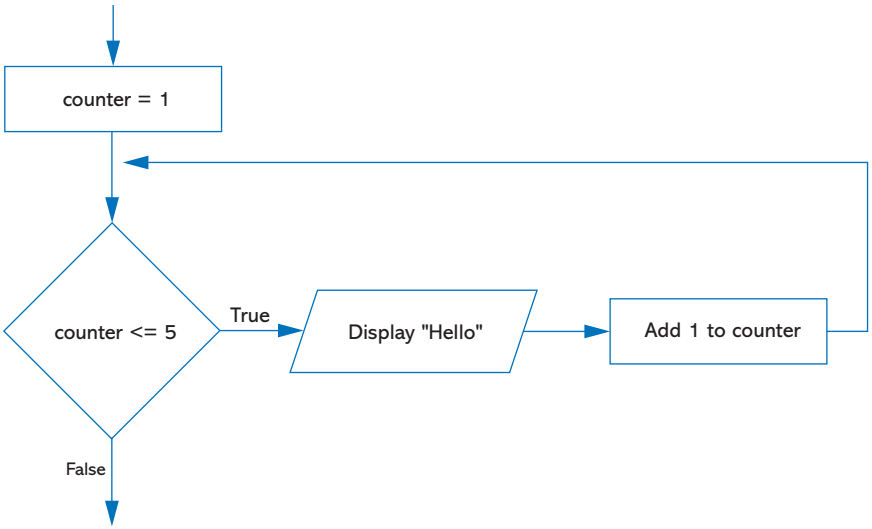
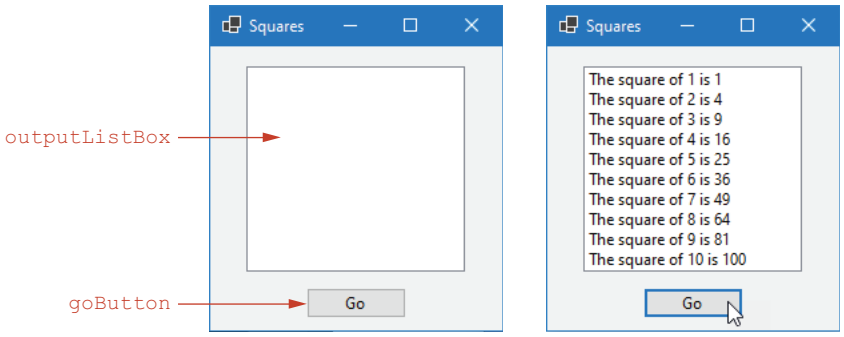


Figure 5-10 Logic of the `for` loop



Let's look at a complete application that uses a `for` loop. In the *Chap05* folder of this book's Student Sample Programs, you will find a project named *Squares*. The purpose of the application is to display the numbers 1–10 and their squares. Figure 5-11 shows the application's form at run time. As shown in the image on the left, the ListBox's name is `outputListBox` and the Button control's name is `goButton`. At run time, when you click the `goButton` control, the `outputListBox` control displays the program's output, as shown in the image on the right.

Figure 5-11 The *Squares* application



Here is the code for the `goButton_Click` event handler:

```
 1 private void goButton_Click(object sender, EventArgs e)
 2 {
 3     // Constant for the maximum number
 4     const int MAX_VALUE = 10;
 5
 6     // Loop counter
 7     int number;
 8
 9     // Display the list of numbers and their squares.
10     for (number = 1; number <= MAX_VALUE; number++)
11     {
12         outputListBox.Items.Add("The square of " +
13             number + " is " + (number * number));
14     }
15 }
```

Let's take a closer look at the code:

- Line 4 declares an `int` constant named `MAX_VALUE`, set to the value 10. This is the maximum number that we will use to calculate a square.
- Line 7 declares an `int` variable named `number`. This variable is used both as a counter variable and in the calculation of squares.
- Line 10 is the beginning of a `for` loop. You can see from this line that the loop works in the following way:
  Initialization: The `number` variable is initialized with the value 1.
  Test: The expression `number <= MAX_VALUE` is tested at the beginning of each iteration.
  Update: The expression `number++` is executed at the end of each iteration.
- Since the `MAX_VALUE` constant is set to the value 10, the `number` variable will be assigned the values 1 through 10 as the loop iterates.
- **Lines 12 and 13:** This statement adds a line to the ListBox showing the current value of the `number` variable, and the square of that value.

The `for` Loop Is a Pretest Loop

Because the `for` loop tests its Boolean expression before it performs an iteration, it is a pretest loop. It is possible to write a `for` loop in such a way that it will never iterate. Here is an example:

```
for (count = 6; count <= 5; count++)
{
   MessageBox.Show("Hello");
}
```

Because the variable count is initialized to a value that makes the Boolean expression false from the beginning, this loop terminates as soon as it begins.

Declaring the Counter Variable in the Initialization Expression

Not only may the counter variable be initialized in the initialization expression, but it may also be declared there. The following code shows an example:

```
for (int count = 1; count <= 5; count++)
{
   MessageBox.Show("Hello");
}
```

In this loop, the count variable is both declared and initialized in the initialization expression. If the variable is used only in the loop, it makes sense to define it in the loop header. This makes the variable's purpose clearer.

When a variable is declared in the initialization expression of a for loop, the scope of the variable is limited to the loop. This means you cannot access the variable in statements outside the loop. For example, the following code would cause a compiler error because the statement in line 6 cannot access the count variable.

```
1 for (int count = 1; count <= 5; count++)
2 {
3     MessageBox.Show("Hello");
4 }
5
6 MessageBox.Show("The value of count is " + count);
```

Other Forms of the Update Expression

In the update expression, the counter variable is typically incremented by 1. This makes it convenient to use the ++ operator in the increment expression. This is not a requirement, however. You can write virtually any expression you wish as the update expression. For example, the following loop increments count by 10:

```
for (int count = 0; count <= 100; count += 10)
{
   MessageBox.Show(count.ToString());
}
```

Notice that in this example the increment expression is count += 10. This means that at the end of each iteration, 10 is added to count. During the first iteration count is set to 0, during the second iteration count is set to 10, during the third iteration count is set to 20, and so forth.

Counting Backward by Decrementing the Counter Variable

Although the counter variable is usually incremented in a count-controlled loop, you can alternatively decrement the counter variable. For example, look at the following code:

```
for (int count = 10; count >= 0; count--)
{
   MessageBox.Show(count.ToString());
}
MessageBox.Show("Blastoff!");
```

In this loop, the `count` variable is initialized with the value 10. The loop iterates as long as `count` is greater than or equal to 0. At the end of each iteration, count is decremented by 1.

During the first iteration `count` is 10, during the second iteration `count` is 9, and so forth. If this were in an actual program, it would display the numbers 10, 9, 8, and so forth, down to 0, and then display Blastoff!

Avoiding Modifying the Counter Variable in the Body of the `for` Loop

Be careful not to place a statement that modifies the counter variable in the body of the `for` loop. All modifications of the control variable should take place in the update expression, which is automatically executed at the end of each iteration. If a statement in the body of the loop also modifies the counter variable, the loop probably will not terminate when you expect it to. The following loop, for example, increments `count` twice for each iteration:

```
for (int count = 1; count <= 10; count++)
{
   MessageBox.Show(count.ToString());
   count++; // Wrong!
}
```

You have seen several examples of the `for` loop. Tutorial 5-3 gives you an opportunity to write one. In the tutorial you will complete an application that uses a `for` loop to convert a series of measurements from the metric system to the English system.

Tutorial 5-3: Using the `for` Loop

VideoNote
Tutorial 5-3: Using the `for` Loop

Your friend Amanda just inherited a European sports car from her uncle. Amanda lives in the United States, and she is afraid she will get a speeding ticket because the car's speedometer works in kilometers per hour. She has asked you to write a program that displays a table of speeds in kilometers per hour with their values converted to miles per hour. The formula for converting kilometers per hour to miles per hour is:

$$MPH = KPH * 0.6214$$

In the formula, *MPH* is the speed in miles per hour and *KPH* is the speed in kilometers per hour.

The table that your program displays should show speeds from 60 kilometers per hour through 130 kilometers per hour, in increments of 10, along with their values converted to miles per hour. The table should look something like this:

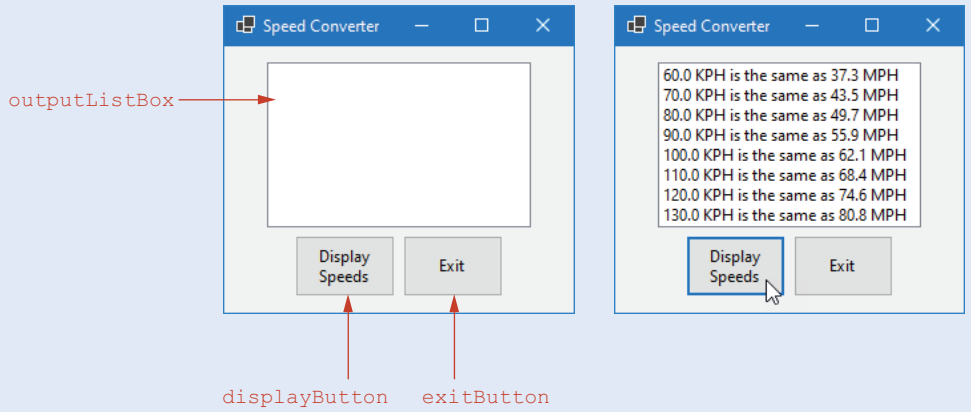
KPH	MPH
60	37.284
70	43.498
80	49.712
etc. . . .	

After thinking about this table of values, you decide that you will write a `for` loop that uses a counter variable to hold the kilometers-per-hour speeds. The counter's starting value will be 60, its ending value will be 130, and you will increase its value by 10 in the

update expression. Inside the loop you will use the counter variable to calculate a speed in miles per hour.

The project, which is named *SpeedConverter*, has already been started for you. It is located in the *Chap05* folder of this book's Student Sample Programs. The application's form, at run time, is shown in Figure 5-12. The image on the left in the figure shows the names of the controls. The image on the right shows how the form appears after the user clicks the *Display Speeds* button.

Figure 5-12 The *Speed Converter* form



Step 1: Start Visual Studio. Open the project named *SpeedConverter* in the *Chap05* folder of this book's Student Sample Programs.

Step 2: Open the Form1 form in the *Designer*.

Step 3: In the *Designer*, double-click the `displayButton` control. This opens the code editor, and you will see an empty event handler named `displayButton_Click`. Complete the `displayButton_Click` event handler by typing the code shown in lines 12–31 in Program 5-3 (at the end of this tutorial). Let's take a closer look at the code:

Lines 13–16: These statements declare the following named constants:

- `START_SPEED`, an `int` constant set to 60. This is the starting speed for the list of conversions and the value with which the loop's counter variable is initialized.
- `END_SPEED`, an `int` constant set to 130. This is the ending speed for the list of conversions. When the counter variable exceeds this value, the loop stops.
- `INTERVAL`, an `int` constant set to 10. This is the amount that you add to loop's counter variable after each iteration.
- `CONVERSION_FACTOR`, a `double` constant set to 0.6214. This is the conversion factor that you use in the formula to convert KPH to MPH.

Lines 19–20: These statements declare the following variables:

- `kph`, an `int` variable to hold the speed in kilometers per hour.
- `mph`, a `double` variable to hold the speed in miles per hour.

Line 23: This is the beginning of a `for` loop that works in the following way:

Initialization: The `kph` variable is initialized with the value of `START_SPEED`, which is 60.

Test: The expression `kph <= END_SPEED` is tested at the beginning of each iteration.

Update: The expression `kph += INTERVAL` is executed at the end of each iteration. This adds the value of `INTERVAL` (which is 10) to the `kph` variable.

As the loop iterates, the `kph` variable is assigned the values 60, 70, 80, and so forth, through 130.

Line 26: This statement converts the value of the `kph` variable to miles per hour and assigns the result to the `mph` variable.

Lines 29–30: This statement adds a line to the `outputListBox` control showing the current value of the `kph` variable and the equivalent value in miles per hour.

Step 4: Switch your view back to the *Designer* and double-click the `exitButton` control. In the code editor you will see an empty event handler named `exitButton_Click`. Complete the `exitButton_Click` event handler by typing the code shown in lines 36–37 in Program 5-3.

Step 5: Save the project. Then, press F5 on the keyboard or click the *Start Debugging* button (▶) on the toolbar to compile and run the application. Click the *Display Speeds* button, and you should see the output shown in the image on the right in Figure 5-12. Click the *Exit* button to close the form.

Program 5-3 Completed Form1 code for the *SpeedConverter* project

```
 1 namespace SpeedConverter
 2 {
 3     public partial class Form1: Form
 4     {
 5         public Form1()
 6         {
 7             InitializeComponent();
 8         }
 9
10         private void displayButton_Click(object sender,
           EventArgs e)
11         {
12             // Constants
13             const int START_SPEED = 60;
14             const int END_SPEED = 130;
15             const int INTERVAL = 10;
16             const double CONVERSION_FACTOR = 0.6214;
17
18             // Variables
19             int kph;        // Kilometers per hour
20             double mph;     // Miles per hour
21
22             // Display the table of speeds.
23             for (kph = START_SPEED; kph <= END_SPEED; kph +=
               INTERVAL)
24             {
25                 // Calculate miles per hour.
26                 mph = kph * CONVERSION_FACTOR;
27
```

```
28                  // Display the conversion.
29                  outputListBox.Items.Add(kph.ToString("n1") +
30                      " KPH is the same as " + mph.ToString("n1") +
                        " MPH");
31              }
32          }
33
34          private void exitButton_Click(object sender, EventArgs e)
35          {
36              // Close the form.
37              this.Close();
38          }
39      }
40  }
```

Checkpoint

5.8 Name the three expressions that appear inside the parentheses in the first line of a `for` loop.

5.9 You want to write a `for` loop that displays *I love to program* 50 times. Assume that you will use a variable named `count` as the counter variable.

a. What initialization expression will you use?

b. What test expression will you use?

c. What update expression will you use?

d. Write the loop.

5.10 What would the following code display?

```
for (int count = 1; count <= 5; count++)
{
  MessageBox.Show(count.ToString());
}
```

5.11 What would the following code display?

```
for (int count = 0; count <= 500; count += 100)
{
  MessageBox.Show(count.ToString());
}
```

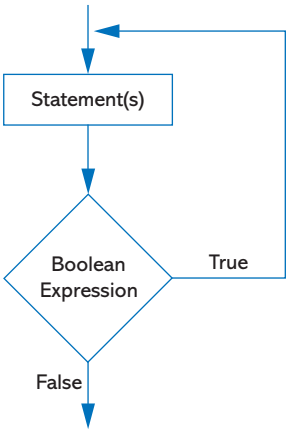
5.4 The do-while Loop

CONCEPT: **The `do-while` loop is a posttest loop, which means it performs an iteration before testing its Boolean expression.**

You have learned that the `while` loop and the `for` are pretest loops, which means they test their Boolean expressions before performing an iteration. The `do-while` loop is a **posttest loop**. This means it performs an iteration before testing its Boolean expression. As a result, the `do-while` loop always performs at least one iteration, even if its Boolean expression is false to begin with. The logic of a `do-while` loop is shown in Figure 5-13.

In the flowchart, one or more statements are executed, and then a Boolean expression is tested. If the Boolean expression is true, the program's execution flows back to the point

Figure 5-13 The logic of a `do-while` loop



just above the first statement in the body of the loop, and this process repeats. If the Boolean expression is false, the program exits the loop.

In code, the `do-while` loop looks something like an inverted `while` loop. Here is the general format of the `do-while` loop:

```
do
{
    statement;
    statement;
    etc.
} while (BooleanExpression);
```

As with the `while` loop, the braces are optional if there is only one statement in the body of the loop. This is the general format of the `do-while` loop with only one conditionally executed statement:

```
do
    statement;
while (BooleanExpression);
```

Notice that a semicolon appears at the very end of the `do-while` statement. This semicolon is required; leaving it out is a common error.

The `do-while` loop is a *posttest* loop. This means it does not test its Boolean expression until it has completed an iteration. As a result, the `do-while` loop always performs at least one iteration, even if the expression is false to begin with. This differs from the behavior of a `while` loop. For example, in the following `while` loop, the statement that calls `MessageBox.Show` will not execute at all:

```
int number = 1;
while (number < 0)
{
    MessageBox.Show(number.ToString());
}
```

But, the statement that calls `MessageBox.Show` in the following `do-while` loop executes one time because the `do-while` loop does not test the expression `number < 0` until the end of the iteration.

```
int number = 1;
do
{
   MessageBox.Show(number.ToString());
} while (number < 0);
```

Checkpoint

5.12 What is a posttest loop?

5.13 What is the difference between the `while` loop and the `do-while` loop?

5.14 How many times will the following loop iterate?

```
int count = 0;
do
{
   MessageBox.Show(count.ToString());
   count++;
} while (count < 0);
```

5.5 Using Files for Data Storage

CONCEPT: When a program needs to save data for later use, it writes the data in a file. The data can be read from the file at a later time.

The programs you have written so far require the user to reenter data each time the program runs because data kept in variables and control properties is stored in RAM and disappears once the program stops running. If a program is to retain data between the times it runs, it must have a way of saving it. Data is saved in a file, which is usually stored on a computer's disk. Once the data is saved in a file, it will remain there after the program stops running. Data that is stored in a file can be retrieved and used at a later time.

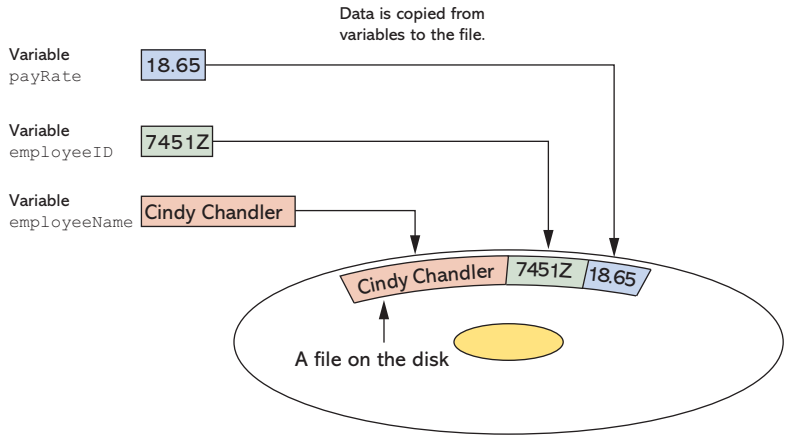
Most of the commercial software that you use on a day-to-day basis store data in files. The following are a few examples:

- **Word processors:** Word processing programs are used to write letters, memos, reports, and other documents. The documents are then saved in files so they can be edited and printed.
- **Image editors:** Image-editing programs are used to draw graphics and edit images, such as the ones that you take with a digital camera. The images that you create or edit with an image editor are saved in files.
- **Spreadsheets:** Spreadsheet programs are used to work with numerical data. Numbers and mathematical formulas can be inserted into the rows and columns of the spreadsheet. The spreadsheet can then be saved in a file for use later.
- **Games:** Many computer games keep data stored in files. For example, some games keep a list of player names with their scores stored in a file. These games typically display the players' names in order of their scores, from highest to lowest. Some games also allow you to save your current game status in a file so you can quit the game and then resume playing it later without having to start from the beginning.
- **Web browsers:** Sometimes when you visit a Web page, the browser stores a small file known as a *cookie* on your computer. Cookies typically contain information about the browsing session, such as the contents of a shopping cart.

Programs that are used in daily business operations rely extensively on files. Payroll programs keep employee data in files, inventory programs keep data about a company's products in files, accounting systems keep data about a company's financial operations in files, and so forth.

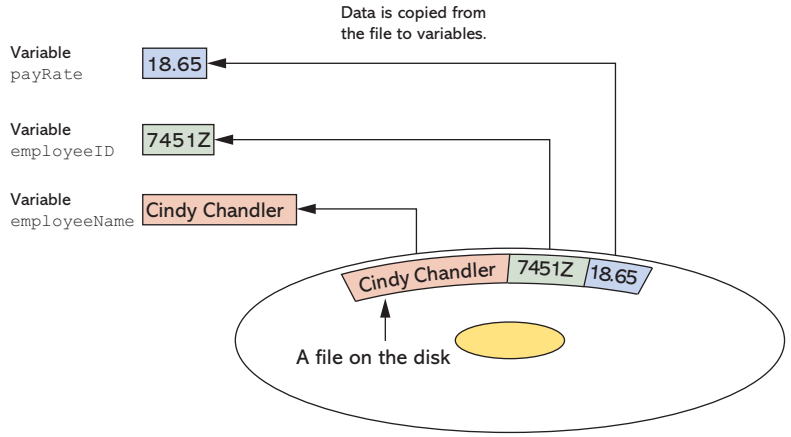
Programmers usually refer to the process of saving data in a file as *writing data* to the file. When a piece of data is written to a file, it is copied from a variable in RAM to the file. This is illustrated in Figure 5-14. The term **output file** is used to describe a file to which data is written. It is called an output file because the program stores output in it.

Figure 5-14 Writing data to a file



The process of retrieving data from a file is known as *reading data* from the file. When a piece of data is read from a file, it is copied from the file into a variable in RAM. Figure 5-15 illustrates this. The term **input file** is used to describe a file from which data is read. It is called an input file because the program gets input from the file.

Figure 5-15 Reading data from a file



In this section, we discuss ways to create programs that write data to files and read data from files. There are always three steps that must be taken when a file is used by a program.

1. **Open the file**—Opening a file creates a connection between the file and the program. Opening an output file usually creates the file on the disk and allows the program to write data to it. Opening an input file allows the program to read data from the file.
2. **Process the file**—In this step, data is either written to the file (if it is an output file) or read from the file (if it is an input file).
3. **Close the file**—When the program is finished using the file, the file must be closed. Closing a file disconnects the file from the program.

Types of Files

In general, there are two types of files: text and binary. A **text file** contains data that has been encoded as text using a scheme such as Unicode. Even if the file contains numbers, those numbers are stored in the file as a series of characters. As a result, the file may be opened and viewed in a text editor such as Notepad. A **binary file** contains data that has not been converted to text. As a consequence, you cannot view the contents of a binary file with a text editor. In this chapter, we work only with text files.

File Access Methods

Most programming languages provide two different ways to access data stored in a file: sequential access and direct access. When you work with a **sequential access file**, you access data from the beginning of the file to the end of the file. If you want to read a piece of data that is stored at the very end of the file, you have to read all the data that comes before it—you cannot jump directly to the desired data. This is similar to the way cassette tape players work. If you want to listen to the last song on a cassette tape, you have to either fast-forward over all of the songs that come before it or listen to them. There is no way to jump directly to a specific song.

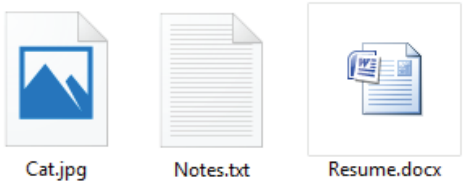
When you work with a **direct access file** (which is also known as a **random access file**), you can jump directly to any piece of data in the file without reading the data that comes before it. This is similar to the way a CD player or an MP3 player works. You can jump directly to any song you want to listen to.

This chapter focuses on sequential access files. Sequential access files are easy to work with, and you can use them to gain an understanding of basic file operations.

Filenames and File Objects

Files on a disk are identified by a **filename**. For example, when you create a document with a word processor and then save the document in a file, you have to specify a filename. When you use a utility such as Windows Explorer to examine the contents of your disk, you see a list of filenames. Figure 5-16 shows how three files named cat.jpg, notes.txt, and resume.docx might be represented in Windows Explorer.

Figure 5-16 Three files

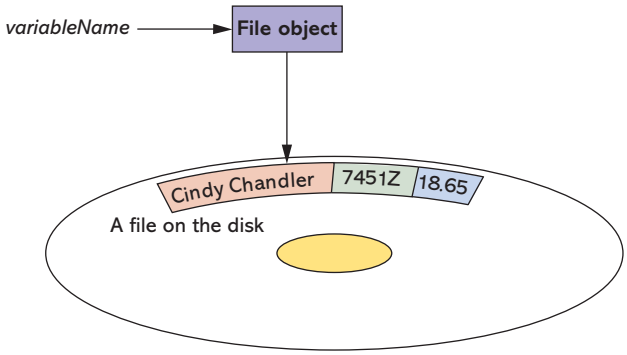


Each operating system has its own rules for naming files. Many systems, including Windows, support the use of **filename extensions**, which are short sequences of characters that appear at the end of a filename and are preceded by a period (which is known as a "dot"). For example, the files depicted in Figure 5-16 have the extensions .jpg, .txt, and .docx. The extension usually indicates the type of data stored in the file. For example, the .jpg extension usually indicates that the file contains a graphic image that is compressed according to the JPEG image standard. The .txt extension usually indicates that the file contains text. The .docx extension usually indicates that the file contains a Microsoft Word document.

In order for a program to work with a file on the computer's disk, the program must create a file object in memory. A **file object** is an object that is associated with a specific file and provides a way for the program to work with that file. In the program, a variable is

linked with the file object. We say that the variable *references* the object. This variable is used to carry out any operations that are performed on the file. This concept is shown in Figure 5-17.

Figure 5-17 A variable referencing a file object that is associated with a file



You will be using two .NET classes to create file objects. When you want to write data to a text file, you use the **`StreamWriter` class**, and when you want to read data from a text file you use the **`StreamReader` class**.

NOTE: In programming terminology, files are considered streams of data. In C#, you use a `StreamWriter` object to open a stream, such as a file, and write data to it. You use a `StreamReader` object to open a stream, such as a file, and read data from it.

Writing Data to a File with a `StreamWriter` Object

You can use the `StreamWriter` class's `WriteLine` method to write a line of text to a file. Let's suppose you want to write a program that creates a text file named courses.txt and writes the names of the courses you are taking to the file. The following code sample shows how you can do this:

```
1 StreamWriter outputFile;
2 outputFile = File.CreateText("courses.txt");
3
4 outputFile.WriteLine("Intro to Computer Science");
5 outputFile.WriteLine("English Composition");
6 outputFile.WriteLine("Calculus I");
7 outputFile.WriteLine("Music Appreciation");
8
9 outputFile.Close();
```

Let's look at each line of code.

Line 1: This statement declares a variable named `outputFile`, which can be used to reference a `StreamWriter` object.

Line 2: In a nutshell, this statement opens the file to which you will be writing data. It does so by calling the `File.CreateText` method, passing the string `"courses.txt"` as an argument. The `File.CreateText` method does the following:

- It creates a text file with the name specified by the argument. If the file already exists, its contents are erased.
- It creates a `StreamWriter` object in memory, associated with the file.
- It returns a reference to the `StreamWriter` object.

Notice that an assignment operator assigns the value returned from the `File.CreateText` method to the `outputFile` variable. This causes the `outputFile` variable to reference the `StreamWriter` object that was created by the method.

After the statement in line 2 executes, the courses.txt file is created on the disk, a `StreamWriter` object associated with the file exists in memory, and the `outputFile` variable references that object.

Line 4: This statement writes the string `"Intro to Computer Science"` to the courses.txt file. It does that by calling the `StreamWriter` class's `WriteLine` method, passing the string that is to be written to the file as an argument. When the `WriteLine` method writes data to a file, it writes a newline character immediately following the data. A **newline character** is an invisible character that specifies the end of a line of text.

Line 5: This statement writes the string `"English Composition"` to the courses.txt file.

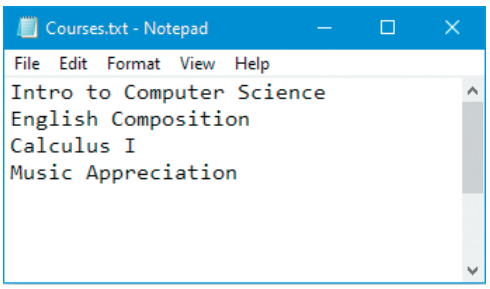
Line 6: This statement writes the string `"Calculus I"` to the courses.txt file.

Line 7: This statement writes the string `"Music Appreciation"` to the courses.txt file.

Line 9: This statement closes the courses.txt file. It does that by calling the `StreamWriter` class's `Close` method.

After this code has executed, we can open the courses.txt file using a text editor and look at its contents. Figure 5-18 shows how the file's contents will appear in Notepad.

Figure 5-18 Contents of the courses.txt file shown in Notepad



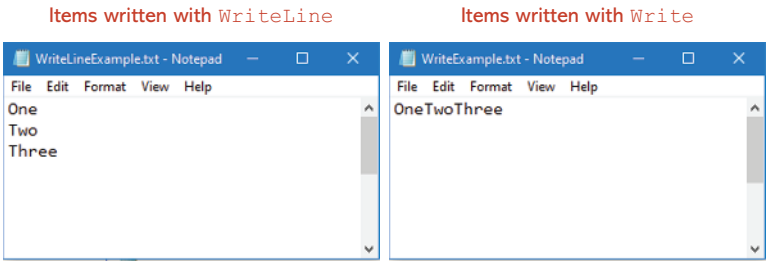
Writing Data with the `Write` Method

Earlier you read that the `StreamWriter` class's `WriteLine` method writes an item of data to a file and then writes a newline character. The newline character specifies the end of a line of text. For example, the following code sample opens a file named Example.txt and then uses the `WriteLine` method to write the strings `"One"`, `"Two"`, and `"Three"` to the file. Because a newline character is written after each string, the strings appear on separate lines when viewed in a text editor. The screen shown on the left in Figure 5-19 shows how the file would appear in Notepad.

```
1 StreamWriter outputFile;
2 outputFile = File.CreateText("Example.txt");
3
4 outputFile.WriteLine("One");
5 outputFile.WriteLine("Two");
6 outputFile.WriteLine("Three");
7
8 outputFile.Close();
```

In some situations, you might want to write an item to a file without a newline character immediately following it. The `StreamWriter` class provides the `Write` method for this

Figure 5-19 Items written with the `WriteLine` and `Write` methods



purpose. It writes an item of data to a text file without writing a newline character. The following code sample demonstrates. The screen shown on the right in Figure 5-19 shows how the resulting file would appear in Notepad.

```
1 StreamWriter outputFile;
2 outputFile = File.CreateText("Example.txt");
3
4 outputFile.Write("One");
5 outputFile.Write("Two");
6 outputFile.Write("Three");
7
8 outputFile.Close();
```

Handling File-Related Exceptions

Unexpected problems can potentially occur when working with files. For example, your program might not have sufficient rights to create a file when it calls the `File.CreateText` method, or the disk might be full when you call the `StreamWriter` class's `WriteLine` method. When unexpected errors such as these occur, an exception is thrown. To handle such exceptions, you can write a `try-catch` statement, with the code that performs file operations placed in the try block. Here is an example:

```
 1 try
 2 {
 3    StreamWriter outputFile;
 4    outputFile = File.CreateText("courses.txt");
 5
 6    outputFile.WriteLine("Intro to Computer Science");
 7    outputFile.WriteLine("English Composition");
 8    outputFile.WriteLine("Calculus I");
 9    outputFile.WriteLine("Music Appreciation");
10
11    outputFile.Close();
12 }
13 catch (Exception ex)
14 {
15    // Display an error message.
16    MessageBox.Show(ex.Message);
17 }
```

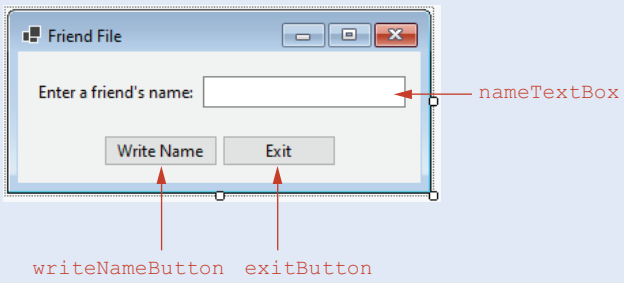
In Tutorial 5-4, you will complete an application that reads input from a TextBox control and writes the input to a file.

Tutorial 5-4:
Writing Data to a Text File

VideoNote
Tutorial 5-4: Writing Data to a Text File

In this tutorial, you complete the *FriendFile* project. The project has already been started for you and is located in the *Chap05* folder of this book's Student Sample Programs. The application's form is shown in Figure 5-20.

Figure 5-20 The *Friend File* form



When you complete the application, it will allow the user to enter a name into the `nameTextBox` control. When the user clicks the `writeNameButton` control, the application opens a text file named Friend.txt, writes the name that was entered into the TextBox control to the file, and then closes the file.

Step 1: Start Visual Studio. Open the project named *FriendFile* in the *Chap05* folder of this book's Student Sample Programs.

Step 2: Open the Form1 form in the *Designer*. The form is shown, along with the names of the important controls, in Figure 5-20. Double-click the `writeNameButton` control. This opens the code editor, and you will see an empty event handler named `writeNameButton_Click`. Complete the `writeNameButton_Click` event handler by typing the code shown in lines 12–34 in Program 5-4. Let's take a closer look at the code:

Line 12: This is the beginning of a `try-catch` statement. The try block appears in lines 14–27, and the catch block appears in lines 31–32. If an exception is thrown by any statement in the try block, the program jumps to the catch block, and line 32 displays an error message.

Line 15: This statement declares a `StreamWriter` variable named `outputFile`. You use this variable to reference a `StreamWriter` object.

Line 18: This statement calls the `File.CreateText` method to create a text file named Friend.txt. The method also creates a `StreamWriter` object in memory associated with the file. The method returns a reference to that object, which is assigned to the `outputFile` variable. As a result, the `outputFile` variable references the `StreamWriter` object. You will be able to use the `outputFile` variable to perform operations on the Friend.txt file.

Line 21: This statement uses the `outputFile` variable to call the `StreamWriter` class's `WriteLine` method. The `nameTextBox` control's Text property is passed as an argument. As a result, the value entered into the TextBox is written to the Friend.txt file.

Line 24: This statement closes the Friend.txt file.

Line 27: This statement displays a message box to let the user know that the name was written to the file.

Step 3: Switch your view back to the *Designer* and double-click the `exitButton` control. In the code editor you will see an empty event handler named `exitButton_Click`. Complete the `exitButton_Click` event handler by typing the code shown in lines 38–39 in Program 5-4.

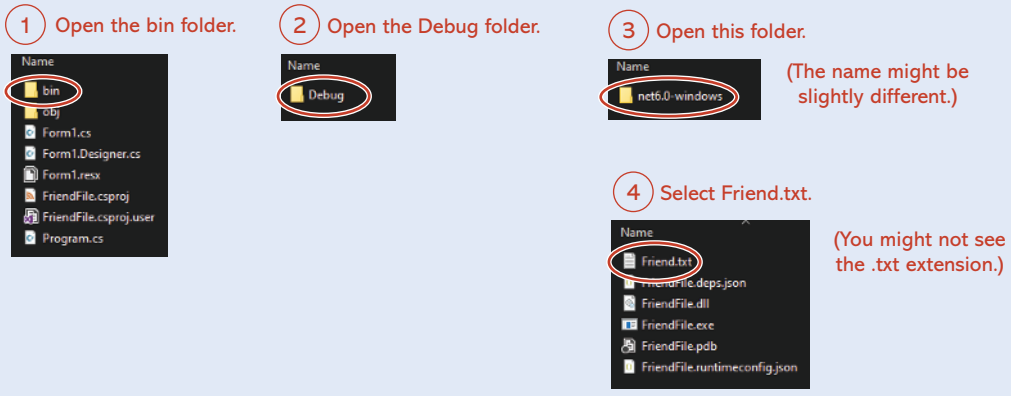
Step 4: Save the project. Then, press F5 on the keyboard or click the *Start Debugging* button ( ) on the toolbar to compile and run the application.

Enter a name into the `nameTextBox` control, and then click the *Write Name* button. You should see a message box appear letting you know that the name was written to the file. Click the *OK* button to dismiss the message box; then click the *Exit* button on the application's form to end the application.

Step 5: Now you will look at the contents of the Friend.txt file that the application created. Click *File* on the menu bar, then click *Open*, and then click *File*...

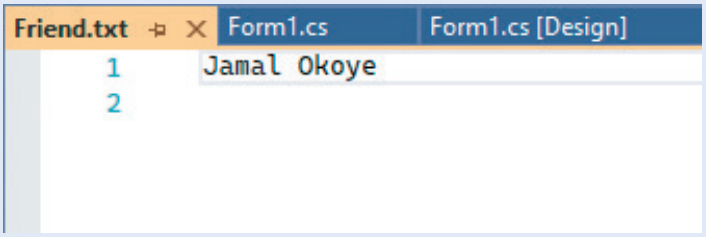
Step 6: You should now see the *Open File* window, viewing the contents of the *Friend File* project folder. As shown in Figure 5-21, open the *bin* folder, then open the *Debug* folder, then open the *netX.X-windows* folder, and then select the file *Friend.txt*. (You might not see the .txt extension, depending on how your system is set up.) Click the *Open* button.

Figure 5-21 Opening the Friend.txt file in the *Open File* window



Step 7: You should now see the contents of the Friend.txt file in Visual Studio, as shown in Figure 5-22. (The example in the figure shows the contents of the file after the user has written *Jamal Okoye* to the file.) When you are finished examining the contents of the file, you can close its tab. We come back to this project in the next tutorial.

Figure 5-22 Sample contents of the Friend.txt file shown in Visual Studio



Program 5-4 Completed Form1 code for the *FriendFile* project

```
1 namespace FriendFile
2 {
3     public partial class Form1: Form
4     {
5         public Form1()
6         {
7             InitializeComponent();
8         }
9
10        private void writeNameButton_Click(object sender,
          EventArgs e)
11        {
12            try
13            {
14                // Declare a StreamWriter variable.
15                StreamWriter outputFile;
16
17                // Create a file and get a StreamWriter object.
18                outputFile = File.CreateText("Friend.txt");
19
20                // Write the friend's name to the file.
21                outputFile.WriteLine(nameTextBox.Text);
22
23                // Close the file.
24                outputFile.Close();
25
26                // Let the user know the name was written.
27                MessageBox.Show("The name was written.");
28            }
29            catch (Exception ex)
30            {
31                // Display an error message.
32                MessageBox.Show(ex.Message);
33            }
34        }
35
36        private void exitButton_Click(object sender, EventArgs e)
37        {
38            // Close the form.
39            this.Close();
40        }
41    }
42 }
```

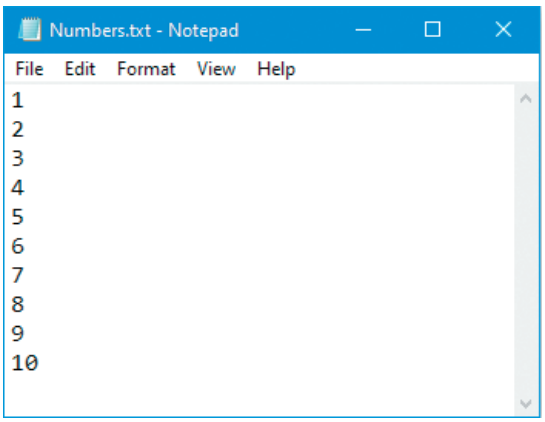
Writing Numeric Data to a Text File

You can use the `StreamWriter` class's `WriteLine` or `Write` method to write numbers (such as `int`s, `double`s, and `decimal`s) to a text file, but the numbers are converted to strings. For example, look at the following sample code (taken from the *NumberFile* project in the *Chap05* folder of the Student Sample Programs):

```
 1 private void writeNumbersButton_Click(object sender, EventArgs e)
 2 {
 3    try
 4    {
 5        // Declare a StreamWriter variable.
 6        StreamWriter outputFile;
 7
 8        // Create a file and get a StreamWriter object.
 9        outputFile = File.CreateText("Numbers.txt");
10
11        // Write the numbers 1 through 10 to the file.
12        for (int count = 1; count <= 10; count++)
13        {
14            outputFile.WriteLine(count);
15        }
16
17        // Close the file.
18        outputFile.Close();
19     }
20     catch (Exception ex)
21     {
22        // Display an error message.
23        MessageBox.Show(ex.Message);
24     }
25 }
```

When this event handler executes, line 9 creates a text file named Numbers.txt, and the loop in lines 12–15 writes the numbers 1–10 to the file. Figure 5-23 shows how the file appears when opened with Notepad.

Figure 5-23 The Numbers.txt file opened in Notepad



Appending Data to an Existing File

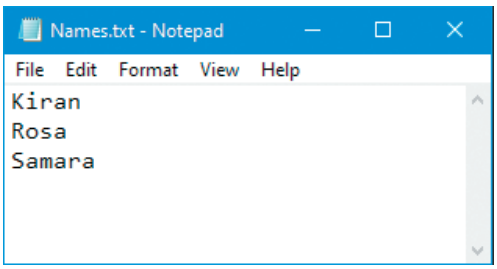
When you call the `File.CreateText` method to open a file and the file specified by the argument already exists, it is erased and a new empty file with the same name is created. For example, when you run the *Friend File* application that you completed in Tutorial 5-4, each time you click the *Write Name* button, the Friend.txt file is erased and a new file is created.

Sometimes you want to preserve an existing file and append new data to its current contents. To append data to an existing file, you open it with the `File.AppendText` method.

It works like the `File.CreateText` method, but the file is not erased if it already exists. Any data written to the file is appended to the file's existing contents.

For example, assume the file Names.txt exists and contains the data shown in Figure 5-24:

Figure 5-24 Names.txt file

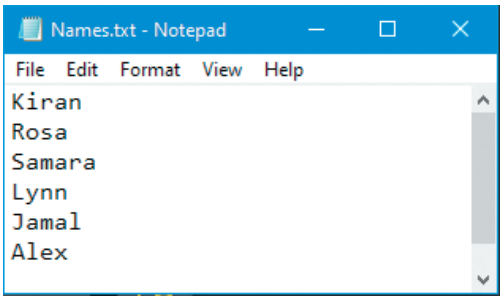


The following code opens the file and appends additional data to its existing contents:

```
1 StreamWriter outputFile;
2 outputFile = File.AppendText("Names.txt");
3
4 outputFile.WriteLine("Lynn");
5 outputFile.WriteLine("Jamal");
6 outputFile.WriteLine("Alex");
7
8 outputFile.Close();
```

After this code executes, the Names.txt file contains the data shown in Figure 5-25:

Figure 5-25 Names.txt file after data has been appended



In Tutorial 5-5, you will modify the code that you wrote in Tutorial 5-4 so the loop appends data to the Friend.txt file.

Tutorial 5-5:
Appending Data to the Friend.txt File

VideoNote
Tutorial 5-5: Appending Data to the Friend.txt File

In this tutorial, you will modify the *FriendFile* project so it appends data to the Friend.txt file. When the user clicks the *Write Name* button, instead of erasing the file's current contents, the application adds the contents of the `nameTextBox` control to the Friend.txt file.

Step 1: If the *FriendFile* project from Tutorial 5-4 is not currently open in Visual Studio, open it now.

Step 2: Open the Form1 form's code in the code editor. You will make modifications to the `writeNameButton_Click` event handler. Program 5-5, at the end of

this tutorial, shows how the event handler code will appear after you make the following changes:

Lines 8 and 9: Change the comments as shown to reflect the way that the Friend.txt file will be opened in line 10.

Line 10: Change this statement so it calls the `File.AppendText` method instead of the `File.CreateText` method.

Lines 21–25: Add the new comments and statements shown in these lines. Line 22 clears the `nameTextBox` control's contents, and line 25 gives the focus to the `nameTextBox` control. This makes the application more convenient for adding several names to the file.

Step 3: Save the project. Then, press F5 on the keyboard or click the *Start Debugging* button (▶) on the toolbar to compile and run the application.

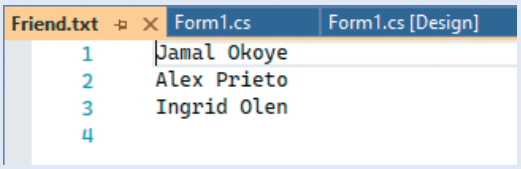
Enter a name into the `nameTextBox` control and then click the *Write Name* button. You should see a message box letting you know that the name was written to the file. When you click the *OK* button to dismiss the message box, notice that the `nameTextBox` is cleared, and the focus is given to the TextBox. Enter another name, and click the *Write Name* button. Again, you see the message box. Click the *OK* button to dismiss the message box and then click the *Exit* button to end the application.

Step 4: Now you will look at the contents of the Friend.txt file. Click *File* on the menu bar, then click *Open*, and then click *File*.

You should see the *Open File* window, viewing the contents of the *Friend File* project folder. Open the *bin* folder, then open the *Debug* folder, and then select the file *Friend.txt*. (You might not see the .txt extension, depending on how your system is set up.) Click the *Open* button.

Step 5: You should now see the contents of the Friend.txt file in Visual Studio. Figure 5-26 shows an example. Notice that the names that you entered were appended to the file each time you clicked the *Write Name* button. When you are finished examining the contents of the file, you can close its tab. We come back to this project in the next tutorial.

Figure 5-26 Sample contents of the Friend.txt file shown in Visual Studio



Program 5-5 Partial code for Form1 in the *FriendFile* project

```
1 private void writeNameButton_Click(object sender, EventArgs e)
2 {
3    try
4    {
5         // Declare a StreamWriter variable.
6         StreamWriter outputFile;
7
8         // Open the Friend.txt file for appending,
9         // and get a StreamWriter object.
10        outputFile = File.AppendText("Friend.txt");
```

```
11
12          // Write the friend's name to the file.
13          outputFile.WriteLine(nameTextBox.Text);
14
15          // Close the file.
16          outputFile.Close();
17
18          // Let the user know the name was written.
19          MessageBox.Show("The name was written.");
20
21          // Clear the nameTextBox control.
22          nameTextBox.Text =  "";
23
24          // Give the focus to the nameTextBox control.
25          nameTextBox.Focus();
26      }
27      catch (Exception ex)
28      {
29          // Display an error message.
30          MessageBox.Show(ex.Message);
31      }
32  }
```

Specifying the Location of an Output File

When you call the `File.CreateText` or `File.AppendText` methods to open a file, you pass the filename as a string argument. If the filename that you pass as an argument does not contain the file's path, the file's location will be the *bin\Debug* folder, under the application's project folder. You saw this in Tutorials 5-4 and 5-5 when you opened the Friend.txt file in Visual Studio.

If you want to open a file in a different location, you can specify a path as well as a filename in the argument that you pass to the `File.CreateText` or `File.AppendText` method. If you specify a path in a string literal, be sure to prefix the string with the `@` character. Here is an example:

```
StreamWriter outputFile;
outputFile = File.CreateText(@"C:\Users\Chris\Documents\Names.txt");
```

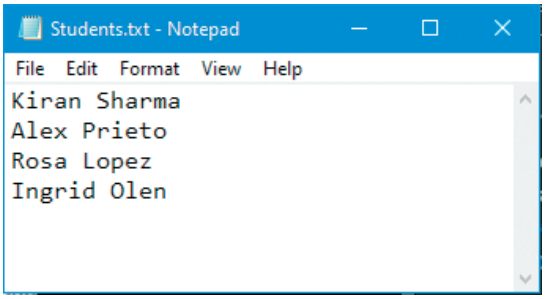
TIP: You can also let the user specify the file location. See Section 5.6 for more information on the SaveFileDialog control.

Reading Data from a File with a `StreamReader` Object

To read data from a text file, you create a `StreamReader` object. You can then use the `StreamReader` class's `ReadLine` method to read a line of text from a file. For example, suppose a file named Students.txt exists and contains the four names shown in Figure 5-27.

Let's suppose you want to write a program that reads the four names from the Students.txt file and displays them in message boxes. The following code sample shows how you can do this. (This code sample is taken from the *StudentNames* project, in the *Chap05* folder of this book's Student Sample Programs.)

Figure 5-27 Contents of the Students.txt file




```
 1 try
 2 {
 3    // Declare a variable to hold an item read from the file.
 4    string studentName;
 5
 6    // Declare a StreamReader variable.
 7    StreamReader inputFile;
 8
 9    // Open the file and get a StreamReader object.
10    inputFile = File.OpenText("Students.txt");
11
12     // Read and display the first name.
13     studentName = inputFile.ReadLine();
14     MessageBox.Show(studentName);
15
16     // Read and display the second name.
17     studentName = inputFile.ReadLine();
18     MessageBox.Show(studentName);
19
20     // Read and display the third name.
21     studentName = inputFile.ReadLine();
22     MessageBox.Show(studentName);
23
24     // Read and display the fourth name.
25     studentName = inputFile.ReadLine();
26     MessageBox.Show(studentName);
27
28     // Close the file.
29     inputFile.Close();
30 }
31 catch (Exception ex)
32 {
33     // Display an error message.
34     MessageBox.Show(ex.Message);
35 }
```

Let's take a closer look at the code.

Line 1: This is the beginning of a `try-catch` statement. An exception will be thrown in the try block if a problem occurs while the file is being opened or while an item is being read from the file. If that happens, the program jumps to the `catch` clause in line 31.

Line 4: This statement declares a `string` variable named `studentName`. Each time we read a line of text from the file, we assign it to this variable.

Line 7: This statement declares a variable named `inputFile` that can be used to reference a `StreamReader` object.

Line 10: This statement opens the file from which we will be reading data. It does so by calling the `File.OpenText` method, passing the string `"Students.txt"` as an argument. The `File.OpenText` method does the following:

- It opens an existing text file with the name specified by the argument. If the file does not exist, an exception is thrown.
- It creates a `StreamReader` object in memory associated with the file.
- It returns a reference to the `StreamReader` object.

Notice that an assignment operator assigns the value returned from the `File.OpenText` method to the `inputFile` variable. This causes the `inputFile` variable to reference the `StreamReader` object that was created by the method.

Line 13: This statement calls the `inputFile.ReadLine` method, which reads a line of text from the file. The line of text is returned as a string from the method and assigned to the `studentName` variable. Since this statement reads the first line of text from the file, the `studentName` variable is assigned the string `"Kiran Sharma"`.

Line 14: This statement displays the contents of the `studentName` variable in a message box.

Lines 17 and 18: The statement in line 17 reads the next line of text from the file and assigns it to the `studentName` variable. After this line executes, the `studentName` variable is assigned the string `"Alex Prieto"`. The statement in line 18 displays the contents of the `studentName` variable in a message box.

Lines 21 and 22: The statement in line 21 reads the next line of text from the file and assigns it to the `studentName` variable. After this line executes, the `studentName` variable is assigned the string `"Rosa Lopez"`. The statement in line 22 displays the contents of the `studentName` variable in a message box.

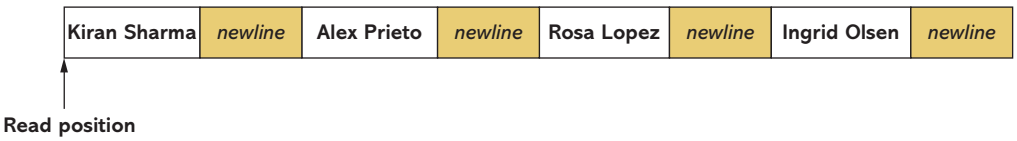
Lines 25 and 26: The statement in line 25 reads the next line of text from the file and assigns it to the `studentName` variable. After this line executes, the `studentName` variable is assigned the string `"Ingrid Olen"`. The statement in line 26 displays the contents of the `studentName` variable in a message box.

Line 29: This statement closes the Students.txt file.

The Read Position

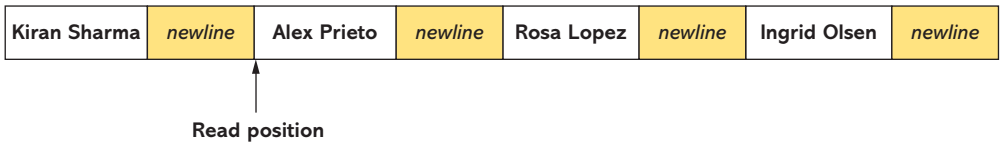
When a program works with an input file, a special value known as a **read position** is internally maintained for that file. A file's read position marks the location of the next item that will be read from the file. When an input file is opened, its read position is initially set to the first item in the file. As items are read from the file, the read position moves forward, toward the end of the file. Let's see how this works in the previous code sample (from the *StudentNames* project). After the statement in line 10 executes, the read position for the Students.txt file is positioned as shown in Figure 5-28.

Figure 5-28 The initial read position



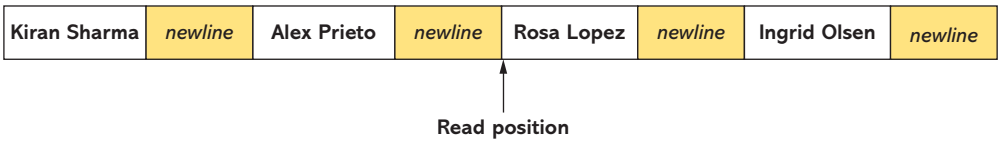
The `ReadLine` method call in line 13 reads an item from the file's current read position and assigns that item to the `studentName` variable. Once this statement executes, the `studentName` variable is assigned the string `"Kiran Sharma"`. In addition, the file's read position is advanced to the next item in the file, as shown in Figure 5-29.

Figure 5-29 Read position after the first `ReadLine` method call



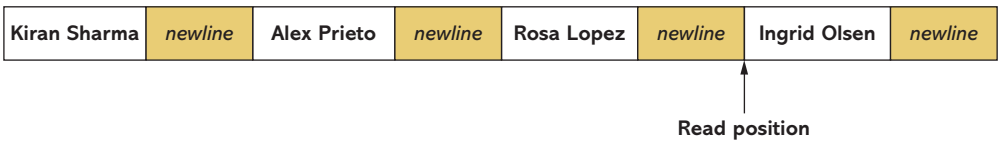
The `ReadLine` method call in line 17 reads an item from the file's current read position and assigns that value to the `studentName` variable. Once this statement executes, the `studentName` variable is assigned the string `"Alex Prieto"`. The file's read position is advanced to the next item, as shown in Figure 5-30.

Figure 5-30 Read position after the second `ReadLine` method call



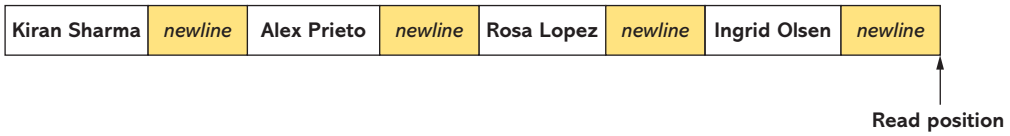
The `ReadLine` method call in line 21 reads an item from the file's current read position and assigns that value to the `studentName` variable. Once this statement executes, the `studentName` variable is assigned the string `"Rosa Lopez"`. The file's read position is advanced to the next item, as shown in Figure 5-31.

Figure 5-31 Read position after the third `ReadLine` method call



The last `ReadLine` method call appears in line 25. It reads an item from the file's current read position and assigns that value to the `studentName` variable. Once this statement executes, the `studentName` variable is assigned the string `"Ingrid Olsen"`. The file's read position is advanced to the end of the file, as shown in Figure 5-32.

Figure 5-32 Read position after the fourth `ReadLine` method call



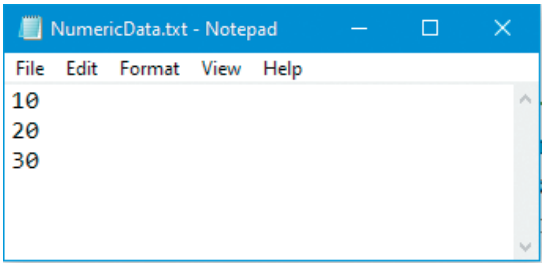
NOTE: Did you notice that the previous code sample read the items in the Students.txt file in sequence, from the beginning of the file to the end of the file? Recall from our discussion at the beginning of the section that this is the nature of a sequential access file.

Reading Numeric Data from a Text File

Remember that when data is stored in a text file, it is encoded as text, using a scheme such as Unicode. Even if the file contains numbers, those numbers are stored in the file as a series of characters. Furthermore, when you read an item from a text file with the `StreamReader` class's `ReadLine` method, that item is returned as a string.

Suppose a text file contains numeric data, such as that shown in Figure 5-33. When we use the `ReadLine` method to read the items from the file, we get the strings `"10"`, `"20"`, and `"30"`. If we need to perform math with these values, we must convert each value from a string to a numeric data type. We can use the `Parse` or `TryParse` families of methods that you already know about to perform this conversion.

Figure 5-33 A text file containing numeric data



Let's suppose you want to write a program that reads the three numbers from the NumericData.txt file shown in Figure 5-33 and displays their total in a message box. The following code sample shows a simple demonstration. (This code sample is taken from the *NumericData* project in the *Chap05* folder of this book's Student Sample Programs.)

```
1  try
2  {
3      // Variables to hold the numbers read from the file
4      // and their total
5      int number1, number2, number3, total;
6
7      // A StreamReader variable.
8      StreamReader inputFile;
9
10     // Open the file and get a StreamReader object.
11     inputFile = File.OpenText("NumericData.txt");
12
13     // Read three numbers from the file.
14     number1 = int.Parse(inputFile.ReadLine());
15     number2 = int.Parse(inputFile.ReadLine());
16     number3 = int.Parse(inputFile.ReadLine());
17
18     // Calculate the total of the numbers.
19     total = number1 + number2 + number3;
20
21     // Display the total.
22     MessageBox.Show("The total is " + total);
23
24     // Close the file.
25     inputFile.Close();
26 }
27 catch (Exception ex)
28 {
29     // Display an error message.
30     MessageBox.Show(ex.Message);
31 }
```

Let's take a closer look at the code.

Line 1: This is the beginning of a `try-catch` statement. Various exceptions can be thrown by the code in the try block, which cause the program to jump to the `catch` clause in line 27.

Line 5: This statement declares the `int` variables `number1`, `number2`, `number3`, and `total`. These variables hold the three values read from the file and their total.

Lines 8–11: After these statements have executed, the NumericData.txt file is opened for reading, and the `inputFile` variable references a `StreamReader` object that is associated with the file.

Line 14: This statement does the following:

- It calls the `inputFile.ReadLine` method to read a line of text from the file.
- The value that is returned from the `inputFile.ReadLine` method (a string) is passed as an argument to the `int.Parse` method.
- The value that is returned from the `int.Parse` method is assigned to the `number1` variable.

After this statement executes, the `number1` variable is assigned the first value read from the file, converted to an `int`. (The `number1` variable is assigned the value 10.)

Line 15: This statement reads the next value from the file, converts it to an `int`, and assigns the result to the `number2` variable. (The `number2` variable is assigned the value 20.)

Line 16: This statement reads the next value from the file, converts it to an `int`, and assigns the result to the `number3` variable. (The `number3` variable is assigned the value 30.)

Line 19: This statement calculates the sum of `number1`, `number2`, and `number3`, and assigns the result to `total`.

Line 22: This statement displays the sum of the numbers in a message box.

Line 25: This statement closes the file.

Reading a File with a Loop and Detecting the End of the File

Quite often a program must read the contents of a file without knowing the number of items that are stored in the file. For example, suppose you need to write a program that displays all the items in a file, but you do not know how many items the file contains. You can open the file and then use a loop to repeatedly read an item from the file and display it. However, an exception will be thrown if the program attempts to read beyond the end of the file. The program needs some way of knowing when the end of the file has been reached so it will not try to read beyond it. The following pseudocode shows the logic:

```
Open the file
While not at the end of the file:
   Read an item from the file
   Display the item
End While
Close the file
```

`StreamReader` objects have a Boolean property named EndOfStream that signals whether the end of the file has been reached. If the file's read position is at the end of the file (and there is no more data to read), the EndOfStream property is set to true. Otherwise, it is set to false. When you need to read all the items in a file without knowing how many items the file contains, you can write a loop that iterates as long as the EndOfStream property is false.

Let's assume `inputFile` references a `StreamReader` object that is associated with a file that is already open. You can write the loop in the following manner:

```
while (inputFile.EndOfStream == false)
{
   // Read an item from the file.
   // Process the item.
}
```

However, most programmers prefer the following logic, which uses the `!` operator:

```
while (!inputFile.EndOfStream)
{
   // Read an item from the file.
   // Process the item.
}
```

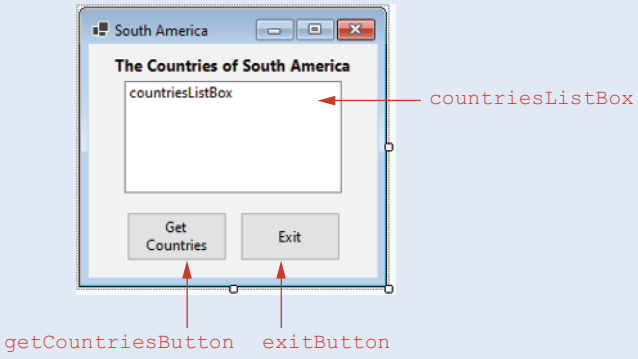
Recall that the `!` operator is the logical NOT operator. When you read the first line of this loop, you naturally think *while NOT at the end of the stream*. In Tutorial 5-6, you will complete an application that uses this technique to display all the items in a file.

Tutorial 5-6:
Using a Loop to Read to the End of a File

VideoNote
Tutorial 5-6: Using a Loop to Read to the End of a File

In this tutorial, you complete the *SouthAmerica* project that is found in the *Chap05* folder of this book's Student Sample Programs. The application's form has already been created and is shown in Figure 5-34. The project also includes an accompanying text file named Countries.txt. The Countries.txt file contains the names of the countries of South America. Figure 5-35 shows the file as it appears in Notepad.

Figure 5-34 The *South America* form

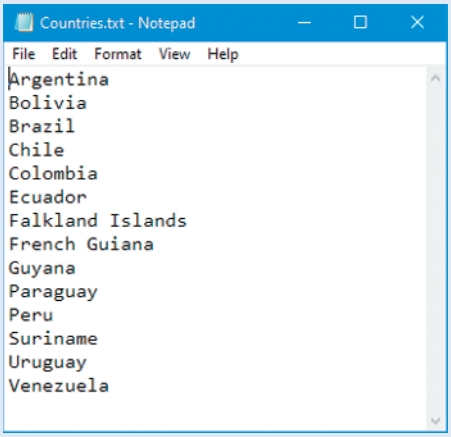


When the completed application runs and the user clicks the *Get Countries* button, the application reads each country name from the file and adds each one to the `countriesListBox` control.

Step 1: Start Visual Studio. Open the project named *SouthAmerica* in the *Chap05* folder of this book's Student Sample Programs.

Step 2: Open the Form1 form in the *Designer*. The form is shown, along with the names of the important controls, in Figure 5-34. Double-click the `getCountriesButton` control. This opens the code editor, and you will see an empty

Figure 5-35 The Countries.txt file



event handler named `getCountriesButton_Click`. Complete the `getCountriesButton_Click` event handler by typing the code shown in lines 12–43 in Program 5-6. Let's take a closer look at the code:

Line 12: This is the beginning of a `try-catch` statement, which handles any exceptions that are thrown while the file is being processed. If an exception is thrown by any statement in the try block, the program jumps to the `catch` clause in line 39.

Line 15: This statement declares the `string` variable `countryName`, which holds the lines of text that are read from the file.

Lines 18–21: After these statements have executed, the Countries.txt file is opened for reading, and the `inputFile` variable references a `StreamReader` object that is associated with the file.

Line 24: This statement clears anything that might be displayed in the `countriesListBox` control. (This prevents the names of the countries from appearing multiple times in the ListBox if the user clicks the *Get Countries* button multiple times.)

Line 27: This is the beginning of a `while` loop that iterates as long as the end of the Countries.txt file has not been reached.

Line 30: This statement reads a line of text from the file and assigns it to the `countryName` variable.

Line 33: This statement adds the contents of the `countryName` variable to the ListBox.

Line 37: This statement closes the file.

Step 3: Switch your view back to the *Designer* and double-click the `exitButton` control. In the code editor you will see an empty event handler named `exitButton_Click`. Complete the `exitButton_Click` event handler by typing the code shown in lines 48–49 in Program 5-6.

Step 4: Save the project. Then, press F5 on the keyboard or click the *Start Debugging* button (▶) on the toolbar to compile and run the application. When the application runs, click the *Get Countries* button. This should fill the ListBox with the names of the countries from the Countries.txt file, as shown in Figure 5-36. Click the *Exit* button to exit the application.

Figure 5-36 The *South America* application displaying the list of countries

Program 5-6 Completed code for Form1 in the *SouthAmerica* project

```
 1 namespace SouthAmerica
 2 {
 3     public partial class Form1: Form
 4     {
 5         public Form1()
 6         {
 7             InitializeComponent();
 8         }
 9
10         private void getCountriesButton_Click(object sender,
           EventArgs e)
11         {
12             try
13             {
14                 // Declare a variable to hold a country name.
15                 string countryName;
16
17                 // Declare a StreamReader variable.
18                 StreamReader inputFile;
19
20                 // Open the file and get a StreamReader object.
21                 inputFile = File.OpenText("Countries.txt");
22
23                 // Clear anything currently in the ListBox.
24                 countriesListBox.Items.Clear();
25
26                 // Read the file's contents.
27                 while (!inputFile.EndOfStream)
28                 {
29                     // Get a country name.
30                     countryName = inputFile.ReadLine();
31
32                     // Add the country name to the ListBox.
33                     countriesListBox.Items.Add(countryName);
34                 }
35
36                 // Close the file.
37                 inputFile.Close();
38             }
39             catch (Exception ex)
```

```
40              {
41                  // Display an error message.
42                  MessageBox.Show(ex.Message);
43              }
44          }
45
46          private void exitButton_Click(object sender, EventArgs e)
47          {
48              // Close the form.
49              this.Close();
50          }
51      }
52  }
```

Calculating a Running Total

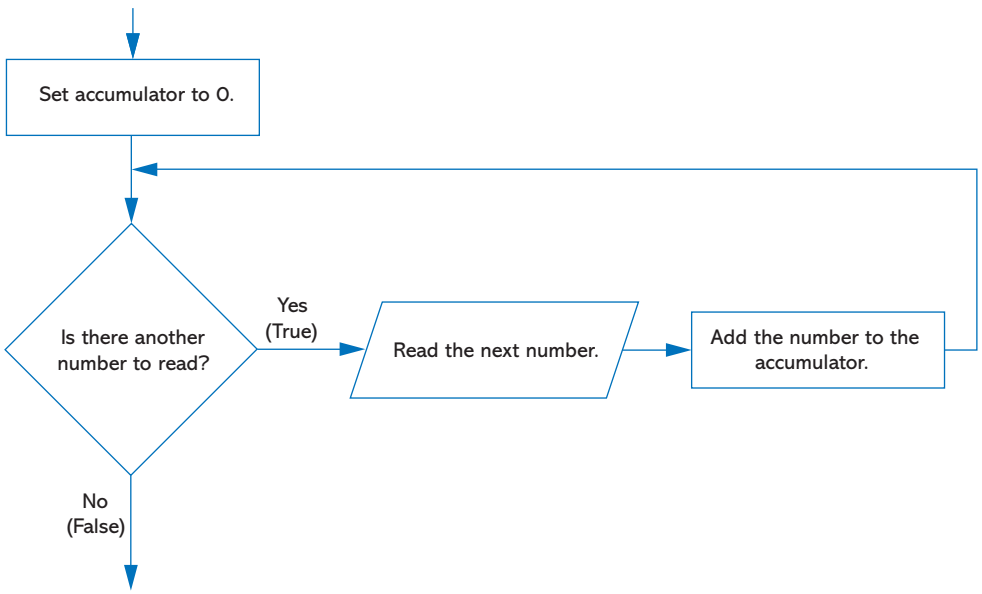
Many programming tasks require you to calculate the total of a series of numbers. In this section you learn how to calculate the total of a series of numbers that are stored in a file. For example, suppose you have a file that contains a business's sales for each day of a week and you need to write a program that calculates the total of all the amounts in the file. The program would read the values in the file and keep a total of the values as they are read.

Programs that calculate the total of a series of numbers typically use two elements:

- A loop that reads each number in the series
- A variable that accumulates the total of the numbers as they are read

The variable that is used to accumulate the total of the numbers is called an **accumulator**. It is often said that the loop keeps a **running total** because it accumulates the total as it reads each number in the series. Figure 5-37 shows the general logic of a loop that calculates a running total.

Figure 5-37 Logic for calculating a running total



When the loop finishes, the accumulator will contain the total of the numbers that were read by the loop. Notice that the first step in the flowchart is to set the accumulator variable to 0. This is a critical step. Each time the loop reads a number, it adds it to the accumulator. If the accumulator starts with any value other than 0, it will not contain the correct total when the loop finishes.

In Tutorial 5-7, you will complete an application that calculates a running total of the values in a file.

Tutorial 5-7:
Calculating a Running Total

VideoNote
Tutorial 5-7: Calculating a Running Total

In this tutorial, you complete the *TotalSales* project that is found in the *Chap05* folder of this book's Student Sample Programs. The application's form has already been created and is shown in Figure 5-38. The project also includes an accompanying text file named Sales.txt. The Sales.txt file contains the amounts shown in Figure 5-39.

Figure 5-38 The *Total Sales* form

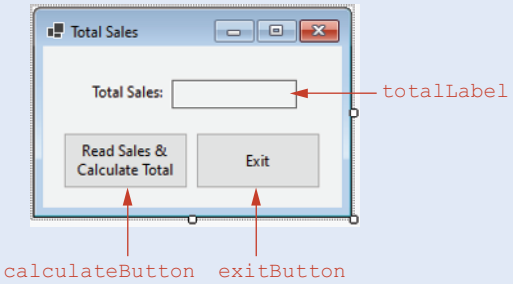
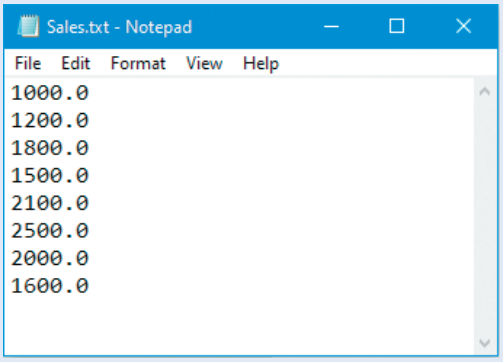


Figure 5-39 Contents of the Sales.txt file



When the completed application runs and the user clicks the *Read Sales & Calculate Total* button, the application calculates the total of the values in the Sales.txt file and displays the total in the `totalLabel` control.

Step 1: Start Visual Studio. Open the project named *TotalSales* in the *Chap05* folder of this book's Student Sample Programs.

Step 2: The Form1 form is shown, along with the names of the important controls, in Figure 5-38. Double-click the `calculateButton` control. This opens the code editor, and you will see an empty event handler named `calculateButton_Click`.

Complete the `calculateButton_Click` event handler by typing the code shown in lines 12–45 in Program 5-7. Let's take a closer look at the code:

Line 12: This is the beginning of a `try-catch` statement, which handles any exceptions that are thrown while the file is being processed. If an exception is thrown by any statement in the try block, the program jumps to the `catch` clause in line 41.

Lines 15–16: These statements declare the `decimal` variables `sales` and `total`. The `sales` variable holds each value that is read from the file, and the `total` variable is used as an accumulator. Notice that the `total` variable is explicitly initialized to 0.

Lines 19–22: After these statements have executed, the Sales.txt file is opened for reading, and the `inputFile` variable references a `StreamReader` object that is associated with the file.

Line 25: This is the beginning of a `while` loop that iterates as long as the end of the Sales.txt file has not been reached.

Line 28: This statement reads a line of text from the file, converts it to a decimal, and assigns the result to the `sales` variable.

Line 31: This statement adds the `sales` variable to the `total` variable.

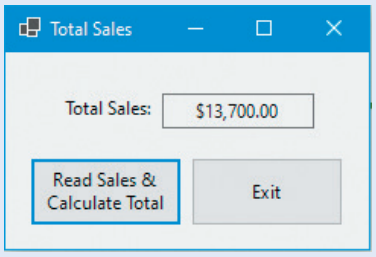
Line 35: This statement closes the file.

Line 38: This statement displays the total, formatted as currency, in the `totalLabel` control.

Step 3: Switch your view back to the *Designer* and double-click the `exitButton` control. In the code editor you will see an empty event handler named `exitButton_Click`. Complete the `exitButton_Click` event handler by typing the code shown in lines 50–51 in Program 5-7.

Step 4: Save the project. Then, press F5 on the keyboard or click the *Start Debugging* button (▶) on the toolbar to compile and run the application. When the application runs, click the *Read Sales & Calculate Total* button. The total sales should be calculated and displayed, as shown in Figure 5-40. Click the *Exit* button to exit the application.

Figure 5-40 The *Total Sales* application displaying the total sales



Program 5-7 Completed code for Form1 in the *TotalSales* project

```
1 namespace TotalSales
2 {
3     public partial class Form1: Form
4     {
5         public Form1()
```

```
 6         {
 7             InitializeComponent();
 8         }
 9
10         private void calculateButton_Click(object sender,
           EventArgs e)
11         {
12             try
13             {
14                 // Variables
15                 decimal sales;        // To hold a sales amount
16                 decimal total = 0m;   // Accumulator, set to 0
17
18                 // Declare a StreamReader variable.
19                 StreamReader inputFile;
20
21                 // Open the file and get a StreamReader object.
22                 inputFile = File.OpenText("Sales.txt");
23
24                 // Read the file's contents.
25                 while (!inputFile.EndOfStream)
26                 {
27                     // Get a sales amount.
28                     sales = decimal.Parse(inputFile.ReadLine());
29
30                     // Add the sales amount to total.
31                     total += sales;
32                 }
33
34                 // Close the file.
35                 inputFile.Close();
36
37                 // Display the total.
38                 totalLabel.Text = total.ToString("C");
39
40             }
41             catch (Exception ex)
42             {
43                 // Display an error message.
44                 MessageBox.Show(ex.Message);
45             }
46         }
47
48         private void exitButton_Click(object sender, EventArgs e)
49         {
50             // Close the form.
51             this.Close();
52         }
53     }
54 }
```

Checkpoint

5.15 What is an output file?

5.16 What is an input file?

5.17 What three steps must be taken by a program when it uses a file?

5.18 What is the difference between a text file and a binary file?

5.19 What are the two types of file access? What is the difference between these two?

5.20 What type of object do you create if you want to write data to a text file?

5.21 What type of object do you create if you want to read data from a text file?

5.22 If you call the `File.CreateText` method and the specified file already exists, what happens to the existing file?

5.23 If you call the `File.AppendText` method and the specified file already exists, what happens to the existing file?

5.24 What is the difference between the `WriteLine` and `Write` methods discussed in this chapter?

5.25 What method do you call to open a text file to read data from it?

5.26 What is a file's read position? Initially, where is the read position when an input file is opened?

5.27 How do you read a line of text from a text file?

5.28 How do you close a file?

5.29 Assume `inputFile` references a `StreamReader` object that is associated with an open file. Which of the following loops is written in the correct general format to read all of the items from the file?

Loop A:

```
while (inputFile.EndOfStream)
{
   // Read an item from the file.
}
```

Loop B:

```
while (!inputFile.EndOfStream)
{
   // Read an item from the file.
}
```

5.6 The OpenFileDialog and SaveFileDialog Controls

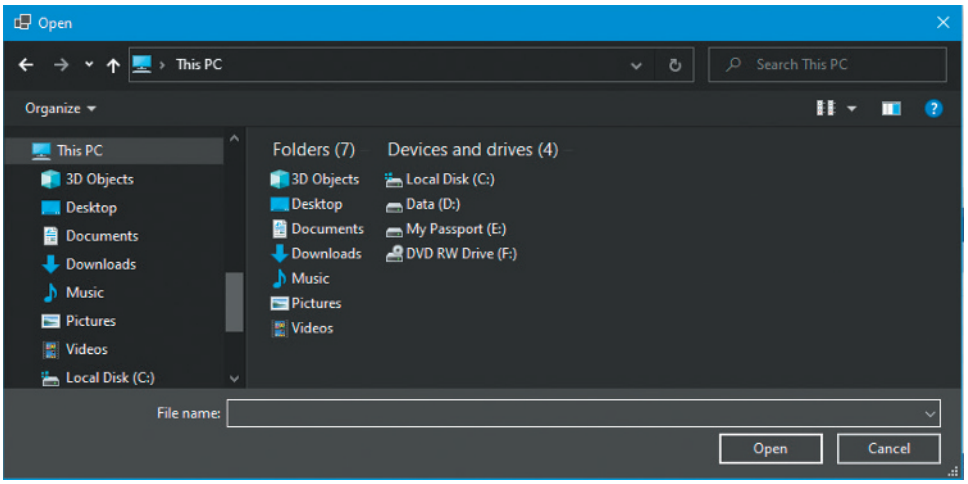
CONCEPT: **The OpenFileDialog and SaveFileDialog controls allow your application to display standard Windows dialog boxes for opening and saving files. These allow the user to easily specify a file's name and location.**

So far, the applications in this chapter that open a file specify the filename as a string literal. Most Windows users, however, are accustomed to using a dialog box to browse their disk for a file to open or for a location to save a file. You can use the OpenFileDialog and SaveFileDialog controls to equip applications with standard Windows dialog boxes for these purposes.

The OpenFileDialog Control

The **OpenFileDialog control** displays a standard Windows *Open* dialog box, such as the one shown in Figure 5-41. The **Open dialog box** is useful in applications that must open an existing file because it allows the user to browse the system and select the file.

Figure 5-41 Windows *Open* dialog box



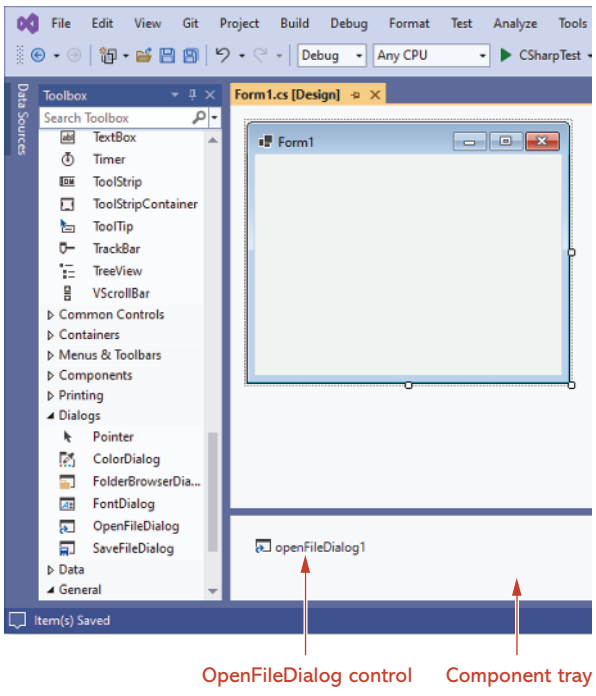
Adding the OpenFileDialog Control to Your Project

To add an OpenFileDialog control to a form, double-click the OpenFileDialog tool under the *Dialogs* group in the *Toolbox* window. When the control is created, it does not appear on the form, but in an area at the bottom of the *Designer* known as the **component tray**. Figure 5-42 shows an example of how an OpenFileDialog control appears in the component tray. The control's default name is OpenFileDialog1. As with other controls, you can change the control's Name property to change its name.

Displaying an Open Dialog Box

In code, you can display an *Open* dialog box by calling the OpenFileDialog control's `ShowDialog` method. For example, assume that we have created an OpenFileDialog

Figure 5-42 An OpenFileDialog control in the component tray



control and changed its name to `openFile`. The following statement calls the control's `ShowDialog` method:

```
openFile.ShowDialog();
```

In most cases, however, you will want to know whether the user clicked the *Open* button or the *Cancel* button to dismiss the *Open* dialog box. If the user clicked the *Open* button, it means that the user has selected a file to open. If the user clicked the *Cancel* button, it means that the user does not want to proceed.

The `ShowDialog` method returns a value that indicates which button the user clicked to dismiss the dialog box. If the user clicked the *Open* button, the value `DialogResult.OK` is returned. If the user clicked the *Cancel* button, the value `DialogResult.Cancel` is returned. Assuming `openFile` is the name of an OpenFileDialog control, the following is an example of an `if-else` statement that calls the `ShowDialog` method and determines whether the user clicked the *Open* button or the *Cancel* button.

```
if (openFile.ShowDialog() == DialogResult.OK)
{
   MessageBox.Show("You clicked the Open button.");
}
else
{
   MessageBox.Show("You clicked the Cancel button.");
}
```

The Filename Property

When the user selects a file with the *Open* dialog box, the file's path and filename are stored in the control's Filename property. Assume `openFile` is the name of an OpenFileDialog control. The following code is an example of how you can display an *Open* dialog box and, if the user clicks the *Open* button to dismiss the dialog box, open the selected file:

```
 1 StreamReader inputFile;
 2
 3 if (openFile.ShowDialog() == DialogResult.OK)
 4 {
 5    // Open the selected file.
 6    inputFile = File.OpenText(openFile.Filename);
 7
 8    // Continue processing the file...
 9 }
10 else
11 {
12    MessageBox.Show("Operation canceled.");
13 }
```

Let's take a closer look at the code. The statement in line 1 declares a `StreamReader` variable named `inputFile`. The `if` statement in line 3 calls the `openFile` control's `ShowDialog` method to display an *Open* dialog box. If the user clicks the *Open* button to dismiss the dialog box, the program continues to line 6, where the name of the selected file is retrieved from the control's Filename property and that file is opened. Otherwise (if the user clicks the *Cancel* button), the program jumps to the `else` clause in line 10.

TIP: When you create an OpenFileDialog control, its Filename property is initially set to the control's default name. For example, if the control's default name is `openFileDialog1`, then the Filename property is also set to *openFileDialog1*. Always be sure to delete the default value of the Filename property.

The InitialDirectory Property

By default, the *Open* dialog box displays the contents of the user's *Documents* directory (or folder). You can specify another directory to be initially displayed by storing its path in the InitialDirectory property. For example, the following code stores the path *C:\Data* in the `openFile` control's InitialDirectory property before displaying an *Open* dialog box:

```
openFile.InitialDirectory = @"C:\Data";
if (openFile.ShowDialog() == DialogResult.OK)
{
   // Continue to process the selected file...
}
else
{
   // The operation was canceled.
}
```

In this example, when the *Open* dialog box is displayed it shows the contents of the directory *C:\Data*.

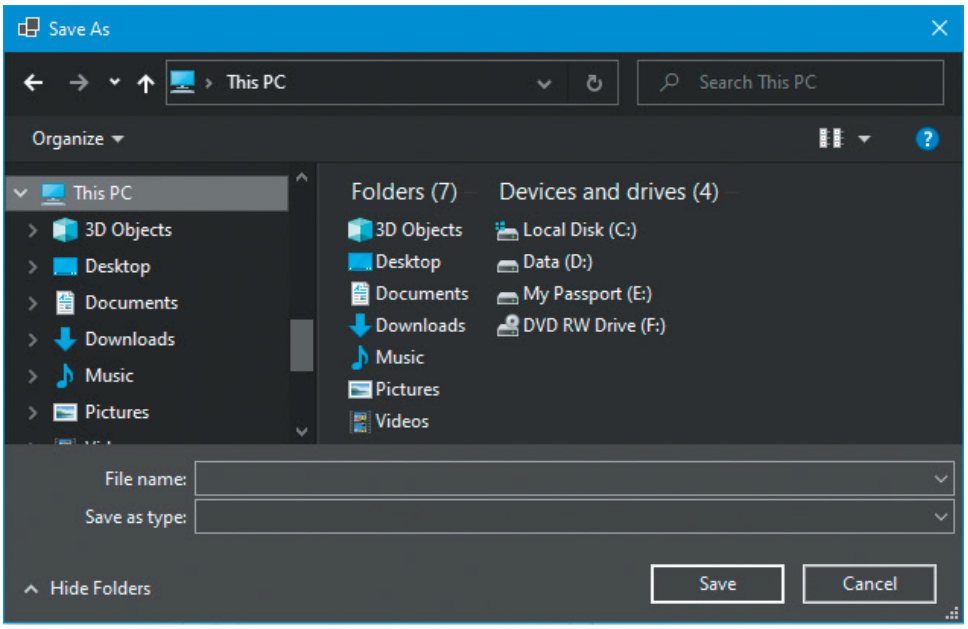
The Title Property

By default, the word *Open* is displayed in an *Open* dialog box's title bar. You can change the default text displayed in the title bar by changing the control's Title property.

The SaveFileDialog Control

The **SaveFileDialog control** displays a standard Windows *Save As* dialog box, such as the one shown in Figure 5-43. The **Save As dialog box** allows the user to browse the system and select a location and name for a file that is about to be saved.

Figure 5-43 Windows *Save As* dialog box



Adding the SaveFileDialog Control to Your Project

The SaveFileDialog control has much in common with the OpenFileDialog control. To add a SaveFileDialog control to a form, double-click the SaveFileDialog tool under the *Dialogs* group in the *Toolbox* window. When the control is created, it appears in the component tray at the bottom of the *Designer*. The control will be given a default name such as saveFileDialog1, but you can change the name with the Name property.

Displaying a *Save As* Dialog Box

In code, you can display a *Save As* dialog box by calling the SaveFileDialog control's `ShowDialog` method. For example, assume that we have created a SaveFileDialog control and changed its name to `saveFile`. The following statement calls the control's `ShowDialog` method:

```
saveFile.ShowDialog();
```

The method returns a value indicating whether the user clicked the *Save* button or the *Cancel* button to dismiss the *Save As* dialog box. If the user clicks the *Save* button, the value `DialogResult.OK` is returned. If the user clicks the *Cancel* button, the value `DialogResult.Cancel` is returned. Assume `saveFile` is the name of a SaveFileDialog control. The following is an example of an `if-else` statement that calls the `ShowDialog` method and determines whether the user clicked the *Save* button or the *Cancel* button:

```
if (saveFile.ShowDialog() == DialogResult.OK)
{
   MessageBox.Show("You clicked the Save button.");
}
else
{
   MessageBox.Show("You clicked the Cancel button.");
}
```

The Filename Property

When the user specifies a location and filename with the *Save As* dialog box, the file's path and filename are stored in the control's Filename property. Assume `saveFile` is the name of a SaveFileDialog control. The following code is an example of how you can display a *Save As* dialog box, and if the user clicks the *Save* button to dismiss the dialog box, open the selected file:

```
 1 StreamWriter outputFile;
 2
 3 if (saveFile.ShowDialog() == DialogResult.OK)
 4 {
 5    // Create the selected file.
 6    outputFile = File.CreateText(saveFile.Filename);
 7
 8    // Write data to the file...
 9 }
10 else
11 {
12     MessageBox.Show("Operation canceled.");
13 }
```

Let's take a closer look at the code. The statement in line 1 declares a `StreamWriter` variable named `outputFile`. The `if` statement in line 3 calls the `saveFile` control's `ShowDialog` method to display a *Save As* dialog box. If the user clicks the *Save* button to dismiss the dialog box, the program continues to line 6, where the selected name and path are retrieved from the control's Filename property, and that file is created. Otherwise (if the user clicks the *Cancel* button), the program jumps to the `else` clause in line 10.

The InitialDirectory Property

By default, the *Save As* dialog box displays the contents of the user's *Documents* directory (or folder). You can specify another directory to be initially displayed by storing its

path in the InitialDirectory property. The following code stores the path *C:\Data* in the `saveFile` control's InitialDirectory property before displaying a *Save As* dialog box:

```
saveFile.InitialDirectory = @"C:\Data";

if (saveFile.ShowDialog() == DialogResult.OK)
{
   // Continue to process the file...
}
else
{
   // The operation was cancelled.
}
```

In this example, when the *Save As* dialog box is displayed, it shows the contents of the directory *C:\Data*.

The Title Property

By default the words *Save As* are displayed in a *Save As* dialog box's title bar. You can change the default text displayed in the title bar by changing the control's Title property.

Checkpoint

5.30 What is the benefit of using an *Open* and/or *Save As* dialog box in an application that works with files?

5.31 What is the purpose of the following OpenFileDialog and SaveFileDialog properties?

InitialDirectory

Title

Filename

5.7 Random Numbers

CONCEPT: **Random numbers are used in a variety of applications. .NET provides the `Random` class that you can use in C# to generate random numbers.**

Random numbers are useful for lots of different programming tasks. The following are just a few examples:

- Random numbers are commonly used in games. For example, computer games that let the player roll dice use random numbers to represent the values of the dice. Programs that show cards being drawn from a shuffled deck use random numbers to represent the face values of the cards.
- Random numbers are useful in simulation programs. In some simulations, the computer must randomly decide how a person, animal, insect, or other living being will behave. Formulas can be constructed in which a random number is used to determine various actions and events that take place in the program.
- Random numbers are useful in statistical programs that must randomly select data for analysis.
- Random numbers are commonly used in computer security to encrypt sensitive data.

.NET provides a class named `Random` that you can use in C# to generate random numbers. First you create an object from the `Random` class with a statement such as this:

```
Random rand = new Random();
```

Let's dissect the statement into two parts. The first part of the statement, appearing on the left side of the = operator, is as follows:

```
Random rand = new Random();
```

This declares a variable named `rand` that can reference a `Random` object.

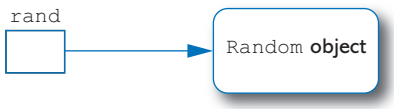
This declares a variable named `rand` that can be used to reference a `Random` object. The second part of the statement, appearing on the right side of the = operator, is as follows:

```
Random rand = new Random();
```

This creates a `Random` object in memory.

The expression `new Random()` causes an object of the `Random` class to be created in memory. The = operator causes the `rand` variable to reference the `Random` object, as illustrated in Figure 5-44. After this statement has executed, we can use the `rand` variable to work with the `Random` object.

Figure 5-44 The `rand` variable references a `Random` object



Using `var` To Declare a `Random` Variable

If you are declaring and initializing a `Random` variable inside of a method, you can use the `var` keyword in your declaration statement. Here is an example:

```
var rand = new Random();
```

The compiler will be able to determine that `rand` is a `Random` reference variable because we are initializing it with a `Random` object. It is also obvious to any person reading this statement that `rand` references a `Random` object.

When declaring and initializing local reference variables in this book, we will use the `var` keyword any time it simplifies the declaration statement without concealing the data type of the object.

NOTE: Remember, the `var` keyword can only be used to declare local variables in a method. It cannot be used to declare fields in a class.

The `Next` Method

Once you have created a `Random` object, you can call its `Next` method to get a random integer number. The following code shows an example:

```
// Declare an int variable.
int number;

// Create a Random object.
var rand = new Random();

// Get a random integer and assign it to number.
number = rand.Next();
```

After this code executes, the `number` variable contains a random integer. If you call the `Next` method with no arguments, as shown in this example, the returned integer is somewhere between 0 and 2,147,483,647. Alternatively, you can pass an argument that specifies an upper limit to the generated number's range. In the following statement, the value assigned to `number` is somewhere between 0 and 99:

```
number = rand.Next(100);
```

You can also specify a lower limit for the range of numbers. In the following statement, the value assigned to `number` is somewhere between 100 and 199:

```
number = rand.Next(100, 200);
```

The following statement shows another example. It assigns a random integer to `number` between −50 and +49:

```
number = rand.Next(-50, 50);
```

The `NextDouble` Method

You can call a `Random` object's `NextDouble` method to get a random floating-point number between 0.0 and 1.0 (not including 1.0). The following code shows an example:

```
// Declare a Double variable.
double number;

// Create a Random object.
var rand = new Random();

// Get a random number and assign it to number.
number = rand.NextDouble();
```

After this code executes, the `number` variable will contain a random floating-point number in the range of 0.0 up to (but not including) 1.0.

In Tutorial 5-8, you will use random numbers to determine whether the heads or tails side of a coin is facing up after the coin has been tossed.

Tutorial 5-8:
Simulating Coin Tosses

VideoNote
Tutorial 5-8: Simulating Coin Tosses

In this tutorial, you create an application that simulates the tossing of a coin. Each time the user tosses the coin, the application uses a `Random` object to get a random integer in the range of 0 through 1. If the random number is 0, it means the tails side of the coin is up, and if the random number is 1, it means the heads side is up. The application displays an image of a coin showing either heads or tails, depending on the value of the random number.

Step 1: Start Visual Studio and begin a new Windows Forms App project named *CoinToss*.

Step 2: Set up the application's form as shown in Figure 5-45. Notice that the form's Text property is set to *Coin Toss*. The names of the controls are shown in the figure. As you place each of the controls on the form, refer to Table 5-1 for the relevant property settings. (Make sure the `headsPictureBox` control's Visible property is set to True, and the `tailsPictureBox` control's Visible property is set to False. This will cause the coin to initially appear heads up when the application runs.)

Figure 5-45 Initial setup of the *Coin Toss* form

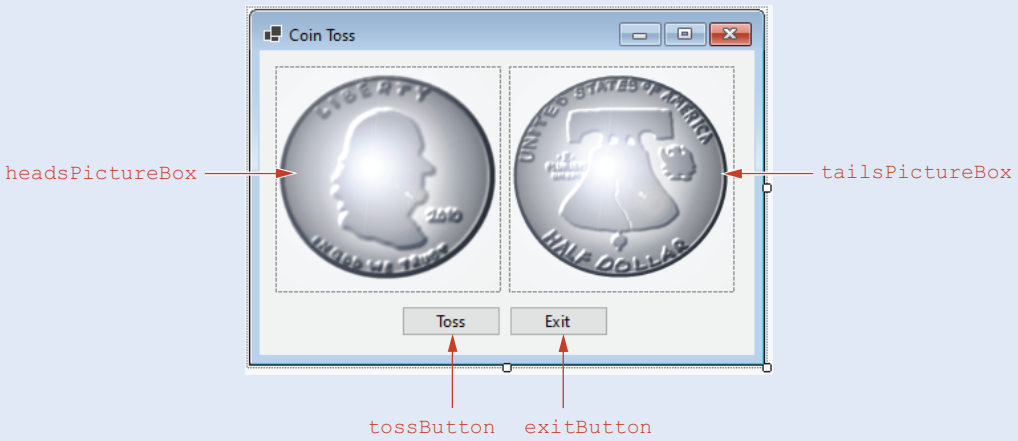
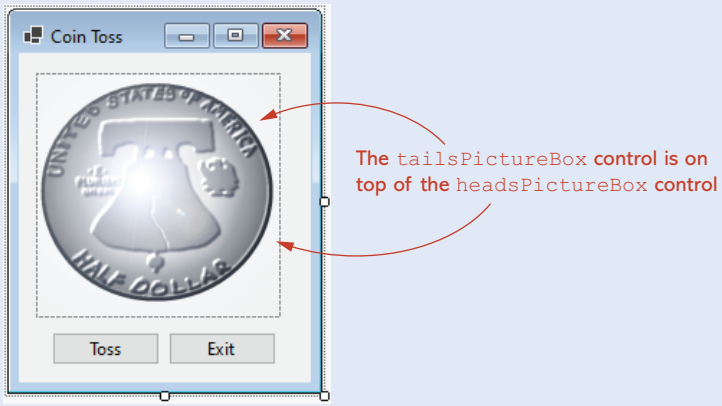


Table 5-1 Control property settings

Control Name	Control Type	Property Settings
`headsPictureBox`	PictureBox	**Image:** Select and import the Heads1.bmp file from the *Chap05* folder of the Student Sample Programs. **Size:** 170, 170 **SizeMode:** Zoom **Visible:** True
`tailsPictureBox`	PictureBox	**Image:** Select and import the Tails1.bmp file from the *Chap05* folder of the Student Sample Programs. **Size:** 170, 170 **SizeMode:** Zoom **Visible:** False
`tossButton`	Button	**Text:** *Toss*
`exitButton`	Button	**Text:** *Exit*

Step 3: After you have set all of the control properties as shown in Table 5-1, move the PictureBox controls so one is on top of the other, as shown in Figure 5-46. (In the figure, the `tailsPictureBox` control is on top, but it really doesn't matter which is on top.) Also, reduce the width of the form and position the button controls as shown in the figure.

Figure 5-46 The controls repositioned and the form size adjusted



Step 4: Now you will create the Click event handlers for the Button controls. At the end of this tutorial, Program 5-8 shows the completed code for the form. You will be instructed to refer to Program 5-8 as you write the event handlers.

In the *Designer*, double-click the `tossButton` control. This opens the code editor, and you will see an empty event handler named `tossButton_Click`. Complete the `tossButton_Click` event handler by typing the code shown in lines 12–34 in Program 5-8. Let's take a closer look at the code:

Line 13: This statement declares an `int` variable named `sideUp`. This variable is used to hold a random number that indicates which side of the coin is up.

Line 16: This statement does the following:

- It declares a variable named `rand` that can be used to reference a `Random` object.
- It creates a `Random` object in memory.
- The = operator causes the `rand` variable to reference the `Random` object.

Line 20: This statement gets a random integer in the range of 0 through 1 and assigns it to the `sideUp` variable. The random integer represents which side of the coin is facing up. The value 0 means that the tails side is facing up, and the value 1 means that the heads side is facing up.

Lines 23–34: This `if-else` statement displays the side of the coin that is facing up. If `sideUp` equals 0, then the statements in lines 26 and 27 display the `tailsPictureBox` control and hide the `headsPictureBox` control. If `sideUp` equals 1, then the statements in lines 32 and 33 display the `headsPictureBox` control and hide the `tailsPictureBox` control.

Step 5: Switch your view back to the *Designer* and double-click the `exitButton` control. In the code editor you will see an empty event handler named `exitButton_Click`. Complete the `exitButton_Click` event handler by typing the code shown in lines 39–40 in Program 5-8.

Step 6: Save the project. Then, press F5 on the keyboard or click the *Start Debugging* button ( ) on the toolbar to compile and run the application. When the application runs, click the *Toss* button several times to simulate several coin tosses. When you are finished, click the *Exit* button to exit the application.

Program 5-8 Completed code for Form1 in the *CoinToss* project

```
 1 namespace CoinToss
 2 {
 3     public partial class Form1: Form
 4     {
 5         public Form1()
 6         {
 7             InitializeComponent();
 8         }
 9
10         private void tossButton_Click(object sender, EventArgs e)
11         {
12             // Variable to indicate which side is up
13             int sideUp;
14
15             // Create a Random object.
16             var rand = new Random();
17
18             // Get a random integer in the range of 0 through 1.
19             // 0 means tails up, 1 means heads up.
20             sideUp = rand.Next(2);
21
22             // Display the side that is up.
23             if (sideUp == 0)
24             {
25                 // Display tails up.
26                 tailsPictureBox.Visible = true;
27                 headsPictureBox.Visible = false;
28             }
29             else
30             {
31                 // Display heads up.
32                 headsPictureBox.Visible = true;
33                 tailsPictureBox.Visible = false;
34             }
35         }
36
37         private void exitButton_Click(object sender, EventArgs e)
38         {
39             // Close the form.
40             this.Close();
41         }
42     }
43 }
```

Random Number Seeds

The numbers that are generated by the `Random` class are not truly random. Instead, they are **pseudorandom numbers** that are calculated by a formula. The formula used to generate random numbers has to be initialized with a value known as a seed value. The **seed value** is used in the calculation that returns the next random number in the series. When a `Random` object is created in memory, it retrieves the system time from the computer's internal clock and uses that as the seed value. The system time is an integer that represents the current date and time, down to a hundredth of a second.

If a `Random` object uses the same seed value each time it is created, it always generates the same series of random numbers. Because the system time changes every hundredth of a second, it is the preferred value to use as the seed in most cases. However, you can specify a different integer value as the seed, if you desire, when you create a `Random` object. Here is an example:

```
var rand = new Random(1000);
```

In this example, the `Random` object that is created uses 1000 as the seed value. Each time a `Random` object is created with this statement, it generates the same series of random numbers. That may be desirable in some applications, when you always want to produce the same set of pseudorandom numbers.

Checkpoint

5.32 What does a `Random` object's `Next` method return?

5.33 What does a `Random` object's `NextDouble` method return?

5.34 Write code that creates a `Random` object and then assigns a random integer in the range of 1 through 100 to the variable `randomNumber`.

5.35 Write code that creates a `Random` object and then assigns a random integer in the range of 100 through 399 to the variable `randomNumber`.

5.36 What does a `Random` object use as its seed value if you do not specify one?

5.37 What happens if the same seed value is used each time a `Random` object is created?

5.8 The Load Event

CONCEPT: **When an application's form loads into memory, an event known as the Load event takes place. You can write an event handler for the Load event, and that handler will execute just before the form is displayed.**

When you run an application, the application's form is loaded into memory and an event known as the **Load event** takes place. The Load event takes place before the form is displayed on the screen. If you want to execute some code at this point, you can write a Load event handler containing the desired code.

To create a Load event handler for a form, simply double-click any area of the form in the *Designer* window where there is no other control. The code editor will open with an empty Load event handler. If the form is named `Form1`, the event handler is named `Form1_Load`. Any code that you write inside the event handler executes when the form's Load event takes place. Here is an example of a Load event handler in a form named `Form1`:

```
private void Form1_Load(object sender, EventArgs e)
{
   MessageBox.Show("Prepare to see the form!");
}
```

Keep in mind that the Load event happens before the form is displayed on the screen. When the application containing this event handler runs, the message box is displayed before the form appears.

Load event handlers are useful for performing setup operations. In Tutorial 5-9, you will complete an application that uses a Load event handler to read items from a text file and add those items to a ListBox control.

Tutorial 5-9:
Creating a Load Event Handler

VideoNote
Tutorial 5-9: Creating a Load Event Handler

In this tutorial, you complete the *LoadEvent* project that is found in the *Chap05* folder of this book's Student Sample Programs. This application is a variation of the *SouthAmerica* project that you created in Tutorial 5-6. This version of the application uses a Load event handler to read the contents of the Countries.txt file and adds those items to a ListBox control.

The application's form has already been created and is shown in Figure 5-47. The project also includes an accompanying text file named Countries.txt. The Countries.txt file contains the names of the countries of South America.

Figure 5-47 The *LoadEvent* project's form

Step 1: Start Visual Studio. Open the project named *LoadEvent* in the *Chap05* folder of this book's Student Sample Programs.

Step 2: The Form1 form is shown, along with the names of the important controls, in Figure 5-47. Double-click any part of the form that does not contain a control. (Be sure not to click the Label control, the ListBox control, or the Button control.) This opens the code editor, and you will see an empty event handler named `Form1_Load`. Complete the `Form1_Load` event handler by typing the code shown in lines 12–40 in Program 5-9. Let's take a closer look at the code:

Line 12: This is the beginning of a `try-catch` statement, which handles any exceptions that are thrown while the file is being processed. If an exception is thrown by any statement in the try block, the program jumps to the `catch` clause in line 36.

Line 15: This statement declares the `string` variable `countryName`, which holds the lines of text that are read from the file.

Lines 18–21: After these statements have executed, the Countries.txt file is opened for reading, and the `inputFile` variable references a `StreamReader` object that is associated with the file.

Line 24: This is the beginning of a `while` loop that iterates as long as the end of the Countries.txt file has not been reached.

Line 27: This statement reads a line of text from the file and assigns it to the `countryName` variable.

Line 30: This statement adds the contents of the `countryName` variable to the ListBox.

Line 34: This statement closes the file.

Step 3: Switch your view back to the *Designer* and double-click the `exitButton` control. In the code editor you will see an empty event handler named `exitButton_Click`. Complete the `exitButton_Click` event handler by typing the code shown in lines 45–46 in Program 5-9.

Step 4: Save the project. Then, press F5 on the keyboard or click the *Start Debugging* button (▶) on the toolbar to compile and run the application. When the application runs, the ListBox should appear filled with the names of the countries from the Countries.txt file, as shown in Figure 5-48. Click the *Exit* button to exit the application.

Figure 5-48 The *Load Event* application displaying the list of countries

Program 5-9 Completed code for Form1 in the *LoadEvent* project

```
1  namespace LoadEvent
2  {
3      public partial class Form1: Form
4      {
5          public Form1()
6          {
7              InitializeComponent();
8          }
9
10         private void Form1_Load(object sender, EventArgs e)
11         {
12             try
13             {
14                 // Declare a variable to hold a country name.
15                 string countryName;
16
17                 // Declare a StreamReader variable.
18                 StreamReader inputFile;
19
20                 // Open the file and get a StreamReader object.
21                 inputFile = File.OpenText("Countries.txt");
22
```

```
23                  // Read the file's contents.
24                  while (!inputFile.EndOfStream)
25                  {
26                      // Get a country name.
27                      countryName = inputFile.ReadLine();
28
29                      // Add the country name to the ListBox.
30                      countriesListBox.Items.Add(countryName);
31                  }
32
33                  // Close the file.
34                  inputFile.Close();
35              }
36              catch (Exception ex)
37              {
38                  // Display an error message.
39                  MessageBox.Show(ex.Message);
40              }
41          }
42
43          private void exitButton_Click(object sender, EventArgs e)
44          {
45              // Close the form.
46              this.Close();
47          }
48      }
49  }
```

Checkpoint

5.38 When does the Load event take place?

5.39 How do you create an event handler for the Load event?

Key Terms

accumulator
binary file
body
component tray
counter variable
decrement
direct access file
file object
filename
filename extensions
increment
initialization expression
input file
iteration
Load event
newline character
Open dialog box
OpenFileDialog control
output file
postfix mode
posttest loop
prefix mode
pretest loop
pseudorandom numbers
random access file
read position
running total
Save As dialog box
SaveFileDialog control
seed value
sequential access file
`StreamReader` class
`StreamWriter` class
test expression
text file
update expression
`while` clause

Review Questions

Multiple Choice

1. A _____ is commonly used to control the number of times that a loop iterates.
   a. counter variable
   b. test expression
   c. `while` clause
   d. controlled variable

2. A(n) _____ loop tests its condition *before* performing an iteration.
   a. preemptive
   b. pretest
   c. infinite
   d. logical

3. The term _____is used to describe a file that data is written to.
   a. input file
   b. output file
   c. saved file
   d. user file

4. The term _____ file is used to describe a file that data is read from.
   a. data
   b. write
   c. read
   d. input

5. A _____ file contains data that has been encoded as text, using a scheme such as Unicode.
   a. text
   b. character
   c. Unicode
   d. system

6. When you work with a _____ file you access data from the beginning of the file to the end of the file.
   a. direct access
   b. random access
   c. sequential access
   d. binary access

7. Files on a disk are identified by a _____.
   a. unique identifier
   b. filename
   c. binary sequencer
   d. file extension

8. A _____ object is an object that is associated with a specific file and provides a way for the program to work with that file.
   a. data
   b. directory
   c. stream
   d. file

9. When a program works with an input file, a special value known as a(n) _____ is internally maintained for that file and marks the location of the next item that will be read from the file.
   a. input locator
   b. accumulator
   c. read position
   d. sequential read value

10. When the user selects a file with the *Open* dialog box, the file's path and filename are stored in the control's _____ property.
   a. Filename
   b. FilePath
   c. Pathname
   d. Text

11. The _____ control displays a standard Windows *Save As* dialog box.
   a. SaveAsDialog
   b. FileDialog
   c. SaveFileDialog
   d. StandardDialog

12. Once you have created a `Random` object, you can call its _____ method to get a random integer number.
   a. `Generate`
   b. `Rand`
   c. `NextInteger`
   d. `Next`

13. When generating random numbers, the _____ value is used in the calculation that returns the next random number in the series.
   a. `Start`
   b. seed
   c. `Next`
   d. sequence

14. When you run an application, the application's form is loaded into memory and an event known as the _____ event takes place.
   a. Startup
   b. Begin
   c. Load
   d. Initialize

True or False

1. The `while` loop is known as a pretest loop, which means it tests its condition *before* performing an iteration.

2. To increment a variable means to increase its value and to decrement a variable means to decrease its value.

3. When a variable is declared in the initialization expression of a `for` loop, the scope of the variable is limited to the loop.

4. The `while` loop always performs at least one iteration, even if its Boolean expression is false to begin with.

5. The term *read file* is used to describe a file that data is read from.

6. To append data to an existing file, you open it with the `File.AppendText` method.
7. As items are read from the file, the read position moves forward, toward the end of the file.
8. The numbers that are generated by the `Random` class are truly random.
9. The Load event takes place after the form is displayed on the screen.

Short Answer

1. What is contained in the body of a loop?
2. Write a programming statement that uses postfix mode to increment a variable named `count`.
3. How many iterations will occur if the test expression of a `for` loop is false to begin with?
4. What are filename extensions? What do they indicate about a file?
5. When an input file is opened, what is its read position initially set to?
6. How can you read all of the items in a file without knowing how many items the file contains?
7. What is a variable that is used to accumulate a total called?
8. By default, the *Open* dialog box displays the contents of the user's *Documents* directory. How can you specify another directory to be initially displayed?
9. Why is the system time the preferred seed value for a `Random` object?
10. What kind of code should be placed in the Load event?

Algorithm Workbench

1. Write a loop that displays your name 10 times.
2. Write a loop that displays all the odd numbers from 1 through 49.
3. Write a loop that displays every fifth number from 0 through 100.
4. Write a code sample that uses a loop to write the numbers from 1 through 10 to a file.
5. Assume that a file named People.txt contains a list of names. Write a code sample that uses a `while` loop to read the file and display its contents in a ListBox control.

Programming Problems

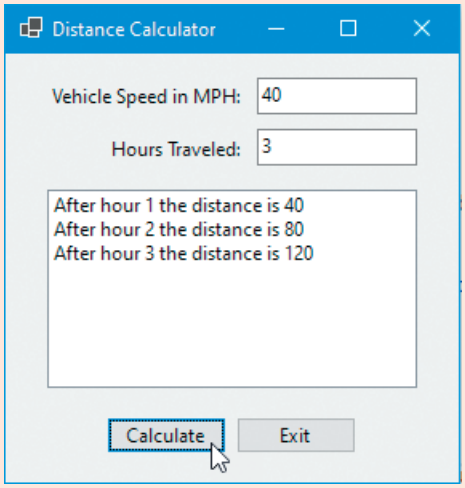
1. **Distance Calculator**

   If you know a vehicle's speed and the amount of time it has traveled, you can calculate the distance it has traveled as follows:

   $$Distance = Speed \times Time$$

   For example, if a train travels 40 miles per hour for 3 hours, the distance traveled is 120 miles. Create an application with a form similar to the one shown in Figure 5-49. The user enters a vehicle's speed and the number of hours traveled into text boxes. When the user clicks the *Calculate* button, the application should use a loop to display in a list box the distance the vehicle has traveled for each hour of that time period.

Figure 5-49 The *Distance Calculator* application



2. **Distance File**

   Modify the *Distance Calculator* program that you wrote for Programming Problem 1 so it writes its output to a file instead of displaying it in a ListBox control. Open the file in Notepad or Visual Studio to confirm the output.

VideoNote
Solving the Celsius to Fahrenheit Table problem

3. **Celsius to Fahrenheit Table**

   Assuming that C is a Celsius temperature, the following formula converts the temperature to a Fahrenheit temperature (*F*):

$$F = \frac{9}{5}C + 32$$

   Create an application that displays a table of the Celsius temperatures 0–20 and their Fahrenheit equivalents. The application should use a loop to display the temperatures in a list box.

4. **Population**

   Create an application that predicts the approximate size of a population of organisms. The application should use text boxes to allow the user to enter the starting number of organisms, the average daily population increase (as a percentage), and the number of days the organisms will be left to multiply. For example, assume the user enters the following values:

   Starting number of organisms: 2
   Average daily increase: 30%
   Number of days to multiply: 10

   The application should display the following table of data in a ListBox control:

Day	Approximate Population
1	2
2	2.6
3	3.38
4	4.394
5	5.7122

Day	Approximate Population
6	7.42586
7	9.653619
8	12.5497
9	16.31462
10	21.209

5. **Pennies for Pay**

   Susan is hired for a job, and her employer agrees to pay her every day. Her employer also agrees that Susan's salary is 1 penny the first day, 2 pennies the second day, 4 pennies the third day, continuing to double each day. Create an application that allows the user to enter the number of days that Susan will work and calculates the total amount of pay she will receive over that period of time.

6. **Ocean Levels**

   Assuming the ocean's level is currently rising at about 1.5 millimeters per year, create an application that displays the number of millimeters that the ocean will have risen each year for the next 10 years. Display the output in a ListBox control.

7. **Tuition Increase**

   At one college the tuition for a full-time student is $6,000 per semester. It has been announced that the tuition will increase by 2 percent each year for the next 5 years. Create an application with a loop that displays the projected semester tuition amount for the next 5 years in a ListBox control.

8. **Dice Simulator**

   Create an application that simulates rolling a pair of dice. When the user clicks a button, the application should generate two random numbers, each in the range of 1 through 6, to represent the value of the dice. Use PictureBox controls to display the dice. (In the Student Sample Programs, in the *Chap05* folder, you will find six images named Die1.bmp, Die2.bmp, Die3.bmp, Die4.bmp, Die5.bmp, and Die6.bmp that you can use in the PictureBoxes.)

9. **Addition Tutor**

   Create an application that generates two random integers, each in the range of 100 through 500. The numbers should be displayed as addition problems on the application's form, such as

   $$247 + 129 = ?$$

   The form should have a text box for the user to enter the problem's answer. When a button is clicked, the application should do the following:

   - Check the user's input and display a message indicating whether it is the correct answer.
   - Generate two new random numbers and display them in a new problem on the form.

10. **Random Number Guessing Game**

    Create an application that generates a random number in the range of 1 through 100 and asks the user to guess what the number is. If the user's guess is higher than the random number, the program should display "Too high, try again." If the user's guess is lower than the random number, the program should display "Too low, try again."

If the user guesses the number, the application should congratulate the user and then generate a new random number so the game can start over.

Optional Enhancement: Enhance the game so it keeps count of the number of guesses that the user makes. When the user correctly guesses the random number, the program should display the number of guesses.

11. Calculating the Factorial of a Number

In mathematics, the notation $n!$ represents the factorial of the nonnegative integer n. The factorial of n is the product of all the nonnegative integers from 1 through n. For example,

$$7! = 1 \times 2 \times 3 \times 4 \times 5 \times 6 \times 7 = 5{,}040$$

and

$$4! = 1 \times 2 \times 3 \times 4 = 24$$

Create an application that lets the user enter a nonnegative integer and then uses a loop to calculate the factorial of that number. Display the factorial in a label or a message box.

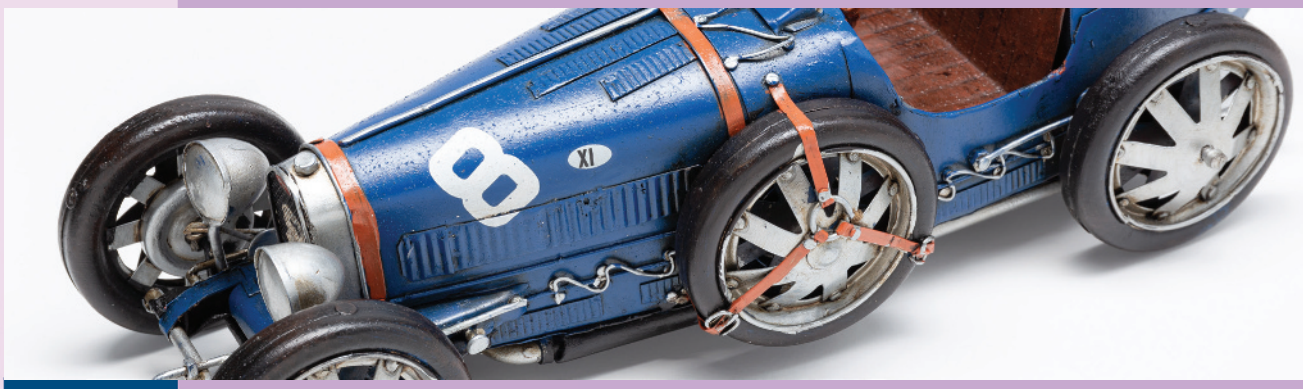
12. Random Number File Writer

Create an application that writes a series of random numbers to a file. Each random number should be in the range of 1 through 100. The application should let the user specify how many random numbers the file will hold and should use a SaveFileDialog control to let the user specify the file's name and location.

13. Random Number File Reader

This exercise assumes you have completed Programming Problem 12, *Random Number File Writer*. Create another application that uses an OpenFileDialog control to let the user select the file that was created by the application that you wrote for Problem 12. This application should read the numbers from the file, display the numbers in a ListBox control, and then display the following data:

- The total of the numbers
- The number of random numbers read from the file



CHAPTER 6

Modularizing Your Code with Methods

TOPICS

6.1 Introduction to Methods
6.2 `void` Methods
6.3 Passing Arguments to Methods
6.4 Passing Arguments by Reference
6.5 Value-Returning Methods
6.6 Debugging Methods

6.1 Introduction to Methods

CONCEPT: **Methods can be used to break a complex program into small, manageable pieces. A `void` method simply executes a group of statements and then terminates. A value-returning method returns a value to the statement that called it.**

In a general sense, a method is a collection of statements that performs a specific task. So far, you have experienced methods in the following two ways:

- You have created event handlers. An event handler is a special type of method that responds to events.
- You have executed predefined .NET methods, such as `MessageBox.Show` and the `TryParse` methods.

In this chapter, you will learn how to create your own methods that can be executed just as you execute .NET methods.

Methods are commonly used to break a problem into small, manageable pieces. Instead of writing one long method that contains all the statements necessary to solve a problem, you can write several small methods that each solve a specific part of the problem. These small methods can then be executed in the desired order to solve the problem. This approach is sometimes called **divide and conquer** because a large problem is divided into several smaller problems that are easily solved. Figures 6-1 and 6-2 illustrate this idea by comparing two programs: one that uses a long, complex event handler containing all the statements necessary to solve a problem and another that divides a problem into smaller problems, each of which are handled by a separate method.

Figure 6-1 Using one long sequence of statements to perform a task

```
namespace Example
{
   public partial class Form1 : Form
   {
       private void myButton_Click(object sender, EventArgs e)
       {
           statement;
           statement;
           statement;
           statement;
           statement;
           statement;
           statement;
           statement;
           statement;
           statement;
           statement;
           statement;
           statement;
           ... and so on.
       }
   }
}
```

In this program the task is performed by one long sequence of statements in an event handler.

Figure 6-2 Using methods to divide and conquer a problem

```
namespace Example
{
   public partial class Form1 : Form
   {
       private void myButton_Click(object sender, EventArgs e)
       {
           Method2();
           Method3();
           Method4();
       }

       private void Method2()
       {
           statement;
           statement;
           statement;
       }

       private void Method3()
       {
           statement;
           statement;
           statement;
       }

       private void Method4()
       {
           statement;
           statement;
           statement;
       }
   }
}
```

Event Handler

Method 2

Method 3

Method 4

In this program the task has been divided into smaller tasks, each of which is performed by a separate method.

In general terms, a program that is broken into smaller units of code, such as methods, is known as a **modularized program**. Modularization tends to simplify code. If a specific task is performed in several places in a program, a method can be written once to perform that task and then be executed any time it is needed. This benefit of using methods is known as **code reuse** because you are writing the code to perform a task once and then reusing it each time you need to perform the task.

void Methods and Value-Returning Methods

In this chapter, you will learn to write two types of methods: `void` methods and value-returning methods. When you call a **void method**, it simply executes the statements it contains and then terminates. When you call a **value-returning method**, it executes the statements that it contains and then it returns a value back to the statement that called it. The `Parse` methods are good examples of value-returning methods. The first type of method that you will learn to write is the `void` method.

6.2 void Methods

CONCEPT: **A `void` method performs a task and then terminates. It does not return a value back to the statement that called it.**

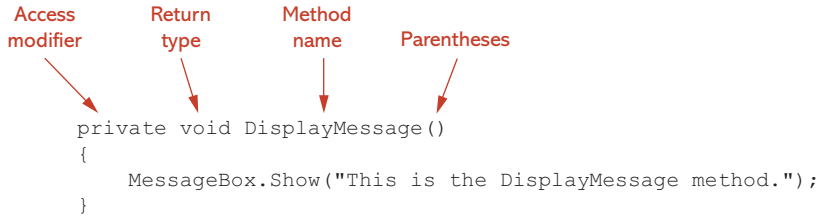
To create a method you write its definition. A method definition has two parts: a header and a body. The **method header**, which appears at the beginning of a method definition, lists several important things about the method, including the method's name. The **method body** is a collection of statements that are performed when the method is executed. These statements are enclosed inside a set of curly braces. Here is an example of a method definition:

```
private void DisplayMessage()
{
    MessageBox.Show("This is the DisplayMessage method.");
}
```

The Method Header

Using the previously shown method definition, Figure 6-3 points out the different parts of the method header, which is the first line.

Figure 6-3 Parts of the method header



Let's take a closer look at the parts identified in the figure:

- **Access modifier**—The keyword `private` is an access modifier. When a method is declared as `private`, it can be called only by code inside the same class as the method. Alternatively, a method that is declared as `public` can be called by code that is outside the class. This is important because some applications have multiple

classes, and unless you specifically intend a method to be available to code outside the class, you should declare it `private`.

- **Return type**—Recall our previous discussion of `void` and value-returning methods. When the keyword `void` appears here, it means that the method is a `void` method, and does not return a value. As you will see later in this chapter, a value-returning method lists a data type here.
- **Method name**—You should give each method a descriptive name. In general, the same rules that apply to variable names also apply to method names. The method in this example is named `DisplayMessage`, so we can easily guess what the method does: It displays a message.
  In this book, we use **Pascal case** for method names. Pascal case is like camelCase (the convention we have been using for variable names), except in a Pascal case name the first character is always uppercase. It is a standard convention among C# programmers to use Pascal case for method names because it differentiates method names from variable and field names.
- **Parentheses**—In the header, the method name is always followed by a set of parentheses. As you will see later in this chapter, you sometimes write declarations inside the parentheses, but for now, the parentheses will be empty.

NOTE: The method header is never terminated with a semicolon.

The Method Body

Beginning at the line after the method header, one or more statements appear inside a set of curly braces (`{ }`). These statements are the method's body and are performed any time the module is executed.

When you write a method definition, Visual Studio automatically indents the statements in the method body. The indentation is not required, but it makes the code easier to read and debug. By indenting the statements in the body of the method, you visually set them apart from the surrounding code. This allows you to tell at a glance what part of the program is part of the method.

Declaring Methods Inside a Class

Methods usually belong to a class, so you must write a method's definition inside the class to which it is supposed to belong. In this chapter, all the methods that you will write will belong to an application's `Form1` class. When you write a method's definition, you write it inside the `Form1` class, as shown in Figure 6-4.

Figure 6-4 Write method definitions inside the `Form1` class

```
namespace CSharpTest
{
    3 references
    public partial class Form1 : Form
    {
        1 reference
        public Form1()
        {
            InitializeComponent();
        }

        Your method definitions will appear
        here, inside the Form1 class.
    }
}
```

Calling a Method

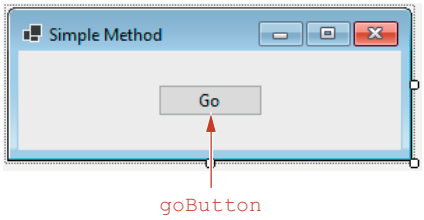
A method executes when it is called. Event handlers are called when specific events take place, but other methods are executed by method call statements. When a method is called, the program branches to that method and executes the statements in its body. Here is an example of a **method call** statement that calls the `DisplayMessage` method we previously examined:

```
DisplayMessage();
```

The statement is simply the name of the method followed by a set of parentheses. Because it is a complete statement, it is terminated with a semicolon.

Let's look at a complete program that uses the `DisplayMessage` method. In the *Chap06* folder of this book's Student Sample Programs is a project named *SimpleMethod*. Figure 6-5 shows the application's form in the Designer, and Program 6-1 shows the form's code.

Figure 6-5 The *SimpleMethod* project's form



Program 6-1 Code for the *SimpleMethod* project's Form1 form

```
 1 namespace SimpleMethod
 2 {
 3     public partial class Form1: Form
 4     {
 5         public Form1()
 6         {
 7             InitializeComponent();
 8         }
 9
10         private void goButton_Click(object sender, EventArgs e)
11         {
12             MessageBox.Show("This is the goButton_Click method.");
13             DisplayMessage();
14             MessageBox.Show("Back in the goButton_Click method.");
15         }
16
17         private void DisplayMessage()
18         {
19             MessageBox.Show("This is the DisplayMessage method.");
20         }
21     }
22 }
```

Let's step through the code. When the user clicks the *Go* button, the `goButton_Click` event handler executes. Inside the event handler the statement in line 12 displays *This is the goButton_Click method* in a message box. Then, line 13 calls the `DisplayMessage` method. As is shown in Figure 6-6, the program jumps to the `DisplayMessage` method and executes the statements in its body. There is only one statement in the body of the `DisplayMessage` method, which is line 19. This statement displays *This is the DisplayMessage method*, and then the method ends. As shown in Figure 6-7, the program

Figure 6-6 Calling the `DisplayMessage` method

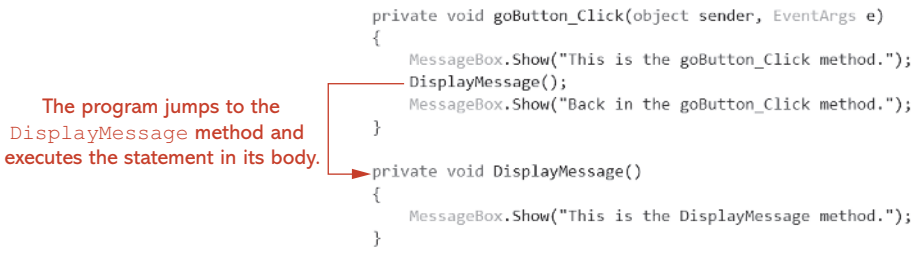
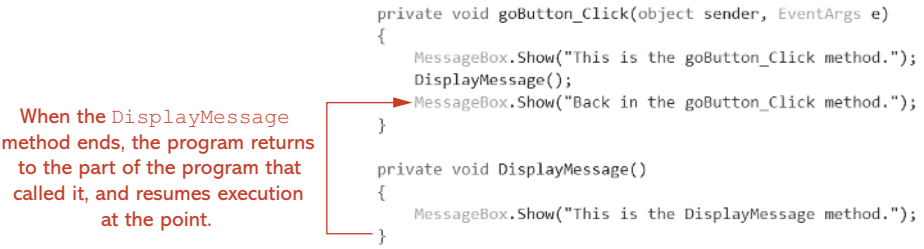


Figure 6-7 The `DisplayMessage` method returns



jumps back to the part of the program that called the `DisplayMessage` method and resumes execution from that point. In this case, the program resumes execution at line 14, which displays *Back in the goButton_Click method*. The `goButton_Click` event handler ends at line 15.

When a method is called, some operations are performed "behind the scenes" so the system will know to where the program should return after the method ends. First, the system saves the memory address of the location to which it should return. This is typically the statement that appears immediately after the method call. This memory location is known as the **return point**. Then, the system jumps to the method and executes the statements in its body. When the method ends, the system jumps back to the return point and resumes execution.

NOTE: When a program calls a method, programmers commonly say that the *control* of the program transfers to that method. This simply means that the method takes control of the program's execution.

In Tutorial 6-1, you will get hands-on practice writing and calling methods.

Tutorial 6-1:
Creating and Calling Methods

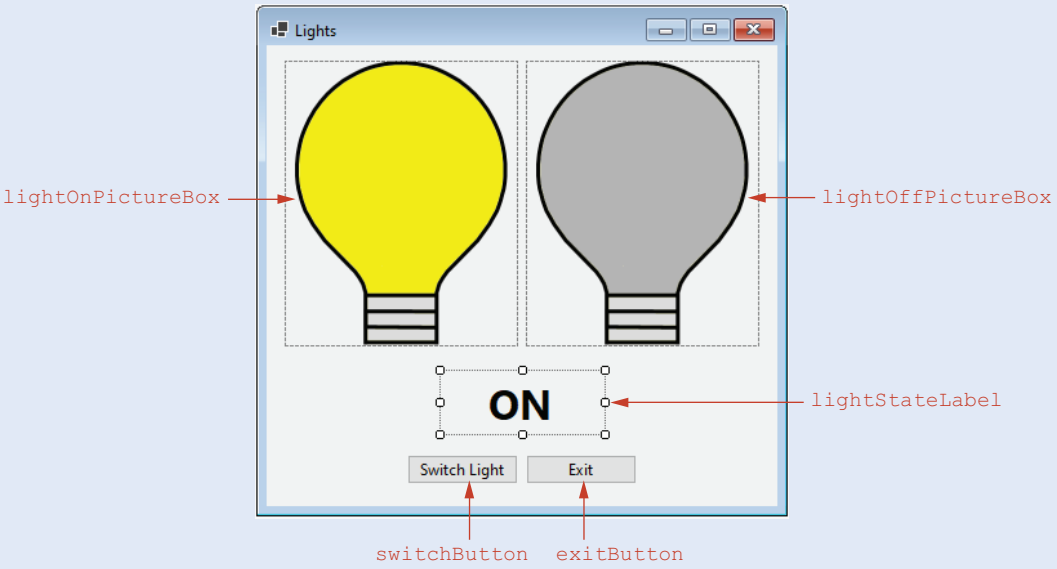
VideoNote
Tutorial 6-1: Creating and Calling Methods

The *Chap06* folder in this book's Student Sample Programs contains a partially created project named *Lights*. In this tutorial, you complete the project so it simulates a light being turned off or on. The project's form, in its initial setup, is shown in Figure 6-8.

Here are some details about specific property settings:

- The `lightOnPictureBox` control's Visible property is initially set to true.
- The `lightOffPictureBox` control's Visible property is initially set to false.

Figure 6-8 The *Lights* project's form in its initial setup



- The `lightStateLabel` displays either *ON* or *OFF* while the application runs to indicate whether the light is on or off. Initially, this control's Text property is set to *ON*.

At run time, when the user clicks the *Switch Light* button, the state of the light is reversed. In other words, if the light is currently on, it will be turned off. If the light is currently off, it will be turned on.

When the light is turned on, the following actions take place:

- The `lightOnPictureBox` control's Visible property is set to `true`.
- The `lightOffPictureBox` control's Visible property is set to `false`.
- The `lightStateLabel` label's Text property is assigned the string `"ON"`.

When the light is turned off, the following actions take place:

- The `lightOffPictureBox` control's Visible property is set to `true`.
- The `lightOnPictureBox` control's Visible property is set to `false`.
- The `lightStateLabel` label's Text property is assigned the string `"OFF"`.

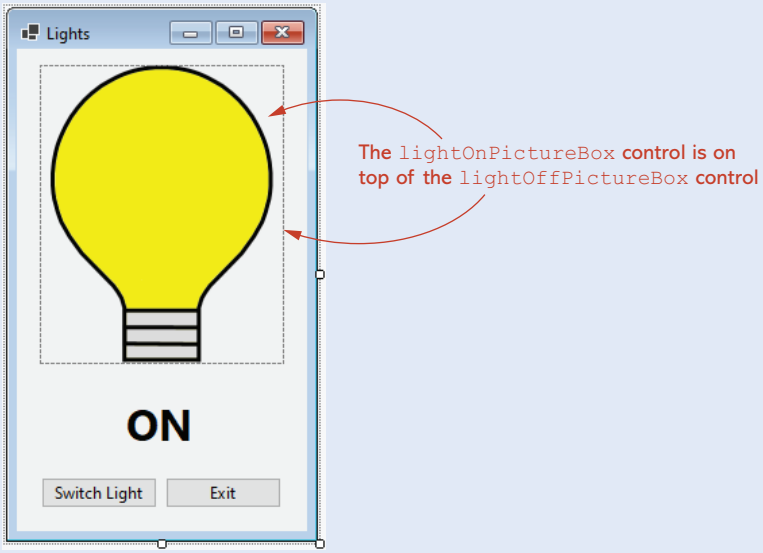
To modularize the code, you create a method named `TurnLightOn` (containing the code to turn the light on), and another method named `TurnLightOff` (containing the code to turn the light off). When you need to turn the light on, you call the `TurnLightOn` method and when you need to turn the light off you call the `TurnLightOff` method.

Step 1: Start Visual Studio. Open the project named *Lights* in the *Chap06* folder of this book's Student Sample Programs.

Step 2: Open the Form1 form in the *Designer*. The form is shown, along with the names of the important controls, in Figure 6-8.

Step 3: Move the PictureBox controls so one is on top of the other, as shown in Figure 6-9. (In the figure, the `lightOnPictureBox` control is on top, but it really does not matter which is on top.) Also, reduce the width of the form and position the button controls as shown in the figure.

Figure 6-9 The controls repositioned and the form size adjusted



Step 4: Program 6-2, at the end of this tutorial, shows the form's completed code. Open the code editor and type the code for the `TurnLightOn` and the `TurnLightOff` methods, shown in lines 10–32 of Program 6-2.

Let's take a closer look at the code. Line 10 is the beginning of a method named `TurnLightOn`. The purpose of this method is to simulate the light turning on. When this method executes, line 13 makes the `lightOnPictureBox` control visible, line 16 makes the `lightOffPictureBox` control invisible, and line 19 sets the `lightStateLabel` control's Text property to `"ON"`.

Line 22 is the beginning of a method named `TurnLightOff`. The purpose of this method is to simulate the light turning off. When this method executes, line 25 makes the `lightOffPictureBox` control visible, line 28 makes the `lightOnPictureBox` control invisible, and line 31 sets the `lightStateLabel` control's Text property to `"OFF"`.

Step 5: Next, you create the Click event handlers for the Button controls. Switch back to the *Designer* and double-click the `switchButton` control. This opens the code editor, and you will see an empty event handler named `switchButton_Click`. Complete the `switchButton_Click` event handler by typing the code shown in lines 36–44 in Program 6-2.

Let's review this code. The `if` statement in line 37 determines whether the `lightOnPictureBox` control is visible. If it is, it means the light is turned on, so the statement in line 39 calls the `TurnLightOff` method to turn the light off. Otherwise, the `else` clause in line 41 takes over, and the `TurnLightOn` method is called on line 43 to turn the light on.

Step 6: Switch your view back to the *Designer* and double-click the `exitButton` control. In the code editor you will see an empty event handler named `exitButton_Click`. Complete the `exitButton_Click` event handler by typing the code shown in lines 49–50 in Program 6-2.

Step 7: Save the project. Then, press F5 on the keyboard, or click the *Start Debugging* button (▶) on the toolbar to compile and run the application. When the application runs, click the *Switch Light* button several times to simulate several coin tosses. When you are finished, click the *Exit* button to exit the application.

Program 6-2 Completed code for Form1 in the *Lights* project

```
 1 namespace Lights
 2 {
 3     public partial class Form1: Form
 4     {
 5         public Form1()
 6         {
 7             InitializeComponent();
 8         }
 9
10         private void TurnLightOn()
11         {
12             // Display the "light on" image.
13             lightOnPictureBox.Visible = true;
14
15             // Hide the "light off" image.
16             lightOffPictureBox.Visible = false;
17
18             // Display the light's state.
19             lightStateLabel.Text = "ON";
20         }
21
22         private void TurnLightOff()
23         {
24             // Display the "light off" image.
25             lightOffPictureBox.Visible = true;
26
27             // Hide the "light on" image.
28             lightOnPictureBox.Visible = false;
29
30             // Display the light's state.
31             lightStateLabel.Text = "OFF";
32         }
33
34         private void switchButton_Click(object sender, EventArgs e)
35         {
36             // Reverse the state of the light.
37             if (lightOnPictureBox.Visible == true)
38             {
39                 TurnLightOff();
40             }
41             else
42             {
43                 TurnLightOn();
44             }
45         }
46
47         private void exitButton_Click(object sender, EventArgs e)
48         {
49             // Close the form.
50             this.Close();
51         }
52     }
53 }
```

Top–Down Design

In this section, we have discussed and demonstrated how methods work. You have seen how the program jumps to a method when it is called and returns to the part of the program that called the method when the method ends. It is important that you understand these mechanical aspects of methods.

Just as important as understanding how methods work is understanding how to use methods to modularize a program. Programmers commonly use a technique known as **top–down design** to break down an algorithm into methods. The process of top–down design is performed in the following manner:

- The overall task that the program is to perform is broken down into a series of subtasks.
- Each subtask is examined to determine whether it can be further broken down into more subtasks. This step is repeated until no more subtasks can be identified.
- Once all of the subtasks have been identified, they are written in code.

This process is called top–down design because the programmer begins by looking at the topmost level of tasks that must be performed and then breaks down those tasks into lower levels of subtasks.

NOTE: The top–down design process is sometimes called **stepwise refinement**.

Checkpoint

6.1 What is the difference between a `void` method and a value-returning method?

6.2 What two parts does a method definition have?

6.3 What does the phrase "calling a method" mean?

6.4 When a `void` method is executing, what happens when the end of the method is reached?

6.5 Describe the steps involved in the top–down design process.

6.3 Passing Arguments to Methods

CONCEPT: **An argument is any piece of data that is passed into a method when the method is called. A parameter is a variable that receives an argument that is passed into a method.**

Sometimes it is useful not only to call a method, but also to send one or more pieces of data into the method. Pieces of data that are sent into a method are known as **arguments**. The method can use its arguments in calculations or other operations.

You are already familiar with how to use arguments in a method call. For example, look at the following statement:

```
MessageBox.Show("Hello");
```

This statement calls the `MessageBox.Show` method and passes the string `"Hello"` as an argument. Here is another example:

```
number = int.Parse(str);
```

Assume that number is an `int` variable and `str` is a `string` variable. This statement calls the `int.Parse` method, passing the `str` variable as an argument.

If you are writing a method and you want it to receive arguments when it is called, you must equip the method with one or more parameter variables. A **parameter variable**, often simply called a parameter, is a special variable that receives an argument when a method is called. Here is an example of a method that has a parameter variable:

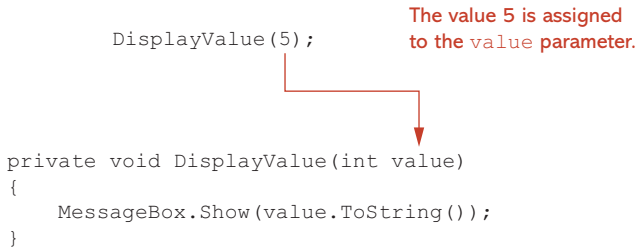
```
private void DisplayValue(int value)
{
    MessageBox.Show(value.ToString());
}
```

Notice the `int` variable declaration that appears inside the parentheses `(int value)`. This is the declaration of a parameter variable, which enables the `DisplayValue` method to accept an `int` value as an argument. Here is an example of a call to the `DisplayValue` method, passing 5 as an argument:

```
DisplayValue(5);
```

This statement executes the `DisplayValue` method. The argument that is listed inside the parentheses is assigned to the method's parameter variable, `value`. This is illustrated in Figure 6-10.

Figure 6-10 Passing the value 5 to the `DisplayValue` method



Inside the `DisplayValue` method, the variable `value` will contain the value of whatever argument was passed into it. If we pass 5 as the argument, the method will display the value 5 in a message box.

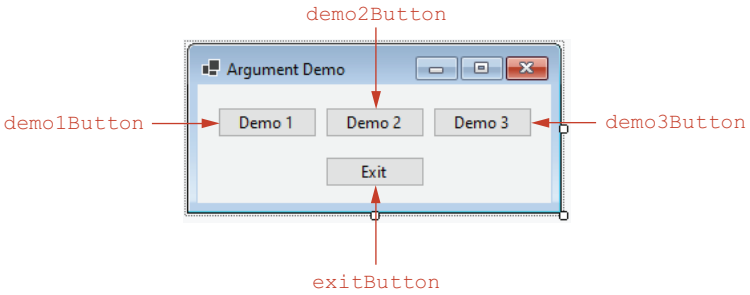
You may also pass the contents of variables and the values of expressions as arguments. For example, the following statements call the `DisplayValue` method with various arguments passed:

```
DisplayValue(x);
DisplayValue(x * 4);
DisplayValue(int.Parse("700"));
```

The first statement is simple. It passes the value of the variable `x` as the argument to the `DisplayValue` method. The second statement is also simple, but it does a little more work: it passes the result of the expression `x * 4` as the argument to the `DisplayValue` method. The third statement does even more work. It passes the value returned from the `int.Parse` method as the argument to the `DisplayValue` method. (The `int.Parse` method is called first, and its return value is passed to the `DisplayValue` method.)

In the *Chap06* folder of this book's Student Sample Programs, you will find a project named *ArgumentDemo* that demonstrates this method. Figure 6-11 shows the application's form, and Program 6-3 shows the form's code.

Figure 6-11 The *ArgumentDemo* project's form



Program 6-3 Code for the *ArgumentDemo* project's Form1 form

```
 1 namespace ArgumentDemo
 2 {
 3     public partial class Form1: Form
 4     {
 5         public Form1()
 6         {
 7             InitializeComponent();
 8         }
 9
10         private void DisplayValue(int value)
11         {
12             MessageBox.Show(value.ToString());
13         }
14
15         private void demo1Button_Click(object sender, EventArgs e)
16         {
17             // Call DisplayValue passing 5 as an argument.
18             DisplayValue(5);
19         }
20
21         private void demo2Button_Click(object sender, EventArgs e)
22         {
23             // Call DisplayValue passing the expression 3 + 5
24             // as an argument.
25             DisplayValue(3 + 5);
26         }
27
28         private void demo3Button_Click(object sender, EventArgs e)
29         {
30             // Use a loop to call DisplayValue 5 times.
31             for (int count = 0; count < 5; count++)
32             {
33                 DisplayValue(count);
34             }
35         }
36
37         private void exitButton_Click(object sender, EventArgs e)
38         {
39             // Close the form.
40             this.Close();
41         }
42     }
43 }
```

The form has four button controls, and a Click event handler has been written for each one. In addition to the event handlers, the code contains the `DisplayValue` method, which we discussed earlier, in lines 10–13.

If you run the application and click the *Demo 1* button, the `demo1Button_Click` event handler executes. Notice that in line 18 the `DisplayValue` method is called, passing 5 as an argument. This causes a message box to appear showing the value 5.

If you click the *Demo 2* button, the `demo2Button_Click` event handler executes. In line 25 the `DisplayValue` method is called, passing the expression `3 + 5` as an argument. This causes a message box to appear showing the value 8.

If you click the *Demo 3* button, the `demo3Button_Click` event handler executes. In line 31 a `for` loop executes five times, each time passing the `count` variable as an argument. This causes a message box to appear five times, showing the values 0 through 4.

NOTE: When calling a method and passing a variable as an argument, simply write the variable name inside the parentheses of the method call. Do *not* write the data type of the argument variable in the method call. For example, the following statement causes an error:

```
DisplayValue(int x); // Error!
```

The method call should appear as follows:

```
DisplayValue(x);     // Correct
```

NOTE: In this text, the values that are passed into a method are called arguments, and the variables that receive those values are called parameters. There are several variations of these terms in use. In some circles these terms are switched in meaning. Also, some call the arguments *actual parameters* and call the parameters *formal parameters*. Others use the terms *actual argument* and *formal argument*. Regardless of which set of terms you use, it is important to be consistent.

In Tutorial 6-2, you complete an application that calls a method and passes an argument to it.

Tutorial 6-2:
Passing an Argument to a Method

VideoNote
Tutorial 6-2: Passing an Argument to a Method

In this tutorial, you complete the *Cards* project in the *Chap06* folder of this book's Student Sample Programs. The project's form, shown in Figure 6-12, has already been created for you. The PictureBox controls show the images of three cards. Each PictureBox control's Visible property is set to False, so they do not initially appear when the application runs. After you complete the application, the user can select a card's name from the ListBox, click the *Show Card* button, and the image of the selected card will appear.

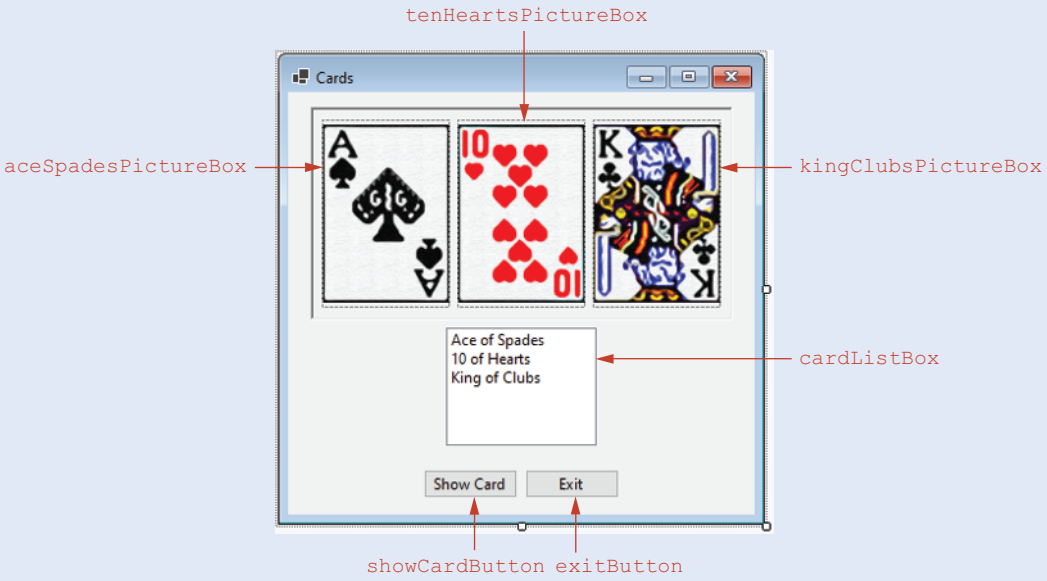
Step 1: Start Visual Studio. Open the project named *Cards* in the *Chap06* folder of this book's Student Sample Programs.

Step 2: Program 6-4, at the end of this tutorial, shows the form's completed code. Open the code editor and type the comments and code for the `ShowCard` method, shown in lines 10–26 of Program 6-4.

The purpose of the `ShowCard` method is to display one of the card PictureBox controls. Let's take a closer look at the code.

Line 12: This is the beginning of the method. The method has a `string` parameter named `card`. When we call the method, we pass the item that the user selected in the ListBox as an argument, and the method displays the specified card.

Figure 6-12 The *Cards* project's form



Lines 14–25: This is a `switch` statement that tests the value of the `card` parameter. If `card` is equal to `"Ace of Spades"`, the program jumps to the `case` statement in line 16 and calls the `ShowAceSpades` method in line 17. If `card` is equal to `"10 of Hearts"`, the program jumps to the `case` statement in line 19 and calls the `ShowTenHearts` method in line 20. If `card` is equal to `"King of Clubs"`, the program jumps to the `case` statement in line 22 and calls the `ShowKingClubs` method in line 23.

Step 3: Type the comments and code for the `ShowAceSpades` method, shown in lines 28–35 of Program 6-4. This method makes the `aceSpadesPictureBox` control visible, and the other PictureBox controls invisible.

Step 4: Type the comments and code for the `ShowTenHearts` method, shown in lines 37–44 of Program 6-4. This method makes the `tenHeartsPictureBox` control visible and the other PictureBox controls invisible.

Step 5: Type the comments and code for the `ShowKingClubs` method, shown in lines 46–53 of Program 6-4. This method makes the `kingClubsPictureBox` control visible, and the other PictureBox controls invisible.

Step 6: Next, you create the Click event handlers for the Button controls. Switch back to the *Designer* and double-click the `showCardButton` control. This opens the code editor, and you will see an empty event handler named `showCardButton_Click`. Complete the `showCardButton_Click` event handler by typing the code shown in lines 57–66 in Program 6-4.

Let's review this code. The `if` statement in line 58 determines whether the user has selected an item in the `cardListBox`. If a value has been selected, line 60 calls the `ShowCard` method. Notice that the item that was selected in the ListBox (converted to a string) is passed as an argument. If the user has not selected an item in the `cardListBox` control, the `else` clause in line 62 takes over, and lines 64–65 display a message box telling the user to select a card.

Step 7: Switch your view back to the *Designer* and double-click the `exitButton` control. In the code editor you will see an empty event handler named `exitButton_Click`. Complete the `exitButton_Click` event handler by typing the code shown in lines 71–72 in Program 6-4.

Step 8: Save the project. Then, press F5 on the keyboard or click the *Start Debugging* button (▶) on the toolbar to compile and run the application. Test the application by selecting each card's name in the ListBox and clicking the *Show Card* button. When you are finished, click the *Exit* button to exit the application.

Program 6-4 Completed code for Form1 in the *Cards* application

```
1  namespace Cards
2  {
3      public partial class Form1: Form
4      {
5          public Form1()
6          {
7              InitializeComponent();
8          }
9
10         // The ShowCard method accepts a string that names
11         // the selected card, and displays that card.
12         private void ShowCard(string card)
13         {
14             switch (card)
15             {
16                 case "Ace of Spades":
17                     ShowAceSpades();
18                     break;
19                 case "10 of Hearts":
20                     ShowTenHearts();
21                     break;
22                 case "King of Clubs":
23                     ShowKingClubs();
24                     break;
25             }
26         }
27
28         // The ShowAceSpades method makes the Ace of Spades
29         // visible and the other cards invisible.
30         private void ShowAceSpades()
31         {
32             aceSpadesPictureBox.Visible = true;
33             tenHeartsPictureBox.Visible = false;
34             kingClubsPictureBox.Visible = false;
35         }
36
37         // The ShowTenHearts method makes the Ten of Hearts
38         // visible and the other cards invisible.
39         private void ShowTenHearts()
40         {
41             tenHeartsPictureBox.Visible = true;
42             aceSpadesPictureBox.Visible = false;
43             kingClubsPictureBox.Visible = false;
44         }
45
46         // The ShowKingClubs method makes the King of Clubs
47         // visible and the other cards invisible.
48         private void ShowKingClubs()
49         {
50             kingClubsPictureBox.Visible = true;
```

```
51              aceSpadesPictureBox.Visible = false;
52              tenHeartsPictureBox.Visible = false;
53          }
54
55          private void showCardButton_Click(object sender,
            EventArgs e)
56          {
57              // If a card is selected in the ListBox, display it.
58              if (cardListBox.SelectedIndex != -1)
59              {
60                  ShowCard(cardListBox.SelectedItem.ToString());
61              }
62              else
63              {
64                  MessageBox.Show("Please select a card from " +
65                                  "the list box.");
66              }
67          }
68
69          private void exitButton_Click(object sender, EventArgs e)
70          {
71              // Close the form.
72              this.Close();
73          }
74      }
75  }
```

Argument and Parameter Data Type Compatibility

When you pass an argument to a method, the argument's data type must be assignment compatible with the receiving parameter's data type. Otherwise, an error occurs when you try to compile the code. We discussed assignment compatibility in Chapter 3. Here is a summary of how it applies to argument passing when using `string`s, `int`s, `double`s, and `decimal`s:

- You can pass only string arguments into `string` parameters.
- You can pass `int` arguments into `int` parameters, but you cannot pass `double` or `decimal` arguments into `int` parameters.
- You can pass either `double` or `int` arguments into `double` parameters, but you cannot pass `decimal` values into `double` parameters.
- You can pass either `decimal` or `int` arguments to `decimal` parameters, but you cannot pass `double` arguments into `decimal` parameters.

Parameter Variable Scope

Recall from Chapter 3 that a variable's scope is the part of the program where the variable may be accessed. A variable is visible only to statements inside the variable's scope. A parameter variable's scope is the method in which the parameter is declared. No statement outside the method can access the parameter variable.

Passing Multiple Arguments

Often it is useful to pass more than one argument to a method. The following code sample shows a method that accepts two arguments. The name of the method is `ShowMax`. It accepts two `int` arguments and displays the value of the argument that is the greatest. If

the arguments are equal, it displays a message saying so. (This method can be found in the *Max* project in the *Chap06* folder of this book's Student Sample Programs.)

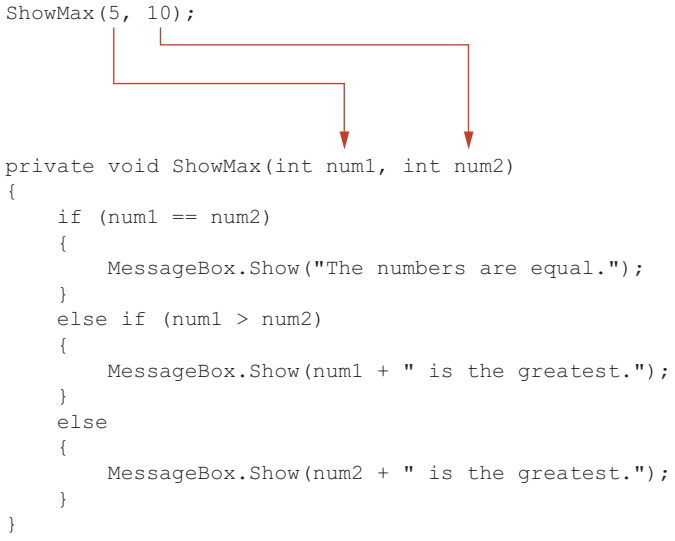
```
 1 private void ShowMax(int num1, int num2)
 2 {
 3     if (num1 == num2)
 4     {
 5         MessageBox.Show("The numbers are equal.");
 6     }
 7     else if (num1 > num2)
 8     {
 9         MessageBox.Show(num1 + " is the greatest.");
10     }
11     else
12     {
13         MessageBox.Show(num2 + " is the greatest.");
14     }
15 }
```

Notice that two parameter variables, `num1` and `num2`, are declared inside the parentheses in the method header (line 1). This is often referred to as a parameter list. Also notice that a comma separates the declarations. Here is an example of a statement that calls the method:

```
ShowMax(5, 10);
```

This statement passes the arguments 5 and 10 to the method. The arguments are passed into the parameter variables according to their positions. In other words, the first argument is passed into the first parameter variable, the second argument is passed into the second parameter variable, and so forth. So, this statement causes 5 to be passed into the `num1` parameter and 10 to be passed into the `num2` parameter. This is illustrated in Figure 6-13.

Figure 6-13 Two arguments passed according to position to a method



Suppose we were to reverse the order in which the arguments are listed in the method call, as shown here:

```
ShowMax(10, 5);
```

This causes 10 to be passed into the `num1` parameter and 5 to be passed into the `num2` parameter. The following code sample shows one more example. This time we are passing variables as arguments.

```
int value1 = 2;
int value2 = 3;
ShowMax(value1, value2);
```

When the `ShowMax` method executes as a result of this code, the `num1` parameter contains 2 and the `num2` parameter contains 3.

NOTE: You have to write the data type for each parameter variable that is declared in a parameter list. For example, a compiler error would occur if the parameter list for the `ShowMax` method were written as shown here:

```
private void ShowMax(int num1, num2) // Error!
```

A data type for both the `num1` and `num2` parameter variables must be listed, as shown here:

```
private void ShowMax(int num1, int num2)
```

Named Arguments

In addition to the conventional approach of positional argument passing (where the first argument is passed into the method's first parameter, the second argument is passed into the method's second parameter, and so forth), C# also allows you to specify which parameter an argument should be passed into. To specify which parameter variable the argument should be passed to, you use the following format to write the argument in the method call:

```
parameterName : value
```

In this format `parameterName` is the name of a parameter variable and `value` is the value being passed to that parameter. An argument that is written using this syntax is known as a **named argument**. To demonstrate, look at the following method:

```
private void ShowName(string firstName, string lastName)
{
    MessageBox.Show(firstName + " " + lastName);
}
```

The following statement shows how the method can be called using named arguments:

```
ShowName(lastName : "Smith", firstName : "Suzanne");
```

This statement specifies that `"Smith"` should be passed into the `lastName` parameter and `"Suzanne"` should be passed into the `firstName` parameter. You get the same results as if you had called the method like this, using positional arguments:

```
ShowName("Suzanne", "Smith");
```

Default Arguments

C# allows you to provide a **default argument** for a method parameter. When a default argument is provided for a parameter, it becomes possible to call the method without explicitly passing an argument into the parameter. Here is an example of a method that has a parameter with a default argument:

```
private void ShowTax(decimal price, decimal taxRate = 0.07m)
{
    // Calculate the tax.
```

```
        decimal tax = price * taxRate;

        // Display the tax.
        MessageBox.Show("The tax is " + tax.ToString("c"));
    }
```

In this method definition, a default argument is provided for the `taxRate` parameter. Notice that the parameter name is followed by an equal sign and a value. The value that follows the equal sign is the default argument. In this case, the value `0.07m` is the default argument for the `taxRate` parameter. Because the `taxRate` parameter has a default argument, we have the option of omitting an argument for it when we call the method. Here is an example:

```
ShowTax(100.0m);
```

This statement calls the `ShowTax` method, passing the value `100.0m` as the argument for the `price` parameter. Because we did not pass an argument into the `taxRate` parameter, its value will be `0.07m`. If we want the `taxRate` parameter to have a different value, we can specify an argument for it when we call the method, as shown here:

```
ShowTax(100.0m, 0.08m);
```

This statement calls the `ShowTax` method, passing `100.0m` as the argument for the `price` parameter and `0.08m` as the argument for the `taxRate` parameter.

Here are some details to keep in mind when using default arguments:

- Default arguments must be literals or constants. You cannot specify a variable as a default argument.
- You can provide default arguments for all of the parameters in a method. However, when only some of the parameters have a default argument (as in the previous example), you must declare the parameters with the default arguments last. For example, a compiler error would occur if we were to write the `ShowTax` method header as shown here:

  ```
  // Illegal method header!
  private void ShowTax(decimal taxRate = 0.07m, decimal price)
  ```

- When a method has several parameters with default arguments and you leave out one of the arguments when you call the method, you have to leave out all the arguments that come after it as well.

Passing Arguments by Value

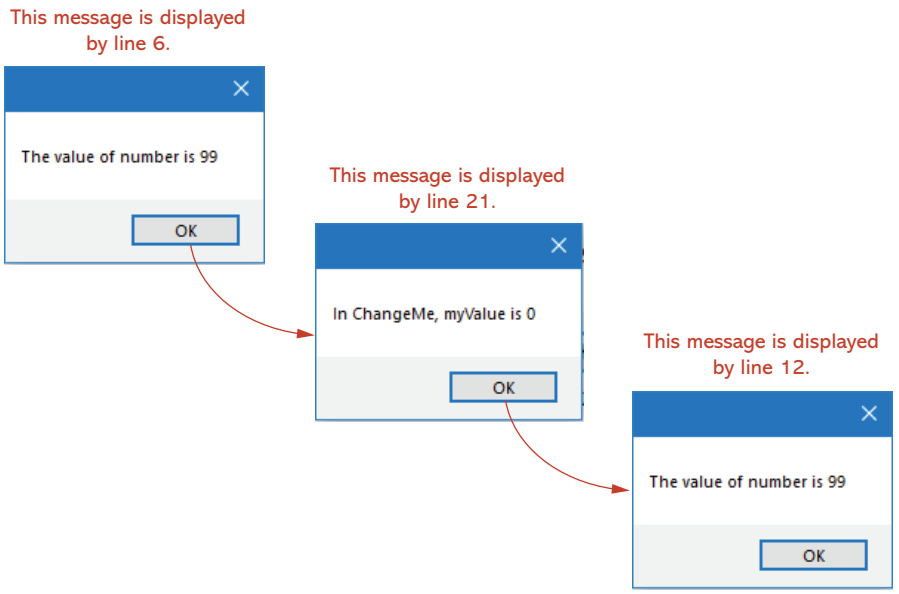
All the example programs that you have looked at so far pass arguments by value. Arguments and parameter variables are separate items in memory. When an argument is **passed by value**, only a copy of the argument's value is passed into the parameter variable. If the contents of the parameter variable are changed inside the method, it has no effect on the argument in the calling part of the program.

For example, the following code comes from the *PassByValue* project in the *Chap06* folder of this book's Student Sample Programs. When you run the application and click the *Go* button, you see the sequence of message boxes shown in Figure 6-14.

```
1 private void goButton_Click(object sender, EventArgs e)
2 {
3     int number = 99;
4
5     // Display the value of number.
6     MessageBox.Show("The value of number is " + number);
7
8     // Call ChangeMe, passing number as an argument.
```

```
 9      ChangeMe(number);
10
11      // Display the value of number again.
12      MessageBox.Show("The value of number is " + number);
13  }
14
15  private void ChangeMe(int myValue)
16  {
17      // Change the value of the myValue parameter.
18      myValue = 0;
19
20      // Display the value of myValue.
21      MessageBox.Show("In ChangeMe, myValue is " + myValue);
22  }
```

Figure 6-14 Sequence of messages displayed by the *PassByValue* project



Inside the `goButton_Click` event handler, a local variable named `number` is declared in line 3 and initialized with the value 99. As a result, the statement in line 6 displays *The value of number is 99*. The `number` variable's value is then passed as an argument to the `ChangeMe` method in line 9. This means that in the `ChangeMe` method, the value 99 is assigned to the `myValue` parameter variable.

Inside the `ChangeMe` method, line 18 assigns the value 0 to the `myValue` parameter variable. This overwrites the value 99 that was passed into the parameter when the method was called. Line 21 displays the message *In ChangeMe, myValue is 0*.

After the `ChangeMe` method finishes, control of the program returns to the `goButton_Click` event handler. When the statement in line 12 executes, the message *The value of number is 99* is displayed. Even though the parameter variable `myValue` was changed in the `ChangeMe` method, the `number` variable in the `goButton_Click` event handler was not modified.

Passing by value works in most situations because arguments are usually sent to methods for informational purposes only. Typically, when you pass a variable as an argument to a method, you want that variable to have the same value before *and* after the method call. Passing an argument by value guarantees that the argument will not be changed by the

method it is passed into. Sometimes, however, you want a method to be able to change the value of a variable that was passed as an argument to it. This requires a slightly different type of argument passing, which is discussed in the next section.

Checkpoint

6.6 What is the purpose of an argument?

6.7 Briefly summarize how assignment compatibility applies to argument passing.

6.8 What is the scope of a parameter variable?

6.9 What is a named argument?

6.10 What does it mean when an argument is passed by value?

6.4 Passing Arguments by Reference

CONCEPT: **When an argument is passed by reference to a method, the method can change the value of the argument in the calling part of the program.**

When you want a method to be able to change the value of a variable that is passed to it as an argument, the variable must be **passed by reference**. In C#, there are two ways to pass an argument by reference:

- You can use a reference parameter in the method.
- You can use an output parameter in the method.

Using Reference Parameters

A **reference parameter** is a special type of parameter variable. When you pass an argument into a reference parameter, the reference parameter does not receive a copy of the argument's value. Instead, it becomes a *reference* to the argument that was passed into it. Anything that is done to the reference parameter is actually done to the argument that it references.

Reference parameters are useful for establishing two-way communication between methods. When a method calls another method and passes an argument by reference, communication between the methods can take place in the following ways:

- The calling method can communicate with the called method by passing an argument.
- The called method can communicate with the calling method by modifying the value of the argument via the reference parameter.

In C#, you declare a reference parameter by writing the `ref` keyword before the parameter variable's data type. For example, look at the following method:

```
private void SetToZero(ref int number)
{
    number = 0;
}
```

Inside the parentheses, the keyword `ref` indicates that `number` is a reference variable. The method assigns 0 to the `number` parameter. Because `number` is a reference parameter, this action is actually performed on the variable that was passed to the method as an argument.

When you call a method that has a reference parameter, you must also write the keyword `ref` before the argument. The following code sample shows an example:

```
int myVar = 99;
SetToZero(ref myVar);
```

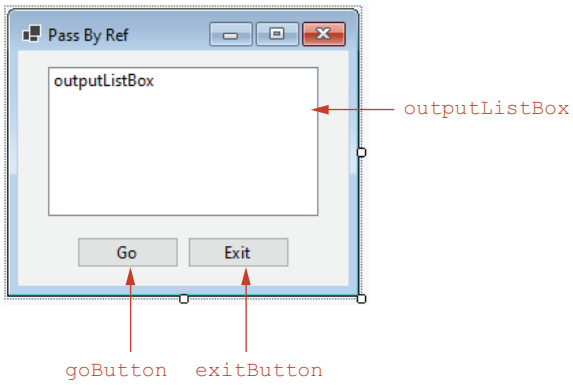
The first statement declares `myVar` as an `int` variable, initialized with the value 99. The second statement calls the `SetToZero` method, passing `myVar` by reference. After the method call, the `myVar` variable is set to the value 0.

When you pass an argument to a `ref` parameter, that argument must already be set to some value. For example, if a variable has not been initialized or assigned a value, you cannot pass it as an argument into a `ref` parameter. The following code sample causes a compiler error:

```
int myVar;             // Declare myVar with no initial value.
SetToZero(ref myVar); // Error! myVar is not set to a value.
```

Let's look at a complete program that uses the `SetToZero` method. In the *Chap06* folder of this book's Student Sample Programs, you will find a project named *PassByRef*. Figure 6-15 shows the project's form, and Program 6-5 shows the form's code.

Figure 6-15 The *PassByRef* project's form



Program 6-5 Code for the *PassByRef* project's Form1 form

```
 1 namespace PassByRef
 2 {
 3     public partial class Form1 : Form
 4     {
 5         public Form1()
 6         {
 7             InitializeComponent();
 8         }
 9
10         // The SetToZero method accepts an int argument
11         // by reference and sets it to zero.
12         private void SetToZero(ref int number)
13         {
14             number = 0;
15         }
16
17         private void goButton_Click(object sender, EventArgs e)
18         {
19             // Declare some local int variables.
20             int x = 99, y = 100, z = 101;
```

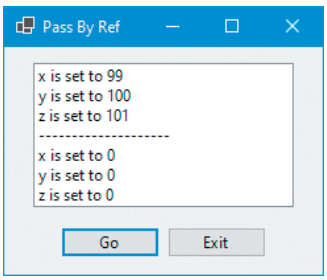
```
21
22              // Display the values in those variables.
23              outputListBox.Items.Clear();
24              outputListBox.Items.Add("x is set to " + x);
25              outputListBox.Items.Add("y is set to " + y);
26              outputListBox.Items.Add("z is set to " + z);
27
28              // Pass each variable to SetToZero.
29              SetToZero(ref x);
30              SetToZero(ref y);
31              SetToZero(ref z);
32
33              // Display the values in those variables again.
34              outputListBox.Items.Add("--------------------");
35              outputListBox.Items.Add("x is set to " + x);
36              outputListBox.Items.Add("y is set to " + y);
37              outputListBox.Items.Add("z is set to " + z);
38          }
39
40          private void exitButton_Click(object sender, EventArgs e)
41          {
42              // Close the form.
43              this.Close();
44          }
45      }
46  }
```

Notice that the form has a ListBox control named `outputListBox`. This ListBox is used to display the program's output. The `SetToZero` method that we previously discussed appears in lines 12–15. The method accepts an `int` argument by reference and assigns the value 0 to the argument.

In the `goButton_Click` method, line 20 declares the `int` variables `x`, `y`, and `z` and initializes them to the values 99, 100, and 101, respectively. Line 23 clears the `outputListBox` control, and lines 24–26 display the values of the `x`, `y`, and `z` variables in the ListBox.

In lines 29–31 `x`, `y`, and `z` variables are passed as arguments, by reference, to the `SetToZero` method. Each time `SetToZero` is called, the variable that is passed as an argument is assigned the value 0. This is shown when the `x`, `y`, and `z` variables are displayed again in lines 35–37. Figure 6-16 shows the application's form after the *Go* button has been clicked.

Figure 6-16 The *Pass By Ref* application's output



Using Output Parameters

An **output parameter** works like a reference parameter. When you pass an argument into an output parameter, the output parameter becomes a reference to the argument that is passed into it. Anything that is done to the output parameter is actually done to the

argument that it references. Output parameters are different from reference parameters in the following ways:

- An argument does not have to be set to a value before it is passed into an output parameter. For example, an uninitialized variable can be passed into an output parameter.
- A method that has an output parameter must set the output parameter to some value before it finishes executing.

In C#, you declare an output parameter by writing the `out` keyword before the parameter variable's data type. For example, we could modify the `SetToZero` method in the following way to make the `number` parameter an output parameter:

```
private void SetToZero(out int number)
{
    number = 0;
}
```

When you call a method that has an output parameter, you must also write the keyword `out` before the argument. The following code sample shows an example:

```
int myVar;
SetToZero(out myVar);
```

The first statement declares `myVar` as an uninitialized `int` variable. The second statement calls the `SetToZero` method, passing `myVar` into the output parameter. After the method call, the `myVar` variable is set to the value 0.

Tutorial 6-3 gives you some experience writing a method that uses an ouput parameter.

Tutorial 6-3:
Using an Output Parameter

VideoNote
Tutorial 6-3: Using an Output Parameter

In this tutorial, you complete the *NorthAmerica* application that is found in the *Chap06* folder of this book's Student Sample Programs. The application's form has already been created and is shown in Figure 6-17. The application also has an accompanying text file named NorthAmerica.txt that is stored in the *Chap06* folder. The NorthAmerica. txt file contains the names of the countries of North America.

Figure 6-17 The *NorthAmerica* project's form

When the completed application runs and the user clicks the *Get Countries* button, the application uses an OpenFileDialog control to let the user select a file. The application reads each country name from the file and adds each one to the `countriesListBox` control.

Step 1: Start Visual Studio. Open the project named *NorthAmerica* in the *Chap06* folder of this book's Student Sample Programs.

Step 2: Open Form1 in the *Designer* and add an OpenFileDialog control to the form. Change the control's name to `openFile` and clear the contents of the control's Filename property.

Step 3: Open the code editor and type the comments and code for the `GetFileName` method, shown in lines 10–23 of Program 6-6. The purpose of the `GetFileName` method is to let the user select the file that should be opened. Let's take a closer look at the code.

Line 13: This is the beginning of the method. The method has a `string` output parameter named `selectedFile`. When we call the method, we pass a `string` variable as an argument. The method lets the user select the file that should be opened and stores its filename and path in the `selectedFile` parameter.

Lines 15–22: This `if` statement calls the `openFile` control's `ShowDialog` method. If the user clicks the *Open* button, the method returns the value `DialogResult.OK`, and line 17 assigns the name of the selected file to the `selectedFile` parameter. If the user clicks the *Cancel* button, line 21 assigns an empty string to the `selectedFile` parameter.

Step 4: Type the comments and code for the `GetCountries` method, shown in lines 25–62 of Program 6-6. In a nutshell, this method accepts a filename as an argument, reads the contents of the specified file, and adds them to the `countriesListBox` control. Here is a more detailed description of each part of the method:

Line 30: This is the beginning of a `try-catch` statement, which handles any exceptions that are thrown while the file is being processed. If an exception is thrown by any statement in the try block, the program will jump to the `catch` clause in line 57.

Line 33: This statement declares the `string` variable `countryName`, which holds the lines of text that are read from the file.

Line 36: This statement declares the `StreamReader` variable `inputFile`.

Line 39: After this statement has executed, the file specified by the `filename` parameter is opened for reading, and the `inputFile` variable references a `StreamReader` object that is associated with the file.

Line 42: This statement clears anything that might be displayed in the `countriesListBox` control.

Line 45: This is the beginning of a `while` loop that iterates as long as the end of the file has not been reached.

Line 48: This statement reads a line of text from the file and assigns it to the `countryName` variable.

Line 51: This statement adds the contents of the `countryName` variable to the ListBox.

Line 55: This statement closes the file.

Step 5: Open the Form1 form in the *Designer*. Double-click the `getCountriesButton` control. This opens the code editor, and you will see an empty event handler named `getCountriesButton_Click`. Complete the `getCountriesButton_Click` event handler by typing the code shown in lines 66–72 in Program 6-6. Let's take a closer look at the code:

Line 66: This statement declares a `string` variable named `filename`.

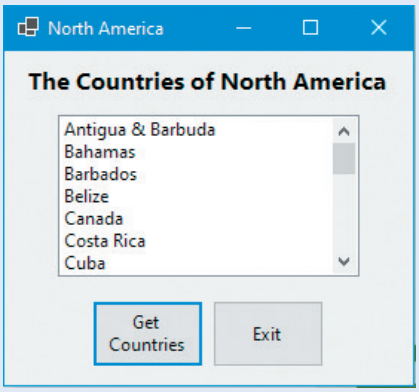
Line 69: This statement calls the `GetFileName` method, passing the `filename` variable as an argument. When the method returns, the `filename` variable contains the name of the file selected by the user.

Line 72: This statement calls the `GetCountries` method, passing the `filename` variable as an argument. The method opens the specified file and fills the `countriesListBox` control with its contents.

Step 6: Switch your view back to the *Designer* and double-click the `exitButton` control. In the code editor you will see an empty event handler named `exitButton_Click`. Complete the `exitButton_Click` event handler by typing the code shown in lines 77–78 in Program 6-6.

Step 7: Save the project. Then, press F5 on the keyboard or click the *Start Debugging* button ( ) on the toolbar to compile and run the application. When the application runs, click the *Get Countries* button. An *Open* dialog box should appear. Navigate to the *Chap06* folder in the Student Sample Programs, select the NorthAmerica.txt file, and click the *Open* button. This should fill the ListBox with the names of the countries from the selected file, as shown in Figure 6-18. Click the *Exit* button to exit the application.

Figure 6-18 The *North America* application displaying the list of countries



Program 6-6 Completed code for Form1 in the *NorthAmerica* project

```
1 namespace NorthAmerica
2 {
3     public partial class Form1 : Form
4     {
5         public Form1()
6         {
7             InitializeComponent();
8         }
9
```

```
10          // The GetFileName method gets a filename from the
11          // user and assigns it to the variable passed as
12          // an argument.
13          private void GetFileName(out string selectedFile)
14          {
15              if (openFile.ShowDialog() == DialogResult.OK)
16              {
17                  selectedFile = openFile.FileName;
18              }
19              else
20              {
21                  selectedFile = "";
22              }
23          }
24
25          // The GetCountries method accepts a filename as an
26          // argument. It opens the specified file and displays
27          // its contents in the countriesListBox control.
28          private void GetCountries(string filename)
29          {
30              try
31              {
32                  // Declare a variable to hold a country name.
33                  string countryName;
34
35                  // Declare a StreamReader variable.
36                  StreamReader inputFile;
37
38                  // Open the file and get a StreamReader object.
39                  inputFile = File.OpenText(filename);
40
41                  // Clear anything currently in the ListBox.
42                  countriesListBox.Items.Clear();
43
44                  // Read the file's contents.
45                  while (!inputFile.EndOfStream)
46                  {
47                      // Get a country name.
48                      countryName = inputFile.ReadLine();
49
50                      // Add the country name to the ListBox.
51                      countriesListBox.Items.Add(countryName);
52                  }
53
54                  // Close the file.
55                  inputFile.Close();
56              }
57              catch (Exception ex)
58              {
59                  // Display an error message.
60                  MessageBox.Show(ex.Message);
61              }
62          }
63
64          private void getCountriesButton_Click(object sender,
            EventArgs e)
65          {
66              string filename;    // To hold the filename
67
```

```
68              // Get the filename from the user.
69              GetFileName(out filename);
70
71              // Get the countries from the file.
72              GetCountries(filename);
73          }
74
75          private void exitButton_Click(object sender, EventArgs e)
76          {
77              // Close the form.
78              this.Close();
79          }
80      }
81  }
```

Checkpoint

6.11 What is a reference parameter?

6.12 How can methods communicate using reference parameters?

6.13 What keyword is used to specify a reference parameter?

6.14 What is an output parameter?

6.5 Value-Returning Methods

CONCEPT: A value-returning method is a method that returns a value to the part of the program that called it.

A value-returning method is like a `void` method in the following ways:

- It contains a group of statements that perform a specific task.
- When you want to execute the method, you call it.

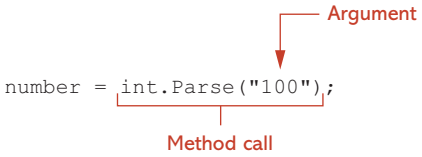
When a value-returning method finishes, however, it returns a value to the statement that called it. The value that is returned from a method can be used like any other value: It can be assigned to a variable, displayed on the screen, used in a mathematical expression (if it is a number), and so forth.

You have already used many of the value-returning .NET methods. For example, the `int.Parse` method accepts a string as an argument and returns the value of the string converted to an `int`. Let's review how that method works. In the following statement, assume `number` is an `int` variable:

```
number = int.Parse("100");
```

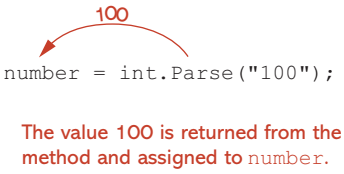
The part of the statement that reads `int.Parse("100")` is a call to the `int.Parse` method, with the string `"100"` passed as an argument. Figure 6-19 illustrates this part of the statement.

Figure 6-19 A statement that calls the `int.Parse` method



Notice that the call to the `int.Parse` method appears on the right side of an = operator. When the method is called, it returns an integer. The integer that is returned is assigned to the `number` variable, as shown in Figure 6-20.

Figure 6-20 The `int.Parse` method returns a value



Writing Your Own Value-Returning Methods

You write a value-returning method in the same way that you write a `void` method, with two exceptions:

- You must specify a data type for a value-returning method. The value that is returned from the method must be of the specified data type.
- A value-returning method must have a `return` statement. The `return` statement causes a value to be returned from the method.

Here is the general format of a value-returning method definition in C#:

```
AccessModifier DataType MethodName(ParameterList)
{
     statement;
     statement;
     etc.
     return  expression;
}
```

- *AccessModifier* is an access modifier such as `private` or `public`.
- *DataType* is the data type of the value that the method returns. We commonly call this the method's **return type**. For example, if the method returns an integer, the word `int` appears here. If the method returns a `double` value, then the word `double` appears here. Likewise, if the method returns a `decimal` value, the word `decimal` appears here.
- *MethodName* is the name of the method.
- *ParameterList* is an optional parameter list. If the method does not accept arguments, then an empty set of parentheses appears.

One of the statements inside the method must be a `return` statement, which takes the following form:

```
return expression;
```

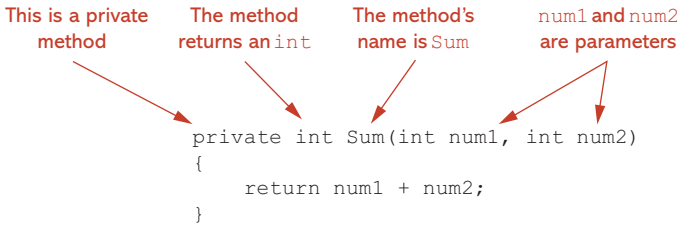
The value of the `expression` that follows the keyword `return` is sent back to the statement that called the method. This can be any value, variable, or expression that has a value (such as a math expression). The value that is returned must be of the same data type as that specified in the method header or a compiler error will occur.

Here is an example of a value-returning method:

```
private int Sum(int num1, int num2)
{
    return num1 + num2;
}
```

Figure 6-21 illustrates the various parts of the method header. Notice that the method returns an `int`, the method's name is `sum`, and the method has two `int` parameters named `num1` and `num2`.

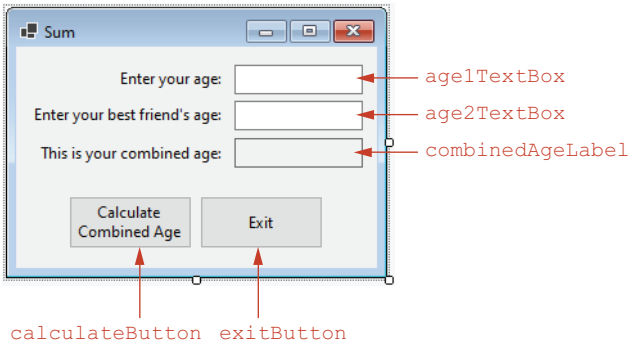
Figure 6-21 Parts of the method header



The purpose of this method is to accept two `int` values as arguments and return their sum. Notice that the `return` statement returns the value of the expression `num1 + num2`. When the `return` statement executes, the method ends its execution and sends the value of `num1 + num2` back to the part of the program that called the method.

Let's look at a complete program that demonstrates the `Sum` method. In the *Chap06* folder of this book's Student Sample Programs, you will find a project named *Sum*. Figure 6-22 shows the application's form, and Program 6-7 shows the form's code. When you run the application, you enter two integers into the `age1TextBox` and `age2TextBox` controls. When you click the *Calculate Combined Age* button, the sum of the two integers is displayed in the `combinedAgeLabel` control.

Figure 6-22 The *Sum* application's form



Program 6-7 Code for the *Sum* application's Form1 form

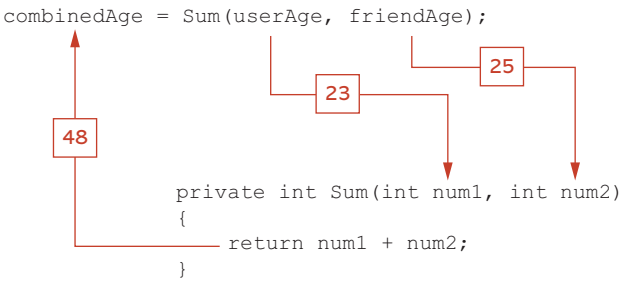
```
 1 namespace Sum
 2 {
 3     public partial class Form1 : Form
 4     {
 5         public Form1()
 6         {
 7             InitializeComponent();
 8         }
 9
10         // The Sum method accepts two int arguments
11         // and returns the sum of the arguments.
12         private int Sum(int num1, int num2)
13         {
14             return num1 + num2;
15         }
16
17         private void calculateButton_Click(object sender,
           EventArgs e)
18         {
19             // Declare variables to hold two ages and their sum.
20             int userAge, friendAge, combinedAge;
21
22             // Get the user's age.
23             if (int.TryParse(age1TextBox.Text, out userAge))
24             {
25                 // Get the best friend's age age.
26                 if (int.TryParse(age2TextBox.Text, out friendAge))
27                 {
28                     // Get the sum of the ages.
29                     combinedAge = Sum(userAge, friendAge);
30
31                     // Display the combined age.
32                     combinedAgeLabel.Text = combinedAge.ToString();
33                 }
34                 else
35                 {
36                     // Display an error message.
37                     MessageBox.Show("Enter an integer for your age.");
38                 }
39             }
40             else
41             {
42                 // Display an error message.
43                 MessageBox.Show("Enter an integer for your age.");
44             }
45         }
46
47         private void exitButton_Click(object sender, EventArgs e)
48         {
49             // Close the form.
50             this.Close();
51         }
52     }
53 }
```

In line 23, the value entered into the `age1TextBox` control is converted to an `int` and stored in the `userAge` variable. In line 26, the value entered into the `age2TextBox` control is converted to an `int` and stored in the `friendAge` variable. Line 29 passes the `userAge` and `friendAge` variables as arguments to the `Sum` method. The sum of the two variables

is returned from the method and assigned to the `combinedAge` variable. In line 32 the value of the `combinedAge` variable is converted to a string and displayed in the `combinedAgeLabel` control.

Let's assume the `userAge` variable is set to the value 23 and the `friendAge` variable is set to the value 25. Figure 6-23 shows how the arguments are passed to the method and how a value is returned from the method.

Figure 6-23 Arguments passed to `Sum` and a value returned



When you call a value-returning method, you usually want to do something meaningful with the value it returns. In line 29 of Program 6-7 the value that is returned from the `Sum` method is assigned to a variable. This is commonly how return values are used, but you can do many other things with them. For example, the following code shows a math expression that uses a call to the `Sum` method:

```
int x = 10, y = 15;
double average;
average = Sum(x, y) / 2.0;
```

In the last statement, the `Sum` method is called with `x` and `y` as its arguments. The method's return value, which is 25, is divided by 2.0. The result, 12.5, is assigned to `average`. Here is another example:

```
int x = 10, y = 15;
MessageBox.Show("The sum is " + Sum(x, y));
```

This code sends the `Sum` method's return value to `MessageBox.Show`, so it can be displayed on the screen. The message *The sum is 25* is displayed. Remember, a value-returning method returns a value of a specific data type. You can use the method's return value anywhere that you can use a regular value of the same data type. This means that anywhere an `int` value can be used, a call to an `int` value-returning method can be used. Likewise, anywhere a `double` value can be used, a call to a `double` value-returning method can be used. The same is true for all other data types.

In Tutorial 6-4, you complete an application that converts a value from one unit of measurement to another. The code will use a value-returning method to perform the conversion.

Tutorial 6-4:
Writing a Value-Returning Method

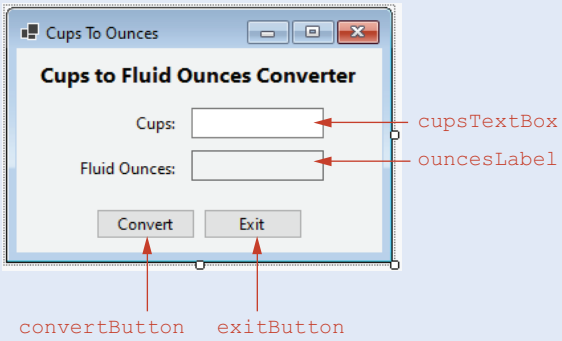
VideoNote
Tutorial 6-4: Writing a Value-Returning Method

Cups and fluid ounces are common units of measurement for food items. Sometimes, when a recipe calls for an item measured in cups, you find that in the grocery store the item is sold in fluid ounces. To know how much you need to purchase for the recipe, you need to convert the required number of cups to fluid ounces. The formula is:

$$Ounces = Cups \times 8$$

In this tutorial, you complete the *CupsToOunces* project that is found in the *Chap06* folder of this book's Student Sample Programs. The project's form has already been created and is shown in Figure 6-24. When you complete the application, you will be able to enter a number of cups into the `cupsTextBox` control, click the *Convert* button, and see the equivalent number of fluid ounces in the `ouncesLabel` control.

Figure 6-24 The *CupsToOunces* project's form



Step 1: Start Visual Studio. Open the project named *CupsToOunces* in the *Chap06* folder of this book's Student Sample Programs.

Step 2: Open the Form1 form in the *Designer*. The form is shown, along with the names of the important controls, in Figure 6-24.

Step 3: Program 6-8, at the end of this tutorial, shows the form's completed code. Open the code editor and type the comments and code for the `CupsToOunces` method shown in lines 10–16 of Program 6-8. The purpose of the method is to accept a number of cups as an argument and return that value converted to fluid ounces. You can see in line 13 that the method has a `double` parameter named `cups`, and in line 15 the method returns the value of `cups` multiplied by 8.

Step 4: Now you will create the Click event handlers for the Button controls. Switch back to the *Designer* and double-click the `convertButton` control. This opens the code editor, and you will see an empty event handler named `convertButton_Click`. Complete the `convertButton_Click` event handler by typing the code shown in lines 20–36 in Program 6-8. Let's review this code:

Line 21: This statement declares the `cups` and `ounces` variables, which hold the number of cups and ounces.

Lines 24–36: The `if` statement in line 24 converts the `cupsTextBox` control's Text property to a `double`, and the result is stored in the `cups` variable. If the conversion is successful, line 27 calls the `CupsToOunces` method, passing `cups` as an argument. The value that is returned from the method is assigned to the `ounces` variable, and in line 30 the value of the `ounces` variable is displayed in the `ouncesLabel` control. If the conversion is not successful, line 35 displays an error message.

Step 5: Switch your view back to the *Designer* and double-click the `exitButton` control. In the code editor you will see an empty event handler named `exitButton_Click`. Complete the `exitButton_Click` event handler by typing the code shown in lines 41–42 in Program 6-8.

Step 6: Save the project. Then, press F5 on the keyboard, or click the *Start Debugging* button (▶) on the toolbar to compile and run the application. When the application runs, enter the value 1 into the `cupsTextBox` control and click the *Convert* button. The application should display 8.0 as the number of fluid ounces. Continue to test the application with other values. When you are finished, click the *Exit* button to exit the application.

Program 6-8 Completed code for Form1 in the *CupsToOunces* project

```
1  namespace CupsToOunces
2  {
3      public partial class Form1 : Form
4      {
5          public Form1()
6          {
7              InitializeComponent();
8          }
9
10         // The CupsToOunces method accepts a number
11         // of cups as an argument and returns the
12         // equivalent number of fluid ounces.
13         private double CupsToOunces(double cups)
14         {
15             return cups * 8.0;
16         }
17
18         private void convertButton_Click(object sender, EventArgs e)
19         {
20             // Variables to hold cups and ounces
21             double cups, ounces;
22
23             // Get the number of cups.
24             if (double.TryParse(cupsTextBox.Text, out cups))
25             {
26                 // Convert the cups to ounces.
27                 ounces = CupsToOunces(cups);
28
29                 // Display the ounces.
30                 ouncesLabel.Text = ounces.ToString("n1");
31             }
32             else
33             {
34                 // Display an error message.
35                 MessageBox.Show("Enter a valid number.");
36             }
37         }
38
39         private void exitButton_Click(object sender, EventArgs e)
40         {
41             // Close the form.
42             this.Close();
43         }
44     }
45 }
```

Boolean Methods

A Boolean method returns either `true` or `false`. You can use a Boolean method to test a condition, and then return either `true` or `false` to indicate whether the condition exists. Boolean methods are useful for simplifying complex conditions that are tested in decision and repetition structures.

For example, suppose you are writing a program that will ask the user to enter a number and then determine whether that number is even or odd. The following code shows how you can make that determination. Assume `number` is an `int` variable containing the number entered by the user.

```
if (number % 2 == 0)
{
    MessageBox.Show("The number is even.");
}
else
{
    MessageBox.Show("The number is odd.");
}
```

The meaning of the Boolean expression being tested by this `if-else` statement is not clear, so let's take a closer look at it:

```
number % 2 == 0
```

This expression uses the `%` operator, which was introduced in Chapter 3. Recall that the `%` operator divides two integers and returns the remainder of the division. So, this code is saying, If the remainder of `number` divided by 2 is equal to 0, then display a message indicating the number is even, or else display a message indicating the number is odd.

Because dividing an even number by 2 always gives a remainder of 0, this logic works. The code would be easier to understand, however, if you could somehow rewrite it to say, If the number is even, then display a message indicating it is even, or else display a message indicating it is odd. As it turns out, this can be done with a Boolean method. In this example, you could write a Boolean method named `IsEven` that accepts an `int` as an argument and returns `true` if the number is even or `false` otherwise. Here is an example how the `IsEven` method might be written:

```
private bool IsEven(int number)
{
    // Local variable to hold true or false
    bool numberIsEven;

    // Determine whether the number is even.
    if (number % 2 == 0)
    {
        numberIsEven = true;
    }
    else
    {
        numberIsEven = false;
    }

    // Return the result.
    return numberIsEven;
}
```

Then you can rewrite the `if-else` statement so it calls the `IsEven` method to determine whether `number` is even:

```
if (IsEven(number))
{
   MessageBox.Show("The number is even.");
}
else
{
   MessageBox.Show("The number is odd.");
}
```

Not only is this logic easier to understand, but now you have a method that you can call in the program any time you need to test a number to determine whether it is even. (The *Chap06* folder of this book's Student Sample Programs has a project named *EvenNumber* that demonstrates the `IsEven` method.)

Using a Boolean Method to Modularize Input Validation

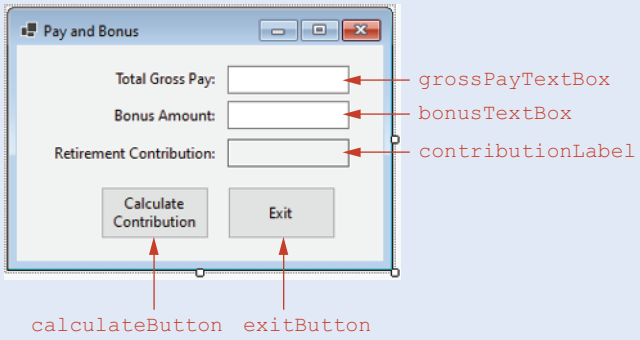
Boolean methods can be useful for modularizing input validation. When a form has multiple TextBox controls, the input validation usually requires multiple, nested `if` statements. In many cases, you can simplify the code by creating a Boolean method that performs all the input validation and returns `true` if all of the input is valid or `false` if any of it is invalid. In Tutorial 6-5, you will complete an application that uses this approach.

Tutorial 6-5:
Modularizing Input Validation with a Boolean Method

VideoNote
Tutorial 6-5: Modularizing Input Validation with a Boolean Method

In addition to regular pay, a company pays its employees an annual bonus. The company also contributes 5 percent of an employee's total compensation (gross pay plus bonus) to a retirement account. In this tutorial, you complete an application that lets you enter an employee's gross pay and bonus amount and calculates the amount of retirement contribution. The project is named *PayAndBonus* and is found in the *Chap06* folder of this book's Student Sample Programs. Figure 6-25 shows the project's form, which has already been created for you.

Figure 6-25 The *PayAndBonus* project's form



Step 1: Start Visual Studio. Open the project named *PayAndBonus* in the *Chap06* folder of this book's Student Sample Programs.

Step 2: Open the Form1 form's code in the code editor. Insert the comment and constant field declaration shown in lines 5 and 6 of Program 6-9. This constant, which is set to the value 0.05, is used in the calculation of the retirement account contribution.

Step 3: Type the comments and code for the `InputIsValid` method, shown in lines 13 through 44 of Program 6-9. Notice in line 17 that the method has reference parameters for gross pay and the bonus amount. The purpose of the method is to get the values entered by the user, convert them to `decimal` values, and assign them to the parameter variables. If the input is successfully converted, the method returns `true`. Otherwise, an error message is displayed and the method returns `false` to indicate that the input is not valid.

Step 4: Open the Form1 form in the *Designer*. Double-click the `calculateButton` control. This opens the code editor, and you will see an empty event handler named `calculateButton_Click`. Complete the `calculateButton_Click` event handler by typing the code shown in lines 48–58 in Program 6-9. Let's take a closer look at the code:

Line 49: This statement declares `decimal` variables to hold the gross pay, bonus amount, and contribution amount.

Lines 51–58: This `if` statement calls the `InputIsValid` method, passing the `grosspay` and `bonus` variables, by reference, as arguments. If the method returns true, the variables will contain the gross pay and bonus amount entered by the user. In that case, line 54 calculates the amount of contribution to the retirement account, and line 57 displays that amount.

Step 5: Switch your view back to the *Designer* and double-click the `exitButton` control. In the code editor you will see an empty event handler named `exitButton_Click`. Complete the `exitButton_Click` event handler by typing the code shown in lines 63–64 in Program 6-9.

Step 6: Save the project. Then, press F5 on the keyboard or click the *Start Debugging* button (▶) on the toolbar to compile and run the application. When the application runs, enter 80000 for the gross pay and 20000 for the bonus amount. Click the *Calculate Contribution* button. The application should display $5,000.00 as the amount of contribution. Try other values if you wish. Click the *Exit* button to exit the application.

Program 6-9 Completed code for Form1 in the *PayAndBonus* project

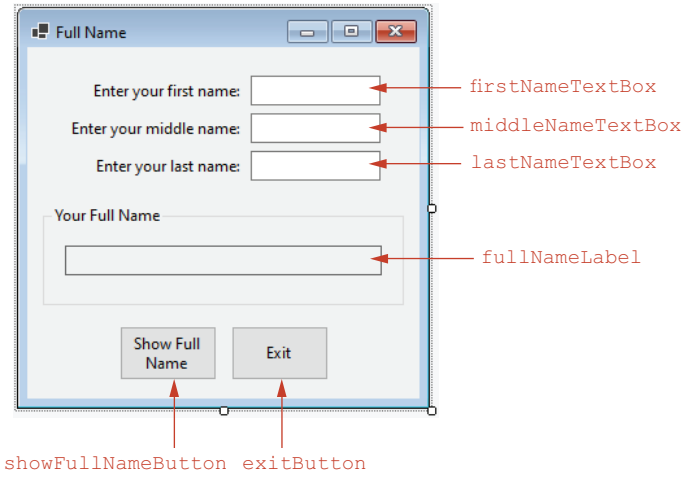
```
 1 namespace PayAndBonus
 2 {
 3     public partial class Form1 : Form
 4     {
 5         // Constant field for the contribution rate
 6         private const decimal CONTRIB_RATE = 0.05m;
 7
 8         public Form1()
 9         {
10             InitializeComponent();
11         }
12
13         // The InputIsValid method converts the user input and
           stores
14         // it in the arguments (passed by reference). If the
           conversion
15         // is successful, the method returns true. Otherwise it
           returns
16         // false.
```

```
17          private bool InputIsValid(ref decimal pay, ref decimal
            bonus)
18          {
19              // Flag variable to indicate whether the input is good
20              bool inputGood = false;
21
22              // Try to convert both inputs to decimal.
23              if (decimal.TryParse(grossPayTextBox.Text, out pay))
24              {
25                  if (decimal.TryParse(bonusTextBox.Text, out bonus))
26                  {
27                      // Both inputs are good.
28                      inputGood = true;
29                  }
30                  else
31                  {
32                      // Display an error message for the bonus.
33                      MessageBox.Show("Bonus amount is invalid.");
34                  }
35              }
36              else
37              {
38                  // Display an error message for gross pay.
39                  MessageBox.Show("Gross pay is invalid.");
40              }
41
42              // Return the result.
43              return inputGood;
44          }
45
46          private void calculateButton_Click(object sender,
            EventArgs e)
47          {
48              // Variables for gross pay, bonus, and contributions
49              decimal grossPay = 0m, bonus = 0m, contributions = 0m;
50
51              if (InputIsValid(ref grossPay, ref bonus))
52              {
53                  // Calculate the amount of contribution.
54                  contributions = (grossPay + bonus) * CONTRIB_RATE;
55
56                  // Display the contribution.
57                  contributionLabel.Text = contributions.
                    ToString("c");
58              }
59          }
60
61          private void exitButton_Click(object sender, EventArgs e)
62          {
63              // Close the form.
64              this.Close();
65          }
66      }
67  }
```

Returning a String from a Method

So far, you've seen examples of methods that return numbers and Boolean values. You can write methods that return any type of data. Let's look at an example program that uses a string-returning method. The *Chap06* folder of this book's Student Sample Programs contains a project named *FullName*. Figure 6-26 shows the project's form, and Program 6-10 shows the form's code. When you run the application, you enter your first name, middle name, and last name into the TextBox controls. When you click the *Show Full Name* button, your full name is displayed in the `fullNameLabel` control.

Figure 6-26 The *FullName* project's form



Program 6-10 Code for the *FullName* project's Form1 form

```
 1 namespace FullName
 2 {
 3     public partial class Form1 : Form
 4     {
 5         public Form1()
 6         {
 7             InitializeComponent();
 8         }
 9
10         // The FullName method accepts arguments for a first
11         // name, a middle name, and a last name. It returns
12         // the full name.
13         private string FullName(string first, string middle,
           string last)
14         {
15             return first + " " + middle + " " + last;
16         }
17
18         private void showFullNameButton_Click(object sender,
           EventArgs e)
19         {
20             // Variables to hold the first, middle, last, and
               full names
21             string first, middle, last, full;
22
```

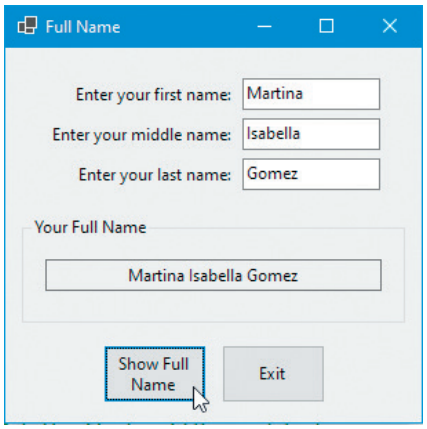
```
23              // Get the first, middle, and last names.
24              first = firstNameTextBox.Text;
25              middle = middleNameTextBox.Text;
26              last = lastNameTextBox.Text;
27
28              // Get the full name.
29              full = FullName(first, middle, last);
30
31              // Display the full name.
32              fullNameLabel.Text = full;
33          }
34
35          private void exitButton_Click(object sender, EventArgs e)
36          {
37              // Close the form.
38              this.Close();
39          }
40      }
41  }
```

Lines 13–16 define a method named `FullName`. Notice the following things about the method:

- Its return type is `string`.
- It has three `string` parameters: `first`, `middle`, and `last`. When we call the method we pass a first name, a middle name, and a last name as arguments.
- In line 15 it returns a string that is the concatenation of the `first`, `middle`, and `last` parameters, with spaces inserted between each.

Figure 6-27 shows an example of the project's form after the user has entered names into each TextBox and clicked the *Show Full Name* button.

Figure 6-27 Example output of the *FullName* project



Checkpoint

6.15 What is a value-returning method? How is it used?

6.16 What is returned by a Boolean method?

6.17 Can a method be written to return any type of data?

6.6 Debugging Methods

CONCEPT: **Visual Studio debugging commands allow you to single-step through applications with methods. The *Step Into* command allows you to single-step through a called method. The *Step Over* command allows you to execute a method call without single-stepping through its lines. The *Step Out* command allows you to execute all remaining lines of a method that you are debugging without stepping through them.**

In Chapter 3, you learned to set a breakpoint in your application's code and to single-step through the code's execution. Let's find out how to step into or step over a method, and step out of a method.

When an application is in break mode, the ***Step Into* command** causes the currently highlighted line (the execution point) to execute. If that line contains a call to a method, the next highlighted line is the first line in that method. In other words, the *Step Into* command allows you to single-step through a method when it is called. Activate the *Step Into* command using one of the following procedures:

- Press the F11 key
- Select *Debug* from the menu bar, and then select *Step Into* from the *Debug* menu
- Click the *Step Into* button ( ) on the *Debug Toolbar*, if the toolbar is visible

Like the *Step Into* command, the ***Step Over* command** causes the currently highlighted line to execute. If the line contains a method call, however, the method is executed without stepping through its statements. Activate the *Step Over* command using one of the following procedures:

- Press the F10 key.
- Select *Debug* from the menu bar, and then select *Step Over* from the *Debug* menu.
- Click the *Step Over* button ( ) on the *Debug Toolbar* if the toolbar is visible.

Use the ***Step Out* command** when single-stepping through a method if you want the remainder of the method to complete execution without single-stepping. After the method has completed, the line following the method call is highlighted, and you may resume single-stepping. Activate the *Step Out* command using one of the following procedures:

- Press the Shift + F11 keys.
- Select *Debug* from the menu bar, and then select *Step Out* from the *Debug* menu.
- Click the *Step Out* button ( ) on the *Debug Toolbar* if the toolbar is visible.

TIP: Visual Studio can be configured in different ways. Under some configurations, the *Step Into* command might be activated by the F8 function key. Similarly, under some configurations the *Step Over* command may be activated by the Shift + F8 keys. Likewise, the *Step Out* command might be activated by the Control + Shift + F8 keys. To find out which keys are used, look carefully at these commands when you click on the *Debug* menu.

In Tutorials 6-6, 6-7, and 6-8, you practice using each of these commands.

Tutorial 6-6:
Practicing the *Step Into* Command

VideoNote
Tutorial 6-6: Practicing the *Step Into* Command

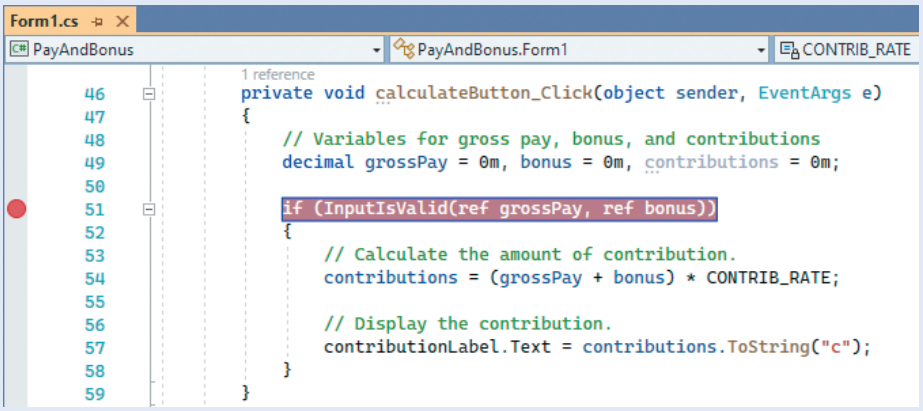
In this tutorial, you use the *PayAndBonus* project from Tutorial 6-5 to practice single-stepping through methods.

Step 1: Open the *PayAndBonus* project from Tutorial 6-5.

Step 2: Open the code editor and set a breakpoint at the line in the `calculateButton_Click` event handler shown in Figure 6-28.

TIP: Set a breakpoint by clicking the mouse while the pointer is positioned in the left margin, next to the line of code. You can also move the text cursor to the line you wish to set as a breakpoint, and then press F9.

Figure 6-28 Location of breakpoint



Step 3: Press the F5 key to run the application in debug mode, enter 80000 for the gross pay and 20000 for the bonus amount.

Step 4: Click the *Calculate Contribution* button. The application enters break mode with the breakpoint line highlighted.

Step 5: Notice that the highlighted line contains a call to the `InputIsValid` method. Press F11 to execute the *Step Into* command.

Step 6: Because you pressed F11, the first line of the `InputIsValid` method is highlighted next. You will now single-step through the statements inside the `InputIsValid` method. Continue pressing the F11 key to single-step through the `InputIsValid` method. When the method's closing brace (`}`) is highlighted, press F11 once more to return to the line that called the method.

Step 7: Now you can exit break mode, and resume normal execution. Perform one of the following procedures:

- Click the ▶ Continue button on the toolbar.
- Press F5.
- Click *Debug* on the menu bar, and then click *Continue*.

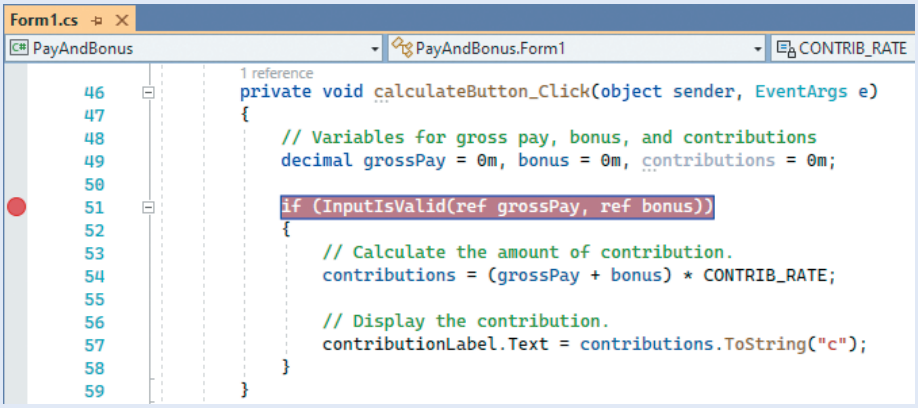
After performing any one of these procedures, the application will exit break mode and resume normal execution. Click the *Exit* button on the application's form to end execution. Leave the project open in Visual Studio. You will use it in the next tutorial.

Tutorial 6-7:
Practicing the *Step Over* Command

VideoNote
Tutorial 6-7: Practicing the *Step Over* Command

Step 1: Make sure the *PayAndBonus* project is still open in Visual Studio from Tutorial 6-6. The breakpoint should still be set as shown in Figure 6-29.

Figure 6-29 Location of breakpoint



Step 2: Press the F5 key to run the application in debug mode, enter 80000 for the gross pay and 20000 for the bonus amount.

Step 3: Click the *Calculate Contribution* button. The application enters break mode with the breakpoint line highlighted.

Step 4: Notice that the highlighted line contains a call to the `InputIsValid` method. Press F10 to execute the *Step Over* command. This executes the method without stepping through it. The next line is now highlighted.

Step 5: Now you can exit break mode, and resume normal execution. Perform one of the following procedures:

- Click the Continue button on the toolbar.
- Press F5.
- Click *Debug* on the menu bar, and then click *Continue*.

After performing any one of these procedures, the application will exit break mode and resume normal execution. Click the *Exit* button on the application's form to end execution. Leave the project open in Visual Studio. You will use it in the next tutorial.

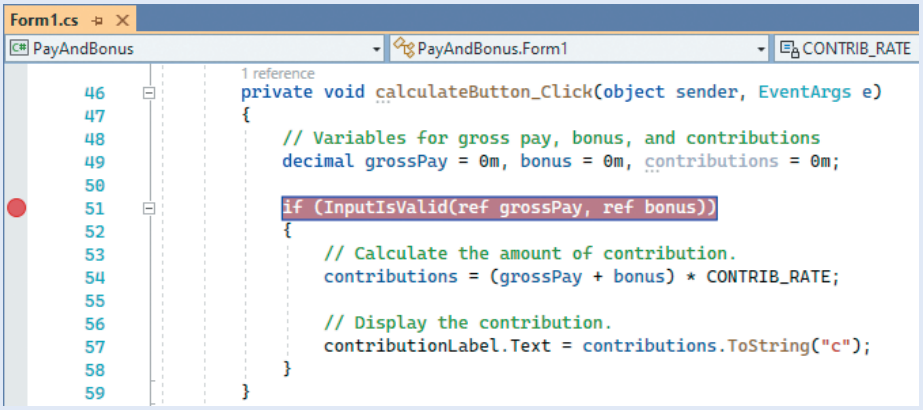
Tutorial 6-8:
Practicing the *Step Out* Command

VideoNote
Tutorial 6-8: Practicing the *Step Out* Command

Step 1: Make sure the *PayAndBonus* project is still open in Visual Studio from Tutorial 6-6. The breakpoint should still be set as shown in Figure 6-30.

Step 2: Press the F5 key to run the application in debug mode, enter 80000 for the gross pay and 20000 for the bonus amount.

Figure 6-30 Location of breakpoint



Step 3: Click the *Calculate Contribution* button. The application enters break mode with the breakpoint line highlighted.

Step 4: Notice that the highlighted line contains a call to the `InputIsValid` method. Press F11 to execute the *Step Into* command.

Step 5: Because you pressed F11, the first line of the `InputIsValid` method is highlighted next. Press the F11 key once again to advance to the next line in the method.

Step 6: Instead of continuing to step through the method, you will now step out of the method. Press the Shift + F11 keys to execute the *Step Out* command. Single-stepping is suspended while the remaining statements in the `InputIsValid` method execute. You are returned to the `calculateButton_Click` event handler at the line containing the method call.

Step 7: Now you can exit break mode, and resume normal execution. Perform one of the following procedures:

- Click the Continue button on the toolbar.
- Press F5.
- Click *Debug* on the menu bar, and then click *Continue*.

After performing any one of these procedures, the application will exit break mode and resume normal execution. Click the *Exit* button on the application's form to end execution.

TIP: When the current execution point does not contain a method call, the *Step Into* and *Step Over* commands perform identically.

Key Terms

arguments
code reuse
default argument
divide and conquer
method body
method call
method header
modularized program
named argument
output parameter
parameter variable
Pascal case
passed by reference
passed by value
reference parameter
return point
return type
Step Into command
Step Out command
Step Over command
stepwise refinement
top–down design
value-returning method
`void` method

Review Questions

Multiple Choice

1. In general terms, a program that is broken into smaller units of code, such as methods, is known as a(n) _____.
   a. object-oriented program
   b. modularized program
   c. procedural program
   d. method-driven program

2. Writing the code to perform a task once and then reusing it each time you need to perform the task is a benefit of using methods called _____.
   a. code reuse
   b. the single-use philosophy
   c. method recycling
   d. code reprocessing

3. When you call a(n) _____, it simply executes the statements it contains and then terminates.
   a. intrinsic method
   b. `empty` method
   c. logical method
   d. `void` method

4. The _____, which appears at the beginning of a method definition, lists several important things about the method, including the method's name.
   a. method title
   b. method description
   c. method header
   d. method declaration

5. The _____ is a collection of statements enclosed inside a set of curly braces that are performed when the method is executed.
   a. method body
   b. method designation
   c. method code
   d. method classification

6. The _____ is the memory address that is saved by the system when a method is called and is the location to which the system should return after a method ends.
   a. calling address
   b. method address
   c. return point
   d. come back position

7. Programmers commonly use a technique known as _____ to break down an algorithm into methods.
   a. prototyping
   b. method modeling
   c. algorithm division
   d. top–down design

8. Pieces of data that are sent into a method are known as _____.
   a. arguments
   b. references
   c. method variables
   d. data entries

9. A(n) _____ is a special variable that receives an argument when a method is called.
   a. reference variable
   b. argument variable
   c. parameter variable
   d. method variable

10. A _____ specifies which parameter an argument should be passed into.
    a. named argument
    b. special argument
    c. constant argument
    d. literal argument

11. When a(n) _____ is provided for a parameter, it becomes possible to call the method without explicitly passing an argument into the parameter.
    a. local argument
    b. empty argument
    c. default argument
    d. expressional argument

12. When an argument is _____, only a copy of the argument's value is passed into the parameter variable.
    a. a named constant
    b. passed by association
    c. passed by reference
    d. passed by value

13. When you want a method to be able to change the value of a variable that is passed to it as an argument, the variable must be _____.
    a. passed by reference
    b. a local variable
    c. passed by value
    d. a named constant

14. A _____ a special type of parameter variable that is useful for establishing two-way communication between methods.
    a. communication variable
    b. reference parameter
    c. method parameter
    d. global variable

15. A(n) _____ can have an uninitialized value passed into it, but it must be set to some value before the method it belongs to finishes executing.
    a. input parameter
    b. reference parameter
    c. output parameter
    d. default parameter

16. A method's _____ is the type of value that the method returns.
    a. data type
    b. return type
    c. value type
    d. method type

True or False

1. Dividing a large problem into several smaller problems that are easily solved is sometimes called divide and conquer.
2. In a Pascal case name, the first character is always uppercase.
3. If a method belongs to a class, then you must write a method's definition inside the class.
4. The contents of variables and the values of expressions cannot be passed as arguments.
5. You do not have to write the data type for each parameter variable that is declared in a parameter list if they are all of the same data type.
6. An output parameter works like a by value parameter.
7. A value-returning method must contain a `return` statement.
8. A Boolean method returns either `yes` or `no`.

Short Answer

1. What do you call a method that executes the statements it contains and then returns a value back to the statement that called it?
2. What is the standard naming convention used among C# programmers for method names? Why?
3. What is another name for the top-down design process?
4. What is a parameter list?
5. How do you specify a named argument?
6. How are output parameters different from reference parameters?
7. How is a value-returning method like a `void` method? How is it different?
8. Can Boolean methods be used to modularize input validation? Why or why not?

Algorithm Workbench

1. Examine the following method header; then write an example call to the method.

```
private void ShowValue()
```

2. The following statement calls a method named `ShowHalf`. The `ShowHalf` method displays a value that is half that of the argument. Write the method.

```
ShowHalf(50);
```

3. Write the method header for a method named `ShowRetailPrice`. The method should include parameter variables for a list price and a markup percentage. Write the method so that the default argument for the markup percentage is set to 50 percent.

4. Examine the following method header; then write an example call to the method.

```
private void ResetValue(ref int value)
```

5. A program contains the following value-returning method:

```
private int Square(int value)
{
    return value * value;
}
```

Write a statement that passes the value `10` as an argument to this method and assigns its return value to the variable `result`.

Programming Problems

1. **Retail Price Calculator**

Create an application that lets the user enter an item's wholesale cost and its markup percentage. It should then display the item's retail price. For example:

- If an item's wholesale cost is $5.00 and its markup percentage is 100 percent, then the item's retail price is $10.00.
- If an item's wholesale cost is $5.00 and its markup percentage is 50 percent, then the item's retail price is $7.50.

The program should have a method named `CalculateRetail` that receives the wholesale cost and the markup percentage as arguments and returns the retail price of the item.

2. **Falling Distance**

When an object is falling because of gravity, the following formula can be used to determine the distance the object falls in a specific time period:

$$d = \frac{1}{2}gt^2$$

The variables in the formula are as follows: d is the distance in meters, g is 9.8, and t is the amount of time in seconds that the object has been falling. Create an application that allows the user to enter the amount of time that an object has fallen and then displays the distance that the object fell. The application should have a method named `FallingDistance`. The `FallingDistance` method should accept an object's falling time (in seconds) as an argument. The method should return the distance in meters that the object has fallen during that time interval.

VideoNote
Solving the Kinetic Energy problem

3. **Kinetic Energy**

In physics, an object that is in motion is said to have kinetic energy. The following formula can be used to determine a moving object's kinetic energy:

$$KE = \frac{1}{2}mv^2$$

In the formula KE is the kinetic energy, m is the object's mass in kilograms, and v is the object's velocity in meters per second. Create an application that allows the user to enter an object's mass and velocity and then displays the object's kinetic energy. The application should have a method named `KineticEnergy` that accepts an object's mass (in kilograms) and velocity (in meters per second) as arguments. The method should return the amount of kinetic energy that the object has.

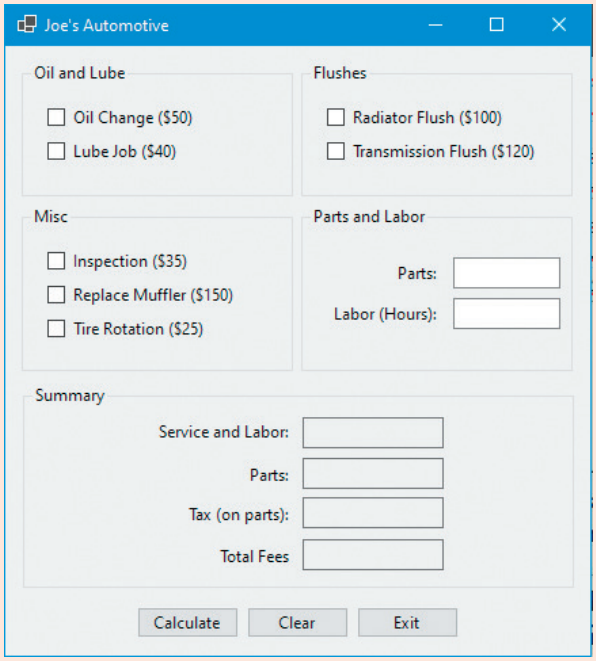
4. **Joe's Automotive**

Joe's Automotive performs the following routine maintenance services:

- Oil change—$50
- Lube job—$40
- Radiator flush—$100
- Transmission flush—$120
- Inspection—$35
- Muffler replacement—$150
- Tire rotation—$25

Joe also performs other nonroutine services and charges for parts and labor ($75 per hour). Create an application that displays the total for a customer's visit to Joe's. The form should resemble the one shown in Figure 6-31.

Figure 6-31 *Automotive* form



The application should have the following value-returning methods:

- `OilChangeCharges`—Returns the charges for an oil change, if selected.
- `LubeCharges`—Returns the charges for a lube job, if selected.
- `RadiatorFlushCharges`—Returns the charges for a radiator flush, if selected.

- `TransmissionFlushCharges`—Returns the charges for a transmission flush, if selected.
- `InspectionCharges`—Returns the charges for an inspection, if selected.
- `MufflerCharges`—Returns the charges for a muffler replacement, if selected.
- `TireRotationCharges`—Returns the charges for a tire rotation, if selected.
- `PartsCharges`—Returns the charges for parts, if applicable.
- `LaborCharges`—Returns the charges for labor, if applicable.
- `TaxCharges`—Returns the taxes on parts, if applicable.
- `TotalCharges`—Returns the total charges.

The application should also have a void method named `ClearForm` that clears all the CheckBoxes, the TextBoxes for parts and labor hours, and the Labels that appear in the *Summary* GroupBox.

5. **Hospital Charges**

   Create an application that calculates the total cost of a hospital stay. The daily base charge is $350. The hospital also charges for medication, surgical fees, lab fees, and physical rehab. The application should accept the following input:

   - The number of days spent in the hospital
   - The amount of medication charges
   - The amount of surgical charges
   - The amount of lab fees
   - The amount of physical rehabilitation charges

   Create and use the following value-returning methods in the application:

   - `CalcStayCharges`—Calculates and returns the base charges for the hospital stay. This is computed as $350 times the number of days in the hospital.
   - `CalcMiscCharges`—Calculates and returns the total of the medication, surgical, lab, and physical rehabilitation charges.
   - `CalcTotalCharges`—Calculates and returns the total charges.

6. **Present Value**

   Suppose you want to deposit a certain amount of money into a savings account and then leave it alone to draw interest for the next 10 years. At the end of 10 years you would like to have $10,000 in the account. How much do you need to deposit today to make that happen? You can use the following formula, which is known as the present-value formula, to find out:

$$P = \frac{F}{(1 + r)^n}$$

   The terms in the formula are as follows:

   - P is the *present value*, or the amount that you need to deposit today.
   - F is the *future value* that you want in the account. (In this case, F is $10,000.)
   - r is the *annual interest rate*.
   - n is the *number of years* that you plan to let the money sit in the account.

   Write a method named `PresentValue` that performs this calculation. The method should accept the future value, annual interest rate, and number of years as arguments. It should return the present value, which is the amount that you need to deposit today. Demonstrate the method in an application that lets the user experiment with different values for the formula's terms.

7. **Prime Numbers**

   A prime number is a number that can be evenly divided by only itself and 1. For example, the number 5 is prime because it can be evenly divided by only 1 and 5. The number 6, however, is not prime because it can be evenly divided by 1, 2, 3, and 6.

Write a Boolean method named `IsPrime` that takes an integer as an argument and returns `true` if the argument is a prime number or `false` otherwise. Use the method in an application that lets the user enter a number and then displays a message indicating whether the number is prime.

TIP: Recall that the `%` operator divides one number by another and returns the remainder of the division. In an expression such as `num1 % num2`, the `%` operator returns 0 if `num1` is evenly divisible by `num2`.

8. **Prime Number List**

   This exercise assumes you have already written the `IsPrime` method in Programming Problem 7. Create another application that uses this method to display all the prime numbers from 1 through 100 in a list box. The program should have a loop that calls the `IsPrime` method.

9. **Rock, Paper, Scissors Game**

   Create an application that lets the user play the game of Rock, Paper, Scissors against the computer. The program should work as follows:

   1. When the program begins, a random number in the range of 1 through 3 is generated. If the number is 1, then the computer has chosen rock. If the number is 2, then the computer has chosen paper. If the number is 3, then the computer has chosen scissors. (Do not display the computer's choice yet.)
   2. The user selects their choice of rock, paper, or scissors. To get this input you can use Button controls, or clickable PictureBox controls displaying some of the artwork that you will find in the student sample files.
   3. The computer's choice is displayed.
   4. A winner is selected according to the following rules:
      - If one player chooses rock and the other player chooses scissors, then rock wins. (Rock smashes scissors.)
      - If one player chooses scissors and the other player chooses paper, then scissors wins. (Scissors cuts paper.)
      - If one player chooses paper and the other player chooses rock, then paper wins. (Paper wraps rock.)
      - If both players make the same choice, the game must be played again to determine the winner.

      Be sure to modularize the program into methods that perform each major task.

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CHAPTER 7

Arrays and Lists

TOPICS

7.1 Value Types and Reference Types
7.2 Array Basics
7.3 Working with Files and Arrays
7.4 Passing Arrays as Arguments to Methods
7.5 Some Useful Array Algorithms
7.6 Advanced Algorithms for Sorting and Searching Arrays
7.7 Two-Dimensional Arrays
7.8 Jagged Arrays
7.9 The `List` Collection
7.10 The ImageList Control

7.1 Value Types and Reference Types

CONCEPT: **The data types in C# and .NET fall into two categories: value types and reference types.**

In this chapter, you will gain more experience working with objects. Specifically, you will work with arrays and collections, which are objects that store groups of data. Before we go into the details of creating and working with those objects, it will be helpful for you to understand how objects are stored in memory. In this section, we discuss the ways that

different types of objects are internally stored by .NET. As a result, you will better understand the concepts presented in this chapter, and chapters to come.

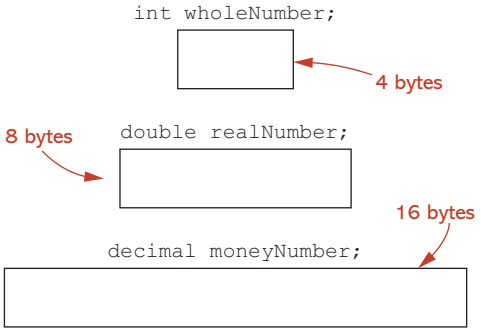
All the data types in C# and .NET fall into two categories: **value types** and **reference types**. Of the C# data types that you have used so far, the following are value types: `int`, `double`, `decimal`, and `bool`. (There are other value types in addition to these, but these are the ones we focus on in this book.)

When you declare a value type variable, the compiler sets aside, or allocates, a chunk of memory that is big enough for that variable. For example, look at the following variable declarations:

```
int wholeNumber;
double realNumber;
decimal moneyNumber;
```

Recall from Chapter 3 that an `int` uses 32 bits of memory (4 bytes), a `double` uses 64 bits of memory (8 bytes), and a `decimal` uses 128 bits of memory (16 bytes). These declaration statements cause memory to be allocated as shown in Figure 7-1.

Figure 7-1 Memory allocated

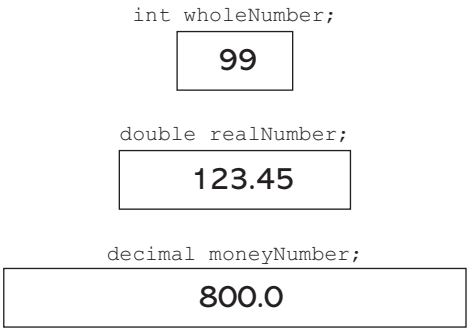


The memory that is allocated for a value type variable is the actual location that will hold any value that is assigned to that variable. For example, suppose we use the following statements to assign values to the variables shown in Figure 7-1:

```
wholeNumber = 99;
realNumber = 123.45;
moneyNumber = 800.0m;
```

Figure 7-2 shows how the assigned values are stored in each variable's memory location.

Figure 7-2 Values assigned to the variables



As you can see from these illustrations, value types are very straightforward. When you are working with a value type, you are using a variable that holds a piece of data.

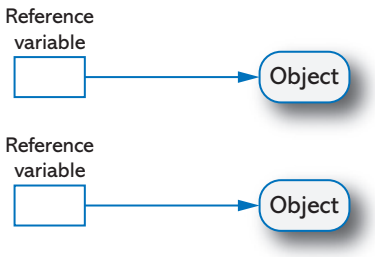
This is different from the way that reference types work. When you are working with a reference type, you are using two things:

- An object that is created in memory
- A variable that references the object

The object that is created in memory holds data of some sort and performs operations of some sort. (Exactly what the data and operations are depends on what kind of object it is.) In order to work with the object in code, you need some way to refer to it. That's where the variable comes in. The variable does not hold an actual piece of data with which your program will work. Instead, it holds a special value known as a **reference**, which links the variable to the object.[1] When you want to work with the object, you use the variable that references it.

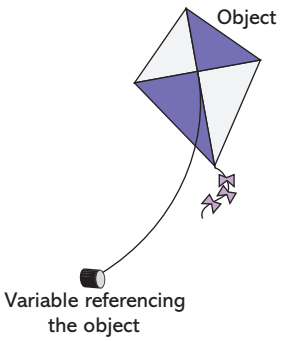
A variable that is used to reference an object is commonly called a **reference variable**. Reference variables can be used only to reference objects. Figure 7-3 illustrates two objects that have been created in memory, each referenced by a variable.

Figure 7-3 Two objects referenced by variables



To understand how reference variables and objects work together, think about flying a kite. In order to fly a kite, you need a spool of string attached to it. When the kite is airborne, you use the spool of string to hold onto the kite and control it. This is similar to the relationship between an object and the variable that references the object. As shown in Figure 7-4, the object is like the kite, and the variable that references the object is like the spool of string.

Figure 7-4 The kite and string metaphor



[1] A reference is similar to a memory address. It is a value that identifies the object's memory location.

Creating a reference type object typically requires the following two steps:

1. You declare a reference variable.
2. You create the object and associate it with the reference variable.

After you have performed these steps, you can use the reference variable to work with the object. Let's look at an example that you have already learned about: creating objects of the `Random` class. Recall from Chapter 5 that the `Random` class allows your program to generate random numbers. Here is an example of how you create an object from the `Random` class:

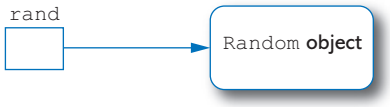
```
Random rand = new Random();
```

Let's look at the different parts of this statement:

- The first part of the statement, appearing on the left side of the = operator, reads `Random rand`. This declares a variable named `rand` that can be used to reference an object of the `Random` type.
- The second part of the statement, appearing on the right side of the = operator, reads `new Random()`. The **new operator** creates an object in memory and returns a reference to that object. So, the expression `new Random()` creates an object from the `Random` class and returns a reference to that object.
- The = operator assigns the reference that was returned from the `new` operator to the `rand` variable.

After this statement executes, the `rand` variable references a `Random` object, as shown in Figure 7-5. The `rand` variable can then be used to perform operations with the object, such as generating random numbers.

Figure 7-5 The `rand` variable referencing a `Random` object



Checkpoint

7.1 Into what two categories do the data types in C# and .NET fall?

7.2 What is the difference in the way you work with value types and reference types?

7.3 How is the relationship between an object and a reference variable similar to a kite and a spool of string?

7.4 Is a variable of the `Random` class a reference type or a value type?

7.2 Array Basics

CONCEPT: **An array allows you to store a group of items of the same data type together in memory. Processing a large number of items in an array is usually easier than processing a large number of items stored in separate variables.**

In the programs you have written so far, you have used variables to store data in memory. The simplest way to store a value in memory is to store it in a variable. Variables work

well in many situations, but they have limitations. For example, they can hold only one value at a time. Consider the following variable declaration:

```
int number = 99;
```

This statement declares an `int` variable named `number`, initialized with the value 99. Consider what happens if the following statement appears later in the program:

```
number = 5;
```

This statement assigns the value 5 to `number`, replacing the value 99 that was previously stored there. Because `number` is an ordinary variable, it can hold only one value at a time.

Because variables hold only a single value, they can be cumbersome in programs that process lists of data. For example, suppose you are asked to write a program that holds the names of 50 employees. Imagine declaring 50 variables to hold all those names:

```
string employee1;
string employee2;
string employee3;
```

and so forth . . .

```
string employee50
```

Then, imagine writing the code to process all 50 names. For example, if you wanted to display the contents of the variables in a ListBox, you would write code such as this:

```
employeeListBox.Items.Add(employee1);   // Display employee 1
employeeListBox.Items.Add(employee2);   // Display employee 2
employeeListBox.Items.Add(employee3);   // Display employee 3
```

and so forth . . .

```
employeeListBox.Items.Add(employee50);  // Display employee 50
```

As you can see, variables are not well suited for storing and processing lists of data. Each variable is a separate item that must be declared and individually processed.

Fortunately, you can use an array as an alternative to a group of variables. An **array** is an object that can hold a group of values that are all the same data type. You can have an array of `int` values, an array of `double` values, and array of `decimal` values, or an array of `string` values, but you cannot store a mixture of data types in an array. Once you create an array, you can write simple and efficient code to process the values that are stored in it.

Arrays are reference type objects. Recall from Section 7.1 that two steps are required to create and use a reference type object:

1. You declare a reference variable.
2. You create the object and associate it with the reference variable.

Suppose you want to create an array that can hold `int` values. Here is an example of how you might declare a reference variable for the array:

```
int[] numbersArray;
```

This statement declares a reference variable named `numbersArray`. Notice that this statement looks like a regular `int` variable declaration except for the set of brackets (`[]`) that appear after the keyword `int`. The expression `int[]` indicates that this variable is a reference to an `int` array. So, we cannot use this variable to store an `int` value. Rather, we can use it to reference an `int` array.

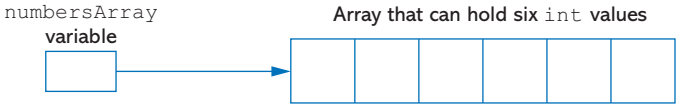
The next step in the process is to create the array object and associate it with the `numbersArray` variable. The following statement shows an example:

```
numbersArray = new int[6];
```

As previously mentioned, the `new` keyword creates an object in memory. The expression that appears after the `new` keyword specifies what type of object to create. In this case, the expression `int[6]` specifies that the object should be an array large enough to hold six `int` values. The number inside the brackets is the array's **size declarator**. It indicates the number of values that the array should be able to hold.

The `new` keyword also returns a reference to the object that it creates. In the previously shown statement, the `new` keyword creates an `int` array and returns a reference to that array. The = operator assigns the reference to the `numbersAray` variable. After this statement executes, the `numbersArray` variable will reference an `int` array that can hold six values. This is shown in Figure 7-6.

Figure 7-6 The `numbersArray` variable referencing an `int` array



In the previous example, we used two statements to (1) declare a reference variable and (2) create an array object. These two steps can be combined into one statement, as shown here:

```
int[] numbersArray = new int[6];
```

You can create arrays of any data. The following are all valid array declarations:

```
double[] temperatures = new double[100];
decimal[] prices = new decimal[50];
string[] nameArray = new string[1200];
```

An array's size declarator must be a nonnegative integer expression. It can be a literal value, as shown in the previous examples, or a variable. It is a preferred practice to use a named constant as a size declarator, however. Here is an example:

```
const int SIZE = 6;
int[] numbersArray = new int[SIZE];
```

This practice can make programs easier to maintain. As you will see later in this chapter, many array-processing techniques require you to refer to the array's size. When you use a named constant as an array's size declarator, you can use the constant to refer to the size of the array in your algorithms. If you ever need to modify the program so the array is a different size, you need only change the value of the named constant.

Using `var` to Declare Arrays

If you are declaring and initializing an array inside of a method, you can use the `var` keyword in your declaration statement. The following code shows an example:

```
const int SIZE = 6;
var numbersArray = new int[SIZE];
```

The compiler will determine that the data type for `numbersArray` is `int[]` because we are initializing it with an `int` array. It is also obvious to any person

reading this statement that `numbersArray` references an `int` array. Here are some other examples:

```
var temperatures = new double[100];

var prices = new decimal[50];

var days = new string[] { "Sunday", "Monday", "Tuesday",
                          "Wednesday", "Thursday", "Friday",
                          "Saturday" };
```

The `var` keyword simplifies the appearance of a declaration statement by reducing the amount of code in the statement. However, the choice of using `var` versus traditional declaration syntax is a matter of personal preference. For now, this book mostly uses traditional declaration syntax. As we progress, however, we will increase our usage of the `var` keyword as we introduce more complex data types.

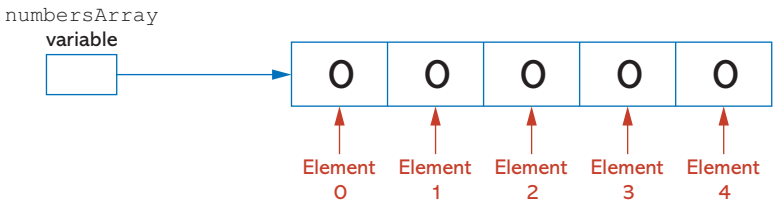
Array Elements and Subscripts

The storage locations in an array are known as **elements**. In memory, an array's elements are located in consecutive memory locations. Each element in an array is assigned a unique number known as a **subscript**. Subscripts are used to identify specific elements in an array. The first element is assigned the subscript 0, the second element is assigned the subscript 1, and so forth. For example, suppose a program has the following declarations:

```
const int SIZE = 5;
var numbersArray = new int[SIZE];
```

As shown in Figure 7-7, the array referenced by `numbersArray` has five elements. The elements are assigned the subscripts 0–4. (Because subscript numbering starts at 0, the subscript of the last element in an array is 1 less than the total number of elements in the array.)

Figure 7-7 Array subscripts



Array Element Default Values

Notice that Figure 7-7 shows each element of the array containing the value 0. When you create a numeric array in C#, its elements are set to the value 0 by default.

Remember, you can create an array to hold any type of value. It is possible to create an array of reference type objects. If you create an array of reference type objects, each element of the array acts as a reference variable. By default, the elements of an array of reference type objects are set to the special value `null`. The value `null` indicates that a reference variable is not set to a valid object and cannot be used for any meaningful purpose.

NOTE: As you will see in Chapter 8, strings are actually reference types, so the default value of a `string` array's elements is `null`.

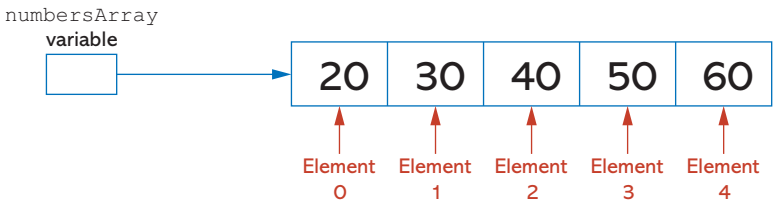
Working with Array Elements

You access the individual elements in an array by using their subscripts. For example, the following code creates an `int` array with five elements and assigns values to each of its elements.

```
const int SIZE = 5;
int[] numbersArray = new int[SIZE];
numbersArray[0] = 20;
numbersArray[1] = 30;
numbersArray[2] = 40;
numbersArray[3] = 50;
numbersArray[4] = 60;
```

This code assigns the value 20 to element 0, the value 30 to element 1, and so forth. Figure 7-8 shows the contents of the array after these statements execute.

Figure 7-8 Values assigned to each element



NOTE: The expression `numbersArray[0]` is pronounced "numbersArray sub zero."

The following code shows another example. It creates a `string` array with three elements and assigns strings to each of its elements.

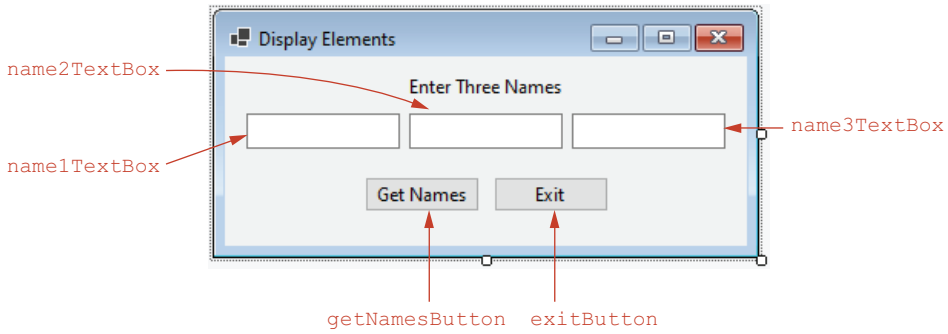
```
const int SIZE = 3;
string[] names = new string[SIZE];
names[0] = "Kris";
names[1] = "Pasha";
names[2] = "Sanjay";
```

The following code sample shows how values can be assigned from TextBox controls to array elements. Assume that an application's form has three TextBox controls named `amount1TextBox`, `amount2TextBox`, and `amount3TextBox` and that the user has entered a numeric value into each one. The following code creates a `decimal` array named `amounts` and assigns each of the TextBox control's input value to an array element.

```
const int SIZE = 3;
decimal[] amounts = new decimal[SIZE];
amounts[0] = decimal.Parse(amount1TextBox.Text);
amounts[1] = decimal.Parse(amount2TextBox.Text);
amounts[2] = decimal.Parse(amount3TextBox.Text);
```

Let's look at a complete program that demonstrates how to assign values to an array and then display the values in the array. In the *Chap07* folder of this book's Student Sample Programs, you will find a project named *DisplayElements*. Figure 7-9 shows the project's form.

Figure 7-9 The *DisplayElements* project's form



Here is the code for the `getNamesButton_Click` event handler:

```
1 private void getNamesButton_Click(object sender, EventArgs e)
2 {
3     // Create an array to hold three strings.
4     const int SIZE = 3;
5     string[] names = new string[SIZE];
6
7     // Get the names.
8     names[0] = name1TextBox.Text;
9     names[1] = name2TextBox.Text;
10    names[2] = name3TextBox.Text;
11
12    // Display the names.
13    MessageBox.Show(names[0]);
14    MessageBox.Show(names[1]);
15    MessageBox.Show(names[2]);
16 }
```

Run the application, enter a name into each of the TextBox controls, and then click the *Get Names* button. The following actions take place:

- In line 5, an array to hold three strings is created.
- In lines 8–10, the names that you entered into the TextBox controls are assigned to the array elements.
- In lines 13–15, each element of the array is displayed in a message box.

The *DisplayElements* project displays the contents of a `string` array. Because the array's elements are strings, we can pass each element directly to the `MessageBox.Show` method without performing a data type conversion. If you want to pass a numeric array element to the `MessageBox.Show` method, however, you will have to call the element's `ToString` method. The following code sample demonstrates:

```
1 // Create an array to hold three integers.
2 const int SIZE = 3;
3 int[] myValues = new int[SIZE];
4
5 // Assign some values to the array elements.
6 myValues[0] = 10;
7 myValues[1] = 20;
8 myValues[2] = 30;
9
10 // Display the array elements.
11 MessageBox.Show(myValues[0].ToString());
12 MessageBox.Show(myValues[1].ToString());
13 MessageBox.Show(myValues[2].ToString());
```

Array Initialization

When you create an array, you can optionally initialize it with a group of values. Here is an example:

```
const int SIZE = 5;
int[] numbersArray = new int[SIZE] { 10, 20, 30, 40, 50 };
```

The series of values inside the braces and separated with commas is called an **initialization list**. These values are stored in the array elements in the order they appear in the list. (The first value, 10, is stored in `numbersArray[0]`, the second value, 20, is stored in `numbersArray[1]`, and so forth.)

When you provide an initialization list, the size declarator can be left out. The compiler determines the size of the array from the number of items in the initialization list. Here is an example:

```
int[] numbersArray = new int[] { 10, 20, 30, 40, 50 };
```

In this example, the compiler determines that the array should have five elements because five values appear in the initialization list.

You can also leave out the `new` operator and its subsequent expression when an initialization list is provided. Here is an example:

```
int[] numbersArray = { 10, 20, 30, 40, 50 };
```

Here are three separate examples that declare and initialize a `string` array named `days`. Each of these examples results in the same array:

```
// Example 1
const int SIZE = 7;
string[] days = new string[SIZE] { "Sunday", "Monday",
                    "Tuesday", "Wednesday", "Thursday",
                    "Friday", "Saturday" };
// Example 2
string[] days = new string[] { "Sunday", "Monday",
                  "Tuesday", "Wednesday", "Thursday",
                  "Friday", "Saturday" };
// Example 3
string[] days = { "Sunday", "Monday", "Tuesday",
                  "Wednesday", "Thursday", "Friday",
                  "Saturday" };
```

Using a Loop to Step through an Array

You can store a number in an `int` variable and then use that variable as a subscript. This makes it possible to use a loop to step through an array, performing the same operation on each element. For example, look at the following code sample:

```
1 // Create an array to hold three integers.
2 const int SIZE = 3;
3 int[] myValues = new int[SIZE];
4
5 // Assign 99 to each array element.
6 for (int index = 0; index < SIZE; index++)
7 {
8       myValues[index] = 99;
9 }
```

Line 3 creates an `int` array named `myValues` with three elements. The `for` loop that starts in line 6 uses an `int` variable named `index` as its counter. The `index` variable is initialized with the value 0 and is incremented after each loop iteration. The loop iterates as long as `index` is less than 3. So, the loop will iterate three times. As it iterates, the `index` variable is assigned the values 0, 1, and 2.

Inside the loop, the statement in line 8 assigns the value 99 to an array element, using the `index` variable as the subscript. This is what happens as the loop iterates:

- The first time the loop iterates, `index` is set to 0, so 99 is assigned to `myValues[0]`.
- The second time the loop iterates, `index` is set to 1, so 99 is assigned to `myValues[1]`.
- The third time the loop iterates, `index` is set to 2, so 99 is assigned to `myValues[2]`.

Invalid Subscripts

When working with an array, it is important that you do not use an invalid subscript. You cannot use a subscript that is less than 0 or greater than the size of the array minus 1. For example, suppose an array has 100 elements. The valid subscripts for the array are the integers 0 through 99. If you try to use any value outside this range, an exception will be thrown at runtime. The following code sample demonstrates how a loop that is not carefully written can cause such an exception to be thrown:

```
1 // Create an array to hold three integers.
2 const int SIZE = 3;
3 int[] myValues = new int[SIZE];
4
5 // Will this loop cause an exception?
6 for (int index = 0; index <= SIZE; index++)
7 {
8     myValues[index] = 99;
9 }
```

Notice that the `for` loop iterates as long as `index` is less than *or equal to* 3. During the loop's last iteration, `index` is set to 3, so the statement in line 8 attempts to make an assignment to `myValues[3]`. There is no element in the array with the subscript 3, so an exception will be thrown.

The Length Property

In C#, all arrays have a Length property that is set to the number of elements in the array. For example, consider an array created by the following statement:

```
double[] temperatures = new double[25];
```

The `temperatures` array's Length property will be set to 25. If we executed the following statement, it would display the message "The temperatures array has 25 elements."

```
MessageBox.Show("The temperatures array has " +
     temperatures.Length + " elements.");
```

The Length property can be useful when processing the entire contents of an array with a loop. The subscript of the last element is always 1 less than the array's Length property. Here is an example:

```
for (int index = 0; index < temperatures.Length; index++)
{
    MessageBox.Show(temperatures[index].ToString());

}
```

NOTE: An array's Length property is read only, so you cannot change its value.

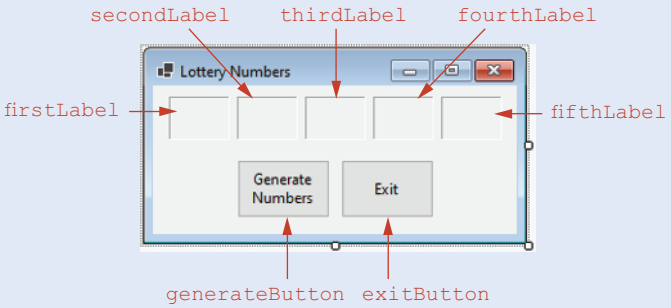
In Tutorial 7-1, you complete an application that generates a set of random numbers similar to those used in lotteries. The numbers will be stored in an array.

Tutorial 7-1:
Using an Array to Hold a List of Random Lottery Numbers

VideoNote
Tutorial 7-1: Using an Array to Hold a List of Random Lottery Numbers

In this tutorial, you complete an application that randomly generates lottery numbers. The application's form is shown in Figure 7-10. When the *Generate Numbers* button is clicked, the application will generate five two-digit integer numbers and store them in an array. The contents of the array will then be displayed in Label controls.

Figure 7-10 The *LotteryNumbers* project's form



Step 1: Start Visual Studio. Open the project named *LotteryNumbers* in the *Chap07* folder of this book's Student Sample Programs.

Step 2: Open the Form1 form in the *Designer*. Double-click the `generateButton` control. This will open the code editor, and you will see an empty event handler named `generateButton_Click`. Complete the `generateButton_Click` event handler by typing the code shown in lines 12–31 in Program 7-1. Let's take a closer look at the code:

Line 13: This statement declares an `int` constant named `SIZE`, set to the value 5. This is used as an array size declarator.

Line 14: This statement creates an `int` array named `lotteryNumbers` with five elements.

Line 17: This statement creates a `Random` object, referenced by a variable named `rand`.

Line 21: This `for` loop uses an `int` variable named `index` as its counter. The `index` variable is initialized with the value 0 and is incremented after each loop iteration. The loop iterates as long as `index` is less than `lotteryNumbers.Length` (which is 5). So, the loop will iterate five times. As it iterates, the `index` variable is assigned the values 0, 1, 2, 3, and 4.

Line 23: This statement gets a random number in the range of 0 through 99 and assigns it to `lotteryNumbers[index]`. The first time the loop iterates, this statement assigns a random number to `lotteryNumbers[0]`. The second time the loop iterates, this statement assigns a random number to `lotteryNumbers[1]`. This continues until the loop is finished. At that time, each element in the array is assigned a random number.

Lines 27–31: These statements display the array elements in the `firstLabel`, `secondLabel`, `thirdLabel`, `fourthLabel`, and `fifthLabel` controls.

Step 3: Switch your view back to the *Designer* and double-click the `exitButton` control. In the code editor you will see an empty event handler named

`exitButton_Click`. Complete the `exitButton_Click` event handler by typing the code shown in lines 36–37 in Program 7-1.

Step 4: Save the project. Then, press F5 on the keyboard, or click the *Start Debugging* button (▶) on the toolbar to compile and run the application. When the application runs, click the *Generate Numbers* button. The application should display a set of random numbers in the Label controls. Click the *Generate Numbers* button several more times to see different sets of random numbers. When you are finished, click the *Exit* button to exit the application.

Program 7-1 Completed code for Form1 in the *LotteryNumbers* project

```
 1 namespace LotteryNumbers
 2 {
 3     public partial class Form1 : Form
 4     {
 5         public Form1()
 6         {
 7             InitializeComponent();
 8         }
 9
10         private void generateButton_Click(object sender,
           EventArgs e)
11         {
12             // Create an array to hold the numbers.
13             const int SIZE = 5;
14             var lotteryNumbers = new int[SIZE];
15
16             // Create a Random object.
17             var rand = new Random();
18
19             // Fill the array with random numbers, in the range
20             // of 0 through 99.
21             for (int index = 0; index < lotteryNumbers.Length;
               index++)
22             {
23                 lotteryNumbers[index] = rand.Next(100);
24             }
25
26             // Display the array elements in the Label controls.
27             firstLabel.Text = lotteryNumbers[0].ToString();
28             secondLabel.Text = lotteryNumbers[1].ToString();
29             thirdLabel.Text = lotteryNumbers[2].ToString();
30             fourthLabel.Text = lotteryNumbers[3].ToString();
31             fifthLabel.Text = lotteryNumbers[4].ToString();
32         }
33
34         private void exitButton_Click(object sender, EventArgs e)
35         {
36             // Close the form.
37             this.Close();
38         }
39     }
40 }
```

Watching for Off-by-One Errors

Because array subscripts start at 0 rather than 1, you have to be careful not to perform an off-by-one error. An **off-by-one error** occurs when a loop iterates one time too many or one time too few. For example, look at the following code sample:

```
1 // Create an array to hold 100 integers.
2 const int SIZE = 100;
3 int[] myValues = new int[SIZE];
4
5 // Assign 99 to each array element.
6 for (int index = 1; index < myValues.Length; index++)
7 {
8     myValues[index] = 99;
9 }
```

The intent of this code is to create an `int` array with 100 elements and assign the value 99 to each element; however, this code has an off-by-one error. During the loop's execution, the `index` variable is assigned the values 1 through 99 when it should be assigned the values 0 through 99. As a result, the first element, which is at subscript 0, is skipped.

Using the `foreach` Loop with Arrays

C# provides a special loop that, in many circumstances, simplifies array processing. It is known as the **`foreach` loop**. When you use the `foreach` loop with an array, the loop automatically iterates once for each element in the array. For example, if you use the `foreach` loop with an eight-element array, the loop will iterate eight times. Because the `foreach` loop automatically knows the number of elements in an array, you do not have to use a counter variable to control its iterations, as with a regular `for` loop.

The `foreach` loop is designed to work with a temporary, read-only variable known as the **iteration variable**. Each time the `foreach` loop iterates, it copies an array element to the iteration variable. For example, the first time the loop iterates, the iteration variable will contain the value of element 0, the second time the loop iterates, the iteration variable will contain the value of element 1, and so forth.

Here is the general format of the `foreach` loop:

```
foreach(Type VariableName in ArrayName)
{
    statement;
    statement;
    etc.
}
```

The statements that appear inside the curly braces are the body of the loop. These are the statements executed each time the loop iterates. As with other control structures, the curly braces are optional if the body of the loop contains only one statement, as shown in the following general format:

```
foreach(Type VariableName in ArrayName)
    statement;
```

Let's take a closer look at the items appearing inside the parentheses:

- *`Type`* is the data type of the values in the array.
- *`VariableName`* is the name of the iteration variable.
- *`in`* is a keyword that must appear after the VariableName.
- *`ArrayName`* is the name of an array.

Suppose we have the following array declaration:

```
int[] numbers = { 3, 6, 9 };
```

We can use the following `foreach` loop to display the contents of the `numbers` array:

```
foreach (int val in numbers)
{
    MessageBox.Show(val.ToString());
}
```

Because the `numbers` array has three elements, this loop will iterate three times. The first time it iterates, `val` will contain the value of `numbers[0]`, so a message box will display the value 3. During the second iteration, `val` will contain the value of `numbers[1]` , so a message box will display the value 6. During the third iteration, `val` will contain the value of `numbers[2]`, so a message box will display the value 9.

Often you will see the `var` keyword used to declare the iteration variable in a `foreach` loop. Here is an example:

```
int[] numbers = { 3, 6, 9 };

foreach (var val in numbers)
{
    MessageBox.Show(val.ToString());
}
```

In this code, the compiler knows that `val` is an `int` because `numbers` is an `int` array.

The `foreach` Loop versus the `for` Loop

When you need to read the values that are stored in an array from the first element to the last element, the `foreach` loop is simpler to use than the `for` loop. With the `foreach` loop, you do not have to be concerned about the size of the array, and you do not have to create a counter variable to hold subscripts; however, because the iteration variable is read only, there are circumstances in which the `foreach` loop is not adequate. You cannot use the `foreach` loop if you need to do any of the following:

- Change the contents of an array element
- Work through the array elements in reverse order
- Access some, but not all, of the array elements
- Simultaneously work with two or more arrays within the loop

In any of these circumstances, you should use the `for` loop to process the array.

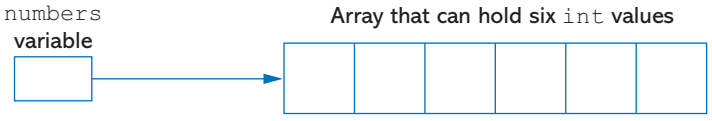
Reassigning an Array Reference Variable

It is possible to reassign an array reference variable to a different array, as demonstrated by the following code sample:

```
1 // Create an array referenced by the numbers variable.
2 int[] numbers = new int[6];
3
4 // Reassign the numbers variable to a new array.
5 numbers = new int[3];
```

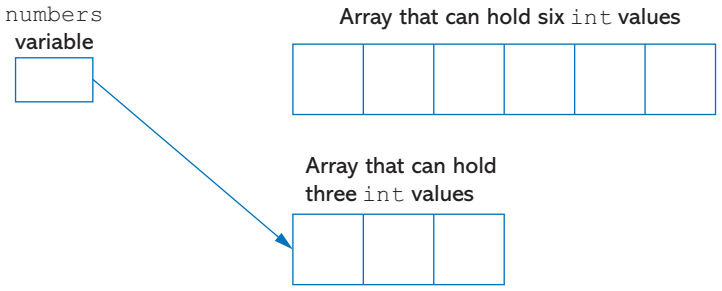
The statement in line 2 creates a six-element `int` array. A reference to the array is assigned to the `numbers` variable. Figure 7-11 shows how the `numbers` variable references the six-element array after this statement executes.

Figure 7-11 The `numbers` variable referencing a six-element array



Then, the statement in line 5 creates a new, three-element `int` array. A reference to the new array is assigned to the `numbers` variable. When line 5 executes, the reference that is currently stored in the `numbers` variable will be replaced by a reference to the three-element array. After this statement executes, the `numbers` variable will reference the three-element array instead of the six-element array. This is illustrated in Figure 7-12.

Figure 7-12 The `numbers` variable referencing a three-element array



Notice in Figure 7-12 that the six-element array still exists in memory, but it is no longer referenced by any variables. Because it is no longer referenced, it cannot be accessed. When an object is no longer referenced, it becomes eligible for garbage collection. **Garbage collection** is a process that periodically runs, removing all unreferenced objects from memory.

Checkpoint

7.5 Write a statement that declares a reference variable named `monthlyPay` for an array that can hold `decimal` values.

7.6 Write a statement so that the `monthlyPay` variable from Checkpoint 7.5 references a `decimal` array that can hold 12 values.

7.7 Combine the statements from Checkpoints 7.5 and 7.6 into a single statement, and use a named constant for a size declarator.

7.8 Write a statement that creates an array of 3 `string` values referenced by a variable named `fullName`. Provide an initialization list for the array using string values for a first, middle, and last name.

7.9 Under what circumstances should you use a `for` loop rather than a `foreach` loop to process data stored in an array?

7.10 What happens when an object such as an array is no longer referenced by a variable?

7.3 Working with Files and Arrays

CONCEPT: **For some problems, files and arrays can be used together effectively. You can easily write a loop that saves the contents of an array to a file, and vice versa.**

Some tasks may require you to save the contents of an array to a file so the data can be used at a later time. Likewise, some situations may require you to read the data from a file into an array. For example, suppose you have a file that contains a set of values and you

want to reverse the order of the values. One technique for doing this is to read the file's values into an array and then write the values in the array back to the file from the end of the array to the beginning.

Writing an Array's Contents to a File

Writing the contents of an array to a file is a straightforward procedure: Open the file and use a loop to step through each element of the array, writing its contents to the file. For example, in the *Chap07* folder of the Student Sample Programs, you will find a project named *ArrayToFile*. When you click the *OK* button, the application writes the contents of an `int` array to a file. The following code shows the Click event handler for the *OK* button.

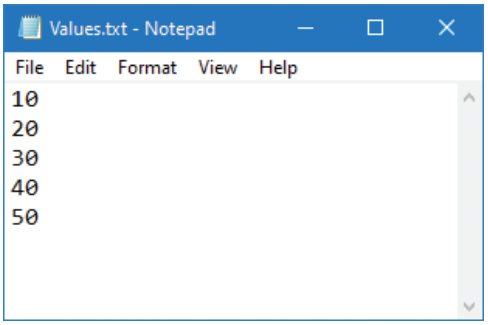
```
1 private void okButton_Click(object sender, EventArgs e)
2 {
3     try
4     {
5         // Create an array with some values.
6         int[] numbers = { 10, 20, 30, 40, 50 };
7
8         // Declare a StreamWriter variable.
9         StreamWriter outputFile;
10
11        // Create the file and get a StreamWriter object.
12        outputFile = File.CreateText("Values.txt");
13
14        // Write the array's contents to the file.
15        for (int index =0; index < numbers.Length; index++)
16        {
17            outputFile.WriteLine(numbers[index]);
18        }
19
20        // Close the file.
21        outputFile.Close();
22
23        // Let the user know it's done.
24        MessageBox.Show("Done");
25    }
26    catch (Exception ex)
27    {
28        // Display an error message.
29        MessageBox.Show(ex.Message);
30    }
31 }
```

The `try-catch` statement handles any file-related errors. Here is a summary of the code inside the try block:

- Line 6 creates an `int` array with five elements, initialized to the values 10, 20, 30, 40, and 50.
- Line 9 declares a `StreamWriter` variable named `outputFile`.
- Line 12 creates a file named Values.txt for writing. After this statement executes, the `outputFile` variable will reference a `StreamWriter` object that is associated with the file.
- Line 15 is the beginning of a `for` loop. The loop iterates once for each element of the array. During the loop's iterations, the `index` variable will be assigned the values 1, 2, 3, 4, and 5.
- Inside the loop, line 17 writes the array element `numbers[index]` to the file.
- Line 21 closes the file.
- Line 24 displays a message box letting the user know the operation is done.

Figure 7-13 shows the contents of the Values.txt file, opened in Notepad, after the *OK* button has been clicked.

Figure 7-13 Contents of the Values.txt file



Reading Values from a File and Storing Them in an Array

Reading the contents of a file into an array is also straightforward: Open the file and use a loop to read each item from the file, storing each item in an array element. The loop should iterate until either the array is filled or the end of the file is reached. For example, in the *Chap07* folder of the Student Sample Programs, you will find a project named *FileToArray*. When you click the *Get Values* button, the application reads values from a file named Values.txt into an `int` array. The contents of the array are then displayed in a list box. The following code shows the Click event handler for the *Get Values* button.

```
1  private void getValuesButton_Click(object sender, EventArgs e)
2  {
3      try
4      {
5          // Create an array to hold items read from the file.
6          const int SIZE = 5;
7          int[] numbers = new int[SIZE];
8
9          // Counter variable to use in the loop
10         int index = 0;
11
12         // Declare a StreamReader variable.
13         StreamReader inputFile;
14
15         // Open the file and get a StreamReader object.
16         inputFile = File.OpenText("Values.txt");
17
18         // Read the file's contents into the array.
19         while (index < numbers.Length && !inputFile.EndOfStream)
20         {
21             numbers[index] = int.Parse(inputFile.ReadLine());
22             index++;
23         }
24
25         // Close the file.
26         inputFile.Close();
27
28         // Display the array elements in the list box.
29         foreach (int value in numbers)
30         {
31             outputListBox.Items.Add(value);
32         }
```

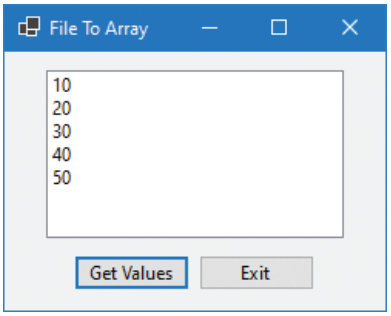
```
33      }
34      catch (Exception ex)
35      {
36          // Display an error message.
37          MessageBox.Show(ex.Message);
38      }
39  }
```

The `try-catch` statement handles any file-related errors. Here is a summary of the code inside the try block:

- Lines 6 and 7 create an `int` array with five elements.
- Line 10 declares an `int` variable named `index`, initialized with the value 0. This variable will be used in a loop to hold subscript values.
- Line 13 declares a `StreamReader` variable named `inputFile`.
- Line 16 opens a file named Values.txt for reading. After this statement executes, the `inputFile` variable references a `StreamReader` object that is associated with the file.
- Line 19 is the beginning of a `while` loop that reads items from the file and assigns them to elements of the `numbers` array. Notice that the loop tests two Boolean expressions connected by the `&&` operator. The first expression is `index < numbers.Length`. The purpose of this expression is to prevent the loop from writing beyond the end of the array. When the array is full, the loop stops. The second expression is `!inputFile.EndOfStream`. The purpose of this expression is to prevent the loop from reading beyond the end of the file. When there are no more values to read from the file, the loop stops.
- Inside the loop, line 21 reads a line of text from the file, converts it to an `int`, and assigns the `int` to `numbers[index]`. Then, line 22 increments `index`.
- Line 26 closes the file.
- The `foreach` loop in lines 29–32 displays the array elements in the `outputListBox` control.

Figure 7-14 shows the application's form after the *Get Values* button has been clicked.

Figure 7-14 The *File To Array* form



7.4 Passing Arrays as Arguments to Methods

CONCEPT: **An array can be passed as an argument to a method. To pass an array, you pass the variable that references the array.**

Sometimes you will want to write a method that accepts an entire array as an argument and performs an operation on the array. For example, the following code shows a method named `ShowArray`. The method accepts an array of strings as an argument and displays each element in a message box.

```
1 private void ShowArray(string[] strArray)
2 {
3     foreach (string str in strArray)
4     {
5         MessageBox.Show(str);
6     }
7 }
```

Notice in line 1 that the method has a parameter variable named `strArray` and that the parameter's data type is `string[]`. The expression `string[]` indicates that this parameter variable is a reference to a `string` array. When you call this method, you must pass a string array as an argument.

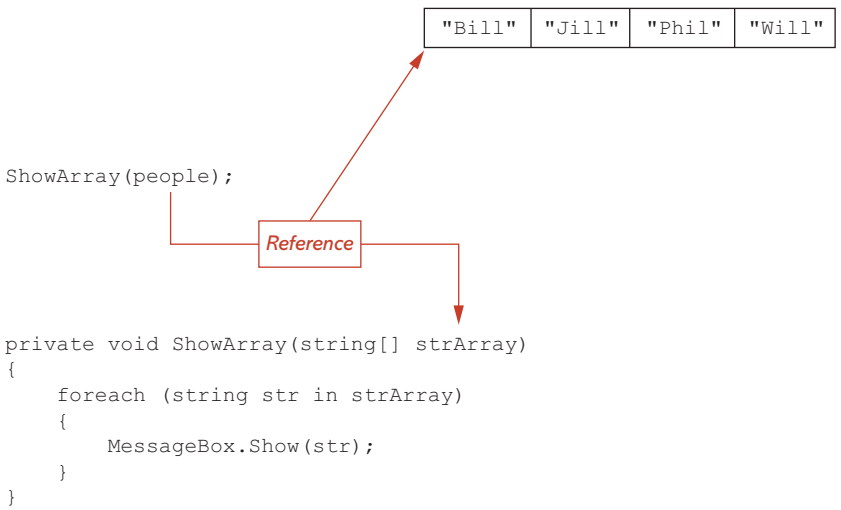
When you call a method and pass an array as an argument, you simply pass the variable that references the array. The following code shows an example of how the `ShowArray` method (previously shown) might be called:

```
1 // Create an array of strings.
2 string[] people = { "Bill", "Jill", "Phil", "Will" };
3
4 // Pass the array to the ShowArray method.
5 ShowArray(people);
```

Line 2 creates an array of strings named `people` and initializes it with four strings. Line 5 calls the `ShowArray` method passing the `people` array as an argument.

Keep in mind that arrays are *always* passed by reference. When you pass an array as an argument, the thing that is passed into the parameter variable is a reference to the array. This is illustrated in Figure 7-15. As shown in the figure, the `people` variable contains a reference to an array. When the `people` variable is passed to the `ShowArray` method, the reference to the array is passed into the `strArray` parameter variable. Figure 7-16 shows that while the `ShowArray` method is executing, the `people` variable and the `strArray` parameter variable reference the same array in memory.

Figure 7-15 An array passed as an argument



Because arrays are always passed by reference, a method that receives an array as an argument has access to the actual array (not a copy of the array). For example, in the *Chap07* folder of the Student Sample Programs, you will find a project named *ArrayArgument*. Figure 7-17 shows the application's form just after the user has clicked the *Go* button.

Figure 7-16 The `people` and `strArray` variables referencing the same array

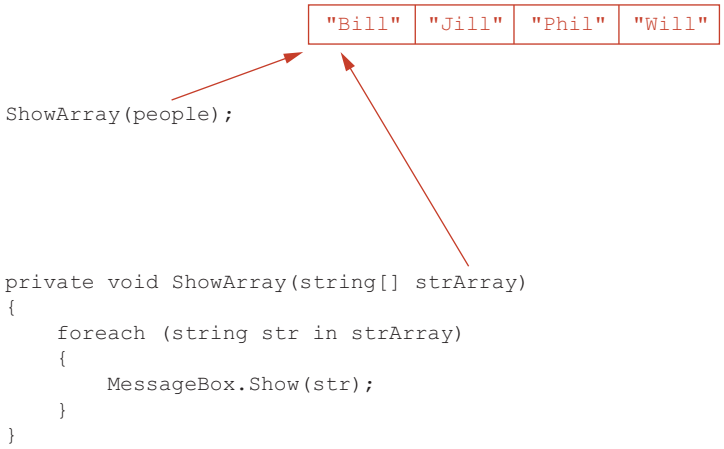
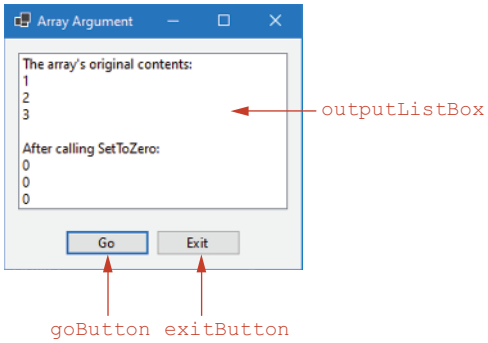


Figure 7-17 The *ArrayArgument* project's output



The following code shows the Click event handler for the *Go* button, and a method named `SetToZero`:

```
1 // Click event handler for the goButton control.
2 private void goButton_Click(object sender, EventArgs e)
3 {
4     // Create an int array.
5     int[] numbers = { 1, 2, 3 };
6
7     // Display the array in the list box.
8     outputListBox.Items.Add("The array's original contents:");
9     foreach (int number in numbers)
10    {
11        outputListBox.Items.Add(number);
12    }
13
14    // Pass the array to the SetToZero method.
15    SetToZero(numbers);
16
17    // Display the array in the list box again.
18    outputListBox.Items.Add("");
19    outputListBox.Items.Add("After calling SetToZero:");
20    foreach (int number in numbers)
21    {
```

```
22          outputListBox.Items.Add(number);
23      }
24  }
25
26  // The SetToZero method accepts an int array as an
27  // argument and sets its elememts to 0.
28  private void SetToZero(int[] iArray)
29  {
30      for (int index = 0; index < iArray.Length; index++)
31      {
32          iArray[index] = 0;
33      }
34  }
```

Let's take a closer look at the `goButton_Click` event handler:

- Line 5 creates an `int` array named `numbers`, initialized with the values 1, 2, and 3.
- Line 8 displays the string `"The array's original contents:"` in the `outputListBox` control.
- The `foreach` loop in lines 9–12 displays the contents of the `numbers` array in the `outputListBox` control. Look at Figure 7-17 and notice that the arrays values are 1, 2, and 3.
- Line 15 calls the `SetToZero` method, passing the `numbers` array as an argument.
- Line 18 displays a blank line in the `outputListBox` control, and line 19 displays the string `"After calling SetToZero:"`.
- The `foreach` loop in lines 9–12 displays the contents of the `numbers` array in the `outputListBox` control. Look at Figure 7-17 and notice that the array's values are now 0, 0, and 0.

As you can see from Figure 7-17, the `SetToZero` method changed the values stored in the `numbers` array. Let's look at the `SetToZero` method:

- Notice in line 28 that the method accepts an `int` array as an argument. The parameter variable's name is `iArray`.
- Line 30 is the beginning of a `for` loop that steps through the array. As the loop iterates, the `index` variable is assigned the values 0, 1, 2, and so forth. The loop iterates as long as `index` is less than `iArray.Length`.
- The statement in line 32 assigns 0 to the array element `iArray[index]`.

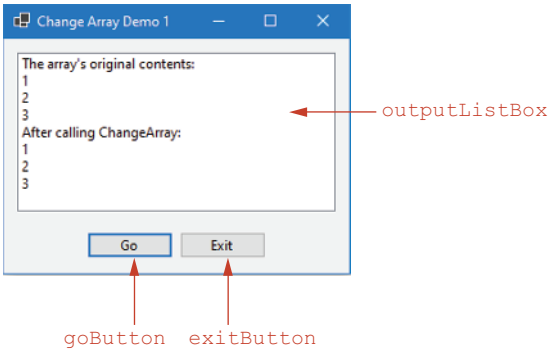
Because the `iArray` parameter is a reference to the array that was passed as an argument, the statement in line 32 assigns 0 to an element of the `numbers` array.

Using `ref` and `out` with Array Parameters

You saw in the previous example that arrays are always passed by reference. When you pass an array as an argument to a method, the method has direct access to the array through its parameter variable. However, the method cannot access the original reference variable that was used to pass the array. For example, in the *Chap07* folder of the Student Sample Programs, you will find a project named *ChangeArrayDemo1*. Figure 7-18 shows the application's form just after the user has clicked the *Go* button. The following code shows the Click event handler for the *Go* button and a method named `ChangeArray`:

```
1  private void goButton_Click(object sender, EventArgs e)
2  {
3      // Create an int array.
4      int[] numbers = { 1, 2, 3 };
5
6      // Display the numbers array's contents.
7      outputListBox.Items.Add("The array's original contents:");
8      foreach (int value in numbers)
```

Figure 7-18 The *ChangeArrayDemo1* project




```
 9      {
10          outputListBox.Items.Add(value);
11      }
12
13      // Pass the numbers array to the ChangeArray method.
14      ChangeArray(numbers);
15
16      // Display the numbers array's contents.
17      outputListBox.Items.Add("After calling ChangeArray:");
18      foreach (int value in numbers)
19      {
20          outputListBox.Items.Add(value);
21      }
22  }
23
24  private void ChangeArray(int[] iArray)
25  {
26      const int NEW_SIZE = 5;
27
28      // Make iArray reference a different array.
29      iArray = new int[NEW_SIZE];
30
31      // Set the new array's elements to 99.
32      for (int index = 0; index < iArray.Length; index++)
33      {
34          iArray[index] = 99;
35      }
36  }
```

Let's take a closer look at the `goButton_Click` event handler:

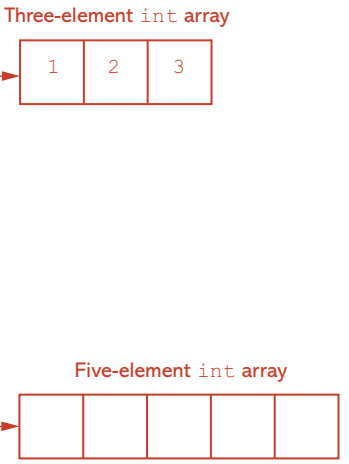
- Line 4 creates an `int` array named `numbers`, initialized with the values 1, 2, and 3.
- Lines 7–11 display the array's contents in the `outputListBox` control. Look at Figure 7-18 and notice that the array's values are 1, 2, and 3.
- Line 14 calls the `ChangeArray` method, passing the `numbers` array as an argument.
- Lines 17–21 display the contents of the `numbers` array in the `outputListBox` control after the `ChangeArray` method has executed. Look at Figure 7-18 and notice that the array's values are still 1, 2, and 3. Apparently, the method did not change the array.

Let's look at the `ChangeArray` method:

- Notice in line 24 that the method accepts an `int` array as an argument. The parameter variable's name is `iArray`. Keep in mind that when we call this method in line 14, passing `numbers` as an argument, the `iArray` parameter and the `numbers` variable reference the same array in memory.
- Line 26 declares an `int` constant named `NEW_SIZE`, set to the value 5.

Figure 7-19 After line 29 executes

```
private void goButton_Click(object sender, EventArgs e)
{
    // Create an int array.
    int[] numbers = { 1, 2, 3 };

    and so forth...

    ChangeArray(numbers);

    and so forth...
}

private void ChangeArray(int[] iArray)
{
    const int NEW_SIZE = 5;

    // Make iArray reference a different array.
    iArray = new int[NEW_SIZE];

    // Set the new array's elements to 99.
    for (int index = 0; index < iArray.Length; index++)
    {
        iArray[index] = 99;
    }
}
```



- Line 29 creates a new `int` array in memory with five elements. A reference to the array is assigned to the `iArray` parameter variable. As shown in Figure 7-19, this causes the `iArray` parameter variable to no longer reference the array that was passed as an argument. Instead, the `iArray` parameter references the new array.
- The `for` loop in lines 32–35 assigns the value 99 to each element of array referenced by `iArray`. This does not affect the `numbers` array.

When you use either the `ref` or `out` keywords with an array parameter, the receiving method not only has access to the array, but it also has access to the reference variable that was used to pass the array. For example, the *ChangeArrayDemo2* project, in the *Chap07* folder of the Student Sample Programs, is identical to the *ChangeArrayDemo1* project, except that the `iArray` parameter is declared with the `ref` keyword in the `ChangeArray` method. The following code shows the Click event handler for the *Go* button, and the `ChangeArray` method.

```
 1 private void goButton_Click(object sender, EventArgs e)
 2 {
 3     // Create an int array.
 4     int[] numbers = { 1, 2, 3 };
 5
 6     // Display the number array's contents.
 7     outputListBox.Items.Add("The array's original contents:");
 8     foreach (int value in numbers)
 9     {
10         outputListBox.Items.Add(value);
11     }
12
13     // Pass the number array to the ChangeArray method.
14     ChangeArray(ref numbers);
15
16     // Display the number array's contents.
17     outputListBox.Items.Add("After calling ChangeArray:");
18     foreach (int value in numbers)
19     {
20         outputListBox.Items.Add(value);
21     }
22 }
```

```
23
24 private void ChangeArray(ref int[] iArray)
25 {
26     const int NEW_SIZE = 5;
27
28     // Make iArray reference a different array.
29     iArray = new int[NEW_SIZE];
30
31     // Set the new array's elements to 99.
32     for (int index = 0; index < iArray.Length; index++)
33     {
34         iArray[index] = 99;
35     }
36 }
```

Notice that in line 24 the `iArray` parameter is declared with the `ref` keyword, and in line 14 the `ref` keyword is used to pass `numbers` as an argument to the `ChangeArray` method. In this code, the `iArray` parameter refers to the `numbers` variable. Anything that is done to the `iArray` parameter is actually done to the `numbers` variable. Figure 7-20 shows how line 29 causes the `numbers` variable to reference the new five-element array.

Figure 7-20 After line 29 executes in the *ChangeArrayDemo2* project

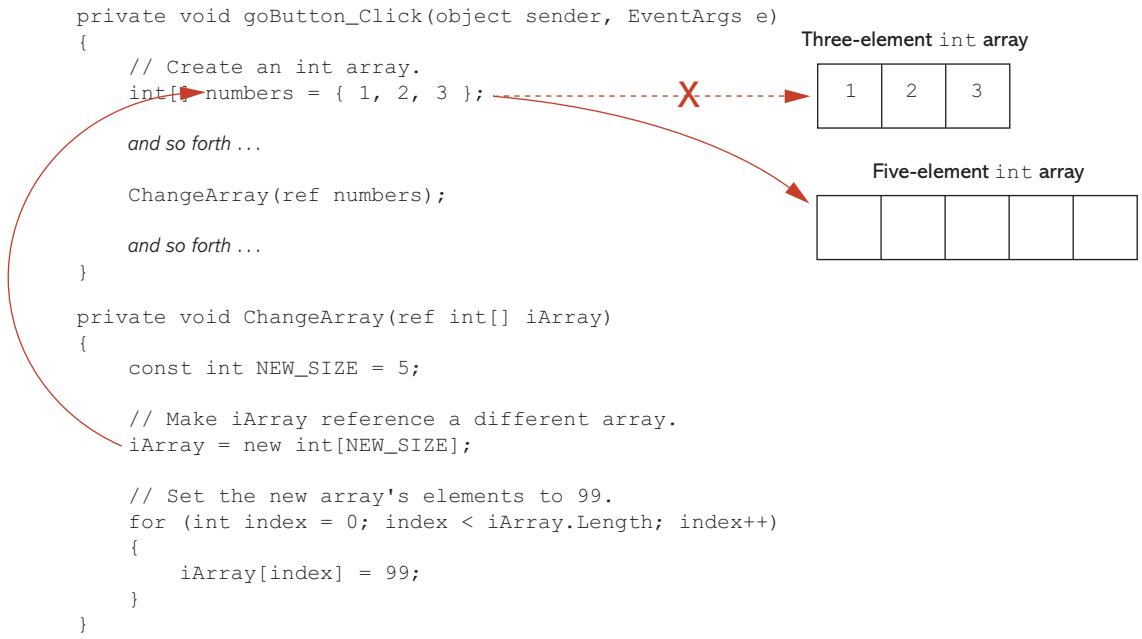


Figure 7-21 shows the application's form just after the user has clicked the *Go* button. Notice from the program's output that after the `ChangeArray` method has been called, the `numbers` variable references a five-element array, and each element's value is 99.

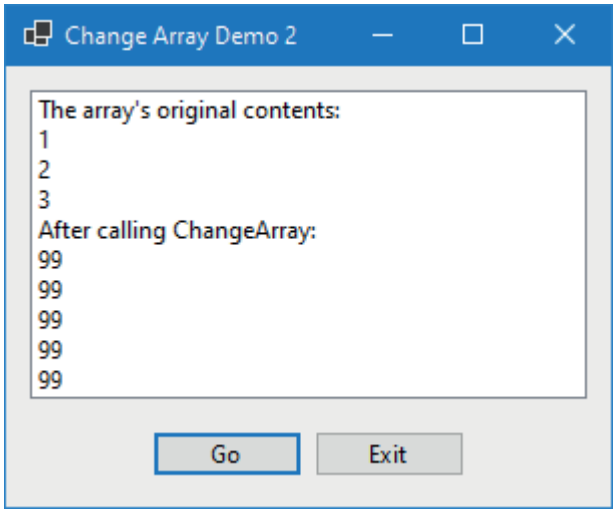
Checkpoint

7.11 When you pass an array as an argument, what is passed into the parameter variable?

7.12 Does a method that receives an array as an argument have access to the actual array or only a copy of the array?

7.13 What is the result when you use either the `ref` or `out` keyword with an array parameter?

Figure 7-21 The *Change Array 1* application



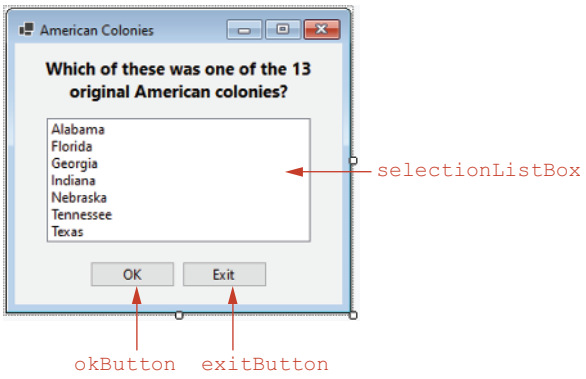
7.5 Some Useful Array Algorithms

The Sequential Search

Programs commonly need to search for data that is stored in an array. Various techniques known as **search algorithms** have been developed to locate a specific item in a larger collection of data, such as an array. In this section, we discuss the simplest of all search algorithms—the sequential search. The **sequential search algorithm** uses a loop to sequentially step through an array, starting with the first element. It compares each element with the value being searched for and stops when the value is found or the end of the array is encountered. If the value being searched for is not in the array, the algorithm unsuccessfully searches to the end of the array.

Let's look at an example. In the *Chap07* folder of the Student Sample Programs, you will find a project named *AmericanColonies*. The application is a game that tests your knowledge of U.S. history. As shown in Figure 7-22, the application's form displays a list of states in a ListBox control. Only one of the states shown in the ListBox was an original American colony. You select the state that you believe was a colony and click the *OK* button to see if you were correct.

Figure 7-22 The *AmericanColonies* project



The following code is taken from the application. It shows a method named `SequentialSearch` and the Click event handler for the *OK* button.

```
 1 // The SequentialSearch method searches a string array
 2 // for a specified value. If the value is found, its
 3 // position is returned. Otherwise, -1 is returned.
 4 private int SequentialSearch(string[] sArray, string value)
 5 {
 6     bool found = false;  // Flag indicating search results
 7     int index = 0;       // Used to step through the array
 8     int position = -1;   // Position of value, if found
 9
10     // Search the array.
11     while (!found && index < sArray.Length)
12     {
13         if (sArray[index] == value)
14         {
15             found = true;
16             position = index;
17         }
18
19         index++;
20     }
21
22     // Return
23     return position;
24 }
25
26 private void okButton_Click(object sender, EventArgs e)
27 {
28     string selection;   // To hold the user's selection
29
30     // Create an array with the colony names.
31     string[] colonies = { "Delaware", "Pennsylvania", "New Jersey",
32                           "Georgia", "Connecticut", "Massachusetts",
33                           "Maryland", "South Carolina", "New
                             Hampshire",
34                           "Virginia", "New York", "North Carolina",
35                           "Rhode Island" };
36
37     if (selectionListBox.SelectedIndex != -1)
38     {
39         // Get the selected item.
40         selection = selectionListBox.SelectedItem.ToString();
41
42         // Determine if the item is in the array.
43         if (SequentialSearch(colonies, selection) != -1)
44         {
45             MessageBox.Show("Yes, that was one of the colonies.");
46         }
47         else
48         {
49            MessageBox.Show("No, that was not one of the colonies.");
50         }
51     }
52 }
```

The `SequentialSearch` method, which begins in line 4, searches a `string` array for a specified value. It accepts a `string` array and a `string` search value as arguments. If the search value is found in the array, the method returns the value's subscript. If the search

value is not found in the array, the method returns –1. Let's take a closer look at the method:

- Line 6 declares a `bool` variable named `found`. The `found` variable is used as a flag. Setting `found` to `false` indicates that the search value has not been found. Setting `found` to `true` indicates that the search value has been found. Notice that `found` is initialized with `false`.
- Line 7 declares an `int` variable named `index` that will be used to step through the elements of the array. Notice that `index` is initialized with the value 0.
- Line 8 declares an `int` variable named `position`. If the search value is found in the array, we save its subscript in the `position` variable. Notice that the `position` variable is initialized with the value −1.
- The `while` loop that begins in line 11 searches the array for the specified value. It iterates as long as `found` is not `true` and `index` is less than the array's length.
- The `if` statement in line 13 determines whether `sArray[index]` is equal to `value`. If this is true, then the search value has been found in the array. In that case, line 15 sets `found` to `true`, and line 16 assigns `index` to `position`.
- Line 19 increments `index`.
- When the loop finishes, line 23 returns the value of the `position` variable. If the search value was found in the array, the `position` variable will contain the value's subscript. If the search value was not found in the array, the `position` variable will still be set to −1.

The Click event handler for the *OK* button begins in line 26. Let's take a closer look at the event handler's code:

- Line 28 declares a `string` variable named `selection`. This variable will hold the item that is selected from the ListBox control.
- Lines 31–35 declare a `string` array named `colonies`. The array is initialized with the names of the U.S. colonies.
- The `if` statement that begins in line 37 determines whether an item has been selected in the `selectionListBox` control. If an item has been selected, the following actions take place:
  - Line 40 gets the selected item and assigns it to the `selection` variable.
  - The `if` statement in line 43 calls the `SequentialSearch` method, passing the `colonies` array and the `selection` variable as arguments. If the value of the `selection` variable is found in the `colonies` array, the method returns a value other than −1, and line 45 displays a message box informing the user that the selected item was one of the colonies. However, if the value of the `selection` variable is not found in the `colonies` array, the method will return −1, and line 49 displays a message box informing the user that the selected item was not one of the colonies.

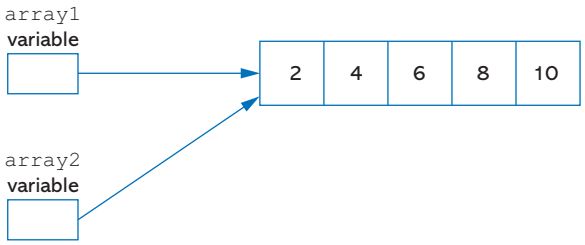
Copying an Array

Because an array is an object, there is a distinction between an array and the variable that references it. The array and the reference variable are two separate entities. This is important to remember when you wish to copy the contents of one array to another. You might be tempted to write something like the following code, thinking that you are copying an array:

```
int[] array1 = { 2, 4, 6, 8, 10 };
int[] array2 = array1; // This does not copy array1.
```

The first statement creates an array referenced by the `array1` variable. The second statement assigns `array1` to `array2`. This does not make a copy of the array referenced by `array1`. Rather, it assigns the reference that is in `array1` to `array2`. After this statement executes, both the `array1` and `array2` variables will reference the same array. This type of assignment operation is called a **reference copy**. Only a reference to the array object is copied, not the contents of the array object. This is illustrated in Figure 7-23.

Figure 7-23 Both `array1` and `array2` referencing the same array



If you want to make a copy of an array, you must create the second array in memory and then copy the individual elements of the first array to the second. This is usually best done with a loop, such as the following:

```
1 const int SIZE = 5;
2 int[] firstArray = { 5, 10, 15, 20, 25 };
3 int[] secondArray = new int[SIZE];
4
5 for (int index = 0; index < firstArray.length; index++)
6 {
7     secondArray[index] = firstArray[index];
8 }
```

The loop in this code copies each element of `firstArray` to the corresponding element of `secondArray`.

Comparing Arrays

You cannot use the == operator to compare two array reference variables and determine whether the arrays are equal. For example, the following code appears to compare two arrays, but in reality it does not:

```
1 int[] firstArray  = { 5, 10, 15, 20, 25 };
2 int[] secondArray = { 5, 10, 15, 20, 25 };
3
4 if (firstArray == secondArray) // This is a mistake.
5 {
6     MessageBox.Show("The arrays are the same.");
7 }
8 else
9 {
10    MessageBox.Show("The arrays are not the same.");
11 }
```

When you use the == operator with reference variables, the operator compares the references that the variables contain, not the contents of the objects referenced by the variables. Because the `firstArray` and `secondArray` variables in this example reference different objects in memory, the result of the Boolean expression `firstArray == secondArray` is false, and the code reports that the arrays are not the same.

To compare the contents of two arrays, you must compare the elements of the two arrays. For example, look at the following code:

```
1 int[] firstArray  = { 2, 4, 6, 8, 10 };
2 int[] secondArray = { 2, 4, 6, 8, 10 };
3 boolean arraysEqual = true; // Flag variable
4 int index = 0;              // To hold array subscripts
5
6 // First determine whether the arrays are the same size.
7 if (firstArray.length != secondArray.length)
```

```
 8 {
 9     arraysEqual = false;
10 }
11
12 // Next determine whether the elements contain the same data.
13 while (arraysEqual && index < firstArray.length)
14 {
15     if (firstArray[index] != secondArray[index])
16     {
17         arraysEqual = false;
18     }
19     index++;
20 }
21
22 if (arraysEqual)
23 {
24     MessageBox.Show("The arrays are equal.");
25 }
26 else
27 {
28     MessageBox.Show("The arrays are not equal.");
29 }
```

This code determines whether `firstArray` and `secondArray` (declared in lines 1 and 2) contain the same values. A Boolean flag variable, `arraysEqual`, is declared and initialized to `true` in line 3. The `arraysEqual` variable used to signal whether the arrays are equal. Another variable, `index`, is declared and initialized to 0 in line 4. The `index` variable is used in a loop to step through the arrays.

First, the `if` statement in line 7 determines whether the two arrays are the same length. If they are not the same length, then the arrays cannot be equal, so the flag variable `arraysEqual` is set to `false` in line 9. Then a `while` loop begins in line 13. The loop executes as long as `arraysEqual` is true and the `index` variable is less than `firstArray.length`. During each iteration, it compares a different set of corresponding elements in the arrays. When it finds two corresponding elements that have different values, the flag variable `arraysEqual` is set to `false`.

After the loop finishes, an `if` statement examines the `arraysEqual` variable in line 22. If the variable is `true`, then the arrays are equal and a message indicating so is displayed in line 24. Otherwise, they are not equal, so a different message is displayed in line 28.

Totaling the Values in an Array

To calculate the total of the values in a numeric array, you use a loop with an accumulator variable. First, the accumulator is initialized with 0. Then, the loop steps through the array, adding the value of each array element to the accumulator.

```
 1 // Create an int array.
 2 int[] units = { 2, 4, 6, 8, 10 };
 3
 4 // Declare and initialize an accumulator variable.
 5 int total = 0;
 6
 7 // Step through the array, adding each element to
 8 // the accumulator.
 9 for (int index = 0; index < units.Length; index++)
10 {
11     total += units[index];
12 }
13
14 // Display the total.
15 MessageBox.Show("The total is " + total);
```

Averaging the Values in an Array

The first step in calculating the average of all the values in a numeric array is to get the total of the values. The second step is to divide the total by the number of elements in the array. The following code shows an example:

```
// Create an array.
double[] scores = { 92.5, 81.6, 65.7, 72.8 }

// Declare and initialize an accumulator variable.
double total = 0.0;

// Declare a variable to hold the average.
double average;

// Step through the array, adding each element to
// the accumulator.
for (int index = 0; index < scores.Length; index++)
{
    total += scores[index];
}

// Calculate the average.
average = total / scores.Length;

// Display the average.
MessageBox.Show("The average is " + average);
```

When this code finishes, the `average` variable will contain the average of the values in the `scores` array. Notice that the last statement, which divides `total` by `scores.length`, is not inside the loop. This statement should execute only once, after the loop has finished its iterations.

Finding the Highest and Lowest Values in an Array

Some programming tasks require you to find the highest value in a set of data. Examples include programs that report the highest sales amount for a given time period, the highest test score in a set of test scores, the highest temperature for a given set of days, and so forth.

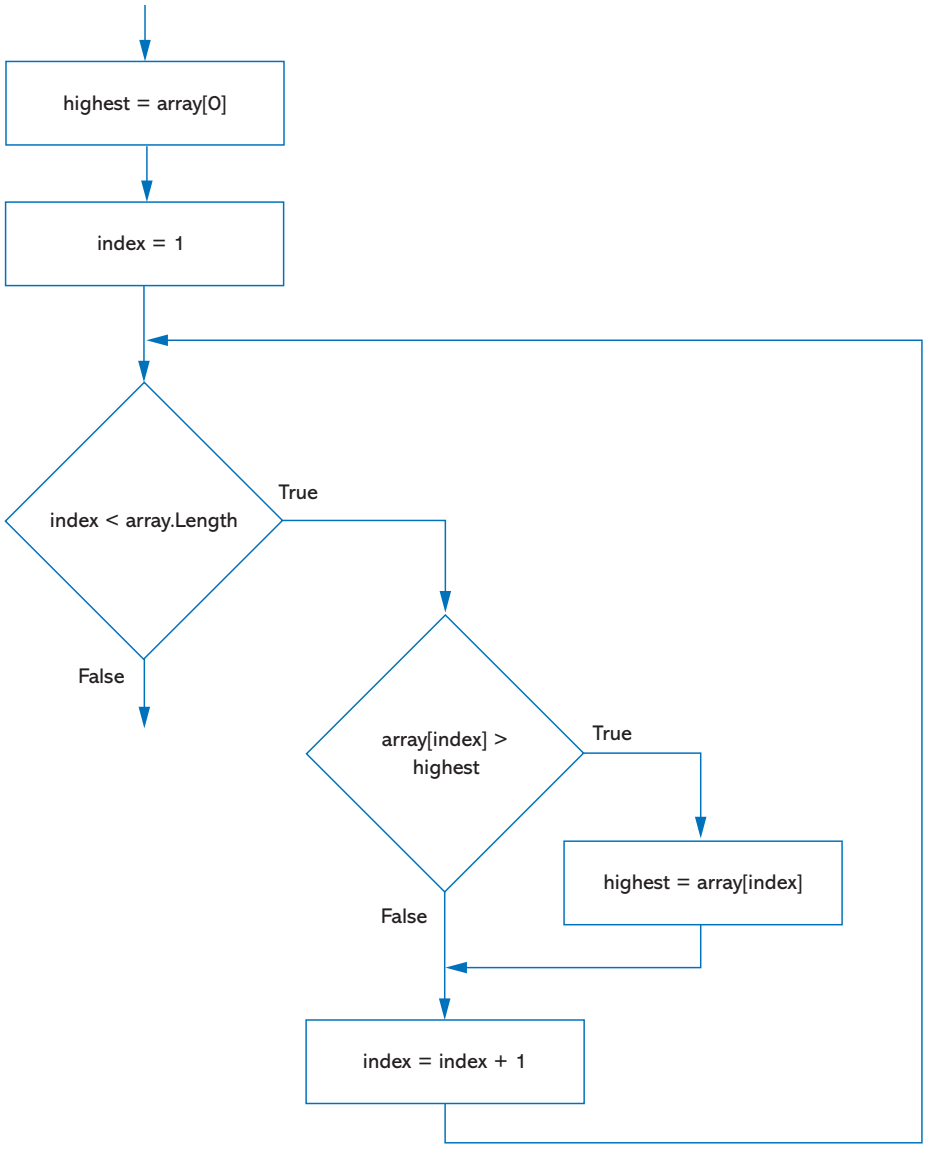
The algorithm for finding the highest value in an array works like this: You create a variable to hold the highest value (the following example names this variable `highest`). Then, you assign the value at element 0 to the `highest` variable. Next, you use a loop to step through the rest of the array elements, beginning at element 1. Each time the loop iterates, it compares an array element to the `highest` variable. If the array element is greater than the `highest` variable, then the value in the array element is assigned to the `highest` variable. When the loop finishes, the `highest` variable will contain the highest value in the array. The flowchart in Figure 7-24 illustrates this logic.

The following code demonstrates this algorithm:

```
// Create an array.
int[] numbers = { 8, 1, 12, 6, 2 };

// Declare a variable to hold the highest value, and
// initialize it with the first value in the array.
int highest = numbers[0];

// Step through the rest of the array, beginning at
// element 1. When a value greater than highest is found,
// assign that value to highest.
for (int index = 1; index < numbers.Length; index++)
```

```
12 {
13     if (numbers[index] > highest)
14     {
15         highest = numbers[index];
16     }
17 }
18
19 // Display the highest value.
20 MessageBox.Show("The highest value is " + highest);
```

Figure 7-24 Flowchart for finding the highest value in an array

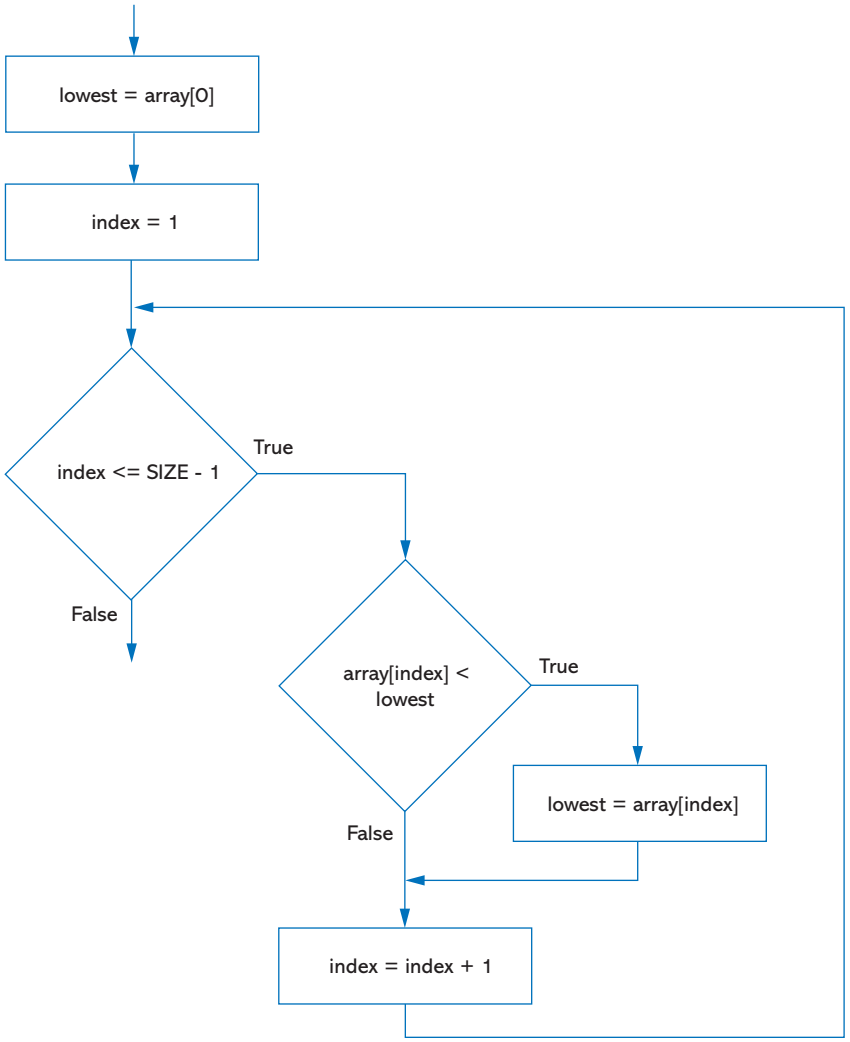


In some programs, you are more interested in finding the lowest value than the highest value in a set of data. For example, suppose you are writing a program that stores several players' golf scores in an array and you need to find the best score. In golf, the lower the score the better, so you need an algorithm that finds the lowest value in the array.

The algorithm for finding the lowest value in an array is very similar to the algorithm for finding the highest score. It works like this: You create a variable to hold the lowest value (the following example names this variable `lowest`). Then, you assign the value at element 0 to the `lowest` variable. Next, you use a loop to step through the rest of the

array elements, beginning at element 1. Each time the loop iterates, it compares an array element to the `lowest` variable. If the array element is less than the `lowest` variable, then the value in the array element is assigned to the `lowest` variable. When the loop finishes, the `lowest` variable contains the lowest value in the array. The flowchart in Figure 7-25 illustrates this logic.

Figure 7-25 Flowchart for finding the lowest value in an array



The following code demonstrates this algorithm:

```
// Create an array.
int[] numbers = { 8, 1, 12, 6, 2 };

// Declare a variable to hold the lowest value, and
// initialize it with the first value in the array.
int lowest = numbers[0];

// Step through the rest of the array, beginning at
// element 1. When a value less than lowest is found,
// assign that value to lowest.
for (int index = 1; index < numbers.Length; index++)
{
    if (numbers[index] < lowest)
```

```
14      {
15          lowest = numbers[index];
16      }
17  }
18
19  // Display the lowest value.
20  MessageBox.Show("The lowest value is " + lowest);
```

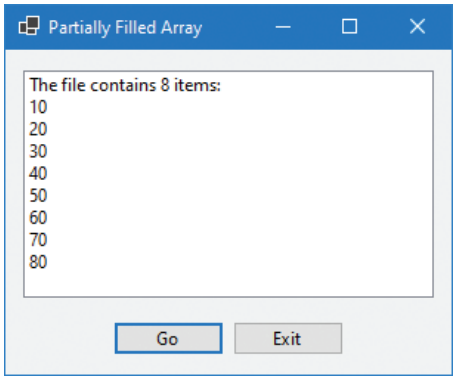
Partially Filled Arrays

Sometimes you need to store a series of items in an array, but you do not know the number of items in the series. As a result, you do not know the exact number of elements needed for the array. One solution is to make the array large enough to hold the largest possible number of items. This can lead to another problem, however. If the actual number of items stored in the array is less than the number of elements, the array will be only partially filled. When you process a partially filled array, you must process only the elements that contain valid data items.

A partially filled array is normally used with an accompanying integer variable that holds the number of items that are actually stored in the array. If the array is empty, then 0 is stored in this variable because there are no items in the array. Each time an item is added to the array, the variable is incremented. When code steps through the array's elements, the value of this variable is used instead of the array's size to determine the maximum subscript.

For example, in the *Chap07* folder of the Student Sample Programs, you will find a project named *PartiallyFilledArray*. When you click the *Go* button, the application reads up to 100 values from a file named Values.txt and stores them in a 100-element `int` array. If the file contains fewer than 100 values, the application will partially fill the array. The contents of the array are then displayed in a list box. Figure 7-26 shows the application's form just after the user has clicked the *Go* button. The following code shows the Click event handler for the *Get Values* button.

Figure 7-26 The *PartiallyFilledArray* project




```
1  private void goButton_Click(object sender, EventArgs e)
2  {
3      try
4      {
5          // Create an array to hold items read from the file.
6          const int SIZE = 100;
7          int[] numbers = new int[SIZE];
8
9          // Variable to hold the number of items stored in
10         // the array
11         int count = 0;
```

```
12
13          // Declare a StreamReader variable.
14          StreamReader inputFile;
15
16          // Open the file and get a StreamReader object.
17          inputFile = File.OpenText("Values.txt");
18
19          // Read the file's contents into the array until the
20          // end of the file is reached, or the array is full.
21          while (!inputFile.EndOfStream && count < numbers.Length)
22          {
23              // Read the next item from the file.
24              numbers[count] = int.Parse(inputFile.ReadLine());
25
26              // Increment count.
27              count++;
28          }
29
30          // Close the file.
31          inputFile.Close();
32
33          // Display the array elements in the list box.
34          outputListBox.Items.Add("The file contains " + count +
35              " items:");
36
37          for (int index = 0; index < count; index++)
38          {
39              outputListBox.Items.Add(numbers[index]);
40          }
41      }
42      catch (Exception ex)
43      {
44          // Display an error message.
45          MessageBox.Show(ex.Message);
46      }
47  }
```

Let's examine the code in detail:

- Line 3 is the beginning of a `try-catch` statement that handles any errors that might result while reading data from the file.
- Line 6 declares a constant, `SIZE`, initialized with the value 100.
- Line 7 declares an `int` array named `numbers` using `SIZE` as the size declarator. As a result, the values array has 100 elements.
- Line 11 declares an `int` variable named `count`, which holds the number of items that are stored in the `numbers` array. Notice that `count` is initialized with 0 because there are no values stored in the array.
- Line 14 declares a `StreamReader` variable named `inputFile`.
- Line 17 opens a file named Values.txt for reading. After this statement executes, the `inputFile` variable references a `StreamReader` object that is associated with the file.
- Line 21 is the beginning of a `while` loop that reads items from the file and assigns them to elements of the `numbers` array. Notice that the loop tests two Boolean expressions connected by the `&&` operator. The first expression is `!inputFile.EndOfStream`. The purpose of this expression is to prevent the loop from reading beyond the end of the file. When there are no more values to read from the file, the loop stops. The second expression is `count < numbers.Length`. The purpose of this expression is to prevent the loop from writing beyond the end of the array. When the array is full, the loop will stop.

- Inside the loop, line 24 reads a line of text from the file, converts it to an `int`, and assigns the `int` to `numbers[index]`.
- Then, line 27 increments the `count` variable. Each time a number is assigned to an array element, the `count` variable is incremented. As a result, the `count` variable holds the number of items that are stored in the array.
- Line 31 closes the file.
- The `for` loop in lines 37–40 displays the array elements in the `outputListBox` control. Rather than stepping through all the elements in the array, however, the loop steps through only the elements that contain values. Notice that the loop iterates as long as `index` is less than `count`. Because `count` contains the number of items stored in the array, the loop stops when the element containing the last valid value has been displayed.

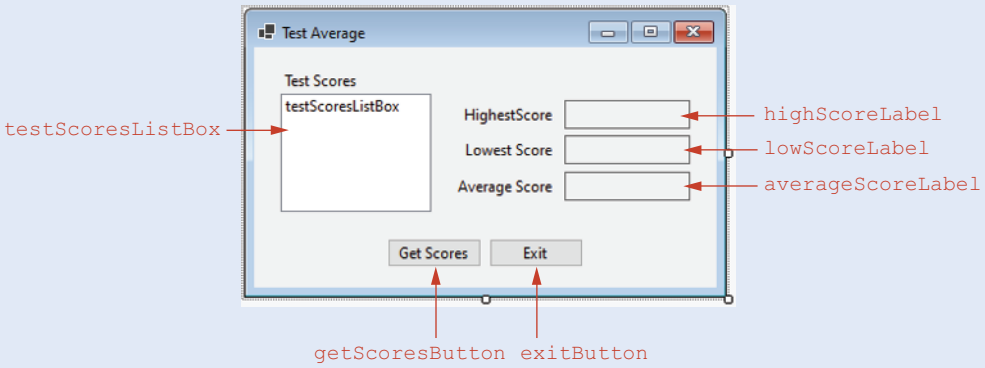
Now that you've seen several algorithms for processing the contents of an array, you should practice writing some of them yourself. Tutorial 7-2 takes you through the process of writing an application that reads data from a file into an `int` array and then determines the highest, lowest, and average values in the array.

Tutorial 7-2:
Processing an Array

VideoNote
Tutorial 7-2: Processing an Array

In this tutorial, you complete an application that reads five test scores from a file and stores the test scores in an array. The application displays the test scores as well as the highest score, the lowest score, and the average score. Figure 7-27 shows the application's form, which has already been created for you. A set of five test scores is stored in a file named TestScores.txt, which has also been created for you.

Figure 7-27 The *TestAverage* project's form



Step 1: Start Visual Studio. Open the project named *TestAverage* in the *Chap07* folder of the Student Sample Programs.

Step 2: Open the Form1 form's code in the code editor. Type the comments and code for the `Average` method, shown in lines 10–29 of Program 7-2. The purpose of the `Average` method is to accept an `int` array as an argument and return the average of the values in the array. This method uses an algorithm similar to the array averaging you saw earlier in this chapter.

Step 3: Type the comments and code for the `Highest` method, shown in lines 31–52 of Program 7-2. The purpose of the `Highest` method is to accept an `int` array as an argument and return the highest value in the array. This method uses an algorithm similar to the algorithm that you saw earlier in this chapter for finding the highest value in an array.

Step 4: Type the comments and code for the `Lowest` method, shown in lines 54–75 of Program 7-2. The purpose of the `Lowest` method is to accept an `int` array as an argument and return the lowest value in the array. This method uses an algorithm similar to the algorithm that you saw earlier in this chapter for finding the lowest value in an array.

Step 5: Now you create the Click event handlers for the Button controls. Switch back to the *Designer* and double-click the `getScoresButton` control. This opens the code editor, and you will see an empty event handler named `getScoresButton_Click`. Complete the `getScoresButton_Click` event handler by typing the code shown in lines 79–123 in Program 7-2. Let's review this code:

Line 79: This is the beginning of a `try-catch` statement that handles any exceptions that are thrown while reading and processing data from the file. If an exception occurs in the try block (lines 81–117), the program jumps to the catch block, and line 122 displays an error message.

Lines 82–88: The following declarations appear in these lines:

- `SIZE`—a constant, set to 5, for the number of test scores
- `scores`—an `int` array that holds the test scores
- `index`—an `int` variable, initialized to 0, that is used in a loop to step through the elements of the scores array
- `highestScore`—an `int` that holds the highest score
- `lowestScore`—an `int` that holds the lowest score
- `averageScore`—a `double` that holds the average score
- `inputFile`—a variable that references the `StreamReader` object that is used to read data from the file

Line 91: After this statement executes, the TestScores.txt file will be opened for reading, and the `inputFile` variable will reference a `StreamReader` object that is associated with the file.

Line 94: This is the beginning of a `while` loop that iterates as long as the end of the TestScores.txt file has not been reached and as long as `index` is less than `scores.Length`. (Recall that `index` starts with the value 0.)

Line 96: This statement reads a line of text from the file and assigns it to the array element `scores[index]`.

Line 97: This statement increments the `index` variable.

Line 101: This statement closes the TestScores.txt file.

Lines 104–107: This `foreach` loop displays the contents of the `scores` array in the `testScoresListBox` control.

Line 110: This statement calls the `Highest` method, passing the `scores` array as an argument. The method returns the highest value in the array, which is assigned to the `highestScore` variable.

Line 111: This statement calls the `Lowest` method, passing the `scores` array as an argument. The method returns the lowest value in the array, which is assigned to the `lowestScore` variable.

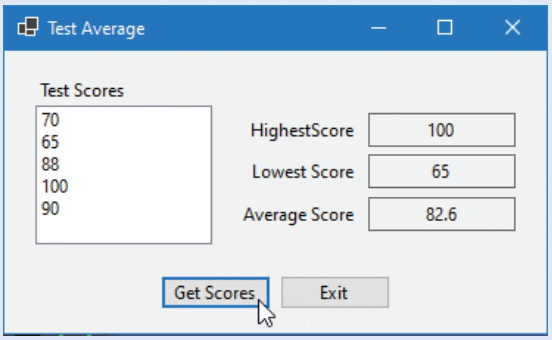
Line 112: This statement calls the `Average` method, passing the `scores` array as an argument. The method returns the average of the values in the array, which is assigned to the `averageScore` variable.

Lines 115–117: These statements display the highest score, lowest score, and average score.

Step 6: Switch your view back to the *Designer* and double-click the `exitButton` control. In the code editor you will see an empty event handler named `exitButton_Click`. Complete the `exitButton_Click` event handler by typing the code shown in lines 128–129 in Program 7-2.

Step 7: Save the project. Then, press F5 on the keyboard or click the *Start Debugging* button (▶) on the toolbar to compile and run the application. When the application runs, click the *Get Scores* button. This should display a set of test scores in ListBox as well as the highest, lowest, and average of the test scores, as shown in Figure 7-28. Click the *Exit* button to exit the application.

Figure 7-28 The *TestAverage* project's form



Program 7-2 Completed code for Form1 in the *TestAverage* project

```
1  namespace TestAverage
2  {
3      public partial class Form1 : Form
4      {
5          public Form1()
6          {
7              InitializeComponent();
8          }
9
10         // The Average method accepts an int array argument
11         // and returns the Average of the values in the array.
12         private double Average(int[] iArray)
13         {
14             int total = 0;   // Accumulator, initialized to 0
15             double average;  // To hold the average
16
17             // Step through the array, adding each element to
18             // the accumulator.
19             for (int index = 0; index < iArray.Length; index++)
20             {
21                 total += iArray[index];
22             }
23
24             // Calculate the average.
25             average = (double)total / iArray.Length;
26
27             // Return the average.
28             return average;
29         }
30
```

```
31          // The Highest method accepts an int array argument
32          // and returns the highest value in that array.
33          private int Highest(int[] iArray)
34          {
35              // Declare a variable to hold the highest value, and
36              // initialize it with the first value in the array.
37              int highest = iArray[0];
38
39              // Step through the rest of the array, beginning at
40              // element 1. When a value greater than highest is
                found,
41              // assign that value to highest.
42              for (int index = 1; index < iArray.Length; index++)
43              {
44                  if (iArray[index] > highest)
45                  {
46                      highest = iArray[index];
47                  }
48              }
49
50              // Return the highest value.
51              return highest;
52          }
53
54          // The Lowest method accepts an int array argument
55          // and returns the lowest value in that array.
56          private int Lowest(int[] iArray)
57          {
58              // Declare a variable to hold the lowest value, and
59              // initialize it with the first value in the array.
60              int lowest = iArray[0];
61
62              // Step through the rest of the array, beginning at
63              // element 1. When a value less than lowest is found,
64              // assign that value to lowest.
65              for (int index = 1; index < iArray.Length; index++)
66              {
67                  if (iArray[index]  < lowest)
68                  {
69                      lowest = iArray[index];
70                  }
71              }
72
73              // Return the lowest value.
74              return lowest;
75          }
76
77          private void getScoresButton_Click(object sender,
            EventArgs e)
78          {
79              try
80              {
81                  // Local variables
82                  const int SIZE = 5;              // Number of tests
83                  int[] scores = new int[SIZE];  // Array of test
                    scores
84                  int index = 0;                   // Loop counter
85                  int highestScore;                // To hold the
                    highest score
```

```
86                  int lowestScore;                // To hold the
                    lowest score
87                  double averageScore;            // To hold the
                    average score
88                  StreamReader inputFile;         // For file
                    input
89
90                  // Open the file and get a StreamReader object.
91                  inputFile = File.OpenText("TestScores.txt");
92
93                  // Read the test scores into the array.
94                  while (!inputFile.EndOfStream && index  <
                    scores.Length)
95                  {
96                      scores[index] = int.Parse(inputFile.
                        ReadLine());
97                      index++;
98                  }
99
100                 // Close the file.
101                 inputFile.Close();
102
103                 // Display the test scores.
104                 foreach (int value in scores)
105                 {
106                     testScoresListBox.Items.Add(value);
107                 }
108
109                 // Get the highest, lowest, and average scores.
110                 highestScore = Highest(scores);
111                 lowestScore = Lowest(scores);
112                 averageScore = Average(scores);
113
114                 // Display the values.
115                 highScoreLabel.Text = highestScore.ToString();
116                 lowScoreLabel.Text = lowestScore.ToString();
117                 averageScoreLabel.Text = averageScore.
                    ToString("n1");
118             }
119             catch (Exception ex)
120             {
121                 // Display an error message.
122                 MessageBox.Show(ex.Message);
123             }
124         }
125
126         private void exitButton_Click(object sender, EventArgs e)
127         {
128             // Close the form.
129             this.Close();
130         }
131     }
132 }
```

7.6 Advanced Algorithms for Sorting and Searching Arrays

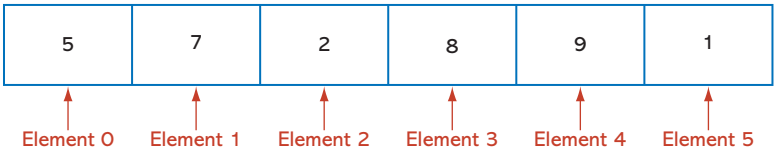
CONCEPT: **A sorting algorithm is used to arrange data into some order. A search algorithm is a method of locating a specific item in a larger collection of data. The selection sort and the binary search are popular sorting and searching algorithms.**

The Selection Sort Algorithm

Often the data in an array must be sorted in some order. Customer lists, for instance, are commonly sorted in alphabetical order. Student grades might be sorted from highest to lowest. Product codes could be sorted so all the products of the same color are stored together. In this section, we explore how to write your own sorting algorithm. A sorting algorithm is a technique for scanning through an array and rearranging its contents in some specific order. The algorithm that we explore is called the selection sort.

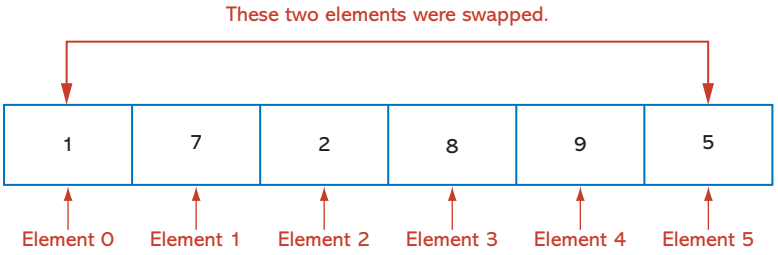
The **selection sort** works like this: The smallest value in the array is located and moved to element 0. Then the next smallest value is located and moved to element 1. This process continues until all the elements have been placed in their proper order. Let's see how the selection sort works when arranging the elements of the array in Figure 7-29.

Figure 7-29 Values in an array



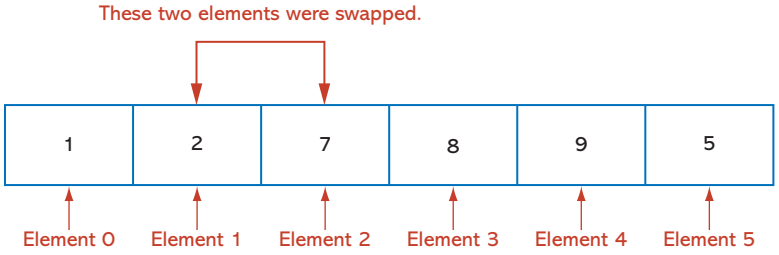
The selection sort scans the array, starting at element 0, and locates the element with the smallest value. Then, the contents of this element are swapped with the contents of element 0. In this example, the 1 stored in element 5 is swapped with the 5 stored in element 0. After the swap, the array appears as shown in Figure 7-30.

Figure 7-30 Values in the array after the first swap



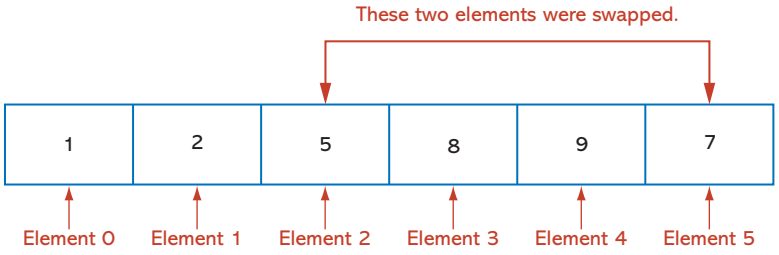
Then, the algorithm repeats the process, but because element 0 already contains the smallest value in the array, it can be left out of the procedure. This time, the algorithm begins the scan at element 1. In this example, the value in element 2 is swapped with the value in element 1. Then, the array appears as shown in Figure 7-31.

Figure 7-31 Values in the array after the second swap



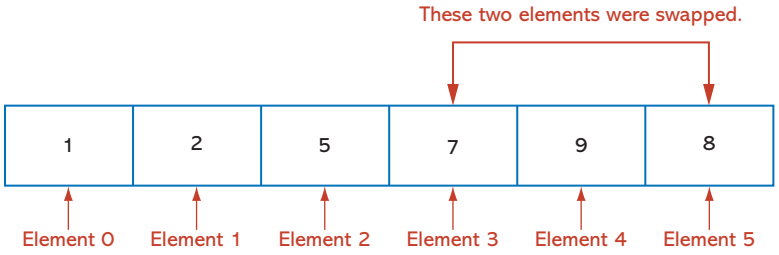
Once again, the process is repeated, but this time the scan begins at element 2. The algorithm will find that element 5 contains the next smallest value. This element's value is swapped with that of element 2, causing the array to appear as shown in Figure 7-32.

Figure 7-32 Values in the array after the third swap



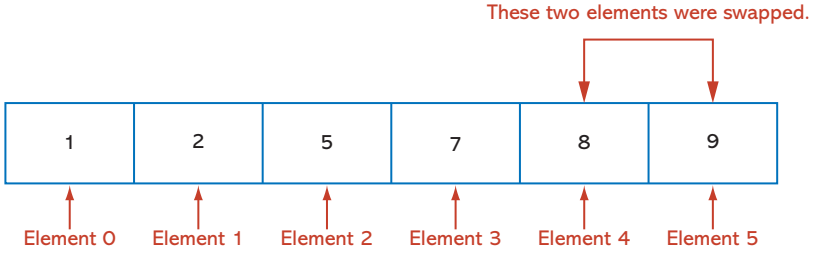
Next, the scanning begins at element 3. Its value is swapped with that of element 5, causing the array to appear as shown in Figure 7-33.

Figure 7-33 Values in the array after the fourth swap



At this point, there are only two elements left to sort. The algorithm finds that the value in element 5 is smaller than that of element 4, so the two are swapped. This puts the array in its final arrangement, as shown in Figure 7-34.

Figure 7-34 Values in the array after the fifth swap



Swapping Array Elements

As you saw in the description of the selection sort algorithm, certain elements are swapped as the algorithm steps through the array. Let's briefly discuss the process of swapping two items in computer memory. Assume we have the following variable declarations:

```
int a = 1;
int b = 9;
```

Suppose we want to swap the values in these variables so the variable `a` contains 9 and the variable `b` contains 1. At first, you might think that we need only assign the variables to each other, like this:

```
// ERROR! The following does NOT swap the variables.
a = b;
b = a;
```

To understand why this does not work, let's step through the code. The first statement is `a = b`. This causes the value 9 to be assigned to `a`. But, what happens to the value 1 that was previously stored in `a`? Remember, when you assign a new value to a variable, the new value replaces any value that was previously stored in the variable. So, the old value, 1, is thrown away. Then the next statement is `b = a`. Since the variable `a` contains 9, this assigns 9 to `b`. After these statements execute, the variables `a` and `b` both contain the value 9.

To successfully swap the contents of two variables, we need a third variable that can serve as a temporary storage location:

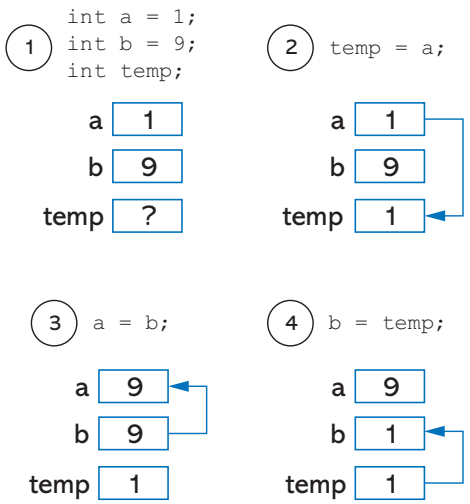
```
int temp;
```

Then we can perform the following steps to swap the values in the variables `a` and `b`:

- Assign the value of `a` to `temp`.
- Assign the value of `b` to `a`.
- Assign the value of `temp` to `b`.

Figure 7-35 shows the contents of these variables as we perform each of these steps. Notice that after the steps are finished, the values in `a` and `b` are swapped.

Figure 7-35 Swapping the values of `a` and `b`



Here is the code for a `Swap` method that we can use to swap to `int` values:

```
1 private void Swap(ref int a, ref int b)
2 {
3     int temp = a;
4     a = b;
5     b = temp;
6 }
```

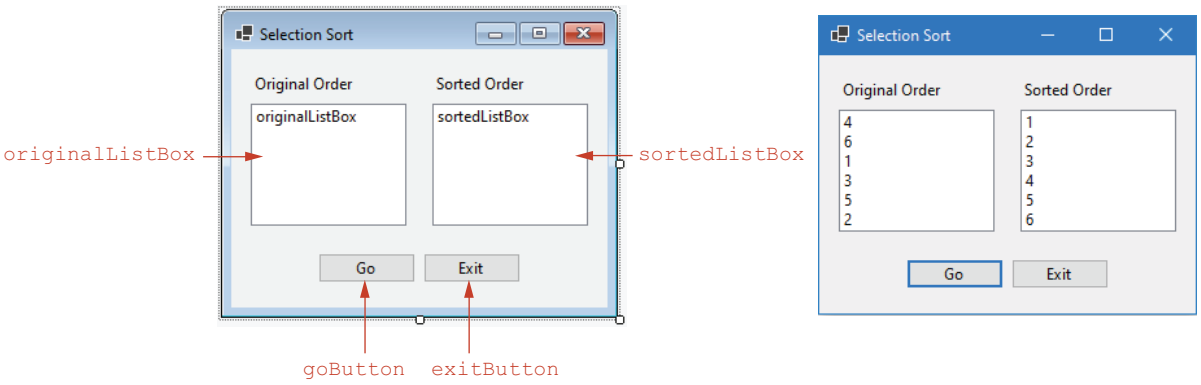
NOTE: It is critical that we use reference parameters in the `Swap` method because the method must be able to change the values of the items that are passed to it as arguments.

Let's look at a complete program that demonstrates the Selection Sort algorithm. In the *Chap07* folder of the Student Sample Programs, you will find a project named *SelectionSort*. Figure 7-36 shows the application's form. On the left, you see the form in the *Designer*, with the names of various controls. On the right, you see the form at run time, after the *Go* button has been clicked. When you click the *Go* button, the following actions take place:

- An `int` array is created, initialized with unsorted values.
- The contents of the array are displayed in the `originalListBox` control.
- The array is passed as an argument to the `SelectionSort` method. The method uses the Selection Sort algorithm to sort the array.
- The contents of the array are displayed in the `sortedListBox` control.

Program 7-3 shows the complete code for the *SelectionSort* project.

Figure 7-36 The *SelectionSort* project's form



Program 7-3 Complete code for Form1 in the *SelectionSort* project

```
1 namespace SelectionSort
2 {
3     public partial class Form1 : Form
4     {
5         public Form1()
6         {
7             InitializeComponent();
8         }
9
```

```
10          // The SelectionSort method accepts an int array as an
            argument.
11          // It uses the Selection Sort algorithm to sort the array.
12          private void SelectionSort(int[] iArray)
13          {
14              int minIndex;   // Subscript of smallest value in
                scanned area
15              int minValue;   // Smallest value in the scanned area
16
17              // The outer loop steps through all the array elements,
18              // except the last one. The startScan variable marks the
19              // position where the scan should begin.
20              for (int startScan = 0; startScan < iArray.Length - 1;
                startScan++)
21              {
22                  // Assume the first element in the scannable area
23                  // is the smallest value.
24                  minIndex = startScan;
25                  minValue = iArray[startScan];
26
27                  // Scan the array, starting at the 2nd element in the
28                  // scannable area, looking for the smallest value.
29                  for (int index = startScan + 1; index < iArray.
                    Length; index++)
30                  {
31                      if (iArray[index] < minValue)
32                      {
33                          minValue = iArray[index];
34                          minIndex = index;
35                      }
36                  }
37
38                  // Swap the element with the smallest value with the
39                  // first element in the scannable area.
40                  Swap(ref iArray[minIndex], ref iArray[startScan]);
41              }
42          }
43
44          // The Swap method accepts two integer arguments, by reference,
45          // and swaps their contents.
46          private void Swap(ref int a, ref int b)
47          {
48              int temp = a;
49              a = b;
50              b = temp;
51          }
52
53          private void goButton_Click(object sender, EventArgs e)
54          {
55              // Create an array of integers.
56              int[] numbers = { 4, 6, 1, 3, 5, 2 };
57
58              // Display the array in original order.
59              foreach (int value in numbers)
60              {
61                  originalListBox.Items.Add(value);
62              }
63
64              // Sort the array.
65              SelectionSort(numbers);
66
```

```
67              // Display the array in sorted order.
68              foreach (int value in numbers)
69              {
70                  sortedListBox.Items.Add(value);
71              }
72          }
73
74          private void exitButton_Click(object sender, EventArgs e)
75          {
76              // Close the form.
77              this.Close();
78          }
79      }
80  }
```

The Binary Search Algorithm

Previously in this chapter, we discussed the sequential search algorithm, which uses a loop to step sequentially through an array, starting with the first element. It compares each element with the value being searched for and stops when the value is found or the end of the array is encountered. If the value being searched for is not in the array, the algorithm unsuccessfully searches to the end of the array.

The advantage of the sequential search is its simplicity: It is very easy to understand and implement. Furthermore, it does not require the data in the array to be stored in any particular order. Its disadvantage, however, is its inefficiency. If the array being searched contains 20,000 elements, the algorithm has to look at all 20,000 elements in order to find a value stored in the last element.

In an average case, an item is just as likely to be found near the beginning of an array as near the end. Typically, for an array of *n* items, the sequential search locates an item in *n*/2 attempts. If an array has 50,000 elements, the sequential search makes a comparison with 25,000 of them in a typical case. This is assuming, of course, that the search item is consistently found in the array. (*n*/2 is the average number of comparisons. The maximum number of comparisons is always *n*.)

When the sequential search fails to locate an item, it must make a comparison with every element in the array. As the number of failed search attempts increases, so does the average number of comparisons. Although the sequential search algorithm is adequate for small arrays, it should not be used on large arrays if speed is important.

The **binary search** is a clever algorithm that is much more efficient than the sequential search. Its only requirement is that the values in the array must be sorted in ascending order. Instead of testing the array's first element, this algorithm starts with the element in the middle. If that element happens to contain the desired value, then the search is over. Otherwise, the value in the middle element is either greater than or less than the value being searched for. If it is greater, then the desired value (if it is in the list) will be found somewhere in the first half of the array. If it is less, then the desired value (again, if it is in the list) will be found somewhere in the last half of the array. In either case, half of the array's elements have been eliminated from further searching.

If the desired value is not found in the middle element, the procedure is repeated for the half of the array that potentially contains the value. For instance, if the last half of the array is to be searched, the algorithm tests *its* middle element. If the desired value is not found there, the search is narrowed to the quarter of the array that resides before or after that element. This process continues until the value being searched for is either found or there are no more elements to test.

Here is the pseudocode for a method that performs a binary search on an array:

```
Method BinarySearch(array, searchValue)
   Set first to 0
   Set last to the last subscript in the array
   Set position to –1
   Set found to false

   While found is not true and first is less than or equal to last
      Set middle to the subscript half way between array[first]and array[last]
      If array[middle] equals searchValue
         Set found to true
         Set position to middle
      Else If array[middle] is greater than searchValue
         Set last to middle –1
      Else
         Set first to middle +1
      End If
   End While

   Return position
End Method
```

This algorithm uses three variables to mark positions within the array: *first*, *last*, and *middle*. The *first* and *last* variables mark the boundaries of the portion of the array currently being searched. They are initialized with the subscripts of the array's first and last elements. The subscript of the element halfway between first and last is calculated and stored in the *middle* variable. If the element in the middle of the array does not contain the search value, the *first* or *last* variable is adjusted so that only the top or bottom half of the array is searched during the next iteration. This cuts the portion of the array being searched in half each time the loop fails to locate the search value.

The following C# method performs a binary search on an integer array. The first parameter, `iArray`, is searched for an occurrence of the number stored in `value`. If the number is found, its array subscript is returned. Otherwise, -1 is returned, indicating the value did not appear in the array.

```
 1 private int BinarySearch(int[] iArray, int value)
 2 {
 3     int first = 0;                // First array element
 4     int last = iArray.Length - 1; // Last array element
 5     int middle;                   // Midpoint of search
 6     int position = -1;            // Position of search value
 7     bool found = false;           // Flag
 8
 9     // Search for the value.
10     while (!found && first <= last)
11     {
12         // Calculate the midpoint.
13         middle = (first + last) / 2;
14
15         // If value is found at midpoint ...
16         if (iArray[middle] == value)
17         {
18             found = true;
19             position = middle;
20         }
21         // else if value is in lower half ...
22         else if (iArray[middle] > value)
```

```
23          {
24              last = middle - 1;
25          }
26          // else if value is in upper half ...
27          else
28          {
29              first = middle + 1;
30          }
31      }
32
33      // Return the position of the item, or -1
34      // if it was not found.
35      return position;
36  }
```

If you want to see a complete application that uses the binary search algorithm, look at the *BinarySearch* project, located in the *Chap07* folder of the Student Sample Programs. It loads a list of names from a file into an array and then performs a binary search to find a specific name in the array.

Checkpoint

7.14 What is a search algorithm?

7.15 What is the purpose of a sorting algorithm?

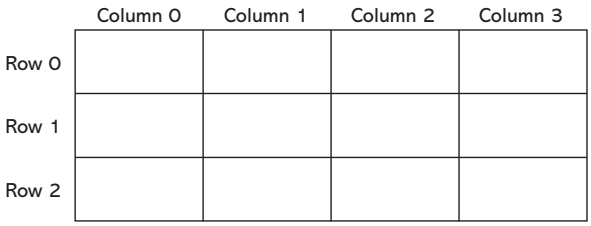
7.16 What is the only requirement of the binary search algorithm?

7.7 Two-Dimensional Arrays

CONCEPT: A two-dimensional array is like several identical arrays put together. It is useful for storing multiple sets of data.

The arrays that you have studied so far are known as one-dimensional arrays. They are called **one-dimensional arrays** because they can hold only one set of data. **Two-dimensional arrays**, which are also called *2D arrays*, can hold multiple sets of data. Think of a two-dimensional array as having rows and columns of elements, as shown in Figure 7-37. This figure shows a two-dimensional array having three rows and four columns. Notice that the rows are numbered 0, 1, and 2, and the columns are numbered 0, 1, 2, and 3. There is a total of 12 elements in the array.

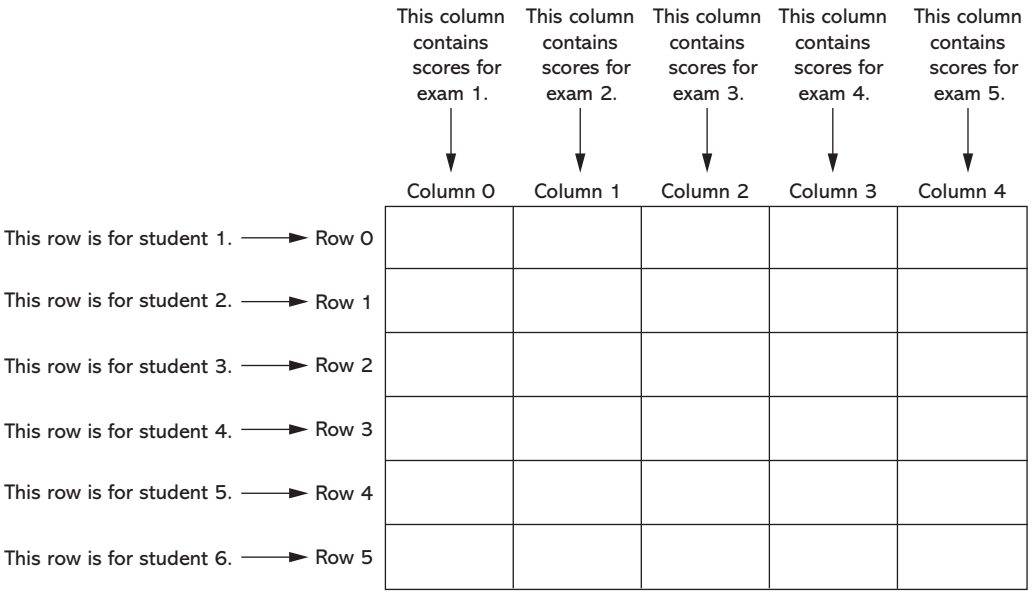
Figure 7-37 A two-dimensional array



Two-dimensional arrays are useful for working with multiple sets of data. For example, suppose you are designing a grade-averaging program for a teacher. The teacher has six students, and each student takes five exams during the semester. One approach would be to create six one-dimensional arrays, one for each student. Each of these arrays would have five elements, one for each exam score. This approach would be cumbersome,

however, because you would have to separately process each of the arrays. A better approach would be to use a two-dimensional array with six rows (one for each student) and five columns (one for each exam score), as shown in Figure 7-38.

Figure 7-38 Two-dimensional array with six rows and five columns



Declaring a Two-Dimensional Array

To declare a two-dimensional array, two size declarators are required: The first one is for the number of rows, and the second one is for the number of columns. Here is an example declaration of a two-dimensional array with three rows and four columns:

```
double[,] scores = new double[3, 4];
```

Notice the comma that appears inside the first set of brackets. This indicates that the `scores` variable references a two-dimensional array. The numbers 3 and 4 are size declarators. The first size declarator specifies the number of rows, and the second size declarator specifies the number of columns. Notice that the size declarators are separated by a comma.

As with one-dimensional arrays, it is best to use named constants as the size declarators. Here is an example:

```
const int ROWS = 3;
const int COLS = 4;
int[,] scores = new int[ROWS, COLS];
```

When processing the data in a two-dimensional array, each element has two subscripts: one for its row and another for its column. In the `scores` array, the elements in row 0 are referenced as follows:

```
scores[0,0]
scores[0,1]
scores[0,2]
scores[0,3]
```

The elements in row 1 are referenced as follows:

```
scores[1,0]
scores[1,1]
scores[1,2]
scores[1,3]
```

And, the elements in row 2 are referenced as follows:

```
scores[2,0]
scores[2,1]
scores[2,2]
scores[2,3]
```

Figure 7-39 illustrates the array with the subscripts shown for each element.

Figure 7-39 Subscripts for each element of the `scores` array

	Column 0	Column 1	Column 2	Column 3
Row 0	scores[0,0]	scores[0,1]	scores[0,2]	scores[0,3]
Row 1	scores[1,0]	scores[1,1]	scores[1,2]	scores[1,3]
Row 2	scores[2,0]	scores[2,1]	scores[2,2]	scores[2,3]

Accessing the Elements in a Two-Dimensional Array

To access one of the elements in a two-dimensional array, you must use two subscripts. For example, suppose we have the following declarations in a program:

```
const int ROWS = 5;
const int COLS = 10;
int[,] values = new int[ROWS, COLS];
```

The following statement assigns the number 95 to `values[2,1]`:

```
values[2,1] = 95;
```

Programs often use nested loops to process two-dimensional arrays. For example, the following code assigns a random number to each element of the `values` array:

```
// Create a Random object.
Random rand = new Random();

// Create a two-dimensional int array.
const int ROWS = 5;
const int COLS = 10;
int[,] values = new int[ROWS, COLS];

// Fill the array with random numbers.
for (int row = 0; row < ROWS; row++)
{
    for (int col = 0; col < COLS; col++)
    {
        values[row, col] = rand.Next(100);
    }
}
```

And the following set of nested loops displays all the elements of the `values` array in a ListBox control named `outputListBox`:

```
// Display the array contents.
for (int row = 0; row < ROWS; row++)
{
    for (int col = 0; col < COLS; col++)
    {
        outputListBox.Items.Add(values[row, col].ToString());
    }
}
```

Implicit Sizing and Initialization of Two-Dimensional Arrays

As with a one-dimensional array, you may provide an initialization list for a two-dimensional array. Recall that when you provide an initialization list for an array, you cannot provide the upper subscript numbers. When initializing a two-dimensional array, you must provide the comma to indicate the number of dimensions. The following is an example of a two-dimensional array declaration with an initialization list:

```
int[,] values = { {1, 2, 3},
                  {4, 5, 6},
                  {7, 8, 9} };
```

Initialization values for each row are enclosed in their own set of braces. In this example, the initialization values for row 0 are `{1, 2, 3}`, the initialization values for row 1 are `{4, 5, 6}`, and the initialization values for row 2 are `{7, 8, 9}`. So, this statement declares an array with three rows and three columns. The values are assigned to the `values` array in the following manner:

```
values[0, 0] is set to 1.
values[0, 1] is set to 2.
values[0, 2] is set to 3.
values[1, 0] is set to 4.
values[1, 1] is set to 5.
values[1, 2] is set to 6.
values[2, 0] is set to 7.
values[2, 1] is set to 8.
values[2, 2] is set to 9.
```

Getting the Length of a Two-Dimensional Array and Its Dimensions

The Length property of a two-dimensional array returns the total number of elements in the array. For example, look at the following code:

```
1 int[,] values = { {1, 2, 3},
2                   {4, 5, 6},
3                   {7, 8, 9} };
4
5 MessageBox.Show("The array has " + values.Length + " elements.");
```

When this code executes, the message box will display *The array has 9 elements*.

Sometimes it's more useful to know the numbers of rows and columns in a two-dimensional array than the total number of elements in the array. For that purpose, arrays have a method named `GetLength`. You pass an integer argument to the method to specify a dimension, and the method returns the length of that dimension. In a two-dimensional array, the rows are considered dimension 0 and the columns are considered dimension 1. To demonstrate, look at the following code snippet:

```
1 int[,] values = { {1, 2, 3},
2                   {4, 5, 6} };
3
4 MessageBox.Show("The array has " + values.GetLength(0) +
5                 " rows and " + values.GetLength(1) +
6                 " columns.");
```

When this code executes, the message box will display *The array has 2 rows and 3 columns*.

The `GetLength` method is useful in loops that step through the elements of a two-dimensional array. For example, the following code uses nested loops to display all the elements of the values array in a ListBox control named `outputListBox`:

```
int[,] values = { {1, 2, 3},
                  {4, 5, 6} };

for (int row = 0; row < values.GetLength(0); row++)
{
    for (int col = 0; col < values.GetLength(1); col++)
    {
        outputListBox.Items.Add(values[row, col].ToString());
    }
}
```

Passing Two-Dimensional Arrays to Methods

When a two-dimensional array is passed to a method, the parameter must be declared as a reference to a two-dimensional array. The following method header shows an example:

```
private void DisplayArray(string[,] array)
```

This method's parameter, `array`, is declared as a reference to a two-dimensional `string` array. Any two-dimensional `string` array can be passed as an argument to the method.

As an example, look at the *Pass2DArray* project, located in the *Chap07* folder of the Student Sample Programs. The project's form has a ListBox named `outputListBox` and a Button named `displayButton`. When you click the button, a two-dimensional array is passed to a method named `DisplayArray`, which displays the contents of the array in the ListBox. Program 7-4 shows the complete code for the application's Form1 form.

Program 7-4 Complete code for Form1 in the *Pass2DArray* project

```
namespace Pass2DArray
{
    public partial class Form1 : Form
    {
        public Form1()
        {
            InitializeComponent();
        }

        private void displayButton_Click(object sender, EventArgs e)
        {
            string[,] names = { { "Amari", "Elena", "Khalid" },
                                { "Sanjay", "Luka", "Olivia" } };

            DisplayArray(names);
        }

        private void DisplayArray(string[,] array)
        {
            for (int row = 0; row < array.GetLength(0); row++)
            {
                for (int col = 0; col < array.GetLength(1); col++)
                {
                    outputListBox.Items.Add(array[row, col]);
                }
            }
```

```
27
28            }
29      }
30 }
```

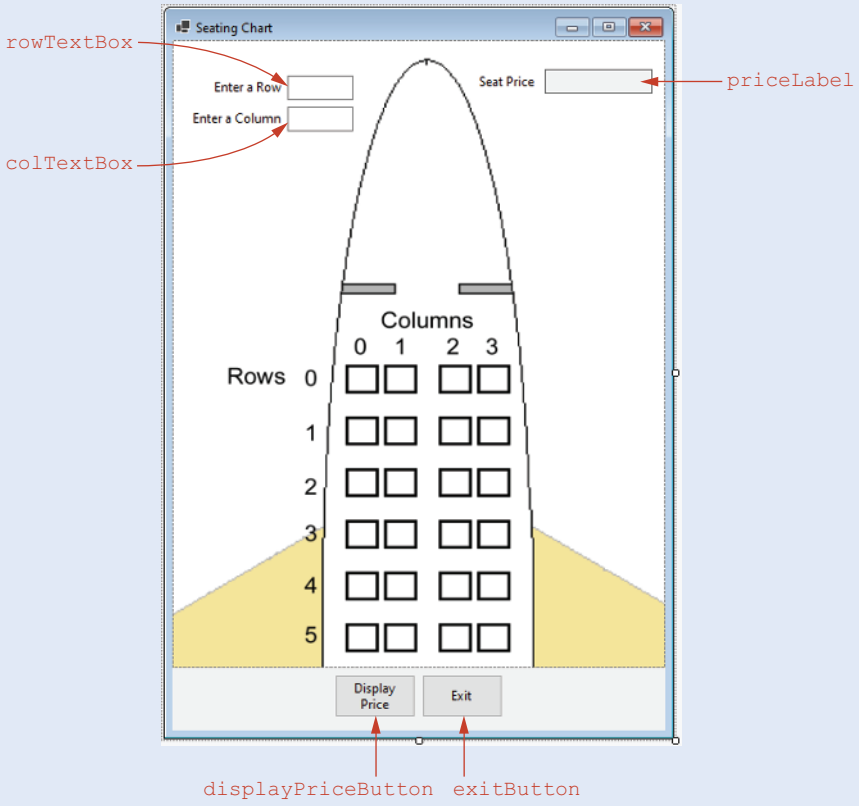
Tutorial 7-3 gives you hands-on practice working with a two-dimensional array.

Tutorial 7-3:
Completing the *SeatingChart* Project

VideoNote
Tutorial 7-3: Completing the *SeatingChart* Project

In this tutorial, you complete the *SeatingChart* project. The application's form, which is shown in Figure 7-40, uses a PictureBox control to display an airplane seating chart that is arranged in rows and columns. When completed, the application allows the user to enter valid row and column numbers in the `rowTextBox` and `colTextBox` text boxes and then click the *Display Price* button. The price of the selected seat will be displayed in the `priceLabel` control. The following table shows the seat prices:

Figure 7-40 The *SeatingChart* project's form



Columns	0	1	2	3
Row 0	$450	$450	$450	$450
Row 1	$425	$425	$425	$425
Row 2	$400	$400	$400	$400
Row 3	$375	$375	$375	$375
Row 4	$375	$375	$375	$375
Row 5	$350	$350	$350	$350

When you write the code for the application, you will create a two-dimensional array to hold these values.

Step 1: Start Visual Studio. Open the project named *SeatingChart* in the *Chap07* folder of this book's Student Sample Programs.

Step 2: Open the Form1 form in the *Designer*. Double-click the `displayPriceButton` control. This opens the code editor, and you will see an empty event handler named `displayPriceButton_Click`. Complete the event handler by typing the code shown in lines 12–68 in Program 7-5. Let's take a closer look at the code:

Line 13: This statement declares two `int` variables, `row` and `col`, to hold the row and column selected by the user.

Lines 16–17: These statements declare `int` constants named `MAX_ROW` and `MAX_COL`, set to the values 5 and 3, respectively. These are used as array size declarators.

Lines 20–26: This statement creates a two-dimensional `decimal` array named `prices`, initialized with the seat prices previously shown.

Line 29: This `if` statement converts the value entered into the `rowTextBox` control to an `int` and stores the result in the `row` variable. If the conversion is successful, the program continues. If the conversion fails, the program jumps to the `else` clause in line 64, and then line 67 displays an error message.

Line 32: This `if` statement converts the value entered into the `colTextBox` control to an `int` and stores the result in the `col` variable. If the conversion is successful, the program continues. If the conversion fails, the program jumps to the `else` clause in line 58, and then line 61 displays an error message.

Line 35: This `if` statement determines whether `row` is in the range of 0 through `MAX_ROW`. If so, the program continues. Otherwise, the program jumps to the `else` clause in line 51, and then lines 54–55 display an error message.

Line 38: This `if` statement determines whether `col` is in the range of 0 through `MAX_COL`. If so, the program continues. Otherwise, the program jumps to the `else` clause in line 44, and then lines 47–48 display an error message.

Lines 41–42: This statement uses `row` and `col` as subscripts to retrieve the selected seat's price from the `prices` array and then displays that value in the `priceLabel` control.

Step 3: Switch your view back to the *Designer* and double-click the `exitButton` control. In the code editor you will see an empty event handler named `exitButton_Click`. Complete the event handler by typing the code shown in lines 73–74 in Program 7-5.

Step 4: Save the project. Then, press F5 on the keyboard or click the *Start Debugging* button ( ) on the toolbar to compile and run the application. When the application runs, experiment by entering row and column numbers for different seats and comparing the displayed price with the table previously shown. When you are finished, click the *Exit* button to end the application.

Program 7-5 Completed code for Form1 in the *SeatingChart* project

```
1 namespace SeatingChart
2 {
3     public partial class Form1 : Form
4     {
```

```
 5         public Form1()
 6         {
 7             InitializeComponent();
 8         }
 9
10         private void displayPriceButton_Click(object sender,
           EventArgs e)
11         {
12             // Variables for the selected row and column
13             int row, col;
14
15             // Constants for the maximum row and column subscripts
16             const int MAX_ROW = 5;
17             const int MAX_COL = 3;
18
19             // Create an array with the seat prices.
20             decimal[,] prices = { {450m, 450m, 450m, 450m},
21                                   {425m, 425m, 425m, 425m},
22                                   {400m, 400m, 400m, 400m},
23                                   {375m, 375m, 375m, 375m},
24                                   {375m, 375m, 375m, 375m},
25                                   {350m, 350m, 350m, 350m}
26                                 };
27
28             // Get the selected row number.
29             if (int.TryParse(rowTextBox.Text, out row))
30             {
31                 // Get the selected column number.
32                 if (int.TryParse(colTextBox.Text, out col))
33                 {
34                     // Make sure the row is within range.
35                     if (row >= 0 && row <= MAX_ROW)
36                     {
37                         // Make sure the column is within range.
38                         if (col >= 0 && col <= MAX_COL)
39                         {
40                             // Display the selected seat's price.
41                             priceLabel.Text =
42                                 prices[row, col].ToString("c");
43                         }
44                         else
45                         {
46                             // Error message for invalid column.
47                             MessageBox.Show("Column must be 0
                               through " +
48                                 MAX_COL);
49                         }
50                     }
51                     else
52                     {
53                         // Error message for invalid row.
54                         MessageBox.Show("Row must be 0 through " +
55                             MAX_ROW);
56                     }
57                 }
58                 else
59                 {
60                     // Display an error message for noninteger
                       column.
```

```
61                      MessageBox.Show("Enter an integer for the
                        column.");
62                  }
63              }
64              else
65              {
66                  // Display an error message for noninteger row.
67                  MessageBox.Show("Enter an integer for the row.");
68              }
69          }
70
71          private void exitButton_Click(object sender, EventArgs
e)
72          {
73              // Close the form.
74              this.Close();
75          }
76      }
77  }
```

Summing All the Elements of a Two-Dimensional Array

To sum all the elements of a two-dimensional array, you can use a pair of nested loops to add the contents of each element to an accumulator. The following code shows an example:

```
1  const int ROWS = 3;
2  const int COLS = 3;
3  int[,] numbers = { {1, 2, 3, 4},
4                     {5, 6, 7, 8},
5                     {9, 10, 11, 12}
6                   };
7
8  int total = 0; // Accumulator, set to 0
9
10 // Sum the array elements.
11 for (int row = 0; row < ROWS; row++)
12 {
13     for (int col = 0; col < COLS; col++)
14     {
15         total += numbers[row, col];
16     }
17 }
18 // Display the sum.
19 MessageBox.Show("The total is " + total);
```

Summing the Rows of a Two-Dimensional Array

Sometimes you may need to calculate the sum of each row in a two-dimensional array. For example, suppose a two-dimensional array is used to hold a set of test scores for a set of students. Each row in the array is a set of test scores for one student. To get the sum of a student's test scores (perhaps so an average may be calculated), you use a loop to add all the elements in one row. The following code shows an example:

```
1  const int ROWS = 3;
2  const int COLS = 3;
```

```
int[,] numbers = { {1, 2, 3, 4},
                   {5, 6, 7, 8},
                   {9, 10, 11, 12}
                 };

int total;  // Accumulator

// Sum each row in the array.
for (int row = 0; row < ROWS; row++)
{
    // Set the accumulator to 0.
    total = 0;

    // Total the row.
    for (int col = 0; col < COLS; col++)
    {
        total += numbers[row, col];
    }

    // Display the row's total.
    MessageBox.Show("The total of row " + row +
                    " is " + total);
}
```

Summing the Columns of a Two-Dimensional Array

Sometimes you may need to calculate the sum of each column in a two-dimensional array. For example, suppose a two-dimensional array is used to hold a set of test scores for a set of students and you wish to calculate the class average for each of the test scores. To do this, you calculate the average of each column in the array. This is accomplished with a set of nested loops. The outer loop controls the column subscript, and the inner loop controls the row subscript. The inner loop calculates the sum of a column, which is stored in an accumulator. The following code demonstrates:

```
const int ROWS = 3;
const int COLS = 4;
int[,] numbers = { {1, 2, 3, 4},
                   {5, 6, 7, 8},
                   {9, 10, 11, 12}
                 };

int total;   // Accumulator

// Sum each column in the array.
for (int col = 0; col < COLS; col++)
{
    // Set the accumulator to 0.
    total = 0;

    // Total the column.
    for (int row = 0; row < ROWS; row++)
    {
        total += numbers[row, col];
    }

    // Display the column's total.
    MessageBox.Show("The total of column " + col +
                    " is " + total);
}
```

Checkpoint

7.17 How many rows and how many columns are in the following array?

```
int[,] values = new decimal[200, 100];
```

7.18 Write a statement that assigns the value 50 to the very last element in the `values` array declared in Checkpoint 7.17.

7.19 Write a declaration for a two-dimensional `int` array initialized with the following table of data:

12	24	32	21	42
99	8	68	32	92
95	34	21	11	7

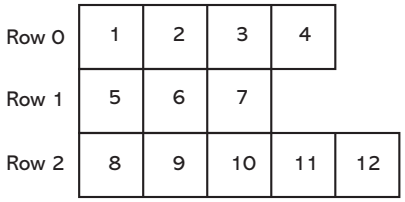
7.8 Jagged Arrays

CONCEPT: A jagged array is similar to a two-dimensional array, but the rows in a jagged array can have different lengths.

In a traditional two-dimensional array, each row has the same number of columns. Mentally, we visualize a two-dimensional array as a rectangular structure. Figure 7-37, previously shown, is an example. For this reason, two-dimensional arrays are sometimes referred to as **rectangular arrays**.

A **jagged array** is similar to a two-dimensional array, but the rows in a jagged array can have different numbers of columns. This is possible because a jagged array is actually an array of arrays. To be more specific, a jagged array is a one-dimensional array, and each element of the array is also a one-dimensional array. Figure 7-41 shows an example. In the figure, row 0 has four columns, row 1 has three columns, and row 2 has five columns.

Figure 7-41 A jagged array



Because a jagged array is an array of arrays, you set it up differently than a two-dimensional array. First you create an array, and then you create each of the arrays that are the elements of the first array. The following code shows an example of how the jagged array in Figure 7-41 might be created and initialized.

```
1  // Create an array of 3 int arrays.
2  int[][] jaggedArray = new int[3][];
3
4  // Create each array that is an element
5  // of the jagged array.
6  jaggedArray[0] = new int[4] { 1, 2, 3, 4 };
7  jaggedArray[1] = new int[3] { 5, 6, 7 };
8  jaggedArray[2] = new int[5] { 8, 9, 10, 11, 12 };
```

Let's take a closer look at the code:

- Line 2 declares an array named `jaggedArray`. Notice that the data type is `int[][]`, with two sets of brackets. This indicates that we are declaring an array of `int` arrays. Also notice that the expression `new int[3][]` uses only one size declarator, specifying the number of rows. The column sizes must be set individually.
- Line 6 creates element 0, which is an `int` array with four columns. The columns are initialized with the values 1, 2, 3, and 4.
- Line 7 creates element 1, which is an `int` array with three columns. The columns are initialized with the values 5, 6, and 7.
- Line 8 creates element 2, which is an `int` array with five columns. The columns are initialized with the values 8, 9, 10, 11, and 12.

To access an item that is stored at a particular row and column in a jagged array, you enclose the row and column subscripts in their own sets of brackets. For example, the following statement displays the value stored at row 1, column 2, of the `jaggedArray` that was previously declared:

```
MessageBox.Show(jaggedArray[1][2].ToString());
```

The following statement shows another example. It assigns the value 99 to row 0, column 3, of `jaggedArray`:

```
jaggedArray[0][3] = 99;
```

A jagged array has a Length property that holds the number of rows, and then each row has its own Length property. You can use a row's Length property to determine the number of columns in that row. For example, the following set of nested loops displays all the values stored in the `jaggedArray` that was previously declared:

```
1 for (int row = 0; row < jaggedArray.Length; row++)
2 {
3     for (int col = 0; col < jaggedArray[row].Length; col++)
4     {
5         MessageBox.Show(jaggedArray[row][col].ToString());
6     }
7 }
```

Checkpoint

7.20 Why are two-dimensional arrays sometimes referred to as rectangular arrays?

7.21 Write a statement that declares a jagged array of `int` values and initialize the columns of each row with the values in the following table of data:

2	4	6		
3	5	7	9	
5	9	11	17	21

7.9 The `List` Collection

CONCEPT: List is a .NET class that is similar to an array. Unlike an array, a List object's size is automatically adjusted to accommodate the number of items being stored in it.

.NET provides a class named **`List`**, which can be used for storing and retrieving items. Once you create a `List` object, you can think of it as a container for holding other objects.

A List object is similar to an array but offers many advantages over an array. Here are a few:

- When you create a List object, you do not have to know the number of items that you intend to store in it.
- A List object automatically expands as items are added to it.
- In addition to adding items to a List, you can remove items as well.
- A List object automatically shrinks as items are removed from it.

Creating a List

Here is an example of how you create a List object that can be used to hold strings:

```
List<string> nameList = new List<string>();
```

This statement creates a List object, referenced by the nameList variable. Notice that in this example the word string is written inside angled brackets <> immediately after the word List. This specifies that the List can hold objects of the string data type. If you try to store any other type of object in this List, an error occurs.

Here is an example of how you create a List object that can be used to hold integers:

```
List<int> numberList = new List<int>();
```

This statement creates a List object, referenced by the numberList variable. Notice that in this example the word int is written inside angled brackets <> immediately after the word List.

Initializing a List

You can optionally initialize a List object when you declare it. Here is an example:

```
List<int> numberList = new List<int>() { 1, 2, 3 };
```

This statement creates a List object that can hold integers and initializes it with the values 1, 2, and 3. Here is an example that creates a List object to hold strings and initializes it with three strings:

```
List<string> nameList = new List<string>() { "Kris",
    "Nina", "Ravi" };
```

Using var to Declare a List

If you are declaring and initializing a List inside of a method, you can use the var keyword in your declaration statement. The following code shows an example:

```
var numberList = new List<int>();
```

The compiler will determine that the data type for numberList is List<int> because we are initializing it with a List<int> object. It is also obvious to any person reading this statement that numberList references a List<int> object.

Here are some other examples:

```
var amounts = new List<double>();

var prices = new List<decimal>();

var nameList = new List<string>() { "Kris", "Nina", "Ravi" };
```

As you can see, the var keyword simplifies and "declutters" the appearance of a List declaration. For that reason, we will use it for List declarations whenever we can.

Adding Items to a List

To add items to an existing List object, you use the Add method. For example, the following statements create a List object and add a series of strings to it:

```
var nameList = new List<string>();
nameList.Add("Kris");
nameList.Add("Nina");
nameList.Add("Ravi");
```

After these statements execute, the nameList object will hold the three strings "Kris", "Nina", and "Ravi".

The items that are stored in a List have a corresponding index. The index specifies the item's location in the List, so it is much like an array subscript. The first item that is added to a List is stored at index 0. The next item that is added to the List is stored at index 1, and so forth. After the previously shown statements execute, "Kris" is stored at index 0, "Nina" is stored at index 1, and "Ravi" is stored at index 2.

The Count Property

A List object has a Count property that holds the number of items stored in the List. For example, the following statement uses the Count property to display the number of items stored in nameList:

```
MessageBox.Show("The List has " + nameList.Count +
                " objects stored in it.");
```

Assuming that nameList holds the strings "Kris", "Nina", and "Ravi", the following statement will be displayed in a message box:

```
The List has 3 objects stored in it.
```

Accessing Items in a List

You can use subscript notation to access the items in a List, just as you can with an array. For example, the following for loop displays the items in the nameList object:

```
for (int index = 0; index < nameList.Count; index++)
{
    MessageBox.Show(nameList[index]);
}
```

Notice that the loop uses the List object's Count property in the test expression to control the number of iterations. Here is an example that reads values from a text file and adds them to a List:

```
 1 // Open the Names.txt file.
 2 StreamReader inputFile = File.OpenText("Names.txt");
 3
 4 // Create a List object to hold strings.
 5 var nameList = new List<string>();
 6
 7 // Read the file's contents.
 8 while (!inputFile.EndOfStream)
 9 {
10     // Read a line and add it to the List.
11     nameList.Add(inputFile.ReadLine());
12 }
```

Let's take a closer look at this code:

- Line 2 opens a file named Names.txt and associates it with a `StreamReader` object that is referenced by the `inputFile` variable.
- Line 5 creates a `List` object, referenced by the `nameList` variable. The object can hold strings.
- The `while` loop that starts in line 8 iterates until the end of the file is reached.
- The statement in line 11 reads a line from the file and adds it to the `nameList` object.

After this code executes, the `nameList` object contains all the lines that were read from the Names.txt file.

You can also use the `foreach` loop to iterate over the items in a `List`, just as you can with an array. Here is an example:

```
foreach (string str in nameList)
{
    MessageBox.Show(str);
}
```

Passing a `List` to a Method

Sometimes you will want to write a method that accepts a `List` as an argument and performs an operation on the `List`. For example, the following code shows a method named `DisplayList`. The method accepts a `List` of strings as an argument and displays each item in `List`.

```
1 private void DisplayList(List<string> sList)
2 {
3     foreach (string str in sList)
4     {
5         MessageBox.Show(str);
6     }
7 }
```

Notice in line 1 that the method has a parameter variable named `sList` and that the parameter's data type is `List<string>`. The parameter variable is a reference to a `List<string>` object. When you call this method, you must pass a `List<string>` object as an argument.

When you call a method and pass a `List<string>` object as an argument, you simply pass the variable that references the `List`. The following code shows an example of how the `DisplayList` method (previously shown) might be called:

```
1 // Create a List of strings.
2 var nameList = new List<string>() { "Kris",
3     "Nina", "Ravi" };
4
5 // Pass the List to the DisplayList method.
6 DisplayList(nameList);
```

The statement in lines 2 and 3 creates a `List` containing the strings `"Kris"`, `"Nina"`, and `"Ravi"`. Line 6 calls the `DisplayList` method, passing the `nameList` object as an argument.

NOTE: `List` objects, like arrays, are always passed by reference.

Removing Items from a List

You can use the RemoveAt method to remove an item at a specific index in a List. The following code shows an example:

```
1 // Create a List of strings.
2 var nameList = new List<string>() { "Kris",
3     "Nina", "Ravi" };
4
5 // Remove the item at index 0.
6 nameList.RemoveAt(0);
```

The statement in lines 2 and 3 creates a List containing the strings "Kris", "Nina", and "Ravi". Then, the statement in line 6 removes the string at index 0. After this statement executes, the List contains the strings "Nina" and "Ravi".

If you know the value of the item that you want to remove from a List, but you do not know the item's index, you can use the Remove method. You pass the item that you want to remove as an argument, and the Remove method searches for that item in the List. If the item is found, it is removed. Here is an example:

```
1 // Create a List of strings.
2 var nameList = new List<string>() { "Kris",
3     "Nina", "Ravi" };
4
5 // Remove "Ravi" from the List.
6 nameList.Remove("Ravi");
```

The statement in lines 2 and 3 creates a List containing the strings "Kris", "Nina", and "Ravi". Then, the statement in line 6 removes "Ravi" from the List. After this statement executes, the List contains the strings "Kris" and "Nina".

The Remove method returns a Boolean value indicating whether the item was actually removed from the List. If the specified item was found in the List and removed, the Remove method returns true. If the item was not found in the List, the Remove method returns false. The following code demonstrates how you can use the value returned from the method:

```
1 // Create a List of strings.
2 var nameList = new List<string>() { "Kris",
3     "Nina", "Ravi" };
4
5 // Remove "Zaria".
6 if (!nameList.Remove("Zaria"))
7 {
8     MessageBox.Show("Zaria was not found.");
9 }
```

The statement in lines 2 and 3 creates a List containing the strings "Kris", "Nina", and "Ravi". Then, the statement in line 6 attempts to remove "Zaria" from the List. The List does not contain the string "Zaria", so the Remove method returns false. The message "Zaria was not found" is displayed. After this code executes, the List still contains the strings "Kris", "Nina", and "Ravi".

NOTE: The Remove method performs a sequential search to locate the specified item. If the List contains a large number of items, its performance will be slow.

If you want to remove all the items from a `List`, you can call the `Clear` method. Here is an example:

```
nameList.Clear();
```

After this statement executes, the `nameList` object is empty.

Inserting an Item

You can use the `Insert` method to insert an item at a specific index in a `List`. The following code shows an example:

```
1 // Create a List of strings.
2 var nameList = new List<string>() { "Kris",
3     "Nina", "Ravi" };
4
5 // Insert an item at index 0.
6 nameList.Insert("Joanne", 0);
```

The statement in lines 2 and 3 creates a `List` containing the strings `"Kris"`, `"Nina"`, and `"Ravi"`. Then, the statement in line 6 inserts the string `"Joanne"` at index 0. After this statement executes, the `List` contains the strings `"Joanne"`, `"Kris"`, `"Nina"`, and `"Ravi"`.

NOTE: An exception will occur if the specified index is less than 0 or greater than the `List` object's Count property.

Searching for Items in a List

Because you can use subscript notation to access the items in a `List`, you can adapt any of the array-processing algorithms that you saw earlier in this chapter so they work with a `List`. For example, you can write code that performs a sequential search, binary search, selection sort, and so forth, on a `List`.

An easy way to search for item in a `List`, however, is to use the `IndexOf` method. The `IndexOf` method accepts a value as an argument, and it searches for that value in the `List`. If the value is found, the method returns its index. If the value is not found, the method returns −1. The following code shows an example:

```
 1 // Create a List of strings.
 2 var nameList = new List<string>() { "Kris",
 3     "Nina", "Ravi" };
 4
 5 // Search for "Nina".
 6 int position = nameList.IndexOf("Nina");
 7
 8 // Was Nina found in the List
 9 if (position != -1)
10 {
11     MessageBox.Show("Nina was found at index " +
12                     position);
13 }
14 else
15 {
16     MessageBox.Show("Nina was not found.");
17 }
```

The statement in lines 2 and 3 creates a `List` containing the strings `"Kris"`, `"Nina"`, and `"Ravi"`. The statement in line 6 calls the `IndexOf` method to search for `"Nina"` in the `List`. The value that is returned from the method is assigned to the `position` variable.

After this statement executes, the `position` variable contains the index of `"Nina"` or −1 if `"Nina"` was not found in the `List`. The `if` statement in lines 9–17 displays one of two possible messages, depending on whether `"Nina"` was found. (If this code were executed, it would display the message "`Nina` was found at index 1".)

There are two additional versions of the `IndexOf` method that allow you to specify the area of the List that should be searched. The following statement shows an example of one of these:

```
position = nameList.IndexOf("Bianca", 2);
```

Notice that two arguments are passed to the `IndexOf` method. The first argument, `"Bianca"`, is the item to search for. The second argument, 2 is the starting index of the search. This specifies that the search should begin at index 2 and end at the last item in the `List`. (The beginning index is included in the search. If you pass an invalid index as an argument, an exception occurs.)

Here is an example of another version of the `IndexOf` method:

```
position = nameList.IndexOf("Bianca", 2, 5);
```

In this example, three arguments are passed to the `IndexOf` method. The first argument, `"Bianca"`, is the item to search for. The second argument, 2 is the starting index of the search. The third argument, 5, is the ending index of the search. This specifies that the search should begin at index 2, and end at index 5. (The beginning and ending indices are included in the search. If either index is invalid, an exception occurs.)

NOTE: The `IndexOf` method performs a sequential search to locate the specified item. If the `List` contains a large number of items, its performance will be slow.

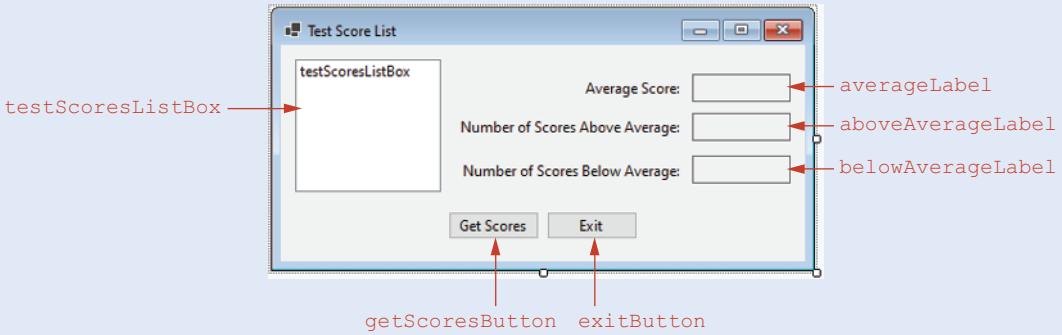
In Tutorial 7-4, you will complete an application that reads the contents of a file into a List, and then performs various operations on the `List`.

Tutorial 7-4:
Completing the *TestScoreList* Project

VideoNote
Tutorial 7-4: Completing the *TestScoreList* Project

In this tutorial, you complete the *TestScoreList* project. The application's form, which is shown in Figure 7-42, has already been created for you. When you complete the application, it will read a set of test scores from a file into a `List`. (The file has also been created for you.) The test scores are displayed in the ListBox control. The average test score is calculated and displayed, as well as the number of above-average test scores and below-average test scores.

Figure 7-42 The *TestScoreList* project's form



Step 1: Start Visual Studio. Open the project named *TestScoreList* in the *Chap07* folder of the Student Sample Programs.

Step 2: Open the Form1 form's code in the code editor. Type the comments and code for the `ReadScores` method, shown in lines 10–33 of Program 7-6. The purpose of the `ReadScores` method is to accept a `List<int>` object as an argument and read the contents of the TestScores.txt file into the list.

Step 3: Type the comments and code for the `DisplayScores` method, shown in lines 35–43 of Program 7-6. The purpose of the `DisplayScores` method is to accept a `List<int>` object as an argument and display its contents in the `testScoresListBox` control.

Step 4: Type the comments and code for the `Average` method, shown in lines 45–63 of Program 7-6. The purpose of the `Average` method is to accept a `List<int>` object as an argument and return the average of the values in the `List`.

Step 5: Type the comments and code for the `AboveAverage` method, shown in lines 65–85 of Program 7-6. The purpose of the `AboveAverage` method is to accept a `List<int>` object as an argument and return the number of above average scores it contains.

Step 6: Type the comments and code for the `BelowAverage` method, shown in lines 87–107 of Program 7-6. The purpose of the `BelowAverage` method is to accept a `List<int>` object as an argument and return the number of below average scores it contains.

Step 7: Next, you create the Click event handlers for the Button controls. Switch back to the *Designer* and double-click the `getScoresButton` control. This opens the code editor, and you will see an empty event handler named `getScoresButton_Click`. Complete the `getScoresButton_Click` event handler by typing the code shown in lines 111–134 in Program 7-6. Let's review this code:

Lines 111–113: These statements declare the following variables:

- `averageScore`—This variable is used to hold the average test score.
- `numAboveAverage`—This variable is used to hold the number of above-average test scores.
- `numBelowAverage`—This variable is used to hold the number of below-average test scores.

Line 116: This statement creates a `List<int>` object, referenced by the `scoresList` variable.

Line 119: This statement calls the `ReadScores` method, passing the `scoresList` object as an argument. After this statement executes, the `scoresList` object contains the test scores that are in the TestScores.txt file.

Line 122: This statement calls the `DisplayScores` method, passing the `scoresList` object as an argument. After this statement executes, the items in the `scoresList` object are displayed in the `testScoresListBox` control.

Line 125: This statement calls the `Average` method, passing the `scoresList` object as an argument. The method returns the average of the values in the `scoresList` object, which is assigned to the `averageScore` variable.

Line 126: This statement displays the average score in the `averageLabel` control.

Line 129: This statement calls the `AboveAverage` method, passing the `scoresList` object as an argument. The method returns the number of above-average scores in the `scoresList` object, which is assigned to the `numAboveAverage` variable.

Line 130: This statement displays the number of above-average scores in the `aboveAverageLabel` control.

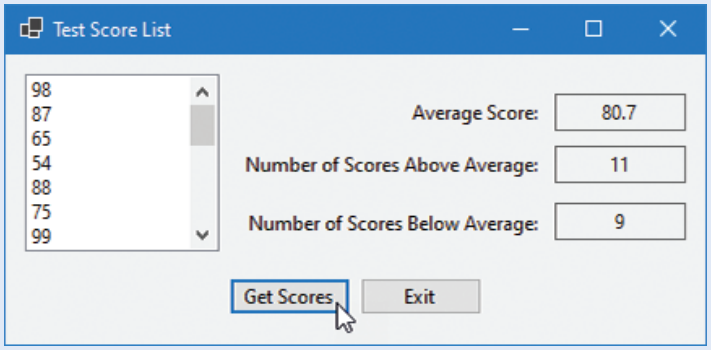
Line 133: This statement calls the `BelowAverage` method, passing the `scoresList` object as an argument. The method returns the number of below-average scores in the `scoresList` object, which is assigned to the `numBelowAverage` variable.

Line 134: This statement displays the number of below-average scores in the `belowAverageLabel` control.

Step 8: Switch your view back to the *Designer* and double-click the `exitButton` control. In the code editor you will see an empty event handler named `exitButton_Click`. Complete the event handler by typing the code shown in lines 150–151 in Program 7-6.

Step 9: Save the project. Then, press F5 on the keyboard or click the *Start Debugging* button (▶) on the toolbar to compile and run the application. When the application runs, click the *Get Scores* button. This should display a set of test scores in the ListBox, as well as the average score, the number of above-average scores, and the number of below-average scores, as shown in Figure 7-43. Click the *Exit* button to exit the application.

Figure 7-43 The *TestScoreList* project's form



Program 7-6 Completed code for Form1 in the *TestScoreList* project

```
1  namespace TestScoreList
2  {
3      public partial class Form1 : Form
4      {
5          public Form1()
6          {
7              InitializeComponent();
8          }
9
10         // The ReadScores method reads the scores from the
11         // TestScores.txt file into the scoresList parameter.
12         private void ReadScores(List<int> scoresList)
```

```
13          {
14              try
15              {
16                  // Open the TestScores.txt file.
17                  StreamReader inputFile = File.
                    OpenText("TestScores.txt");
18
19                  // Read the scores into the list.
20                  while (!inputFile.EndOfStream)
21                  {
22                      scoresList.Add(int.Parse(inputFile.
                        ReadLine()));
23                  }
24
25                  // Close the file.
26                  inputFile.Close();
27              }
28              catch (Exception ex)
29              {
30                  // Display an error message.
31                  MessageBox.Show(ex.Message);
32              }
33          }
34
35          // The DisplayScores method displays the contents of the
36          // scoresList parameter in the ListBox control.
37          private void DisplayScores(List<int> scoresList)
38          {
39              foreach (int score in scoresList)
40              {
41                  testScoresListBox.Items.Add(score);
42              }
43          }
44
45          // The Average method returns the average of the values
46          // in the scoresList parameter.
47          private double Average(List<int> scoresList)
48          {
49              int total = 0;      // Accumulator
50              double average;     // To hold the average
51
52              // Calculate the total of the scores.
53              foreach (int score in scoresList)
54              {
55                  total += score;
56              }
57
58              // Calculate the average of the scores.
59              average = (double)total / scoresList.Count;
60
61              // Return the average.
62              return average;
63          }
64
65          // The AboveAverage method returns the number of
66          // above average scores in scoresList.
67          private int AboveAverage(List<int> scoresList)
68          {
69              int numAbove = 0;       // Accumulator
70
```

```
71              // Get the average score.
72              double avg = Average(scoresList);
73
74              // Count the number of above average scores.
75              foreach (int score in scoresList)
76              {
77                  if (score > avg)
78                  {
79                      numAbove++;
80                  }
81              }
82
83              // Return the number of above average scores.
84              return numAbove;
85          }
86
87          // The BelowAverage method returns the number of
88          // below average scores in scoresList.
89          private int BelowAverage(List<int> scoresList)
90          {
91              int numBelow = 0;       // Accumulator
92
93              // Get the average score.
94              double avg = Average(scoresList);
95
96              // Count the number of below average scores.
97              foreach (int score in scoresList)
98              {
99                  if (score < avg)
100                 {
101                     numBelow++;
102                 }
103             }
104
105             // Return the number of below average scores.
106             return numBelow;
107         }
108
109         private void getScoresButton_Click(object sender,
            EventArgs e)
110         {
111             double averageScore;    // To hold the average score
112             int numAboveAverage;    // Number of above average
                scores
113             int numBelowAverage;    // Number of below average
                scores
114
115             // Create a List to hold the scores.
116             var scoresList = new List<int>();
117
118             // Read the scores from the file into the List.
119             ReadScores(scoresList);
120
121             // Display the scores.
122             DisplayScores(scoresList);
123
124             // Display the average score.
125             averageScore = Average(scoresList);
126             averageLabel.Text = averageScore.ToString("n1");
127
```

```
            // Display the number of above average scores.
            numAboveAverage = AboveAverage(scoresList);
            aboveAverageLabel.Text = numAboveAverage.ToString();

            // Display the number of below average scores.
            numBelowAverage = BelowAverage(scoresList);
            belowAverageLabel.Text = numBelowAverage.ToString();
        }

        private void exitButton_Click(object sender, EventArgs
e)
        {
            // Close the form.
            this.Close();
        }
    }
}
```

Checkpoint

7.22 Write a statement that initializes a `List` with 4 values of the `double` data type.

7.23 Write a statement that adds a new value to the `List` object created in Checkpoint 7.22.

7.24 Write a statement that clears the contents of the `List` object created in Checkpoint 7.22.

7.25 Is it possible to write code that performs a sequential search, binary search, selection sort, and so forth, on a `List`? Why or why not?

7.10 The ImageList Control

CONCEPT: The ImageList control allows you to store a collection of images. At run time, you can retrieve an image from an ImageList control and display it in a PictureBox control.

The **ImageList control** is a container that can hold multiple images. As its name implies, it is a list of images. You can use an index to retrieve an image from an ImageList control and display the image in a PictureBox control.

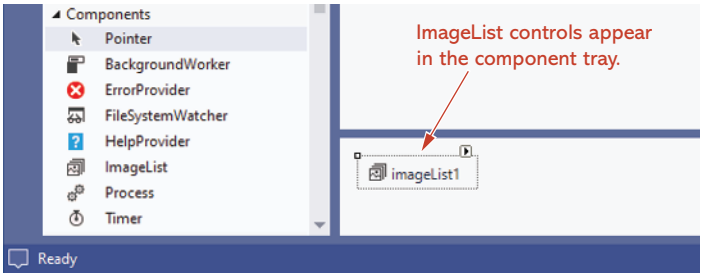
There are a few guidelines that you should follow as you plan to use an ImageList control in an application:

- All the images stored in an ImageList control should be the same size.
- The images stored in an ImageList control can be no more than 256 by 256 pixels in size.
- All the images stored in an ImageList control should be in the same format (.bmp, .jpg, etc.)

Although these guidelines might seem restrictive, keep in mind that ImageList controls are designed to store small images such as icons or thumbnails. They also work well in game programs that display images such as cards.

You will find the ImageList control in the *Components* section of the *Toolbox*. When you double-click the ImageList tool in the *Toolbox*, an ImageList control is created in the component tray area, at the bottom of the *Designer*. (The ImageList control does not appear on the form.) Figure 7-44 shows an example. When you create ImageList controls, they are given default names such as `imageList1`, `imageList2`, and so forth.

Figure 7-44 An ImageList control



After you create an ImageList control, you should set its ImageSize property to the size of the images that you plan to store in the control. The default size is 16,16. You can set the ImageSize property to any value from 0,0 through 256,256. (If the value of the ImageSize property does not match the size of the images that are stored in the control, the images will appear distorted when you display them.)

Then, you can use the Images property to add images to the control. In the *Properties* window, click the ellipses button (...) that appears next to the Images property. This displays the *Images Collection Editor* window shown in Figure 7-45. Click the *Add* button, and an *Open* dialog box appears. Use the dialog box to locate and select the image file (or multiple image files) that you want to add to the ImageList control.

Figure 7-45 The *Images Collection Editor* window

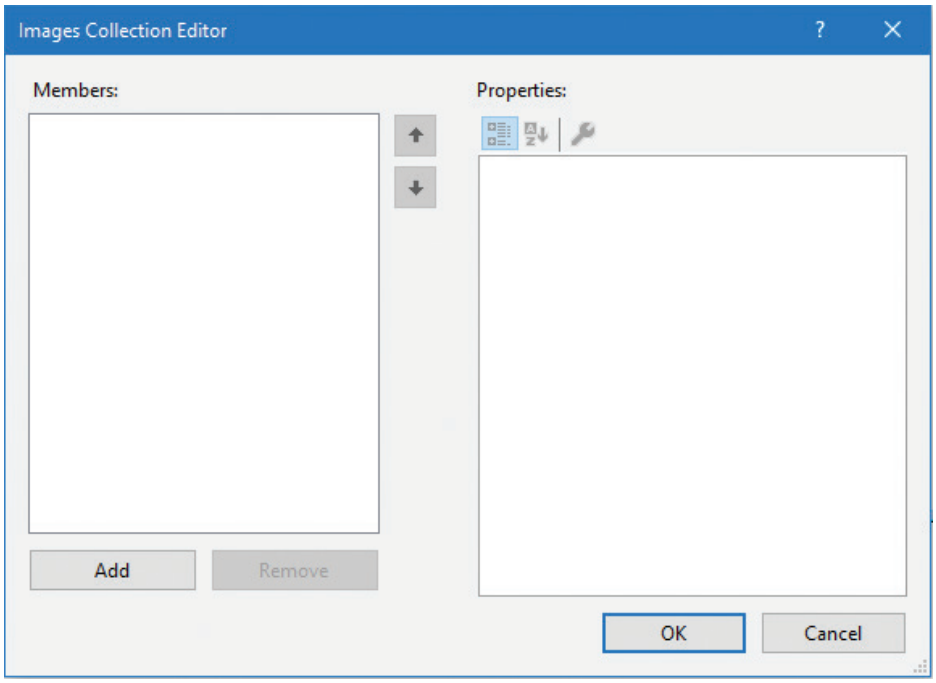
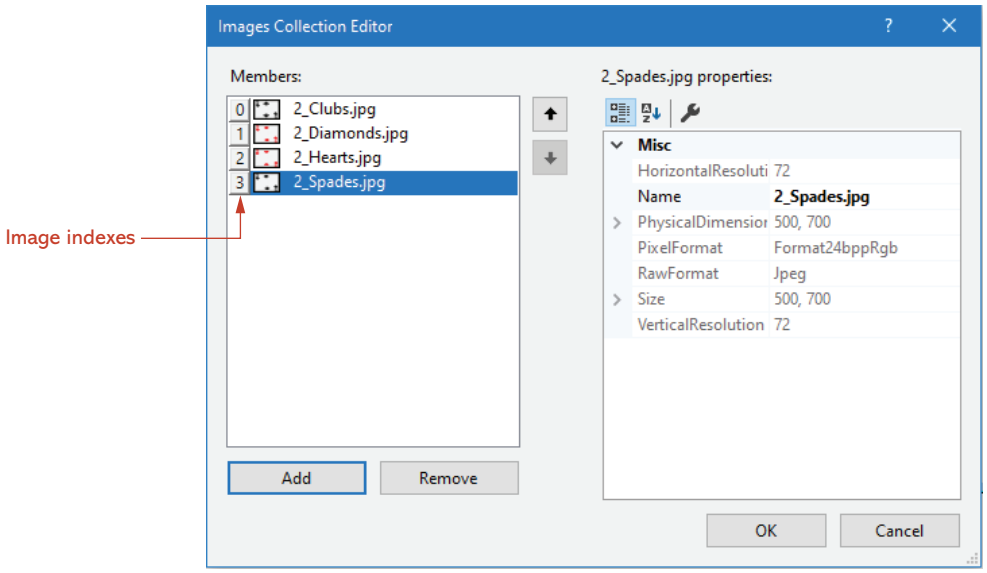


Figure 7-46 shows the *Images Collection Editor* window after four images have been added. The names of the images, which are shown in the *Members* list, are 2_Clubs.bmp, 2_Diamonds.bmp, 2_Hearts.bmp, and 2_Spades.bmp. Notice that an index value appears

next to each image's name. You will use the index value later to retrieve images from the control. When you have added all the images that you want, click the *OK* button.

Figure 7-46 Four images added



If you know the index value for a particular image, you can retrieve that image from the ImageList control and display it in a PictureBox. The following code shows an example. Assume `myImageList` is the name of an ImageList control and `myPictureBox` is the name of a PictureBox control.

```
myPictureBox.Image = myImageList.Images[5];
```

This statement gets the image at index 5 from `myImageList` and assigns it to the `myPictureBox` control's Image property. As a result, the image is displayed in the PictureBox.

In code, you can determine the number of images that are stored in an ImageList control by getting the value of the control's `Images.Count` property. The following code shows an example. Assume `myImageList` is the name of an ImageList control and `numberImages` is the name of an `int` variable.

```
numberImages = myImageList.Images.Count;
```

In Tutorial 7-5, you complete an application that randomly selects images from an ImageList control and displays them in a PictureBox.

Tutorial 7-5:
Completing the *RandomCard* Project

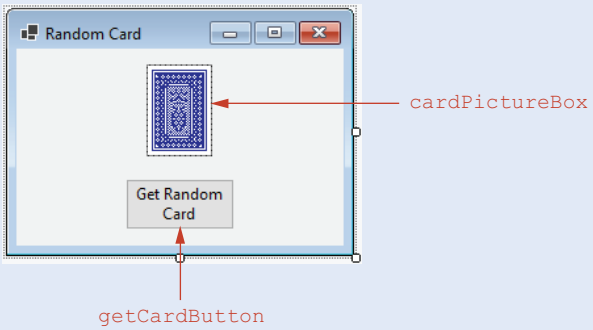
VideoNote
Tutorial 7-5: Completing the *RandomCard* Project

In this tutorial, you complete the *RandomCard* project, which randomly displays images of poker cards selected from an ImageList control. The application's form, which has already been created for you, is shown in Figure 7-47. Notice that the PictureBox control is already displaying the image of a card's back.

Step 1: Start Visual Studio. Open the project named *RandomCard* in the *Chap08* folder of the Student Sample Programs.

Step 2: Make sure the Form1 form is open in the *Designer*. Scroll down in the *Toolbox* until you see the *Components* section, and double-click the *ImageList*

Figure 7-47 The *RandomCard* project's form



tool. This should create an ImageList control named `imageList1`, which you should see in the component tray.

Step 3: Change the name of the `imageList1` control to `cardImageList`.

Step 4: Each of the images that you will add to the `cardImageList` control are 50 by 70 pixels in size. Change the `cardImageList` control's ImageSize property to 50, 70.

Step 5: Next, you add images to the `cardImageList` control. The images that you add are in the *Images/Cards/Poker Small* folder of the Student Sample Programs. In the *Properties* window, click the ellipses button (...) that appears next to the Images property. This displays the *Images Collection Editor* window.

Click the *Add* button, and an *Open* dialog box appears. Use the dialog box to select as many of the card image files as you want to add to the ImageList control. (You can use the Shift or Ctrl key while clicking filenames to select multiple files.) When you have selected all the images that you want to add, click the *OK* button.

Step 6: Now you create the Click event handler for the `getCardButton` control. Double-click the `getCardButton` control in the *Designer*. This opens the code editor, and you will see an empty event handler named `getCardButton_Click`. Complete the event handler by typing the code shown in lines 12–19 in Program 7-7. Let's review this code.

Line 13: This statement creates a `Random` object, referenced by the `rand` variable.

Line 16: This statement declares an `int` variable named `index`. The variable is initialized with a random number that is returned from the `rand.Next` method. Notice that `cardImageList.Images.Count` is passed as an argument to the method. As a result, we get a random number in the range of 0 up to, but not including, the number of images in the control.

Line 19: This statement uses the `index` variable to retrieve an image from the `cardImageList` control and assigns it to the `cardPictureBox` control's Image property. As a result, the image is displayed in the PictureBox.

Step 7: Save the project. Then, press F5 on the keyboard or click the *Start Debugging* button ( ) on the toolbar to compile and run the application. When the application runs, click the *Get Random Card* button several times to see the images that are randomly selected. When you are finished, click the *Exit* button to exit the application.

Program 7-7 Completed code for Form1 in the *RandomCard* project

```
 1 namespace RandomCard
 2 {
 3     public partial class Form1 : Form
 4     {
 5         public Form1()
 6         {
 7             InitializeComponent();
 8         }
 9
10         private void getCardButton_Click(object sender, EventArgs e)
11         {
12             // Create a Random object.
13             Random rand = new Random();
14
15             // Get a random index.
16             int index = rand.Next(cardImageList.Images.Count);
17
18             // Display a card.
19             cardPictureBox.Image = cardImageList.Images[index];
20         }
21     }
22 }
```

Checkpoint

7.26 How you can determine the number of images that are stored in an ImageList control?

7.27 Under which section is the ImageList control located in the *Toolbox*?

7.28 There are a few guidelines that you should follow as you plan to use an ImageList control in an application. What are they?

7.29 How do you add images to an ImageList control?

7.30 What value determines which image from the ImageList will be displayed in a PictureBox?

Key Terms

array
binary search
elements
`foreach` loop
garbage collection
ImageList control
initialization list
iteration variable
jagged array
`List`
`new` operator
off-by-one error
one-dimensional arrays
rectangular arrays
reference
reference copy
reference types
reference variable
search algorithms
selection sort
sequential search algorithm
size declarator
subscript
two-dimensional arrays
value types

Review Questions

Multiple Choice

1. The memory that is allocated for a _____ variable is the actual location that will hold any value that is assigned to that variable.
   a. reference type
   b. general type
   c. value type
   d. framework type

2. A variable that is used to reference an object is commonly called a(n) _____.
   a. reference variable
   b. resource variable
   c. object variable
   d. component variable

3. When you want to work with an object, you use a variable that holds a special value known as a(n) _____ to link the variable to the object.
   a. union
   b. reference
   c. object linker
   d. data coupling

4. The _____creates an object in memory and returns a reference to that object.
   a. = operator
   b. object allocator
   c. reference variable
   d. `new` operator

5. A(n) _____ is an object that can hold a group of values that are all of the same data type.
   a. array
   b. collection
   c. container
   d. set

6. The _____ indicates the number of values that the array should be able to hold.
   a. allocation limit
   b. size declarator
   c. data type
   d. compiler

7. The storage locations in an array are known as _____.
   a. elements
   b. sectors
   c. pages
   d. blocks

8. Each element in an array is assigned a unique number known as a(n) _____.
   a. element identifier
   b. subscript
   c. index
   d. sequencer

9. When you create an array, you can optionally initialize it with a group of values called a(n) _____.

a. default value group
b. initialization list
c. defined set
d. value list

10. In C#, all arrays have a _____ that is set to the number of elements in the array.

a. Limit property
b. Size property
c. Length property
d. Maximum property

11. A(n) _____ occurs when a loop iterates one time too many or one time too few.

a. general error
b. logic error
c. loop count error
d. off-by-one error

12. C# provides a special loop that, in many circumstances, simplifies array processing. It is known as the _____.

a. `for` loop
b. `foreach` loop
c. `while` loop
d. `do-while` loop

13. The `foreach` loop is designed to work with a temporary, read-only variable that is known as the _____.

a. element variable
b. loop variable
c. index variable
d. iteration variable

14. _____ is a process that periodically runs, removing all unreferenced objects from memory.

a. Systematic reallocation
b. Memory cleanup
c. Garbage collection
d. Object maintenance

15. Various techniques known as _____ have been developed to locate a specific item in a larger collection of data, such as an array.

a. seek functions
b. request methods
c. traversal procedures
d. search algorithms

16. The _____ uses a loop to step through an array, starting with the first element, searching for an item.

a. sequential search algorithm
b. top–down method
c. ascending search algorithm
d. basic search function

17. A(n) _____ is a type of assignment operation that copies a reference to an array and not the contents of the array.
    a. object copy
    b. reference copy
    c. double reference
    d. parallel copy

18. The _____ is a clever algorithm that is much more efficient than the sequential search.
    a. linear search
    b. bubble sort
    c. binary search
    d. selection sort

19. A _____ is similar to a two-dimensional array, but the rows can have different numbers of columns.
    a. one-dimensional array
    b. columnar array
    c. jagged array
    d. split row array

20. .NET provides a class named _____, which can be used for storing and retrieving items.
    a. `Matrix`
    b. `Database`
    c. `Container`
    d. `List`

21. The _____ is a container that can hold multiple images.
    a. ImageList control
    b. GroupBox control
    c. PictureBox control
    d. ComboBox control

True or False

1. When you are working with a value type, you are using a variable that holds a piece of data.
2. Reference variables can be used only to reference objects.
3. Individual variables are well suited for storing and processing lists of data.
4. Arrays are reference type objects.
5. You can store a mixture of data types in an array.
6. When you create a numeric array in C#, its elements are set to the value 0 by default.
7. The subscript of the last element will always be one less than the array's Length property.
8. You use the == operator to compare two array reference variables and determine whether the arrays are equal.
9. A jagged array is similar to a two-dimensional array, but the rows in a jagged array can have different numbers of columns.
10. When you create a `List` object, you do not have to know the number of items that you intend to store in it.
11. ImageList controls are designed to store small images such as icons or thumbnails.

Short Answer

1. How much memory is allocated by the compiler when you declare a value type variable?
2. What type of variable is needed to work with an object in code?
3. What two steps are typically required for creating a reference type object?
4. Are variables well suited for processing lists of data? Why or why not?
5. What value is returned by the Length property of an array?
6. What can cause an off-by-one error when working with an array?
7. How do you keep track of elements that contain data in a partially filled array?
8. Briefly describe the selection sort algorithm.
9. How is the binary search more efficient that the sequential search algorithm?
10. What advantages does a `List` have over an array?
11. What is the result if the value of the ImageSize property does not match the size of the images that are stored in the ImageList control?

Algorithm Workbench

1. Assume `names` is a variable that references an array of 20 `string` values. Write a `foreach` loop that displays each of the elements of the array in a ListBox control.
2. The variables `numberArray1` and `numberArray2` reference arrays that have 100 elements each. Write code that copies the values from `numberArray1` to `numberArray2`.
3. Write code for a sequential search that determines whether the value `-1` is stored in an array with a reference variable named `values`. The code should display a message indicating whether the value was found.
4. Write a declaration statement that creates a two-dimensional array referenced by a variable named `grades`. The array should store `int` values using 18 rows and 12 columns.
5. Write code that sums each column in the array in Question 4.
6. Create a `List` object that uses the binary search algorithm to search for the string `"A"`. Display a message box indicating whether the value was found.
7. Write a statement that retrieves the image stored at index 0 from an ImageList control named `slideShowImageList` and displays it in a PictureBox control named `slideShowPictureBox`.

Programming Problems

VideoNote
Solving the Total
Sales Problem

1. **Total Sales**

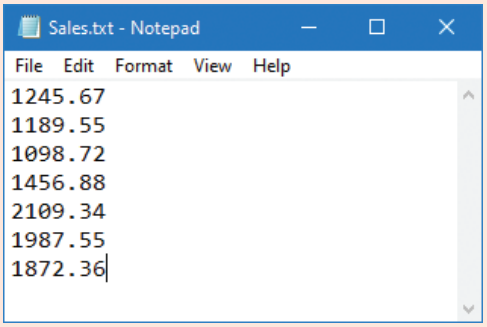
   In the *Chap07* folder of the Student Sample Programs, you will find a file named Sales.txt. Figure 7-48 shows the file's contents displayed in Notepad. Create an application that reads this file's contents into an array, displays the array's contents in a ListBox control, and calculates and displays the total of the array's values.

2. **Sales Analysis**

   Modify the application that you created in Programming Exercise 1 so it also displays the following:

   - The average of the values in the array
   - The largest value in the array
   - The smallest value in the array

Figure 7-48 The Sales.txt file



3. **Charge Account Validation**

   In the *Chap07* folder of the Student Sample Programs, you will find a file named ChargeAccounts.txt. The file contains a list of a company's valid charge account numbers. There are a total of 18 charge account numbers in the file, and each one is a 7-digit number, such as `5658845`.

   Create an application that reads the contents of the file into an array or a `List`. The application should then let the user enter a charge account number. The program should determine whether the number is valid by searching for it in the array or `List` that contains the valid charge account numbers. If the number is in the array or `List`, the program should display a message indicating the number is valid. If the number is not in the array or `List`, the program should display a message indicating the number is invalid.

4. **Driver's License Exam**

   The local driver's license office has asked you to create an application that grades the written portion of the driver's license exam. The exam has 20 multiple-choice questions. Here are the correct answers:

1. B	2. D	3. A	4. A	5. C
6. A	7. B	8. A	9. C	10. D
11. B	12. C	13. D	14. A	15. D
16. C	17. C	18. B	19. D	20. A

   Your program should store these correct answers in an array. The program should read the student's answers for each of the 20 questions from a text file and store the answers in another array. (Create your own text file to test the application.) After the student's answers have been read from the file, the program should display a message indicating whether the student passed or failed the exam. (A student must correctly answer 15 of the 20 questions to pass the exam.) It should then display the total number of correctly answered questions, the total number of incorrectly answered questions, and a list showing the question numbers of the incorrectly answered questions.

5. **World Series Champions**

   In the *Chap07* folder of the Student Sample Programs, you will find the following files:

   - Teams.txt—This file contains a list of several Major League baseball teams in alphabetical order. Each team listed in the file has won the World Series at least once.

- WorldSeriesWinners.txt—This file contains a chronological list of the World Series' winning teams from 1903 through 2012. (The first line in the file is the name of the team that won in 1903, and the last line is the name of the team that won in 2012. Note that the World Series was not played in 1904 or 1994.)

Create an application that displays the contents of the Teams.txt file in a ListBox control. When the user selects a team in the ListBox, the application should display the number of times that team has won the World Series in the time period from 1903 through 2012.

TIP: Read the contents of the WorldSeriesWinners.txt file into a `List` or an array. When the user selects a team, an algorithm should step through the list or array counting the number of times the selected team appears.

6. **Name Search**

   In the *Chap07* folder of the Student Sample Programs, you will find a file named popular_names.txt. This file contains a list of the 400 most popular names given to children born in the United States from the year 2000 through 2009.

   Create an application that reads the contents of the file into an array or a `List`. The user should be able to enter a name, and the application should display a message indicating whether the name was among the most popular.

7. **Population Data**

   In the *Chap07* folder of the Student Sample Programs, you will find a file named USPopulation.txt. The file contains the midyear population of the United States, in thousands, during the years 1950 through 1990. The first line in the file contains the population for 1950, the second line contains the population for 1951, and so forth.

   Create an application that reads the file's contents into an array or a `List`. The application should display the following data:

   - The average annual change in population during the time period
   - The year with the greatest increase in population during the time period
   - The year with the least increase in population during the time period

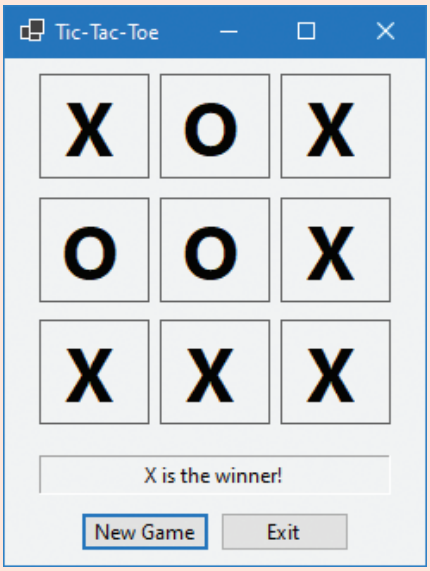
8. **Tic-Tac-Toe Simulator**

   Create an application that simulates a game of tic-tac-toe. Figure 7-49 shows an example of the application's form. The form shown in the figure uses eight large Label controls to display the Xs and Os.

   The application should use a two-dimensional `int` array to simulate the game board in memory. When the user clicks the *New Game* button, the application should step through each element of the array, performing the following:

   - A random number in the range of 0 through 1 should be stored in the element. The number 0 represents the letter O, and the number 1 represents the letter X.
   - Each time a value is stored in an element, the application should check to see if the game has been won.
   - If the game has been won, the application should display a message indicating whether player X won or player Y won. The application will then stop stepping through the array.
   - If the application steps through the entire array and no winner is detected, a message should be displayed indicating the game ended in a tie.

Figure 7-49 The *Tic-Tac-Toe* application



9. **Jagged Array of Exam Scores**

   Dr. Hunter teaches three sections of her Intro to Computer Science class. She has 12 students in section 1, 8 students in section 2, and 10 students in section 3. In the *Chap07* folder of the Student Sample Programs, you will find the following files:

   - Section1.txt—This file contains the final exam scores for each student in section 1. (There are 12 integer scores in the file.)
   - Section2.txt—This file contains the final exam scores for each student in section 2. (There are 8 integer scores in the file.)
   - Section3.txt—This file contains the final exam scores for each student in section 3. (There are 10 integer scores in the file.)

   Create an application that reads these three files and stores their contents in a jagged array. The array's first row should hold the exam scores for the students in section 1, the second row should hold the exam scores for the students in section 2, and the third row should hold the exam scores for the students in section 3.

   The application should display each section's exam scores in a separate ListBox control and then use the jagged array to determine the following:

   - The average exam score for each individual section
   - The average exam score for all the students in the three sections
   - The highest exam score among all three sections and the section number in which that score was found
   - The lowest exam score among all three sections and the section number in which that score was found

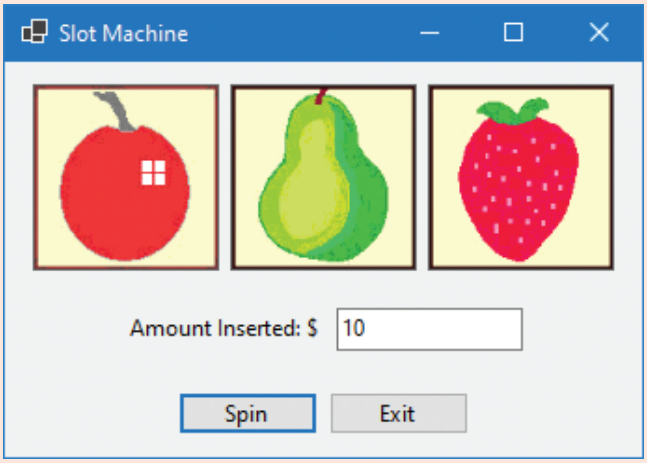
10. **Slot Machine Simulation**

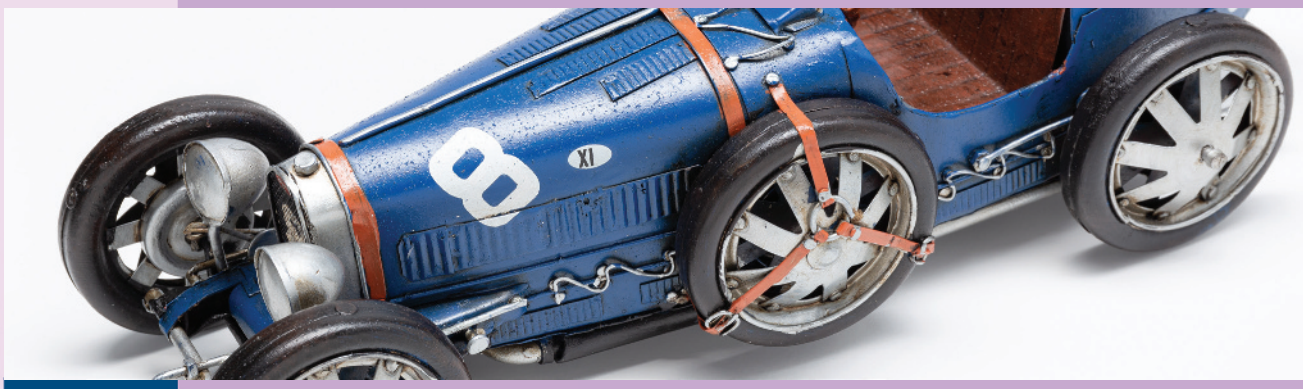
    A slot machine is a gambling device into which the user inserts money and then pulls a lever (or presses a button). The slot machine then displays a set of random images. If two or more of the images match, the user wins an amount of money that the slot machine dispenses back to the user. Create an application that simulates a slot machine. Figure 7-50 shows an example of how the form should look at run time.

The application should let the user enter into a TextBox the amount of money they are inserting into the machine. When the user clicks the *Spin* button, the application should display three randomly selected symbols. (Slot machines traditionally display fruit symbols. You will find a set of fruit symbols in the *Images\Fruit Symbols* folder of the Student Sample Programs.)

If none of the randomly displayed images match, the program should inform the user that they have won $0. If two of the images match, the program should inform the user that they have won two times the amount entered. If three of the images match, the program should inform the user that they have won three times the amount entered. When the user clicks the Exit button to exit the application, the program should display the total amount of money entered into the slot machine and the total amount won.

Figure 7-50 *Slot Machine* application





CHAPTER 8

Text Processing

TOPICS

8.1 Introduction
8.2 Interpolated Strings
8.3 Working with Characters
8.4 Working with Substrings
8.5 Manipulating Strings
8.6 The `StringBuilder` Class
8.7 Tokenizing Strings
8.8 Method Chaining
8.9 The `String.Format` Method

8.1 Introduction

Many of the programs that you have written so far have worked with strings, but only in a limited way. The operations that you have performed with strings so far have primarily involved only input and output. For example, you have read strings as input from the keyboard and from files, and written strings as output to the screen and to files.

There are many types of programs that not only read strings as input and write strings as output, but also perform operations on strings. Word processing programs, for example, manipulate large amounts of text, and thus work extensively with strings. Email programs and search engines are other examples of programs that perform operations on

strings. C# and .NET provide a wide variety of tools and programming techniques that you can use to examine and manipulate strings. We will look at many of these in this chapter.

8.2 Interpolated Strings

CONCEPT: Interpolated strings are a special type of string literal that allow you to format values in a variety of ways.

Quite often you will need to insert values into a string. For example, suppose you have a `decimal` variable named `sales`, and you want to display a string such as "Our sales are *XXXX* for the day.", where *XXXX* is the value of the `sales` variable. You need to insert the value of `sales` into the string. One way to accomplish this is with string concatenation. Here is an example:

```
decimal sales = 8500.0m;
MessageBox.Show("Our sales are " + sales + " for the day.");
```

This code snippet would display the string "Our sales are 8500.0 for the day." in a message box. A better way to accomplish this, however, is with an interpolated string. An **interpolated string** allows you to insert values into a string and format those values in specific ways. Here is an example:

```
decimal sales = 8500.0m;
MessageBox.Show($"Our sales are {sales} for the day.");
```

An interpolated string is a string literal that begins with a `$` character, immediately followed by an opening quotation mark. In this example, the interpolated string is

```
$"Our sales are {sales} for the day."
```

Inside the interpolated string, `{sales}` is a placeholder for the `sales` variable. When the statement is executed, the placeholder is replaced with the value of the `sales` variable. As a result, the code displays "Our sales are 8500.0 for the day." in a message box. The following code shows an example that displays "The temperature is 45.7." in a message box:

```
double temperature = 45.7;
MessageBox.Show($"The temperature is {temperature}.");
```

Here is an example that displays "You have earned 900 points." in a Label control:

```
int points = 900;
pointsLabel.Text = $"You have earned {points} points.";
```

NOTE: There can be no space between the `$` and the opening quotation mark in an interpolated string.

The following code snippet shows an interpolated string that uses two placeholders:

```
string firstName = "Zaria";
string lastName = "Romano";
nameLabel.Text = $"{lastName}, {firstName}";
```

When this code executes, the string "Romano, Zaria" will be displayed in the `nameLabel` control.

TIP: In many situations, you can use an interpolated string as an alternative to calling a variable's `ToString` method. For example, the following code calls the `size` variable's `ToString` method and displays the resulting value in a message box:

```
int size = 4;
MessageBox.Show(size.ToString());
```

The following code performs the same operation, using an interpolated string:

```
int size = 4;
MessageBox.Show($"{size}");
```

Let's look at another example. In the following code, assume `testScoreLabel` is a Label control:

```
double testScore = 87.5;
testScoreLabel.Text = $"{testScore}";
```

This code uses an interpolated string to convert the `testScore` variable to a string, and the resulting value is assigned to the Label control's Text property.

Placeholder Expressions

In the previous examples, we used placeholders to display the values of variables. In addition to variable names, placeholders can contain any valid C# expression. Here is an example:

```
double celsius = 12.25;
MessageBox.Show($"Fahrenheit: {(celsius * 1.8) + 32}.");
```

In this example `{(celsius * 1.8) + 32}` is the placeholder. When the statement is executed, the placeholder is replaced with the value of the expression `(celsius * 1.8) + 32`. This code displays "Fahrenheit: 54.05." in a message box.

Formatting Values

Placeholders in an interpolated string can include a **format specifier** that causes the placeholder's value to be formatted when it is displayed. For example, if you want a number to be formatted as currency, you will include the `C` format specifier, as shown here:

```
decimal amount = 123456789.45678m;
balanceLabel.Text = $"The balance is {amount:C}.";
```

Notice that the placeholder expression and the format specifier are separated by a colon. In this example, the value of the `amount` variable will be formatted as currency. When this code executes, the following string will be displayed in the `balanceLabel` control:

```
The balance is $123,456,789.46.
```

Several format specifiers are supported. Table 8-1 lists a few of them. Notice that these are the same format specifiers that you can use with the `ToString()` method, as discussed in Chapter 3.

Table 8-1 Format specifiers

Format Specifier	Description
`N`	Number format: The number is formatted with comma separators and a decimal point. By default, two digits display to the right of the decimal point. Negative values are displayed with a leading minus sign. An example is −2,345.67.

Table 8-1 Format specifiers (*continued*)

Format Specifier	Description
F	Fixed-point format: The number is formatted with no comma separators and a decimal point. By default, two digits display to the right of the decimal point. Negative values are displayed with a leading minus (−) sign. An example is −2345.67.
E	Exponential scientific format: The number is formatted with a single digit to the left of the decimal point. The letter e appears in front of the exponent, and the exponent has a leading + or − sign. By default, six digits display to the right of the decimal point, and a leading minus sign is used if the number is negative. An example is −2.345670e+003.
C	Currency format: Formats the number to include a leading currency symbol (such as $), comma separators, and a decimal point. By default, two digits display to the right of the decimal point. Negative values are surrounded by parentheses. An example of a negative value is ($2,345.67).
P	Percent format: Causes the number to be multiplied by 100 and displayed with a trailing space and % sign. By default, two digits display to the right of the decimal point. Negative values are displayed with a leading minus sign. For example, the number 0.125 would be formatted as 12.5 % and the number −0.2345 would be formatted as −23.45 %.

NOTE: The format specifiers listed in Table 8-7 are not case sensitive. For example, the placeholder `{amount:N}` is the same as `{amount:n}`.

Specifying the Precision

Each numeric format specifier can optionally be followed by an integer that indicates the number of digits to display after the decimal point. For example, the format `N3` displays three digits after the decimal point. Table 8-2 shows a variety of numeric formatting examples, based on the North American locale.

Table 8-2 Numeric formatting examples (North American locale)

Number	Format Specifier	Formatting Result
12.3	N3	12.300
12.348	N2	12.35
1234567.1	N	1,234,567.10
123456.0	F2	123456.00
123456.0	E3	1.235e+005
.234	P	23.40 %
–1234567.8	C	($1,234,567.80)
1234.567	C0	$1,235

Here is an example:

```
double number = 12345.6789;
MessageBox.Show($"The number is {number:N3}");
```

In this example, the placeholder `{number:N3}` specifies that the value of `number` should be formatted as a number with 3 decimal places. When this code executes, the following string will be displayed:

```
The number is 12,345.679
```

Rounding

Rounding can occur when the number of digits you have specified after the decimal point in the format specifier is smaller than the precision of the numeric value. Suppose, for example, that we use the `N2` format specifier on the value 1.235. The resulting value would be 1.24. If the next digit after the last displayed digit is 5 or higher, the last displayed digit is rounded *away from zero*.

Specifying a Minimum Field Width

A format specifier can include a **minimum field width**, which is the minimum number of spaces that should be used to display the value. The following example displays a floating-point number in a field that is 20 spaces wide:

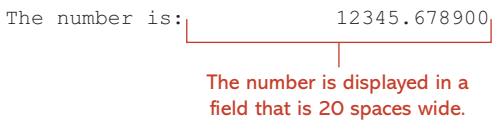
```
double number = 12345.6789;
MessageBox.Show($"The number is:{number,20}");
```

Notice that the placeholder expression and the **width specifier** are separated by a comma. The placeholder `{number,20}` specifies that the value of the `number` variable should appear in a field that is a minimum of 20 spaces wide. When this code executes, the following string will be displayed:

```
The number is:        12345.678900
```

The field width is illustrated in Figure 8-1.

Figure 8-1 The number occupies a field that is 20 spaces wide



In this example, the number 12345.678900 uses only 12 spaces, but it is inserted into a field that is 20 spaces wide. When this is the case, the number will be right-justified in the field. If a value is too large to fit in the specified field width, the field is automatically enlarged to accommodate it. The following example displays a floating-point number in a field that is only one space wide:

```
double number = 12345.6789;
MessageBox.Show($"The number is:{number,1}");
```

The value of the `number` variable requires more than one space, however, so the field width is expanded to accommodate the entire number. When this code executes, the following string will be displayed:

```
The number is:12345.6789
```

If you want a number to be left-justified within its field, you insert a minus sign (–) before the field width specifier. Here is an example:

```
double number = 12345.6789;
MessageBox.Show($"***{number,-20}***");
```

When this code executes, the following string is displayed:

```
***12345.6789          ***
```

You can include both a width specifier and a format specifier, as shown here:

```
double number = 12345.6789;
MessageBox.Show($"The number is:{number,20:C}");
```

In this example, the placeholder `{number,20:C}` specifies that the value of the `number` variable should appear in a field that is a minimum of 20 spaces wide, formatted as currency. When this code executes, the following string will be displayed:

```
The number is:           $12,345.68
```

Checkpoint

8.1 Assume the following variable declaration exists in a program:

```
double number = 1234567.456;
```

Write an interpolated string that formats the value of the `number` variable as:

```
1,234,567.46
```

8.2 Assume the following variable declaration exists in a program:

```
double number = 123.456;
```

Write an interpolated string that rounds the value of the `number` variable to one decimal place, in a field that is 10 spaces wide. (Do not use comma separators.)

8.3 Assume the following variable declaration exists in a program:

```
int number = 123456;
```

Write an interpolated string that formats the value of the `number` variable in a field that is 10 spaces wide, with comma separators.

8.4 Assume the following variable declaration exists in a program:

```
double number = 123456.789;
```

Write an interpolated string that formats the value of the `number` variable as follows:

- rounded to two decimal places
- formatted with comma separators
- left-justified in a field that is 20 spaces wide

8.3 Working with Characters

CONCEPT: **Some programming tasks require that you manipulate strings at a detailed level. C# and .NET provide tools that let you work with individual characters and sets of characters within strings.**

Sometimes the data with which a program must work comes in the form of text. Word processors, text messaging programs, e-mail applications, Web browsers, and spell-checkers are just a few examples of programs that work extensively with text. The earlier chapters in this book have demonstrated some simple text-processing techniques, such as comparing strings and converting strings to other data types. Sometimes, however, you need to operate on strings at a more detailed level. Some operations require that you access or manipulate the individual characters in a string.

For example, you have probably used programs or websites that require you to set up a password that meets certain requirements. Some systems require that passwords have a minimum length and contain at least one uppercase letter, at least one lowercase letter, and at least one numeric digit. These requirements are intended to prevent ordinary words from being used as passwords and thus make the passwords more secure. When a new password is created, the system has to examine each of the password's characters to determine whether it meets the requirements. In the next section, you will see an example

of an algorithm that performs this very operation. First, however, we discuss the `char` data type and the process of retrieving the individual characters in a string.

The `char` Data Type

So far in this book, we have used the `string` data type to store text. C# also provides the `char` data type, which is used to store individual characters. A variable of the `char` data type can hold only one character at a time. Here is an example of how you might declare a `char` variable:

```
char letter;
```

This statement declares a `char` variable named `letter`, which can store one character. In C#, **character literals** are enclosed in single quotation marks. Here is an example showing how we would assign a character to the `letter` variable:

```
letter = 'g';
```

This statement assigns the character `'g'` to the `letter` variable. Because `char` variables can hold only one character, they are not compatible with strings. For example, you cannot assign a string to a `char` variable, even if the string contains only one character. The following statement, for example, will not compile because it attempts to assign a string literal to a `char` variable:

```
letter = "g";  // ERROR! Cannot assign a string to a char
```

It is important that you do not confuse character literals, which are enclosed in single quotation marks, with string literals, which are enclosed in double quotation marks.

When you need to convert a `char` variable to a string, you can call its `ToString` method. For example, the following statement displays the value of the `letter` variable in a message box:

```
MessageBox.Show(letter.ToString());
```

The following statement shows another example. Assume that `letter` is a `char` variable and `outputLabel` is the name of a Label control:

```
outputLabel.Text = letter.ToString();
```

In many situations you can use an interpolated string to convert the variable to a string. For example, the following statement displays the value of the `letter` variable in a message box:

```
MessageBox.Show($"{letter}");
```

The following statement assigns the value of the `letter` variable to the `outputLabel` control's Text property:

```
outputLabel.Text = $"{letter}";
```

Retrieving the Individual Characters in a String

C# allows you to access the individual characters in a string using subscript notation. This makes it possible to work with a string as if it were an array of characters. You use subscript 0 to access the first character, subscript 1 to access the second character, and so forth. The subscript of the last character is 1 less than the string's length. The following code shows an example:

```
1 // Declare a string and a char.
2 string name = "Jacob";
3 char letter;
4
5 // Get the first character (at position 0).
6 letter = name[0];
7
8 // Display the character.
9 MessageBox.Show(letter.ToString());
```

In this code, line 2 declares a `string` variable named `name`, initialized with the string `"Jacob"`. Line 3 declares a `char` variable named `letter`. Line 6 gets the character at position 0 in `name` (the first character in the string) and assigns it to the `letter` variable. Line 9 displays the value of the `letter` variable in a message box. If this code were executed, a message box would appear showing the character J.

NOTE: As with arrays, an exception occurs if you attempt to use an invalid subscript with a string. String subscripts must be at least 0, and they must be less than the length of the string.

The following code sample shows how a loop can be used to step through the characters in a string. Notice that in the `for` loop, the `index` variable has a starting value of 0 and is incremented after each iteration, and the loop iterates as long as `index` is less than `name.Length`. If this code were executed, a series of message boxes would appear displaying the characters J, a, c, o, and b.

```
 1 // Declare a string and a char.
 2 string name = "Jacob";
 3 char letter;
 4
 5 // Display the characters in the string.
 6 for (int index = 0; index < name.Length; index++)
 7 {
 8     letter = name[index];
 9     MessageBox.Show(letter.ToString());
10 }
```

Keep in mind that subscripts provide read-only access to the characters in a string. You cannot use a subscript to change the value of a character. For example, the following code will not compile because the second statement attempts to use a subscript expression to change the value of the first character in the `name` variable:

```
string name = "Jill";
name[0] = 'B';          // ERROR! This will not work!
```

You can also use the `foreach` loop to retrieve the individual characters in a string. The following code shows an example. If this code were executed, a series of message boxes would appear displaying the characters J, a, c, o, and b.

```
 1 // Declare a string and a char.
 2 string name = "Jacob";
 3 char letter;
 4
 5 // Display the characters in the string.
 6 foreach (char letter in name)
 7 {
 9     MessageBox.Show(letter.ToString());
10 }
```

Character Testing and Conversion Methods

The `char` data type provides several methods for testing the value of a character. Some of the methods are listed in Table 8-3. Note that each of the methods listed in the table returns a Boolean value of `true` or `false`.

Table 8-3 Some of the character testing methods

Method	Description

`char.IsDigit(ch)`	The argument *ch* is a character. The method returns `true` if *ch* is a digit (0 through 9) or `false` otherwise.
	Example:
	`string str = "12345";`
	`if (char.IsDigit(str[0]))`
	`{`
	`  MessageBox.Show("Character 0 is " +`
	`  "a digit.");`
	`}`

`char.IsDigit(str, index)`	The argument *str* is a string, and *index* is the position of a character within *str*. The method returns `true` if the specified character is a digit (0 through 9) or `false` otherwise.
	Example:
	`string str = "12345";`
	`if (char.IsDigit(str, 0))`
	`{`
	`  MessageBox.Show("Character 0 is " +`
	`  "a digit.");`
	`}`

`char.IsLetter(ch)`	The argument *ch* is a character. The method returns `true` if *ch* is an alphabetic letter or `false` otherwise.
	Example:
	`string str = "Hello World";`
	`if (char.IsLetter(str[0]))`
	`{`
	`  MessageBox.Show("Character 0 is " +`
	`  "a letter.");`
	`}`

`char.IsLetter(str, index)`	The argument *str* is a string, and *index* is the position of a character within *str*. The method returns `true` if the specified character is an alphabetic letter or `false` otherwise.
	Example:
	`string str = "Hello World";`
	`if (char.IsLetter(str, 0))`
	`{`
	`  MessageBox.Show("Character 0 is " +`
	`  "a letter.");`
	`}`

`char.IsLetterOrDigit(ch)`	The argument *ch* is a character. The method returns `true` if *ch* is either an alphabetic letter or a numeric digit. Otherwise, the method returns `false`.
	Example:
	`string str = "Hello World";`
	`if (char.IsLetterOrDigit(str[0]))`
	`{`
	`  MessageBox.Show("Character 0 is " +`
	`  "either a letter " +`
	`  "or a digit.");`
	`}`

(*continued*)

Table 8-3 Some of the character testing methods (*continued*)

Method	Description

`char.IsLetterOrDigit(str, index)`	The argument `str` is a string, and `index` is the position of a character within `str`. The method returns `true` if the specified character is either an alphabetic letter or a numeric digit. Otherwise, the method returns `false`.
	Example:
	`string str = "12345";`
	`if (char.IsLetterOrDigit(str, 0))`
	`{`
	`  MessageBox.Show("Character 0 is " +`
	`  "either a letter " +`
	`  "or a digit.");`
	`}`

`char.IsLower(ch)`	The argument `ch` is a character. The method returns `true` if `ch` is a lowercase letter or `false` otherwise.
	Example:
	`string str = "hello world";`
	`if (char.IsLower(str[0]))`
	`{`
	`  MessageBox.Show("Character 0 is " +`
	`  "lowercase.");`
	`}`

`char.IsLower(str, index)`	The argument `str` is a string, and `index` is the position of a character within `str`. The method returns `true` if the specified character is a lowercase letter or `false` otherwise.
	Example:
	`string str = "hello world";`
	`if (char.IsLower(str, 0))`
	`{`
	`  MessageBox.Show("Character 0 is " +`
	`  "lowercase.");`
	`}`

`char.IsPunctuation(ch)`	The argument `ch` is a character. The method returns `true` if `ch` is categorized as a punctuation mark or `false` otherwise.
	Example:
	`string str = "Hello!";`
	`if (char.IsPunctuation(str[5]))`
	`{`
	`  MessageBox.Show("Character 5 is a " +`
	`  "punctuation mark.");`
	`}`

`char.IsPunctuation(str, index)`	The argument `str` is a string, and `index` is the position of a character within `str`. The method returns `true` if the specified character is categorized as a punctuation mark or `false` otherwise.
	Example:
	`string str = "Hello!";`
	`if (char.IsPunctuation(str, 5))`
	`{`
	`  MessageBox.Show("Character 5 is a " +`
	`  "punctuation mark.");`
	`}`

Table 8-3 Some of the character testing methods (*continued*)

Method	Description

`char.IsUpper(ch)`	The argument `ch` is a character. The method returns `true` if `ch` is an uppercase letter or `false` otherwise.
	Example:
	`string str = "Hello World";`
	`if (char.IsUpper(str[0]))`
	`{`
	`  MessageBox.Show("Character 0 is " +`
	`  "uppercase.");`
	`}`

`char.IsUpper(str, index)`	The argument `str` is a string, and `index` is the position of a character within `str`. The method returns `true` if the specified character is an uppercase letter or `false` otherwise.
	Example:
	`string str = "Hello World";`
	`if (char.IsUpper(str, 0))`
	`{`
	`  MessageBox.Show("Character 0 is " +`
	`  "uppercase.");`
	`}`

`char.IsWhiteSpace(ch)`	The argument `ch` is a character. The method returns `true` if `ch` is a white-space character or `false` otherwise. (White-space characters are the space, tab, linefeed, carriage-return, formfeed, vertical-tab, and newline characters.)
	Example:
	`string str = "  ";`
	`if (char.IsWhiteSpace(str[0]))`
	`{`
	`  MessageBox.Show("Character 0 is " +`
	`  "whitespace.");`
	`}`

`char.IsWhiteSpace(str, index)`	The argument `str` is a string, and `index` is the position of a character within `str`. The method returns `true` if the specified character is a white-space character or `false` otherwise. (White-space characters are the space, tab, linefeed, carriage-return, formfeed, vertical-tab, and newline characters.)
	Example:
	`string str = "  ";`
	`if (char.IsWhiteSpace(str, 0))`
	`{`
	`  MessageBox.Show("Character 0 is " +`
	`  "whitespace.");`
	`}`

In Tutorial 8-1, you use several of the character testing methods in an application that validates passwords.

Tutorial 8-1:
Completing the *PasswordValidation* Project

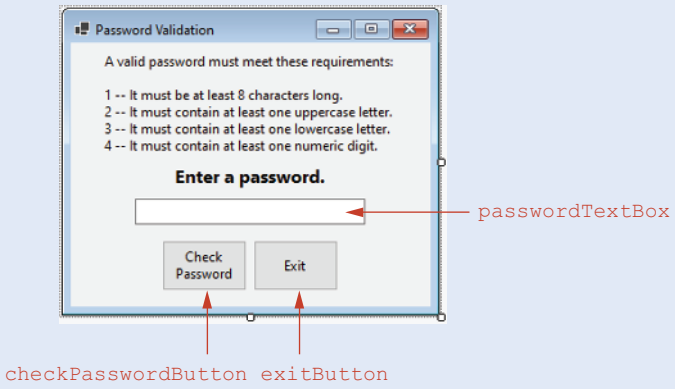
VideoNote
Tutorial 8-1: Completing the *PasswordValidation* Project

Many password-protected systems allow users to set up their own passwords. When a user creates a password, the system examines the password to determine whether it meets the minimum requirements. If it does not, the system rejects the password and requires the user to create another, more secure, password.

In this tutorial, you complete the *PasswordValidation* project. The application's form, which has already been created for you, is shown in Figure 8-2. When the application is complete, the user will enter a password and then click the *Check Password* button. The application will check the password to make sure it meets the following requirements:

- The password must be at least eight characters long.
- The password must contain at least one uppercase character.
- The password must contain at least one lowercase character.
- The password must contain at least one numeric digit.

Figure 8-2 The *PasswordValidation* project's form



Step 1: Start Visual Studio. Open the project named *PasswordValidation* in the *Chap08* folder of the Student Sample Programs.

Step 2: Open the Form1 form's code in the code editor. Type the comments and code for the `NumberUpperCase` method, shown in lines 10–27 of Program 8-1. The purpose of the `NumberUpperCase` method is to accept a string as an argument and return the number of uppercase letters contained in the string.

Step 3: Type the comments and code for the `NumberLowerCase` method, shown in lines 29–46 of Program 8-1. The purpose of the `NumberLowerCase` method is to accept a string as an argument and return the number of lowercase letters contained in the string.

Step 4: Type the comments and code for the `NumberDigits` method, shown in lines 48–65 of Program 8-1. The purpose of the `NumberDigits` method is to accept a string as an argument and return the number of numeric digits contained in the string.

Step 5: Next, you create the Click event handlers for the Button controls. Switch back to the *Designer* and double-click the `checkPasswordButton` control. This opens the code editor, and you will see an empty event handler named `checkPasswordButton_Click`. Complete the event handler by typing the code shown in lines 69–86 in Program 8-1. Let's review this code:

Lines 69–72: The statement in line 69 declares a constant for the minimum password length. Line 72 declares the `password` variable and initializes it with the value entered by the user.

Line 75: The `if-else` statement that begins in line 75 evaluates a compound Boolean expression. In plain English, the statement should be interpreted like this:

If the password's length is at least 8 and
the number of uppercase letters in the password is at least 1 and
the number of lowercase letters in the password is at least 1 and
the number of numeric digits in the password is at least 1,
Then the password is valid.

Else
The password does not meet the requirements.

Step 6: Switch your view back to the *Designer* and double-click the `exitButton` control. In the code editor, you will see an empty event handler named `exitButton_Click`. Complete the event handler by typing the code shown in lines 91–92 in Program 8-1.

Step 7: Save the project. Then, press F5 on the keyboard or click the *Start Debugging* button (▶) on the toolbar to compile and run the application. When the application runs, test various passwords to confirm that the application works properly. When you are finished, click the *Exit* button to exit the application.

Program 8-1 Completed code for Form1 in the *PasswordValidation* project

```
1  namespace PasswordValidation
2  {
3     public partial class Form1: Form
4     {
5         public Form1()
6         {
7             InitializeComponent();
8         }
9
10        // The NumberUpperCase method accepts a string argument
11        // and returns the number of uppercase letters it contains.
12        private int NumberUpperCase(string str)
13        {
14            int upperCase = 0; // The number of uppercase letters
15
16            // Count the uppercase characters in str.
17            foreach (char ch in str)
18            {
19               if (char.IsUpper(ch))
```

```
            {
               upperCase++;
            }
      }

      // Return the number of uppercase characters.
      return upperCase;
   }

   // The NumberLowerCase method accepts a string argument
   // and returns the number of lowercase letters it contains.
   private int NumberLowerCase(string str)
   {
      int lowerCase = 0; // The number of lowercase letters

      // Count the lowercase characters in str.
      foreach (char ch in str)
      {
         if (char.IsLower(ch))
         {
             lowerCase++;
         }
      }

      // Return the number of lowercase characters.
      return lowerCase;
   }

   // The NumberDigits method accepts a string argument
   // and returns the number of numeric digits it contains.
   private int NumberDigits(string str)
   {
      int digits = 0; // The number of digits

      // Count the digits in str.
      foreach (char ch in str)
      {
         if (char.IsDigit(ch))
         {
             digits++;
         }
      }

      // Return the number of digits.
      return digits;
   }

   private void checkPasswordButton_Click(object sender, EventArgs e)
   {
      const int MIN_LENGTH = 8; // Password's minimum length

      // Get the password from the TextBox.
      string password = passwordTextBox.Text;

      // Validate the password.
      if (password.Length >= MIN_LENGTH &&
          NumberUpperCase(password) >= 1 &&
```

```
77              NumberLowerCase(password) >= 1 &&
78              NumberDigits(password) >= 1)
79          {
80              MessageBox.Show("The password is valid.");
81          }
82          else
83          {
84              MessageBox.Show("The password does not meet " +
85                  "the requirements.");
86          }
87      }
88
89      private void exitButton_Click(object sender, EventArgs e)
90      {
91          // Close the form.
92          this.Close();
93      }
94  }
95 }
```

Character Case Conversion

The `char` data type also provides the `ToLower` and `ToUpper` methods listed in Table 8-4 for converting the case of a character. Each method accepts a `char` argument and returns a `char` value.

Table 8-4 Character case conversion methods

Method	Description
`char.ToLower(ch)`	The argument *ch* is a character. The method returns the lowercase equivalent of *ch*.
`char.ToUpper(ch)`	The argument *ch* is a character. The method returns the uppercase equivalent of *ch*.

If the `ToLower` method's argument is an uppercase character, the method returns the lowercase equivalent. For example, look at the following code. The statement in line 2 assigns the character `'a'` to the `letter` variable.

```
1  string str = "ABC";
2  char letter = char.ToLower(str[0]);
```

If the argument is already lowercase, the `ToLower` method returns it unchanged.

If the `ToUpper` method's argument is a lowercase character, the method returns the uppercase equivalent. For example, look at the following code. The statement in line 2 assigns the character `'A'` to the `letter` variable.

```
1  string str = "abc";
2  char letter = char.ToUpper(str[0]);
```

If the argument is already uppercase, the `ToUpper` method returns it unchanged.

Checkpoint

8.5 Declare a `char` variable named `letter` and assign the letter `'A'` to the variable.

8.6 Write a statement that displays the value of a `char` variable named `letterGrade` in a message box.

8.7 Write a statement that declares a `char` variable named `lastLetter` and stores the last character of a string named `alphabet` in the variable.

8.8 Write a `foreach` loop that displays each character of a `string` variable named `serialNumber` in a message box.

8.9 Write the first line of an `if` statement that calls the `char.IsPuncuation` method and passes the last character of a `string` variable named `sentence` as an argument.

8.10 Write the first line of an if statement that calls the `char.IsUpper` method, passing a string variable named `sentence` and the index value for the first character of the `sentence` variable as arguments.

8.11 Write a statement that calls the `char.ToUpper` method, passes a `char` variable named `lowercase` as an argument, and stores the result in a `char` variable named `uppercase`.

8.12 What value does the `char.ToLower` function return if the argument is already lowercase?

8.4 Working with Substrings

CONCEPT: **A substring is a string within a string. .NET provides various methods for working with and manipulating substrings.**

Searching for Substrings

Some tasks require you to search for a specific string of characters within a string. A string within a string is called a **substring**. Objects of the `string` data type have several methods that allow you to search for substrings. Table 8-5 summarizes the `Contains`, `StartsWith`, and `EndsWith` methods. Each of the methods in Table 8-5 returns a Boolean value indicating whether the substring was found.

Table 8-5 Some of the substring-searching methods

Method	Description

str`.Contains(`*substring*`)`	*str* is the name of a `string` variable or is a string expression. The argument *substring* is also a string. The method returns `true` if *str* contains the string *substring* or `false` otherwise.
	Example:
	`// The following code displays`
	`// "ice was found."`
	`string str = "chocolate ice cream";`
	`if (str.Contains("ice"))`
	`{`
	`  MessageBox.Show("ice was found.");`
	`}`

Table 8-5 Some of the substring-searching methods (*continued*)

Method	Description

str.Contains(*ch*)	*str* is the name of a string variable or is a string expression. The argument *ch* is a character.
	The method returns true if *str* contains the character *ch* or false otherwise.
	Example:
	`// The following code displays`
	`// "b was found."`
	`string str = "abcd";`
	`if (str.Contains('b'))`
	`{`
	`  MessageBox.Show("b was found.");`
	`}`

str.StartsWith(*substring*)	*str* is the name of a string variable or is a string expression. The argument *substring* is also a string. The method returns true if *str* starts with the string *substring* or false otherwise.
	Example:
	`// The following code displays`
	`// "The string starts with choc."`
	`string str = "chocolate ice cream";`
	`if (str.StartsWith("choc"))`
	`{`
	`  MessageBox.Show("The string starts " +`
	`  "with choc.");`
	`}`

str.EndsWith(*substring*)	*str* is the name of a string variable or is a string expression. The argument *substring* is also a string. The method returns true if *str* ends with the string *substring* or false otherwise.
	Example:
	`// The following code displays`
	`// "The string ends with cream."`
	`string str = "chocolate ice cream";`
	`if (str.EndsWith("cream"))`
	`{`
	`  MessageBox.Show("The string ends " +`
	`  "with cream.");`
	`}`

The methods shown in Table 8-5 let you know whether a specified substring is found within a string. Sometimes you also want to know the position of the substring. When that is the case, you can use one of the IndexOf or LastIndexOf methods shown in Table 8-6. Note that each method in Table 8-6 returns an int.

Sometimes you need to retrieve a specific set of characters from a string. For example, suppose a string contains a US telephone number, such as "(919)555-1212" and you are interested in getting the area code. You need a way to retrieve the characters in positions 1 through 3. You can use the Substring method, as shown in Table 8-7. Note that the Substring method returns a string.

Table 8-6 Methods for getting a character or substring's position

Method	Description

str.IndexOf(*substring*)	*str* is the name of a string variable or is a string expression. The argument *substring* is also a string. If *substring* is found in *str*, the method returns the integer position of *substring*'s first occurrence. If *substring* is not found in *str*, the method returns −1.
	Example:
	`// The following code displays "10"`
	`string str = "chocolate ice cream";`
	`int position = str.IndexOf("ice");`
	`if (position != -1)`
	`{`
	`  MessageBox.Show(position.ToString());`
	`}`
	`else`
	`{`
	`  MessageBox.Show("ice was not found.");`
	`}`

str.IndexOf(*substring*, *start*)	*str* is the name of a string variable or is a string expression. The argument *substring* is a string, and the argument *start* is an integer. The method searches *str* for *substring*, starting at the position specified by *start* and going to the end of *str*. If *substring* is found in this portion of *str*, the method returns the integer position of its first occurrence. If *substring* is not found, the method returns −1.
	Example:
	`// The following code displays "2"`
	`string str = "cocoa beans";`
	`int position = str.IndexOf("co", 2);`
	`if (position != -1)`
	`{`
	`  MessageBox.Show(position.ToString());`
	`}`
	`else`
	`{`
	`  MessageBox.Show("co was not found.");`
	`}`

str.IndexOf(*substring*, *start*, *count*)	*str* is the name of a string variable or is a string expression. The argument *substring* is a string, the argument *start* is an integer, and the argument *count* is also an integer. The method searches *str* for *substring*, starting at the position specified by *start* and continuing for *count* characters. If *substring* is found in this portion of *str*, the method returns the integer position of its first occurrence. If *substring* is not found, the method returns −1.

Table 8-6 Methods for getting a character or substring's position (*continued*)

Method	Description

	Example:
	`// The following code displays "6"`
	`string str = "xx oo xx oo xx";`
	`int position = str.IndexOf("xx", 3, 8);`
	`if (position != -1)`
	`{`
	`  MessageBox.Show(position.ToString());`
	`}`
	`else`
	`{`
	`  MessageBox.Show("xx was not found.");`
	`}`

`str.IndexOf(ch)`	*str* is the name of a `string` variable or is a string expression. The argument *ch* is a character. If *ch* is found in *str*, the method returns the integer position of *ch*'s first occurrence. If *ch* is not found in *str*, the method returns −1.
	Example:
	`// The following code displays "2"`
	`string str = "chocolate ice cream";`
	`int position = str.IndexOf('o');`
	`if (position != -1)`
	`{`
	`  MessageBox.Show(position.ToString());`
	`}`
	`else`
	`{`
	`  MessageBox.Show("o was not found.");`
	`}`

`str.IndexOf(ch, start)`	*str* is the name of a `string` variable or is a string expression. The argument *ch* is a character, and the argument *start* is an integer. The method searches *str* for *ch*, starting at the position specified by *start* and going to the end of *str*. If *ch* is found in this portion of *str*, the method returns the integer position of its first occurrence. If *ch* is not found, the method returns −1.
	Example:
	`// The following code displays "4"`
	`string str = "chocolate ice cream";`
	`int position = str.IndexOf('o', 3);`
	`if (position != -1)`
	`{`
	`  MessageBox.Show(position.ToString());`
	`}`
	`else`
	`{`
	`  MessageBox.Show("o was not found.");`
	`}`

(*continued*)

Table 8-6 Methods for getting a character or substring's position (*continued*)

Method	Description

str.IndexOf(*ch*, *start*, *count*)	*str* is the name of a string variable or is a string expression. The argument *ch* is a character, the argument *start* is an integer, and the argument *count* is also an integer. The method searches *str* for *ch*, starting at the position specified by *start* and continuing for *count* characters. If *ch* is found in this portion of *str*, the method returns the integer position of its first occurrence. If *substring* is not found, the method returns −1.
	Example:
	`// The following code displays "12"`
	`string str = "chocolate ice cream";`
	`int position = str.IndexOf('e', 10, 4);`
	`if (position != -1)`
	`{`
	`  MessageBox.Show(position.ToString());`
	`}`
	`else`
	`{`
	`  MessageBox.Show("e was not found.");`
	`}`

str.LastIndexOf(*substring*)	*str* is the name of a *string* variable or is a string expression. The argument *substring* is also a string. If *substring* is found in *str*, the method returns the integer position of *substring*'s last occurrence. If *substring* is not found in *str*, the method returns −1.
	Example:
	`// The following code displays "11".`
	`string str = "blue green blue";`
	`int position = str.LastIndexOf("blue");`
	`if (position != -1)`
	`{`
	`  MessageBox.Show(position.ToString());`
	`}`
	`else`
	`{`
	`  MessageBox.Show("blue was not found.");`
	`}`

str.LastIndexOf(*substring*, *start*)	*str* is the name of a string variable or is a string expression. The argument *substring* is a string, and the argument *start* is an integer. The method searches *str* for *substring*, starting at the position specified by *start* and proceeding backward toward the beginning of *str*. If *substring* is found in this portion of *str*, the method returns its integer position. If *substring* is not found, the method returns −1.

Table 8-6 Methods for getting a character or substring's position (*continued*)

Method	Description

	Example:
	`// The following code displays "6".`
	`string str = "xx oo xx oo xx";`
	`int position = str.LastIndexOf("xx", 10);`
	`if (position != -1)`
	`{`
	`  MessageBox.Show(position.ToString());`
	`}`
	`else`
	`{`
	`  MessageBox.Show("xx was not found.");`
	`}`

str.LastIndexOf(*substring*, *start*, *count*)	*str* is the name of a `string` variable or is a string expression. The argument *substring* is a string, the argument *start* is an integer, and the argument *count* is also an integer. The method searches *str* for *substring*, starting at the position specified by *start* and proceeding backward toward the beginning of *str* for *count* characters. If *substring* is found in this portion of *str*, the method returns its integer position. If *substring* is not found, the method returns −1.
	Example:
	`// The following code displays "6".`
	`string str = "oo xx oo xx oo";`
	`int position = str.LastIndexOf("oo", 10, 8);`
	`if (position != -1)`
	`{`
	`  MessageBox.Show(position.ToString());`
	`}`
	`else`
	`{`
	`  MessageBox.Show("oo was not found.");`
	`}`

str.LastIndexOf(*ch*)	*str* is the name of a `string` variable or is a string expression. The argument *ch* is a character. If *ch* is found in *str*, the method returns the integer position of *ch*'s last occurrence. If *ch* is not found in *str*, the method returns −1.
	Example:
	`// The following code displays "14".`
	`string str = "chocolate ice cream";`
	`int position = str.LastIndexOf('c');`
	`if (position != -1)`
	`{`
	`  MessageBox.Show(position.ToString());`
	`}`
	`else`
	`{`
	`  MessageBox.Show("c was not found.");`
	`}`

(*continued*)

Table 8-6 Methods for getting a character or substring's position (*continued*)

Method	Description

str.LastIndexOf(*ch*, *start*)	*str* is the name of a `string` variable or is a string expression. The argument *ch* is a character, and the argument *start* is an integer. The method searches *str* for *ch*, starting at the position specified by *start* and proceeding backward toward the beginning of *str*. If *ch* is found in this portion of *str*, the method returns its integer position. If *ch* is not found, the method returns −1.
	Example:
	`// The following code displays "12".`
	`string str = "chocolate ice cream";`
	`int position = str.LastIndexOf('e', 14);`
	`if (position != -1)`
	`{`
	`  MessageBox.Show(position.ToString());`
	`}`
	`else`
	`{`
	`  MessageBox.Show("e was not found.");`
	`}`

str.LastIndexOf(*ch*, *start*, *count*)	*str* is the name of a `string` variable or is a string expression. The argument *ch* is a character, the argument *start* is an integer, and the argument *count* is also an integer. The method searches *str* for *ch*, starting at the position specified by *start* and proceeding backward for *count* characters. If *ch* is found in this portion of *str*, the method returns its integer position. If *substring* is not found, the method returns −1.
	Example:
	`// The following code displays "12".`
	`string str = "chocolate ice cream";`
	`int position = str.LastIndexOf('e', 14, 8);`
	`if (position != -1)`
	`{`
	`  MessageBox.Show(position.ToString());`
	`}`
	`else`
	`{`
	`  MessageBox.Show("e was not found.");`
	`}`

Table 8-7 The `Substring` method

Method	Description

str.Substring(*start*)	*str* is the name of a `string` variable or is a string expression. The argument *start* is an integer that specifies a position in *str*. The method returns a string containing the characters beginning at *start*, continuing to the end of *str*.
	Example:
	`// The following code displays "beans".`
	`string str = "cocoa beans";`
	`MessageBox.Show(str.Substring(6));`

Table 8-7 The `Substring` method (*continued*)

Method	Description

str`.Substring(`*start*`,`*count*`)`	*str* is the name of a `string` variable or is a string expression. The argument *start* is an integer that specifies a position in *str*, and *count* is an integer that specifies a number of characters. The method returns a string containing the characters beginning at *start* and continuing for *count* characters.
	Example:
	`// The following code displays "cocoa".`
	`string str = "cocoa beans";`
	`MessageBox.Show(str.Substring(0, 5));`

Checkpoint

8.13 Write a statement that calls the `StartsWith` method of a string named `dessert`, passes the string "strawberry" as an argument, and stores the result in a `Boolean` variable named `found`.

8.14 What value is returned by the `IndexOf` and `LastIndexOf` methods if the substring being searched for is not found?

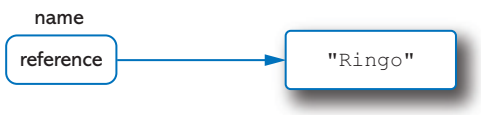
8.5 Manipulating Strings

CONCEPT: **Although `string` objects cannot be modified, there are several methods that create a modified copy of a `string` object.**

In C#, a `string` variable is a reference variable. Recall from chapter 7 that a reference variable holds a reference to an object in memory. When a string is assigned to a `string` variable, the `string` variable will hold a reference to a `string` object that exists in memory. For example, the following statement declares a `string` variable named `name`, and initializes it with the string "Ringo". After this statement executes, the `name` variable will hold a reference to a `string` object containing "Ringo". This is illustrated in Figure 8-3.

```
string name = "Ringo";
```

Figure 8-3 A **`string`** variable holds a reference to a **`string`** object



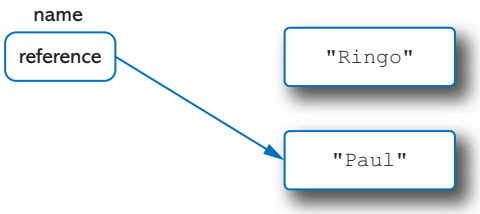
In C#, `string` objects are **immutable**, which means that once a `string` object is created, you cannot change the object's value. This might seem counterintuitive because you are allowed to write code such as this:

```
string name = "Ringo";
name = "Paul";
```

The first statement declares a `string` variable named `name` and initializes it with the string "Ringo". As previously shown in Figure 8-3, the `name` variable will reference a `string`

object containing "Ringo". Then, the second statement assigns a new string, "Paul", to `name`. This statement does not change the value of the `string` object that `name` currently references. Instead, it creates a new `string` object containing "Paul" and assigns the new object to `name`. After this statement executes, the `name` variable will hold a reference to a `string` object containing "Paul". This is illustrated in Figure 8-4.

Figure 8-4 A different object is assigned to the **`string`** variable



As you can see in Figure 8-4, the object containing "Ringo" is no longer referenced by any variables, so it cannot be accessed. Eventually, it will be removed from memory.

Methods for Manipulating Strings

Although you cannot change the contents of a `string` object, you can get a modified copy of a `string` object.

Table 8-8 describes several methods that `string` objects have for modifying the contents of a string in the following ways:

- The `Insert` method inserts a string into another string.
- The `Remove` methods remove specified characters from a string.
- The `ToLower` method converts a string to all lowercase characters.
- The `ToUpper` method converts a string to all uppercase characters.
- The `Trim` method removes all leading and trailing spaces from a string. (**Leading spaces** are spaces that appear at the beginning of a string, and **trailing spaces** are spaces that appear at the end of a string.)
- The `TrimStart` method removes all leading spaces from a string.
- The `TrimEnd` method removes all trailing spaces from a string.

The methods described in Table 8-8 return a modified copy of the calling `string` object. For example, look at the following code, which demonstrates the `Remove` method:

```
1 string str1 = "sunshine";
2 string str2 = str1.Remove(3);
3 MessageBox.Show(str1);
4 MessageBox.Show(str2);
```

Line 1 declares a `string` variable named `str1`, initialized with the string "sunshine". Line 2 calls the `str1.Remove` method, passing 3 as an argument. The method returns a copy of `str1` with all the characters from position 3 to the end of the string removed. (The method returns the string "sun".) The string that is returned is assigned to the `str2` variable. Line 3 displays the value reference by the `str1` variable, which is "sunshine", and line 4 displays the value reference by the `str2` variable, which is "sun".

Table 8-8 Methods for modifying a string

Method	Description

`str.Insert(start, strItem)`	*str* is the name of a `string` variable or is a string expression. The *start* argument is an integer that specifies a position in *str*, and the *strItem* argument is a string that is to be inserted. The method returns a string containing a copy of *str* with *strItem* inserted into it beginning at the position specified by *start*.
	Example:
	`// The following displays "New York City".`
	`string str1 = "New City";`
	`string str2 = str1.Insert(4, "York ");`
	`MessageBox.Show(str2);`

`str.Remove(start)`	*str* is the name of a `string` variable or is a string expression. The *start* argument is an integer that specifies a position in *str*. The method returns a string containing a copy of *str* with all the characters from the position specified by *start* to the end removed.
	Example:
	`// The following displays "blue".`
	`string str1 = "blueberry";`
	`string str2 = str1.Remove(4);`
	`MessageBox.Show(str2);`

`str.Remove(start, count)`	*str* is the name of a `string` variable or is a string expression. The *start* argument is an integer that specifies a position in *str*, and *count* is an integer that specifies a number of characters. The method returns a string containing a copy of *str* with *count* characters removed, beginning at the position specified by *start*.
	Example:
	`// The following displays "jelly doughnuts".`
	`string str1 = "jelly filled doughnuts";`
	`string str2 = str1.Remove(6, 7);`
	`MessageBox.Show(str2);`

`str.Replace(string1, string2)`	*str* is the name of a `string` variable or is a string expression. This method returns a copy of *str* in which all occurrences of *string1* have been replaced with *string2*.
	Example:
	`// The following displays "7 days ago"`
	`string str1 = "7 years ago";`
	`string str2 = str1.replace("years", "days");`
	`MessageBox.Show(str2);`

`str.ToCharArray()`	*str* is the name of a `string` variable or is a string expression. This method returns a `char` array containing the individual characters of *str*.
	Example:
	`// After this code executes, the arr array`
	`// will contain the elements 'A', 'B', and`
	`// 'C'.`
	`string str = "ABC";`
	`char[] arr = str.ToCharArray();`

(*continued*)

Table 8-8 Methods for modifying a string (*continued*)

Method	Description

str.ToLower()	*str* is the name of a string variable or is a string expression. The method returns a string containing a copy of *str* converted to lowercase.
	Example:
	`// The following displays "abc".`
	`string str1 = "ABC";`
	`string str2 = str1.ToLower();`
	`MessageBox.Show(str2);`

str.ToUpper()	*str* is the name of a string variable or is a string expression. The method returns a string containing a copy of *str* converted to uppercase.
	Example:
	`// The following displays "ABC".`
	`string str1 = "abc";`
	`string str2 = str1.ToUpper();`
	`MessageBox.Show(str2);`

str.Trim()	*str* is the name of a string variable or is a string expression. The method returns a string containing a copy of *str* with all leading and trailing spaces removed.
	Example:
	`// The following displays ">Hello<".`
	`string str1 = "   Hello   ";`
	`string str2 = str1.Trim();`
	`MessageBox.Show(">" + str2 + "<");`

str.Trim*Start*()	*str* is the name of a string variable or is a string expression. The method returns a string containing a copy of *str* with all leading spaces removed.
	Example:
	`// The following displays ">Hello<".`
	`string str1 = "   Hello";`
	`string str2 = str1.TrimStart();`
	`MessageBox.Show(">" + str2 + "<");`

str.TrimEnd()	*str* is the name of a string variable or is a string expression. The method returns a string containing a copy of *str* with all trailing spaces removed.
	Example:
	`// The following displays ">Hello<".`
	`string str1 = "Hello   ";`
	`string str2 = str1.TrimEnd();`
	`MessageBox.Show(">" + str2 + "<");`

In Tutorials 8-2 and 8-3, you use several of the string methods that we have discussed in applications that format and unformat US telephone numbers.

Tutorial 8-2:
Completing the *TelephoneFormat* Project

VideoNote
Tutorial 8-2: Completing the *Telephone Format* Project

Telephone numbers in the United States are commonly formatted to appear in the following manner:

```
(XXX)XXX-XXXX
```

In the format, `X` represents a digit. The three digits that appear inside the parentheses are the area code. The three digits following the area code are the prefix, and the four digits after the hyphen are the line number. Here is an example:

```
(919)555-1212
```

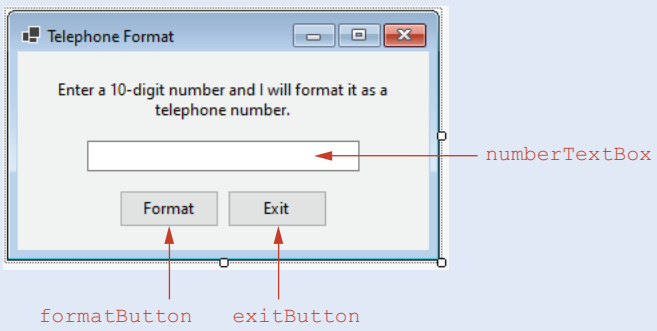
Although the parentheses and the hyphen make the number easier for people to read, those characters are unnecessary for processing by a computer. In a computer system, a telephone number is commonly stored as an unformatted series of digits, as shown here:

```
9195551212
```

Programs that work with telephone numbers sometimes need to unformat numbers that have been entered by the user. This means that the parentheses and the hyphen must be removed prior to storing the number in a file or processing it in some other way. In addition, such programs need the ability to format a number so it contains the parentheses and the hyphen before displaying it on the screen or printing it on paper.

In this tutorial, you complete the *TelephoneFormat* project. The application's form, which has already been created for you, is shown in Figure 8-5. When the application is complete, you will be able to enter a string of 10 digits into the `numberTextBox` control and click the *Format* button to see the string of digits formatted as a telephone number.

Figure 8-5 The *TelephoneFormat* project's form



Step 1: Start Visual Studio. Open the project named *TelephoneFormat* in the *Chap08* folder of the Student Sample Programs.

Step 2: Open the Form1 form's code in the code editor. Type the comments and code for the `IsValidNumber` method, shown in lines 10–40 of Program 8-2. The method accepts a string as an argument and returns `true` if the string is 10 characters long and contains only digits. Otherwise, the method returns `false`. You use this method to make sure that the user has entered a valid string of digits. Let's take a closer look at the code.

Line 15: This statement declares the constant `VALID_LENGTH`, initialized with the value 10, to represent the length of a valid string.

Line 16: This statement declares a `bool` variable named `valid`, initialized with the value `true`. The `valid` variable is used as a flag to indicate whether the `str` parameter is a valid number.

Line 19: This `if` statement determines whether the `str` parameter's length is valid. If so, the program continues at line 22. Otherwise, the program jumps to the `else` clause in line 32, and line 35 assigns `false` to the `valid` variable.

Line 22: This `foreach` loop steps through each of the characters in the `str` parameter. If the statement in line 26 determines that a character is *not* a digit, then the statement in line 28 assigns `false` to the `valid` variable.

Line 39: This statement returns the value of the `valid` variable.

Step 3: Type the comments and code for the `TelephoneFormat` method, shown in lines 42–54 of Program 8-2. The `TelephoneFormat` method accepts a string as an argument, passed by reference, and inserts the parentheses and the hyphen at the proper locations so it appears in the form `(XXX)XXX-XXXX`. Let's take a closer look at the code.

Line 44: Notice that the `ref` keyword is used in the declaration of the `str` parameter variable.

Line 47: This statement calls the `Insert` method to insert `"("` at position 0. All the characters in the string are automatically shifted right one space to accommodate the inserted character.

Line 50: This statement calls the `Insert` method to insert `")"` at position 4, shifting the characters that previously appeared beginning at position 4 to the right one space.

Line 53: This statement calls the `Insert` method to insert `"-"` at position 8, shifting the characters that previously appeared beginning at position 8 to the right one space. After this statement executes, the string referenced by `str` is formatted as `(XXX)XXX-XXXX`.

Step 4: Next, you create the Click event handlers for the Button controls. Switch back to the *Designer* and double-click the `formatButton` control. This opens the code editor, and you will see an empty event handler named `formatButton_Click`. Complete the event handler by typing the code shown in lines 58–72 in Program 8-2. Let's review this code:

Line 59: Notice the expression on the right side of the `=` operator is `numberTextBox.Text.Trim()`. This expression calls the `Trim` method on the Text property of the `numberTextBox` control. The method returns a copy of the `numberTextBox` control's Text property with all leading and trailing spaces removed. The result is assigned to the `input` variable.

Line 63: This `if` statement calls the `IsValidNumber` method, passing the `input` variable as an argument. If the value of the `input` variable is a valid number, the method returns `true` and the program continues to line 65. In line 65 the `TelephoneFormat` method is called, passing `input` by reference as an argument, and then line 66 displays the value of the `input` variable in a message box.

If the value of the `input` variable is not a valid number, the `IsValidNumber` method returns `false`, and the program jumps to the `else` clause in line 68. Then, line 71 displays an error message.

Step 5: Switch your view back to the *Designer* and double-click the `exitButton` control. In the code editor, you will see an empty event handler named

`exitButton_Click`. Complete the event handler by typing the code shown in lines 77–78 in Program 8-2.

Step 6: Save the project. Then, press F5 on the keyboard or click the *Start Debugging* button (▶) on the toolbar to compile and run the application. When the application runs, test various input values to confirm that the application works properly. For example, if you enter `9195551212`, the application should display `(919)555-1212`, and if you enter `abc5551212`, the application should display `"Invalid input"`. When you are finished, click the *Exit* button to exit the application.

Program 8-2 Completed code for Form1 in the *TelephoneFormat* project

```
1  namespace TelephoneFormat
2  {
3      public partial class Form1: Form
4      {
5          public Form1()
6          {
7              InitializeComponent();
8          }
9
10         // The IsValidNumber method accepts a string and
11         // returns true if it contains 10 digits, or false
12         // otherwise.
13         private bool IsValidNumber(string str)
14         {
15             const int VALID_LENGTH = 10; // Length of a valid string
16             bool valid = true; // Flag to indicate validity
17
18             // Check the string's length.
19             if (str.Length == VALID_LENGTH)
20             {
21                 // Check each character in str.
22                 foreach (char ch in str)
23                 {
24                     // If this character is not a digit, the
25                     // string is not valid.
26                     if (!char.IsDigit(ch))
27                     {
28                         valid = false;
29                     }
30                 }
31             }
32             else
33             {
34                 // Incorrect length.
35                 valid = false;
36             }
37
38             // Return the status.
39             return valid;
40         }
41
42         // The TelephoneFormat method accepts a string argument
43         // by reference and formats it as a telephone number.
44         private void TelephoneFormat(ref string str)
```

```
45          {
46              // First, insert the left paren at position 0.
47              str = str.Insert(0, "(");
48
49              // Next, insert the right paren at position 4.
50              str = str.Insert(4, ")");
51
52              // Next, insert the hyphen at position 8.
53              str = str.Insert(8, "-");
54          }
55
56          private void formatButton_Click(object sender, EventArgs e)
57          {
58              // Get a trimmed copy of the user's input.
59              string input = numberTextBox.Text.Trim();
60
61              // If the input is a valid number, format it
62              // and display it.
63              if (IsValidNumber(input))
64              {
65                  TelephoneFormat(ref input);
66                  MessageBox.Show(input);
67              }
68              else
69              {
70                  // Display an error message.
71                  MessageBox.Show("Invalid input");
72              }
73          }
74
75          private void exitButton_Click(object sender, EventArgs e)
76          {
77              // Close the form.
78              this.Close();
79          }
80      }
81  }
```

Tutorial 8-3:
Completing the *TelephoneUnformat* Project

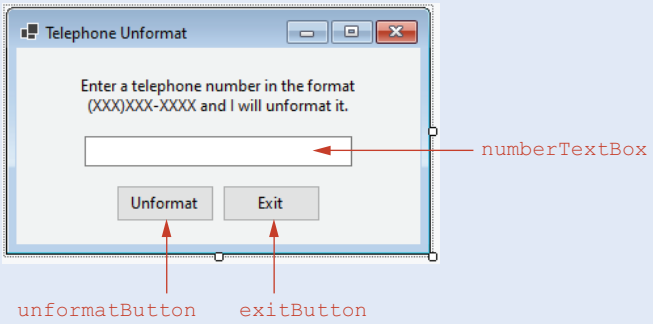
VideoNote
Tutorial 8-3: Completing the *Telephone-Unformat* Project

In this tutorial, you complete the *TelephoneUnformat* project. The application's form, which has already been created for you, is shown in Figure 8-6. When the application is complete, you will be able to enter a telephone number in the format `(XXX)XXX-XXXX` into the `numberTextBox` control and click the *Unformat* button to see it with the parentheses and the hyphen removed.

Step 1: Start Visual Studio. Open the project named *TelephoneUnformat* in the *Chap08* folder of the Student Sample Programs.

Step 2: Open the Form1 form's code in the code editor. Type the comments and code for the `IsValidFormat` method, shown in lines 10–34 of Program 8-3. The method accepts a string as an argument and returns `true` if the string is formatted as `(XXX)XXX-XXXX`. Otherwise, the method returns `false`. You use

Figure 8-6 The *TelephoneUnformat* project's form



this method to make sure that the user has entered a properly formatted telephone number. Let's take a closer look at the code:

Line 18: This statement declares the constant `VALID_LENGTH`, initialized with the value 13, to represent the length of a properly formatted telephone number.

Line 19: This statement declares a `bool` variable named `valid`, initialized with the value `true`. The `valid` variable is used as a flag to indicate whether the `str` parameter is properly formatted.

Lines 22–23: The `if-else` statement that begins in these lines determines whether the `str` parameter is properly formatted. In plain English, the statement should be interpreted like this:
If the string's length is 13 and the character at position 0 is `"("` *and the character at position 4 is* `")"` *and the character at position 8 is* `"-"` *Then*

Set valid to true.

Else

Set valid to false.

After the `if-else` statement executes, the `valid` variable will be set to either `true` or `false`, indicating whether `str` is properly formatted.

Line 33: This statement returns the value of the `valid` variable.

Step 3: Type the comments and code for the `Unformat` method, shown in lines 36–55 of Program 8-3. The `Unformat` method accepts a string as an argument, passed by reference. The method assumes that the string is formatted as `(XXX) XXX-XXXX`. When the method is finished, the parentheses and the hyphen will be removed from the string. Let's take a closer look at the code:

Line 44: This statement deletes the character at position 0, which is the `"("` character. All the remaining characters are automatically shifted left by one position to occupy the space left by the deleted character.

Line 49: This statement deletes the character at position 3, which is the `")"` character. The characters that previously appeared beginning at position 4 are automatically shifted left to occupy the space left by the deleted character.

Line 54: This statement deletes character at position 6, which is the hyphen. The characters previously appearing to the right of the hyphen are automatically moved left by one position. After this statement executes, the string referenced by `str` will be unformatted, appearing simply as a string of digits.

Step 4: Next, you create the Click event handlers for the Button controls. Switch back to the *Designer* and double-click the `unformatButton` control. This opens the code editor, and you will see an empty event handler named `unformatButton_Click`. Complete the event handler by typing the code shown in lines 59–73 in Program 8-3. Let's review this code:

Line 60: Notice the expression on the right side of the = operator is `numberTextBox.Text.Trim()`. This expression calls the `Trim` method on the Text property of the `numberTextBox` control. The method returns a copy of the `numberTextBox` control's Text property with all leading and trailing spaces removed. The result is assigned to the `input` variable.

Line 64: This `if` statement calls the `IsValidFormat` method, passing the `input` variable as an argument. If the value of the `input` variable is properly formatted, the method returns `true`, and the program continues to line 66. In line 66 the `Unformat` method is called, passing `input` by reference as an argument, and then line 67 displays the value of the `input` variable in a message box.

If the value of the `input` variable is not properly formatted, the `IsValidFormat` method returns `false`, and the program jumps to the `else` clause in line 69. Then, line 72 displays an error message.

Step 5: Switch your view back to the *Designer* and double-click the `exitButton` control. In the code editor you will see an empty event handler named `exitButton_Click`. Complete the event handler by typing the code shown in lines 78–79 in Program 8-3.

Step 6: Save the project. Then, press F5 on the keyboard or click the *Start Debugging* button (▶) on the toolbar to compile and run the application. When the application runs, test various input values to confirm that the application works properly. For example, if you enter `(919)555-1212`, the application should display `9195551212`, and if you enter `919-555-1212`, the application should display `"Invalid input"`. When you are finished, click the *Exit* button to exit the application.

Program 8-3 Completed code for Form1 in the *TelephoneUnformat* project

```
1  namespace TelephoneUnformat
2  {
3      public partial class Form1: Form
4      {
5          public Form1()
6          {
7              InitializeComponent();
8          }
9
10         // The IsValidFormat method accepts a string argument
11         // and determines whether it is properly formatted as
12         // a US telephone number in the following manner:
13         // (XXX)XXX-XXXX
14         // If the argument is properly formatted, the method
15         // returns true, otherwise false.
16         private bool IsValidFormat(string str)
17         {
18             const int VALID_LENGTH = 13; // Length of a valid string
19             bool valid;                  // Flag to indicate
                                            validity
20
21             // Determine whether str is properly formatted.
```

```
22              if (str.Length == VALID_LENGTH && str[0] == '(' &&
23                  str[4] == ')' && str[8] == '-')
24              {
25                  valid = true;
26              }
27              else
28              {
29                  valid = false;
30              }
31
32              // Return the value of valid.
33              return valid;
34          }
35
36          // The unformat method accepts a string, by reference,
37          // assumed to contain a telephone number formatted in
38          // this manner: (XXX)XXX-XXXX.
39          // The method unformats the string by removing the
40          // parentheses and the hyphen.
41          private void Unformat(ref string str)
42          {
43              // First, delete the left paren at position 0.
44              str = str.Remove(0, 1);
45
46              // Next, delete the right paren. Because of the
47              // previous deletion it is now located at
48              // position 3.
49              str = str.Remove(3, 1);
50
51              // Next, delete the hyphen. Because of the
52              // previous deletions it is now located at
53              // position 6.
54              str = str.Remove(6, 1);
55          }
56
57          private void unformatButton_Click(object sender, EventArgs e)
58          {
59              // Get a trimmed copy of the user's input.
60              string input = numberTextBox.Text.Trim();
61
62              // If the input is properly formatted, unformat it
63              // and display it.
64              if (IsValidFormat(input))
65              {
66                  Unformat(ref input);
67                  MessageBox.Show(input);
68              }
69              else
70              {
71                  // Display an error message.
72                  MessageBox.Show("Invalid input");
73              }
74          }
75
76          private void exitButton_Click(object sender, EventArgs e)
77          {
78              // Close the form.
79              this.Close();
80          }
81      }
82  }
```

Checkpoint

8.15 A program has two `string` variables named `str1` and `str2`. Write a statement that trims the leading and trailing whitespace characters from the `str1` variable and assigns the result to the `str2` variable.

8.16 A program has two `string` variables named `vegetable` and `veggie`. Write a statement that assigns an all lowercase copy of the `vegetable` variable to the `veggie` variable.

8.17 A program has two `string` variables named `str1` and `str2`. Write a statement that inserts the string `"www"` into `str1` at index 5 and assigns the result to the `str2` variable.

8.18 A program has two `string` variables named `str1` and `str2`. Write a statement that removes the first 6 characters from `str1` and assigns the result to the `str2` variable.

8.19 A program has two `string` variables named `str1` and `str2`. Write a statement that replaces all occurrences of `"dollars"` with `"euros"` in `str1` and assigns the result to the `str2` variable.

8.6 The `StringBuilder` Class

CONCEPT: **The `StringBuilder` class allows you to create objects that are like strings, but mutable.**

`StringBuilder` is a .NET class that allows you to create objects that are similar to `string` objects. `StringBuilder` objects are **mutable**, however. This means you can change the contents of a `StringBuilder` object without creating a new object in memory. You can change specific characters, insert characters, delete characters, and perform other operations. A `StringBuilder` object will grow or shrink in size, as needed, to accommodate the changes.

The fact that string objects are immutable is rarely a problem, but you might consider using `StringBuilder` objects if your program needs to make a lot of changes to one or more strings. This will improve the program's performance by reducing the number of `string` objects that must be created and then removed when they are no longer being used.

The `StringBuilder` class is in the `System.Text` namespace in .NET, so you will need to write the following statement at the top of any program that uses the class:

```
using System.Text;
```

Without this statement, the C# compiler will not be able to find the `StringBuilder` class.

Capacity and Length

When a `StringBuilder` object is created, a block of memory is allocated for the string that will be stored in the object. The number of characters that this block of memory can hold is the object's **capacity**. By default, a `StringBuilder` object's capacity is 16 characters. This initial capacity of 16 characters is a starting point, however. If you store a string that is larger than 16 characters in the `StringBuilder` object, the object's capacity will automatically grow to accommodate the string.

A StringBuilder object's length is the length of the string that is stored in the object. For example, suppose we create a StringBuilder object and assign it the string "Kiran". The object's length would be 5, and its capacity would be 16.

Creating a StringBuilder Object

There are several different ways to create a StringBuilder object. Here is an example of how you would create an empty StringBuilder object:

```
StringBuilder sb = new StringBuilder();
```

After this statement executes, the sb variable will reference a StringBuilder object whose value is set to String.Empty. If the declaration appears inside a method, you can use the var keyword, as shown here:

```
var sb = new StringBuilder();
```

You can initialize a StringBuilder object with a string when you create it. Here is an example:

```
var name = new StringBuilder("Trinidad");
```

After this statement executes, the name variable will reference a StringBuilder object whose value is "Trinidad".

As previously mentioned, a StringBuilder object's default capacity is 16 characters. If you wish, you can specify the object's capacity when you create it. Here is an example:

```
var sb = new StringBuilder(100);
```

This statement creates a StringBuilder object with a capacity of 100, and whose value is set to String.Empty. Here is an example of initializing a StringBuilder object with a string and specifying its capacity:

```
var name = new StringBuilder("Trinidad", 100);
```

You will not usually need to specify a StringBuilder object's capacity because the capacity automatically expands to accommodate the size of the string that is assigned to the object. However, if you've written some code that causes a StringBuilder object to frequently increase its capacity, you might improve the code's performance by giving the StringBuilder object a large capacity.

Retrieving a StringBuilder Object's String Value

You can use the ToString method to retrieve a StringBuilder object's string value. Here is an example that displays the value of a StringBuilder object in a message box:

```
var sb = new StringBuilder("Berkeley");
MessageBox.Show(sb.ToString());
```

Here is an example that assigns a StringBuilder object's value to a Label control's Text property:

```
var sb = new StringBuilder("Berkeley");
outputLabel.Text = sb.ToString();
```

You can also use an interpolated string to retrieve a StringBuilder object's value, as shown in the following code:

```
var sb = new StringBuilder("Berkeley");
MessageBox.Show($"{sb}");
outputLabel.Text = $"{sb}";
```

StringBuilder Properties

Table 8-9 describes three of the StringBuilder class's properties.

Table 8-9 StringBuilder properties

Property	Description

Capacity	An int property that gets or sets the object's capacity.
	Example:
	`var sb = new StringBuilder();`
	`sb.Capacity = 100;`
	`MessageBox.Show($"The capacity is {sb.Capacity}.");`

Length	An int property that gets or sets the object's length. If you set the Length property to a value less than the object's current length, the object's value is truncated. If you set the Length property to a value greater than the object's current length, extra NULL characters are added to the end of the object's current value. (The NULL character is ASCII code 0.)
	Example:
	`var city = new StringBuilder("Cupertino");`
	`MessageBox.Show($"The length of {city} is {city.Length}.");`
	`city.Length = 5;`
	`MessageBox.Show($"The length of {city} is {city.Length}.");`
	`// The code above displays:`
	`// The length of Cupertino is 9.`
	`// The length of Cuper is 5.`

MaxCapacity	Read-only int property. Gets the maximum amount that the object's capacity can be set to.
	Example:
	`var sb = new StringBuilder();`
	`MessageBox.Show($"The max capacity is {sb.MaxCapacity}.");`

Using Indexes with a StringBuilder Object

As with regular strings, you can access the individual characters in a StringBuilder object using subscript notation. This makes it possible to work with a string as if it were an array of characters. You use subscript 0 to access the first character, subscript 1 to access the second character, and so forth. The subscript of the last character is 1 less than the string's length. For example, the following code uses a for loop to convert each character in a StringBuilder to uppercase:

```
var name = new StringBuilder("Anjali");
for (int i = 0; i < name.Length; i++)
{
    name[i] = char.ToUpper(name[i]);
}
```

You can also use an index to change an individual character in a StringBuilder. For example, the following code changes the first character of name to 'D'. After the code executes, the value of name will be "Dom":

```
var name = new StringBuilder("Tom");
name[0] = 'D';
```

NOTE: You cannot use the `foreach` loop to step through the characters of a `StringBuilder` unless you use the `ToString` method to convert the `StringBuilder` to a regular string. The following code shows an example:

```
var name = new StringBuilder("Anjali");
foreach (char ch in name.ToString())
{
    MessageBox.Show($"{ch}");
}
```

Appending Data to a `StringBuilder`

You cannot use the assignment operator (=) to assign a value to a `StringBuilder` object. Instead, you use the `Append` method to append data to a `StringBuilder` object. There are several different versions of the `Append` method that allow you to append strings, characters, numbers, and other forms of data to an existing `StringBuilder` object's value. Let's look at some examples of how to use the `Append` method.

Appending Strings

You can use the `Append` method to append a string to an existing `StringBuilder` object. Here is an example:

```
var city = new StringBuilder();
city.Append("Boston");
```

The first statement creates an empty `StringBuilder` object named `city`. The second statement appends the string "Boston" to the `city` object's existing value. After this code executes, the `city` object's value will be "Boston".

Here is an example where we append a string to a `StringBuilder` that already contains a value:

```
var sb = new StringBuilder("abc");
sb.Append("def");
```

The first statement creates a `StringBuilder` object named `sb`., initialized with the string "abc". The second statement appends the string "def" to the `sb` object's existing value. After this code executes, the `sb` object's value will be "abcdef".

You can also append an interpolated string to a `StringBuilder` object. Here is an example:

```
double temperature = 45.7;
var status = new StringBuilder();
status.Append($"The temperature is {temperature}.");
```

After this code executes, the `status` object's value will be "The temperature is 45.7.".

If you want to append part of a string (a substring) to a `StringBuilder` object, you pass three arguments to the `Append` method: the string that contains the substring, the starting index of the substring, and the number of characters in the substring. Here is an example:

```
int start = 2;
int count = 3;
var sb = new StringBuilder();
sb.Append("abcdef", start, count);
```

After this code executes, the `sb` object's value will be "cde".

Appending Characters and Character Arrays

You can use the `Append` method to append a character to an existing `StringBuilder` object. Here is an example:

```
var sb = new StringBuilder("wxy");
sb.Append('z');
```

The first statement creates an empty `StringBuilder` object named `sb`, initialized with the string "wxy". The second statement appends the character 'z' to the `sb` object's existing value. After this code executes, the `sb` object's value will be "wxyz".

If you want to append multiple copies of a character to a `StringBuilder` object, you pass the character and the number of copies as arguments. Here is an example:

```
var sb = new StringBuilder();
sb.Append('*', 5);
```

After this code executes, the `sb` object's value will be "*****".

If you pass an array of characters to the `Append` method, the array's contents will be appended, as a string, to the `StringBuilder`'s value. Here is an example:

```
char[] letters = {'d', 'e', 'f'};
var sb = new StringBuilder("abc");
sb.Append(letters);
```

After this code executes, the `sb` object's value will be "abcdef".

If you want to append a range of elements from a character array to a `StringBuilder` object, you pass three arguments to the `Append` method: a character array, the starting index of the range, and the number of characters to append. Here is an example:

```
char[] letters = {'a', 'b', 'c', 'd', 'e', 'f'};
int start = 2;
int count = 3;
var sb = new StringBuilder();
sb.Append(letters, start, count);
```

After this code executes, the `sb` object's value will be "cde".

Appending Numbers and Booleans

You can use the `Append` method to append the string representation of a number to an existing `StringBuilder` object. Here is an example:

```
int number = 99;
var sb = new StringBuilder("Points earned: ");
sb.Append(number);
```

After this code executes, the `sb` object's value will be "Points earned: 99".

The `Append` method can accept numbers of any data type. Here is an example that appends a decimal to a `StringBuilder`:

```
decimal price = 14.99m;
var sb = new StringBuilder("The price is $");
sb.Append(price);
```

After this code executes, the `sb` object's value will be "The price is $14.99".

You can also use the `Append` method to append the string representation of a `bool` value (which is either "True" or "False") to an existing `StringBuilder` object. Here is an example:

```
bool status = true;
```

```
var sb = new StringBuilder();
sb.Append(status);
```

After this code executes, the `sb` object's value will be "True".

Appending `StringBuilder` Objects

You can use the `Append` method to append the value of a `StringBuilder` object to another `StringBuilder` object. Here is an example:

```
var firstName = new StringBuilder("Lucero");
var lastName = new StringBuilder("Navarro");
var fullName = new StringBuilder();
fullName.Append(firstName);
fullName.Append(" ");
fullName.Append(lastName);
```

After this code executes, the `fullName` object's value will be "Lucero Navarro".

Inserting Data into a `StringBuilder`

You use the `Insert` method to insert data at a specific position in a `StringBuilder` object. There are several different versions of the `Insert` method that allow you to insert strings, characters, numbers, and other forms of data into an existing `StringBuilder` object's value. Each version of the `Insert` method uses this general format:

```
sb.Insert(position, dataToInsert);
```

In the general format, *`sb`* is the name of a `StringBuilder` object, *`position`* is the integer index in the `StringBuilder` where the data will be inserted, and *`dataToInsert`* is the data to insert. Here is an example of how you would insert a string into an existing `StringBuilder` object:

```
var sentence = new StringBuilder("The glass is half full.");
sentence.Insert(4, "red ");
```

This code inserts the string "red " into the `StringBuilder` object at index 4. After the code executes, the `sentence` object's value will be "The red glass is half full."

The `Insert` method can insert numbers of any data type. Here is an example of how you would insert the value of an `int` into an existing `StringBuilder` object:

```
var sb = new StringBuilder("We drove 40 miles.");
sb.Insert(11, 0);
```

This code converts the integer 0 to a string, and then inserts it into the `StringBuilder` object at index 11. After the code executes, the `sb` object's value will be "We drove 400 miles." The following example shows a `decimal` value being inserted into a `StringBuilder` object:

```
var sb = new StringBuilder("The balance is.");
decimal amount = 100m;
sb.Insert(15, amount);
```

This code converts the `decimal` variable amount to a string, and then inserts it into the `StringBuilder` object at index 15. After the code executes, the `sb` object's value will be "The balance is 100." The following example shows a `double` value being inserted into a `StringBuilder` object:

```
var sb = new StringBuilder("<>");
double amount = 3.14;
sb.Insert(1, amount);
```

This code converts the `double` variable amount to a string, and then inserts it into the `StringBuilder` object at index 1. After the code executes, the `sb` object's value will be "3.14".

When you insert a `bool` value into a `StringBuilder` object, the value is converted to its string representation (either "True" or "False") and then inserted at the specified position. Here is an example:

```
bool status = true;
var sb = new StringBuilder("Status: ");
sb.Insert(sb.Length, status);
```

After this code executes, the `sb` object's value will be "Status: True".

Removing Characters from a `StringBuilder`

You can use the `Remove` method to remove a range of characters from a `StringBuilder` object. Here is the general format how you call the method:

```
sb.Remove(start, count);
```

In the general format, *`sb`* is the name of a `StringBuilder` object, *`start`* is the index of the first character in the range that is to be removed, and *`count`* is the number of characters to remove. Here is an example:

```
var sb = new StringBuilder("64 gigabytes of memory");
sb.Remove(3, 4);
```

This code removes four characters, starting at index 3, from sb. After this code executes, sb will contain the string `"64 bytes of memory"`.

Replacing Characters and Substrings in a `StringBuilder`

You can use the `Replace` method to replace all occurrences of a specified character or string within a `StringBuilder` with another character or string. To replace all occurrences of a specific character, use the following general format of the `Replace` method:

```
sb.Replace(ch, newCh);
```

In the general format, *`sb`* is the name of a `StringBuilder` object, *`ch`* is the character to be replaced, and *`newCh`* is the character that you want to replace *`ch`* with. After the method executes, all occurrences of *`oldCh`* will be replaced with *`newCh`*. Here is an example:

```
var sb = new StringBuilder("YYZZQQYY");
sb.replace('Y', 'b');
```

The code replaces every occurrence of the character `'Y'` with the character `'b'`. After the code executes, `sb` will contain the string `"bbZZQQbb"`.

To replace all occurrences of a substring, use the following general format of the `Replace` method:

```
sb.Replace(substring, newSubstring);
```

In the general format, *`sb`* is the name of a `StringBuilder` object, *`substring`* is the substring to be replaced, and *`newSubstring`* is the string that you want to replace *`substring`* with. After the method executes, all occurrences of *`substring`* will be replaced with *`newSubstring`*. Here is an example:

```
var sb = new StringBuilder("yes maybe yes maybe yes maybe");
sb.replace("yes", "no");
```

The code replaces every occurrence of the substring `"yes"` with the string `"no"`. After the code executes, `sb` will contain the string `"no maybe no maybe no maybe"`.

Sometimes you don't want to replace every occurrence of a substring or character in a StringBuilder, but every occurrence within a region of the StringBuilder. To replace all occurrences of a character within a specified region of the StringBuilder, use the following general format of the Replace method:

```
sb.Replace(ch, newCh, start, count);
```

In the general format, *sb* is the name of a StringBuilder object, *ch* is the character to be replaced, *newCh* is the string that you want to replace *ch* with, *start* is the starting index of the region within the StringBuilder, and *count* is the length of the region. After the method executes, all occurrences of *ch* within the specified region will be replaced with *newCh*. Here is an example:

```
var sb = new StringBuilder("zzzzzzzzzz");
sb.Replace('z', 'a', 2, 5);
```

The code replaces every occurrence of the character 'z' with the character 'a' within the region starting at index 2 and spanning 5 characters. After the code executes, sb will contain the string "zzaaaaazzz".

To replace all occurrences of a substring within a specified region of the StringBuilder, use the following general format of the Replace method:

```
sb.Replace(substring, newSubstring, start, count);
```

In the general format, *sb* is the name of a StringBuilder object, *substring* is the substring to be replaced, *newSubstring* is the string that you want to replace *substring* with, *start* is the starting index of the region within the StringBuilder, and *count* is the length of the region. After the method executes, all occurrences of *substring* within the specified region will be replaced with *newSubstring*. Here is an example:

```
var sb = new StringBuilder("hello hello hello hello");
sb.Replace("hello", "hi", 5, 13);
```

The code replaces every occurrence of the substring "hello" with the string "hi" within the region starting at index 5 and spanning 13 characters. After the code executes, sb will contain the string "hello hi hi hello".

Other StringBuilder Methods

Table 8-10 describes some of the other StringBuilder methods.

Table 8-10 Miscellaneous **StringBuilder** methods

sb.Clear();	*sb* is the name of a StringBuilder object. The method removes all the characters stored in the object.
	Example:
	`// Assume sb is a StringBuilder.`
	`sb.Clear();`

sb.CopyTo(*startFrom*, *array*, *startTo*, *count*);	*sb* is the name of a StringBuilder object. The method copies specified characters from *sb* to a character array. The *startFrom* argument is an integer specifying the starting index of the characters to be copied from *sb*. The *array* argument is the character array. The *startTo* argument is an integer specifying the starting index in the array where the characters will be copied. The *count* argument is an integer specifying the number of characters to copy.

(*continued*)

Table 8-10 Miscellaneous **`StringBuilder`** methods (*continued*)

	Example:
	`// After the following code executes, arr will contain`
	`// the elements 'C', 'D', 'E', '*', '*', '*'.`
	`var sb = new StringBuilder("ABCDEFGH");`
	`char[] arr = new char[6] {'*', '*', '*', '*', '*', '*',};`
	`sb.CopyTo(2, arr, 0, 3);`

sb`.EnsureCapacity(`*capacity*`);`	*sb* is the name of a `StringBuilder` object. The method ensures that the capacity of *sb* is at least the value specified by the integer argument *capacity*.
	Example:
	`// The following code sets the capacity of sb to 99.`
	`var sb = new StringBuilder();`
	`sb.EnsureCapacity(99);`

sb1`.Equals(`*sb2*`);`	*sb1* and *sb2* are the names of two `StringBuilder` objects. The method returns `true` if the two objects contain the same characters, and their Capacity and MaxCapacity properties are equal. Otherwise, the method returns `false`.
	Example:
	`// The following code displays "The two are equal."`
	`var sb1 = new StringBuilder("ABC");`
	`var sb2 = new StringBuilder("ABC");`
	`if (sb1.Equals(sb2))`
	`  MessageBox.Show("The two are equal.");`

In Tutorial 8-4 you will complete an application that uses a `StringBuilder` object to play a word game.

Tutorial 8-4
Completing the *WordGame* Project

VideoNote
Tutorial 8-4: Completing the *WordGame* Project

In this tutorial you will complete a project that plays a "Mad Libs" style word game. The application will let the user enter the following items:

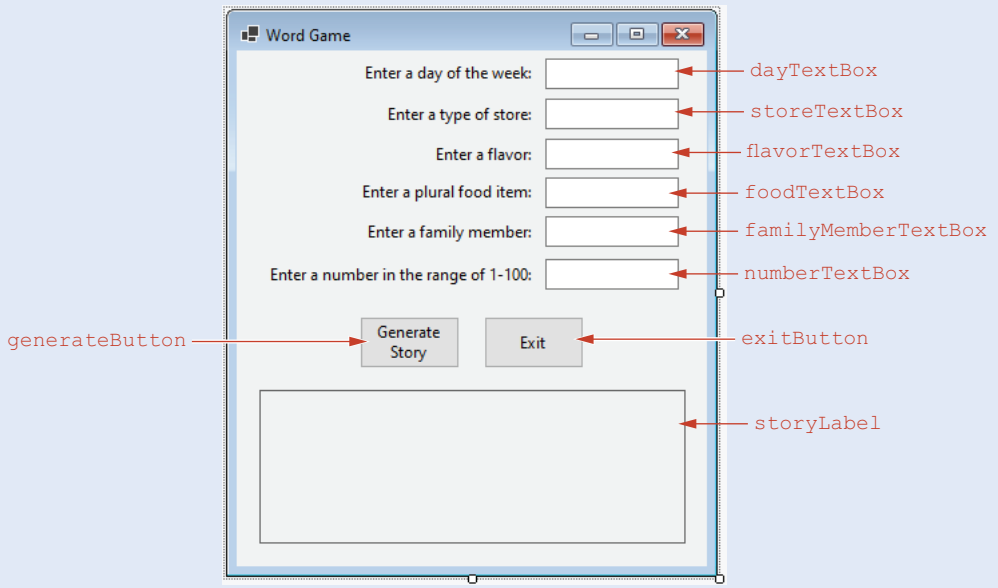
- A day of the week
- A type of store
- A flavor
- A food item, plural
- A family member
- A number in the range of 1 through 100.

After these items have been entered, the user can click a button to generate the following story with the user's input inserted at the appropriate locations.

Every <*day-of-week*> we walk to the local <*store*> and buy <*flavor*> <*food-item-plural*>. On the way back home visit our <*family-member*>, who is <*number*> years old.

The application's form, which has already been created for you, is shown in Figure 8-7.

Figure 8-7 The *WordGame* project's form



Step 1: Start Visual Studio. Open the project named *WordGame* in the *Chap08* folder of the Student Sample Programs.

Step 2: Open the Form1 form's code in the code editor. Insert the `using System.Text;` statement shown in line 1 of Program 8-4. This statement is necessary because the `StringBuilder` class is part of the `System.Text` namespace in .NET.

Step 3: Write the comments and code for the `NumberIsValid` method, shown in lines 12–24 of Program 8-4. The method examines the contents of the `numberTextBox` control to make sure it contains an integer in the range of 1–100. If the control contains a valid number, the method returns `true`. Otherwise, it returns `false`.

Step 4: Next, you create the Click event handlers for the Button controls. Switch back to the *Designer* and double-click the `generateButton` control. This opens the code editor, and you will see an empty event handler named `generateButton_Click`. Complete the event handler by typing the code shown in lines 28–49 in Program 8-4. Let's review this code:

Lines 28–32: This statement creates a `StringBuilder` object named `story` and initializes it with a string that contains our story. Notice the string contains the following values at various locations within the string:

- `<day-of-week>`
- `<store>`
- `<flavor>`
- `<food-item-plural>`
- `<family-member>`
- `<number>`

These values will act as placeholders for the items that the user will input. To generate the story, the application will replace each of these placeholders with the corresponding value that was entered by the user.

Line 25: This `if` statement calls the `NumberIsValid` method to validate the contents of the `numberTextBox` control. If the control contains a valid value, the statements in lines 37–43 will be executed. Otherwise, the program will jump to the `else` clause in line 45.

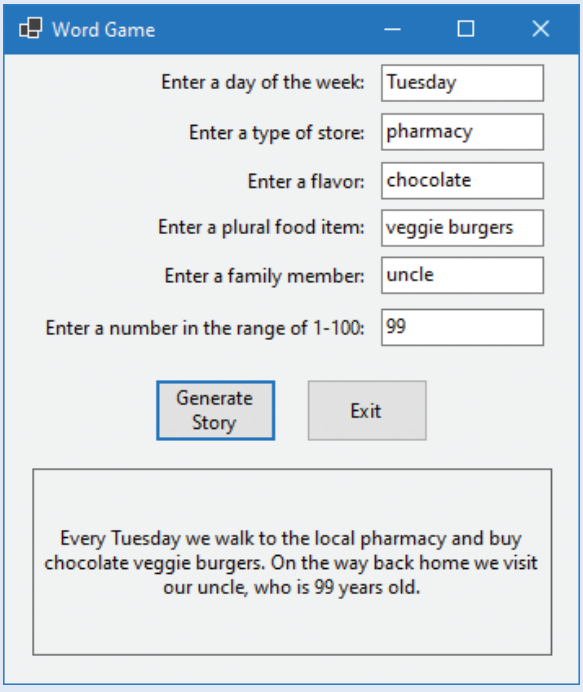
Lines 37–43: These statements replace the placeholders in the story object with the values entered by the user. Line 37 replaces the `<day-of-week>` placeholder with the contents of the `dayTextBox` control, line 38 replaces the `<store>` placeholder with the contents of the `storeTextBox` control, and so forth.

Lines 47–48: These statements execute if the user entered an invalid value in the `numberTextBox` control. Line 47 displays an error message, and line 48 gives the focus to the `numberTextBox` control.

Step 5: Switch back to the *Designer* and double-click the `exitButton` control. In the code editor you will see an empty event handler named `exitButton_Click`. Complete the event handler by typing the code shown in lines 54–55 in Program 8-4.

Step 6: Save the project. Then, press F5 on the keyboard or click the Start Debugging button (▶) on the toolbar to compile and run the application. When the application runs, enter various input values to confirm that the application works properly. Figure 8-8 shows an example of the application's form after the user has entered some sample input.

Figure 8-8 Sample input values for the *WordGame* application



Program 8-4 Completed code for Form1 in the *WordGame* project

```
1 using System.Text;
2
3 namespace WordGame
```

```
 4 {
 5     public partial class Form1: Form
 6     {
 7         public Form1()
 8         {
 9             InitializeComponent();
10         }
11
12         // The NumberIsValid method returns true if the numberTextBox
13         // control contains a number in the range of 1-100.
           Otherwise it
14         // returns false.
15         private bool NumberIsValid()
16         {
17             bool status = false;
18             int number;
19
20             if (int.TryParse(numberTextBox.Text, out number))
21                 if (number >= 1 && number <= 100)
22                     status = true;
23             return status;
24         }
25
26         private void generateButton_Click(object sender, EventArgs e)
27         {
28             var story = new StringBuilder(
29                 "Every <day-of-week> we walk to the local <store> " +
30                 "and buy <flavor> <food-item-plural>. On the way " +
31                 "back home we visit our <family-member>, who is " +
32                 "<number> years old.");
33
34             // If the number of years is valid, generate the story.
35             if (NumberIsValid())
36             {
37                 story.Replace("<day-of-week>", dayTextBox.Text);
38                 story.Replace("<store>", storeTextBox.Text);
39                 story.Replace("<flavor>", flavorTextBox.Text);
40                 story.Replace("<food-item-plural>", foodTextBox.Text);
41                 story.Replace("<family-member>", familyMember
                   TextBox.Text);
42                 story.Replace("<number>", numberTextBox.Text);
43                 storyLabel.Text = story.ToString();
44             }
45             else
46             {
47                 MessageBox.Show("Enter an integer in the range of
                   1-100.");
48                 numberTextBox.Focus();
49             }
50         }
51
52         private void exitButton_Click(object sender, EventArgs e)
53         {
54             // Close the form.
55             this.Close();
56         }
57     }
58 }
```

Checkpoint

8.20 The `StringBuilder` class is mutable. What does this mean?

8.21 In a program that makes lots of changes to strings, would it be more efficient to use string objects or `StringBuilder` objects? Why?

8.22 Write a statement that declares a `StringBuilder` object named `city`, initialized with the string `"Regensburg"`.

8.23 You wish to add a string to the end of the existing contents of a `StringBuilder` object. What method do you use?

8.24 You wish to insert a string into the existing contents of a `StringBuilder` object. What method do you use?

8.25 You wish to delete a specific character from the existing contents of a `StringBuilder` object. What method do you use?

8.26 You wish to change a specific character in a `StringBuilder` object. What method do you use?

8.7 Tokenizing Strings

CONCEPT: **Tokenizing a string is a process of breaking a string down into smaller string components, which are called tokens. In C#, `string` objects have a `Split` method that tokenizes the contents of the string.**

Sometimes, a string contains a series of words or other items of data separated by spaces or other characters. For example, look at the following string:

```
"peach raspberry strawberry vanilla"
```

This string contains the following four items of data: peach, raspberry, strawberry, and vanilla. In programming terms, items such as these are known as **tokens**. Notice that a space appears between the items. The character that separates tokens is known as a **delimiter**. Here is another example:

```
"17;92;81;12;46;5"
```

This string contains the following tokens: 17, 92, 81, 12, 46, and 5. Notice that a semicolon appears between each item. In this example, the semicolon is used as a delimiter. Some programming problems require you to read a string that contains a list of items and then extract all the tokens from the string for processing. For example, look at the following string that contains a date:

```
"3-22-2023"
```

The tokens in this string are 3, 22, and 2023, and the delimiter is the hyphen character. Perhaps a program needs to extract the month, day, and year from such a string.

The process of breaking a string into tokens is known as **tokenizing**. In C#, `string` objects have a method named `Split` that is used to tokenize the string. When you call a `string` object's `Split` method, the method extracts tokens from the string and returns them as an array of strings. Each element in the array is one of the tokens.

We discuss two ways that you can call the `Split` method. The first way is to pass the value `null` as an argument. When you pass `null` as an argument to the `Split` method,

the method tokenizes the string using white-space characters as delimiters. (White-space characters are the space, tab, linefeed, carriage-return, formfeed, vertical-tab, and newline characters.) The method returns a `string` array, with each element of the array containing one of the tokens. The following code shows an example:

```
 1 // Create a String to tokenize.
 2 string str = "one two three four";
 3
 4 // Get the tokens from the string.
 5 string[] tokens = str.Split(null);
 6
 7 // Display each token.
 8 foreach (string s in tokens)
 9 {
10     MessageBox.Show(s);
11 }
```

Let's take a closer look at the code:

- Line 2 declares a `string` variable named `str`, initialized with the string `"one two three four"`.
- Line 5 declares a `string` array named `tokens`. The `tokens` array is initialized with the array that is returned from the `str.Split(null)` method call. After this statement executes, the `tokens` array will have four elements, containing the strings `"one"`, `"two"`, `"three"`, and `"four"`.
- The `foreach` loop in lines 8–11 displays each of the `tokens` array's elements in message boxes. When the loop executes, four message boxes will be displayed, one after the other, showing the strings `"one"`, `"two"`, `"three"`, and `"four"`.

The second way that we call the `Split` method is to pass a `char` array as an argument. Each element of the `char` array is used as a delimiter. The following code shows an example:

```
 1 // Create a String to tokenize.
 2 string str = "one;two;three;four";
 3
 4 // Create an array of delimiters.
 5 char[] delim = { ';' };
 6
 7 // Get the tokens from the string.
 8 string[] tokens = str.Split(delim);
 9
10 // Display each token.
11 foreach (string s in tokens)
12 {
13     MessageBox.Show(s);
14 }
```

Let's take a closer look at the code:

- Line 2 declares a `string` variable named `str`, initialized with the string `"one;two;three;four"`.
- Line 5 declares a `char` array named `delim`. Notice that the initialization list contains only one character. The `delim` array has one element, containing the `';'` character.
- Line 8 declares a `string` array named `tokens`. The `tokens` array is initialized with the array that is returned from the `str.Split(delim)` method call. After this statement executes, the `tokens` array will have four elements, containing the strings `"one"`, `"two"`, `"three"`, and `"four"`.
- The `foreach` loop in lines 11–14 displays each of the `tokens` array's elements in message boxes. When the loop executes, four message boxes will be displayed, one after the other, showing the strings `"one"`, `"two"`, `"three"`, and `"four"`.

Some situations require that you use multiple characters as delimiters in the same string. For example, look at the following e-mail address:

```
joe@gaddisbooks.com
```

This string uses two delimiters: `@` (the *at* symbol) and `.` (the period). To extract the tokens from this string, we must specify both characters as delimiters. Here is an example:

```
 1 // Create a String to tokenize.
 2 string str = "joe@gaddisbooks.com";
 3
 4 // Create an array of delimiters.
 5 char[] delim = { '@', '.' };
 6
 7 // Get the tokens from the string.
 8 string[] tokens = str.Split(delim);
 9
10 // Display each token.
11 foreach (string s in tokens)
12 {
13     MessageBox.Show(s);
14 }
```

Let's take a closer look at the code:

- Line 2 declares a `string` variable named `str`, initialized with the string `"joe@gaddisbooks.com"`.
- Line 5 declares a `char` array named `delim`, initialized with the characters `'@'` and `'.'`.
- Line 8 declares a `string` array named `tokens`. The `tokens` array is initialized with the array that is returned from the `str.Split(delim)` method call. After this statement executes, the `tokens` array will have three elements, containing the strings `"joe"`, `"gaddisbooks"`, and `"com"`.
- The `foreach` loop in lines 11–14 displays each of the `tokens` array's elements in message boxes. When the loop executes, four message boxes will be displayed, one after the other, showing the strings `"joe"`, `"gaddisbooks"`, and `"com"`.

Trimming a String before Tokenizing

When you are tokenizing a string that was entered by the user and you are using characters other than white spaces as delimiters, you probably want to trim the string before tokenizing it. Otherwise, if the user enters leading white-space characters, they will become part of the first token. Likewise, if the user enters trailing white-space characters, they will become part of the last token. For example, look at the following code:

```
 1 // Create a string with leading and trailing whitespaces.
 2 String str = "   one;two;three   ";
 3
 4 // Create a char array containing the semicolon.
 5 char[] delim = { ';' };
 6
 7 // Get the tokens from the string.
 8 string[] tokens = str.Split(delim);
 9
10 // Display each token.
11 foreach (string s in tokens)
12 {
13     MessageBox.Show($"*{s}*");
14 }
```

This code produces three message boxes, displaying the following output:

```
*   one*
*two*
*three   *
```

Notice that the first token contains the leading spaces and the last token contains the trailing spaces. To prevent leading and/or trailing white-space characters from being included in the first and last tokens, use the `Trim` method to remove them. Here is the same code, modified to use the `Trim` method:

```
// Create a string with leading and trailing white spaces.
String str = "   one;two;three   ";

// Create a char array containing the semicolon.
char[] delim = { ';' };

// Trim the string.
str = str.Trim();

// Get the tokens from the string.
string[] tokens = str.Split(delim);

// Display each token.
foreach (string s in tokens)
{
    MessageBox.Show($"*{s}*");
}
```

This code produces three message boxes, displaying the following output:

```
*one*
*two*
*three*
```

In Tutorial 8-5, you complete an application that reads and tokenizes data from a file that has been exported by Microsoft Excel.

Tutorial 8-5:
Completing the *CSVReader* Project

VideoNote
Tutorial 8-5: Completing the *CSVReader* Project

A professor keeps her students' test scores in a Microsoft Excel spreadsheet. Figure 8-9 shows a set of five test scores for five students. Each column holds a test score, and each row represents the scores for one student.

Figure 8-9 Microsoft Excel spreadsheet

	A	B	C	D	E	F
1	87	79	91	82	94	
2	72	79	81	74	88	
3	94	92	81	89	96	
4	77	56	67	81	79	
5	79	82	85	81	90	
6						

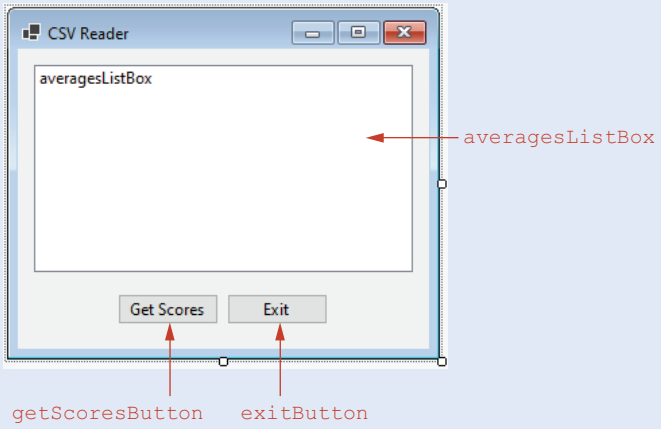
Suppose the professor wants you to write a C# application that reads the test scores from the Excel spreadsheet and performs some operation with them. Excel, like many commercial applications, has the ability to export data to a text file. When the data in a spreadsheet is exported, each row is written to a line, and the values of the cells are separated by commas. For example, when the data shown in Figure 8-9 is exported, it will be written to a text file in the following format:

```
87,79,91,82,94
72,79,81,74,88
94,92,81,89,96
77,56,67,81,79
79,82,85,81,90
```

This is called the **comma separated value, or CSV**, file format. When you save a spreadsheet in this format, Excel saves it to a file with the *.csv* extension. In a C# application, you can read a line from the file into a `string` variable and then use the `Split` method to extract the cell values as tokens.

To learn how this is done, you will complete the *CSVReader* project. The application's form, which has already been created for you, is shown in Figure 8-10. A file named Grades.csv, which contains the test scores shown in Figure 8-9, has also been created for you. When you run the completed application and click the *Get Scores* button, it will read the test scores from the Grades.csv file. Each student's average test score will be calculated and displayed in the `averagesListBox` control.

Figure 8-10 The *CSVReader* project's form



Step 1: Start Visual Studio. Open the project named *CSVReader* in the *Chap08* folder of the Student Sample Programs.

Step 2: With Form1 open in the *Designer*, double-click the `getScoresButton` control. This opens the code editor, and you will see an empty event handler named `getScoresButton_Click`. Complete the event handler by typing the code shown in lines 12–63 in Program 8-5. Let's review this code:

Line 12: This is the beginning of a `try-catch` statement. The try block appears in lines 14–57, and the catch block appears in lines 61–62. If an exception is thrown by any statement in the try block, the program will jump to the catch block, and line 62 will display an error message.

Lines 14–18: These statements declare the following variables:

- `inputFile`, a `StreamReader` variable that will be used to read file input
- `line`, a `string` variable to hold a line that has been read from the file
- `count`, an `int` that will be used to count the number of students as a loop iterates (Notice that the `count` variable is initialized with the value 0.)
- `total`, an `int` that will hold the total of a student's test scores
- `average`, a `double` that will hold the average of a student's test scores

Line 21: This statement creates a `char` array named `delim`. The array has one element containing the character `','`.

Line 24: This statement opens the Grades.csv file. After the statement executes, the `inputFile` variable references a `StreamReader` object that is associated with the file.

Line 26: This is the beginning of a `while` loop that reads the contents of the Grades.csv file.

Line 29: This statement increments the `count` variable, which keeps count of the number of students. This statement will cause `count` to be set to 1 during the loop's first iteration, 2 during the loop's 2nd iteration, and so forth.

Line 32: This statement reads a line from the Grades.csv file and assigns it to the `line` variable.

Line 35: This statement tokenizes the `line` variable, using the `','` character as a delimiter. After this statement executes, the tokens are stored in the `tokens` array's elements. (The first test score is stored in `tokens[0]`, the second test score is stored in `tokens[1]`, etc.)

Line 38: This statement sets the `total` variable to 0. This is necessary because `total` is about to be used as an accumulator in calculating the total of the test scores that are in the `tokens` array.

Lines 42–45: This `foreach` loop steps through the elements of the `tokens` array. In line 44 each element is converted to an `int` and then added to the `total` variable. After the loop finishes, the total variable will contain the total of the test scores that are in the `tokens` array.

Line 49: This statement calculates the average of the test scores and assigns the result to the `average` variable. Note that the `(double)` cast operator is used to prevent integer division.

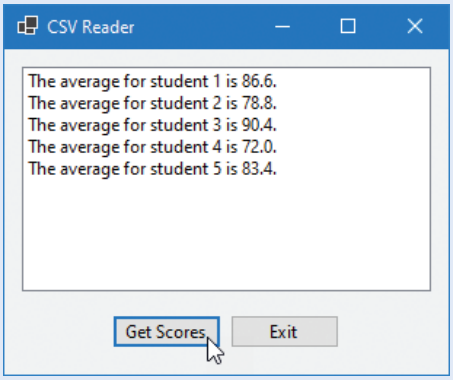
Lines 52–53: This statement displays the average, along with the student number, in the `averageListBox` control.

Line 57: This statement closes the Grades.csv file.

Step 3: Switch your view back to the *Designer* and double-click the `exitButton` control. In the code editor you see an empty event handler named `exitButton_Click`. Complete the event handler by typing the code shown in lines 68–69 in Program 8-5.

Step 4: Save the project. Then press F5 on the keyboard or click the *Start Debugging* button (▶) on the toolbar to compile and run the application. When the application runs, click the *Get Scores* button. The student averages should appear as shown in Figure 8-11. When you are finished examining the averages, click the *Exit* button to exit the application.

Figure 8-11 Test averages displayed



Program 8-5 Completed code for Form1 in the *CSVReader* project application

```
namespace CSVReader
{
    public partial class Form1: Form
    {
        public Form1()
        {
            InitializeComponent();
        }

        private void getScoresButton_Click(object sender,
        EventArgs e)
        {
            try
            {
                StreamReader inputFile; // To read the file
                string line;            // To hold a line from
                                           the file
                int count = 0;          // Student counter
                int total;              // Accumulator
                double average;         // Test score average

                // Create a delimiter array.
                char[] delim = {','};

                // Open the CSV file.
                inputFile = File.OpenText("Grades.csv");

                while (!inputFile.EndOfStream)
                {
                    // Increment the student counter.
                    count++;

                    // Read a line from the file.
                    line = inputFile.ReadLine();

                    // Get the test scores as tokens.
                    string[] tokens = line.Split(delim);

                    // Set the accumulator to 0.
                    total = 0;

                    // Calculate the total of the
                    // test score tokens.
                    foreach (string str in tokens)
                    {
                        total += int.Parse(str);
                    }

                    // Calculate the average of these
                    // test scores.
                    average = (double)total/tokens.Length;

                    // Display the average.
                    averagesListBox.Items.Add(
                        $"The average for student {count} is
                        {average:n1}.");
                }

                // Close the file.
                inputFile.Close();
            }
```

```
59              catch (Exception ex)
60              {
61                  // Display an error message.
62                  MessageBox.Show(ex.Message);
63              }
64          }
65
66          private void exitButton_Click(object sender, EventArgs e)
67          {
68              // Close the form.
69              this.Close();
70          }
71      }
72  }
```

Checkpoint

8.27 The following string contains three tokens. What are they? What character is the delimiter?

```
"apples pears bananas"
```

8.28 Look at the following code:

```
string str = "one two three four";
string[] tokens = str.Split(" ");
int x = tokens.Length;
string first = tokens[0];
```

What value will be stored in `x`? What value will the `first` variable reference?

8.29 Look at the following string:

```
"/home/rjones/mydata.txt"
```

Write code using the `Split` method that can be used to extract the following tokens from the string: home, rjones, mydata, and txt.

8.30 Look at the following string:

```
"dog$cat@bird%squirrel"
```

Write code using the `Split` method that can be used to extract the following tokens from the string: dog, cat, bird, and squirrel.

8.31 Declare a `char` array named `delimiters` and initialize it with the comma and semicolon characters.

8.32 What delimiter is used to separate data in a spreadsheet that has been exported to a text file with the .csv file extension?

8.8 Method Chaining

CONCEPT: **Method chaining is a coding technique that, in some situations, can combine several statements into a single statement, and can eliminate the need for temporary variable declarations.**

Suppose you need to manipulate a string in multiple ways. For example, suppose `str` is a `string` variable and we need to perform the following operations on it:

- Trim leading and trailing whitespace characters
- Convert to uppercase
- Replace all remaining spaces with underscore characters

The following code shows an example of how we might perform each step:

```
1 string temp1 = str.Trim();
2 string temp2 = temp1.ToUpper();
3 string result = temp2.Replace(' ', '-');
```

The statement in line 1 calls `str`'s `Trim` method, which returns a copy of `str` with leading and trailing whitespace trimmed. The copy is assigned to a variable named `temp1`. Then, line 2 calls `temp1`'s `ToUpper` method, which returns a copy of `temp1` converted to uppercase characters. The copy is assigned to a variable named `temp2`. Then, line 3 calls `temp2`'s `Replace` method, which returns a copy of `temp2` with all space characters replaced with a hyphen. The copy is assigned to a variable named `result`.

Notice that the code requires three statements, and it declares three variables: `temp1` holds the string returned from the `Trim` method, `temp2` holds the string returned from the `ToUpper` method, and `result` holds the string returned from the `Replace` method. Once this code has executed, however, we only need the `result` variable. The `temp1` and `temp2` variables are needed only temporarily.

This code can be simplified with a technique known as **method chaining**. Method chaining allows you to combine this type of code into a single statement and eliminate the declaration of the temporary variables. Here is an example of how we can rewrite the previously shown code with method chaining:

```
string result = str.Trim().ToUpper().Replace(' ', '-');
```

In this statement, `str.Trim()` is called first. It returns a copy of `str` that has been trimmed. The trimmed string's `ToUpper` method is called next. It returns a copy of the trimmed string that has been converted to uppercase. This trimmed, uppercase string's `Replace` method is called next. It returns a copy of the trimmed, uppercase string in which all spaces have been replaced with hyphens. This string is assigned to the `result` variable.

Method chaining can be used in any scenario where you must call a method to get an object, then use that object to call another method, etc. For example, suppose `str` is a `string`, and you want to convert it to lowercase and insert "#" at its beginning. Here is an example of the code using method chaining:

```
string result = str.ToLower().Insert(0 "#");
```

In this statement, `str.ToLower` is called first. It returns a copy of `str` that has been converted to lowercase. That string's `Insert` method is called. It returns a copy of the lowercase string in which "#" has been inserted at index 0. This is the last value that is returned from the method chain, so it is assigned to `result`.

The following code shows another example:

```
int number = 199;
char[] carr = number.ToString().Insert(0, "$").ToCharArray();
```

The first statement declares an `int` variable named `number`, initialized with the value 199. The second statement declares a `char` array variable named `carr` and initializes it with the value returned from a method chain. In the method chain, `number.ToString` is called first. This returns the string "199". That string's `Insert` method is then called. It returns a copy of the string in which "$" has been inserted at index 0. That string's `ToCharArray` method is then called. It returns a `char` array whose elements contain the characters of the string. (The array's elements will contain the characters `'$'`, `'1'`, `'9'`, and `'9'`.) The array is assigned to the `carr` variable.

Many of the StringBuilder methods, such as Append, Clear, Insert, Remove, and Replace, return a reference to the StringBuilder object that called them. For example, look at the following statement and assume that sb is a StringBuilder object:

```
sb.Replace("abc", "xyz");
```

This statement replaces all occurrences of "abc" with "xyz". When the method is finished, it returns a reference to the sb object. This makes it possible for us to use method chaining. Here is an example:

```
var sb = new StringBuilder("$$$red green blue");
sb.Remove(0, 3).Replace(' ', '-').Insert(0, '[').Append(']');
```

The first statement creates a StringBuilder object named sb, initialized with the string "$$$red green blue". The second statement is a method chain that performs the following:

- It first calls sb.Remove(0, 3). This removes the first three characters from sb and returns a reference to the sb object.
  - Then, that reference is used to call Replace(' ', '-'). This replaces all the spaces in sb with hyphens, and returns a reference to sb.
    - That reference is then used to call Insert(0, '['). This inserts the '[' character at the beginning of sb's string, and returns a reference to sb.
      - Then, that reference is used to call Append(']'). This appends the ']' character to sb.

After the method chain completes all these method calls, the sb object will contain the string "[red-green-blue]".

To improve readability, long method chain statements are sometimes written across multiple lines. Here is an example:

```
string result = str.Trim()
.ToUpper()
.Replace(' ', '-');
```

Checkpoint

8.33 Assume str is a string variable that has already been initialized with a value. Rewrite the following code with a method chain. (Your code should use only the variables str and result.)

```
string t1 = str.Insert(0, ">");
string t2 = t1.ToUpper();
string result = t2.Replace('*', '#');
```

8.34 After the following code executes, what value will the result variable be assigned? What value will str have?

```
string str = " abc def ";
string result = str.Trim().ToUpper().Replace(' ', '-');
```

8.9 The String.Format Method

CONCEPT: The String.Format method allows you to format and insert values into a string.

Before string interpolation was introduced in the C# language, the **String.Format method** was used to create formatted strings. The String.Format method is still used, and it appears in a lot of C# code examples that are posted on the web. For those reasons, we will briefly cover it in this section.

The `String.Format` method allows you to insert values into a string and format those values in specific ways. The method is used in the following general format:

```
String.Format(FormatString, Arg0, Arg1, Arg2, . . .)
```

The first argument is known as the ***format string***. It is the string that you want to insert values into. The arguments that follow the format string are known as argument 0, argument 1, argument 2, and so forth. These are values that will be inserted into the format string. The method returns the resulting string with the specified values inserted into it.

Here is an example:

```
decimal sales = 1000m;
string message = String.Format("Our sales are {0} for the day.",
                               sales);
```

In this example, the `String.Format` method takes two arguments:

- `"Our sales are {0} for the day."` is the format string
- The `sales` variable is argument 0

When this code executes, the value of `sales` will be inserted into the format string. But where will it be inserted? Notice the special sequence `{0}` that appears inside the format string. This is known as a format item. A **format item** is a placeholder for a value that will be inserted into the format string. The number that appears inside the braces is an index that identifies the argument to be inserted. 0 means argument 0, 1 means argument 1, and so forth. So, the string that is returned from the `String.Format` method will be:

```
Our sales are 1000 for the day.
```

Here is another example:

```
double temp1 = 72.5, temp2 = 83.7;
string message = String.Format
("The temperatures are {0} and {1} degrees", temp1, temp2);
```

In this example, argument 0 is the `temp1` variable and argument 1 is the `temp2` variable. When this code executes, the string that is returned from the `String.Format` method will be:

```
The temperatures are 72.5 and 83.7 degrees.
```

Formatting the Arguments

In a format item, you can optionally include a **format specifier** that causes the argument to be formatted in a specific way. For example, if you want a number to be formatted as currency, you would include the `C` format specifier, as shown here:

```
decimal amount = 123456789.45678m;
string message = String.Format("The balance is {0:C}", amount);
```

Notice that the index and the format specifier are separated by a colon. When this code executes, the string that is returned from the `String.Format` method will be:

```
The balance is $123,456,789.46
```

Several format specifiers are supported. Table 8–11 lists a few of them. Notice that these are the same format specifiers that you can use with the `ToString()` method, as discussed in Chapter 3.

Table 8–11 Format specifiers

Format Specifier	Description
N	Number format: The number is formatted with comma separators and a decimal point. By default, two digits display to the right of the decimal point. Negative values are displayed with a leading minus sign. An example is −2,345.67.
F	Fixed-point format: The number is formatted with no comma separators and a decimal point. By default, two digits display to the right of the decimal point. Negative values are displayed with a leading minus (–) sign. An example is −2345.67.
E	Exponential scientific format: The number is formatted with a single digit to the left of the decimal point. The letter e appears in front of the exponent, and the exponent has a leading + or + sign. By default, six digits display to the right of the decimal point, and a leading minus sign is used if the number is negative. An example is −2.345670e + 003.
C	Currency format: Formats the number to include a leading currency symbol (such as $), comma separators, and a decimal point. By default, two digits display to the right of the decimal point. Negative values are surrounded by parentheses. An example of a negative value is ($2,345.67).
P	Percent format: Causes the number to be multiplied by 100 and displayed with a trailing space and % sign. By default, two digits display to the right of the decimal point. Negative values are displayed with a leading minus sign. For example, the number 0.125 would be formatted as 12.5% and the number −0.2345 would be formatted as −23.45%

NOTE: The format specifiers listed in Table 8–11 are not case sensitive. For example, the format item `{0:N}` is the same as `{0:n}`.

Specifying a Minimum Field Width

A format item can include a **minimum field width**, which is the minimum number of spaces that should be used to display the value. The following example displays a floating-point number in a field that is 20 spaces wide:

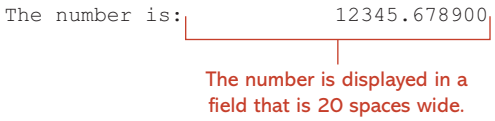
```
double number = 12345.6789;
string message = String.Format("The number is:{0,20}", number);
```

Notice that the index and the **width specifier** are separated by a comma. The format item `{0,20}` specifies that argument 0 should appear in a field that is a minimum of 20 spaces wide. When this code executes, the string that is returned from the `String.Format` method will be:

```
The number is:        12345.678900
```

The field width is illustrated in Figure 8–12.

Figure 8–12 The number occupies a field that is 20 spaces wide



In this example, the number 12345.678900 uses only 12 spaces, but it is inserted into a field that is 20 spaces wide. When this is the case, the number will be right-justified in the field. If a value is too large to fit in the specified field width, the field is automatically enlarged to accommodate it. The following example displays a floating-point number in a field that is only one space wide:

```
double number = 12345.6789;
string message = String.Format("The number is:{0,1}", number);
```

The value of the `number` variable requires more than one space; however, the field width is expanded to accommodate the entire number. When this code executes, the string that is returned from the `String.Format` method will be:

```
The number is:12345.6789
```

If you want a number to be left-justified within its field, you insert a minus sign (–) before the field width specifier. Here is an example:

```
double number = 12345.6789;
string message = String.Format("***{0,-20}***", number);
```

When this code executes, the string that is returned from the `String.Format` method will be:

```
***12345.6789          ***
```

You can include both a width specifier and a format specifier, as shown here:

```
double number = 12345.6789;
string message = String.Format("The number is:{0,20:C}", number);
```

In this example, the format item `{0,20:C}` specifies that argument 0 should appear in a field that is a minimum of 20 spaces wide, formatted as currency. When this code executes, the string that is returned from the `String.Format` method will be:

```
The number is:          $12,345.68
```

Specifying the Precision

Each numeric format specifier can optionally be followed by an integer that indicates the number of digits to display after the decimal point. For example, the format `N3` displays three digits after the decimal point. Table 8–12 shows a variety of numeric formatting examples, based on the North American locale.

Here is an example:

```
double number = 12345.6789;
string message = String.Format("The number is {0:N3}", number);
```

In this example, the format item `{0:N3}` specifies that argument 0 should appear in a field that is a formatted as a number with three decimal places. When this code executes, the string that is returned from the `String.Format` method will be:

```
The number is 12,345.679
```

Table 8–12 Numeric formatting examples (North American locale)

Number	Format Specifier	Formatting Result
12.3	N3	12.300
12.348	N2	12.35
1234567.1	N	1,234,567.10
123456.0	F2	123456.00
123456.0	E3	1.235e + 005
.234	P	23.40%
−1234567.8	C	($1,234,567.80)

Rounding

Rounding can occur when the number of digits you have specified after the decimal point in the format string is smaller than the precision of the numeric value. Suppose, for example, that we use the N2 format specifier on the value 1.235. The resulting value would be 1.24. If the next digit after the last displayed digit is 5 or higher, the last displayed digit is rounded *away from zero*.

Checkpoint

8.35 Assume the following variable declaration exists in a program:

```
double number = 1234567.456;
```

Write a statement that uses the String.Format method to convert the value of the number variable to a string, formatted as:1,234,567.46

8.36 Assume the following variable declaration exists in a program:

```
double number = 123.456;
```

Write a statement that uses the String.Format method to convert the value of the number variable to a string, rounded to one decimal place, in a field that is 10 spaces wide. (Do not use comma separators.)

8.37 Assume the following variable declaration exists in a program:

```
int number = 123456;
```

Write a statement that uses the String.Format method to convert the value of the number variable to a string, in a field that is 10 spaces wide, with comma separators.

8.38 Assume the following variable declaration exists in a program:

```
double number = 123456.789;
```

Write a statement that uses the String.Format method to convert the value of the number variable to a string, left-justified, with comma separators, in a field that is 20 spaces wide, rounded to two decimal places.

Key Terms

capacity
character literals
comma separated value (CSV)
delimiter
format item
format specifier
format string
immutable
interpolated string
leading spaces
method chaining
minimum field width
mutable
substring
tokenizing
tokens
trailing spaces
width specifier

Review Questions

Multiple Choice

1. In C#, _____ are enclosed in single quotation marks.
   a. strings
   b. enumerators
   c. tokens
   d. character literals

2. The `char` data type provides the _____ for converting the case of a character.
   a. `Lowercase` and `Uppercase` methods
   b. `ToLower` and `ToUpper` methods
   c. `IsDigit` and `IsLetter` methods
   d. `IsWhiteSpace` and `IsPunctuation` methods

3. A string within a string is called a(n) _____.
   a. character
   b. inner string
   c. substring
   d. thread

4. The _____ of a `string` object allow you to search for substrings.
   a. `TrimStart` and `TrimEnd` methods
   b. `IndexOf` and `LastIndexOf` methods
   c. `IsWhiteSpace` and `IsPunctuation` methods
   d. `Contains`, `StartsWith`, and `EndsWith` methods

5. When you want to know the position of the substring, you can use one of the _____ of a `string` object.
   a. `TrimStart` or `TrimEnd` methods
   b. `IndexOf` or `LastIndexOf` methods
   c. `ToLower` or `ToUpper` methods
   d. `StartsWith` or `EndsWith` methods

6. The _____ of a `string` object can be used to retrieve a specific set of characters from a string.
   a. `Substring` method
   b. `IndexOf` method
   c. `TrimStart` and `TrimEnd` methods
   d. `StartsWith` and `EndsWith` methods

7. _____ are spaces that appear at the beginning of a string.
   a. Leading spaces
   b. Primary spaces
   c. Starting spaces
   d. Empty spaces

8. _____ are spaces that appear at the end of a string.
   a. Blank spaces
   b. Secondary spaces
   c. Ending spaces
   d. Trailing spaces

9. A series of words or other items of data contained in a string and separated by spaces or other characters are known as _____.
   a. substrings
   b. elements
   c. characters
   d. tokens

10. The character that separates tokens is known as _____.
    a. partition
    b. literal
    c. delimiter
    d. symbol

11. The process of breaking a string into tokens is known as _____.
    a. extracting
    b. tokenizing
    c. delimiting
    d. parsing

12. The _____ file format is commonly used to export spreadsheet data to a text file.
    a. spreadsheet data volume, or SDV
    b. comma separated value, or CSV
    c. extensible markup language, or XML
    d. portable document format, or PDF

True or False

1. You cannot store a string in a variable of the `char` data type.
2. C# allows you to access the individual characters in a string using subscript notation.
3. When you call a `string` object's `Split` method, the method divides the string into two substrings and returns them as an array of strings.

Short Answer

1. What method can be used to convert a `char` variable to a string?
2. List the method you would use to determine whether each of the following is true or false for the value of a character:
   a. numeric digit
   b. alphabetic letter
   c. alphabetic letter or numeric digit
   d. punctuation mark
   e. white-space character
   f. uppercase letter
   g. lowercase letter

3. List the method you would use to determine each of the following about the value of a `string` object:
   a. starts with the substring "VENI"
   b. contains the substring "VIDI"
   c. ends with the substring "VICI"
4. Briefly describe each of the following `string` object methods:
   a. `Insert`
   b. `Remove`
   c. `ToLower`
   d. `ToUpper`
   e. `Trim`
   f. `TrimStart`
   g. `TrimEnd`
5. What characters are used as delimiters when you pass `null` as an argument to the `Split` method of a `string` object?

Algorithm Workbench

1. Write a method that accepts a string as an argument and checks it for proper capitalization and punctuation. The method should determine if the string begins with an uppercase letter and ends with a punctuation mark. The method should return `true` if the string meets the criteria; otherwise it should return `false`.
2. Write a method that accepts a string as an argument and displays its contents backward. For instance, if the string argument is `"gravity"`, the method should display `"ytivarg"`.
3. Assume the `double` variable `number` contains the value 12345.6789. Write an interpolated string that formats the number as 12345.7.
4. Assume the `double` variable `number` contains the value 12345.6789. Write an interpolated string that formats the number as 12,345.68.
5. Assume the `int` variable `number` contains the value 1234567. Write an interpolated string that formats the number as 1,234,567.
6. Why should you use `StringBuilder` objects instead of `string` objects in a program that makes lots of changes to strings?

Programming Problems

1. **Word Counter**

   Create an application with a method that accepts a string as an argument and returns the number of words it contains. For instance, if the argument is `"Four score and seven years ago,"` the method should return the number 6. The application should let the user enter a string, and then it should pass the string to the method. The number of words in the string should be displayed.

2. **Average Number of Letters**

   Modify the program you wrote for Problem 1 (Word Counter) so it also displays the average number of letters in each word.

3. **Sentence Capitalizer**

   Create an application with a method that accepts a string as an argument and returns a copy of the string with the first character of each sentence capitalized. For instance, if the argument is `"hello. my name is Joe. what is your name?"` the

method should return the string `"Hello. My name is Joe. What is your name?"` The application should let the user enter a string and then pass it to the method. The modified string should be displayed.

4. **Vowels and Consonants**

   Create an application with a method that accepts a string as an argument and returns the number of vowels that the string contains. The application should have another method that accepts a string as an argument and returns the number of consonants that the string contains. The application should let the user enter a string, and should display the number of vowels and the number of consonants it contains.

5. **Most Frequent Character**

   Create an application that lets the user enter a string and displays the character that appears most frequently in the string.

6. **Word Separator**

   Create an application that accepts as input a sentence in which all the words are run together but the first character of each word is uppercase. Convert the sentence to a string in which the words are separated by spaces and only the first word starts with an uppercase letter. For example, the string `"StopAndSmellTheRoses."` would be converted to "Stop and smell the roses."

7. **Pig Latin**

   Create an application that accepts a sentence as input and converts each word to "Pig Latin." In one version, to convert a word to Pig Latin you remove the first letter and place that letter at the end of the word. Then you append the string `"ay"` to the word. Here is an example:

   English: I SLEPT MOST OF THE NIGHT

   Pig Latin: IAY LEPTSAY OSTMAY FOAY HETAY IGHTNAY

VideoNote
Solving the Sum of Numbers in a String Problem

8. **Sum of Numbers in a String**

   Create an application that lets the user enter a string containing series of numbers separated by commas. Here is an example of valid input:

   ```
   7,9,10,2,18,6
   ```

   The program should calculate and display the sum of all the numbers.

9. **Alphabetic Telephone Number Translator**

   Many companies use telephone numbers like 555-GET-FOOD so the number is easier for their customers to remember. On a standard telephone, the alphabetic letters are mapped to numbers in the following fashion:

   A, B, and C = 2

   D, E, and F = 3

   G, H, and I = 4

   J, K, and L = 5

   M, N, and O = 6

   P, Q, R, and S = 7

   T, U, and V = 8

   W, X, Y, and Z = 9

   Create an application that lets the user enter a 10-character telephone number in the format `XXX-XXX-XXXX`. The application should display the telephone number with

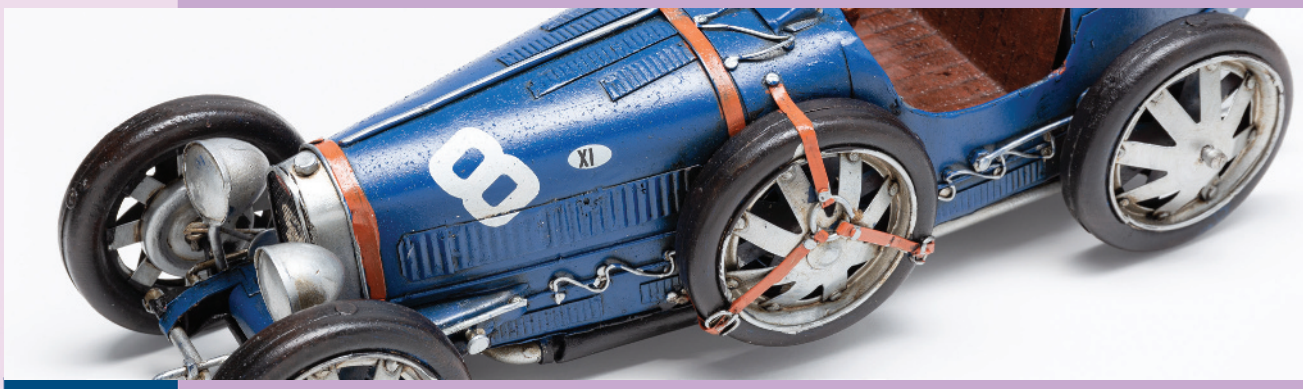
any alphabetic characters that appeared in the original translated to their numeric equivalent. For example, if the user enters 555-GET-FOOD, the application should display 555-438-3663.

10. Morse Code Converter

Design a program that asks the user to enter a string and then converts that string to Morse code. Morse code is a code where each letter of the English alphabet, each digit, and various punctuation characters are represented by a series of dots and dashes. Table 8-13 shows part of the code.

Table 8–13 Morse code

Character	Code	Character	Code	Character	Code	Character	Code
space	*space*	6	-....	G	--.	Q	--.-
comma	--..--	7	--...	H	....	R	.-.
period	.-.-.-	8	---..	I	..	S	...
?	..--..	9	----.	J	.---	T	-
0	-----	A	.-	K	-.-	U	..-
1	.----	B	-...	L	.-..	V	...-
2	..---	C	-.-.	M	---	W	.--
3	...--	D	-..	N	-.	X	-..-
4	....-	E	.	O	---	Y	-.--
5	.....	F	..-.	P	.--.	Z	--..



CHAPTER

9 Structured Data Types

TOPICS

9.1 Introduction
9.2 Structures
9.3 The `DateTime` and `TimeSpan` Structures
9.4 Enumerated Types
9.5 Dictionaries
9.6 Tuples

9.1 Introduction

Now that you have studied the fundamentals of programming using Visual C#, you can use the topics presented in this chapter to perform more advanced operations. First, we discuss structures, which allow you to encapsulate several variables into a single item. After that we discuss enumerated types, which are data types that you can create, consisting of specified values. Last, we discuss dictionaries, which are data structures for storing items known as key–value pairs. With a dictionary, you use a key to locate a specific value.

9.2 Structures

CONCEPT: **C# allows you to group several variables together into a single item known as a structure.**

So far, you have created applications that keep data in individual variables. Sometimes, a relationship exists between different items of data. For example, a used-car dealer's inventory system might use the variables shown in the following declaration statements:

```
string make;     // The car's make
int year;        // The car's year model
double mileage; // The car's mileage
```

All these variables are related because they will hold data about the same car. Their declaration statements, however, do not make it clear that they belong together. To create a relationship between variables, C# gives you the ability to package them together into a single item known as a structure. A **structure** is a data type you can create that contains one or more variables known as **fields**. The fields can be of different data types.

Before a structure can be used, it must be declared. Here is the general format that we use for declaring a structure:

```
struct StructureName
{
    Field Declarations
}
```

The first line of the structure declaration begins with the keyword `struct`, followed by the structure name. A set of braces appears next, and the braces contain one or more field declarations. Here is an example of a structure declaration:

```
1 struct Automobile
2 {
3     public string make;
4     public int year;
5     public double mileage;
6 }
```

The name of this structure is `Automobile`. In this book, we always begin structure names with an uppercase letter. This is not required, but many programmers follow this practice because it helps distinguish structure names from variable names.

Lines 3, 4, and 5 declare three fields. Line 3 declares a `string` field named `make`, line 4 declares an `int` field named `year`, and line 5 declares a `double` field named `mileage`. Notice that each declaration begins with the keyword `public`. The keyword `public` is an access modifier that specifies the field can be directly accessed by statements outside the structure.

NOTE: Structures in C# are capable of much more than we discuss in this book. For example, in addition to declaring fields, you can also write methods inside of structures. We use structures for their simplest purpose only: to encapsulate a set of variables into a single item. If you find yourself using a structure for more than this, you should consider using a class instead. We discuss classes in Chapter 10.

In Chapter 2, we discussed the way that C# code is organized in namespaces, classes, and methods. Structure declarations can be written in a variety of places. For example, a structure declaration can appear in these locations:

- Outside the application's namespace
- Inside the application's namespace
- Inside a class
- Inside another structure

In this book, we always declare structures inside an application's namespace but not inside of a class. Figure 9-1 shows where, in a form's code, that we typically write structure declarations. In the figure, we have written the declaration of the `Automobile` structure inside the application's namespace but not inside the `Form1` class.

Keep in mind that a structure declaration does not create anything in memory. It simply tells the C# compiler what the structure is made of. Before you can use the structure to store data, you must create an **instance** of the structure in memory. (In programming terms, an *instance* and an *object* are the same thing. When you create an instance of a structure, you are creating an object.)

At the beginning of Chapter 7, we discussed value types and reference types. Structures are value types, so creating an instance of a structure is as simple as declaring a variable. For example, suppose we want to create an instance of the `Automobile` structure and we want the name of the instance to be `sportsCar`. We would write this:

```
Automobile sportsCar;
```

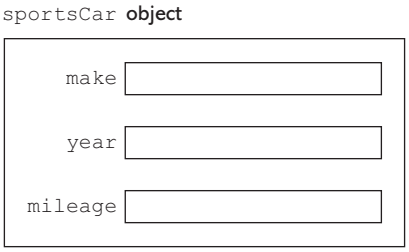
After this statement executes, an instance of the `Automobile` structure exists in memory. The name of the instance is `sportsCar`. As illustrated in Figure 9-2, the `sportsCar` object has three fields: `make`, `year`, and `mileage`.

Figure 9-1 Where we typically write structure declarations

```
namespace Example
{
    struct Automobile
    {
        public string make;
        public int year;
        public double mileage;
    }

    public partial class Form1 : Form
    {
        public Form1()
        {
            InitializeComponent();
        }
    }
}
```

We will write structure declarations here.

Figure 9-2 The `sportsCar` object as an instance of the `Automobile` structure

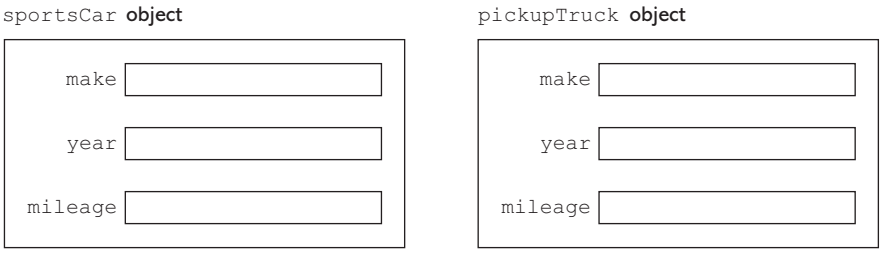


You can create multiple instances of a structure with a declaration statement, as shown here:

```
Automobile sportsCar, pickupTruck;
```

This statement creates two instances of the `Automobile` structure. The objects are named `sportsCar` and `pickupTruck`. As illustrated in Figure 9-3, each object has its own `make`, `year`, and `mileage` fields.

Figure 9-3 Two instances of the `Automobile` structure



Using the `new` Operator to Create Structure Instances

When you create a structure object with a simple declaration statement, as previously shown, the object's fields are uninitialized, and if you attempt to use any of them before assigning them a value, a compiler error occurs. As an alternative, you can use the `new` operator to create an instance of a structure, as shown here:

```
Automobile sportsCar = new Automobile();
```

This is the recommended technique for creating structure instances because the `new` operator not only creates the instance in memory, but it also initializes the object's fields with the default value of 0. (If the structure contains any fields that are reference variables, they are initialized with the special value `null`.) Here is an example of creating two instances of the `Automobile` structure using the `new` operator:

```
Automobile sportsCar = new Automobile();
Automobile pickupTruck = new Automobile();
```

If you know the values that you want to store in a structure's fields at the time you instantiate the structure, you can use **object initializer syntax** to initialize the object's fields. Here is an example:

```
Automobile truck = new Automobile
{
    make = "Dodge Ram",
    year = 2014,
    mileage = 85000.0
};
```

This statement creates an instance of the `Automobile` structure in memory. The object's name is `truck`. The object's fields will be initialized with the following values: the `make`

field will be initialized with the string "Dodge Ram," the `year` field will be initialized with the value 2014, and the `mileage` field will be initialized with the value 85000.0.

Using `var` to Declare Structure Objects

If you are using the `new` operator to initialize a structure object inside of a method, you can use the `var` keyword in your declaration statement. Here is an example:

```
var sportsCar = new Automobile();
```

The compiler will be able to determine that `sportsCar` is an `Automobile` object because we are initializing it with an `Automobile` object. It is also obvious to any person reading this statement that `sportsCar` is an `Automobile` object.

When declaring local structure objects in this book, we will use the `var` keyword any time it simplifies the declaration statement without concealing the data type of the object.

NOTE: Remember, the `var` keyword can only be used to declare local variables in a method. It cannot be used to declare fields in a class.

Accessing a Structure's Fields

Once you have created an instance of a structure, you can access its fields using the dot operator (a period). For example, suppose an application contains the `Automobile` structure declaration previously shown and the following code appears in a method:

```
1 var sportsCar = new Automobile();
2 sportsCar.make = "Ford Mustang";
3 sportsCar.year = 1965;
4 sportsCar.mileage = 67500.0;
```

Line 1 creates an instance of the `Automobile` structure in memory. The object's name is `sportsCar`. Line 2 assigns the string `"Ford Mustang"` to the `sportsCar` object's `make` field. Line 3 assigns the value 1965 to the `sportsCar` object's `year` field. Line 4 assigns the value 67,500.0 to the `sportsCar` object's `mileage` field.

The following code shows another example using the same `sportsCar` object. These statements display the values of the object's fields in message boxes.

```
1 MessageBox.Show(sportsCar.make);
2 MessageBox.Show(sportsCar.year.ToString());
3 MessageBox.Show(sportsCar.mileage.ToString());
```

The following is another example. This statement displays a message such as `"1965 Ford Mustang with 67500 miles."`

```
MessageBox.Show(
    $"{sportsCar.year} {sportsCar.make} with {sportsCar.mileage} miles.");
```

Assigning One Structure Object to Another

You can use the assignment operator (=) to assign one structure object to another. For example, assume that `car1` and `car2` are both instances of the `Automobile` structure. The following statement assigns `car1` to `car2`:

```
car2 = car1;
```

After this statement executes, the `car2` object's fields contain the same values as the `car1` object's fields.

Passing Structure Objects to Methods

As with other types of objects, you can pass a structure object as an argument to a method. The following code shows a method named `DisplayAuto` that has been written to accept an instance of the `Automobile` structure as an argument:

```
1 private void DisplayAuto(Automobile auto)
2 {
3    MessageBox.Show($"{auto.year} {auto.make} with {auto.mileage} miles.");
4 }
```

Notice in line 1 that the method has a parameter variable named `auto`, and its data type is `Automobile`. When we call this method, we pass an `Automobile` object as an argument, as shown in the following code:

```
1 // Create an instance of the Automobile structure.
2 var sportsCar = new Automobile();
3
4 // Assign values to the object's fields.
5 sportsCar.make = "Chevy Corvette";
6 sportsCar.year = 1970;
7 sportsCar.mileage = 50000.0;
8
9 // Display the object's fields.
10 DisplayAuto(sportsCar);
```

In line 10 the `sportsCar` object is passed as an argument to the `DisplayAuto` method. Inside the `DisplayAuto` method, the `auto` parameter contains a copy of the `sportsCar` object.

Structure objects can be passed by value or by reference. Normally, structure objects are passed by value. The parameter variable contains a copy of the argument, and any changes that are made to the parameter do not affect the original argument. If the receiving method needs to change the contents of the original argument, however, the `ref` or `out` keyword can be used in the parameter declaration. The following code shows a method that uses a reference parameter of the `Automobile` type. Assume that the application's form has TextBox controls named `makeTextBox`, `yearTextBox`, and `mileageTextBox`.

```
1 private void GetData(ref Automobile auto)
2 {
3    try
4    {
5        // Get the data from the TextBoxes.
6        auto.make = makeTextBox.Text;
7        auto.year = int.Parse(yearTextBox.Text);
8        auto.mileage = double.Parse(mileageTextBox.Text);
9    }
10   catch (Exception ex)
11   {
12       // Display the exception message.
13       MessageBox.Show(ex.Message);
14   }
15 }
```

Notice that the `auto` parameter is declared with the `ref` keyword. The statements in lines 6–8 get data from the TextBox controls and assign that data to the fields of the `auto` parameter. Since `auto` is a `ref` parameter, the values are actually assigned to the object that is passed as an argument to the method.

When we call this method, we pass an `Automobile` object by reference, as shown here:

```
1 // Create an instance of the Automobile structure.
2 var car = new Automobile();
3
4 // Get data for the object.
5 GetData(ref car);
```

After the method executes, the `car` object contains the data that was entered into the TextBox controls.

Returning a Structure Object from a Method

Just as methods can be written to return an `int`, `string`, `decimal`, `double`, or other data type, they can also be written to return a structure. Let's look at an example. The following method returns an `Automobile` object:

```
1  private Automobile GetVintageFord()
2  {
3      var auto = new Automobile
4      {
5          make = "Ford Galaxy",
6          year = 1968,
7          mileage = 40000.0
8      };
9
10     return auto;
11 }
```

Notice in the method header, in line 1, the method's return type is `Automobile`. That means the method returns an `Automobile` object. In lines 3 through 8, an `Automobile` object named `auto` is created and initialized with data. Then, line 10 returns the `auto` object.

The object that is returned from the method can be assigned to an `Automobile` variable. For example, look at the following code:

```
Automobile car;
car = GetVintageFord();
```

The first statement declares an `Automobile` variable named `car`. The second statement calls the `GetVintageFord` method and assigns its return value to `car`. After this code executes, `car.make` will be set to `"Ford Galaxy"`, `car.year` will be set to 1968, and `car.mileage` will be set to 40000.0.

TIP: Suppose you are declaring a local structure variable and using a method's return value to initialize it, as shown here:

```
Automobile car = GetVintageFord();
```

You might be tempted to use the `var` keyword in this statement, as shown here:

```
var car = GetVintageFord();
```

The `var` keyword is not recommended in a statement such as this because the method name does not reveal the data type of the object that it is returning. Although the C# compiler will process this statement as intended, a person who is reading this code might have trouble understanding the data type of the `car` variable. Remember, your code should be self-documenting. Any time you use the `var` keyword to declare a variable, make sure the data type can be clearly understood from the initialization value.

Comparing Structure Objects

You cannot perform comparison operations directly on structure objects. For example, assume that `sportsCar` and `raceCar` are instances of the `Automobile` structure. The following statement will cause an error:

```
if (sportsCar == raceCar) // Error!
```

In order to compare two structure objects, you must compare the individual fields, as shown in the following code:

```
if (sportsCar.make == raceCar.make &&
    sportsCar.year == raceCar.year &&
    sportsCar.mileage == raceCar.mileage)
{
    MessageBox.Show("The two are equal.");
}
```

Arrays of Structure Objects

Structure objects can be stored in an array. For example, assume the `Automobile` structure previously shown exists in an application. The following code creates an array of five `Automobile` objects:

```
const int SIZE = 5;
Automobile[] cars = new Automobile[SIZE];
```

Alternatively, if the array is declared inside a method, you can use the `var` keyword as shown here:

```
const int SIZE = 5;
var cars = new Automobile[SIZE];
```

When you create a structure array, each element of the array is a structure instance and the fields of each instance are initialized to 0. (If any field is a reference variable, it is initialized to the value `null`.)

Each element of a structure array may be accessed through a subscript. For example, `cars[0]` is the first object in the array, `cars[1]` is the second, and so forth. To access a field of any element, simply place the dot operator and field name after the subscript. For example, the following expression refers to the `mileage` member of `cars[2]`:

```
cars[2].mileage
```

The following `for` loop steps through the `cars` array, displaying the data stored in each element:

```
for (int index = 0; index < cars.Length; index++)
{
    MessageBox.Show(cars[index].year + " " +
        cars[index].make + " with " +
        cars[index].mileage + " miles.");
}
```

You can also use the `foreach` loop to iterate over all of the elements in a structure array, as shown in the following code:

```
foreach (Automobile aCar in cars)
{
    MessageBox.Show(aCar.year + " " +
        aCar.make + " with " +
        aCar.mileage + " miles.");
}
```

Storing Structure Objects in a `List`

In Chapter 7, we discussed the `List` class, which is a container for storing a collection of objects. Here is an example of how you would create a `List` that can hold `Automobile` objects:

```
var carList = new List<Automobile>();
```

This statement creates a `List` object, referenced by the `carList` variable. Notice that the word `Automobile` is written inside angled brackets, `<>`, immediately after the word `List`. This specifies that the `List` can hold only objects of the `Automobile` data type.

To add a structure object to a `List`, you use the `Add` method. The following code shows an example:

```
 1 // Create a List to hold Automobile objects.
 2 var carList = new List<Automobile>();
 3
 4 // Create an instance of the Automobile structure.
 5 var sportsCar = new Automobile();
 6
 7 // Assign values to the object's fields.
 8 sportsCar.make = "Chevy Corvette";
 9 sportsCar.year = 1970;
10 sportsCar.mileage = 50000.0;
11
12 // Add the object to the List.
13 carList.Add(sportsCar);
```

The statement in line 2 creates a `List` named `carList` that can hold `Automobile` objects. Line 5 creates an instance of the `Automobile` structure named `sportsCar`, and lines 8–10 assign values to the object's fields. Line 13 adds the object to the `List`.

Keep in mind that structure instances are value type objects, and when you add a value type object to a `List`, the `List` will contain a copy of the object. For example, look at the following code:

```
 1 // Create a List to hold Automobile objects.
 2 var carList = new List<Automobile>();
 3
 4 // Create an instance of the Automobile structure.
 5 var sportsCar = new Automobile();
 6
 7 // Assign values to the object's fields.
 8 sportsCar.make = "Chevy Corvette";
 9 sportsCar.year = 1970;
10 sportsCar.mileage = 50000.0;
11
12 // Add the object to the List.
13 carList.Add(sportsCar);
14
15 // Assign new values to the object's fields.
16 sportsCar.make = "Ford Mustang";
17 sportsCar.year = 1965;
18 sportsCar.mileage = 67500.0;
19
20 // Add the object to the List.
21 carList.Add(sportsCar);
```

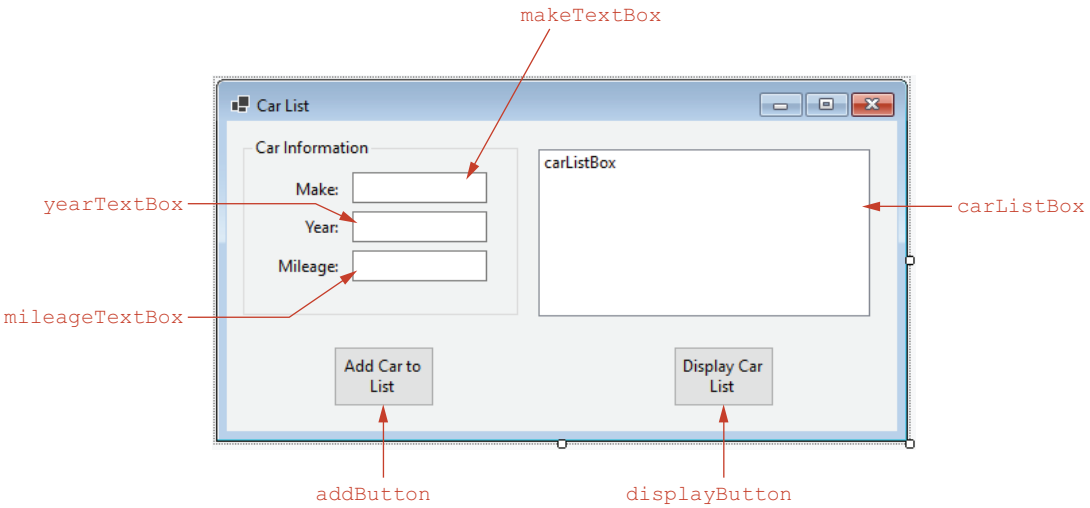
The statement in line 2 creates a `List` named `carList` that can hold `Automobile` objects. Lines 5–10 create an `Automobile` object named `sportsCar` and assign values to the object's fields. Line 13 adds the `sportsCar` object to the `List`. When line 13 executes, a

copy of the `sportsCar` object is made and stored in the `List`. (This happens because the `sportsCar` object is a value type object.) At this point, two instances of the `Automobile` structure exist: the `sportsCar` object and the copy of the `sportsCar` object that is in the `List`.

Lines 16–18 assign new values to the `sportsCar` object's fields, and then line 21 adds the `sportsCar` object to the `List` again. Once more, a copy of the `sportsCar` object is made and added to the `List`. At this point, three instances of the `Automobile` structure exist: the `sportsCar` object and the two objects that are in the `List`.

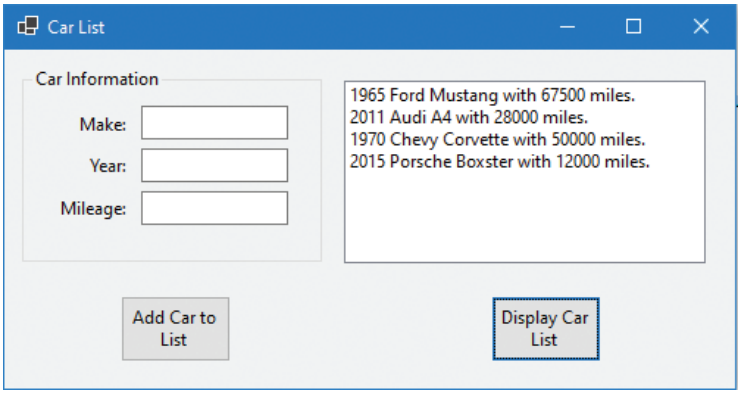
Let's look at a complete program that demonstrates how structure objects can be added to a `List`. In the *Chap09* folder of this book's Student Sample Programs, you will find a project named *CarList*. Figure 9-4 shows the application's form in the *Designer*.

Figure 9-4 The `CarList` project's form



When you run the application, you can enter data about a car into the TextBox controls. When you click the `addButton` control, that data is assigned to an `Automobile` object and then added to a `List`. You can do this as many times as you wish. When you click the `displayButton` control, the data from each object in the `List` is displayed in the `carListBox` control. Figure 9-5 shows an example of the application's form after four objects have been added to the `List` and the `displayButton` control has been clicked.

Figure 9-5 The `CarList` project's form with data displayed



Program 9-1 shows the application's `Form1` code. Let's take a closer look:

Lines 3–8: The `Automobile` structure is declared in these lines.

Line 13: This statement creates a `List` that can hold `Automobile` objects. Notice that the `List` is declared as a field in the `Form1` class. All the methods in the class can access it.

Lines 23–37: The `GetData` method appears in these lines. The method accepts an `Automobile` object, by reference, as an argument. The data that has been entered into the form's TextBox controls is assigned to the object's fields.

Lines 39–57: The `addButton_Click` event handler appears in these lines. Here is a summary of the event handler's code:

- Line 42 creates the `car` object, which is an instance of the `Automobile` structure.
- Line 45 calls the `GetData` method, passing the `car` object, by reference, as an argument. After this statement executes, the `car` object's fields contain the data that was entered into the form's TextBoxes by the user.
- Line 48 adds a copy of the `car` object to the `carList`.
- Lines 51–53 clear the contents of the TextBoxes.
- Line 56 gives the focus to the `makeTextBox` control.

Lines 59–71: The `displayButton_Click` event handler appears in these lines. Here is a summary of the event handler's code:

- Line 62 clears the ListBox control's current contents.
- The `foreach` loop in lines 65–70 displays data about each object in the `carList`.
  - The statement in lines 68–69 adds the results of an interpolated string to the `carListBox` control. This displays a line of information about a car.

Program 9-1 Code for Form1 in the *CarList* project

```
 1 namespace CarList
 2 {
 3     struct Automobile
 4     {
 5         public string make;
 6         public int year;
 7         public double mileage;
 8     }
 9
10     public partial class Form1 : Form
11     {
12         // Create a List as a field.
13         private List<Automobile> carList = new List<Automobile>();
14
15         public Form1()
16         {
17             InitializeComponent();
18         }
19
20         // The GetData method gets the data entered
21         // by the user and assigns it to the parameter
22         // object's fields.
23         private void GetData(ref Automobile auto)
24         {
25             try
26             {
27                 // Get the data from the TextBoxes.
28                 auto.make = makeTextBox.Text;
29                 auto.year = int.Parse(yearTextBox.Text);
30                 auto.mileage = double.Parse(mileageTextBox.Text);
```

```
31              }
32              catch (Exception ex)
33              {
34                  // Display the exception message.
35                  MessageBox.Show(ex.Message);
36              }
37          }
38
39          private void addButton_Click(object sender, EventArgs e)
40          {
41              // Create an instance of the Automobile structure.
42              var car = new Automobile();
43
44              // Get the data entered by the user.
45              GetData(ref car);
46
47              // Add the car object to the List.
48              carList.Add(car);
49
50              // Clear the TextBoxes.
51              makeTextBox.Clear();
52              yearTextBox.Clear();
53              mileageTextBox.Clear();
54
55              // Reset the focus.
56              makeTextBox.Focus();
57          }
58
59          private void displayButton_Click(object sender, EventArgs e)
60          {
61              // Clear the ListBox's current contents.
62              carListBox.Items.Clear();
63
64              // Display the car info in the ListBox.
65              foreach (Automobile car in carList)
66              {
67                  // Display the car info in the ListBox.
68                  carListBox.Items.Add(
69                     $"{car.year} {car.make} with {car.mileage}");
70              }
71          }
72      }
73  }
```

In Tutorial 9-1, you complete a phone book application that uses a `List` of structure objects.

Tutorial 9-1:
Completing the *Phonebook* Project

VideoNote
Tutorial 9-1: Completing the *Phonebook* Project

In this tutorial, you complete the *Phonebook* project. When the application is complete, it will let you select a person's name from a ListBox control, and then it will display that person's phone number. Figure 9-6 shows an example of how the form will appear at run time. In the figure, Alex Prieto has been selected in the ListBox control, and Alex's phone number is displayed in a Label control. The application's form has already been created for you and is shown in Figure 9-7 with the names of some of its controls.

Figure 9-6 The *Phonebook* project's form with a name selected

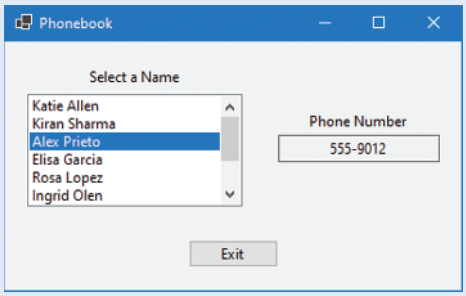
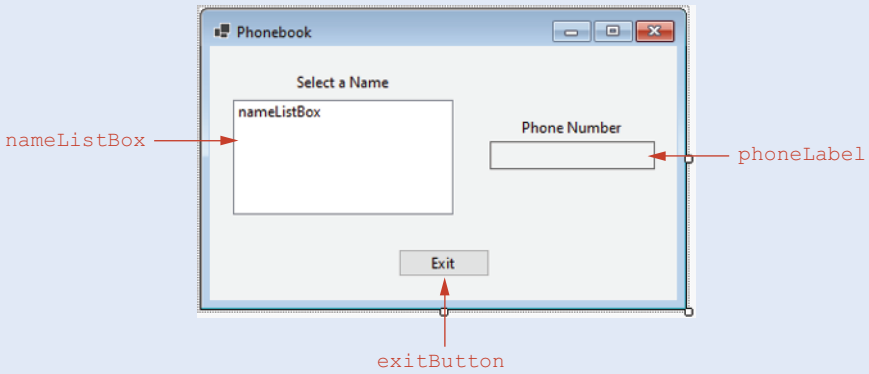
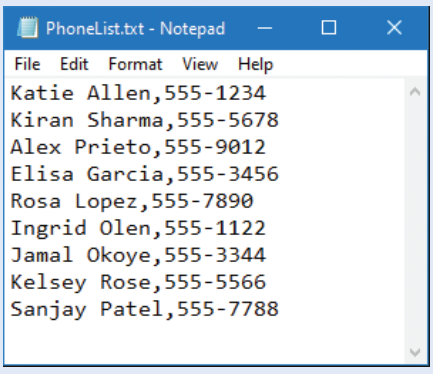


Figure 9-7 The *Phonebook* project's form in the *Designer*



Before you start writing code, let's go over a summary of how the application will work. A file named PhoneList.txt, containing several names and corresponding phone numbers, has been created for you and is included with the project. Its contents are shown in Figure 9-8. Notice that each line in the file contains a name, followed by a comma, followed by a phone number.

Figure 9-8 The contents of the PhoneList.txt file



When the application starts, the form's Load event handler calls a method named `ReadFile`. The `ReadFile` method reads the contents of the PhoneList.txt file. Each line that is read from the file will be tokenized, using the comma character as a delimiter. This will result in two `string` tokens: one containing a person's name and the other

containing a phone number. The tokens are stored in an instance of the following `PhoneBookEntry` structure:

```
struct PhoneBookEntry
{
    public string name;
    public string phone;
}
```

The `PhoneBookEntry` object is then added to a `List`. When the `ReadFile` method is finished, the `List` will contain a `PhoneBookEntry` object for each line in the PhoneList.txt file.

The Load event handler then calls another method named `DisplayNames`. The `DisplayNames` method steps through the `List`, getting the `name` field of each object and adding the name to the ListBox control. When the `DisplayNames` method is finished, the ListBox control will display all the names that are contained in the `List`.

When the user clicks a name in the ListBox control, a SelectedIndexChanged event occurs. You are to write an event handler that responds to this event. The event handler gets the index of the selected item in the ListBox control, and it uses that index to retrieve an object from the `List`. The object contains the phone number for the selected name, which is displayed in a Label control.

Step 1: Start Visual Studio. Open the project named *Phonebook* in the *Chap09* folder of the Student Sample Programs.

Step 2: Open the Form1 form's code in the code editor.

Write the declaration of the `PhoneBookEntry` structure, shown in lines 3–7. Make sure you write the declaration inside the `Phonebook` namespace, exactly as shown in Program 9-2.

Step 3: Next you write the comment and the declaration for `phoneList`, which appears in lines 11–12 in Program 9-2. Notice that `phoneList` is declared as a field in the `Form1` class. It is available to all the methods in the class.

Step 4: Write the comments and code for the `ReadFile` method, shown in lines 19–60 of Program 9-2. The `ReadFile` method opens the PhoneList.txt file. The `while` loop that begins in line 39 iterates until the end of the file is reached. In each iteration of the loop, the following lines of code are executed:

Line 42: This statement reads a line from the file and assigns it to the `line` variable.

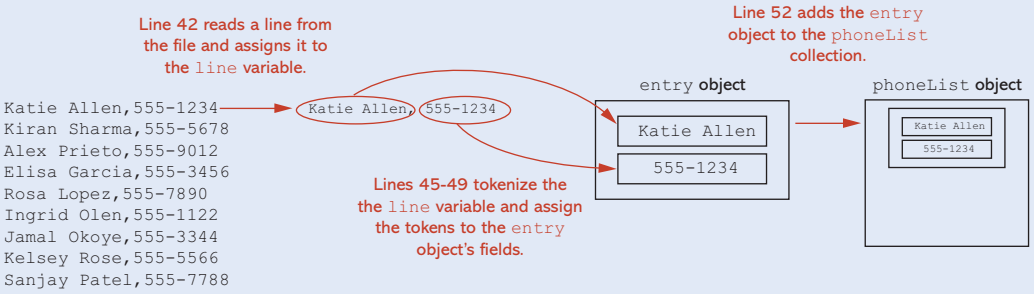
Line 45: This statement tokenizes the `line` variable, using the comma character as a delimiter. The tokens are assigned to the `tokens` array. (After this statement executes, `tokens[0]` will contain a name and `tokens[1]` will contain a phone number.)

Lines 48–49: These statements assign the tokens to the `name` and `phone` fields of the `entry` object (which is an instance of the `PhoneBookEntry` structure).

Line 52: This statement adds the `entry` object to the `phoneList` collection.

Figure 9-9 illustrates the steps performed in these lines of code. After the loop is finished, `phoneList` will contain a `PhoneBookEntry` object for each line in the PhoneList.txt file.

Figure 9-9 A summary of lines 42–52



Step 5: Write the comments and code for the `DisplayNames` method, shown in lines 62–70 of Program 9-2. The `foreach` loop in line 66 steps through all the objects in `phoneList`. For each object in `phoneList`, line 68 adds the `name` field to the `nameListBox` control.

Note that the names are added to the ListBox control in the same order that they appear in `phoneList`. For example,

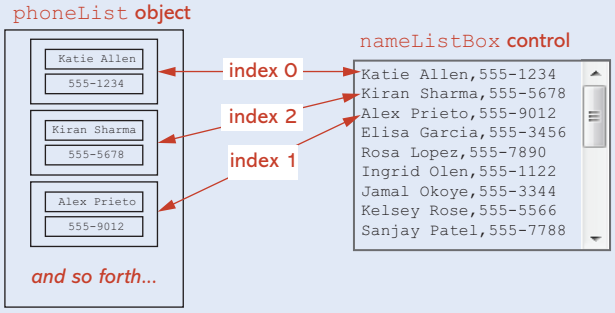
The name "Katie Allen" is at index 0 in the ListBox, and the object containing "Katie Allen" is at index 0 in `phoneList`.

The name "Kiran Sharma" is at index 1 in the ListBox, and the object containing "Kiran Sharma" is at index 1 in `phoneList`.

Figure 9-10 illustrates how the objects in `phoneList` and the names in the `nameListBox` control are related by their indexes. You take advantage of this relationship in Step 7 when you write the code to retrieve an object from `phoneList`.

Step 6: Next, you create the form's Load event handler. Switch back to the *Designer* and double-click any part of the form that does not contain a control. This opens the code editor, and you will see an empty event handler named `Form1_Load`. Complete the event handler by typing the code shown in lines 74–78 in Program 9-2. The statement in line 75 calls the `ReadFile` method, and line 78 calls the `DisplayNames` method.

Figure 9-10 Objects in `phoneList` and names in `nameListBox` as related by their indexes



NOTE: When the items in two data structures are related by their indexes, it is said that a **parallel relationship** exists between the data structures.

Step 7: Now you create the SelectedIndexChange event handler for the `nameListBox` control. Switch back to the *Designer* and double-click the `nameListBox` control. This opens the code editor, and you will see an empty event handler named

nameListBox_SelectedIndexChanged. Complete the event handler by typing the code shown in lines 83–87 in Program 9-2.

We mentioned in Step 5 that the names in the ListBox and the objects in phoneList are related by their indexes. When the user selects a name from the ListBox, all we have to do is get that item's index and then use that same index to retrieve the corresponding object from phoneList. That is exactly what happens in this event handler. Line 84 gets the index of the selected item in the nameListBox control and assigns it to the index variable. Line 87 uses the index variable to get an object from phoneList. The retrieved object's phone field is assigned to phoneLabel.Text.

Step 8: Switch your view back to the *Designer* and double-click the exitButton control. In the code editor you will see an empty event handler named exitButton_Click. Complete the event handler by typing the code shown in lines 92–93 in Program 9-2.

Step 9: Save the project. Then, press on the keyboard or click the *Start Debugging* button () on the toolbar to compile and run the application. When the application runs, select various names in the ListBox to confirm that the application works properly. When you are finished, click the *Exit* button to exit the application.

Program 9-2 Completed code for Form1 in the *Phonebook* project

```
 1 namespace Phonebook
 2 {
 3    struct PhoneBookEntry
 4    {
 5        public string name;
 6        public string phone;
 7    }
 8
 9    public partial class Form1 : Form
10    {
11        // Field to hold a list of PhoneBookEntry objects.
12        private List<PhoneBookEntry> phoneList = new
          List<PhoneBookEntry>();
13
14        public Form1()
15        {
16            InitializeComponent();
17        }
18
19        // The ReadFile method reads the contents of the
20        // PhoneList.txt file and stores it as PhoneBookEntry
21        // objects in the phoneList.
22        private void ReadFile()
23        {
24            try
25            {
26                StreamReader inputFile; // To read the file
27                string line; // To hold a line from the file
28
29                // Create an instance of the PhoneBookEntry
                  structure.
30                var entry = new PhoneBookEntry();
31
32                // Create a delimiter array.
33                char[] delim = {','};
34
```

```
35                // Open the PhoneList file.
36                inputFile = File.OpenText("PhoneList.txt");
37
38                // Read the lines from the file.
39                while (!inputFile.EndOfStream)
40                {
41                    // Read a line from the file.
42                    line = inputFile.ReadLine();
43
44                    // Tokenize the line
45                    string[] tokens = line.Split(delim);
46
47                    // Store the tokens in the entry object.
48                    entry.name = tokens[0];
49                    entry.phone = tokens[1];
50
51                    // Add the entry object to the List.
52                    phoneList.Add(entry);
53                }
54            }
55            catch (Exception ex)
56            {
57                // Display an error message.
58                MessageBox.Show(ex.Message);
59            }
60        }
61
62        // The DisplayNames method displays the list of names
63        // in the namesListBox control.
64        private void DisplayNames()
65        {
66            foreach (var entry in phoneList)
67            {
68                nameListBox.Items.Add(entry.name);
69            }
70        }
71
72        private void Form1_Load(object sender, EventArgs e)
73        {
74            // Read the PhoneList.txt file.
75            ReadFile();
76
77            // Display the names.
78            DisplayNames();
79        }
80
81        private void nameListBox_SelectedIndexChanged(object
          sender, EventArgs e)
82        {
83            // Get the index of the selected item.
84            int index = nameListBox.SelectedIndex;
85
86            // Display the corresponding phone number.
87            phoneLabel.Text = phoneList[index].phone;
88        }
89
90        private void exitButton_Click(object sender, EventArgs e)
91        {
92            // Close the form.
93            this.Close();
94        }
95    }
96 }
```

Checkpoint

9.1 When a structure is declared, what is created in memory?

9.2 What does the keyword `public` specify about a field declared in a structure?

9.3 When you create an instance of a structure, what are you creating?

9.4 What effect does the `new` operator have when creating an instance of a structure?

9.5 Once you have created an instance of a structure, how can you access its fields?

9.6 Suppose an application contains a structure named `Engine` and that `motor1` and `motor2` are instances of the `Engine` structure. Write a statement that assigns `motor1` to `motor2`.

9.7 Write a statement that creates an array named `motors` that can hold 100 `Engine` objects.

9.8 What causes a parallel relationship to exist between data structures?

9.3 The `DateTime` and `TimeSpan` Structures

CONCEPT: **In .NET, the `DateTime` structure stores a date, a time, or both. The `TimeSpan` structure stores information about a time interval.**

The **`DateTime` structure**, which is defined in .NET, allows you to create objects that store data about an instant in time. A `DateTime` object stores both date and time. Internally, the date and time values are stored in the following properties:

- Month—an integer in the range of 1–12
- Day—an integer in the range of 1–31
- Year—an integer in the range of 1–9999
- Hour—an integer in the range of 0–23
- Minute—an integer in the range of 0–59
- Second—an integer in the range of 0–59
- Millisecond—an integer in the range of 0–999

Notice that the Hour property is set according to the 24-hour clock. For example, the hour 1 is 1:00 AM, and the hour 13 is 1:00 PM.

There are several ways to create an instance of the `DateTime` structure. The following statement creates a `DateTime` instance named `dt`, set to the default date January 1, 0001 and the default time 12:00:00 AM, 0 milliseconds:

```
DateTime dt = new DateTime();
```

If the instance you are creating is declared in a method, you can alternatively use the `var` keyword, as shown here:

```
var dt = new DateTime();
```

You can change the individual date and time properties as shown in the following code sample:

```
var dt = new DateTime();
dt.Month = 12;
dt.Day = 15;
dt.Year = 2019;
dt.Hour = 13;
```

```
dt.Minute = 30;
dt.Second = 0;
dt.Millisecond = 0;
```

In this example, the `dt` instance is set to the date December 15, 2019 and the time 1:00:00 PM.

When you create a `DateTime` instance, you can optionally initialize it with a specific date, time, or both. Here are three general formats:

```
DateTime instanceName  =  new DateTime(Year, Month, Day);
DateTime instanceName  =  new DateTime(Year, Month, Day, Hour,
                          Minute, Second);
DateTime instanceName  =  new DateTime(Year, Month, Day, Hour,
                          Minute, Second, Millisecond);
```

The values for *Year*, *Month*, *Day*, *Hour*, *Minute*, *Second*, and *Millisecond* are integers. The following example creates a `DateTime` instance named `dt`, set to the date November 5, 2000. The Hour, Minute, Second, and Millisecond properties are set to the default value of 0, which is midnight:

```
var dt = new DateTime(2000, 11, 5);
```

The following example creates a `DateTime` instance named `dt`, set to the date November 5, 2000, and the time 8:30 AM. The Second property is set to 15, and the Millisecond property is set to the default value of 0:

```
var dt = new DateTime(2000, 11, 5, 8, 30, 15);
```

The following example creates a `DateTime` instance named `dt`, set to the date November 5, 2000, and the time 8:30 AM. The Second property is set to 15, and the Millisecond property is set to 500:

```
var dt = new DateTime(2000, 11, 5, 8, 30, 15, 500);
```

Displaying a `DateTime` Object

You can retrieve any of a `DateTime` object's individual properties and use them in your program's output. For example, the following code sample displays a message using the `DateTime` object's Month and Year properties:

```
var dt = new DateTime(2000, 11, 5);
MessageBox.Show(dt.Month.ToString() + ", " + dt.Year.ToString());
```

In many cases, the easiest way to display a `DateTime` object is to call its `ToString` method. You can use format strings to format it as a short date, long date, short time, long time, and so on. Table 9-1 lists the most commonly used format strings for dates and times.

Table 9-1 Common **`DateTime`** format strings used with the **`ToString`** method

Format String	Description
`d`	Shows the date only, in **short date** format which shows the month, day, and year. An example is *8/23/2024*.
`D`	Shows the date only, in **long date** format which shows the day of the week, month, day, and year. An example is *Friday, August 23, 2024*.
`t`	Shows the time only, in **short time** format which shows the hours and minutes. An example is *3:22 PM*.
`T`	Shows the time only, in **long time** format which shows the hours, minutes, seconds, and an AM/PM indicator. An example is *3:22:00 PM*.
`F`	Shows both the full date and time in **long date** and **long time** format. An example is *Friday, August 23, 2024 3:22:00 PM*.

The following code sample shows an example of the short date format. When this code runs, the string *11/5/2000* will be displayed in a message box:

```
var dt = new DateTime(2000, 11, 5, 8, 30, 15, 500);
MessageBox.Show(dt.ToString("d"));
```

The following code sample shows an example of the long date format. When this code runs, the string *Sunday, November 5, 2000* will be displayed in a message box:

```
var dt = new DateTime(2000, 11, 5, 8, 30, 15, 500);
MessageBox.Show(dt.ToString("D"));
```

The following code sample shows an example of the short time format. When this code runs, the string *8:30 AM* will be displayed in a message box:

```
var dt = new DateTime(2000, 11, 5, 8, 30, 15, 500);
MessageBox.Show(dt.ToString("t"));
```

The following code sample shows an example of the long time format. When this code runs, the string *8:30:15 AM* will be displayed in a message box:

```
var dt = new DateTime(2000, 11, 5, 8, 30, 15, 500);
MessageBox.Show(dt.ToString("T"));
```

The following code sample shows an example of the full date and time format. When this code runs, the string *Sunday, November 5, 2000 8:30:15 AM* will be displayed in a message box:

```
var dt = new DateTime(2000, 11, 5, 8, 30, 15, 500);
MessageBox.Show(dt.ToString("F"));
```

The `DateTime` structure also supports the following methods for formatting:

- `ToShortDateString()`—Returns the date formatted as a short date, which is the same as `ToString("d")`
- `ToLongDateString()`—Returns the date formatted as a long date, which is the same as `ToString("D")`
- `ToShortTimeString()`—Returns the time formatted as a short time, which is the same as `ToString("t")`
- `ToLongTimeString()`—Returns the time formatted as a long time, which is the same as `ToString("D")`

Getting the Current Date and Time from the System

The expression `DateTime.Now` gives you the current date and time from the system. Here is an example of how you can assign the system's current date and time to a `DateTime` variable:

```
var currentDateTime = DateTime.Now;
```

The expression `DateTime.Today` gives you the current date, but not the time. Here is an example of how you can assign the system's current date to a `DateTime` variable:

```
var currentDate = DateTime.Today;
```

After this statement is executed, the `currentDate` object will be set to the current date, but the object's time properties will be set to the default value of 12:00 AM.

Converting Strings to `DateTime` Structures

You are already familiar with the `TryParse` method, and how to use it to convert strings to numeric data types. You can also use the `TryParse` method to convert a string to a `DateTime` object. Here is the general format:

```
DateTime.TryParse(string, out targetVariable)
```

In the general format, *string* is the string that you want to convert, and *targetVariable* is the name of a DateTime object. The method tries to convert the *string* argument to a DateTime object. If the conversion is successful, the converted value is assigned to the *targetVariable*, and the method returns the Boolean value true to indicate that the conversion was successful. If the conversion is not successful, the method stores the default date January 1, 0001 and the default time 12:00:00 AM in the *targetVariable* and returns the Boolean value false to indicate that the *string* could not be converted.

Here are some examples of strings that can be successfully converted to a DateTime object with the TryParse method:

- "4/20/2024" is converted to 4/20/2024 12:00:00 AM
- "2024-4-20" is converted to 4/20/2024 12:00:00 AM
- "20 April 2024" is converted to 4/20/2024 12:00:00 AM
- "4/20/2024 3:45:50 PM" is converted to 4/20/2024 3:45:50 PM
- "4/20/2024 3:45:50.500 PM" is converted to 4/20/2024 3:45:50 PM, 500 milliseconds

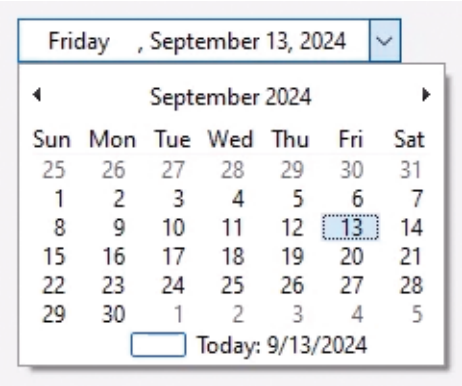
You can use the DateTime.TryParse method to convert dates and/or times that have been entered into a TextBox control. The following code sample shows an example. Assume dateTimeTextBox is a TextBox control:

```
var dt = new DateTime();
if (DateTime.TryParse(dateTimeTextBox.Text, out dt))
{
    MessageBox.Show("Successful Conversion");
}
else
{
    MessageBox.Show("Failed Conversion");
}
```

The DateTimePicker Control

The **DateTimePicker control** (found in the Toolbox's *Common Controls* group) provides an easy way for the user to select dates. Figure 9-11 shows an example. It is a drop-down control that displays a small scrollable calendar showing an entire month. The user picks a date from the calendar and your program retrieves that date as a DateTime object from the control's Value property. Initially, the Value property is set to the current date, and the current date will appear selected in the control's calendar.

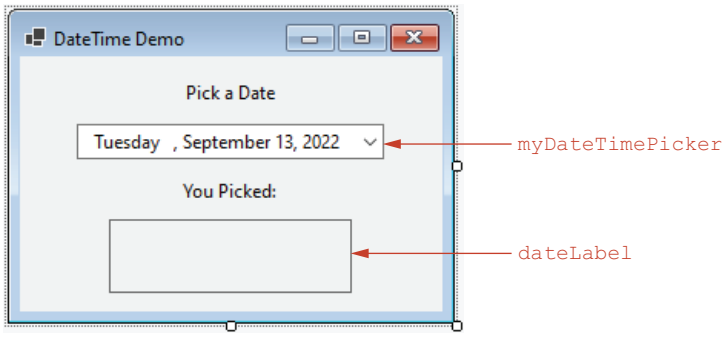
Figure 9-11 The DateTimePicker control



When the user selects a date from a DateTimePicker control, a ValueChanged event is triggered. If you write a ValueChanged event handler for the control, the event handler will execute any time a date is selected. To generate a code template for the ValueChanged event handler, simply double-click the DateTimePicker control in the *Designer* window.

Figure 9-12 shows the Form1 form from the *DateTimeDemo* project in the Student Sample Programs folder for this chapter. As shown in Figure 9-13, the user selects a date from the `myDateTimePicker` control's drop-down calendar, and then the selected date is displayed in the `dateLabel` control. The application's code is shown in Program 9-3.

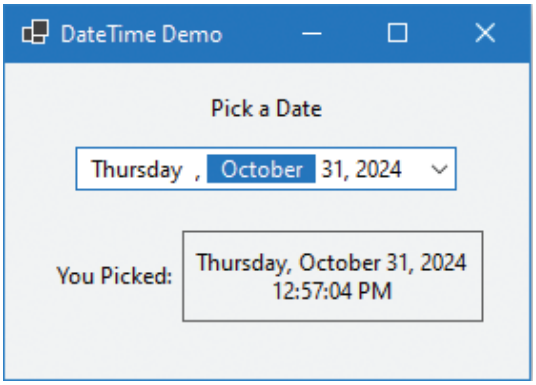
Figure 9-12 The *DateTimeDemo* project's form



Program 9-3 The *DateTime Demo* application's Form1 code

```
1  namespace DateTimeDemo
2  {
3      public partial class Form1 : Form
4      {
5          public Form1()
6          {
7              InitializeComponent();
8          }
9
10         private void myDateTimePicker_ValueChanged(object sender,
           EventArgs e)
11         {
12             // Get the date selected by the user.
13             DateTime selected = myDateTimePicker.Value;
14
15             // Display the date.
16             dateLabel.Text = selected.ToString("F");
17         }
18     }
19 }
```

Figure 9-13 The *DateTime Demo* application



Comparing DateTime Objects

You can use C#'s relational operators to compare two DateTime objects, and determine whether one of the objects is later than, earlier than, or equal to the other object. When used with DateTime objects, the relational operators work like this:

- The expression obj1 > obj2 will be true if obj1 is later than obj2.
- The expression obj1 < obj2 will be true if obj1 is earlier than obj2.
- The expression obj1 == obj2 will be true if obj1 is the same as obj2.
- The expression obj1 != obj2 will be true if obj1 is not the same as obj2.
- The expression obj1 >= obj2 will be true if obj1 is later than, or the same as, obj2.
- The expression obj1 <= obj2 will be true if obj1 is earlier than, or the same as, obj2.

For example, suppose we have the following structure declaration in a program:

```
1 struct Person
2 {
3     public string name;
4     public DateTime birthdate;
5 }
```

This structure holds a person's name, as a string, and their birthdate, as a DateTime object. The following code sample creates two instances of the Person structure and then determines which person is the oldest:

```
 1 // Create the person1 object.
 2 var person1 = new Person();
 3 person1.name = "Emily";
 4 person1.birthdate = new DateTime(1996, 10, 7);
 5
 6 // Create the person2 object.
 7 var person2 = new Person();
 8 person2.name = "Dave";
 9 person2.birthdate = new DateTime(1994, 4, 12);
10
11 // Determine who is the oldest.
12 if (person1.birthdate > person2.birthdate)
13 {
14     MessageBox.Show($"{person2.name} is older than {person1.name}");
15 }
16 else if (person2.birthdate > person1.birthdate)
17 {
18     MessageBox.Show($"{person1.name} is older than {person2.name}");
19 }
20 else if (person1.birthdate == person2.birthdate)
21 {
22     MessageBox.Show("They are the same age.");
23 }
```

The TimeSpan Structure

The **TimeSpan structure** allows you to create objects that store data about an amount of time, such as five days, or 20 hours.

A TimeSpan object stores time values in the following properties:

- Days—an integer, positive or negative
- Hours—an integer in the range of −23 through 23
- Minutes—an integer in the range of −59 through 59
- Seconds—an integer in the range of −59 through 59
- Milliseconds—an integer in the range of −999 through 999

There are several ways to create an instance of the `TimeSpan` structure. The following statement creates a `TimeSpan` instance named `tspan`, set to the default value of 0 days, 0 hours, 0 minutes, 0 seconds, and 0 milliseconds:

```
TimeSpan tspan = new TimeSpan();
```

If the instance you are creating is declared in a method, you can alternatively use the `var` keyword, as shown here:

```
var tspan = new TimeSpan();
```

You can change the individual properties as shown in the following code sample:

```
var tspan = new TimeSpan();
tspan.Days = 15;
tspan.Hours = 6;
tspan.Minutes = 30;
tspan.Seconds = 10;
tspan.Milliseconds = 500;
```

In this example, the `tspan` object is set to 15 days, 6 hours, 30 minutes, 10 seconds, and 500 milliseconds.

When you create a `TimeSpan` object, you can optionally initialize it with specific values. Here are three general formats:

```
TimeSpan instanceName = new TimeSpan(Hours, Minutes, Seconds);
TimeSpan instanceName = new TimeSpan(Days, Hours, Minutes, Seconds);
TimeSpan instanceName = new TimeSpan(Days, Hours, Minutes, Seconds,
                                     Milliseconds);
```

In the general formats, the values for *`Days`*, *`Hours`*, *`Minutes`*, *`Seconds`*, and *`Milliseconds`* are integers.

Performing Math with `DateTime` and `TimeSpan` Objects

You can perform math with `DateTime` and `TimeSpan` objects in the following ways:

- You can add a `TimeSpan` to a `DateTime` object to get another `DateTime` object that is later than the first object. For example, suppose we have a `DateTime` object that is set to April 5, and we have a `TimeSpan` object that is set to 10 days. If we add the `TimeSpan` object to the `DateTime` object, we will get another `DateTime` object that is set to April 15.
- You can subtract one `DateTime` object from another `DateTime` object to get a `TimeSpan` object representing the amount of time between the two `DateTime` objects. For example, suppose we have a `DateTime` object that is set to January 20, and we subtract from it another `DateTime` object that is set to January 15. The result would be a `TimeSpan` object that is set to 10 days.

Here is an example of how we can create a `TimeSpan` object that represents 10 days, and then add that object to a `DateTime` object:

```
1 // Create a DateTime object for August 8, 2024.
2 DateTime dt = new DateTime(2024, 8, 15);
3
4 // Create a TimeSpan object for 10 days.
5 TimeSpan tspan = new TimeSpan(10, 0, 0, 0);
6
7 // Add the TimeSpan object to the DateTime object.
8 DateTime futureDate = dt + tspan;
```

After this code is executed, the `futureDate` object will be set to August 25, 2024.

The following code sample shows how we can subtract one `DateTime` object from another `DateTime` object. The result is a `TimeSpan` object representing the amount of time between the two `DateTime` objects:

```
// Create two DateTime objects.
DateTime date1 = new DateTime(2024, 8, 15);
DateTime date2 = new DateTime(2024, 8, 30);

// Get the amount of time between the two dates.
TimeSpan tspan = date2 - date1;
```

After this code is executed, the `tspan` object will be set to 15 days.

Table 9-2 shows several methods that you can also use to mathematically adjust the value of a `DateTime` object.

Table 9-2 `DateTime` math methods

Method	Description
`Add`	General Format: *`dtObject`*`.Add(`*`tspan`*`)` In the general format, *`dtObject`* is a `DateTime` object and *`tspan`* is a `TimeSpan` object. The `TimeSpan` object is added to the `DateTime` object and a new `DateTime` object containing the result is returned.
`AddYears`	General Format: *`dtObject`*`.AddYears(`*`years`*`)` In the general format, *`dtObject`* is a `DateTime` object and *`years`* is an `int` specifying a number of years. The number of years is added to the `DateTime` object and a new `DateTime` object containing the result is returned.
`AddMonths`	General Format: *`dtObject`*`.AddMonths(`*`months`*`);` In the general format, *`dtObject`* is a `DateTime` object and *`months`* is an `int` specifying a number of months. The number of months is added to the `DateTime` object and a new `DateTime` object containing the result is returned.
`AddDays`	General Format: *`dtObject`*`.AddDays(`*`days`*`)` In the general format, *`dtObject`* is a `DateTime` object and *`days`* is an `int` specifying a number of days. The number of days is added to the `DateTime` object and a new `DateTime` object containing the result is returned.
`AddHours`	General Format: *`dtObject`*`.AddHours(`*`hours`*`)` In the general format, *`dtObject`* is a `DateTime` object and *`hourss`* is an `int` specifying a number of hours. The number of hours is added to the `DateTime` object and a new `DateTime` object containing the result is returned.
`AddMinutes`	General Format: *`dtObject`*`.AddMinutes(`*`minutes`*`)` In the general format, *`dtObject`* is a `DateTime` object and *`minutes`* is an `int` specifying a number of minutes. The number of minutes is added to the `DateTime` object and a new `DateTime` object containing the result is returned.
`AddSeconds`	General Format: *`dtObject`*`.AddSeconds(`*`seconds`*`)` In the general format, *`dtObject`* is a `DateTime` object and *`seconds`* is an `int` specifying a number of seconds. The number of seconds is added to the `DateTime` object and a new `DateTime` object containing the result is returned.

(*Continued*)

Table 9-2 `DateTime` math methods (*Continued*)

Method	Description
`AddMilliseconds`	General Format: *`dtObject`*`.AddMilliseconds(`*`milliseconds`*`)` In the general format, *`dtObject`* is a `DateTime` object and *`milliseconds`* is an `int` specifying a number of milliseconds. The number of milliseconds is added to the `DateTime` object and a new `DateTime` object containing the result is returned.
`Subtract` (*version 1*)	General Format: *`dtObject1`*`.Subtract(`*`dtObject2`*`)` In the general format, *`dtObject1`* is a `DateTime` object and *`dtObject2`* is another `DateTime` object. The *`dtObject2`* object is subtracted from the *`dtObject1`* object and a new `DateTime` object containing the result is returned.
`Subtract` (*version 2*)	General Format: *`dtObject`*`.Subtract(`*`tspan`*`)` In the general format, *`dtObject`* is a `DateTime` object and *`tspan`* is a `TimeSpan` object. The *`tspan`* object is subtracted from the *`dtObject`* object and a new `DateTime` object containing the result is returned.

Checkpoint

9.9 If you create a `DateTime` object without initializing it, what default date and time will it hold?

9.10 Give an example of instantiating a `DateTime` object named `dt`, initialized with the date March 22, 1965.

9.11 Give an example of how you would display a `DateTime` object named `dt` in short date format, in a message box.

9.12 Give an example of instantiating a `DateTime` object named `dt`, initialized with the system's current date and time.

9.13 Give an example of instantiating a `DateTime` object named `dt`, initialized with the system's current date, but not the time.

9.14 Given the following declarations, would the expression `date1 > date2` be true or false?

```
DateTime date1 = new DateTime(1977, 3, 22);
DateTime date2 = new DateTime(1978, 3, 22);
```

9.15 What data does a `TimeSpan` object hold?

9.16 After the following code is executed, what date will be stored in the `date2` object?

```
DateTime date1 = new DateTime(1977, 3, 22);
TimeSpan ts = new TimeSpan(3, 0, 0, 0);
DateTime date2 = date1 + ts;
```

9.4 Enumerated Types

CONCEPT: An enumerated data type is a programmer-defined data type. It consists of predefined constants, known as enumerators, that represent integer values.

Sometimes, in a program, you need a way to represent values that cannot be stored in memory in a straightforward manner. For example, suppose you are writing a program that works with the days of the week (Sunday, Monday, Tuesday, etc.), and you need

some way to represent each day of the week in memory. One solution is to let the integers 0 through 6 represent the days of the week; 0 could represent Sunday, 1 could represent Monday, and so forth. Although this approach will work, it has some drawbacks.

For example, it may not be clear to anyone else reading the code that the values 0 through 6 represent the days of the week. Also, you might decide that Sunday is the first day of the week, whereas someone else assumes that Monday is the first day of the week. When that other person sees the value 0, they might think that it represents Monday.

A better solution for dealing with this type of data is to create an enumerated data type. An **enumerated data type** is a data type that you can create. When you create an enumerated data type, you specify a set of symbolic values that belong to that data type. Here is an example of an enumerated data type declaration:

```
enum Day
{
   Sunday, Monday, Tuesday, Wednesday,
   Thursday, Friday, Saturday
}
```

An enumerated data type declaration begins with the keyword `enum`, followed by the name of the type, followed by a list of identifiers inside braces. The example declaration creates an enumerated data type named `Day`. The identifiers `Sunday`, `Monday`, `Tuesday`, `Wednesday`, `Thursday`, `Friday`, and `Saturday`, which are listed inside the braces, are known as **enumerators**. They represent the values that belong to the `Day` data type.

NOTE: Notice that the enumerators are not enclosed in quotation marks; therefore, they are not strings. Enumerators must be legal C# identifiers.

The enumerators are constants that represent integer values. When you declare an enumerated type, the enumerators are assigned integer values, starting with 0. For example, in the `Day` data type, the `Day.Sunday` enumerator is assigned the value 0, the `Day.Monday` enumerator is assigned the value 1, and so forth.

An `enum` declaration can appear in any of the following places:

- Outside the application's namespace
- Inside the application's namespace
- Inside a class

In this book we always write `enum` declarations in the same region that we write structure declarations: inside an application's namespace but not inside a class.

Once you have created an enumerated data type in your program, you can declare variables of that type. For example, the following statement declares `workDay` as a variable of the `Day` type:

```
Day workDay;
```

We refer to this as an `enum` variable. Because `workDay` is a variable of the `Day` type, the values that we can assign to it are the `Day` type's enumerators. For example, the following statement assigns the value `Day.Wednesday` to the `workDay` variable:

```
workDay = Day.Wednesday;
```

Notice that we assigned `Day.Wednesday` instead of just `Wednesday`. The name `Day.Wednesday` is the fully qualified name of the `Day` type's `Wednesday` enumerator. You have to use an enumerator's fully qualified name because it is possible to have the same enumerator appear in multiple enumerated types. Here is another example:

```
Day weekendDay = Day.Saturday;
```

This statement declares a `Day` variable named `weekendDay` and initializes it with the value `Day.Saturday`.

The following code shows another example. (Assume the application's form has a radio button control named `mondayRadioButton`.)

```
1 Day selectedDay;
2 if (mondayRadioButton.Checked)
3 {
4   selectedDay = Day.Monday;
5 }
```

In this code, line 1 declares a `Day` variable named `selectedDay`. The `if` statement in line 2 determines whether the `mondayRadioButton` control is selected. If it is, the statement in line 4 assigns `Day.Monday` to the `selectedDay` variable.

You can make comparisons with `enum` variables and enumerators. The following code shows an example. (Assume `selectedDay` is a `Day` variable.)

```
1 if (selectedDay == Day.Wednesday)
2 {
3   MessageBox.Show("Halfway through the week!");
4 }
```

The `if` statement in line 1 determines whether the `selectedDay` variable is equal to `Day.Wednesday`. If it is, line 3 displays a message box.

The following code shows how an `enum` variable can be tested in a `switch` statement. (Assume `selectedDay` is a `Day` variable.)

```
 1 switch (selectedDay)
 2 {
 3     case Day.Sunday:
 4         MessageBox.Show("Rest.");
 5         break;
 6     case Day.Monday:
 7         MessageBox.Show("Back to work.");
 8         break;
 9     case Day.Tuesday:
10         MessageBox.Show("Just a regular work day.");
11         break;
12     case Day.Wednesday:
13         MessageBox.Show("Halfway through the week.");
14         break;
15     case Day.Thursday:
16         MessageBox.Show("Almost there.");
17         break;
18     case Day.Friday:
19         MessageBox.Show("Last day!");
20         break;
21     case Day.Saturday:
22         MessageBox.Show("Sleep late today.");
23         break;
24 }
```

In line 1 the `switch` statement tests the `selectedDay` variable. Depending on the value of the variable, the program branches to the appropriate `case` statement.

Converting an Enumerator or `enum` Variable to a String

Enumerators and `enum` variables have a `ToString` method. When you call an enumerator's `ToString` method, it returns the name of the enumerator as a string. For example, the following code will display the string `"Sunday"` in a message box:

```
MessageBox.Show(Day.Sunday.ToString());
```

When you call an `enum` variable's `ToString` method, it returns the name of the value that the variable contains, as a string. For example, the following code will display the string `"Thursday"` in a message box:

```
Day today = Day.Thursday;
MessageBox.Show(today.ToString());
```

When you use the `+` operator with a string and an `enum` variable, the `enum` variable's `ToString` method is implicitly called. Here is an example:

```
Day today = Day.Thursday;
MessageBox.Show("Today is " + today);
```

This code will display the string `"Today is Thursday"` in a message box.

You can also use an interpolated string to convert an `enum` variable to a string, as shown here:

```
Day today = Day.Thursday;
MessageBox.Show($"{today}");
```

This code displays the string `"Thursday"` in message box. The following code shows another example that displays `"Today is Thursday"` in a message box:

```
Day today = Day.Thursday;
MessageBox.Show($"Today is {today}");
```

In Tutorial 9-2, you complete an application that uses an enumerated type to represent the colors of the spectrum.

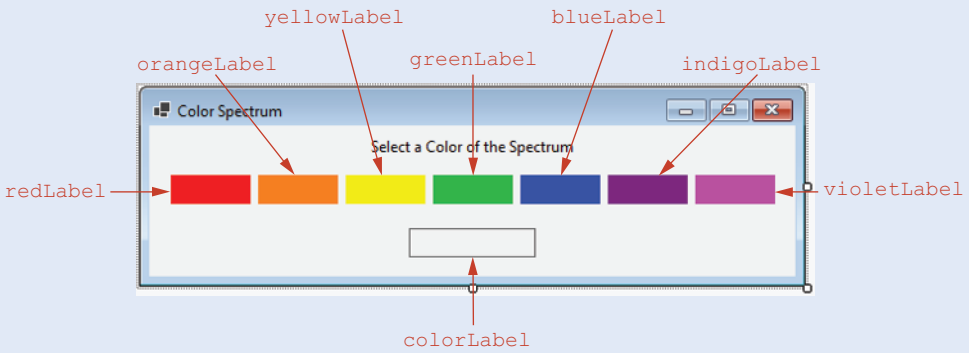
Tutorial 9-2:
Completing the *ColorSpectrum* Project

VideoNote
Tutorial 9-2: Completing the *Color Spectrum* project

The mnemonic ROY G BIV is commonly used to help remember the following sequence of colors of the visible spectrum: red, orange, yellow, green, blue, indigo, and violet. In this tutorial you complete the *ColorSpectrum* project, which shows these colors. The application will display the name of a color if you click it.

The application's form, which has already been created for you, is shown in Figure 9-14. The colors are actually a set of Label controls with their BackColor properties set to the appropriate color.

Figure 9-14 The *ColorSpectrum* project's form



Step 1: Start Visual Studio. Open the project named *ColorSpectrum* in the *Chap09* folder of the Student Sample Programs.

Step 2: Open the Form1 form's code in the code editor. Write the declaration of the `Spectrum` enumerated type, shown in lines 3–7 of Program 9-4. Make sure you write the declaration inside the `ColorSpectrum` namespace, exactly as

shown in Program 9-4. Notice that the enumerators in the `Spectrum` type represent the colors with which you will be working.

Step 3: Write the comments and code for the `DisplayColor` method, shown in lines 16–21 of Program 9-4. Notice in line 18 that this method has a parameter variable of the `Spectrum` type. When we call this method, we pass a `Spectrum` enumerator as an argument. In line 20, the name of the enumerator is displayed in the `colorLabel` control.

Step 4: Next, you create the Click event handlers for the Label controls. Switch back to the *Designer* and double-click the `redLabel` control. This opens the code editor, and you will see an empty event handler named `redLabel_Click`. Complete the event handler by typing the statement shown in line 25 in Program 9-4. This statement calls the `DisplayColor` method, passing the `Spectrum.Red` eumerator as an argument.

Step 5: Repeat this process for the rest of the Label controls, writing the remaining Click event handlers shown in Program 9-4.

Step 6: Save the project. Then, press F5 on the keyboard, or click the *Start Debugging* button (▶) on the toolbar to compile and run the application. When the application runs, click the colors shown on the form to confirm that the application works properly. When you are finished, click the *Exit* button to exit the application.

Program 9-4 Completed code for Form1 in the *ColorSpectrum* project

```
 1 namespace ColorSpectrum
 2 {
 3     enum Spectrum
 4     {
 5         Red, Orange, Yellow, Green,
 6         Blue, Indigo, Violet
 7     }
 8
 9     public partial class Form1 : Form
10     {
11         public Form1()
12         {
13             InitializeComponent();
14         }
15
16         // The DisplayColor method displays the
17         // name of a color.
18         private void DisplayColor(Spectrum color)
19         {
20             colorLabel.Text = color.ToString();
21         }
22
23         private void redLabel_Click(object sender, EventArgs e)
24         {
25             DisplayColor(Spectrum.Red);
26         }
27
28         private void orangeLabel_Click(object sender, EventArgs e)
29         {
30             DisplayColor(Spectrum.Orange);
```

```
31          }
32
33          private void yellowLabel_Click(object sender, EventArgs e)
34          {
35              DisplayColor(Spectrum.Yellow);
36          }
37
38          private void greenLabel_Click(object sender, EventArgs e)
39          {
40              DisplayColor(Spectrum.Green);
41          }
42
43          private void blueLabel_Click(object sender, EventArgs e)
44          {
45              DisplayColor(Spectrum.Blue);
46          }
47
48          private void indigoLabel_Click(object sender, EventArgs e)
49          {
50              DisplayColor(Spectrum.Indigo);
51          }
52
53          private void violetLabel_Click(object sender, EventArgs e)
54          {
55              DisplayColor(Spectrum.Violet);
56          }
57      }
58  }
```

Getting an Enumerator's or `enum` Variable's Integer Value

You cannot assign an enumerator directly to an `int` variable, but you can convert an enumerator to its underlying integer type by using a cast operator. Here is an example:

```
int value = (int)Day.Friday;
```

This statement declares an `int` variable named `value` and initializes it with the integer value of `Day.Friday`. After this statement executes, `value` will equal 5. Here is another example:

```
Day workDay = Day.Monday;
int value = (int)workDay;
```

The first statement declares a `Day` variable named `workDay`, initialized with `Day.Monday`. The second statement declares an `int` variable named `value` and initializes it with the integer value of the `workDay` variable. After this statement executes, `value` will equal 1.

Specifying Integer Values for Enumerators

By default, the enumerators in an enumerated data type are assigned the integer values 0, 1, 2, and so forth. If this is not appropriate, you can specify the values to be assigned, as in the following example:

```
enum Water { Freezing = 32, Boiling = 212 }
```

In this example, the `Water.Freezing` enumerator is assigned the integer value 32 and the `Water.Boiling` enumerator is assigned the integer value 212.

The integer values that you assign to enumerators do not have to be unique. For example, the following code shows an enumerated type named `MonthDays`. Its enumerators are `January`, `February`, `March`, and so forth. Notice that the values assigned to the

enumerators are the days of the months (January has 31 days, February has 28 days, etc.). This data type might appear in an application that uses the days of each month.

```
1 enum MonthDays
2 {
3       January = 31, February = 28, March = 31,
4       April = 30, May = 31, June = 30,
5       July = 31, August = 31, September = 30,
6       October = 31, November = 30, December = 31
7 }
```

Comparing Enumerators and `enum` Variables

Previously you saw that enumerators and `enum` variables can be compared using the equality operator (==). You can also compare enumerators and `enum` variables with the other relational operators. For example, using the `Day` data type we have been discussing, the following expression is true:

```
Day.Friday > Day.Monday
```

The expression is true because the enumerator `Day.Friday` is stored in memory as 5 and the enumerator `Day.Monday` is stored as 1. The following code displays the message `"Friday is greater than Monday."`

```
1 if (Day.Friday > Day.Monday)
2 {
3   MessageBox.Show("Friday is greater than Monday.");
4 }
```

The following code shows another example that compares two `enum` variables. This code displays the message `"Friday is greater than Monday."`

```
1 Day day1 = Day.Friday;
2 Day day2 = Day.Monday;
3
4 if (day1 > day2)
5 {
6     MessageBox.Show($"{day1} is greater than {day2}.");
7 }
```

Enumerators and `enum` variables can be compared directly with integer values. For example, the following code displays the message `"Sunday is equal to zero."`

```
1 if (Day.Sunday == 0)
2 {
3     MessageBox.Show("Sunday is equal to zero.");
4 }
```

Using an `enum` Variable to Step through an Array's Elements

Because enumerators represent integer values, they can be used in a loop to step through the elements of an array. For example, look at the following code:

```
1 decimal[] sales = {1000, 2000, 3000, 4000, 5000, 6000, 7000};
2
3 for (Day dayCount = Day.Sunday; dayCount <= Day.Saturday; dayCount++)
4 {
5     MessageBox.Show($"Sales for {dayCount} were
                      {sales[(int)dayCount]:c}.");
6 }
```

Line 1 creates a `decimal` array named `sales`. The `for` loop that begins in line 3 uses an `enum` variable name `dayCount` as its counter variable. Notice the following about the loop:

- The `dayCount` variable is initialized with the value `Day.Sunday`.
- The loop iterates as long as `dayCount` is less than or equal to `Day.Saturday`.
- At the end of each iteration, `dayCount` is incremented.

In line 5 the expression `(int)dayCount` is used as an array subscript. As the loop executes, it displays the following messages:

- `"Sales for Sunday were $1,000.00"`
- `"Sales for Monday were $2,000.00"`
- `"Sales for Tuesday were $3,000.00"`
- `"Sales for Wednesday were $4,000.00"`
- `"Sales for Thursday were $5,000.00"`
- `"Sales for Friday were $6,000.00"`
- `"Sales for Saturday were $7,000.00"`

Checkpoint

9.17 Look at the following declaration:

```
enum Flower { Rose, Daisy, Petunia }
```

a. What is the name of the data type?

b. In memory, what value will be stored for the enumerator `Flower.Rose`? For `Flower.Daisy`? For `Flower.Petunia`?

c. Write a statement that declares a variable of this enumerated type. The variable should be named `flora`. Initialize the variable with the `Flower.Petunia` enumerator.

9.18 What method do you use to display the string value of a enumerator?

9.19 How can you get the integer value of an enumerator?

9.5 Dictionaries

CONCEPT: **A dictionary is an object that stores a collection of data. Each element in a dictionary has two parts: a key and a value. Each key is associated with a specific value and can be used to locate that value.**

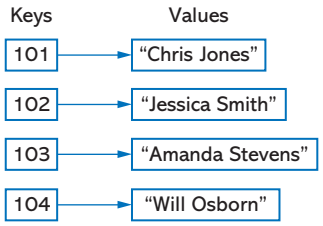
When you hear the word "dictionary," you probably think about a large book such as the Merriam-Webster dictionary, containing words and their definitions. If you want to know the meaning of a particular word, you locate it in the dictionary to find its definition.

In programming, a **dictionary** is an object that stores a collection of data. Each element that is stored in a dictionary has two parts: a **key** and a **value**. In fact, dictionary elements are commonly referred to as **key–value pairs**. When you want to retrieve a specific value from a dictionary, you use the key that is associated with that value. This is similar to the process of looking up a word in the Merriam-Webster dictionary, where the words are keys and the definitions are values.

For example, suppose each employee in a company has an ID number, and we want to write a program that lets us look up an employee's name by entering that

employee's ID number. We could create a dictionary in which each element contains an employee ID number as the key and that employee's name as the value. This is illustrated in Figure 9-15. If we know an employee's ID number, then we can retrieve that employee's name.

Figure 9-15 Key–value pairs



Another example would be a program that lets us enter a person's name and gives us that person's phone number. The program could use a dictionary in which each element contains a person's name as the key and that person's phone number as the value. If we know a person's name, then we can retrieve that person's phone number.

The `Dictionary` Class

When you hear the word "dictionary," you might think about a large book such as the Merriam-Webster dictionary, containing words and their definitions. If you want to know the meaning of a particular word, you locate it in the dictionary to find its definition.

In C# programming, a **dictionary** is an object that stores a collection of data. Each element that is stored in a dictionary has two parts: a key and a value. In fact, dictionary elements are commonly referred to as key–value pairs. When you want to retrieve a specific value from a dictionary, you use the key that is associated with that value. This is similar to the process of looking up a word in the Merriam-Webster dictionary, where the words are keys and the definitions are values.

For example, suppose each employee in a company has an ID number, and we want to write a program that lets us look up an employee's name by entering that employee's ID number. We could create a dictionary in which each element contains an employee ID number as the key, and that employee's name as the value. If we know an employee's ID number, then we can retrieve that employee's name.

Another example would be a program that lets us enter a person's name and gives us that person's phone number. The program could use a dictionary in which each element contains a person's name as the key, and that person's phone number as the value. If we know a person's name, then we can retrieve that person's phone number.

.NET provides a **`Dictionary` class** that you can use to implement a dictionary container in your C# programs. First, you declare a `Dictionary` object. Here is the general format:

```
Dictionary<KeyType, ValueType> DName = new Dictionary<KeyType, ValueType>();
```

where:

- *`KeyType`* is the data type of each element's key
- *`ValueType`* is the data type of each element's value
- *`DName`* is the name of the `Dictionary`

Here is an example of how you might declare a `Dictionary` to hold employee ID numbers (as `int`s) and their corresponding employee names (as `string`s):

```
Dictionary<int, string> employees = new Dictionary<int, string>();
```

This statement declares a `Dictionary` named `employees`. Inside the angled brackets `<>`, the first data type (`int`) is the type of each key, which in this case are the employee ID numbers. The second data type (`string`) is the type of the corresponding values, which in this case are the employee names. So, each element of this `Dictionary` will be a key–value pair where the key is an `int` and the value is a `string`.

If you are declaring a `Dictionary` variable locally inside a method, you can simplify the declaration with the `var` keyword, as shown here:

```
var employees = new Dictionary<int, string>();
```

The keys in a `Dictionary` must be unique. No two elements in a `Dictionary` can have the same key value.

Initializing a `Dictionary`

You can initialize a `Dictionary` with an initialization list. The following code shows an example:

```
var employees = new Dictionary<int, string>()
{
   {101, "Rosa Lopez"},
   {102, "Carmen Cabera"},
   {103, "Sanjay Patel"},
   {104, "Will Osborn"}
};
```

This code creates a `Dictionary` named `employees`, and initializes it as follows:

- The first element is `{101, "Rosa Lopez"}`. In this element, 101 is the key and "Rosa Lopez" is the value.
- The second element is `{102, "Carmen Cabera"}`. In this element, 102 is the key and "Carmen Cabera" is the value.
- The third element is `{103, "Sanjay Patel"}`. In this element, 103 is the key and "Sanjay Patel" is the value.
- The fourth element is `{104, "Will Osborn"}`. In this element, 104 is the key and "Will Osborn" is the value.

Notice the following aspects of the initialization list:

- The initialization list is enclosed in a set of curly braces `{}`;
- Each element is enclosed in its own set of curly braces, with the key and the value separated by a comma;
- The elements are separated by commas;
- After the last curly brace, the declaration ends with a semicolon.

As previously mentioned, the keys in a `Dictionary` must be unique. If two or more elements in an initialization list have the same key, an exception will occur at runtime.

Adding Elements to an Existing `Dictionary`

You can add an element to a `Dictionary` with an assignment statement in the following general format:

```
dictionaryName[key] = value;
```

In the general format, *`dictionaryName`* is the name of the `Dictionary`, and *`key`* is a key. If *`key`* already exists in the `Dictionary`, its associated value will be changed to *`value`*. If the *`key`* does not exist, it will be added to the `Dictionary`, along with *`value`* as its associated value. The following code shows an example:

```
var employees = new Dictionary<int, string>();
employees[110] = "Beth Young";
```

```
employees[111] = "Jamal Ramzi";
employees[112] = "Ravi Burman";
```

The first statement creates an empty `Dictionary` named `employees`. The statements that follow then add the following elements to the `Dictionary`:

- Key = 110, Value = "Beth Young"
- Key = 111, Value = "Jamal Ramzi"
- Key = 112, Value = "Ravi Burman"

Remember that the keys in a `Dictionary` must be unique. If you assign a new value to an existing key, the new value will replace the previously stored value. For example, look at the following code:

```
var employees = new Dictionary<int, string>();
employees[110] = "Beth Young";
employees[110] = "Jamal Ramzi";
```

The first statement creates an empty `Dictionary` named `employees`. The second statement adds an element with the key 110 and the value "Beth Young." The third statement assigns the value "Jamal Ramzi" to the element with the key 110.

Adding Elements with the `Add` Method

In addition to using the `[]` operator, you can also use the **Add method** to add an element to a `Dictionary`. Here is the general format:

```
dictionaryName.Add(key, value);
```

In the general format, *`dictionaryName`* is the name of the `Dictionary`, and *`key`* is a key. If the *`key`* does not exist, it will be added to the `Dictionary`, along with *`value`* as its associated value. If *`key`* already exists in the `Dictionary`, an exception will be thrown. The following code shows an example:

```
var employees = new Dictionary<int, string>();
employees.Add(110, "Beth Young");
employees.Add(111, "Jamal Ramzi");
employees.Add(112, "Ravi Burman");
```

The first statement creates an empty `Dictionary` named `employees`. The statements that follow then add the following elements to the `Dictionary`:

- Key = 110, Value = "Beth Young"
- Key = 111, Value = "Jamal Ramzi"
- Key = 112, Value = "Ravi Burman"

Retrieving Values from a `Dictionary`

To retrieve a value from a `Dictionary`, you can use an expression in the following general format:

```
dictionaryName[key]
```

In the general format, *`dictionaryName`* is the variable that references the `Dictionary`, and *`key`* is a key. If the key exists in the `Dictionary`, the expression returns the value that is associated with the key. If the key does not exist, an exception will occur. The following code snippet demonstrates. Assume this code is in an application that has a ListBox control named `emplyeesListBox`.

```
var employees = new Dictionary<int, string>();

// Add elements to the Dictionary
employees[101] = "Ingrid Olen";
employees[102] = "Sanjay Patel";
```

```
employees[103] = "Rosa Lopez";
employees[104] = "Will Osborn";

// Display the elements
for (int key = 101; key < 105; key++)
{
    employeesListBox.Items.Add(employees[key]);
}
```

Removing Elements

You can remove an element from a `Dictionary` using the `Remove` method. You call the `Remove` method, passing the key that is associated with the element you want to remove. The following code shows an example:

```
var employees = new Dictionary<int, string>();

// Add elements to the Dictionary
employees[101] = "Ingrid Olen";
employees[102] = "Sanjay Patel";
employees[103] = "Rosa Lopez";
employees[104] = "Will Osborn";

// Delete the employee with the ID 102.
employees.Remove(102);
```

This code creates a `Dictionary` with four elements. Then, it deletes the element with the key 102 (and the value "Sanjay Patel"). The `Dictionary` then contains only three elements.

The `Remove` method returns `true` if the specified element was successfully removed, or `false` if it was not found. Here is an example of using the returned value to determine whether an element was removed:

```
if (!employees.Remove(102))
{
    messageLabel.Text = "Element not found";
}
```

Iterating over a `Dictionary` with the `foreach` Loop

When an element is stored in a `Dictionary`, it is stored as an object of the `KeyValuePair` type. The `KeyValuePair` type is a `struct` that has two properties: `Key` and `Value`. The element's key is stored in the `Key` property, and the element's value is stored in the `Value` property.

You can use the `foreach` loop to iterate over all of the `KeyValuePair` elements in a `Dictionary`. The following code shows an example of how to display all of the elements in a `Dictionary`. Assume this code is in an application that has a ListBox control named `empListBox`.

```
var employees = new Dictionary<int, string>()
{
   {101, "Ingrid Olen"},
   {102, "Sanjay Patel"},
   {103, "Rosa Lopez"},
   {104, "Will Osborn"}
};
// Display each element.
foreach (KeyValuePair<int, string> element in employees)
{
   empListBox.Items.Add($"ID: {element.Key} Name: {element.Value}");
}
```

In the `foreach` loop header, we are declaring `element` as the iteration variable:

```
KeyValuePair<int, string> element
```

This declaration specifies that `element` is a `KeyValuePair` object, and that its `Key` property is an `int`, and its `Value` property is a `string`. As long as we understand these characteristics of the `element` variable, we can simplify its declaration by using the `var` keyword:

```
foreach (var element in employees)
{
   empListBox.Items.Add($"ID: {element.Key} Name: {element.Value}");
}
```

The `var` key word tells the compiler to determine the variable's data type from the context in which it is being used.

The `ContainsKey`, `ContainsValue`, and `Contains` Methods

You can use the `ContainsKey` method to determine whether a specific key exists in a dictionary. The `ContainsKey` method accepts a key as its argument, and returns `true` if the key was found, or `false` otherwise.

```
if (!employees.ContainsKey(103))
{
   MessageBox.Show("Employee 103 was not found.");
}
```

You can use the `ContainsValue` method to determine whether a specific value exists in a dictionary. The `ContainsValue` method accepts a value as its argument, and returns `true` if the value was found, or `false` otherwise.

```
if (!employees.ContainsValue("Chris Jones"))
{
   MessageBox.Show("Chris Jones was not found.");
}
```

You can use the `Contains` method to determine whether a specific element exists as a `KeyValuePair` in a dictionary. The `Contains` method accepts a `KeyValuePair` as its argument, and returns `true` if the element was found, or `false` otherwise.

```
var element = new KeyValuePair<int, string>(101, "Ingrid Olen");

if (!employees.Contains(element))
{
   MessageBox.Show("Ingrid Olen was not found.");
}
```

Retrieving a Value with the `TryGetValue` Method

When you call the `TryGetValue` method, you pass two arguments: (1) the key that you are searching for, and (2) an output variable in which you want to store the value associated with the key. Here is the general format of how the `TryGetValue` method is called:

```
dictionaryName.TryGetValue(key, out targetVariable)
```

In the general format, *`dictionaryName`* is the name of a `Dictionary`, *`key`* is the key that you are searching for, and *`targetVariable`* is the name of a variable. The method searches for `key` in the `Dictionary`, and if it is found, the value associated with *`key`* is stored in the *`targetVariable`*, and the method returns `true` to indicate that the element was found. If *`key`* was not found, the method stores a default value in the *`targetVariable`* and returns `false` to indicate the `key` was not found. The following code shows an example:

```
// Create a Dictionary with employee information
var employees = new Dictionary<int, string>()
```

```
{
   {101, "Ingrid Olen"},
   {102, "Sanjay Patel"},
   {103, "Rosa Lopez"},
   {104, "Will Osborn"}
};

// Variable to hold an employee's name
string name;

// Search for employee 104
if (employees.TryGetValue(104, out name))
{
   MessageBox.Show(name);
}
else
{
   MessageBox.Show("Employee 104 was not found.");
}
```

The `Count` Property and the `ElementAt` Method

`Dictionary` objects have a `Count` property that is set to the number of elements in the `Dictionary`. The `Count` property can be used along with the `ElementAt` method to iterate over the elements in a `Dictionary` using a `for` loop. Here is the general format of the `ElementAt` method:

```
dictionaryName.ElementAt(index)
```

In the general format, *`dictionaryName`* is a `Dictionary` and *`index`* is a position within the `Dictionary`, with 0 being the first element. The method returns the element at the specified position as a `KeyValuePair` object. If the index is out of bounds, the method throws an exception. The following code shows an example. Assume this code is in an application that has a ListBox control named `empListBox`.

```
var employees = new Dictionary<int, string>()
{
   {101, "Ingrid Olen"},
   {102, "Sanjay Patel"},
   {103, "Rosa Lopez"},
   {104, "Will Osborn"}
};

for (int i = 0; i < employees.Count; i++)
{
    // Get a KeyValuePair element.
    var element = employees.ElementAt(i);

    // Display the element.
    empListBox.Items.Add($"{element.Key} {element.Value}");
}
```

The `Clear` Method

The `Clear` method deletes all the elements in a `Dictionary`, leaving the `Dictionary` empty. The method's general format is:

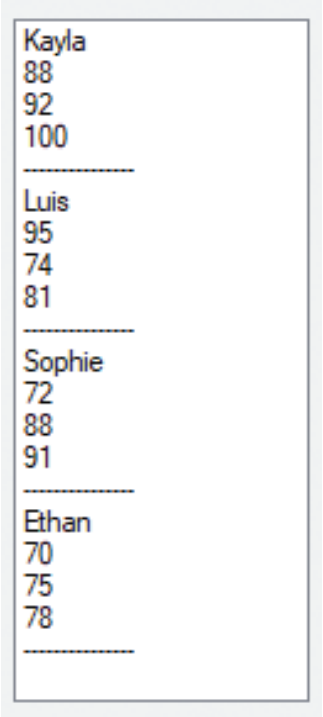
```
dictionaryName.Clear();
```

In the general format, *`dictionaryName`* is the name of a `Dictionary`.

Storing Lists as Values in a Dictionary

In the following code snippet, we create a `Dictionary` in which the keys are student names (as strings), and the values are `List`s containing test scores. Assume this code is in an application that has a ListBox control named `scoresListBox`. The output of the code (the contents of the ListBox) is shown in Figure 9-16.

Figure 9-16 Student names and test scores




```
// Create Lists to hold test scores
var student1scores = new List<int>() {88, 92, 100};
var student2scores = new List<int>() {95, 74, 81};
var student3scores = new List<int>() {72, 88, 9};
var student4scores = new List<int>() {70, 75, 78};

// Create a Dictionary to hold student names and test scores.
var testScores = new Dictionary<string, List<int>>();

// Add each student's name and test scores to the dictionary.
testScores["Kayla"] = student1scores;
testScores["Luis"] = student2scores;
testScores["Sophie"] = student3scores;
testScores["Ethan"] = student4scores;

// Display each student's test score.
foreach (var student in testScores)
{
    scoresListBox.Items.Add(student.Key);
    foreach (int score in student.Value)
    {
        scoresListBox.Items.Add(score);
    }
    scoresListBox.Items.Add("----------");
}
```

Let's take a closer look at the example code:

- Lines 2–5 create 4 `Lists`, each holding a different student's test scores.
- Line 8 defines a `Dictionary` named `testScores`. Notice that the `Dictionary`'s type is `Dictionary<string, List<int>>`. This indicates that the keys will be of the `string` type, and the values will be of the `List<int>` type.
- Lines 11–14 add the following elements to the `Dictionary`:
  - Key = "Kayla," Value = `student1Scores`, which is a `List<int>` object
  - Key = "Luis," Value = `student2Scores`, which is a `List<int>` object
  - Key = "Sophie," Value = `student3Scores`, which is a `List<int>` object
  - Key = "Ethan," Value = `student4Scores`, which is a `List<int>` object
- The `foreach` loop that begins in line 17 iterates over each element in the `Dictionary`. Remember that the iteration variable, `student`, is an object of the `KeyValuePair` data type. The `student` object's `Key` property is the element's key, and its `Value` property is the element's value. So, the expression `student.Key` is a string, and the expression `student.Value` is a `List<int>` object. The `foreach` loop that begins in line 20 iterates over each element of the `student.Value` list, adding it to the ListBox control.

The previous code snippet illustrates the process of storing `Lists` as values in a `Dictionary`, but we can simplify the program. When we declare the `Dictionary`, we can use an initialization list that also creates the `List` objects. That will allow us to eliminate the `List` declarations in lines 2–5. The following code snippet shows the program with these changes.

```
 1 var testScores = new Dictionary<string, List<int>>()
 2 {
 3     {"Kayla", new List<int>() {88, 92, 100}},
 4     {"Luis", new List<int>() {95, 74, 81}},
 5     {"Sophie", new List<int>() {72, 88, 91}},
 6     {"Ethan", new List<int>() {70, 75, 78}}
 7 };
 8
 9 // Display each student's test score.
10 foreach (var student in testScores)
11 {
12     scoresListBox.Items.Add(student.Key);
13     foreach (int score in student.Value)
14     {
15         scoresListBox.Items.Add(score);
16     }
17     scoresListBox.Items.Add("——————————");
18 }
```

In Tutorial 9-3, you will complete an application that uses a `Dictionary` to store people's names and birthdays.

Tutorial 9-3:
Storing Names and Birthdays in a `Dictionary`

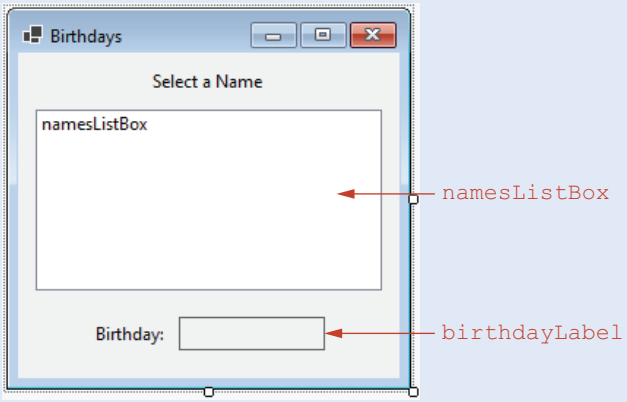
VideoNote
Tutorial 9-3: Storing Names and Birthdays in a `Dictionary`

In this tutorial, you will complete an application that stores names and birthdays in a `Dictionary`. Each entry in the dictionary uses a person's name as the key, and that person's birthday as the value. Figure 9-17 shows the application's form. When the application is finished, it will display all the names in the ListBox. When a name is clicked in the ListBox, that person's birthday will be retrieved from the `Dictionary` and displayed in the `birthdayLabel` control.

Step 1: Open the project named *Birthdays* in the *Chap09* folder of this book's Student Sample Programs, available for download at www.pearsonhighered.com/gaddis.

Step 2: Open the Form1 form in the Designer. The form is shown, along with the names of the important controls, in Figure 9-17. Press the F7 key to open the form's source code in the code editor. Program 9-5 shows how the code will appear after you have completed it.

Figure 9-17 The *Birthdays* project's Form1 form



Step 3: Write the `Dictionary` declaration that appears in lines 5–14 of Program 9-5. This is the declaration of a `Dictionary` named `birthdays`, initialized with the names and birthdays of five people.

NOTE: We cannot use the `var` keyword to declare the `Dictionary` because we are declaring it as a field in the class.

Step 4: Now you will write the `Form1_Load` event handler. Switch your view back to the Form1 form in the Designer and double-click any part of the form that does not contain a control. (Be sure not to click the Button, Label or ListBox controls.) This opens the code editor, and you will see an empty `Form1_Load` event handler. Complete the `Form1_Load` event handler by typing the code shown in lines 23–27 in Program 9-5. This code iterates over the `Dictionary` elements, displaying the names in the ListBox.

Step 5: Switch your view back to the *Designer* and double-click the `namesListBox` control. In the code editor you will see an empty event handler named `namesListBox_SelectedIndexChanged`. This is the event handler that executes each time an item in the ListBox is clicked. Complete the event handler by writing the code shown in lines 32–39 in Program 9-5. Let's take a closer look at the code:

Lines 33: This statement gets the name that was selected in the ListBox, and assigns it to the `name` variable.

Line 36: This statement gets a value from the `Dictionary`, using the `name` variable as the key. The value that is retrieved is assigned to the `birthday` variable.

Line 39: This statement displays the value of the `birthday` variable in the `birthdayLabel` control.

Step 6: Save the project, then compile and run the application. Experiment by clicking different names in the ListBox and observe the birthdays that are displayed. Close the application when you are finished.

Program 9-5 Complete Form1 code for the *Birthdays* project

```
1 namespace Birthdays
2 {
3     public partial class Form1 : Form
```

```
    {
        // Dictionary with names and birthdays
        Dictionary<string, string> birthdays =
            new Dictionary<string, string>()
            {
                {"Cameron", "4/17/1998"},
                {"Rami", "5/7/1996"},
                {"Amira", "1/22/1997"},
                {"Lola", "5/5/1995"},
                {"Samir", "10/4/1999"}
            };

        public Form1()
        {
            InitializeComponent();
        }

        private void Form1_Load(object sender, EventArgs e)
        {
            // Display the names in the ListBox.
            foreach (var element in birthdays)
            {
                namesListBox.Items.Add(element.Key);
            }
        }

        private void namesListBox_SelectedIndexChanged(object
        sender, EventArgs e)
        {
            // Get the selected name.
            string name = namesListBox.SelectedItem.ToString();

            // Get the selected person's birthday.
            string birthday = birthdays[name];

            // Display the selected person's birthday.
            birthdayLabel.Text = birthday;
        }
    }
}
```

In Tutorial 9-4, you will complete an application that uses a `Dictionary` to simulate a deck of cards. The application allows the user to deal a hand of cards from the deck.

Tutorial 9-4:
Using a `Dictionary` to Simulate a Deck of Cards

VideoNote

Tutorial 9-4: Using a `Dictionary` to Simulate a Deck of Cards

In some games involving poker cards, the cards are assigned numeric values. For example, in the game of Blackjack, the cards are given the following numeric values:

- Numeric cards are assigned the value they have printed on them. For example, the value of the 2 of spades is 2, and the value of the 5 of diamonds is 5.
- Jacks, queens, and kings are valued at 10.
- Aces are valued at either 1 or 11, depending on the player's choice.

In this tutorial, you will complete an application that uses a `Dictionary` to simulate a standard deck of poker cards, where the cards are assigned numeric values similar to those used in Blackjack. (In the program, you will assign the value 1 to all aces.) The key–value pairs use the name of the card as the key, and the card's numeric value as the value. For example, the key–value pair for the queen of hearts is

```
"Queen of Hearts", 10
```

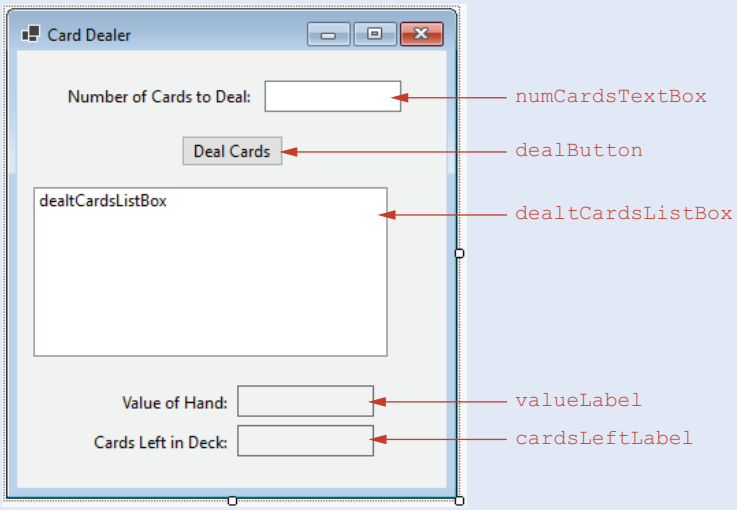
And the key–value pair for the 8 of diamonds is

```
"8 of Diamonds", 8
```

The application's form is shown in Figure 9-18, and the completed code for the application is shown in Program 9-6. The completed application will work like this:

- The user enters the number of cards to deal into the `numCardsTextBox` control.
- The user clicks the `dealButton` control to deal a hand of that many cards from the deck. To simulate dealing the cards, the application will randomly select elements from the `Dictionary`. The randomly selected elements will be removed from the `Dictionary` so they can't be dealt more than once.
- The names of the randomly selected cards will be displayed in the `dealtCardsListBox` control.
- The total numeric value of the hand will be displayed in the `valueLabel` control.
- The number of cards that are left in the deck will be displayed in the `cardsLeftLabel` control.

Figure 9-18 The *CardDealer* project's Form1 form



Step 1: Open the project named *CardDealer* in the *Chap09* folder of this book's Student Sample Programs, available for download at www.pearsonhighered.com/gaddis.

Step 2: Open the Form1 form in the Designer. The form is shown, along with the names of the important controls, in Figure 9-18. Press the F7 key to open the form's source code in the code editor.

Notice that the `Dictionary` has already been declared as a private field in the `Form1` class. The code in lines 6 through 35 of Program 9-6 creates a `Dictionary` named deck, initialized with key–value pairs representing the cards in a standard poker deck. (The blank lines that appear in lines 14, 21, and 28 were inserted to make the code easier to read.)

Step 3: Switch your view back to the *Designer* and double-click any part of the form that does not contain a control. (Be sure not to click any of the Label, ListBox, TextBox, or Button controls.) This opens the code editor, and you will see an empty event handler named `Form1_Load`. Complete the `Form1_Load` event handler by typing the code shown in lines 44–45 in Program 9-6. This code displays the number of elements in the `Dictionary` in the `cardsLeftLabel` control.

Step 4: Switch your view back to the *Designer* and double-click the `dealButton` control. In the code editor, you will see an empty event handler named `dealButton_Click`. Complete the `dealButton_Click` event handler by typing the code shown in lines 50–107 in Program 9-6. Let's take a closer look at the code:

Lines 50 and 51: These statements declare two local `int` variables: `numCards` and `handValue`. The `numCards` variable will hold the number of cards that the user wants to deal, and the `handValue` variable will hold the value of a dealt hand of cards.

Line 52: This statement declares a `Random` object for generating random numbers.

Lines 55–58: These statements clear the contents of the `dealtCardsListBox` and the `valueLabel` controls.

Line 61: This `if` statement attempts to convert `numCardsTextBox.Text` to an `int`. If the conversion is successful, the result is stored in the `numCards` variable, and the program continues executing at line 63. If the conversion is not successful, the program jumps to the `else` clause in line 103, and line 106 displays the message "Enter an integer."

Line 64: This `if` statement determines whether the user entered a valid number of cards, which must be greater than 0, and no more than the number of cards in the deck. If the number is valid, the program continues executing at line 66. Otherwise, the program jumps to the `else` clause in line 89.

There are two possible reasons the program might reach the `else` clause in line 89: Either there are no more cards in the deck (the `Dictionary` has 0 elements), or the user entered an invalid number of cards. If there are no more cards in the deck, the statement in line 94 displays the message "The deck is empty!". Otherwise, the statement in lines 98 and 99 displays an error message.

Line 67: The `for` loop that begins at this line iterates once for each card that is to be dealt. (The number of iterations is equal to the number of cards that the user entered.)

Line 70: This statement gets a random integer that is at least 0, but no more than `deck.Count` –1, and assigns it to the `randNumber` variable.

Line 73: This statement gets an element from the `Dictionary` as a `KeyValuePair` object, using `randNumber` as an index. It is assigned to the `card` variable.

Line 76: This statement displays the `card` object's Key property, which holds the name of the card, in the `dealtCardListBox` control.

Line 79: This statement adds the `card` object's Value property, which holds the card's numeric value, to the `handValue` variable.

Line 80: This statement displays the `handValue` variable in the `valueLabel` control.

Line 83: This statement removes the card from the deck. This is done so the card will not be dealt again.

Line 87: This statement displays the number of cards that are left in the deck, in the `cardsLeftLabel` control.

Step 5: Save the project, then compile and run the application. Experiment by dealing different numbers of cards until the deck is empty. Each time you empty the deck, you will need to close the application and rerun it to start with a fresh deck.

Program 9-6 Complete Form1 code for the *CardDealer* project

```
 1 namespace CardDealer
 2 {
 3     public partial class Form1: Form
 4     {
 5         // Dictionary to simulate a deck of cards
 6         private Dictionary<string, int> deck = new
           Dictionary<string, int>()
 7         {
 8             // Spades
 9             {"Ace of Spades",1}, {"2 of Spades",2}, {"3 of
               Spades",3},
10             {"4 of Spades",4}, {"5 of Spades",5}, {"6 of
               Spades",6},
11             {"7 of Spades",7}, {"8 of Spades",8}, {"9 of
               Spades",9},
12             {"10 of Spades",10}, {"Jack of Spades",10}, {"Queen
               of Spades",10},
13             {"King of Spades", 10},
14
15             // Hearts
16             {"Ace of Hearts",1}, {"2 of Hearts",2}, {"3 of
               Hearts",3},
17             {"4 of Hearts",4}, {"5 of Hearts",5}, {"6 of
               Hearts",6},
18             {"7 of Hearts",7}, {"8 of Hearts",8}, {"9 of
               Hearts",9},
19             {"10 of Hearts",10}, {"Jack of Hearts",10}, {"Queen
               of Hearts",10},
20             {"King of Hearts", 10},
21
22             // Clubs
23             {"Ace of Clubs",1}, {"2 of Clubs",2}, {"3 of
               Clubs",3},
24             {"4 of Clubs",4}, {"5 of Clubs",5}, {"6 of
               Clubs",6},
25             {"7 of Clubs",7}, {"8 of Clubs",8}, {"9 of
               Clubs",9},
26             {"10 of Clubs",10}, {"Jack of Clubs",10}, {"Queen of
               Clubs",10},
27             {"King of Clubs", 10},
28
29             // Diamonds
30             {"Ace of Diamonds",1}, {"2 of Diamonds",2}, {"3 of
               Diamonds",3},
31             {"4 of Diamonds",4}, {"5 of Diamonds",5}, {"6 of
               Diamonds",6},
32             {"7 of Diamonds",7}, {"8 of Diamonds",8}, {"9 of
               Diamonds",9},
33             {"10 of Diamonds",10}, {"Jack of Diamonds",10},
               {"Queen of Diamonds",10},
```

```
34              {"King of Diamonds", 10}
35          };
36
37          public Form1()
38          {
39              InitializeComponent();
40          }
41
42          private void Form1_Load(object sender, EventArgs e)
43          {
44              // Update the number of cards left in the deck.
45              cardsLeftLabel.Text = deck.Count.ToString();
46          }
47
48          private void dealButton_Click(object sender,
            EventArgs e)
49          {
50              int numCards = 0; // Number of cards to deal
51              int handValue = 0; // Value of the dealt cards
52              var rand = new Random(); // Random number
                generator
53
54              // Clear the ListBox
55              dealtCardsListBox.Items.Clear();
56
57              // Clear the hand value.
58              valueLabel.Text = "";
59
60              // Get the number of cards to deal.
61              if (int.TryParse(numCardsTextBox.Text, out numCards))
62              {
63                  // Make sure the number is within range.
64                  if (numCards > 0 && numCards <= deck.Count)
65                  {
66                      // Deal the cards.
67                      for (int count = 0; count < numCards;
                        count++)
68                      {
69                          // Get a random number.
70                          int randNumber = rand.Next(0, deck.
                            Count);
71
72                          // Get the card at the random
                            position.
73                          var card = deck.ElementAt(randNumber);
74
75                          // Display the card.
76                          dealtCardsListBox.Items.Add(card.Key);
77
78                          // Update and display the hand value.
79                          handValue += card.Value;
80                          valueLabel.Text = handValue.ToString();
81
82                          // Remove the selected card from the
                            deck.
83                          deck.Remove(card.Key);
84                      }
85
86                      // Update the number of cards left in the
                        deck.
```

```
87                      cardsLeftLabel.Text = deck.Count.
                        ToString();
88                  }
89                  else
90                  {
91                      // Error messages for number out of range
92                      if (deck.Count == 0)
93                      {
94                          MessageBox.Show("The deck is empty!");
95                      }
96                      else
97                      {
98                          MessageBox.Show("Enter a number in the
                            range of " +
99                          $"1 through {deck.Count}.");
100                     }
101                 }
102             }
103             else
104             {
105                 // Error message for noninteger value
106                 MessageBox.Show("Enter an integer.");
107             }
108         }
109     }
110 }
```

Checkpoint

9.20 Each element that is stored in a `Dictionary` has two parts. What are they?

9.21 Write a statement that declares a `Dictionary` named `myDictionary`. The keys in `myDictionary` should be `int`s, and the values should be `string`s.

9.22 Suppose an empty `Dictionary` named `employee` has been created. What does the following statement do?

```
employee[543] = "Joanne Manchester";
```

9.23 Describe two ways in which you can retrieve an element with a particular key in a `Dictionary`.

9.24 What two properties does a `KeyValuePair` object have?

9.6 Tuples

CONCEPT: A tuple is a collection of multiple items. The items in a tuple can be of the same data type, or of different data types.

A **tuple** is a container that can hold multiple pieces of data. The items that are stored in a tuple can be of the same data type, or different data types. You can create a tuple by enclosing a comma-separated list of items in a set of parentheses, as shown here:

```
var product = (123, "USB Cable", 9.99m);
```

This statement declares a variable named `product` and initializes it with a tuple that contains three items. The first item is 123 (an `int`), the second item is "USB Cable" (a `string`), and the third item is 9.99m (a `decimal`). Here is another example:

```
var sequence = (1, 2, 3, 4);
```

This statement declares a variable named `sequence` and initializes it with a tuple that contains the integers 1, 2, 3, and 4.

The data items that are stored in a tuple are known as fields. There is no limit to the number of fields that a tuple may contain. By default, the names of the fields are `Item1`, `Item2`, `Item3`, and so on. You can use these field names to access a tuple's fields, as shown in the following code:

```
1 var product = (123, "USB Cable", 9.99m);
2 MessageBox.Show(product.Item1.ToString());
3 MessageBox.Show(product.Item2);
4 MessageBox.Show(product.Item3.ToString());
```

After line 1 executes, the `product` variable will be assigned a tuple with three fields: the `Item1` field is the integer 123, the `Item2` fields is the string "USB Cable", and the `Item3` field is the decimal 9.99m. Line 2 displays the value of the `Item1` field in a message box. Line 3 displays the value of the `Item2` field in a message box. Line 4 displays the value of the `Item3` field in a message box.

The following code shows another example:

```
var numbers = (1, 2);
MessageBox.Show($"{numbers.Item1} {numbers.Item2}");
```

After line 1 executes, the `numbers` variable will be assigned a tuple with two fields: the `Item1` field is the integer 1, and the `Item2` field is the integer 2. The second line displays both fields in a message box.

Declaring a Tuple Variable

The previous examples show how a tuple variable can be declared with the `var` keyword. If you do not have an initialization value, or you want to declare the tuple variable at the class level, you cannot use the `var` keyword. In such cases, you must explicitly provide the data type in the declaration statement. For example, the following statement declares a tuple variable named `t`:

```
(int, string) t;
```

The data type in this statement is `(int, string)`. The `t` variable has two fields: `Item1` is an `int` and `Item2` is a `string`. We can assign any tuple that is compatible with this type to `t`. Here is an example:

```
t = (100, "Olivia");
```

A compiler error will occur if we try to assign an incompatible tuple to a tuple variable. For example, the following statement causes an error:

```
t = (100, "Olivia", "olivia@example.com"); // Error!
```

This statement tries to assign a `(int, string, string)` tuple to the variable `t`. An error occurs because the data type of `t` is `(int, string)`.

Here is an example of a statement that explicitly declares the data type of a tuple variable and initializes the variable with a tuple:

```
(int, string) t = (100, "Olivia");
```

This statement declares the variable `t`. The data type of `t` is `(int, string)`, and it is initialized with the tuple `(100, "Olivia")`.

NOTE: Like structures, tuples are value types.

Returning a Tuple from a Method

A method can be written to return a tuple. This is one of the most common uses of tuples because it allows a method to return more than one value. The following code shows an example.

```
1 private (int, string, decimal) getProduct()
2 {
3       return (123, "USB Cable", 9.99m);
4 }
```

This code defines a method named `getProduct`. Notice in line 1, the method's return type is `(int, string, decimal)`. This specifies that the method returns a tuple consisting of an `int`, a `string`, and a `decimal`. Then, line 3 returns the tuple `(123, "USB Cable", 9.99m)`. The following statement shows how we might call the method and assign its return value to a variable:

```
var product = getProduct();
```

This statement declares a variable named `product` and initializes it with the tuple that is returned from the `getProduct` method. After this statement executes, the `product` variable will be assigned the tuple `(123, "USB Cable", 9.99m)`. Alternatively, we could have written the statement without the `var` keyword, as shown here:

```
(int, string, decimal) product = getProduct();
```

Let's look at another example. Assume an application has two TextBoxes named `firstNameTextBox` and `lastNameTextBox`. The following code defines a method named `getName` that gets the user's input from the TextBoxes and returns a tuple containing those values:

```
1 private (string, string) getName()
2 {
3       return (firstNameTextBox.Text, lastNameTextBox.Text);
4 }
```

In line 1, the method's return type is `(string, string)`. This specifies that the method returns a tuple consisting of two `string`s. Line 3 creates a tuple containing the Text properties of the two TextBoxes, and returns that tuple. The following statement shows how we might call the method and assign its return value to a variable:

```
var name = getName();
```

After this statement executes, the `name` variable will be assigned the tuple that is returned from the `getName` method. This statement is equivalent to the following statement:

```
(string, string) name = getName();
```

Named Fields

By default, a tuple's fields are named `Item1`, `Item2`, and so on. You can specify your own names for the fields, however. There are two ways to specify **named fields**: in a tuple initialization expression, or in the data type of a tuple declaration. The following statement shows an example of a tuple initialization expression that specifies field names:

```
var name = (FirstName: "Anjali", LastName: "Kumar");
```

This statement declares a variable named `name` and initializes it with a tuple with two fields. The first field is named `FirstName`, and its value is "Anjali". The second field is

named `LastName` and its value is "Kumar". The following statement shows how we can use the field names to access the data stored in the tuple:

```
MessageBox.Show($"{name.FirstName} {name.LastName}");
```

This statement displays "Anjali Kumar" in a message box.

The following statement shows an example of declaring a tuple variable with named fields:

```
(double Fahrenheit, double Celsius) temperature;
```

This statement declares a tuple variable named `temperature`. The tuple has two fields. The first field is a `double` named `Fahrenheit`, and the second field is a `double` named `Celsius`. Here is an example of how we might assign a tuple value to the variable and use its field names:

```
temperature = (212.0, 100.0);
MessageBox.Show($"{temperature.Fahrenheit} {temperature.Celsius}");
```

The first statement assigns the tuple `(212.0, 100.0)` to the `temperature` variable. The second statement displays "212 100" in a message box.

Here is an example of a method that returns a tuple with named fields:

```
1 private (int ID, string Description, decimal Price) getProduct()
2 {
3       return (123, "USB Cable", 9.99m);
4 }
```

This code defines a method named `getProduct` that returns a tuple with named fields. The first field is an `int` named `ID`. The second field is a `string` named `Description`. The third field is a `decimal` named `Price`. In the following code we call the method and display the field values of the tuple that it returns.

```
1 var product = getProduct();
2 MessageBox.Show($"{product.ID}");
3 MessageBox.Show($"{product.Description}");
4 MessageBox.Show($"{product.Price}");
```

The default field names `Item1`, `Item2`, and so on can still be used with tuples that named field. For example, look at the following code.

```
1 var name = (FirstName: "Anjali", LastName: "Kumar");
2 MessageBox.Show($"{name.FirstName} {name.LastName}");
3 MessageBox.Show($"{name.Item1} {name.Item2}");
```

In this code, line 1 assigns a tuple with names fields to the `name` variable. Line 2 uses the tuple's custom field names and line 3 uses the tuple's default field names. Both statements perform the same operation.

Deconstructing a Tuple

Look at the following method:

```
1 private (int, string) getInfo()
2 {
3       return (100, "Olivia");
4 }
```

This method returns a tuple with two fields. Suppose we call the method to get the tuple that it returns, and we want to extract the tuple's fields into separate variables. Here is one approach:

```
1 var info = getInfo();
2 int number = info.Item1;
3 string name = info.Item2;
```

In this code, line 1 gets a tuple from the `getInfo` method and assigns the tuple to the `info` variable. Line 2 assigns the tuple's `Item1` field to the `number` variable, and line 2 assigns the tuple's `Item2` field to the `name` variable. After this code executes, `number` will be assigned the value 100 and `name` will be assigned the value "Olivia".

This multi-step approach can be simplified with a technique known as **deconstruction**. A deconstruction statement allows you to extract all of a tuple's fields into separate variables with a single statement. The following code shows how we can use deconstruction to rewrite the previously shown code:

```
(int number, string name) = getInfo();
```

Inside the parentheses, we have two variable declarations: `int number` and `string name`. The variables are initialized with the fields of the tuple that is returned from the `getInfo` method. After this statement executes, the `number` variable will be assigned 100 and the `name` variable will be assigned "Olivia".

You can also write a deconstruction statement that uses variables that have already been declared. Here is an example:

```
int number;
string name;
(number, name) = getInfo();
```

Let's look at one more example. The following code declares and initializes a tuple variable named `info`. Then, it deconstructs the tuple and assigns its fields to the `city` and `country` variables:

```
// Create a tuple.
var info = ("Jakarta", "Indonesia");

// Deconstruct the tuple.
(string city, string country) = info;
```

After this code executes, the `city` variable will be assigned the value "Jakarta" and the `country` variable will be assigned the value "Indonesia".

Tuple Assignment

You can assign one tuple to another as long as both tuples have the same number of fields, and the data types of the tuple's corresponding fields are compatible. In other words,

- `Item1` of the tuple on the left side of the = operator must be compatible with `Item1` of the tuple on the right side of the = operator.
- `Item2` of the tuple on the left side of the = operator must be compatible with `Item2` of the tuple on the right side of the = operator.
- And so on.

For example, the following code shows a tuple assignment that will work:

```
1 // Initialize a tuple variable.
2 var tuple1 = (99, "macaroni", 4.99m);
3
4 // Declare a tuple variable.
5 (int, string, decimal) tuple2;
6
7 // Assign tuple1 to tuple2.
8 tuple2 = tuple1;
```

Line 2 declares a variable named `tuple1` and initializes it with a tuple. Line 5 declares a tuple variable named `tuple2` but does not initialize the variable. Line 8 assigns the value of `tuple1` to `tuple2`. This assignment will work because the following conditions are met:

- Both tuples have the same number of fields.
- In both tuples, `Item1` is an `int`.

- In both tuples, `Item2` is a `string`.
- In both tuples, `Item3` is a `decimal`.

After line 8 executes, both `tuple1` and `tuple2` will contain the same values.

The following code has a tuple assignment statement that will *not* work. Can you spot the error?

```
1 // Initialize a tuple variable.
2 var tuple1 = (99, "macaroni", 4.99m, "gluten free");
3
4 // Declare a tuple variable.
5 (int, string, decimal) tuple2;
6
7 // Error! This will not work.
8 tuple2 = tuple1;
```

This code will cause a compiler error because `tuple1` and `tuple2` have different numbers of fields (`tuple1` has four fields, and `tuple2` has three fields).

The following code will also cause a compiler error. Can you spot the error?

```
1 // Initialize a tuple variable.
2 var tuple1 = (99m, "macaroni", 4.99m);
3
4 // Declare a tuple variable.
5 (int, string, decimal) tuple2;
6
7 // Error! This will not work.
8 tuple2 = tuple1;
```

In this code, `tuple1.Item1` is a `decimal`, but `tuple2.Item1` is an `int`. In C#, a `decimal` cannot be implicitly converted to an `int`, so this code will cause a compiler error.

In a tuple assignment, the field names of the tuples do not have to match. The following code shows an example.

```
1 // Initialize a tuple variable.
2 var tuple1 = (ID: 99, Name: "macaroni", Price: 4.99m);
3
4 // Declare a tuple variable.
5 (int Number, string Desc, decimal Cost) tuple2;
6
7 // Assign tuple1 to tuple2.
8 tuple2 = tuple1;
```

In this example, the field names of `tuple1` are `ID`, `Name`, and `Price`. The field names of `tuple2` are `Number`, `Desc`, and `Cost`. The assignment statement in line 8 will work, however, because the data types of the corresponding fields are the same.

Changing the Value of a Field

You can change the value of a tuple's field by assigning a new value to it. The following code shows an example:

```
1 var point = (100, 200);
2 // The following statement displays 100, 200
3 MessageBox.Show($"{point.Item1}, {point.Item2}");
4
5 point.Item1 = 0;
6 point.Item2 = 0;
7 // The following statement displays 0, 0
8 MessageBox.Show($"{point.Item1}, {point.Item2}");
```

This code creates a tuple in line 1 containing the values 100 and 200. Lines 5 and 6 change the values of its fields to 0 and 0. The following example shows the same code, except with named fields:

```
1 var point = (X: 100, Y: 200);
2 // The following statement displays 100, 200
3 MessageBox.Show($"{point.X}, {point.Y}");
4
5 point.X = 0;
6 point.Y = 0;
7 // The following statement displays 0, 0
8 MessageBox.Show($"{point.X}, {point.Y}");
```

Comparing Tuples for Equality and Inequality

You can use the == and != operators to compare tuples. The == operator determines whether two tuples are equal, and the != operator determines whether two tuples are not equal. In order to use these operators, however, the two tuples must be comparable. Two tuples are comparable if both tuples have the same number of fields, and the data types of the tuple's corresponding fields are comparable. For example the tuples in the following code are comparable:

```
var tuple1 = (10, "Cheese pizza", 9.99m);
var tuple2 = (20, "Ramen noodles", 4.99m);
```

The tuples in the following code are not comparable, however, because they do not have the same number of fields:

```
var tuple1 = (10, "Cheese pizza", 9.99m, "gluten free");
var tuple2 = (20, "Ramen noodles", 4.99m);
```

The tuples in the following code are not comparable because their `Item1` fields cannot be compared (you cannot compare an `int` with a `string`):

```
var tuple1 = (10, "Cheese pizza", 9.99m);
var tuple2 = ("20", "Ramen noodles", 4.99m);
```

If you write a statement that tries to compare two incomparable tuples, a compiler error will occur.

If two tuples are comparable, you can use the == and != operators to compare them. The following code shows an example:

```
var tuple1 = (10, "Cheese pizza", 9.99m);
var tuple2 = (20, "Ramen noodles", 4.99m);
if (tuple1 == tuple2)
    MessageBox.Show("The two are the same.");
else
    MessageBox.Show("The two are not the same.");
```

This code will display the message "The two are not the same.". Here is another example:

```
var tuple1 = (10, "Cheese pizza", 9.99m);
var tuple2 = (20, "Ramen noodles", 4.99m);
if (tuple1 != tuple2)
    MessageBox.Show("Those are different foods.");
else
    MessageBox.Show("Pizza and noodles are the same?");
```

This code will display the message "Those are different foods.".

NOTE: The == and != operators ignore field names when comparing tuples. Two tuples can be equal even if they have different field names.

Inferred Field Names

If a tuple does not have named fields, and it is initialized with the values of named variables, the C# compiler will infer field names from the names of the variables. For example, look at the following code:

```
1 int id = 99;
2 string name = "Olivia";
3 var info = (id, name);
4 MessageBox.Show($"{info.id} {info.name}");
```

Line 1 declares a variable named `id`, initialized with the value 99. Line 2 declares a variable named `name`, initialized with the value "Olivia". Line 3 declares a variable named `info`, and initializes it with the tuple `(id, name)`. Because we did not assign names to the tuple's fields, the compiler assigns the name `id` to the first field and `name` to the second field. Line 4 displays the values of `info.id` and `info.name` in a message box. The message that is displayed is "99 Olivia".

Keep in mind that inferred names are assigned to a tuple's fields only when you have not assigned your own names to the fields. Also, a name will not be inferred to a field if the inferred name duplicates a default name, such as `Item1`.

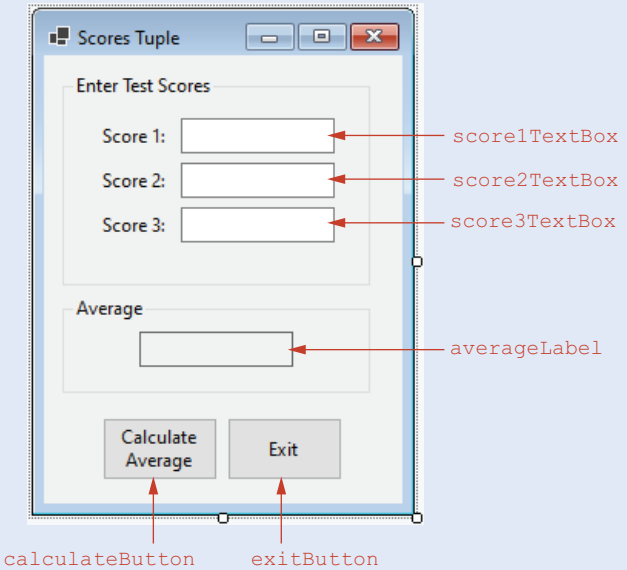
In Tutorial 9-5 you will write code that returns a tuple from a method, and then deconstructs the tuple to work with the values it contains.

Tutorial 9-5:
Completing the *ScoresTuple* Project

VideoNote
Tutorial 9-5: Completing the *ScoresTuple* Project

In this tutorial, you will complete the *ScoresTuple* project. When the project is complete, it will use a method to get three test scores and return them in a tuple. The tuple will be deconstructed and the average of the scores will be calculated and displayed. The application's form, which has already been created for you, is shown in Figure 9-19.

Figure 9-19 The *ScoresTuple* project's form



Step 1: Start Visual Studio. Open the project named *ScoresTuple* in the Chap09 folder of the Student Sample Programs.

Step 2: Open the Form1 form in the Designer. The form is shown, along with the names of the important controls, in Figure 9-19. Press the F7 key to open the form's source code in the code editor. Program 9-7 shows how the code will appear after you have completed it.

Step 3: Write the comments and code for the `getScores` method that appears in lines 10–37 of Program 9-7. Let's take a closer look at the code:

Line 12: This is the method header. Notice the method's return type is `(int, int, int)`. This means the method returns a tuple containing three `int` values.

Line 15: This statement declares three variables, `s1`, `s2`, and `s3`, to hold the test scores.

Lines 18–23: The try block gets the values from the three TextBoxes, converts them to integers, and assigns the converted values to the variables `s1`, `s2`, and `s3`. If any of the values cannot be converted, the `int.Parse` method will throw an exception and the program will jump to the `catch` clause in line 24. Otherwise, the program will continue to line 36.

Lines 24–33: The `catch` clause will handle any exceptions that are thrown by the `int.Parse` method in the try block. Line 27 displays an error message and then lines 30–32 assign 0 to the `s1`, `s2`, and `s3` variables.

Line 36: This statement returns a tuple containing the `s1`, `s2`, and `s3` variables.

Step 4: Switch your view back to the Designer and double-click the `calculateButton` control. In the code editor, you will see an empty event handler named `calculateButton_Click`. Complete the `calculateButton_Click` event handler by typing the code shown in lines 41–49 in Program 9-7. Let's take a closer look at the code:

Line 43: This statement calls the `getScores` method. The method returns a tuple containing three integers as fields. The tuple is deconstructed, and its fields are assigned to the `score1`, `score2`, and `score3` variables.

Line 46: This statement calculates the average of the `score1`, `score2`, and `score3` variables, and assigns the result to the `average` variable.

Line 49: This statement displays the value of the `average` variable, rounded to 1 decimal place, in the `averageLabel` control.

Step 5: Save the project, then compile and run the application. Experiment by entering various sets of values in the TextBox controls and calculating their average. Close the application when you are finished.

Program 9-7 Completed Form1 code for the *ScoresTuple* project

```
1 namespace ScoresTuple
2 {
3     public partial class Form1: Form
4     {
```

```
 5         public Form1()
 6         {
 7             InitializeComponent();
 8         }
 9
10         // The getScores method gets the test scores that were
11         // entered into the TextBoxes and returns them in a
           tuple.
12         private (int, int, int) getScores()
13         {
14             // Variables to hold the test scores.
15             int s1 = 0, s2 = 0, s3 = 0;
16
17             // Get the user's input.
18             try
19             {
20                 s1 = int.Parse(score1TextBox.Text);
21                 s2 = int.Parse(score2TextBox.Text);
22                 s3 = int.Parse(score3TextBox.Text);
23             }
24             catch
25             {
26                 // Display an error message.
27                 MessageBox.Show("Check your input. Invalid data
                   was entered.");
28
29                 // Set all the scores to 0.
30                 s1 = 0;
31                 s2 = 0;
32                 s3 = 0;
33             }
34
35             // Return the socres in a tuple.
36             return (s1, s2, s3);
37         }
38
39         private void calculateButton_Click(object sender,
           EventArgs e)
40         {
41             // Get the test scores as a tuple and deconstruct
42             // the tuple.
43             (int score1, int score2, int score3) = getScores();
44
45             // Calculate the avarage.
46             double average = (score1 + score2 + score3)/3.0;
47
48             // Display the average, rounded to 1 decimal place.
49             averageLabel.Text = average.ToString("n1");
50         }
51
52         private void exitButton_Click(object sender, EventArgs e)
53         {
54             // Close the form.
55             this.Close();
56         }
57     }
58 }
```

Checkpoint

9.25 Look at the following statement:

```
var info = (1, 2, "foo");
```

What is the data type of the `info` variable?

a. `[int, int, string]`

b. `(int, int, string)`

c. `(string, int, int)`

d. `tuple(Item1, Item2, Item3)`

9.26 Look at the following declaration statement:

```
(string, double, decimal) info;
```

Which of the following statements correctly assign a tuple to `info`?

a. `info = ["foo", 1.5, 99m];`

b. `info = "foo", 1.5, 99m;`

c. `info = ("foo", 1.5, 99m);`

d. `info = new("foo", 1.5, 99m);`

9.27 Look at the following statement:

```
var info = (1, 2, "foo");
```

What are the default names for `info`'s fields?

a. `Item0, Item1, Item2`

b. `[0], [1], [2]`

c. `info.1, info.2 info.3`

d. `Item1, Item2, Item3`

9.28 Look at the following method:

```
private (string A, string B, string C) getInfo()
{
   return ("Athos", "Porthos", "Aramis");
}
```

What does the following code display?

```
var info = getInfo();
MessageBox.Show(info.Item1);
```

a. `Athos`

b. `Porthos`

c. `Aramis`

d. `Athos Porthos Aramis`

9.29 Look at the following statement:

```
(int Uno, int Dos) info = (1, 2);
```

What does the following code display?

```
MessageBox.Show($"{info.Item1}");
```

a. `Uno`

b. `Dos`

c. `1`

d. `2`

9.30 Look at the following statement:

```
(int Uno, int Dos) info = (1, 2);
```

What does the following code display?

```
MessageBox.Show($"{info.Dos}");
```

a. `Dos = 1`

b. `Dos`

c. `1`

d. `2`

9.31 Look at the following method:

```
private (string A, string B, string C) getInfo()
{
   return ("Athos", "Porthos", "Aramis");
}
```

What does the following code display?

```
(string x, string y, string z) = getInfo();
MessageBox.Show(y);
```

a. `Athos`

b. `Porthos`

c. `Aramis`

d. `Athos Porthos Aramis`

Key Terms

`DateTime` Structure
`DateTimePicker` control
deconstruction
Dictionary
enumerated data type
enumerators
field
instance
key–value pair
named field
parallel relationship
structure
`TimeSpan` structure
tuple

Review Questions

Multiple Choice

1. A _____ is a data type you can create that contains one or more variables known as fields.
   a. structure
   b. collection
   c. volume
   d. list

2. Before you can use a structure to store data, you must create a(n) _____ of the structure in memory.
   a. copy
   b. instance
   c. declaration
   d. reference

3. When the items in two data structures are related by their indexes, it is said that a _____ exists between the data structures.
   a. binary union
   b. parallel relationship
   c. unilateral bond
   d. virtual connection

4. A program uses a `DateTime` object's `ToString` method to display its value, and the output is 3/17/1999. Which format string was used with the `ToString` method?
   a. `"D"`
   b. `"d"`
   c. `"F"`
   d. `"MDY"`

5. Which expression returns the current date and time from the system?
   a. `DateTime.Today`
   b. `DateTime.SysDateTime`
   c. `DateTime.Current`
   d. `DateTime.Now`

6. Which expression returns the current date, but not the time, from the system?
   a. `DateTime.Today`
   b. `DateTime.SysDateTime`
   c. `DateTime.Current`
   d. `DateTime.Now`

7. Which of the following is not a property of a `TimeSpan` object?
   a. Days
   b. Months
   c. Hours
   d. Minutes

8. When you create a(n) _____, you specify a set of symbolic values that belong to that data type
   a. abstract data type
   b. symbolic data type
   c. enumerated data type
   d. cryptic data type

9. _____ are constants that represent integer values.
   a. Literals
   b. Enumerators
   c. Constants
   d. Tokens

10. You can use the _____ method to determine whether a key exists in a dictionary.
    a. `KeyIn`
    b. `ContainsKey`
    c. `KeyMatch`
    d. `KeyExists`

11. You can use the _____ method to determine whether a value exists in a dictionary.
    a. `ValueIn`
    b. `ContainsValue`
    c. `ValueMatch`
    d. `ValueExists`

12. You use _____ to remove an element from a dictionary.
    a. the `Remove` method
    b. the `Erase` method
    c. the `Delete` method
    d. the `Del` method

13. The _____ property holds the number of elements in a dictionary.
    a. Size
    b. Len
    c. Elements
    d. Count

14. Assuming `food` is a `Dictionary`, what does the following statement do?

```
food[101] = "Cheesburger";
```

    a. After the statement executes, the `Dictionary` will have an element with the key 101 and the value "Cheeseburger."
    b. After the statement executes, the `Dictionary` will have an element with the key "Cheeseburger" and the value 101.
    c. After the statement executes, the `Dictionary` will have 101 elements, each with the key "Cheeseburger."
    d. After the statement executes, the element at position 101 will have the key "Cheeseburger."

15. Look at the following statement:

```
var info = ("past", "present", "future");
```

    What type of object is being assigned to the `info` variable?
    a. A `struct`
    b. A tuple
    c. A list
    d. A dictionary

16. What is displayed by the following code:

```
var info = ("past", "present", "future");
MessageBox.Show(info.Item1)'
```

    a. present
    b. future
    c. past
    d. past present future

True or False

1. The fields contained in a structure must be of the same data type.
2. Before you can use the structure to store data, you must create an instance of the structure in memory.
3. Structure objects can be passed into a method only by reference.
4. You cannot perform comparison operations directly on structure objects.
5. Enumerators must be enclosed in quotation marks.
6. An `enum` declaration can only appear inside the application's namespace.
7. The integer values that you assign to enumerators do not have to be unique.
8. You can compare enumerators and `enum` variables with relational operators.

9. The `DateTime.TryParse` method will successfully convert the string "8/22/1990" to a `DateTime` object.
10. You can use relational operators to compare two `DateTime` objects and determine which one is earlier or later.
11. You can store duplicate keys in a `Dictionary`.
12. A `Dictionary` object's `Remove` method returns an integer indicating the number of elements that were removed.
13. A `KeyValuePair` object has three properties: Index, Key, and Value.
14. You cannot change the value of a tuple's field.
15. A tuple can hold values of different data types.

Short Answer

1. List the places that a structure can be declared in code.
2. Assume an application contains a structure named `Fruit`. Write a statement that demonstrates how you would create a `List` that can hold `Fruit` objects.
3. What will the following code display?

```
var dct = new Dictionary<string, int>()
{
   {"Monday", 1},
   {"Tuesday", 2},
   {"Wednesday", 3}
};
MessageBox.Show(dct["Tuesday"].ToString());
```

4. How do you delete an element from a dictionary?
5. How do you determine the number of elements that are stored in a dictionary?
6. Can enumerators be used in a loop to step through the elements of an array? Why or why not?
7. Why should you use the `new` operator when creating instances of a structure?
8. Suppose a tuple contains 12 items. What is the default field name for the first item? What is the default field name for the last item?

Algorithm Workbench

1. Look at the following structure declaration:

```
struct Engine
{
  public int cylinders;
  public int horsepower;
  public int torque;
}
```

   Write a method that accepts two `Engine` structures as arguments, determines if the two structures are equal, and returns the Boolean value `true` if the structures are equal or `false` if the structures are not equal.

2. Declare an enumerated data type named `Direction` with enumerators for North, South, East, and West.
3. Give an example of instantiating a `DateTime` object named `dt`, initialized with the date August 22, 1965.

4. `date1` and `date2` are `DateTime` objects containing two different dates. Write code that displays the values of these two objects in chronological order.

5. Given the following declaration, write code that uses a `TimeSpan` object to add five days to the `date1` object and assigns the result to a second `DateTime` object named `date2`.

```
var date1 = new DateTime(1977, 3, 22);
```

6. Write a statement that defines a `Dictionary` named `food`. The keys should be `string`s, and the values should be `int`s.

7. Suppose a program declares a `Dictionary` as follows:

```
var customers = new Dictionary<int, string>();
```

Write statements that use the `Dictionary` class's `Add` method to insert the following elements:

```
9001, "Jen Williams"
9002, "Frank Smith"
9003, "Geri Rose"
```

8. Suppose a program has the following `Dictionary` definition:

```
var customers = new Dictionary<int, string>();
```

Write a `for` loop that iterates over the `Dictionary`. Each element's key should be added to a ListBox named `customerListBox`.

9. Suppose a program has the following `Dictionary` definition:

```
var customers = new Dictionary<int, string>();
```

Write an `if` statement that determines whether the key 999 exists in the `Dictionary`. If the key exists, display the message "Customer 999 found" in a message box. Otherwise, display the message "Customer 999 not found."

10. Write a statement that creates a tuple and assigns that tuple to a local variable named `tup`. The tuple should contain the following fields:
    - A field named `ID` that has the value `456`
    - A field named `City` that has the value `"Jakarta"`

11. Suppose a tuple named `info` has been declared and initialized. The tuple has two fields. The first field is a `string` and the second field is an `int`. Write a statement that deconstructs the `info` tuple and assigns its first field to a variable named `country`, and its second field to a variable named `population`.

Programming Problems

1. **Weather Data**

In the student sample program files for this chapter, you will find a text file named weather.txt. The file contains weather data for the month of January, 2018. Each line in the file contains the following four pieces of data for one day:

- Date
- Precipitation
- High temperature
- Low temperature

The data stored in each line is formatted in the following way:

```
Date;Precipitation;HighTemp;LowTemp
```

Notice that the data items are separated by semicolons. The first line in the file looks like this:

```
1/1/2018;0;29;10
```

The data stored in this line are:

- Date = 1/1/2018
- Precipitation = 0 inches
- High temperature = 29
- Low temperature = 10

Design a structure that has fields for the four pieces of data. In an application, read the file and store each day's data in an instance of the structure. The structure instances should be stored in a `List`.

The application should let the user select a date using a DateTimePicker control. The control should be configured so it only allows the user to select dates in January, 2018. When the user selects a date, the application should display the weather data for that date.

TIP: Use the `Split` method that we discussed in Chapter 8 to tokenize each line into the four pieces of data.

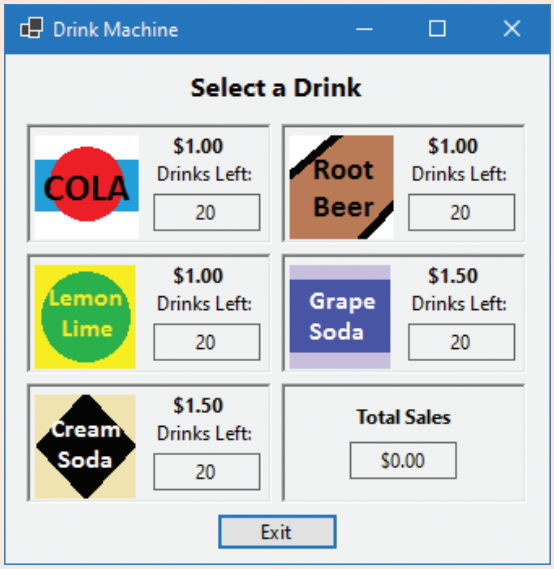
2. **Drink Vending Machine Simulator**

Create an application that simulates a soft-drink vending machine. The application should let the user select one of the following soft drinks:

- Cola ($1.00 each)
- Root Beer ($1.00 each)
- Lemon Lime Soda ($1.00 each)
- Grape Soda ($1.50 each)
- Cream Soda ($1.50 each)

Figure 9-20 shows an example of the application's form at run time. (The images that are displayed in the PictureBox controls are in the *Images\Drink Machine* folder of the Student Sample Programs.) When the application starts, the vending machine will have 20 of each type of soft drink. Each time the user selects a drink, the application should subtract 1 from the quantity of the selected drink. It should also update and display the total amount of sales. If the user selects a drink that is sold out, a message should be displayed indicating so.

Figure 9-20 Drink vending machine simulator



In the application's code, create a structure that has fields for the following data:

Drink name
Drink cost
Number of drinks in machine

The program should create an array of five structure objects. Each element of the array should keep data for a specific type of soft drink.

3. **Magic Dates Revisited**

   Recall from Chapter 4's Programming Challenges that the date June 10, 1960, is special because when it is written in the following format, the month times the day equals the year:

   6/10/60

   In that Programming Challenge, you were asked to create an application that lets the user enter a month (in numeric form), a day, and a two-digit year, and then determine whether the month times the day equals the year. If so, it should display a message saying the date is magic. Otherwise, it should display a message saying the date is not magic. Recreate that application using a DateTimePicker control for the user input.

4. **Time Clock**

   Create an application that serves as a time clock that employees can use to clock in and clock out of work. When the application starts, the employee can click a button to clock in. At some later time, the employee can click a button to clock out, and the application will calculate and display the amount of time that the employee worked.

5. **Course Information**

   Write a program that creates a `Dictionary` containing course numbers and the room numbers of the rooms where the courses meet. The `Dictionary` should have the following key–value pairs:

VideoNote
Solving the Course Information Problem

Course Number (Key)	Room Number (Value)
CS101	3004
CS102	4501
CS103	6755
NT110	1244
CM241	1411

   The program should also create a `Dictionary` containing course numbers and the names of the instructors that teach each course. The `Dictionary` should have the following key–value pairs:

Course Number (Key)	Instructor (Value)
CS101	Haynes
CS102	Alvarado
CS103	Rich
NT110	Burke
CM241	Lee

The program should also create a `Dictionary` containing course numbers and the meeting times of each course. The `Dictionary` should have the following key–value pairs:

Course Number (Key)	Meeting Time (Value)
CS101	8:00 AM
CS102	9:00 AM
CS103	10:00 AM
NT110	11:00 AM
CM241	1:00 PM

The program should let the user enter a course number, and then it should display the course's room number, instructor, and meeting time.

6. **Capital Quiz**

   Write a program that creates a `Dictionary` containing the U.S. states as keys and their capitals as values. (Use the Internet to get a list of the states and their capitals.) The program should then randomly quiz the user by displaying the name of a state and asking the user to enter that state's capital. The program should keep a count of the number of correct and incorrect responses. (As an alternative to the U.S. states, the program can use the names of countries and their capitals.)

7. **File Encryption and Decryption**

   Write a program that uses a `Dictionary` to assign "codes" to each letter of the alphabet. For example:

   ```
   var codes = new Dictionary<char, char>()
    { {'A', '%'}, {'a', '9'}, {'B', '@'}, {'b', '#'}, etc...};
   ```

   Using this example, the letter A would be assigned the symbol %, the letter "a" would be assigned the number 9, the letter B would be assigned the symbol @, and so forth. The program should open a specified text file, read its contents, and then use the `Dictionary` to write an encrypted version of the file's contents to a second file. Each character in the second file should contain the code for the corresponding character in the first file. Write a second program that opens an encrypted file and displays its decrypted contents on the screen.

8. **Word Frequency**

   Write a program that reads the contents of a text file. The program should create a dictionary in which the keys are the individual words found in the file and the values are the number of times each word appears. For example, if the word "the" appears 128 times, the dictionary would contain an element with "the" as the key and 128 as the value. The program should either display the frequency of each word or create a second file containing a list of each word and its frequency.

9. **Name and Email Addresses**

   Write a program that keeps names and email addresses in a `Dictionary` as key–value pairs. The program should let the user look up a person's email address, add a new name and email address, change an existing email address, and delete an existing name and email address. The program should save the contents of the `Dictionary` to a file when the user exits the program. Each time the program starts, it should retrieve the information from the file and restore the `Dictionary`.

10. World Series Winners

In this chapter's source code folder (available on the Computer Science Portal at www.pearsonhighered.com/gaddis), you will find a text file named WorldSeriesWinners.txt. This file contains a chronological list of the World Series' winning teams from 1903 through 2009. The first line in the file is the name of the team that won in 1903, and the last line is the name of the team that won in 2009. (Note the World Series was not played in 1904 or 1994. There are entries in the file indicating this.)

Write a program that reads this file and creates a `Dictionary` in which the keys are the names of the teams, and each key's associated value is the number of times the team has won the World Series. The program should also create a `Dictionary` in which the keys are the years, and each key's associated value is the name of the team that won that year.

The program should let the user pick a year in the range from 1903 through 2009. It should then display the name of the team that won the World Series that year, and the number of times that team has won the World Series.

11. Word Index

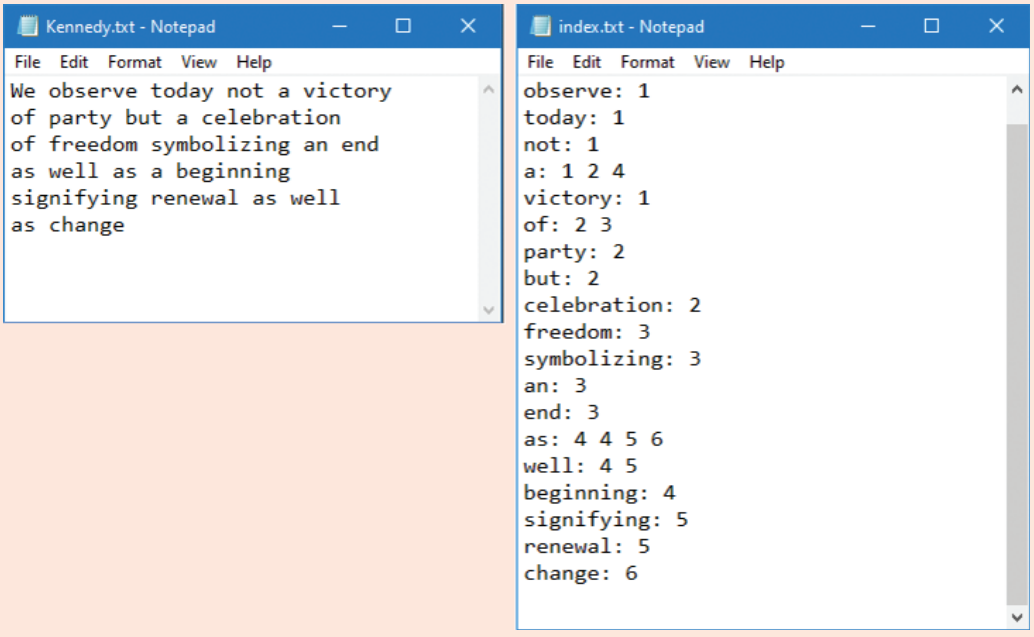
Write a program that reads the contents of a text file. The program should create a `Dictionary` in which the key–value pairs are described as follows:

- Key—The keys are the individual words found in the file.
- Values—Each value is a `List` that contains the line numbers in the file where the word (the key) is found.

For example, suppose the word "robot" is found in lines 7, 18, 94, and 138. The `Dictionary` would contain an element in which the key was the string "robot," and the value was a list containing the numbers 7, 18, 94, and 138.

Once the `Dictionary` is built, the program should create another text file, known as a word index, listing the contents of the `Dictionary`. The word index file should contain an alphabetical listing of the words that are stored as keys in the `Dictionary`, along with the line numbers where the words appear in the original file. Figure 9-21 shows an example of an original text file (Kennedy.txt) and its index file (index.txt).

Figure 9-21 Example of original file and index file



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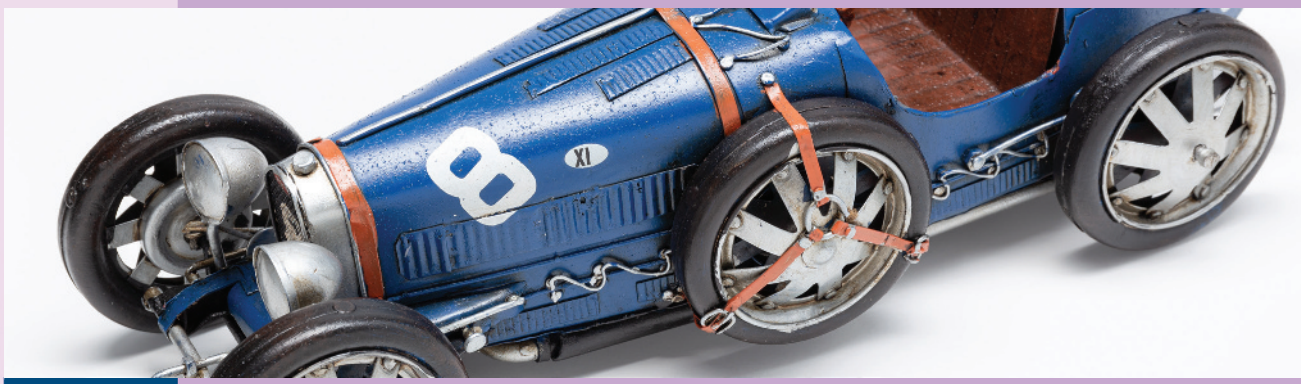


CHAPTER 10

Introduction to Classes

TOPICS

10.1 Introduction to Classes
10.2 Passing an Object to a Method
10.3 Properties
10.4 Auto-Properties
10.5 Parameterized Constructors and Method Overloading
10.6 Storing Class Type Objects in Arrays and `Lists`
10.7 Finding the Classes and Their Responsibilities in a Problem
10.8 Application of Classes: Creating Multiple Forms in a Project
10.9 Static Class Members

10.1 Introduction to Classes

CONCEPT: **A class is the blueprint for an object. It specifies the fields and methods a particular type of object has. From the class, one or more objects may be created.**

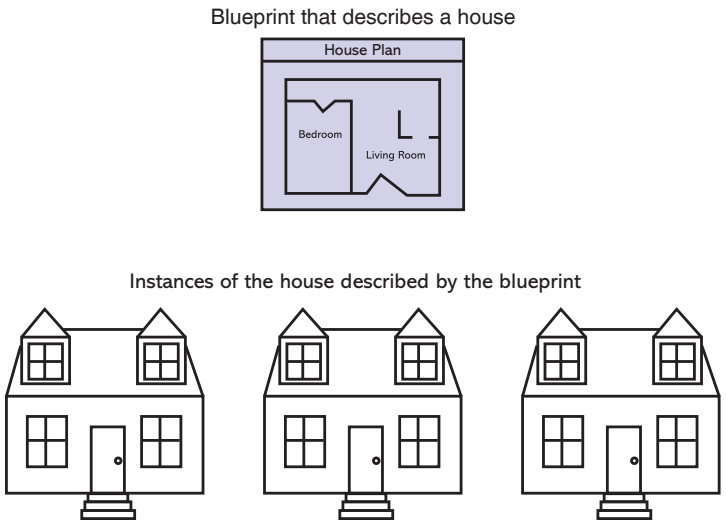
As you have worked through this book, you have used objects extensively in all the programs that you have written. Some objects, such as the controls that you place on a form, are visual. Other objects, such as `Random` objects, arrays, `List` objects,

and ImageList controls, cannot be seen on the screen but exist in memory and perform important tasks.

We mentioned in Chapter 1 that objects do not just magically appear in your program. Before a specific type of object can be used, that object has to be created in memory. And, before an object can be created in memory, you must have a class for the object.

A class is code that describes a particular type of object. It specifies the data that an object can hold (fields and properties) and the actions that an object can perform (methods). You can think of a class as a code "blueprint" that can be used to create a particular type of object. It serves a similar purpose as the blueprint for a house. The blueprint itself is not a house but is a detailed description of a house. When we use it to build an actual house, we could say we are building an instance of the house described by the blueprint. If we so desire, we can build several identical houses from the same blueprint. Each house is a separate instance of the house described by the blueprint. This idea is illustrated in Figure 10-1.

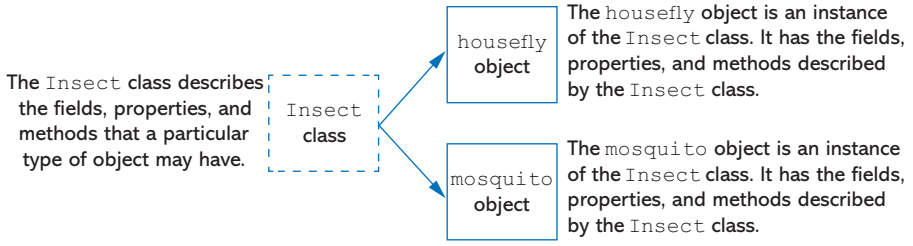
Figure 10-1 A blueprint and houses built from it



So, a class is not an object but a description of an object. When the program is running, it can use the class to create, in memory, as many objects of a specific type as needed. Each object that is created from a class is called an **instance** of the class.

For example, Jessica is an entomologist (someone who studies insects), and she also enjoys writing computer programs. She designs a program to catalog different types of insects. As part of the program, she creates a class named `Insect`, which specifies fields, properties, and methods for holding and manipulating data common to all types of insects. The `Insect` class is not an object but a specification that objects may be created from. Next, she writes programming statements that create a `housefly` object, which is an instance of the `Insect` class. The `housefly` object is an entity that occupies computer memory and stores data about a housefly. It has the fields, properties, and methods specified by the `Insect` class. Then she writes programming statements that create a `mosquito` object. The `mosquito` object is also an instance of the `Insect` class. It has its own area in memory and stores data about a mosquito. Although the `housefly` and `mosquito` objects are two separate entities in the computer's memory, they were both created from the `Insect` class. This means that each object has the fields, properties, and methods described by the `Insect` class. This is illustrated in Figure 10-2.

Figure 10-2 The `housefly` and `mosquito` objects as instances of the `Insect` class



Creating a Class

You create a class by writing a **class declaration**. This is the general format of a class declaration:

```
class ClassName
{
  Member declarations go here...
}
```

The first line of a class declaration is known as the **class header**. It starts with the word `class`, followed by the name of the class. The same rules for naming variables apply to naming classes. Most programmers follow the convention of beginning class names with an uppercase letter. This helps to easily distinguish class names from variable names when reading code.

Following the class header is an opening curly brace. Next, you write the class's **member declarations**. These are the statements that define the class's fields, properties, and methods. A closing curly brace appears at the end of the class declaration.

Let's look at a simple example. Suppose we are writing a program to simulate the tossing of a coin. In the program, we need to repeatedly toss a coin and each time determine whether it landed heads up or tails up. First, we write a class named `Coin` that can perform the behaviors of the coin. The following code sample shows the `Coin` class. (Note that this class is only part of the application's code. In a moment, you will see where it should appear in the project, but for now, we concentrate only on this class.)

```
 1 class Coin
 2 {
 3     // Field to represent the side facing up;
 4     private string sideUp;
 5
 6     // Constructor
 7     public Coin()
 8     {
 9         sideUp = "Heads";
10     }
11
12     // The toss method simulates tossing the coin.
13     public void Toss()
14     {
15         // Create a Random object.
16         var rand = new Random();
17
18         // Use a random number to determine
19         // the side of the coin is facing up.
20         // 0 = Heads, 1 = Tails
21         if (rand.Next(2) == 0)
22         {
23             sideUp = "Heads";
24         }
```

```
25          else
26          {
27              sideUp = "Tails";
28          }
29      }
30
31      // The GetSideUp method returns the value
32      // of the sideUp field.
33      public string GetSideUp()
34      {
35          return sideUp;
36      }
37  }
```

The first line is the class header. It specifies that the name of the class is `Coin`. The curly braces that appear in lines 2 and 37 enclose the contents of the class. Let's take a closer look at the code inside the class:

Line 4: This statement declares a field named `sideUp`. The `sideUp` field is a `string` variable that indicates which side of the coin is facing up. If the heads side is facing up, the string `"Heads"` is assigned to this field. If the tails side is facing up, the string `"Tails"` is assigned to this field.

Notice that the field declaration begins with the keyword `private`. The keyword `private` is an access modifier that specifies the field cannot be directly accessed by statements outside the class.

By using the `private` access modifier, a class can hide its data from code outside the class. When a class's fields are hidden from outside code, the data is protected from accidental corruption. It is a common practice to make all a class's fields private and to provide access to those fields through methods only.

Lines 7–10: This code defines a special method known as a constructor. A **constructor** is a method that is automatically executed when an object is created. In most cases, a constructor is used to initialize an object's fields with starting values. It is called a constructor because it helps construct an object.

This constructor performs a simple task. In line 9, it assigns the string `"Heads"` to the `sideUp` field. As a result, any time we create an object of the `Coin` class, that object's `sideUp` field will initially be assigned the string `"Heads"`.

Notice the following about the constructor header in line 7:

- The name of the constructor is the same as the name of the class. In this case, the name of the constructor is `Coin`.
- The header does not specify a return type—not even `void`.
- The constructor header begins with the `public` access modifier. In most cases, a class's constructor is public. In this book we always use the `public` access modifier with constructors.

Lines 13–29: This code defines a `void` method named `Toss`. Notice that the `public` access modifier is used in the method header in line 13. Because the method is public, it can be called from code outside the `Coin` class.

The purpose of the `Toss` method is to simulate the tossing of the coin. When the method is called, line 16 creates a `Random` object, referenced by the `rand` variable. The `if` statement in line 21 gets a random number in the range of 0 through 1. If the number is 0, then line 23 assigns the string `"Heads"` to the `sideUp` field. Otherwise, line 27 assigns the string `"Tails"` to the `sideUp` field.

Lines 33–36: This code defines a method named `GetSideUp`. Notice that the `public` access modifier is used in line 33, which means that the method can be called from code outside the `Coin` class. Also notice that the method returns a `string`.

The purpose of the `GetSideUp` method is to return a string indicating which side of the coin is facing up. Notice that in line 35 the value of the `sideUp` field is returned.

Creating an Object

Remember, a class is not an object but a description of an object. The `Coin` class specifies what a `Coin` object is made of, but it does not create a `Coin` object in memory. To create a `Coin` object, we must write a statement such as this:

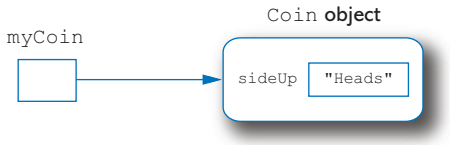
```
Coin myCoin = new Coin();
```

At this point in your studies, you have created plenty of objects, so this type of statement will be familiar to you. For example, `Random` objects, arrays, and `Lists` are all created this way. Let's look at the different parts of this statement:

- The first part of the statement, appearing on the left side of the = operator, reads `Coin myCoin`. This declares a variable named `myCoin` that can be used to reference an object of the `Coin` class.
- The second part of the statement, appearing on the right side of the = operator, reads `new Coin()`. This expression creates an instance of the `Coin` class and calls the class's constructor. The `new` operator returns a reference to the object.
- The = operator assigns the reference that was returned from the `new` operator to the `myCoin` variable.

After this statement executes, the `myCoin` variable will reference a `Coin` object, as shown in Figure 10-3. Notice in the figure that the object's `sideUp` field is set to `"Heads"`, as a result of the constructor.

Figure 10-3 The **myCoin** variable references a **Coin** object



NOTE: Classes are reference types. When you create an object from a class, you use a reference variable to reference that object. When you want to work with the object, you use the variable that references it.

Once we have created a `Coin` object, we can perform operations with it. For example, assuming the variable `myCoin` references a `Coin` object, the following statement calls the object's `Toss` method:

```
myCoin.Toss();
```

And, the following statement displays the side of the coin that is facing up:

```
MessageBox.Show(myCoin.GetSideUp());
```

Using `var` to Declare a Class Reference Variable

If you are declaring and initializing a class reference variable inside of a method, you can use the `var` keyword in your declaration statement. Here is an example:

```
var myCoin = new Coin();
```

The compiler will be able to determine that `myCoin` is a `Coin` reference variable because we are initializing it with a `Coin` object. It is also obvious to any person reading this statement that `myCoin` references a `Coin` object.

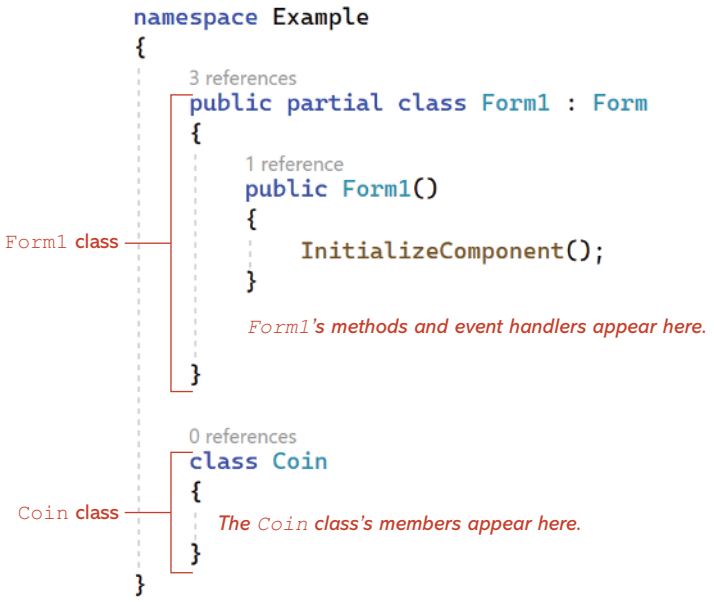
When declaring and initializing local class reference variables in this book, we will use the `var` keyword any time it simplifies the declaration statement without concealing the data type of the object.

NOTE: Remember, the `var` keyword can only be used to declare local variables in a method. It cannot be used to declare fields in a class.

Where to Write Class Declarations

In C#, you have some flexibility in choosing where to write class declarations. When you start writing your own classes in a project, it is possible to write them in the same file that contains the form's class. Figure 10-4 shows an example of how the `Coin` class can be written in the same file as a project's `Form1` class. Notice in the figure that the `Coin` class is written inside the project namespace. This is not required, but it is a good idea since the class is part of the project. (It does not matter if the class is written before or after the `Form1` class.)

Figure 10-4 The **`Form1`** class and the **`Coin`** class in the same source code file



Although this approach might be acceptable for small classes, it is recommended that you write each class in its own separate file. Doing so makes your code more organized and helps keep your source code files to a manageable size. It also makes it easy for you to reuse classes in other projects. In this book, we always store classes in their own files.

Visual Studio automates the process of adding a new class file to a project. When you are ready to write a new class, follow these steps to create a source file for the class:

1. With the project open in Visual Studio, click *Project* on the menu bar; then select *Add Class. . . .* This is shown in Figure 10-5.
2. The *Add New Item* window, shown in Figure 10-6, should appear. Make sure *Class* is selected as the type of item. Notice in the figure that *Class1.cs* appears in the

Figure 10-5 Selecting *Add Class. . .* on the *Project* menu

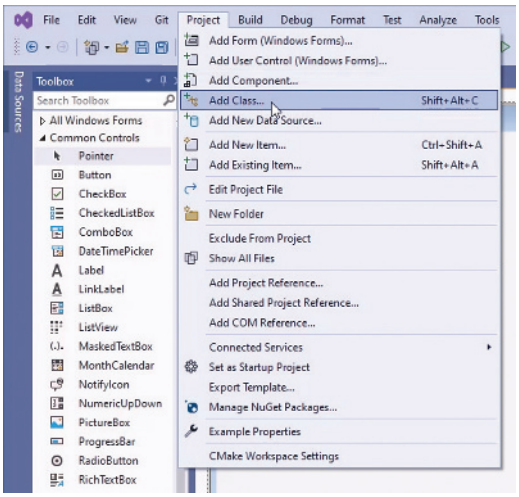
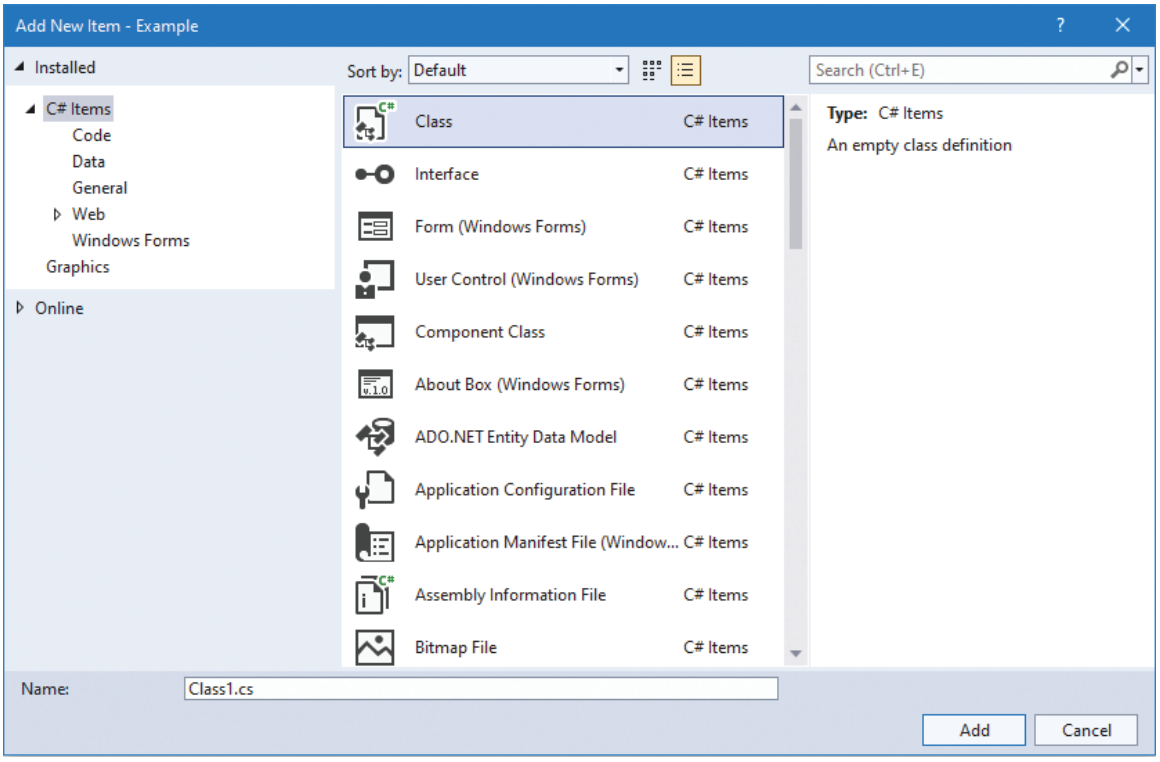


Figure 10-6 The *Add New Item* window



Name text box. This is the default filename that Visual Studio provides. Change the name to match the name of the class that you are creating. For example, if you are creating a class named `Coin`, you change the name to *Coin.cs*. (Be sure that the filename ends with the .cs extension.)

3. Click the *Add* button.

After performing these steps, the specified source code file will be created in the project and will be displayed in the code editor. The source code file contains several `using` statements and an empty class declaration that you can edit. Figure 10-7 shows an example. You will also see an entry for the new source code file in the *Solution Explorer*, as shown in Figure 10-8.

NOTE: Notice in Figure 10-7 that Visual Studio automatically adds the `internal` keyword to the class header. This means that the class is accessible only to code inside this project. In this book we won't be writing any code that needs to access our classes from outside their projects, but it's okay to leave the `internal` keyword in the class header.

Figure 10-7 A new class file displayed in the code editor

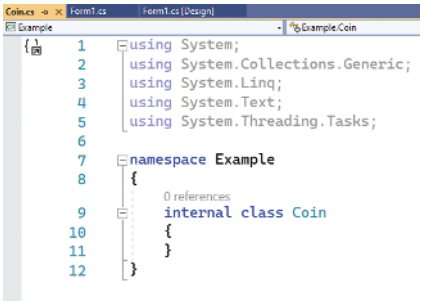
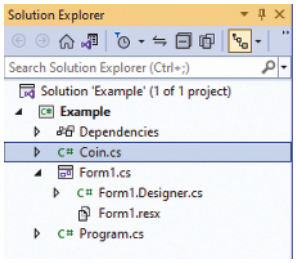


Figure 10-8 *Solution Explorer* window



Tutorial 10-1 leads you through the process of creating the `Coin` class in an application that uses it to simulate a coin that can be tossed.

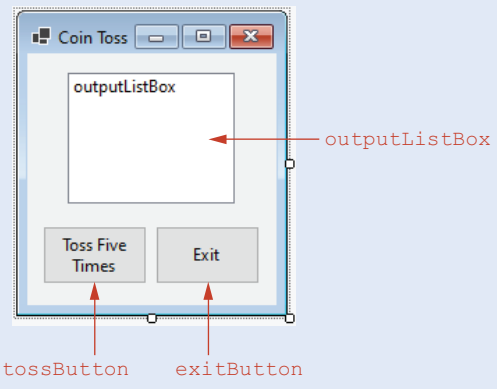
Tutorial 10-1:
Creating and Using the `Coin` Class

VideoNote
Tutorial 10-1: Creating and Using the `Coin` Class

In this tutorial, you complete the *CoinToss* project. The application's form, which has already been created for you, is shown in Figure 10-9. When you run the completed application, you can click the *Toss Five Times* button and the application will simulate a coin being tossed five times. The results of each coin toss are displayed in the list box.

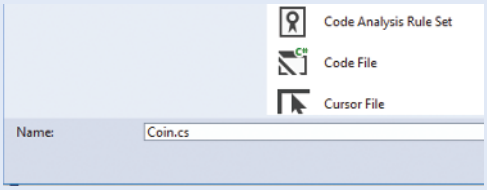
Step 1: Start Visual Studio. Open the project named *CoinToss* in the *Chap10* folder of the Student Sample Programs.

Figure 10-9 The *CoinToss* project's form



Step 2: Click *Project* on the Visual Studio menu bar; then select *Add Class*. . . . The *Add New Item* window should appear. (See Figure 10-6 for an example.) Make sure *Class* is selected as the type of item. Notice that the default filename *Class1.cs* appears in the *Name* text box. Change the name to *Coin.cs*, as shown in Figure 10-10, and then click the *Add* button. This adds a source code file named Coin.cs to the project.

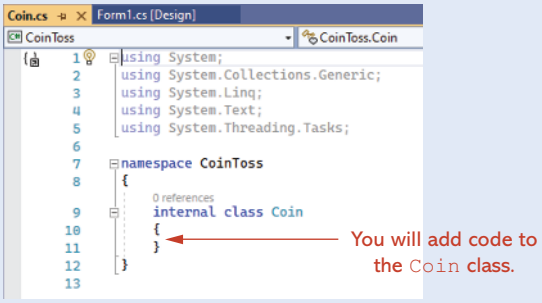
Figure 10-10 The filename changed to Coin.cs



Step 3: The Coin.cs file should now be displayed in the code editor, as shown in Figure 10-11. Notice that the file contains an empty `Coin` class. Complete the code for the `Coin` class by typing lines 11–44 in Program 10-1.

Step 4: Next, create the Click event handlers for the Button controls. Click the tab that reads *Form1.cs [Design]* in the area just above the code editor to switch your view to the Form1 form in the Designer. (Alternatively, you can double-click the *Form1.cs* entry in the *Solution Explorer*.)

Figure 10-11 The Coin.cs file in the code editor



Step 5: Double-click the `tossButton` control. This opens the Form1.cs file in the code editor, and you will see an empty event handler named `tossButton_Click`. Complete the event handler by typing the code shown in lines 12–26 in Program 10-2. Let's review this code:

Line 13: This statement does the following:

- It declares a `Coin` reference variable named `myCoin`.
- It creates a `Coin` object in memory and calls its constructor.
- It assigns a reference to the `Coin` object to the `myCoin` variable.

After this statement executes, the `myCoin` variable will reference a `Coin` object. The object's `sideUp` field will be set to `"Heads"`.

Line 16: This statement clears the `outputListBox` control.

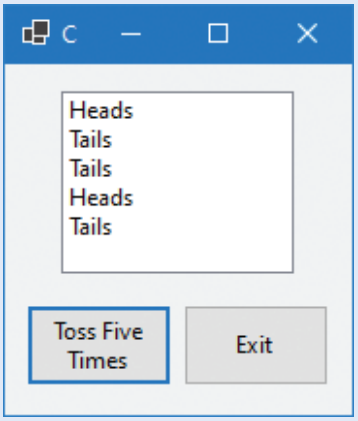
Lines 19–26: This `for` loop iterates five times, simulating five tosses of a coin. During each iteration, the following actions take place:

- Line 22 calls the `myCoin.Toss()` method.
- Line 25 calls the `myCoin.GetSideUp()` method to get the side of the coin that is facing up. The result of the method call is displayed in the `outputListBox` control.

Step 6: Switch your view back to the Form1 form in the *Designer* and double-click the `exitButton` control. In the code editor you will see an empty event handler named `exitButton_Click`. Complete the event handler by typing the code shown in lines 31–32 in Program 10-2.

Step 7: Save the project. Then, press F5 on the keyboard or click the *Start Debugging* button (▶) on the toolbar to compile and run the application. When the application runs, click the *Toss Five Times* button and view the results. Figure 10-12 shows an example of the application's output. Click the button as many times as you wish to see different, random results. Click the *Exit* button when you are finished.

Figure 10-12 Example output of the *CoinToss* application



Program 10-1 Completed code for the Coin.cs file in the *CoinToss* project

```
 1 using System;
 2 using System.Collections.Generic;
 3 using System.Linq;
 4 using System.Text;
 5 using System.Threading.Tasks;
 6
 7 namespace CoinToss
 8 {
 9     internal class Coin
10     {
11         // Field to represent the side facing up.
12         private string sideUp;
13
14         // Constructor
15         public Coin()
16         {
17             sideUp = "Heads";
18         }
```

```
19
20          // The toss method simulates tossing the coin.
21          public void Toss()
22          {
23              // Create a Random object.
24              var rand = new Random();
25
26              // Use a random number to determine
27              // the side of the coin that is facing up.
28              // 0 = Heads, 1 = Tails
29              if (rand.Next(2) == 0)
30              {
31                  sideUp = "Heads";
32              }
33              else
34              {
35                  sideUp = "Tails";
36              }
37          }
38
39          // The GetSideUp method returns the value
40          // of the sideUp field.
41          public string GetSideUp()
42          {
43              return sideUp;
44          }
45      }
46  }
```

Program 10-2 Completed code for Form1 in the *CoinToss* project

```
 1 namespace CoinToss
 2 {
 3     public partial class Form1 : Form
 4     {
 5         public Form1()
 6         {
 7             InitializeComponent();
 8         }
 9
10         private void tossButton_Click(object sender, EventArgs e)
11         {
12             // Create a Coin object.
13             var myCoin = new Coin();
14
15             // Clear the ListBox.
16             outputListBox.Items.Clear();
17
18             // Toss the coin five times.
19             for (int count = 0; count < 5; count++)
20             {
21                 // Toss the coin.
22                 myCoin.Toss();
23
24                 // Display the side that is up.
25                 outputListBox.Items.Add(myCoin.GetSideUp());
26             }
27         }
```

```
28
29          private void exitButton_Click(object sender, EventArgs e)
30          {
31              // Close the form.
32              this.Close();
33          }
34      }
35  }
```

Scope of Fields

Recall from Chapter 3 that a variable's scope is the part of a program where the variable may be accessed by its name. A variable's name is visible only to statements inside the variable's scope. The location of a variable's declaration determines the variable's scope.

In this chapter, you have seen variables declared as fields in a class. A field can be accessed by any method in the same class as the field. If a field is declared with the `public` access specifier, it can also be accessed by code outside the class.

Shadowing

A method can have a parameter or a local variable with the same name as a field in the same class. When this happens, the name of the parameter or local variable **shadows** the name of the field. This means that the field name is hidden by the name of the parameter or local variable. As a result, the field cannot be accessed within the method.

Initializing Fields at Declaration

A field can be initialized in its declaration statement in the same way that a local variable can be initialized: after the name of the field, you write an = operator, followed by a value. For example, the following code shows an alternate version of the `Coin` class. In this version, we initialize the `sideUp` field in its declaration statement (see line 4). Because the `sideUp` field is initialized at its declaration, we have left out the constructor. This version of the class works identically to the previous version.

```
 1 class Coin
 2 {
 3     // Field to represent the side facing up;
 4     private string sideUp = "Heads";
 5
 6     // The toss method simulates tossing the coin.
 7     public void Toss()
 8     {
 9         // Create a Random object.
10         var rand = new Random();
11
12         // Use a random number to determine
13         // the side of the coin is facing up.
14         // 0 = Heads, 1 = Tails
15         if (rand.Next(2) == 0)
16         {
17             sideUp = "Heads";
18         }
```

```
19          else
20          {
21              sideUp = "Tails";
22          }
23      }
24
25      // The GetSideUp method returns the value
26      // of the sideUp field.
27      public string GetSideUp()
28      {
29          return sideUp;
30      }
31  }
```

NOTE: Suppose a class's fields are initialized at their declarations, and the class also has a constructor that assigns values to the fields. When the class is instantiated, the fields will first be initialized to the values listed in their declaration statements. Then the constructor will execute, assigning its values to the fields.

Uninitialized Fields

If a field is uninitialized and it is not assigned a value by a class constructor, it is automatically set to a default value that is determined by the field's data type. Table 10-1 shows the default values.

Table 10-1 Default values for uninitialized fields

Data Type	Default Value
Integer types	0
Floating-point types	0.0
`bool`	`false`
`char`	`'\0'`
`string`	`null`
Reference types	`null`
`struct`	All the members of the `struct` are set to the same default values shown in this table.
`enum`	0

Recall that C# will not allow you to use an uninitialized local variable. This is not true with class fields, however. Because an uninitialized field is set to a default value, the C# compiler will allow you to use it without first assigning a value to it. For this reason, you must be careful not to perform an operation with a `null` reference type field.

Checkpoint

10.1 How is a class like a blueprint?

10.2 Briefly describe the process of writing a class declaration.

10.3 What is a constructor?

10.4 List the three steps for adding a new class file to a Visual C# project.

10.2 Passing an Object to a Method

CONCEPT: **When a class object is passed as an argument to a method, a reference to the object is passed into the method's parameter variable. As a result, the parameter variable references the object.**

When you are developing applications that work with objects, you often need to write methods that accept objects as arguments. For example, the following code shows a method named `ShowCoinStatus` that accepts a `Coin` object as an argument:

```
1 private void ShowCoinStatus(Coin coin)
2 {
3     MessageBox.Show($"The coin is {coin.GetSideUp()} up.");
4 }
```

The following code sample shows how we might create a `Coin` object and then pass it as an argument to the `ShowCoinStatus` method:

```
var myCoin = new Coin();
ShowCoinStatus(myCoin);
```

Because classes are reference types, objects that are instances of a class are *always* passed by reference. When you pass an object that is an instance of a class as an argument, the thing that is passed into the parameter variable is a reference to the object. As a result, the method that receives the object as an argument has access to the actual object (not a copy of the object). For example, look at the following `Flip` method:

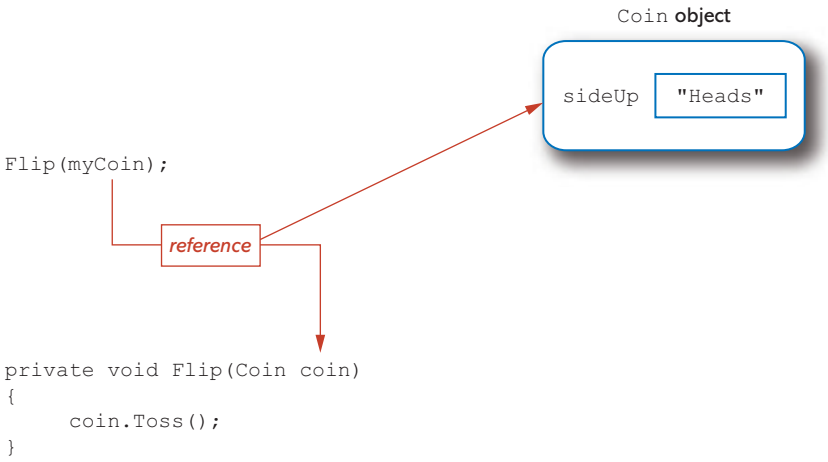
```
1 private void Flip(Coin coin)
2 {
3     coin.Toss();
4 }
```

This method accepts a `Coin` object as an argument, and it calls the object's `Toss` method. The following code demonstrates the method being called:

```
1  // Create a Coin object.
2  var myCoin = new Coin();
3
4  // This will display "Heads".
5  MessageBox.Show(myCoin.GetSideUp());
6
7  // Pass the object to the Flip method.
8  Flip(myCoin);
9
10 // This might display "Heads", or it might
11 // display "Tails".
12 MessageBox.Show(myCoin.GetSideUp());
```

The statement in line 2 creates a `Coin` object, referenced by the variable `myCoin`. Line 5 displays the value of the `myCoin` object's `sideUp` field. Because the object's constructor set the `sideUp` field to `"Heads"`, we know that line 5 will display the string `"Heads"`. Line 8 calls the `Flip` method, passing the `myCoin` object as an argument. Figure 10-13 shows how this actually passes a reference to the object into the method. Inside the `Flip` method, the `myCoin` object's `Toss` method is called. Line 12 displays the value of the `myCoin` object's `sideUp` field again. This time, we cannot predict whether "Heads" or "Tails" will be displayed because the `myCoin` object's `Toss` method has been called.

Figure 10-13 Example output of the *CoinToss* application



Checkpoint

10.5 When you pass an object that is an instance of a class as an argument, what is passed into the parameter variable?

10.3 Properties

CONCEPT: **A property is a class member that holds a piece of data about an object. Properties are implemented as special methods that set and get the values of corresponding fields.**

A **property** is a special type of class member that allows an object to store and retrieve a piece of data. You are already familiar with properties because you have used them extensively when creating forms. GUI controls have properties that determine their characteristics. For example, a Label control's Text property determines the text that is displayed by the control.

In code, you work with properties in the same ways that you work with variables. For example, assume `messageLabel` is the name of a Label control. You can assign a specific value to the control's Text property, as shown here:

```
messageLabel.Text = "Hello";
```

Also, you can assign a property to a variable. The following statement initializes the variable `str` with the `messageLabel` control's Text property:

```
string str = messageLabel.Text;
```

From these examples, you can see that a property is a class member that behaves like a public field. A property is not a public field, however. It is a special set of methods, known as **accessors**, which work in conjunction with a private field. The private field, which is known as the property's **backing field**, holds any data that is assigned to the property. The accessors allow code outside the class to get the property's value and assign values to the property.

Let's look at an example of a simple class that has a property. Suppose you want to create a class named `Pet` that represents the family pet. The class will have a Name property to hold the pet's name. The following code shows how the `Pet` class might be written:

```
1 class Pet
2 {
3     // Field for the pet's name
4     private string _name;
5
6     // Constructor
7     public Pet()
8     {
9         _name = "";
10    }
11
12    // Name property
13    public string Name
14    {
15        get
16        {
17            return _name;
18        }
19
20        set
21        {
22            _name = value;
23        }
24    }
25 }
```

Let's take a closer look at the code inside the class:

Line 4: This statement declares a private `string` field named `_name`. The `_name` field is the backing field for the Name property. When a value is assigned to the Name property, it is stored in the `_name` field.

Notice that we have started the field's name with an underscore character. This is not a requirement, but some programmers begin the names of backing fields with an underscore to eliminate confusion between the field name and the property name. By following this practice, when you are reading the code and you see a variable name that begins with an underscore, you know immediately that it is a backing field for a property.

Lines 7–10: This is the class constructor. When an instance of the `Pet` class is created, this constructor assigns an empty string to the `_name` field.

Line 13: This is the beginning of a property declaration. It specifies three things about the property:

- `public`—The property is public, so it can be used by code outside the class. (Most properties are declared as public.)
- `string`—The property's data type is `string`.
- `Name`—The name of the property is Name.

Line 14: An opening curly brace appears in this line. The corresponding closing curly brace appears in line 24. Inside these braces are the Name property's accessors.

Lines 15–18: This is the property's `get` accessor. You can think of the `get` accessor as a method that returns the property's value. You can see that line 17 is a `return` statement, returning the value of the `_name` field. The `get` accessor is executed any time the property is read.

Lines 20–23: This is the property's `set` accessor. The purpose of the `set` accessor is to set the property to a value. The `set` accessor has an implicit parameter named `value`.

It is "implicit" because it is not declared, as the parameters in a regular method are. The **value parameter** is automatically created by the compiler, and its data type is the same as that of the property. In this case, the `value` parameter's data type is `string`.

Any time a value is assigned to a property, the property's `set` accessor is executed, and the value being assigned is passed into the `value` parameter. Line 22 assigns the `value` parameter to the `_name` field.

The following code shows how you can use the `Pet` class's Name property:

```
 1 // Create two Pet objects.
 2 var myDog = new Pet();
 3 var myCat = new Pet();
 4
 5 // Set their Name properties.
 6 myDog.Name = "Fido";
 7 myCat.Name = "Sylvester";
 8
 9 // Display their names.
10 MessageBox.Show($"My dog's name is {myDog.Name}.");
11 MessageBox.Show($"My cat's name is {myCat.Name}.");
```

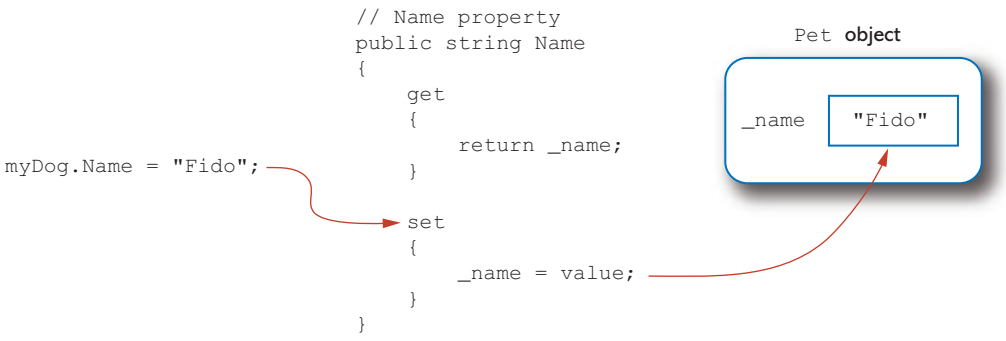
Let's take a closer look at the code:

Line 2: This statement creates a `Pet` object, calls the `Pet` class constructor, and assigns a reference to the object to the `myDog` variable. After this statement executes, the `myDog` variable references a `Pet` object.

Line 3: This statement creates a `Pet` object, calls the `Pet` class constructor, and assigns a reference to the object to the `myCat` variable. After this statement executes, the `myCat` variable references a `Pet` object.

Line 6: This statement sets the `myDog` object's Name property to `"Fido"`. When this statement executes, the Name property's `set` accessor is executed, and the string `"Fido"` is passed into the `value` parameter. The `value` parameter is then assigned to the `myDog` object's `_name` field. This process is illustrated in Figure 10-14.

Figure 10-14 Setting the **myDog** object's Name property to **"Fido"**



Line 7: This statement sets the `myCat` object's Name property to `"Sylvester"`. When this statement executes, the Name property's `set` accessor is executed, and the string `"Sylvester"` is passed into the `value` parameter. The `value` parameter is then assigned to the `myCat` object's `_name` field.

Line 10: This statement displays the message "My dog's name is Fido" in a message box. When the `myDog` object's Name property is retrieved, the `get` accessor returns the value of the object's `_name` field.

Line 11: This statement displays the message "My cat's name is Sylvester" in a message box. When the `myCat` object's Name property is retrieved, the `get` accessor returns the value of the object's `_name` field.

When an accessor contains only a single statement, many programmers prefer to write the entire accessor in one line of code. For example, the Name property in the `Pet` class could have been written like this:

```
// Name property
public string Name
{
   get { return _name; }
   set { _name = value; }
}
```

This concise style of writing accessors reduces the number of lines code in a class declaration without sacrificing the code's readability. From this point forward in the book, we will use this style when an accessor has only one statement.

Properties versus Public Fields

At this point, you might be wondering why you should go to the trouble of creating a property when it would be easier to simply create a public field. For example, in the `Pet` class, couldn't we have just declared a public `string` field named `Name`? If the field is public, it can be directly accessed by code outside the class without the need for accessors.

It is possible to take that approach, but it is not recommended. Class fields are almost always declared `private` in order to protect them from accidental corruption. When code outside the class needs to access a field, it does so through public methods (or, in the case of a property, through accessors). This ensures that the object owning the fields is in control of all changes being made to them.

Passing Properties as Arguments

You can pass a property as an argument to a method. For example, line 8 in the following code sample passes a `Pet` object's Name property as an argument to the `MessageBox.Show` method:

```
1 // Create a Pet object.
2 var myDog = new Pet();
3
4 // Set the object's Name property.
5 myDog.Name = "Fido";
6
7 // Display the Name property.
8 MessageBox.Show(myDog.Name);
```

When you pass a property as an argument to a method, there is one restriction: Properties can be passed only by value. If you try to pass a property to a `ref` or an `out` parameter, an error occurs.

Referring to a Property inside the Class

Sometimes a class needs to work with its own properties. In other words, a method in a class needs to work with a property in the same class. When this is the case, the method can either work with the property by its name, or with the property's backing field. For example, look at the following `Car` class:

```
1 class Car
2 {
3     // Field for the car's speed
4     private int _speed;
5
```

```
 6     // Constructor
 7     public Car()
 8     {
 9         Speed = 0;
10     }
11
12     // Speed property
13     public int Speed
14     {
15         get { return _speed; }
16         set { _speed = value; }
17     }
18
19     // Accelerate method
20     public void Accelerate()
21     {
22         Speed += 10;
23     }
24
25     // Brake method
26     public void Brake()
27     {
28         Speed -= 10;
29     }
30 }
```

This class has one property named Speed, defined in lines 13–17. In line 9, the constructor initializes the Speed property to 0. The class also has a method named `Accelerate` (in lines 20–23), and another method named `Brake` (in lines 26–29). Notice that in line 22 the `Accelerate` method adds 10 to the Speed property, and in line 28 the `Brake` method subtracts 10 from the Speed property. In this class, the constructor, the `Accelerate` method, and the `Brake` method work with the Speed property by its name.

Alternatively, the methods can work with the `_speed` field, which is the backing field for the Speed property. Here is another version of the `Car` class that uses this approach:

```
 1 class Car
 2 {
 3     // Field for the car's speed
 4     private int _speed;
 5
 6     // Constructor
 7     public Car()
 8     {
 9         _speed = 0;
10     }
11
12     // Speed property
13     public int Speed
14     {
15         get { return _speed; }
16         set { _speed = value; }
17     }
18
19     // Accelerate method
20     public void Accelerate()
21     {
22         _speed += 10;
23     }
24
25     // Brake method
26     public void Brake()
```

```
27      {
28          _speed -= 10;
29      }
30  }
```

Notice that in lines 9, 22, and 28 the class modifies the backing field for the Speed property. In this class this approach has the same effect as working with the property by its name.

NOTE: In some classes, a property's accessors might do more work than simply setting or getting the value of a backing field. For example, a `set` accessor might perform data validation, or a `get` accessor might return the results of a calculation. When this is the case, working with a property's name might produce different results than working directly with the property's backing field. In such cases, you must decide which approach produces the results you desire.

Read-Only Properties

Sometimes it is useful to make a property read-only. A **read-only property** can be read, but it cannot be modified. If you need to make a property read-only, take these two steps:

- Use the **`readonly` keyword** in the declaration of the backing field.
- Do not write a `set` accessor for the property.

As an example, look at the following version of the `Pet` class, in which the Name property is read-only:

```
1  class Pet
2  {
3      // Field for the pet's name
4      private readonly string _name;
5
6      // Constructor
7      public Pet()
8      {
9          _name = "Fido";
10     }
11
12     // Name property
13     public string Name
14     {
15         get
16         {
17             return _name;
18         }
19     }
20 }
```

First, notice in line 4 that the `readonly` keyword appears in the declaration of the `_name` field. This is the backing field for the Name property. When a field is declared as `readonly`, it can be assigned a value in only two places: in the field's declaration statement (as an initialization value) or in the class's constructor. If an assignment to the field is made in any other place, a compiler error will occur.

Also notice that the Name property has a `get` accessor but does not have a `set` accessor. Therefore, if we create an object of the `Pet` class, we can get the value of the Name property, but an error will occur if we try to assign a value to the Name property.

In line 9 of this class, the constructor assigns the string `"Fido"` to the Name property's backing field. Any time we create an instance of this class, the instance's Name property will be set to `"Fido"`. Because the property is read-only, we cannot change its value.

Private `set` Accessors

Sometimes you want a property to be read-only to all code that exists outside the class, but you want methods inside the class to be able to change the value of the property. When this is the case, you cannot use the `readonly` keyword to declare the property's backing field. This is because a `readonly` field can only be assigned a value in its declaration statement, or in the class constructor. If you want any method in a class, other than the constructor, to change a property's value, the property's backing field cannot be declared as `readonly`.

To make a property read-only to code outside the class, but still allow methods inside the class to change the property, you should give the property a private `set` accessor. The following version of the `Car` class shows an example:

```
 1 class Car
 2 {
 3     // Field for the car's speed
 4     private int _speed;
 5
 6     // Constructor
 7     public Car()
 8     {
 9         Speed = 0;
10     }
11
12     // Speed property
13     public int Speed
14     {
15         get { return _speed; }
16         private set { _speed = value; }
17     }
18
19     // Accelerate method
20     public void Accelerate()
21     {
22         Speed += 10;
23     }
24
25     // Brake method
26     public void Brake()
27     {
28         Speed -= 10;
29     }
30 }
```

Notice in line 16 that the Speed property's `set` accessor is declared private. As a result, only the methods inside the `Car` class, such as the `Accelerate` and `Brake` methods, can change the value of the property. To code outside the class, the Speed property is read-only.

Avoiding Stale Data

Sometimes the value of a property is dependent on other data. For example, look at the following `Circle` class:

```
1 class Circle
2 {
3     // Field
4     private double _diameter;
5
6     // Constructor
```

```
 7      public Circle()
 8      {
 9          _diameter = 0.0;
10      }
11
12      // Diameter property
13      public double Diameter
14      {
15          get { return _diameter; }
16          set { _diameter = value; }
17      }
18
19      // Radius property (read-only)
20      public double Radius
21      {
22          get { return _diameter / 2; }
23      }
24  }
```

This class has two properties: Diameter (defined in lines 13–17) and Radius (defined in lines 20–23). Notice that the Radius property does not have its own backing field, as the Diameter property does.

Rather, the Radius property's `get` accessor returns the value of a calculation. The radius is not stored in a field because a circle's radius is dependent on the circle's diameter. If we store both the diameter and the radius in their own fields, the radius field will become incorrect as soon as the diameter field changes.

When a field's value is tightly dependent on other data and that field is not updated when the other data is changed, it is said that the field has become **stale**. When designing a class, you should take care not to store in a field any calculated data that can potentially become stale. Instead, provide a read-only property, or a method, that returns the value of the calculation.

Checkpoint

10.6 What is a property?

10.7 When you pass a property as an argument to a method, there is one restriction. What is it?

10.8 How do you make a property read only?

10.9 What will happen if you try to assign a value to a read-only property?

10.10 What is stale data?

10.4 Auto-Properties

CONCEPT: An auto-property is a property that is implemented with a simplified declaration syntax, not requiring the explicit declaration of a backing field.

Sometimes a property simply gets and sets the value of a backing field, without performing any other operation. For example, recall the original version of the `Pet` class that we discussed earlier in this section. The class has a Name property that gets and sets the value of a backing field named `_name`. Here is the relevant code from the `Pet` class:

```
private string _name;
public string Name
{
    get
    {
        return _name;
    }
    set
    {
        _name = value;
    }
}
```

This code can be greatly simplified if we implement the property as an **auto-property**. Auto-properties allow you to create simple properties without declaring a backing field, and without writing code to get and set the property's value. Here is an example of how the Name property in the `Pet` class can be written as an auto-property:

```
public string Name
{
    get;
    set;
}
```

When the property is written this way, the C# compiler automatically creates a hidden backing field, as well as the code for the `get` and `set` methods. In fact, most programmers prefer to write an even shorter version of the property, like this:

```
public string Name { get; set; }
```

If you want to initialize an auto-property to a particular value, you can do so in the following manner:

```
public string Name { get; set; } = "Fido";
```

This code creates a Name property, initialized with the string `"Fido"`.

If an auto-property is uninitialized and it is not assigned a value by a class constructor, it is automatically set to a default value that is determined by the property's data type. Table 10-2 shows the default values.

Table 10-2 Default values for uninitialized properties

Data Type	Default Value
Integer types	0
Floating-point types	0.0
`bool`	`false`
`char`	`'\0'`
`string`	`null`
Reference types	`null`
`struct`	All the members of the `struct` are set to the same default values shown in this table.
`enum`	0

To illustrate how auto-properties can simplify code, compare the original version of the `Pet` class, containing 25 lines of code, with the version shown here, containing only five lines of code.

```
1 class Pet
2 {
3    // Name property
4    public string Name { get; set; } = "";
5 }
```

Read-Only Auto-Properties

If you leave out the `set` keyword in an auto-property, the property becomes read-only. Here is an example of how we can make the `Pet` class's Name property read-only:

```
1 class Pet
2 {
3     // Name property
4     public string Name { get; } = "Fido";
5 }
```

In this example, the Name property is initialized with the value "Fido". When we create an instance of this class, we can get the Name property's value, but we cannot set its value.

There are only two places where the compiler will allow you to assign a value to a read-only auto-property:

- In the property's declaration, as an initialization value
- In the class's constructor

Here is an example of the `Pet` class, showing the Name property being assigned a value in the constructor:

```
 1 class Pet
 2 {
 3     // Constructor
 4     public Pet()
 5     {
 6        Name = "Fido";
 7     }
 8
 9     // Name property
10     public string Name { get; }
11 }
```

Using Private `set` Accessors with Auto-Properties

A read-only auto-property can only be assigned a value in its declaration, or in the class constructor. If you want any method in a class, other than the constructor, to change an auto-property's value, the property will need to have a private `set` accessor. Recall from our discussion earlier in this chapter that when a property has a private `set` accessor, methods in the same class as the property can change the property's value, but code outside the class cannot.

For example, look at the following version of the `Car` class. In this version, the Speed property is implemented as an auto-property with a private `set` accessor.

```
1 internal class Car
2 {
3     // Constructor
4     public Car()
5     {
```

```
 6          Speed = 0;
 7      }
 8
 9      // Speed property
10      public int Speed { get; private set; }
11
12      // Accelerate method
13      public void Accelerate()
14      {
15          Speed += 10;
16      }
17
18      // Brake method
19      public void Brake()
20      {
21          Speed -= 10;
22      }
23  }
```

Notice in line 10 that the Speed property's `set` accessor is declared private. As a result, only the methods inside the `Car` class, such as the `Accelerate` and `Brake` methods, can change the value of the property. To code outside the class, the Speed property is read-only.

Tutorial 10-2 leads you through the process of creating a class with properties. You use the class in a simple application that creates an object of the class and tests its properties.

Tutorial 10-2:
Creating and Using the `CellPhone` Class

VideoNote

Tutorial 10-2: Creating and Using the `CellPhone` Class

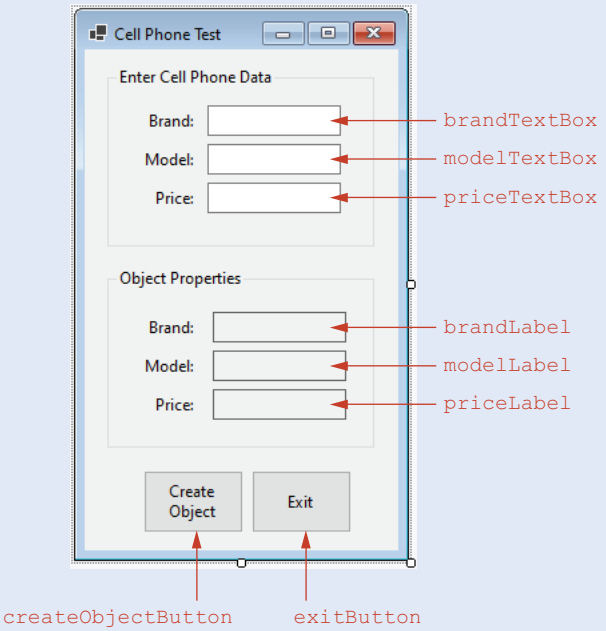
Suppose you work as a programmer for a company that sells cell phones and wireless service. Your department is creating an application to manage the company's inventory of cells phones. You have been asked to create a class that represents a cell phone. The class should have the following public properties:

- Brand—a `string` property that will store the phone's brand name
- Model—a `string` property that will store the phone's model
- Price—a `decimal` property that will store the phone's retail price

The class should also have a constructor that initializes the properties. The Brand and Model properties will be initialized with empty strings, and the Price property will be initialized with the value 0.

You will create the `CellPhone` class in a project that will create an object of the class and test the object's properties. The project is named *CellPhoneTest* and has already been started for you; it is located in the *Chap10* folder of the Student Sample Programs. Figure 10-15 shows the application's form. At run time, you enter sample cell phone data into the text boxes and then click the *Create Object* button. The application creates an object of the `CellPhone` class and assigns the data from the text boxes to the object's properties. It then reads and displays the values of the object's properties in the `brandLabel`, `modelLabel`, and `priceLabel` controls.

Figure 10-15 The *CellPhoneTest* project's form



Step 1: Start Visual Studio. Open the project named *CellPhoneTest* in the *Chap10* folder of the Student Sample Programs.

Step 2: Click *Project* on the Visual Studio menu bar and then select *Add Class. . . .* The *Add New Item* window should appear. Make sure *Class* is selected as the type of item. Change the default filename to *CellPhone.cs* and then click the *Add* button. This adds a source code file named CellPhone.cs to the project.

Step 3: The CellPhone.cs file should now be displayed in the code editor. Complete the code for the `CellPhone` class by typing lines 11–26 in Program 10-3.

Step 4: Open the Form1.cs file in the code editor. Type the comments and code for the `GetPhoneData` method, shown in lines 10–34 of Program 10-4. Notice in line 13 that the method has a `CellPhone` parameter named `phone`. When we call this method, we pass a `CellPhone` object to it as an argument.

The purpose of the `GetPhoneData` method is to read the data that the user has entered into the form's text boxes and store that data in the `CellPhone` object's properties. Let's review this code:

Line 16: This statement declares a local `decimal` variable named `price`. This variable temporarily holds the value that the user entered for the phone's price while we perform input validation on that value.

Line 19: This statement assigns the `brandTextBox` control's Text property to the `phone` object's Brand property.

Line 22: This statement assigns the `modelTextBox` control's Text property to the `phone` object's Model property.

Lines 25–33: The `if` statement in line 25 calls the `decimal.TryParse` method to convert the `priceTextBox` control's Text property to a `decimal`, and it stores the result in the `price` variable. If the conversion is successful, line 27 assigns the `price` variable to the `phone` object's Price property. If the conversion is not successful, line 32 displays an error message.

You might be wondering why we need the `price` variable in this code. Why not simply pass the `phone.Price` property as the second argument to the `decimal.TryParse` method in line 25? The reason is that the second argument passed to the `decimal.TryParse` method is passed to an `out` parameter. Recall that you cannot pass a property to a `ref` or an `out` parameter. So, we pass the `price` variable, and if the conversion is successful, we assign the `price` variable to the `phone.Price` property.

Step 5: Next, you create the Click event handlers for the Button controls. Switch your view to the Form1 form in the Designer. Double-click the `createObjectButton` control. This opens the Form1.cs file in the code editor, and you will see an empty event handler named `createObjectButton_Click`. Complete the event handler by typing the code shown in lines 38–47 in Program 10-4. Let's review this code:

Line 39: This statement creates a `CellPhone` object in memory, referenced by a variable named `myPhone`.

Line 42: This statement calls the `GetPhoneData` method, passing the `myPhone` object as an argument.

Lines 45–47: These statements display the values of the `myPhone` object's Brand, Model, and Price properties in the `brandLabel`, `modelLabel`, and `priceLabel` controls.

Step 6: Switch your view back to the Form1 form in the *Designer* and double-click the `exitButton` control. In the code editor you will see an empty event handler named `exitButton_Click`. Complete the event handler by typing the code shown in lines 52–53 in Program 10-4.

Step 7: Save the project. Then, press F5 on the keyboard or click the *Start Debugging* button (▶) on the toolbar to compile and run the application. When the application runs, enter some sample data in the TextBox controls and click the *Create Object* button. You should see the values that you entered displayed in the `brandLabel`, `modelLabel`, and `priceLabel` controls. Click the *Exit* button when you are finished.

Program 10-3 Completed code for the CellPhone.cs file in the *CellPhoneTest* project

```
 1 using System;
 2 using System.Collections.Generic;
 3 using System.Linq;
 4 using System.Text;
 5 using System.Threading.Tasks;
 6
 7 namespace CellPhoneTest
 8 {
 9     class CellPhone
10     {
11         // Constructor
12         public CellPhone()
13         {
14             Brand = "";
15             Model = "";
16             Price = 0m;
17         }
18
```

```
        // Brand property
        public string Brand { get; set; }

        // Model property
        public string Model { get; set; }

        // Price property
        public decimal Price { get; set; }
    }
}
```

Program 10-4 Completed code for Form1 in the *CellPhoneTest* project

```
namespace CellPhoneTest
{
    public partial class Form1 : Form
    {
        public Form1()
        {
            InitializeComponent();
        }

        // The GetPhoneData method accepts a CellPhone object
        // as an argument. It assigns the data entered by the
        // user to the object's properties.
        private void GetPhoneData(CellPhone phone)
        {
            // Temporary variable to hold the price.
            decimal price;

            // Get the phone's brand.
            phone.Brand = brandTextBox.Text;

            // Get the phone's model.
            phone.Model = modelTextBox.Text;

            // Get the phone's price.
            if (decimal.TryParse(priceTextBox.Text, out price))
            {
                phone.Price = price;
            }
            else
            {
                // Display an error message.
                MessageBox.Show("Invalid price");
            }
        }

        private void createObjectButton_Click(object sender,
        EventArgs e)
        {
            // Create a CellPhone object.
            var myPhone = new CellPhone();

            // Get the phone data.
            GetPhoneData(myPhone);

            // Display the phone data.
            brandLabel.Text = myPhone.Brand;
```

```
46              modelLabel.Text = myPhone.Model;
47              priceLabel.Text = myPhone.Price.ToString("c");
48          }
49
50          private void exitButton_Click(object sender,
            EventArgs e)
51          {
52              // Close the form.
53              this.Close();
54          }
55      }
56  }
```

Checkpoint

10.11 Rewrite the following property as an auto-property:

```
private int _id;
private int ID
{
     get
     {
         return _id;
     }
     set
     {
         id = value;
     }
}
```

10.12 How do you declare a read-only auto-property?

10.5 Parameterized Constructors and Method Overloading

CONCEPT: **A constructor that accepts arguments is known as a parameterized constructor. A class can have multiple versions of the same method, which are known as overloaded methods.**

Constructors can accept arguments in the same way as other methods. A constructor that accepts arguments is known as a **parameterized constructor** because it has parameter variables. When a class has a parameterized constructor, you can pass initialization values to the constructor when you create an object.

For example, look at the following `Pet` class:

```
1 class Pet
2 {
3     // Constructor
4     public Pet(string petName)
5     {
6         Name = petName;
7     }
```

```
 8
 9      // Name property
10      public string Name { get; set; }
11  }
```

Notice in line 4 that the constructor has a `string` parameter named `petName`, which is assigned to the Name property in line 6. When we create an instance of this class, we need to pass an argument to the constructor as shown here:

```
var myPet = new Pet("Whiskers");
```

This statement passes the string `"Whiskers"` as an argument to the `Pet` class constructor. A `Pet` object will be created, and its Name property will be set to `"Whiskers"`.

In Tutorial 10-3, you write a class that simulates a bank account. The class's constructor will accept an argument that specifies the account's starting balance.

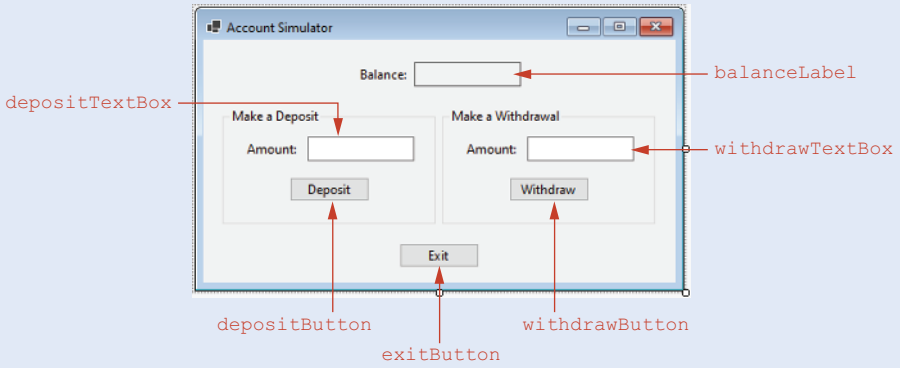
Tutorial 10-3:
Creating and Using the `BankAccount` Class

VideoNote
Tutorial 10-3: Creating and Using the `BankAccount` Class

In this tutorial, you write a class named `BankAccount` that simulates a bank account. When you create an instance of the class, you pass the account's starting balance as an argument to the constructor. The class will have a `Deposit` method that adds an amount to the balance and a `Withdraw` method that subtracts an amount from the balance. The class will also have a read-only Balance property that reports the account's balance.

The `BankAccount` class is part of the *AccountSimulator* project, which has already been started for you. Figure 10-16 shows the application's form. When the completed application runs, it creates a `BankAccount` object with a starting balance of $1,000.00. The balance is displayed in the `balanceLabel` control. If you want to make a deposit, you can enter the amount into the `depositTextBox` and click the *Deposit* button. If you want to make a withdrawal, you can enter the amount into the `withdrawTextBox` and click the *Withdraw* button. Each time you perform one of these actions, the new account balance is displayed.

Figure 10-16 The *AccountSimulator* project's form



Step 1: Start Visual Studio. Open the project named *AccountSimulator* in the *Chap10* folder of the Student Sample Programs.

Step 2: Click *Project* on the Visual Studio menu bar and then select *Add Class*. . . . The *Add New Item* window should appear. Make sure *Class* is selected as the type of item. Change the default filename to *BankAccount.cs* and then click the *Add* button. This adds a source code file named BankAccount.cs to the project.

Step 3: The BankAccount.cs file should now be displayed in the code editor. Complete the code for the `BankAccount` class by typing lines 11–30 in Program 10-5. Let's take a closer look at the code:

Lines 12–15: These lines are the parameterized class constructor. When an instance of the class is created, an argument specifying the account's starting balance is passed into the `startingBalance` parameter. In line 14, the `startingBalance` parameter is assigned to the Balance property.

Line 18: This is the declaration of the Balance property. Notice that the property has a private `set` accessor. This means the property may be set by methods in the `BankAccount` class, but is read-only to code outside the class.

Lines 21–24: This is the `Deposit` method, which accepts a `decimal` argument and adds the argument to the Balance property.

Lines 27–30: This is the `Withdraw` method, which accepts a `decimal` argument, and subtracts the argument from the Balance property.

Step 4: Open the Form1.cs file in the code editor. Type the comment and declaration shown in lines 5 and 6 of Program 10-6. The statement in line 6 does the following:

- It creates an object of the `BankAccount` class, passing the value 1000 as an argument to the constructor.
- It assigns a reference to the object to a field named `account`.

Because the `account` variable is declared as a field, all the methods in the `Form1` class have access to it.

Step 5: Next, you create a Load event handler for the form. The Load event handler displays the account balance in the `balanceLabel` control. Switch your view to the Form1 form in the *Designer* and double-click any part of the form that does not contain a control. This opens the code editor, and you will see an empty event handler named `Form1_Load`. Complete the event handler by typing the code shown in lines 15 and 16 in Program 10-6. Notice that line 16 uses the `ToString` method to format the `account` object's Balance property as currency and assigns the result to the `balanceLabel` control's Text property.

Step 6: Now you create the Click event handlers for the Button controls. Switch your view back to the Form1 form in the *Designer* and double-click the `depositButton` control. This opens the code editor, and you will see an empty event handler named `depositButton_Click`. Complete the event handler by typing the code shown in lines 21–39 in Program 10-6. Let's look at the code:

Line 21: This statement declares a local `decimal` variable named `amount`. This holds the amount that the user wishes to deposit.

Lines 24–39: The `if` statement in line 24 calls the `decimal.TryParse` method to convert the `depositTextBox` control's Text property to a `decimal`, and it stores the result in the `amount` variable. If the conversion is successful, the following actions take place:

- Line 27 calls the `account` object's `Deposit` method, passing `amount` as an argument.
- Line 30 displays the account balance, formatted as currency.
- Line 33 clears the `depositTextBox` control.

If the conversion is not successful, line 38 displays an error message.

Step 7: Switch your view back to the Form1 form in the *Designer* and double-click the `withdrawButton` control. This opens the code editor, and you will see an empty event handler named `withdrawButton_Click`. Complete the event handler by typing the code shown in lines 44–62 in Program 10-6. Let's look at the code:

Line 44: This statement declares a local `decimal` variable named `amount`. This holds the amount that the user wishes to withdraw.

Lines 47–62: The `if` statement in line 47 calls the `decimal.TryParse` method to convert the `withdrawTextBox` control's Text property to a `decimal`, and it stores the result in the `amount` variable. If the conversion is successful, the following actions take place:

- Line 50 calls the `account` object's `Withdraw` method, passing `amount` as an argument.
- Line 53 displays the account balance, formatted as currency.
- Line 56 clears the `withdrawTextBox` control.

If the conversion is not successful, line 61 displays an error message.

Step 8: Switch your view back to the Form1 form in the *Designer* and double-click the `exitButton` control. In the code editor, you will see an empty event handler named `exitButton_Click`. Complete the event handler by typing the code shown in lines 67–68 in Program 10-6.

Step 9: Save the project. Then, press F5 on the keyboard or click the *Start Debugging* button (▶) on the toolbar to compile and run the application. When the application runs, experiment by depositing and withdrawing various amounts. Click the *Exit* button when you are finished.

Program 10-5 Completed code for the BankAccount.cs file in the *AccountSimulator* project

```
 1 using System;
 2 using System.Collections.Generic;
 3 using System.Linq;
 4 using System.Text;
 5 using System.Threading.Tasks;
 6
 7 namespace AccountSimulator
 8 {
 9     internal class BankAccount
10     {
11         // Constructor
12         public BankAccount(decimal startingBalance)
13         {
14             Balance = startingBalance;
15         }
16
17         // Balance property
18         public decimal Balance { get; private set; }
19
20         // Deposit method
21         public void Deposit(decimal amount)
22         {
23             Balance += amount;
24         }
25
```

```
26          // Withdraw method
27          public void Withdraw(decimal amount)
28          {
29              Balance -= amount;
30          }
31      }
32  }
```

Program 10-6 Completed code for Form1 in the *AccountSimulator* project

```
 1  namespace AccountSimulator
 2  {
 3      public partial class Form1 : Form
 4      {
 5          // BankAccount field with a $1000 starting balance
 6          private BankAccount account = new BankAccount(1000);
 7
 8          public Form1()
 9          {
10              InitializeComponent();
11          }
12
13          private void Form1_Load(object sender, EventArgs e)
14          {
15              // Display the starting balance.
16              balanceLabel.Text = account.Balance.ToString("c");
17          }
18
19          private void depositButton_Click(object sender,
            EventArgs e)
20          {
21              decimal amount; // To hold the amount of deposit
22
23              // Convert the amount to a decimal.
24              if (decimal.TryParse(depositTextBox.Text, out
                amount))
25              {
26                  // Deposit the amount into the account.
27                  account.Deposit(amount);
28
29                  // Display the new balance.
30                  balanceLabel.Text = account.Balance.
                    ToString("c");
31
32                  // Clear the text box.
33                  depositTextBox.Clear();
34              }
35              else
36              {
37                  // Display an error message.
38                  MessageBox.Show("Invalid amount");
39              }
40          }
41
42          private void withdrawButton_Click(object sender,
            EventArgs e)
43          {
44              decimal amount; // To hold the amount of withdrawal
45
```

```
46              // Convert the amount to a decimal.
47              if (decimal.TryParse(withdrawTextBox.Text, out
                amount))
48              {
49                  // Withdraw the amount from the account.
50                  account.Withdraw(amount);
51
52                  // Display the new balance.
53                  balanceLabel.Text = account.Balance.
                    ToString("c");
54
55                  // Clear the text box.
56                  withdrawTextBox.Clear();
57              }
58              else
59              {
60                  // Display an error message.
61                  MessageBox.Show("Invalid amount");
62              }
63          }
64
65          private void exitButton_Click(object sender, EventArgs e)
66          {
67              // Close the form.
68              this.Close();
69          }
70      }
71  }
```

Overloaded Methods

Sometimes you need different ways to perform the same operation. For example, the `BankAccount` class that you wrote in Tutorial 10-3 has a `Deposit` method that accepts a `decimal` argument. Suppose you are using the class in an application that needs to pass a `double` variable to the method instead of a `decimal` variable. In this application, before you call the `Deposit` method, you have to convert the `double` variable to a `decimal` so it can be passed to the method. It would be nice if the `Deposit` method could accept either a `decimal` argument or a `double` argument. Then, it would be unnecessary to convert a `double` variable before passing it to the method. This can be accomplished with overloading.

When a method is **overloaded**, it means that multiple methods in the same class have the same name but use different types of parameters. Here is an example of how we might overload the `Deposit` method inside the `BankAccount` class:

```
1  // Deposit method
2  public void Deposit(decimal amount)
3  {
4      Balance += amount;
5  }
6
7  // Deposit method
8  public void Deposit(double amount)
9  {
10     Balance += (decimal)amount;
11 }
```

The first `Deposit` method (in lines 2–5) accepts a `decimal` argument, which is added to the Balance property field in line 4. The second `Deposit` method (in lines 8–11) accepts a `double` argument. Line 10 uses a cast operator to convert its value to a `decimal`, and assigns the result to the Balance property.

When you write a call to the `Deposit` method, the compiler must determine which one of the overloaded methods you intended to call. The process of matching a method call with the correct method is known as **binding**. When an overloaded method is called, the compiler uses the method's name and parameter list to determine to which method to bind the call. If we call the `Deposit` method and pass a `decimal` argument, the version of the method that has a `decimal` parameter is called. Likewise, if we call the `Deposit` method and pass a `double` argument, the version of the method that has a `double` parameter is called.

The compiler uses a method's signature to distinguish it from other methods of the same name. A method's **signature** consists of the method's name and the data type and argument kind (by value, `ref`, or `out`) of the method's parameters, from left to right. For example, here are the signatures of the `Deposit` methods that we previously showed:

```
Deposit(decimal)
Deposit(double)
```

Note that the method's return type is *not* part of the signature. For this reason, you cannot overload methods by giving them different return types.

Overloaded Constructors

Constructors can also be overloaded, which means that a class can have more than one constructor. The rules for overloading constructors are the same for overloading other methods: Each version of the constructor must have a different parameter list. As long as each constructor has a unique signature, the compiler can tell them apart.

Here is an example of how we might overload the `BankAccount` class constructor:

```
 1 // Constructor
 2 public BankAccount()
 3 {
 4     Balance = 0;
 5 }
 6
 7 // Constructor
 8 public BankAccount(decimal startingBalance)
 9 {
10     Balance = startingBalance;
11 }
```

The first constructor (lines 2–5) is a **parameterless constructor**, which means it accepts no arguments. (Parameterless constructors are also known as **no-arg constructors**.) The following statement calls the parameterless constructor as it creates a `BankAccount` object:

```
var account = new BankAccount();
```

The second constructor (lines 8–11) accepts a `decimal` argument. The following statement calls this constructor as it creates a `BankAccount` object:

```
var account = new BankAccount(500m);
```

The following statement also calls the second constructor (lines 8–11) because an integer value can be implicitly converted to a `decimal`:

```
var account = new BankAccount(500);
```

The following code will cause an error, however, because there is no constructor that accepts a `double` argument:

```
// Error
var account = new BankAccount(500.0);
```

Default Constructors

It is perfectly legal to write a class without any constructors. If you write a class with no constructor whatsoever, the compiler provides a **default constructor** that takes no arguments. It is equivalent to a parameterless constructor that has no statements in its body. For example, suppose we have written the following class:

```
class Circle
{
    public int CenterX { get; set; }
    public int CenterY { get; set; }
}
```

The compiler will provide a default constructor that makes this class equivalent to:

```
class Circle
{
    public Circle()
    {
    }

    public int CenterX { get; set; }
    public int CenterY { get; set; }
}
```

Checkpoint

10.13 What is a parameterized constructor?

10.14 What does it mean when a method is overloaded?

10.15 What is the process of matching a method call with the correct method called?

10.16 What happens if you write a class with no constructor whatsoever?

10.17 Describe the purpose of the default constructor.

10.6 Storing Class Type Objects in Arrays and `Lists`

CONCEPT: **You can store a collection of class type objects in an array or a `List`.**

Arrays of Class Type Objects

Objects that are instances of a class can be stored in an array. For example, assume the `CellPhone` class previously shown exists in an application. The following code creates a `CellPhone` array with four elements:

```
const int SIZE = 4;
CellPhone[] phones = new CellPhone[SIZE];
```

Alternatively, if you are declaring the array inside a method, you can use the `var` keyword as shown here:

```
const int SIZE = 4;
var phones = new CellPhone[SIZE];
```

Although this code creates an array, it does not yet contain any objects. When you create an array of a class type, each element of the array is a reference variable. By default, each element will be initialized with the value `null`. The next step is to create the objects that each element will reference. This can be done one element at a time, as shown here:

```
phones[0] = new CellPhone();
phones[1] = new CellPhone();
phones[2] = new CellPhone();
phones[3] = new CellPhone();
```

Or, it can be done with a loop, as shown here:

```
for (int index = 0; index < phones.Length; index++)
{
   phones[index] = new CellPhone();
}
```

Alternatively, you can initialize the array elements in the declaration statement:

```
CellPhone[] phones =
{
    new CellPhone(), new CellPhone(),
    new CellPhone(), new CellPhone()
};
```

The following shows another example. This code creates a `BankAccount` array and initializes its elements with references to four `BankAccount` objects.

```
BankAccount[] accounts =
{
    new BankAccount(1000),
    new BankAccount(2000),
    new BankAccount(3000),
    new BankAccount(4000)
};
```

Notice the arguments that are passed to the `BankAccount` constructor for each object. If the following code were executed, it would display "The balance is 1000", "The balance is 2000", and so forth:

```
for (int index = 0; index < accounts.Length; index++)
{
   MessageBox.Show($"The balance is {accounts[index].Balance}");
}
```

The following code performs the same operation using a `foreach` loop:

```
foreach (var acct in accounts)
{
   MessageBox.Show($"The balance is {acct[index].Balance}");
}
```

Arrays Hold References to Class Type Objects

Keep in mind that when you store class type objects in an array, the array holds references to those objects. For example, assume the following `Point` class exists in an application:

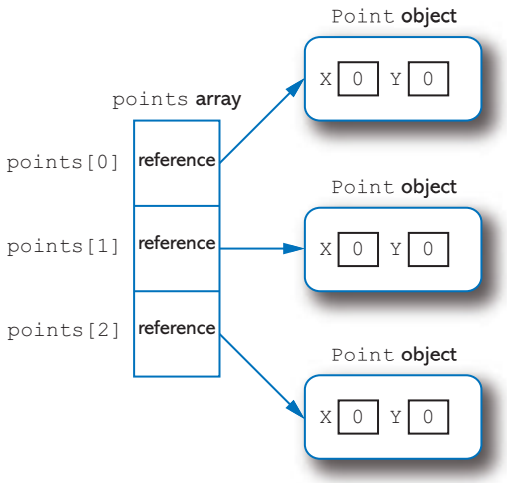
```
class Point
{
    // Constructor
```

```
        public Point()
        {
            X = 0;
            Y = 0;
        }

        // Properties
        public int X { get; set; }
        public int Y { get; set; }
    }
```

The following code creates an array of three `Point` objects. As shown in Figure 10-17, each element of the array is a reference to an object.

```
const int SIZE = 3;
var points = new Point[SIZE];
for (int index = 0; index < points.Length; index++)
{
     points[index] = new Point();
}
```

Figure 10-17 An array of objects



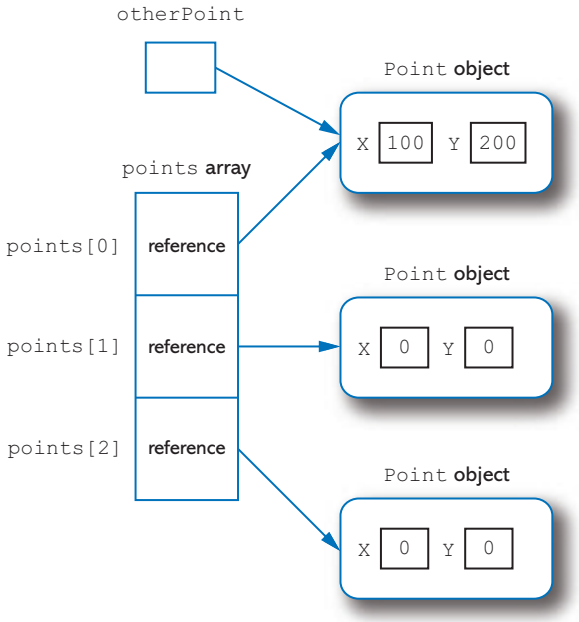
Because the array elements are references to objects, you must be aware that other references to the same objects might exist. For example, look at the following code:

```
var otherPoint = points[0];
otherPoint.X = 100
otherPoint.Y = 200
```

This code assigns `points[0]` to a variable named `otherPoint`. As a result, `points[0]` and `otherPoint` will reference the same object in memory. When the X and Y properties of `otherPoint` are changed, they are also changed for `points[0]`. This is illustrated in Figure 10-18.

As you can see from this example, it is possible to write code that inadvertently modifies an object in an array because it is using an extraneous reference to the same object. In Chapter 11, you will learn about immutability, and how to design classes so this type of accidental data modification can be prevented.

Figure 10-18 points[0] and otherPoint reference the same object



Lists of Class Type Objects

In Chapter 7, we discussed the List class, which is a container for storing a collection of objects. Here is an example of how you would create a List that can hold CellPhone objects:

```
List<CellPhone> phoneList = new List<CellPhone>();
```

Alternatively, if you are declaring the List inside a method, you can use the var keyword as shown here:

```
var phoneList = new List<CellPhone>();
```

This statement creates a List object, referenced by the phoneList variable. Notice that the word CellPhone is written inside angled brackets, <>, immediately after the word List. This specifies that the List can hold only objects of the CellPhone class type.

To add an object to a List, you use the Add method. The following code shows an example:

```
 1 // Create a List to hold CellPhone objects.
 2 var phoneList = new List<CellPhone>();
 3
 4 // Create an instance of the CellPhone class.
 5 var myPhone = new CellPhone();
 6
 7 // Assign values to the object's properties.
 8 myPhone.Brand = "Acme Electronics";
 9 myPhone.Model = "M1000";
10 myPhone.Price = 199;
11
12 // Add the object to the List.
13 phoneList.Add(myPhone);
```

The statement in line 2 creates a List named phoneList that can hold CellPhone objects. Line 5 creates an instance of the CellPhone class, referenced by the myPhone variable. Lines 8–10 assign values to the object's properties. Line 13 adds the object to the List.

`List`s Hold References to Class Type Objects

Keep in mind that when you store class type objects in a `List`, the `List` holds references to those objects. Because the `List` elements are references to objects, you must be aware that other references to the same objects might exist elsewhere in your application. This makes it possible to write code that inadvertently modifies an object in a `List`. In Chapter 11, you will learn about immutability, and how to design classes so this type of accidental data modification can be prevented.

In Tutorial 10-4, you complete an application that uses a `List` to hold a collection of `CellPhone` objects.

Tutorial 10-4:
Completing the *CellPhoneInventory* Project

VideoNote
Tutorial 10-4: Completing the *CellPhone-Inventory* Project

In this tutorial, you complete the *CellPhoneInventory* project. Figure 10-19 shows the application's form, which has already been created for you. When you run the completed application, you can enter data about a cell phone into the TextBox controls. When you click the *Add Phone* button, that data is assigned to a `CellPhone` object's properties and then the object is added to a `List`. You can do this as many times as you wish. Each time you add a cell phone, its brand and model is displayed in the list box. If you select a phone in the list box, the application displays that phone's price.

Figure 10-19 The *CellPhoneInventory* project's form

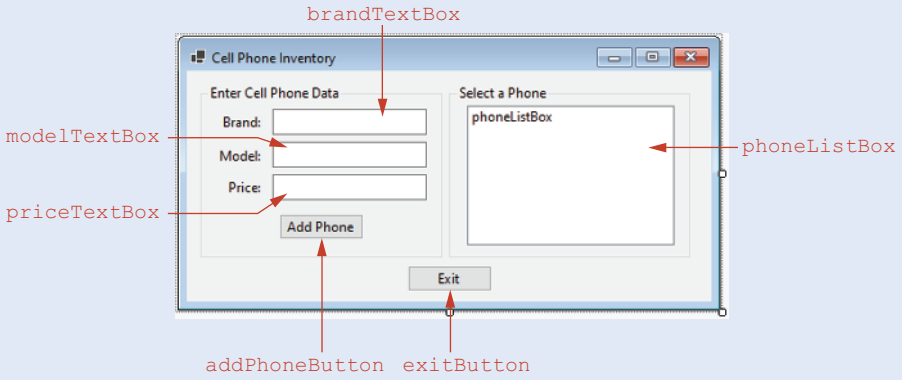
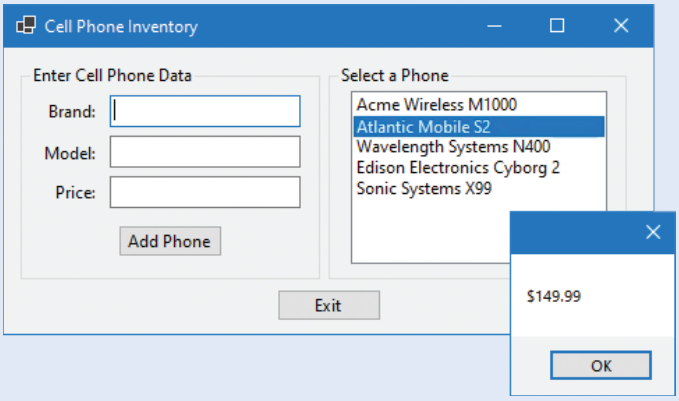


Figure 10-20 shows an example of the application at run time. In the figure, five cell phones have been added, and *Atlantic Mobile S2* has been selected in the list box. The selected phone's price is displayed in the message box.

Figure 10-20 The *CellPhoneInventory* project running



Step 1: Start Visual Studio. Open the project named *CellPhoneInventory* in the *Chap10* folder of the Student Sample Programs. Notice that the `CellPhone` class has already been added to the project for you in the file CellPhone.cs.

Step 2: Open the Form1.cs file in the code editor. Write the comment and the declaration for `phoneList`, which appears in lines 7 and 8 in Program 10-7. Notice that `phoneList` is declared as a field in the `Form1` class. It will be available to all the methods in the class.

Step 3: Type the comments and code for the `GetPhoneData` method, shown in lines 15–39 of Program 10-7. Notice that the method accepts a `CellPhone` object as an argument. The purpose of the `GetPhoneData` method is to read the data that the user has entered into the form's text boxes and store that data in the argument object's properties. (This method might look familiar. We used it in Tutorial 10-2.)

Step 4: Switch your view to the Form1 form in the *Designer* and double-click the `addPhoneButton` control. This opens the code editor, and you will see an empty event handler named `addPhoneButton_Click`. Complete the event handler by typing the code shown in lines 43–61 in Program 10-7. Let's look at the code:

Line 44: This statement creates a `CellPhone` object in memory, referenced by a variable named `myPhone`.

Line 47: This statement calls the `GetPhoneData` method, passing the `myPhone` object as an argument. After the method executes, the `myPhone` object's properties are set to the data entered by the user.

Line 50: This statement adds the `myPhone` object to the `phoneList`.

Line 53: This statement adds an entry to the `phoneListBox` control displaying the `myPhone` object's Brand and Model properties.

Lines 56–58: These statements clear the contents of the TextBox controls.

Line 61: This statement sets the focus to the `brandTextBox` control.

Note that each `CellPhone` object that is added to the `phoneList` has a corresponding item in the ListBox control. An object in the `phoneList` and its corresponding item in the ListBox share the same index. For example,

- The `CellPhone` object at index 0 in `phoneList` corresponds to the item at index 0 in the ListBox.
- The `CellPhone` object at index 1 in `phoneList` corresponds to the item at index 1 in the ListBox.

Recall from Chapter 8 that when the items in two data structures are related by their indexes, it is said that a parallel relationship exists between the data structures.

Step 5: Next, you create the SelectedIndexChange event handler for the `phoneListBox` control. Switch back to the *Designer* and double-click the `phoneListBox` control. This opens the code editor, and you will see an empty event handler named `phoneListBox_SelectedIndexChange`. Complete the event handler by typing the code shown in lines 66–70 in Program 10-7.

We mentioned in Step 4 that the items in the list box and the objects in the `phoneList` are related by their indexes. When the user selects an item from the list box, all we have to do is get that item's index and then use that same index to retrieve the corresponding object from the `phoneList`. That is exactly what happens in this event handler. Line 67 gets the index of the selected item in the

`phoneListBox` control and assigns it to the `index` variable. Line 70 uses the `index` variable to get an object from `phoneList` and display its Price property in a message box.

Step 6: Switch your view back to the *Designer* and double-click the `exitButton` control. In the code editor you will see an empty event handler named `exitButton_Click`.

Complete the event handler by typing the code shown in lines 75 and 76 in Program 10-7.

Step 7: Save the project. Then, press F5 on the keyboard or click the *Start Debugging* button (▶) on the toolbar to compile and run the application. When the application runs, enter some sample data in the TextBox controls and click the *Add Phone* button. Repeat this with several different sets of data. Each time you add a phone, you should see its brand and model displayed in the list box. After you have entered data for several phones, click the entries that appear in the list box to see each phone's price. Click the *Exit* button when you are finished.

Program 10-7 Completed code for Form1 in the *CellPhoneInventory* project

```
1  using System.Windows.Forms;
2
3  namespace CellPhoneInventory
4  {
5      public partial class Form1 : Form
6      {
7          // List to hold CellPhone objects
8          List<CellPhone> phoneList = new List<CellPhone>();
9
10         public Form1()
11         {
12             InitializeComponent();
13         }
14
15         // The GetPhoneData method accepts a CellPhone object
16         // as an argument. It assigns the data entered by the
17         // user to the object's properties.
18         private void GetPhoneData(CellPhone phone)
19         {
20             // Temporary variable to hold the price.
21             decimal price;
22
23             // Get the phone's brand.
24             phone.Brand = brandTextBox.Text;
25
26             // Get the phone's model.
27             phone.Model = modelTextBox.Text;
28
29             // Get the phone's price.
30             if (decimal.TryParse(priceTextBox.Text, out price))
31             {
32                 phone.Price = price;
33             }
34             else
35             {
36                 // Display an error message.
37                 MessageBox.Show("Invalid price");
```

```
38              }
39          }
40
41          private void addPhoneButton_Click(object sender,
            EventArgs e)
42          {
43              // Create a CellPhone object.
44              var myPhone = new CellPhone();
45
46              // Get the phone data.
47              GetPhoneData(myPhone);
48
49              // Add the CellPhone object to the List.
50              phoneList.Add(myPhone);
51
52              // Add an entry to the list box.
53              phoneListBox.Items.Add($"{myPhone.Brand} {myPhone.
                Model}");
54
55              // Clear the TextBox controls.
56              brandTextBox.Clear();
57              modelTextBox.Clear();
58              priceTextBox.Clear();
59
60              // Reset the focus.
61              brandTextBox.Focus();
62          }
63
64          private void phoneListBox_SelectedIndexChanged(object
            sender, EventArgs e)
65          {
66              // Get the index of the selected item.
67              int index = phoneListBox.SelectedIndex;
68
69              // Display the selected item's price.
70              MessageBox.Show($"{phoneList[index].Price:c}");
71          }
72
73          private void exitButton_Click(object sender, EventArgs e)
74          {
75              // Close the form.
76              this.Close();
77          }
78      }
79  }
```

Checkpoint

10.18 When you create an array of a class type, with what value will each element of the array be initialized?

10.19 How can you initialize an array of a class type with references to objects of the class?

10.20 How do you specify the class type that a `List` can hold?

10.7 Finding the Classes and Their Responsibilities in a Problem

When developing an object-oriented program, one of your first tasks is to identify the classes that you will need to create. Typically, your goal is to identify the different types of real-world objects that are present in the problem and then create classes for those types of objects within your application.

Over the years, software professionals have developed numerous techniques for finding the classes in a given problem. One simple and popular technique involves the following steps:

1. Get a written description of the problem domain.
2. Identify all the nouns (including pronouns and noun phrases) in the description. Each of these is a potential class.
3. Refine the list to include only the classes that are relevant to the problem.

Let's take a closer look at each of these steps.

Writing a Description of the Problem Domain

The **problem domain** is the set of real-world objects, parties, and major events related to the problem. If you adequately understand the nature of the problem you are trying to solve, you can write a description of the problem domain yourself. If you do not thoroughly understand the nature of the problem, you should have an expert write the description for you.

For example, suppose we are programming an application that the manager of Joe's Automotive Shop will use to print service quotes for customers. Here is a description that an expert, perhaps Joe himself, might have written:

> Joe's Automotive Shop services foreign cars and specializes in servicing cars made by Mercedes Benz, Porsche, and BMW. When a customer brings a car to the shop, the manager gets the customer's name, address, and telephone number. The manager then determines the make, model, and year of the car and gives the customer a service quote. The service quote shows the estimated parts charges, estimated labor charges, sales tax, and total estimated charges.

The problem domain description should include any of the following:

- Physical objects such vehicles, machines, or products
- Any role played by a person, such as manager, employee, customer, teacher, student, and so forth
- The results of a business event, such as a customer order or, in this case, a service quote
- Record-keeping items, such as customer histories and payroll records

Identify All of the Nouns

The next step is to identify all of the nouns and noun phrases. (If the description contains pronouns, include them, too.) Here's another look at the previous problem domain description. This time the nouns and noun phrases appear in bold.

> **Joe's Automotive Shop** services **foreign cars** and specializes in servicing **cars** made by **Mercedes Benz**, **Porsche**, and **BMW**. When a **customer** brings a **car** to the **shop**, the **manager** gets the **customer**'s **name**, **address**, and **telephone number**. The **manager** then determines the **make**, **model**, and **year** of the **car** and gives the **customer** a **service quote**. The **service quote** shows the **estimated parts charges**, **estimated labor charges**, **sales tax**, and **total estimated charges**.

Notice that some of the nouns are repeated. The following list shows all the nouns without duplicating any of them:

address
BMW
car
cars
customer
estimated labor charges
estimated parts charges
foreign cars
Joe's Automotive Shop
make
manager
Mercedes Benz
model
name
Porsche
sales tax
service quote
shop
telephone number
total estimated charges
year

Refining the List of Nouns

The nouns that appear in the problem description are merely candidates to become classes. It might not be necessary to make classes for them all. The next step is to refine the list to include only the classes that are necessary to solve the particular problem at hand. We look at the common reasons that a noun can be eliminated from the list of potential classes.

1. **Some of the nouns really mean the same thing.**
   In this example, the following sets of nouns refer to the same thing:
   - **car, cars**, and **foreign cars**

   These all refer to the general concept of a car.
   - **Joe's Automotive Shop** and **shop**

   Both of these refer to the company "Joe's Automotive Shop."

We can settle on a single class for each of these. In this example, we arbitrarily eliminate **cars** and **foreign cars** from the list and use the word **car**. Likewise, we eliminate **Joe's Automotive Shop** from the list and use the word **shop**. The updated list of potential classes is as follows:

address
BMW
car
cars
customer

Because **car, cars**, and **foreign cars** mean the same thing in this problem, we have eliminated **cars** and **foreign cars**. Also, because **Joe's Automotive Shop** and **shop** mean the same thing, we have eliminated **Joe's Automotive Shop**.

estimated labor charges
estimated parts charges
~~foreign cars~~
~~Joe's Automotive Shop~~
make
manager
Mercedes Benz
model
name
Porsche
sales tax
service quote
shop
telephone number
total estimated charges
year

2. **Some nouns might represent items that we do not need to be concerned with in order to solve the problem.**

   A quick review of the problem description reminds us of what our application shovuld do: print a service quote. In this example, we can eliminate two unnecessary classes from the list:

   - We can cross **shop** off the list because our application needs to be concerned only with individual service quotes. It does not need to work with or determine any companywide information. If the problem description asked us to keep a total of all the service quotes, then it would make sense to have a class for the shop.
   - We will not need a class for the **manager** because the problem statement does not direct us to process any information about the manager. If there were multiple shop managers, and the problem description had asked us to record which manager generated each service quote, then it would make sense to have a class for the manager.

The updated list of potential classes at this point is as follows:

address
BMW
~~car~~
~~cars~~
customer
estimated labor charges
estimated parts charges
foreign cars
~~Joe's Automotive Shop~~
make
manager
Mercedes Benz
model
name
Porsche

Our problem description does not direct us to process any information about the **shop** or any information about the **manager**, so we have eliminated those from the list.

sales tax
service quote
shop
telephone number
total estimated charges
year

3. **Some of the nouns might represent objects, not classes.**

   We can eliminate **Mercedes, Porsche,** and **BMW** as classes because, in this example, they all represent specific cars and can be considered instances of a **car** class. At this point, the updated list of potential classes is as follows:

   address
   ~~BMW~~
   car
   cars
   customer
   estimated labor charges
   estimated parts charges
   ~~foreign cars~~
   ~~Joe's Automotive Shop~~
   make
   ~~manager~~
   ~~Mercedes Benz~~
   model
   name
   ~~Porsche~~
   sales tax
   service quote
   ~~shop~~
   telephone number
   total estimated charges
   year

   We have eliminated Mercedes Benz, Porsche, and BMW, because they are all instances of a car class. That means that these nouns identify objects, not classes.

4. **Some of the nouns might represent simple values that can be assigned to a variable and do not require a class.**

   A class contains data attributes and methods. Data attributes are the fields and properties that define the object's state. Methods are actions or behaviors that can be performed by an object of the class. If a noun represents a type of item that would not have any identifiable data attributes or methods, then it can probably be eliminated from the list. To help determine whether a noun represents an item that would have data attributes and methods, ask the following questions about it:

   - Would you use a group of related values to represent the item's state?
   - Are there any obvious actions to be performed by the item?

   If the answers to both of these questions are no, then the noun probably represents a value that can be stored in a simple variable. If we apply this test to each noun that remains in our list, we can conclude that the following are probably not classes: **address, estimated labor charges, estimated parts charges, make, model, name, sales tax, telephone number,**

total estimated charges, and **year**. These are all simple string or numeric values that can be stored in variables. Here is the updated list of potential classes:

~~address~~
~~BMW~~
car
~~cars~~
customer
~~estimated labor charges~~
~~estimated parts charges~~
~~foreign cars~~

We have eliminated address, estimated labor charges, estimated parts charges, make, model, name, sales tax, telephone number, total estimated charges, and year as classes because they represent simple values that can be stored in variables.

NOTE: Some object-oriented designers take note of whether a noun is plural or singular. Sometimes a plural noun will indicate a class and a singular noun will indicate an object.

~~Joe's Automotive Shop~~
~~make~~
~~manager~~
~~Mercedes Benz~~
~~model~~
~~name~~
~~Porsche~~
~~sales tax~~
service quote
~~shop~~
~~telephone number~~
~~total estimated charges~~
~~year~~

As you can see from the list, we have eliminated everything except **car, customer**, and **service quote**. This means that in our application, we need classes to represent cars, customers, and service quotes. Ultimately, we will write a `Car` class, a `Customer` class, and a `ServiceQuote` class.

Identifying a Class's Responsibilities

Once the classes have been identified, the next task is to identify each class's responsibilities. A class's *responsibilities* are as follows:

- The things that the class is responsible for knowing
- The actions that the class is responsible for doing

When you have identified the things that a class is responsible for knowing, then you have identified the class's fields and/or properties. Likewise, when you have identified the actions that a class is responsible for doing, you have identified its methods.

It is often helpful to ask these questions: In the context of this problem, what must the class know? What must the class do? The first place to look for the answers is in the description of the problem domain. Many of the things that a class must know and do will be mentioned. Some class responsibilities, however, might not be directly mentioned

in the problem domain, so brainstorming is often required. Let's apply this methodology to the classes we previously identified from our problem domain.

The `Customer` class

In the context of our problem domain, what must the `Customer` class know? The description directly mentions the following items:

- The customer's name
- The customer's address
- The customer's telephone number

These are all values that can be represented as strings and stored as properties. The `Customer` class can potentially know many other things. One mistake that can be made at this point is to identify too many things that an object is responsible for knowing. In some applications, a `Customer` class might know the customer's e-mail address. This particular problem domain does not mention that the customer's e-mail address is used for any purpose, so we should not include it as a responsibility.

Now let's identify the class's methods. In the context of our problem domain, what must the `Customer` class do? The only obvious actions are these:

- Create and initialize an object of the `Customer` class.
- Get and set the customer's name.
- Get and set the customer's address.
- Get and set the customer's telephone number.

From this list we can see that the `Customer` class will have a constructor as well as fields and properties for the data attributes. The following code shows how the `Customer` class might be written:

```
 1 class Customer
 2 {
 3     // Constructor
 4     public Customer(string name, string address, string phone)
 5     {
 6         Name = name;
 7         Address = address;
 8         Phone = phone;
 9     }
10
11     // Properties
12     public string Name { get; set; }
13     public string Address { get; set; }
14     public string Phone { get; set; }
15 }
```

The `Car` Class

In the context of our problem domain, what must an object of the `Car` class know? The following items are all data attributes of a car and are mentioned in the problem domain:

- The car's make
- The car's model
- The car's year

Now let's identify the class's methods. In the context of our problem domain, what must the `Car` class do? Once again, the only obvious actions are the standard set of methods that we will find in most classes. Specifically, the actions are these:

- Create and initialize an object of the `Car` class.
- Get and set the car's make.

- Get and set the car's model.
- Get and set the car's year.

The following code shows how the `Car` class might be written:

```
 1 class Car
 2 {
 3     // Constructor
 4     public Car(string make, string model, int year)
 5     {
 6         Make = make;
 7         Model = model;
 8         Year = year;
 9     }
10
11     // Properties
12     public string Make { get; set; }
13     public string Model { get; set; }
14     public int Year { get; set; }
15 }
```

The `ServiceQuote` Class

In the context of our problem domain, what must an object of the `ServiceQuote` class know? The problem domain mentions the following items:

- The estimated parts charges
- The estimated labor charges
- The sales tax
- The total estimated charges

Careful thought and a little brainstorming reveals that two of these items are the results of calculations: sales tax and total estimated charges. Furthermore, to calculate the sales tax, the class must also know the sales tax rate.

Now let's identify the class's methods. In the context of our problem domain, what must the `ServiceQuote` class do? Once again, the only obvious actions are the standard set of methods that we will find in most classes. Specifically, the actions are these:

- Create and initialize an object of the `ServiceQuote` class.
- Get and set the estimated parts charges.
- Get and set the estimated labor charges.
- Get and set the sales tax rate.
- Get the sales tax.
- Get the total estimated charges.

The following code shows how the `ServiceQuote` class might be written:

```
 1 internal class ServiceQuote
 2 {
 3     // Constructor
 4     public ServiceQuote(decimal pcharges, decimal lcharges,
                           decimal txrate)
 5     {
 6         PartsCharges = pcharges;
 7         LaborCharges = lcharges;
 8         TaxRate = txrate;
 9     }
10
11     // Properties
12     public decimal PartsCharges { get; set; }
13     public decimal LaborCharges { get; set; }
14     public decimal TaxRate { get; set; }
```

```
15
16      // Read-only properties
17      public decimal SalesTax
18      {
19          get { return PartsCharges * TaxRate; }
20      }
21
22      public decimal TotalCharges
23      {
24          get
25          {
26              return PartsCharges + LaborCharges +
27                     (PartsCharges * TaxRate);
28          }
29      }
30  }
```

This Is Only the Beginning

You should look at the process that we have discussed in this section merely as a starting point. It's important to realize that designing an object-oriented application is an iterative process. It may take you several attempts to identify all the classes that you will need and determine all their responsibilities. As the design process unfolds, you will gain a deeper understanding of the problem, and consequently you will see ways to improve the design.

Checkpoint

10.21 What is a problem domain?

10.22 When designing an object-oriented application, who should write a description of the problem domain?

10.23 How do you identify the potential classes in a problem domain description?

10.24 What are a class's responsibilities?

10.25 What two questions should you ask to determine a class's responsibilities?

10.26 Will all a class's actions always be directly mentioned in the problem domain description?

10.8 Application of Classes: Creating Multiple Forms in a Project

CONCEPT: **A Visual C# project can have multiple forms. Each form has its own class that can be instantiated and displayed on the screen.**

The applications you have created so far have used only one form, named Form1. The Form1 form is displayed when the application runs, and when the Form1 form closes, the application ends. You are not limited to one form in a project, however. You may create multiple forms in a project to use as dialog boxes, to display error messages, and so forth. Then you can display these forms as they are needed.

Every form in a Visual C# project has a class. For example, if a project has a form named Form1, then the project has a class named `Form1`, which is stored in a file named Form1.cs. When you add additional forms to a project, you add additional classes, which are stored in their own files. When you create event handlers for a specific form's controls, you write them as methods in that form's class.

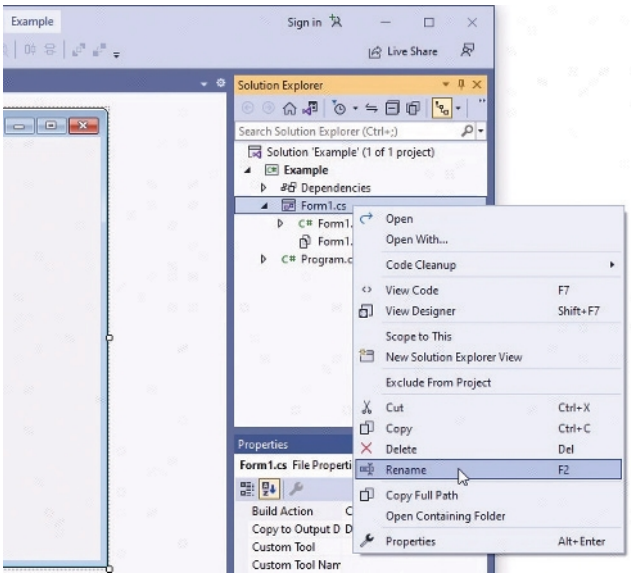
Renaming the Form1 Form

When you add forms to a Visual C# project, they are given default names such as Form1, Form2, and so forth. If you have only one form in a project, there is no compelling reason to change the form's name. However, when you have multiple forms in a project, you should give each form a meaningful name that describes its purpose.

Before you add a new form to a project, it is a good idea to change the name of the Form1 form. In this book, we always change the name of the Form1 form to MainForm because it is usually the main form in an application. To change the form's name, you use the *Solution Explorer* to change the name of the Form1.cs file to MainForm.cs. When you do this, Visual Studio automatically changes the name of the Form1 form to MainForm. Here is the procedure to follow:

1. Right-click the Form1.cs entry that appears in the *Solution Explorer*. The pop-up menu shown in Figure 10-21 should appear.

Figure 10-21 Right-clicking the form file in the *Solution Explorer*



2. Select *Rename* from the pop-up menu.
3. In the *Solution Explorer*, the form's filename should become highlighted. Type the new name, which is *MainForm.cs*, and press Enter. (Be sure to type the .cs extension.)
4. The dialog box shown in Figure 10-22 will appear next. Click *Yes* to rename the form.

Figure 10-22 Clicking *Yes* to rename the form

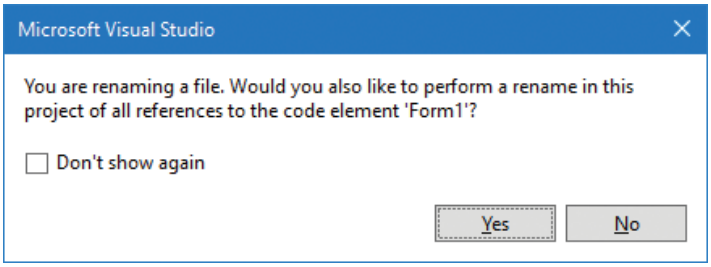
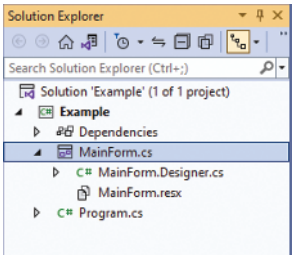


Figure 10-23 shows an example of the Solution Explorer after the Form1.cs file has been renamed MainForm.cs.

Figure 10-23 Form file renamed as MainForm.cs

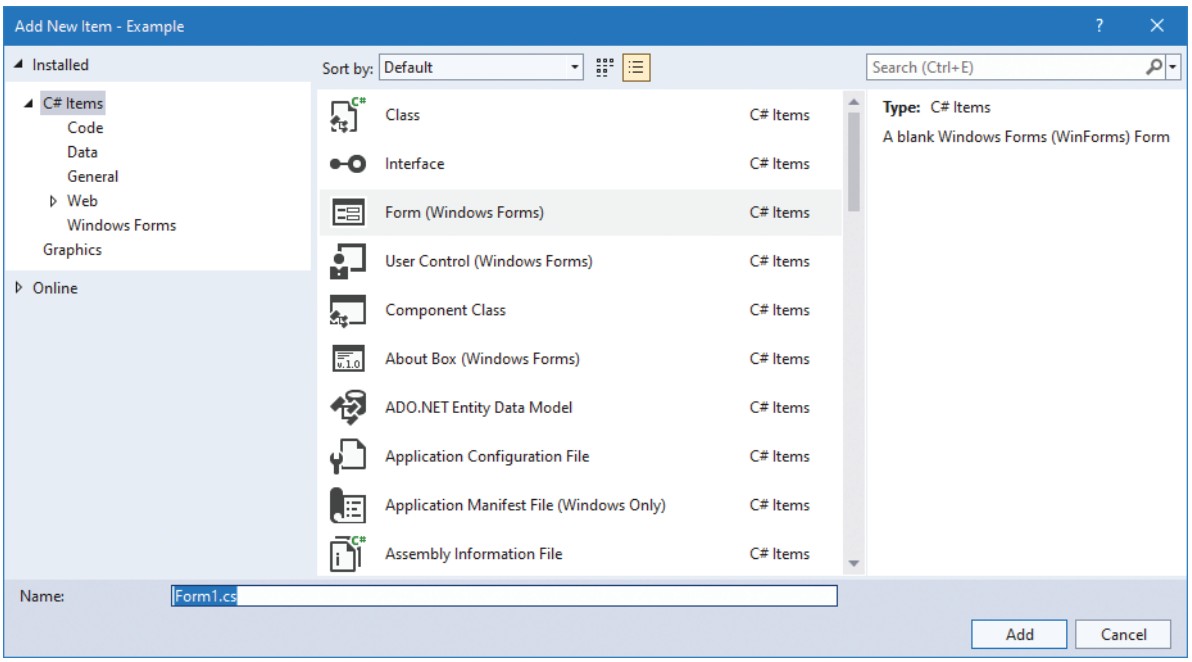


Adding a New Form to a Project

Follow these steps to add a new form to a project:

1. Click *Project* on the Visual Studio menu bar, and then select *Add Form (Windows Form)...* from the *Project* menu. The *Add New Item* window, shown in Figure 10-24, should appear.

Figure 10-24 *Add New Item* window

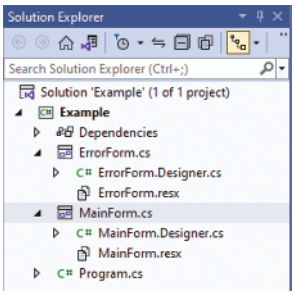


2. Near the bottom of the *Add New Item* window, a *Name* text box appears, where you can specify the new form's file name. Initially, a default name will appear here. (Notice that in Figure 10-24 the default name Form1.cs appears. The actual name that appears on your screen may be different.) Change the default name that is displayed in the *Name* text box to a more descriptive name. For example, if you wish to name the new form ErrorForm, enter *ErrorForm.cs* in the *Name* text box. (Make sure you specify the *.cs* extension with the file name that you enter.)

3. Click the *Add* button.

After completing these steps, a new blank form is added to your project. The new form is displayed in the *Designer,* and an entry for the new form's file appears in the *Solution Explorer*. Figure 10-25 shows an example of the *Solution Explorer* with two form files: *ErrorForm.cs* and *MainForm.cs*. Once you have added a form to a project, you can place any controls on it that you desire and write the necessary event handlers for it.

Figure 10-25 *Solution Explorer* window showing two forms



Switching between Forms and Form Code

In Visual Studio, you can easily switch your view to another form by double-clicking the form's entry in the *Solution Explorer* window. The form is then displayed in the *Designer*. You can also use the tabs that appear at the top of the *Designer* to display different forms or their code. For example, look at Figure 10-26. It shows the tabs that appear for a project with two forms: ErrorForm and MainForm. The tabs that display the *[Design]* designator cause a form to be displayed in the *Design* window. The tabs that appear without the designator cause a form's code to be displayed in the code editor.

When you open a form in the code editor, you will see several `using` directives, a namespace declaration, and the form's class. Figure 10-27 shows an example. The file in the figure contains the code for a form named ErrorForm. Notice that the form's class is named `ErrorForm`.

Figure 10-26 *Designer* tabs

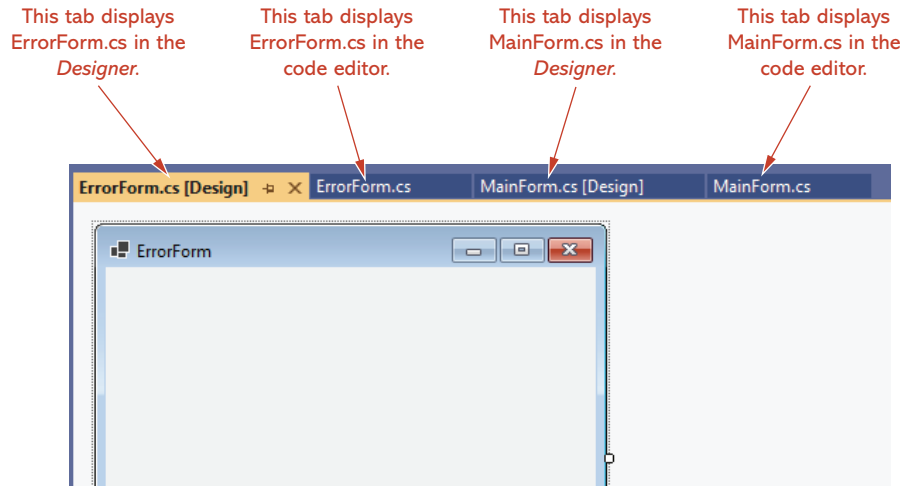
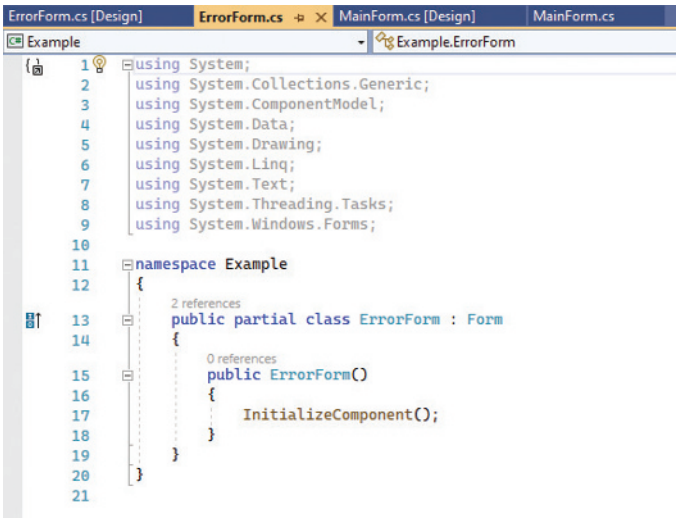


Figure 10-27 A source code file containing a form class



Removing a Form

If you wish to remove a form from a project and delete its file from the disk, follow these steps.

1. Right-click the form's entry in the *Solution Explorer* window.
2. On the pop-up menu, click *Delete*.

If you wish to remove a form from a project but do not want to delete its file from the disk, follow these steps.

1. Right-click the form's entry in the *Solution Explorer* window.
2. On the pop-up menu click *Exclude From Project*.

Displaying a Form

In your application's code, the first step in displaying a form is to create an instance of the form's class. For example, suppose a project has a form named ErrorForm. The following statement creates an instance of the `ErrorForm` class:

```
var myErrorForm = new ErrorForm();
```

This statement declares a reference variable named `myErrorForm`. It also creates an object of the `ErrorForm` class in memory and assigns a reference to the object to the `myErrorForm` variable. After this statement executes, you will be able to use the `myErrorForm` variable to perform operations with the form.

Creating an instance of a form's class does not display the form on the screen. The next step is to call the form's `ShowDialog` method. Here is an example:

```
myErrorForm.ShowDialog();
```

The ShowDialog method displays a form on the screen, and it gives that form the focus. This means that control of the application transfers to the form. When the user closes the form, control of the application returns to the point where the `ShowDialog` method was called, and execution resumes.

Now that we've covered the basic concepts of creating and displaying a form, go through the steps in Tutorial 10-5. In the tutorial, you create a simple application that has two forms.

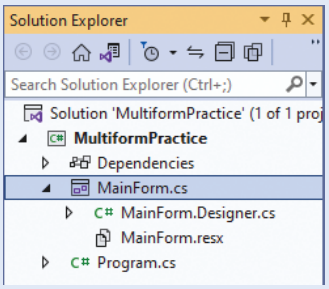
Tutorial 10-5:
Creating an Application with Two Forms

VideoNote
Tutorial 10-5: Creating an Application with Two Forms

Step 1: Start Visual Studio. Create a new Windows Forms App project named *MultiformPractice*.

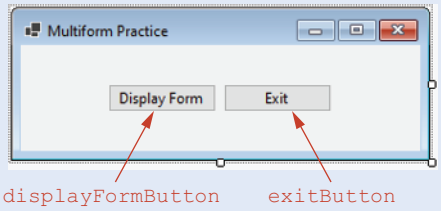
Step 2: In the *Solution Explorer* window, rename the *Form1.cs* file to *MainForm.cs*. (Right-click *Form1.cs* and then select *Rename* from the pop-up menu.) Changing the form's file name to *MainForm.cs* changes the form's name to MainForm. The *Solution Explorer* window should appear as shown in Figure 10-28.

Figure 10-28 *Solution Explorer* after changing *Form1.cs* to *MainForm.cs*



Step 3: In the *Designer*, set up the MainForm form as shown in Figure 10-29.

Figure 10-29 The MainForm form



Step 4: Perform the following to create another form named MessageForm in the project:

- Click *Project* on the menu bar and then select *Add Form (Windows Form)*. . . .
- The *Add New Item* window will appear. Enter *MessageForm.cs* as the name.
- Click the *Add* button.

As shown in Figure 10-30, a new form named MessageForm will appear in the *Designer*. Notice that an entry for *MessageForm.cs* appears in the *Solution Explorer*.

Step 5: In the *Designer*, set up the MessageForm as shown in Figure 10-31.

Step 6: Next, you will create the Click event handler for the `closeButton` control. In the *Designer*, double-click the `closeButton` control. This displays the MessageForm.cs file in the code editor, and you will see an empty event handler named `closeButton_Click`. Complete the event handler by typing the code shown in lines 22–23 in Program 10-8. When the user clicks the `closeButton` control, this event handler closes the form.

Figure 10-30 MessageForm added to the project

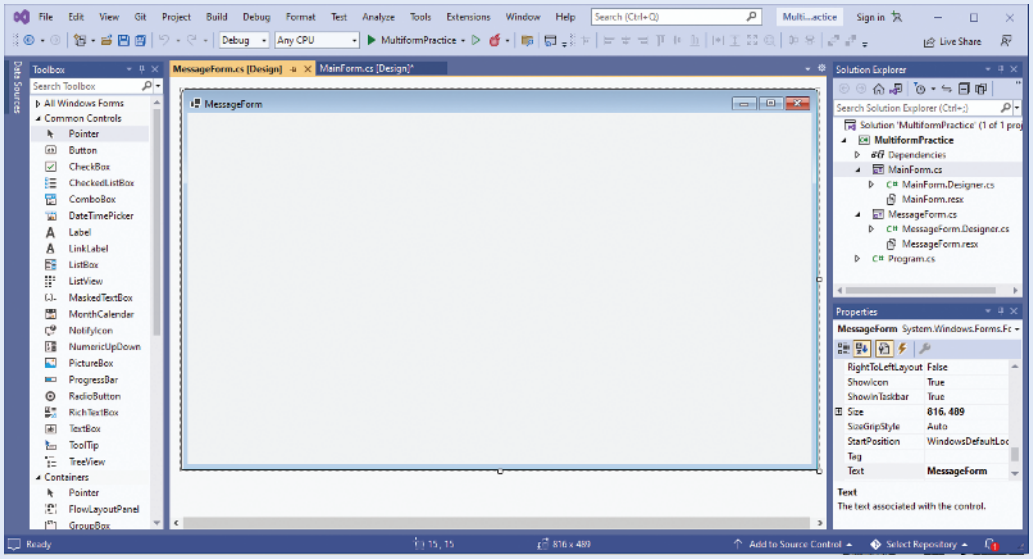
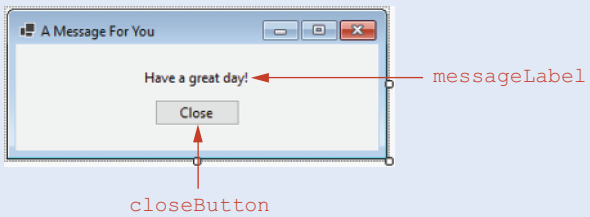


Figure 10-31 MessageForm



Step 7: Use the tabs at the top of the *Designer* to switch to *MainForm.cs* [*Design*]. This brings up the MainForm in the *Designer*.

Step 8: Now you will create the Click event handler for the `displayFormButton` control. Double-click the `displayFormButton` control. This displays the MainForm.cs file in the code editor, and you will see an empty event handler named `displayFormButton_Click`. Complete the event handler by typing the code shown in lines 12–16 in Program 10-9. Let's review this code:

Line 13: This statement does the following:

- It declares a `MessageForm` reference variable named `myMessageForm`.
- It creates a `MessageForm` object in memory.
- It assigns a reference to the `MessageForm` object to the `myMessageForm` variable.

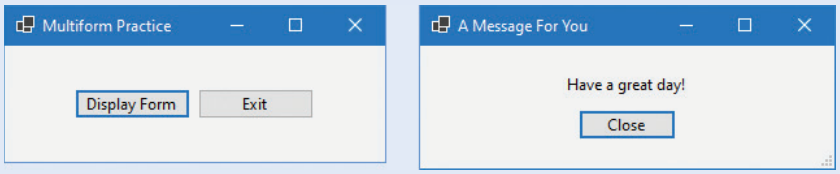
After this statement executes, the `myMessageForm` variable will reference a `MessageForm` object.

Line 16: This statement displays the MessageForm on the screen and transfers control of the application to the form.

Step 9: Switch your view back to the MainForm form in the *Designer* and double-click the `exitButton` control. In the code editor you will see an empty event handler named `exitButton_Click`. Complete the event handler by typing the code shown in lines 21–22 in Program 10-9.

Step 10: Save the project. Then, press F5 on the keyboard or click the *Start Debugging* button (▶) on the toolbar to compile and run the application. When the application runs, the MainForm should appear, as shown on the left in Figure 10-32. Click the *Display Form* button. The MessageForm form should appear, as shown on the right in Figure 10-32.

Figure 10-32 The MainForm and the MessageForm forms displayed



Step 11: On the MessageForm, click the *Close* button. This should close the MessageForm. Next, click the *Exit* button on the MainForm to end the application.

Program 10-8 Completed code for the MessageForm form

```
 1 using System;
 2 using System.Collections.Generic;
 3 using System.ComponentModel;
 4 using System.Data;
 5 using System.Drawing;
 6 using System.Linq;
 7 using System.Text;
 8 using System.Threading.Tasks;
 9 using System.Windows.Forms;
10
11 namespace MultiformPractice
12 {
13     public partial class MessageForm : Form
14     {
15         public MessageForm()
16         {
17             InitializeComponent();
18         }
19
20         private void closeButton_Click(object sender, EventArgs e)
21         {
22             // Close the form.
23             this.Close();
24         }
25     }
26 }
```

Program 10-9 Completed code for the MainForm form

```
1 namespace MultiformPractice
2 {
3     public partial class MainForm : Form
4     {
5         public MainForm()
6         {
7             InitializeComponent();
8         }
9
```

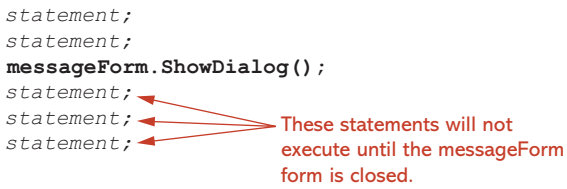
```
10          private void displayFormButton_Click(object sender,
            EventArgs e)
11          {
12              // Create an instance of the MessageForm class.
13              var myMessageForm = new MessageForm();
14
15              // Display the form.
16              myMessageForm.ShowDialog();
17          }
18
19          private void exitButton_Click(object sender, EventArgs e)
20          {
21              // Close the form.
22              this.Close();
23          }
24      }
25  }
```

Modal and Modeless Forms

A form can be either modal or modeless. The `ShowDialog` method displays a form in modal fashion. When a **modal form** is displayed, no other form in the application can receive the focus until the modal form is closed. The user must close the modal form before they can work with any other form in the application. A **modeless form**, on the other hand, allows the user to switch focus to another form while it is displayed. The user does not have to close a modeless form to switch focus to another form. If you want to display a form in modeless fashion, call its `Show method`.

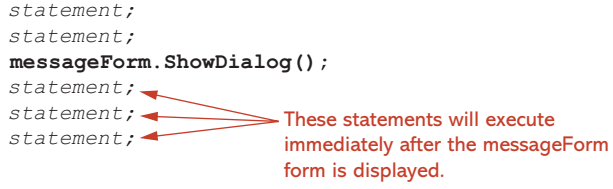
There is another important difference between modal and modeless forms. When you call the `ShowDialog` method to display a modal form, control of the program is transferred to the form, and no statements appearing after the `ShowDialog` method call will execute until the form is closed. Figure 10-33 illustrates this concept.

Figure 10-33 Execution of statements after displaying a modal form



When you call the `Show` method to display a modeless form, however, the statements that appear after the `Show` method call continue to execute while the form is displayed. The application does not wait until the modeless form is closed before executing these statements. Figure 10-34 illustrates this concept.

Figure 10-34 Execution of statements after displaying a modeless form



TIP: Modeless forms are typically used in special situations. In most multiform applications, you will want to display forms in modal fashion.

Accessing Controls on a Different Form

The controls that you place on a form have private access by default. This means that a control that exists on a particular form can be accessed only by code that is written in that form's class. With private access, you cannot write code in one form that accesses a control on another form.

You can change the access of a control by changing its **Modifiers property** in the *Properties* window. If you change a control's Modifiers property to Public, the control can be accessed by code outside the form's class.

For example, suppose an application has a form named `GreetingsForm`, and `GreetingsForm` has a Label control named `messageLabel`. Furthermore, let's assume that the `messageLabel` control's Modifiers property has been set to Public. The following code shows how you can create an instance of `GreetingsForm`, assign a value to the `messageLabel` control's Text property, and then display the form:

```
1  var greetingsForm = new GreetingsForm();
2  greetingsForm.messageLabel.Text = "Good day!"
3  greetingsForm.ShowDialog();
```

The statement in line 1 creates an instance of the `GreetingsForm` class, referenced by the `greetingsForm` variable. At this point, the form exists in memory, but it has not been displayed on the screen. The statement in line 2 assigns the string `"Good day!"` to the `messageLabel` control's Text property. Notice that the control's name is preceded by `greetingsForm` followed by a dot. This indicates that the control is not on the current form, but on the form that is referenced by the `greetingsForm` variable. The statement in line 3 calls the form's `ShowDialog` method to display the form on the screen. When the form appears on the screen, the `messageLabel` control will display the text *Good day!*

In Tutorial 10-6, you get a chance to create a multiform application in which code on one form creates an instance of another form and assigns values to controls on that form.

Tutorial 10-6:
Accessing a Control on a Different Form

VideoNote
Tutorial 10-6: Accessing a Control on a Different Form

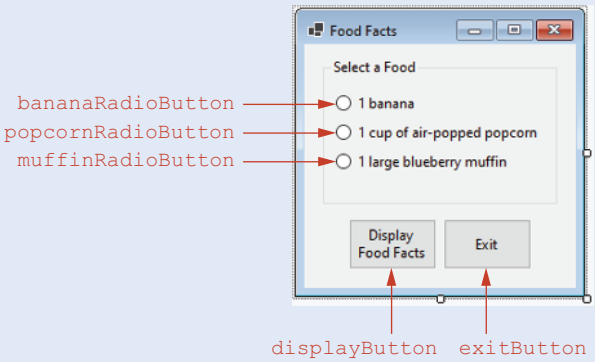
In this tutorial, you will create an application that allows the user to select a food from the application's main form, and then display a second form that shows the selected food's nutritional information.

Step 1: Start Visual Studio. Create a new Windows Forms App project named *FoodFacts*.

Step 2: In the *Solution Explorer* window, rename the *Form1.cs* file to *MainForm.cs*. (Right-click *Form1.cs*, and then select *Rename* from the pop-up menu.) Changing the form's file name to *MainForm.cs* changes the form's name to MainForm.

Step 3: In the *Designer*, set up the MainForm form as shown in Figure 10-35.

Figure 10-35 The MainForm form



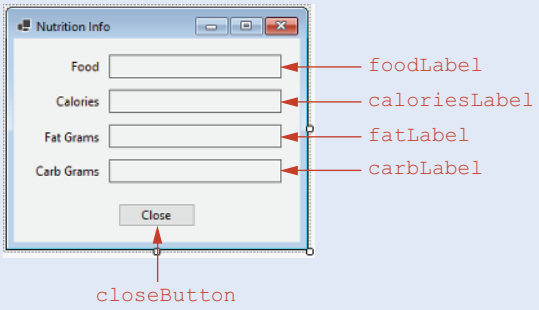
Step 4: Perform the following steps to create another form named NutritionForm in the project:

- Click *Project* on the menu bar, then select *Add Form (Windows Form)*. . .
- The *Add New Item* window will appear. Enter *NutritionForm.cs* as the name.
- Click the *Add* button.

Step 5: In the *Designer*, set up the NutritionForm form with the controls shown in Figure 10-36. The `foodLabel`, `caloriesLabel`, `fatLabel`, and `carbLabel` controls have the following property settings:

- AutoSize is set to *False*.
- BorderStyle is set to *Fixed3D*.
- Modifiers is set to Public.

Figure 10-36 The NutritionForm form



Step 6: Now, you will create the Click event handler for the `closeButton` control. In the *Designer*, double-click the `closeButton` control. This will display the NutritionForm.cs file in the code editor, and you will see an empty event handler named `closeButton_Click`. Complete the event handler by typing the code shown in lines 22 through 23 in Program 10-10. When the user clicks the `closeButton` control, this event handler will close the form.

Step 7: Use the tabs at the top of the *Designer* to switch to *MainForm.cs* [*Design*]. This brings up the MainForm in the *Designer*.

Step 8: Now, you will create the Click event handler for the `displayButton` control. Double-click the `displayButton` control. This will display the MainForm.cs file in the code editor, and you will see an empty event handler named

`displayButton_Click`. Complete the event handler by typing the code shown in lines 12 through 39 in Program 10-11. Let's review this code:

Line 13: This statement creates an instance of the `NutritionForm` class, referenced by the `nutriForm` variable. Keep in mind that although the form has been created in memory, it has not yet been displayed on the screen.

Line 16: This `if` statement determines whether the `bananaRadioButton` control is selected. If so, the statements in lines 18 through 21 use the `nutriForm` variable to assign values to the Label controls on the NutritionForm form. The values that are assigned are the nutritional values for a banana.

Line 23: If the `bananaRadioButton` control is not selected, this `else if` clause determines whether the `popcornRadioButton` control is selected. If so, the statements in lines 25 through 28 use the `nutriForm` variable to assign values to the Label controls on the NutritionForm form. The values that are assigned are the nutritional values for one cup of air-popped popcorn.

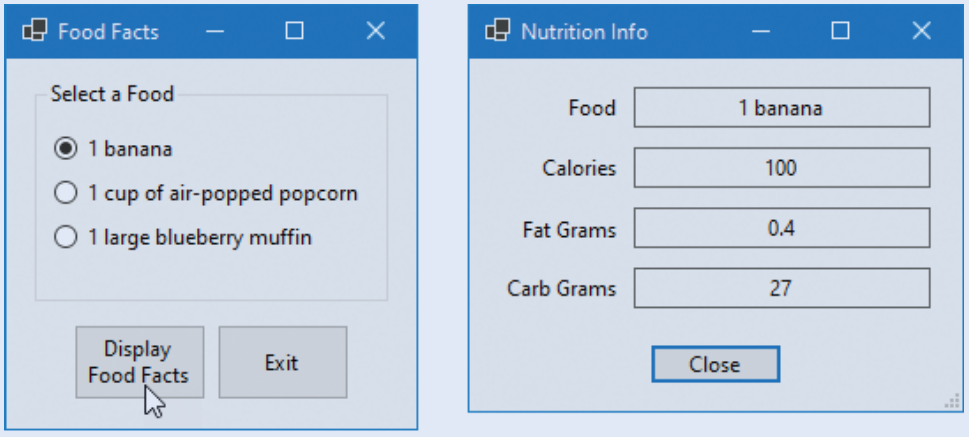
Line 30: If neither the `bananaRadioButton` nor the `popcornRadioButton` controls are selected, this `else if` clause determines whether the `muffinRadioButton` control is selected. If so, the statements in lines 32 through 35 use the `nutriForm` variable to assign values to the Label controls on the NutritionForm form. The values that are assigned are the nutritional values for one large blueberry muffin.

Line 39: This statement displays the NutritionForm on the screen, and transfers control of the application to the form.

Step 9: Switch your view back to the MainForm form in the *Designer* and double-click the `exitButton` control. In the code editor, you will see an empty event handler named `exitButton_Click`. Complete the event handler by typing the code shown in lines 44 through 45 in Program 10-11.

Step 10: Save the project. Then, press F5 on the keyboard, or click the *Start Debugging* button (▶) on the toolbar to compile and run the application. When the application runs, the MainForm should appear, as shown on the left in Figure 10-37. With the *1 banana* radio button selected, click the *Display Food Facts* button. The NutritionForm form should appear, as shown on the right in the figure.

Figure 10-37 The MainForm and the NutritionForm forms displayed at run time



Step 11: On the NutritionForm, click the *Close* button. This should close the NutritionForm. Try selecting the other radio buttons on the MainForm form and clicking

the *Display Food Facts* button to see each item's nutritional information. When you are finished, click the *Exit* button on the MainForm form to end the application.

Program 10-10 Completed code for the NutritionForm form

```
 1 using System;
 2 using System.Collections.Generic;
 3 using System.ComponentModel;
 4 using System.Data;
 5 using System.Drawing;
 6 using System.Linq;
 7 using System.Text;
 8 using System.Threading.Tasks;
 9 using System.Windows.Forms;
10
11 namespace FoodFacts
12 {
13     public partial class NutritionForm : Form
14     {
15         public NutritionForm()
16         {
17             InitializeComponent();
18         }
19
20         private void closeButton_Click(object sender, EventArgs e)
21         {
22             // Close the form.
23             this.Close();
24         }
25     }
26 }
```

Program 10-11 Completed code for the MainForm form

```
 1 namespace FoodFacts
 2 {
 3     public partial class MainForm : Form
 4     {
 5         public MainForm()
 6         {
 7             InitializeComponent();
 8         }
 9
10         private void displayButton_Click(object sender,
           EventArgs e)
11         {
12             // Create an instance of the NutritionForm class.
13             var nutriForm = new NutritionForm();
14
15             // Find the selected radio button.
16             if (bananaRadioButton.Checked)
17             {
18                 nutriForm.foodLabel.Text = "1 banana";
19                 nutriForm.caloriesLabel.Text = "100";
20                 nutriForm.fatLabel.Text = "0.4";
21                 nutriForm.carbLabel.Text = "27";
22             }
```

```
23              else if (popcornRadioButton.Checked)
24              {
25                  nutriForm.foodLabel.Text = "1 cup air-popped
                    popcorn";
26                  nutriForm.caloriesLabel.Text = "31";
27                  nutriForm.fatLabel.Text = "0.4";
28                  nutriForm.carbLabel.Text = "6";
29              }
30              else if (muffinRadioButton.Checked)
31              {
32                  nutriForm.foodLabel.Text = "1 large blueberry
                    muffin";
33                  nutriForm.caloriesLabel.Text = "385";
34                  nutriForm.fatLabel.Text = "9";
35                  nutriForm.carbLabel.Text = "67";
36              }
37
38              // Display the NutritionForm.
39              nutriForm.ShowDialog();
40          }
41
42          private void exitButton_Click(object sender, EventArgs e)
43          {
44              // Close the form.
45              this.Close();
46          }
47      }
48  }
```

Checkpoint

10.27 What steps should you follow to change a form's name?

10.28 What steps should you follow to add a new form to a project?

10.29 How do you remove a form from a project and delete its file from the disk?

10.30 What is the first step in displaying a form?

10.31 What happens after a form that has been displayed with the `ShowDialog` method is closed by the user?

10.9 Static Class Members

CONCEPT: **A static class member belongs to the class, not objects instantiated from the class.**

A Quick Review of Instance Fields and Instance Methods

Recall that each instance of a class has its own set of fields, which are known as instance fields. You can create several instances of a class and store different values in each instance's fields. For example, the `Coin` class that we created earlier in this chapter has a

field named `sideUp`. When you call the class's `Toss` method, the `sideUp` field, which is a string, is randomly set to either "Heads" or "Tails." Let's say that `myCoin` references an instance of the `Coin` class and we execute the following statement:

```
myCoin.Toss();
```

This statement stores a value in the `sideUp` field that belongs to the instance that is referenced by `myCoin`. You can think of instance fields as belonging to a specific instance of a class.

You will also recall that classes may have instance methods as well. When you call an instance method, it performs an operation on a specific instance of the class. For example, assuming that `myCoin` references an instance of the `Coin` class, look at the following statement:

```
status = myCoin.GetSideUp();
```

This statement calls the `GetSideUp` method, which returns the value of the `sideUp` field that belongs to a specific instance of the `Coin` class: the one referenced by `myCoin`. Both instance fields and instance methods are associated with a specific instance of a class, and they cannot be used until an instance of the class is created.

Static Members

It is possible to create a field, property, or method that does not belong to any instance of a class. Such members are known as **static fields**, **static properties**, and **static methods**. When a value is stored in a static field or **static property**, it is not stored in an instance of the class. In fact, an instance of the class doesn't even have to exist for values to be stored in the class's static fields and static properties. Likewise, static methods do not operate on the fields or properties that belong to any instance of the class. Instead, they can operate only on static fields and static properties. You can think of static fields, static properties, and static methods as belonging to the class instead of an instance of the class. In this section, we will take a closer look at static members. First we will examine static fields.

Static Fields

When a field is declared with the **`static` keyword**, there will be only one copy of the field in memory, regardless of the number of instances of the class that might exist. A single copy of a class's static field is shared by all instances of the class. For example, the `Countable` class shown in the following code snippet uses a static field to keep count of the number of instances of the class that are created.

```
 1  class Countable
 2  {
 3     // Static field
 4     private static int instanceCount = 0;
 5
 6     // Constructor
 7     public Countable()
 8     {
 9        instanceCount++;
10     }
11
12     // Method to get the value of instanceCount
13     public int GetInstanceCount()
14     {
15        return instanceCount;
16     }
17  }
```

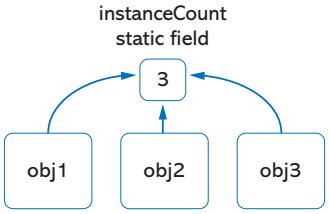
Line 4 declares a static field named `instanceCount`. A static field is created by placing the `static` keyword after the access specifier and before the field's data type. Notice that we have explicitly initialized the `instanceCount` field with the value 0. This initialization takes place only once, regardless of the number of instances of the class that are created.

Next, look at the constructor in lines 7–10. The constructor uses the `++` operator to increment the `instanceCount` field. Each time an instance of the `Countable` class is created, the constructor will be called and the `instanceCount` field will be incremented. As a result, the `instanceCount` field will contain the number of instances of the `Countable` class that have been created. The `getInstanceCount` method, in lines 28–31, returns the value in `instanceCount`. The following code snippet demonstrates this class:

```
1  // Create three Countable objects.
2  var obj1 = new Countable();
3  var obj2 = new Countable();
4  var obj3 = new Countable();
5
6  // Get the number of instances.
7  int numObjects = obj3.GetInstanceCount();
```

This code creates three instances of the `Countable` class, referenced by the variables `obj1`, `obj2`, and `obj3`. Although there are three instances of the class, there is only one copy of the static field. This is illustrated in Figure 10–38. In line 7 the code calls the `getInstanceCount` method to retrieve the number of instances that have been created. (The `numObjects` variable will be assigned the value 3.) Although the program calls the `getInstanceCount` method from `obj3`, the same value would be returned from any of the objects.

Figure 10–38 All instances share the static `instanceCount` field



NOTE: C# automatically stores 0 in all uninitialized value-type static fields (or `false` in uninitialized static `boolean` fields), and `null` in all uninitialized reference-type static fields. The `instanceCount` field in the `Countable` class is explicitly initialized so it is clear to anyone reading the code that the field starts with the value 0.

Static Properties

A property can also be static. When a property is declared with the `static` keyword, there will be only one copy of the property's value in memory, regardless of the number of instances of the class that might exist. All instances of the class will share that one copy of the property's data. For example, the following code snippet shows another version of the `Countable` class. This version of the class uses a static property to keep count of the number of instances of the class that are created.

```
1  class Countable
2  {
3     // Static property
4     public static int InstanceCount { get; set; } = 0;
```

```
 5
 6      // Constructor
 7      public Countable()
 8      {
 9          InstanceCount++;
10      }
11  }
```

Line 4 declares a static property named `InstanceCount`. The property is implemented as an auto-property, and initialized with the value 0. This initialization takes place only once, regardless of the number of instances of the class that are created.

When a class has a static property, the code outside the class cannot use an instance of the class to access that property. For example, the following code snippet will cause an error.

```
var obj = new Countable();            // Create an instance of Countable
int numObjects = obj.InstanceCount; // ERROR!
```

Instead of using an instance of the class to access a static property, you use the class itself. Here is an example:

```
var obj = new Countable();
int numObjects = Countable.InstanceCount;
```

When this code executes, the `numObjects` variable will be assigned the value 1. Another example follows. When this code executes, the `numObjects` variable will be assigned the value 3.

```
1  // Create three Countable objects.
2  var obj1 = new Countable();
3  var obj2 = new Countable();
4  var obj3 = new Countable();
5
6  // Get the number of instances.
7  int numObjects = Countable.InstanceCount;
```

If you explicitly declare the backing field for a static property, keep in mind that both the property and the backing field must be declared `static`. Here is another example of the `Countable` class. In this version of the class, the `InstanceCount` property is not implemented as an auto-property. Its backing field is explicitly declared. Notice that in line 4 the `_instanceCount` backing field is declared `static` and in line 7 the `InstanceCount` property is also declared `static`.

```
 1  class Countable
 2  {
 3      // Backing field for the InstanceCount property
 4      private static int _instanceCount = 0;
 5
 6      // InstanceCount property
 7      public static int InstanceCount
 8      {
 9          get { return _instanceCount; }
10          set { _instanceCount = value; }
11      }
12
13      // Constructor
14      public Countable()
15      {
16          InstanceCount++;
17      }
18  }
```

Static Methods

When a class contains a static method, you don't have to create an instance of the class in order to execute the method. The following code snippet shows an example of a class with static methods.

```
 1  class Metric
 2  {
 3     // The following method converts miles to kilometers.
 4     public static double MilesToKilometers(double m)
 5     {
 6         return m * 1.609;
 7     }
 8
 9     // The following method converts kilometers to miles.
10     public static double KilometersToMiles(double k)
11     {
12         return k / 1.609;
13     }
14  }
```

A static method is created by placing the key word `static` after the access specifier in the method header. The `Metric` class has two static methods: `MilesToKilometers` and `KilometersToMiles`. Because they are declared as `static`, they belong to the class and may be called without any instances of the class being in existence. You simply write the name of the class before the dot operator in the method call. Here is an example:

```
double kilometers = Metric.MilesToKilometers(10.0);
double miles = Metric.KilometersToMiles(1609.0);
```

The first statement calls the `MilesToKilometers` method, passing the value 10.0 as an argument. The value that is returned from the method is assigned to the `kilometers` variable. The second statement calls the `KilometersToMiles` method, passing the value 1609.0 as an argument. The value that is returned from the method is assigned to the `miles` variable. Notice that both the methods are called directly from the `Metric` class, not from an instance of the class.

Static methods are convenient for many tasks because they can be called directly from the class, as needed. They are most often used to create utility classes that perform operations on data but have no need to collect and store data. The `Metric` class is a good example. It is used as a container to hold methods that convert miles to kilometers and vice versa but is not intended to store any data.

The only limitation that static methods have is that they cannot refer to nonstatic members of the class. This means that any method called from a static method must also be static. It also means that if the method uses any of the class's fields or properties, they must be static as well.

Static Classes

You've seen that the `static` keyword can be used to declare fields, properties, and methods as static. You can also use the `static` keyword to declare a class as static. If you declare a class as static, then all the members of the class must also be static. If a **static class** contains a nonstatic field, property, or method, a compiler error will occur.

If a class is declared `static`, the compiler will not allow you to instantiate the class. Also, static classes cannot have constructors. This restriction makes sense because constructors are responsible for initializing instances, and because static classes cannot be instantiated, there's no reason for them to have constructors.

In Tutorial 10-7, you will write a static class that logs messages to a text file as the application runs.

Tutorial 10-7:
Creating a Simple Logging Utility with a Static Class

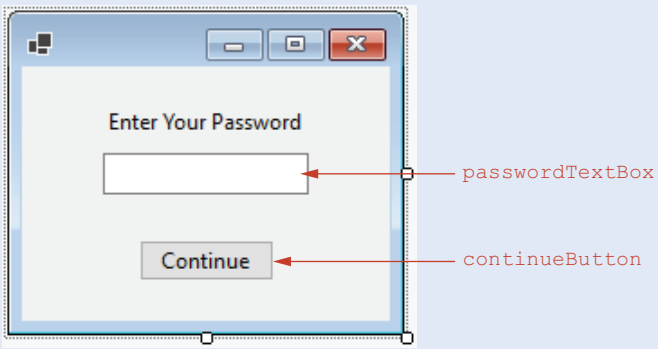
VideoNote
Tutorial 10-7: Creating a Simple Logging Utility with a Static Class

A *logging utility* is a class or set of classes that record information about a program's activity to a log file. Log files are used for various purposes. For example, an *audit log* records information about how a program is used, who uses it, when the program is used, etc. A *debugging log* records diagnostic messages that can be used to track down and resolve errors in a program's code.

In this tutorial, you will write a static class that works as a simple logging utility. The class will have methods to open a log file, write messages to the log file, and close the log file. You will then insert statements into an application that uses the logging utility to record information describing various events that take place during the program's execution.

The application that you will modify is the *CredentialLogger* project. Its form is shown in Figure 10–39. The user enters a password into the `passwordTextBox` control and then clicks the `continueButton` control to check the password. If the password matches any of the passwords found in an array, the message "Password accepted" is displayed. Otherwise, the message "Password not found" is displayed.

Figure 10–39 The *CredentialLogger* application's Form1 form



Step 1: Start Visual Studio. Open the project named *CredentialLogger* in the *Chap10* folder of the Student Sample Programs.

Step 2: In this step you will add a static class named `Log` to the project. Click *Project* on the Visual Studio menu bar; then select *Add Class*. . . . The *Add New Item* window should appear with the *Class* template already selected in the center pane. Change the name of the file to Log.cs, and then click the *Add* button. This adds a source code file named Log.cs to the project.

Step 3: Program 10-12 shows how the contents of the Log.cs file should appear when you have completed it. Write the code shown in bold in the program. (Don't overlook the `static` keyword in line 5.)

Let's review the code in the `Log` class:

- The `WriteLog` method appears in lines 12–22. This method opens a log file, appends a string to the log file, and closes the log file. The method accepts two arguments: `filename`, which is the name of the log file, and `str`, which is the string to write to the log file.
- The `ClearLogFile` method appears in lines 25–32. It has one parameter: `filename`, which is the name of a log file. This method clears the contents of the specified log file.

Step 4: Switch your view back to the Form1 form in the Designer and press F7 on the keyboard. (Alternatively, you can click *View* on the menu bar, and then click *Code*.) This opens the Form1.cs file in the code editor. Insert the comment and field declaration that appears in lines 9 and 10 of Program 10-13. The statement in line 10 declares a private `string` field named `logName`. It is initialized with the filename of the log file.

Step 5: Insert the lines that are shown in bold in the `continueButton_Click` event handler. These lines of code make entries in the log file at various points in the program's execution.

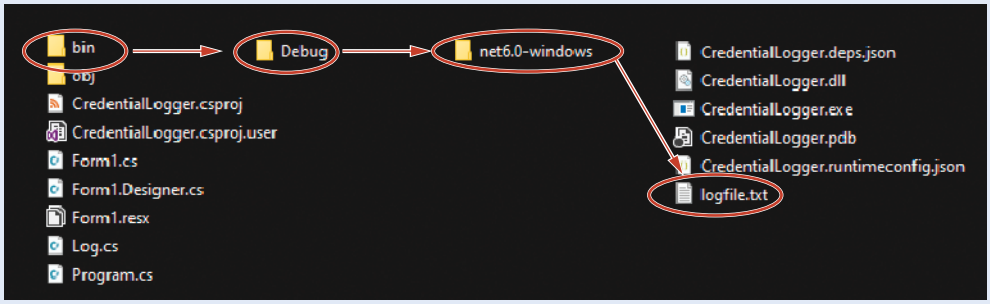
Step 6: Switch your view back to the Form1 form in the Designer and double-click any part of the form that does not contain a control. (Be sure not to click the Button, Label, or TextBox controls.) This opens the code editor, and you will see an empty `Form1_Load` event handler. Complete the `Form1_Load` event handler by typing the code shown in lines 54–55 in Program 10-13. This code clears the contents of the log file each time the application starts up.

Step 7: Save the project and run the application. Experiment by entering different passwords, some valid and some invalid, and clicking the `continueButton` control each time that you enter one. After you have done this a few times, close the application.

Step 8: Now you will open the log file and examine its contents. In Visual Studio, click *File*, then click *Open*, then click *File*. . . . A dialog will appear allowing you to browse the contents of the project directory. As shown in Figure 10–40, double-click *bin*, then double-click *Debug*. At this point you will see a folder with a name similar to *net6.0-windows*. Although the name might be slightly different on your system, double-click it. Then double-click *logfile.txt*. The log file will appear in the editor. The contents of the file will depend on the actions that you took while the program was running, but an example is shown here:

```
 1  continueButton clicked.
 2  The user entered: asdfasf
 3  valid = False
 4  continueButton clicked.
 5  The user entered: 7GxjUb
 6  7GxjUb found in the array.
 7  valid = True
 8  continueButton clicked.
 9  The user entered: xkxkxk
10  valid = False
```

Figure 10–40 Opening the log file



Program 10-12 The `Log` class

```
 1 using System;
 2 using System.Collections.Generic;
 3 using System.Linq;
 4 using System.Text;
 5 using System.Threading.Tasks;
 6
 7 namespace CredentialLogger
 8 {
 9     internal static class Log
10     {
11         // Write a message to the log file.
12         public static void WriteLog(string filename, string
           str)
13         {
14             // Open the file.
15             StreamWriter logFile = File.AppendText(filename);
16
17             // Write the log entry.
18             logFile.WriteLine(str);
19
20             // Close the log file.
21             logFile.Close();
22         }
23
24         // Clear an existing log file.
25         public static void ClearLogFile(string filename)
26         {
27             // Create a new log file.
28             StreamWriter logFile = File.CreateText(filename);
29
30             // Close the log file.
31             logFile.Close();
32         }
33     }
34 }
```

Program 10-13 The completed Form1 code

```
 1 namespace CredentialLogger
 2 {
 3     public partial class Form1: Form
 4     {
 5         // Array containing valid passwords
 6         private string[] passwords = {
 7             "7GxjUb", "rJ96qd", "6AaPb7", "4PnmSX"};
 8
 9         // Log file name
10         private string logName = "logfile.txt";
11
12         public Form1()
13         {
14             InitializeComponent();
15         }
16
17         private void continueButton_Click(object sender,
           EventArgs e)
18         {
```

```
19              // Flag for validity
20              bool valid = false;
21
22              // Make an entry in the log file.
23              Log.WriteLog(logName, "continueButton clicked.");
24              Log.WriteLog(logName, $"The user entered:
                {passwordTextBox.Text}");
25
26              // Check the password
27              foreach (string password in passwords)
28              {
29                  if (passwordTextBox.Text == password)
30                  {
31                      valid = true;
32
33                      // Make an entry in the log file.
34                      Log.WriteLog(logName, $"{password} found in
                        the array.");
35                  }
36              }
37
38              // Make an entry in the log file.
39              Log.WriteLog(logName, $"valid = {valid}");
40
41              // Display the results
42              if (valid)
43              {
44                  MessageBox.Show("Password accepted");
45              }
46              else
47              {
48                  MessageBox.Show("Password not found");
49              }
50          }
51
52          private void Form1_Load(object sender, EventArgs e)
53          {
54              // Clear the log file.
55              Log.ClearLogFile(logName);
56          }
57      }
58  }
```

Checkpoint

10.32 Class `A` has a public static field named `value`. A program creates three instances of class `A`. How many copies of the `value` field exist in memory?

10.33 A class named `MyClass` has a public static `int` field named `value`. There are no instances of the class in existence. Write a statement that assigns 10 to the `value` field.

10.34 A class named `MyClass` has a public static `void` method named `Go`. The `Go` method has no parameters. Write a statement that calls the `Go` method.

10.35 Class `A` has a public static method named `Work`, and an instance field named `idNumber`. Is it possible for the `Work` method to assign a value to the `idNumber` field?

Key Terms

accessors
auto-properties
backing field
binding
class declaration
class header
constructor
default constructor
instance
member declarations
modal form
modeless form
Modifiers property
overloaded
parameterized constructor
parameterless constructor
problem domain
property
read-only property
shadows
signature
stale
static class
static field
`static` keyword
static method
static property
`value` parameter

Review Questions

Multiple Choice

1. Each object that is created from a class is called a(n) _____ of the class.
   a. Reference
   b. Example
   c. Instance
   d. Event

2. You create a class by writing a(n) _____.
   a. Blueprint
   b. Class declaration
   c. Initialization list
   d. Object name

3. The first line of a class declaration is known as the _____.
   a. Class preface
   b. Class title
   c. Class header
   d. Class directive

4. The class's _____ are the statements that define the class's fields, properties, and methods.
   a. Data agents
   b. Body statements
   c. Private definitions
   d. Member declarations

5. A(n) _____ is a method that is automatically executed when an object is created.
   a. Opener
   b. Loader
   c. Constructor
   d. Assembler

6. A(n) _____ is a special type of class member that allows an object to store and retrieve a piece of data.
   a. Property
   b. Asset
   c. Inserter/retriever
   d. Accessory

7. A special set of methods, known as _____, work in conjunction with a private field and allow code outside the class to get the property's value, and assign values to the property.
   a. Accessors
   b. Imitators
   c. Intermediates
   d. Directives

8. The private field, which is known as the property's _____, holds any data that is assigned to the property.
   a. Private data
   b. Backing field
   c. Holding value
   d. Masked variable

9. The _____ parameter of the `set` accessor is automatically created by the compiler, and its data type is the same as that of the property.
   a. Property
   b. Value
   c. Member
   d. Detail

10. A _____ can be read, but it cannot be modified.
    a. Limited property
    b. Temporary value
    c. Field value
    d. Read-only property

11. When the value of an item is dependent on other data and that item is not updated when the other data is changed, what has the value become?
    a. Bitter
    b. Stale
    c. Asynchronous
    d. Moldy

12. A constructor that accepts arguments is known as a(n) _____.
    a. Argumentative constructor
    b. Changeable constructor
    c. Parameterized constructor
    d. Secondary constructor

13. When a method is _____, it means that multiple methods in the same class have the same name but use different types of parameters.
    a. Emulated
    b. Versioned
    c. Threaded
    d. Overloaded

14. The process of matching a method call with the correct method is known as _____.

a. Sorting
b. Matching
c. Styling
d. Binding

15. A method's _____ consists of the method's name and the data type and argument kind (by value, `ref`, or `out`) of the method's parameters, from left to right.

a. Appearance
b. Signature
c. Identifier
d. Footprint

16. A _____ is a constructor that accepts no arguments.

a. Parameterless constructor
b. Basic constructor
c. Primary constructor
d. Passive constructor

17. If you write a class with no constructor whatsoever, the compiler will provide a(n) _____.

a. Default constructor
b. Constructor list
c. Parameterized method
d. Error message

18. The _____ is the set of real-world objects, parties, and major events related to the problem.

a. Critical path
b. Problem domain
c. Solution set
d. Operation focus

19. A class's responsibilities are _____.

a. The objects created from the class
b. Things the class knows
c. Actions the class performs
d. Both b and c

20. The _____ displays a form on the screen, and it gives that form the focus.

a. `Show` method
b. `ShowDialog` method
c. `Clear` method
d. `Focus` method

21. This type of method cannot access any nonstatic member variables in its own class.

a. Instance
b. Void
c. Static
d. Nonstatic

22. The compiler will not allow you to instantiate this type of class.

a. Base
b. Private
c. Static
d. Nonstatic

True or False

1. Objects that are instances of a class are always passed by value.
2. Class declarations must be written inside the project namespace.
3. A class is an object.
4. It is a common practice to make all a class's fields private and to provide access to those fields through methods.
5. The same rules for naming variables apply to naming classes.
6. If you need to make a property read only, you simply do not write a `set` accessor for the property.
7. If you try to pass a property to a `ref` or an `out` parameter, an error will occur.
8. Class fields are almost always declared `public` in order to make their values easily accessible to code outside of the class.
9. The `get` accessor can be thought of as a method that returns the class property's value.
10. Constructors can accept arguments in the same way as other methods.
11. It is legal to write a class without any constructors.
12. Objects that are instances of a class can be stored in an array.
13. The objects of a class can be stored in an array, but not in a `List`.
14. One way to find the classes needed for an object-oriented program is to identify all the verbs in a description of the problem domain.
15. Every form in a Visual C# project has a class.
16. By default, a control's Modifiers property is set to Public.
17. A static member method may refer to nonstatic fields of the same class, but only after an instance of the class has been defined.
18. All the members of a static class must be static as well.

Short Answer

1. When a method receives an object as an argument, does the method have access to the actual object or a copy of the object?
2. What are the advantages of storing classes in their own files?
3. How is a constructor used?
4. What is the difference between a class and an instance of a class?
5. What convention do most programmers follow when naming classes?
6. What is the `value` parameter? How is it created?
7. What is executed any time a class property is read?
8. What is executed any time a value is assigned to a class property?
9. How can you protect class fields from accidental corruption?
10. What is stale data?
11. Is it possible to pass initialization values to the constructor when you create an object? If so, how?

12. How does the compiler distinguish a method from other methods of the same name?
13. What do you call the constructor that is provided by the compiler, if no constructor is specified when a class is written?
14. In Visual Studio, how you can you switch your view to another form?
15. How do you remove a form from a project but keep its file on the disk?
16. What property of a control must you change if you want to access that control from code outside the form's class?
17. What is the difference between a modal form and a modeless form?
18. A class has a public static field named `name`. A program creates three instances of the class. How many copies of the `name` field exist in memory?

Algorithm Workbench

1. Write a statement that creates an instance of the `Transcript` class.
2. Write the accessors for a property named `CustomerNumber` that assigns a string value to the `_customerNumber` field.
3. Suppose that an application declares an array of class objects with the following statement:

   ```
   Employee[] employees = new Employee[9];
   ```

   Write a loop that creates 10 instances of the class and assigns them to the elements of the array.
4. Look at the following description of a problem domain:

   The bank offers the following types of accounts to its customers: savings accounts, checking accounts, and money market accounts. Customers are allowed to deposit money into an account (thereby increasing its balance), withdraw money from an account (thereby decreasing its balance), and earn interest on the account. Each account has an interest rate.

   Assume that you are writing a program that will calculate the amount of interest earned for a bank account.

   a. Identify the potential classes in this problem domain.
   b. Refine the list to include only the necessary class or classes for this problem.
   c. Identify the responsibilities of the class or classes.
5. A class named `MyClass` has a public static `int` field named `unitsOnHand`. There are no instances of the class in existence. Write a statement that assigns 10 to the `unitsOnHand` field.
6. A class named `MyClass` has a public static `void` method named `Reset`. The `Reset` method has no parameters. Write a statement that calls the `Reset` method.

Programming Problems

VideoNote
Solving the Pet Class Problem

1. **`Pet` Class**

   Create a class named `Pet` (similar to the one discussed in this chapter), which has the following properties:

   - Name—The Name property holds the name of a pet.
   - Type—The Type property holds the type of animal that a pet is. Example values are "Dog", "Cat", and "Bird".
   - Age—The Age property holds the pet's age.

Demonstrate the class in an application that creates an object of the class and lets the user enter the name, type, and age of their pet. This data should be stored in the object. Retrieve the pet's Name, Type, and Age properties and display their values on the screen.

2. **Trivia Game**

   Create a simple trivia game for two players that works like this:

   - Starting with player 1, each player gets a turn at answering 5 trivia questions. (There should be a total of 10 questions.) When a question is displayed, 4 possible answers are also displayed. Only one of the answers is correct, and if the player selects the correct answer, that player earns a point.
   - After answers have been selected for all the questions, the program displays the number of points earned by each player and declares the player with the highest number of points the winner.

   Here is an example of the information for a trivia question and its four possible answers:

   Question: "How long is a year on Mars?",

   Possible Answer #1: "550 Earth days",

   Possible Answer #2: "498 Earth days",

   Possible Answer #3: "126 Earth days",

   Possible Answer #4: "687 Earth days"

   Correct Answer: 4

   To create this application, write a `Question` class to hold the data for a trivia question. The `Question` class should have properties for the following data:

   - A trivia question
   - Possible answer 1
   - Possible answer 2
   - Possible answer 3
   - Possible answer 4
   - The number of the correct answer (1, 2, 3, or 4)

   The `Question` class also should have an appropriate constructor. Your application should have a `List` containing 10 `Question` objects, one for each trivia question. Make up your own trivia questions on the subject or subjects of your choice for the objects.

3. **`Employee` Class**

   Write a class named `Employee` that has the following properties:

   - **Name**—The Name property holds the employee's name.
   - **IdNumber**—The IdNumber property holds the employee's ID number.
   - **Department**—The Department property holds the name of the department in which the employee works.
   - **Position**—The Position property holds the employee's job title.

   The class should have the following overloaded constructors:

   - A constructor that accepts the following values as arguments and assigns them to the appropriate properties: employee's name, employee's ID number, department, and position.
   - A constructor that accepts the following values as arguments and assigns them to the appropriate properties: employee's name and ID number. The Department and Position properties should be assigned an empty string (`""`).
   - A parameterless constructor that assigns empty strings (`""`) to the Name, Department, and Position properties, and 0 to the IdNumber property.

In an application, create three `Employee` objects to hold the following data:

Name	ID Number	Department	Position
Susan Meyers	47899	Accounting	Vice President
Mark Jones	39119	IT	Programmer
Joy Rogers	81774	Manufacturing	Engineer

The application should store this data in the three objects and display the data for each employee on the screen.

4. `RetailItem` Class

Write a class named `RetailItem` that holds data about an item in a retail store. The class should have the following properties:

- **Description**—The Description property should hold a brief description of the item.
- **UnitsOnHand**—The UnitsOnHand property should hold the number of units currently in inventory.
- **Price**—The Price property should hold the item's retail price.

Write a constructor that accepts arguments for each property.

The application should create an array of three `RetailItem` objects containing the following data:

	Description	Units on Hand	Price
Item 1	Jacket	12	$59.95
Item 2	Jeans	40	$34.95
Item 3	Shirt	20	$24.95

The application should have a loop that steps through the array, displaying each element's properties.

5. Dorm and Meal Plan Calculator

A university has the following dormitories:

Allen Hall	$1,500 per semester
Pike Hall	$1,600 per semester
Farthing Hall	$1,800 per semester
University Suites	$2,500 per semester

The university also offers the following meal plans:

7 meals per week	$ 600 per semester
14 meals per week	$1,200 per semester
Unlimited meals	$1,700 per semester

Create an application with two forms. The main form should allow the user to select a dormitory and a meal plan. The application should show the total charges on the second form.

6. E-Mail Address Book

Create an application with a class named `PersonEntry`. The `PersonEntry` class should have properties for a person's name, e-mail address, and phone number. Also, create a text file that contains the names, e-mail addresses, and phone numbers for at least five people. When the application starts, it should read the data from the file and create a `PersonEntry` object for each person's data. The `PersonEntry` objects should be added to a List, and each person's name should be displayed in a list box on the application's main form. When the user selects a name from the list

box, a second form should appear displaying that person's name, e-mail address, and phone number.

7. **Area Class**

   Write a class that has three overloaded static methods for calculating the areas of the following geometric shapes:

   - Circles
   - Rectangles
   - Cylinders

   Here are the formulas for calculating the area of the shapes.

   Area of a circle: $Area = \pi r^2$

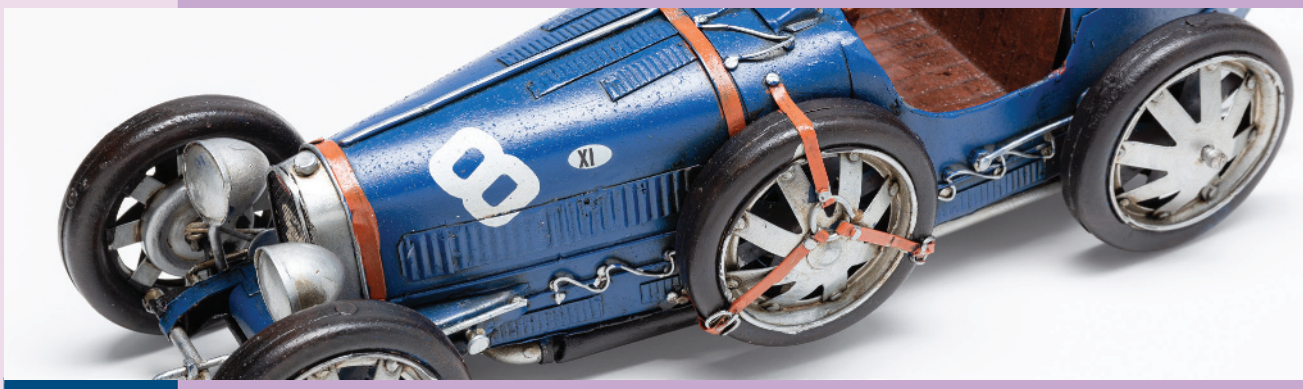
   where p is `Math.PI` and r is the circle's radius

   Area of a rectangle: $Area = Width * Length$

   Area of a cylinder: $Area = \pi r^2 h$

   where p is `Math.PI`, r is the radius of the cylinder's base, and h is the cylinder's height

   Because the three methods are to be overloaded, they should each have the same name, but different parameter lists. Demonstrate the class in a complete application.



CHAPTER 11

More about Classes

TOPICS

11.1 Returning Objects from Methods
11.2 Writing a `ToString` Method
11.3 Working with `null`
11.4 The `this` Reference Variable
11.5 Copying Objects
11.6 Object Initializer Syntax and Init-Only Properties
11.7 Mutable and Immutable Classes
11.8 Aggregation and Composition
11.9 Passing and Returning Anonymous Objects
11.10 Class Collaboration

11.1 Returning Objects from Methods

CONCEPT: **When a method returns an object that is an instance of a class, it returns a reference to that object.**

Just as methods can be written to return an `int`, `double`, `decimal`, or other data type, they can also be written to return an object. When a method returns an object that is an instance of a class, it returns a reference to that object. For example, suppose we have the following class for storing contact information in an application:

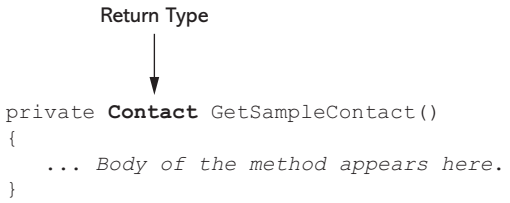
```
class Contact
{
    public Contact(string name, string email)
    {
        Name = name;
        Email = email;
    }
    public string Name { get; set; }
    public string Email { get; set; }
}
```

Suppose we also have the following method in the same application:

```
private Contact GetSampleContact()
{
    var contact = new Contact("Jane Doe", "jane@example.com");
    return contact;
}
```

The purpose of the `GetSampleContact` method is to create a `Contact` object that contains sample information, and then return a reference to that object. Notice that the method has a return data type of `Contact`, as shown in Figure 11-1.

Figure 11-1 The method's return type



Here is an example of how we might call the `GetSampleContact` method:

```
var sampleContact = GetSampleContact();
```

This statement calls the `GetSampleContact` method, which returns a reference to a `Contact` object. The reference is assigned to a variable named `sampleContact`, as illustrated in Figure 11-2. The reference is assigned to the `sampleContact` variable as shown in Figure 11-3.

Figure 11-2 The method returns a reference to a `Contact` object

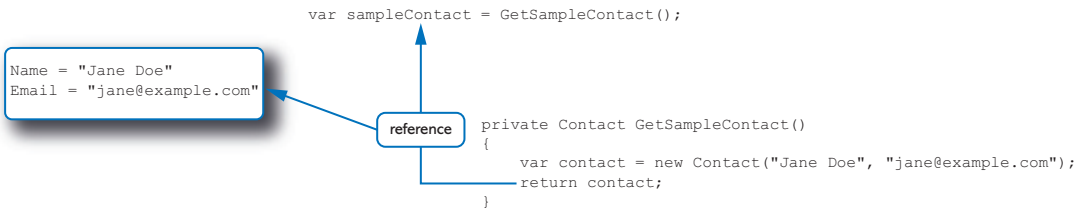
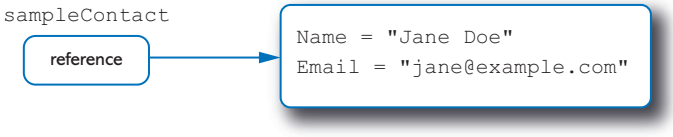
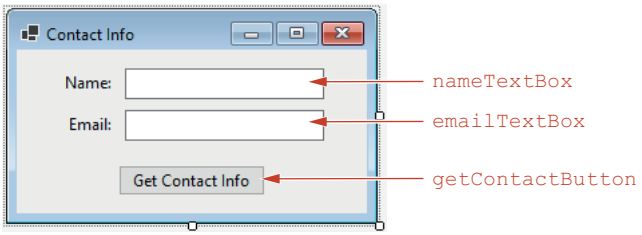


Figure 11-3 The `sampleContact` variable references the `Contact` object



Let's look at a complete application that demonstrates these concepts. In the *Chap11* folder of this book's Student Sample Programs, you will find a project named *ReturnObject*. Figure 11-4 shows the application's form in the *Designer*, and Program 11-1 shows the application's `Form1` code.

Figure 11-4 The *ReturnObject* application's form



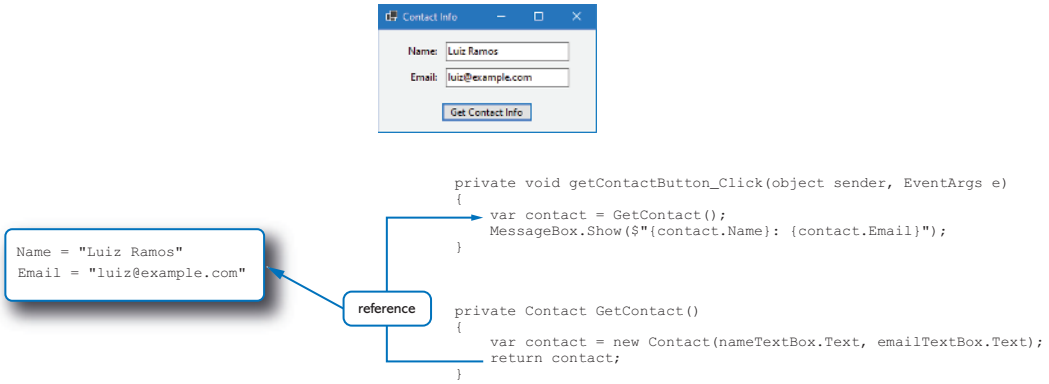
Program 11-1 Code for Form1 in the *ReturnObject* project

```
 1 namespace ReturnObject
 2 {
 3     public partial class Form1 : Form
 4     {
 5         public Form1()
 6         {
 7             InitializeComponent();
 8         }
 9
10         private void getContactButton_Click(object sender, EventArgs e)
11         {
12             var contact = GetContact();
13             MessageBox.Show($"{contact.Name}: {contact.Email}");
14         }
15
16         private Contact GetContact()
17         {
18            var contact = new Contact(nameTextBox.Text,
              emailTextBox.Text);
19            return contact;
20         }
21     }
22
23     class Contact
24     {
25         public Contact(string name, string email)
26         {
27             Name = name;
28             Email = email;
29         }
30
31         public string Name { get; set; }
32         public string Email { get; set; }
33
34     }
35 }
```

First, look at the `GetContact` method in lines 16–20. The statement in line 18 creates a `Contact` object, assigning the contents of `nameTextBox.Text` to the object's Name property and `emailTextBox.Text` to the Email property. Then, line 19 returns a reference to the object.

When you run the application, you can enter a person's name and email address into the TextBox controls, and then click the `getContactButton` control. In the button's Click event handler, line 12 calls the `GetContact` method and assigns the method's return value to a variable named `contact`. This process is illustrated in Figure 11-5. Then, a message box is displayed, showing the name and address that are stored in the object.

Figure 11-5 The `GetContact` method returns a reference to a `Contact` object



Checkpoint

11.1 Assume `Student` is the name of a class, and the following is a method header. What does the method return?

```
private Student GetStudent()
```

11.2 Writing a `ToString` Method

CONCEPT: **Most classes can benefit from having a `ToString` method which is implicitly called under certain circumstances. Typically, the method returns a string that represents the state of an object.**

When you write a class in C#, it is automatically given a method named `ToString`. The `ToString` method returns a string containing the class's fully qualified name. A **fully qualified name** is typically the name of the class's namespace, followed by a dot (.), followed by the name of the class. For example, if you look back at Program 11-1 in the previous section, you will see that the `Contact` class is in a namespace named `ReturnObject`. Therefore, the fully qualified name of the `Contact` class is `ReturnObject.Contact`.

As another example, suppose we have the following class for storing student information:

```
class Student
{
    // Constructor
    public Student(string id, string name)
    {
        ID = id;
        Name = name;
    }

    // Properties
    public string ID { get; set; }
    public string Name { get; set; }
}
```

Assume the `Student` class is in a namespace named `StudentRoster`. The following code creates an instance of the `Student` class and then calls that object's `ToString` method, displaying its return value in a message box:

```
var student = new Student("74566-CS", "Asha Singh");
MessageBox.Show(student.ToString());
```

The message box will display a string containing the class's fully qualified name, which is:

```
StudentRoster.Student
```

If you want a class's `ToString` method to return a different value, you can write a custom `ToString` method in the class. This is known as **overriding** the method. (You will learn more about overriding in Chapter 12.) When you override the `ToString` method, you use the following general format:

```
public override string ToString()
{
    // Body of the method appears here.
}
```

For example, we might want the `Student` class to have a `ToString` method that returns a string showing the current values of the object's properties, as shown here:

```
class Student
{
    // Constructor
    public Student(string id, string name)
    {
        ID = id;
        Name = name;
    }

    // Properties
    public string ID { get; set; }
    public string Name { get; set; }

    // ToString method
    public override string ToString()
    {
        return $"Student ID: {ID} Student Name: {Name}";
    }
}
```

The following code demonstrates:

```
var student = new Student("74566-CS", "Asha Singh");
MessageBox.Show(student.ToString());
```

When this code executes, the following text will be displayed in a message box:

```
Student ID: 74566-CS Student Name: Asha Singh
```

Implicitly Calling the `ToString` Method

An object's `ToString` method is automatically called any time the object is implicitly converted to a string. For example, if the string concatenation operator is used on an object, the object's `ToString` method will be called. The following code demonstrates:

```
var student = new Student("74566-CS", "Asha Singh");
MessageBox.Show("Student Info -> " + student);
```

In the last line of code, notice that we are using the string concatenation operator (`+`) on the `student` object. As a result, the `student` object's `ToString` method will be called, and the string that is returned will be used in the concatenation expression. When this code executes, the following text will be displayed in a message box:

```
Student Info -> Student ID: 74566-CS Student Name: Asha Singh
```

The `ToString` method is also called when an object is used in an interpolated string's placeholder expression. The following code demonstrates:

```
var student = new Student("74566-CS", "Asha Singh");
MessageBox.Show($"Student Info -> {student}");
```

When this code executes, the following text will be displayed in a message box:

```
Student Info -> Student ID: 74566-CS Student Name: Asha Singh
```

Tutorial 11-1 leads you through the process of adding a `ToString` method to an existing class. You will add a `ToString` method to the `CellPhone` class that was introduced in Chapter 10.

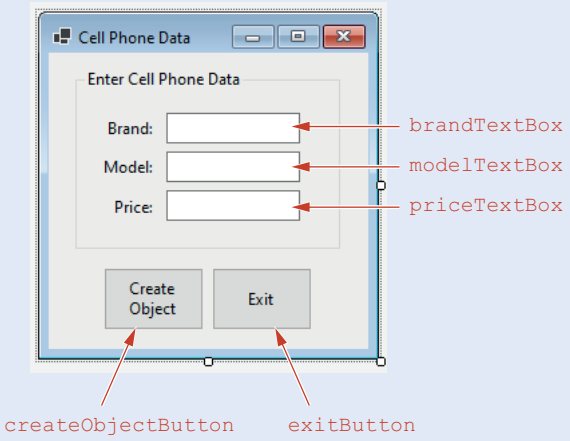
Tutorial 11-1:
Adding a `ToString` Method to a Class

VideoNote
Tutorial 11-1: Adding a `ToString` Method to a Class

In Chapter 10, Tutorial 10-2, you created a `CellPhone` class that holds data about a cell phone. The class has three properties (Brand, Model, and Price) and a constructor. In this tutorial you will add a `ToString` method to the class, and then you will complete an application that uses the `ToString` method to display a `CellPhone` object's data.

The project you will complete in this tutorial is named *CellPhoneData*. It is in the Chapter 11 folder of the Student Sample Programs. Figure 11-6 shows the application's form. The completed application will allow you to enter sample cell phone data into the text boxes and then click the *Create Object* button. The application will then create a `CellPhone` object and use the object's `ToString` method to display its data in a message box.

Figure 11-6 The *CellPhoneData* application's form



Step 1: Start Visual Studio. Open the project named *CellPhoneData* in the *Chap11* folder of the Student Sample Programs.

Step 2: Open the CellPhone.cs file in the code editor. Write the comment and the definition of the `ToString` method, which appears in lines 30–33 in Program 11-2. This method returns an interpolated string containing the values of the object's Brand, Model, and Price properties. Notice that the Price property is formatted as currency.

Step 4: Switch your view to the Form1 form in the *Designer* and double-click the `createObjectButton` control. This opens the code editor, and you will see an empty event handler named `createObjectButton_Click`. Complete the event handler by typing the code shown in lines 38–45 in Program 11-3. Let's look at the code:

Line 39: This statement creates a `CellPhone` object in memory, referenced by a variable named `myPhone`.

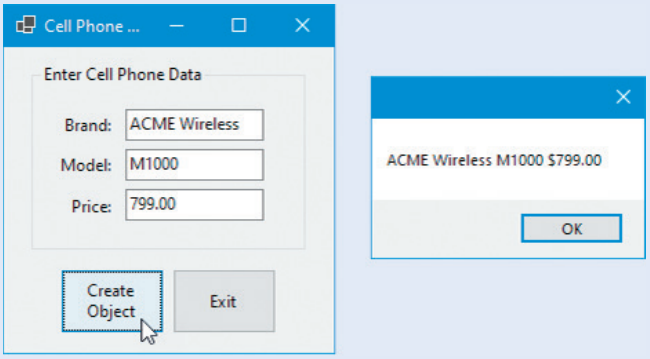
Line 42: This statement calls the `GetPhoneData` method, passing the `myPhone` object as an argument. (The `GetPhoneData` method is defined in lines 13–34.) After the method executes, the `myPhone` object's properties are set to the data entered by the user.

Line 45: This statement displays in a message box, the string that is returned from the `myPhone.ToString()` method.

Step 5: Switch your view back to the *Designer* and double-click the `exitButton` control. In the code editor you will see an empty event handler named `exitButton_Click`. Complete the event handler by typing the code shown in lines 50 and 51 in Program 11-3.

Step 6: Save the project. Then, press F5 on the keyboard or click the *Start Debugging* button (▶) on the toolbar to compile and run the application. When the application runs, enter some sample data into the TextBox controls and click the *Create Object* button. The application will use the data you entered to create a `CellPhone` object. It will then display, in a message box, the values of the object's Brand, Model, and Price properties. Figure 11-7 shows an example of the application's output. When you are ready to end the application, click the OK button to dismiss the message box, then click the *Exit* button.

Figure 11-7 Example output of the *CellPhoneData* application



Program 11-2 Completed code for the CellPhone.cs file in the *CellPhoneData* project

```
 1 using System;
 2 using System.Collections.Generic;
 3 using System.Diagnostics;
 4 using System.Linq;
 5 using System.Reflection;
 6 using System.Text;
 7 using System.Threading.Tasks;
 8
 9 namespace CellPhoneData
10 {
11     internal class CellPhone
12     {
13         // Constructor
14         public CellPhone()
15         {
```

```
            Brand = "";
            Model = "";
            Price = 0m;
        }

        // Brand property
        public string Brand { get; set; }

        // Model property
        public string Model { get; set; }

        // Price property
        public decimal Price { get; set; }

        public override string ToString()
        {
            return $"{Brand} {Model} {Price:c}";
        }
    }
}
```

Program 11-3 Completed code for Form1 in the *CellPhoneData* project

```
namespace Cell_Phone_Data
{
    public partial class Form1: Form
    {
        public Form1()
        {
            InitializeComponent();
        }

        // The GetPhoneData method accepts a CellPhone object
        // as an argument. It assigns the data entered by the
        // user to the object's properties.
        private void GetPhoneData(CellPhone phone)
        {
            // Temporary variable to hold the price.
            decimal price;

            // Get the phone's brand.
            phone.Brand = brandTextBox.Text;

            // Get the phone's model.
            phone.Model = modelTextBox.Text;

            // Get the phone's price.
            if (decimal.TryParse(priceTextBox.Text, out price))
            {
                phone.Price = price;
            }
            else
            {
                // Display an error message.
                MessageBox.Show("Invalid price");
            }
        }

        private void createObjectButton_Click(object sender,
        EventArgs e)
        {
```

```
38              // Create a CellPhone object.
39              var myPhone = new CellPhone();
40
41              // Get the phone data.
42              GetPhoneData(myPhone);
43
44              // Display the phone data.
45              MessageBox.Show(myPhone.ToString());
46          }
47
48          private void exitButton_Click(object sender, EventArgs e)
49          {
50              // Close the form.
51              this.Close();
52          }
53      }
54  }
```

Checkpoint

11.2 Assume the following class exists in a namespace named `Two_D_Space`:

```
class Point
{
    public int X { get; set; }
    public int Y { get; set; }

    public Point(int x, int y)
    {
        X = x;
        Y = y;
    }
}
```

After the following code executes, what value will be assigned to the `position` variable?

```
var point = new Point(199, 249);
string position = point.ToString();
```

11.3 Assume the following class exists in a namespace named `Two_D_Space`:

```
class Point
{
    public int X { get; set; }
    public int Y { get; set; }

    public Point(int x, int y)
    {
        X = x;
        Y = y;
    }

    public override string ToString()
    {
        return $"({X},{Y})";
    }
}
```

After the following code executes, what value will be assigned to the `position` variable?

```
var point = new Point(199, 249);
string position = point.ToString();
```

11.4 Assume the following class exists in a namespace named `Two_D_Space`:

```
class Point
{
    public int X { get; set; }
    public int Y { get; set; }

    public Point(int x, int y)
    {
        X = x;
        Y = y;
    }

    public override string ToString()
    {
        return $"({X},{Y})";
    }
}
```

After the following code executes, what value will be assigned to the `position` variable?

```
var point = new Point(199, 249);
string position = $"{point}";
```

11.3 Working with `null`

CONCEPT: **In C#, `null` is a special value that represents no value. When `null` is assigned to a reference variable, it indicates that the variable references no object.**

If a reference type variable is assigned the value null, it means the variable refers to no object. When a reference variable is set to `null`, you cannot use it to perform an operation that requires the existence of an object. In fact, there are only a few operations that can be performed with the value `null` without throwing an exception:

- You can initialize a reference type variable to the value `null`.
- You can assign `null` to an existing reference type variable.
- You can check a reference type variable to determine whether it is set to `null`.
- You can pass `null` as an argument to a reference type parameter in a method.
- You can write a method that returns `null`.

For example, assume the following `Contact` class exists in a program:

```
class Contact
{
    public Contact(string name, string email)
    {
        Name = name;
        Email = email;
    }

    public string Name { get; set; }
    public string Email { get; set; }
}
```

The following code demonstrates how a `null` reference can cause an error.

```
1 var contact = new Contact("Alysha", "alisha@example.com");
2 contact = null;
3 contact.Email = "admin@example.com";
```

After the statement in line 1 executes, the `contact` variable will reference a `Contact` object. But then, the statement in line 2 assigns `null` to the `contact` variable. After this statement executes, `contact` will no longer reference the object. Then, the statement

in line 3 attempts to use the contact variable to set the Email property. Because the contact variable is null, this statement will throw an exception.

The error that occurs in the previously shown code can be avoided if we make sure we do not assign null to an existing reference variable. However, there are more subtle ways that a reference variable can be set to null. For example, look at the following method definition:

```
private void DisplayContact(Contact contact)
{
    MessageBox.Show($"{contact.Name}: {contact.Email}");
}
```

This method accepts a reference to a Contact object as an argument and displays the object's Name and Email properties. However, the code will throw an exception if a null reference is passed as an argument to the method. To prevent this from happening we can add a simple if statement that makes sure the contact parameter is not set to null before we use it for any operations:

```
private void DisplayContact(Contact contact)
{
    if (contact != null)
        MessageBox.Show($"{contact.Name}: {contact.Email}");
}
```

It is also possible for a method to return a null reference. For example, the following method accepts a Contact object as its argument, converts the object's Name and Email properties to uppercase, and then returns a reference to the object. Can you see what happens if null is passed as an argument into the method?

```
1 private Contact Uppercase(Contact contact)
2 {
3    if (contact != null)
4    {
5        contact.Name = contact.Name.ToUpper();
6        contact.Email = contact.Email.ToUpper();
7    }
8
9    return contact;
10 }
```

The if statement in line 3 determines whether the contact parameter is not null. If it is not null, lines 5 and 6 convert the object's Name and Email properties to uppercase. Then, line 9 returns the value of the contact parameter. However, if contact is equal to null, the statements in lines 5 and 6 are skipped and line 9 returns the null reference. (In a nutshell, if null is passed as an argument into the method, the method will return null.)

This can potentially lead to problems. For example, in the following code assume that myContact is a Contact variable:

```
var ucaseContact = Uppercase(myContact);
MessageBox.Show($"{ucaseContact.Name}: {ucaseContact.Email}");
```

The first statement passes myContact to the Uppercase method. The method returns a reference that is assigned to the ucaseContact variable. Can you see what happens if myContact is set to null? The Uppercase method will return null, which means the ucaseContact variable will be assigned null. When the second statement executes, an exception will be thrown. To prevent this from happening we can add a simple if statement that makes sure the value returned from the Uppercase method is not null before we use it for any operations:

```
var ucaseContact = Uppercase(myContact);
if (ucaseContact != null)
    MessageBox.Show($"{ucaseContact.Name}: {ucaseContact.Email}");
```

Garbage Collection

When an object is no longer needed, it should be removed from memory so the memory can be used for other purposes. Fortunately, you do not have to do the work of removing objects from memory after you are finished using them. The .NET runtime system periodically performs a process known as **garbage collection**, which automatically removes unreferenced objects from memory. For example, look at the following code:

```
 1 // Create a Contact object.
 2 var contact1 = new Contact("Alysha", "alysha@example.com");
 3
 4 // Reference the same object with another variable.
 5 var contact2 = contact1;
 6
 7 // Nullify the contact1 variable.
 8 contact1 = null;
 9
10 // The object is still referenced by contact2. Nullify the
11 // contact2 variable so it no longer references the object.
12 contact2 = null;
```

Line 2 creates a `Contact` object, referenced by the `contact1` variable. Line 5 declares the `contact2` variable and initializes it with the value of the `contact1` variable. At this point, both the `contact1` and `contact2` variables reference the same object in memory. This is illustrated in Figure 11-8.

Line 8 assigns `null` to `contact1`. This removes the address of the object from the `contact1` variable, causing it to no longer reference the object. Figure 11-9 illustrates this. The object is still accessible, however, because it is referenced by the `contact2` variable. Then, the statement in line 12 assigns `null` to `contact2`. This removes the object's address from `contact2`, causing it to no longer reference the object. Figure 11-10 illustrates this. Because the object is no longer accessible, it will be removed from memory the next time the garbage collector process runs.

Figure 11-8 After line 2, both `contact1` and `contact2` reference the same object in memory

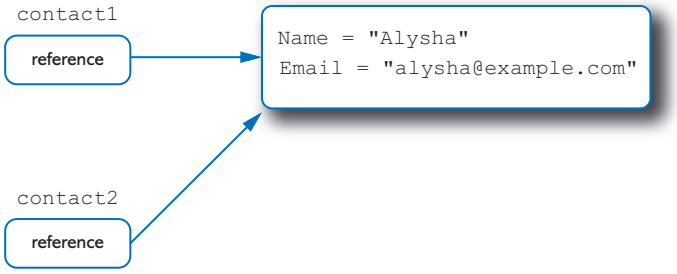


Figure 11-9 After line 8 only `contact2` references the object

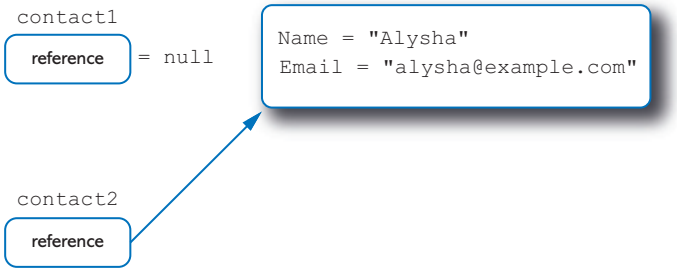
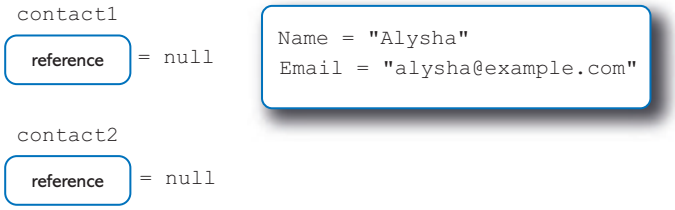


Figure 11-10 After line 12 the object is no longer referenced



Watch for `null` Strings as Fields or Properties

The `string` data type is a reference type. If a class has an uninitialized `string` as a field or property, it will be set to `null` by default. Because of this, you must be careful that you do not use a `null string` field to call a `string` method or access one of the `string`'s properties (such as Length). For example, look at the `FullName` class shown here:

```
 1 // This class is unsafe because the string properties are uninitialized.
 2 class FullName
 3 {
 4     public string FirstName { get; set; }
 5     public string MiddleName { get; set; }
 6     public string LastName { get; set; }
 7
 8     public int Length
 9     {
10         get
11         {
12             return FirstName.Length +
13                    MiddleName.Length +
14                    LastName.Length;
15         }
16     }
17
18     public override string ToString()
19     {
20         return $"{FirstName} {MiddleName} {LastName}";
21     }
22 }
```

The `FullName` class has three string properties: FirstName, MiddleName, and LastName. The properties are not initialized, and the class does not have a programmer-defined constructor. As a result, these properties will be set to the default value of `null` when an instance of the class is created. Notice that the class also has a Length property that returns the sum of the lengths of FirstName, MiddleName, and LastName. The property's `get` accessor will throw an exception if it is called while any of these `string` properties are set to `null`.

We can fix the class by initializing the `string` properties to an empty string in lines 4 through 6:

```
public string FirstName { get; set; } = "";
public string MiddleName { get; set; } = "";
public string LastName { get; set; } = "";
```

Another way to fix the class is to provide a constructor that assigns empty strings to the `string` properties:

```
public FullName()
{
    FirstName = "";
    MiddleName = "";
    LastName = "";
}
```

Nullable Value Types

By default, the value `null` can be assigned only to reference type variables. You cannot assign `null` to variables of value types such as `int`, `double`, and `decimal`. For example, the following code will cause an error at compile-time:

```
int total = null; // Error!
```

However, C# allows you to declare a **nullable value type** variable by appending a question mark to its data type. For example, the following statement declares a nullable `int` variable and initializes it with the value `null`:

```
int? total = null;
```

This statement declares a variable named `total` that can be assigned any `int` value, as well as `null`. Here is an example that declares and initializes a nullable `decimal` variable:

```
decimal? amount = null;
```

You can also make a `struct` nullable by appending a question mark to its structure name. For example, recall that `DateTime` is a .NET `struct`. The following statement declares a nullable `DateTime` structure named `anniversary`:

```
DateTime? anniversary = null;
```

Beyond these examples, we do not use nullable value types in this book. However, the following situation describes a simple way in which nullable value types might be used. Sometimes it is useful to return `null` from a method to indicate that an error has occurred. For example, consider the following method:

```
private double divide(double num1, double num2)
{
    return num1 / num2;
}
```

The `divide` method takes two arguments, `num1` and `num2`, and returns the result of `num1` divided by `num2`. However, an error will occur if `num2` is equal to zero because division by zero is not possible. In C#, an exception is thrown when an integer or a `decimal` is divided by zero. When a floating-point number is divided by zero, the result is either the special value `Infinity` or the special value `NaN` (not a number).

To prevent division by zero, we can modify the method to determine whether `num2` is equal to 0 before we perform the division operation. If `num2` is equal to 0, we simply return `null`. Here is the modified code:

```
1  private double? divide(double num1, double num2)
2  {
3      double? result;
4
5      if (num2 == 0)
6          result = null;
7      else
8          result = num1 / num2;
9      return result;
10 }
```

Notice that the method's return value is `double?`, and the data type of the local variable `result` is also `double?`. The following code demonstrates how to call the `divide` method and use its return value to determine whether an error has occurred.

```
double a = 9;
double b = 0;
double? quotient;
quotient = divide(a, b);
if (quotient == null)
    MessageBox.Show("Cannot divide by zero.");
else
    MessageBox.Show($"{a} divided by {b} is {quotient}.");
```

Checkpoint

11.5 Assume a reference variable refers to an object. If the variable is assigned `null`, does it still refer to the object?

11.6 If a reference variable is set to `null`, what happens when you try to use the variable to call a method or set a property?

11.7 If a class has an uninitialized field that is a `string`, what is the field's default value?

11.8 What does the .NET garbage collector do?

11.9 Will the following statement cause an error, or not?

```
int value = null;
```

11.10 Will the following statement cause an error, or not?

```
double? value = null;
```

11.4 The this Reference Variable

CONCEPT: **The `this` keyword is the name of a reference variable that an object can use to refer to itself. It is available to all nonstatic methods.**

The `this` keyword is the name of a reference variable that an object can use to refer to itself. To demonstrate, let's look at a modified version of the `Coin` class that was introduced in Chapter 10:

```
 1 class Coin
 2 {
 3     // Field to represent the side facing up;
 4     private string sideUp = "Heads";
 5
 6     // The toss method simulates tossing the coin.
 7     public void Toss()
 8     {
 9         // Create a Random object.
10         var rand = new Random();
11
12         // Use a random number to determine
13         // the side of the coin is facing up.
14         // 0 = Heads, 1 = Tails
15         if (rand.Next(2) == 0)
```

```
16          {
17              this.sideUp = "Heads";
18          }
19          else
20          {
21              this.sideUp = "Tails";
22          }
23      }
24
25      // The GetSideUp method returns the value
26      // of the sideUp field.
27      public string GetSideUp()
28      {
29          return this.sideUp;
30      }
31  }
```

Recall that the `Coin` class has a `sideUp` field that is set to either `"Heads"` or `"Tails"`. When the `Toss` method executes, it generates a random number to determine which value to assign to `sideUp`. Notice that line 17 assigns `"Heads"` to `this.sideUp`, and line 21 assigns `"Tails"` to `this.sideUp`. The `this` reference variable refers to the object that is calling the method, so the expression `this.sideUp` means *this object's `sideUp` field*.

The following `Contact` class shows an example that uses the `this` keyword to access the object's properties:

```
1  class Contact
2  {
3      public Contact(string name, string email)
4      {
5          this.Name = name;
6          this.Email = email;
7      }
8
9      public string Name { get; set; }
10     public string Email { get; set; }
11 }
```

This class has two properties, Name and Email, and a constructor that accepts arguments for the properties. Notice the use of `this` in lines 5 and 6. Line 5 assigns the `name` parameter to this object's Name property, and line 6 assigns the `email` parameter to this object's Email property.

Using `this` to Overcome Shadowing

One common use of the `this` keyword is to overcome the shadowing of a field name by a parameter name. If a method's parameter variable has the same name as a field in the same class, then the parameter name shadows the field name. This means that the field name is hidden by the name of the parameter variable. As a result, the field cannot be accessed within the method.

For example, look at the following `Counter` class. The class has a field named `value`, a constructor that assigns its argument to the `value` field, a method named `increment` that increments the `value` field, and a method named `getValue` that returns the value of the `value` field.

```
1  class Counter
2  {
3      private int value;
4
5      public Counter(int v)
```

```
 6      {
 7          value = v;
 8      }
 9
10      public void increment()
11      {
12          value++;
13      }
14
15      public int getValue()
16      {
17          return value;
18      }
19  }
```

Notice that the constructor uses a parameter named `v` to accept an argument that is assigned to the `value` field. Sometimes it is difficult (and even time-consuming) to think of a good parameter name that is different from a field name. To avoid this problem, many programmers give parameters the same names as the fields to which they correspond, and then use the `this` keyword to refer to the field names. For example, the `Counter` class's constructor could be rewritten as follows:

```
public Counter(int value)
{
    this.value = value;
}
```

Although the parameter name `value` shadows the field name `value`, the `this` keyword overcomes the shadowing. Because this is a reference to the calling object, the expression `this.value` refers to the calling object's `value` field.

Using `this` to Call an Overloaded Constructor from Another Constructor

You already know that a class's constructor is automatically called when an instance of the class is created. You cannot call a constructor explicitly, as you do other methods. However, there is an exception to this rule: You can use the `this` keyword to call one constructor from another constructor in the same class. For example, look at the following `Circle` class:

```
 1  class Circle
 2  {
 3      public Circle(double r)
 4      {
 5          Radius = r;
 6      }
 7
 8      public Circle(): this(1.0)
 9      {
10      }
11
12      public double Radius {get; set;}
13  }
```

The class has a Radius property and two constructors. The first constructor, in lines 3 through 6, accepts an argument that is assigned to the Radius property. The second constructor, in lines 8 through 10, is a parameterless constructor. Notice the following notation that appears at the end of the constructor header in line 8:

```
: this(1.0)
```

This notation calls the first constructor, passing the value 1.0 as an argument. As a result, any time the parameterless constructor is executed, the Radius property will be set to 1.0. The following statement demonstrates:

```
var circle = new Circle();
MessageBox.Show($"The circle's radius is {circle.Radius}.");
```

This code will display the message *The circle's radius is 1*.

As another example, look at the following version of the `Contact` class:

```
 1 class Contact
 2 {
 3     public Contact(string name, string email)
 4     {
 5         Name = name;
 6         Email = email;
 7     }
 8
 9     public Contact(string name): this(name, "")
10     {
11     }
12
13     public string Name { get; set; }
14     public string Email { get; set; }
15 }
```

This version of the `Contact` class has two constructors. The first constructor, in lines 3 through 7, has two parameters, `name` and `email`. The second constructor, in lines 9 through 11, has only one parameter, `name`. Notice that the second constructor simply calls the first constructor passing `name` and an empty string as arguments. As a result, any time you pass only one argument to the constructor, that argument will be assigned to the Name property and an empty string will be assigned to the Email property.

Checkpoint

11.11 Suppose the `Employee` class has two properties: Name and ID. Which of the following constructors correctly uses the `this` keyword?

a.
```
public Employee(string name, string id)
{
    Name = this.name;
    ID = this.id;
}
```

b.
```
public Employee(string name, string id)
{
    this.Name = name;
    this.ID = id;
}
```

c.
```
public Employee(string name, string id)
{
    this(Name) = name;
    this(ID) = id;
}
```

d.
```
public Employee(string name, string id)
{
    this:Name = name;
    this:ID = id;
}
```

11.5 Copying Objects

CONCEPT: **A copy constructor is a special constructor that copies the data from one object and stores that data in a new object.**

You cannot make a copy of an object with a simple assignment statement, as you would with a primitive variable. Once again, suppose we have the following `Contact` class in an application:

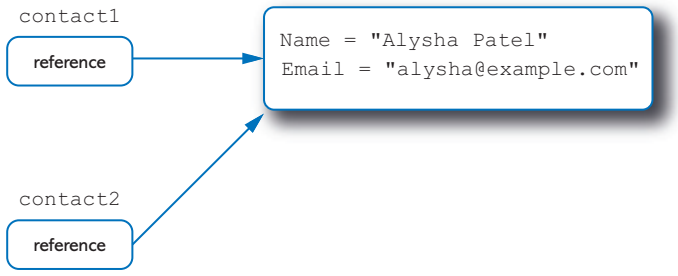
```
class Contact
{
    public Contact()
    {
        Name = "";
        Email = "";
    }

    public Contact(string name, string email)
    {
        Name = name;
        Email = email;
    }

    public string Name { get; set; }
    public string Email { get; set; }
}
```

Suppose that in the same application we also have the following code:

```
var contact1 = new Contact("Alysha Patel", "alysha@example.com");
var contact2 = contact1;
```

The first statement declares the `contact1` variable, referencing a new `Contact` object. The second statement declares the `contact2` variable and assigns it the value of `contact1`. This does not make a copy of the object referenced by `contact1`. Rather, it makes a copy of the reference that is stored in `contact1` and assigns that reference to `contact2`. After this statement executes, both the `contact1` and `contact2` variables will refer to the same object in memory. This is illustrated in Figure 11-11.

Figure 11-11 Both variables reference the same object



This type of assignment statement can lead to problems if you assume that you are making a copy of the object. When two reference variables refer to the same object, either variable can be used to change the object's contents. The following code demonstrates:

```
1 var contact1 = new Contact("Alysha Patel", "alysha@example.com");
2 var contact2 = contact1;
3
4 contact2.Name = "Jane Doe";
5 contact2.Email = "jane@example.com";
6
7 MessageBox.Show($"{contact1.Name}: {contact1.Email}");
8 MessageBox.Show($"{contact2.Name}: {contact2.Email}");
```

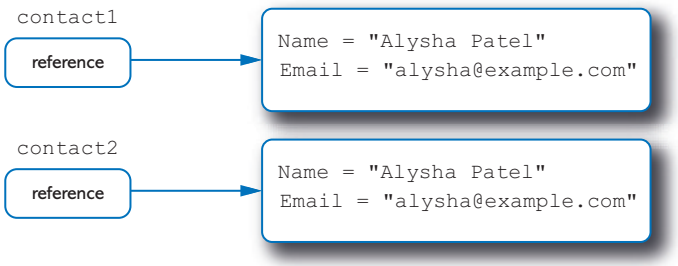
- Line 1 declares the `contact1` variable and initializes it with a new `Contact` object. The object's Name property is set to "Alysha Patel" and its Email property set to "alysha@example.com".
- Line 2 declares the `contact2` variable and initializes it with the `contact1` variable. After this statement executes, `contact1` and `contact2` will reference the same object in memory.
- Line 4 uses the `contact2` variable to change the object's Name property to "Jane Doe".
- Line 5 uses the `contact2` variable to change the object's Email property to "jane@example.com".
- Line 7 uses the `contact1` variable to display the object's Name and Email properties. The message box will display "Jane Doe: jane@example.com".
- Line 8 uses the `contact2` variable to display the object's Name and Email properties. Because `contact2` refers to the same object as `contact1`, this message box will also display "Jane Doe: jane@example.com".

If you want to make a copy of an object, you must create a new object and then set the new object's fields and properties to the same values as the fields and properties of the object that is being copied. Here is an example:

```
var contact1 = new Contact("Alysha Patel", "alysha@example.com");
var contact2 = new Contact();
contact2.Name = contact1.Name;
contact2.Email = contact1.Email;
```

After this code executes, `contact1` and `contact2` will refer to two different objects in memory. The two object's properties will be set to the same values. This is shown in Figure 11-12.

Figure 11-12 The variables refer to different objects



Copy Constructors

A copy constructor is a constructor that accepts an object of the same class as an argument. It makes the object that is being created a copy of the object that was passed as an argument. For example, look at the following version of the `Contact` class:

```
1  class Contact
2  {
3      // Constructor
4      public Contact()
5      {
6          Name = "";
7          Email = "";
8      }
9
10     // Constructor
11     public Contact(string name, string email)
12     {
13         Name = name;
14         Email = email;
15     }
16
```

```
17      // Copy constructor
18      public Contact(Contact object2)
19      {
20          Name = object2.Name;
21          Email = object2.Email;
22      }
23
24      // Properties
25      public string Name { get; set; }
26      public string Email { get; set; }
27  }
```

This version of the class has a copy constructor, shown in lines 18 through 22. Notice that the constructor accepts a `Contact` object as an argument. The parameter variable `object2` will reference the object that was passed as an argument. The constructor copies the values that are in `object2`'s Name and Email properties to the Name and Email properties of the object that is being created.

The following code demonstrates the copy constructor. It creates a `Contact` object referenced by the variable `contact1`. Then it creates another `Contact` object referenced by the variable `contact2`. The object referenced by `contact2` is a copy of the object referenced by `contact1`.

```
var contact1 = new Contact("Alysha Patel", "alysha@example.com");
var contact2 = new Contact(contact1);
```

Tutorial 11-2 leads you through the process of adding a copy constructor to an existing class. You will add a copy constructor to the `CellPhone` class that you modified in Tutorial 11-1.

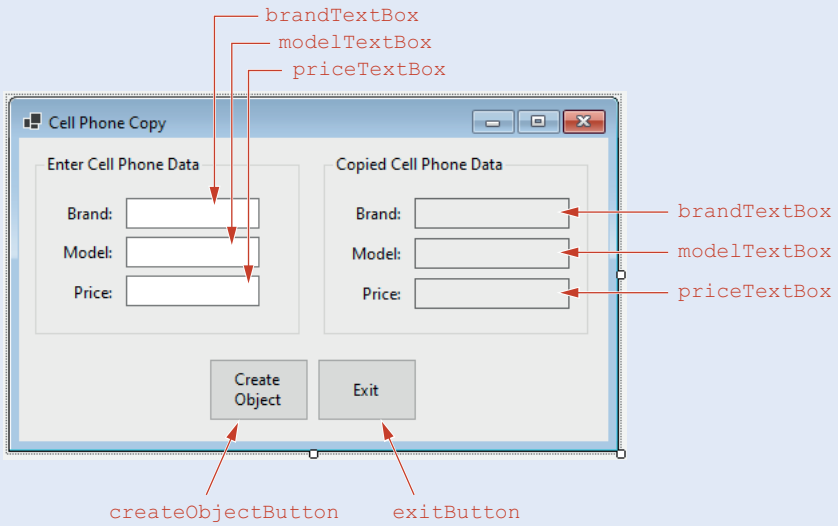
Tutorial 11-2:
Writing a Copy Constructor

VideoNote
Tutorial 11-2: Writing a Copy Constructor

In Tutorial 11-1 you added a `ToString` method to the `CellPhone` class. In this tutorial you will add a copy constructor to the class, and then you will complete an application that uses the copy constructor to make a copy of a `CellPhone` object.

The project you will complete in this tutorial is named *CellPhoneCopy*. It is in the *Chap11* folder of the Student Sample Programs. Figure 11-13 shows the application's form. At run time, you enter sample cell phone data into the text boxes and then click the *Create Object* button. The application creates a `CellPhone` object and then uses the object's copy constructor to create a copy. The application displays the properties of the copied object on the form.

Figure 11-13 The *CellPhoneCopy* project's form



Step 1: Start Visual Studio. Open the project named *CellPhoneCopy* in the *Chap11* folder of the Student Sample Programs.

Step 2: Open the CellPhone.cs file in the code editor. Write the comment and the definition of the copy constructor, which appears in lines 19–25 in Program 11-4. The copy constructor accepts a `CellPhone` object as its argument. It assigns that object's Brand, Model, and Price properties to the corresponding properties of the object that is being created.

Step 4: Switch your view to the Form1 form in the *Designer* and double-click the `createObjectButton` control. This opens the code editor, and you will see an empty event handler named `createObjectButton_Click`. Complete the event handler by typing the code shown in lines 38–50 in Program 11-5. Let's look at the code:

Line 39: This statement creates a `CellPhone` object in memory, referenced by a variable named `myPhone`.

Line 42: This statement calls the `GetPhoneData` method, passing the `myPhone` object as an argument. (The `GetPhoneData` method is defined in lines 13–34.) After the method executes, the `myPhone` object's properties are set to the data entered by the user.

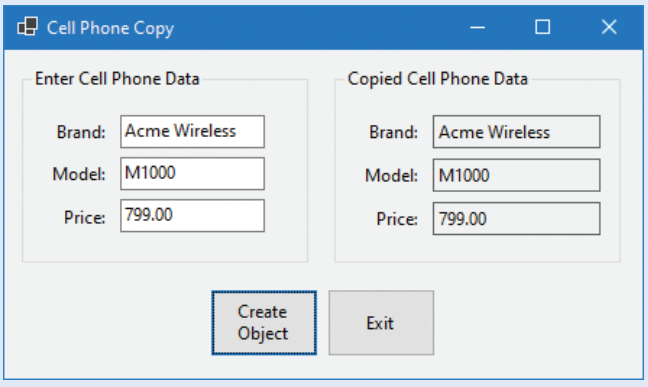
Line 45: This statement creates a second `CellPhone` object, calling the `CellPhone` class's copy constructor. The `myPhone` object is passed as an argument to the copy constructor. A copy of the `myPhone` object is created and referenced by the variable `phone2`.

Lines 48–50: These statements display the `phone2` object's Brand, Model, and Price properties in the `brandLabel`, `modelLabel`, and `priceLabel` controls.

Step 5: Switch your view back to the *Designer* and double-click the `exitButton` control. In the code editor you will see an empty event handler named `exitButton_Click`. Complete the event handler by typing the code shown in lines 55 and 56 in Program 11-5.

Step 6: Save the project. Then, press F5 on the keyboard or click the *Start Debugging* button ( ) on the toolbar to compile and run the application. When the application runs, enter some sample data into the TextBox controls and click the *Create Object* button. The application will use the data you entered to create a `CellPhone` object. It will then create a copy of the CellPhone object, and display the copy's Brand, Model, and Price properties. Figure 11-14 shows an example of the application's output. When you are ready to end the application, click the *OK* button to dismiss the message box, then click the *Exit* button.

Figure 11-14 Example output of the *CellPhoneCopy* application



Program 11-4 Completed code for the CellPhone.cs file in the *CellPhoneCopy* application

```
 1 using System;
 2 using System.Collections.Generic;
 3 using System.Linq;
 4 using System.Text;
 5 using System.Threading.Tasks;
 6
 7 namespace CellPhoneCopy
 8 {
 9     internal class CellPhone
10     {
11         // Constructor
12         public CellPhone()
13         {
14             Brand = "";
15             Model = "";
16             Price = 0m;
17         }
18
19         // Copy constructor
20         public CellPhone(CellPhone object2)
21         {
22             Brand = object2.Brand;
23             Model = object2.Model;
24             Price = object2.Price;
25         }
26
27         // Brand property
28         public string Brand { get; set; }
29
30         // Model property
31         public string Model { get; set; }
32
33         // Price property
34         public decimal Price { get; set; }
35
36         // ToString method
37         public override string ToString()
38         {
39             return $"{Brand} {Model} {Price:c}";
40         }
41     }
42 }
```

Program 11-5 Completed code for Form1 in the *CellPhoneCopy* application

```
 1 namespace Cell_Phone_Copy
 2 {
 3     public partial class Form1 : Form
 4     {
 5         public Form1()
 6         {
 7             InitializeComponent();
 8         }
 9
10         // The GetPhoneData method accepts a CellPhone object
11         // as an argument. It assigns the data entered by the
12         // user to the object's properties.
```

```
13          private void GetPhoneData(CellPhone phone)
14          {
15              // Temporary variable to hold the price.
16              decimal price;
17
18              // Get the phone's brand.
19              phone.Brand = brandTextBox.Text;
20
21              // Get the phone's model.
22              phone.Model = modelTextBox.Text;
23
24              // Get the phone's price.
25              if (decimal.TryParse(priceTextBox.Text, out price))
26              {
27                  phone.Price = price;
28              }
29              else
30              {
31                  // Display an error message.
32                  MessageBox.Show("Invalid price");
33              }
34          }
35
36          private void createObjectButton_Click(object sender,
            EventArgs e)
37          {
38              // Create a CellPhone object.
39              var myPhone = new CellPhone();
40
41              // Get the phone data.
42              GetPhoneData(myPhone);
43
44              // Make a copy of myPhone.
45              var phone2 = new CellPhone(myPhone);
46
47              // Display the phone data.
48              brandLabel.Text = phone2.Brand;
49              modelLabel.Text = phone2.Model;
50              priceLabel.Text = phone2.Price.ToString();
51          }
52
53          private void exitButton_Click(object sender, EventArgs e)
54          {
55              // Close the form.
56              this.Close();
57          }
58      }
59  }
```

Checkpoint

11.12 Assume `Employee` is a class. After the following code executes, will `employee1` and `employee2` reference two different objects in memory, or the same object?

```
var employee1 = new Employee();
var employee2 = employee1;
```

11.13 Assume `Person` is a class with two properties: FirstName and LastName. Which of the following constructors is a copy constructor?

a.
```
public Person(string first, string last)
{
    FirstName = first;
    LastName = last;
}
```

b.
```
public Person(Person object2)
{
    FirstName = object2.FirstName;
    LastName = object2.LastName;
}
```

11.6 Object Initializer Syntax and Init-Only Properties

CONCEPT: **Object initializer syntax can be used to initialize an object's properties without requiring the use of a constructor. An init-only property can be assigned a value only in the property's declaration statement, by a constructor, or with an object initializer.**

Look at the following `Circle` class:

```
1 class Circle
2 {
3     public int CenterX { get; set; }
4     public int CenterY { get; set; }
5 }
```

This class has two properties, CenterX and CenterY. Suppose you want to create an instance of this class and initialize its properties with the values 100 and 50, respectively. First, you create the instance of the class, and then you set the individual properties, as shown here:

```
1 var circ = new Circle();
2 circ.CenterX = 100;
3 circ.CenterY = 50;
```

Code such as this is common because quite often you know the values that you want to store in an object's properties at the time you create the object. Because this type of code is common, C# provides a syntax that you can use to initialize an object's properties at the time you create the object. The syntax is known as object initializer syntax. Here is an example of how you can use the syntax to create an instance of the `Circle` class and initialize its properties:

```
var circ = new Circle() { CenterX = 100, CenterY = 50 };
```

This statement creates an instance of the `Circle` class named `circ`. Notice the assignment statements inside the curly braces. These are known as **object initializers**. The object's CenterX property is initialized with the value 100, and the CenterY property is initialized with the value 50.

The previous example shows an object initialization statement that is written on one line of code. Quite often, they are written across multiple lines, as shown here:

```
var circ = new Circle()
{
    CenterX = 100,
    CenterY = 50
};
```

When object initialization syntax is used to instantiate a class, the class's constructor is always called first. Then, the properties are initialized as specified by the object initializers. In the previous example, the `Circle` class didn't have a programmer-defined constructor, so the default constructor was called. Then, the properties were initialized.

Let's look at another example. Suppose we have the following `Person` class:

```
 1 internal class Person
 2 {
 3     public Person()
 4     {
 5         Name = "";
 6         Email = "";
 7     }
 8
 9     public string Name { get; set; }
10     public string Email { get; set; }
11 }
```

The `Person` class has two properties, Name and Email. It also has a parameterless constructor that initializes both properties to empty strings. Here is an example of how you can use object initialization syntax to create an instance of the class and initialize its properties:

```
var person = new Person()
{
    Name = "Ravi Burman",
    Email = "ravi@example.com"
};
```

When this code executes, the `Person` class's constructor is called first. It initializes the Name and Email properties with empty strings. Then, the object initializers are applied, and the object's Name property is assigned "Ravi Burman" and the Email property is assigned "ravi@example.com."

Let's look at one more example. Suppose we have the following `Person` class:

```
 1 class Person
 2 {
 3     public Person (string n, string e)
 4     {
 5         Name = n;
 6         Email = e;
 7         PhoneNumber = "";
 8     }
 9
10     public string Name { get; set; }
11     public string Email { get; set; }
12     public string PhoneNumber { get; set; }
13 }
```

This class has three properties: Name, Email, and PhoneNumber. The constructor, however, has only two parameters: `n` and `e`. These parameters are assigned to the Name and Email properties in lines 5 and 6. Line 7 assigns an empty string to the PhoneNumber property. So, when we instantiate the class, we can use the constructor to set values for Name and Email, but not PhoneNumber. To set the PhoneNumber property, we have to write a separate statement, as shown in this example:

```
var person = new Person("Ravi Burman", "ravi@example.com");
person.PhoneNumber = "555-1212";
```

Alternatively, we could apply an object initializer to the PhoneNumber property, like this:

```
Person person = new Person("Ravi Burman", "ravi@example.com")
{
    PhoneNumber = "555-1212"
};
```

After this statement executes, the person object's Name property is initialized with the string `"Ravi Burman"`, the Email property is initialized with the string `"ravi@example.com"`, and the PhoneNumber property is assigned `"555-1212"`.

Init-Only Properties

In Chapter 10 you learned that to make a property read-only, you leave out the property's `set` accessor. (If the property has an explicitly defined backing field, you also declare the backing field as `readonly`.) If you leave out the `set` accessor for a property, however, you will not be able to use object initialization syntax to initialize that property. To use object initialization syntax on a property, the property must be settable.

To make object initialization syntax possible with read-only properties, C# provides the **`init` accessor**. An `init` accessor is like a `set` accessor, but it allows a property to be assigned a value only during the construction phase of the object. This means the property can be initialized in its declaration statement, by a constructor, or by an object initializer. When a property has an `init` accessor, it is known as an **init-only property**.

Here is a version of the `Circle` class that uses init-only properties:

```
1 class Circle
2 {
3     public int CenterX { get; init; }
4     public int CenterY { get; init; }
5 }
```

We can now use object initialization syntax to create an instance of the `Circle` class and initialize its properties:

```
var circ = new Circle() { CenterX = 100, CenterY = 50 };
```

However, a compiler error will occur if we try to change the value of the object's CenterX or CenterY properties, as shown here:

```
var circ = new Circle() { CenterX = 100, CenterY = 50 };
circ.CenterX = 0; // ERROR! Property cannot be set.
```

The second statement in this example will cause a compiler error because we cannot change the value of an object's init-only properties after the object has been created.

NOTE: If a property has an `init` accessor, it cannot also have a `set` accessor.

Here is another example:

```
1 internal class Person
2 {
3     public Person()
4     {
5         Name = "";
6         Email = "";
7     }
8
```

```
 9      public string Name { get; init; }
10      public string Email { get; init; }
11  }
```

This version of the `Person` class has two init-only properties, Name and Email. It also has a parameterless constructor that initializes both properties to empty strings. Here is an example of how you can use object initialization syntax to create an instance of the class and initialize its properties:

```
var person = new Person()
{
    Name = "Carmen Martina",
    Email = "carmen@example.com"
};
```

When this code executes, the `Person` class's constructor is called first. It initializes the Name and Email properties with empty strings. Then, the object initializers are applied, and the object's Name property is assigned "Carmen Martina" and the Email property is assigned "carmen@example.com."

Using `init` with Fully Defined Properties

The previous examples have demonstrated how the `init` keyword can be used with auto-properties. You can also use `init` with fully defined properties that have an explicitly declared backing field. Simply substitute the `init` keyword for the `set` keyword. Here is an example:

```
 1  class Pet
 2  {
 3      // Backing field for the Name property
 4      private readonly string _name;
 5
 6      // Constructor
 7      public Pet()
 8      {
 9         Name = "";
10      }
11
12      // Name property
13      public string Name
14      {
15         get
16         {
17             return _name;
18         }
19         init
20         {
21            _name = value;
22         }
23      }
24  }
```

The `Pet` class has an init-only Name property that is fully defined with a `readonly` backing field. Notice that the `init` accessor, in lines 19–22, is written just like a `set` accessor, except the keyword `init` is used instead of `set`. We can use object initialization syntax to create an instance of the class and initialize its Name property, as shown here:

```
var pet = new Pet() { Name = "Whiskers" };
```

When this statement executes, the `Pet` class's parameterless constructor is called first. It assigns an empty string to the Name property. Then, the object initializer assigns "Whiskers" to the Name property.

Checkpoint

11.14 The following class has been defined in an application:

```
class Point
{
    public int X { get; set; }
    public int Y { get; set; }
}
```

Rewrite the following code as a single statement that uses object initialization syntax:

```
var point = new Point();
point.X = 5;
point.Y = 10;
```

11.15 The following class has been defined in an application:

```
class Person
{
    public Person()
    {
        Name = "unknown";
    }

    public string Name { get; set; }
}
```

When the following code executes, what will be displayed in the message box?

```
var person = new Person() { Name = "JoJo" };
MessageBox.Show(person.Name);
```

a. `unknown`

b. `JoJo`

c. `null`

d. *an empty string*

11.16 The following class has been defined in an application:

```
class Person
{
    public string Name { get; }
    public string Email { get; }
}
```

Why will the following statement cause an error?

```
var person = new Person() { Name = "JoJo", Email = "jojo@example.com" };
```

11.17 Rewrite the following class so the Name and Email properties are init-only:

```
class Person
{
    public string Name { get; }
    public string Email { get; }
}
```

11.7 Mutable and Immutable Classes

CONCEPT: **A mutable class allows the values of its properties and/or fields to be changed. An immutable class does not allow the value of its properties and/or fields to be changed after they have been initialized.**

A class that allows its properties and/or fields to be changed is known as a **mutable class**. A **mutable object** is an instance of a mutable class. An **immutable class** is a class that does not allow its properties and/or fields to be changed once they are initialized with values. An **immutable object** is an instance of an immutable class. The internal state of an immutable object cannot be changed once the object is created.

Safety is the primary reason for writing immutable classes. If a program contains more than one reference to a mutable object, all those references can potentially modify the object. This can lead to bugs that are difficult to find. An immutable object, however, cannot be changed, so it is safe to have multiple variables referencing it. For that reason, many programmers prefer to make their classes immutable whenever they can. If you want to be sure that a C# class is immutable, you should follow these guidelines:

- Make all properties init-only or read-only.
- Make all fields `readonly`. This includes the backing fields for any properties that are not auto-properties.
- The class should initialize all its fields and properties. This can be done either with an initialization value in the declaration statement, or with a constructor.
- The class should not have a method or `get` accessor that returns a reference to a mutable field or property. Instead, the method or `get` accessor should make a copy of the mutable field or property and return a reference to the copy. Any changes that are made to the copy will not affect the original object.
- If a constructor, or any other method in the class, accepts a reference to a mutable object, that reference should not be assigned to a field or property. Instead, the constructor should make a copy of the mutable object and store a reference to the copy.

Throughout this chapter we have looked at various versions of the `Contact` class. Now let's look at an immutable version of the `Contact` class:

```
 1 class Contact
 2 {
 3     // Constructor
 4     public Contact(string name, string email)
 5     {
 6         Name = name;
 7         Email = email;
 8     }
 9
10     // Properties
11     public string Name { get; init; }
12     public string Email { get; init; }
13 }
```

Notice the following things about the class:

- The auto-properties in lines 11–12 are init-only.
- The class has one constructor, shown in lines 6–10. The constructor accepts arguments that are assigned to the Name and Email properties. Because this is the class's only constructor, you *must* use it when creating a `Contact` object.

When you create an instance of the immutable `Contact` class, you must pass the initial values for the object's properties to the constructor. Once the object is created and initialized, its properties cannot be changed.

The following example is an immutable class with a field, and no properties:

```
1 class Circle
2 {
3     private readonly double radius;
4
5     public Circle(double r)
6     {
7         radius = r;
8     }
9
10    public double getRadius()
11    {
12        return radius;
13    }
14
15    public double getDiameter()
16    {
17        return radius * 2;
18    }
19 }
```

The `Circle` class has a `readonly` field named `radius` (line 3) and a parameterized constructor (lines 5–8). The constructor accepts an argument that is assigned to the `radius` field. Recall that a `readonly` field can be assigned a value only in its declaration statement or in the class constructor. When you create an instance of the immutable `Circle` class, you must pass to the constructor an initial value for the object's `radius` field. Once the object is created and initialized, the `radius` field cannot be changed.

Here is an example of an immutable class that has a property with a backing field:

```
1 class Pet
2 {
3     // Backing field for the Name property.
4     private readonly string _name;
5
6     // Constructor
7     public Pet(string n)
8     {
9         _name = n;
10    }
11
12    // Name property
13    public string Name
14    {
15        get { return _name; }
16    }
17 }
```

The `Pet` class has a Name property with a `readonly` backing field named `_name`. The Name property is implemented as a read-only property because it does not have a set accessor. When you create an instance of the `Pet` class, you must pass to the constructor an initial value for the Name property. Once the object is created and initialized, the Name property cannot be changed.

Here is an example of an immutable class that has init-only auto-properties, and no constructor:

```
1 class Employee
2 {
3     public string Name { get; init; } = "";
4     public string IdNumber { get; init; } = "";
5 }
```

When you create an instance of the immutable `Employee` class, you can use object initializer syntax to initialize the properties, as shown here:

```
var employee = new Employee()
{
    Name = "Joanne Manchester",
    IdNumber = "701"
};
```

Once the object is created and initialized, the Name and IdNumber properties cannot be changed.

Immutable Classes with Mutable Objects as Fields or Properties

When working with reference variables, it's important to remember what they contain. A reference variable does not contain an object, but rather a reference to an object. A reference is similar to a memory address, so a reference variable indicates where in memory an object is located. Often we say that a reference variable points to an object.

A class can have fields and properties that are references. When you declare a reference field as `readonly`, and then initialize it with an object reference, it means the field cannot refer to any other object. Similarly, once a read-only reference property is initialized with an object reference, it cannot refer to any other object. For example, look at the following class:

```
1 class Person
2 {
3     private readonly string name;
4
5     public Person(string n)
6     {
7         name = n;
8     }
9
10    public void ChangeName(string n)
11    {
12        name = n; // This is an ERROR.
13    }
14
15    public string GetName()
16    {
17        return name;
18    }
19 }
```

Line 3 declares `name` as a `readonly string` field. The field is initialized by the constructor in line 7. Once the constructor has initialized the field, it cannot be changed during the life of the object. But then, inside the `ChangeName` method, line 12 tries to assign a new value to the `name` field. This line will cause a compiler error because the `name` field is `readonly` and cannot be changed after the constructor has initialized it.

Although the contents of a `readonly` reference field or a read-only reference property cannot be changed, the contents of the object that the field or property refers to can be changed, if that object is mutable. Let's see an example. First, look at the following mutable `City` class that holds information about a city:

```
1 // Mutable City Class
2 class City
3 {
4     // Constructor
5     public City(string name, double pop)
6     {
```

```
7         Name = name;
8         Population = pop;
9     }
10
11    // Copy constructor
12    public City(City object2)
13    {
14        Name =object2.Name;
15        Population = object2.Population;
16    }
17
18    // Properties
19    public string Name { get; set; }
20    public double Population { get; set; }
21 }
```

The `City` class has two auto-properties: Name and Population. Notice that both properties are mutable because they each have a `set` accessor. The class also has two constructors: a parameterized constructor that accepts arguments for name and population, as well as a copy constructor.

Now look at the following `Country` class that holds data about a country. This class is supposed to be immutable, but it has some problems.

```
1 // Country class - not so immutable
2 class Country
3 {
4     // Constructor
5     public Country(string name, City cap)
6     {
7         Name = name;
8         Capital = cap;
9     }
10
11    // Copy constructor
12    public Country(Country object2)
13    {
14        Name = object2.Name;
15        Capital = object2.Capital;
16    }
17
18    // Properties
19    public string Name { get; init; }
20    public City Capital { get; init; }
21 }
```

The class has two init-only auto-properties: Name and Capital. Notice in line 20 that the Capital property is a `City` reference. The parameterized constructor in lines 5–9, accepts two arguments: `name`, which is a `string`, and `cap`, which is a reference to a `City` object. Line 7 assigns `name` to the Name property and line 8 assigns `cap` to the Capital property.

The first reason that the `Country` class is not truly immutable because the Capital property returns a reference to a `City` object, and the `City` object is mutable. That means code outside the class can get a reference to the Capital property's object and make changes to it. Here is an example:

```
1 // Create a Country object.
2 var country = new Country("Norway", new City("Oslo", 681000.0));
3
4 // Display the country's info.
5 MessageBox.Show($"The capital of {country.Name} is {country.
  Capital.Name}.");
6
7 // Get the City object for the capital.
```

```
 8 var capCity = country.Capital;
 9
10 // Change the capital city's name.
11 capCity.Name = "Bergen";
12
13 // Display the country's info again.
14 MessageBox.Show($"The capital of {country.Name} is {country.
   Capital.Name}.");
```

Let's take a closer look at the code. The statement in line 2 creates a `Country` object, referenced by the `country` variable. The object's properties are initialized with the following data:

- The Name property is set to "Norway".
- The Capital property is initialized with a `City` object with its Name property set to "Oslo" and its Population property set to 681000.0.

The statement in line 5 displays the name of the country and the name of its capital city in a message box. The message box will display the following string: "The capital of Norway is Oslo."

Then, line 8 assigns the `country` object's Capital property to a variable named `capCity`. Remember, the Capital property is a reference to a `City` object. After this statement executes, the `country` object's Capital property *and* the `capCity` variable will both reference the same object in memory! Figure 11-15 shows the references. This is a problem because the `capCity` variable can now be used to change the contents of the `country` object's Capital property.

That is precisely what happens in line 11. The statement assigns the string `"Bergen"` to the `capCity` object's Name property. After this statement executes, the `country` object's Capital.Name property will also equal "Bergen". This is shown in Figure 11-16.

The statement in line 14 once again displays the name of the country and the name of its capital city in a message box. This time, the message box displays the following string: "The capital of Norway is Bergen."

Figure 11-15 `capCity` references the same object as the `country` object's Capital property

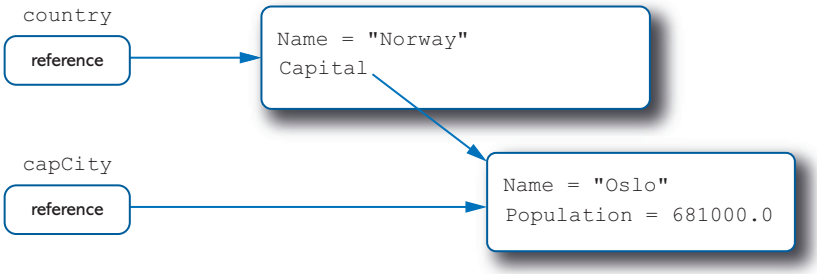
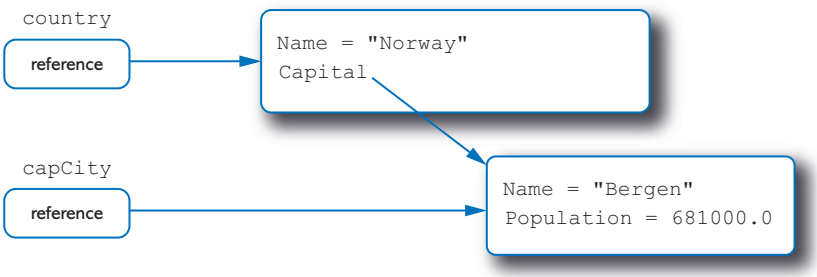


Figure 11-16 The `country` object's Capital.Name property has been changed



There is also a problem with the `Country` class's parameterized constructor. For your convenience, here is the constructor's code:

```
4 // Constructor
5 public Country(string name, City cap)
6 {
7     Name = name;
8     Capital = cap;
9 }
```

When this constructor is called, a reference to an existing `City` object will be passed as an argument into the `cap` parameter. In line 8, the `cap` parameter is assigned to the Capital property of the object that is being constructed. This can cause a problem because the Capital property will now reference a mutable object that was created by code outside the class. If the outside code still has a reference to the object, it can use that reference to make changes to the object.

There is also a problem with the `Country` class's copy constructor. For your convenience, here is the copy constructor's code:

```
11 // Copy constructor
12 public Country(Country object2)
13 {
14     Name = object2.Name;
15     Capital = object2.Capital;
16 }
```

In line 15, the Capital property of the object that being copied is assigned to the Capital property of the object that is being constructed. After the copy constructor executes, the Capital properties of both the object being copied and the object being constructed will reference the same object in memory. As a result, any changes made by one `Country` object will affect the other.

How do we fix the `Country` class so it is truly immutable? One solution is to rewrite the `City` class so it is immutable. It is safe for a program to have multiple references to an immutable object because none of the references can be used to change the object. But assuming we cannot rewrite the `City` class, our solution is to rewrite the `Country` class so it adheres to the following guidelines:

- The class should not have a method, or `get` accessor, that returns a reference to a mutable field or property. Instead, the method or `get` accessor should make a copy of the mutable field or property and return a reference to the copy. Any changes that are made to the copy will not affect the original object.
- If a constructor, or any other method in the class, accepts a reference to a mutable object, that reference should not be assigned to a field or property. Instead, the constructor should make a copy of the mutable object and store a reference to the copy.

Here is the corrected version of the `Country` class:

```
 1 // Immutable Country Class
 2 class Country
 3 {
 4     // Backing field for the Capital property
 5     private readonly City _capital;
 6
 7     // Constructor
 8     public Country(string name, City cap)
 9     {
10         Name = name;
11         _capital = new City(cap);
12     }
13
```

```
14      // Copy constructor
15      public Country(Country object2)
16      {
17          Name = object2.Name;
18          _capital = new City(object2.Capital);
19      }
20
21      // Name property
22      public string Name { get; init; }
23
24      // Capital property
25      public City Capital
26      {
27          get { return new City(_capital); }
28      }
29  }
```

First, we need to make sure the Capital property's `get` accessor does not simply return a reference to the property's underlying `City` object. Instead, it must return a *copy* of the `City` object. So, we rewrote the Capital property and provided our own code in the `get` accessor. This meant we also had to explicitly declare a backing field for the property. Line 5 declares the `_capital` field as `private` and `readonly`. Then, the code for the Capital property appears in lines 25–28. Notice that the `get` accessor, in line 27, does not simply return the `_capital` field, but it uses the `City` class's copy constructor to make a copy of the object that `_capital` refers to. It then returns a reference to the copy.

Next, the parameterized constructor in lines 8–12 has been rewritten. Line 10 assigns the `name` parameter to the Name property. Notice, however, that line 11 does not simply assign the `cap` parameter to the `_capital` field. Instead, it uses the `City` class's copy constructor to make a copy of the object that `cap` refers to. A reference to the copy is assigned to the `_capital` field.

Last, the copy constructor in lines 15–19 has been rewritten. Line 17 assigns `object2`'s Name property to the Name property of the object that is being constructed. Then, line 18 uses the `City` class's copy constructor to make a copy of `object2`'s Capital property. A reference to the copy is assigned to the `_capital` field.

Now, let's revisit our test code using the truly immutable `Country` class:

```
1  // Create a Country object.
2  var country = new Country("Norway", new City("Oslo", 681000.0));
3
4  // Display the country's info.
5  MessageBox.Show($"The capital of {country.Name} is {country.
                   Capital.Name}.");
6
7  // Get the City object for the capital.
8  var capCity = country.Capital;
9
10 // Change the capital city's name.
11 capCity.Name = "Bergen";
12
13 // Display the country's info again.
14 MessageBox.Show($"The capital of {country.Name} is {country.
                   Capital.Name}.");
```

As before, the statement in line 2 creates a `Country` object and assigns its object reference to the `country` variable. The country name is "Norway", the capital name is "Oslo", and the capital's population is 681000.0.

The statement in line 5 displays the following string in a message box: "The capital of Norway is Oslo."

Line 8 assigns the `country` object's Capital property to a variable named `capCity`. After this statement executes, the `capCity` variable will reference a copy of the `country` object's Capital property. Figure 11-17 shows the references.

The statement in line 11 assigns the string `"Bergen"` to the `capCity` object's Name property. However, because `capCity` refers to a copy of the Capital property, it does not affect the original data stored in the `Country` object. This is shown in Figure 11-18 and is proven when the statement in line 14 displays the `Country` object's data again, showing that it has not changed.

Figure 11-17 `capCity` references a copy of the `country` object's Capital property

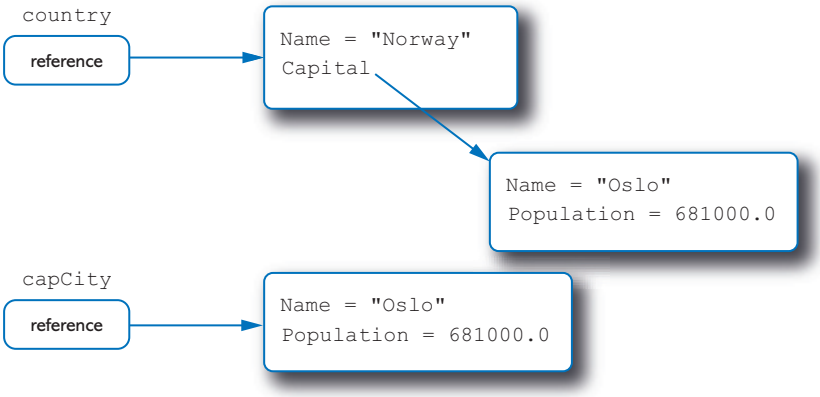
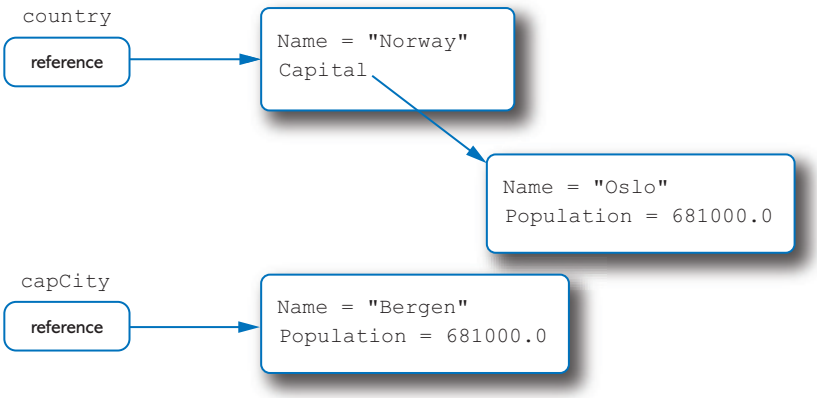


Figure 11-18 The `country` object's Capital.Name property has not been changed



NOTE: In case you are wondering why we didn't rewrite the Name property, it's because the Name property is a `string`, and `strings` are immutable in C#. It is safe to implement `strings` and other immutable types as auto-properties.

Checkpoint

11.18 If you have a reference to an immutable object, can you change the object's internal state?

11.19 How should an immutable class's field variables be declared?

11.20 How should an immutable class's auto-properties be declared?

11.21 Can an immutable class's properties have `set` accessors?

11.22 Suppose you are writing an immutable class that has a mutable object as a field or property. Should the immutable class have a method that returns a reference to the mutable field or property?

11.23 If an immutable class's constructor accepts a reference to a mutable object as an argument, should the reference be stored in a field or property?

11.8 Aggregation and Composition

CONCEPT: An aggregate class is a class that has an instance of another class as a field or property. A composite class is an aggregate class that is responsible for creating and managing the lifetimes of the objects it contains.

In real life, objects are frequently made of other objects. A house, for example, is made of door objects, window objects, wall objects, and much more. It is the combination of all these objects that makes a house object.

When designing software, it often makes sense to create an object from other objects. For example, suppose we have the following `Address` class to store a person's address:

```
1 class Address
2 {
3     // Constructor
4     public Address(string street, string city, string state, string zip)
5     {
6         Street = street;
7         City = city;
8         State = state;
9         Zip = zip;
10    }
11
12    // Properties
13    public string Street { get; init; }
14    public string City { get; init; }
15    public string State { get; init; }
16    public string Zip { get; init; }
17 }
```

The class has init-only properties for street address, city, state, and ZIP code (lines 13–16). It also has a parameterized constructor that initializes the properties (lines 4–9). We can use the `Address` class anywhere we need to store a person's address. For example, suppose we need a class that stores a customer's name and address. Here is an example of how we can write a `Customer` class that has an `Address` object as one of its properties:

```
1 class Customer
2 {
3     // Constructor
4     public Customer(string name, Address address)
5     {
6         Name = name;
7         MailingAddress = address;
8     }
9
10    // Properties
11    public string Name { get; init; }
12    public Address MailingAddress { get; init; }
13 }
```

The `Customer` class has an init-only `Name` property that is a string (line 11) and an init-only `MailingAddress` property that is an `Address` object (line 12). Notice that the constructor (lines 4–8) has two parameters. The `name` parameter is a `string`, and the `address` parameter references an `Address` object. Line 6 assigns the `name` parameter to the `Name` property, and line 7 assigns the `address` parameter to the `MailingAddress` property.

Suppose we need to create a `Customer` object containing the following information:

Name: Gregory Chang
Street Address: 123 Imaginary Lane
City: Woodfield
State: NY
ZIP Code: 14910

The following code shows an example:

```
// Create an Address object.
var customerAddress = new Address("123 Imaginary Lane", "Woodfield",
"NY", "14910");
// Create a Customer object.
var customer = new Customer("Gregory Chang", customerAddress);
```

Notice that we first created an `Address` object containing the customer's address. Then we created a `Customer` object, passing the customer's name and the `Address` object as arguments. Once this code has executed, the `customer` variable will reference a `Customer` object with the following properties:

- The `customer.Name` property contains the customer's name.
- The `customer.MailingAddress` property is an `Address` object that contains its own properties, as summarized here:
  - The `customer.MailingAddress.Street` property contains the customer's street address.
  - The `customer.MailingAddress.City` property contains the customer's city.
  - The `customer.MailingAddress.State` property contains the customer's state.
  - The `customer.MailingAddress.Zip` property contains the customer's ZIP code.

The following code shows how we might retrieve the `Customer` object's properties. Assume this code is used in an application that has Label controls named `nameLabel`, `streetLabel`, `cityLabel`, and so on:

```
nameLabel.Text = customer.Name;
streetLabel.Text = customer.MailingAddress.Street;
cityLabel.Text = customer.MailingAddress.City;
stateLabel.Text = customer.MailingAddress.State;
zipLabel.Text = customer.MailingAddress.Zip;
```

Notice the way we use dot notation to access the properties of the `MailingAddress` object. For example, if we want to get the customer's ZIP code, we need to access the `customer.MailingAddress.Zip` property. This indicates that `Zip` is a property of the `MailingAddress` object, which is a property of the `customer` object.

When a class contains an instance of another class as a field or property, it is said there is a **has-a relationship** between the classes. For example, there is a has-a relationship between the `Customer` class and the `Address` class because a `Customer` has an `Address`.

Tutorial 11-3 leads you through the process of creating an aggregate class. You will use the class in a simple application that creates an object of the class and displays its properties.

Tutorial 11-3:
Creating an Aggregate Class

VideoNote
Tutorial 11-3: Creating an Aggregate Class

In this tutorial, you will create a `Course` class that represents a course you are taking in college. The `Course` class will be an aggregate class with the following read-only properties:

- `CourseName`—A string to hold the name of the course
- `CourseInstructor`—An instance of the `Instructor` class
- `CourseTextBook`—An instance of the `TextBook` class

Notice that one of the properties is an instance of the `Instructor` class. The `Instructor` class will have the following read-only properties:

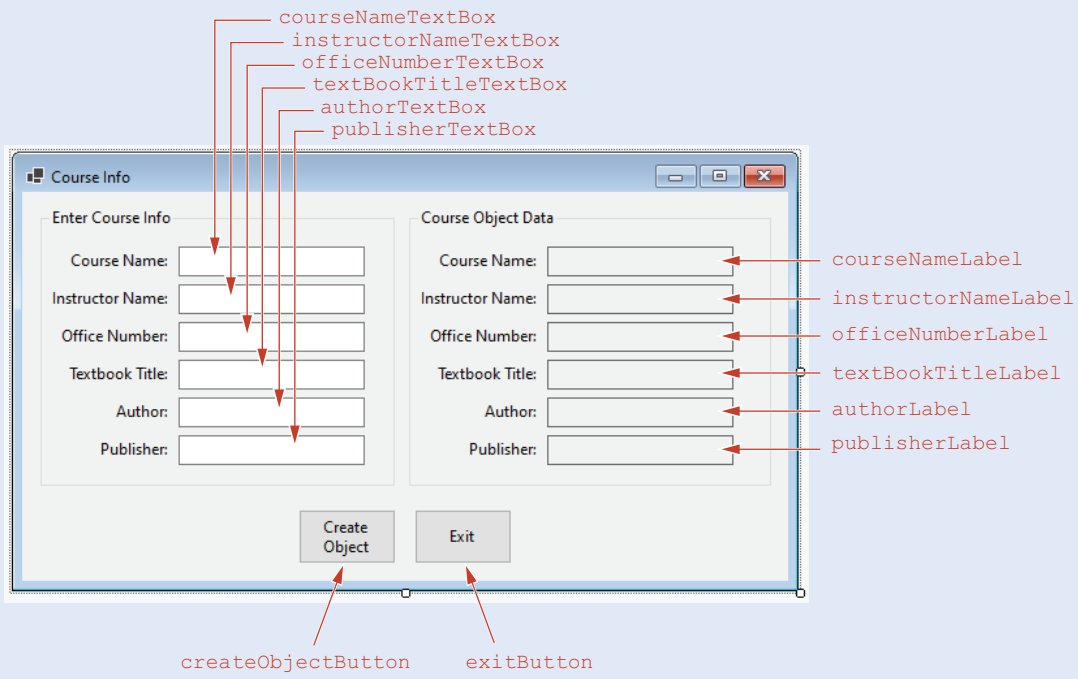
- `Name`—A string to hold the instructor's name
- `OfficeNumber`—A string to hold the instructor's office number

Also notice that one of the properties is an instance of the `TextBook` class. The `TextBook` class will have the following read-only properties:

- `Title`—A string to hold the textbook's title
- `Author`—A string to hold the author's name
- `Publisher`—A string to hold the publisher's name

In this tutorial you will write the classes and complete an application that demonstrates them. A project named *CourseInfo* has already been started for you; it is in the *Chap11* folder of the Student Sample Programs. Figure 11-19 shows the application's form. When the project is complete, you will be able to enter sample course data into the text boxes on the left side of the form, and then click the *Create Object* button. The application will use the input data to create an instance of the `Instructor` class, an instance of the `TextBook` class, and an instance of the `Course` class. It will then read and display the values of the object's properties in the labels on the right side of the form.

Figure 11-19 The *CourseInfo* project's form



Step 1: Start Visual Studio. Open the project named *CourseInfo* in the *Chap11* folder of the Student Sample Programs.

Step 2: Now you will write the `Instructor` class. Click *Project* on the Visual Studio menu bar and then select *Add Class...* . The *Add New Item* window should appear. Make sure *Class* is selected as the type of item. Change the default filename to *Instructor.cs* and then click the *Add* button. This adds a source code file named Instructor.cs to the project.

Step 3: The Instructor.cs file should now be displayed in the code editor. Complete the code for the `Instructor` class by typing lines 11–27 in Program 11-6.

Step 4: Now you will write the `TextBook` class. Click *Project* on the Visual Studio menu bar and then select *Add Class...* . The *Add New Item* window should appear. Make sure *Class* is selected as the type of item. Change the default filename to *TextBook.cs* and then click the *Add* button. This adds a source code file named TextBook.cs to the project.

Step 5: The TextBook.cs file should now be displayed in the code editor. Complete the code for the `TextBook` class by typing lines 11–30 in Program 11-7.

Step 6: Now you will write the `Course` class. Click *Project* on the Visual Studio menu bar and then select *Add Class...* . The *Add New Item* window should appear. Make sure *Class* is selected as the type of item. Change the default filename to *Course.cs* and then click the *Add* button. This adds a source code file named Course.cs to the project.

Step 7: The Course.cs file should now be displayed in the code editor. Complete the code for the `Course` class by typing lines 11–36 in Program 11-8.

Step 8: Next, you create the Click event handlers for the Button controls. Switch your view to the Form1 form in the Designer. Double-click the `createObjectButton` control. This opens the Form1.cs file in the code editor, and you will see an empty event handler named `createObjectButton_Click`. Complete the event handler by typing the code shown in lines 12–31 in Program 11-9. Let's review this code:

Lines 13–14: This statement creates an `Instructor` object. After the statement executes, the `instructor` variable will reference an `Instructor` object that is initialized with the values the user entered into the `instructorNameTextBox` and `officeNumberTextBox` controls.

Lines 17–19: This statement creates a `TextBook` object. After the statement executes, the `textbook` variable will reference a `TextBook` object that is initialized with the values the user entered into the `textBookTitleTextBox`, `authorTextBox`, and `publisherTextBox` controls.

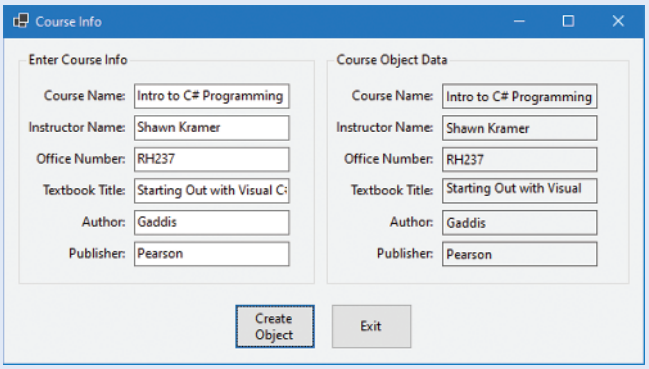
Lines 22–23: This statement creates a `Course` object. After the statement executes, the `course` variable will reference a `Course` object that is initialized with the value the user entered into the `courseNameTextBox` control, the object referenced by the `instructor` variable, and the object reference by the `textbook` variable.

Lines 26–31: These statements display the properties of the `Course` object, including the properties of its `Instructor` and `TextBook` objects, in the label controls on the right side of the application's form.

Step 9: Switch your view back to the Form1 form in the Designer and double-click the `exitButton` control. In the code editor you will see an empty event handler named `exitButton_Click`. Complete the event handler by typing the code shown in lines 36–37 in Program 11-9.

Step 10: Save the project. Then, press F5 on the keyboard or click the Start Debugging button (▶) on the toolbar to compile and run the application. When the application runs, enter some sample data into the TextBox controls and click the *Create Object* button. The application will use the data you entered to create a `Course` object. It will then display the properties of the `Course` object, including the properties of its `Instructor` and `TextBook` objects, in the label controls on the right side of the application's form. Figure 11-20 shows an example of the application's output. Click the *Exit* button when you are finished.

Figure 11-20 Example output of the *Course Info* application



Program 11-6 Completed code for the Instructor.cs file in the *Course Info* application

```
 1 using System;
 2 using System.Collections.Generic;
 3 using System.Linq;
 4 using System.Text;
 5 using System.Threading.Tasks;
 6
 7 namespace CourseInfo
 8 {
 9     internal class Instructor
10     {
11         // Constructor
12         public Instructor(string name, string office)
13         {
14             Name = name;
15             OfficeNumber = office;
16         }
17
18         // Parameterless constructor
19         public Instructor()
20         {
21             Name = "";
22             OfficeNumber = "";
23         }
24
```

```
25      // Properties
26      public string Name { get; }
27      public string OfficeNumber { get; }
28      }
29  }
```

Program 11-7 Completed code for the TextBook.cs file in the *Course Info* application

```
 1  using System;
 2  using System.Collections.Generic;
 3  using System.Linq;
 4  using System.Text;
 5  using System.Threading.Tasks;
 6
 7  namespace CourseInfo
 8  {
 9      internal class TextBook
10      {
11          // Constructor
12          public TextBook(string title, string author, string pub)
13          {
14              Title = title;
15              Author = author;
16              Publisher = pub;
17          }
18
19          // Parameterless constructor
20          public TextBook()
21          {
22              Title = "";
23              Author = "";
24              Publisher = "";
25          }
26
27          // Properties
28          public string Title { get; }
29          public string Author { get; }
30          public string Publisher { get; }
31      }
32  }
```

Program 11-8 Completed code for the Course.cs file in the *Course Info* application

```
 1  using System;
 2  using System.Collections.Generic;
 3  using System.Linq;
 4  using System.Text;
 5  using System.Threading.Tasks;
 6
 7  namespace CourseInfo
 8  {
 9      internal class Course
10      {
11          // Constructor
```

```
12          public Course(string name, Instructor instructor,
            TextBook textBook)
13          {
14              // Initialize the CourseName property.
15              if (name != null)
16                  CourseName = name;
17              else
18                  CourseName = "";
19
20              // Initialize the Instructor property.
21              if (instructor != null)
22                  CourseInstructor = instructor;
23              else
24                  CourseInstructor = new Instructor();
25
26              // Initialize the TextBook property.
27              if (textBook != null)
28                  CourseTextBook = textBook;
29              else
30                  CourseTextBook = new TextBook();
31          }
32
33          // Properties
34          public string CourseName { get; }
35          public Instructor CourseInstructor { get; }
36          public TextBook CourseTextBook { get; }
37      }
38  }
```

Program 11-9 Completed code for Form1 in the *Course Info* application

```
 1 namespace CourseInfo
 2 {
 3     public partial class Form1 : Form
 4     {
 5         public Form1()
 6         {
 7             InitializeComponent();
 8         }
 9
10         private void createObjectButton_Click(object sender,
           EventArgs e)
11         {
12             // Create an Instructor object.
13             var instructor = new Instructor(instructorNameTextBox.Text,
14              officeNumberTextBox.Text);
15
16             // Create a TextBook object.
17             var textbook = new TextBook(textBookTitleTextBox.Text,
18              authorTextBox.Text,
19              publisherTextBox.Text);
20
21             // Create a Course object.
22             var course = new Course(courseNameTextBox.Text,
23                                     instructor, textbook);
24
25             // Display the Course object's data.
26             courseNameLabel.Text = course.CourseName;
```

```
27              instructorNameLabel.Text = course.CourseInstructor.
                Name;
28              officeNumberLabel.Text = course.CourseInstructor.
                OfficeNumber;
29              textBookTitleLabel.Text = course.CourseTextBook.Title;
30              authorLabel.Text = course.CourseTextBook.Author;
31              publisherLabel.Text = course.CourseTextBook.Publisher;
32          }
33
34          private void exitButton_Click(object sender, EventArgs e)
35          {
36              // Close the form.
37              this.Close();
38          }
39      }
40  }
```

Composition: A Strong Form of Aggregation

We have demonstrated how an object can contain other objects. In our first example we demonstrated a `Customer` object that contains an `Address` object. Then, in Tutorial 11-3, we demonstrated a `Course` object that contains an `Instructor` object and a `TextBook` object. In both examples, the existence of the inner objects does not depend on the existence of the outer object. To illustrate this, look at the following code which creates instances of the `Address` and `Customer` classes:

```
1  // Create an Address object.
2  var myAddress = new Address("123 Imaginary Lane", "Woodfield",
3  "NY", "14910");
4
5  // Create a Customer object.
6  var customer = new Customer("Gregory Chang", myAddress);
7
8  // Nullify the customer variable.
9  customer = null;
10
11 // We can still access the Address object.
12 MessageBox.Show(myAddress.Street);
```

The statement in lines 2–3 creates an `Address` object that is referenced by the `myAddress` variable. Line 6 creates a `Customer` object that contains the `Address` object. The `Customer` object is referenced by the `customer` variable. At this point, both the `myAddress` variable and the `customer.MailingAddress` property reference the same `Address` object. This is illustrated in Figure 11-21.

Then, line 9 assigns `null` to the `customer` variable. This means that the `Customer` object is no longer referenced and will be removed from memory by the garbage collector. This is shown in Figure 11-22. However, the `Address` object is still referenced by the `myAddress` variable, so it will not be removed from memory. In fact, we can still use the `myAddress` variable to access the object. Line 12 displays the `myAddress` object's Street property in a message box.

In this example, a `Customer` object contains an `Address` object, but the `Customer` object does not own the `Address` object. This means the existence of the `Address` object is not dependent on the existence of the `Customer` object. We can delete the `Customer` object from memory, and if the `Address` object is still referenced, it will continue to exist.

Figure 11-21 The `Address` object is referenced twice

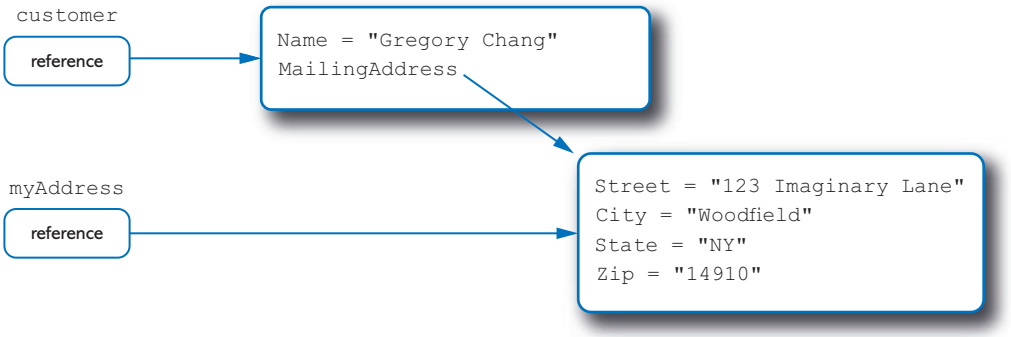
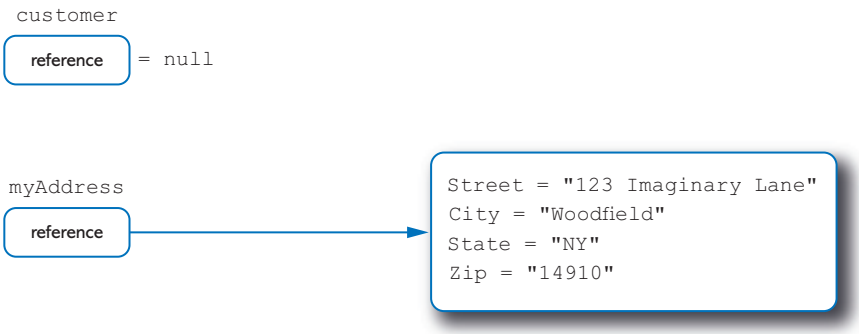


Figure 11-22 The `customer` variable is set to `null`



Composition

There is a special type of aggregation known as **composition**, in which the lifetime of the inner class objects is dependent on the lifetime of the outer class objects. This means:

- The outer class is responsible for creating its instances of the inner class.
- When an instance of the outer class is destroyed, its instances of the inner class are also destroyed.

To demonstrate composition, let's look at a modified version of the `Customer` class:

```
1 class Customer
2 {
3     // Constructor
4     public Customer(string name, string street, string city,
5                     string state, string zip)
6     {
7         Name = name;
8         MailingAddress = new Address(street, city, state, zip);
9     }
10
11    // Properties
12    public string Name { get; init; }
13    public Address MailingAddress { get; init; }
14 }
```

Can you spot the differences between this version of the `Customer` class and the previous version? Here they are:

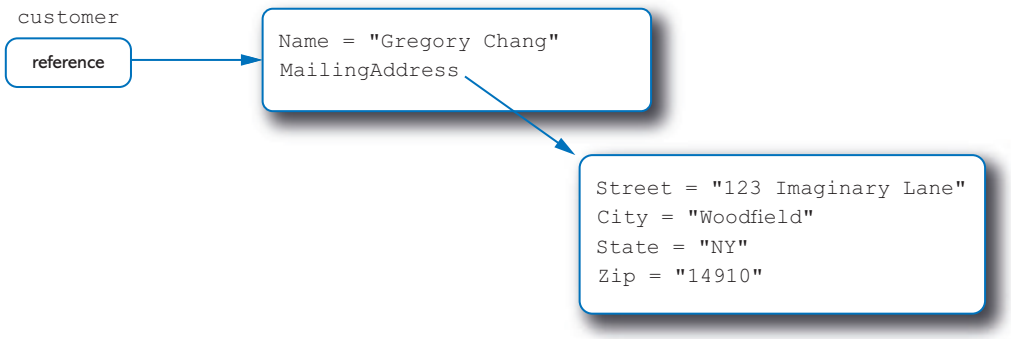
- In this version, the constructor does not accept an `Address` object as an argument. This constructor accepts `string` arguments for the customer's name, street address, city, state, and ZIP code.
- In line 8, this constructor uses the `new` operator to create an instance of the `Address` class, passing the necessary arguments to its constructor.

When you use this version of the `Customer` class, you do not create the `Address` object independently. You simply create a `Customer` object, passing all the customer's information to the `Customer` class's constructor. The following code shows an example:

```
var customer = new Customer("Gregory Chang", "123 Imaginary Lane",
                            "Woodfield", "NY", "14910");
```

When this statement executes, the `Customer` class constructor will create the `Address` object and assign it to the `MailingAddress` property. This is shown in Figure 11-23.

Figure 11-23 The `Customer` object owns the `Address` object



As you can see in figure, the `Address` object is referenced only by the `customer.MailingAddress` property. When the `Customer` object is removed from memory, the `Address` object will no longer be referenced, so it will be removed as well.

Checkpoint

11.24 What makes composition different from aggregation?

11.9 Passing and Returning Anonymous Objects

CONCEPT: An anonymous object is an object that is not referenced.

The `new` operator creates an object, calls its constructor, and returns a reference to that object. For example, look at the following expression:

```
new Contact("Jane Doe", "jane@example.com")
```

This is known as a **new expression**. It creates a `Contact` object in memory, passes arguments to the class constructor, and returns a reference to the object. When we create an object, we usually want to keep a reference to that object, so we write `new` expressions on the right-hand side of an = operator. Here is an example:

```
var contact = new Contact("Jane Doe", "jane@example.com");
```

When this statement executes, the reference that is returned from the `new` operator is assigned to the `contact` variable. We can subsequently use the `contact` variable to work with the object.

What would happen if we used a `new` expression to create an object, but we didn't assign the object reference to a variable? Here is an example of such a statement:

```
new Contact("Jane Doe", "jane@example.com");
```

This statement is simply a `new` expression with a semicolon at the end. It creates a `Contact` object in memory, but the reference that is returned from the expression is thrown away. The object that is created is known as an **anonymous object**, which means that it is not referenced. The next time the garbage collector runs, the object will be deleted from memory.

There are two situations in which anonymous objects are commonly used:

- To create an object just for the purpose of passing that object to a method.
- To create an object just for the purpose of returning that object from a method.

Let's look at each of these situations.

Passing an Anonymous Object as an Argument to a Method

Suppose we have the following `Point` class in a program:

```
 1 class Point
 2 {
 3     // Constructor
 4     public Point(int x, int y)
 5     {
 6         X = x;
 7         Y = y;
 8     }
 9
10     // Properties
11     public int X { get; }
12     public int Y { get; }
13 }
```

This class stores a set of *X,Y* coordinates. It has two properties named X and Y, as well as a parameterized constructor that initializes the properties. Suppose we also have the following method in the same program:

```
private void DisplayPoint(Point p)
{
   MessageBox.Show($"{p.X}, {p.Y}");
}
```

The `DisplayPoint` method accepts a `Point` object as its argument and displays the object's X and Y properties. Here is an example of how we might call the method:

```
var point = new Point(100, 100);
DisplayPoint(point);
```

In this code we first create a `Point` object, and then we pass that object to the `DisplayPoint` method. If we don't need to use the `Point` object for any other purpose, we can combine the two steps as shown here:

```
DisplayPoint(new Point(100, 100));
```

In this statement, the `new` expression creates an anonymous `Point` object. The expression returns a reference to that object, and the reference is passed to the `DisplayPoint` method. In the `DisplayPoint` method, the reference is assigned to the parameter `p`. When that happens, the object is no longer anonymous because `p` references it.

Returning an Anonymous Object from a Method

Recall the `GetSampleContact` method that we discussed earlier in this chapter. The method, which is shown here, returns a `Contact` object that contains sample data:

```
1 private Contact GetSampleContact()
2 {
3     var contact = new Contact("Jane Doe", "jane@example.com");
4     return contact;
5 }
```

The statement in line 3 creates a `Contact` object and assigns its reference to the `contact` variable. Then the statement in line 4 returns the object. Because the `Contact` object is created only so it can be returned from the method, we can combine the two steps as shown here:

```
1 private Contact GetSampleContact()
2 {
3     return new Contact("Jane Doe", "jane@example.com");
4 }
```

This version of the method returns a reference to the anonymous object that is created in line 3. The code that called the method will receive the reference. For example, the following statement calls the method and assigns the reference that it returns to the `sampleContact` variable. When that happens, the object is no longer anonymous because `sampleContact` references it.

```
var sampleContact = GetSampleContact();
```

Checkpoint

11.25 What is an anonymous object?

11.26 Name two situations discussed in this section in which anonymous objects are commonly used.

11.10 Class Collaboration

CONCEPT: It is common for classes to interact, or collaborate, with each other to perform their operations.

In an object-oriented application it is common for objects of different classes to collaborate. This simply means that objects interact with each other. Sometimes one object will need the services of another object in order to perform its operations. For example, let's say an object needs to generate random numbers and save those random numbers to a file. The object would use the services of a `Random` object to generate the numbers, and then use the services of a `StreamWriter` object to write the numbers to a file. In this example, the object is collaborating with objects created from .NET classes. The objects that you create from your own classes can also collaborate with each other.

In Tutorial 11-4 you will write two classes: `Stock` and `StockPurchase`. The `Stock` class represents a company's stock, and the `StockPurchase` class simulates a stock purchase. The `StockPurchase` class is responsible for calculating the cost of the stock purchase. To do that, it must use the `Stock` class to get the price per share of the stock.

Tutorial 11-4:
Class Collaboration

VideoNote
Tutorial 11-4: Class Collaboration

In this tutorial you will complete an application that calculates the cost of purchasing a number of shares of stock. In the project, you will write two classes: `Stock` and `StockPurchase`.

The `Stock` class is an immutable class that represents a company's stock. It will have the following init-only properties:

- `Symbol`—A string that holds the trading symbol for the company's stock. This is a short series of characters that are used to identify the stock on the stock exchange. For example, the XYZ Company's stock might have the trading symbol XYZ.
- `SharePrice`—A double that holds the current price-per-share of the stock.

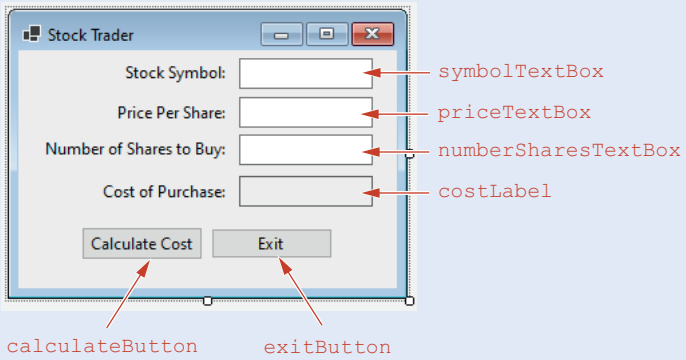
The `StockPurchase` class is an immutable class that calculates the cost of the purchasing a specified number of shares of stock. To perform the calculation, the `StockPurchase` class must use the `Stock` class to get the stock's price per share. The `StockPurchase` class will have the following init-only properties:

- `StockToPurchase`—A `Stock` object representing the stock that is being purchased.
- `Shares`—The number of shares being purchased.

The `StockPurchase` class will also have a method named `GetCost` that will return the cost of the purchase.

A project named *StockTrader* has already been started for you; it is in the *Chap11* folder of the Student Sample Programs. The application's form is shown in Figure 11-24. At run time, you enter the stock symbol, the price per share, and the number of shares to buy. When you click the *Calculate Cost* button, the application uses the input data to create an instance of the `Stock` class, and an instance of the `StockPurchase` class. It then displays the cost of the purchase.

Figure 11-24 The *StockTrader* project's form



Step 1: Start Visual Studio. Open the project named *StockTrader* in the *Chap11* folder of the Student Sample Programs.

Step 2: Now you will write the `Stock` class. Click *Project* on the Visual Studio menu bar and then select *Add Class...* . The *Add New Item* window should appear. Make sure *Class* is selected as the type of item. Change the default filename to *Stock.cs* and then click the *Add* button. This adds a source code file named Stock.cs to the project.

Step 3: The Stock.cs file should now be displayed in the code editor. Complete the code for the `Stock` class by typing lines 11–18 in Program 11-10.

Lines 12–16: This is the parameterized constructor. As arguments, it accepts the stock symbol and the stock's price-per-share. These values are used to initialize the `Symbol` and `SharePrice` properties.

Lines 19–23: This is the parameterless constructor. It assigns an empty string to the `Symbol` property and 0 to the `SharePrice` property.

Line 26: This statement declares the init-only `Symbol` property.

Line 27: This statement declares the init-only `SharePrice` property.

Step 4: Now you will write the `StockPurchase` class. Click *Project* on the Visual Studio menu bar and then select *Add Class...*. The *Add New Item* window should appear. Make sure *Class* is selected as the type of item. Change the default filename to *StockPurchase.cs* and then click the *Add* button. This adds a source code file named StockPurchase.cs to the project.

Step 3: The StockPurchase.cs file should now be displayed in the code editor. Complete the code for the `StockPurchase` class by typing lines 11–33 in Program 11-11.

Lines 12–16: This is the parameterized constructor. As arguments, it accepts a reference to a `Stock` object and the stock's price-per-share. These values are used to initialize the `StockToPurchase` and `Shares` properties. (Notice that we are assigning the `stock` parameter to the `StockToPurchase` property without making a copy of the `Stock` object that `stock` references. This is acceptable because `Stock` is an immutable class.)

Lines 19–23: This is the parameterless constructor. It assigns `null` to the `StockToPurchase` property and 0 to the `Shares` property.

Line 26: This statement declares the init-only `StockToPurchase` property. This property is a `Stock` object representing the stock that is being purchased.

Line 27: This statement declares the init-only `Shares` property. This is the number of shares being purchased.

Lines 30–33: This is the `GetCost` method. It returns the cost of the purchase, which is the number of shares multiplied by the stock's share price. Line 32 gets the share price from the `StockToPurchase` object's `SharePrice` property.

Step 4: Next, you create the Click event handlers for the Button controls. Switch your view to the Form1 form in the Designer. Double-click the `calculateButton` control. This opens the Form1.cs file in the code editor, and you will see an empty event handler named `calculateButton_Click`. Complete the event handler by typing the code shown in lines 12–38 in Program 11-12. Let's review this code.

Lines 12–13: These lines declare the variables `price` and `numShares`. They will hold the price-per-share and the number of shares to buy, respectively.

Line 16: This `if` statement tries to convert `priceTextBox.Text` to a `decimal`. If the conversion is successful, the result is stored in the `price` variable, and the program continues executing at line 19. If the conversion is not successful, the program jumps to the `else` clause in line 35, and line 37 displays the message "Enter a valid price."

Line 19: This statement creates a `Stock` object, referenced by the `stock` variable. The object's Symbol property will be set to the value that has been

entered into the `symbolTextBox` control, and its SharePrice property will be set to the value of the `price` variable.

Line 22: This `if` statement tries to convert `numSharesTextBox.Text` to an `int`. If the conversion is successful, the result is stored in the `numShares` variable, and the program continues executing at line 25. If the conversion is not successful, the program jumps to the `else` clause in line 30, and line 32 displays the message "Enter a valid number of shares."

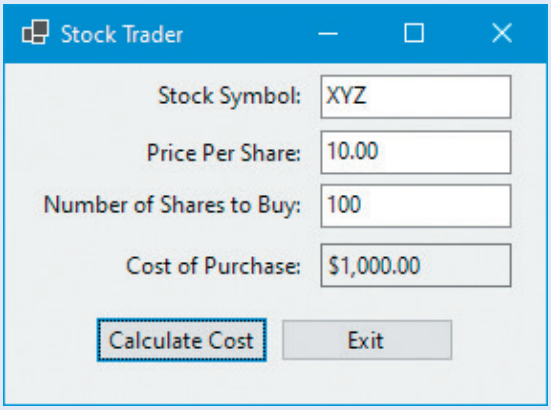
Line 25: This statement creates a `StockPurchase` object, referenced by the `stockPurchase` variable. The object's StockToPurchase property will be set to the object referenced by the stock variable, and its Shares property will be set to the value of the `numShares` variable.

Line 28: This statement calls the `StockPurchase` object's `GetCost` method and converts its return value to a string that is formatted as currency. The resulting value is assigned to `costLabel.Text`, which displays the value in the `costLabel` control.

Step 5: Switch your view back to the *Designer* and double-click the `exitButton` control. In the code editor you will see an empty event handler named `exitButton_Click`. Complete the event handler by typing the code shown in lines 43 and 44 in Program 11-12.

Step 6: Save the project. Then, press F5 on the keyboard or click the *Start Debugging* button (▶) on the toolbar to compile and run the application. When the application runs, enter some sample data into the TextBox controls and click the *Calculate Cost* button. The application will use the data you entered to create a `Stock` object and a `StockPurchase` object. It will then display, in the `costLabel` control, the value of the number of shares multiplied by the price-per-share. Figure 11-25 shows an example of the application's output. When you are ready to end the application, click the OK button to dismiss the message box, then click the *Exit* button.

Figure 11-25 Example output of the *Stock Trader* application



Program 11-10 Completed code for the Stock.cs file in the *Stock Trader* application

```
1 using System;
2 using System.Collections.Generic;
3 using System.Linq;
4 using System.Text;
5 using System.Threading.Tasks;
6
```

```
 7 namespace StockTrader
 8 {
 9     internal class Stock
10     {
11         // Constructor
12         public Stock(string symbol, decimal price)
13         {
14             Symbol = symbol;
15             SharePrice = price;
16         }
17
18         // Constructor
19         public Stock()
20         {
21             Symbol = "";
22             SharePrice = 0m;
23         }
24
25         // Properties
26         public string Symbol { get; init; }
27         public decimal SharePrice { get; init; }
28     }
29 }
```

Program 11-11 Completed code for the StockPurchase.cs file in the *Stock Trader* application

```
 1 using System;
 2 using System.Collections.Generic;
 3 using System.Linq;
 4 using System.Text;
 5 using System.Threading.Tasks;
 6
 7 namespace StockTrader
 8 {
 9     internal class StockPurchase
10     {
11         // Constructor
12         public StockPurchase(Stock stock, int shares)
13         {
14             StockToPurchase = stock;
15             Shares = shares;
16         }
17
18         // Constructor
19         public StockPurchase()
20         {
21             StockToPurchase = null;
22             Shares = 0;
23         }
24
25         // Properties
26         public Stock StockToPurchase { get; init; }
27         public int Shares { get; init; }
28
29         // The GetCost method returns the cost of the purchase.
30         public decimal GetCost()
31         {
```

```
32              return Shares * StockToPurchase.SharePrice;
33          }
34      }
35  }
```

Program 11-12 Completed code for Form1 in the *Stock Trader* application

```
 1  namespace StockTrader
 2  {
 3      public partial class Form1 : Form
 4      {
 5          public Form1()
 6          {
 7              InitializeComponent();
 8          }
 9
10          private void calculateButton_Click(object sender,
            EventArgs e)
11          {
12              decimal price;          // Price-per-share
13              int numShares;          // Number of shares to buy
14
15              // Validate the price entered by the user.
16              if (decimal.TryParse(priceTextBox.Text, out price))
17              {
18                  // Create a Stock object.
19                  var stock = new Stock(symbolTextBox.Text, price);
20
21                  // Validate the number of shares.
22                  if (int.TryParse(numberSharesTextBox.Text, out
                    numShares))
23                  {
24                      // Create a StockPurchase object.
25                      var stockPurchase = new StockPurchase(stock,
                        numShares);
26
27                      // Display the cost of the purchase.
28                      costLabel.Text = stockPurchase.GetCost().
                        ToString("c");
29                  }
30                  else
31                  {
32                      MessageBox.Show("Enter a valid number of
                        shares.");
33                  }
34              }
35              else
36              {
37                  MessageBox.Show("Enter a valid price.");
38              }
39          }
40
41          private void exitButton_Click(object sender, EventArgs e)
42          {
43              // Close the form.
44              this.Close();
45          }
46      }
47  }
```

Key Terms

anonymous object
composition
fully qualified name
garbage collection
has-a relationship
immutable class
immutable object
`init` accessor
init-only property
mutable class
mutable object
new expression
nullable value type
object initializers

Review Questions

Multiple Choice

1. When an object is passed as an argument to a method, this is actually passed.
   a. A copy of the object
   b. The name of the object
   c. A reference to the object
   d. None of these; you cannot pass an object

2. If a method has a parameter that is a reference variable, you should make sure the parameter is not equal to _____ before using it in an operation.
   a. `false`
   b. `0`
   c. `null`
   d. `Nan`

3. If you write this method for a class, it will be automatically called it any time you concatenate an object of the class with a string.
   a. `ToString`
   b. `PlusString`
   c. `StringConvert`
   d. `ConcatString`

4. Making an instance of one class a field in another class is called _____.
   a. Nesting
   b. Class fielding
   c. Aggregation
   d. Concatenation

5. A class whose internal state cannot be changed is known as a(n) _____ class.
   a. Foundational
   b. Concrete
   c. Mutable
   d. Immutable

6. The _____ keyword can be used by an object to refer to itself.
   a. `callingObject`
   b. `this`
   c. `me`
   d. `instance`

7. The .NET runtime system periodically performs this process, which automatically removes unreferenced objects from memory.
   a. Memory cleansing
   b. Memory deallocation
   c. Garbage collection
   d. Object expungement

True or False

1. When an object is passed as an argument to a method, the method can access the argument.
2. A method cannot return a reference to an object.
3. You can write a method that returns `null`.
4. Reference variables can be initialized with `null`, but you cannot assign `null` to a reference variable after it has been initialized.
5. If you try to check a reference variable to determine whether it is equal to `null`, an exception will be thrown.
6. You can pass `null` as an argument to a method.
7. If an immutable class has a mutable object as a field or property, it is permissible for a method in the immutable class to return a reference to the mutable field or property.
8. If an immutable class's constructor accepts a reference to a mutable object, that reference should not be stored in a field.
9. If a reference field is declared as `readonly`, it means the internal state of the object that the variable refers to cannot be changed.
10. When you declare a reference field as `readonly`, and then initialize it with a reference to an object, the reference variable cannot refer to any other object.

Short Answer

1. Describe the difference in the way value type variables and reference type variables are passed as arguments to a method.
2. A "has a" relationship can exist between classes. What does this mean?
3. What happens if you attempt to call a method using a reference variable that is set to `null`?
4. Is it advisable or not advisable to write a method that returns a reference to an object that is a private field? What is the exception to this?
5. What is the `this` keyword?
6. Under what circumstances does an object become a candidate for garbage collection?
7. Assume a class named `Child` exists. The following `Parent` class has an instance of the `Child` class as a field. Is this an example of composition?

```
class Parent
{
    private readonly Child child;

    public Parent(Child c)
    {
        child = c;
    }
}
```

8. Assume a class named `Child` exists. The following `Parent` class has an instance of the `Child` class as a field. Is this an example of composition?

```
class Parent
{
    private readonly Child child;

    public Parent()
    {
```

```
            child = new Child();
        }
    }
```

9. Is the following class mutable or immutable?

```
class Person
{
     public string FirstName { get; set; }
     public string LastName { get; set; }

     public Person(string fname, string lname)
     {
          FirstName = fname;
          LastName = lname;
     }
}
```

Algorithm Workbench

1. Consider the following class.

```
class Contact
{
    public string Name { get; init; }
    public string Email { get; init; }

    public Contact(string name, string email)
    {
        Name = name;
        Email = email;
    }
}
```

   a. Write a `ToString` method for the class. The method should return a string containing the Name property, followed by a space, followed by the Email property.
   b. Write a copy constructor for the class.

2. Assume a program uses a class named `Employee`. The program has the following method that displays employee information:

```
private void DisplayEmployee(Employee employee)
{
    MessageBox.Show($"{employee.Name} {employee.ID}");
}
```

   As it is currently written, the method will throw an exception if `null` is passed to it as an argument. Rewrite the method so it will not throw an exception if `null` is passed to it as an argument.

3. Assume a program uses a class named `Employee`. The program has the following method that returns an `Employee` object:

```
private Employee GetEmployeeObject()
{
    var employee = new Employee("", "");
    return employee;
}
```

   Rewrite the method in such a way that it does not declare the `employee` variable. The `return` statement should return an anonymous object.

4. Write the definition of a class named `Line` that has an `int` property named Length and a `string` property named Color. Both properties should be init-only properties. Do not write a constructor in the class.

5. Look at the following class:

```
class Circle
{
    public int Radius { get; init; }
    public string Color { get; init; }
}
```

Write a statement that uses object initializer syntax to create an instance of the `Circle` class named `myCircle`, with the Radius property set to 100, and the Color property set to "Blue".

Programming Challenges

1. **`Month` Class**

Write an immutable class named `Month`. The class should have the following properties:

- An `int` property named MonthNumber that holds the number of the month. For example, January would be 1, February would be 2, and so forth.
- A `string` property named MonthName that holds the name of the month. For example, if MonthNumber is 1, MonthName would be "January". If MonthNumber is 2, MonthName would be "February", and so on.

In addition, provide the following methods:

- A parameterless constructor that sets the MonthNumber property to 1.
- A constructor that accepts the number of the month as an argument. It should set the MonthNumber property to the value passed as the argument. If a value less than 1 or greater than 12 is passed, the constructor should set MonthNumber to 1. The constructor should also set the MonthName property to the correct corresponding value.
- A constructor that accepts the name of the month, such as "January" or "February" as an argument. It should set the MonthNumber property to the correct corresponding value.
- A copy constructor.
- A `ToString` method that returns the value of the MonthName property.
- A `GreaterThan` method that accepts a `Month` object as an argument. If the calling object's MonthNumber property is greater than the argument's MonthNumber property, this method should return `true`. Otherwise, it should return `false`.
- A `LessThan` method that accepts a `Month` object as an argument. If the calling object's MonthNumber property is less than the argument's MonthNumber property, this method should return `true`. Otherwise, it should return `false`.

2. **Car Instrument Simulator**

For this assignment, you will create a set of classes that work together to simulate a car's fuel gauge and odometer. The classes you will design are the following:

- The `FuelGauge` Class: This class will simulate a fuel gauge. Its responsibilities are as follows:
  - — To know the car's current amount of fuel, in gallons.
  - — To report the car's current amount of fuel, in gallons.
  - — To be able to increment the amount of fuel by 1 gallon. This simulates putting fuel in the car. (The car can hold a maximum of 15 gallons.)
  - — To be able to decrement the amount of fuel by 1 gallon if the amount of fuel is greater than 0 gallons. This simulates burning fuel as the car runs.

- The `Odometer` Class: This class will simulate the car's odometer. Its responsibilities are as follows:
  - To know the car's current mileage.
  - To report the car's current mileage.
  - To be able to increment the current mileage by 1 mile. The maximum mileage the odometer can store is 999,999 miles. When this amount is exceeded, the odometer resets the current mileage to 0.
  - To be able to work with a `FuelGauge` object. It should decrease the `FuelGauge` object's current amount of fuel by 1 gallon for every 24 miles traveled. (The car's fuel economy is 24 miles per gallon.)

Demonstrate the classes by creating instances of each. Simulate filling the car up with fuel, and then run a loop that increments the odometer until the car runs out of fuel. During each loop iteration, display the car's current mileage and amount of fuel.

3. Parking Ticket Simulator

For this assignment you will design a set of classes that work together to simulate a police officer issuing a parking ticket. You should design the following classes:

- The `ParkedCar` Class: This class should simulate a parked car. The class's responsibilities are as follows:
  - To know the car's make, model, color, license number, and the number of minutes that the car has been parked.
- The `ParkingMeter` Class: This class should simulate a parking meter. The class's only responsibility is as follows:
  - To know the number of minutes of parking time that has been purchased.
- The `ParkingTicket` Class: This class should simulate a parking ticket. The class's responsibilities are as follows:
  - To report the make, model, color, and license number of the illegally parked car
  - To report the amount of the fine, which is $25 for the first hour or part of an hour that the car is illegally parked, plus $10 for every additional hour or part of an hour that the car is illegally parked
  - To report the name and badge number of the police officer issuing the ticket
- The `PoliceOfficer` Class: This class should simulate a police officer inspecting parked cars. The class's responsibilities are as follows:
  - To know the police officer's name and badge number
  - To examine a `ParkedCar` object and a `ParkingMeter` object, and determine whether the car's time has expired
  - To issue a parking ticket (generate a `ParkingTicket` object) if the car's time has expired

Create an application that demonstrates how these classes work together.

4. Playlist Application

Create an application that creates a playlist. The application will use the following classes:

- A `Song` class that has two properties: Title and Artist. Both properties are `string`s.
- A `PlayList` class that keeps `Song` objects in a `List`. The class should have methods for adding `Song` objects to the `List` and removing `Song` objects from the list.

The application should let the user enter the information for a song (the title and artist), and that song will then be added to the playlist. The user should be able to see the songs in the playlist (perhaps in a ListBox) and remove songs that are no longer wanted.

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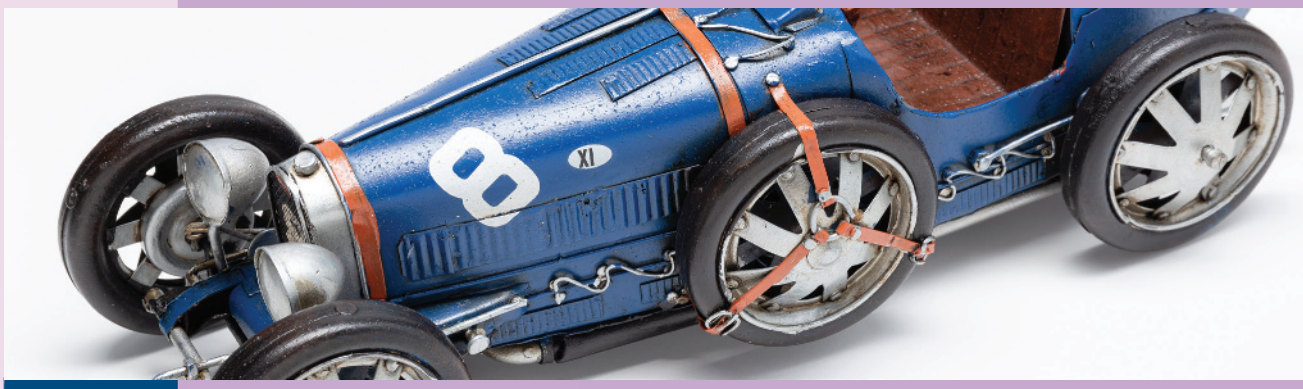


CHAPTER 12

Inheritance, Polymorphism, and Interfaces

TOPICS

12.1 Inheritance
12.2 Polymorphism
12.3 Abstract Classes
12.4 Extension Methods
12.5 Interfaces

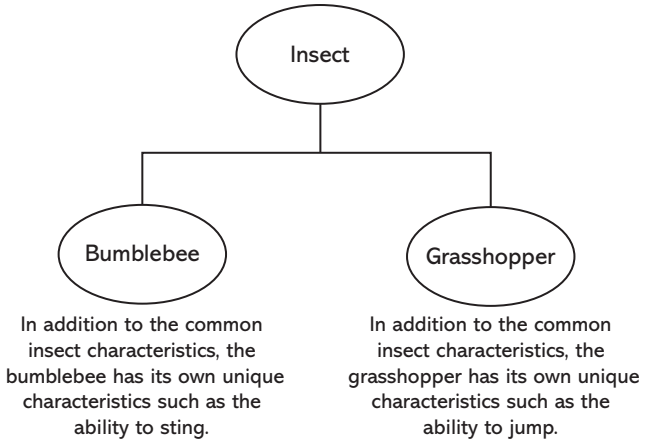
12.1 Inheritance

CONCEPT: **Inheritance allows a new class to extend an existing class. The new class inherits the members of the class it extends.**

Generalization and Specialization

In the real world, you can find many objects that are specialized versions of other more general objects. For example, the term *insect* describes a very general type of creature with numerous characteristics. Because grasshoppers and bumblebees are insects, they have all the general characteristics of an insect. In addition, they have special characteristics of their own. For example, the grasshopper has its jumping ability, and the bumblebee has its stinger. Grasshoppers and bumblebees are specialized versions of insects. This is illustrated in Figure 12-1.

Figure 12-1 Bumblebees and grasshoppers, specialized versions of insects



Inheritance and the "Is a" Relationship

When one object is a specialized version of another object, there is an **"is a" relationship** between them. For example, a grasshopper is an insect. Here are a few other examples of the "is a" relationship:

- A poodle is a dog.
- A car is a vehicle.
- A flower is a plant.
- A rectangle is a shape.
- A football player is an athlete.

When an "is a" relationship exists between objects, it means that the specialized object has all of the characteristics of the general object, plus additional characteristics that make it special. In object-oriented programming, inheritance is used to create an "is a" relationship among classes. This allows you to extend the capabilities of a class by creating another class that is a specialized version of it.

Inheritance involves a base class and a derived class. The **base class** is the general class and the **derived class** is the specialized class. You can think of the derived class as an extended version of the base class. The derived class inherits fields, properties, and methods from the base class without any of them having to be rewritten. Furthermore, new fields, properties, and methods may be added to the derived class, and that is what makes it a specialized version of the base class.

NOTE: Base classes are sometimes called **superclasses**, and derived classes are sometimes called **subclasses**. Either set of terms is correct. For consistency, this text uses the terms *base class* and *derived class*.

Let's look at an example of how inheritance can be used. Suppose we are developing an application that a car dealership can use to manage its inventory of used cars. The dealership's inventory includes three types of automobiles: cars, pickup trucks, and sport-utility vehicles (SUVs). Regardless of the type, the dealership keeps the following data about each automobile:

- Make
- Year model
- Mileage
- Price

Each type of vehicle that is kept in inventory has these general characteristics plus its own specialized characteristics. For cars, the dealership keeps the following additional data:

- Number of doors (2 or 4)

For pickup trucks, the dealership keeps the following additional data:

- Drive type (two-wheel drive or four-wheel drive)

And, for SUVs, the dealership keeps the following additional data:

- Passenger capacity

In designing this program, one approach would be to write the following three classes:

- A `Car` class with properties for the make, year model, mileage, price, and the number of doors
- A `Truck` class with properties for the make, year model, mileage, price, and the drive type
- A `SportUtility` class with properties for the make, year model, mileage, price, and the passenger capacity

This is an inefficient approach, however, because all three classes have a large number of common properties. As a result, the classes would contain a lot of duplicated code. In addition, if we discover later that we need to add more common attributes, we would have to modify all three classes.

A better approach would be to write an `Automobile` base class to hold all the general data about an automobile and then write derived classes for each specific type of automobile. The following code shows the `Automobile` class:

```
 1 class Automobile
 2 {
 3     // Constructor
 4     public Automobile()
 5     {
 6         Make = "";
 7         Model = "";
 8         Mileage = 0;
 9         Price = 0m;
10     }
11
12     // Properties
13     public string Make { get; set; }
14     public string Model { get; set; }
15     public int Mileage { get; set; }
16     public decimal Price { get; set; }
17 }
```

The Make, Model, Mileage, and Price properties are declared in lines 13–16. The parameterless constructor, in lines 4–10, initializes the Make and Model properties with empty strings, and the Mileage and Price properties with 0.

The `Automobile` class is a complete class from which we can create objects. However, the `Automobile` class holds only general data about an automobile. It does not hold any of the specific pieces of data that the dealership wants to keep about cars, pickup trucks, and SUVs. To hold data about those specific types of automobiles, we write derived classes that inherit from the `Automobile` class. The following code shows the `Car` class, which is derived from the `Automobile` class:

```
1 class Car: Automobile
2 {
3     // Constructor
4     public Car()
```

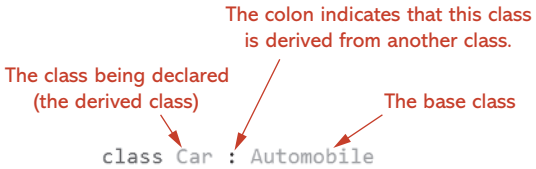
```
5      {
6          Doors = 0;
7      }
8
9      // Doors property
10     public int Doors { get; set; }
11 }
```

Take a closer look at the class header in line 1:

```
class Car: Automobile
```

This line of code has some new notation. After the name of the class, `Car`, a colon appears, followed by the name of another class, `Automobile`. This line indicates that we are defining a class named `Car`, and it is derived from the `Automobile` class. The `Car` class is the derived class and the `Automobile` class is the base class. Figure 12-2 illustrates this notation. If we want to express the relationship between the `Car` class and the `Automobile` class, we can say that a `Car` is an `Automobile`.

Figure 12-2 Inheritance notation in the class header



Because the `Car` class is derived from the `Automobile` class, the `Car` class inherits all the `Automobile` class's members, except its constructor. (The purpose of the base class's constructor is to create an instance of the base class, so it makes sense that the derived class does not inherit the base class's constructor.)

NOTE: Although the derived class inherits the base class's private members, it cannot directly access them. Only methods in the base class can directly access the base class's private members.

The `Car` class has a parameterless constructor that appears in lines 4–7. In line 9, the constructor assigns the value 0 to the Doors property. Here is a summary of what happens when an object of the `Car` class is created:

- The base class constructor executes first. In this case, the `Automobile` class's constructor is called, and it assigns empty strings to the Make and Model properties and 0 to the Mileage and Price properties.
- The derived class constructor executes next. In this case, 0 is assigned to the Doors property.

Before going any further, let's look at a complete application that demonstrates the `Automobile` and `Car` classes. In the *Chap12* folder of the Student Sample Programs you will find a project named *CarDemo*. This project contains a form named Form1, the `Automobile` class, and the `Car` class. Figure 12-3 shows the Form1 form as it appears in the *Designer*.

When you run the application, enter sample car data into the text boxes and then click the *Create Object* button. The application will create an object of the `Car` class and assign the data from the text boxes to the object's properties. It will then read and display the values of the object's properties in the `makeLabel`, `modelLabel`, `mileageLabel`, `priceLabel`, and `doorsLabel` controls. Figure 12-4 shows an example of the application's form, at run time, after the user has entered some data and clicked the *Create Object* button.

Figure 12-3 The *CarDemo* project's Form1 form

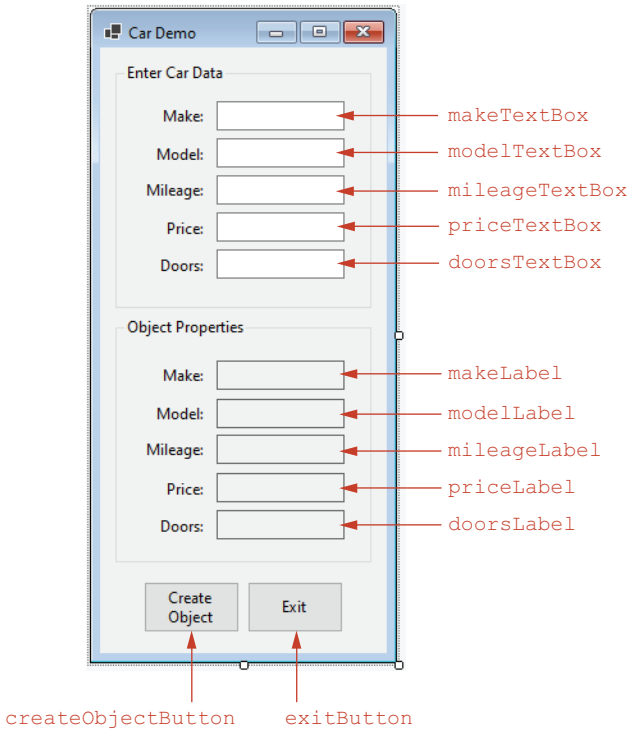
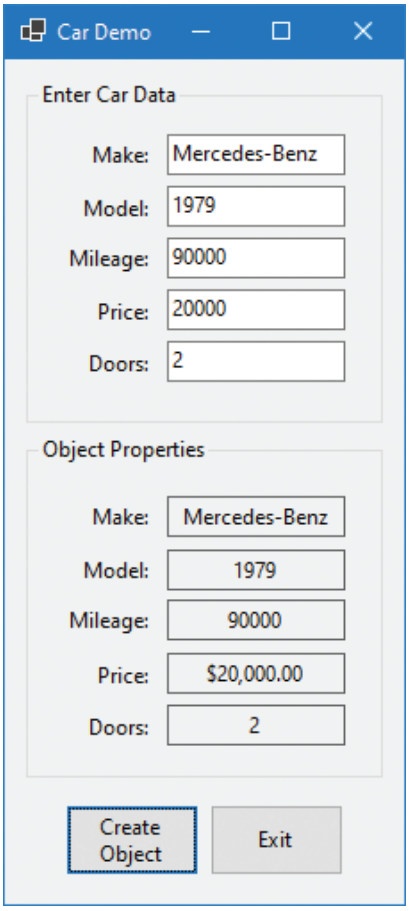


Figure 12-4 The *CarDemo* application



Program 12-1 shows the code in the Form1.cs file.

Program 12-1 Form1 code in the *CarDemo* project

```
namespace CarDemo
{
    public partial class Form1 : Form
    {
        public Form1()
        {
            InitializeComponent();
        }

        // The GetCarData method accepts a Car object as an
        // argument. It assigns the data entered by the
        // user to the object's properties.
        private void GetCarData(Car car)
        {
            // Temporary variables to hold mileage, price,
            // and number of doors
            int mileage;
            decimal price;
            int doors;

            // Get the car's make.
            car.Make = makeTextBox.Text;

            // Get the car's model.
            car.Model = modelTextBox.Text;

            // Get the car's mileage.
            if (int.TryParse(mileageTextBox.Text, out mileage))
            {
                car.Mileage = mileage;

                // Get the car's price.
                if (decimal.TryParse(priceTextBox.Text, out price))
                {
                    car.Price = price;

                    // Get the number of doors.
                    if (int.TryParse(doorsTextBox.Text, out doors))
                    {
                        car.Doors = doors;
                    }
                    else
                    {
                        // Display an error message.
                        MessageBox.Show("Invalid number of doors");
                    }
                }
                else
                {
                    // Display an error message.
                    MessageBox.Show("Invalid price");
                }
            }
            else
            {
                // Display an error message.
                MessageBox.Show("Invalid mileage");
```

```
58            }
59        }
60
61        private void createObjectButton_Click(object sender,
          EventArgs e)
62        {
63            // Create a Car object.
64            var myCar = new Car();
65
66            // Get the car data.
67            GetCarData(myCar);
68
69            // Display the car data.
70            makeLabel.Text = myCar.Make;
71            modelLabel.Text = myCar.Model;
72            mileageLabel.Text = myCar.Mileage.ToString();
73            priceLabel.Text = myCar.Price.ToString("c");
74            doorsLabel.Text = myCar.Doors.ToString();
75        }
76
77        private void exitButton_Click(object sender, EventArgs e)
78        {
79            // Close the form.
80            this.Close();
81        }
82    }
83 }
```

Let's take a closer look at the code:

Lines 13–59: This is the code for the `GetCarData` method. The purpose of the method is to accept a `Car` object as an argument, get the car data entered by the user (performing input validation on the mileage, price, and doors), and store that data in the `Car` object's properties.

Line 64: This statement creates an instance of the `Car` class, referenced by the `myCar` variable.

Line 67: This statement calls the `GetCarData` method, passing the `myCar` object as an argument. After the method executes, the `myCar` object's properties will contain the data entered by the user.

Lines 70–74: These statements display the `myCar` object's properties.

It's possible to derive more classes from the `Automobile` class. For example, suppose we need a `Truck` class and a `SportUtility` class. Here is the code for the `Truck` class:

```
1  class Truck: Automobile
2  {
3      // Constructor
4      public Truck()
5      {
6          Drive = "";
7      }
8
9      // Drive property
10     public string Drive { get; set; }
11 }
```

The `Truck` class has a `string` property named Drive. This property holds a value such as "Four-Wheel Drive" or "Two-Wheel Drive." The constructor, which appears in lines 4–7, assigns an empty string to the Drive property.

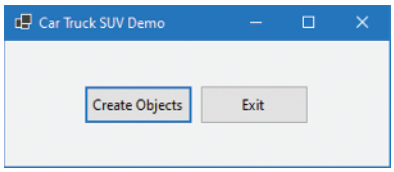
Now, let's look at the `SportUtility` class, which is also derived from the `Automobile` class:

```
1 class SportUtility: Automobile
2 {
3     // Constructor
4     public SportUtility()
5     {
6         Passengers = 0;
7     }
8
9     // Passengers property
10    public int Passengers { get; set; }
11 }
```

The `SportUtility` class has an `int` property named Passengers. This property holds the vehicle's passenger capacity. The constructor, which appears in lines 4–7, assigns an empty string to the Passengers property.

Now, let's look at a program that demonstrates all four of these classes. In the *Chap12* folder of the Student Sample Programs, you will find a project named *CarTruckSUVDemo*. This project contains a form named Form1, the `Automobile` class, the `Car` class, the `Truck` class, and the `SportUtility` class. Figure 12-5 shows the Form1 form at run time, and Program 12-2 shows the code in the Form1.cs file.

Figure 12-5 The *CarTruckSUVDemo* project's Form1 form



Program 12-2 Form1 code in the *CarTruckSUVDemo* project

```
1 namespace CarTruckSUVDemo
2 {
3     public partial class Form1 : Form
4     {
5         public Form1()
6         {
7             InitializeComponent();
8         }
9
10        private void createObjectsButton_Click(object sender,
          EventArgs e)
11        {
12            // Create a Car object for a used 2001 BMW with 70,000
13            // miles, priced at $15,000, with 4 doors.
14            var myCar = new Car();
15            myCar.Make = "BMW";
16            myCar.Model = "2001";
17            myCar.Mileage = 70000;
18            myCar.Price = 15000;
19            myCar.Doors = 4;
20
21            // Create a Truck object for a used 2002 Toyota
22            // pickup with 40,000 miles, priced at $12,000,
23            // with four wheel drive.
```

```
24              var myTruck = new Truck();
25              myTruck.Make = "Toyota";
26              myTruck.Model = "2002";
27              myTruck.Mileage = 40000;
28              myTruck.Price = 12000;
29              myTruck.Drive = "Four Wheel Drive";
30
31              // Create a SportUtility object for a used 2000
32              // Volvo with 30,000 miles, priced at $15,500,
33              // with a passenger capacity of 5.
34              var mySUV = new SportUtility();
35              mySUV.Make = "Volvo";
36              mySUV.Model = "2000";
37              mySUV.Mileage = 30000;
38              mySUV.Price = 15500;
39              mySUV.Passengers = 5;
40
41              // Display data about the car.
42              MessageBox.Show($"{myCar.Model} {myCar.Make} with " +
43                  $"{myCar.Mileage} miles, priced at {myCar.Price:c} " +
44                  $"with {myCar.Doors} doors.");
45
46              // Display data about the truck.
47              MessageBox.Show($"{myTruck.Model} {myTruck.Make} with " +
48                  $"{myTruck.Mileage} miles, priced at {myTruck.
                    Price:c} " +
49                  $"with {myTruck.Drive}.");
50
51              // Display data about the SUV.
52              MessageBox.Show($"{mySUV.Model} {mySUV.Make} with " +
53                  $"{mySUV.Mileage} miles, priced at {mySUV.Price:c} " +
54                  $"with {mySUV.Passengers} passengers.");
55          }
56
57          private void exitButton_Click(object sender, EventArgs e)
58          {
59              // Close the form.
60              this.Close();
61          }
62      }
63  }
```

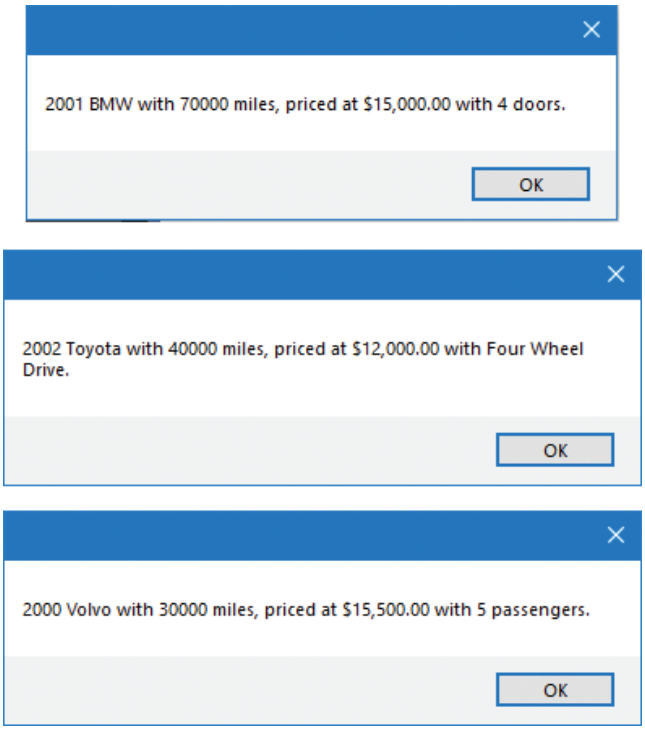
Let's step through the code in the `createObjectsButton_Click` event handler. Line 14 creates an instance of the `Car` class, referenced by the `myCar` variable. Lines 15–19 assign values to the objects Make, Model, Mileage, Price, and Doors properties.

Line 24 creates an instance of the `Truck` class, referenced by the `myTruck` variable. Lines 25–29 assign values to the objects Make, Model, Mileage, Price, and Drive properties.

Line 34 creates an instance of the `SportUtility` class, referenced by the `mySUV` variable. Lines 35–39 assign values to the objects Make, Model, Mileage, Price, and Passengers properties.

Lines 42–44 display the properties of the `myCar` object in the topmost message box shown in Figure 12-6. Lines 47–49 display the properties of the `myTruck` object in the middle message box shown in Figure 10-6. Lines 52–54 display the properties of the `mySUV` object in the bottom message box shown in Figure 12-6.

Figure 12-6 Message boxes displayed by the *CarTruckSUVDemo* project



In Tutorial 12-1, you create a base class for a savings account, and a derived class for a certificate of deposit.

Tutorial 12-1:
Creating and Testing the `SavingsAccount` and `CDAccount` Classes

VideoNote
Tutorial 12-1: Creating and Testing the `SavingsAccount` and `CDAccount` Classes

Bank Financial Systems, Inc., develops financial software for banks and credit unions. The company is developing a new system that manages customer accounts. One of your tasks is to develop a class that represents a savings account. The data that must be held by an object of this class is

- The account number
- The interest rate
- The account balance

You must also develop a class that represents a certificate of deposit (CD) account. The data that must be held by an object of this class is

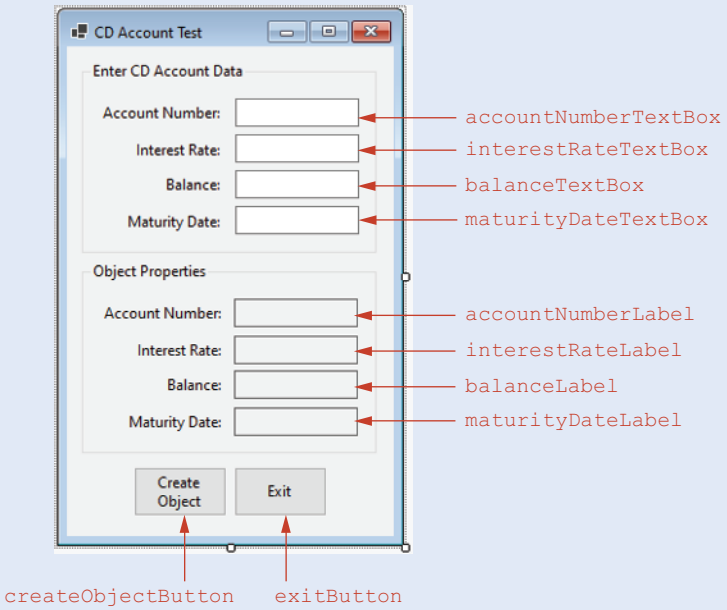
- The account number
- The interest rate
- The account balance
- The account maturity date

As you analyze these requirements, you realize that a CD account is really a specialized version of a savings account. The class that represents a CD will hold all the same data as the class that represents a savings account, plus an extra property for the maturity date. You decide to create a `SavingsAccount` class to represent a savings account and

then create a class that is derived from `SavingsAccount`, named `CDAccount`, to represent a CD account.

To test the classes, you will use them in an application that lets the user enter data about a CD account, creates an object of the `CDAccount` class, sets the object's properties to the data that the user entered, and then displays the object's data. The application's form, which has already been created for you, is shown in Figure 12-7.

Figure 12-7 The *CDAccountTest* project's form



Step 1: Start Visual Studio. Open the project named *CDAccountTest* in the *Chap12* folder of the Student Sample Programs.

Step 2: Click *Project* on the Visual Studio menu bar and then select *Add Class. . . .* The *Add New Item* window should appear. Make sure *Class* is selected as the type of item. Change the default filename to *SavingsAccount.cs* and then click the *Add* button. This adds a source code file named SavingsAccount.cs to the project.

Step 3: The SavingsAccount.cs file should now be displayed in the code editor. Complete the code for the `SavingsAccount` class by typing lines 11–22 in Program 12-3. The class code is very straightforward. It declares properties named AccountNumber, InterestRate, and Balance, as well as a parameterless constructor that initializes the properties to default values.

Step 4: Click *Project* on the Visual Studio menu bar and then select *Add Class. . . .* The *Add New Item* window should appear. Make sure *Class* is selected as the type of item. Change the default filename to *CDAccount.cs* and then click the *Add* button. This adds a source code file named CDAccount.cs to the project.

Step 5: The CDAccount.cs file should now be displayed in the code editor. Complete the code for the `CDAccount` class as follows:

- At the end of the class header type: `SavingsAccount` to indicate that this class is derived from the `SavingsAccount` class. (See line 9 in Program 12-4.)

- Type the code shown in lines 11–18 in Program 12-4. The class code is very straightforward. It declares a property named MaturityDate as well as a parameterless constructor that initializes the property to an empty string.

Step 6: Open the Form1.cs file in the code editor. Type the comments and code for the `GetCDData` method, shown in lines 10–47 of Program 12-5. Notice in line 13 that the method has a `CDAccount` parameter named `account`. When we call this method, we pass a `CDAccount` object to it as an argument.

The purpose of the `GetCDData` method is to read the data that the user has entered into the form's text boxes and store that data in the `account` object's properties. The values that have been entered for the interest rate and the balance are validated.

Step 7: Next, you create the Click event handlers for the Button controls. Switch your view to the Form1 form in the *Designer*. Double-click the `createObjectButton` control. This opens the Form1.cs file in the code editor, and you will see an empty event handler named `createObjectButton_Click`. Complete the event handler by typing the code shown in lines 51–61 in Program 12-5. Let's review this code:

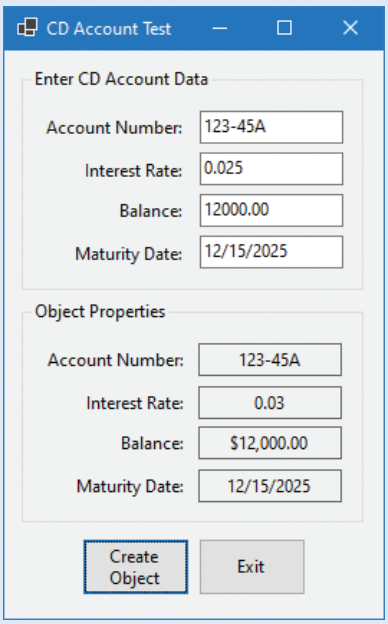
Line 52: This statement creates a `CDAccount` object in memory, referenced by a variable named `myAccount`.

Line 55: This statement calls the `GetCDData` method, passing the `myAccount` object as an argument. After the method executes, the `myAccount` object's properties will be set to the values entered by the user.

Lines 58–61: These statements display the values of the `myAccount` object's properties.

Step 8: Switch your view back to the Form1 form in the *Designer* and double-click the `exitButton` control. In the code editor you will see an empty event handler named `exitButton_Click`. Complete the event handler by typing the code shown in lines 66–67 in Program 12-5.

Figure 12-8 The *CD Account Test* application



NOTE: The MaturityDate property in the `CDAccount` class is implemented as a `string`. If you prefer, you can use the `DateTime` data type that we discussed in Chapter 9.

Step 9: Save the project. Then, press ... on the keyboard or click the *Start Debugging* button (▶) on the toolbar to compile and run the application. When the application runs, enter some sample data in the TextBox controls and click the *Create Object* button. You should see the values that you entered displayed in the *Object Properties* group box. Figure 12-8 shows an example. Click the *Exit* button when you are finished.

Program 12-3 Completed code for the SavingsAccount.cs file in the *CDAccountTest* project

```
 1 using System;
 2 using System.Collections.Generic;
 3 using System.Linq;
 4 using System.Text;
 5 using System.Threading.Tasks;
 6
 7 namespace CD_Account_Test
 8 {
 9     internal class SavingsAccount
10     {
11         // Constructor
12         public SavingsAccount()
13         {
14             AccountNumber = "";
15             InterestRate = 0;
16             Balance = 0;
17         }
18
19         // Properties
20         public string AccountNumber { get; set; }
21         public decimal InterestRate { get; set; }
22         public decimal Balance { get; set; }
23     }
24 }
```

Program 12-4 Completed code for the CDAccount.cs file in the *CDAccountTest* project

```
 1 using System;
 2 using System.Collections.Generic;
 3 using System.Linq;
 4 using System.Text;
 5 using System.Threading.Tasks;
 6
 7 namespace CD_Account_Test
 8 {
 9     class CDAccount : SavingsAccount
```

```
10      {
11          // Constructor
12          public CDAccount()
13          {
14              MaturityDate = "";
15          }
16
17          // MaturityDate property
18          public string MaturityDate { get; set; }
19      }
20  }
```

Program 12-5 Completed code for Form1 in the *CDAccountTest* project

```
 1  namespace CDAccountTest
 2  {
 3      public partial class Form1 : Form
 4      {
 5          public Form1()
 6          {
 7              InitializeComponent();
 8          }
 9
10          // The GetCDData method accepts a CDAccount object
11          // as an argument. It assigns the data entered by
12          // the user to the object's properties.
13          private void GetCDData(CDAccount account)
14          {
15              // Temporary variables to hold interest rate
16              // and balance
17              decimal interestRate;
18              decimal balance;
19
20              // Get the account number.
21              account.AccountNumber = accountNumberTextBox.Text;
22
23              // Get the maturity date.
24              account.MaturityDate = maturityDateTextBox.Text;
25
26              // Get the interest rate.
27              if (decimal.TryParse(interestRateTextBox.Text, out
                interestRate))
28              {
29                  account.InterestRate = interestRate;
30
31                  // Get the balance.
32                  if (decimal.TryParse(balanceTextBox.Text, out
                    balance))
33                  {
34                      account.Balance = balance;
35                  }
36                  else
37                  {
38                      // Display an error message.
39                      MessageBox.Show("Invalid balance");
40                  }
41              }
42              else
```

```
43            {
44                  // Display an error message.
45                  MessageBox.Show("Invalid interest rate");
46            }
47        }
48
49        private void createObjectButton_Click(object sender,
          EventArgs e)
50        {
51            // Create a CDAccount object.
52            var myAccount = new CDAccount();
53
54            // Get the CD account data.
55            GetCDData(myAccount);
56
57            // Display the CD account data.
58            accountNumberLabel.Text = myAccount.AccountNumber;
59            interestRateLabel.Text = myAccount.InterestRate.
              ToString("n2");
60            balanceLabel.Text = myAccount.Balance.ToString("c");
61            maturityDateLabel.Text = myAccount.MaturityDate;
62        }
63
64        private void exitButton_Click(object sender, EventArgs e)
65        {
66            // Close the form.
67            this.Close();
68        }
69    }
70 }
```

Base Class and Derived Class Constructors

When you create an instance of a derived class, the base class constructor is executed first, and then the derived class constructor is executed. In all the examples that we have discussed so far, the base class and the derived class have parameterless constructors. When you create an instance of a derived class, by default the base class's parameterless constructor is automatically executed.

But what happens if the base class's constructor has parameters? Or, what happens if the base class has multiple, overloaded constructors and you want to make sure a specific one is called? In either of these situations, the derived class constructor must explicitly call the base class constructor.

For example, look at the following `Rectangle` class:

```
1  class Rectangle
2  {
3     // Constructor
4      public Rectangle()
5      {
6          Length = 0;
7          Width = 0;
8      }
9
10     // Constructor
11     public Rectangle(int length, int width)
```

```
12      {
13          Length = length;
14          Width = width;
15      }
16
17      // Properties
18      public int Length { get; set; }
19      public int Width { get; set; }
20  }
```

This class holds data about a rectangle. It has Length and Width properties to store a rectangle's length and width. It also has two constructors: a parameterless constructor (lines 8–12) that initializes the Length and Width properties to 0 and a constructor that accepts arguments for the Length and the Width properties (lines 12–15).

The following `Box` class is derived from the `Rectangle` class:

```
 1  class Box: Rectangle
 2  {
 3      // Constructor
 4      public Box()
 5      {
 6          Height = 0;
 7      }
 8
 9      // Constructor
10      public Box(int length, int width, int height)
11         :base(length, width)
12      {
13          Height = height;
14      }
15      // Height property
16      public int Height { get; set; }
17  }
```

The `Box` class holds data about a rectangular box, which has a length, width, and height. The class inherits the Length and Width properties from the `Rectangle` class, and it has its own Height property. Notice that the `Box` class has two constructors: a parameterless constructor (lines 7–10) and a parameterized constructor (lines 13–17).

If you create an instance of the `Box` class, calling its parameterless constructor, the `Rectangle` class's parameterless constructor will be executed. Here is an example:

```
Box myBox = new Box();
```

After this statement executes, the `myBox` object's Length, Width, and Height properties will be set to 0.

Now, let's look at the `Box` class's parameterized constructor. The following code appears in line 10:

```
public Box(int length, int width, int height)
```

This constructor has three parameters: `length`, `width`, and `height`. Notice the following code in line 11:

```
: base(length, width)
```

The `base` keyword refers to the base class. This is an explicit call to the base class's parameterized constructor, passing `length` and `width` as arguments. To understand how this works, consider the following statement:

```
Box myBox = new Box(100, 200, 300);
```

This statement creates an instance of the `Box` class, passing 100 into the `length` parameter, 200 into the `width` parameter, and 300 into the `height` parameter. The code in line 14 of the `Box` class calls the `Rectangle` class's parameterized constructor, passing `length` and `width` as arguments. After the `Rectangle` class's constructor executes, the `Box` class constructor resumes execution. The resulting `Box` object's Length property will be set to 100, its Width property will be set to 200, and its Height property will be set to 300.

Summary of Constructor Issues in Inheritance

- When you create an instance of a derived class, the base class constructor is executed first and then the derived class constructor is executed.
- When you create an instance of a derived class, by default the base class's parameterless constructor is automatically executed.
- If you want a parameterized constructor in the base class to execute, you must explicitly call it from the derived class's constructor. You do this by writing the notation: `base(`*`parameterList`*`)` in the derived class's constructor header.
- If the base class does not have a parameterless constructor, the derived class constructor must use the notation: `base(`*`parameterList`*`)` to call one of the base class's parameterized constructors.

Protected Members

Until now you have used two access modifiers within a class: `public` and `private`. You know that `public` members of a class can be accessed by outside the class, as well as code inside the class. Class members that are `private` can only be accessed by code inside the same class. Even in an inheritance relationship, the private members of a base class cannot be accessed by code in a derived class.

For example, suppose we have the following `Person` class:

```
 1 class Person
 2 {
 3     // Field for ID Number
 4     private int idNumber;
 5
 6     // Constructor
 7     public Person(int id, string name)
 8     {
 9         idNumber = id;
10         Name = name;
11     }
12
13     // Name property
14     public string Name { get; set; }
15
16     // ToString method
17     public override string ToString()
18     {
19         return $"{idNumber} {Name}";
20     }
21 }
```

Line 4 declares a `private` field named `idNumber`. The field is assigned a value by the constructor in line 9. In line 19 the `ToString` method returns a string that includes the value of the field.

Because the `idNumber` field in the `Person` class is `private`, it cannot be accessed by code outside the class. For example, look at the following `Student` class, which is derived from the `Person` class:

```
1 internal class Student : Person
2 {
3     // Constructor
4     public Student(int id, string name, string major) : base(id, name)
5     {
6         Major = major;
7     }
8
9     // Major property
10    public string Major { get; set; }
11
12    // ChangeID method
13    public void ChangeID(int newID)
14    {
15        idNumber = newID; // ERROR! idNumber is private.
16    }
17
18    // ToString method
19    public override string ToString()
20    {
21        return base.ToString() + " " + Major;
22    }
23 }
```

Notice that this class has a method named `ChangeID` in lines 13–16. In this method, line 15 assigns a new value to the `idNumber` field. This statement will cause a compiler error, however, because the `idNumber` field is declared `private` in the base class.

When a member of a base class needs to be private to all outside code except derived classes, you can declare that member as `protected`. A **`protected` member** of a class may be directly accessed by methods of the same class or methods of a derived class. Here is an example of how we can declare the `idNumber` field in the `Person` class as protected:

```
1 class Person
2 {
3     // Field for ID Number
4     protected int idNumber;
5
6     // Constructor
7     public Person(int id, string name)
8     {
9         idNumber = id;
10        Name = name;
11    }
12
13    // Name property
14    public string Name { get; set; }
15
16    // ToString method
17    public override string ToString()
18    {
19        return $"{idNumber} {Name}";
20    }
21 }
```

In this version of the class, the `protected` access modifier is used in the declaration of the `idNumber` field in line 4. Now, the `Student` class will be able to directly access the field.

Restricting Inheritance with the `sealed` Keyword

You can use the **`sealed` keyword** to make sure that a class is not inherited from, or that a method is not overridden. Here is an example of a `sealed` class:

```
sealed class YourClass
{
    // Member declarations
}
```

Because `YourClass` is sealed, a compiler error will occur if any other class tries to use it as a base class. Although a sealed class cannot be inherited from, it can inherit from a base class. Here is an example:

```
class MyClass
{
    // Member declarations
}

sealed class YourClass: MyClass
{
    // Member declarations
}
```

In this example, `YourClass` inherits from `MyClass`, but no class can inherit from `YourClass`.

Checkpoint

12.1 In this section, we discussed base classes and derived classes. Which is the general class and which is the specialized class?

12.2 What does it mean to say there is an "is a" relationship between two objects?

12.3 What does a derived class inherit from its base class?

12.4 Look at the following code, which is the first line of a class declaration. What is the name of the base class? What is the name of the derived class?

```
class Canary: Bird
```

12.5 Briefly summarize the constructor issues in inheritance.

12.6 When a class member is declared as `protected`, what code may access it?

12.7 What is the difference between private members and protected members?

12.2 Polymorphism

CONCEPT: **Polymorphism allows derived classes to have methods with the same names as methods in their base classes. It gives the ability for a program to call the correct method, depending on the type of object that is used to call it.**

The term **polymorphism** refers to an object's ability to take different forms. It is a powerful feature of object-oriented programming. In this section, we look at two essential ingredients of polymorphic behavior:

1. The ability to define a method in a base class and then define a method with the same name in a derived class. When a derived class method has the same name as a base class method, the derived class method overrides the base class method.

2. The ability to call the correct version of an overridden method, depending on the type of object that is used to call it. If a derived class object is used to call an overridden method, then the derived class's version of the method is the one that executes. If a base class object is used to call an overridden method, then the base class's version of the method is the one that executes.

Perhaps the best way to describe polymorphism is to demonstrate it, so let's consider a simple example. Look at the following code for the `Animal` class:

```
1 class Animal
2 {
3     // Constructor
4     public Animal(string species)
5     {
6         Species = species;
7     }
8
9     // Species property
10    public string Species { get; set; }
11
12    // MakeSound method
13    public virtual void MakeSound()
14    {
15        MessageBox.Show("Grrrrrrr");
16    }
17 }
```

Line 10 declares the Species property. The constructor in lines 4–7 takes an argument that is assigned to the Species property.

The class has a method named `MakeSound`, in lines 13–16. The method simply displays `"Grrrrrrr"` in a message box. Notice the `virtual` keyword that appears in the method header. The `virtual` keyword declares that a derived class is allowed to override this method.

Here is an example of code that uses an instance of the `Animal` class:

```
1  Animal myAnimal = new Animal("regular animal");
2  MessageBox.Show($"The species is {myAnimal.Species}.");
3  myAnimal.MakeSound();
```

The statement in line 1 creates an object of the `Animal` class, passing the string `"regular animal"` to the constructor. The object is referenced by the `myAnimal` variable. Line 2 displays "The species is regular animal" in a message box. Line 3 calls the `myAnimal` object's `MakeSound` method, which displays "Grrrrrrr" in a message box.

Next, look at the `Dog` class, which is derived from the `Animal` class:

```
1 class Dog: Animal
2 {
3     // Constructor
4     public Dog(string name): base("Dog")
5     {
6         Name = name;
7     }
8
9     // Name property
10    public string Name { get; set; }
11
12    // MakeSound method
13    public override void MakeSound()
```

```
14      {
15          MessageBox.Show("Woof! Woof!");
16      }
17  }
```

Line 10 declares the Name property. The constructor in lines 4–7 takes an argument for the dog's name. Notice that the constructor calls the base class constructor, passing the string `"Dog"` as an argument. The `Animal` class constructor assigns the string `"Dog"` to the Species property.

Although the `Dog` class is derived from the `Animal` class, the `Animal` class's `MakeSound` method is inadequate for the `Dog` class. So, the `Dog` class overrides the `Animal` class's `MakeSound` method. The `Dog` class's `MakeSound` method, in lines 13–16, displays `"Woof! Woof!"` in a message box, which is more appropriate for a dog. Notice the `override` keyword that appears in the method header. The `override` keyword declares that this method overrides a method in the base class.

Here is an example of code that uses an instance of the `Dog` class:

```
1 Dog myDog = new Dog("Fido");
2 MessageBox.Show($"The species is {myDog.Species}.");
3 MessageBox.Show($"The animal's name is {myDog.Name}.");
4 myDog.MakeSound();
```

The statement in line 1 creates an object of the `Dog` class, passing the string `"Fido"` to the constructor. The object is referenced by the `myDog` variable. Line 2 displays "The species is Dog" in a message box. Line 3 displays "The animal's name is Fido" in a message box. Line 4 calls the `myDog` object's `MakeSound` method. When we use a `Dog` object to call the `MakeSound` method, the version of the method that is in the `Dog` class is the one that executes. So, the statement in line 4 displays "Woof! Woof!" in a message box.

Next, look at the `Cat` class, which is also derived from the `Animal` class:

```
1  class Cat: Animal
2  {
3      // Constructor
4      public Cat(string name): base("Cat")
5      {
6          Name = name;
7      }
8
9      // Name property
10     public string Name { get; set; }
11
12     // MakeSound method
13     public override void MakeSound()
14     {
15         MessageBox.Show("Meow");
16     }
17 }
```

Line 10 declares the Name property. The constructor in lines 4–7 takes an argument for the cat's name. Notice that the constructor calls the base class constructor, passing the string `"Cat"` as an argument. The `Animal` class constructor assigns the string `"Cat"` to the Species property.

Although the `Cat` class is derived from the `Animal` class, the `Animal` class's `MakeSound` method is inadequate for the `Cat` class. So, the `Cat` class overrides the `Animal` class's `MakeSound` method. The `Cat` class's `MakeSound` method, in lines 13–16, displays `"Meow"` in a message box, which is more appropriate for a cat. Notice the `override` keyword that appears in the method header.

Here is an example of code that uses an instance of the `Cat` class:

```
1 Cat myCat = new Cat("Kitty");
2 MessageBox.Show($"The species is {myCat.Species}.");
3 MessageBox.Show($"The animal's name is {myCat.Name}.");
4 myCat.MakeSound();
```

The statement in line 1 creates an object of the `Cat` class, passing the string `"Kitty"` to the constructor. The object is referenced by the `myCat` variable. Line 2 displays "The species is Cat" in a message box. Line 3 displays "The animal's name is Kitty" in a message box. Line 4 calls the `myCat` object's `MakeSound` method. When we use a `Cat` object to call the `MakeSound` method, the version of the method that is in the `Cat` class is the one that executes. So, the statement in line 4 displays "Meow" in a message box.

Because of the "is a" relationship between a base class and a derived class, an object of the `Dog` class is not just a `Dog` object. It is also an `Animal` object. (A dog is an animal.) Because of this relationship, we can use an `Animal` class variable to reference a `Dog` object. For example, look at the following code:

```
1 Animal myAnimal = new Dog("Fido");
2 MessageBox.Show($"The species is {myAnimal.Species}.");
3 myAnimal.MakeSound();
```

Line 1 declares `myAnimal` as a variable of the `Animal` type. It also creates an object of the `Dog` class and assigns a reference to the `Dog` object to the `myAnimal` variable. After this statement executes, we will have an `Animal` variable referencing a `Dog` object. This assignment is legal because a `Dog` object is also an `Animal` object. Line 2 displays "The species is Dog" in a message box. Line 3 calls the `myAnimal` object's `MakeSound` method, which displays "Woof! Woof!" in a message box.

Similarly, we can use an `Animal` variable to reference a `Cat` object, as shown in the following code:

```
1 Animal myAnimal = new Cat("Kitty");
2 MessageBox.Show($"The species is {myAnimal.Species}.");
3 myAnimal.MakeSound();
```

The statement in line 2 displays "The species is Cat" in a message box. Line 3 calls the `myAnimal` object's `MakeSound` method, which displays "Meow" in a message box.

Overriding Properties

Properties in a base class can be overridden in the same way that methods can be overridden. In the base class, you write the `virtual` keyword in the property declaration, as shown here:

```
1  class Rectangle
2  {
3      // Properties
4      public int Length { get; set; }
5      public int Width { get; set; }
6
7      public virtual int Area
8      {
9          get { return Length * Width; }
10     }
11 }
```

This `Rectangle` class declares Length and Width properties, and a virtual read-only Property named Area (notice the `virtual` keyword in line 7). The Area property returns the area of the rectangle, which is calculated as Length * Width.

Suppose we need a `Box` class to represent a box that has a length, a width, and a height. We could derive the `Box` class from the `Rectangle` class, and in the `Box` class we could add a Height property to account for the box's height. However, in the `Box` class, we will have to override the Area property because the area of a three-dimensional box is calculated differently than the area of a two-dimensional rectangle. Here is an example:

```
1 class Box: Rectangle
2 {
3     // Properties
4     public int Height { get; set; }
5
6     public override int Area
7     {
8         get { return base.Area * Height; }
9     }
10 }
```

The `Box` class's Height property appears in line 4. In lines 6–9, we override the Area property (notice the `override` keyword in line 6). In line 8, we use the expression `base.Area` to get the value of the base class's Area property (which gives us Length * Width), and we multiply that value by Height. The result is the area of the box.

Base Class Reference Variables Know about Base Class Members Only

A base class reference variable can reference an object of any class that is derived from the base class; however, there is a limit to what the base class variable can do with those objects. A base class reference variable knows only about the members that are declared in the base class. If the derived class introduces additional methods, properties, or fields, a base class reference variable cannot access them. For example, look at the following code:

```
1 Animal myAnimal = new Dog("Fido");
2 MessageBox.Show($"The species is {myAnimal.Species}.");
3 MessageBox.Show($"The animal's name is {myAnimal.Name}."); // ERROR!
4 myAnimal.MakeSound();
```

Line 3 in this code will not compile. The `myAnimal` variable is an `Animal` reference variable, and the `Animal` class does not have a Name property. Even though the `myAnimal` variable is referencing a `Dog` object, it cannot access the object's Name property.

The "Is a" Relationship Does Not Work in Reverse

It is important to understand that the "is a" relationship does not work in reverse. Although the statement "a dog is an animal" is true, the statement "an animal is a dog" is not true. This is because all dogs are animals, but not all animals are dogs. So, the following statement will not compile:

```
Dog myDog = new Animal("Dog");
```

You cannot assign an `Animal` reference to a `Dog` variable. This makes sense because `Dog` objects have capabilities that go beyond those of an `Animal` object.

NOTE: Interestingly, the C# compiler will allow you to make such an assignment if you use a cast operator, as shown here:

```
Dog myDog = (Dog) new Animal("Dog");
```

However, an exception will be thrown at run time when the assignment takes place.

Passing Objects to Base Class Parameters

Polymorphism gives you a great deal of flexibility when designing applications. For example, look at the following method:

```
private void ShowAnimalInfo(Animal animal)
{
    MessageBox.Show($"Species: {animal.Species}");
    animal.MakeSound();
}
```

This method displays information about an animal. Because it has an `Animal` variable as its parameter, you can pass an `Animal` object to the method when you call it. The method then displays the object's Species property and calls its `MakeSound` method.

The `ShowAnimalInfo` method works with an `Animal` object, but what if you also need methods that display the same information about `Dog` objects and `Cat` objects? Do you need to write additional methods for each of these types? Because of polymorphism, the answer is *no*. In addition to `Animal` objects, you can also pass `Dog` objects or `Cat` objects as arguments to the `ShowAnimalInfo` method previously shown.

In Tutorial 12-2, you complete an application that uses the `Animal`, `Dog`, and `Cat` classes to demonstrate polymorphism.

Tutorial 12-2:
Completing the *Polymorphism* Application

VideoNote
Tutorial 12-2: Completing the *Polymorphism* Project

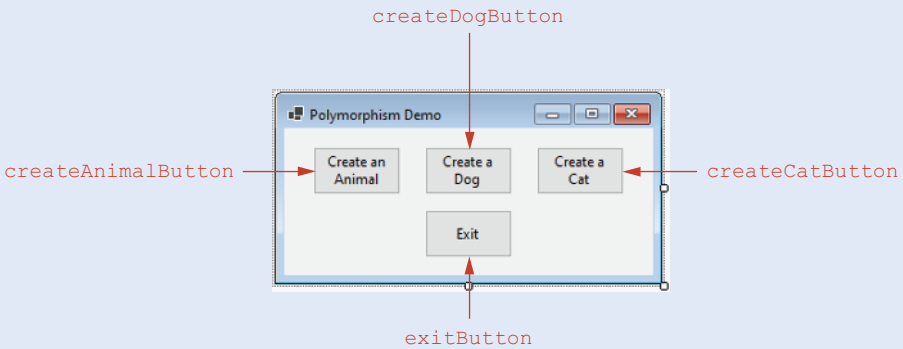
In this tutorial you complete the *Polymorphism* project, which demonstrates the `Animal`, `Dog`, and `Cat` classes shown in this section. The project also incorporates the `ShowAnimalInfo` method previously described.

The project's form, which has already been created for you, is shown in Figure 12-9. When you run the completed application, clicking the *Create an Animal* button creates an object of the `Animal` class and passes the object to the `ShowAnimalInfo` method. Clicking the *Create a Dog* button creates an object of the `Dog` class and passes that object to the `ShowAnimalInfo` method. Clicking the *Create a Cat* button creates an object of the `Cat` class and passes that object to the `ShowAnimalInfo` method.

Step 1: Start Visual Studio. Open the project named *Polymorphism* in the *Chap12* folder of the Student Sample Programs.

Step 2: Click *Project* on the Visual Studio menu bar and then select *Add Class*. . . . The *Add New Item* window should appear. Make sure *Class* is selected as the type

Figure 12-9 The *Polymorphism* project's form



of item. Change the default filename to *Animal.cs* and then click the *Add* button. This adds a source code file named Animal.cs to the project.

Step 3: The Animal.cs file should now be displayed in the code editor. Complete the code for the `Animal` class by typing lines 11–24 in Program 12-6.

Step 4: Click *Project* on the Visual Studio menu bar and then select *Add Class*. . . . The *Add New Item* window should appear. Make sure *Class* is selected as the type of item. Change the default filename to *Dog.cs* and then click the *Add* button. This adds a source code file named Dog.cs to the project.

Step 5: The Dog.cs file should now be displayed in the code editor. Add the `: Animal` notation to the class header, shown in line 9 of Program 12-7. Then, complete the code for the `Dog` class by typing lines 11–24 in Program 12-7.

Step 6: Click *Project* on the Visual Studio menu bar and then select *Add Class*. . . . The *Add New Item* window should appear. Make sure *Class* is selected as the type of item. Change the default filename to *Cat.cs*, and then click the *Add* button. This adds a source code file named Cat.cs to the project.

Step 7: The Cat.cs file should now be displayed in the code editor. Add the `: Animal` notation to the class header, shown in line 9 of Program 12-8. Then, complete the code for the `Cat` class by typing lines 11–24 in Program 12-8.

Step 8: Open the Form1.cs file in the code editor. Type the comments and code for the `ShowAnimalInfo` method shown in lines 10–17 of Program 12-9.

Step 9: Next, you create the Click event handlers for the Button controls. Switch your view to the Form1 form in the Designer. Double-click the `createAnimalButton` control. This opens the Form1.cs file in the code editor, and you will see an empty event handler named `createAnimalButton_Click`. Complete the event handler by typing the code shown in lines 21–22 in Program 12-9.

Let's review this code. Line 21 creates an object of the `Animal` class, passing `"Regular animal"` as an argument to the constructor. Line 22 passes the object to the `ShowAnimalInfo` method.

Step 10: Switch your view to the Form1 form in the *Designer*. Double-click the `createDogButton` control. This will open the Form1.cs file in the code editor, and you will see an empty event handler named `createDogButton_Click`. Complete the event handler by typing the code shown in lines 27–29 in Program 12-9.

Let's review this code. Line 27 creates an object of the `Dog` class, passing `"Fido"` as an argument to the constructor. Line 28 displays the object's Name property in a message box. Line 29 passes the object to the `ShowAnimalInfo` method.

Step 11: Switch your view to the Form1 form in the Designer. Double-click the `createCatButton` control. This opens the Form1.cs file in the code editor, and you will see an empty event handler named `createCatButton_Click`. Complete the event handler by typing the code shown in lines 34–36 in Program 12-9.

Let's review this code. Line 34 creates an object of the `Cat` class, passing `"Kitty"` as an argument to the constructor. Line 35 displays the object's Name property in a message box. Line 36 passes the object to the `ShowAnimalInfo` method.

Step 12: Switch your view back to the Form1 form in the *Designer* and double-click the `exitButton` control. In the code editor you will see an empty event handler

named `exitButton_Click`. Complete the event handler by typing the code shown in lines 41–42 in Program 12-9.

Step 13: Save the project. Then, press ... on the keyboard or click the *Start Debugging* button( ) on the toolbar to compile and run the application. When the application runs, click the *Create an Animal* button. You should see the following messages displayed in message boxes:

Species: Regular animal

Grrrrrrr

Next, click the *Create a Dog* button. You should see the following messages displayed in message boxes:

The dog's name is Fido.

Species: Dog

Woof! Woof!

Next, click the *Create a Cat* button. You should see the following messages displayed in message boxes:

The cat's name is Kitty.

Species: Cat

Meow

Click the *Exit* button when you are finished.

Program 12-6 Completed code for the Animal.cs file in the *Polymorphism* project

```
 1 using System;
 2 using System.Collections.Generic;
 3 using System.Linq;
 4 using System.Text;
 5 using System.Threading.Tasks;
 6
 7 namespace Polymorphism
 8 {
 9     internal class Animal
10     {
11         // Constructor
12         public Animal(string species)
13         {
14             Species = species;
15         }
16
17         // Species property
18         public string Species { get; set; }
19
20         // MakeSound method
21         public virtual void MakeSound()
22         {
23             MessageBox.Show("Grrrrrrr");
24         }
25     }
26 }
```

Program 12-7 Completed code for the Dog.cs file in the *Polymorphism* project

```
 1 using System;
 2 using System.Collections.Generic;
 3 using System.Linq;
 4 using System.Text;
 5 using System.Threading.Tasks;
 6
 7 namespace Polymorphism
 8 {
 9     internal class Dog : Animal
10     {
11         // Constructor
12         public Dog(string name): base("Dog")
13         {
14             Name = name;
15         }
16
17         // Name property
18         public string Name { get; set; }
19
20         // MakeSound method
21         public override void MakeSound()
22         {
23             MessageBox.Show("Woof! Woof!");
24         }
25     }
26 }
```

Program 12-8 Completed .code for the Cat.cs file in the *Polymorphism* project

```
 1 using System;
 2 using System.Collections.Generic;
 3 using System.Linq;
 4 using System.Text;
 5 using System.Threading.Tasks;
 6
 7 namespace Polymorphism
 8 {
 9     internal class Cat : Animal
10     {
11         // Constructor
12         public Cat(string name): base("Cat")
13         {
14             Name = name;
15         }
16
17         // Name property
18         public string Name { get; set; }
19
20         // MakeSound method
21         public override void MakeSound()
22         {
23             MessageBox.Show("Meow");
24         }
25     }
26 }
```

Program 12-9 Completed code for Form1 in the *Polymorphism* project

```
 1 namespace Polymorphism
 2 {
 3     public partial class Form1 : Form
 4     {
 5         public Form1()
 6         {
 7             InitializeComponent();
 8         }
 9
10         // The ShowAnimalInfo method accepts an Animal
11         // object as an argument. It displays the object's
12         // species and calls its MakeSound method.
13         private void ShowAnimalInfo(Animal animal)
14         {
15             MessageBox.Show($"Species: {animal.Species}");
16             animal.MakeSound();
17         }
18
19         private void createAnimalButton_Click(object sender,
           EventArgs e)
20         {
21             var myAnimal = new Animal("Regular animal");
22             ShowAnimalInfo(myAnimal);
23         }
24
25         private void createDogButton_Click(object sender,
           EventArgs e)
26         {
27             var myDog = new Dog("Fido");
28             MessageBox.Show($"The dog's name is {myDog.Name}.");
29             ShowAnimalInfo(myDog);
30         }
31
32         private void createCatButton_Click(object sender,
           EventArgs e)
33         {
34             var myCat = new Cat("Kitty");
35             MessageBox.Show($"The cat's name is {myCat.Name}.");
36             ShowAnimalInfo(myCat);
37         }
38
39         private void exitButton_Click(object sender, EventArgs e)
40         {
41             // Close the form.
42             this.Close();
43         }
44     }
45 }
```

Preventing Method Overriding with the `sealed` Keyword

When a method is declared with the `sealed` keyword, it cannot be overridden in a subclass. The following class definitions show an example:

```
class MyClass
{
    public virtual void Message()
```

```
    {
        // Code goes here
    }
}
class YourClass: MyClass
{
    public sealed void Message()
    {
        // Code goes here
    }
}
```

In this example, `YourClass` inherits from `MyClass` and overrides the `Message` method. The `YourClass` class is not sealed so other classes may inherit from it. However, in `YourClass`, the `Message` method is sealed, so it cannot be overridden in any class that inherits from `YourClass`. This technique can be used to make sure that a particular base class method is used by subclasses and not a modified version of it.

Checkpoint

12.8 Look at the following class definitions:

```
class Vegetable
{
    public virtual void message()
    {
       MessageBox.Show("I'm a vegetable.");
    }
}

class Potato: Vegetable
{
    public override void message()
    {
       MessageBox.Show("I'm a potato.");
    }
}
```

Given these class declarations, what will the following code display?

```
Vegetable v = new Potato();
Potato p = new Potato();
v.message();
p.message();
```

12.9 Does the "is a" relationship work in reverse? Why or why not?

12.3 Abstract Classes

CONCEPT: An abstract class serves as a base class but is not instantiated itself. An abstract method has no body and must be overridden in a derived class.

Sometimes it does not make sense to instantiate a base class. For example, consider a factory that manufactures airplanes. The factory does not make a generic airplane but makes two specific types of airplanes: a prop-driven plane and a commuter jet. The computer software that catalogs the planes might use a class named `Airplane`. The `Airplane` class

has members representing the common characteristics of all airplanes. In addition, the software has a class named `PropPlane` and a class named `CommuterJet`. These classes represent the two specific airplane models the factory manufactures and are derived from the `Airplane` class. The `Airplane` class is never instantiated but is used as a base class for the other classes.

A class that is not intended to be instantiated but is to be used only as a base class is called an **abstract class**. An abstract class serves as a starting point, providing some members for its derived classes. To declare a class as abstract, you use the `abstract` keyword in the class header. Here is the general format:

```
abstract class ClassName
{
    // Member declarations
}
```

The primary difference between an abstract class and a regular class (which is sometimes called a **concrete class**) is that the abstract class cannot be instantiated. A statement that tries to use the `new` operator to instantiate an abstract class will not compile.

Abstract classes can also contain abstract methods. An **abstract method** is a method that appears in a base class but expects to be overridden in a derived class. An abstract method has only a header and no body. In the header, the `abstract` keyword appears before the return type. Here is a very simple example of an abstract class that contains an abstract method:

```
abstract class Person
{
    public abstract void DoSomething();
}
```

When an abstract method appears in a class, it must be overridden in any class that is derived from the class. In this example, if a class is derived from the `Person` class, the derived class must override the `DoSomething` method.

Abstract classes can also contain abstract properties. An **abstract property** is a property that appears in a base class but expects to be overridden in a derived class. In the property header, the `abstract` keyword appears before the property type. Here is a very simple example of an abstract class that contains an abstract method:

```
abstract class Person
{
    public abstract string JobTitle
    {
        get;
        set;
    }
}
```

Notice that the abstract property shows `get` and `set` accessors, but it does not specify what those accessors do. When an abstract property appears in a class, it must be overridden in any class that is derived from the class. In this example, if a class is derived from the `Person` class, the derived class must override the `JobTitle` property.

NOTE: If you want to create an abstract read-only property, leave out the `set` accessor.

In Tutorial 12-3, you complete an application that uses an abstract base class with an abstract property.

Tutorial 12-3:
Completing the *ComputerScienceStudent* Project

VideoNote
Tutorial 12-3: Completing the *ComputerScienceStudent* Project

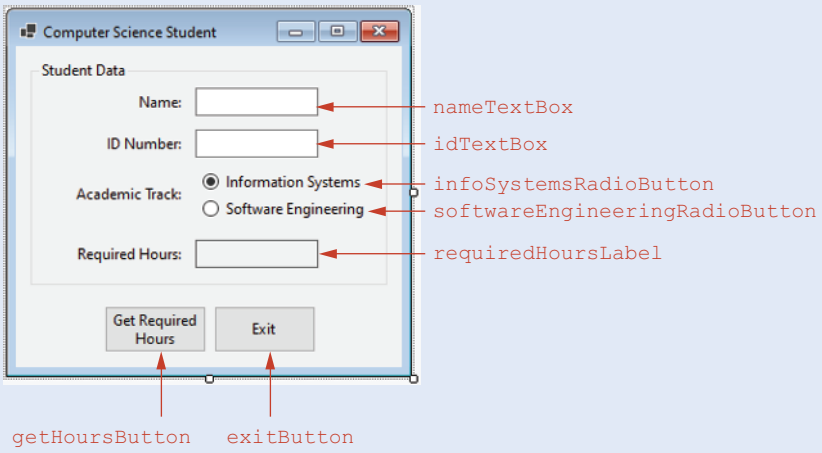
In this tutorial, you write an abstract class named `Student`. The `Student` class holds data that is common to all students, but it does not hold all the data needed for students of specific majors. The `Student` class is intended to be a base class that can be derived by other classes that represent students of specific majors.

The `Student` class has a Name property to hold a student's name and an ID property to hold a student's ID number. It also has an abstract read-only property named RequiredHours. The purpose of the RequiredHours property is to hold the number of required hours for a specific major. Any class that is derived from the `Student` class must override the RequiredHours property.

You also write a class named `CompSciStudent`, which is derived from the Student class. The `CompSciStudent` class has a field named `AcademicTrack`, which holds the name of the student's academic track. It also overrides the RequiredHours property to calculate and return the number of hours required for a computer science major.

You will demonstrate the classes in the *ComputerScienceStudent* Project. The application's form, which has already been created for you, is shown in Figure 12-10. When you run the completed application, you will enter a student's name and ID number into the text boxes, and you will select an academic track using the radio buttons. When you click the *Get Required Hours* button, the application will create a `CompSci Student` object, initialized with the data you entered. It will then get the value of the RequiredHours property and display it on the form.

Figure 12-10 The *ComputerScienceStudent* project's form



Step 1: Start Visual Studio. Open the project named *ComputerScienceStudent* in the *Chap12* folder of the Student Sample Programs.

Step 2: Click *Project* on the Visual Studio menu bar and then select *Add Class. . .* . The *Add New Item* window should appear. Make sure *Class* is selected as the type of item. Change the default filename to *Student.cs* and then click the *Add* button. This adds a source code file named Student.cs to the project.

Step 3: The Student.cs file should now be displayed in the code editor. Add the `abstract` keyword to the class header, shown in line 9 of Program 12-10. Next, complete the code for the `Student` class by typing lines 11–23 in Program 12-10.

Step 4: Click *Project* on the Visual Studio menu bar and then select *Add Class*. . . . The *Add New Item* window should appear. Make sure *Class* is selected as the type of item. Change the default filename to *CompSciStudent.cs* and then click the *Add* button. This adds a source code file named CompSciStudent.cs to the project.

Step 5: The CompSciStudent.cs file should now be displayed in the code editor. Add the `: Student` notation to the class header, shown in line 9 of Program 12-11. Then, complete the code for the `CompSciStudent` class by typing lines 11–30 in Program 12-11.

Notice that the RequiredHours property, in lines 27–30, overrides the abstract RequiredHours property in the base class. This property calculates and returns the number of required hours.

Step 6: Now you will create the Click event handlers for the Button controls. Switch your view to the Form1 form in the *Designer*. Double-click the `getHoursButton` control. This opens the Form1.cs file in the code editor, and you will see an empty event handler named `getHoursButton_Click`. Complete the event handler by typing the code shown in lines 12–33 in Program 12-12. Let's review this code:

Line 13: This statement declares three local `string` variables: `name`, `id`, and `track`. They will be used to hold user input.

Lines 16–17: These statements get the name and ID number entered by the user and assign those values to the `name` and `id` variables.

Lines 20–27: This `if-else` statement determines which radio button is selected and assigns a value to the `track` variable.

Lines 30: This statement creates a `CompSciStudent` object, referenced by the variable `csStudent`, and initializes it with the data entered by the user.

Lines 33: This statement gets the value of the `csStudent` object's RequiredHours property and displays it in the `requiredHoursLabel` control.

Step 7: Switch your view back to the Form1 form in the *Designer* and double-click the `exitButton` control. In the code editor you will see an empty event handler named `exitButton_Click`. Complete the event handler by typing the code shown in lines 38–39 in Program 12-12.

Step 8: Save the project. Then, press ... on the keyboard or click the *Start Debugging* button ( ) on the toolbar to compile and run the application. When the application runs, enter some sample data in the TextBox controls, select a value with the radio buttons, and click the *Get Required Hours* button. You should see the value 120.0 displayed as the number of required hours. Click the *Exit* button when you are finished.

Program 12-10 Completed code for the Student.cs file in the *ComputerScience-Student* project

```
1 using System;
2 using System.Collections.Generic;
3 using System.Linq;
4 using System.Text;
5 using System.Threading.Tasks;
```

```
 6
 7 namespace ComputerScienceStudent
 8 {
 9    internal abstract class Student
10     {
11        // Constructor
12        public Student(string name, string id)
13        {
14            Name = name;
15            ID = id;
16        }
17
18        // Concrete Properties
19        public string Name { get; set; }
20        public string ID { get; set; }
21
22        // RequiredHours property (abstract)
23        public abstract double RequiredHours { get; }
24     }
25 }
```

Program 12-11 Completed code for the CompSciStudent.cs file in the *ComputerScienceStudent* project

```
 1 using System;
 2 using System.Collections.Generic;
 3 using System.Linq;
 4 using System.Text;
 5 using System.Threading.Tasks;
 6
 7 namespace ComputerScienceStudent
 8 {
 9     internal class CompSciStudent : Student
10     {
11         // Constants
12         private const double MATH_HOURS = 20;
13         private const double CS_HOURS = 40;
14         private const double GEN_HOURS = 60;
15
16         // Constructor
17         public CompSciStudent(string name, string id, string
           track)
18             : base(name, id)
19         {
20             AcademicTrack = track;
21         }
22
23         // AcademicTrack property
24         public string AcademicTrack { get; set; }
25
26         // RequiredHours property
27         public override double RequiredHours
28         {
29             get { return MATH_HOURS + CS_HOURS + GEN_HOURS; }
30         }
31     }
32 }
```

Program 12-12 Completed code for Form1 in the *ComputerScienceStudent* project

```
 1 namespace ComputerScienceStudent
 2 {
 3     public partial class Form1 : Form
 4     {
 5         public Form1()
 6         {
 7             InitializeComponent();
 8         }
 9
10         private void getHoursButton_Click(object sender,
           EventArgs e)
11         {
12             // Variables to hold input
13             string name, id, track;
14
15             // Get the student's name and ID.
16             name = nameTextBox.Text;
17             id = idTextBox.Text;
18
19             // Get the student's academic track.
20             if (infoSystemsRadioButton.Checked)
21             {
22                 track = "Information Systems";
23             }
24             else
25             {
26                 track = "Software Engineering";
27             }
28
29             // Create a CompSciStudent object.
30             var csStudent = new CompSciStudent(name, id, track);
31
32             // Display the student's required hours.
33             requiredHoursLabel.Text = csStudent.RequiredHours.
               ToString("n1");
34         }
35
36         private void exitButton_Click(object sender, EventArgs e)
37         {
38             // Close the form.
39             this.Close();
40         }
41     }
42 }
```

Checkpoint

12.10 What is the purpose of an abstract class?

12.11 If a class is abstract, what cannot be done with the class?

12.12 If a class is derived from a base class that has an abstract method, what must the derived class do?

12.13 What must be done with an abstract property before it can be used?

12.14 How can you create an abstract read-only property?

12.4 Extension Methods

CONCEPT: An extension method is a static method that extends an existing class or data type, including classes and types in .NET.

Suppose you want to extend a class, but you cannot inherit from it because it is sealed. This might be the case when the class you want to extend is part of .NET, or was written by someone other than you. In this situation, you can use extension methods to extend the class. An **extension method** is a method that you can add onto a class without modifying the class or inheriting from it.

Extension methods can be applied to classes, `struct`s, and interfaces (which are discussed later in this chapter). They can even be applied to primitive types like `int` and `double`. In fact, let's look at how we can apply an extension method to the primitive `int` data type. Suppose we want `int` variables to have a method that tells us whether the variable is odd or even. For example, we would like to be able to do something like this:

```
int number = 4;
if (number.IsEven())
{
    MessageBox.Show("The number is even.");
}
```

To accomplish this, we need to define an extension method for the `int` type named `IsEven`. The extension method will return `true` if the number it is evaluating is even, or `false` otherwise.

Here are the key things to know about implementing an extension method:

- An extension method is a static method, stored in a static class. You can write multiple extension methods in the same static class.
- It is recommended to put the extension method's class in its own namespace, separate from your project's namespace.
- An extension always has at least one parameter. The method's first parameter must be an object of the class or type that the method is extending. The declaration of the first parameter must begin with the keyword `this`.

Here's an example of how we might implement the `IsEven` extension method previously shown:

```
1  namespace ExtensionMethods
2  {
3      public static class Extensions
4      {
5          public static bool IsEven(this int i)
6          {
7              return (i % 2) == 0;
8          }
9      }
10 }
```

This code defines a namespace named `ExtensionMethods`, and inside that namespace is a static class named `Extensions`. (Any program that uses the extension method will need to have the statement `using ExtensionMethods;` in order to access the method.) In the

`Extensions` class, lines 5–8 define the static `IsEven` method. Take a closer look at the parameter declaration in line 5:

```
this int i
```

The `this` keyword signals the compiler that the method is an extension method. The parameter `i` is an `int`, so the compiler knows that we are defining an extension method for the `int` data type. When the method executes, the `i` parameter will reference the `int` variable that the method is working on. For example, look at the following code:

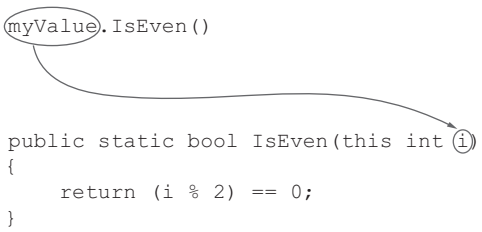
```
int myValue = 4;
if (myValue.IsEven())
{
   MessageBox.Show("The number is even.");
}
```

In this code, when the `IsEven` method executes, the `myValue` variable will implicitly be passed to the `i` parameter. This is illustrated in Figure 12-11. Here is another example:

```
int num = 4;
if (num.IsEven())
{
   MessageBox.Show("The number is even.");
}
```

In this code, when the `IsEven` method executes, the `num` variable will be implicitly passed to the `i` parameter.

Figure 12-11 `myValue` is passed implicitly into the `i` parameter



NOTE: An extension method can have multiple parameters. The `this` keyword appears only before the first parameter, however.

Extension methods must have at least one parameter, preceded by the `this` keyword, to refer to the object that the method is working on. If you need to explicitly pass other arguments into an extension method, just declare additional parameters after the first parameter. Here is an example:

```
public static bool IsGreaterThan(this int a, int b)
{
    return a > b;
}
```

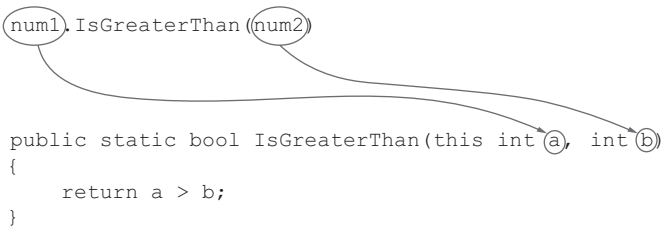
This method has two parameters: `a` and `b`. Because `a` is preceded by the `this` keyword, we know that it will implicitly refer to the `int` that the method is working on. The `b` parameter will hold whatever value is explicitly passed into the method. Here is an example of how we might call the method:

```
int num1 = 5, num2 = 4;
if (num1.IsGreaterThan(num2))
```

```
{
    MessageBox.Show("num1 is greater than num2.");
}
```

In this code, when the `IsGreaterThan` method executes, the `num1` variable will be passed to the `a` parameter, and the `num2` variable will be passed to the `b` parameter. This is illustrated in Figure 12-12.

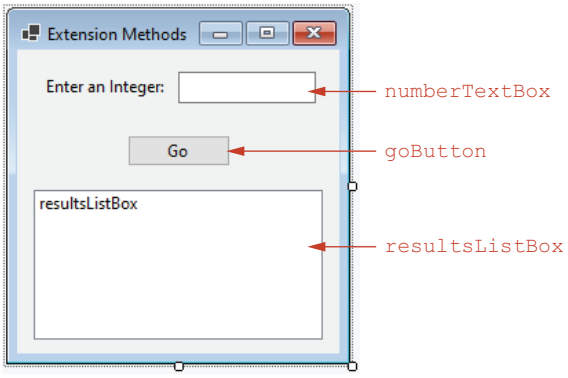
Figure 12-12 `num1` is passed implicitly into the `a` parameter, and `num2` is passed explicitly into the `b` parameter



NOTE: In an extension method, only the first parameter begins with the `this` keyword.

Let's look at a complete application that demonstrates extension methods. In the *Chap12* folder of this book's *Student Sample Programs*, you will find a project named *ExtensionMethods*. Figure 12-13 shows the application's form in the *Designer*. When the application runs, the user enters an integer into the `numberTextBox` control and then clicks the `goButton` control. The application displays information about the number in the `resultsListBox` control.

Figure 12-13 The *ExtensionMethods* form



In the project you will find three extension methods defined inside a static class named `Extensions`, as shown in Program 12-13. The class is in a namespace named `ExtensionMethods`.

Program 12-13 The `Extensions` class

```
1 using System;
2 using System.Collections.Generic;
3 using System.Linq;
4 using System.Text;
5 using System.Threading.Tasks;
```

```
 6
 7 namespace ExtensionMethods
 8 {
 9     internal static class Extensions
10     {
11         public static bool IsEven(this int i)
12         {
13             return (i % 2) == 0;
14         }
15
16         public static bool IsMultipleOf(this int a, int b)
17         {
18             return (a % b) == 0;
19         }
20
21         public static bool IsWholeNumber(this double d)
22         {
23             return (d - (int)d) == 0;
24         }
25     }
26 }
```

Let's take a closer look at each of the extension methods.

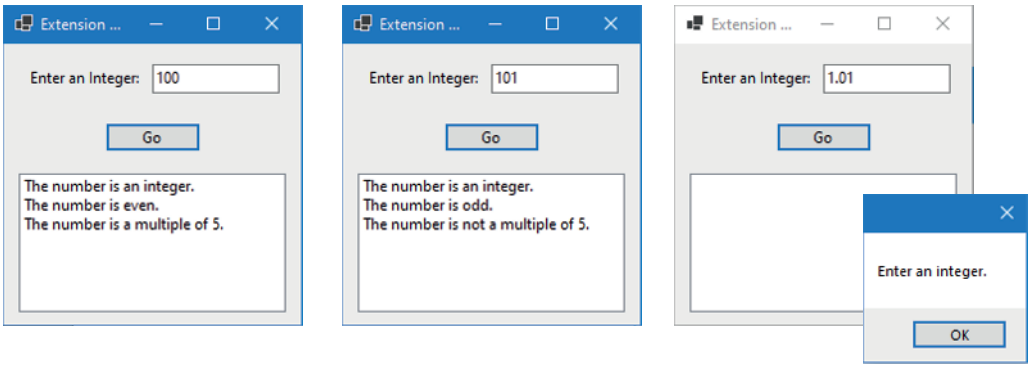
- The `IsEven` extension method is defined in lines 11–14. The purpose of this method is to determine whether an integer is even. Looking at the parameter declaration `this int i`, we know that the method extends the `int` data type. When the method executes, the `i` parameter will reference the `int` variable that the method is working on. In line 13, the method determines whether the remainder of `i` divided by 2 is 0. If so, the method returns `true`. Otherwise, it returns `false`.
- The `IsMultipleOf` extension method is defined in lines 16–19. The purpose of this method is to determine whether one integer is a multiple of another integer. Looking at the first parameter declaration `this int a`, we know that the method extends the `int` data type. The method also has a second parameter declaration: `int b`. When we call this method, we have to pass an argument into the `b` parameter. In line 18, the method determines whether the remainder of `a` divided by `b` is 0. If so, the method returns `true`. Otherwise, it returns `false`.
- The `IsWholeNumber` extension method is defined in lines 21–24. The purpose of this method is to determine whether a `double` is a whole number. Looking at the parameter declaration `this double d`, we know that the method extends the `double` data type. When the method executes, the `d` parameter will reference the `double` variable that the method is working on. In line 23, the method determines whether `d` minus `(int)d` is equal to 0. If so, the method returns `true`. Otherwise, it returns `false`.

Program 12-14 shows the Form1 code. In the `goButton_Click` method, line 13 clears the `resultsListBox` control. Then, line 16 gets the contents of `numberTextBox`, converts it to a `double`, and assigns that value to the `double` variable `dnum`. The `if` statement in line 17 uses the `dnum` variable to call the `IsWholeNumber` extension method. If the method returns `true`, it continues at line 19. Otherwise, in line 44 the message "Enter an integer." is displayed in a message box.

If the user entered an integer, line 19 displays "The number is an integer." in the `resultsListBox`. Line 20 gets the contents of `numberTextBox`, converts it to an `int`, and assigns that value to the `int` variable `inum`. The `if` statement in line 23 uses the `inum` variable to call the `IsEven` method. If the method returns `true`, the message "The number is even." is displayed in `resultsListBox`. Otherwise, the message "The number is even." is displayed in `resultsListBox`.

The `if` statement in line 33 uses the `inum` variable to call the `IsMultipleOf` method, passing 5 as an argument. If the method returns `true`, the message "The number is a multiple of 5." is displayed in `resultsListBox`. Otherwise, the message "The number is not a multiple of 5." is displayed in `resultsListBox`. Figure 12-14 shows three different samples of the application's output, with different input provided in each sample.

Figure 12-14 Sample output of the *Extension Methods* application



Program 12-14 The Form1 code for the *Integer Extension Methods* application

```
 1 namespace ExtensionMethods
 2 {
 3     public partial class Form1 : Form
 4     {
 5         public Form1()
 6         {
 7             InitializeComponent();
 8         }
 9
10         private void goButton_Click(object sender, EventArgs e)
11         {
12             // Clear the ListBox.
13             resultsListBox.Items.Clear();
14
15             // Make sure the number is an integer
16             double dnum = double.Parse(numberTextBox.Text);
17             if (dnum.IsWholeNumber())
18             {
19                 resultsListBox.Items.Add("The number is an integer.");
20                 int inum = int.Parse(numberTextBox.Text);
21
22                 // Is the number even?
23                 if (inum.IsEven())
24                 {
25                     resultsListBox.Items.Add("The number is even.");
26                 }
27                 else
28                 {
29                     resultsListBox.Items.Add("The number is odd.");
30                 }
31
```

```
32                  // Is the number a multiple of 5?
33                  if (inum.IsMultipleOf(5))
34                  {
35                      resultsListBox.Items.Add("The number is a
                        multiple of 5.");
36                  }
37                  else
38                  {
39                      resultsListBox.Items.Add("The number is not a
                        multiple of 5.");
40                  }
41              }
42              else
43              {
44                  MessageBox.Show("Enter an integer.");
45              }
46          }
47      }
48  }
```

In Tutorial 12-4 you will write an extension method that extends the `string` data type.

Tutorial 12-4
Creating Extension Methods

VideoNote
Tutorial 12-4: Creating Extension Methods

In the United States, it is common for dates to be written in the format *mm/dd/yyyy*, where *mm* is the month, *dd* is the day, and *yyyy* is the year. In Europe, it is more common for dates to be written in the format *dd/mm/yyyy*.

In this tutorial, you will complete an application that converts dates written in the U.S. format to the European format. The application's form is shown in Figure 12-15. To complete the application, you will write a `string` extension method that performs the conversion.

Step 1: Start Visual Studio. Open the project named *EuropeanDates* in the *Chap12* folder of the *Student Sample Programs*.

Step 2: In this step, you will add a static class named `Extensions` to the project. Click *Project* on the Visual Studio menu bar; then select *Add Class...*. The *Add New Item* window should appear with the Class template already selected in the center pane. Change the name of the file to Extensions.cs, and then click the *Add* button. This adds a source code file named Extensions.cs to the project, as shown in Figure 12-16.

Figure 12-15 The *EuropeanDates* project's Form1 form

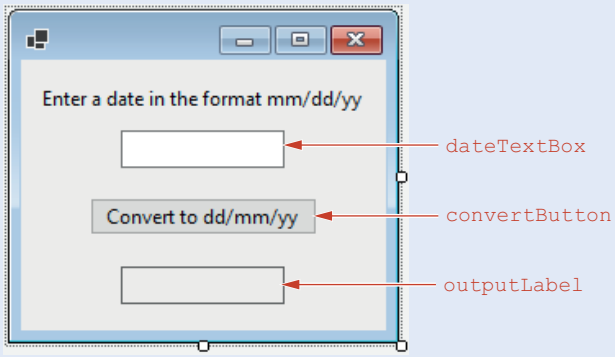
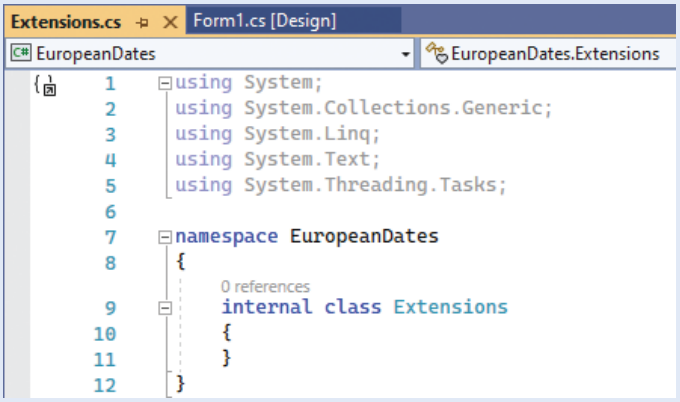


Figure 12-16 The `Extensions` class



Step 3 Program 12-15 shows how the contents of the Extensions.cs file should appear when you are finished modifying it. Perform the following steps to complete the file.

- By default, the class is in the project namespace which is `EuropeanDates`. Change the name of the namespace to `ExtensionMethods`, as shown in line 7 of Program 12-15.
- Insert the `static` keyword in line 9. Extension methods must be stored in a static class.
- Write the code for the `EuroDate` extension method in lines 11–23 of Program 12-15.

Let's review the code in the `Eurodate` extension method:

- When the method is called, the string that was used to call the method is passed into the `str` parameter.
- Line 13 declares a local `string` named `euroFormat` to hold the result of the conversion.
- Line 14 declares a `char` array named `delim` with one element: the `'/'` character. We will tokenize the `str` parameter using this character as the delimiter.
- Line 15 calls the `Split` method to tokenize the `str` parameter, using the `'/'` character in the `delim` array as the delimiter. The method returns the tokens that were found as a `string` array, which is assigned to the `tokens` variable. (We covered the `Split` method in Chapter 8.)
- We are assuming that `str` contains a string in the format *mm/dd/yyyy*. If so, the `Split` method will return a `string` array with three elements. Element 0 will contain the month, element 1 will contain the day, and element 2 will contain the year. If `str` contains any other string, then the `Split` method will return other tokens, depending on the contents of `str`.
- The `if` statement in line 18 determines whether the `tokens` array has three elements. If so, the statement in line 19 reassembles the tokens into a European-formatted date string, and assigns it to the `euroFormat` variable. If the `tokens` array does not have three elements, we know that the user didn't enter a properly formatted date string, so we assign the original value of `str` to `euroFormat`.
- Line 22 returns `euroFormat`.

Step 4: Switch your view back to the Form1 form in the Designer and double-click the `convertButton` control. This opens the Form1.cs file in the code editor with an empty event handler named `convertButton_Click`. Complete the event handler by typing the code shown in lines 14–18 of Program 12-16.

Step 5: Save the project and run the application. Experiment by entering different dates, as well as invalid values, to see how the application handles the input. Figure 12-17 shows two examples of the application's form at runtime. The screenshot on the left shows an example with a valid date converted to European format, and the screenshot on the right shows an example with invalid input entered. Close the application when you are finished.

Program 12-15 The `Extensions` class

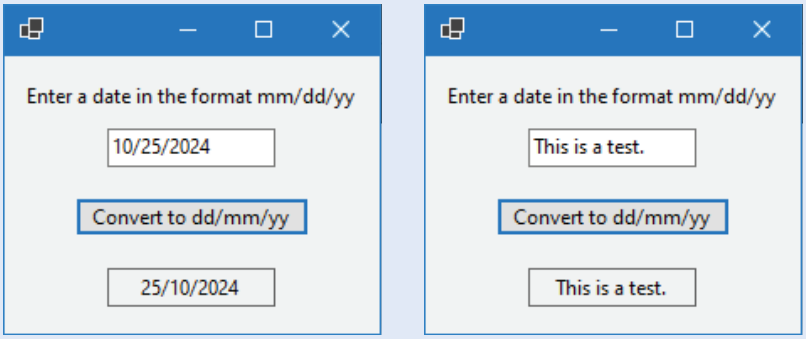
```
 1 using System;
 2 using System.Collections.Generic;
 3 using System.Linq;
 4 using System.Text;
 5 using System.Threading.Tasks;
 6
 7 namespace ExtensionMethods
 8 {
 9   internal static class Extensions
10   {
11      public static string EuroDate(this string str)
12      {
13         string euroFormat;                    // To hold the result
14         char[] delim = { '/' };               // Token delimiter
15         string[] tokens = str.Split(delim);   // Tokenize the string
16
17         // Validate the number of tokens and return the result.
18         if (tokens.Length == 3)
19             euroFormat = tokens[1] + "/" + tokens[0] + "/" +
               tokens[2];
20         else
21              euroFormat = str;
22         return euroFormat;
23      }
24   }
25 }
```

Program 12-16 The Form1 code

```
 1 using ExtensionMethods;
 2
 3 namespace EuropeanDates
 4 {
 5     public partial class Form1 : Form
 6     {
 7         public Form1()
 8         {
 9             InitializeComponent();
10         }
11
12         private void convertButton_Click(object sender, EventArgs e)
```

```
13          {
14              // Get the input entered by the user.
15              string input = dateTextBox.Text;
16
17              // Display the converted output.
18              outputLabel.Text = input.EuroDate();
19          }
20      }
21  }
```

Figure 12-17 The application's form at runtime



Checkpoint

12.15 If you want to extend a class, why would you choose to write an extension method instead of a derived class?

12.16 Can an extension method be nonstatic?

12.17 What type of class must an extension method be written in?

12.18 How does the compiler know that a method is an extension method?

12.19 What is the first parameter of an extension method?

12.5 Interfaces

CONCEPT: An interface specifies behavior for a class.

An **interface** is like a class that contains only abstract methods and/or abstract properties. An interface cannot be instantiated. Instead, it is implemented by other classes. When a class **implements** an interface, the class must provide all the methods and properties that are specified by the interface.

An interface declaration looks similar to a class, except the keyword `interface` is used instead of the keyword `class`, and the methods that are specified in an interface typically have no bodies, only headers that are terminated by semicolons. Also, any properties that are specified in an interface are abstract, and do not usually contain

code other than the keywords `get`, `set`, and `init`. Here is the general format of an interface definition:

```
interface InterfaceName
{
     Method headers and/or Properties...
}
```

The following code snippet shows an example of an interface named `IDisplayable`. In line 3, the interface specifies a `void` method named `display()`.

```
1  interface IDisplayable
2  {
3     void Display();
4  }
```

Notice that the `Display` method header in line 3 does not have an access modifier. This is because all methods and properties declared in an interface are implicitly `public`.

Any class that implements the `IDisplayable` interface must provide an implementation of the `Display` method (with the exact signature specified by the interface, and with the same return type). The `Person` class shown in the following code snippet is an example:

```
1 class Person : IDisplayable
2 {
3    // Constructor
4    public Person(string n)
5    {
6       Name = n;
7    }
8
9    // Name property
10   public string Name { get; set; }
11
12   // Display method, required by IDisplayable
13   public void Display()
14   {
15      MessageBox.Show($"My name is {Name}.");
16   }
17 }
```

The first line of the class declaration is written as:

```
class Person: IDisplayable
```

After the name of the class, `Person`, a colon appears, followed by the name of the interface `IDisplayable`. This line indicates that we are defining a class named `Person`, and it implements the `IDisplayable` interface. (This is the same syntax as declaring a class that inherits from another class.) Because the `Person` class implements the `IDisplayable` interface, it must provide an implementation of the interface's `Display` method. This is done in lines 13 through 16 of the `Person` class. The following code snippet demonstrates the `Person` class.

```
1  // Create an instance of the Person class.
2  var p = new Person("Antonio");
3
4  // Call the object's Display method.
5  p.Display();
```

NOTE: It is a common practice to begin the name of an interface with "`I`".

Properties in Interfaces

It is important to remember that an interface specifies methods and properties that a class must have, but an interface never specifies how a class must implement those methods and properties. So, properties that are listed in an interface do not usually contain code other than the keywords `get`, `set`, and `init`. When a class implements an interface that has a property, the class must provide that property with the accessors that are listed in the interface. For example, look at the following interface:

```
1  interface IInventory
2  {
3      decimal Cost { get; set; }
4  }
```

This interface is named `IInventory` and it specifies a `decimal` property named Cost, with `get` and `set` accessors. Any class that implements this interface must provide a `decimal` property named Cost, with `get` and `set` accessors. Here is an example:

```
1  class Item: IInventory
2  {
3      // Constructor
4      public Item(string id, decimal cost)
5      {
6          ID = id;
7          Cost = cost;
8      }
9
10     // Properties
11     public string ID { get; set; }
12     public decimal Cost { get; set; }
13 }
```

This class is named `Item`, and it implements the `IInventory` interface. Notice in line 12 the Cost property is implemented, as specified in the `IInventory` interface: its data type is `decimal` and it has `get` and `set` accessors. A compiler error will occur if the `Item` class does not have a Cost property that matches the specifications shown in the `IInventory` interface.

Read-Only Interface Properties

You've learned that a property becomes read-only if you leave out its `set` accessor. This is not necessarily the case with interface properties, however. For example, look at the following version of the `IInventory` interface:

```
1  interface IInventory
2  {
3      decimal Cost { get; }
4  }
```

In this version of the interface, the Cost property has a `get` accessor, but no `set` accessor. When a class implements this interface, it must provide a Cost property that has *at least* a `get` accessor. The property in the class can also have a `set` accessor. For example, the following `Widget` class will compile with no errors, even though its Cost property has both a `get` and a `set` accessor:

```
1  class Widget: IInventory
2  {
3      public decimal Cost { get; set; }
4  }
```

So, omitting the `set` accessor in an interface property does not guarantee that the property will be read-only in the classes that implement the interface. If you want to be sure

that an interface property is implemented as read-only, specify an `init` accessor in the interface. Here is an example:

```
1  interface IInventory
2  {
3      decimal Cost { get; init; }
4  }
```

Any class implementing this version of the `IInventory` interface will have to implement the Cost property as an init-only property. Here is an example of the `Widget` class that correctly implements our revised `IInventory` interface:

```
1  class Widget: IInventory
2  {
3      public decimal Cost { get; init; }
4  }
```

An Interface Is a Contract

When a class implements an interface, it is agreeing to provide all the methods and/or properties that are specified by the interface. It is often said that an interface is like a contract, and when a class implements an interface it must adhere to the contract.

Creating an Interface in Visual Studio

To write the code for a new interface in Visual Studio, follow these steps to create the interface's source code file and add it to your project:

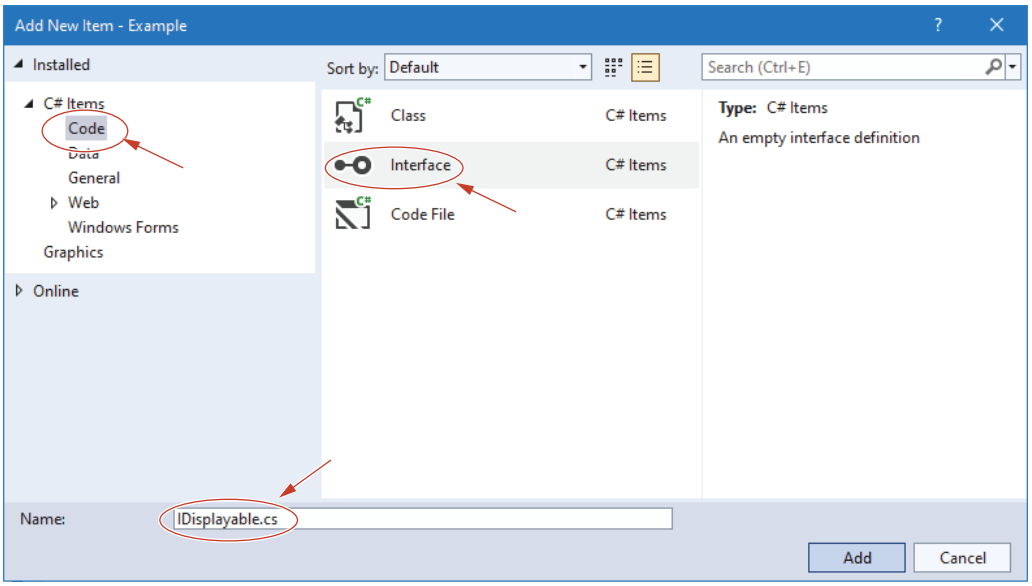
1. With the project open in Visual Studio, click *Project* on the menu bar; then select *Add New Item. . . .*
2. The *Add New Item* window will appear. As shown in Figure 12-18, select *Code* in the left pane, select ***Interface*** in the center pane, and change the default name that appears in the *Name* box at the bottom of the window to the name you wish to give the interface. (Be sure to keep the .cs file extension as part of the file name.)
3. Click the *Add* button to create the new source code file in the project. Program 12-17 shows an example. Notice that an empty interface is automatically created (lines 9–11) and placed in the project's namespace.

Program 12-17 Example of a new interface source code file

```
1 using System;
2 using System.Collections.Generic;
3 using System.Linq;
4 using System.Text;
5 using System.Threading.Tasks;
6
7 namespace InterfaceDemo
8 {
9     internal interface IDisplayable
10    {
11    }
12 }
```

In Tutorial 12-5 you will go through the steps of writing an interface and then creating two classes that each implement the interface.

Figure 12-18 Add New Item window



Tutorial 12-5:
Writing and Implementing an Interface

VideoNote
Tutorial 12-5: Writing and Implementing an Interface

Board games, and other physical games, commonly have devices that are "tossed" in order to get a random value. For example, a six-sided die can be tossed to get a random value in the range of 1–6. A coin can be tossed to get one of two values: heads or tails. You can think of dice and coins as random-number generators for physical games. When used in games, dice and coins both have the following behaviors in common:

- They can be tossed
- Tossing them generates a random value

In this tutorial, you complete the CoinDice application. You will write a `Die` class to simulate a six-sided die, and a `Coin` class to simulate a coin. You will also write an interface named `ITossable` that both the `Die` class and the `Coin` class will implement. The `ITossable` interface will specify the following members:

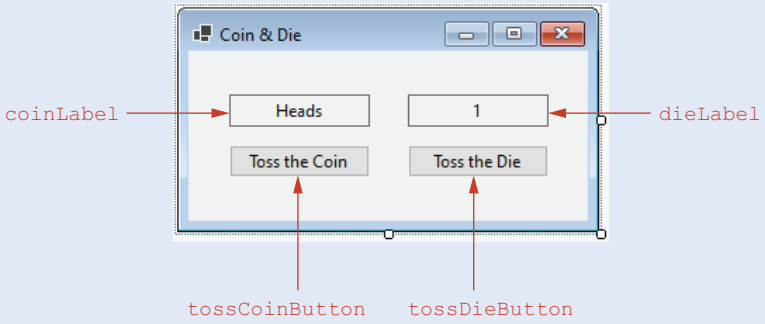
- A method named `Toss`. The `Toss` method will cause the implementing object to generate a random value.
- An `int` property named `Value`. This property will hold the most recent value that the implementing object generated.

The application's form, which has already been created for you, is shown in Figure 12-19. When you run the completed application, you can click the *Toss the Coin* button and the application will simulate a coin being tossed, displaying either *Heads* or *Tails* in the `coinLabel` control. You can click the *Toss the Die* button and the application will simulate a 6-sided die being tossed, displaying its value in the `dieLabel` control.

Step 1: Start Visual Studio. Open the project named CoinDice in the *Chap12* folder of the Student Sample Programs.

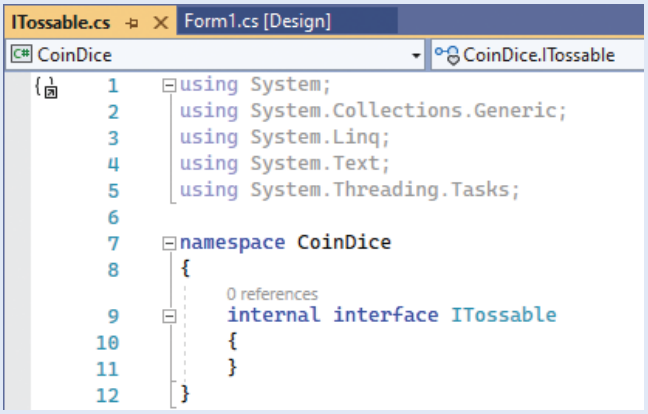
Step 2: In this step, you will add an interface named `ITossable` to the project. Click *Project* on the Visual Studio menu bar; then select *Add New Item*.... The *Add New Item* window should appear. (See Figure 12-18 for an example.) Make

Figure 12-19 The *CoinDice* application's form



sure *Interface* is selected as the type of item (in the center pane). Notice that the default filename Interface1.cs appears in the *Name* text box. Change the name to ITossable.cs, and then click the *Add* button. This adds a source code file named ITossable.cs to the project, as shown in Figure 12-20.

Figure 12-20 The `ITossable` interface



Step 3: Program 12-18 shows the completed `ITossable` interface. Add the code shown in bold, in lines 11 and 12. Here is a summary of the code that you are adding to the interface:

- Line 11 specifies an `int` property named `Value`, with a getter and a setter.
- Line 12 specifies a `void` method named `Toss()`.

Program 12-18 The completed `ITossable` interface

```
 1 using System;
 2 using System.Collections.Generic;
 3 using System.Linq;
 4 using System.Text;
 5 using System.Threading.Tasks;
 6
 7 namespace CoinDice
 8 {
 9     internal interface ITossable
10     {
11         int Value { get; set; }
```

```
12          void Toss();
13      }
14  }
```

Step 4: In this step, you will add the `Die` class to the project. Click *Project* on the Visual Studio menu bar; then select *Add Class*. . . . The *Add New Item* window should appear. Make sure *Class* is selected as the type of item (in the center pane). Notice that the default filename Class1.cs appears in the *Name* text box. Change the name to Die.cs, and then click the *Add* button. This adds a source code file named Die.cs to the project.

Step 5: The `Die` class will implement the `ITossable` interface. As shown in line 9 of Program 12-19, add the notation: `ITossable` to the class header.

Step 6: Complete the class by adding the code shown in bold in lines 11–17 in Program 12-19. Here is a summary of the code that you are adding to the interface:

- Line 11 is an `int` property named `Value`, with a getter and a setter, and a default value of 1. This satisfies one of the requirements of the `ITossable` interface.
- Lines 13–17 is a `void` method named `Toss()`. The method generates a random number in the range of 1–6. This method also satisfies one of the requirements of the `ITossable` interface.

Program 12-19 The completed `Die` class

```
1  using System;
2  using System.Collections.Generic;
3  using System.Linq;
4  using System.Text;
5  using System.Threading.Tasks;
6
7  namespace CoinDice
8  {
9      internal class Die: ITossable
10     {
11       public int Value { get; set; } = 1;
12
13       public void Toss()
14       {
15           var rand = new Random();
16           Value = rand.Next(6) + 1;
17       }
18     }
19 }
```

Step 7: In this step you will add the `Coin` class to the project. Click *Project* on the Visual Studio menu bar; then select *Add Class*. . . . The *Add New Item* window should appear. Make sure *Class* is selected as the type of item (in the center pane). Notice that the default filename Class1.cs appears in the *Name* text box. Change the name to Coin.cs, and then click the *Add* button. This adds a source code file named Coin.cs to the project.

Step 8: The `Coin` class will implement the `ITossable` interface. As shown in line 9 of Program 12-20, add the notation: `ITossable` to the class header.

Step 9: Complete the class by adding the code shown in bold in lines 11–25 of Program 12-20. Here is a summary of the code that you are adding to the interface:

- Line 11 is an `int` property named `Value`, with a getter and a setter, and a default value of 1. This satisfies one of the requirements of the `ITossable` interface.
- Lines 13–17 is a `void` method named `Toss()`. The method generates a random number in the range of 0–1. This method also satisfies one of the requirements of the `ITossable` interface.
- Lines 19–25 is a method named `HeadsOrTails()`. The method returns the string "Heads" if the `Value` property is set to 0, or "Tails" otherwise.

Program 12-20 The completed `Coin` class

```
1 using System;
2 using System.Collections.Generic;
3 using System.Linq;
4 using System.Text;
5 using System.Threading.Tasks;
6
7 namespace CoinDice
8 {
9     internal class Coin: ITossable
10     {
11       public int Value { get; set; } = 0;
12
13       public void Toss()
14       {
15           var rand = new Random();
16           Value = rand.Next(2);
17      }
18
19       public string HeadsOrTails()
20       {
21           if (Value == 0)
22               return "Heads";
23           else
24               return "Tails";
25       }
26    }
27 }
```

Step 10: Open the Form1.cs file in the code editor. Type the comment and declarations shown in lines 5 through 7 of Program 12-21. The statement in line 6 creates an object of the `Coin` class and assigns a reference to that object to a field named `coin`. The statement in line 7 creates an object of the `Die` class and assigns a reference to that object to a field named `die`. Because the `coin` and `die` variables are declared as fields, all the methods in the `Form1` class have access to them.

Step 11: Switch your view back to the Form1 form in the Designer and double-click the `tossCoinButton` control. This opens the Form1.cs file in the code editor, and you will see an empty event handler named `tossCoinButton_Click`. Complete the event handler by typing the code shown in lines 16–20 in Program 12-21. Let's review this code:

- Line 17 calls the `coin` object's `Toss` method.

- Line 20 calls the `coin` object's `HeadOrTails` method, which returns either "Heads" or "Tails." The value that is returned is displayed in the `coinLabel` control.

Step 12: Switch your view back to the Form1 form in the Designer and double-click the `tossDieButton` control. This opens the Form1.cs file in the code editor, and you will see an empty event handler named `tossDieButton_Click`. Complete the event handler by typing the code shown in lines 25–29 in Program 12-21. Let's review this code:

- Line 26 calls the `die` object's `Toss` method.
- Line 29 gets the `die` object's `Value` property, converts it to a string, and displays its value in the `dieLabel` control.

Step 13: Save the project. Then, press F5 on the keyboard or click the Start Debugging button (▶) on the toolbar to compile and run the application. When the application runs, experiment by clicking the buttons several times and watching the values that are displayed for the die and the coin.

Program 12-21 Completed code for the Form1.cs file in the CoinDice application

```
 1 namespace CoinDice
 2 {
 3     public partial class Form1 : Form
 4     {
 5         // Fields for the coin and the die
 6         private Coin coin = new Coin();
 7         private Die die = new Die();
 8
 9         public Form1()
10         {
11             InitializeComponent();
12         }
13
14         private void tossCoinButton_Click(object sender,
           EventArgs e)
15         {
16             // Toss the coin.
17             coin.Toss();
18
19             // Display the side that is up.
20             coinLabel.Text = coin.HeadsOrTails();
21         }
22
23         private void tossDieButton_Click(object sender, EventArgs e)
24         {
25             // Toss the die.
26             die.Toss();
27
28             // Display the side that is up.
29             dieLabel.Text = die.Value.ToString();
30         }
31     }
32 }
```

Implementing Multiple Interfaces

You might be wondering why we need both abstract classes and interfaces, since they are so similar to each other. One reason is that a class can inherit from only one base class, but C# allows a class to implement multiple interfaces. When a class implements multiple interfaces, it must provide the methods and properties specified by all the interfaces.

To specify multiple interfaces in a class declaration, simply list the names of the interfaces, separated by commas, after the colon. Here is an example of a class that implements multiple interfaces:

```
class MyClass: IDisplayable, IComparable, IRelatable
{
}
```

The name of the class is `MyClass`, and it implements three interfaces: `IDisplayable`, `IComparable`, and `IRelatable`.

Declaring a Derived Class That Implements an Interface

When you are declaring a class that inherits from a base class and implements an interface, the name of the base class must appear first after the colon in the derived class header. If the name of the base class does not appear first, an error will occur at compile-time.

For example, suppose we have a class named `Person` and an interface named `IDisplayable`. Here is the declaration of a class named `Customer` that is derived from the `Person` class and implements the `IDisplayable` interface:

```
class Customer: Person, IDisplayable
{
}
```

The following code shows the declaration of a class named `Customer` that is derived from the `Person` class and implements three interfaces: `IDisplayable`, `IComparable`, and `IRelatable`:

```
class Customer: Person, IDisplayable, IComparable, IRelatable
{
}
```

Polymorphism and Interfaces

Just as you can create reference variables of a class type, C# allows you to create reference variables of an interface type. An interface reference variable can reference any object that implements that interface, regardless of its class type. This is another example of polymorphism. For example, recall the `ITossable` interface and the `Die` and `Coin` classes that you wrote in Tutorial 12-5. Look at the following code snippet:

```
1   ITossable tossable = new Die();
2   tossable.Toss();
3   MessageBox.Show(tossable.Value.ToString());
```

Line 1 creates a `Die` object and assigns that object to the `tossable` variable. Even though `tossable` is an `ITossable` variable, it can reference `Die` objects because the `Die` class implements the `ITossable` interface. Line 2 calls the `tossable` object's `Toss` method, and line 3 displays the `tossable` object's `Value` property in a message box.

The following code snippet demonstrates the same steps can be taken with a `Coin` object:

```
1   ITossable tossable = new Coin();
2   tossable.Toss();
3   MessageBox.Show(tossable.Value.ToString());
```

Although an `ITossable` variable can reference objects of any class that implements the `ITossable` interface, there is a limit to what the variable can do with those objects. Recall that the `ITossable` interface specifies only two things: a property named `Value` and a method named `Toss`. So, an `ITossable` variable can be used only to access the `Value` property and call the `Toss` method, regardless of the type of object the variable references. For example, look at the following code:

```
1   ITossable tossable = new Coin();           // This works
2   tossable.Toss();                           // This works
3   MessageBox.Show(tossable.HeadsOrTails());  // ERROR!
```

The last statement in this code snippet causes a compiler error because it attempts to use the `tossable` variable to call the `HeadsOrTails` method. Even though the `tossable` variable references a `Coin` object, the variable does not know about any of the object's methods or properties other than the ones declared in the `ITossable` interface.

The ability to reference an object with an interface variable is particularly useful when passing objects to methods. For example, suppose a business sells digital songs and movies, for download, on their website. The business' e-commerce software uses two classes: `DigitalSong` and `DigitalMovie`. Each of these classes implements the following `IRetail` interface:

```
1  interface IRetail
2  {
3     decimal RetailPrice { get; set; }
4  }
```

The `IRetail` interface specifies a `decimal` property named `RetailPrice`. So, the `DigitalSong` and `DigitalMovie` classes must each have this property. Here is the `DigitalSong` class:

```
1  class DigitalSong: IRetail
2  {
3     public string Title { get; set; }        // Song title
4     public string Artist { get; set; }       // Name of artist
5     public decimal RetailPrice { get; set; } // Retail Price
6  }
```

And here is the `DigitalMovie` class:

```
1  class DigitalMovie: IRetail
2  {
3      public string Title { get; set; }        // Movie title
4      public int RunningTime { get; set; }     // Length in minutes
5      public decimal RetailPrice { get; set; } // Retail Price
6  }
```

Suppose we need a method that displays the retail price of either a digital song or a digital movie. We could write two separate methods, one that has a `DigitalSong` parameter, and another that has a `DigitalMovie` parameter. However, since we need only the object's retail price, a better approach would be to write one method that has an `IRetail` parameter. Because an `IRetail` variable can reference either a `DigitalSong` object or a `DigitalMovie` object, we can use the variable to retrieve the retail price of either type of object. Here is an example of such a method:

```
1  static void DisplayRetail(IRetail item)
2  {
3     MessageBox.Show(item.RetailPrice.ToString());
4  }
```

The following code snippet shows how we can pass either a `DigitalSong` object or a `DigitalMovie` object as an argument to the `DisplayRetail` method:

```
// Create a DigitalSong object.
DigitalSong song = new DigitalSong()
{
   Title = "Wipe The Tears From My Eyes",
   Artist = "Billy Nelson",
   RetailPrice = 1.0m
};

// Create a DigitalMovie object.
DigitalMovie movie = new DigitalMovie()
{
   Title = "It Came From Planet X",
   RunningTime = 102,
   RetailPrice = 14.95m
};

// Display each object's retail price.
DisplayRetail(song);
DisplayRetail(movie);
```

Arrays and `Lists` of Interface References

Polymorphism allows us to create arrays and `Lists` of interface references. The following code snippet creates an array of `IRetail` objects that is initialized with both `DigitalSong` and `DigitalMovie` objects:

```
IRetail[] items = {
    new DigitalSong()
    {
        Title = "Wipe The Tears From My Eyes",
        Artist = "Billy Nelson",
        RetailPrice = 1.0m
    },
    new DigitalSong()
    {
        Title = "Sunshine On My shoulder",
        Artist = "Joe Looney",
        RetailPrice = 0.99m
    },
    new DigitalMovie()
    {
        Title = "It Came From Planet X",
        RunningTime = 102,
        RetailPrice = 14.95m
    },
    new DigitalMovie()
    {
        Title = "A Night In Paris",
        RunningTime = 124,
        RetailPrice = 16.95m
    }
};
```

In the `RetailItems` array, elements 0 and 1 are `DigitalSong` objects, and elements 2 and 3 are `DigitalMovie` objects. The following code snippet shows how we can use a `foreach` loop to iterate over the array elements, passing each one to the `DisplayRetail` method that was previously shown:

```
  foreach (IRetail item in items)
  {
```

```
    DisplayRetail(item);
}
```

The following code snippet creates a `List` of `IRetail` objects that is initialized with both `DigitalSong` and `DigitalMovie` objects:

```
var itemList = new List<IRetail>
{
    new DigitalSong()
    {
        Title = "Wipe The Tears From My Eyes",
        Artist = "Billy Nelson",
        RetailPrice = 1.0m
    },
    new DigitalSong()
    {
        Title = "Sunshine On My shoulder",
        Artist = "Joe Looney",
        RetailPrice = 0.99m
    },
    new DigitalMovie()
    {
        Title = "It Came From Planet X",
        RunningTime = 102,
        RetailPrice = 14.95m
    },
    new DigitalMovie()
    {
        Title = "A Night In Paris",
        RunningTime = 124,
        RetailPrice = 16.95m
    }
};
```

The following code snippet shows how we can use a `foreach` loop to iterate over the `List` elements, passing each one to the `DisplayRetail` method that was previously shown:

```
foreach (IRetail item in itemList)
{
    DisplayRetail(item);
}
```

Restrictions of Interfaces

Interfaces are restricted in several ways. Here are some important limitations to keep in mind:

- You cannot declare any fields within an interface. Only properties and methods are allowed.
- Properties that are declared in an interface are abstract, so they cannot contain any code other than the `get` and/or `set` keywords.
- Methods that are declared in an interface are abstract, so they cannot contain any code.
- All the members of an interface are public, so you cannot list an access modifier with any property or method in an interface.
- Interfaces cannot have a constructor.
- You cannot declare a class, structure, enumeration or other interfaces inside of an interface.
- Interfaces can only inherit from other interfaces. An interface cannot inherit from a class.

Checkpoint

12.20 What is the purpose of an interface?

12.21 How is an interface similar to an abstract class?

12.22 Write the first line of a class named `Customer`, which implements an interface named `IRelatable`.

12.23 Write the first line of a class named `Employee`, which implements interfaces named `IPayable` and `IListable`.

Key Terms

abstract class
abstract method
abstract property
base class
concrete class
derived class
extension method
implements
interface
"is a" relationship
polymorphism
`protected` member
subclasses
superclasses

Review Questions

Multiple Choice

1. When one object is a specialized version of another object, there is an _____ between them.
   a. "is a" relationship
   b. Innate association
   c. Inherent union
   d. Unbreakable union

2. In an inheritance relationship, the _____ is the general class.
   a. Derived class
   b. Base class
   c. Dependent class
   d. Child class

3. In an inheritance relationship, the _____ is the specialized class.
   a. Base class
   b. Primary class
   c. Derived class
   d. Parent class

4. Base classes are sometimes called _____.
   a. Megaclasses
   b. Primitive classes
   c. Starter classes
   d. Superclasses

5. Derived classes are sometimes called _____.
   a. Refined classes
   b. Subclasses
   c. Child classes
   d. Neoclasses

6. The _____ refers to the base class.
   a. `friend` keyword
   b. `this` keyword
   c. `base` keyword
   d. `class` keyword

7. The term _____ refers to an object's ability to take different forms.
   a. Multi-instance
   b. By referencing
   c. Polymorphism
   d. Oligopoly

8. When a derived class method has the same name as a base class method, it is often said that the derived class method _____ the base class method.
   a. Terminates
   b. Cancels out
   c. Overrides
   d. Short circuits

9. The _____ declares that a derived class is allowed to override a method.
   a. `void` keyword
   b. `protected` keyword
   c. `base` keyword
   d. `virtual` keyword

10. The _____ declares that this method overrides a method in the base class.
    a. `override` keyword
    b. `class` keyword
    c. `virtual` keyword
    d. `base` keyword

11. A class that is not intended to be instantiated, but used only as a base class, is called a(n) _____.
    a. Dummy class
    b. Subclass
    c. Virtual class
    d. Abstract class

12. To declare a class as abstract, you use the _____ in the class header.
    a. `abstract` keyword
    b. `base` keyword
    c. `void` keyword
    d. `virtual` keyword

13. A regular, nonabstract class is sometimes called a _____.
    a. True class
    b. Model class
    c. Concrete class
    d. Real class

14. A(n) _____ is a method that appears in a base class but expects to be overridden in a derived class.
    a. Abstract method
    b. Virtual method
    c. Concrete method
    d. Base method

15. A(n) _____ is a property that appears in a base class but expects to be overridden in a derived class.
    a. Virtual property
    b. Concrete property
    c. Base property
    d. Abstract property

16. _____ allows a base class reference variable to reference a derived class object.
    a. Polymorphism
    b. Inheritance
    c. Generalization
    d. Specialization

17. This type of method extends a class without modifying the class or inheriting from it.
    a. Submethod
    b. Add-on method
    c. Extension method
    d. Sealed method

18. The first parameter declaration in an extension method always begins with the _____keyword.
    a. `extension`
    b. `this`
    c. `static`
    d. `ext`

19. When a class implements an interface, it must _____.
    a. Provide only methods that are not listed in the interface
    b. Provide all of the methods that are listed in the interface, with the exact signatures and return types specified
    c. Not have a constructor
    d. Be an abstract class

True or False

1. The base class inherits fields, properties, and methods from the derived class.

2. Polymorphism allows a class variable of the base class type to reference objects of either the base class or the derived class types.

3. Properties in a base class cannot be overridden in the same way that methods can be overridden.

4. A base class reference variable can reference an object of any class that is derived from the base class.

5. A statement that tries to use the `new` operator to instantiate an abstract class will not compile.

6. A class that is not intended to be instantiated, but used only as a base class, is called a concrete class.

7. When an abstract property appears in a class, it must be overridden in any class that is derived from the class.
8. To write an extension method, you must inherit from the class that you are extending.
9. An extension method never has a parameter.
10. Extension methods must be static
11. A class may only implement one interface.
12. All members of an interface are implicitly public.

Short Answer

1. What does a derived class inherit from its base class?
2. Look at the following code, which is the first line of a class declaration. What is the name of the base class? What is the name of the derived class?

```
class Tiger : Felis
```

3. Can methods in the derived class directly access the base class's private members?
4. When you create an instance of a derived class, which constructor is called first?
5. In what kind of situation would you want to use an abstract class instead of a base class?
6. What is a primary difference between an abstract class and a regular class?
7. Can abstract classes also contain abstract properties?
8. How does the compiler know that a method is an extension method?
9. Consider the following extension method:

```
public static bool DoSomething(this string a, int b)
{
    // Code goes here
}
```

   What class or data type does the method extend?
10. Why is it often said that an interface is like a contract?
11. What is the only code allowed in properties that are listed in an interface?

Algorithm Workbench

1. Write the first line of the definition for a `Poodle` class. The class should be derived from the `Dog` class.
2. Look at the following class declarations:

```
class Plant
{
    public virtual void Message()
    {
        MessageBox.Show("I'm a plant.");
    }
}
class Tree : Plant
{
    public override void Message()
```

```
    {
        MessageBox.Show("I'm a tree.");
    }
}
```

Given these class definitions, what will the following code display?

```
Plant p = new Tree();
p.Message();
```

3. Write a parameterized constructor for a base class named `Movie` with a `string` field named `_title` and an `int` field named `_runningTime`.
4. A class named `Wave` has a virtual property named Frequency. A class named `Sound` is derived from the `Wave` class. Write example code showing how the Frequency property might appear in the `Sound` class.
5. Create an abstract class called `Star`. Include an abstract method named `SolarMasses` that returns a value of the `double` data type.
6. Consider the following extension method:

```
public static bool IsGood(this int a, int b)
{
    // Code goes here
}
```

In the following code, what value is passed to the `a` parameter of the `IsGood` method?

```
int number = 2;
if (number.IsGood(99))
{
    // Code goes here
}
```

7. Write the first line of the definition for a `SmartPhone` class. The class should extend the `MobileDevice` class, and it should implement the `ICallable` and `ITextable` interfaces.
8. Write an interface named `INameable` that specifies the following methods:

```
void SetName(string n)
string GetName()
```

Programming Problems

VideoNote
Solving the Employee and Production-Worker Classes Problem

1. **`Employee` and `ProductionWorker` Classes**

   Create an `Employee` class that has properties for the following data:

   - Employee name
   - Employee number

   Next, create a class named `ProductionWorker` that is derived from the `Employee` class. The `ProductionWorker` class should have properties to hold the following data:

   - Shift number (an integer, such as 1, 2, or 3)
   - Hourly pay rate

   The workday is divided into two shifts: day and night. The Shift property will hold an integer value representing the shift that the employee works. The day shift is shift 1 and the night shift is shift 2.

Create an application that creates an object of the `ProductionWorker` class and lets the user enter data for each of the object's properties. Retrieve the object's properties and display their values.

2. **`ShiftSupervisor` Class**

   In a particular factory, a shift supervisor is a salaried employee who supervises a shift. In addition to a salary, the shift supervisor earns a yearly bonus when their shift meets production goals. Create a `ShiftSupervisor` class that is derived from the `Employee` class you created in Programming Problem 1. The `ShiftSupervisor` class should have a property that holds the annual salary and a property that holds the annual production bonus that a shift supervisor has earned. Demonstrate the class in an application.

3. **`TeamLeader` Class**

   In a particular factory, a team leader is an hourly paid productionworker that leads a small team. In addition to hourly pay, team leaders earn a fixed monthly bonus. Team leaders are required to attend a minimum number of hours of training per year. Design a `TeamLeader` class that is derived from the `ProductionWorker` class you created in Programming Problem 1. The `TeamLeader` class should have properties for the monthly bonus amount, the required number of training hours, and the number of training hours that the team leader has attended. Demonstrate the class in an application.

4. **`Person` and `Customer` Classes**

   Design a class named `Person` with properties for holding a person's name, address, and telephone number. Next, design a class named `Customer`, which is derived from the `Person` class. The `Customer` class should have a property for a customer number and a Boolean property indicating whether the customer wishes to be on a mailing list. Demonstrate an object of the `Customer` class in a simple application.

5. **`PreferredCustomer` Class**

   A retail store has a preferred customer plan where customers can earn discounts on all their purchases. The amount of a customer's discount is determined by the amount of the customer's cumulative purchases in the store as follows:

   - When a preferred customer spends $500, they get a 5 percent discount on all future purchases.
   - When a preferred customer spends $1,000, they get a 6 percent discount on all future purchases.
   - When a preferred customer spends $1,500, they get a 7 percent discount on all future purchases.
   - When a preferred customer spends $2,000 or more, they get a 10 percent discount on all future purchases.

   Design a class named `PreferredCustomer`, which is derived from the `Customer` class you created in Programming Problem 4. The `PreferredCustomer` class should have properties for the amount of the customer's purchases and the customer's discount level. Demonstrate the class in a simple application.

6. **`Ship`, `CruiseShip`, and `CargoShip` Classes**

   Start a new project and add the following classes to it:

   A `Ship` class that the following members:

   - A property for the name of the ship (a string).
   - A property for the year that the ship was built (a string).
   - A constructor
   - A `ToString` method that displays the ship's name and the year it was built.

A `CruiseShip` class that is derived from the `Ship` class. The `CruiseShip` class should have the following members:

- A property for the maximum number of passengers (an int).
- A constructor
- A `ToString` method that overrides the `ToString` method in the base class. The `CruiseShip` class's `ToString` method should display only the ship's name and the maximum number of passengers.

A `CargoShip` class that extends the `Ship` class. The `CargoShip` class should have the following members:

- A property for the cargo capacity in tonnage (an int).
- A constructor
- A `ToString` method that overrides the `ToString` method in the base class. The `CargoShip` class's `ToString` method should display only the ship's name and the ship's cargo capacity.

The application should demonstrate the classes by creating a `Ship` array. Assign various `Ship`, `CruiseShip`, and `CargoShip` objects to the array elements. The program should then step through the array, calling each object's `ToString` method.

7. **`String` Extension Methods**

Write the following extension methods and demonstrate them in an application:

- A `string` extension method that converts a `string` to an array of `char`.
- A `string` extension method that takes a `string` formatted as a date (such as *mm/dd/yyyy*) and breaks it into a three-element `string` array containing the month, day, and year.
- A `string` extension method that formats a 10-character `string` as a telephone number, such as `(xxx)xxx-xxxx`.
- A `string` extension method that returns the `string` backwards.
- A `string` extension method that counts the number of words in a string.

8. **`Crypto` Interface Project**

Encryption can be described as the process of writing data in a secret code. Decryption is the process of restoring encrypted data to its original state. If you want to send a secret message to someone, you use an encryption algorithm to encrypt the message before you send it. The person receiving the encrypted message must use a decryption algorithm to convert the message to its original contents.

Although not very secure, one of the simplest types of encryption algorithms is the substitution algorithm. A substitution algorithm takes a string and substitutes each character with some other predefined character. For example, each occurrence of the letter 'A' might be substituted with the letter 'M', and each occurrence of the letter 'B' might be substituted with the letter 'W'. To decrypt the message, the substitutions must happen in reverse.

In this programming problem, you will create a project and add the following interface:

```
interface Crypto
{
    string Encrypt(string str);
    string Decrypt(string str);
}
```

The `Crypto` interface specifies two methods: `encrypt` and `decrypt`. The intention of the `encrypt` method is to accept a string argument and return an encrypted version of that string. The intention of the `decrypt` method is to accept an encrypted string as its argument and return a decrypted version of that string.

Next, you will write two classes named `Cipher1` and `Cipher2`. Each of these classes must implement the `Crypto` interface. The `Cipher1` class will use a substitution algorithm to encrypt and decrypt strings, and the `Cipher2` class will use a different substitution algorithm to encrypt and decrypt strings. (You must design the algorithm for each class.)

Once you have written the interface and the classes, create a user interface that allows the user to enter a message. The application should then perform the following:

- Encrypt the message using the `Cipher1` object and display the encrypted message that is returned from the `Cipher1` object's `Encrypt` method.
- Decrypt the message using the `Cipher1` object and display the decrypted message that is returned from the `Cipher1` object's `Dencrypt` method. This should be the same as the original message.
- Encrypt the message using the `Cipher2` object and display the encrypted message that is returned from the `Cipher2` object's `Encrypt` method.
- Decrypt the message using the `Cipher2` object and display the decrypted message that is returned from the `Cipher2` object's `Decrypt` method. This should be the same as the original message.

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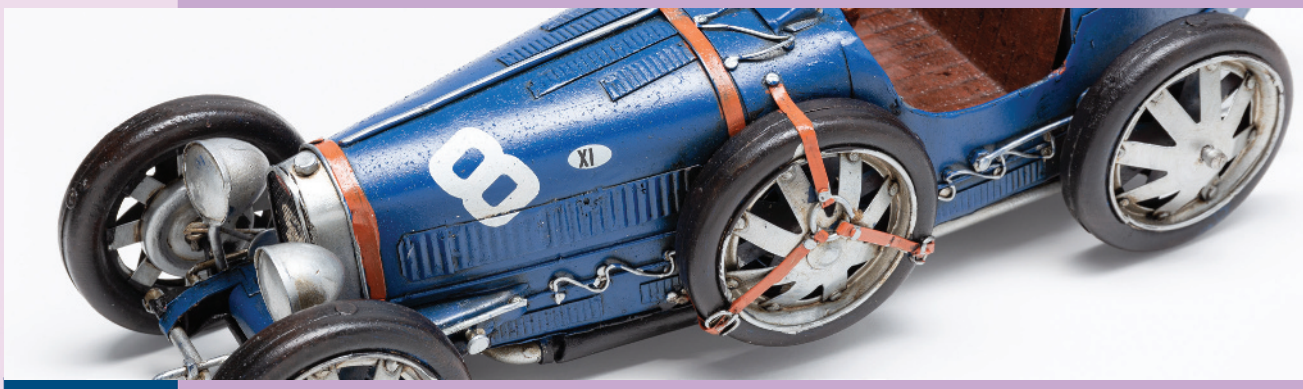


CHAPTER 13 Databases

TOPICS

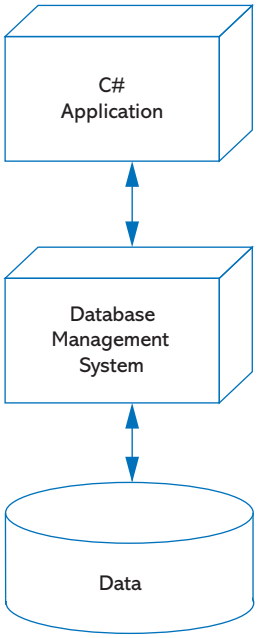
13.1 Introduction to Database Management Systems
13.2 Tables, Rows, and Columns
13.3 Creating a Database in Visual Studio
13.4 The DataGridView Control
13.5 Connecting to an Existing Database and Using Details View Controls
13.6 More about Data-Bound Controls
13.7 Selecting Data with the SQL `Select` Statement

13.1 Introduction to Database Management Systems

CONCEPT: **A database management system (DBMS) is software that manages large collections of data.**

If an application needs to store only a small amount of data, traditional files work well. These types of files are not practical, however, when a large amount of data must be stored and manipulated. Many businesses keep hundreds of thousands—or even millions—of data items in files. When a traditional file contains this much data, simple operations such as searching, inserting, and deleting become cumbersome and inefficient.

Figure 13-1 A C# application interacting with a DBMS, which manipulates data



When developing applications that work with an intensive amount of data, most developers prefer to use a database management system. A **database management system** (**DBMS**) is software that is specifically designed to store, retrieve, and manipulate large amounts of data in an organized and efficient manner. Once the data is stored using the database management system, applications may be written in C# or other languages to communicate with the DBMS. Rather than retrieving or manipulating the data directly, applications can send instructions to the DBMS. The DBMS carries out those instructions and sends the results back to the application, as Figure 13-1 illustrates.

Although Figure 13-1 is simplified, it illustrates the layered nature of an application that works with a DBMS. The topmost layer of software, which—in this case—is written in C#, interacts with the user. It also sends instructions to the next layer of software, the DBMS. The DBMS works directly with the data and sends the results of operations back to the application.

Suppose a company keeps all its product records in a database. The company has a C# application that allows the user to look up information on any product by entering its product ID number. The C# application instructs the DBMS to retrieve the record for the product with the specified product ID number. The DBMS retrieves the product record and sends the data back to the C# application. The C# application displays the data to the user.

The advantage of this layered approach to software development is that the C# programmer does not need to know about the physical structure of the data. The programmer needs to know only how to interact with the DBMS. The DBMS handles the actual reading of, writing of, and searching for data.

SQL Server Express Edition

There are numerous DBMSs in use today, and Visual C# can interact with many of them. Some of the more popular ones are Microsoft SQL Server, Oracle, DB2, MySQL, and Microsoft Access. In this book, we use Microsoft SQL Server Express Edition because it is an entry-level system designed to work with Visual Studio.

Checkpoint

13.1 What is a database management system (DBMS)?

13.2 Why do most businesses use a DBMS to store their data instead of creating their own text files to hold the data?

13.3 When developing a C# application that uses a DBMS to store and manipulate data, why doesn't the programmer need to know specific details about the physical structure of the data?

13.2 Tables, Rows, and Columns

CONCEPT: **Data that is stored in a database is organized into tables, rows, and columns.**

A database management system stores data in a **database**. The data that is stored in a database is organized into one or more tables. Each **table** holds a collection of related data. The data that is stored in a table is then organized into rows and columns. A **row** is a complete set of information about a single item. The data that is stored in a row is divided into columns. Each **column** holds an individual piece of information about the item.

For example, suppose we are developing a phone book application and we want to store a list of names and phone numbers in a database. We initially store the following list:

Katie Allen	555-1234
Kiran Sharma	555-5678
Alex Prieto	555-9012
Rosa Lopez	555-3456
Ingrid Olen	555-7890
Jamal Ramzi	555-1122
Carmen Cabera	555-3344
Sanjay Patel	555-5566
Kelsey Rose	555-2345

Think about how this data would appear if we stored it as rows and columns in a spreadsheet. We would put the names in one column and the phone numbers in another column. Each row, then, would contain data about one person. Figure 13-2 shows how the third row contains the name and phone number for Alex Prieto.

Figure 13-2 The table

This row contains data about one person:
Name: Alex Prieto
Phone: 555-9012

Name	Phone
Katie Allen	555-1234
Kiran Sharma	555-5678
Alex Prieto	555-9012
Rosa Lopez	555-3456
Ingrid Olen	555-7890
Jamal Ramzi	555-1122
Carmen Cabera	555-3344
Sanjay Patel	555-5566
Kelsey Rose	555-2345

When we create a database table to hold this information, we organize it in a similar manner. We give the table a name, such as `Person`. In the table, we create a column for the names and a column for the phone numbers. Each column in a table must have a name, so we can name our columns `Name` and `Phone`, respectively.

Column Data Types

When you create a database table, you must specify a data type for the columns. The data types that you can choose from are not C# data types, however. They are the data types that are provided by the DBMS. In this book, we are using Microsoft SQL Server, so we will select from the data types provided by that DBMS. Table 13-1 lists a few of the Microsoft SQL Server data types and shows the C# data type with which each is generally compatible.

NOTE: Table 13-1 shows the data types that you will most often use while learning. These are only a small number of the data types provided by Microsoft SQL Server, however.

Table 13-1 A few of the Microsoft SQL Server data types

SQL Server Data Type	Description	Corresponding C# or .NET Framework Data Type
`bit`	True/False values	`bool`
`datetime`	A date and a time	`DateTime`
`decimal(`*t*`, `*d*`)`	A decimal value with *t* total digits and *d* digits appearing after the decimal point	`decimal`
`float`	Real numbers	`double`
`int`	An integer number	`int`
`money`	Values that represent currency	`decimal`
`nchar(`*n*`)`	A fixed-length Unicode string with a maximum length of *n* characters	`string`
`nvarchar(`*n*`)`	A variable-length Unicode string with a maximum length of *n* characters	`string`

NOTE: The `nchar(`*n*`)` and `nvarchar(`*n*`)` data types are both used to store strings. An `nchar(`*n*`)` column is fixed in length and will always use *n* characters of space in the database. An `nvarchar(`*n*`)` column is variable in length, and the amount of space it uses in the database will be enough to accommodate the piece of data it holds, up to *n* characters.

Primary Keys

Most database tables have a **primary key**, which is a column that can be used to identify a specific row. The column that is designated as the primary key must hold a unique value for each row. Here are some examples:

- A table stores employee data, and one of the columns holds employee ID numbers. Because each employee's ID number is unique, this column can be used as the primary key.
- A table stores product data, and one of the columns holds the product number. Because each product has a unique product number, this column can be used as the primary key.
- A table stores invoice data, and one of the columns holds invoice numbers. Each invoice has a unique invoice number, so this column can be used as a primary key.

Identity Columns

Sometimes, the data that you want to store in a table does not contain any unique items that can be used as a primary key. For example, in the `Person` table that we previously described, neither the `Name` column nor the `Phone` column contains unique data. Two people can have the same name, so it is possible that a name might appear more than once in the `Name` column. Also, multiple people can share the same phone number, so it is possible that a phone number might appear more than once in the `Phone` column. Consequently, you cannot use the `Name` column or the `Phone` column as a primary key.

In a case such as this, it is necessary to create an identity column specifically to serve as the primary key. An **identity column** is a column that contains unique values that are generated by the DBMS. Identity columns typically contain integers. Each time a new row is added to the table, the DBMS automatically assigns a unique value to that row's identity column.

For example, when designing the `Person` table that we previously discussed, we could create an `int` column named `PersonID` and designate that column as an identity column. (As a result, the DBMS assigns a unique integer value to the `PersonID` column for each row.) Then we could designate the `PersonID` column as the table's primary key. Figure 13-3 shows an example of the `Person` table after we have created it and entered data into it.

Figure 13-3 The `Person` table with data entered

PersonID	Name	Phone
1	Katie Allen	555-1234
2	Kiran Sharma	555-5678
3	Alex Prieto	555-9012
4	Rosa Lopez	555-3456
5	Ingrid Olen	555-7890
6	Jamal Ramzi	555-1122
7	Carmen Cabera	555-3344
8	Sanjay Patel	555-5566
9	Kelsey Rose	555-2345

Allowing Null Values

If a column contains no data, it is said to be null. Sometimes it is okay for a column to be left empty. Some columns, however, such as primary keys, must contain a value. When you are designing a table, you can specify whether a column is allowed to be null. If a particular column is not allowed to be null, any time you add a row of data to the table, the DBMS will require that a value be provided for the column. Leaving the column empty results in an error.

Checkpoint

13.4 Describe each of the following terms:

a. database

b. table

c. row

d. column

13.5 Match the Microsoft SQL Server data type with the compatible C# data type.

1. `bit`	a. `double`
2. `datetime`	b. `int`
3. `float`	c. `bool`
4. `int`	d. `DateTime`
5. `money`	e. `string`
6. `nchar`	f. `decimal`

13.6 What is the purpose of a primary key?

13.7 What is an identity column?

13.8 If a particular column is not allowed to be null, what is the result if the column is left empty?

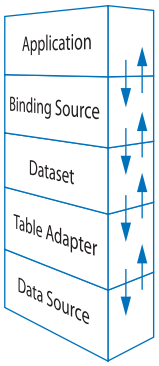
13.3 Creating a Database in Visual Studio

CONCEPT: **A .NET application uses several components, arranged in layers, to connect to a database. Visual Studio provides tools that allow you to create a database and configure the various components that an application needs to connect to it.**

Connecting an Application to a Database

A .NET application requires a set of special components that let it connect to, and work with, a database. The components are layered one on top of the other, as shown in Figure 13-4.

Figure 13-4 The components used by an application to connect to a database



Here is a summary of the layers shown in the figure:

- **Data Source**—As its name implies, a **data source** is a source of data with which the application can work. Data sources are usually databases, but they can also be Excel spreadsheets, XML data, text files, or Web services.
- **Table Adapter**—A **table adapter** connects to a data source and can retrieve data from a table in a data source. It can also update the table in the data source.
- **Dataset**—A **dataset** gets a copy of a table from the table adapter and keeps the copy of the table in memory. Instead of working directly with the database, the application works with the dataset, modifying rows, deleting rows, adding new rows, etc. Then, the dataset can instruct the table adapter to write the changes back to the database.
- **Binding Source**—A **binding source** is a component that can connect user interface controls directly to a dataset.

Fortunately, Visual Studio provides wizards that make it easy to create and configure the necessary components. In Tutorial 13-1, you start a new C# project and use Visual Studio to perform the following:

- Create an SQL Server database
- Design a table in the database
- Add data to the table

Tutorial 13-1:
Starting the *PhoneBook* Project and Creating the Phonelist.mdf Database

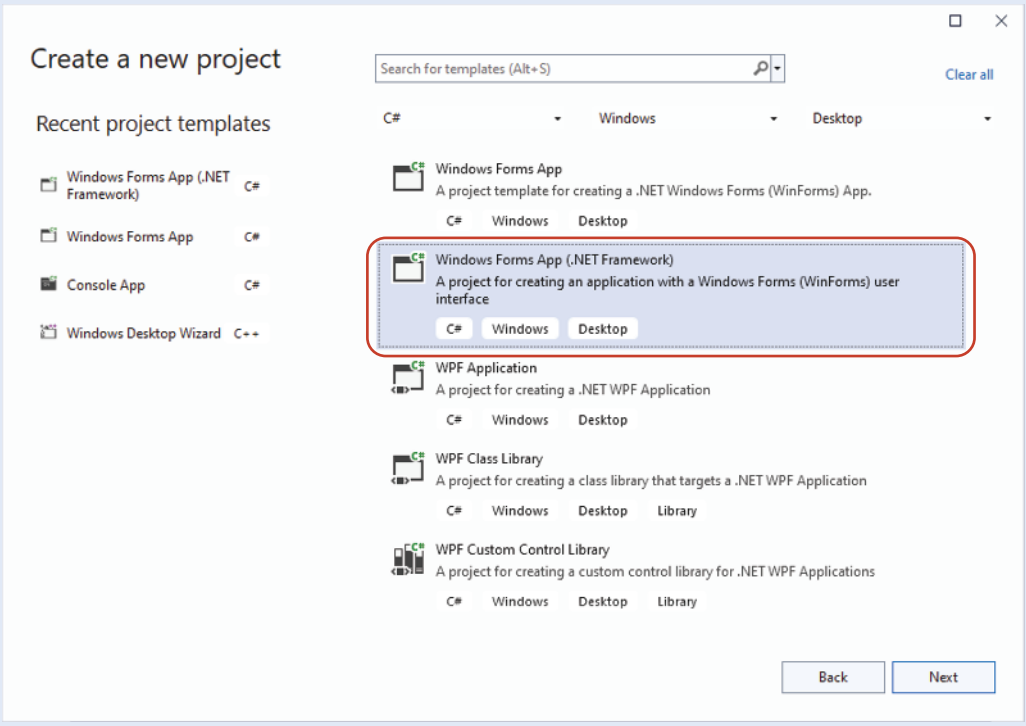
VideoNote
Tutorial 13-1: Starting the *PhoneBook* Project and Creating the Phonelist.mdf Database

NOTE: In this chapter's tutorials, we will use the ***Windows Forms App (.NET Framework)*** project type, instead of the *Windows Forms App* project type that you have been using up to this point. Some of the visual tools for creating database connections have not yet been implemented in the *Windows Forms App* project type, so be sure to pick the ***Windows Forms App (.NET Framework)*** project type.

In the next two tutorials, you create an application that uses a database to store a list of names and phone numbers. In this tutorial, you start the application by creating the database. In the next tutorial, you connect the application to the database, add a dataset, and add a control to the form that allows the user to interact with the database.

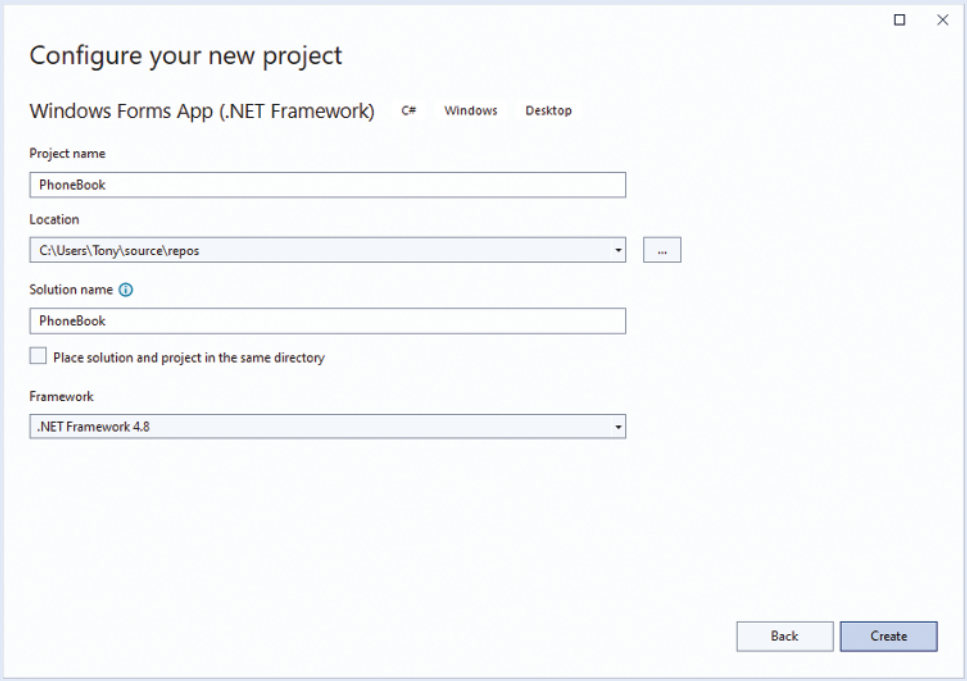
Step 1: Start Visual Studio and create a new project. In the *Create a New Project* window, be sure to select the Windows Forms App (.NET Framework) project type, as shown in Figure 13-5.

Figure 13-5 Select the *Windows Forms App (.NET Framework)* project type



Step 2: The *Configure your project* window appears next. Change the name of the project to *PhoneBook* and make sure the highest version of the .NET Framework is selected in the *Framework* dropdown list. Figure 13-6 shows an example. Then, click the *Create* button. A new project will be created.

Figure 13-6 The *Configure your project* Window

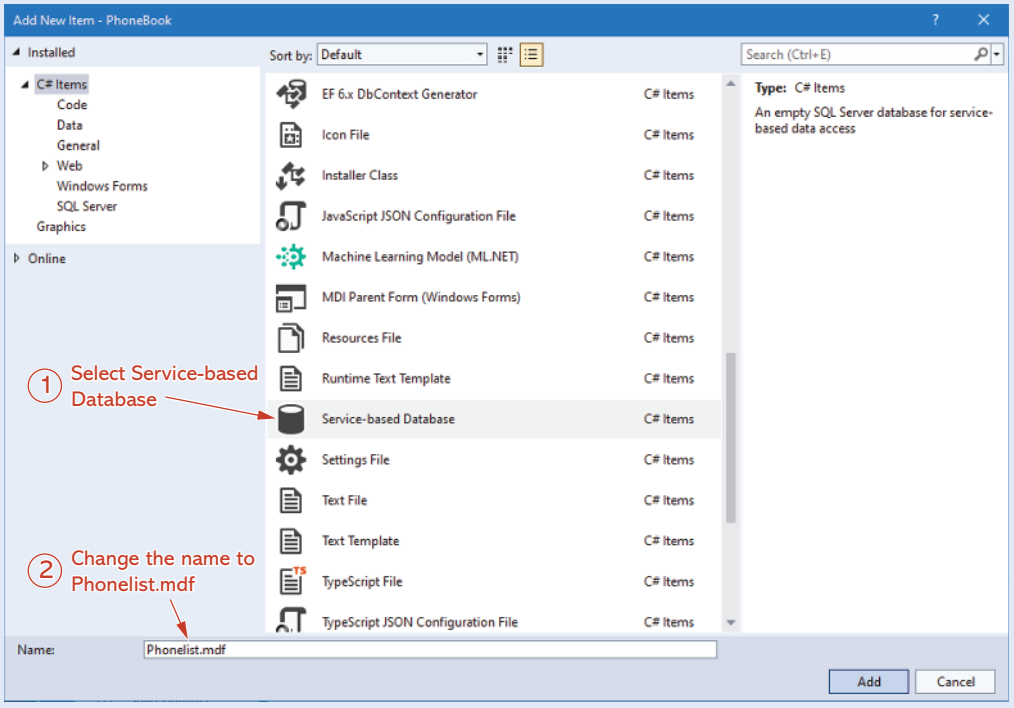


Step 3: Change the Form1 form's Text property to *Phone Book*.

Step 4: Next you create a new database and add it to the project. Click *Project* on the Visual Studio menu bar and then select *Add New Item*. . . .

Step 5: The *Add New Item* window will appear. Scroll down the list of items and select *Service-based Database*, as shown in Figure 13-7. You also need to

Figure 13-7 The *Add New Item* window

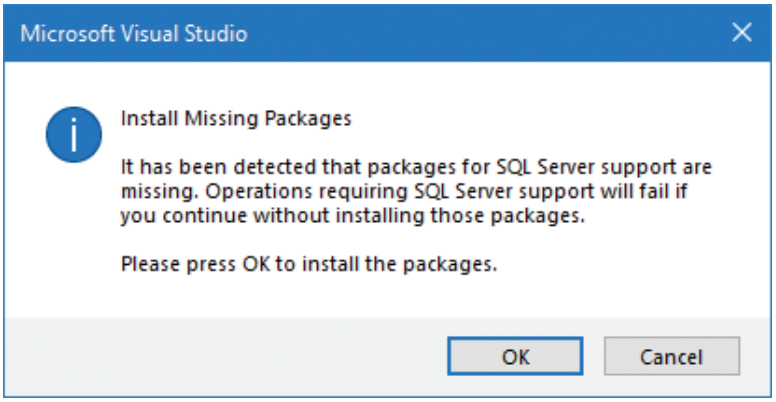


specify the database's name in the *Name* text box at the bottom of the window. A default name, such as *Database1.mdf* appears there. Change the name to *Phonelist.mdf*. (Be sure to keep the .mdf file extension.)

Step 6: Click the *Add* button in the *Add New Item* window. This adds an empty SQL Server database named Phonelist.mdf to your project.

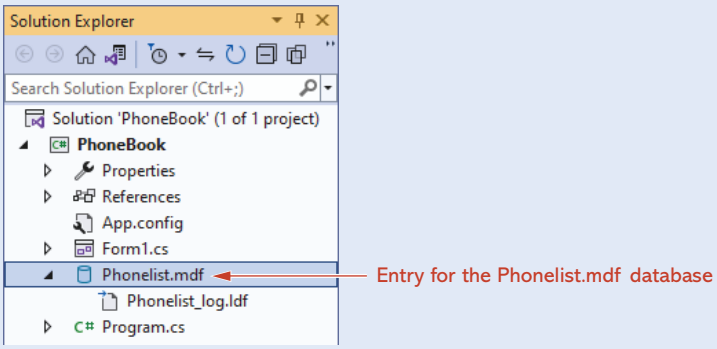
NOTE: If this is your first time creating a database in Visual Studio, you will see the window shown in Figure 13-8, prompting you to install the missing SQL Server packages. Click the OK button and follow the prompts to install the files.

Figure 13-8 Prompt to install missing packages



Step 7: Look at the *Solution Explorer*, shown in Figure 13-9, and notice that it now contains an entry for the Phonelist.mdf database. Double-click the entry for Phonelist.mdf database. Notice that a new window named *Server Explorer* is now open in the area where the *Toolbox* is located. The *Server Explorer* is shown in Figure 13-10.

Figure 13-9 *Solution Explorer* entry for the Phonelist.mdf database



Step 8: Now you will add a table to the Phonelist.mdf database. In the *Server Explorer*, make sure the Phonelist.mdf entry is expanded, as shown in Figure 13-10. Right-click on *Tables*, and as shown in Figure 13-11, click *Add New Table* on the menu.

This displays the *Table Designer*, as shown in Figure 13-12. You use the *Table Designer* to specify the names and data types of the table's columns.

Figure 13-10 *Server Explorer*

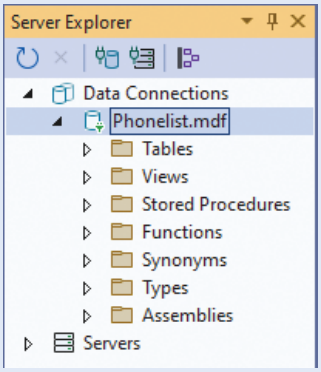


Figure 13-11 Adding a table to the Phonelist.mdf database

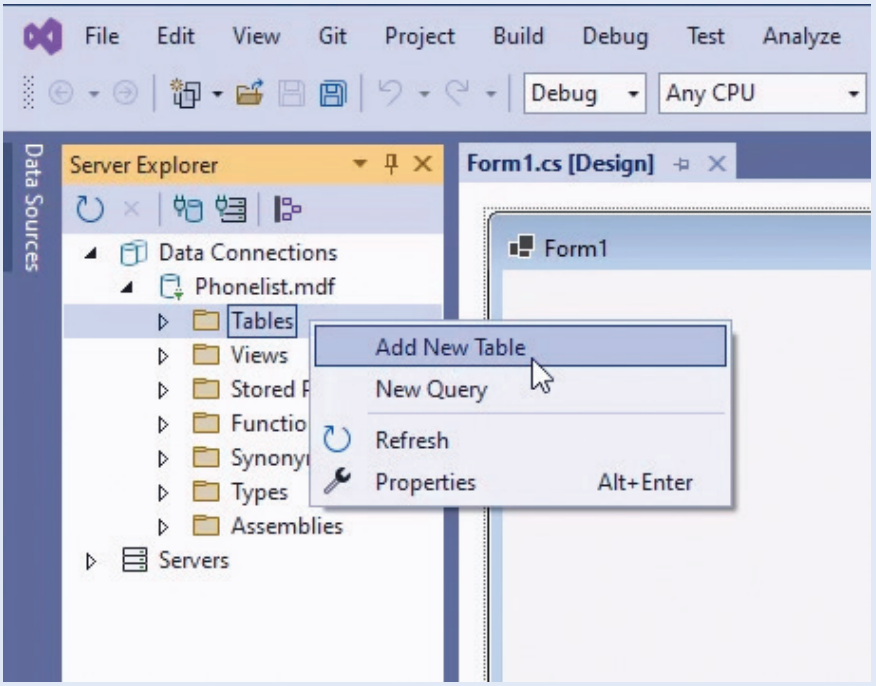
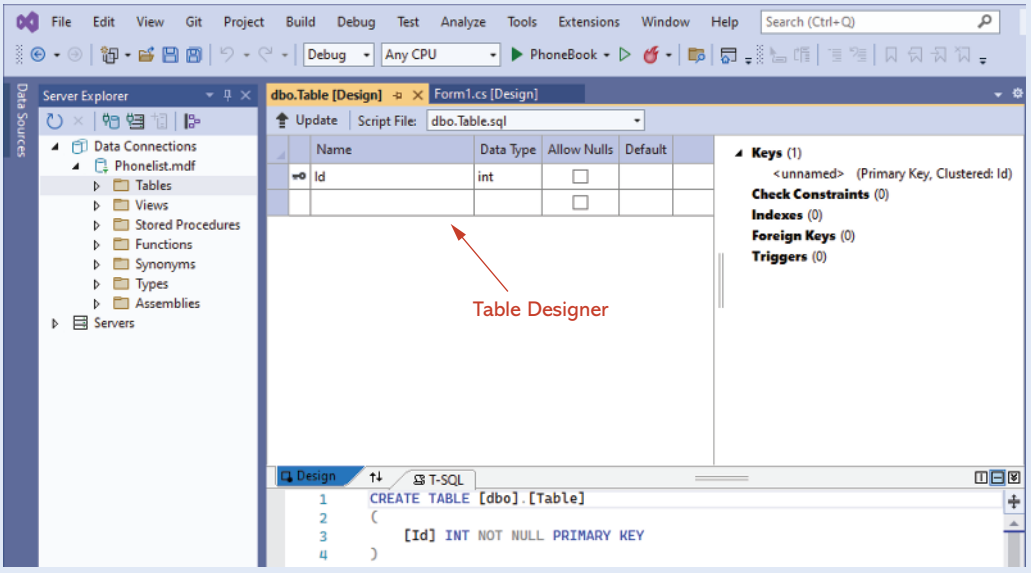


Figure 13-12 The *Table Designer*

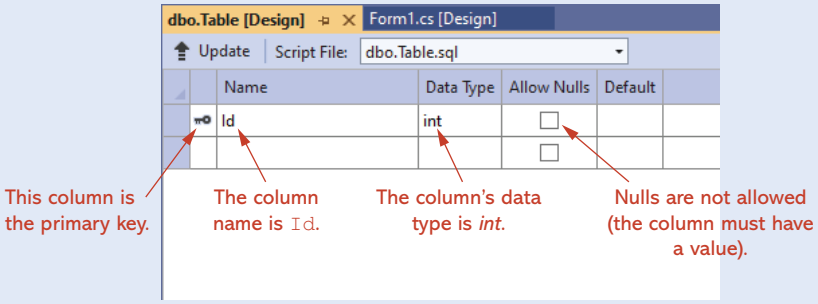


Step 9: Look closely at the *Table Designer* and notice that a column named `Id` has automatically been added to the table. Figure 13-13 shows how to read the information about the column in the *Table Designer*:

- The key icon ( ) that appears to the left of the name indicates that the column is the table's primary key.
- The name of the column is `Id`.
- The column's data type is `int`.
- The *Allow Nulls* checkbox is not checked. This means that the column must have a value (it cannot be null).

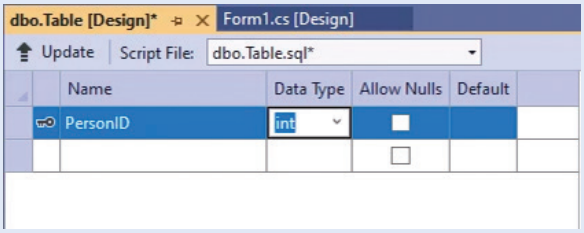
We want to keep this column, but we are going to make some changes to it. In the next step, we will change the column's name to `PersonID`, and we will make it an identity column (so the database will automatically generate unique integer values for it).

Figure 13-13 The `Id` column that is automatically added to the table



Step 10: Click on the column name (`Id`) in the *Table Designer*, and change the name to `PersonID`. Press the Tab key. The *Table Designer* should appear as shown in Figure 13-14.

Figure 13-14 The column name changed to `PersonID`



Step 11: With the column still selected, notice that the *Properties* window is showing the column's properties. Scroll down until you see *Identity Specification*. Double-click *Identity Specification*, and you should see the (*Is Identity*), *Identity Increment*, and *Identity Seed* properties appear below it, as shown in Figure 13-15.

Step 12: Change the (*Is Identity*) property to *True*. This designates the column as an identity column. Notice that the *Identity Increment* and *Identity Seed* properties are both now set to *1*, as shown in Figure 13-16. Leave these values as they are.

Step 13: In the *Table Designer*, add another column named `Name`. Select `nvarchar(50)` as the data type and remove the check under *Allow Nulls*.

Figure 13-15 The column's *Identity Specification* properties displayed

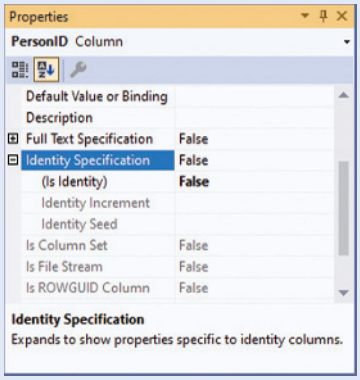
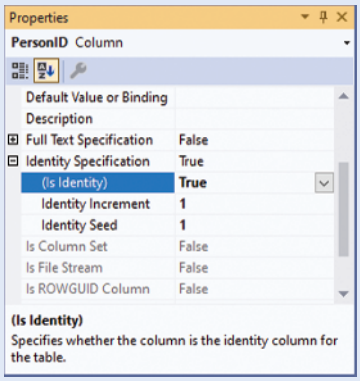
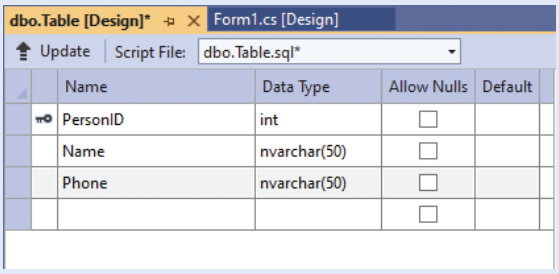


Figure 13-16 The *Identity Specification* properties set to the desired values



Step 14: Add another column named `Phone`. Select *`nvarchar(50)`* as the data type and remove the check under *Allow Nulls*. The *Table Designer* should now appear as shown in Figure 13-17.

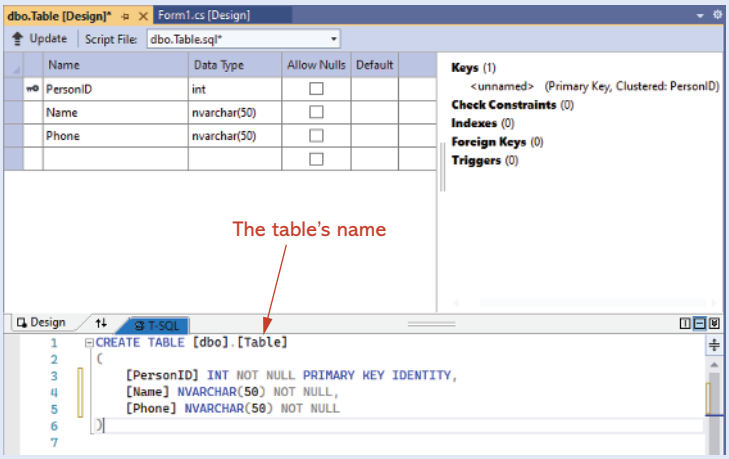
Figure 13-17 The `PersonID`, `Name`, and `Phone` columns created in the *Table Designer*



Step 15: You are finished designing the table, but you need to change the table's name. When you create a new table in the *Server Explorer*, it is given a default name, such as `Table`. Notice at the bottom of the screen (below the *Table Designer*), you see an area labeled *T-SQL*. As shown in Figure 13-18, the first statement in this window shows the name of the table. Change the first statement so it reads as follows:

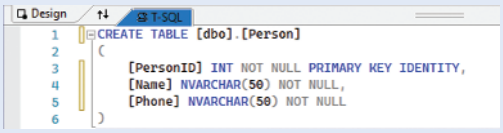
```
CREATE TABLE [dbo].[Person]
```

Figure 13-18 The *T-SQL* window



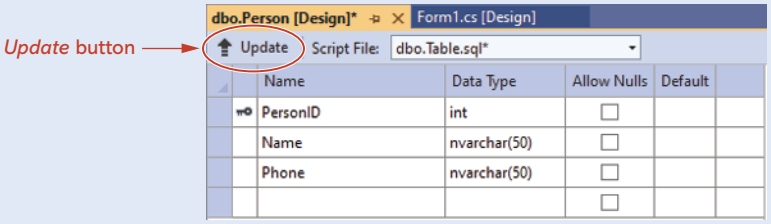
The *T-SQL* window should now appear as shown in Figure 13-19.

Figure 13-19 The table named changed to `Person`



Step 16: Although you changed the table's name in the *T-SQL* window, the change does not take effect until you update the database. Click the *Update* button that appears in the upper left corner of the *Table Designer*, as shown in Figure 13-20.

Figure 13-20 The *Update* button to update the database



Step 17: The *Preview Database Updates* window, shown in Figure 13-21, will appear next. Click the *Update Database* button. After a moment, you should see confirmation that the update was successful at the bottom of the screen, similar to that shown in Figure 13-22. Close the *Table Designer* window.

Step 18: The next step is to enter data into the table. In the *Server Explorer*, expand the *Tables* entry and then right-click the *Person* entry. You will see the pop-up menu shown in Figure 13-23. Click *Show Table Data* in the menu.

Figure 13-21 The *Preview Database Updates* window

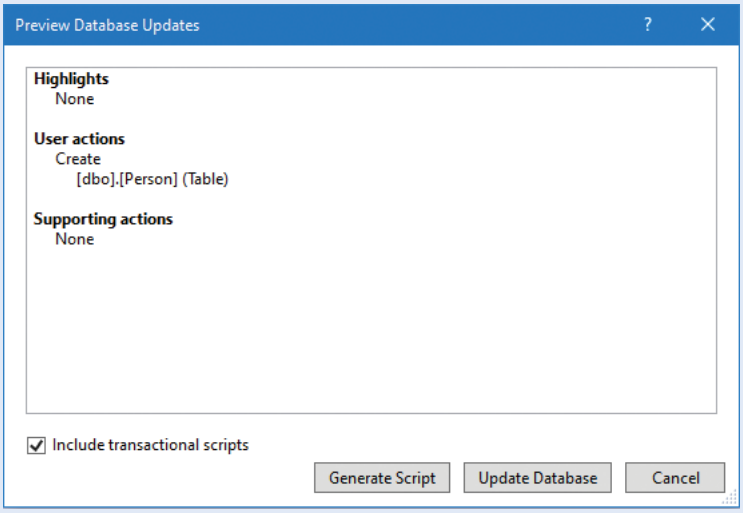
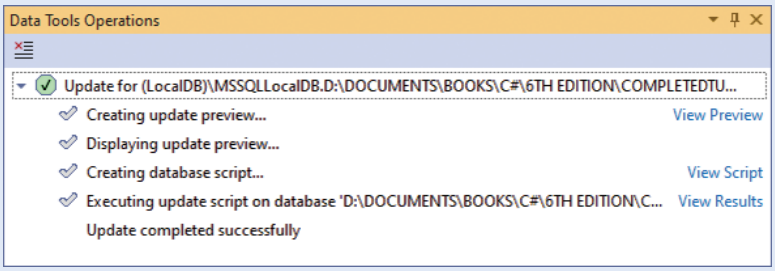
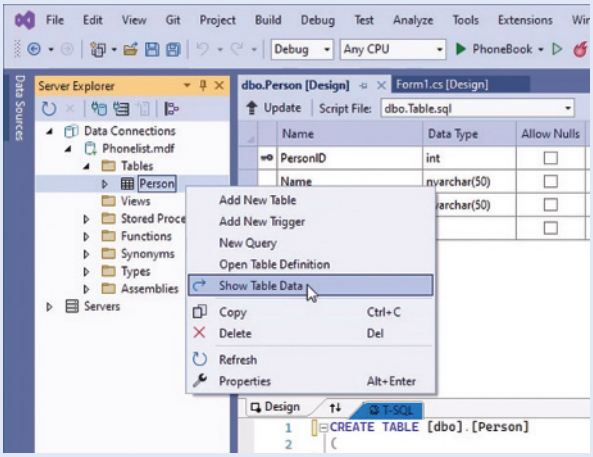


Figure 13-22 Confirmation that the database was updated



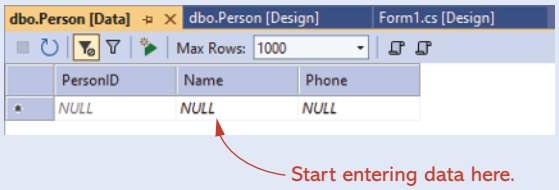
NOTE: If you do not see the entry for the `Person` table in the *Server Explorer*, click the *Refresh* button ( ) in the upper-left corner of the window.

Figure 13-23 Clicking *Show Table Data*



Step 19: The table should now be opened and ready for input, as shown in Figure 13-24. Enter several names and phone numbers into the `Name` and the `Phone` columns.

Figure 13-24 Ready to enter data into the `Person` table



(Figure 13-25 shows several examples.) As you enter data, keep the following in mind:

- Do not enter values for the `PersonID` column. The system automatically generates a value for the `PersonID` column each time you add a new row.
- You must enter values for the `Name` and `Phone` columns. Recall that you removed the check from *Allow Nulls* for these columns when you designed the table. If you try to move away from a row without specifying values for these columns, you will see an error message.
- Exclamation points will appear next to the columns as you enter data. These simply mean that the data has not been saved to the database. When you move away from the row, the data is saved and the exclamation points disappear.

Figure 13-25 shows the table with sample data entered.

Figure 13-25 Sample data entered into the `Person` table

PersonID	Name	Phone
1	Katie Allen	555-1234
2	Kiran Sharma	555-5678
3	Alex Prieto	555-9012
4	Rosa Lopez	555-3456
5	Ingrid Olen	555-7890
6	Jamal Ramzi	555-1122
7	Carmen Cabera	555-3344
8	Sanjay Patel	555-5566
9	Kelsey Rose	555-2345
NULL	NULL	NULL

Step 20: Once you have entered all the data, click *File* on the Visual Studio menu and then click *Save All*. The rows are saved to the Phonelist.mdf database.

Step 21: Close the table so the Form1 form is displayed in the *Designer*.

Let's review what you have done up to this point:

- Created an SQL Server database named Phonelist.mdf
- Designed and created a table named `Person`
- Entered sample data into the table

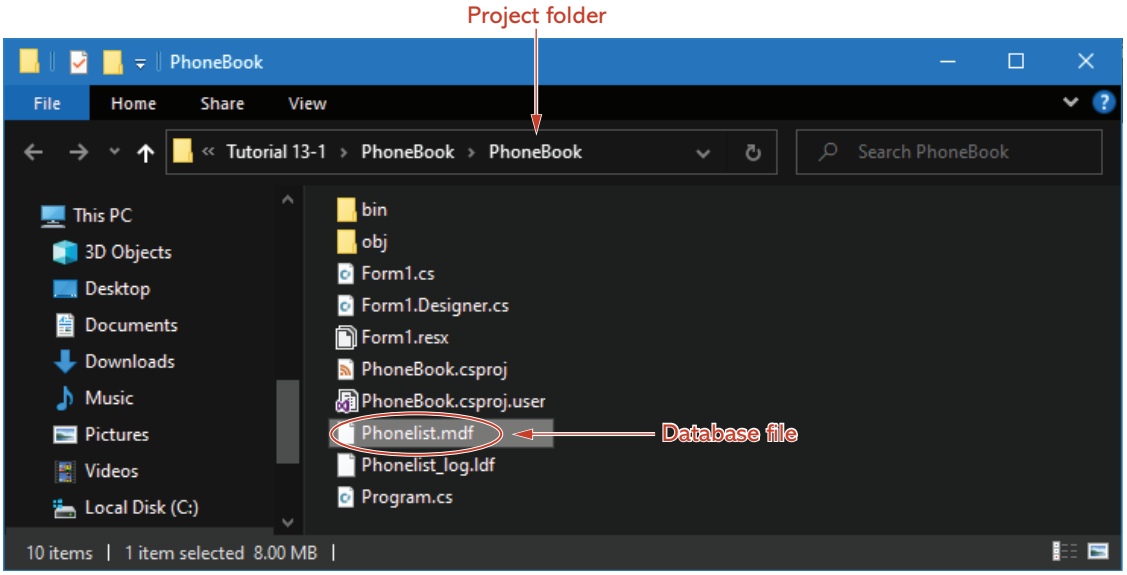
In the next tutorial, you connect the Phonelist.mdf database to the application's form and add a control to the form that lets the user interact with the database. If you intend to continue to the next tutorial at this time, leave the project open in Visual Studio.

The Database File's Location

When you use Visual Studio to create a database, as you did in Tutorial 13-1, the database file will be created in the project folder. For example, Figure 13-26 shows how the Phonelist.mdf database file was created in the *Phone Book* application's project folder.

NOTE: When you create an SQL Server database, you will also see a file that ends with the .LDF extension. For example, in Figure 13-26 you see a file named Phonelist_log.LDF. This is a transaction log file. The SQL Server DBMS uses it to keep a log of all the operations that you perform on the database.

Figure 13-26 Location of the Phonelist.mdf database file



Checkpoint

13.9 List each of the .NET components that allow an application to connect to, and work with, a database in the order they are layered. Begin the list with the application as the topmost layer.

13.10 What is the relationship between the table adapter and data source components?

13.11 What component is used to connect interface controls to a dataset?

13.4 The DataGridView Control

CONCEPT: **A DataGridView control can display a database table in a scrollable grid.**

A **data-bound control** is a user interface control that is connected to a data source. For example, a data-bound control can be connected to a column in a database table. Data-bound controls automatically display data from the data source and can be used to change the data that they are bound to. One of the simplest and most powerful

data-bound controls is the DataGridView control. A **DataGridView control** can display an entire database table in a scrollable grid on an application's form.

In Tutorial 13-2, you continue working on the *PhoneBook* project that you started in Tutorial 13-1. You connect the application's form to the Phonelist.mdf database and add a DataGridView control that can be used to view and update the `Person` table.

Tutorial 13-2:
Completing the *PhoneBook* Project

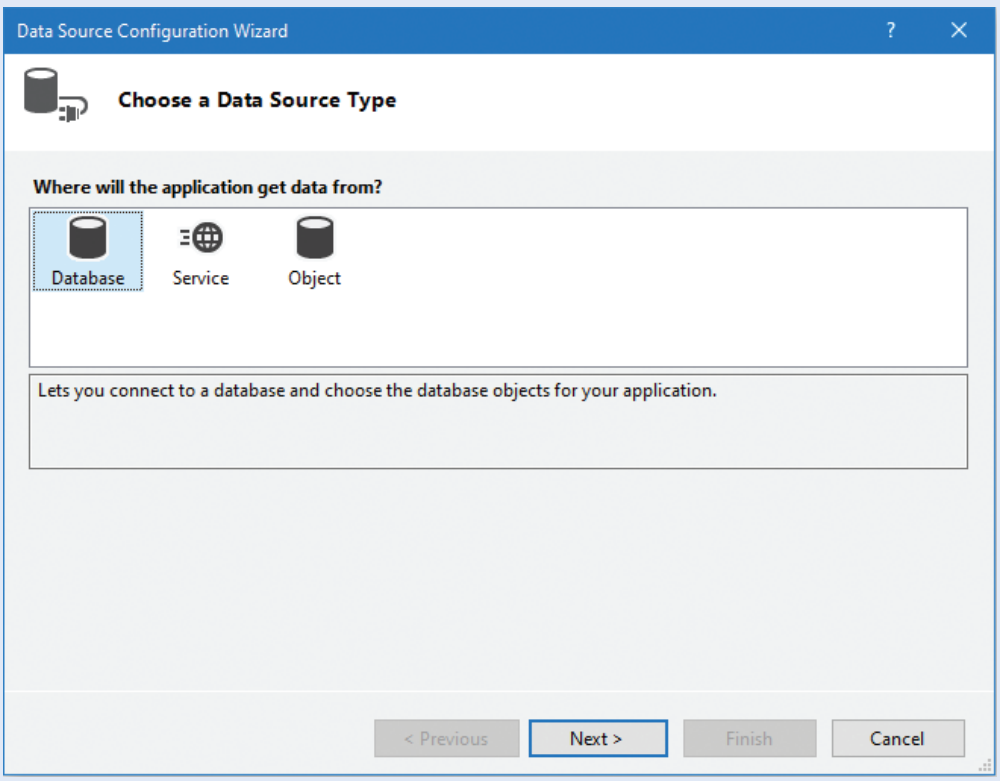
VideoNote
Tutorial 13-2: Completing the *PhoneBook* Project

In this tutorial, you complete the *PhoneBook* project. You add the Phonelist.mdf database to the project as a data source, create a dataset that is connected to the `Person` table, and add a DataGridView control that lets the user interact with the database.

Step 1: Make sure the *PhoneBook* project is open in Visual Studio from the previous tutorial.

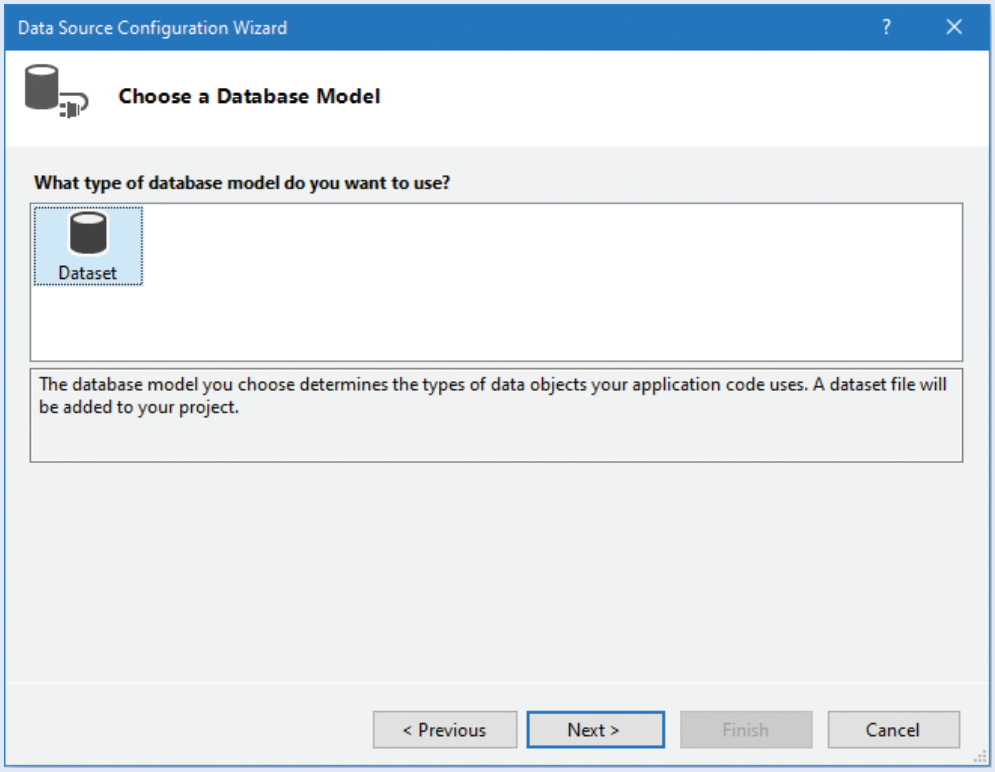
Step 2: Click *Project* on the Visual Studio menu and then click *Add New Data Source*. . . . You should see the *Data Source Configuration Wizard*, as shown in Figure 13-27.

Figure 13-27 The *Data Source Configuration Wizard—Choose a Data Source Type*



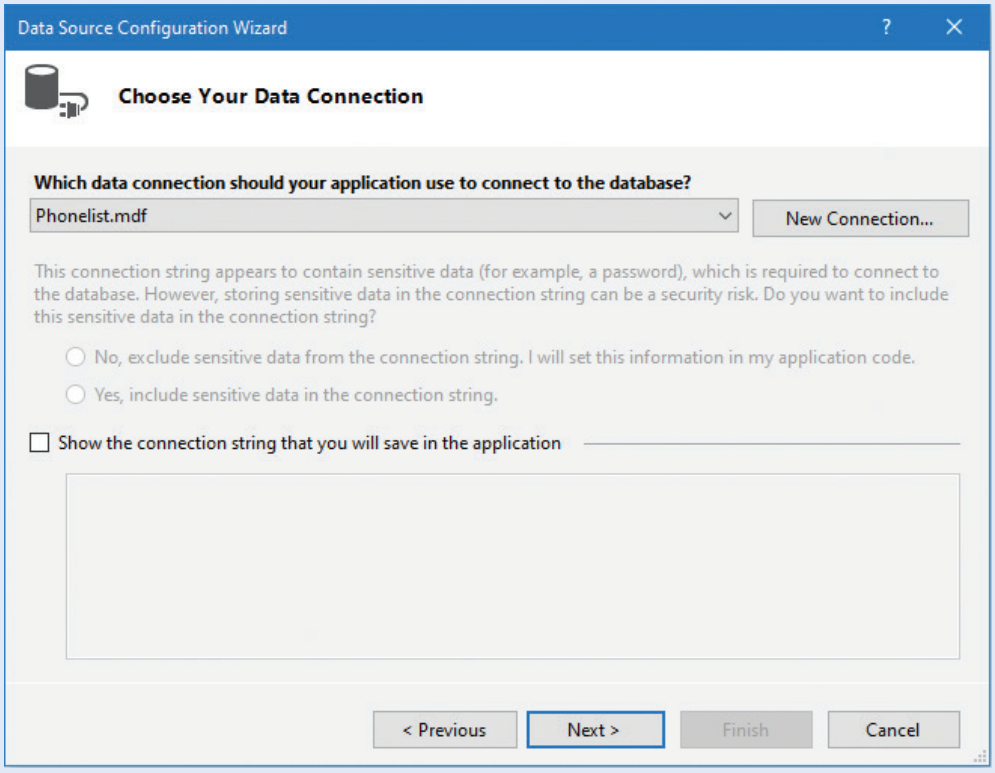
Step 3: The data source you are adding is a database, so make sure *Database* is selected and then click *Next* >. The window now appears, as shown in Figure 13-28.

Figure 13-28 The *Data Source Configuration Wizard—Choose a Database Model*



Step 4: Make sure *Dataset* is selected, and click *Next* >. The window now appears, as shown in Figure 13-29.

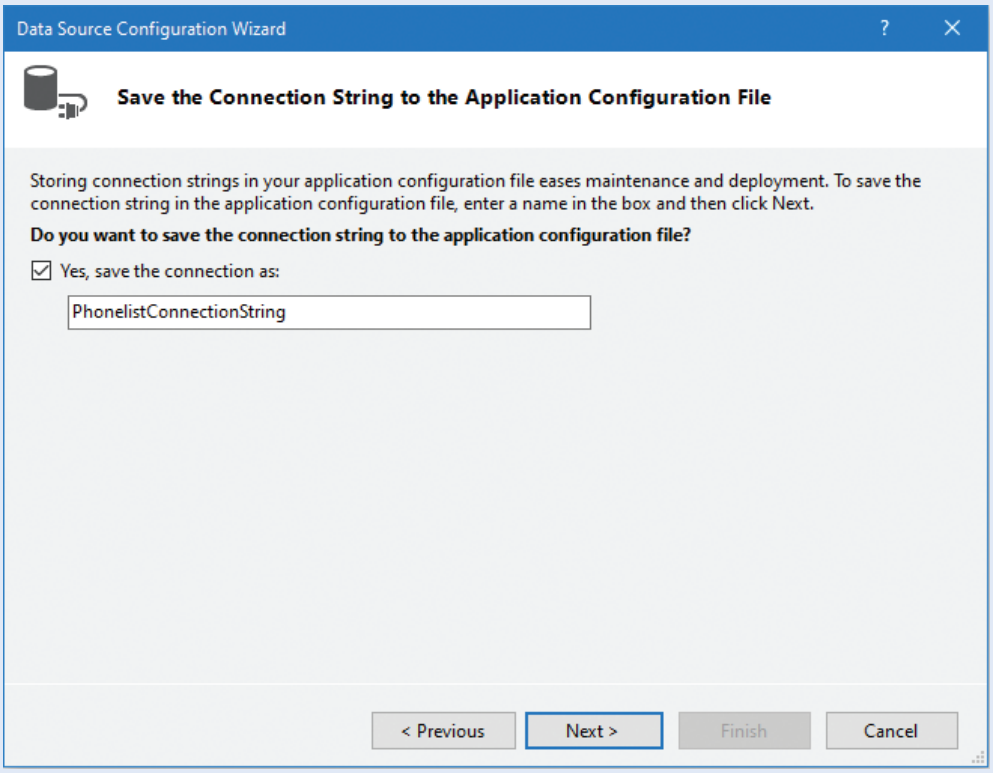
Figure 13-29 The *Data Source Configuration Wizard—Choose Your Data Connection*



Step 5: In this window, you are asked to choose a data connection, and the Phonelist .mdf database should be automatically selected, as shown in Figure 13-29. Simply click *Next* >. The window now appears, as shown in Figure 13-30.

Step 6: In this window, you are asked if you want to save a connection string. Leave the *Yes* box checked and click *Next* >.

Figure 13-30 The *Data Source Configuration Wizard—Save the Connection String*



Step 7: In the window that appears now, you are asked to choose the database objects that you want to include in the dataset. Expand the *Tables* entry, and place a check next to `Person`, as shown in Figure 13-31.

At the bottom of the window you can specify a name for the dataset. The default name `PhonelistDataSet` is already provided. Because you selected the `Person` table as the database object, change the dataset name to `PersonDataSet`. Click *Finish*.

Here is a summary of what you just did:

- You added the Phonelist.mdf database as a data source to the application.
- You created a dataset component that is connected to the `Person` table. The name of the dataset component is `PersonDataSet`.

Step 8: Visual Studio provides a *Data Sources* window that lets you see all the data sources in the current project. Click *View* on the Visual Studio menu bar, then select *Other Windows*, then click *Data Sources*. The *Data Sources* window should be displayed, as shown in Figure 13-32. Notice that the window shows the name of the dataset, which is `PersonDataSet`. Expand the `PersonDataSet` entry and you will see the name of the table to which the dataset is connected, which is `Person`.

Figure 13-31 The *Data Source Configuration Wizard—Choose Your Database Objects*

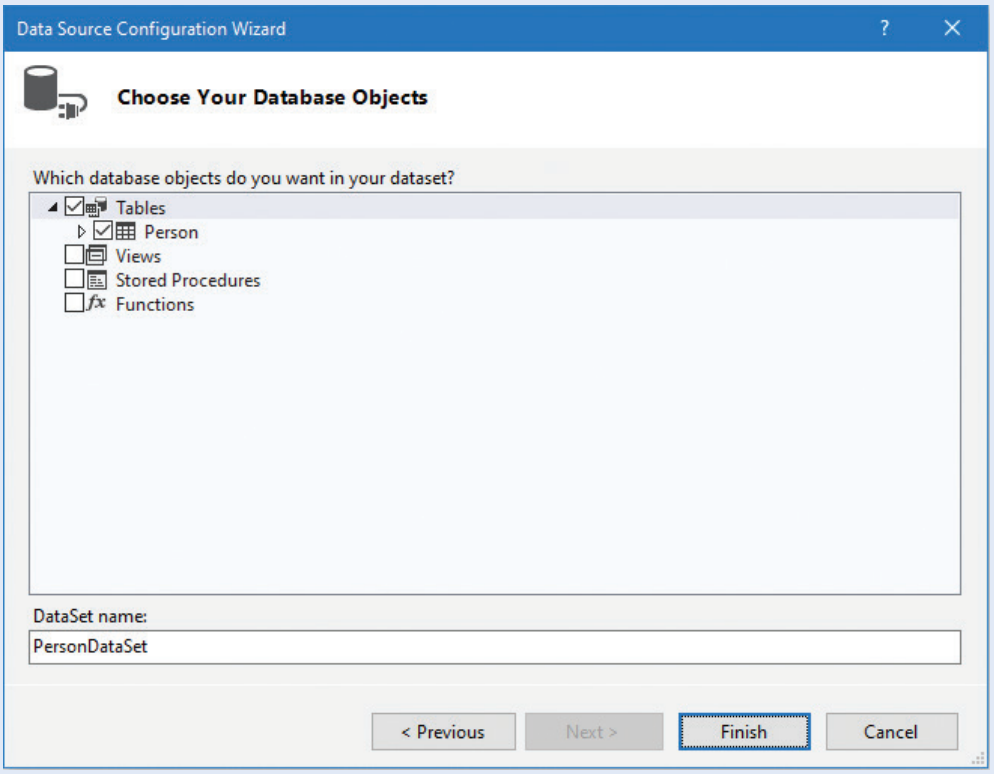
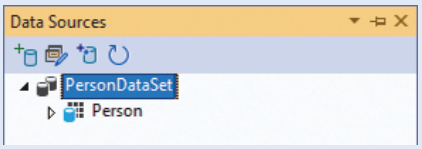
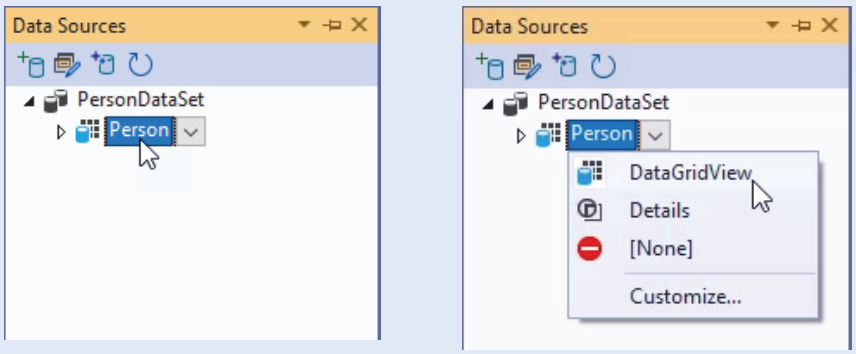


Figure 13-32 The *Data Sources* window



Step 9: Next you select DataGridView as the default data-bound control for the `Person` table. In the *Data Sources* window, click the entry for the `Person` table, as shown in the image on the left in Figure 13-33. Then, click the down arrow ( ), and select *DataGridView*, as shown in the image on the right.

Figure 13-33 Selecting DataGridView as the data-bound control for the `Person` table



Step 10: Now you add a DataGridView control to the form. As shown in Figure 13-34, click and drag the `Person` table from the *Data Sources* window onto the form. As shown in Figure 13-35, this creates a DataGridView control and a navigation bar on the form. (Adjust the size of the form and the size and position of the DataGridView control so they appear similar to Figure 13-35.)

Figure 13-34 Dragging the `Person` table onto the form

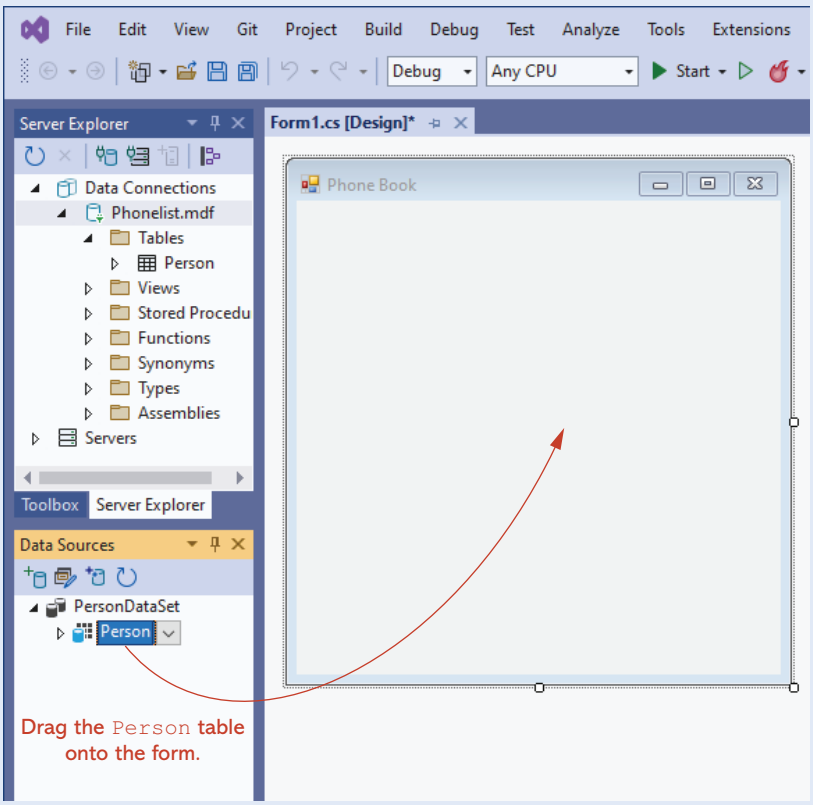
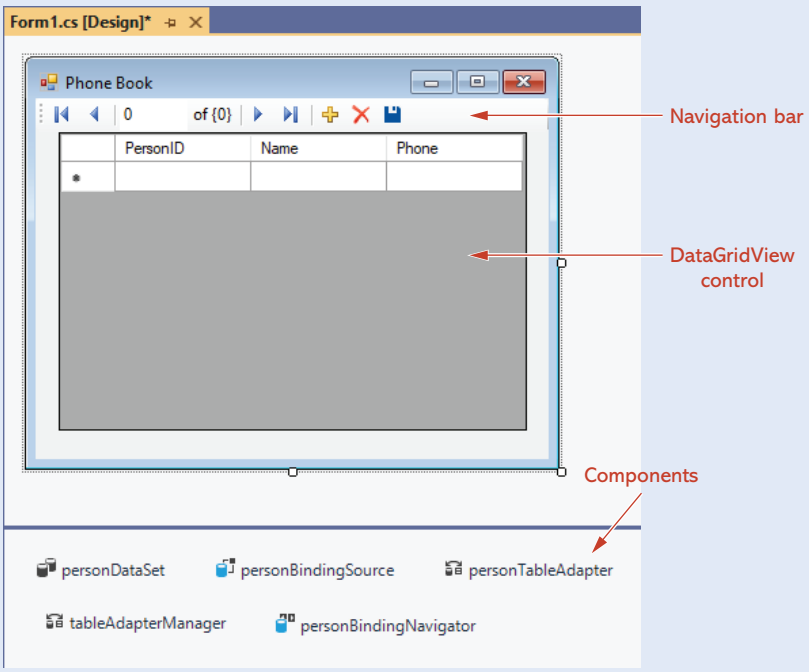


Figure 13-35 The DataGridView control placed on the form



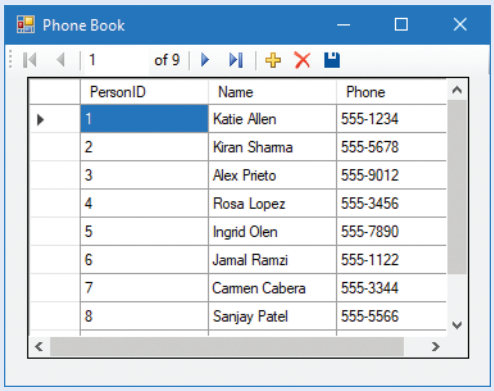
Notice that the component tray (the gray area at the bottom of the *Designer*) now contains several items. These are components that were automatically created when you placed the DataGridView control on the form. Here's a summary of the components:

- `personTableAdapter`—This is the table adapter. It gets data directly from the data source, which in this case is the Person.mdf database.
- `tableAdapterManager`—In many applications, the database has multiple tables from which we want to get data. A table adapter manager is a component that manages multiple tables.
- `personDataSet`—This is the dataset component. It gets a copy of the table from the table adapter and keeps it in memory. The application works with the dataset instead of working directly with the database.
- `personBindingSource`—This component connects the DataGridView control to the data source, and provides a channel of communication between the two.
- `personBindingNavigator`—This is the navigation bar.

Step 11: Save and run the application. The form will appear similar to Figure 13-36, with the rows of the `Person` table displayed in the DataGridView control. You can use the DataGridView control in the following ways:

- To change the value of a cell, click the cell with the mouse and then change its data as necessary.
- To select a row, click the button that appears next to it, along the left side of the control.
- To delete a row, select it; then either click the *Delete* button ( ) on the navigation bar or press Delete on the keyboard.

Figure 13-36 The *Phone Book* application running



- To add a new row, scroll to the bottom of the grid, where you will see an empty row. Enter the new data in the empty row.
- The changes that you make affect only the in-memory copy of the table that is held in the dataset. To save the changes to the database, click the *Save* button ( ) on the navigation bar.
- To sort the rows on a column value, click the column heading.

Step 12: Use the DataGridView control to make several changes to the dataset. For example, change a value in an existing row, add a new row, and delete an existing row. Be sure to click the *Save* button ( ) on the navigation bar to save your changes.

Step 13: Close the application and then rerun it to verify that your changes were saved. When you are finished, close the application.

Auto-Generated Code

When you place a data-bound control, such as the DataGridView, on a form, Visual Studio does a lot of work behind the scenes to make the control functions. In fact, you completed the application in Tutorial 13-2 without writing a single line of code! If you open the form in the code editor, however, you will see that Visual Studio generated some code, as shown in Program 13-1.

Program 13-1 Form1 code in the *Phone Book* application

```
 1 using System;
 2 using System.Collections.Generic;
 3 using System.ComponentModel;
 4 using System.Data;
 5 using System.Drawing;
 6 using System.Linq;
 7 using System.Text;
 8 using System.Threading.Tasks;
 9 using System.Windows.Forms;
10
11 namespace PhoneBook
12 {
13  public partial class Form1 : Form
14  {
15     public Form1()
16     {
17         InitializeComponent();
18     }
19
20     private void personBindingNavigatorSaveItem_Click(object
       sender, EventArgs e)
21     {
22         this.Validate();
23     this.personBindingSource.EndEdit();
24     this.tableAdapterManager.UpdateAll(this.personDataSet);
25
26     }
27
28     private void Form1_Load(object sender, EventArgs e)
29     {
30          // TODO: This line of code loads data into the...
31          this.personTableAdapter.Fill(this.personDataSet.Person);
32
33    }
34  }
35 }
```

Visual Studio adds an event handler in lines 20–26 that executes when the user clicks the *Save* button on the navigation bar. In a nutshell, the statements in this method apply any changes that have been made to the DataGridView control to the dataset and then save the dataset to the database.

Visual Studio also adds a Load event handler for the form, in lines 28–33. In line 31, the event handler calls the table adapter's `Fill` method, passing a reference to the `Person` table (contained in the dataset) as an argument. This statement causes the table adapter to load data from the database into the dataset.

Checkpoint

13.12 What do you call a user interface control that is connected to a data source?

13.13 List the components that are automatically created when you place a DataGridView control on a form.

13.14 What is the purpose of the table adapter's `Fill` method?

13.15 What kind of information is displayed in the *Data Sources* window? How is it displayed?

13.5 Connecting to an Existing Database and Using Details View Controls

CONCEPT: **You can easily connect an application to an existing database. The Details view controls are an alternative to the DataGridView control for interacting with a database.**

In Tutorial 13-1, you used Visual Studio to create a database from scratch and populate its table with data. More often, programmers must create applications that connect to existing databases. In Tutorial 13-3, you create an application and connect it to a database that is provided in the Student Sample Program files. The database is named ProductDB.mdf, and you will find it in the *Chap13* folder. The database has one table, named `Product`. Figure 13-37 shows the data that is stored in the `Product` table.

Figure 13-37 The `Product` table in the ProductDB.mdf database

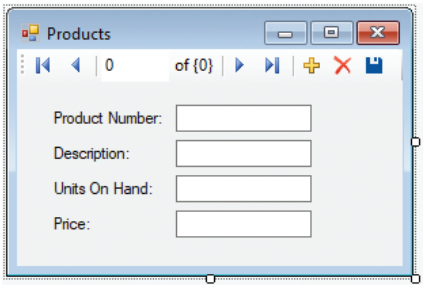
Product_Number	Description	Units_On_Hand	Price
10-01	Oxford Cloth Shirt	100	39.9500
10-02	Poplin Shirt	150	38.5000
10-03	Plaid Flannel Shirt	80	34.7500
10-04	Canvas Shirt	200	32.9500
10-05	Cotton Shirt	250	28.9500
20-01	Chino Classic Fit Pants	100	28.9500
20-02	Chino Relaxed Fit Pants	100	28.9500
20-03	Denim Jeans	275	32.9500
20-04	Wool Trousers	75	78.9500
30-01	Warm-Up Jacket	125	59.9500
30-02	Fleece Pullover	150	28.9500
30-03	Parka	75	199.9500

Here are some things to know about the columns in the `Product` table:

- The `Product_Number` column is the primary key. Its data type is `nchar(5)`, and nulls are not allowed.
- The `Description` column's data type is `nvarchar(50)`, and nulls are not allowed.
- The `Units_On_Hand` column's data type is `int`, and nulls are not allowed.
- The `Price` column's data type is `money`, and nulls are not allowed.

In the tutorial, you also learn how to create a Details view. A **Details view** is a set of individual controls that are bound to the columns in a single row. Rather than showing multiple rows at once, a Details view lets the user see one row at a time. Figure 13-38 shows an example of a form with a set of Details view controls. In the figure, the TextBox controls are bound to the columns of the `Product` table. Notice that the form also has a navigation bar. The navigation bar works just like the one that appears with a DataGridView control. It can be used to move forward and backward in the table, add new rows, delete the currently displayed row, and save the changes that have been made.

Figure 13-38 Details view



Notice in Figure 13-38 that each TextBox control has a Label control next to it, identifying the column. When you create a Details view, these Label controls are created automatically. As you can see in the figure, the underscores in the column name are replaced by spaces in the Label controls.

Tutorial 13-3:
Creating the *Products* Application and Using a Details View

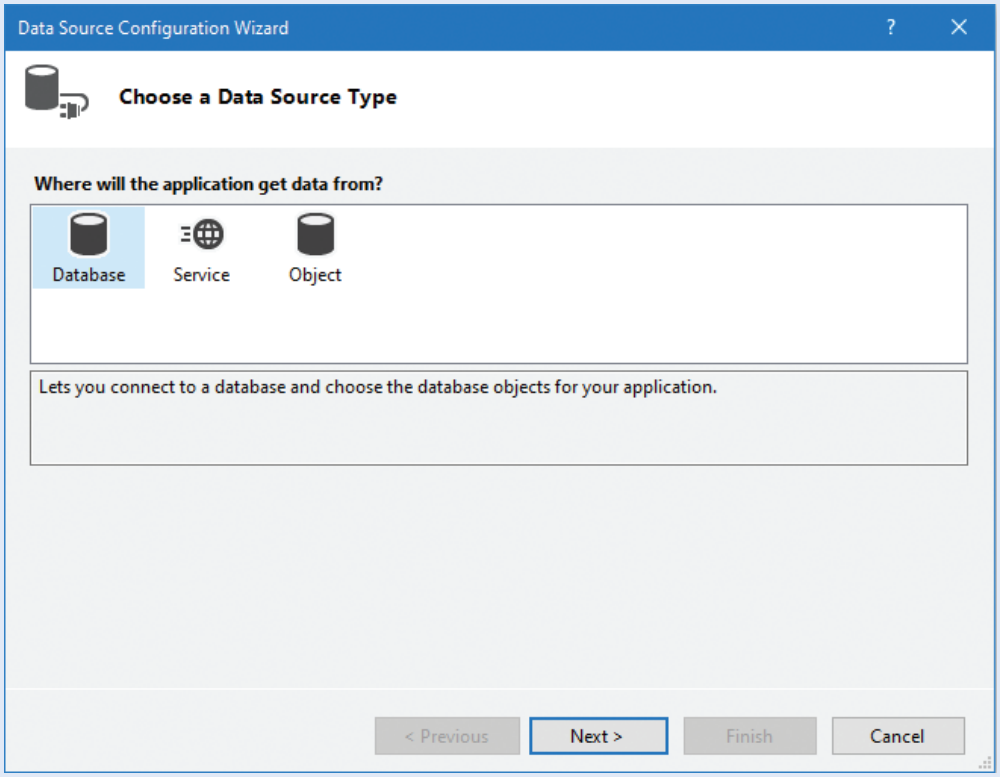
VideoNote
Tutorial 13-3: Creating the *Products* Application and Using a Details View

Step 1: Start Visual Studio. Create a new *Windows Forms App (.NET Framework)* project named *Products*.

Step 2: Change the Form1 form's Text property to *Products*.

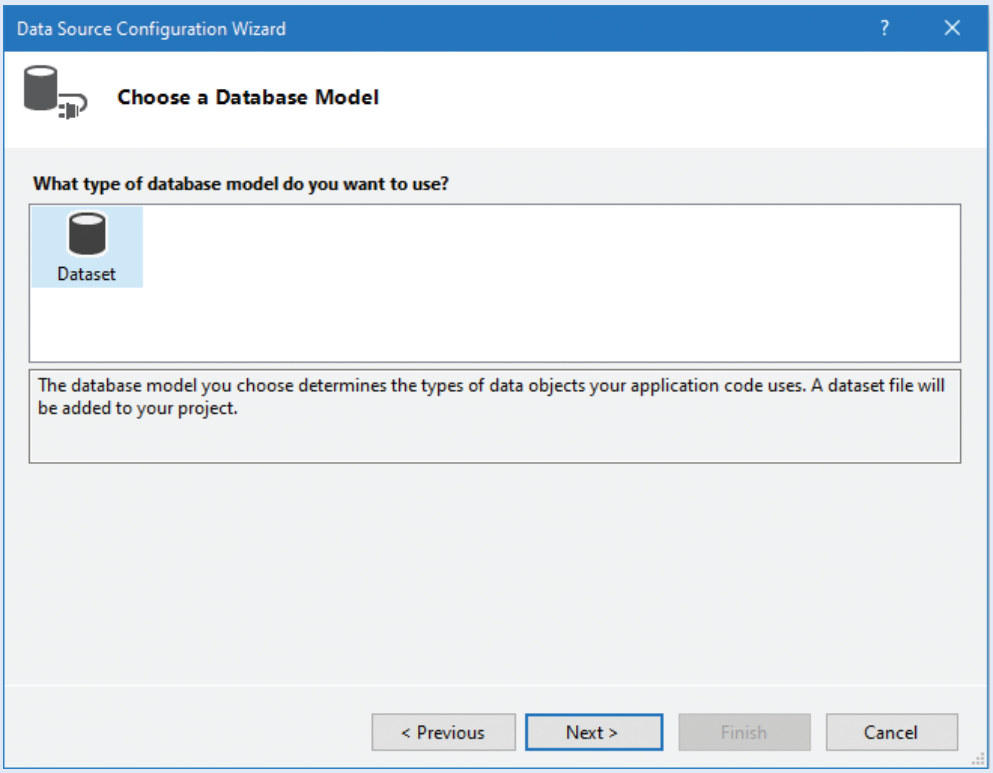
Step 3: Click *Project* on the Visual Studio menu bar and then click *Add New Data Source*. . . . You should see the *Data Source Configuration Wizard*, as shown in Figure 13-39. Make sure *Database* is selected and then click *Next* >.

Figure 13-39 The *Data Source Configuration Wizard—Choose a Data Source Type*



Step 4: The window now appears as shown in Figure 13-40. Make sure *Dataset* is selected, and click *Next >*.

Figure 13-40 The *Data Source Configuration Wizard—Choose a Database Model*



Step 5: The window now appears as shown in Figure 13-41. In this window, you are asked to choose a data connection. Because you have not previously created any data connections in this project, none are available for you to choose from. So, you will have to create a new data connection. Click the *New Connection* button.

Step 6: You see the *Add Connection* dialog box next, as shown in Figure 13-42.

Perform the following:

- Look *carefully* at the selected data source. Make sure *Microsoft SQL Server Database File (SqlClient)* is selected. If not, click the *Change. . .* button and select *Microsoft SQL Server Database File*.
- As indicated in the figure, click the *Browse. . .* button. Navigate to the *Chap13* folder in the Student Sample Programs and select the *ProductDB.mdf* file.

Figure 13-43 shows how the dialog box should appear now. At this point, you can optionally click the *Test Connection* button to test the connection to the database. You should see the message *Test Connection Succeeded*.

Step 7: Click the *OK* button on the *Add Connection* dialog box to close the dialog box.

Step 8: You should be back at the *Data Source Configuration Wizard*, as shown in Figure 13-44. As shown in the figure, ProductDB.mdf should be selected as the data connection. Click the *Next >* button to continue.

Figure 13-41 The *Data Source Configuration Wizard—Choose Your Data Connection*

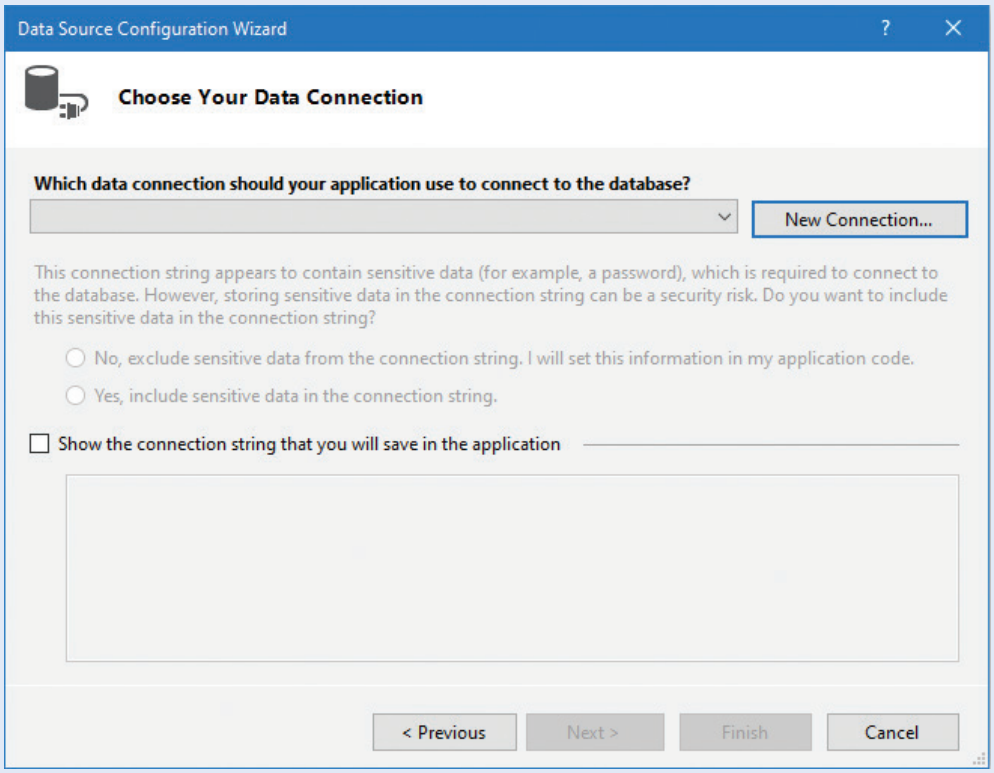
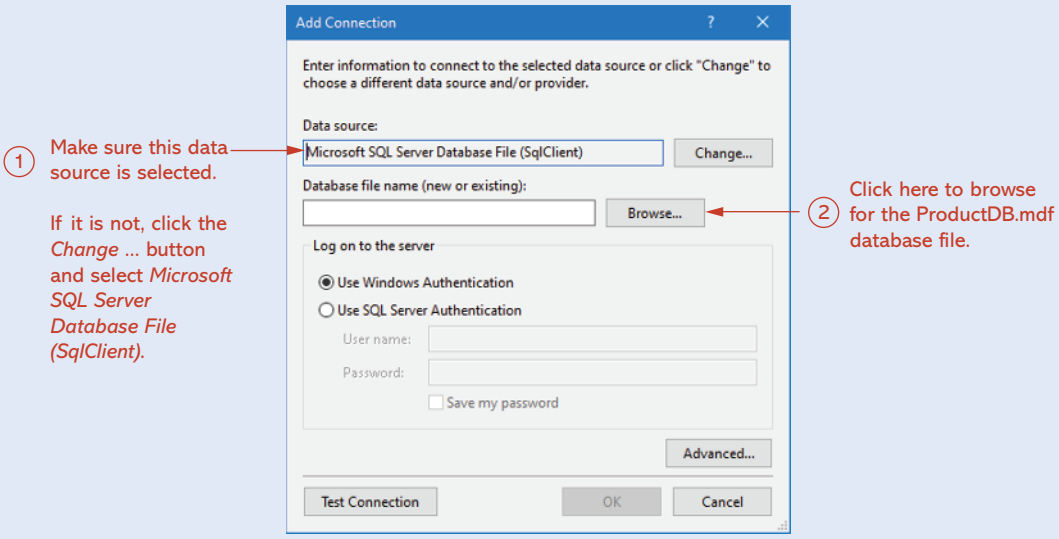


Figure 13-42 The *Add Connection* dialog box



Step 9: You should see the dialog box shown in Figure 13-45. The message is telling you that the database file is located outside the project folder, and you are being asked if you want to copy it into the project. Unless your instructor has told you to do otherwise, click *Yes*. That will make it easier for you to copy the complete project to submit to your instructor.

Figure 13-43 The data source and database file selected

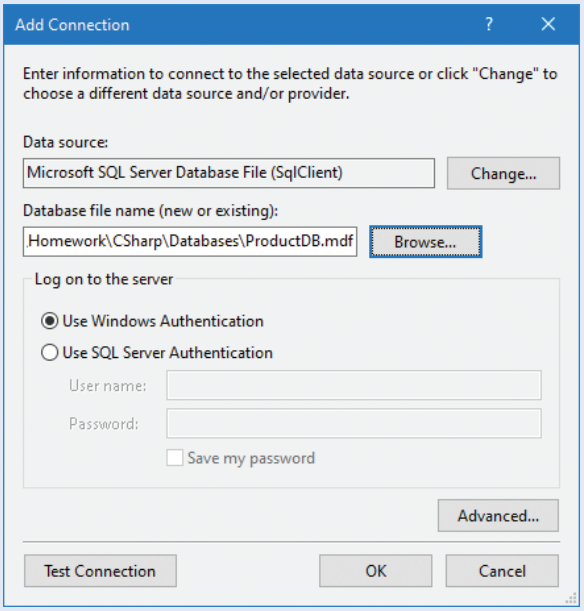
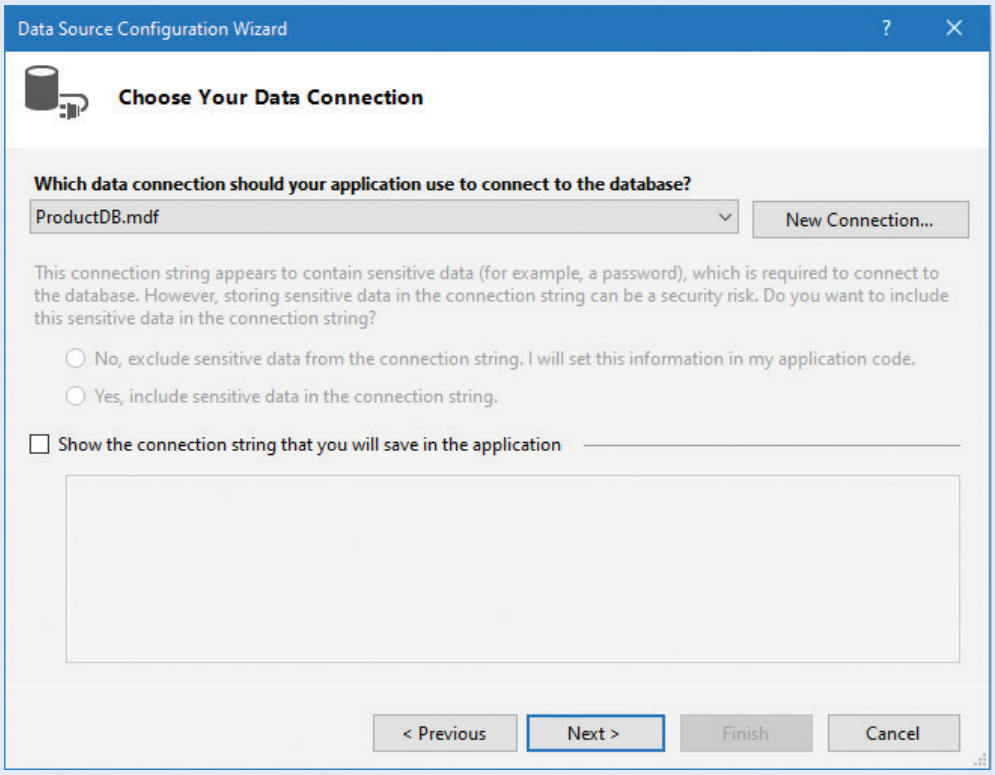


Figure 13-44 The *Data Source Configuration Wizard* with ProductDB.mdf selected



Step 10: The *Data Source Configuration Wizard* should now appear as shown in Figure 13-46. Click the *Next* > button to continue.

Step 11: The *Data Source Configuration Wizard* should now appear, as shown in Figure 13-47. You are asked to choose the database objects that you want to

Figure 13-45 Message about copying the database to the project

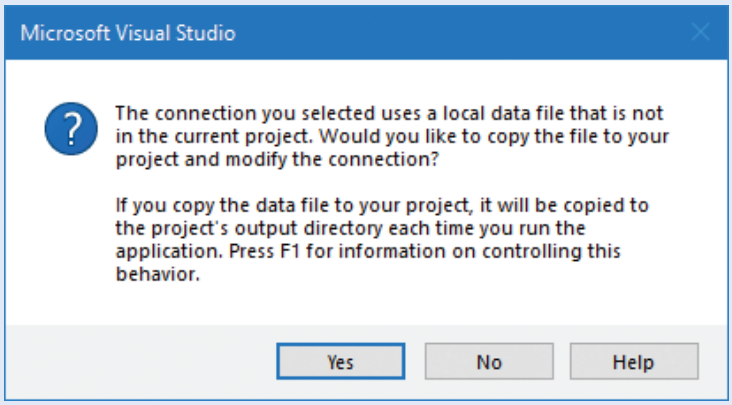
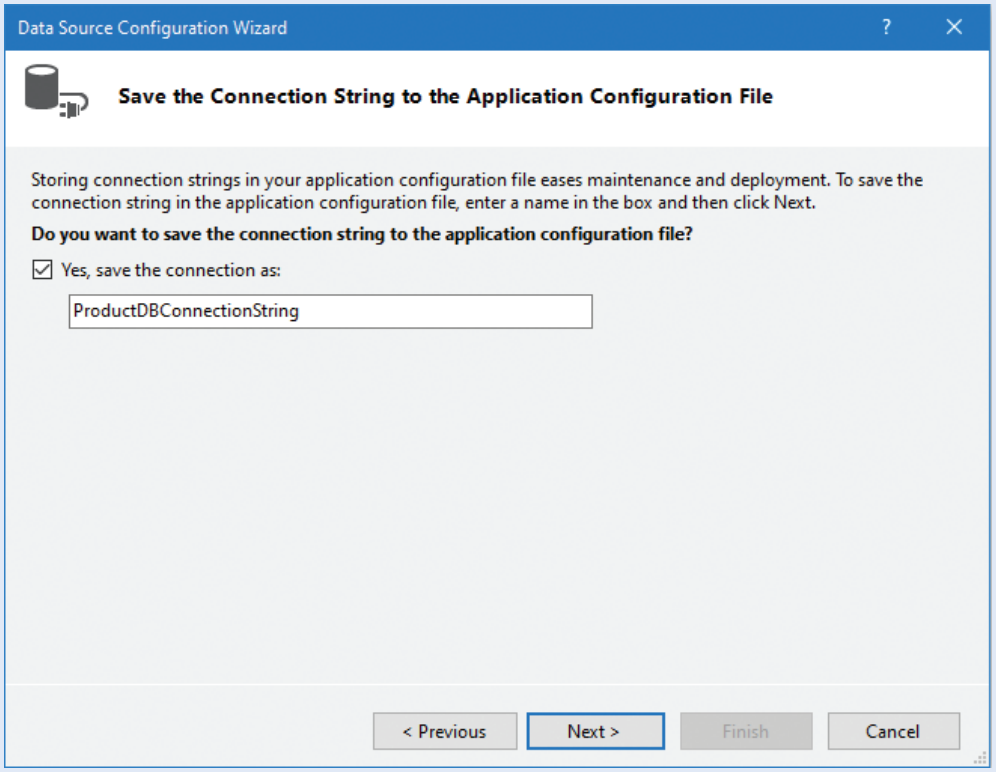


Figure 13-46 The *Data Source Configuration Wizard—Save the Connection String*



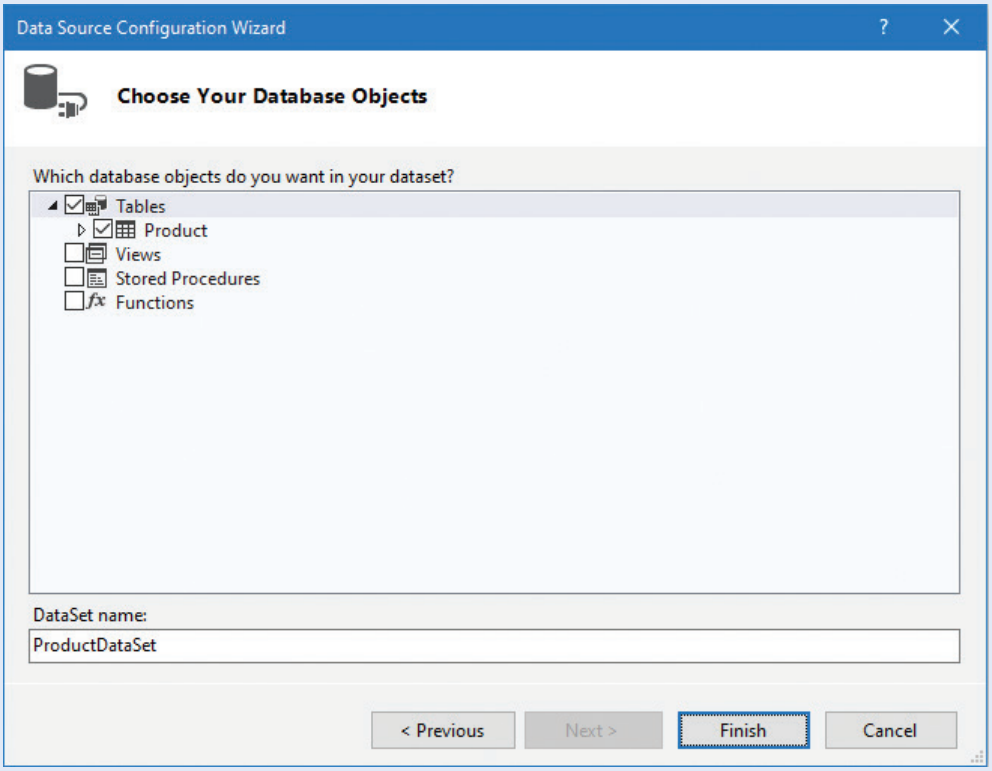
include in the dataset. Expand the *Tables* entry and place a check next to `Product`, as shown in Figure 13-47.

At the bottom of the window, you can specify a name for the dataset. The default name `ProductDBDataSet` is already provided. Because you selected the `Product` table as the database object, change the dataset name to `ProductDataSet`. Click *Finish*.

Here is a summary of what you just did:

- You added the ProductDB.mdf database as a data source to the application.
- You created a dataset component that is connected to the `Product` table. The name of the dataset component is `ProductDataSet`.

Figure 13-47 The *Data Source Configuration Wizard—Choose Your Database Objects*



Step 12: Next you will display the *Data Sources* window. Click *View* on the Visual Studio menu bar, then select *Other Windows*, and then click *Data Sources*. The *Data Sources* window should be displayed, as shown in Figure 13-48. Notice that the window shows the name of the dataset, which is `ProductDataSet`. Expand the entry for `ProductDataSet` and you will see the name of the table that the dataset is connected to, which is `Product`.

Figure 13-48 The *Data Sources* window

Step 13: Now you will select *Details* as the default data-bound control for the `Product` table. In the *Data Sources* window, click the entry for the `Product` table, as shown in the image on the left in Figure 13-49. Then, click the down arrow ( ), and select *Details*, as shown in the image on the right.

Step 14: Next you add a Details view onto the form. As shown in Figure 13-50, click and drag the `Product` table from the *Data Sources* window onto the form. As shown in Figure 13-51, this creates a set of Details view controls, complete with a navigation bar, on the form. (Adjust the size of the form and the size and position of the controls so they appear similar to Figure 13-51.)

Figure 13-49 Selecting *Details* as the data-bound control for the `Product` table

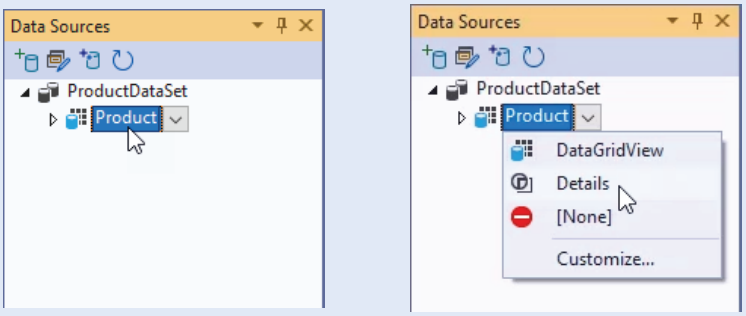


Figure 13-50 Dragging the `Product` table onto the form

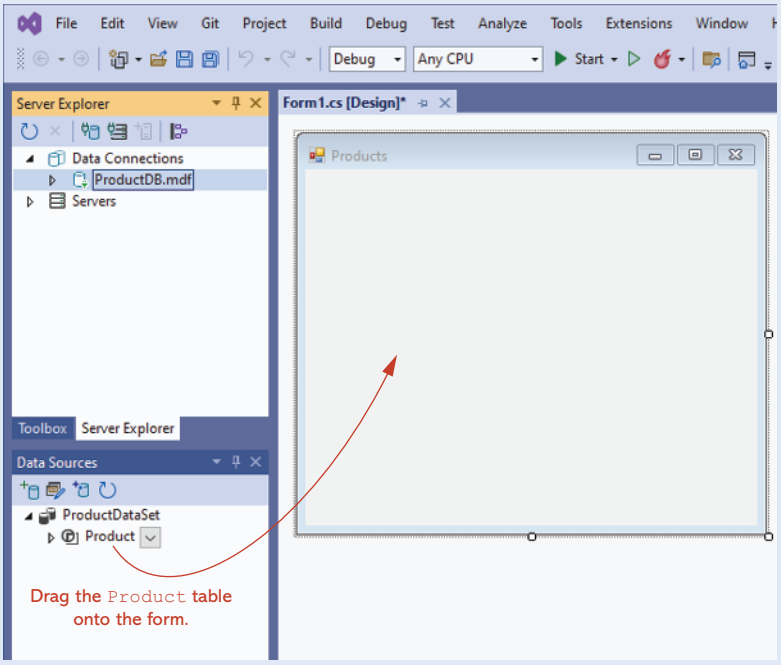
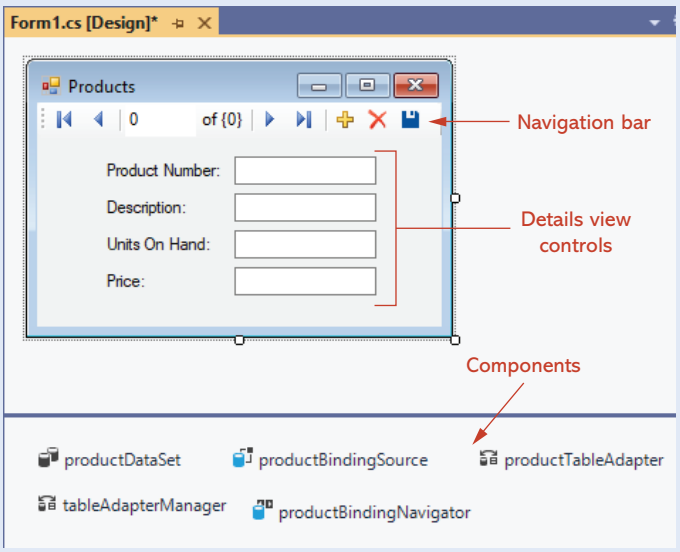


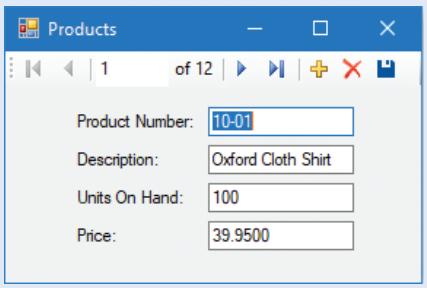
Figure 13-51 Placing the Details view controls on the form



Step 15: Save and run the application. The form will appear similar to Figure 13-52, with data from the first row of the `Product` table displayed. You can use the form in the following ways:

- On the navigation bar, the *Move next* button ( ) moves your view to the next row, and the *Move last* button ( ) moves your view to the last row.
- The *Move previous* button ( ) moves your view to the previous row, and the *Move first* button ( ) moves your view to the first row.
- The values of a row are displayed in text boxes. To change a value, simply click the text box with the mouse and then change its data as necessary.
- To add a new row, click the *Add* button ( ). An empty row will be created and displayed. Enter data for the new row into the text boxes.
- To delete the row that is currently displayed, click the *Delete* button ( ) on the navigation bar.
- The changes that you make affect only the in-memory copy of the table that is held in the dataset. To save the changes to the database, click the *Save* button ( ) on the navigation bar.

Figure 13-52 The *Phone Book* application running



Step 16: Use the Details view controls to make several changes to the dataset. For example, change a value in an existing row, add a new row, and delete an existing row. Be sure to click the *Save* button ( ) on the navigation bar to save your changes. Close the application when you are finished.

Copies of the Database at Run Time

Recall that in Step 9 of Tutorial 13-3, Visual Studio displayed a message indicating that you had selected a database file located outside the project folder, and you were asked if you wanted to copy the file into the project. (See Figure 13-45.) You answered *Yes*, and, as a result, the database file was copied into the project folder.

Then, when you ran the application for the first time, Visual Studio copied the database file from the project folder to the project's output folder, which is the *bin\Debug* folder. Each subsequent time the application runs, it connects to the copy of the database in the output folder instead of the project folder.

Suppose you make a change to the application, such as repositioning a control, and you run it again. Because you modified the application, Visual Studio rebuilds the project (recompiles it). Visual Studio also copies the database file, once again, from the project folder to the output folder. As a side effect, any changes that you previously made to the database will be lost!

You can get around this behavior by answering *No* to the dialog box shown in Figure 13-45. That causes the application to always connect to the database in its external location. However, if you are being graded on the project, your instructor may have trouble running it because the database file will be missing. So, it is normally a good idea to answer *Yes* to the dialog box in Figure 13-45, as long as you are aware that rebuilding the project causes the database in the output folder to be overwritten.

Checkpoint

13.16 How can you create a form that shows the columns of only a single row of a database in an application?

13.17 What happens when you drag a table from the *Data Sources* window onto a form?

13.18 Where is a project's output folder located?

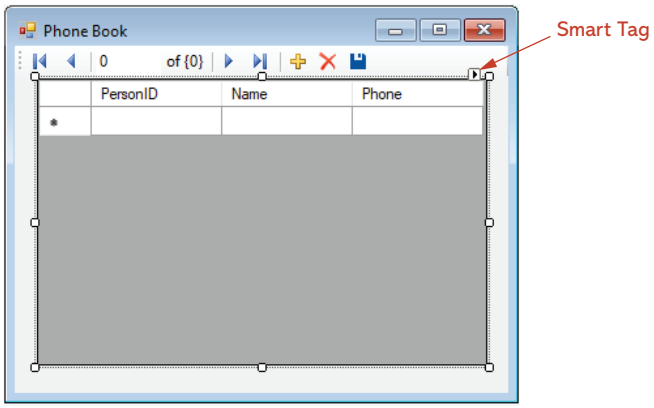
13.6 More about Data-Bound Controls

CONCEPT: **The DataGridView control and the Details view may be customized in various ways. Other controls, such a list boxes, can be data-bound to allow interaction with a database.**

Customizing the DataGridView Control

In the *Designer*, if you select a DataGridView control, you will see a small arrow in the upper-right corner of the control's bounding box. Figure 13-53 shows an example. This is called a smart tag. When you click on the smart tag, a tasks panel will pop up, giving you a number of options that you can perform with the DataGridView control. Figure 13-54 shows an example of the tasks panel.

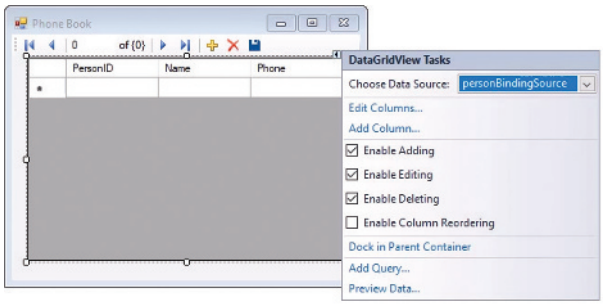
Figure 13-53 A DataGridView control's smart tag



Notice in the figure that the tasks panel has the following check boxes:

- *Enable Adding*—When this item is checked, the user may add rows in the DataGridView control. This item is checked by default.
- *Enable Editing*—When this item is checked, the user may change the contents of rows in the DataGridView control. This item is checked by default.

Figure 13-54 A DataGridView control's tasks panel

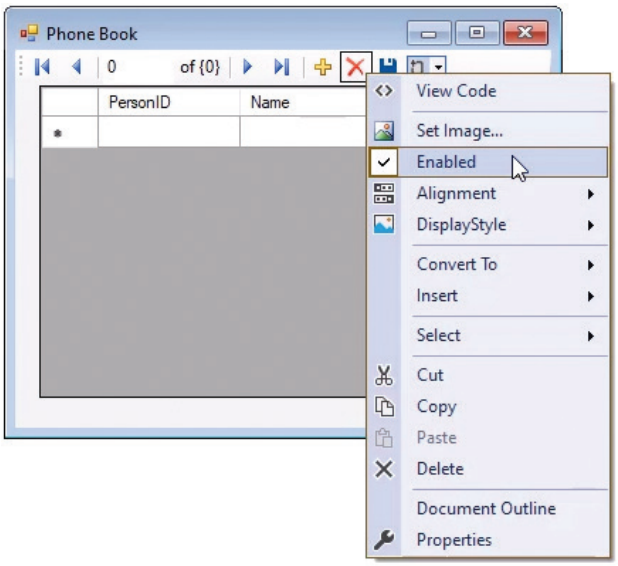


- *Enable Deleting*—When this item is checked, the user may delete rows from the DataGridView control. This item is checked by default.
- *Enable Column Reordering*—When this item is checked, the user may click and drag columns, while the application is running, to rearrange them in the DataGridView control. This item is not checked by default.

If you do not want the user to be able to add, edit, or delete rows in the DataGridView control, simply remove the appropriate checkmarks.

Keep in mind that the navigation bar has buttons that allow the user to add, delete, and save the data in the DataGridView control. These buttons will still work, even after you have disabled adding, editing, and deleting in the DataGridView control's tasks panel. To disable a button on the navigation bar, right-click it in the *Designer*. This will display the pop-up menu shown in Figure 13-55. Notice in the figure that *Enabled* is checked. Click *Enabled* to remove the checkmark, thus disabling the button. When a button is disabled, it will appear grayed-out on the navigation bar. Figure 13-56 shows an example with the *Add*, *Delete*, and *Save* buttons disabled.

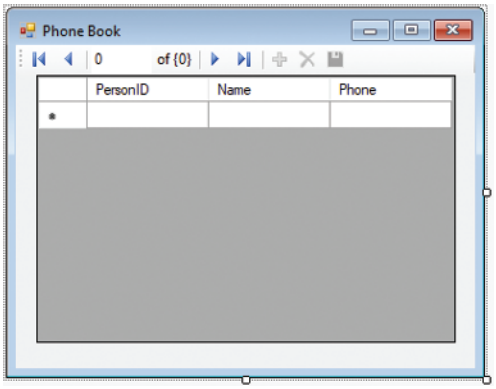
Figure 13-55 Disabling the *Delete* button



Customizing the Details View

In Tutorial 13-3, you saw how several controls are created automatically when you create Details view for a table. For example, TextBoxes were created and bound to the

Figure 13-56 The *Add*, *Delete*, and *Save* buttons disabled (grayed-out)



individual columns. Label controls that identify the columns were also created, as was a navigation bar.

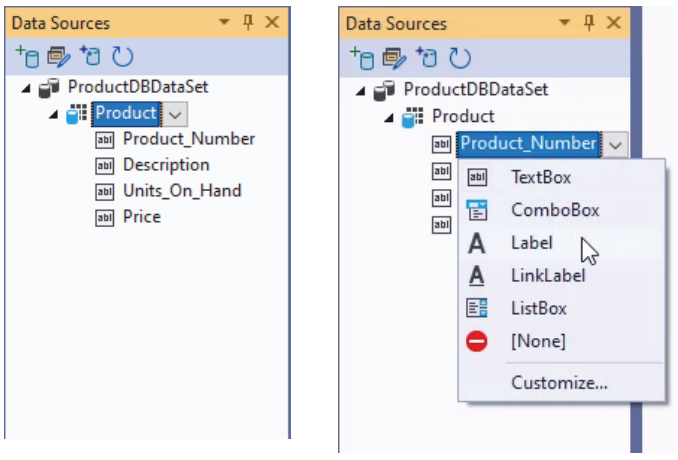
When you create a Details view, the type of control that a column is automatically bound to is determined by the column's data type, as follows:

- Columns containing character data are bound to TextBox controls by default.
- Numeric columns are bound to TextBox controls by default.
- Bit columns are bound to CheckBox controls.
- Datetime columns are bound to DateTimePicker controls, which allow the user to select a date from a small pop-up calendar.

It is possible to change the type of control that a column is bound to in a Details view. For example, recall that the `Product` table in Tutorial 13-3 has a column named `Product_Number`. The `Product_Number` column contains character data, so a TextBox control was created for it when you placed the Details view on the form. However, the `Product_Number` column is the primary key for the table, so you might not want the user to be able to change it. A Label control might be a better option. That way, the user can see the product number but cannot change it.

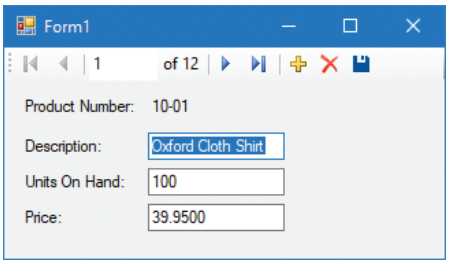
Before you place a Details view on a form, you can select the type of control to which each column will be bound. In the *Data Sources* window, expand the table entry as shown in the image on the left in Figure 13-57. Then, select a column and click the down arrow

Figure 13-57 Selecting the type of data-bound control for a column



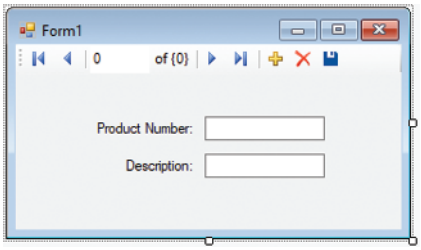
( ) that appears next to it, as shown in the image on the right in the figure. A menu will pop up that shows the types of controls that you can select. In the image on the right in Figure 13-57, we are selecting the Label control for the `Product_Number` column. Then, when we create the Details view, it appears as shown in Figure 13-58.

Figure 13-58 Details view with the product number displayed in a Label control



When you drag an entire table from the *Data Sources* window and drop it onto a form, the Details view that is created contains data-bound controls for all the columns. Sometimes you might not want all the columns displayed, however. Alternatively, you can drag the individual columns, one at a time, from the *Data Sources* window onto the form. Figure 13-59 shows an example of a Details view that displays only the product number and description. We created this Details view by dragging the `Product_Number` column and then the `Description` column from the *Data Sources* window onto the form.

Figure 13-59 Product number and description displayed



Binding Columns to ListBox Controls

A list box can be a convenient way to look up data in a table. You can bind a column to a ListBox control, which causes all the values in that column to be displayed in the list box. When the user selects an item from the list box, the rest of the columns from the selected row can be displayed.

To bind a ListBox control to a column, you must set two of the control's properties: DataSource and DisplayMember. The DataSource property identifies the table from which the ListBox will get its data. The DisplayMember property identifies the column. In Tutorial 13-4, you use this technique to create a data lookup form for the ProductDB.mdf database.

Tutorial 13-4:
Creating the *ProductLookup* Project

VideoNote
Tutorial 13-4: Creating the *ProductLookup* Project

Step 1: Start Visual Studio. Create a new *Windows Forms App (.NET Framework)* project named *ProductLookup*.

Step 2: Change the Form1 form's Text property to *Product Lookup*.

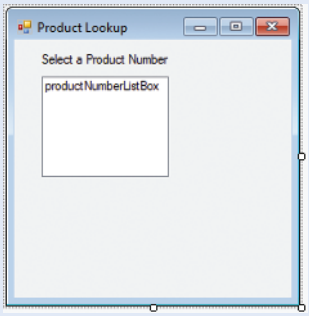
Step 3: Perform the following steps to connect the application to the ProductDB.mdf database, and select the `Product` table as the dataset:

- From the *Project* menu, select *Add New Data Source*. . . .
- Select *Database* and click *Next* >.
- Select *Dataset* and click *Next* >.
- Click *New Connection*. . . .
- In the *Add Connection* window, make sure *Microsoft SQL Server Database File (SqlClient)* is the selected data source.
- In the *Add Connection* window, click *Browse*, and go to the *Chap13* folder in the Student Sample Programs. Select the *ProductDB.mdf* file. Click *OK*.
- Click *Next* >, and when asked if you want to copy the file to the project, click *Yes*.
- Click *Next* >.
- Select the `Product` table as the database object, and change the name of the dataset to ProductDataSet. Click *Finish*.

NOTE: This is the same procedure that you performed in Steps 3–11 in Tutorial 13-3. If you want more detailed guidance, go back to that tutorial to see the actual screens.

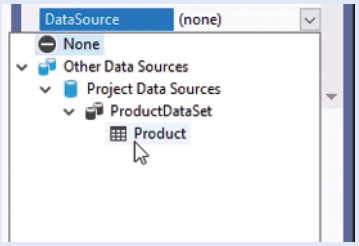
Step 4: Create a ListBox control named `productNumberListBox`. Just above the ListBox, place a Label control that displays "Select a Product Number". The form should look similar to Figure 13-60.

Figure 13-60 ListBox and Label created



Step 5: Select the ListBox control. In the *Properties* window, select the DataSource property and then click the down arrow ( ) that appears next to it. In the list that pops up, expand *Other Data Sources*, then expand *Project Data Sources*, then expand *ProductDataSet*, and then select *Product*. This is shown in Figure 13-61.

Figure 13-61 `Product` table selected in the DataSource property



Step 6: Select the ListBox control's DisplayMember property and then click the down arrow ( ) that appears next to it. In the list that pops up, select `Product_Number`.

Step 7: In the *Data Sources* window, set the `Product` table's view to *Details*, as shown in Figure 13-62.

Step 8: In the *Data Sources* window, expand the `Product` table so the column names are displayed, as shown in Figure 13-63.

Figure 13-62 Select *Details*

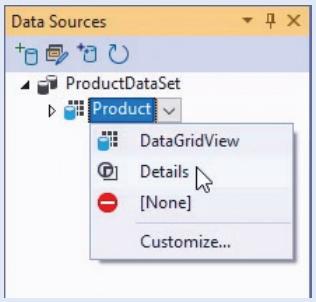
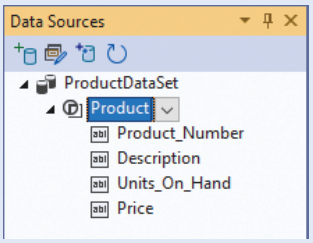
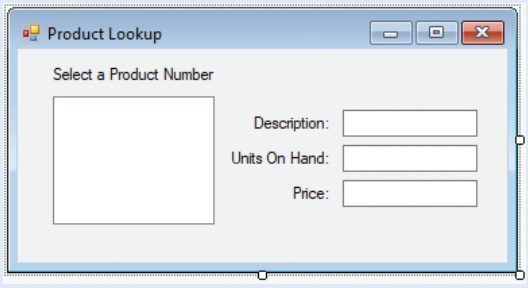


Figure 13-63 Column names displayed



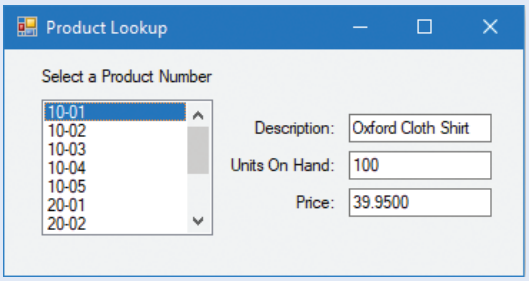
Step 9: One at a time, drag the `Description`, `Units_On_Hand`, and `Price` columns from the *Data Sources* window onto the form. Place them as shown in Figure 13-64. Resize the form so it appears similar to that shown in the figure.

Figure 13-64 Column names displayed



Step 10: Save the project and run the application. The application's form should appear similar to Figure 13-65. Notice that the list box contains a list of all the product numbers in the `Product` table. When you select a product number in the list box, that product's description, units on hand, and price are displayed in the TextBox controls. When you are finished, exit the application.

Figure 13-65 The *Product Lookup* application running



Having Data-Bound Controls on Multiple Forms

You have seen that when you place a data-bound control on a form, Visual Studio automatically creates a set of components that allow the form to connect to the database. These components include a dataset. When you place data-bound controls on multiple forms, keep in mind that each form has its own dataset. When you make changes to the dataset in one form, those changes do not automatically appear in other forms that have their own dataset. Consider the following scenario:

- Form1 displays a view of a database, perhaps in a DataGridView control.
- Some code in Form1 displays Form2.
- Form2 also displays a view of the same database, allowing the user to make changes and save those changes to the database. The user closes Form2.
- Form1 is still displaying its original copy of the data. Its dataset does not automatically update, so Form1 does not show the changes that were made by Form2.

To make sure that a dataset contains a current copy of the data, you need to write code that causes the table adapter to read the database table and fill the dataset. You do this by calling the table adapter's `Fill` method. In Tutorial 13-5, you learn how to do this by creating an application that correctly handles the scenario previously discussed.

Tutorial 13-5:
Creating the *MultiformProducts* Project

VideoNote
Tutorial 13-5: Creating the *MultiformProducts* Project

Step 1: Start Visual Studio. Create a new *Windows Forms App (.NET Framework)* project named *MultiformProducts*.

Step 2: Change the Form1 form's Text property to *Products*.

Step 3: In the *Solution Explorer*, change the name of Form1.cs to *MainForm.cs*. This changes the name of the Form1 form to MainForm.

Step 4: Perform the following steps to connect the application to the ProductDB.mdf database and select the `Product` table for the dataset:

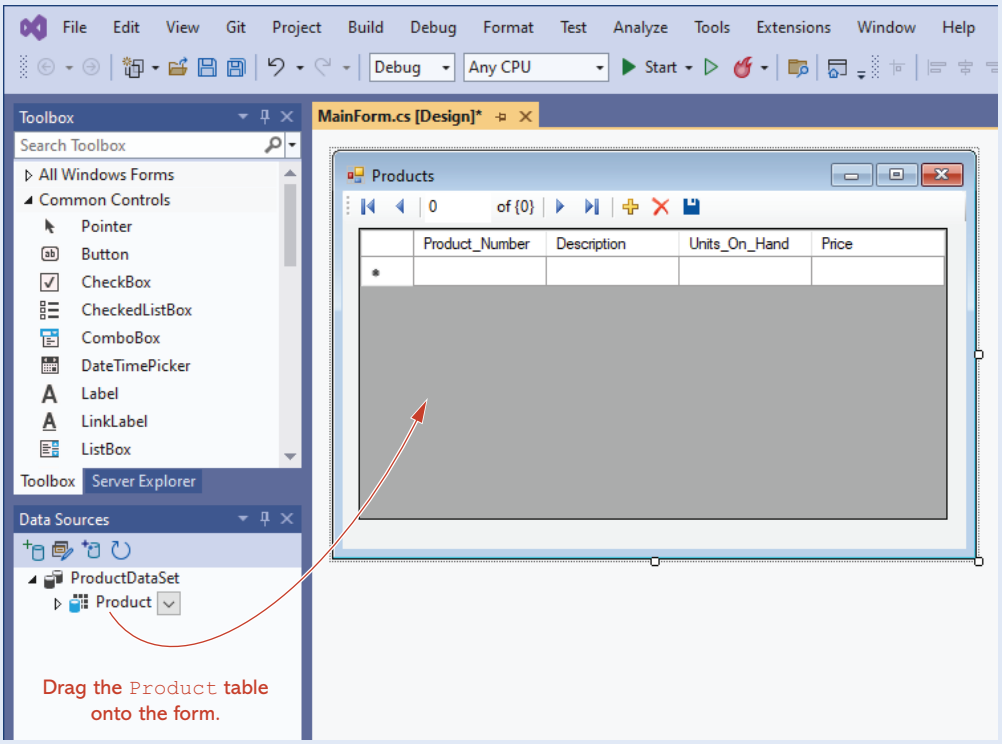
- From the *Project* menu, select *Add New Data Source*. . . .
- Select *Database* and click *Next* >.
- Select *Dataset* and click *Next* >.
- Click *New Connection*. . . .
- In the *Add Connection* window, make sure *Microsoft SQL Server Database File (SqlClient)* is the selected data source.

- In the *Add Connection* window, click *Browse* and go to the *Chap13* folder in the Student Sample Programs. Select the *ProductDB.mdf* file. Click *OK*.
- Click *Next* >, and when asked if you want to copy the file to the project, click *Yes*.
- Click *Next* >.
- Select the `Product` table as the database object, and change the name of the dataset to `ProductDataSet`. Click *Finish*.

NOTE: This is the same procedure that you performed in Steps 3–11 in Tutorial 13-3. If you want more detailed guidance, go back to that tutorial to see the actual screens.

Step 5: Open the *Data Sources* window and, as shown in Figure 13-66, drag a DataGridView control onto the form, bound to the `Product` table. After creating the DataGridView control, adjust size of the form and the control as shown in Figure 13-67.

Figure 13-66 Create a DataGridView control on the form



Step 6: Create the Button controls shown in Figure 13-68. (You write Click event handlers for these buttons later.)

Step 7: Add another form named DetailsForm to the project. As a reminder, these are the steps:

- Click *Project* on the Visual Studio menu bar and then select *Add Form (Windows Form)*. . . .
- In the *Add New Item* window, enter *DetailsForm.cs* as the name and click *Add*.

Figure 13-67 The form and DataGridView control resized

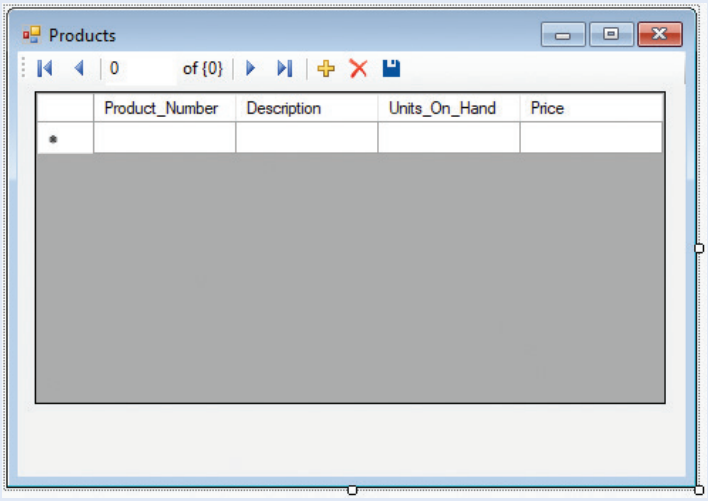
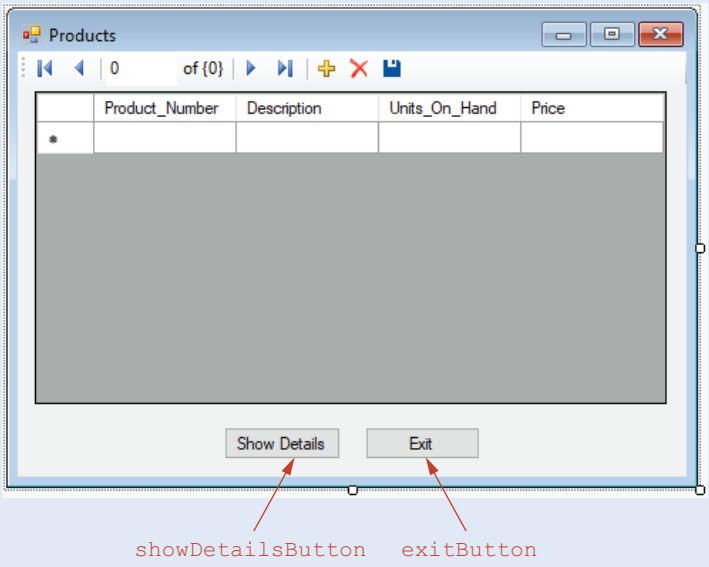


Figure 13-68 Button controls placed on the MainForm



Step 8: Change the form's Text property to *Details*.

Step 9: In the *Data Sources* window, change the `Product` table's default view to *Details*, as shown in Figure 13-69. Then, as shown in Figure 13-70, drag a Details view onto the form.

Figure 13-69 Changing the `Product` table's default view to *Details*

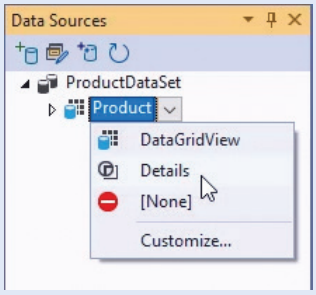
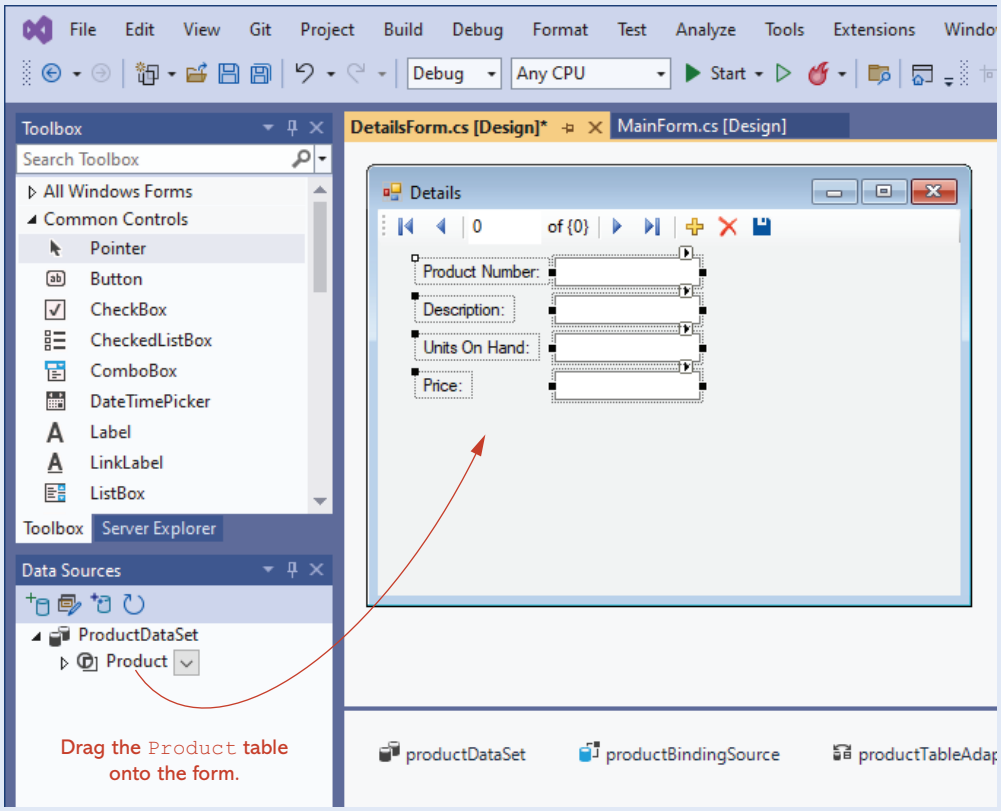
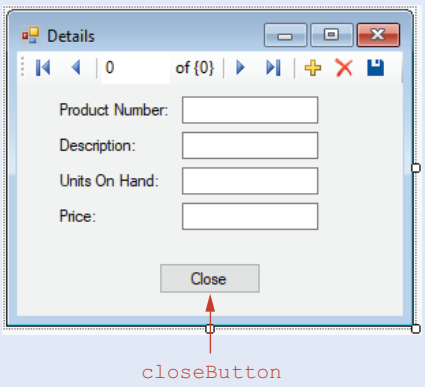


Figure 13-70 Creating a Details view on the form



Step 10: Resize the DetailsForm form and create the Button control shown in Figure 13-71.

Figure 13-71 Form resized and *Close* button placed



Step 11: Double-click the `closeButton` control, and complete its Click event handler, as shown in lines 37–38 of Program 13-2.

Step 12: Switch your view back to the MainForm form in the *Designer*. Double-click the `showDetailsButton` control, and complete its Click event handler, as shown in lines 37–44 of Program 13-3. Let's take a closer look at the code:

Line 38: This statement creates an instance of the `DetailsForm` class, referenced by the `details` variable.

Line 41: This statement displays the DetailsForm form. The user will be able to make changes to the product data using the DetailsForm and save those changes to the database.

Line 44: This statement calls the table adapter's `Fill` method, passing the dataset as an argument. As a result, the dataset will be filled with the current contents of the `Product` table, and the DataGridView control's contents will be updated.

Step 13: Switch your view back to the MainForm form in the *Designer*. Double-click the `exitButton` control, and complete its Click event handler, as shown in lines 49 and 50 of Program 13-3.

Step 14: Save the project and run the application. The MainForm should show the `Product` table in the DataGridView control. Click the *Show Details* button to display the DetailsForm.

In the DetailsForm, make some changes to the data. For example, change the value of an existing piece of data, insert a new row, or delete an existing row. After making the changes, be sure to click the *Save* button on the navigation bar. Then, click the *Close* button to close the DetailsForm.

Back at the MainForm, browse the data in the DataGridView control to confirm that it shows the changes you just made. When you are finished, exit the application.

Program 13-2 Completed code for the DetailsForm

```
 1 using System;
 2 using System.Collections.Generic;
 3 using System.ComponentModel;
 4 using System.Data;
 5 using System.Drawing;
 6 using System.Linq;
 7 using System.Text;
 8 using System.Threading.Tasks;
 9 using System.Windows.Forms;
10
11 namespace MultiformProducts
12 {
13   public partial class DetailsForm : Form
14   {
15      public DetailsForm()
16      {
17          InitializeComponent();
18      }
19
20      private void productBindingNavigatorSaveItem_Click(object
        sender, EventArgs e)
21      {
22          this.Validate();
23          this.productBindingSource.EndEdit();
24          this.tableAdapterManager.UpdateAll(this.productDataSet);
25
26      }
27
28      private void DetailsForm_Load(object sender, EventArgs e)
```

```
29      {
30          // TODO: This line of code loads data into the...
31          this.productTableAdapter.Fill(this.productDataSet.
            Product);
32
33      }
34
35      private void closeButton_Click(object sender, EventArgs e)
36      {
37          // Close the form.
38          this.Close();
39      }
40   }
41 }
```

Program 13-3 Completed code for the MainForm form

```
 1 using System;
 2 using System.Collections.Generic;
 3 using System.ComponentModel;
 4 using System.Data;
 5 using System.Drawing;
 6 using System.Linq;
 7 using System.Text;
 8 using System.Threading.Tasks;
 9 using System.Windows.Forms;
10
11 namespace MultiformProducts
12 {
13   public partial class MainForm : Form
14   {
15      public MainForm()
16      {
17          InitializeComponent();
18      }
19
20      private void productBindingNavigatorSaveItem_Click(object
        sender, EventArgs e)
21      {
22          this.Validate();
23          this.productBindingSource.EndEdit();
24          this.tableAdapterManager.UpdateAll(this.productDataSet);
25
26      }
27
28      private void MainForm_Load(object sender, EventArgs e)
29      {
30          // TODO: This line of code loads data into the...
31          this.productTableAdapter.Fill(this.productDataSet.
            Product);
32
33      }
34
35      private void showDetailsButton_Click(object sender,
        EventArgs e)
36      {
37          // Create an instance of the DetailsForm.
38          DetailsForm details = new DetailsForm();
39
40          // Display the form.
```

```
41              details.ShowDialog();
42
43              // Update the dataset.
44              this.productTableAdapter.Fill(this.productDataSet.
                Product);
45          }
46
47          private void exitButton_Click(object sender, EventArgs e)
48          {
49              // Close the form.
50              this.Close();
51          }
52      }
53  }
```

Checkpoint

13.19 What happens when a control's smart tag is clicked in the *Designer*?

13.20 When you create a Details view, what determines the type of control to which a column is bound?

13.21 Which two properties must be set to bind a ListBox control to a column?

13.22 What is the name of the table adapter method that populates the dataset with a current copy of the data?

13.7 Selecting Data with the SQL `Select` Statement

CONCEPT: SQL is a standard language that most DBMS support. The `Select` statement is used in SQL to retrieve data from a database.

SQL, which stands for **structured query language**, is a standard language for working with database management systems. It was originally developed by IBM in the 1970s. Since then, SQL has been adopted by almost all database software vendors as the language of choice for interacting with their DBMS.

SQL consists of several keywords. You use the keywords to construct statements, which are also known as **queries**. These statements, or queries, are submitted to the DBMS and are instructions directing the DBMS to carry out operations on its data. In this section, you learn how to construct simple SQL statements and then pass them to the SQL Server DBMS.

NOTE: Although SQL is a language, you do not use it to write applications. It is intended only as a standard means of interacting with a DBMS. You still need a general programming language, such as C#, to write an application for the ordinary user.

In SQL you use the **`Select` statement** to retrieve the rows in a table. As its name implies, the `Select` statement allows you to select specific rows. We start with a very simple form of the statement, as shown here:

```
Select Columns From Table
```

In the general form, *Columns* is one or more column names, and *Table* is a table name. Here is an example Select statement that we might execute on the ProductDB.mdf database:

```
Select Description From Product
```

This statement retrieves the Description column for every row in the Product table. You can specify more than one column in a Select statement by separating the column names with commas. Here is an example:

```
Select Description, Price From Product
```

This statement retrieves the Description column and the Price column for every row in the Product table. If you wish to retrieve every column in a table, you can use the * character instead of listing column names. Here is an example:

```
Select * From Product
```

This statement retrieves every column for every row in the Product table. SQL statements are free form, which means that tabs, newlines, and spaces between the keywords are ignored. For example, the statement

```
Select * From Product
```

works the same as

```
Select
    *
From
    Product
```

In addition, SQL keywords and table names are case insensitive. The previous statement could be written as

```
SELECT * FROM PRODUCT
```

Specifying a Search Criteria with the Where Clause

Sometimes you want to retrieve every row in a table. In many situations, however, you want to narrow the list down to only a few selected rows in the table. That is where the Where clause comes in. The Where clause can be used with the Select statement to specify the search criteria. When you use the Where clause, only the rows that meet the search criteria are returned. The general format of a Select statement with a Where clause is

```
Select Columns From Table Where Criteria
```

In the general format, Criteria is a conditional expression. Here is an example of a Select statement that uses the Where clause:

```
Select * From Product Where Price > 20.00
```

The first part of the statement, Select * From Product, specifies that we want to see every column. The Where clause specifies that we want only the rows in which the contents of the Price column are greater than 20.00. As a result, the Select statement retrieves only the rows in which Price is greater than 20.00.

SQL supports the relational operators listed in Table 13-2 for writing conditional expressions in a Where clause.

Notice that the equal to and not equal to operators in SQL are different from those in C#. The equal to operator is one equal sign, not two equal signs. The not equal to operator is <>.

Table 13-2 SQL relational operators

Operator	Meaning
>	Greater than
<	Less than
>=	Greater than or equal to
<=	Less than or equal to
=	Equal to
<>	Not equal to

Let's look at a few more examples of the Select statement. The following statement could be used to retrieve the product numbers and prices of all the items that are priced at $28.95:

```
Select Product_Number, Price From Product Where Price = 28.95
```

The following Select statement retrieves all the columns from only the rows where the description is "Denim Jeans":

```
Select * From Product Where Description = 'Denim Jeans'
```

If you look carefully at the previous statement, you will notice another difference between SQL syntax and C# syntax. In SQL, string literals are enclosed in single quotes, not double quotes.

TIP: If you need to include a single quote as part of a string, simply write two single quotes in its place. For example, suppose you wanted to search the Product table for Katy's Wool Cap. You could use the following statement:

```
Select * From Product Where Description = 'Katy''s Wool Cap'
```

String comparisons in SQL are case sensitive. If you ran the following statement against the Product table, you would not get any results:

```
Select * From Product Where Description = 'denim jeans'
```

However, you can use the Lower() function to convert a string to lowercase. Here is an example:

```
Select * From Product Where Lower(Description) = 'denim jeans'
```

This statement converts the Description column to all lowercase before performing the comparison. As a result, it will return all rows where Description equals "denim jeans" regardless of case. You could use the Upper function, which converts a string to uppercase, to achieve the same results:

```
Select * From Product Where Upper(Description) = 'DENIM JEANS'
```

Using the Like Operator

Sometimes searching for an exact string will not yield the results you want. For example, suppose the Product table contains the rows shown in Figure 13-72 and we want a list of all the shirts.

Figure 13-72 The `Product` table in the ProductDB.mdf database

Product_Number	Description	Units_On_Hand	Price
10-01	Oxford Cloth Shirt	100	39.9500
10-02	Poplin Shirt	150	38.5000
10-03	Plaid Flannel Shirt	80	34.7500
10-04	Canvas Shirt	200	32.9500
10-05	Cotton Shirt	250	28.9500
20-01	Chino Classic Fit Pants	100	28.9500
20-02	Chino Relaxed Fit Pants	100	28.9500
20-03	Denim Jeans	275	32.9500
20-04	Wool Trousers	75	78.9500
30-01	Warm-Up Jacket	125	59.9500
30-02	Fleece Pullover	150	28.9500
30-03	Parka	75	199.9500

The following statement will not work. Can you see why?

```
Select * From Product Where Description = 'Shirt'
```

This statement will search for rows where the `Description` column is equal to the string `"Shirt"`. Unfortunately, it will find none because there are no rows in which the Description column is equal to `"Shirt"`. However, the word `"Shirt"` does appear in the `Description` column of some of the rows. For example, in Figure 13-72 the first row's `Description` column is `"Oxford Cloth Shirt"`, the second row's `Description` column is `"Poplin Shirt"`, and so forth.

In order to find all the shirts, we need to search for rows where `"Shirt"` appears as a substring in the `Description` column. You can perform just such a search using the `Like` operator. Here is an example of how to use it.

```
Select * From Product Where Description Like '%Shirt%'
```

The `Like` operator is followed by a string that contains a character pattern. In this example, the character pattern is `'%Shirt%'`. The `%` symbol is used as wildcard character. It represents any sequence of zero or more characters. The pattern `'%Shirt%'` specifies that the string `"Shirt"` must appear with any sequence of characters before or after it. So, the statement previously shown would return all rows in which the `Description` column contains the string `"Shirt"`.

Likewise, the following statement will result in all the rows where the `Description` column starts with the word `"Chino"`:

```
Select * From Product Where Description Like 'Chino%'
```

The underscore character (_) is also used as a wildcard. Unlike the `%` character, the underscore represents a single character. For example, look at the following statement:

```
Select * From Product Where Product_Number Like '2_-0_'
```

This statement will result in all the rows where the `Product_Number` column begins with `"2"`, followed by any one character, followed by `"-0"`, followed by any one character.

You can use the `Not` operator to disqualify a character pattern in a search criteria. For example, suppose you want a list of all the items that are not shirts. The following statement will yield just those results:

```
Select * From Product Where Description Not Like '%Shirt%'
```

Using `And` and `Or`

You can use the `And` and `Or` logical operators to specify multiple search criteria in a `Where` clause. For example, look at the following statement:

```
Select * From Product Where Price > 20.00 And Price < 30.00
```

The And operator requires that both search criteria be true in order for a row to be qualified as a match. The only rows that will be returned from this statement are those where the Price column contains a value that is greater than 20.00 and less than 30.00.

Here's an example that uses the Or operator:

```
Select * From Product
Where Description Like '%Shirt%' or Product_Number Like '10-%'
```

The Or operator requires that either of the search criteria be true in order for a row to be qualified as a match. This statement searches for rows where the Description column contains the string "Shirt" at any position or where the Product_Number column starts with "10-".

Sorting the Results of a Select Query

If you wish to sort the results of a Select query, you can use the Order By clause. Here is an example:

```
Select * From Product Order By Price
```

This statement produces a list of all the rows in the Product table, ordered by the Price column. The list will be sorted in ascending order on the Price column, meaning that the lowest-priced coffees will appear first.

Here's a Select query that uses both a Where clause and an Order By clause:

```
Select * From Product
Where Price > 20.00
Order By Price
```

This statement produces a list of all the rows in the Product table where the Price column contains a value greater than 20.00, listed in ascending order by price.

If you want the list sorted in descending order (from highest to lowest), use the Desc operator, as shown here:

```
Select * From Product
Where Price > 20.00
Order By Price Desc
```

Table Adapter Queries

A **table adapter query** is an SQL statement that is stored in a table adapter and can be executed simply by calling a method. For example, all table adapters contain a query that fills a dataset with data from a table. That query can be executed by calling the table adapter's Fill method. When you place a data-bound control, such as a DataGridView control, on a form, a Load event handler that calls the table adapter's Fill method is automatically created for the form.

Let's look at an example. In Visual Studio, open the *MultiformProducts* project that you created in Tutorial 13-5, and open source code for the MainForm form. Look at the form's Load event handler, which is shown here (the comment that appears in line 3 has been shortened to fit on the page):

```
1 private void MainForm_Load(object sender, EventArgs e)
2 {
3     // TODO: This line of code loads data into...
4     this.productTableAdapter.Fill(this.productDataSet.Product);
5
6 }
```

The statement in line 4 calls the `productTableAdapter`'s `Fill` method, passing the dataset's `Product` table as an argument. The `Fill` method fills the dataset table with rows that are returned from an SQL statement. Now let's look at the SQL statement that is executed by the `Fill` method. In the *Solution Explorer*, you should see an entry named *ProductDataSet.xsd*, as shown in Figure 13-73. This is the **schema definition file** that describes the contents of the `productDataSet`. Double-click the *ProductDataSet.xsd* entry to open it in an editor window, as shown in Figure 13-74. The area at the top of the diagram shows the columns that are included in the dataset, and the area at the bottom shows the table adapter query methods.

Figure 13-73 Schema definition file entry in the *Solution Explorer*

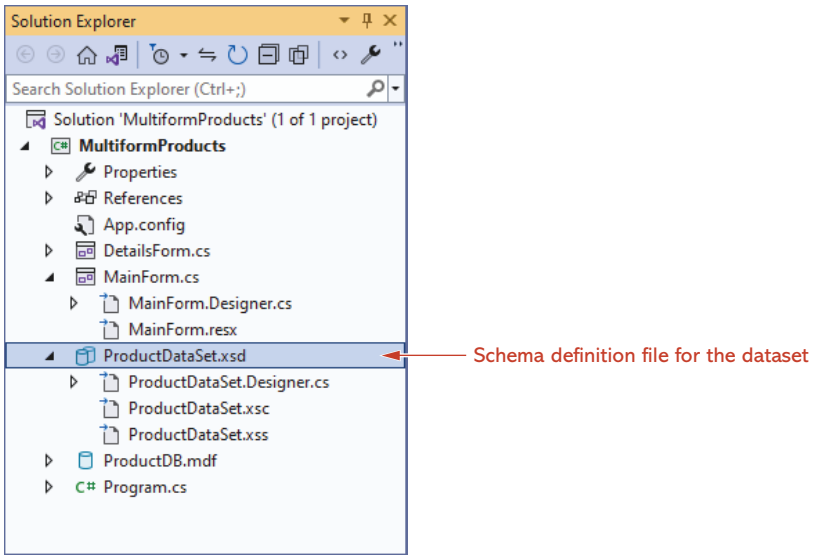
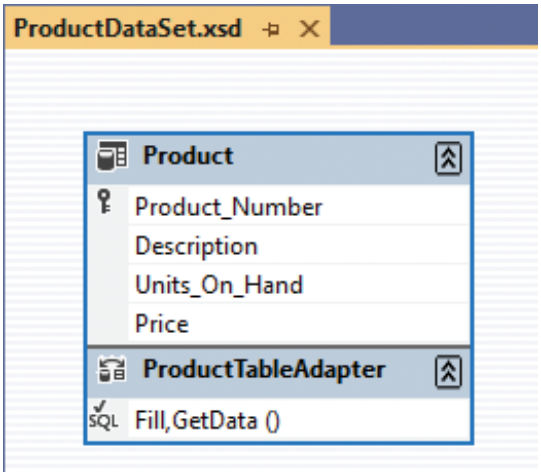


Figure 13-74 ProductDataSet.xsd opened for editing



Right-click the area that reads *Fill, GetData()* and, as shown in Figure 13-75, select *Configure* from the pop-up menu. This displays the *TableAdapter Configuration Wizard* window shown in Figure 13-76. Notice the SQL query that appears in *TableAdapter Configuration Wizard* window. The dataset is filled with the rows that are returned by this `Select` statement. If you were to change this `Select` statement, for instance, by adding a `Where` clause and then by clicking the *Finish* button, the data that is initially displayed in the DataGridView control would change accordingly. When you are finished examining the window, click the *Cancel* button to close it.

Figure 13-75 Select *Configure* from the pop-up menu

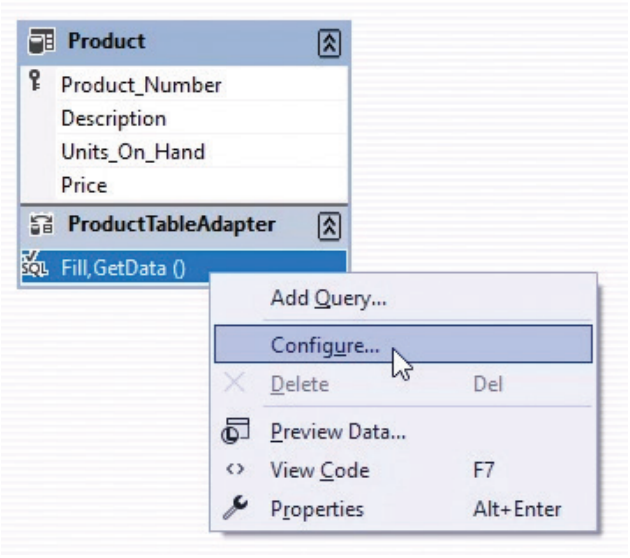
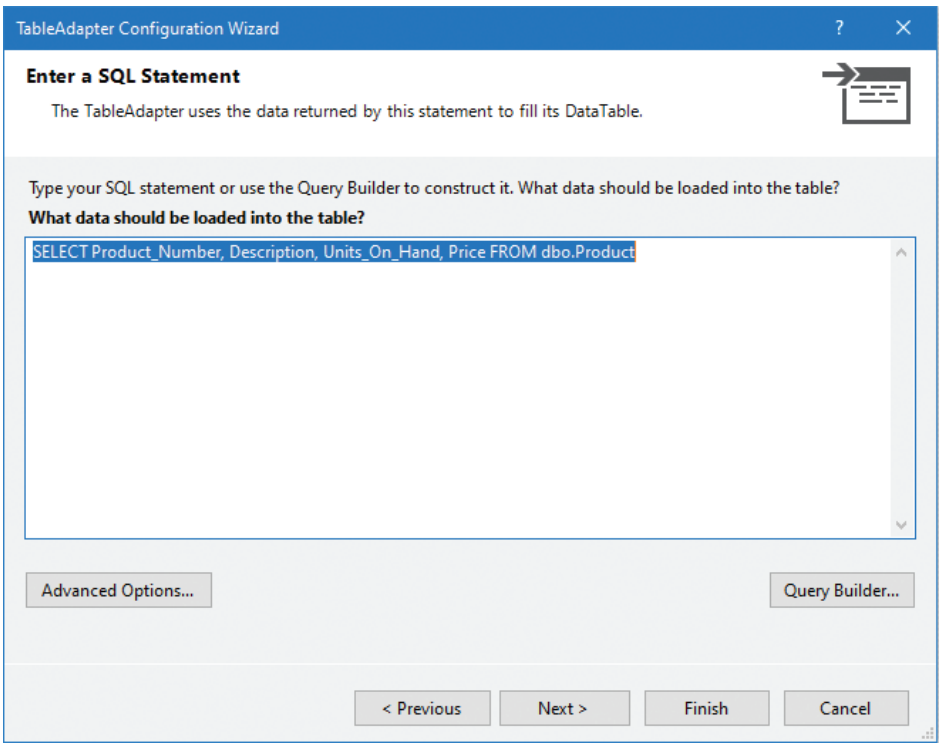


Figure 13-76 *TableAdapter Configuration Wizard*



NOTE: Notice that the table name in the `Select` statement is preceded with the `dbo.` prefix. The prefix `dbo` stands for database owner and is part of an SQL Server naming convention. In our example programs, the `dbo` prefix is optional.

Adding New Table Adapter Queries

You can add your own SQL queries to a table adapter and call them from your C# code. In Tutorial 13-6, you add several queries to a table adapter, and then you create buttons on the application's form that executes those queries.

Tutorial 13-6:
Creating the *ProductQueries* Project

VideoNote
VideoNote: Tutorial 13-6: Creating the *ProductQueries* Project

In this tutorial, you create an application that displays the `Product` table from the ProductDB.mdf database in a DataGridView control. You also add two SQL queries to the table adapter. The queries perform the following:

- Return all the rows, sorted by price in ascending order
- Return only the rows with more than 100 units on hand

Then, you create buttons on the applications form that execute the queries.

Step 1: Start Visual Studio. Create a new *Windows Forms App (.NET Framework)* project named *ProductQueries*.

Step 2: Change the Form1 form's Text property to *Product Queries*.

Step 3: Perform the following steps to connect the application to the ProductDB.mdf database and select the `Product` table for the dataset:

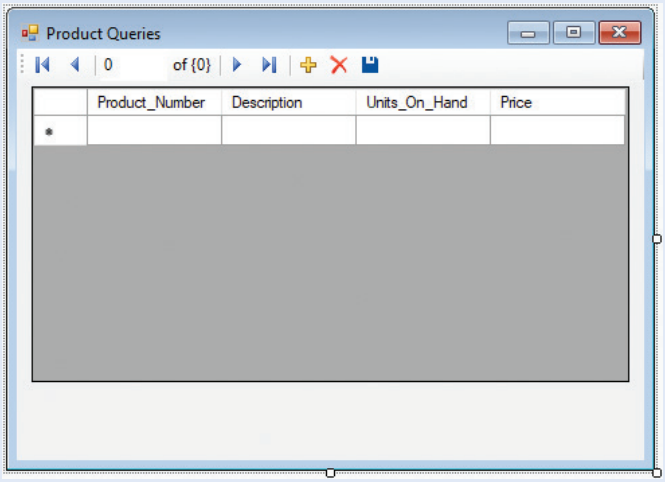
- From the *Project* menu, select *Add New Data Source*. . . .
- Select *Database* and click *Next* >.
- Select *Dataset* and click *Next* >.
- Click *New Connection*. . . .
- In the *Add Connection* window, make sure *Microsoft SQL Server Database File (SqlClient)* is the selected data source.
- In the *Add Connection* window, click *Browse* and go to the *Chap13* folder in the Student Sample Programs. Select the *ProductDB.mdf* file. Click *OK*.
- Click *Next* >, and when asked if you want to copy the file to the project, click *Yes*.
- Click *Next* >.
- Select the `Product` table as the database object, and change the name of the dataset to `ProductDataSet`. Click *Finish*.

NOTE: This is the same procedure that you performed in Steps 3–11 in Tutorial 13-3. If you want more detailed guidance, go back to that tutorial to see the actual screens.

Step 4: Open the *Data Sources* window and drag a DataGridView control onto the form, bound to the `Product` table. After creating the DataGridView control, adjust the size of the form and the control, as shown in Figure 13-77.

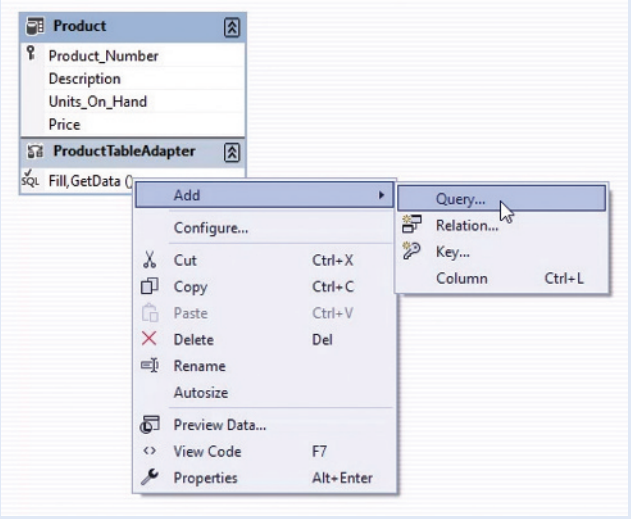
Step 5: Now you will add the first query, which sorts the data in the dataset by price. In the *Solution Explorer*, double-click the *ProductDataSet.xsd* entry. This opens the dataset schema description. Right-click the area that reads *ProductTableAdapter*. As shown in Figure 13-78, select *Add*, and then click *Query*. . . .

Figure 13-77 The application's form



NOTE: It is possible that you will see *Add Query. . .* on the pop-up menu instead of *Add*. If that is the case, simply click *Add Query. . .*.

Figure 13-78 Adding a query to the table adapter



Step 6: The *TableAdapter Query Configuration Wizard* appears next, as shown in Figure 13-79. Make sure *Use SQL Statements* is selected, and then click *Next >*.

Step 7: The wizard now prompts you to choose a query type, as shown in Figure 13-80. Make sure *SELECT which returns rows* is selected, and then click *Next >*.

Step 8: The wizard now prompts you to *Specify a SQL SELECT statement*. A statement is already shown that returns every row in the `Product` table. Add the `Order By Price` clause to the end of the statement, as shown in Figure 13-81. Click *Next >*.

Figure 13-79 *TableAdapter Query Configuration Wizard—Choose a Command Type*

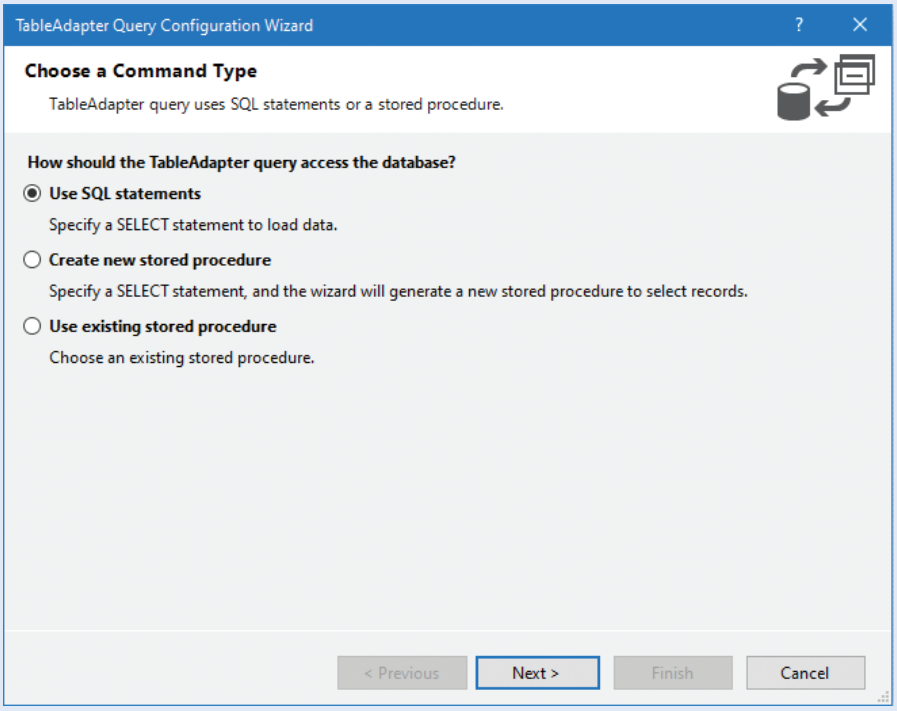
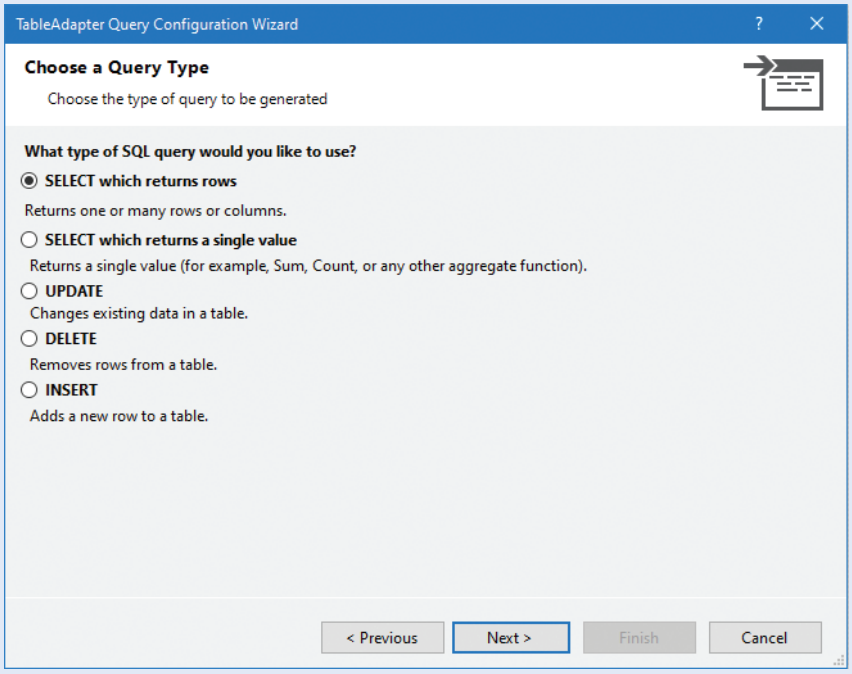
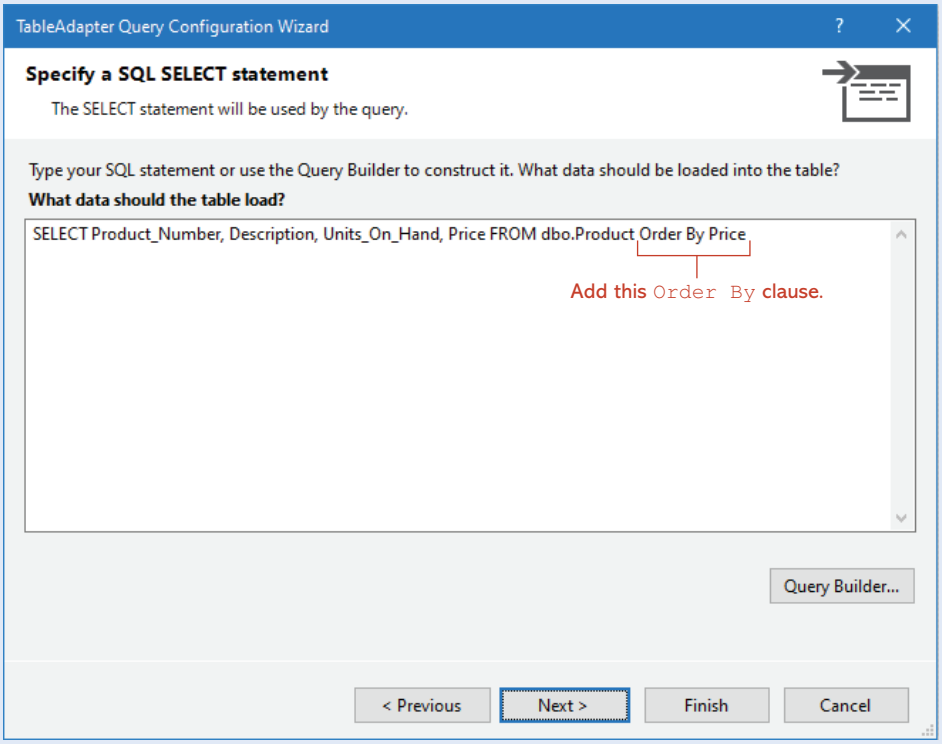


Figure 13-80 *TableAdapter Query Configuration Wizard—Choose a Query Type*



Step 9: The wizard now prompts you to *Choose Methods to Generate*. We are interested only in generating a method to fill the dataset with the results of our SQL statement. Make the following selections:

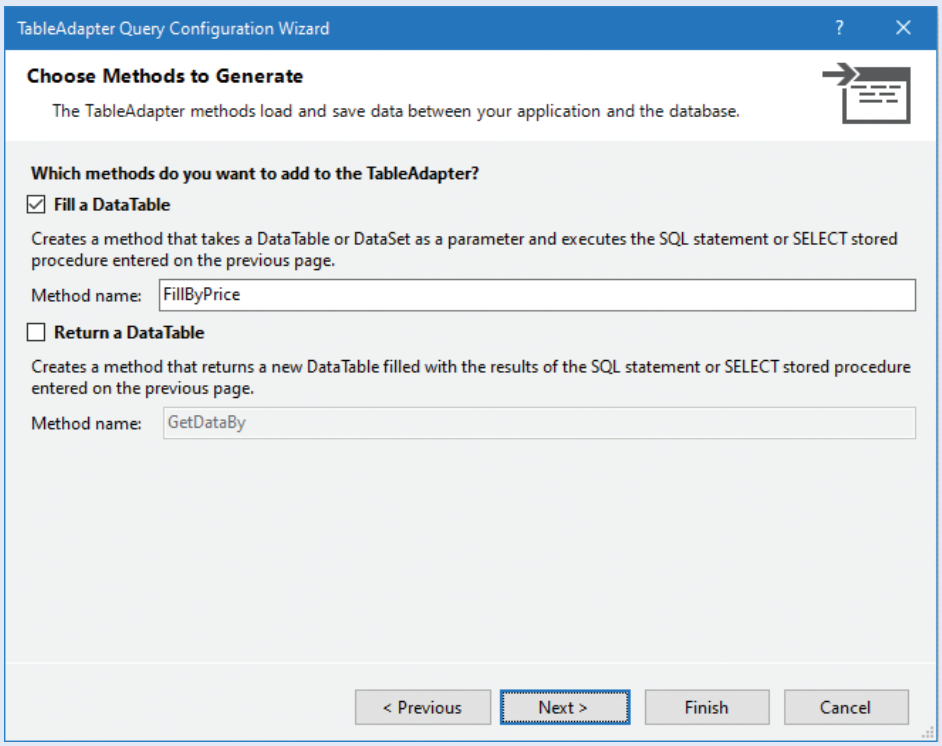
Figure 13-81 *TableAdapter Query Configuration Wizard—Specify a SQL SELECT statement*



- Make sure *Fill a DataTable* is checked.
- Under *Fill a DataTable*, enter `FillByPrice` as the method name. This will be the name of the method that executes the SQL statement.
- Make sure *Return a DataTable* is not checked.

The window should now appear as shown in Figure 13-82. Click *Next >*.

Figure 13-82 *TableAdapter Query Configuration Wizard—Choose Methods to Generate*



Step 10: The wizard now shows you the results, as shown in Figure 13-83. Click the *Finish* button.

The schema definition file (ProductDataSet.xsd) should now appear as shown in Figure 13-84. Notice that the method `FillByPrice()` now appears in the area at the bottom of the diagram.

NOTE: After you have created a table adapter query, you can edit it by right-clicking its entry in the dataset schema definition and then clicking *Configure . . .* from the pop-up menu.

Figure 13-83 *TableAdapter Query Configuration Wizard—Wizard Results*

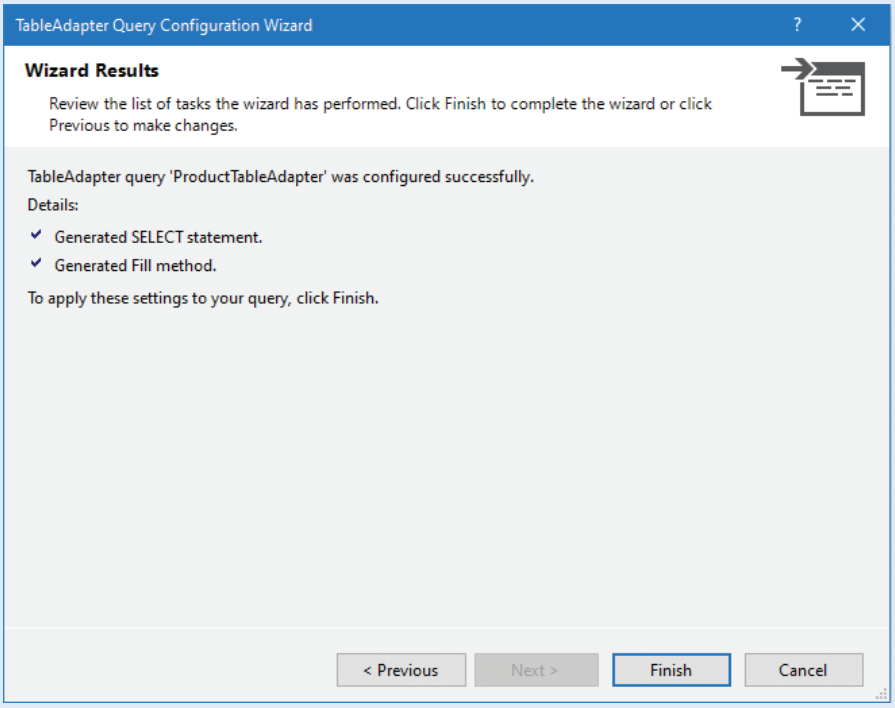
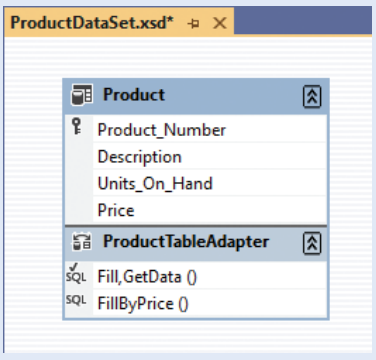


Figure 13-84 The `FillByPrice()` method appearing in the schema definition file

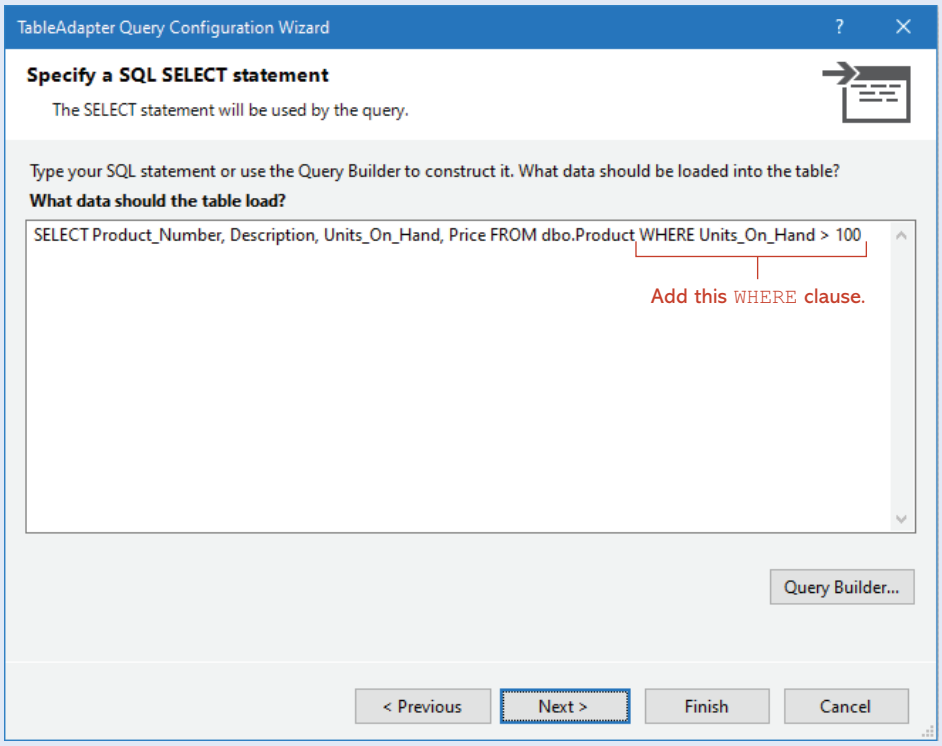


Step 11: Next you add the second query, which returns only the rows with units on hand greater than 100. Perform the following steps (most of these are the same as those you performed to add the previous query):

- In the schema definition, right-click the area that reads *ProductTableAdapter*.

- In the pop-up menu, select *Add*, and then click *Query*. . . . (Or, if you see *Add Query*. . ., then select that.)
- The *TableAdapter Query Configuration Wizard* appears next. Make sure *Use SQL Statements* is selected and then click *Next* >.
- The wizard now prompts you to choose a query type. Make sure *SELECT which returns rows* is selected and then click *Next* >.
- The wizard now prompts you to *Specify a SQL SELECT statement*. A statement is already shown that returns every row in the `Product` table. Add the `WHERE Units_On_Hand > 100` clause to the end of the statement, as shown in Figure 13-85. Click *Next* >.

Figure 13-85 *TableAdapter Query Configuration Wizard—Specify a SQL SELECT statement*



- The wizard now prompts you to *Choose Methods to Generate*. Make the following selections:

  Make sure *Fill a DataTable* is checked.
  Under *Fill a DataTable*, enter `FillByUnits` as the method name. This will be the name of the method that executes the SQL statement.
  Make sure *Return a DataTable* is not checked.
- The window should now appear as shown in Figure 13-86. Click *Next* >.
- The wizard now shows you the results. Click the *Finish* button.

The schema definition file (ProductDataSet.xsd) should now appear as shown in Figure 13-87. Notice that the two methods you created, `FillByPrice()` and `FillByUnits()`, now appear in the area at the bottom of the diagram.

Step 12: Now you add Button controls to the application's form that execute the SQL queries. Open Form1 in the *Designer* and add the two Button controls shown in Figure 13-88.

Figure 13-86 *TableAdapter Query Configuration Wizard—Choose Methods to Generate*

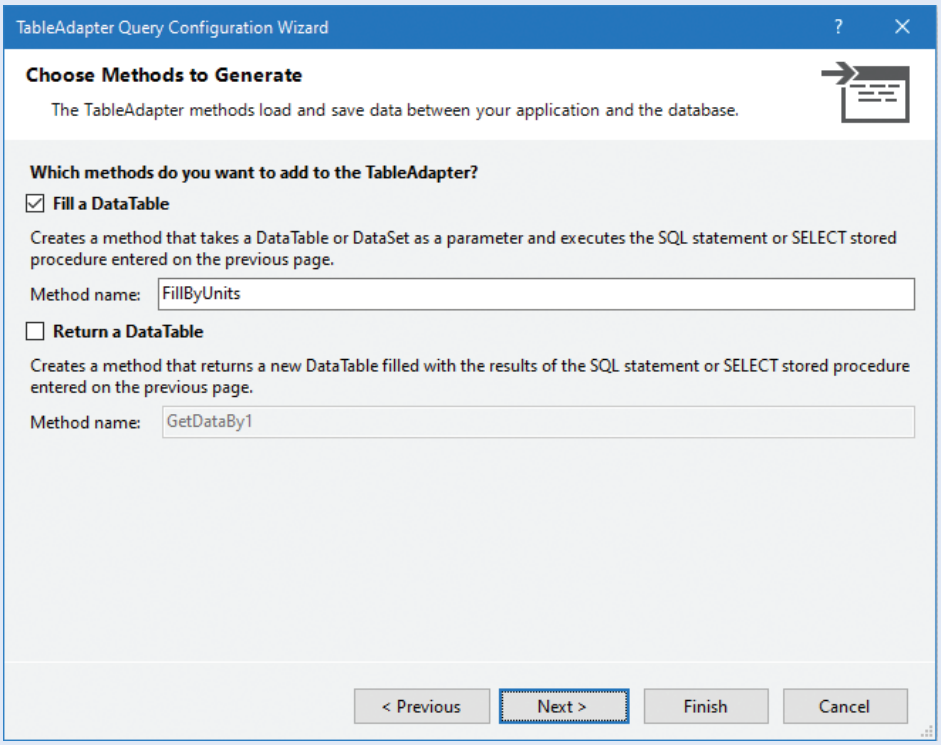
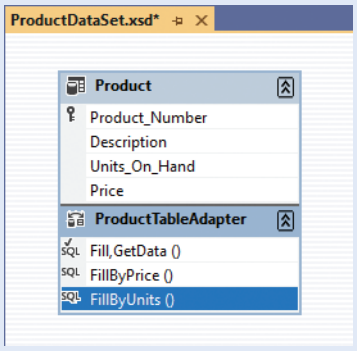


Figure 13-87 The `FillByPrice()` and `FillByUnits()` methods as they appear in the schema definition file

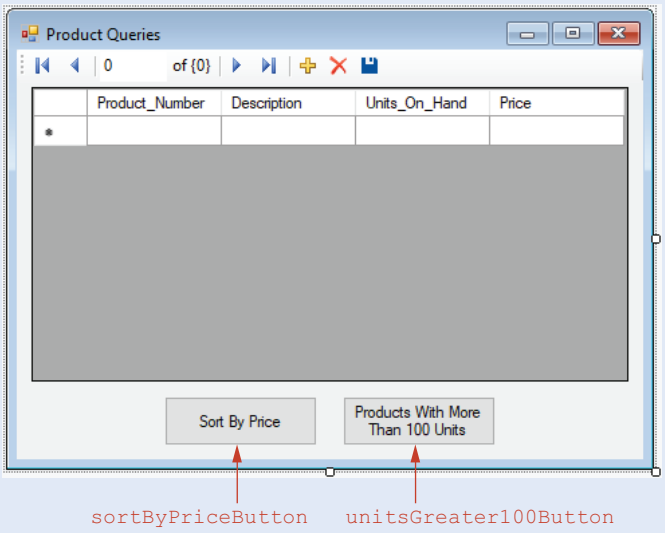


Step 13: Double-click the `sortByPriceButton` control. Complete `sortByPriceButton_Click` the event handler by typing the code in line 37 in Program 13-4. This statement calls the table adapter's `FillByPrice` method.

Step 14: Switch your view back to Form1 in the *Designer* and double-click the `unitsGreater100Button` control. Complete `unitsGreater100Button_Click` event handler by typing the code in line 42 in Program 13-4. This statement calls the table adapter's `FillByUnits` method.

Step 15: Save the project, and then run the application. When the form appears, it should show all the `Product` table's rows in the DataGridView control. When you click the *Sort By Price* button, the data should appear sorted by price.

Figure 13-88 Button controls added to the form



When you click the *Products With More Than 100 Units* button, you should see only the rows where `Units_On_Hand` is greater than 100. When you are finished, close the application's form to exit. (If you plan to immediately continue to the next tutorial, leave the project opened in Visual Studio. You continue to develop this application in the next tutorial.)

Program 13-4 Completed Form1 code in the *ProductQueries* project

```
 1 using System;
 2 using System.Collections.Generic;
 3 using System.ComponentModel;
 4 using System.Data;
 5 using System.Drawing;
 6 using System.Linq;
 7 using System.Text;
 8 using System.Threading.Tasks;
 9 using System.Windows.Forms;
10
11 namespace ProductQueries
12 {
13    public partial class Form1 : Form
14    {
15       public Form1()
16       {
17           InitializeComponent();
18       }
19
20       private void productBindingNavigatorSaveItem_
         Click(object sender, EventArgs e)
21       {
22           this.Validate();
23           this.productBindingSource.EndEdit();
24           this.tableAdapterManager.UpdateAll(this.productDataSet);
25
26       }
27
28       private void Form1_Load(object sender, EventArgs e)
29       {
```

```
30            // TODO: This line of code loads data into the...
31            this.productTableAdapter.Fill(this.productDataSet.
              Product);
32
33        }
34
35        private void sortByPriceButton_Click(object sender,
          EventArgs e)
36        {
37            this.productTableAdapter.FillByPrice
              (this.productDataSet.Product);
38        }
39
40        private void unitsGreater100Button_Click(object sender,
          EventArgs e)
41        {
42            this.productTableAdapter.FillByUnits
              (this.productDataSet.Product);
43        }
44    }
45 }
```

SQL Math Functions

SQL provides several functions for performing calculations. For example, the `Avg` function calculates the average value in a particular column. Here is an example `Select` statement using the `Avg` function:

```
Select Avg(Price) From Product
```

This statement produces a single value: the average of all the values in the `Price` column. Because we did not use a `Where` clause, it uses all the rows in the `Product` table in the calculation. Here is an example that calculates the average price of all the items having a product number that begins with `"20"`:

```
Select Avg(Price)From Product Where Product_Number LIKE '20%'
```

Another of the mathematical functions is `Sum`, which calculates the sum of a column's values. The following statement, which is probably not very useful, calculates the sum of the values in the `Price` column:

```
Select Sum(Price) From Product
```

The `Min` and `Max` functions determine the minimum and maximum values found in a column. The following statement gives the minimum value in the `Price` column:

```
Select Min(Price) From Product
```

The following statement gives the maximum value in the `Price` column:

```
Select Max(Price) From Product
```

The `Count` function can be used to determine the number of rows in a table, as demonstrated by the following statement:

```
Select Count(*) From Product
```

The `*` simply indicates that you want to count entire rows. Here is another example, which gives the number of items with a price greater than 50.0:

```
Select Count(*) From Product Where Price > 50.0
```

In Tutorial 13-7, you will add another query to the table adapter in the *Product Queries* application. The new query will display average price of all the items in the database.

Tutorial 13-7:
Enhancing the *ProductQueries* Project

VideoNote
Tutorial 13-7: Enhancing the *ProductQueries* Project

In this tutorial, you add a query to the *ProductQueries* project that returns the average price of all the items in the `Product` table. You also add a button to the application's form that executes the query and displays its result.

Step 1: Make sure the *ProductQueries* project is open in Visual Studio from the previous tutorial.

Step 2: In the *Solution Explorer*, double-click the *ProduceDataSet.xsd* entry. This opens the dataset schema description. Right-click the area that reads *ProductTableAdapter*. In the pop-up menu, select *Add* and then click *Query*. . . . (Or, if you see *Add Query*. . . on the pop-up menu, select it.)

Step 3: The *TableAdapter Query Configuration Wizard* appears next. Make sure *Use SQL Statements* is selected and then click *Next* >.

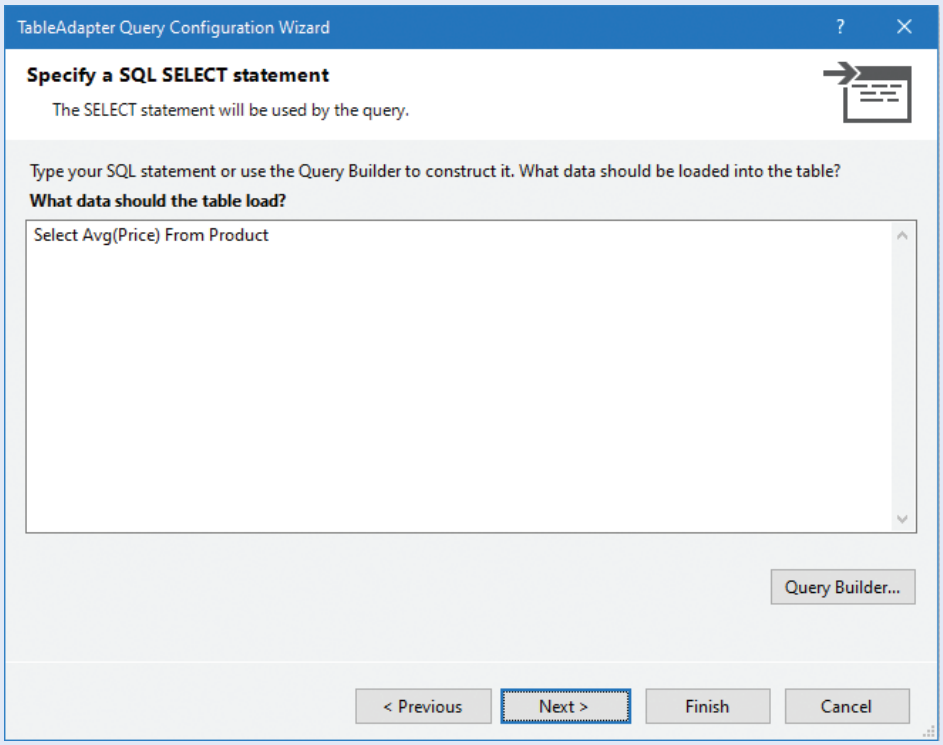
Step 4: The wizard now prompts you to choose a query type. Make sure *SELECT which returns a single value* is selected and then click *Next* >.

Step 5: The wizard now prompts you to *Specify a SQL SELECT statement*. Delete the statement currently shown in the window and replace it with the following:

```
Select Avg(Price) From Product
```

The window should appear as shown in Figure 13-89. Click *Next* >.

Figure 13-89 *TableAdapter Query Configuration Wizard—Specify a SQL SELECT statement*



Step 6: The wizard now prompts you to *Choose Function Name*. The name that you specify in this window will be the name of the method that executes the query. As shown in Figure 13-90, change the name to `AveragePrice` and then click *Next* >.

Figure 13-90 *TableAdapter Query Configuration Wizard—Choose Function Name*

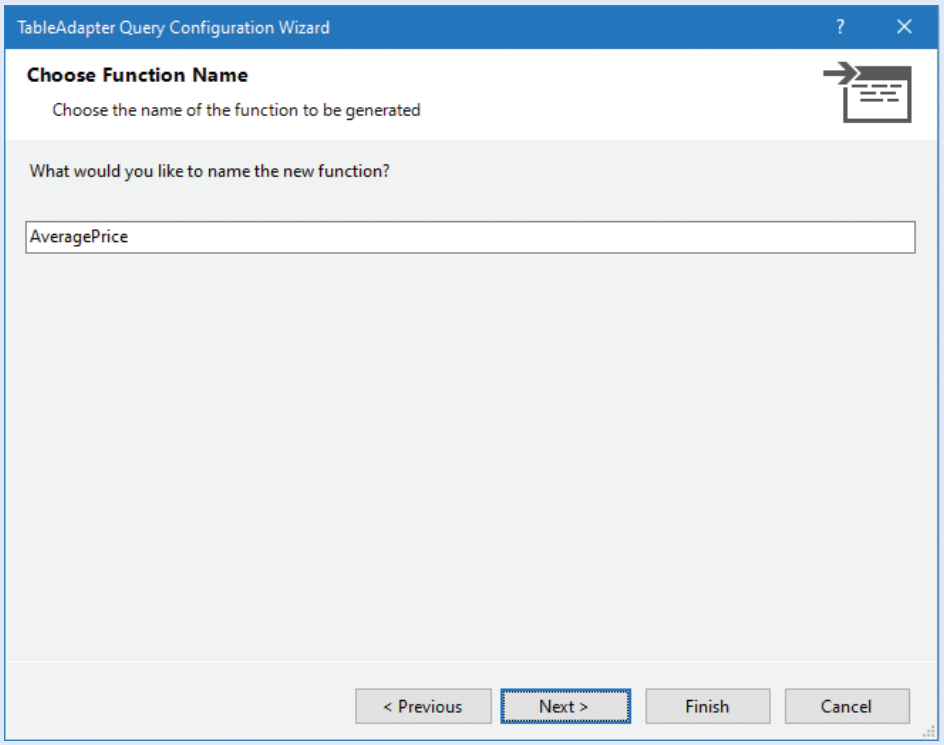
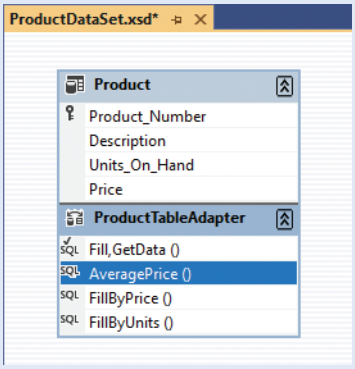


Figure 13-91 The `AveragePrice()` method as it appears in the schema definition file



Step 7: The wizard now shows you the results. Click the *Finish* button.

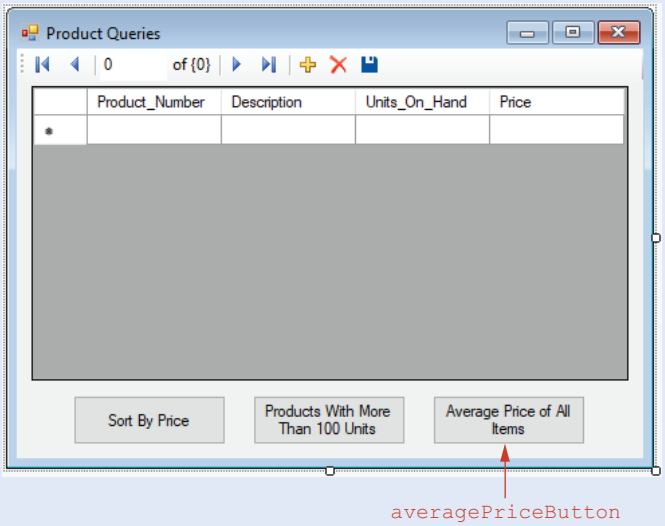
The schema definition file (ProductDataSet.xsd) should now appear, as shown in Figure 13-91. Notice that the `AveragePrice` method appears, along with the other methods that you previously created.

Step 8: Next you add a Button control to the application's form that executes the table adapter's `AveragePrice` method. Open *Form1* in the *Designer* and the `averagePriceButton` control shown in Figure 13-92.

Step 9: Double-click the `averagePriceButton` control. Complete the `averagePriceButton_Click` event handler by typing the code in lines 47–54 in Program 13-5. Let's take a closer look at the code:

Line 48: This statement declares a local `decimal` variable named `averagePrice`.

Figure 13-92 The averagePriceButton control added to the form



Line 51: This statement calls the table adapter's AveragePrice method, which returns the result of the Select statement that you wrote in Step 4. The value that is returned is assigned to the averagePrice variable. Notice that we had to use a cast operator to explicitly convert the value to a decimal.

Line 54: This statement displays a message box showing the average price of all items in the table.

Step 10: Save the project and then run the application. Click the *Average Price of All Items* button. You should see a message box displaying a message indicating the average price of all the items. When you are finished, close the application's form to exit.

Program 13-5 Completed Form1 code in the *ProductQueries* project

```
 1 using System;
 2 using System.Collections.Generic;
 3 using System.ComponentModel;
 4 using System.Data;
 5 using System.Drawing;
 6 using System.Linq;
 7 using System.Text;
 8 using System.Threading.Tasks;
 9 using System.Windows.Forms;
10
11 namespace ProductQueries
12 {
13    public partial class Form1: Form
14    {
15       public Form1()
16       {
17           InitializeComponent();
18       }
19
20       private void productBindingNavigatorSaveItem_
         Click(object sender, EventArgs e)
21       {
```

```
22            this.Validate();
23            this.productBindingSource.EndEdit();
24            this.tableAdapterManager.UpdateAll(this.productDataSet);
25
26        }
27
28        private void Form1_Load(object sender, EventArgs e)
29        {
30            // TODO: This line of code loads data into the...
31            this.productTableAdapter.Fill(this.productDataSet.
              Product);
32
33        }
34
35        private void sortByPriceButton_Click(object sender,
          EventArgs e)
36        {
37            this.productTableAdapter.FillByPrice(this.productDa-
              taSet.Product);
38        }
39
40        private void unitsGreater100Button_Click(object sender,
          EventArgs e)
41        {
42            this.productTableAdapter.FillByUnits
              (this.productDataSet.Product);
43        }
44
45        private void averagePriceButton_Click(object sender,
          EventArgs e)
46        {
47            // Declare a variable to hold the average price.
48            decimal averagePrice;
49
50            // Get the average price.
51            averagePrice = (decimal) this.productTableAdapter.
              AveragePrice();
52
53            // Display the average price.
54            MessageBox.Show($"Average price of all items:
              {averagePrice:c}");
55        }
56    }
57 }
```

Query Parameters

SQL queries can accept arguments. Just as with regular methods, arguments are passed into parameter variables. In an SQL statement, a parameter variable begins with the `@` symbol. The following statement shows an example:

```
Select * From Product Where Price < @priceValue
```

This statement retrieves all the rows in which the `Price` column is less than the value of the `priceValue` parameter. Here is another example:

```
Select * From Product
Where Price < @priceValue And Units_On_Hand < @unitsValue
```

This statement retrieves all the rows in which the Price column is less than the value of the @priceValue parameter and the Units_On_Hand column is less than the value of the @unitsValue parameter.

When you call the table adapter method for an SQL query, you have to pass arguments for any parameters that are used in the query. In Tutorial 13-8, you create an application that gets a value from the user and passes that value as an argument to an SQL query.

Tutorial 13-8:
Creating the *ProductSearch* Project

VideoNote
Tutorial 13-8: Creating the *ProductSearch* Project

In this tutorial, you create an application that displays the Product table from the ProductDB.mdf database in a DataGridView control. The user will be able to enter a value into a text box and search for all items that contain that value in their description.

Step 1: Start Visual Studio. Create a new *Windows Forms App (.NET Framework)* project named *ProductSearch*.

Step 2: Change the Form1 form's Text property to *Product Search*.

Step 3: Perform the following steps to connect the application to the ProductDB.mdf database and select the Product table for the dataset:

- From the *Project* menu, select *Add New Data Source*. . . .
- Select *Database* and click *Next* >.
- Select *Dataset* and click *Next* >.
- Click *New Connection*. . . .
- In the *Add Connection* window, make sure *Microsoft SQL Server Database File (SqlClient)* is the selected data source.
- In the *Add Connection* window, click *Browse* and go to the *Chap13* folder in the Student Sample Programs. Select the *ProductDB.mdf file*. Click *OK*.
- Click *Next* >, and when asked if you want to copy the file to the project, click *Yes*.
- Click *Next* >.
- Select the Product table as the database object, and change the name of the dataset to ProductDataSet. Click *Finish*.

NOTE: This is the same procedure that you performed in Steps 3–11 in Tutorial 13-3. If you want more detailed guidance, go back to that tutorial to see the actual screens.

Step 4: Open the *Data Sources* window and drag a DataGridView control onto the form, bound to the Product table.

Step 5: Adjust the size of the form and the DataGridView control, as shown in Figure 13-93. Then, place the group box, text box, and buttons as shown in the figure. Here is a summary of what the Button controls will do when the application is completed:

- The searchButton control will get the value that has been entered into the searchTextBox and pass that as an argument to an SQL query. The query will return all rows in the Product table that

contain the value that was passed as an argument. Those rows will be displayed in the DataGridView control.

- The `showAllButton` will display all the rows of the `Product` table in the DataGridView control.

Step 6: In the *Solution Explorer*, double-click the *ProduceDataSet.xsd* entry. This opens the dataset schema description. Right-click the area that reads *ProductTableAdapter*. In the pop-up menu, select *Add* and then click *Query*. . . . (Or, if you see *Add Query*. . . on the pop-up menu, select it.)

Step 7: The *TableAdapter Query Configuration Wizard* appears next. Make sure *Use SQL Statements* is selected and then click *Next* >.

Step 8: The wizard now prompts you to choose a query type. Make sure *SELECT which returns rows* is selected and then click *Next* >.

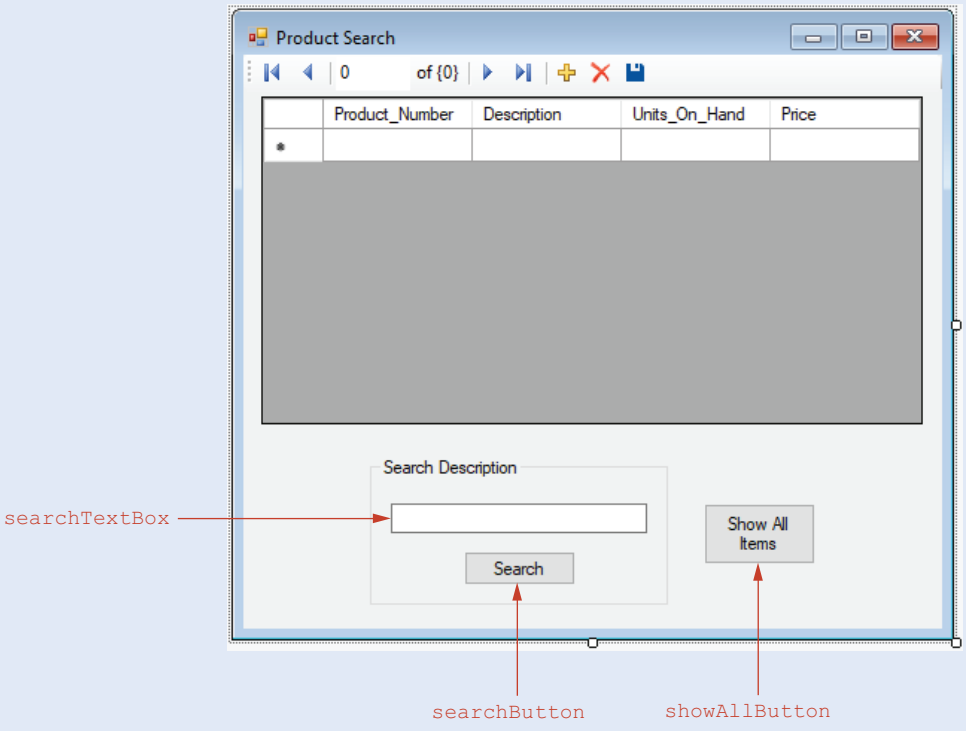
Step 9: The wizard now prompts you to *Specify a SQL SELECT statement*. Modify the default statement that is provided as shown here:

```
Select Product_Number, Description, Units_On_Hand, Price
From Product
Where Description Like '%' + @value + '%'
```

Notice that a parameter query named `@value` is used in the `Where` clause. Also, notice the use of the `+` operator for string concatenation. When this query executes, it will return all rows in which the `Description` column contains the value specified by the `@value` parameter. The window at this point should appear as shown in Figure 13-94.

Click *Next* >.

Figure 13-93 The application's form



Step 10: The wizard now prompts you to *Choose Methods to Generate*. Make the following selections:

- Make sure *Fill a DataTable* is checked.
- Under *Fill a DataTable*, enter `SearchDesc` as the method name. This will be the name of the method that executes the SQL statement.
- Make sure *Return a DataTable* is not checked.

The window should now appear as shown in Figure 13-95. Click *Next >*.

Step 11: The wizard now shows you the results. Click the *Finish* button.

The schema definition file (ProductDataSet.xsd) should now appear as shown in Figure 13-96. Notice that the `SearchDesc(@value)` now appears in the area at the bottom of the diagram.

Step 12: Open Form1 in the *Designer* and double-click the `searchButton` control. Complete the `searchButton_Click` event handler by typing the code shown in lines 37 and 38 in Program 13-6.

The statement in these lines calls the table adapter's `SearchDesc` method. The first argument is the dataset's `Product` table, and the second argument is the `searchTextBox` control's Text property. The second argument will be passed into the SQL query's `@value` parameter.

Step 13: Switch your view back to Form1 in the *Designer* and double-click the `showAllButton` control. Complete the `showAllButton_Click` event handler by typing the code shown in line 43 in Program 13-6. The statement this line calls the table adapter's `Fill` method. Recall that the `Fill` method fills the dataset table with all of the rows from the table.

Figure 13-94 *TableAdapter Query Configuration Wizard—Specify a SQL SELECT statement*

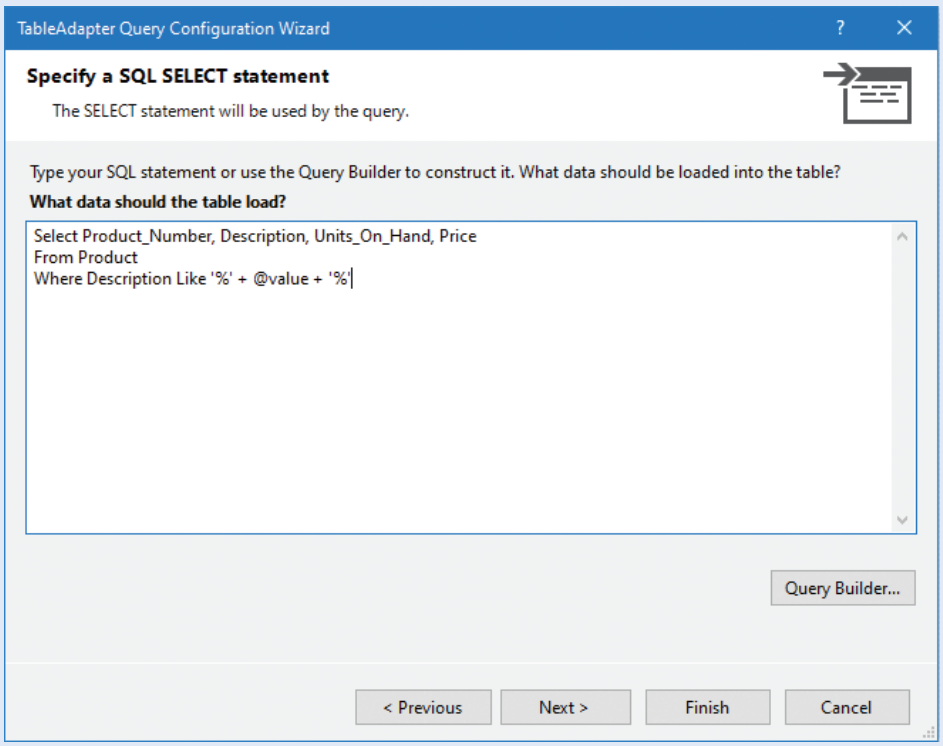


Figure 13-95 *TableAdapter Query Configuration Wizard—Choose Methods to Generate*

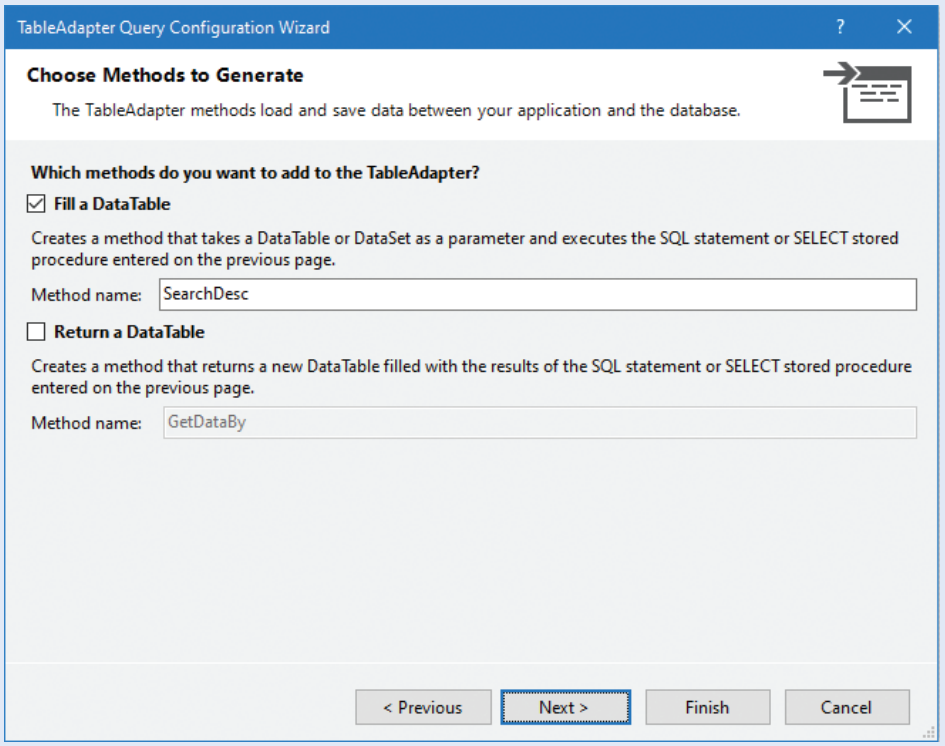
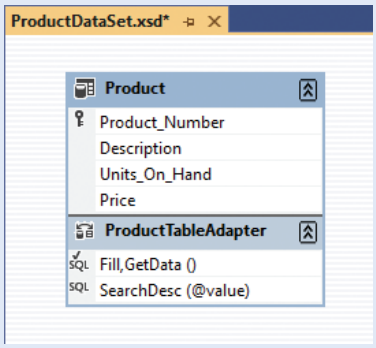


Figure 13-96 The `SearchDesc` method appearing in the schema definition file



Step 14: Save the project and then run the application. Enter a search term, such as *Shirt* or *Pants*, in the text box and click the *Search* button. You should see all the rows that contain that search term displayed in the DataGridView control. To reset the DataGridView control so it displays all of the rows, click the *Show All Items* button. Experiment with several search terms. When you are finished, close the application.

Program 13-6 Completed Form1 code in the *ProductSearch* project

```
1 using System;
2 using System.Collections.Generic;
3 using System.ComponentModel;
4 using System.Data;
5 using System.Drawing;
```

```
 6 using System.Linq;
 7 using System.Text;
 8 using System.Threading.Tasks;
 9 using System.Windows.Forms;
10
11 namespace ProductSearch
12 {
13    public partial class Form1 : Form
14    {
15       public Form1()
16       {
17          InitializeComponent();
18       }
19
20       private void productBindingNavigatorSaveItem_
         Click(object sender, EventArgs e)
21       {
22          this.Validate();
23          this.productBindingSource.EndEdit();
24          this.tableAdapterManager.UpdateAll(this.productDataSet);
25
26       }
27
28       private void Form1_Load(object sender, EventArgs e)
29       {
30          // TODO: This line of code loads data into the...
31          this.productTableAdapter.Fill(this.productDataSet.
            Product);
32
33       }
34
35       private void searchButton_Click(object sender, EventArgs e)
36       {
37          this.productTableAdapter.SearchDesc(
38             this.productDataSet.Product, searchTextBox.Text);
39       }
40
41       private void showAllButton_Click(object sender, EventArgs e)
42       {
43          this.productTableAdapter.Fill(this.productDataSet.Product);
44       }
45    }
46 }
```

Checkpoint

13.23 In SQL, what is the purpose of the `Select` statement?

13.24 What are the instructions for the DBMS to carry out operations on its data called?

13.25 How are the relational operators used in SQL different from those used in C#?

13.26 How do you write a single quote as part of a string in SQL?

13.27 What is the purpose of the `Like` operator in SQL?

13.28 How is the underscore character different from the `%` character in SQL?

13.29 How do you sort the results of a `Select` query in SQL?

13.30 In an SQL statement, with what symbol does a query parameter begin?

Key Terms

binding source
column
database
database management system (DBMS)
data-bound control
DataGridView control
dataset
data source
Details view
identity column
primary key
queries
row
schema definition file
`Select` statement
structured query language (SQL)
table
table adapter
table adapter query

Review Questions

Multiple Choice

1. A(n) _____ is software that is specifically designed to store, retrieve, and manipulate large amounts of data in an organized and efficient manner.
   a. Software development kit (SDK)
   b. Database management system (DBMS)
   c. Application programming interface (API)
   d. Driver development kit (DDK)

2. A _____ is a collection of tables that are stored using a database management system.
   a. Data source
   b. Dataset
   c. Database
   d. Spreadsheet

3. A _____ holds a collection of related data that is organized into rows and columns.
   a. Table
   b. Grid
   c. Dataset
   d. Worksheet

4. A _____ is a complete set of information about a single item in a table of a database.
   a. Data source
   b. Dataset
   c. Column
   d. Row

5. A(n) _____ holds an individual piece of information about the item in a row of a table in a database.
   a. Column
   b. Volume
   c. Record
   d. Entry

6. A _____ is a unique column value that can be used to identify a specific row in a table of a database.
   a. Data cell
   b. Primary key
   c. Prime column
   d. Lookup entry

7. A(n) _____ is a column that contains unique values that are generated by the DBMS.
   a. Generated column
   b. Pseudo-primary key
   c. Identity column
   d. Automatic entry

8. A _____ is a source of data with which the application can work.
   a. Database communicator
   b. Dataset
   c. Data source
   d. Parser

9. A _____ connects to a data source and can retrieve and update data from a table in a data source.
   a. Source connection
   b. Dataset
   c. Binding source
   d. Table adapter

10. A _____ gets a copy of a table from the table adapter and keeps the copy of the table in memory.
    a. Connection
    b. Dataset
    c. Read-only table
    d. Table adapter

11. A _____ is a component that can connect user interface controls directly to a dataset.
    a. Controller
    b. Source editor
    c. Binding source
    d. Component adapter

12. A _____ is a user interface control that is connected to a data source.
    a. DBMS component
    b. Data-bound control
    c. Source control
    d. Database control

13. A _____ can display an entire database table in a scrollable grid on an application's form.
    a. Scrollable data control
    b. Table control
    c. SmartGrid control
    d. DataGridView control

14. A _____ is a set of individual controls that are bound to the columns in a single row.

a. Data-bound control set
b. Data collection
c. Details view
d. Column view

15. In a Visual Studio project, the *bin\Debug* folder is known as the _____.

a. Input folder
b. Output folder
c. Build folder
d. Home folder

16. In the *Designer*, the _____ that appears in the upper-right corner of a control's bounding box can be clicked to display a tasks panel, which provides a number of options that can be performed with the control.

a. Double arrow
b. Smart tag
c. Resizing handle
d. Name

17. The _____ identifies the table from which to get data.

a. DataSource property
b. Table property
c. DataBound property
d. DataTable property

18. The _____ identifies the column of the table from which to display data.

a. DisplayData property
b. DataOutput property
c. ShowColumn property
d. DisplayMember property

19. _____, which stands for _____, is a standard language for working with database management systems.

a. SQL, structured query language
b. DBL, database language
c. XML, extensible markup language
d. HTML, hypertext markup language

20. _____ are instructions for the DBMS to carry out operations on its data.

a. Command strings
b. Queries
c. Build statements
d. Data tokens

21. In SQL, you use the _____ to retrieve the rows in a table.

a. `Select` statement
b. `Like` operator
c. `%` symbol
d. `Where` clause

22. A(n) _____ is an SQL statement that is stored in a table adapter and can be executed simply by calling a method.

a. Automated query
b. Dataset query
c. Table adapter query
d. Structured data query

23. The _____ describes the contents of a dataset, and its name appears in the *Solution Explorer* ending with the .xsd file extension.
    a. Schema definition file
    b. Data markup file
    c. Table description file
    d. DBMS protocol file

True or False

1. When developing applications that work with an intensive amount of data, most developers prefer to use traditional files.
2. The DBMS works directly with the data and sends the results of operations back to the application.
3. The data that is stored in a database is organized into one or more tables.
4. Each column in a table must have a name.
5. When you create a database table, the data types that you can choose from are C# data types.
6. When working with the data in a database table, the column that is designated as the primary key must hold a unique value for each row.
7. If a column in a database table contains no data, it is said to be null.
8. An application works directly with a database, modifying rows, deleting rows, adding new rows, and so forth.
9. Rather than showing multiple rows at once, a Details view lets the user see one row at a time.
10. SQL is a language that can be used to write data intensive applications.
11. String comparisons in SQL are case sensitive.

Short Answer

1. In what situation do traditional files work well for storing data?
2. Briefly describe the layered nature of an application that works with a DBMS.
3. What is the data that is stored in a row of a table in a database divided into?
4. Are the data types used when creating a database table C# data types? If not, what kind of data types are they?
5. How do you create a primary key for a table with columns that could potentially contain the same values?
6. What type of columns in a database table must always contain a value?
7. Instead of working with the database directly, with what component does an application work?
8. What control can be used to display an entire database table in a scrollable grid on an application's form?
9. What is a Details view?
10. In SQL, what statement do you use to retrieve specific rows from a table in a database?
11. How do you sort the results of a `Select` query in descending order in SQL?

Algorithm Workbench

1. Look at the following SQL statement:

```
Select Name From Employee
```

   a. What is the name of the table from which this statement is retrieving data?
   b. What is the name of the column that is being retrieved?

For Questions 2–8, assume that a database has a table named `Stock`*, with the following columns:*

Column Name	Type
Trading_Symbol	nchar(10)
Company_Name	nchar(25)
Num_Shares	int
Purchase_Price	money
Selling_Price	money

2. Write a `Select` statement that returns all the columns from every row in the table.
3. Write a `Select` statement that returns only the `Trading_Symbol` column from every row in the table.
4. Write a `Select` statement that returns the `Trading_Symbol` column and the `Num_Shares` column from every row in the table.
5. Write a `Select` statement that returns the `Trading_Symbol` column only from the rows where `Purchase_Price` is greater than $25.00.
6. Write a `Select` statement that returns all the columns from the rows where `Trading_Symbol` starts with `"SU"`.
7. Write a `Select` statement that returns the `Trading_Symbol` column only from the rows where `Selling_Price` is greater than `Purchase_Price` and `Num_Shares` is greater than 100.
8. Write a `Select` statement that returns the `Trading_Symbol` column and the `Num_Shares` column only from the rows where `Selling_Price` is greater than `Purchase_Price` and `Num_Shares` is greater than 100. The results should be sorted by the `Num_Shares` column in ascending order.

Programming Problems

VideoNote
Solving the Personnel Database Problem

1. **Personnel Database**

   Use Visual Studio to create a database named Personnel.mdf. The database should have a table named `Employee`, with columns for employee ID, name, position, and hourly pay rate. The employee ID should be the primary key. Insert at least five sample rows of data into the `Employee` table. Create an application that displays the `Employee` table in a DataGridView control.

2. **Multiform Personnel Database Application**

   Create an application that connects to the Personnel.mdf database that you created in Programming Problem 1. The application's main form should display the `Employee` table in a DataGridView control. The main form should also have a button that, when clicked, displays a second form. The second form should display the `Employee` table in a Details view. Make sure that when the second form is closed, the main form refills the dataset so the most current data is displayed.

3. **Hourly Pay Sorter**

   Create an application that connects to the Personnel.mdf database that you created in Programming Problem 1. The application's form should display the `Employee` table in a DataGridView control. The form should also have the following controls:

   - A button that, when clicked, sorts the data in ascending order by hourly pay rate.
   - A button that, when clicked, sorts the data in descending order by hourly pay rate.

4. **Employee Search**

   Create an application that connects to the Personnel.mdf database that you created in Programming Problem 1. The application's form should display the `Employee` table in a DataGridView control. The application should let the user specify a name in a text box and then search for that name in the `Employee` table. The application should display any rows that contain a full or partial match of the specified name.

5. **Highest and Lowest Pay Rate**

   Create an application that connects to the Personnel.mdf database that you created in Programming Problem 1. The application's form should display the `Employee` table in a DataGridView control. The form should also have the following controls:

   - A button that, when clicked, displays a message indicating the highest (maximum) pay rate in the table.
   - A button that, when clicked, displays a message indicating the lowest (minimum) pay rate in the table.

6. **Population Database**

   In the *Chap13* folder of the Student Sample Programs, you will find a database file named PopulationDB.mdf. The database has a table named `City`. The `City` table has the following columns:

Column Name	Data Type
`City`	`nvarchar(50)` *Primary key*
`Population`	`float`

   The `City` column stores the name of a city and the `Population` column stores the population of that city. The database has 20 rows already entered.

   Create an application that connects to the PopulationDB.mdf database and allows the user to perform the following:

   - Use data-bound controls to add new rows to the database, change existing rows, and delete rows.
   - Sort the list of cities by population, in ascending order.
   - Sort the list of cities by population, in descending order.
   - Sort the list of cities by name.
   - Get the total population of all the cities.
   - Get the average population of all the cities.
   - Get the highest population.
   - Get the lowest population.

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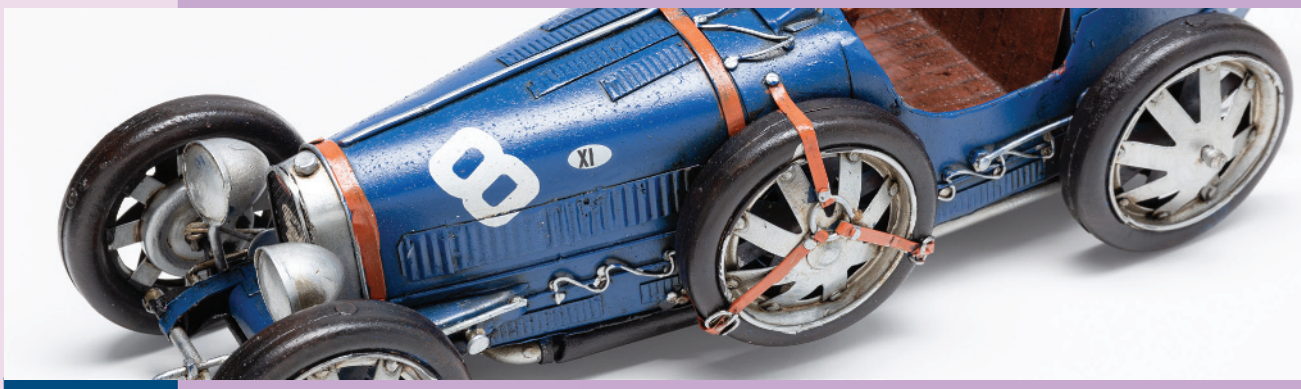


CHAPTER 14

Delegates, Anonymous Methods, and Lambda Expressions

TOPICS

14.1 Delegates
14.2 Anonymous Methods
14.3 Lambda Expressions
14.4 Built-In Delegates: `Action`, `Func`, and `Predicate`

14.1 Delegates

CONCEPT: **A delegate is an object that works like a reference variable that is referring to a method.**

You have learned that reference variables refer to objects. When you create an object in memory, you can assign that object to a reference variable, and then use the reference variable to work with the object. In this section you will learn about delegates, which allow you to create reference variables that can refer to methods. When a delegate refers to a method, you can use the delegate to call the method.

For demonstration purposes, let's look at a simple example of declaring and using a delegate. Suppose we have the following method inside of a class:

```
private void DisplayMessage(string str)
{
    MessageBox.Show(str);
}
```

Notice the following things about the method:

- The return type is `void`
- The method has one parameter, a string

Suppose you want to assign this method to a delegate, and then use that delegate to call the method. First, you must declare a **delegate type**. In this step you are not creating an actual delegate, but you are describing the method that the delegate can refer to. You do this by specifying the method's return type and its parameters. Here is an example:

```
delegate void MyDelegate(string str);
```

This declaration begins with the keyword `delegate`, followed by code that looks like a method header, terminated with a semicolon. This code declares that we are creating a delegate type named `MyDelegate`, that can reference any `void` method that has one parameter, a `string`. (The name of the parameter is irrelevant.)

Once the delegate type is declared, you can use it to create an actual delegate. (This step is often described as creating an instance of the delegate type.) The delegate can reference any method that has a return type and parameter list matching that of the delegate type. Here is an example, using the `MyDelegate` type we previously created:

```
MyDelegate Display = DisplayMessage;
```

This statement creates a delegate named `Display`, and it assigns the `DisplayMessage` method to the delegate. As a result, you can use the `Display` delegate to call the `DisplayMessage` method. Here is an example:

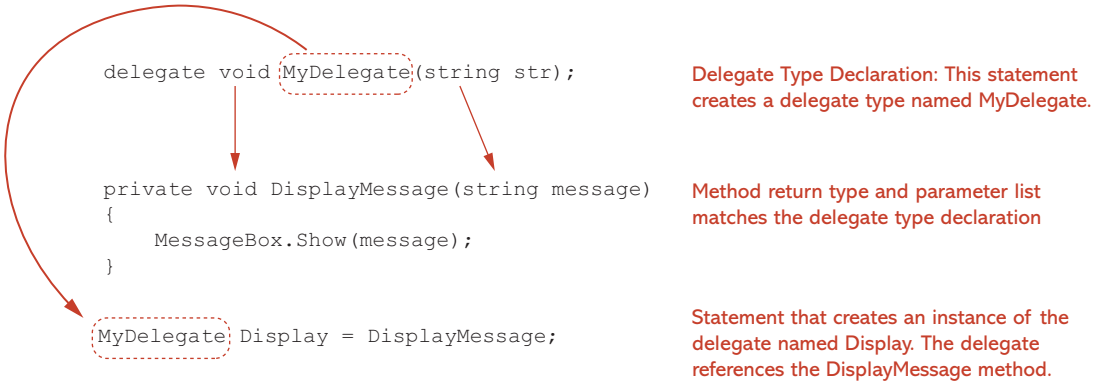
```
Display("Hello World");
```

To summarize the creation and use of a delegate:

1. Declare a delegate type.
2. Create an instance of the delegate type and make it reference a method.
3. Use the delegate to call the method.

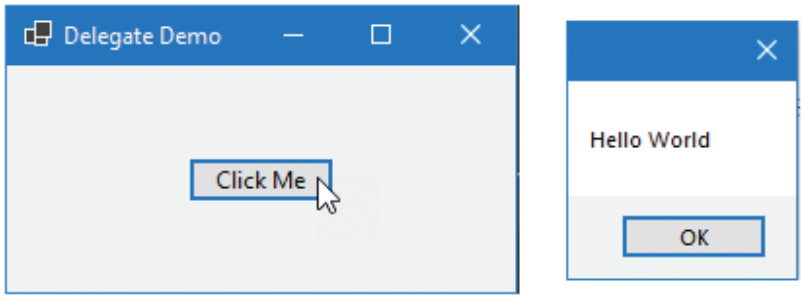
Figure 14-1 illustrates how a delegate type declaration, a method, and a statement that creates an instance of the delegate type work together.

Figure 14-1 Creating and using a delegate



Let's look at a complete application that demonstrates this code. Program 14-1 shows the Form1 code for the DelegateDemo application. Figure 14-2 shows the application's form, and the message box that appears when the user clicks the Button control.

Figure 14-2 The DelegateDemo application



Program 14-1 Form1 code for the DelegateDemo application

```
 1 using System;
 2
 3 namespace DelegateDemo
 4 {
 5     delegate void MyDelegate(string str);
 6
 7     public partial class Form1 : Form
 8     {
 9         public Form1()
10         {
11             InitializeComponent();
12         }
13
14         private void myButton_Click(object sender, EventArgs e)
15         {
16             // Assign the DisplayMessage method to a delegate.
17             MyDelegate Display = DisplayMessage;
18
19             // Use the delegate to call the method.
20             Display("Hello World");
21         }
22
23         private void DisplayMessage(string message)
24         {
25             MessageBox.Show(message);
26         }
27     }
28 }
```

In the application's code, line 5 declares the delegate type. Notice that the delegate type is declared inside the application's namespace, but it is not inside of any class. This is a common practice, although it is possible to declare a delegate type inside of a class. Line 17 creates an instance of the delegate, and makes it reference the `DisplayMessage` method. (The `DisplayMessage` method appears in lines 23–26.)

Reassigning a Delegate

A delegate can be reassigned to another method, as long as the method's return type and parameter list match that of the delegate type. For example, suppose we have the following methods in a class:

```
private void DisplayString(string str)
{
    MessageBox.Show(str);
}

private void DisplayLength(string str)
{
    int len = str.Length;
    MessageBox.Show(len.ToString());
}
```

The `DisplayString` method accepts a `string` and displays it in a message box, and the `DisplayLength` method accepts a `string` and displays its length in a message box. Both methods return `void` and have one `string` parameter. The following delegate type is compatible with both methods:

```
delegate void WorksWithAString(string str);
```

The following code demonstrates how we can create one delegate to work with both methods:

```
 1  // Create a string.
 2  string mystring = "Abracadabra";
 3
 4  // Create a delegate named StringOperation, and assign the
 5  // DisplayString method to it.
 6  WorksWithAString StringOperation = DisplayString;
 7
 8  // Call the method that StringOperation references.
 9  StringOperation(mystring);
10
11  // Assign the DisplayLength method to the StringOperation delegate.
12  StringOperation = DisplayLength;
13
14  // Call the method that StringOperation references.
15  StringOperation(mystring);
```

The statement in line 6 creates a delegate named `StringOperation` and assigns the `DisplayString` method to it. When line 9 executes, it is the `DisplayString` method that is called. Then, line 12 assigns the `DisplayLength` method to the `StringOperation` delegate. As a result, when line 15 executes, it is the `DisplayLength` method that is called.

Passing a Delegate to a Method

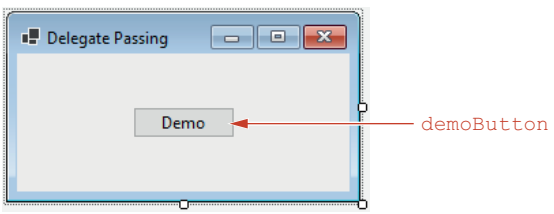
Delegates can be passed as arguments to methods. This means that you can write a method that accepts another method as an argument. Program 14-2 shows the code for the DelegatePassing application, and Figure 14-3 shows the application's form.

Program 14-2 Form1 code for the DelegatePassing application

```
 1 namespace DelegatePassing
 2 {
 3     delegate void DoSomething(ref double data);
 4
 5     public partial class Form1 : Form
 6     {
 7         public Form1()
 8         {
 9             InitializeComponent();
10         }
11
```

```
12          private void SetToZero(ref double data)
13          {
14              data = 0;
15          }
16
17          private void MakeTwice(ref double data)
18          {
19              data *= 2;
20          }
21
22          private void CutInHalf(ref double data)
23          {
24              data /= 2;
25          }
26
27          private void ShowResult(ref double myvalue, DoSomething
                                    MyMethod)
28          {
29              // Perform the action on the data.
30              MyMethod(ref myvalue);
31
32              // Display the data
33              MessageBox.Show(myvalue.ToString());
34          }
35
36          private void demoButton_Click(object sender, EventArgs e)
37          {
38              // A value to work with
39              double value = 100.0;
40
41              // Create a delegate to reference MakeTwice.
42              DoSomething operation = MakeTwice;
43
44              // Pass value and the delegate to ShowResult.
45              ShowResult(ref value, operation);
46
47              // Make the delegate reference the CutInHalf method.
48              operation = CutInHalf;
49
50              // Pass value and the delegate to ShowResult.
51              ShowResult(ref value, operation);
52
53              // Make the delegate reference the SetToZero method.
54              operation = SetToZero;
55
56              // Pass value and the delegate to ShowResult.
57              ShowResult(ref value, operation);
58          }
59      }
60  }
```

Figure 14-3 The DelegatePassing application's form



Let's take a closer look at the program. First, notice the code for the `SetToZero`, `MakeTwice`, and `CutInHalf` methods in lines 12 through 25. Each of these methods returns `void` and has a `double` reference parameter.

In line 3 we declare a delegate type named `DoSomething` that specifies a `void` return type and a `double` reference parameter. So, the delegate type is compatible with `SetToZero`, `MakeTwice`, and `CutInHalf` methods.

The `ShowResult` method appears in lines 27 through 34. Look at the method's parameter list. The first parameter is a `double` reference variable name `myvalue`. The second parameter is a `DoSomething` delegate named `MyMethod`. When we call the `ShowResult` method, we have to pass a `double` (by reference), and a `DoSomething` delegate.

Inside the `ShowResult` method, line 30 calls the method that is referenced by the `MyMethod` delegate, passing the `myvalue` variable as an argument. Then line 33 displays the value of `myvalue` in a message box.

The `demoButton_Click` event handler, shown in lines 36 through 58, performs the following:

- Line 39 declares a `double` variable named `value`, initialized with 100.0.
- Line 42 creates a delegate named `operation` and assigns the `MakeTwice` method to it.
- Line 45 calls the `ShowResult` method, passing the `value` variable and the `operation` delegate as arguments. As a result, the value 200 is displayed in a message box.
- Line 48 assigns the `CutInHalf` method to the `operation` delegate.
- Line 51 calls the `ShowResult` method, passing the `value` variable and the `operation` delegate as arguments. As a result, the value 100 is displayed in a message box.
- Line 54 assigns the `SetToZero` method to the `operation` delegate.
- Line 57 calls the `ShowResult` method, passing the `value` variable and the `operation` delegate as arguments. As a result, the value 0 is displayed in a message box.

Multicast Delegates

In the previous examples, you saw how a delegate can be used to reference a method. When you invoke a delegate, the method that the delegate references is executed. A delegate can also reference multiple methods at the same time. A delegate that references multiple methods is known as a **multicast delegate**. When you invoke a multicast delegate, each of the methods referenced by the delegate will be executed in the order in which they were added to the delegate. You use the += operator to add a method to a delegate. For example, suppose you have the following methods:

```
private void DisplayString(string str)
{
    MessageBox.Show(str);
}

private void DisplayLength(string str)
{
    int len = str.Length;
    MessageBox.Show(len.ToString());
}
```

Suppose you also have the following delegate type declaration:

```
delegate void WorksWithAString(string str);
```

The following code demonstrates how we can create a multicast delegate to invoke both methods:

```
1 // Create a string.
2 string mystring = "Abracadabra";
```

```
 3
 4 // Create a delegate referencing the DisplayString method.
 5 WorksWithAString StringOperation = DisplayString;
 6
 7 // Add the DisplayLength method to the delegate.
 8 StringOperation += DisplayLength;
 9
10 // Call the methods that StringOperation references.
11 StringOperation(mystring);
```

The statement in line 5 creates a delegate named `StringOperation` and assigns the `DisplayString` method to it. Then, the statement in line 8 uses the `+=` operator to add the `DisplayLength` method to the delegate. When line 11 executes, the `DisplayString` method is executed first, followed by the `DisplayLength` method.

Here are some other points to keep in mind regarding multicast delegates:

- You can use the `-=` operator to remove a method from a multicast delegate. Using the previously shown example, we could remove the `DisplayLength` method from the `StringOperation` delegate with this statement:

  ```
  StringOperation -= DisplayLength;
  ```

- If a multicast delegate references methods that return a value (non-`void` methods), the delegate will return only the value that is returned from the last method that is executed. All other values returned in the invocation chain will be lost. In practice, it is best if multicast delegates reference only `void` methods.

Checkpoint

14.1 What is a delegate?

14.2 Suppose you want to use a delegate to call a method. What three steps do you take?

14.3 What is the purpose of a delegate type?

14.4 Once a delegate has been assigned to a method, can it be reassigned?

14.5 What is a multicast delegate?

14.2 Anonymous Methods

CONCEPT: A delegate can refer to an anonymous method, which is a method that has no name.

An **anonymous method** is a method that has no name. In C#, delegates give you the ability to create anonymous methods. For example, suppose we have the following delegate type declaration:

```
delegate void WorksWithAString(string str);
```

Given this type declaration, the following code creates a delegate that references an anonymous method:

```
WorksWithAString Display = delegate(string str)
{
    MessageBox.Show(str);
};
```

This code creates a delegate named `Display`. The `Display` delegate is of the `WorksWithAString` type. The `Display` delegate references an anonymous method that has a `string` parameter, so when we call the method, we must pass a `string` argument to it, as shown here:

```
Display("Calling the anonymous method.");
```

This statement uses the `Display` delegate to call the anonymous method, passing the string "Calling the anonymous method" as an argument. The anonymous method displays the string in a message box.

Let's look at a complete application that demonstrates anonymous methods. Program 14-3 shows the Form1 code for the AnonymousMethod application.

Program 14-3 Form1 code for the AnonymousMethod application

```
 1 namespace AnonymousMethod
 2 {
 3     delegate int IntCalculator(int a, int b);
 4
 5     public partial class Form1 : Form
 6     {
 7         public Form1()
 8         {
 9             InitializeComponent();
10         }
11
12         private void demoButton_Click(object sender, EventArgs e)
13         {
14             int result;
15
16             IntCalculator Sum = delegate (int a, int b)
17             {
18                 return a + b;
19             };
20
21             IntCalculator Multiply = delegate (int a, int b)
22             {
23                 return a * b;
24             };
25
26             result = Sum(1, 2);
27             MessageBox.Show($"1 + 2 = {result}");
28
29             result = Multiply(2, 4);
30             MessageBox.Show($"2 * 4 = {result}");
31         }
32     }
33 }
```

- In line 3, we declare a delegate type named `IntCalculator` that specifies an `int` return type and two `int` parameters.
- In lines 16 through 19, we create an `IntCalculator` delegate named `Sum`, referencing an anonymous method that accepts two `int` arguments and returns their sum.
- In lines 21 through 24, we create another `IntCalculator` delegate named `Multiply`, referencing an anonymous method that accepts two `int` arguments and returns their product.

- In line 26, we call the method referenced by the `Sum` delegate, passing the arguments 1 and 2, and assigning the return value to the `result` variable. In line 27, we display the `result` variable in a message box.
- In line 29, we call the method referenced by the `Multiply` delegate, passing the arguments 2 and 4, and assigning the return value to the `result` variable. In line 30, we display the `result` variable in a message box.

Checkpoint

14.6 What is an anonymous method?

14.7 Given the following delegate type declaration, which of the following code segments (a or b) uses an anonymous method?

```
delegate void DoSomething(string str);
```

a)

```
void ShowStr(string str)
{
    MessageBox.Show(str);
}
DoSomething Display = ShowStr;
```

b)

```
DoSomething Display = delegate(string str)
{
    MessageBox.Show(str);
};
```

14.3 Lambda Expressions

CONCEPT: A lambda expression is a concise expression that creates an anonymous method.

A **lambda expression** is a shortcut technique for writing an anonymous method. Here is the general format of a simple lambda expression that accepts one argument, and returns a value:

```
parameter  =>  expression
```

In this general format, the lambda expression begins with a parameter variable, followed by the **lambda operator** (=>), followed by an expression that has a value. Here is an example:

```
x => x * x
```

The `x` that appears on the left side of the `=>` operator is the name of a parameter variable, and the expression `x * x` that appears on the right side of the `=>` operator is the value that is returned.

This lambda expression works like a method that has a parameter variable named `x`, and it returns the value of `x * x`.

For example, suppose we have the following delegate type declaration:

```
delegate int IntOperation(int x);
```

The name of the delegate type is `IntOperation`, and it specifies a return type of `int`, and one `int` parameter. Also, suppose that we want to create a delegate of the `IntOperation` type that references a method that returns the square of `x`. The following code shows how we can use a lambda expression to create the delegate:

```
IntOperation Square = x => x * x;
```

On the left side of the = operator, we declare an `IntOperation` delegate named `Square`. On the right side of the = operator, we have a lambda expression that creates an anonymous method with the following characteristics:

- The parameter `x` that is used in the lambda expression represents the argument that is passed to the method. We do not have to specify the data type of `x` because the compiler will determine it. Because the `IntOperation` delegate type specifies an `int` parameter, the `x` parameter in the lambda expression will automatically become an `int`.
- The expression `x * x` is the value that is returned from the method.

The following table shows the lambda expression on the left, and equivalent code on the right that explicitly creates an anonymous method. You can see that the lambda expression is much more concise than the anonymous method declaration.

Lambda Expression	Anonymous Method

`IntOperation Square = x => x * x;`	`IntOperation Square = delegate(int x)`
	`{`
	`  return x * x;`
	`};`

Let's look at a complete application that demonstrates the use of a lambda expression. Program 14-4 shows the Form1 code for the LambdaDemo application.

Program 14-4 Form1 code for the LamdaDemo application

```
1  namespace LambdaDemo
2  {
3      delegate int IntOperation(int x);
4
5      public partial class Form1 : Form
6      {
7          public Form1()
8          {
9              InitializeComponent();
10         }
11
12         private void demoButton_Click(object sender, EventArgs e)
13         {
14             int result;
15             IntOperation Square = x => x * x;
16
17             result = Square(5);
18             MessageBox.Show($"5 squared is  {result}");
19         }
20     }
21 }
```

- In line 3, we declare a delegate type named `IntOperation` that specifies an `int` return type and an `int` parameter.
- In line 15, we declare an `IntOperation` delegate named `Square`, referencing an anonymous method with an `int` parameter named `x`. The method returns the value of `x * x`.
- In line 17, we call the method referenced by the `Square` delegate, passing the argument 5, and assigning the return value to the `result` variable. In line 18, we display the `result` variable in a message box.

Lambda Expressions with Multiple Parameters

If a delegate type specifies multiple parameters, any lambda expression that you use with the delegate must also have multiple parameters. For example, consider this delegate type declaration:

```
delegate void IntCalculator(int a, int b);
```

To use more than one parameter in a lambda expression, simply write a comma-separated list and enclose the list in parentheses. Here is an example:

```
IntCalculator Sum = (a, b) => a + b;
```

This lambda expression has two parameters, `a` and `b`. The expression returns the value of `a + b`.

Let's look at another application that demonstrates lambda expressions. Program 14-5 shows the Form1 code for the LambdaDemo2 application. This application uses lambda expressions to perform the same operation as the AnonymousMethod application shown in Program 14-3.

Program 14-5 Form1 code for the LamdaDemo2 application

```
 1 namespace LambdaDemo2
 2 {
 3     delegate int IntCalculator(int a, int b);
 4
 5     public partial class Form1 : Form
 6     {
 7         public Form1()
 8         {
 9             InitializeComponent();
10         }
11
12         private void demoButton_Click(object sender, EventArgs e)
13         {
14             int result;
15             IntCalculator Sum = (a, b) => a + b;
16             IntCalculator Multiply = (a, b) => a * b;
17
18             result = Sum(1, 2);
19             MessageBox.Show($"1 + 2 =  {result}");
20
21             result = Multiply(2, 4);
22             MessageBox.Show($"2 * 4 =  {result}");
23         }
24     }
25 }
```

- In line 3, we declare a delegate type named `IntCalculator` that specifies an `int` return type and two `int` parameters.
- In line 15, we declare an `IntCalculator` delegate named `Sum`, referencing an anonymous method with two `int` parameters named `a` and `b`. The method returns the sum of `a` and `b`.
- In line 16, we declare another `IntCalculator` delegate named `Multiply`, referencing an anonymous method with two `int` parameters named `a` and `b`. The method returns the sum of `a` and `b`.
- In line 18, we call the method referenced by the `Sum` delegate, passing the arguments 1 and 2, and assigning the return value to the `result` variable. In line 19, we display the `result` variable in a message box.
- In line 21, we call the method referenced by the `Multiply` delegate, passing the arguments 2 and 4, and assigning the return value to the `result` variable. In line 22, we display the `result` variable in a message box.

Lambda Expressions with No Parameters

If a delegate type specifies no parameters, any lambda expression that you use with the delegate must also have no parameters. For example, consider this delegate type declaration:

```
delegate void TestMessage();
```

Simply write a set of empty parentheses as the parameter list, as shown here:

```
TestMessage Display = () => MessageBox.Show("Testing, 1, 2, 3...");
```

When this lambda expression is invoked, it displays the string `"Testing, 1, 2, 3..."` in a message box.

Lambda Expressions That Do Not Return a Value

If a delegate type specifies a `void` return type, any lambda expression that you use with the delegate will also be `void`. For example, consider this delegate type declaration:

```
delegate void DisplayInt(int x);
```

The following statement creates a delegate of the `DisplayInt` type named `Display`:

```
DisplayInt Display = x => MessageBox.Show(x.ToString());
```

This lambda expression has a parameter, `x`. When the expression is invoked, it displays the value of `x` in a message box.

Explicitly Declaring a Parameter's Data Type

You do not have to specify the data type of a lambda expression's parameter because the compiler is normally able to determine it from the delegate type declaration. However, C# allows you to explicitly declare the data type of a parameter, if you wish. For example, suppose we have the following delegate type declaration:

```
delegate int IntOperation(int x);
```

The lambda expression in the following statement explicitly declares the data type of the x parameter:

```
IntOperation Square = (int x) => x * x;
```

Note that the parameter declaration (on the left side of the = > operator) must be enclosed in parentheses. Here is another example, involving two parameters:

```
delegate void IntCalculator(int a, int b);
IntCalculator Sum = (int a, int b) => a + b;
```

Using Multiple Statements in the Body of a Lambda Expression

You can write multiple statements in the body of a lambda expression, but if you do, you must enclose the statements in a set of curly braces, and you must write a `return` statement if the expression returns a value. For example, consider this delegate type declaration:

```
delegate int IntOperation(int x);
```

The following statement creates a delegate of the `IntOperation` type named `TimesTwo`:

```
IntOperation TimesTwo = (x) => {
     int a = x * 2;
     return a;
};
```

Accessing Outer Variables within a Lambda Expression

A lambda expression can access the local variables that are declared in the method that contains the lambda expression, as long as those variables are in scope and have been assigned a value. The following code demonstrates. Notice that the lambda expression in line 9 accesses the local variable `factor`, which has been assigned the value 10:

```
1 private void demoButton_Click(object sender, EventArgs e)
2 {
3
4    int factor = 10;
5    int result;
6
7    // Create a delegate that multiplies its
8    // argument by factor.
9    IntOperation Multiplier = x => x * factor;
10
11   result = Multiplier(5);
12   MessageBox.Show(result.ToString());
13 }
```

Passing Lambda Expressions as Arguments

You can use a lambda expression anywhere that a compatible delegate is expected. For example, if you are calling a method that expects a delegate to be passed as an argument, you can pass a compatible lambda expression instead. The code in Program 14-6 shows an example.

Program 14-6 Form1 code for the LamdaPassingDemo application

```
1 namespace LambdaPassingDemo
2 {
3     delegate void IntRefOperation(ref int x);
4
5     public partial class Form1 : Form
6     {
7         public Form1()
8         {
9             InitializeComponent();
10        }
11
12        private void ArrayStepper(int[] array, IntRefOperation op)
```

```
13          {
14              for (int index = 0; index < array.Length; index++)
15              {
16                  op(ref array[index]);
17              }
18          }
19
20          private void demoButton_Click(object sender, EventArgs e)
21          {
22              // Create an int array with 4 elements.
23              int[] numbers = { 10, 20, 30, 40 };
24
25              // Display the contents of the array.
26              ArrayStepper(numbers, (ref int x) => MessageBox.
                Show(x.ToString()));
27
28              // Set each array element to 0.
29              ArrayStepper(numbers, (ref int x) => x = 0);
30
31              // Display the contents of the array.
32              ArrayStepper(numbers, (ref int x) => MessageBox.
                Show(x.ToString()));
33          }
34      }
35  }
```

- Line 3 declares a delegate type named `IntRefOperation` that has an `int` reference parameter.
- Line 12 is the header for a method named `ArrayStepper`. The method has two parameters: an `int` array named `array`, and an `IntRefOperation` delegate named `op`.
- Line 23 creates an `int` array named `numbers`, initialized with the values 10, 20, 30, 40.
- Line 26 calls the `ArrayStepper` method passing, as arguments, the `numbers` array and the following lambda expression:

  ```
  (ref int x) => MessageBox.Show(x.ToString())
  ```

  Because the `ArrayStepper` method expects the second argument to be an `IntRefOperation` delegate, this lambda expression is converted to an `IntRefOperation` delegate. As a result, the `ArrayStepper` method displays each of the array elements in a message box.
- Line 29 calls the `ArrayStepper` method passing, as arguments, the `numbers` array and the following lambda expression:

  ```
  (ref int x) => x = 0
  ```

  Because the `ArrayStepper` method expects the second argument to be an `IntRefOperation` delegate, this lambda expression is converted to an `IntRefOperation` delegate. As a result, the `ArrayStepper` method call sets each array element to 0.
- Line 32 performs the same operation as line 26: it displays each element of the array.

Using Lambda Expressions to Define Class Methods

Lambda expressions offer a quick and concise way to define simple methods within a class declaration. For example, look at the following `SimpleMath` class:

```
1 class SimpleMath
2 {
3    public int Add(int a, int b) => a + b;
4    public int Sub(int a, int b) => a - b;
```

```
5     public int Mult(int a, int b) => a * b;
6     public double Div(int a, int b) => (double) a / b;
7 }
```

This class uses lambda expressions to define four methods named `Add`, `Sub`, `Mult`, and `Div`. Compare this code with the following class declaration which uses regular method definitions:

```
1 class SimpleMath
2 {
3     public int Add(int a, int b)
4     {
5         return a + b;
6     }
7
8     public int Sub(int a, int b)
9     {
10        return a - b;
11    }
12
13    public int Mult(int a, int b)
14    {
15        return a * b;
16    }
17
18    public double Div(int a, int b)
19    {
20        return (double) a / b;
21    }
22 }
```

As you can see, the first version of the class, which takes up only seven lines of code, is much more concise than the second version, which takes up 22 lines of code.

Checkpoint

14.8 What is a lambda expression?

14.9 Look at the following expression and answer the questions that follow:

```
x => x * 2
```

a. What is the lambda operator?

b. What is the lambda expression's parameter?

c. What does the lambda expression return?

14.10 Is the following code correct, or does it contain an error? If it contains an error, what is the error?

```
delegate int IntOperation(int a);
IntOperation Sum = (a, b) => a + b;
```

14.11 What variables that are declared outside of a lambda expression can be accessed by the lambda expression?

14.4 Built-In Delegates: Action, Func, and Predicate

CONCEPT: **.NET provides delegate types that simplify the process of creating certain types of delegates.**

So far you've learned that implementing a delegate is a two-step process: first you declare a delegate type, and then you create an instance of the delegate. .NET provides some built-in delegate types that, in many situations, simplify the process of creating delegates. We will look at three of these:

- `Action`—Invokes a `void` method that accepts 0 to 16 arguments
- `Func`—Invokes a method that accepts 0 to 16 arguments and returns a value
- `Predicate`—Invokes a method that accepts one argument and returns a `boolean` value

When you create an `Action`, `Func`, or `Predicate` delegate, you do not have to declare the delegate type because these types are already declared in .NET. Let's take a closer look at each type.

`Action` Delegates

You can use an **`Action` delegate** to invoke a `void` method that takes 0 to 16 arguments. To use an `Action` delegate, you do not have to declare the delegate type because it is built into .NET. You simply create an instance of the delegate. Here is an example of how to use a lambda expression to create an `Action` delegate that accepts no arguments:

```
Action DisplayTest = () => MessageBox.Show("Testing, 1, 2, 3...");
```

We could use an anonymous method to create an equivalent `Action` delegate, as shown here:

```
Action DisplayTest = delegate()
{
    MessageBox.Show("Testing, 1, 2, 3...");
};
```

In both of these examples, the name of the delegate is `DisplayTest`. When this delegate is invoked, it displays the string `"Testing, 1, 2, 3..."` in a message box. Regardless of whether we used a lambda expression or an anonymous method to create the delegate, we would invoke the delegate like this:

```
DisplayTest();
```

Let's look at a complete application that demonstrates this `Action` delegate. Program 14-7 shows the Form1 code for the *ActionDelegateDemo* application.

Program 14-7 Form1 code for the *ActionDelegateDemo* application

```
 1 namespace ActionDelegateDemo
 2 {
 3     public partial class Form1 : Form
 4     {
 5         public Form1()
 6         {
 7             InitializeComponent();
 8         }
 9
10         private void demoButton_Click(object sender, EventArgs e)
11         {
12             // Create an Action delegate.
13             Action DisplayTest = () => MessageBox.Show
               ("Testing, 1, 2, 3...");
14
15             // Invoke the delegate.
16             DisplayTest();
```

```
17          }
18      }
19  }
```

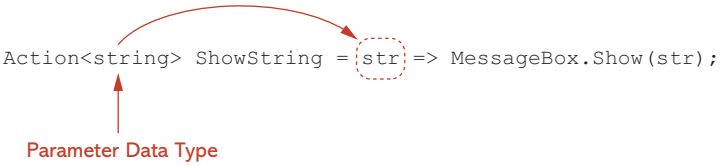
Passing Arguments to an `Action` *Delegate*

If you want to pass arguments to an `Action` delegate, you must specify the data type of each of the delegate's parameters. Here is an example of how to create an `Action` delegate that has one parameter, a `string`:

```
Action<string> ShowString = str => MessageBox.Show(str);
```

In this statement, the delegate type is `Action<string>`. As previously mentioned, this specifies that we are creating an `Action` delegate that has one parameter, a `string`. Notice that on the right side of the `=` operator, the lambda expression has a parameter named `str`. The data type of the `str` parameter is `string`. This is illustrated in Figure 14-4.

Figure 14-4 An `Action` delegate with one parameter (lambda expression version)

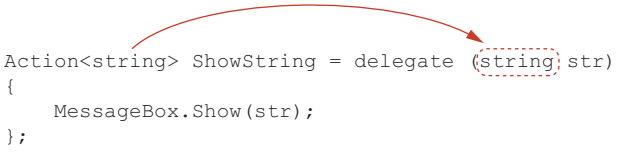


We could use an anonymous method to create an equivalent `Action<string>` delegate, as shown here:

```
Action<string> ShowString = delegate(string str)
{
    MessageBox.Show(str);
};
```

Notice that in the anonymous method, we have to specify the parameter data type twice, as illustrated in Figure 14-5.

Figure 14-5 An `Action` Delegate with one parameter (anonymous method version)



As previously mentioned, an `Action` delegate can take 0 to 16 arguments. Here is an example of using a lambda expression to create an `Action` delegate that has two parameters:

```
Action<int, int> ShowSum = (a, b) => {
    int result = a + b;
    MessageBox.Show(result.ToString());
};
```

In this statement, the delegate type is `Action<int, int>`. This specifies that we are creating an `Action` delegate that has two parameters, both of which are `int`s. Notice that on the right side of the `=` operator, the lambda expression has two parameters named `a` and `b`. Both of these parameters are `int`s.

Here is another example:

```
Action<string, int> ShowMessage = (str, x) => {
    MessageBox.Show(str + x.ToString());
};
```

In this statement, the delegate type is `Action<string, int>`. This specifies that we are creating an `Action` delegate that has two parameters: the first parameter is a string, and the second parameter is an `int`. Notice that on the right side of the = operator, the lambda expression has two parameters named `str` and `x`. The `str` parameter is a `string` and the `x` parameter is an `int`.

An `Action` delegate can have up to 16 parameters. You simply list the data types of each parameter inside the brackets, in the order that the parameters are declared. For example, the following code creates an `Action` delegate with four parameters:

```
Action<int, double, string> ShowValues = (a, b, c) => {
    MessageBox.Show($"{a} {b} {c}");
};
```

In this code, the parameter `a` is an `int`, the parameter `b` is a `double`, and the parameter `c` is a `string`.

Func Delegates

You can use a **Func delegate** to invoke a value-returning method that takes 0 to 16 arguments. Here is an example of how to use a lambda expression to create a `Func` delegate that accepts no arguments and returns a `string`:

```
Func<string> GetString = () => "This is a test.";
```

We could use an anonymous method to create an equivalent `Action` delegate, as shown here:

```
Func<string> GetString = delegate()
{
    return "This is a test.";
};
```

In both of these examples, the name of the delegate is `GetString`. When this delegate is invoked, it returns the string `"This is a test."`. Here is an example of how we might invoke the method:

```
MessageBox.Show(GetString());
```

This statement will display the string `"This is a test."` in a message box.

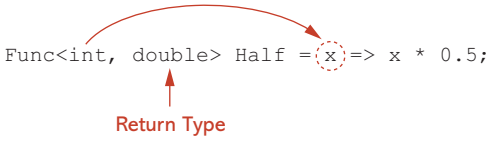
Passing Arguments to a Func Delegate

If you want to pass arguments to a `Func` delegate, you must specify the data type of each of the delegate's parameters as well as the data type of the value that is returned from the delegate. Here is an example of how to create a `Func` delegate that returns a `double`, and has one parameter, an `int`:

```
Func<int, double> Half = x => x * 0.5;
```

Inside the brackets, `int` is the data type of the parameter `x` and `double` is the delegate's return type. This is illustrated in Figure 14-6.

Figure 14-6 A **Func** delegate with one parameter (lambda expression version)

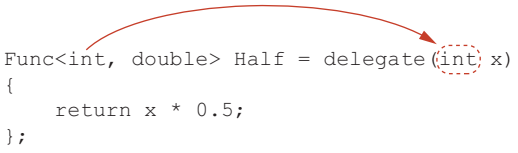


We could use an anonymous method to create an equivalent Func<int, double> delegate, as shown here:

```
Func<int, double> Half = delegate(int x)
{
    return x * 0.5;
};
```

Notice that in the anonymous method we specify the parameter data type twice, as illustrated in Figure 14-7.

Figure 14-7 A **Func** delegate with one parameter (anonymous method version)



In both of these examples, the name of the delegate is Half. When this delegate is invoked, it accepts an int argument and it returns half of the argument's value as a double. Here is an example of how we might invoke the delegate (assume result is a double that has already been declared):

```
result = Half(5);
```

This statement will assign the value 2.5 to the result variable. Let's look at a complete application that demonstrates this Func delegate. Program 14-8 shows the Form1 code for the *FuncDelegateDemo* application. When the application runs and the user clicks the demoButton control, the statement in line 15 displays *Half of 25 is 12.5* in a message box.

Program 14-8 Form1 code for the *FuncDelegateDemo* application

```
 1 namespace FuncDelegateDemo
 2 {
 3     public partial class Form1 : Form
 4     {
 5         public Form1()
 6         {
 7             InitializeComponent();
 8         }
 9
10         private void demoButton_Click(object sender, EventArgs e)
11         {
12             Func<int, double> Half = x => x * 0.5;
13             double result = Half(25);
14
15             MessageBox.Show($"Half of 25 is {result}");
16         }
17     }
18 }
```

A Func delegate can have up to 16 parameters. You simply list the data types of each parameter inside the brackets, in the order that the parameters are declared, followed by the delegate's return type. For example, the following code creates a Func delegate that has two parameters (an int and a double) and returns a decimal:

```
Func<int, double, decimal> Sum = (a, b) => (decimal)a + (decimal)b;
```

In this code, the parameter `a` is an `int`, the parameter `b` is a `double`, and the delegate's return type is `decimal`.

`Predicate` Delegates

You can use a **`Predicate` delegate** to invoke a method that accepts one argument and returns a `boolean` value. Here is an example of a `Predicate` delegate that has an `int` parameter:

```
Predicate<int> IsEven = x => x % 2 == 0;
```

We could use an anonymous method to create an equivalent `Predicate` delegate, as shown here:

```
Predicate <int> IsEven = delegate(int x)
{
    return x % 2 == 0;
};
```

In both of these examples, the name of the delegate is `IsEven`. It accepts an `int` argument and returns `true` if the argument is an even number. Otherwise it returns `false`. Here is an example of how we might invoke the method:

```
if (IsEven(num))
    MessageBox.Show("The number is even.");
```

This statement will display the string `"The number is even."` in a message box if the `num` variable contains an even number.

`List` Methods That Work with Delegates

The `List` collection that we discussed in Chapter 7 provides several methods that accept a delegate as an argument. Each of these methods uses the delegate to perform an operation on the elements of the list. Table 14-1 describes these methods.

Table 14-1 **`List`** methods that accept delegates as arguments

Method	Description
`Exists`	**Determines whether an element exists in the `List`** The `Exists` method accepts a `Predicate` delegate as its argument. The `Predicate` must accept an argument of the same data type as the `List` elements. The `Exists` method iterates over the `List`, passing each element as an argument to the `Predicate`. The `Exists` method returns `true` if the `Predicate` returned `true` for any of the elements. Otherwise, the `Exists` method returns `false`.
`Find`	**Finds the first element for which a condition is true** The `Find` method accepts a `Predicate` delegate as its argument. The `Predicate` must accept an argument of the same data type as the `List` elements. The `Find` method returns the first element that, when passed to the `Predicate`, causes the `Predicate` to return `true`.
`FindAll`	**Finds all the elements for which a condition is true** The `FindAll` method accepts a `Predicate` delegate as its argument. The `Predicate` must accept an argument of the same data type as the `List` elements. The `FindAll` method returns a `List` of all the elements that, when passed to the `Predicate`, causes the `Predicate` to return `true`.

Table 14-1 `List` methods that accept delegates as arguments *(continued)*

Method	Description
`FindIndex`	**Finds the index of the first element for which a condition is true** The `FindIndex` method accepts a `Predicate` delegate as its argument. The `Predicate` must accept an argument of the same data type as the `List` elements. The `FindIndex` method returns the index of the first element that, when passed to the `Predicate`, causes the `Predicate` to return `true`.
`FindLast`	**Finds the last element for which a condition is true** The `FindLast` method accepts a `Predicate` delegate as its argument. The `Predicate` must accept an argument of the same data type as the `List` elements. The `FindLast` method returns the last element that, when passed to the `Predicate`, causes the `Predicate` to return `true`.
`FindLastIndex`	**Finds the index of the last element for which a condition is true** The `FindLastIndex` method accepts a `Predicate` delegate as its argument. The `Predicate` must accept an argument of the same data type as the `List` elements. The `FindLastIndex` method returns the index of the last element that, when passed to the `Predicate`, causes the `Predicate` to return `true`.
`ForEach`	**Performs an action on each element of the `List`** The `ForEach` method accepts an `Action` delegate as its argument. The `Action` delegate must accept one argument. The argument's data type must match the data type of the `List` elements. The method iterates over the `List`, passing each element as an argument to the delegate.
`RemoveAll`	**Removes all elements for which a condition is true** The `RemoveAll` method accepts a `Predicate` delegate as its argument. The `Predicate` must accept an argument of the same data type as the `List` elements. The `RemoveAll` method removes all the elements that, when passed to the `Predicate`, causes the `Predicate` to return `true`.
`TrueForAll`	**Determines whether a condition is true for all elements in the `List`** The `TrueForAll` method accepts a `Predicate` delegate as its argument. The `Predicate` must accept an argument of the same data type as the `List` elements. The `TrueForAll` method returns true if all the `List` elements, when passed to the `Predicate`, cause the `Predicate` to return `true`. Otherwise, the `TrueForAll` method returns `false`.

The methods listed in Table 14-1 are straightforward, but let's look at a couple of them. First, the `ForEach` method. Here is the general format:

```
listName.ForEach(ActionDelegate)
```

In the general format, *`listName`* is the name of a `List`. The *`ActionDelegate`* argument is an `Action` delegate that accepts one argument. The `ForEach` method iterates over the entire `List`, passing each element as an argument to *`ActionDelegate`*. Let's look at a complete application that demonstrates the `ForEach` method. Program 14-9 shows the Form1 code for the *ForEachDemo1* application.

Program 14-9 Form1 code for the *ForEachDemo1* application

```
1 namespace ForEachDemo1
2 {
3     public partial class Form1 : Form
```

```
4     {
5         public Form1()
6         {
7             InitializeComponent();
8         }
9
10        private void demoButton_Click(object sender, EventArgs e)
11        {
12            // Create a list of integers.
13            var numbers = new List<int>() { 1, 2, 3, 4, 5 };
14
15            // Create an Action delegate that displays an integer.
16            Action<int> Display = x => MessageBox.Show(x.ToString());
17
18            // Invoke the Display delegate for each element of the List.
19            numbers.ForEach(Display);
20        }
21    }
22 }
```

Let's take a closer look at the code:

- The code in line 13 creates a `List` of integers named `numbers`, and initializes the `List` with the values 1, 2, 3, 4, and 5.
- Line 16 creates an `Action` delegate named `Display` with an `int` parameter named `x`. When the delegate is invoked, the `x` parameter's value is displayed in a message box.
- Line 19 calls the numbers `List`'s `ForEach` method, passing the `Display` delegate as an argument. The `ForEach` method iterates over the `List`, passing each element as an argument to the `Display` delegate. As a result, five message boxes will appear (one after the other), with each showing the value of an element.

Remember, if a method accepts a delegate as an argument, you can pass a compatible lambda expression to the method. We can simplify the code in Program 14-9 by passing a lambda expression to the `List`'s `ForEach` method. This allows us to eliminate the delegate declaration that appears in line 20. Program 14-10 shows the revised code.

Program 14-10 Form1 code for the *ForEachDemo2* application

```
1 namespace ForEachDemo2
2 {
3     public partial class Form1 : Form
4     {
5         public Form1()
6         {
7             InitializeComponent();
8         }
9
10        private void demoButton_Click(object sender, EventArgs e)
11        {
12            // Create a list of integers.
13            var numbers = new List<int>() { 1, 2, 3, 4, 5 };
14
15            // Display each element of the List.
16            numbers.ForEach(x => MessageBox.Show(x.ToString()));
17        }
18    }
19 }
```

This program works the same as the code previously shown in Program 14-9. In line 16, we pass the following lambda expression as an argument to the ForEach method:

```
x => MessageBox.Show(x.ToString())
```

When the application is compiled, this lambda expression will be converted to an Action delegate and passed to the ForEach method.

Let's also look at the RemoveAll method. Here is the general format:

```
listName.RemoveAll(PredicateDelegate)
```

In the general format, *listName* is the name of a List. The *PredicateDelegate* argument is a Predicate delegate. The RemoveAll method iterates over the entire List, passing each element as an argument to *PredicateDelegate*. If the *PredicateDelegate* returns true, the element is removed from the List. If the *PredicateDelegate* returns false, no action is taken for that element. Let's look at a complete application that demonstrates the RemoveAll method. Program 14-11 shows the Form1 code for the *RemoveAllDemo* application.

Program 14-11 Form1 code for the *RemoveAllDemo* application

```
 1 namespace RemoveAllDemo
 2 {
 3     public partial class Form1 : Form
 4     {
 5         public Form1()
 6         {
 7             InitializeComponent();
 8         }
 9
10         private void demoButton_Click(object sender, EventArgs e)
11         {
12             // Create a list of integers.
13             var numbers = new List<int>() { -3, -2, -1, 0, 1, 2 };
14
15             // Remove all the elements containing a negative value.
16             numbers.RemoveAll(x => x < 0);
17
18             // Display each element of the List.
19             numbers.ForEach(x => MessageBox.Show(x.ToString()));
20         }
21     }
22 }
```

Let's take a closer look at the code:

- The code in line 13 creates a List of integers named numbers, and initializes the List with the values –3, –2, –1, 0, 1, and 2.
- Line 16 calls the List's RemoveAll method, passing the following lambda expression as an argument:

  ```
  x => x < 0
  ```

  This lambda expression is converted by the compiler to a Predicate delegate that returns true if the argument x is less than 0, or false otherwise. As a result, line 16 removes all the List's elements that contain a negative number. Once this completes, the List will contain the values 0, 1, and 2.
- Line 19 calls the numbers List's ForEach method, passing a lambda expression that displays its argument in a message box. As a result, three message boxes will appear (one after the other), showing the values 0, 1, and 2.

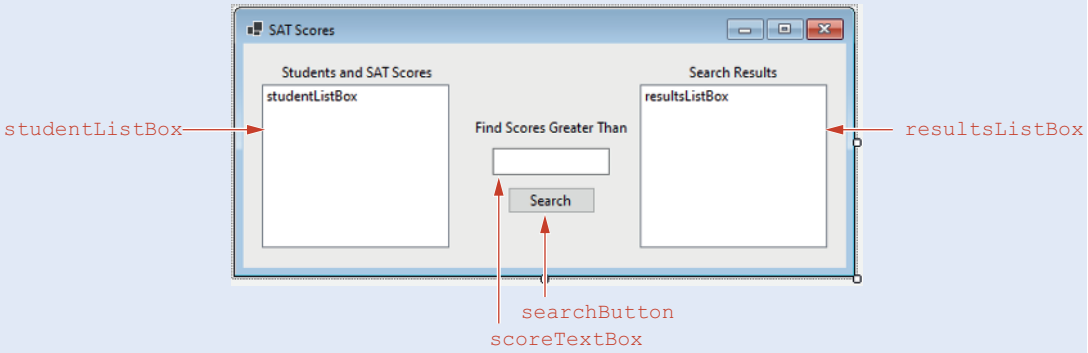
In Tutorial 14-1, you complete an application that uses the `FindAll` method to find all the values in a List that are greater than a specified value.

Tutorial 14-1:
Working with `List` Methods and Lambdas

VideoNote
Tutorial 14-1: Working with `List` Methods and Lambdas

In this tutorial you complete the *SATScores* project. The application's form is shown in Figure 14-8. When the application runs, it displays a list of students and their SAT scores in the `studentListBox`. Once you have completed the tutorial, the application will allow you to search for students with an SAT score greater than a specified minimum.

Figure 14-8 The *SATScores* Project's Form



The application stores each student's name and SAT score in a `StudentScore` class, which has already been written and is included in the project. The `StudentScore` class is shown in Program 14-12. Here is a summary of the class:

- Lines 12–16 is the constructor, which accepts arguments for the student's name and SAT score. These values are assigned to the Name and Score properties.
- Line 19 is the declaration of the Name property, a `string`. This property holds a student's name.
- Line 20 is the declaration of the Score property, an `int`. This property holds a student's SAT score.

Step 1: Open the project named *SATScores* in the *Chap14* folder of this book's Student Sample Programs.

Step 2: Open the Form1 form in the Designer. The form is shown, along with the names of the important controls, in Figure 14-8.

Step 3: Now, you will create the Click event handler for the `searchButton` control. At the end of this tutorial, Program 14-13 shows the completed code for the form. You will be instructed to refer to Program 14-13 as you write the event handlers.

Note: The code in lines 5–16 of Program 14-13 has already been written for you. This code creates a `List` named `studentList` and initializes it with eight `StudentScore` objects.

In the *Designer*, double-click the `searchButton` control. This will open the code editor, and you see an empty event handler named `searchButton_Click`. Complete the `searchButton_Click` event handler by

typing the code shown in lines 34–54 in Program 14-13. Let's take a closer look at the code:

Line 34: This statement declares a local variable named `minScore`. We will use the variable to hold the score that the user enters into the `scoreTextBox` control.

Line 37: This `if` statement tries to convert `scoreTextBox.Text` to an `int`. If the conversion is successful, the result is stored in the `minScore` variable, and the program continues executing at line 39. If the conversion is not successful, the program jumps to the `else` clause in line 51, and line 53 displays an error message.

Line 40: This statement clears the contents of the `resultsListBox` control.

Line 43: This statement performs multiple steps. (1) It declares a variable named `results`. (2) It calls the `studentList` object's `FindAll` method. (3) It assigns the value that is returned from the `FindAll` method to `results`.

It can be seen from Table 14-1 that the `FindAll` method accepts a `Predicate` delegate as its argument. The method returns a `List` of all the elements that, when passed to the `Predicate`, causes the `Predicate` to return `true`. In this statement, we pass the following lambda expression as the `Predicate`:

```
s => s.Score > minScore
```

The expression works like this: `s` is a `StudentScore` object. The `Predicate` returns `true` if `s.Score` is greater than `minScore`. Otherwise, the `Predicate` returns `false`. So, the `FindAll` method will return a `List` of all the `StudentScore` objects that have a `Score` property greater than `minScore`.

Lines 46–49: This `foreach` statement displays all the elements of `results`.

Step 4: Save the project. Then, compile and run the application. Experiment by searching for students with scores that are greater than various values. When you are finished, close the application.

Program 14-12 The **`StudentScore`** class

```
 1 using System;
 2 using System.Collections.Generic;
 3 using System.Linq;
 4 using System.Text;
 5 using System.Threading.Tasks;
 6
 7 namespace SATScores
 8 {
 9     internal class StudentScore
10     {
11         // Constructor
12         public StudentScore(string n, int s)
13         {
14             Name = n;
15             Score = s;
16         }
17
18         // Properties
19         public string Name { set; get; }
20         public int Score { set; get; }
21     }
22 }
```

Program 14-13 Form1 code for the *SATScores* project

```
 1 namespace SATScores
 2 {
 3     public partial class Form1 : Form
 4     {
 5         // List of students and their SAT scores
 6         private List<StudentScore> studentList = new
           List<StudentScore>()
 7         {
 8             new StudentScore("Katie Allen", 1275),
 9             new StudentScore("Carmen Cabera", 1060),
10             new StudentScore("Rosa Lopez", 1155),
11             new StudentScore("Ingrid Olen", 1310),
12             new StudentScore("Sanjay Patel", 975),
13             new StudentScore("Jamal Ramzi", 1260),
14             new StudentScore("Kelsey Rose", 1055),
15             new StudentScore("Asha Singh", 920)
16         };
17
18         public Form1()
19         {
20             InitializeComponent();
21         }
22
23         private void Form1_Load(object sender, EventArgs e)
24         {
25             // Populate the studentListBox
26             foreach (var student in studentList)
27             {
28                 studentListBox.Items.Add($"{student.Name}
                   ({student.Score})");
29             }
30         }
31
32         private void searchButton_Click(object sender,
           EventArgs e)
33         {
34             int minScore;
35
36             // Get the minimum score
37             if (int.TryParse(scoreTextBox.Text, out minScore))
38             {
39                 // Clear the resultsListBox
40                 resultsListBox.Items.Clear();
41
42                 // Get the students with a score greater than
                   minScore.
43                 var results = studentList.FindAll(s => s.Score
                   > minScore);
44
45                 // Display those students.
46                 foreach (var student in results)
47                 {
48                     resultsListBox.Items.Add($"{student.Name}
                       ({student.Score})");
49                 }
50             }
```

```
51              else
52              {
53                  MessageBox.Show("Enter an integer score.");
54              }
55          }
56      }
57  }
```

Checkpoint

14.12 What is an `Action` delegate?

14.13 What is a `Func` delegate?

14.14 What is a `Predicate` delegate?

Key Terms

`Action` delegate
anonymous method
delegate
delegate type
`Func` delegate
lambda expression
lambda operator
multicast delegate
`Predicate` delegate

Review Questions

Multiple Choice

1. You can use a delegate to reference a(n) _____.
   a. Object
   b. Variable
   c. Method
   d. Structure

2. Before you can create a delegate, you must declare _____.
   a. A delegate type
   b. A delegate variable
   c. The delegate's lifetime
   d. The delegate's scope

3. A delegate can reference _____.
   a. Any method whatsoever
   b. Any `void` method
   c. Any method that has a parameter list matching that of the delegate type.
   d. Any method that has a return type and parameter list matching that of the delegate type.

4. A delegate that references multiple methods is known as a(n) _____.
   a. Compound delegate
   b. Multicast delegate
   c. Complex delegate
   d. Error! Delegates cannot refer to more than one method.

5. A(n) _____ method has no name.
   a. Untyped
   b. Anonymous
   c. Hidden
   d. Protected

6. A lambda expression is a shortcut technique for writing a(n) _____.
   a. Extension method
   b. `if-else` statement
   c. Anonymous method
   d. Unsupervised delegate

7. _____ is the lambda operator.
   a. `=>`
   b. `->`
   c. `<=`
   d. `<=>`

8. If a lambda expression contains multiple statements in its body, those statements must be enclosed in _____.
   a. Curly braces `{ }`
   b. Square brackets `[]`
   c. Parentheses `()`
   d. Angled brackets `<>`

9. This type of delegate invokes a `void` method that accepts 0 to 16 arguments.
   a. `Func`
   b. `Predicate`
   c. `Action`
   d. `Automaton`

10. This type of delegate invokes a method that accepts one argument and returns a `boolean` value.
    a. `Func`
    b. `Predicate`
    c. `Action`
    d. `Automaton`

11. This type of delegate invokes a method that accepts 0 to 16 arguments and returns a value.
    a. `Func`
    b. `Predicate`
    c. `Action`
    d. `Automaton`

True or False

1. You can use a delegate to call a method.
2. Once a delegate has been assigned, it cannot be reassigned.
3. You can use a delegate to pass a method as an argument to another method.
4. A delegate is compatible with any method that has a return type and parameter list matching that of the delegate's type.
5. A delegate cannot refer to an anonymous method.

6. An anonymous method is a shortcut technique for writing a lambda expression.
7. You can assign a lambda expression to a delegate.
8. Writing a lambda expression usually requires more code than writing an anonymous method.
9. A lambda expression can be passed as an argument to a method.

Short Answer

1. Suppose you have already declared a delegate type, and you are ready to create an instance of the delegate. What types of methods can the delegate refer to?
2. What is an anonymous method?
3. In a lambda expression, what appears on the left side of the lambda operator?
4. When must a lambda expression have a return statement?
5. What variables outside of a lambda expression can the lambda expression access?
6. Describe each of the following predefined delegate types: `Action`, `Func`, and `Predicate`.

Algorithm Workbench

1. Write a delegate type declaration named `DisplayInteger` that is compatible with this method:

```
private void ShowValue(int i)
{
    MessageBox.Show(i.ToString());
}
```

2. Write a statement that creates an instance of the delegate type that you declared in question 1, named `ShowIntValue`. The delegate should reference the `ShowValue` method.
3. Write a statement that uses the delegate that you created in question 2 to call the `ShowValue` method. Assume `number` is an `int` variable. Pass `number` as an argument to the method.
4. Suppose you have two methods named `MethodA` and `MethodB`. The delegate `MyDelegate` already refers to `MethodA`. Write a statement that adds `MethodB` to the delegate. (The delegate should refer to both `MethodA` and `MethodB`.)
5. Look at this delegate type declaration:

```
delegate void WorksWithAnInt(int i);
```

   Write the code to create an instance of the delegate type named `ShowInt`. The code should also create an anonymous method that displays the value or the parameter `i` in a message box.

6. Write a lambda expression that has a parameter named `x`, and returns the value of `x` times 2.
7. Write code that does the following:
   - Uses a lambda expression that accepts two `int` arguments, `a` and `b`, and returns the value of `a * b`.
   - Assigns the lambda expression to a delegate named `Multiply`.
   - Uses the `Multiply` delegate to call the lambda expression, passing the values 2 and 10 as arguments. The result should be assigned to an `int` variable named `product`.

8. Write a lambda expression that accepts no arguments and displays "Hello World" in a message box.

9. Given the following class declaration:

```
class MyClass
{
   public int MyMethod(int a, int b) => a * b;
}
```

What value will be stored in `x` after the following code executes?

```
MyClass obj = new MyClass();
int x = obj.MyMethod(2, 4);
```

10. Write code that assigns a lambda expression to a `Predicate` delegate named `LessThan100`. The delegate should have one `int` parameter and return `true` if the parameter is less than 100. Otherwise, the delegate should return `false`.

Programming Challenges

VideoNote
Solving the List Manipulator Problem

1. **List Manipulator**

   In the student sample program files for this chapter, you will find a text file named random.txt. The file contains 100 random integers. Write an application that reads the contents of the file into a `List`. Then, use the `List` methods discussed in this chapter to perform the following:

   - Use the `RemoveAll` method to remove all the `List` elements that contain a negative number.
   - Use the `FindAll` method to create a second `List` containing all the values that are within the range of 1 through 10. Display these values in a ListBox.

2. **Prime Number Generation**

   A positive integer greater than 1 is said to be *prime* if it has no divisors other than 1 and itself. A positive integer greater than 1 is *composite* if it is not prime. Create an application that asks the user to enter an integer greater than 1, and then displays all of the prime numbers that are less than or equal to the number entered. The program should work as follows:

   - Once the user has entered a number, the program should populate a `List` with all of the integers from 2, up through the value entered.
   - The application should then use the `FindAll` method to find all of the elements that contain a prime number.
   - The application should display all of the elements that were returned from the `FindAll` method in a ListBox.

3. **House Price Analysis**

   In the student sample program files for this chapter, you will find a text file named house_prices.csv. The file contains the following data about houses for sale in a city:

   - Price
   - Number of bedrooms
   - Number of bathrooms
   - Square feet of living space

   The data stored in each line is formatted in the following way:

   Price,Bedrooms,Bathrooms,SquareFeet

   Notice that the data items are separated by commas. The first line in the file looks like this:

   ```
   221900,3,1,1180
   ```

The data stored in this line is:

- Price = $221,900
- Bedrooms = 3
- Bathrooms = 1
- Square feet = 1,180

Design a class that has properties for the four pieces of data. In an application, read the file and store each house's data in an instance of the class. The class instances should be stored in a `List`.

The application should let the user search the `List` for the prices of houses with the number of bedrooms within a range, the number of bathrooms within a range, and the square feet of living space within a range. For example, all houses with two to four bedrooms, two to three bathrooms, and 1500–2500 square feet of living space. The application should display the data for each house that matches the search criteria in a ListBox.

Here are some tips:

- Use the `Split` method that we discussed in Chapter 8 to tokenize each line of the file into the four pieces of data.
- Use the `FindAll` method discussed earlier in this chapter to search the `List`.

4. English Surnames

In the student sample program files for this chapter, you will find a text file named surnames.txt. The file contains 14,674 English surnames. Each line in the file has one surname. Write a program that reads the names from the file into a `List`. The program should let the user perform the following searches:

- Find all the names that are longer than a specified length.
- Find all the names that are shorter than a specified length.
- Find all the names that begin with a specified string of characters. (For example, find all the names that begin with "All".)
- Determine whether a specified name exists in the `List`.

5. Dow Jones Industrial Average Highs

In the student sample program files for this chapter, you will find a text file named DJIA.csv. The file contains the following daily data about the Dow Jones Industrial Average for the year 2017:

- Date
- Closing Value
- Opening Value
- Highest Value for the Day
- Lowest Value for the Day

The data stored in each line is formatted in the following way:

```
Date,ClosingValue,OpeningValue,High,Low
```

Notice that the data items are separated by commas. The first line in the file looks like this:

```
3-Jan-17,19881.76,19872.86,19938.53,19775.93
```

The data stored in this line is:

- Date = January 3, 2017
- Closing Value = 19,881.76
- Opening Value = 19,872.86
- High Value for this date = 19,938.53
- Low Value for this date = 19,775.93

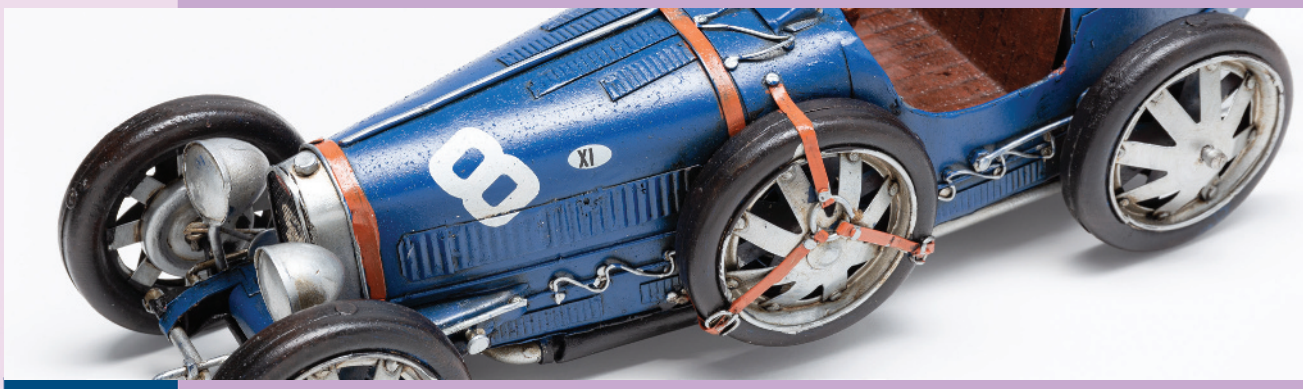
Create a class that has properties for the five pieces of data. In an application, read the file and store each day's data in an instance of the class. The class instances should be stored in a `List`.

The application should let the user search the `List` for:

- The first date that the closing value was at a specified value, or greater
- All the dates that the closing value was at or greater than a specified value

Here are some tips:

- Use the `Split` method that we discussed in Chapter 8 to tokenize each line of the file into the four pieces of data.
- Use the `Find` and `FindAll` methods discussed earlier in this chapter to search the `List`.



CHAPTER

15 Language-Integrated Query (LINQ)

TOPICS

15.1 Introduction to LINQ
15.2 LINQ Method Syntax
15.3 More LINQ Extension Methods
15.4 Using LINQ to Query a Database

15.1 Introduction to LINQ

CONCEPT: **LINQ (Language-Integrated Query) is a query language that is built into C# and can be used to query data from many sources.**

In Chapter 13 you learned about SQL (Structured Query Language), which allows you to query the data in a database. .NET also provides another querying language named **LINQ**, which stands for **Language-Integrated Query**. While SQL allows you to query the data in a database, .NET provides several LINQ tools that allow you to query many types of data. In this section, we will use a tool called **LINQ to Objects**, which allows you to query data that are stored in collections, such as arrays and lists.

A statement that uses a LINQ query is written in the following general format:

```
var resultVariable = LINQquery;
```

In the general format, *resultVariable* is a variable that we are declaring to hold the results of the LINQ query, and *LINQquery* is the LINQ query. Notice that instead of specifying a data type for *resultVariable*, we are using the `var` key word in the declaration. The C# compiler will automatically determine the data type for the variable.

Let's look at an example. Suppose we have declared the following array of integers:

```
int[] numbers = { 4, 104, 2, 102, 1, 101, 3, 103 };
```

If we want to query this array to get all the values that are greater than 100, we can write the following statement:

```
var results = from item in numbers
              where item > 100
              select item;
```

This statement declares a variable named `results`, which will hold the results of the LINQ query. On the right side of the = operator is the LINQ query. The = operator will assign the results of the LINQ query to the `results` variable. The LINQ query reads:

```
from item in numbers
where item > 100
select item
```

This query will return all the items in the `numbers` array that are greater than 100. To understand how the query works, let's look at each line of code in the query. The first line reads:

```
from item in numbers
```

This expression works like a `foreach` loop that starts like this:

```
foreach(item in numbers)
```

In the code, `item` is the name of an iteration variable, and `numbers` is the name of an array. The query iterates once for each element of the `numbers` array. Each time it iterates, it copies an array element to the `item` variable. The first time the query iterates, the `item` variable will contain the value of element 0, the second time the query iterates, the `item` variable will contain the value of element 1, and so forth.

The second and third lines of the query read as follows:

```
where item > 100
select item
```

This code is similar to an SQL `Select` statement, except that in LINQ the `where` clause appears before the `select` clause. The `where` clause specifies a Boolean search criterion (in this case, `item > 100`), and the `select` clause causes the value of the `item` variable to be retrieved if the search criterion is true. In plain English, this expression says "select each item that is greater than 100."

After the statement executes, we can use a loop to examine the values in the `results` variable. For example, the following code segment shows how we can add the values to a list box named `resultsListBox`. After the code executes, the list box will contain the values 104, 102, 101, and 103.

```
int[] numbers = { 4, 104, 2, 102, 1, 101, 3, 103 };
var results = from item in numbers
              where item > 100
              select item;
foreach (var value in results)
```

```
{
   resultsListBox.Items.Add(value);
}
```

You can also use LINQ to query the data in a `List`, as shown in the following code snippet:

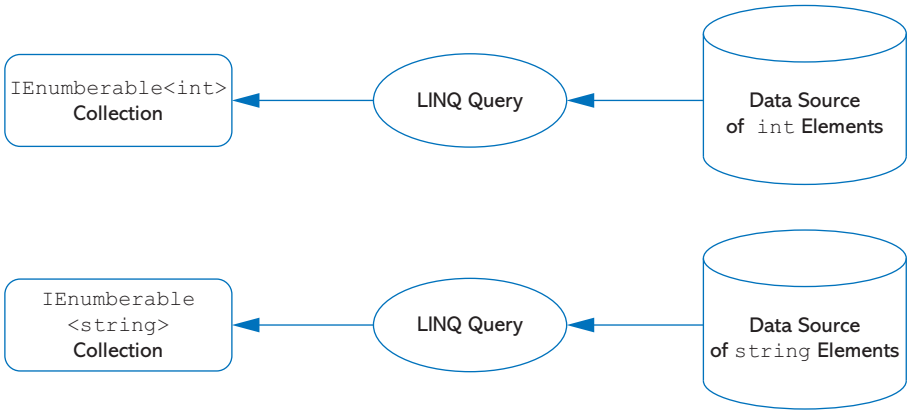
```
var numbers = new List<int>() { 4, 104, 2, 102, 1, 101, 3, 103 };

var results = from item in numbers
              where item > 100
              select item;
foreach (var value in results)
{
    resultsListBox.Items.Add(value);
}
```

The `IEnumberable<T>` Interface

You use LINQ queries to search for, and retrieve, specific pieces of data from a source of data. The source can be an in-memory collection, such as an array or a `List`, or it can be a larger repository of data, such as a database. When you execute a LINQ query, the query returns its results as an **`IEnumerable<T>`** collection, where *`T`* is the data type of the individual elements in the source of data.

For example, if the query's data source is an array of `int`s, the query will return an `IEnumerable<int>` collection. Or, if the query's data source is a `List` of `string`s, the query will return an `IEnumerable<string>` collection. This is illustrated in Figure 15-1.

Figure 15-1 LINQ queries return their results as an `IEnumberable<T>` collection



`IEnumerable<T>` is an interface that is defined in the .NET Framework. An `IEnumerable<T>` collection is simply a collection, like a `List`, that you can iterate over. So, querying data with LINQ is usually a process of executing a query on a data source, getting the results as a collection, and then iterating over the collection to perform some operation with its elements.

At this point, you would have perhaps realized that in the two previous LINQ examples, you did not see the word `IEnumerable` in any of the code. Here is the previously shown code that queries an `int` array:

```
int[] numbers = { 4, 104, 2, 102, 1, 101, 3, 103 };
var results = from item in numbers
              where item > 100
              select item;
```

In this statement, the `results` variable is an `IEnumerable<int>` collection, but instead of declaring it as `IEnumerable<int>`, we used the `var` keyword. Recall that the `var` keyword causes the compiler to determine a variable's data type from its initialization value. So, in this example the data type of the `results` variable will be the data type of the value that is returned from the LINQ expression, which is `IEnumerable<int>`. It is the same as if we had written:

```
IEnumerable<int> results = from item in numbers
                           where item > 100
                           select item;
```

As you can see, the `var` keyword simplifies the code. For that reason, most programmers prefer to use it.

More about the `select` Operator

LINQ queries begin with a `from` clause, and they usually end with a `select` clause. The `from` clause iterates over a collection of data elements, assigning each element to an iteration variable. The `select` clause determines what will be included in the results of the query, and it returns the results as an `IEnumerable<T>` collection. The `select` clause also determines what data type *T* will be. For example, look at the following code snippet:

```
1  int[] numbers = { 4, 104, 2, 102, 1, 101, 3, 103 };
2
3  var results = from item in numbers
4                select item;
5
6  foreach (var value in results)
7  {
8     MessageBox.Show(value);
9  }
```

The LINQ query that appears in lines 3 and 4 is very simple. Since there is no `where` clause to filter the data, it retrieves every item in the `numbers` array. The data types of the various variables used in the statement will be as follows:

- The query is searching an `int` array, so in line 3 the data type of the `item` variable will be `int`.
- In line 4, the `select` statement is selecting an `int` (the `item` variable), so the query will return an `IEnumerable<int>` collection. This also means that in line 3 the data type of the `results` variable will be `IEnumerable<int>`.
- In line 6, the data type of the `value` variable will be `int`.

Here is an example that works with a `string` array:

```
1  string[] names = { "Joe", "Jill", "John", "Jennifer" };
2
3  var results = from item in names
4                where  String.Compare(item, "Joe") < 0
5                select item;
6
7  foreach (var value in results)
8  {
9     MessageBox.Show(value);
10 }
```

In this example, the LINQ query returns all strings in the `names` array that are less than `"Joe"`. The query will return an `IEnumerable<string>` collection. So, the data types in the statement will be as follows:

- The query is searching a `string` array, so in line 3 the data type of the `item` variable will be `string`.

- In line 5, the `select` statement is selecting a `string` (the `item` variable), so the query will return an `IEnumerable<string>` collection. This also means that in line 3 the data type of the `results` variable will be `IEnumerable<string>`.
- In line 7, the data type of the `value` variable will be `string`.

Let's look at another example. Suppose we have the following `Contact` class that stores a person's name and phone number:

```
 1  class Contact
 2  {
 3     // Properties
 4     public string Name { get; set; }
 5     public string PhoneNumber { get; set; }
 6
 7     // Constructor
 8     public Contact(string n, string ph)
 9     {
10         Name = n;
11         PhoneNumber = ph;
12     }
13  }
```

Suppose we want to search an array of `Contact` objects for a specific person's phone number. The following code snippet shows an example of how we might search for Kiran Sharma's phone number:

```
 1  Contact[] contacts =
 2  {
 3      new Contact("Katie Allen", "555-1234"),
 4      new Contact("Kiran Sharma", "555-5678"),
 5      new Contact("Alex Prieto", "555-9012")
 6  };
 7
 8  var results = from item in contacts
 9                where item.Name == "Kiran Sharma"
10                select item;
11
12  foreach (var value in results)
13  {
14     MessageBox.Show(value.PhoneNumber);
15  }
```

In this example, the LINQ query searches the `contacts` array for all `Contact` objects with their `Name` property set to `"Kiran Sharma"`. The query will return a collection that is an `IEnumerable<Contact>` collection. So, the data types in the statement will be as follows:

- The query is searching an array of `Contact` objects, so in line 8, `item` will be a `Contact` object.
- In line 10, the `select` statement is selecting a `Contact` object (`item`), so the query will return an `IEnumerable<Contact>` collection. This also means that in line 8, the data type of the `results` variable will be `IEnumerable<Contact>`.
- In line 12, the data type of the `value` variable will be `Contact`.

Sometimes we want the select clause to "shape" the data that is returned from a query in a particular way. For example, suppose we have the following `RetailItem` class that stores a retail item's description and price:

```
1 class RetailItem
2 {
3     // Constructor
4     public RetailItem(string d, decimal p)
```

```
5     {
6         Description = d;
7         Price = p;
8     }
9
10    // Properties
11    public string Description { get; set; }
12    public decimal Price { get; set; }
13 }
```

Suppose we want to search an array of `RetailItem` objects for all items with a price greater than 100, and we want the query to return only the description of the item. The following code snippet shows an example:

```
1  RetailItem[] retailItems =
2  {
3      new RetailItem("Headphones", 199.99m),
4      new RetailItem("Color Printer", 99.99m),
5      new RetailItem("Security Camera", 129.99m),
6      new RetailItem("Bluetooth Speaker", 79.99m)
7  };
8
9  var results = from item in retailItems
10               where item.Price > 100m
11               select item.Description;
12
13 foreach (var value in results)
14 {
15     MessageBox.Show(value);
16 }
```

In this example, the LINQ query searches the `retailItems` array for all `RetailItem` objects with a `Price` property greater than 100. Notice that in line 11, the `select` statement selects `item.Description`, which is a `string`. So the query will return an `IEnumerable<string>` collection. This also means that in line 9, the data type of the `results` variable will be `IEnumerable<string>`.

Using Multiple `where` Clauses

You can use multiple `where` clauses in a LINQ expression to specify multiple search criteria. The following code snippet shows an example:

```
1  RetailItem[] retailItems =
2  {
3      new RetailItem("Headphones", 199.99m),
4      new RetailItem("Color Printer", 99.99m),
5      new RetailItem("Security Camera", 129.99m),
6      new RetailItem("Bluetooth Speaker", 79.99m)
7  };
8
9  var results = from item in retailItems
10               where item.Price > 90m
11               where item.Price < 120m
12               select item.Description;
13
14 foreach (var value in results)
15 {
16     MessageBox.Show(value);
17 }
```

Notice that the LINQ query has two `where` clauses in lines 10 and 11. The first `where` clause causes the query to retrieve the items with a price that is greater than 90. Then,

from that set of results, the second `where` clause causes the query to retrieve the items with a price that is less than 120. We would get the same results if we had written the query like this:

```
var results = from item in retailItems
              where item.Price > 90m && item.Price < 120m
              select item.Description;
```

The `orderby` Clause

If you wish to sort the results of a LINQ query, you can use the `orderby` clause. The following code snippet shows an example.

```
 1  int[] numbers = { 4, 104, 2, 102, 1, 101, 3, 103 };
 2
 3  var results = from item in numbers
 4                orderby item
 5                select item;
 6
 7  foreach (var value in results)
 8  {
 9      MessageBox.Show(value);
10  }
```

The `orderby item` clause in line 4 causes the query results to be sorted in ascending order (lowest to highest). The loop in lines 7–10 will display the values 1, 2, 3, 4, 101, 102, 103, and 104 in message boxes.

Here is an example that uses the previously shown `RetailItem` class:

```
 1  RetailItem[] retailItems =
 2  {
 3      new RetailItem("Headphones", 199.99m),
 4      new RetailItem("Color Printer", 99.99m),
 5      new RetailItem("Security Camera", 129.99m),
 6      new RetailItem("Bluetooth Speaker", 79.99m)
 7  };
 8
 9  var results = from item in retailItems
10                where item.Price < 150m
11                orderby item.Price
12                select item.Description;
13
14  foreach (var value in results)
15  {
16      MessageBox.Show(value);
17  }
```

The query in lines 9–12 searches the `retailItems` array and retrieves the descriptions of all the `RetailItem` objects with a price less than 150. The `orderby` clause in line 11 sorts the results by their `Price` property, so the descriptions will be displayed in order of price, from lowest to highest.

If you want to sort the results of a query in descending order (highest to lowest), use the `descending` keyword, as shown in line 3 in the following code snippet.

```
1  var results = from item in retailItems
2                where item.Price < 150m
3                orderby item.Price descending
4                select item.Description;
5
```

In Tutorial 15-1, you will use LINQ to query objects that are stored in a `List`.

Tutorial 15-1:
Completing the *StudentRoster* Project

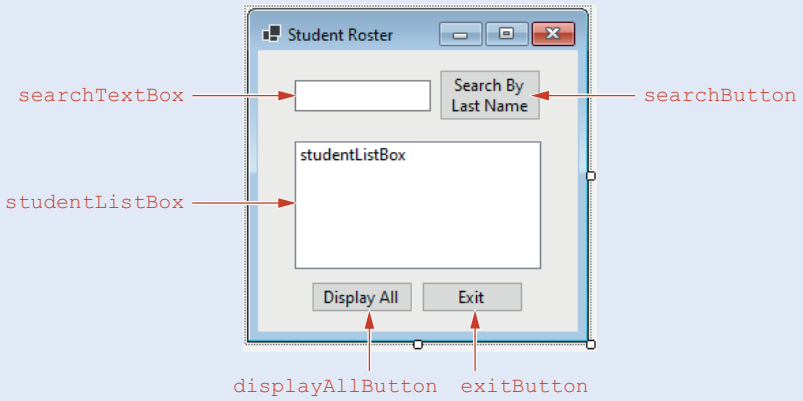
VideoNote
Tutorial 15-1: Completing the *StudentRoster* Project

In this tutorial, you will complete the *StudentRoster* project. The application displays a list of student names in a ListBox control. Figure 15-2 shows the application's form, which has already been created for you. The user can click the `displayAllButton` and all of the student names will be displayed in the ListBox. The user can also search for students by entering a last name in the `searchTextBox` control and then clicking the `searchButton` control.

The application stores each student's first and last name in a `Student` class, which has already been written and is included in the project. The `Student` class is shown in Program 15-1. Here is a summary of the class:

- Lines 12–16 is the constructor, which accepts `string` two arguments. The `last` argument is assigned to the LastName property and the `first` argument is assigned to the FirstName property.
- Line 19 is the declaration of the LastName property, a `string`. This property holds a student's last name.
- Line 21 is the declaration of the FirstName property, a `string`. This property holds a student's first name.
- Lines 23–26 is the declaration of the FullName property, a read-only `string`. This property returns the student's first and last name as a `string`.

Figure 15-2 The Form1 form



Step 1: Open the project named *StudentRoster* in the *Chap15* folder of this book's *Student Sample Programs*.

Step 2: Open the *Form1* form in the Designer. The form is shown, along with the names of the important controls, in Figure 15-2.

Step 3: Now, you will create the Click event handlers for the Button controls. At the end of this tutorial, Program 15-2 shows the completed code for the form. You will be instructed to refer to Program 15-2 as you write the event handlers.

Note: The code in lines 5–17 of Program 15-2 has already been written for you. This code creates a `List` named `students` and initializes it with 10 `Student` objects.

In the *Designer*, double-click the `displayAllButton` control. This will open the code editor, and you see an empty event handler named `displayAllButton_Click`. Complete the `displayAllButton_Click` event handler by typing the code shown in lines 26–39 in Program 15-2. Let's take a closer look at the code:

Lines 28–30: This is a LINQ query that gets all of the `Student` objects in the `students` list, sorted by last name. The results are assigned to the `results` variable as an `IEnumerable<Student>` collection.

Line 33: This statement clears the contents of the `studentListBox` control.

Lines 36–39: This `foreach` loop iterates over each of the `Student` objects in the `results` collection. Each time it iterates, it gets the `FullName` property from a `Student` object and adds it to the `studentListBox` control.

Step 4: Switch your view back to the *Designer* and double-click the `searchButton` control. In the code editor, you will see an empty event handler named `searchButton_Click`. Complete the `searchButton_Click` event handler by typing the code shown in lines 44–58 in Program 15-2. Let's take a closer look at the code:

Lines 46–49: This is a LINQ query that gets all of the `Student` objects in the `students` list with the LastName property matching the contents of the `searchTextBox`, sorted by first name. The results are assigned to the `results` variable as an `IEnumerable<Student>` collection.

Line 52: This line clears the contents of the `studentListBox` control.

Lines 55–58: This `foreach` loop iterates over each of the `Student` objects in the `results` collection. Each time it iterates, it gets the `FullName` property from a `Student` object and adds it to the `studentListBox` control.

Step 5: Switch your view back to the *Designer* and double-click the `exitButton` control. In the code editor, you will see an empty event handler named `exitButton_Click`. Complete the `exitButton_Click` event handler by typing the code shown in lines 63–64 in Program 15-2.

Step 6: Save the project, then compile and run the application. First, click the *Display All* button to display all the student names. Then, enter *Adams* in the TextBox and click the *Search By Last Name* button. The ListBox should display *Adams, James* and *Adams, Susan*.

Delete the contents of the TextBox control and enter *Lopez*. Click the *Search By Last Name* button. The ListBox should display *Lopez, Nora* and *Lopez, Rosa*.

Experiment by searching for other last names and observing the results that are displayed in the ListBox control. When you are finished, click the *Exit* button to close the application. We will continue to work with this project in the next tutorial.

Program 15-1 The `Student` Class

```
1 using System;
2 using System.Collections.Generic;
3 using System.Linq;
4 using System.Text;
```

```
 5 using System.Threading.Tasks;
 6
 7 namespace StudentRoster
 8 {
 9     internal class Student
10     {
11         // Constructor
12         public Student(string last, string first)
13         {
14             LastName = last;
15             FirstName = first;
16         }
17
18         // Properties
19         public string LastName { get; set; }
20
21         public string FirstName { get; set; }
22
23         public string FullName
24         {
25             get { return $"{LastName}, {FirstName}"; }
26         }
27     }
28 }
```

Program 15-2 Complete Form1 code for the *StudentRoster* project

```
 1 namespace StudentRoster
 2 {
 3     public partial class Form1 : Form
 4     {
 5         List<Student> students = new List<Student>()
 6         {
 7             new Student("Adams", "Susan"),
 8             new Student("Adams", "James"),
 9             new Student("Watson", "Earl"),
10             new Student("Lopez", "Rosa"),
11             new Student("Lopez", "Nora"),
12             new Student("Patel", "Sanjay"),
13             new Student("Patel", "Nisha"),
14             new Student("Campbell", "Patrick"),
15             new Student("Gonzalez", "Nina"),
16             new Student("Anderson", "Ashley")
17         };
18
19         public Form1()
20         {
21             InitializeComponent();
22         }
23
24         private void displayAllButton_Click(object sender,
           EventArgs e)
25         {
26             // Get all the Student objects in the List,
27             // ordered by last name.
28             var results = from student in students
29                           orderby student.LastName
30                           select student;
```

```
31
32              // Clear the ListBox.
33              studentListBox.Items.Clear();
34
35              // Display the results in the ListBox.
36              foreach (Student s in results)
37              {
38                  studentListBox.Items.Add(s.FullName);
39              }
40          }
41
42          private void searchButton_Click(object sender, EventArgs e)
43          {
44              // Get all the Student objects with the specified
45              // last name, ordered by first name.
46              var results = from student in students
47                            where student.LastName ==
                              searchTextBox.Text
48                            orderby student.FirstName
49                            select student;
50
51              // Clear the ListBox.
52              studentListBox.Items.Clear();
53
54              // Display the results in the ListBox.
55              foreach (Student s in results)
56              {
57                  studentListBox.Items.Add(s.FullName);
58              }
59          }
60
61          private void exitButton_Click(object sender, EventArgs e)
62          {
63              // Close the application.
64              this.Close();
65          }
66      }
67  }
```

Checkpoint

15.1 Assume values is an `int` array. What does the following LINQ query do?

```
var results = from item in values
              where item > 10
              select item;
```

15.2 Look again at the query shown in Checkpoint 15.1. What data type will the `results` variable be of?

15.3 In a LINQ query, what does the `from` clause do?

15.4 In a LINQ query, what does the `select` clause do?

15.5 How do you sort the results of a LINQ query in ascending order? How do you sort the results in descending order?

15.2 LINQ Method Syntax

CONCEPT: **You can write a LINQ query using either query syntax or method syntax. If you use query syntax, the C# compiler translates your queries into method calls that invoke extension methods that are defined in the .NET Framework. If you use method syntax, you are calling those extension methods directly.**

There are two ways to write a LINQ query: you can use **query syntax** or **method syntax**. All the previous examples were written in query syntax, which resembles an SQL query. Queries that are written in method syntax take the form of method calls.

The .NET Framework defines a set of LINQ extension methods that work with all collections in the framework. The extension methods correspond to the various clauses that appear in a LINQ query. For example, there is a `Where` method, there is an `OrderBy` method, and there is a `Select` method. When you compile a C# program that contains a query written in query syntax, the compiler translates the query into method syntax. If you wish, you can perform LINQ queries by calling the LINQ extension methods directly, instead of using query syntax.

The `Where` Method

This is the general format of the `Where` method:

```
collection.Where(FuncDelegate)
```

In the general format, *`collection`* is the collection you are searching and *`FuncDelegate`* is a `Func` delegate (or lambda expression) that has one parameter and returns a `bool` value. The `Where` method uses the `Func` delegate to determine which elements will be included in the query results. Each element of the *`collection`* will be passed as an argument to the `Func` delegate. If the `Func` delegate returns `true`, the element will be included. If the `Func` delegate returns `false`, the element will not be included.

The `Where` method returns its query results as an `IEnumerable<T>` collection. The elements in the collection will be of the same data type as the elements of the collection that is being searched for. So, if the `Where` method executes on an `int` array, the method will return its query results as an `IEnumerable<int>` collection.

The following code snippet shows a simple example.

```
1   int[] numbers = { 4, 104, 2, 102, 1, 101, 3, 103 };
2
3   var results = numbers.Where(item => item > 100);
4
5   foreach (var value in results)
6   {
7      MessageBox.Show(value.ToString());
8   }
```

In this example, the LINQ query appears in line 3. The lambda expression that we are passing to the query has one parameter named `item`. The lambda expression returns `true` if `item` is greater than 100, or `false` otherwise. So, the LINQ query returns all the elements of the `numbers` array that are greater than 100.

The following code snippet shows an example using the `RetailItem` class shown earlier in this chapter.

```
1  RetailItem[] retailItems =
2  {
```

```
3       new RetailItem("Headphones", 199.99m),
4       new RetailItem("Color Printer", 99.99m),
5       new RetailItem("Security Camera", 129.99m),
6       new RetailItem("Bluetooth Speaker", 79.99m)
7   };
8
9    var results = retailItems.Where(item => item.Price < 100m);
10
11   foreach (var value in results)
12   {
13      MessageBox.Show(value.Price.ToString());
14   }
```

In this example, the LINQ query in line 9 returns all the elements of the `retailItems` array with a price that is less than 100.

The `OrderBy` and `OrderByDescending` Methods

If you want to sort the elements of a collection in ascending order, you can use the `OrderBy` method. This is the general format of the `OrderBy` method:

```
collection.OrderBy(FuncDelegate)
```

In the general format, *`collection`* is the collection you are searching for and *`FuncDelegate`* is a `Func` delegate (or lambda expression) that has one parameter and returns a value. Each element of the *`collection`* will be passed as an argument to the `Func` delegate. The delegate returns the field of the element that the results should be sorted on. The `OrderBy` method returns the query results as an `IEnumerable<T>` collection.

The following code shows an example of sorting all the elements of an array of `RetailItem` objects:

```
1   RetailItem[] retailItems =
2   {
3       new RetailItem("Headphones", 199.99m),
4       new RetailItem("Color Printer", 99.99m),
5       new RetailItem("Security Camera", 129.99m),
6       new RetailItem("Bluetooth Speaker", 79.99m)
7   };
8
9    var results = retailItems.OrderBy(item => item.Price);
10
11   foreach (var value in results)
12   {
13      MessageBox.Show(value.Price.ToString());
14   }
```

Take a closer look at the lambda expression we are passing to the `OrderBy` method on line 9:

```
item => item.Price
```

The `item` parameter receives an element of the array, and then returns that element's `Price` property. As a result, the query will give us the elements of the array sorted by their `Price` property, in ascending order.

If you want to see results sorted in descending order, you cannot use the `OrderBy` method. Instead, you use the `OrderByDescending` method, as shown here:

```
var results = retailItems.OrderByDescending(item => item.Price);
```

Quite often you will need to combine calls to the extension methods in a LINQ query. For example, suppose you have a `RetailItem` array, and you want to get all the items with a price that is less than 100, and you want the results sorted by price in ascending order.

This requires that you call both the `Where` method and the `OrderBy` method. The following code snippet shows one way to accomplish this:

```
1  RetailItem[] retailItems =
2  {
3      new RetailItem("Headphones", 199.99m),
4      new RetailItem("Color Printer", 99.99m),
5      new RetailItem("Security Camera", 129.99m),
6      new RetailItem("Bluetooth Speaker", 79.99m)
7  };
8
9  var results = retailItems.Where(item => item.Price < 100m);
10 var results2 = results.OrderBy(item => item.Price);
11
12 foreach (var value in results2)
13 {
14     MessageBox.Show(value.Price);
15 }
```

In this example, line 9 executes the `Where` method on the `retailItems` array. It gets all the elements that have a `Price` value less than 100. The results of that query are returned as an `IEnumberable<RetailItem>` collection and assigned to the `results` variable. Then, the statement in line 10 executes the `OrderBy` method on the `results` collection. It returns the elements of the collection, sorted by their `Price` property in ascending order. The results are returned in another `IEnumberable<RetailItem>` collection and assigned to the `results2` variable. So, the `results2` variable refers to the collection containing the final results.

Rather than executing multiple, separate queries, LINQ supports a technique known as method chaining that allows you to call all the necessary extension methods in one statement. Here is an example:

```
1  RetailItem[] retailItems =
2  {
3      new RetailItem("Headphones", 199.99m),
4      new RetailItem("Color Printer", 99.99m),
5      new RetailItem("Security Camera", 129.99m),
6      new RetailItem("Bluetooth Speaker", 79.99m)
7  };
8
9  var results = retailItems.Where(item => item.Price < 100m)
10                           .OrderBy(item => item.Price);
11
12 foreach (var value in results)
13 {
14     MessageBox.Show(value.Price);
15 }
```

In this example, the `Where` method is called first to get all the elements that have a `Price` value less than 100. The `Where` method returns its results as an `IEnumberable<RetailItem>` collection. Then, the `OrderBy` method is called on the collection that was returned from the `Where` method. The `IEnumberable<RetailItem>` collection that is returned from the `OrderBy` method is assigned to the `results` variable.

The `Select` Method

You probably noticed that in the previous method syntax examples, we did not use the `Select` method. That's because the `Select` method is optional when you are writing queries in method syntax. You only need to use it if you want to reshape the data that are returned from the query. Here is the general format of the `Select` method:

```
collection.Select(FuncDelegate)
```

In the general format, *collection* is the collection you are searching for and *FuncDelegate* is a `Func` delegate (or lambda expression) that has one parameter and returns a value. Each element of the *collection* will be passed as an argument to the `Func` delegate. The delegate returns the data that should be included in the search results. For example, look at the following code snippet:

```
1  string[] names = { "Jake", "Allison", "Will", "Sara" };
2  var results = names.Select(item => item.ToUpper());
```

The query in line 2 returns an `IEnumerable<string>` collection containing all the elements of the `names` collection, converted to uppercase. Here is another example:

```
1  string[] names = { "Bill", "Allison", "Will", "Sara" };
2
3  var results = names.Where(item => item.Contains("ill"))
4                     .Select(item => item.ToUpper());
```

The query in line 2 returns an `IEnumerable<string>` collection containing the strings `"BILL"` and `"WILL"`. The following is one more example:

```
 1  RetailItem[] retailItems =
 2  {
 3      new RetailItem("Headphones", 199.99m),
 4      new RetailItem("Color Printer", 99.99m),
 5      new RetailItem("Security Camera", 129.99m),
 6      new RetailItem("Bluetooth Speaker", 79.99m)
 7  };
 8
 9  var results = retailItems.Where(item => item.Price < 150m)
10                           .OrderBy(item => item.Price)
11                           .Select(item => item.Description);
```

The query in lines 9 and 10 returns an `IEnumerable<string>` collection containing the `Description` property of each item that that is priced less than 150, sorted by price in ascending order.

In Tutorial 15-2, you will use method syntax to rewrite existing LINQ queries that use query syntax.

Tutorial 15-2:
Modifying the *StudentRoster* Project

VideoNote
Tutorial 15-2: Modifying the *StudentRoster* Project

In this tutorial, you modify the *StudentRoster* application that you completed in Tutorial 15-1. In that tutorial, you used query syntax to write two LINQ queries that retrieve data from a List of Student objects. In this tutorial, you rewrite the queries using method syntax. Program 15-3 shows the completed Form1 source code, as it will appear after you make the changes in this tutorial.

Step 1: If Visual Studio is not already running, launch it and open the *StudentRoster* project that you completed in Tutorial 15-1.

Step 2: Open the Form1.cs file in the Code Editor. Find the following LINQ query in the `displayAllButton_Click` event handler:

```
var results = from student in students
              orderby student.LastName
              select student;
```

The LINQ query is written in query syntax. Replace the query with the following statement, which is written in method syntax:

```
var results = students.OrderBy(stu => stu.LastName);
```

Step 3: Find the following LINQ query in the `searchButton_Click` event handler:

```
var results = from student in students
              where student.LastName == searchTextBox.Text
              orderby student.FirstName
              select student;
```

The LINQ query is written in query syntax. Replace the query with the following statement, which is written in method syntax:

```
var results = students.Where(stu => stu.LastName ==
                                    searchTextBox.Text)
                      .OrderBy(stu => stu.FirstName);
```

Step 4: Once you have completed Steps 2 and 3, the Form1 source code should appear as shown in Program 15-3. Save the project. Then, press F5 on the keyboard or click the Start Debugging button (▶) on the toolbar to compile and run the application.

The application should work exactly as it did in Tutorial 15-1. Perform the following steps to test it:

- Click the *Display All* button to display all of the student names.
- Enter *Adams* in the TextBox and click the *Search By Last Name* button. The ListBox should display *Adams, James* and *Adams, Susan*.
- Delete the contents of the TextBox control and enter *Lopez*. Click the *Search By Last Name* button. The ListBox should display *Lopez, Nora* and *Lopez, Rosa*.
- Experiment by searching for other last names and observing the results that are displayed in the ListBox control. When you are finished, click the *Exit* button to close the application.

Program 15-3 Complete Form1 code for the *StudentRoster* project

```
1  namespace StudentRoster
2  {
3      public partial class Form1 : Form
4      {
5          List<Student> students = new List<Student>()
6          {
7              new Student("Adams", "Susan"),
8              new Student("Adams", "James"),
9              new Student("Watson", "Earl"),
10             new Student("Lopez", "Rosa"),
11             new Student("Lopez", "Nora"),
12             new Student("Patel", "Sanjay"),
13             new Student("Patel", "Nisha"),
14             new Student("Campbell", "Patrick"),
15             new Student("Gonzalez", "Nina"),
16             new Student("Anderson", "Ashley")
17         };
18
19         public Form1()
```

```
20          {
21              InitializeComponent();
22          }
23
24          private void displayAllButton_Click(object sender,
            EventArgs e)
25          {
26              // Get all the Student objects in the List,
27              // ordered by last name.
28              var results = students.OrderBy(stu => stu.LastName);
29
30              // Clear the ListBox.
31              studentListBox.Items.Clear();
32
33              // Display the results in the ListBox.
34              foreach (Student s in results)
35              {
36                  studentListBox.Items.Add(s.FullName);
37              }
38          }
39
40          private void searchButton_Click(object sender,
            EventArgs e)
41          {
42              // Get all the Student objects with the specified
43              // last name, ordered by first name.
44              var results = students.Where(stu => stu.LastName ==
45                                           searchTextBox.Text)
46                                           .OrderBy(stu => stu.
                                             FirstName);
47
48              // Clear the ListBox.
49              studentListBox.Items.Clear();
50
51              // Display the results in the ListBox.
52              foreach (Student s in results)
53              {
54                  studentListBox.Items.Add(s.FullName);
55              }
56          }
57
58          private void exitButton_Click(object sender, EventArgs e)
59          {
60              // Close the application.
61              this.Close();
62          }
63      }
64  }
```

Checkpoint

15.6 Convert the following query to method syntax:

```
var results = from item in retailItems
              where item.Price < 100.0m
              orderby item.Price
              select item.Description;
```

15.3 More LINQ Extension Methods

CONCEPT: **There are numerous LINQ extension methods in .NET that perform useful operations but cannot be invoked with LINQ query syntax. You must use LINQ method syntax to call them.**

There are several LINQ extension methods defined in .NET that cannot be invoked using LINQ query syntax. They can be called only in queries that are written in method syntax. In this section, we will look at many of those extension methods.

Quantifier Methods

A **quantifier method** is a method that returns `true` or `false` to indicate whether one or more elements in a collection match a condition. LINQ provides the quantifier methods listed in Table 15-1.

Table 15-1 LINQ quantifier methods

Method	Description
`All`	Summary: Returns `true` if all the elements of a collection match a condition. Returns `false` otherwise. General format: *`collection`*`.All(`*`FuncDelegate`*`)` Description: In the general format, *`collection`* is the collection you are searching and *`FuncDelegate`* is a `Func` delegate (or lambda expression) that has one parameter and returns a `bool` value. Each element of the *`collection`* will be passed as an argument to the `Func` delegate. If the `Func` delegate returns `true` for all the elements, the `All` method returns `true`. Otherwise, the `All` method returns `false`.
`Any`	Summary: Returns `true` if any of the elements of a collection match a condition. Returns `false` otherwise. General format: *`collection`*`.Any(`*`FuncDelegate`*`)` Description: In the general format, *`collection`* is the collection you are searching and *`FuncDelegate`* is a `Func` delegate (or lambda expression) that has one parameter and returns a `bool` value. Each element of the *`collection`* will be passed as an argument to the `Func` delegate. If the `Func` delegate returns `true` for any of the elements, the `Any` method returns `true`. Otherwise, the `Any` method returns `false`.
`Contains`	Summary: Returns `true` if a collection contains a particular element. Returns `false` otherwise. General format: *`collection`*`.Contains(`*`value`*`)` Description: In the general format, *`collection`* is the collection you are searching and *`value`* is a value that you are searching for.

Each of the methods in Table 15-1 takes, as an argument, a `Func` delegate (or lambda expression) that has one parameter and returns a `bool` value. Let's look at some examples that use the `RetailItem` class presented earlier. The following code snippet demonstrates the `All` and `Any` methods:

```
1  RetailItem[] retailItems =
2  {
3      new RetailItem("Headphones", 199.99m),
4      new RetailItem("Color Printer", 99.99m),
```

```
5       new RetailItem("Security Camera", 129.99m),
6       new RetailItem("Bluetooth Speaker", 79.99m)
7   };
8
9   if (retailItems.All(item => item.Price < 200m))
10      MessageBox.Show("All items are less than $200.");
11
12  if (retailItems.Any(item => item.Price < 100m))
13      MessageBox.Show("Some items are less than $100");
```

The `if` statement in 9 uses the `All` method to determine whether all the elements of the `retailItems` array have a `Price` value that is less than 200. Then, the `if` statement in 12 uses the `Any` method to determine whether any of the elements of the `retailItems` array have a `Price` value that is less than 100.

The `Contains` method, which was first mentioned in Chapter 8, returns `true` if a collection contains a specified value. Otherwise it returns `false`. The following code snippet demonstrates:

```
1  string[] names = { "New York", "London", "Paris", "Munich" };
2
3  if (names.Contains("London"))
4      MessageBox.Show("London is one of the cities.");
```

Aggregation Methods

An **aggregation method** is a method that calculates a single value from a collection of values. LINQ provides the aggregation methods listed in Table 15-2.

Table 15-2 LINQ aggregation methods

Method	Description
`Average`	Summary: Returns the average of a collection of numeric values. General format: *`collection`*`.Average()` General format: *`collection`*`.Average(`*`FuncDelegate`*`)` Description: In the first general format, *`collection`* is a numeric collection. The method returns the average of the values in the *`collection`*. In the second general format, *`collection`* is a collection and *`FuncDelegate`* is a `Func` delegate (or lambda expression) that has one parameter and returns a numeric value. Each element of the *`collection`* will be passed as an argument to the `Func` delegate. The `Average` method returns the average of the values that are returned from the *`FuncDelegate`*.
`Count`	Summary: Returns the number of elements in a collection, or the number of elements that match a condition. General format: *`collection`*`.Count()` General format: *`collection`*`.Count(`*`FuncDelegate`*`)` Description: In the first general format, *`collection`* is a collection. The method returns the number of elements in the *`collection`*. In the second general format, *`collection`* is a collection and *`FuncDelegate`* is a `Func` delegate (or lambda expression) that has one parameter and returns a `bool` value. Each element of the *`collection`* will be passed as an argument to the `Func` delegate. The `Count` method returns the number of elements that caused *`FuncDelegate`* to return `true`.

(*Continued*)

Table 15-2 LINQ aggregation methods (*Continued*)

Method	Description
`Max`	Summary: Returns the largest value from a collection of numeric values, or the largest value that is returned from a delegate that performs an operation on each element of a collection. General format: *`collection`*`.Max()` General format: *`collection`*`.Max(`*`FuncDelegate`*`)` Description: In the first general format, *`collection`* is a collection of numeric values. The method returns the largest value in the *`collection`*. In the second general format, *`collection`* is a collection and *`FuncDelegate`* is a `Func` delegate (or lambda expression) that has one parameter and returns a value. Each element of the *`collection`* will be passed as an argument to the `Func` delegate. The `Max` method returns the largest value that is returned from the *`FuncDelegate`*.
`Min`	Summary: Returns the smallest value from a collection of numeric values, or the smallest value that is returned from a delegate that performs an operation on each element of a collection. General format: *`collection`*`.Min()` General format: *`collection`*`.Min(`*`FuncDelegate`*`)` Description: In the first general format, *`collection`* is a collection of numeric values. The method returns the smallest value in the *`collection`*. In the second general format, *`collection`* is a collection and *`FuncDelegate`* is a `Func` delegate (or lambda expression) that has one parameter and returns a value. Each element of the *`collection`* will be passed as an argument to the `Func` delegate. The `Min` method returns the smallest value that is returned from the *`FuncDelegate`*.
`Sum`	Summary: Returns the sum of the elements of a numeric collection, or the sum of the values that are returned from a delegate that performs an operation on each element of a collection. General format: *`collection`*`.Sum()` General format: *`collection`*`.Sum(`*`FuncDelegate`*`)` Description: In the first general format, *`collection`* is a collection of numeric values. The method returns the sum of the values in the *`collection`*. In the second general format, *`collection`* is a collection and *`FuncDelegate`* is a `Func` delegate (or lambda expression) that has one parameter and returns a numeric value. Each element of the *`collection`* will be passed as an argument to the `Func` delegate. The `Sum` method returns the sum of the values that are returned from the *`FuncDelegate`*.

The following code snippet demonstrates the simplest form of each of these methods:

```
1  int[] numbers = { 1, 2, 3, 4, 5 };
2
3  MessageBox.Show($"Average: {numbers.Average()}");
4  MessageBox.Show($"Count: {numbers.Count()}");
5  MessageBox.Show($"Max: {numbers.Max()}");
6  MessageBox.Show($"Min: {numbers.Min()}");
7  MessageBox.Show($"Sum: {numbers.Sum()}");
```

You can get more focused results from each of these methods by passing an argument. The following code snippet demonstrates this:

```
RetailItem[] retailItems =
{
    new RetailItem("Headphones", 199.99m),
    new RetailItem("Color Printer", 99.99m),
    new RetailItem("Security Camera", 129.99m),
    new RetailItem("Bluetooth Speaker", 79.99m)
};

// Get the average price.
var avg = retailItems.Average(item => item.Price);
MessageBox.Show($"Average Price: {avg}");

// Get the number of items priced less than $100.
var lessThan100 = retailItems.Count(item => item.Price < 100m);
MessageBox.Show($"Items less than $100: {lessThan100}");

// Get the highest price.
var mostExpensive = retailItems.Max(item => item.Price);
MessageBox.Show($"Most expensive price: {mostExpensive}");

// Get the lowest price.
var leastExpensive = retailItems.Min(item => item.Price);
MessageBox.Show($"Least expensive price: {leastExpensive}");

// Get the sum of all the prices.
var sum = retailItems.Sum(item => item.Price);
MessageBox.Show($"Sum of the prices: {sum}");
```

Element Methods

An **element method** is a method that returns a specific element from a collection. LINQ provides the element methods listed in Table 15-3.

Table 15-3 LINQ element methods

Method	Description
`ElementAt`	Summary: Returns the element at a specific position in a collection. General format: *`collection`*`.ElementAt(`*`index`*`)` Description: In the general format, *`collection`* is a collection of numeric values and *`index`* is a position within the collection, with 0 being the first element. The method returns the element at the specified position. If the index is out of bounds, the method throws an `ArgumentOutOfRangeException`.
`ElementAtOrDefault`	Summary: Returns the element at a specific position in a collection or, if the position is invalid, the method returns a default value. General format: *`collection`*`.ElementAtOrDefault(`*`index`*`)` Description: In the general format, *collection* is a collection of numeric values and *`index`* is a position within the collection, with 0 being the first element. The method returns the element at the specified position. If the index is out of bounds, the method returns the default value for the element's data type (0 for `int`, `null` for reference types, etc.).

(*Continued*)

Table 15-3 LINQ element methods (*Continued*)

Method	Description
`First`	Summary: Returns the first element of a collection or the first element that matches a condition. General format: *`collection`*`.First()` General format: *`collection`*`.First(`*`FuncDelegate`*`)` Description: In the first general format, *`collection`* is a collection. The method returns the first element in the collection. If the collection is empty, the method throws an `InvalidOperationException`. In the second general format, *`collection`* is a collection and *`FuncDelegate`* is a `Func` delegate (or lambda expression) that has one parameter and returns a `bool` value. Each element of the *`collection`* will be passed as an argument to the `Func` delegate. The `First` method returns the first element that caused *`FuncDelegate`* to return `true`.
`FirstOrDefault`	Summary: Returns the first element of a collection or the first element that matches a condition. If the collection is empty, or no elements match the condition, the method returns a default value. General format: *`collection`*`.FirstOrDefault()` General format: *`collection`*`.FirstOrDefault(`*`FuncDelegate`*`)` Description: In the first general format, *`collection`* is a collection. The method returns the first element in the collection. If the collection is empty, the method returns the default value for the element's data type (0 for `int`, `null` for reference types, etc.). In the second general format, *`collection`* is a collection and *`FuncDelegate`* is a `Func` delegate (or lambda expression) that has one parameter and returns a `bool` value. Each element of the *`collection`* will be passed as an argument to the `Func` delegate. The `FirstOrDefault` method returns the first element that caused *`FuncDelegate`* to return `true`. If the collection is empty, or no elements match the condition, the method returns the default value for the element's data type (0 for `int`, `null` for reference types, etc.).
`Last`	Returns the last element of a collection or the last element that matches a condition. General format: *`collection`*`.Last()` General format: *`collection`*`.Last(`*`FuncDelegate`*`)` Description: In the first general format, *`collection`* is a collection. The method returns the last element in the collection. If the collection is empty, the method throws an `InvalidOperationException`. In the second general format, *`collection`* is a collection and *`FuncDelegate`* is a `Func` delegate (or lambda expression) that has one parameter and returns a `bool` value. Each element of the *`collection`* will be passed as an argument to the `Func` delegate. The `Last` method returns the last element that caused *`FuncDelegate`* to return `true`.

Table 15-3 LINQ element methods (*Continued*)

Method	Description
`LastOrDefault`	Summary: Returns the last element of a collection or the last element that matches a condition. If the collection is empty, or no elements match the condition, the method returns a default value. General format: *`collection`*`.LastOrDefault()` General format: *`collection`*`.LastOrDefault(`*`FuncDelegate`*`)` Description: In the first general format, *`collection`* is a collection. The method returns the last element in the collection. If the collection is empty, the method returns the default value for the element's data type (0 for `int`, `null` for reference types, etc.). In the second general format, *`collection`* is a collection and *`FuncDelegate`* is a `Func` delegate (or lambda expression) that has one parameter and returns a `bool` value. Each element of the *`collection`* will be passed as an argument to the `Func` delegate. The `FirstOrDefault` method returns the last element that caused *`FuncDelegate`* to return `true`. If the collection is empty, or no elements match the condition, the method returns the default value for the element's data type (0 for `int`, `null` for reference types, etc.).
`Single`	Summary: Returns the single element of a collection, or the single element that matches a condition. General format: *`collection`*`.Single()` General format: *`collection`*`.Single(`*`FuncDelegate`*`)` Description: In the first general format, *`collection`* is a collection. If the collection contains one element, the method returns that element. If the collection is empty, or has more than one element, the method throws an `InvalidOperationException`. In the second general format, *`collection`* is a collection and *`FuncDelegate`* is a `Func` delegate (or lambda expression) that has one parameter and returns a `bool` value. Each element of the *`collection`* will be passed as an argument to the `Func` delegate. The `Single` method returns the single element that caused *`FuncDelegate`* to return `true`. If there are no elements that satisfy the condition, or more than one element satisfies the condition, the method throws an `InvalidOperationException`.
`SingleOrDefault`	Summary: Returns the single element of a collection, or the single element that matches a condition. If there is no element that matches the condition, the method returns a default value. If more than one element matches the condition, the method throws an exception. General format: *`collection`*`.SingleOrDefault()` General format: *`collection`*`.SingleOrDefault(`*`FuncDelegate`*`)` Description: In the first general format, *`collection`* is a collection. If the collection contains one element, the method returns that element. If the collection is empty, the method returns the default value for the element's data type (0 for `int`, `null` for reference types, etc.). If the collection has more than one element, the method throws an `InvalidOperationException`.

(*Continued*)

Table 15-3 LINQ element methods (*Continued*)

Method	Description
	In the second general format, *collection* is a collection and *FuncDelegate* is a `Func` delegate (or lambda expression) that has one parameter and returns a `bool` value. Each element of the *collection* will be passed as an argument to the `Func` delegate. The `SingleOrDefault` method returns the single element that caused *FuncDelegate* to return `true`. If the collection is empty, the method returns the default value for the element's data type (0 for `int`, `null` for reference types, etc.). If the collection has more than one element that caused *FuncDelegate* to return `true`, the method throws an `InvalidOperationException`.

The following code snippet demonstrates each of these methods:

```
RetailItem[] retailItems =
{
    new RetailItem("Headphones", 199.99m),
    new RetailItem("Color Printer", 99.99m),
    new RetailItem("Security Camera", 129.99m),
    new RetailItem("Bluetooth Speaker", 79.99m)
};

// Get the element at index 2.
var element = retailItems.ElementAt(2);
MessageBox.Show($"At index 2: {element.Description}");

// Try to get an invalid element.
element = retailItems.ElementAtOrDefault(4);
if (element == null)
    MessageBox.Show("Position 4 is invalid.");

// Get the first element where Price < 100.
element = retailItems.First(item => item.Price < 100m);
MessageBox.Show($"First item less than $100: {element.Description}");

// Get the first element where Price > 200. (Invalid)
element = retailItems.FirstOrDefault(item => item.Price < 100m);
if (element == null)
    MessageBox.Show("No items greater than $200.");

// Get the last element where Price < 100.
element = retailItems.Last(item => item.Price < 100m);
MessageBox.Show($"Last item less than $100: {element.Description}");

// Get the last element where Price > 200. (Invalid)
element = retailItems.LastOrDefault(item => item.Price < 100m);
if (element == null)
    MessageBox.Show("No items greather than $200.");

// Get the one element where Price == 79.99.
element = retailItems.Single(item => item.Price == 79.99m);
MessageBox.Show($"The only item at $79.99: {element.Description}");

```

```
40  // Get the one element where Price == 200. (Invalid)
41  element = retailItems.SingleOrDefault(item => item.Price == 200m);
42  if (element == null)
43      MessageBox.Show("No items priced at $200.");
```

Generation Methods

A **generation method** is a method that generates a new collection of values. LINQ provides the generation methods listed in Table 15-4.

Table 15-4 LINQ generation methods

Method	Description
`DefaultIfEmpty`	Summary: Returns the elements of a collection, as they currently exist, unless the collection is empty. If the collection is empty, the method returns a collection containing one default value. General format: *`collection`*`.DefaultIfEmpty()` General format: *`collection`*`.DefaultIfEmpty(`*`defaultValue`*`)` Description: In the first general format, *`collection`* is a collection. If the collection contains one or more elements, the method returns a copy of the collection. If the collection is empty, the method returns a collection containing one element set to the default value for the element's data type (0 for `int`, `null` for reference types, etc.). In the second general format, *`collection`* is a collection and *`defaultValue`* is the value to use as the default value. If the collection contains one or more elements, the method returns a copy of the collection. If the collection is empty, the method returns a collection containing one element set to *`defaultValue`*.
`Enumerable.Empty`	Summary: Returns an empty `IEnumerable<`*`T`*`>` collection of a specified data type. General format: `Enumerable.Empty<`*`T`*`>()` Description: In the general format, *`T`* is the desired data type. The method returns an empty `IEnumerable<`*`T`*`>` collection.
`Enumerable.Range`	Summary: Returns an `IEnumerable<int>` collection containing a specified number of sequential integers that start at a specific value. General format: `Enumerable.Range(`*`start, size`*`)` Description: In the general format, *`start`* is the starting value of the sequence and *`size`* is the number of elements to generate. The method returns an `IEnumerable<int>` collection.
`Enumerable.Repeat`	Summary: Returns a collection of a specified data type and size, containing one value that is repeated for each element. General format: `Enumerable.Repeat<`*`T`*`>(`*`value, size`*`)` Description: In the general format, *`T`* is the desired data type, *`value`* is the value to repeat, and *`size`* is the number of elements to generate. The method returns an `IEnumerable<`*`T`*`>` collection.

The following code snippet demonstrates the `DefaultIfEmpty` and `Enumerable.Empty` methods. In this code, both `numbers` and `nums` will reference empty `IEnumerable<int>` collections. The `defaultNums` collection will have one element, set to the value 99:

```
// Create an empty int collection.
var numbers = Enumerable.Empty<int>();

// Create another empty int collection.
var nums = numbers.DefaultIfEmpty();

// Create an int collection with 1 element
// set to the value 99.
var defaultNums = numbers.DefaultIfEmpty(99);
```

The following statement demonstrates the `Enumerable.Range` method. The `numbers` collection will have five elements set to the values 100, 101, 102, 103, and 104:

```
var numbers = Enumerable.Range(100, 5);
```

The following statement demonstrates the `Enumerable.Repeat` method. The `numbers` collection will have 10 `int` elements with each element set to the value 99:

```
var numbers = Enumerable.Repeat<int>(99, 10);
```

The following statement also demonstrates the `Enumerable.Repeat` method. The `retailItems` collection will have three `RetailItem` elements. Each element will have a Description property set to `"unknown"` and a Price property set to 0:

```
var retailItems =
    Enumerable.Repeat<RetailItem>(new RetailItem("unknown", 0m), 3);
```

Set Methods

A **set method** is a method that performs a mathematical set operation on two collections of values. (The `Distinct` method works with only one collection.) LINQ provides the set methods listed in Table 15-5.

Table 15-5 LINQ Set methods

Method	Description
`Distinct`	Summary: Returns a copy of a collection with duplicates removed. General format: *`collection`*`.Distinct()` Description: In the general format, *`collection`* is a collection. The method returns copy of *`collection`* with duplicates removed.
`Except`	Summary: Returns a collection that contains the elements of the first collection that are not in the second collection. General format: *`collection1`*`.Except(`*`collection2`*`)` Description: In the general format, *`collection1`* and *`collection2`* are collections. The method returns an `IEnumerable<`*`T`*`>` collection containing the elements of *`collection1`* that do not appear in *`collection2`*. This is the mathematical difference of the two sets.
`Intersect`	Summary: Returns a collection that contains only the elements that are found in two collections. General format: *`collection1`*`.Intersect(`*`collection2`*`)` Description: In the general format, *`collection1`* and *`collection2`* are collections. The method returns an `IEnumerable<`*`T`*`>` collection containing only elements that appear in both collections. This is the mathematical intersection of the two sets.

Table 15-5 LINQ Set methods (*Continued*)

Method	Description
`Union`	Summary: Returns a collection that contains all the distinct elements of two sets. General format: *`collection1`*`.Union(`*`collection2`*`)` Description: In the general format, *`collection1`* and *`collection2`* are collections. The method returns an `IEnumerable<`*`T`*`>` collection containing the distinct elements of both collections. This is the mathematical union of the two sets.

NOTE: The methods in Table 15-5 work as described with primitive data types. If you want to use them with collections of class objects, however, the methods will work with the objects' references, not the data that the objects contain. If you want to use these methods with collections of objects, based on the data that the objects contain, you will have to implement a custom comparator for those objects. We do not discuss custom comparators in this book.

The following code snippet demonstrates the `Distinct` method. The `result` collection will have four elements set to the values 1, 2, 3, and 4:

```
1   int[] set1 = { 1, 2, 2, 3, 3, 3, 4, 4, 4, 4 };
2   var result = set1.Distinct();
```

The following code snippet demonstrates the `Except` method. The `result` collection will have two elements set to the values 1 and 2:

```
1   int[] set1 = { 1, 2, 3, 4 };
2   int[] set2 = { 3, 4, 5, 6 };
3   var result = set1.Except(set2);
```

The following code snippet demonstrates the `Intersect` method. The `result` collection will have two elements set to the values 3 and 4:

```
1   int[] set1 = { 1, 2, 3, 4 };
2   int[] set2 = { 3, 4, 5, 6 };
3   var result = set1.Intersect(set2);
```

The following code snippet demonstrates the `Union` method. The `result` collection will have six elements set to the values 1, 2, 3, 4, 5, and 6:

```
1   int[] set1 = { 1, 2, 3, 4 };
2   int[] set2 = { 3, 4, 5, 6 };
3   var result = set1.Union(set2);
```

Partitioning Methods

A **partitioning method** is a method that divides a collection into two sections and returns one of the sections. LINQ provides the generation methods listed in Table 15-6.

The following code snippet demonstrates the `Skip` method. The `result` collection will have four elements set to the values 4, 5, 6, and 7:

```
1   int[] numbers = { 1, 2, 3, 4, 5, 6, 7 };
2   var result = numbers.Skip(3);
```

Table 15-6 LINQ partitioning methods

Method	Description
`Skip`	Summary: Skips a specified number of elements, starting at the first element of a collection. The remaining elements are returned. General format: *`collection`*`.Skip(`*`count`*`)` Description: In the general format, *`collection`* is a collection. The method returns an `IEnumerable<`*`T`*`>` collection containing a copy of *`collection`* with the first *`count`* elements skipped.
`SkipWhile`	Summary: Skips elements, starting at the first element, that match a condition. Returns the remaining elements. General format: *`collection`*`.SkipWhile(`*`FuncDelegate`*`)` Description: In the general format, *`collection`* is a collection and *`FuncDelegate`* is a `Func` delegate (or lambda expression) that has one parameter and returns a `bool` value. Each element of the *`collection`* will be passed as an argument to the `Func` delegate. Starting at the beginning of the collection, the elements that cause the `Func` delegate to return `true` are skipped. As soon as an element is found that causes the `Func` delegate to return `false`, the `SkipWhile` method returns an `IEnumerable<`*`T`*`>` collection containing a copy of that element and all the elements after it.
`Take`	Summary: Returns the specified number of elements from the collection, starting at the first element. General format: *`collection`*`.Take(`*`count`*`)` Description: In the general format, *`collection`* is a collection and *`count`* is an integer. The method returns an `IEnumerable<`*`T`*`>` collection containing the first *`count`* elements of the *`collection`*. If *`collection`* contains less than *`count`* elements, then all of the elements of *`collection`* are returned.
`TakeWhile`	Summary: Returns elements, starting at the first element, that match a condition. General format: *`collection`*`.TakeWhile(`*`FuncDelegate`*`)` Description: In the general format, *`collection`* is a collection and *`FuncDelegate`* is a `Func` delegate (or lambda expression) that has one parameter and returns a `bool` value. Each element of the *`collection`* will be passed as an argument to the `Func` delegate. Returns an `IEnumerable<`*`T`*`>` collection that contains all the elements from the beginning of *`collection`* up to the first element that causes the `Func` delegate to return `false`.

The following code snippet demonstrates the `SkipWhile` method. The `result` collection will have three elements set to the values 5, 6, and 7:

```
1   int[] numbers = { 1, 2, 3, 4, 5, 6, 7 };
2   var result = numbers.SkipWhile(item => item < 5);
```

The following code snippet demonstrates the `Take` method. The `result` collection will have three elements set to the values 1, 2, and 3:

```
1   int[] numbers = { 1, 2, 3, 4, 5, 6, 7 };
2   var result = numbers.Take(3);
```

The following code snippet demonstrates the `TakeWhile` method. The `result` collection will have four elements set to the values 1, 2, 3, and 4:

```
1   int[] numbers = { 1, 2, 3, 4, 5, 6, 7 };
2   var result = numbers.TakeWhile(item => item < 5);
```

Miscellaneous Methods

LINQ provides the miscellaneous methods listed in Table 15-7.

Table 15-7 Miscellaneous LINQ methods

Method	Description
`Concat`	Summary: Returns an `IEnumerable<T>` collection that is the concatenation of two collections. General format: *`collection1`*`.Concat(`*`collection2`*`)` Description: In the general format, *`collection1`* and *`collection2`* are collections. The method returns an `IEnumerable<T>` collection containing the elements of *`collection1`* followed by the elements of *`collection2`*.
`SequenceEqual`	Summary: Compares two collections and returns `true` if they are equal, or `false` otherwise. General format: *`collection1`*`.SequenceEqual(`*`collection2`*`)` Description: In the general format, *`collection1`* and *`collection2`* are collections. If the two collections contain the same values and the same number of elements, the method returns `true`. Otherwise, the method returns `false`. *Note: This method works as described with primitive data types. If you want to compare two collections of class objects, however, the method will compare the objects' references, not the data that the objects contain. If you want to perform a comparison of two collections of objects, based on the data that the objects contain, you will have to implement a custom comparator for those objects. We do not discuss custom comparators in this book.*

The following code snippet demonstrates the `Concat` method. The `result` collection will have six elements set to the values 1, 2, 3, 4, 5, and 6:

```
1  int[] numbers1 = { 1, 2, 3 };
2  int[] numbers2 = { 4, 5, 6 };
3  var result = numbers1.Concat(numbers2);
```

The following code snippet demonstrates the `SequenceEqual` method. The code will display both the messages `"numbers1 and numbers2 are the same"` and `"numbers1 and numbers3 are NOT the same"`.

```
1  int[] numbers1 = { 1, 2, 3 };
2  int[] numbers2 = { 1, 2, 3 };
3  int[] numbers3 = { 4, 5, 6 };
4
5  if (numbers1.SequenceEqual(numbers2))
6      MessageBox.Show("numbers1 and numbers2 are the same");
7
8  if (!numbers1.SequenceEqual(numbers3))
9      MessageBox.Show("numbers1 and numbers3 are NOT the same");
```

In Tutorial 15-3, you will use LINQ methods to perform set operations on two lists of data.

Tutorial 15-3:
Completing the *TeamPlayer* Project

VideoNote
Tutorial 15-3: Completing the *TeamPlayer* Project

In this tutorial, you complete the *TeamPlayer* project. The application creates two lists: one that holds the names of students on the baseball team, and another that holds the names of students on the basketball team. When you complete the application, it will perform the following set operations on the two lists:

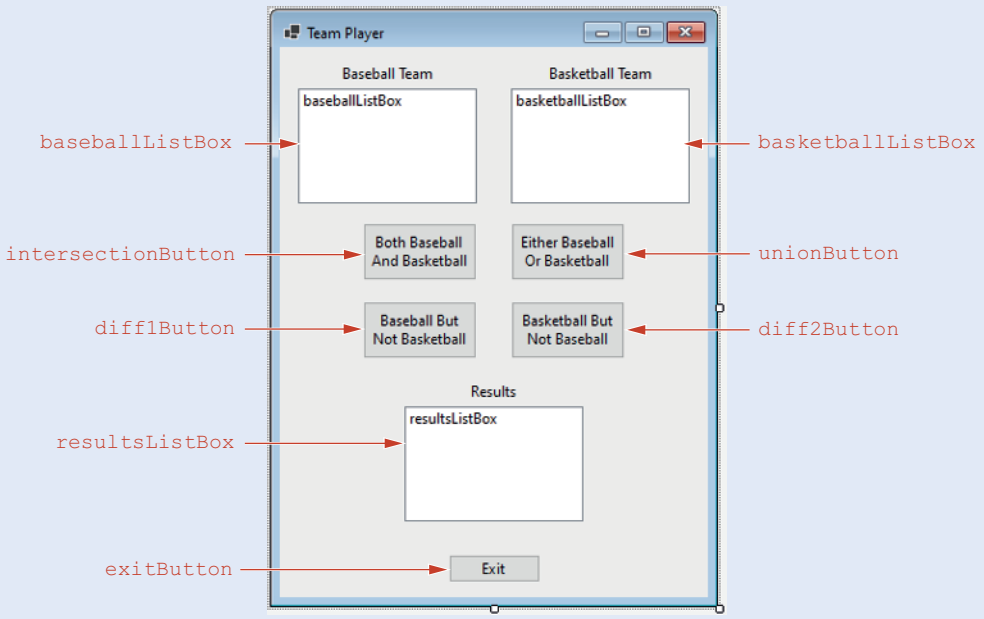
- It finds the intersection of the lists to display the names of students who play both sports.
- It finds the union of the lists to display the names of students who play either sport.
- It finds the difference of the baseball and basketball lists to display the names of students who play baseball but not basketball.
- It finds the difference of the baseball list from the basketball list to display the names of students who play basketball but not baseball. It also finds the difference of the basketball list from the baseball list to display the names of students who play baseball but not basketball.

Figure 15-3 shows the application's form, which has already been created for you. Program 15-4 shows the application's code as it will appear when you have completed this tutorial.

Step 1: Open the project named *TeamPlayer* in the *Chap15* folder of this *book's Student Sample Programs*.

Step 2: Open the Form1 form in the Designer. The form is shown, along with the names of the important controls, in Figure 15-3.

Figure 15-3 The *TeamPlayer* project's Form1 form



Step 3: Double-click any part of the form that does not contain a control. (Be sure not to click any of the Label controls, ListBox controls, or Button controls.) This opens the code editor, and you will see an empty event handler named

`Form1_Load`. You will also notice that we have already declared two lists as fields in the Form1 class. The baseball list is declared in lines 6-9, and the basketball list is declared in lines 12–15. These lists are initialized with the names of the baseball players and the basketball players, respectively.

Complete the `Form1_Load` event handler by typing the code shown in lines 24–40 in Program 15-4. Let's take a closer look at the code:

Line 25: This statement calls the `OrderBy` extension method on the `baseball` list to get the elements of the list sorted in ascending order.

Line 28: This statement calls the `OrderBy` extension method on the `basketball` list to get the elements of the list sorted in ascending order.

Lines 31–34: This `foreach` loop displays the sorted names of the baseball players in the `baseballListBox` control.

Lines 37–40: This `foreach` loop displays the sorted names of the basketball players in the `basketballListBox` control.

Step 4: Switch your view back to the *Designer* and double-click the `intersectionButton` control. In the code editor, you will see an empty event handler named `intersectionButton_Click`. Complete the `intersectionButton_Click` event handler by typing the code shown in lines 45–56 in Program 15-4. Let's take a closer look at the code:

Lines 46 and 47: This statement calls the `Intersect` extension method to get the elements that are in both the `baseball` and `basketball` lists, sorted in ascending order.

Line 50: This statement clears the contents of the `resultsListBox` control.

Lines 53–56: This `foreach` loop displays the results of the statement that was executed in lines 46 and 47, in the `resultsListBox` control.

Step 5: Switch your view back to the *Designer* and double-click the `unionButton` control. In the code editor you will see an empty event handler named `unionButton_Click`. Complete the `unionButton_Click` event handler by typing the code shown in lines 61–72 in Program 15-4. Let's take a closer look at the code:

Lines 62 and 63: This statement calls the `Union` extension method to get the elements that are in either the `baseball` or `basketball` lists, sorted in ascending order.

Line 66: This statement clears the contents of the `resultsListBox` control.

Lines 69–72: This `foreach` loop displays the results of the statement that was executed in lines 62 and 63, in the `resultsListBox` control.

Step 6: Switch your view back to the *Designer* and double-click the `diff1Button` control. In the code editor, you will see an empty event handler named `diff1Button_Click`. Complete the `diff1Button_Click` event handler by typing the code shown in lines 77–89 in Program 15-4. Let's take a closer look at the code:

Lines 79 and 80: This statement calls the `Except` extension method to get the elements that are in the `baseball` list, but not the `basketball` list, sorted in ascending order.

Line 83: This statement clears the contents of the `resultsListBox` control.

Lines 86–89: This `foreach` loop displays the results of the statement that was executed in lines 79 and 80, in the `resultsListBox` control.

Step 7: Switch your view back to the *Designer* and double-click the `diff2Button` control. In the code editor, you will see an empty event handler named `diff2Button_Click`. Complete the `diff2Button_Click` event handler by typing the code shown in lines 94–106 in Program 15-4. Let's take a closer look at the code:

Lines 96 and 97: This statement calls the `Except` extension method to get the elements that are in the `basketball` list, but not the `baseball` list, sorted in ascending order.

Line 100: This statement clears the contents of the `resultsListBox` control.

Lines 103–106: This `foreach` loop displays the results of the statement that was executed in lines 96 and 97, in the `resultsListBox` control.

Step 8: Switch your view back to the *Designer* and double-click the `exitButton` control. In the code editor, you will see an empty event handler named `exitButton_Click`. Complete the `exitButton_Click` event handler by typing the code shown in lines 111 and 112 in Program 15-4.

Step 9: Save the project. Then, press F5 on the keyboard or click the Start Debugging button (▶) on the toolbar to compile and run the application. Click each of the buttons to test the code. The program output should match the following:

Both Baseball and Basketball	Either Baseball or Basketball	Baseball But Not Basketball	Basketball But Not Baseball
Alicia	Aida	Ada	Eva
Carmen	Alicia	Jodi	Sarah
	Carmen		
	Eva		
	Jodi		
	Sarah		

When you are finished, click the Exit button to close the application.

Program 15-4 Completed Form1 code for the *Team Player* application

```
1  namespace TeamPlayer
2  {
3      public partial class Form1 : Form
4      {
5          // Baseball team members
6          List<string> baseball = new List<string>()
7          {
8              "Jodi", "Carmen", "Alicia", "Aida"
9          };
10
11         // Basketball team members
12         List<string> basketball = new List<string>()
13         {
14             "Eva", "Carmen", "Alicia", "Sarah"
15         };
16
17         public Form1()
18         {
19             InitializeComponent();
20         }
```

```
21
22          private void Form1_Load(object sender, EventArgs e)
23          {
24              // Get the baseball players sorted in ascending order.
25              var baseballPlayers = baseball.OrderBy(str => str);
26
27              // Get the basketball players sorted in ascending order.
28              var basketballPlayers = basketball.OrderBy(str => str);
29
30              // Display the baseball players.
31              foreach (string player in baseballPlayers)
32              {
33                  baseballListBox.Items.Add(player);
34              }
35
36              // Display the basketball players.
37              foreach (string player in basketballPlayers)
38              {
39                  basketballListBox.Items.Add(player);
40              }
41          }
42
43          private void intersectionButton_Click(object sender,
            EventArgs e)
44          {
45              // Get the elements that are in both lists, sorted.
46              var result = baseball.Intersect(basketball)
47                                   .OrderBy(str => str);
48
49              // Clear the Results ListBox.
50              resultsListBox.Items.Clear();
51
52              // Display the results.
53              foreach (string player in result)
54              {
55                  resultsListBox.Items.Add(player);
56              }
57          }
58
59          private void unionButton_Click(object sender, EventArgs e)
60          {
61              // Get the elements that are in either list,
                sorted.
62              var result = baseball.Union(basketball)
63                                   .OrderBy(str => str);
64
65              // Clear the Results ListBox.
66              resultsListBox.Items.Clear();
67
68              // Display the results.
69              foreach (string player in result)
70              {
71                  resultsListBox.Items.Add(player);
72              }
73          }
74
75          private void diff1Button_Click(object sender, EventArgs e)
```

```
        {
            // Get the elements that are in the baseball list
            // but not in the basketball list, sorted.
            var result = baseball.Except(basketball)
                                 .OrderBy(str => str);

            // Clear the Results ListBox.
            resultsListBox.Items.Clear();

            // Display the results.
            foreach (string player in result)
            {
                resultsListBox.Items.Add(player);
            }
        }

        private void diff2Button_Click(object sender, EventArgs e)
        {
            // Get the elements that are in the basketball list
            // but not in the baseball list, sorted.
            var result = basketball.Except(baseball)
                                   .OrderBy(str => str);

            // Clear the Results ListBox.
            resultsListBox.Items.Clear();

            // Display the results.
            foreach (string player in result)
            {
                resultsListBox.Items.Add(player);
            }
        }

        private void exitButton_Click(object sender, EventArgs e)
        {
            // Close the form.
            this.Close();
        }
    }
}
```

Checkpoint

15.7 What is the difference between the `ElementAt` method and the `ElementAtOrDefault` method?

15.8 What happens if you call the `Single` method on a collection that has more than one element?

15.9 Suppose you need an `IEnumerable<int>` collection containing sequential integers, starting at a specific value. What method would you call to create the collection?

15.10 Suppose you need to remove all duplicates from a collection of primitive values. What method would you call?

15.11 If you need to get a specific number of elements from the beginning of a collection, what method would you use?

15.12 What method would you use to compare two collections of a primitive type to determine whether they are equal?

15.4 Using LINQ to Query a Database

CONCEPT: **By using an Object Relational Mapping tool, or ORM, you can use LINQ to query a database.**

NOTE: In this section's tutorials, we will use the ***Windows Forms App (.NET Framework)*** project type, instead of the *Windows Forms App* project type. Some of the visual tools discussed in this section have not yet been implemented in the *Windows Forms App* project type, so be sure to pick the ***Windows Forms App (.NET Framework)*** project type when starting a new project in this section.

Previously in this chapter, we discussed LINQ as a tool for querying data in arrays and lists. Using a tool called **LINQ to SQL**, you can also use LINQ to query databases. LINQ to SQL is an **Object Relational Mapping**, or **ORM** tool. An ORM tool is software that converts a database into objects. The programmer can then write LINQ queries that work with these objects, and behind the scenes, LINQ to SQL translates the LINQ queries into SQL queries.

There are three types of objects that we will work with in this section: data context objects, table objects, and entity objects. Let's take a closer look at each of these.

Data Context Objects

A **data context object** represents a database. It has the ability to connect to a database, translate LINQ queries into SQL queries, and execute those SQL queries on the database. When you set up LINQ to SQL to work with a specific database, Visual Studio will automatically generate a class in your project that is derived from the `DataContext` class in the .NET Framework. In your code, you can create an object that is an instance of the derived class. This object is the data context object that represents the database.

Table Objects

A **table object** represents a table in a database. A table object is an instance of the `Table` class in the .NET Framework. It is essentially an `IEnumerable<T>` collection that contains the rows of a database table as its elements. In code, you can iterate over the elements of a table object in the same way that you can iterate over a `List` or an array. You can also execute LINQ queries on table objects.

When Visual Studio generates a derived `DataContext` class to represent a specific database, the derived `DataContext` class will contain table objects as properties, each representing a table in the database. For example, if you are working with a database that contains two tables, the object that represents the database will have two table properties, one for each table in the database.

Entity Objects

An **entity object** represents a row in a table. It has properties that represent the columns in the row. For example, let's say that we are working with a table, and each row in the table has the following three columns:

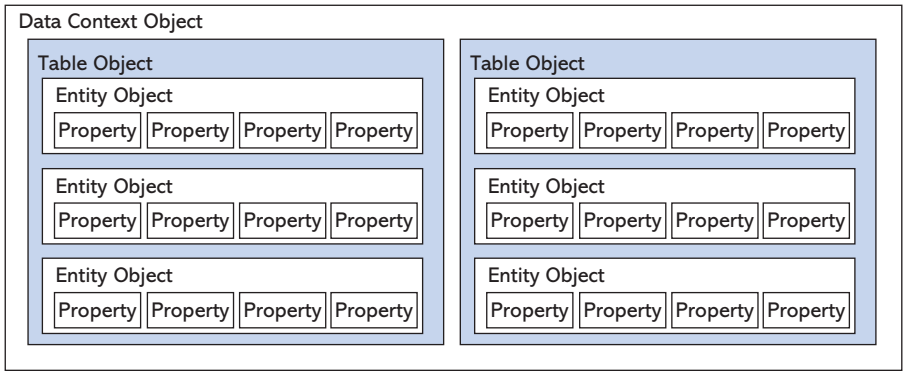
- An `int` column named `EmployeeID`
- An `nvarchar(50)` column named `Name`
- A `money` column named `PayRate`

An entity object that represents this row will have the following three properties:

- An `int` property named `EmployeeID`
- A `string` property named `Name`
- A `decimal` property named `PayRate`

Figure 15-4 illustrates a data context object that contains two table objects. Each table object is a collection of entity objects. This corresponds to a database that has two tables.

Figure 15-4 Data context, table, and entity objects



Installing LINQ to SQL

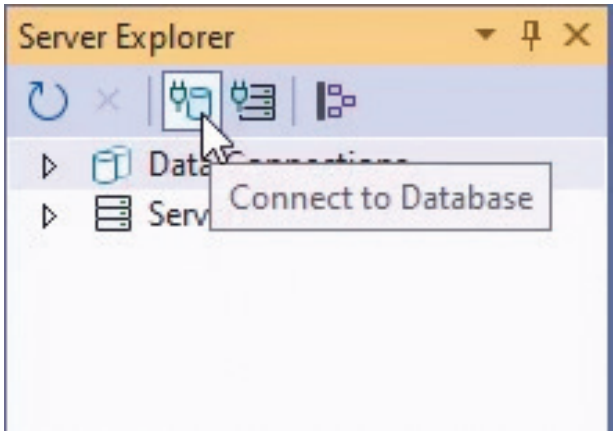
LINQ to SQL is not automatically installed with Visual Studio. Before you can use it, you must use the Visual Studio Installer to download and install it. See Appendix E for instructions.

Setting Up LINQ to SQL to Work with a Database

To set up LINQ to SQL to work with a database, you should typically follow these steps:

1. **Add a connection to the database that you want to work with.**
   There are different ways to accomplish this. One way is to open the *Server Explorer* and click the *Connect to Database* icon, as shown in Figure 15-5. This causes the *Add Connection* window to appear, as shown in Figure 15-6, which lets you browse your file system and select the desired database file.

Figure 15-5 Connect to Database



2. **Add a DBML file to the project.**
   In this step, you create a **DBML file** in your project. **DBML** stands for **Database Markup Language**. Visual Studio uses the DBML file to automatically generate the classes needed for LINQ to SQL. Here are the steps for adding a DBML file to the project:
   - Click *Project* on the menu bar, then click *Add New Item*....
   - The *Add New Item* window appears next. As shown in Figure 15-7, select *Data* in the left pane, select *LINQ to SQL Classes* in the center pane.
   - In the *Name* text box, change the default name of the DBML file to something more descriptive, and click the *Add* button to add the file to the project.
3. **Use the Object Relational Designer to select the database tables you want to work with.**
   The **Object Relational Designer**, shown in Figure 15-8, will appear as soon as you add the DBML file to your project. The Object Relational Designer is a visual design tool that you use to create the classes needed for LINQ to SQL to work with a particular database. In the simplest case, you drag a table from the Server Explorer and drop it into the Object Relational Designer. The LINQ to SQL classes needed to represent the table will automatically be created in the project.

Figure 15-6 The *Add Connection* window

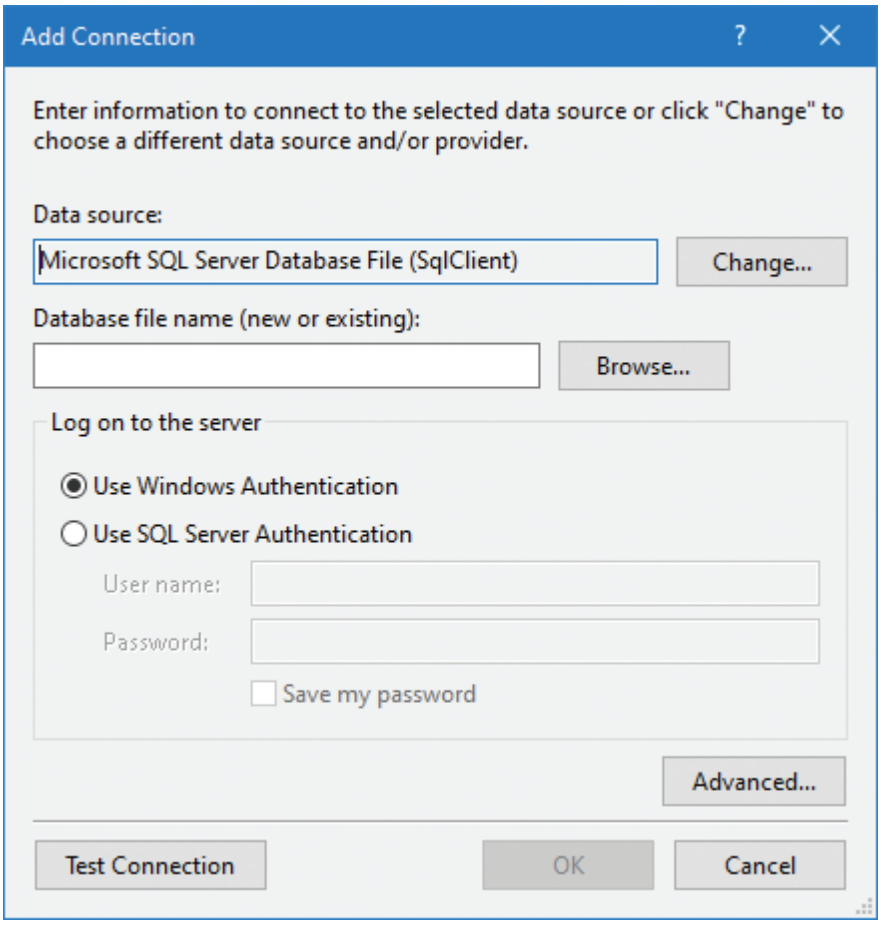


Figure 15-7 The *Add New Item* window

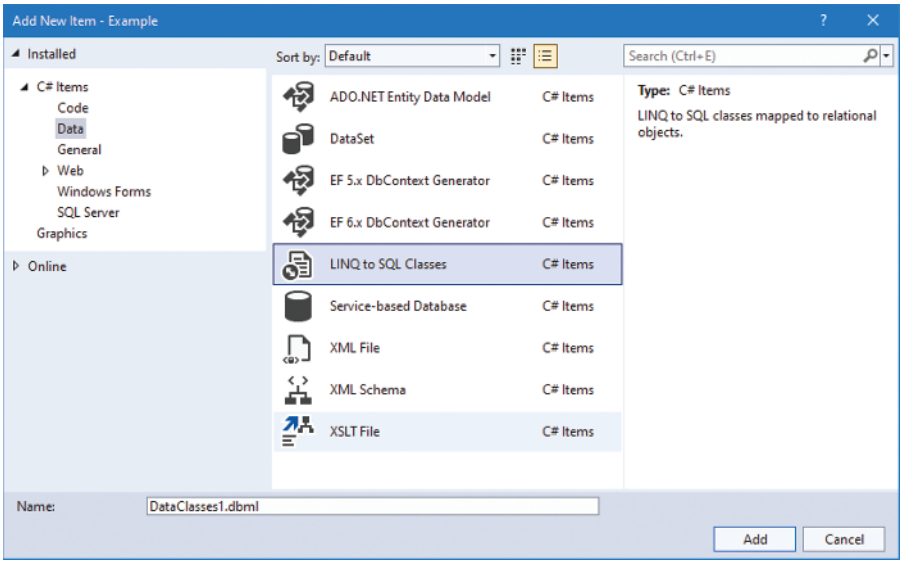
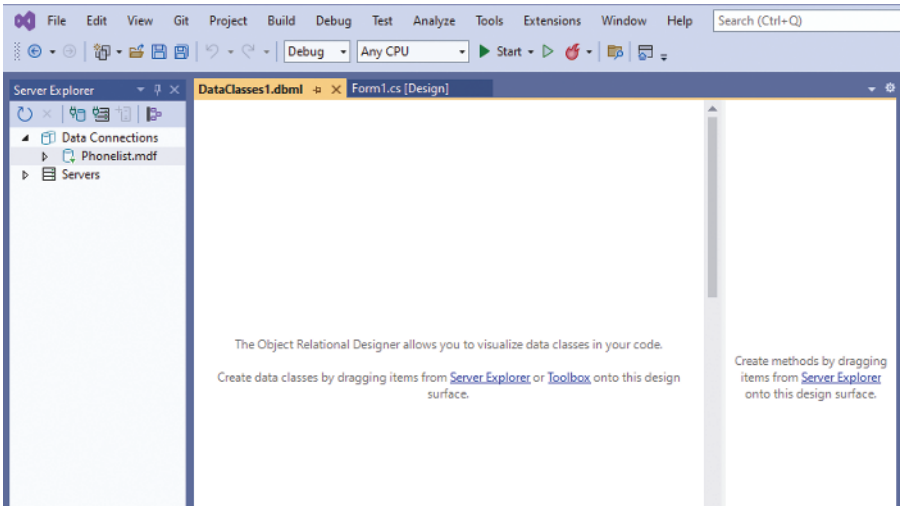


Figure 15-8 The *Object Relational Designer* window



Once you have performed these steps, the following classes will automatically be created in your project:

- A data context class will be created to represent the database. This class is derived from the `DataContext` class in the .NET Framework.
- The data context class contains one or more table objects (as properties) that represent the tables in the database. Each table object is an instance of the `Table` class in the .NET Framework. Each table object is a collection of entity objects that represent the rows in the table.
- The entity objects contain properties that represent the columns in the table.

LINQ to SQL Naming Conventions

When Visual Studio generates a data context class, the naming convention is straightforward: The name of the data context class is based on the name of the DBML file. For example, if the DBML file is named Personnel.dbml, then the data context class will be named `PersonnelDataContext`.

The naming convention for table objects and their associated entity classes is not quite as simple. By default, a table object's name is the plural version of the database table's name, and the associated entity class's name is the singular version of the database table's name. For example, suppose our database table is named `Employee`. Visual Studio will name the table object `Employees` and the entity class `Employee`. This means that the `Employees` object (a table object) will be a collection of `Employee` objects (entity objects).

Let's look at another example. Suppose a database contains the inventory of a used car dealer. The database is named UsedCars.mdf, and it has a table named `Car`. Each row in the `Car` table contains data about a car in the inventory. We are creating an application to work with the database, so our DBML file is named UsedCars.dbml. Visual Studio will use the following names for our LINQ to SQL classes:

- The data context class will be named `UsedCarsDataContext`.
- The `UsedCarsDataContext` class will have a table property named `Cars` (the plural version of the database table name).
- The entity class for the table will be named `Car` (the singular version of the database table name).

Working with the Data Context Object

Once you have used the Object Relational Designer to generate the desired LINQ to SQL classes, you are ready to write code that works with the database. The first step is to create an instance of the data context class. For example, suppose we are working with the used car database previously described. To create an instance of the `UsedCarsDataContext` class, we would write a statement such as this:

```
UsedCarsDataContext db = new UsedCarsDataContext();
```

This statement creates a `UsedCarsDataContext` object named `db`. We can use the `db` object to access the database's table, as shown in the following code snippet. Assume the database table has a column named `Mileage`. This code iterates over each row in the table, adding the `Mileage` column to a ListBox.

```
foreach(Car c in db.Cars)
{
    mileageListBox.Items.Add(c.Mileage.ToString());
}
```

In Tutorial 15-4, you will complete an application by connecting it to a database, creating the LINQ to SQL classes needed to represent the database, and writing code that iterates over a table in the database.

Tutorial 15-4:
Completing the *PhonelistNames* Project

VideoNote
Tutorial 15-4: Completing the *PhonelistNames* Project

In this tutorial, you will complete an application that uses the Phonelist database that was originally introduced in Chapter 13. Recall that the Phonelist database has a table named `Person` with the following columns:

- `PersonID` – an `int` identity column containing a unique ID for each row
- `Name` – an `nvarchar(50)` column containing a person's name
- `Phone` – an `nvarchar(50)` column containing a person's phone number

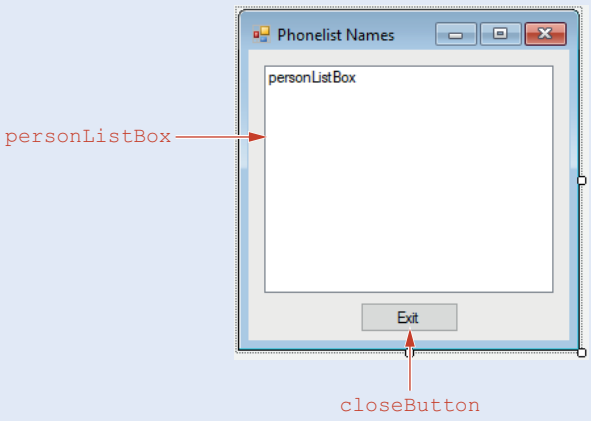
Our application will be simple. It will use a LINQ query to retrieve all the names in the `Person` table, and display those names in a ListBox named `personListBox`.

Step 1: If you have not already done so, install the LINQ to SQL component (as described in Appendix E).

Step 2: Open the project named *PhonelistNames* in the *Chap15* folder of this book's *Student Sample Programs*.

Step 3: Open the Form1 form in the Designer. The form is shown, along with the names of the important controls, in Figure 15-9.

Figure 15-9 The *PhonelistNames* project Form1 form



Step 4: Now you will create a connection to the Phonelist.mdf database, which is in the *Chap15* folder of this book's *Student Sample Programs*.

Perform the following steps to connect the application to the Phonelist.mdf database:

- If you do not see the *Server Explorer* window, click *View* on the menu bar, and then click *Server Explorer*.
- In the *Server Explorer* window, click the *Connect to Database* icon as previously shown in Figure 15-4.
- The *Add Connection* window should appear next, as previously shown in Figure 15-7. Make sure *Microsoft SQL Server Database File (SqlClient)* is selected under *Data Source*. Then click the *Browse...* button to select the database file. Navigate to the *Chap15\Databases* folder of the *Student Sample Programs* and select the Phonelist.mdf file.
- Click the *Test Connection* button to test the connection. You should see the message *Test Connection Succeeded*. Click the *OK* button to complete the connection.

Step 5: Now you will add the LINQ to SQL classes to the project. Perform the following steps:

- Click *Project* on the menu bar, then click *Add New Item...*.
- The *Add New Item* window appears next. As shown in Figure 15-10, select *Data* in the left pane, select *LINQ to SQL Classes* in the center pane, and change the name of the file to *Phonelist.dbml*. Click the *Add* button to add the classes to the project.

Step 6: Now you will create the classes you need to access the database.

- The *Object Relational Designer* should be open, as shown in Figure 15-11. In the *Server Explorer* window, expand the Phonelist.mdf entry, and then expand the *Tables* Entry, as shown in Figure 15-12.

Figure 15-10 The Add New Item window

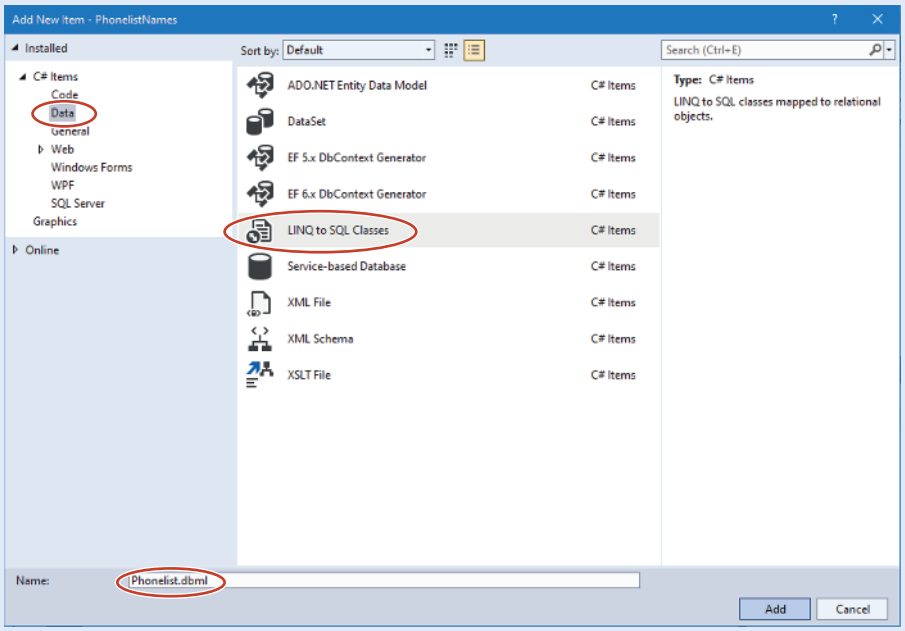
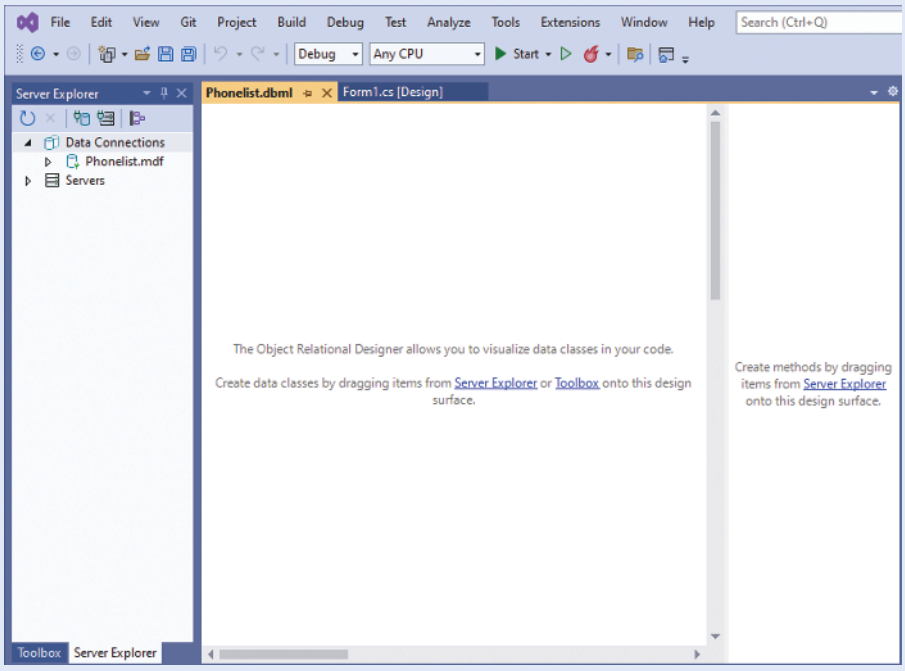


Figure 15-11 The Object Relational Designer



- Drag the `Person` table from the *Server Explorer* window and drop it in the *Object Relational Designer*. You will see the window shown in Figure 15-13. Click *Yes*.
- The *Object Relational Designer* now shows a visual representation of a class named `Person`, as shown in Figure 15-14. Notice that the class has properties named PersonID, Name, and Phone. This class is the entity class that represents a row in the `Person` table.

Figure 15-12 Expand Tables in the Server Explorer

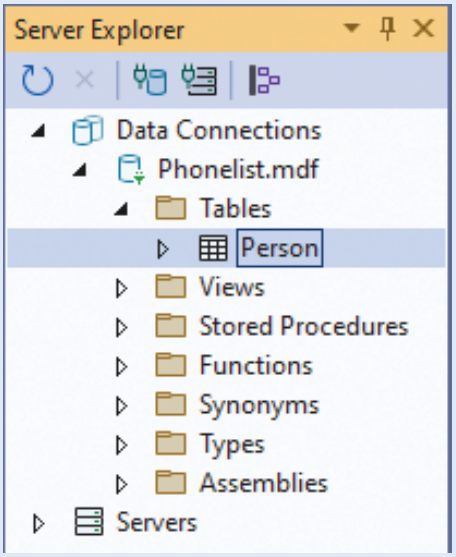


Figure 15-13 Message about copying the database to the project

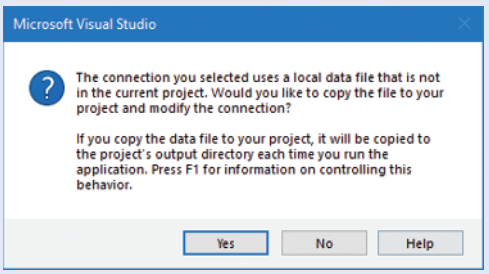
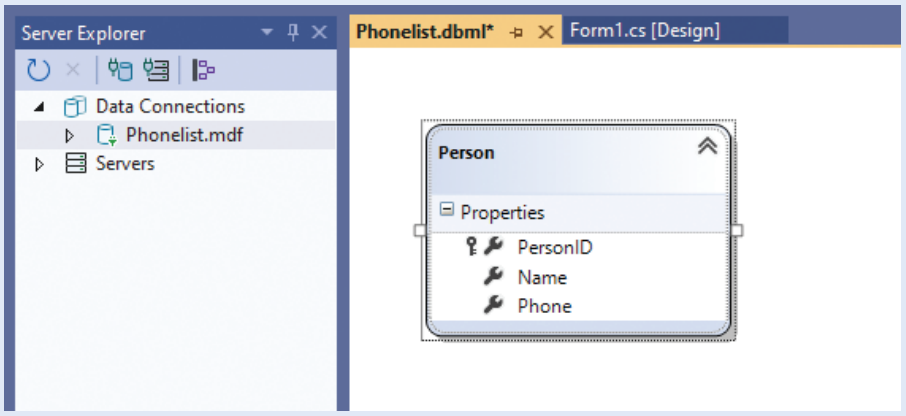


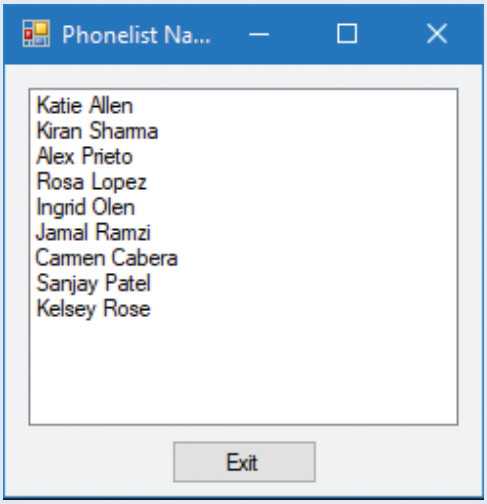
Figure 15-14 The `Person` Class Created in the Object Relational Designer



At this point, your project now has a data context class named `PhonelistDataContext`. The `PhonelistDataContext` class contains table object property named `Persons`. (Notice the pluralization of the name.) The `Persons` object is a collection of `Person` objects. Each `Person` object represents a row in the database table.

Step 7: Open the Form1 form in the Designer. The form, along with the names of the important controls, was previously shown in Figure 15-15.

Figure 15-15 The application's form at runtime



Step 8: Double-click any part of the form that does not contain a control. (Be sure not to click the ListBox control or the Button control.) This opens the code editor, and you will see an empty event handler named `Form1_Load`.

Complete the `Form1_Load` event handler by typing the code shown in lines 22–29 in Program 15-5. Let's take a closer look at the code:

Line 23: This statement creates a `PhoneListDataContext` object named `db`. We chose the name `db` because the object represents the database.

Lines 26–29: This `foreach` loop iterates over each element in the `Persons` collection. For each element, it displays the Name property in the ListBox.

Step 9: Switch your view back to the Form1 form in the *Designer* and double-click the `exitButton` control. In the code editor, you will see an empty event handler named `exitButton_Click`. Complete the `exitButton_Click` event handler by typing the code shown in lines 34 and 35 in Program 15-5.

Step 10: Save the project. Then, press F5 on the keyboard or click the Start Debugging button ( ) on the toolbar to compile and run the application. The application's form should appear as shown in Figure 15-15. When you are finished, click the *Close* button to close the application.

Program 15-5 Completed Form1 code for the *PhonelistNames* project

```
1 using System;
2 using System.Collections.Generic;
3 using System.ComponentModel;
4 using System.Data;
5 using System.Drawing;
6 using System.Linq;
7 using System.Text;
8 using System.Threading.Tasks;
```

```
 9 using System.Windows.Forms;
10
11 namespace PhonelistNames
12 {
13     public partial class Form1 : Form
14     {
15         public Form1()
16         {
17             InitializeComponent();
18         }
19
20         private void Form1_Load(object sender, EventArgs e)
21         {
22             // Create a Data Context object.
23             var db = new PhonelistDataContext();
24
25             // Display all the names in the ListBox.
26             foreach (Person item in db.Persons)
27             {
28                 personListBox.Items.Add(item.Name);
29             }
30         }
31
32         private void exitButton_Click(object sender, EventArgs e)
33         {
34             // Close the form.
35             this.Close();
36         }
37     }
38 }
```

Writing LINQ to SQL Queries

In Tutorial 15-4, you used a simple `foreach` loop to iterate over a table object. However, the point of creating the LINQ to SQL classes is to use LINQ queries with the database! Using LINQ to retrieve data from a database is just as easy as using LINQ to retrieve data from a `List` or an array. Let's continue to use the Phonelist database (from Tutorial 15-4) as our example. The following code snippet creates a data context object and then uses a LINQ query to retrieve all the phone numbers from the `Persons` table object. The results are sorted by name.

```
var db = new PhonelistDataContext();
var results = from person in db.Persons
              orderby person.Name
              select person.Phone;
foreach(string name in results)
{
    resultsListBox.Items.Add(name);
}
```

The following code shows another example. This code retrieves the phone numbers of all the rows in which the Name property contains “Smith”:

```
var db = new PhonelistDataContext();
var results = from person in db.Persons
              where person.Name.Contains("Smith")
              select person.Phone;
```

```
foreach(string ph in results)
{
    resultsListBox.Items.Add(ph);
}
```

In Tutorial 15-5, you will complete an application by connecting it to a database, creating the LINQ to SQL classes needed to represent the database, and writing various LINQ queries to retrieve data from the database.

Tutorial 15-5:
Completing the *ProductReports* Project

VideoNote
Tutorial 15-5: Completing the *ProductReports* Project

In this tutorial, you complete an application that uses the ProductDB database that was originally introduced in Chapter 13. Recall that the ProductDB database has a table named `Product` with the following columns:

- `Product_Number`–an `nchar(5)` column, primary key, containing a product number
- `Description`–an `nvarchar(50)` column containing a product description
- `Units_On_Hand`–an `int` column containing the number of units on hand
- `Price`–a `money` column containing the product's price

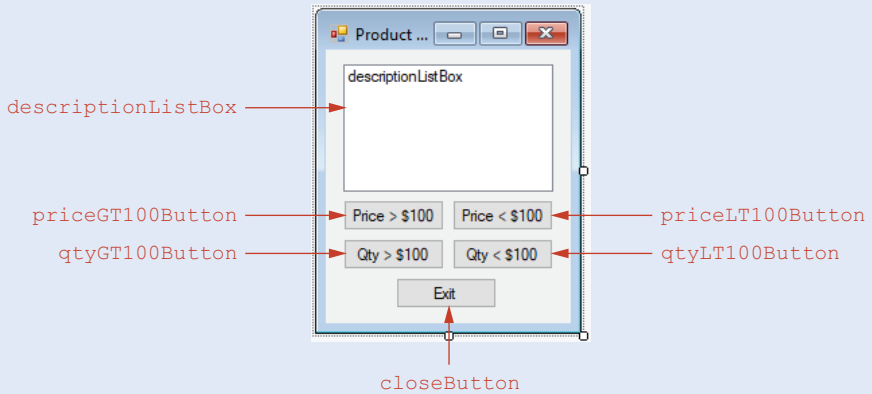
The application's form is shown in Figure 15-16. The Button controls perform the following actions:

- The `priceGT100Button` executes a LINQ query to get all the items from the database with `Price` greater than $100.
- The `qtyGT100Button` executes a LINQ query to get all the items from the database with `Units_On_Hand` greater than 100.
- The `priceLT100Button` executes a LINQ query to get all the items from the database with `Price` less than $100.
- The `qtyLT100Button` executes a LINQ query to get all the items from the database with `Units_On_Hand` less than 100.

Step 1: Open the project named *ProductReports* in the *Chap15* folder of this book's *Student Sample Programs*.

Step 2: Open the Form1 form in the Designer. The form is shown, along with the names of the important controls, in Figure 15-16.

Figure 15-16 The *ProductReports* Project's Form1 Form



Step 3: Now you will create a connection to the ProductDB.mdf database, which is in the *Chap15* folder of this book's *Student Sample Programs*.

Perform the following steps to connect the application to the ProductDB.mdf database:

- If you do not see the *Server Explorer* window, click *View* on the menu bar, and then click *Server Explorer*.
- In the *Server Explorer* window, click the *Connect to Database* icon as previously shown in Figure 15-4.
- The *Add Connection* window should appear next, as previously shown in Figure 15-7. Make sure *Microsoft SQL Server Database File (SqlClient)* is selected under *Data Source*. Then click the *Browse…* button to select the database file. Navigate to the *Chap15\Databases* folder of the *Student Sample Programs* and select the ProductDB.mdf file.
- Click the *Test Connection* button to test the connection. You should see the message *Test Connection Succeeded*. Click the *OK* button to complete the connection.

Step 4: Now you will add the LINQ to SQL classes to the project. Perform the following steps:

- Click *Project* on the menu bar, then click *Add New Item…*.
- The *Add New Item* window appears next. Select *Data* in the left pane, select *LINQ to SQL Classes* in the center pane, and change the name of the file to *ProductDB.dbml*. Click the *Add* button to add the classes to the project.

Step 5: Now you will use the *Object Relational Designer* to create the classes you need to access the database.

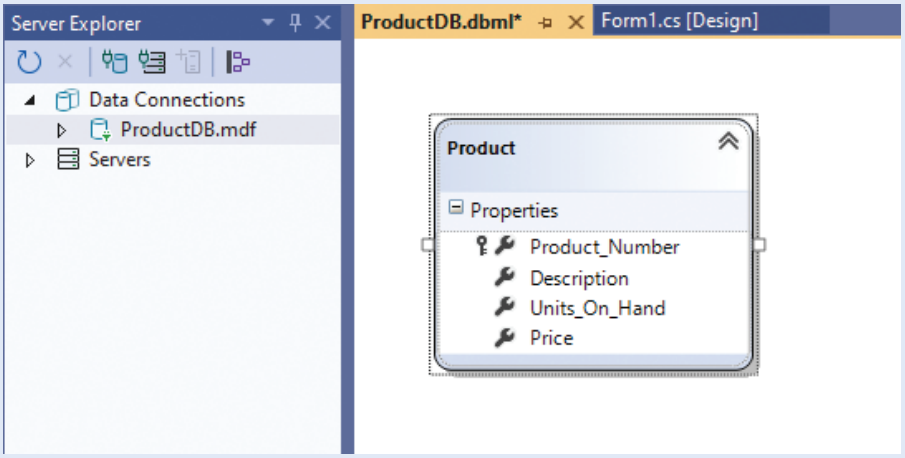
- The *Object Relational Designer* should be open. In the *Server Explorer* window, expand the ProductDB.mdf entry, and then expand the *Tables* Entry.
- Drag the `Product` table from the *Server Explorer* window and drop it in the *Object Relational Designer*. You will see a window displaying a message indicating that you have selected a database file located outside the project folder, and you are asked if you want to copy the file into the project. Click *Yes*.
- The *Object Relational Designer* now shows a visual representation of a class named `Product`, as shown in Figure 15-17. Notice that the class has properties named Product_Number, Description, Units_On_Hand, and Price. This class is the entity class that represents a row in the `Product` table.

At this point, your project now has a data context class named `ProductDBDataContext`. The `ProductDBDataContext` class contains table object property named `Products`. (Notice the pluralization of the name.) The `Products` object is a collection of `Product` objects. Each `Product` object represents a row in the database table.

Step 6: Open the Form1 form in the Designer. The form, along with the names of the important controls, was previously shown in Figure 15-16.

Step 7: Double-click the `priceGT100Button` control. This opens the code editor, and you will see an empty event handler named `priceGT100Button_Click`.

Figure 15-17 The `Product` class created in the Object Relational Designer



Complete the `priceGT100Button_Click` event handler by typing the code shown in lines 22–37 in Program 15-6. Let's take a closer look at the code:

Line 23: This statement creates a `ProductDBDataContext` object named `db`. We chose the name `db` because the object represents the database.

Lines 26–28: This statement executes a LINQ query that returns the descriptions of all the items in the `db.Products` collection with a price greater than 100. The results of the query are assigned to the `results` variable.

Line 31: This statement clears the contents of the `descriptionListBox` control.

Lines 34–37: This `foreach` loop iterates over the query results and adds each description to the `descriptionListBox` control.

Step 8: Open the Form1 form in the Designer and double-click the `priceLT100Button` control. This opens the code editor, and you will see an empty event handler named `priceLT100Button_Click`.

Complete the `priceLT100Button_Click` event handler by typing the code shown in lines 42–57 in Program 15-6. Let's take a closer look at the code:

Line 43: This statement creates a `ProductDBDataContext` object named `db`. We chose the name `db` because the object represents the database.

Lines 46–48: This statement executes a LINQ query that returns the descriptions of all the items in the `db.Products` collection with a price less than 100. The results of the query are assigned to the `results` variable.

Line 51: This statement clears the contents of the `descriptionListBox` control.

Lines 54–57: This `foreach` loop iterates over the query results and adds each description to the `descriptionListBox` control.

Step 9: Open the Form1 form in the Designer and double-click the `qtyGT100Button` control. This opens the code editor, and you will see an empty event handler named `qtyGT100Button_Click`.

Complete the `qtyGT100Button_Click` event handler by typing the code shown in lines 62–77 in Program 15-6. Let's take a closer look at the code:

Line 63: This statement creates a `ProductDBDataContext` object named `db`. We chose the name `db` because the object represents the database.

Lines 66–68: This statement executes a LINQ query that returns the descriptions of all the items in the `db.Products` collection with units on hand greater than 100. The results of the query are assigned to the `results` variable.

Line 71: This statement clears the contents of the `descriptionListBox` control.

Lines 74–77: This `foreach` loop iterates over the query results and adds each description to the `descriptionListBox` control.

Step 10: Open the Form1 form in the Designer and double-click the `qtyLT100Button` control. This opens the code editor, and you will see an empty event handler named `qtyLT100Button_Click`.

Complete the `qtyLT100Button_Click` event handler by typing the code shown in lines 82–97 in Program 15-6. Let's take a closer look at the code:

Line 83: This statement creates a `ProductDBDataContext` object named `db`. We chose the name `db` because the object represents the database.

Lines 86–88: This statement executes a LINQ query that returns the descriptions of all the items in the `db.Products` collection with units on hand less than 100. The results of the query are assigned to the `results` variable.

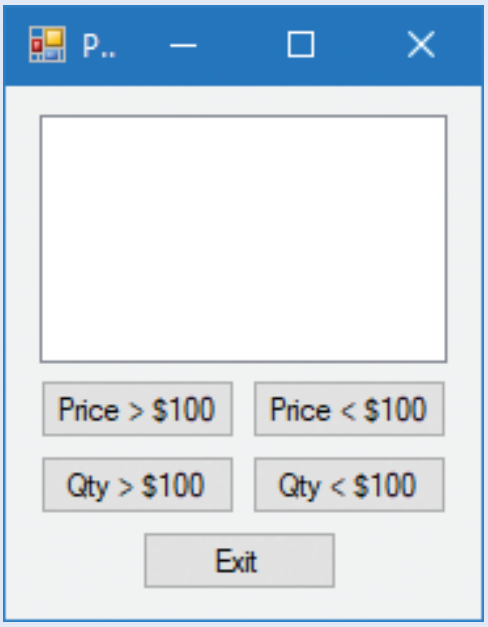
Line 91: This statement clears the contents of the `descriptionListBox` control.

Lines 94–97: This `foreach` loop iterates over the query results and adds each description to the `descriptionListBox` control.

Step 11: Open the Form1 form in the Designer and double-click the `exitButton` control. This opens the code editor, and you will see an empty event handler named `exitButton_Click`. Complete the `exitButton_Click` event handler by typing the code shown in lines 102 and 103 in Program 15-6.

Step 12: Save the project. Then, press F5 on the keyboard or click the Start Debugging button ( ) on the toolbar to compile and run the application. The application's form should appear as shown in Figure 15-18. Click each of the buttons to test them. When you are finished, click the *Close* button to close the application.

Figure 15-18 The application's form at runtime



Program 15-6 Completed Form1 code for the *Product Reports* application

```
 1 using System;
 2 using System.Collections.Generic;
 3 using System.ComponentModel;
 4 using System.Data;
 5 using System.Drawing;
 6 using System.Linq;
 7 using System.Text;
 8 using System.Threading.Tasks;
 9 using System.Windows.Forms;
10
11 namespace ProductReports
12 {
13     public partial class Form1 : Form
14     {
15         public Form1()
16         {
17             InitializeComponent();
18         }
19
20         private void priceGT100Button_Click(object sender,
           EventArgs e)
21         {
22             // Create a data context object.
23             var db = new ProductDBDataContext();
24
25             // Get the products with a price > 100.
26             var results = from product in db.Products
27                           where product.Price > 100m
28                           select product.Description;
29
30             // Clear the ListBox.
31             descriptionListBox.Items.Clear();
32
33             // Display those products' descriptions.
34             foreach (string description in results)
35             {
36                 descriptionListBox.Items.Add(description);
37             }
38         }
39
40         private void priceLT100Button_Click(object sender,
           EventArgs e)
41         {
42             // Create a data context object.
43             var db = new ProductDBDataContext();
44
45             // Get the products with a price < 100.
46             var results = from product in db.Products
47                           where product.Price < 100m
48                           select product.Description;
49
50             // Clear the ListBox.
51             descriptionListBox.Items.Clear();
52
53             // Display those products' descriptions.
54             foreach (string description in results)
```

```
55              {
56                  descriptionListBox.Items.Add(description);
57              }
58          }
59
60          private void qtyGT100Button_Click(object sender,
            EventArgs e)
61          {
62              // Create a data context object.
63              var db = new ProductDBDataContext();
64
65              // Get the products with units on hand > 100.
66              var results = from product in db.Products
67                            where product.Units_On_Hand > 100
68                            select product.Description;
69
70              // Clear the ListBox.
71              descriptionListBox.Items.Clear();
72
73              // Display those products' descriptions.
74              foreach (string description in results)
75              {
76                  descriptionListBox.Items.Add(description);
77              }
78          }
79
80          private void qtyLT100Button_Click(object sender,
            EventArgs e)
81          {
82              // Create a data context object.
83              var db = new ProductDBDataContext();
84
85              // Get the products with units on hand < 100.
86              var results = from product in db.Products
87                            where product.Units_On_Hand < 100
88                            select product.Description;
89
90              // Clear the ListBox.
91              descriptionListBox.Items.Clear();
92
93              // Display those products' descriptions.
94              foreach (string description in results)
95              {
96                  descriptionListBox.Items.Add(description);
97              }
98          }
99
100         private void exitButton_Click(object sender, EventArgs e)
101         {
102             // Close the form.
103             this.Close();
104         }
105     }
106 }
```

Using LINQ Queries with Data-Bound Controls

You learned in Chapter 13 that a data-bound control is a user interface control that is connected to a data source. Data-bound controls automatically display data from the data source and can be used to change the data that they are bound to. One of the simplest and most powerful data-bound controls is the DataGridView control. A DataGridView control can display an entire database table in a scrollable grid on an application's form.

Data-bound controls can also be used to display the results of a LINQ query. These are the steps to follow if you want to use LINQ to query a database table, and then display those results in a data-bound control:

1. In Visual Studio, connect the project to the database and use the Object Relational Designer to create an entity class for the table you wish to query.
2. Add a new data source to the project, and select Object as the data source type.
3. Select the entity class as the data object.
4. In the Designer, open a form and create the desired data-bound control (such as a DataGridView).
5. In code, write a LINQ query that returns a collection of objects that instances of the entity class you selected in step 3.
6. Assign the results of the LINQ query to the data-bound control's DataSource property.

You will practice these steps in Tutorial 15-6. In the tutorial, you create an application that uses LINQ to query a database table, and displays the query results in a DataGridView control.

Tutorial 15-6:
Creating the *QueryDataSource* Project

VideoNote
Tutorial 15-6: Creating the *QueryDataSource* Project

In this tutorial, you will create an application that uses a DataGridView control to display the results of a LINQ query that is executed on the ProductDB database.

Step 1: Launch Visual Studio and start a new *Windows Desktop App (.NET Framework)* project named *QueryDataSource*. Set the Form1 form's Text property to *Products*.

Step 2: Create a connection to the ProductDB.mdf database, which is in the *Chap15* folder of this book's *Student Sample Programs*.

Perform the following steps to create the connection:

- If you do not see the *Server Explorer* window, click *View* on the menu bar, and then click *Server Explorer*.
- In the *Server Explorer* window, click the *Connect to Database* icon as previously shown in Figure 15-4.
- The *Add Connection* window should appear next, as previously shown in Figure 15-7. Make sure *Microsoft SQL Server Database File (SqlClient)* is selected under *Data Source*. Then click the *Browse...* button to select the database file. Navigate to the *Chap15\Databases* folder of the *Student Sample Programs* and select the ProductDB.mdf file.

- Click the *Test Connection* button to test the connection. You should see the message *Test Connection Succeeded*. Click the *OK* button to complete the connection.

Step 3: In this step, you will add the LINQ to SQL classes to the project, for which you need to perform the following steps:

- Click *Project* on the menu bar, then click *Add New Item...*.
- The *Add New Item* window appears next. Select *Data* in the left pane, select *LINQ to SQL Classes* in the center pane, and change the name of the file to *ProductDB.dbml*. Click the *Add* button to add the classes to the project.

Step 4: Now you will use the *Object Relational Designer* to create the classes you need to access the database.

- The *Object Relational Designer* should be open. In the *Server Explorer* window, expand the ProductDB.mdf entry, and then expand the *Tables* Entry.
- Drag the `Product` table from the *Server Explorer* window and drop it in the *Object Relational Designer*. You will see a window displaying a message indicating that you have selected a database file located outside the project folder, and you would be asked if you want to copy the file into the project. Click *Yes*.
- The *Object Relational Designer* now shows a visual representation of a class named `Product`. Notice that the class has properties named Product_Number, Description, Units_On_Hand, and Price. This class is the entity class that represents a row in the `Product` table.

At this point, your project now has a data context class named `ProductDBDataContext`. The `ProductDBDataContext` class contains table object property named `Products`. (Notice the pluralization of the name.) The `Products` object is a collection of `Product` objects. Each `Product` object represents a row in the database table.

Step 5: Click the *Save All* button ( ) to save the project. Do not skip this step! The entity class that you created in Step 4 must be saved before continuing to the next step.

Step 6: If the *Data Sources* window is not already opened, open it by selecting *View* > *Other Windows...* > *Data Sources*.

Step 7: In the *Data Sources* window, click the *Add New Data Source* button, as shown in Figure 15-19. This displays the *Choose a Data Source Type* window shown in Figure 15-20. Select *Object* and then click *Next*.

Step 8: The *Select the Data Objects* window, shown in Figure 15-21, appears next. Look at the figure carefully. Expand the *QueryDataSource* entries as shown in the figure and select *Product*. This action selects the `Product` class (the entity class) that you created in Step 4. Click *Finish*. The Data Sources window should now appear as shown in Figure 15-22. Notice that the `Product` class is shown as a data source.

Step 9: Open the Form1 form in the Designer.

Step 10: In the *Data Sources* window, click the down arrow that appears next to `Product`, and select *DataGridView*, as shown in Figure 15-23. This selects the DataGridView control as the default data-bound control for the `Product` data source.

Figure 15-19 *Add New Data Source*

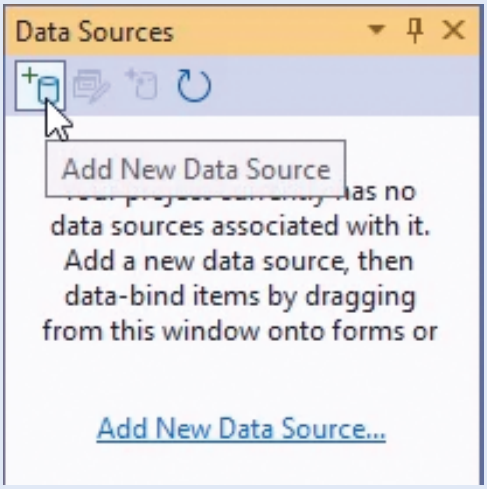
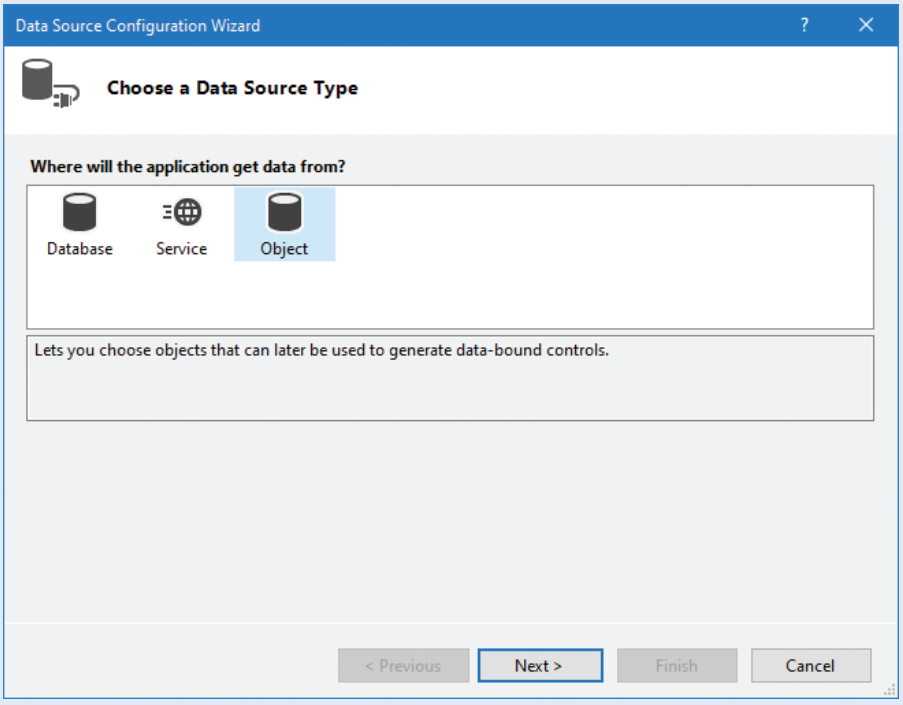


Figure 15-20 *Choose a Data Source Type*



Step 11: Click and drag the `Product` data source from the *Data Sources* window onto the form, as shown in Figure 15-24. This creates a DataGridView control on the form. Notice in the Properties window that the control's default name is `productDataGridView`. We will keep this as the control's name.

Adjust the size of the DataGridView control so you can see all the columns (Product_Number, Description, Units_On_Hand, and Price).

Step 12: Double-click any part of the form that does not contain a control. (Be sure not to click any part of the DataGridView control or its navigation bar.) This opens the code editor, and you will see an empty event handler named `Form1_Load`.

Figure 15-21 *Select the Data Objects*

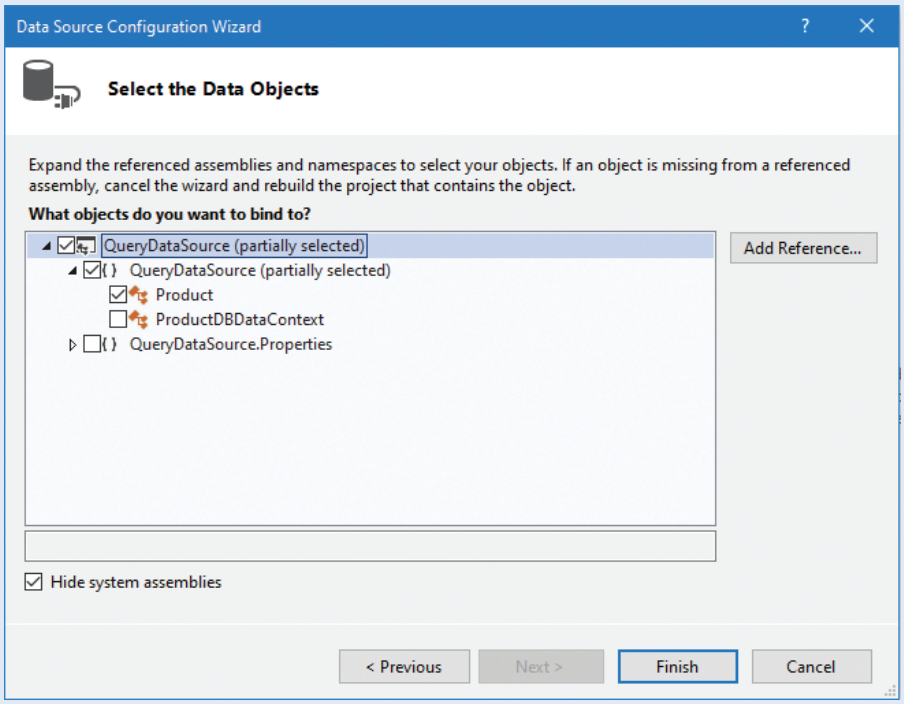


Figure 15-22 The updated Data Sources window

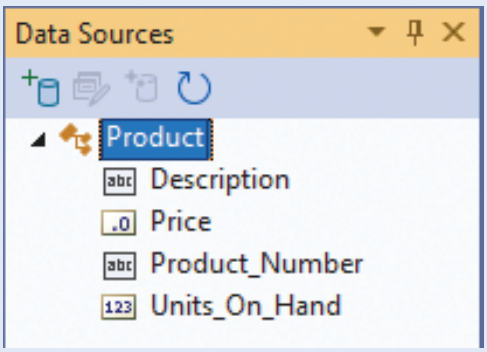


Figure 15-23 Select *DataGridView*

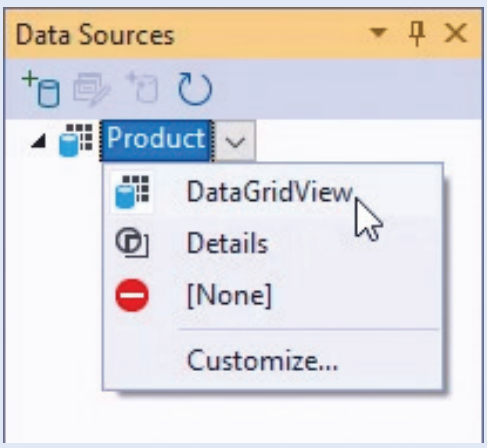
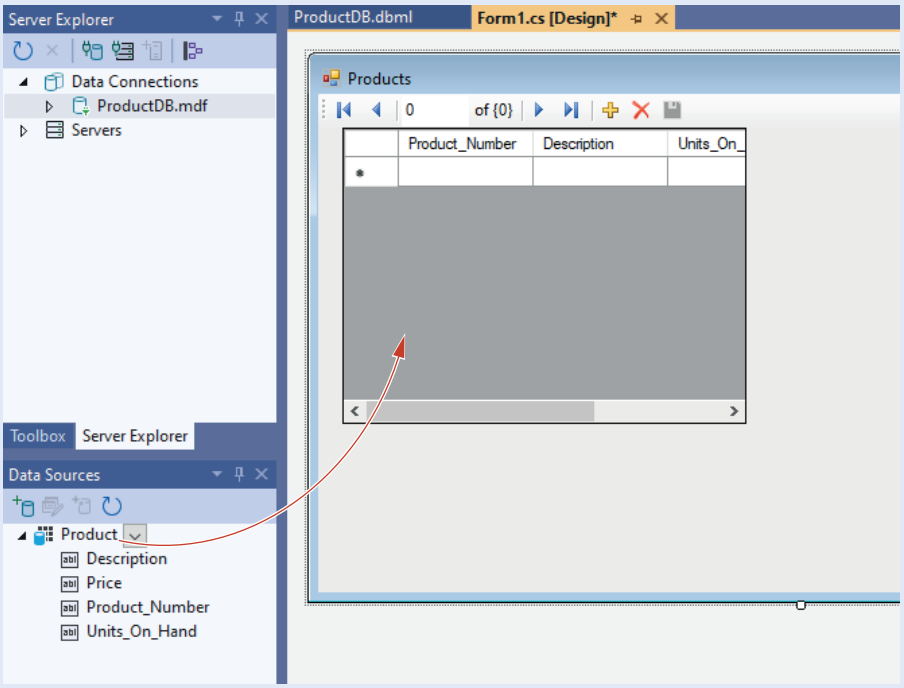


Figure 15-24 Creating the DataGridView control



Complete the `Form1_Load` event handler by typing the code shown in lines 22–32 in Program 15-7. Let's take a closer look at the code:

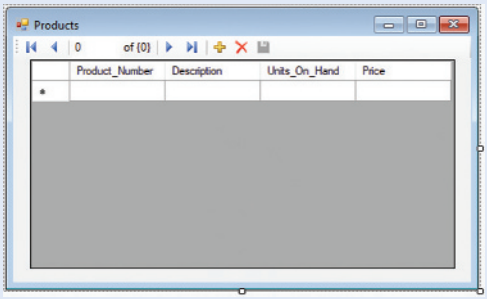
Line 23: This statement creates a data context object named `db`.

Lines 27 and 28: This statement executes a LINQ query on the `db.Products` table collection. It retrieves all of the `Product` objects from the table. At the database level, this causes all the rows in the `Product` table to be retrieved. They are returned from the query as a collection of `Product` objects, and assigned to the `results` variable.

Line 32: This statement assigns the results variable to DataGridView control's DataSource property. As a result, the contents of the results collection will be displayed in the DataGridView.

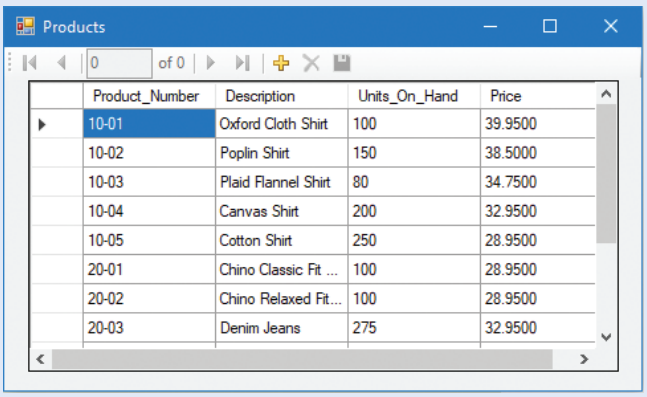
Step 13: Open the Form1 form in the Designer and adjust the size of the form and the position of the DataGridView control to appear similar to Figure 15-25.

Figure 15-25 The final appearance of the form



Step 14: Save the project. Then, press F5 on the keyboard or click the Start Debugging button ( ) on the toolbar to compile and run the application. The application's form should appear as shown in Figure 15-26. When you are finished, click the *Close* button to close the application.

Figure 15-26 The application's form at runtime



Program 15-7 Completed Form1 code for the *QueryDataSource* project

```
 1 using System;
 2 using System.Collections.Generic;
 3 using System.ComponentModel;
 4 using System.Data;
 5 using System.Drawing;
 6 using System.Linq;
 7 using System.Text;
 8 using System.Threading.Tasks;
 9 using System.Windows.Forms;
10
11 namespace QueryDataSource
12 {
13     public partial class Form1 : Form
14     {
15         public Form1()
16         {
17             InitializeComponent();
18         }
19
20         private void Form1_Load(object sender, EventArgs e)
21         {
22             // Create a data context object.
23             var db = new ProductDBDataContext();
24
25             // Get all the Product objects from the
26             // Products collection.
27             var results = from product in db.Products
28                           select product;
29
30             // Assign the results of the query to the
31             // DataGridView control.
32             productDataGridView.DataSource = results;
33         }
34     }
35 }
```

Updating, Inserting, and Deleting Rows with LINQ

In addition to retrieving rows from a database table, LINQ also allows you to update rows, insert rows, and delete rows. In the following examples, we will assume that we are working with the ProductDB database. Our data context class is named `ProductDBDataContext`, and it has a table property named Products. Our entity class is named `Product`.

The first example, shown below, gets the product with Product_Number set to 10-04. It changes the product's price to 99.99, and then updates the database with the new row.

```
1  var db = new ProductDBDataContext();
2
3  Product prod = db.Products.FirstOrDefault(item =>
4                       item.Product_Number == "10-04");
5
6  if (prod != null)
7  {
8      prod.Price = 99.99m;
9      db.SubmitChanges();
10 }
```

Let's take a closer look at the code. Line 1 creates a data context object. Lines 3–4 use the `FirstOrDefault` extension method to get the first item in the `db.Products` collection that has its Product_Number property set to "10-04". If the item is found, it will be assigned to the `prod` variable. If it is not found, the `prod` variable will be set to `null`. If `prod` is not set to `null`, the statement in line 8 assigns 99.99 to its Price property, and the statement in line 9 calls the `db.SubmitChanges()` to update the database with the new data.

The following code shows another example. In this example, the LINQ query in lines 3–5 returns all of the products that have a Product_Number property containing the string "10-". The `foreach` loop in lines 7–10 iterates over all the products that were returned from the query, changing their Units_On_Hand property to 0. The statement in line 8 calls the `db.SubmitChanges()` to update the database with the new data.

```
1  var db = new ProductDBDataContext();
2
3  var results = from prod in db.Products
4                where prod.Product_Number.Contains("10-")
5                select prod;
6
7  foreach(Product p in results)
8  {
9      p.Units_On_Hand = 0;
10 }
11
12 db.SubmitChanges();
```

The following example shows a new row being inserted into the database table:

```
1  var db = new ProductDBDataContext();
2
3  var prod = new Product();
4  prod.Product_Number = "40-01";
5  prod.Description = "Hiking Boots";
6  prod.Units_On_Hand = 10;
7  prod.Price = 299m;
8
9  db.Products.InsertOnSubmit(prod);
10 db.SubmitChanges();
```

In this code, lines 3–7 create an instance of the `Product` entity class and assign values to the object's properties. Line 9 calls the `db.Products.InsertOnSubmit` method, passing the object as an argument. When the `db.SubmitChanges()` method is called in line 10, the object is inserted into the database.

The following example shows how to delete an item from a database table:

```
1  var db = new ProductDBDataContext();
2
3  Product prod = db.Products.FirstOrDefault(item =>
4                       item.Product_Number == "10-04");
5
6  if (prod != null)
7  {
8      db.Products.DeleteOnSubmit(prod);
9      db.SubmitChanges();
10 }
```

Lines 3–4 use the `FirstOrDefault` extension method to get the first item in the `db.Products` collection that has its Product_Number property set to "10-04". If the item is found, it will be assigned to the `prod` variable. If it is not found, the `prod` variable will be set to `null`. If `prod` is not set to `null`, the statement in line 8 calls the `db.Products.DeleteOnSubmit` method, passing the `prod` object as an argument. When the `db.SubmitChanges()` method is called in line 9, the object is deleted from the database.

Checkpoint

15.13 What is a data context object?

15.14 What are table objects and entity objects?

15.15 How are the columns in a row represented in LINQ to SQL?

15.16 What are the three general steps you take to set up LINQ to SQL to work with a database?

15.17 Suppose your application is setup and connected to a database with LINQ to SQL. You have created a new instance of an entity class and assigned values to its properties. What two methods must you call if you want to insert the new entity object into the database table as a row?

Key Terms

aggregation methods
data context object
Database Markup Language (DBML)
DBML file
element methods
entity object
generation methods
`IEnumerable<T>` interface
Language-Integrated Query (LINQ)
LINQ to Objects
LINQ to SQL
method syntax
Object Relational Designer
Object Relational Mapping (ORM)
partitioning methods
quantifier methods
query syntax
set methods
table object

Review Questions

Multiple Choice

1. If a LINQ query's source of data is a `List<int>`, the query will return a collection of the _____ type.

   a. `List<int>`
   b. `array of int`
   c. `IEnumerable<int>`
   d. `IQueryResults<int>`

2. In a LINQ query, the _____ clause iterates over a collection of elements.

   a. `from`
   b. `for`
   c. `select`
   d. `where`

3. In a LINQ query, the _____ clause determines what will be included in the query results.

   a. `from`
   b. `for`
   c. `select`
   d. `where`

4. The _____ clause causes the results of a LINQ query to be sorted.

   a. `sort`
   b. `orderby`
   c. `order`
   d. `sortby`

5. You can write a LINQ query using either query syntax or _____.

   a. Pseudocode
   b. SQL syntax
   c. Procedural syntax
   d. Method syntax

6. In LINQ to SQL, a(n) _____ object represents a database.

   a. Table
   b. Entity
   c. Data context
   d. Row

7. In LINQ to SQL, a(n) _____ object represents a table in a database.

   a. Table
   b. Entity
   c. Data context
   d. Row

8. In LINQ to SQL, a(n) _____ object represents a row in a database.

   a. Entity
   b. Data context
   c. Row
   d. Record

9. The _____ is a visual design tool that you use to create the classes needed for LINQ to SQL to work with a particular database.
   a. Database Designer
   b. Object Relational Designer
   c. LINQ Designer
   d. Table Designer

10. To update an existing row in a database table, you call the data context object's _____ method.
   a. `Update`
   b. `Commit`
   c. `Reload`
   d. `SubmitChanges`

True or False

1. The `var` keyword makes code more difficult to read, but it improves performance by a factor of 2.
2. In a LINQ query, the `from` clause iterates over a collection of elements.
3. You can use more than one `where` clause in a LINQ query.
4. The C# compiler translates all LINQ queries that are written in method syntax to query syntax.
5. The data context object has table objects as properties that represent the tables in the database.
6. A table object is a collection of entity objects that represent the rows in the table.
7. One limitation of LINQ to SQL is that query results cannot be displayed in a data-bound control.
8. One limitation of LINQ to SQL is that you cannot insert or delete rows in a database table. Tables are essentially read-only.

Short Answer

1. In the following code, assume `values` is an array of `doubles`. What is the data type of the `results` variable?

```
var results = from item in nums
              where item < 0
              select item;
```

2. What are the two syntaxes that you can use to write LINQ queries?
3. What extension method do you call if you want to order a query's results in descending order?
4. In LINQ to SQL, how are databases, tables, rows, and columns represented?
5. Describe the process of inserting a new row into a table with LINQ to SQL.

Algorithm Workbench

1. Assume `nums` is a `List` of integers. Write a LINQ query that gets all the values from the `List` that are less than 0. Assign the results to a variable named `results`.
2. Assume `nums` is a `List` of integers. Write a LINQ query that gets all the values from the `List` sorted in ascending order. Assign the results to a variable named `results`.
3. Assume `nums` is a `List` of integers. Write a LINQ query that gets all the values from the `List` sorted in descending order. Assign the results to a variable named `results`.

4. Rewrite the following query using method syntax:

```
var results = from item in values
              where item > 0 && item < 10
              orderby item
              select item;
```

For Questions 5–8, assume that a database has a table named `Stock`*, with the following columns:*

Column Name	Type
Trading_Symbol	nchar(10)
Company_Name	nchar(25)
Num_Shares	int
Purchase_Price	money
Selling_Price	money

Also assume that a data context object has been created with the following statement:

```
StockDataContext db = new StockDataContext();
```

5. Write a LINQ to SQL query that returns all the columns from every row in the table.
6. Write a LINQ to SQL query that returns only the `Trading_Symbol` column from every row in the table.
7. Write a LINQ to SQL query that returns the `Company_Name` column only from the rows where `Purchase_Price` is greater than $50.00.
8. Write a LINQ to SQL query that returns the `Trading_Symbol` column only from the rows where `Selling_Price` is greater than `Purchase_Price` and `Num_Shares` is greater than 100.

Programming Problems

VideoNote
Solving the Unique Words Problem

1. **Unique Words**

   Create an application that opens a specified text file and then displays a list of all the unique words found in the file. *Hint: Use a LINQ method to remove all duplicate words.*

2. **Text File Analysis**

   Create an application that reads the contents of two text files and uses LINQ extension methods to perform the following:

   - It should display a list of all the unique words contained in both files.
   - It should display a list of the words that appear in both files.
   - It should display a list of the words that appear in the first file but not the second.
   - It should display a list of the words that appear in the second file but not the first.
   - It should display a list of the words that appear in either the first or second file but not both.

   Hint: Use set operations to perform these analyses. Also, see Chapter 8 for a discussion of string tokenizing.

3. **Gas Prices**

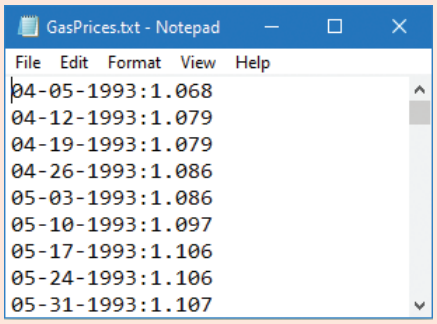
In the student sample program files for this chapter, you will find a text file named GasPrices.txt. The file contains the weekly average prices for a gallon of gas in the US, beginning on April 5, 1993, and ending on August 26, 2013. Figure 15-27 shows an example of the first few lines of the file's contents:

Each line in the file contains the average price for a gallon of gas on a specific date. Each line is formatted in the following way:

```
MM-DD-YYYY:Price
```

MM is the two-digit month, *DD* is the two-digit day, and *YYYY* is the four-digit year. *Price* is the average price per gallon of gas on the specified date.

Figure 15-27 The GasPrices.txt file



For this assignment, you need to create one or more applications that read the contents of the file and use LINQ to perform the following calculations:

Average Price Per Year: Calculate the average price of gas per year, for each year in the file. (The file's data start in April, 1993 and ends in August, 2013. Use the data that are present for the years 1993 and 2013.)

Average Price Per Month: Calculate the average price for each month in the file.

Highest and Lowest Prices Per Year: For each year in the file, determine the date and amount for the lowest price, and the highest price.

List of Prices, Lowest to Highest: Generate a text file that lists the dates and prices, sorted from the lowest price to the highest.

List of Prices, Highest to lowest: Generate a text file that lists the dates and prices, sorted from the highest price to the lowest.

You can create one application to perform all of these calculations, or you can create different applications, one for each calculation. Regardless of the approach that you take, you should read the contents of the GasPrices.txt file, and extract its data into one or more collections appropriate for your algorithm.

Hint: See Chapter 8 for a discussion of string tokenizing.

4. **Product Search**

In the *Chap15* folder of the *Student Sample Programs*, you will find a copy of the ProductDB database file that you worked with in Chapter 13. The database has a table named Product. The Product table has the following columns:

- `Product_Number - nchar(5)`, primary key, nulls not allowed
- `Description - nvarchar(50)`, nulls not allowed
- `Units_On_Hand - int`, nulls not allowed
- `Price` – money, nulls not allowed

Write an application that connects to the database and uses LINQ to SQL to let the user perform the following searches:

- Search for a product number
- Search for products with descriptions containing specified text

5. **Product Units on Hand Search**

   Write an application that connects to the ProductDB database (described in Programming Problem 4). The application should use LINQ to SQL to interact with the database. It should initially display all the products in the database sorted by units on hand, in ascending order. It should allow the user to search for products with more units on hand than a specified amount, and less units on hand than a specified amount.

6. **Product Price Search**

   Write an application that connects to the ProductDB database (described in Programming Problem 4). The application should use LINQ to SQL to interact with the database. It should initially display all the products in the database sorted by price, in ascending order. It should allow the user to specify a minimum and a maximum price, and then display the products within that range of prices.

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APPENDIX

A C# Primitive Data Types

Data Type	Description
`bool`	A variable to store the Boolean values `true` and `false`.
`byte`	An unsigned 8-bit integer to store values in the range of 0 through 255.
`sbyte`	A signed 8-bit integer to store values in the range of −128 through 127.
`char`	A 16-bit variable to hold a Unicode character.
`decimal`	A 128-bit variable to hold real numbers, rounded to 28 digits of precision, in the range of -7.9×10^{28} to 7.9×10^{28}.
`double`	A 64-bit variable to hold real numbers, rounded to 15 digits of precision, in the range of $\pm 5.0 \times 10^{-324}$ to $\pm 1.7 \times 10^{308}$.
`float`	A 32-bit variable to hold real numbers rounded to 15 digits of precision, in the range of -3.4×10^{38} to $+3.4 \times 10^{38}$.
`int`	A signed 32-bit integer to store values in the range of −2,147,483,648 through 2,147,483,647.
`uint`	An unsigned 32-bit integer to store values in the range of 0 through 4,294,967,295.
`long`	A signed 64-bit integer to store values in the range of −9,223,372,036,854,775,808 through 9,223,372,036,854,775,807.
`ulong`	An unsigned 64-bit integer to store values in the range of 0 through 18,446,744,073,709,551,615.
`short`	A signed 16-bit integer to store values in the range of −32,768 through 32,767.
`ushort`	An unsigned 16-bit integer to store values in the range of 0 through 65,535.
`string`	A variable to hold strings.

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APPENDIX B

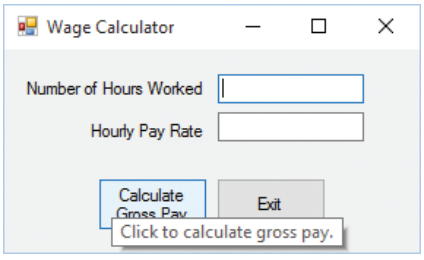
Additional User Interface Controls

The chapters in this textbook have introduced you to the fundamental Visual C# controls. There are many more controls available in Visual C#, and this appendix introduces you to several of them. The examples discussed in this appendix can be found in the Student Sample Program Files.

ToolTips

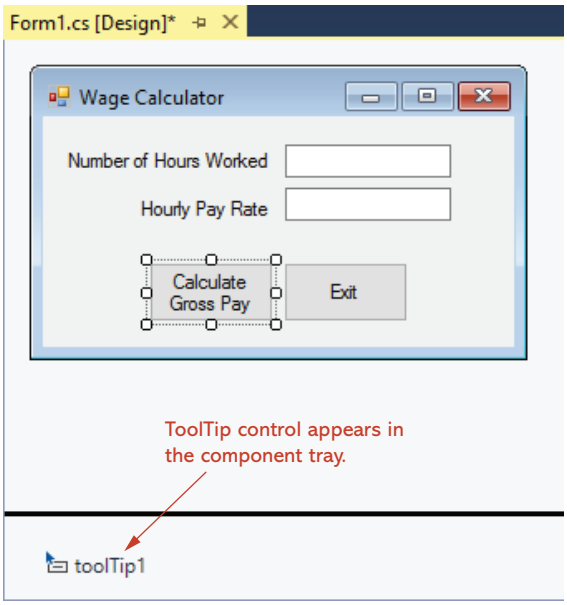
A **ToolTip** is a small box displayed when the user holds the mouse cursor over a control. The box shows a short description of what the control does. Most Windows applications use ToolTips as a way of providing immediate and concise help to the user. Figure B-1 shows an example.

Figure B-1 A ToolTip displayed



The **ToolTip control** allows you to create ToolTips for other controls on a form. Place a ToolTip control in your application by double-clicking the *ToolTip* icon in the *Toolbox*, just as you place other controls. When you do so, a ToolTip control appears in the component tray, the resizable area at the bottom of the *Designer* that holds invisible controls. Figure B-2 shows an example.

Figure B-2 ToolTip control



When you add a ToolTip control to a form, a new property is added to all the other controls on that form. The new property is named ToolTip on *ToolTipControl*, where *ToolTipControl* is the name of the ToolTip control. For example, suppose you add a ToolTip control to a form and keep the default name ToolTip1. The new property added to the other controls is named ToolTip on ToolTip1.

The ToolTip on *ToolTipControl* property holds the string that is displayed as the control's ToolTip. Any text that you enter into a control's ToolTip on *ToolTipControl* property is displayed as that control's ToolTip when the user holds the mouse cursor over the control.

Other ToolTip Properties

ToolTip controls have other properties that affect their behavior. The InitialDelay property determines the amount of time, in milliseconds, that elapses between the user pointing the mouse at a control and the ToolTip's appearance. The default setting is 500. (One millisecond is one-thousandth second, so 500 milliseconds is one-half second.)

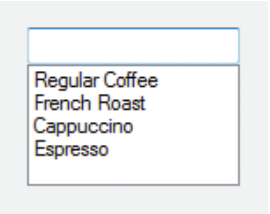
The AutoPopDelay property is also a measure of time in milliseconds. It determines how long a ToolTip remains on the screen once it is displayed. The default setting is 5000. The ReshowDelay property holds the number of milliseconds that elapse between the displaying of different ToolTips as the user moves the mouse from control to control. The default setting is 100.

You can set these properties individually or set them all at once with the AutomaticDelay property. When you store a value in the AutomaticDelay property, InitialDelay is set to the same value, AutoPopDelay is set to 10 times the value, and ReshowDelay is set to one-fifth the value.

Combo Boxes

A **combo box** is like a list box that has been combined with a text box. Figure B-3 shows an example of one style of combo box known as a simple combo box.

Figure B-3 A simple combo box



Combo boxes and list boxes are similar in the following ways:

- They both display a list of items to the user.
- They both have Items, Items.Count, SelectedIndex, SelectedItem, and Sorted properties.
- They both have `Items.Add` and `Items.Clear` methods.
- All these properties and methods work the same with combo boxes and list boxes.

Additionally, a combo box has a rectangular area that works like a text box. The user may either select an item from the combo box's list or type text into the combo box's text input area.

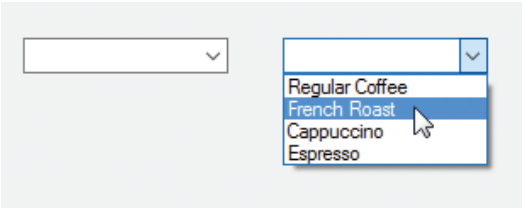
Like a text box, the combo box has a Text property. If the user types text into the combo box, the text is stored in the Text property. Also, when the user selects an item from the combo box's list, the item is copied to the Text property.

Combo Box Styles

There are three different styles of combo boxes: the drop-down combo box, the simple combo box, and the drop-down list combo box. You can select a combo box's style with its DropDownStyle property. Let's look at each of the styles in more detail.

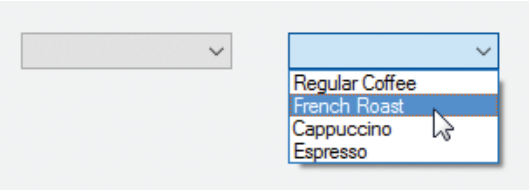
- **Drop-Down Combo Box.** This is the default setting for the combo box DropDownStyle property. At run time, a drop-down combo box like the one shown on the left in Figure B-4 appears. When the user clicks the down arrow, a list drops down as shown in the image on the right in the figure. The user may either select an item from the list or type input into the text box that appears at the top of the control. The item that is selected or the text that is entered is assigned to the combo box's Text property.

Figure B-4 A drop-down combo box



- **Simple Combo Box.** This is the style of combo box shown in Figure B-3. With the simple style of combo box, the list of items does not drop down but is always displayed. As with the drop-down combo box, this style allows the user to select an item from the list or type text directly into the text box area. When typing, the user is not restricted to the items that appear in the list. The item that is selected or the text that is entered is assigned to the combo box's Text property.
- **Drop-Down List Combo Box.** With the drop-down list combo box style, the user may not type text directly into the combo box. An item must be selected from the list. At run time, a drop-down list combo box appears, as shown in the image on the left in Figure B-5. When the user clicks the down arrow, a list of items appears, as shown in the image on the right in the figure. The item that is selected is assigned to the combo box's Text property.

Figure B-5 A drop-down list combo box



Getting the User's Input from a Combo Box

As with the list box, you can determine which item has been selected from a combo box's list by retrieving the value in the SelectedIndex or SelectedItem property. If the user has typed text into the combo box's text area, however, you cannot use the SelectedIndex or SelectedItem property to get the text. The best way to get the user's input is with the Text property, which contains either the user's text input or the item selected from the list.

List Boxes versus Combo Boxes

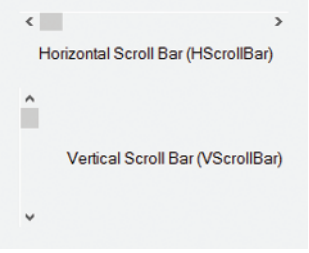
The following guidelines will help you decide when to use a list box and when to use a combo box:

- Use a drop-down or simple combo box when you want to provide the user a list of items from which to select but do not want to limit the user's input to the items on the list.
- Use a list box or a drop-down list combo box when you want to limit the user's selection to a list of items. The drop-down list combo box generally takes less space than a list box (because the list doesn't appear until the user clicks the down arrow), so use it when you want to conserve space on the form.

Scroll Bars

Scroll bars provide a visual way to adjust a value within a range of values. These types of controls display a slider that may be dragged along a track. Visual C# provides a horizontal scroll bar control named HScrollBar and a vertical scroll bar control named VScrollBar. Figure B-6 shows examples of each of these controls. You can find these controls in the *Toolbox*, in the *All Windows Forms* group.

Figure B-6 Horizontal and vertical scroll bars



Here is a summary of the important properties of each of these controls:

- The Value property is an integer value that is adjusted as the user moves the control's slider. (The default value is 0.)
- The Minimum property is the lower limit of the scrollable range. (The default value is 0.)

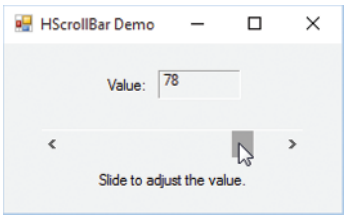
- The Maximum property is the upper limit of the scrollable range. (The default value is 100.)
- The LargeChange property is the integer amount by which the Value property changes when the user clicks the scroll bar area that lies to either side of the slider. This is also the amount by which the Value property changes when the user presses the Page Up or Page Down key on the keyboard while the control has the focus. (The default value is 10.)
- The SmallChange property is the integer amount by which the Value property changes when the user clicks one of the arrows that appear at either end of a scroll bar control. (The default value is 1.)

When a horizontal scroll bar's slider is moved toward its left side, the Value property is decreased. When the slider is moved toward the scroll bar's right side, the Value property is increased. When a vertical scroll bar's slider is moved toward its top, the Value property is decreased. When the slider is moved toward the scroll bar's bottom, the Value property is increased.

When the user moves the slider on a scroll bar control, a Scroll event occurs. If you write a Scroll event handler for the control, the event handler will execute any time the slider is moved. To generate a code template for the Scroll event handler, simply double-click the scroll bar control in the *Designer*.

Figure B-7 shows the form in an example application that demonstrates the HScrollBar control. The HScrollBar control is named `hScrollBar`, and the label that displays the value is named `valueLabel`. The form's code is shown in Program B-1. This project can be found in the *Appendix B* folder in the Student Sample Programs. You will also find a similar project that demonstrates the VScrollBar control.

Figure B-7 *HScrollBar Demo* application form



Program B-1 Form1 code in the *HScrollBar Demo* application

```
 1 using System;
 2 using System.Collections.Generic;
 3 using System.ComponentModel;
 4 using System.Data;
 5 using System.Drawing;
 6 using System.Linq;
 7 using System.Text;
 8 using System.Threading.Tasks;
 9 using System.Windows.Forms;
10
11 namespace HScrollBar_Demo
12 {
13     public partial class Form1 : Form
14     {
15         public Form1()
16         {
17             InitializeComponent();
18         }
```

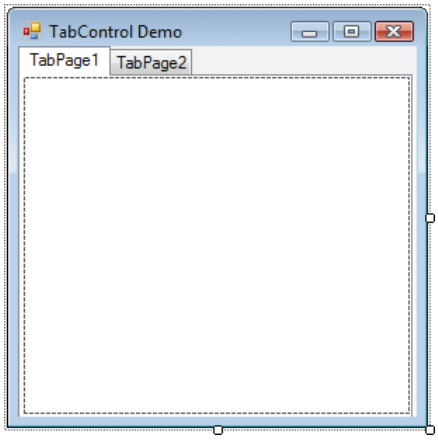
```
19
20          private void Form1_Load(object sender, EventArgs e)
21          {
22              hScrollBar.Value = 0;
23              hScrollBar.Minimum = 0;
24              hScrollBar.Maximum = 100;
25              hScrollBar.LargeChange = 10;
26              hScrollBar.SmallChange = 1;
27
28              // Display the scroll bar's initial value
29              // in the label control.
30              valueLabel.Text = hScrollBar.Value.ToString();
31          }
32
33         private void hScrollBar_Scroll(object sender, ScrollEventArgs e)
34          {
35              // Display the scroll bar value.
36              valueLabel.Text = hScrollBar.Value.ToString();
37          }
38      }
39 }
```

Using a TabControl to Organize a Form

A TabControl allows you to create a user interface that is made of multiple pages, with each page containing its own set of controls. The TabControl appears as a container on a form, with one or more tabs positioned along its top edge. Each tab represents a different page, known as a TabPage. When the user clicks a tab, the control displays that page. You can find the TabControl in the *Toolbox*, in the *Containers* group.

When you insert a new TabControl, it will contain two TabPage controls named TabPage1 and TabPage2. This is shown in Figure B-8. Keep in mind that a TabControl is a container that contains TabPage controls. When you are working with a TabControl in the *Designer*, you can select the TabControl (the container) or you can select the individual TabPage controls that it contains. When you work with a TabControl for the first time, you should practice selecting each of the controls in the group so you know at all times which one you are working with.

Figure B-8 A TabControl with two TabPages



Each of the TabPage controls has its own set of properties that can be changed in the *Properties* window. For example, to change the text that is displayed on a TabPage's tab, you change that TabPage control's Text property.

As previously mentioned, a TabControl contains two TabPage controls when first inserted in a form. To add more TabPages, select the TabControl and then select its TabPages property. (Click the ellipses button ... that appears next to the TabPages property window.) This opens the *TabPage Collection Editor*, shown in Figure B-9. This window allows you to add new TabPages, remove existing TabPages, and edit each TabPage's properties.

Figure B-9 *TabPage Collection Editor* window

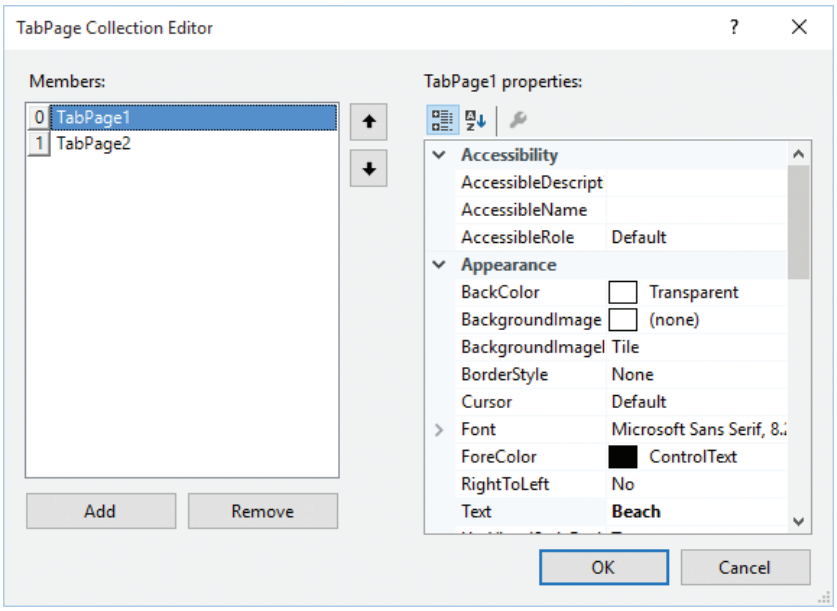
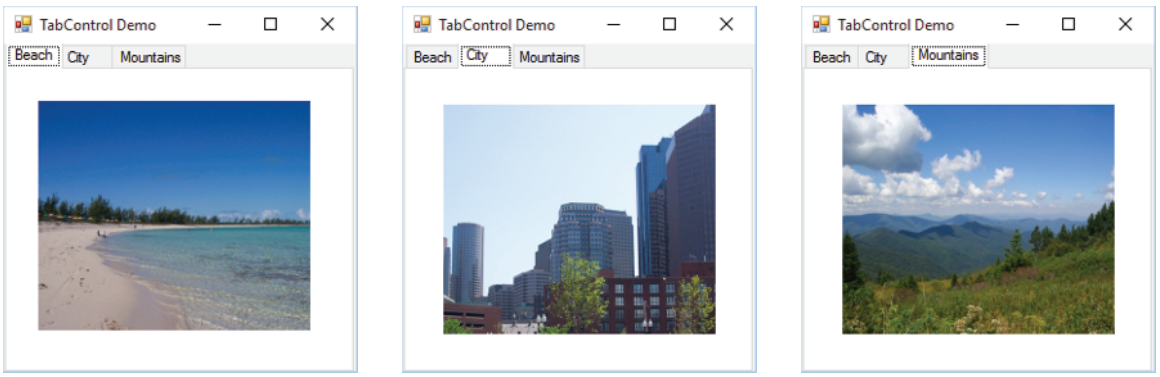


Figure B-10 shows an example application named *TabControl Demo* that can be found in the Student Sample Programs. The application's form has a TabControl with three TabPages. Each TabPage contains a PictureBox control displaying an image. (There is no code in the application.)

Figure B-10 *TabControl Demo* application



The WebBrowser Control (.NET Framework Projects)

The WebBrowser control is available in *Windows Forms App (.NET Framework)* projects. It is found in the *Common Controls* group in the *Toolbox*, and allows you to display a Web page on an application's form. The control has a property named Url that can be set to a Web page's URL (Uniform Resource Locator). At run time, that Web page is displayed in the control.

At design time, you can use the *Properties* window to set the Url property. You simply type a valid URL such as www.gaddisbooks.com into the property's value box. If you want to set the Url property in code, you must create a Uri object (Uniform Resource Identifier) and assign that object to the property. Here is an example:

```
WebBrowser1.Url = new Uri("http://www.gaddisbooks.com");
```

Alternatively, you can call the control's *Navigate* method to display a Web page, as shown here:

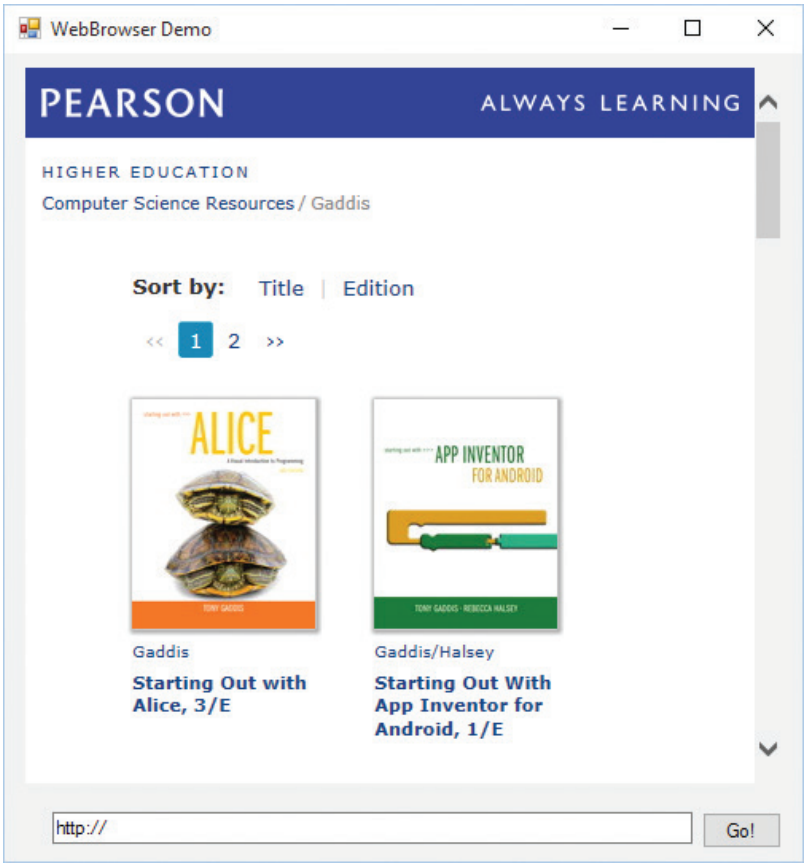
```
WebBrowser1.Navigate(New Uri("http://www.gaddisbooks.com"));
```

In either of these approaches, an exception will be thrown if an invalid Web address is used.

When a Web page has finished loading, a DocumentCompleted event occurs. If you want to perform some action after a page has loaded, you can write a handler for this event. (Just double-click the WebBrowser control in the *Designer* window to create a code template for the DocumentCompleted event handler.)

Figure B-11 shows the *WebBrowser Demo* application in the Student Sample Programs. The application's form has a WebBrowser control named `WebBrowser1`, a TextBox control named `urlTextBox`, and a Button control named `goButton`. When the user clicks the `goButton` button, the application sets the `WebBrowser1` control's Url property to the address that has been typed into the `urlTextBox` text box. The Form1 form's code is shown in Program B-2.

Figure B-11 *WebBrowser Demo* application



Program B-2 Form1 code in the *WebBrowser Demo* application

```
 1 using System;
 2 using System.Collections.Generic;
 3 using System.ComponentModel;
 4 using System.Data;
 5 using System.Drawing;
 6 using System.Linq;
 7 using System.Text;
 8 using System.Threading.Tasks;
 9 using System.Windows.Forms;
10
11 namespace WebBrowser_Demo
12 {
13     public partial class Form1 : Form
14     {
15         public Form1()
16         {
17             InitializeComponent();
18         }
19
20         private void goButton_Click(object sender, EventArgs e)
21         {
22             try
23             {
24                 WebBrowser1.Url = new Uri(urlTextBox.Text);
25             }
26             catch (Exception ex)
27             {
28                 // Error message for an invalid Web address.
29                 MessageBox.Show(ex.Message);
30             }
31         }
32     }
33 }
```

The ErrorProvider Component

The ErrorProvider component (found in the *Components* group in the *Toolbox*) allows you to indicate that the user has entered an invalid value by displaying a blinking error icon ( ) next to a specific control on the application's form. When the user hovers the mouse pointer over the icon, an error message is displayed as a ToolTip.

When you insert an ErrorProvider component, it appears in the component tray at the bottom of the *Designer* window with a default name such as `ErrorProvider1`. In code, when the user enters an invalid value with a specific control, you call the ErrorProvider component's `SetError` method. Here is the general format for calling the method:

```
ErrorProviderName.SetError(ControlName, ErrorMessage);
```

In the general format, *`ErrorProviderName`* is the name of the ErrorProvider component, *`ControlName`* is the name of the control that you want to display the error icon next to, and *`ErrorMessage`* is the error message to associate with the error. Here is an example:

```
ErrorProvider1.SetError(payRateTextBox, "Invalid pay rate");
```

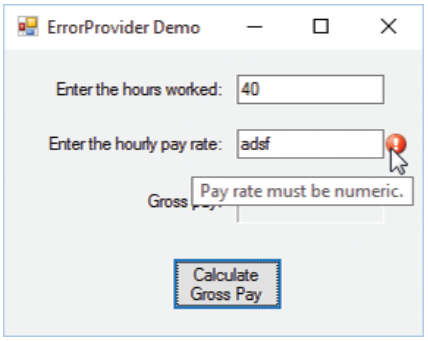
This statement uses the `ErrorProvider1` component to display an error icon next to the `payRateTextBox` control. When the user hovers the mouse pointer over the error icon, the message *Invalid pay rate* is displayed as a ToolTip.

The error icon will remain displayed next to the specified control until you call the `SetError` method again, passing the same control name as the first argument and an empty string as the second argument. Here is an example:

```
ErrorProvider1.SetError(payRateTextBox, "");
```

Figure B-12 shows the *ErrorProvider Demo* application in the Student Sample Programs. The user enters a number of hours in the `hoursTextBox` control, a numeric pay rate in the `payRateTextBox` control, and then clicks the `calcButton` control to calculate gross pay. If a nonnumeric value is entered for either the hours or the pay rate, an ErrorProvider component displays an error icon next to the control containing the invalid value. In Figure B-12, the user has entered an invalid value for the pay rate. The code for Form1 is shown in Program B-3.

Figure B-12 *ErrorProvider Demo* application



Program B-3 Form1 code in the *ErrorProvider Demo* application

```
 1 using System;
 2 using System.Collections.Generic;
 3 using System.ComponentModel;
 4 using System.Data;
 5 using System.Drawing;
 6 using System.Linq;
 7 using System.Text;
 8 using System.Threading.Tasks;
 9 using System.Windows.Forms;
10
11 namespace ErrorProvider_Demo
12 {
13     public partial class Form1 : Form
14     {
15         public Form1()
16         {
17             InitializeComponent();
18         }
19
20         private void calcButton_Click(object sender, EventArgs e)
21         {
22             // Variables for hours, pay rate, and gross pay
23             decimal hours, payRate, grossPay;
24
25             // Clear any existing errors.
26             ErrorProvider1.SetError(hoursTextBox, "");
27             ErrorProvider1.SetError(payRateTextBox, "");
28
29             // Get values and calculate gross pay.
30             if (decimal.TryParse(hoursTextBox.Text, out hours))
```

```
31              {
32                 if (decimal.TryParse(payRateTextBox.Text, out payRate))
33                  {
34                      // Calculate the gross pay.
35                      grossPay = hours * payRate;
36
37                      // Display the gross pay.
38                      grossPayLabel.Text = grossPay.ToString("c");
39                  }
40                  else
41                  {
42                      // Invalid pay rate
43                      ErrorProvider1.SetError(payRateTextBox,
44                          "Pay rate must be numeric.");
45                  }
46              }
47              else
48              {
49                  // Invalid hours
50                  ErrorProvider1.SetError(hoursTextBox,
51                      "Hours must be numeric.");
52              }
53          }
54      }
55 }
```

Using the SelectionStart and SelectionLength Properties to Select Text in a TextBox

TextBox controls have two properties, SelectionStart and SelectionLength, which you can use to make the process of correcting invalid input more convenient for the user. When the user enters an invalid value, you can display an error message and then use these properties to automatically select the invalid input for the user. Then, the user can immediately retype the input without having to use the mouse to select the TextBox.

The SelectionStart and SelectionLength properties can be used in code to automatically select the text in a text box. The SelectionStart property holds the position of the first selected character in the text box. The SelectionLength property holds the number of characters that are selected. For example, assume that `nameTextBox` is a text box and look at the following code.

```
nameTextBox.Focus();
nameTextBox.SelectionStart = 0;
nameTextBox.SelectionLength = 5;
```

The first statement gives `nameTextBox` the focus. The second statement establishes that the first character in `nameTextBox` (which is at position 0) is the first selected character. The next statement establishes that five characters will be selected. Together, the statements cause the first five characters in `nameTextBox` to be selected.

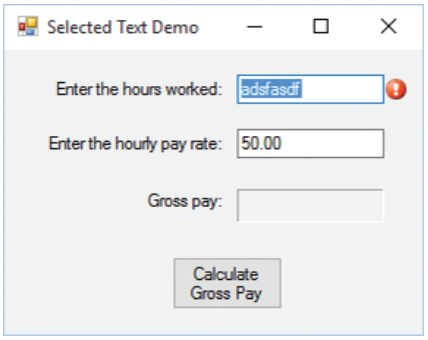
So, how do you use similar code to select all the text in a TextBox? You use the TextBox's Length property to get the length of the text and assign that value to the TextBox's SelectionLength property. Here is an example:

```
nameTextBox.Focus();
nameTextBox.SelectionStart = 0;
nameTextBox.SelectionLength = nameTextBox.Text.Length;
```

After these statements execute, all the contents of the `nameTextBox` control are selected. When the user types a key, that keystroke immediately erases all the selected text.

Figure B-13 shows the *Selected Text Demo* application in the Student Sample Program files. This is a modified version of the *ErrorProvider Demo* application previously shown. In this version, after the ErrorProvider displays an error icon, the content of the TextBox containing the invalid input is automatically selected. In the figure, the user has entered an invalid value for the hours. The application's code is shown in Program B-4.

Figure B-13 *Selected Text Demo* application



Program B-4 Form1 code in the *Selected Text Demo* application

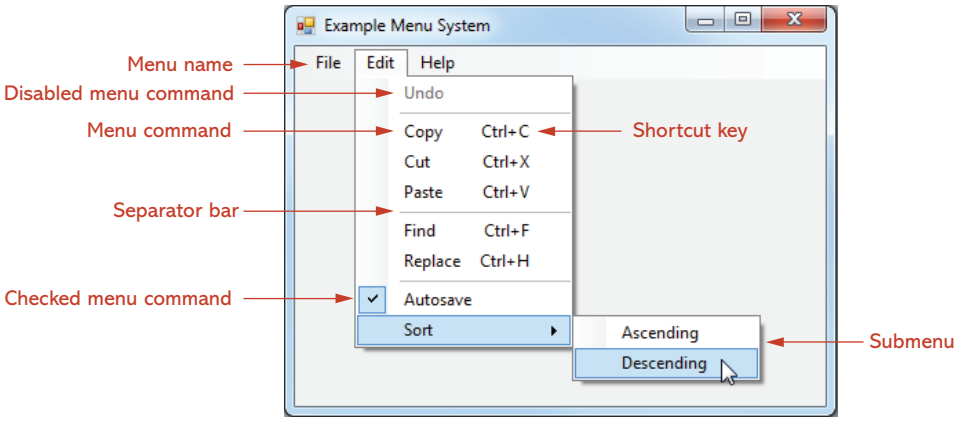
```
 1 using System;
 2 using System.Collections.Generic;
 3 using System.ComponentModel;
 4 using System.Data;
 5 using System.Drawing;
 6 using System.Linq;
 7 using System.Text;
 8 using System.Threading.Tasks;
 9 using System.Windows.Forms;
10
11 namespace Selected_Text_Demo
12 {
13     public partial class Form1 : Form
14     {
15         public Form1()
16         {
17             InitializeComponent();
18         }
19
20         private void calcButton_Click(object sender, EventArgs e)
21         {
22             // Variables for hours, pay rate, and gross pay
23             decimal hours, payRate, grossPay;
24
25             // Clear any existing errors.
26             ErrorProvider1.SetError(hoursTextBox, "");
27             ErrorProvider1.SetError(payRateTextBox, "");
28
29             // Get values and calculate gross pay.
30             if (decimal.TryParse(hoursTextBox.Text, out hours))
31             {
32                 if (decimal.TryParse(payRateTextBox.Text, out
                   payRate))
33                 {
34                     // Calculate the gross pay.
35                     grossPay = hours * payRate;
```

```
36
37                      // Display the gross pay.
38                      grossPayLabel.Text = grossPay.ToString("c");
39                  }
40                  else
41                  {
42                      // Invalid pay rate
43                      ErrorProvider1.SetError(payRateTextBox,
44                          "Pay rate must be numeric.");
45
46                      // Select the invalid input.
47                      payRateTextBox.Focus();
48                      payRateTextBox.SelectionStart = 0;
49                      payRateTextBox.SelectionLength =
50                          payRateTextBox.Text.Length;
51                  }
52              }
53              else
54              {
55                  // Invalid hours
56                  ErrorProvider1.SetError(hoursTextBox,
57                      "Hours must be numeric.");
58
59                  // Select the invalid input.
60                  hoursTextBox.Focus();
61                  hoursTextBox.SelectionStart = 0;
62                  hoursTextBox.SelectionLength =
63                      hoursTextBox.Text.Length;
64              }
65          }
66      }
67  }
```

Creating a Menu System

A **menu system** is a collection of commands organized in one or more drop-down menus. In Visual Studio, the **menu designer** allows you to visually create a custom menu system for any form in an application. Before you learn how to use the menu designer, you must learn about the typical components of a menu system. Look at the Example Menu System shown in Figure B-14.

Figure B-14 Example menu system



The menu system in the figure consists of the following items:

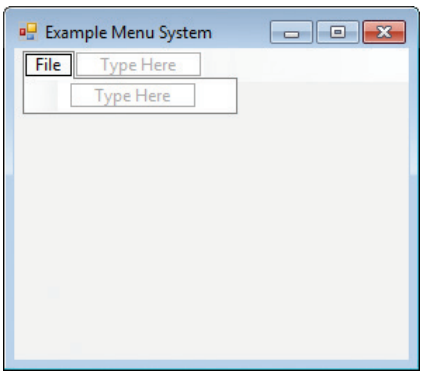
- **Menu names.** Each drop-down menu has a name. The menu names are listed on a menu strip that appears just below the form's title bar. The menu names in Figure B-14 are *File*, *Edit*, and *Help*. The user may activate a menu by clicking the menu name. In the figure, the *Edit* menu has been activated.
- **Menu command.** Menus have commands. The user selects a command by clicking it, entering its access key, or entering its shortcut key.
- **Shortcut key.** A shortcut key is a key or combination of keys that cause a menu command to execute. Shortcut keys are shown on a menu to the right of their corresponding commands. For example, in Figure B-14, Ctrl+C is the shortcut key for the *Copy* command.
- **Disabled menu command.** You can cause a menu command to be disabled when you do not want the user to select it. A disabled menu command appears in dim lettering (grayed out) and cannot be selected. In Figure B-14, the *Undo* command is disabled.
- **Checked menu command.** A checked menu command is usually one that turns an option on or off. A checkmark appears to the left of the command, indicating the option is turned on. When no checkmark appears to the left of the command, the option is turned off. The user toggles a checked menu command each time he or she selects it. In Figure B-14, *Autosave* is a checked menu command.
- **Submenu.** Some of the commands on a menu are actually the names of submenus. You can tell when a command is the name of a submenu because a right arrow appears to its right. Activating the name of a submenu causes the submenu to appear. For example, in Figure B-14, clicking the *Sort* command causes a submenu to appear.
- **Separator bar.** A separator bar is a horizontal bar used to separate groups of commands on a menu. In Figure B-14, separator bars are used to separate the *Copy*, *Cut*, and *Paste* commands into one group, the *Find* and *Replace* commands into another group, and the *Sort* command in a box by itself. Separator bars are used only as visual aids and cannot be selected by the user.

The MenuStrip Control

An application's menu system is constructed with a **MenuStrip control**. When your form is displayed in the *Designer*, find the *Menus & Toolbars* section of the *Toolbox* window and double-click the *MenuStrip* icon. A MenuStrip control will appear in the component tray at the bottom of the *Designer*, with a default name of MenuStrip1.

When the MenuStrip control is selected, you will see the words *Type Here* displayed in a strip at the top of the form. This is the **menu designer**, a tool that allows you to visually edit the contents of the menu. You simply click inside this strip and type the names of the items that you want to appear in the menu. Figure B-15 shows an example where a *File* menu has been added.

Figure B-15 Inserting text into a menu item



Each item that you create in a menu system is a **ToolStripMenuItem object**. When you select a menu item in the menu designer, you see its properties listed in the *Properties* window. The text that you typed for the item in the menu designer will appear in the object's Text property.

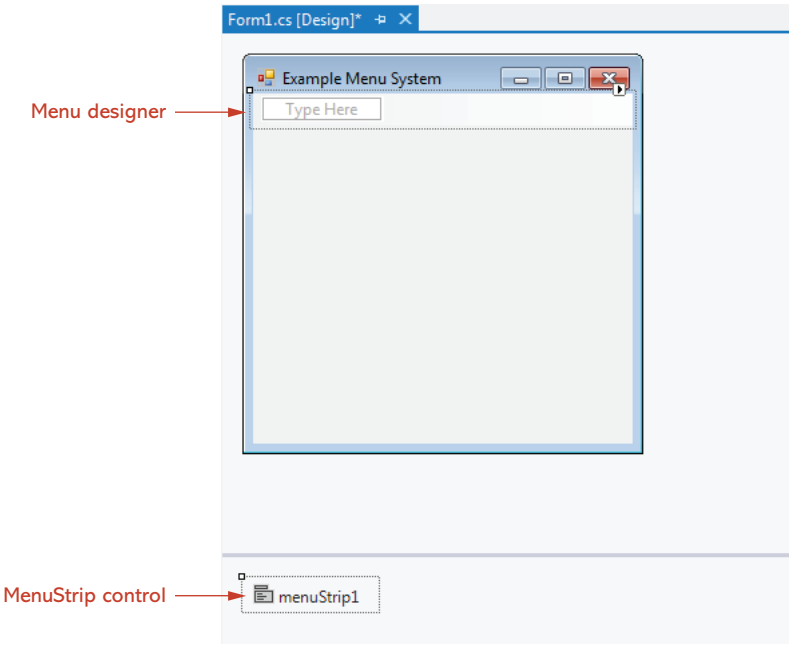
ToolStripMenuItem objects are given default names that are based on the text that you typed for the object in the menu designer. For example, if you type the text "File" for a menu item, the object's name will be `fileToolStripMenuItem`. For another example, if you type the text "Help" for a menu item, the object's name will be `helpToolStripMenuItem`.

ToolStripMenuItem objects also respond to events. You can make a menu functional by writing Click event procedures for its objects.

How to Use the Menu Designer

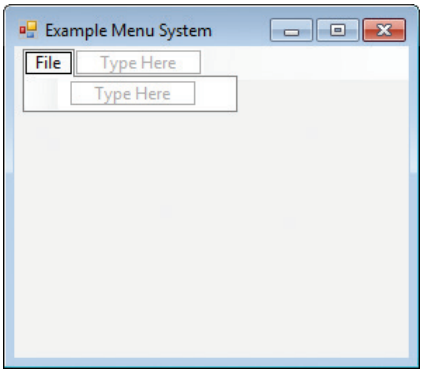
Once you have placed a MenuStrip control in a form's component tray, you can use the menu designer to create menu items. Start the menu designer by selecting the MenuStrip control. Figure B-16 shows a form with a MenuStrip control selected in the component tray, and the menu designer started. The menu designer appears on the form in the location that the menu system will appear.

Figure B-16 MenuStrip control selected and menu designer started



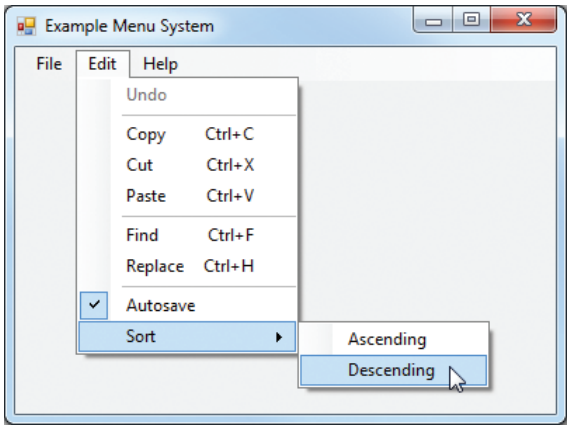
Notice in Figure B-16 that the words *Type Here* appear in a small box in the menu designer. This marks the position of the first menu item. A ToolStripMenuItem object is automatically created when you type text into the box. The text you type is stored in the item's Text property and is displayed on the menu strip. Figure B-17 shows the menu designer after the word *File* has been typed as the text for the first menu item.

Figure B-17 MenuStrip object with *File* as its text



Notice that the menu designer now shows two new *Type Here* boxes, one below and one to the right of the first object. Simply click in one of the boxes to select it and then type the text that you wish to appear at that position. Figure B-18 shows the menu designer with a more complete menu system. The menu system has *File*, *Edit*, and *Help* menus. The *Edit* menu is displayed.

Figure B-18 Menu designer with many items



Shortcut Keys

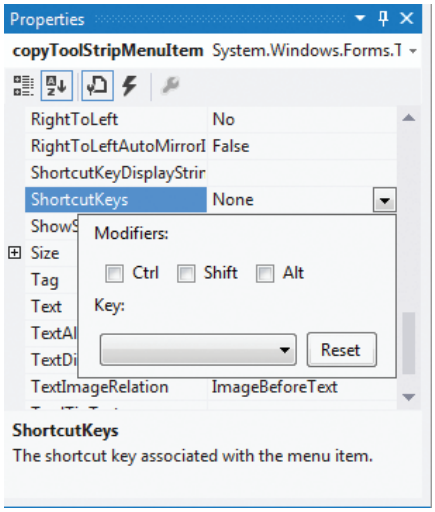
As previously stated, a shortcut key is a key or combination of keys that cause a menu command to execute. Table B-1 lists some commonly used shortcut keys in Windows applications.

Table B-1 Some commonly used shortcut keys in Windows applications

Shortcut Key	Command
Control + S	Save
Control + P	Print
Control + C	Copy
Control + X	Cut
Control + V	Paste

Shortcut keys are shown on a menu to the right of their corresponding commands. To create a shortcut key for a menu item, click the down arrow that appears next to the **ShortcutKeys property** in the *Properties* window. A dialog appears, as shown in Figure B-19. The *Key* drop-down list shows all the available shortcut keys and allows you to select a key from the list. The dialog also allows you to select the Control, Shift, or Alt key (or any combination of these). For example, if you want to assign Control+S as a shortcut key, you would select the *S* key in the drop-down list and place a check next to *Ctrl*.

Figure B-19 The ShortcutKeys property



You must also make sure that the **ShowShortcut property** is set to *True*. When set to *False*, the item's shortcut key is not displayed.

Checked Menu Items

Some programs have menu items that simply turn a feature on or off. For example, suppose you are creating an application that functions as an alarm clock and you want the user to be able to turn the alarm on or off with a menu item. A common approach would be to have a checked menu item for the alarm. When a checkmark appears next to the menu item, it indicates that the alarm is on. When the checkmark is not displayed next to the menu item, it indicates that the alarm is off. When the user clicks the menu item, it toggles its state between on and off. This type of menu item is called a checked menu item.

To give a menu item the ability to become checked or unchecked when it is clicked by the user, you set the item's **CheckOnClick property** to *True*. You can then set the **Checked property** to either *True* or *False* to specify how the item should initially appear when the application runs. If you set the Checked property to *True*, the item will appear with a checkmark next to it. If you set the Checked property to *False*, no checkmark will be shown.

In code, you can use the Checked property to determine whether a menu item is checked. If the Checked property is set to *True*, it means the item is checked. If the Checked property is set to *False*, it means the item is unchecked. The following code shows an example. This code tests the Checked property of a menu item named `alarmToolStripMenuItem`. If the item is checked, a message box is displayed.

```
if (alarmToolStripMenuItem.Checked)
{
    MessageBox.Show("WAKE UP!");
}
```

Disabled Menu Items

A disabled menu item appears dimmed, or grayed out, and may not be selected by the user. You may disable a menu item by setting its Enabled property to *False*. For example, applications that provide *Cut*, *Copy*, and *Paste* commands usually disable the *Paste* command until something is cut or copied. So, the *Paste* menu item's Enabled property can be set to False at design time (in the *Properties* window) and then set to `true` in code after the *Cut* or *Copy* command has been used. Assuming that the *Paste* menu item is named `pasteToolStripMenuItem`, the following code enables it:

```
pasteToolStripMenuItem.Enabled = true;
```

Separator Bars

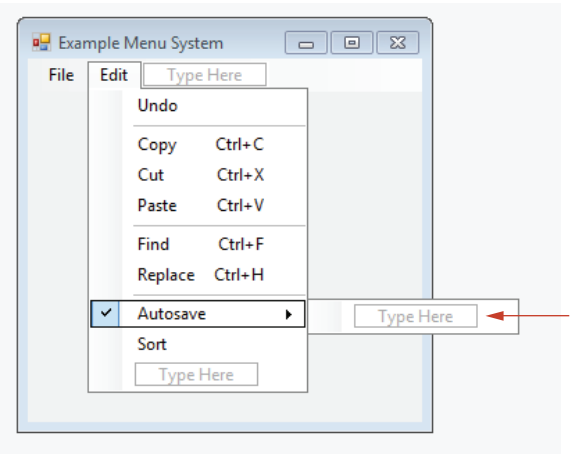
You can insert a separator bar into a menu in either of the following ways:

- Right-click an existing menu item. On the pop-up menu that appears, select *Insert* and then select *Separator*. A separator bar is inserted above the menu item.
- Type a hyphen (-) as a menu item's Text property.

Submenus

When an existing menu item is selected in the menu designer, a *Type Here* box is displayed to its right. Figure B-20 shows an example. This box allows you to create a submenu item. When you create a submenu, a right arrow (▸) is automatically displayed next to the menu item that is the parent of the submenu.

Figure B-20 Creating a submenu



Inserting Menu Items in an Existing Menu

If you need to insert a new menu item above an existing menu item, start the menu designer and then right-click the existing menu item. On the pop-up menu that appears, select *Insert* and then select *MenuItem*. A new menu item is inserted above the existing menu item. If you need to insert a new menu item at the bottom of an existing menu, start the menu designer and simply select the desired menu or submenu. A *Type Here* box automatically appears at the bottom.

Deleting Menu Items

To delete a menu item, start the menu designer and perform one of the following procedures:

- Right-click the menu item you wish to delete. On the pop-up menu, select *Delete*.
- Select the menu item you wish to delete and then press Delete on the keyboard.

Rearranging Menu Items

You can move a menu item by clicking and dragging. Simply select it in the menu designer and drag it to the desired location.

ToolStripMenuItem Click Events

You do not have to write code to display a menu or a submenu. When the user clicks a menu item that displays a menu or a submenu, Visual Studio automatically causes the menu or submenu to appear.

If a menu item does not have a menu or submenu to display, you make it functional by providing a Click event handler for it. For example, assume a menu system has a *File* menu with an *Exit* command, which causes the application to end. The menu item for the *Exit* command is named `exitToolStripMenuItem`. Here is the code for the object's Click event handler:

```
private void exitToolStripMenuItem_Click(object sender, EventArgs e)
{
    // Close the form.
    this.Close();
}
```

To write a Click event handler for a menu item, start the menu designer and then double-click the desired menu item. An empty event handler is created.

Standard Menu Items

Although not all applications have identical menu systems, it is standard for most applications to have the following menu items:

- A *File* menu as the leftmost item on the menu strip.
- An *Exit* command on the *File* menu. This command ends the application.
- A *Help* menu as the rightmost item on the menu strip.
- An *About* command on the *Help* menu. This command displays a window giving basic information about the application.

You should always add these items to your menu systems because most Windows users expect to see them. You should also assign shortcut keys to the most commonly used commands. Study the menu system in an application such as Microsoft Word or Microsoft Excel to become familiar with a typical menu design.

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APPENDIX C

ASCII/Unicode Characters

The following table lists the ASCII (American Standard Code for Information Interchange) character set, which is the same as the first 127 Unicode character codes. This group of character codes is known as the *Latin Subset of Unicode*. The code columns show character codes and the character columns show the corresponding characters. For example, the code 65 represents the letter *A*. Note that the first 31 codes, and code 127, represent control characters that are not printable.

Code	Character	Code	Character	Code	Character	Code	Character	Code	Character
0	NUL	26	SUB	52	4	78	N	104	h
1	SOH	27	Escape	53	5	79	O	105	i
2	STX	28	FS	54	6	80	P	106	j
3	ETX	29	GS	55	7	81	Q	107	k
4	EOT	30	RS	56	8	82	R	108	l
5	ENQ	31	US	57	9	83	S	109	m
6	ACK	32	(Space)	58	:	84	T	110	n
7	BEL	33	!	59	;	85	U	111	O
8	Backspace	34	”	60	<	86	V	112	p
9	HTab	35	#	61	=	87	W	113	q
10	LineFeed	36	$	62	>	88	X	114	r
11	VTab	37	%	63	?	89	Y	115	s
12	FormFeed	38	&	64	@	90	Z	116	t
13	CR	39	'	65	A	91	[	117	u
14	SO	40	(	66	B	92	\	118	v
15	SI	41	)	67	C	93	]	119	w
16	DLE	42	*	68	D	94	^	120	x
17	DC1	43	+	69	E	95	_	121	y
18	DC2	44	,	70	F	96	'	122	z
19	DC3	45	-	71	G	97	a	123	{
20	DC4	46	.	72	H	98	b	124	|
21	NAK	47	/	73	I	99	c	125	}
22	SYN	48	0	74	J	100	d	126	~
23	ETB	49	1	75	K	101	e	127	DEL
24	CAN	50	2	76	L	102	f		
25	EM	51	3	77	M	103	g		

APPENDIX D

Answers to Checkpoint Questions

Chapter 1

1.1 A program is a set of instructions that a computer follows to perform a task.

1.2 Hardware is all the physical devices, or components, that a computer is made of.

1.3 The central processing unit (CPU), main memory, secondary storage devices, input devices, and output devices

1.4 The CPU

1.5 Main memory

1.6 Secondary storage

1.7 Input device

1.8 Output device

1.9 Operating system

1.10 Utility program

1.11 Application software

1.12 1 byte

1.13 A bit

1.14 Binary

1.15 It is an encoding scheme that uses a set of 128 numeric codes to represent the English letters, various punctuation marks, and other characters. These numeric codes are used to store characters in a computer's memory. (ASCII stands for the American Standard Code for Information Interchange.)

1.16 Unicode

1.17 Digital data is data that is stored in binary, and a digital device is any device that works with binary data.

1.18 Machine language

1.19 Main memory, or RAM

1.20 The fetch-decode-execute cycle

1.21 It is an alternative to machine language. Instead of using binary numbers for instructions, assembly language uses short words that are known as mnemonics.

1.22 A high-level language

1.23 Syntax

1.24 A compiler

1.25 An interpreter

1.26 A syntax error

1.27 The part of the computer with which the user interacts

1.28 A command line interface requires the user to type commands. If a command is typed correctly, it is executed and the results are displayed. If a command is not typed correctly, an error message is displayed.

1.29 The program

1.30 A program that responds to events that are caused by the user, such as the clicking of a mouse

1.31 A program component that contains data and performs operations

1.32 Object oriented

1.33 An object can (1) store data in fields or properties and (2) perform operations called methods.

1.34 Controls

1.35 To perform some task

1.36 Code that describes a particular type of object

1.37 A collection of classes and other code that can be used, along with a programming language such as C#, to create programs for the Windows operating system

1.38 When a program needs specific objects that cannot be found in the .NET framework

1.39
1. *Understand the program's purpose*
2. *Design the graphical user interface (GUI)*
3. *Design the program's logic*
4. *Write the code*
5. *Correct syntax errors*
6. *Test the program and correct logic errors*

1.40 A set of well-defined logical steps that must be taken to perform a task

1.41 An algorithm that is written out in this manner, in plain English statements

1.42 A diagram that graphically depicts the steps of an algorithm

1.43
- Oval—terminal symbol
- Parallelogram—input or output
- Rectangle—processing

1.44 The *Solution Explorer* window allows you to navigate among the files in a Visual C# project.

1.45 When you are creating a Visual C# application, you use the *Properties* window to examine and change a control's properties.

1.46 The standard toolbar contains buttons that execute frequently used commands.

1.47 The *Toolbox* is a window that allows you to select the controls that you want to use in an application's user interface. The toolbar contains buttons that execute frequently used Visual Studio commands.

1.48 A small rectangular box that pops up when you hover the mouse pointer over a button on the toolbar or in the *Toolbox* for a few seconds

1.49 A Visual C# project consists of several files. You can think of a project as a collection of files that belong to a single application.

1.50 A solution is a container that holds one or more Visual C# projects. If you are developing applications for a large organization, you might find it convenient to store several related projects together in the same solution.

Chapter 2

2.1 An empty form in the *Designer*

2.2 It is enclosed by a thin dotted line called a bounding box.

2.3 To resize an object's bounding box using the mouse

2.4 A name

2.5 To display the properties of the currently selected object

2.6 Causes the properties to be displayed in alphabetical order

2.7 Causes the properties to be displayed in groups

2.8 The text that is displayed in the form's title bar

2.9 The form's width and height in pixels

2.10 A scrollable list of controls that you can add to a form

2.11 To add a control to a form, you simply find it in the *Toolbox* and then double-click it.

2.12 What the button will do when it is clicked

2.13 The first character must be one of the letters *a–z* or *A–Z* or an underscore character. After the first character, you may use the letters *a–z* or *A–Z*, the digits 0–9, or underscores. The name cannot contain spaces.

2.14 The camelCase naming convention

2.15 Source code

2.16 If you want your application to perform any meaningful actions you must write code.

2.17 The application's start-up code

2.18 Code that is associated with the Form1 form

2.19 Namespaces, classes, and methods

2.20 A container that holds classes

2.21 Braces (`{}`)

2.22 Click the tab for the desired window.

2.23 Double-click the button control in the *Designer*.

2.24 An event that occurs when an application is running and the user clicks a control

2.25 `MessageBox.Show`

2.26 Data that is literally written into a program

2.27 Double quotation marks

2.28 Press the F5 key on the keyboard, or click the *Start Debugging* button on the toolbar.

2.29 The *Common Controls* group

2.30 The Text property

2.31 The Font property

2.32 None

2.33 Click the down-arrow button that appears next to the property's value. Select the desired value from the list.

2.34 The AutoSize property

2.35 The TextAlign property

2.36 Place a Label control on the form at the location where you want the result to be displayed. Erase the contents of the Label control's Text property. In the application's code, you write the necessary statements to perform the calculation and then you store the result of the calculation in the Label control's Text property.

2.37 It clears the text that is displayed in a Label control.

2.38 To display a graphic image on a form

2.39 In the *Common Controls* group

2.40 Use its Image property to specify the image that it will display.

2.41 Normal

2.42 The image is uniformly resized to fit in the PictureBox without losing its original aspect ratio.

2.43 Create a Click event handler for the PictureBox control that displays the image.

2.44 Yes, if a control's Visible property is set to False, the control can still be seen at design time, but during run time, the control will not be visible.

2.45 Comments are brief notes that are placed in a program's source code to explain how parts of the program work.

2.46 A line comment begins with two forward slashes and appears on one line. A block comment starts with a forward slash and an asterisk and ends with an asterisk followed by a forward slash and can occupy multiple lines.

2.47 Be careful not to reverse the beginning and ending symbols, and don't forget the ending symbol.

2.48 Programmers commonly use blank lines and indentations in their code to create a sense of visual organization; this makes the code easier to understand and to read.

2.49 `this.Close();`

2.50 It is underlined with a jagged line in the code editor.

2.51 A description of the error will pop up in a *ToolTip* window.

2.52 A window appears that displays the errors.

Chapter 3

3.1 A TextBox control

3.2 Retrieve the contents of the control's Text property

3.3 A string

3.4 Assign an empty string (`""`) to the Text property.

3.5 To store a value in memory

3.6 `string myFavoriteFood = "pizza";`

3.7 a. String
b. Real number
c. String
d. Integer

3.8 a. Legal
b. Illegal; spaces are not allowed in variable names.
c. Legal
d. Illegal; first character of the variable name cannot be a number.

3.9 The string `"Hello!"`

3.10 While the event is executing; then it is deleted.

3.11 The statement will result in an error.

3.12 Yes, `string` variables can be assigned only string values.

3.13 `string name, city, state;`

3.14 a. `decimal`
b. `int`
c. `double`
d. `decimal`
e. `int`

3.15 b, variables of the `int` data type can be assigned only whole-number values.

3.16 `int dollars = (int)deposit;`

3.17 6

3.18 `string`, `double`, `int`, and `decimal`

3.19 `length` and `width`

3.20 1. Perform any operations that are enclosed in parentheses.
2. Perform any multiplications, divisions, or modulus operations as they appear from left to right.
3. Perform any additions or subtractions as they appear from left to right.

3.21
```
decimal pricePerFoot = 2.99m;
decimal boardLength = 10.5;
decimal totalCost = boardLength * pricePerFoot;
```
or
```
decimal pricePerFoot = 2.99m;
double boardLength = 10.5;
decimal totalCost = (decimal)boardLength * pricePerFoot;
```

3.22 9

3.23 70

3.24 2.5

3.25 a. `count += 1;`
b. `amount -= 5;`
c. `radius *= 10;`
d. `length /= 2;`

3.26 `int.Parse`

3.27 a. `decimal.Parse("90.5");`
b. `decimal.Parse(grandTotal);`
c. `decimal.Parse("50");`
d. `decimal.Parse(priceTextBox.Text);`

3.28 The result will be an error because numeric values cannot be assigned to strings.

3.29 a. `MessageBox.Show(grandTotal.ToString());`
b. `MessageBox.Show(highScore.ToString());`
c. `MessageBox.Show(sum.ToString());`
d. `MessageBox.Show(width.ToString());`

3.30 `resultLabel.Text = result.ToString();`

3.31 `MessageBox.Show(salary.ToString("c"));`

3.32 a. `"P"` or `"p"` for percent format
b. `"E"` or `"e"` for exponential scientific format
c. `"C"` or `"c"` for currency format
d. `"N"` or `"n"` for number format
e. `"F"` or `"f"` for fixed-point scientific format

3.33 12

3.34 `"d5"`

3.35 `millimeters.ToString("f4");`

3.36 An unexpected error that occurs while a program is running, such as user input that is not in the correct format

3.37 Click the *Stop Debugging* button or by pressing Shift + F5.

3.38 Statements that can potentially throw an exception

3.39 If a statement in the try block throws an exception

3.40 Assign a name to the exception object when writing the `catch` clause of a `try-catch` statement and then use the Message property of the exception object in the catch block to display the default error message in a message box.

3.41
```
try
{
    int value1; // To hold value 1.
    int value2; // To hold value 2.
    int sum;    // To hold the sum.

    // Get the values as input from the user.
    value1 = int.Parse(value1TextBox.Text);
    value2 = int.Parse(value2TextBox.Text);

    // Calculate the sum of value 1 and value 2.
    sum = value1 + value2;

    // Display the result in the Label control.
    sumLabel.Text = sum.ToString();
}
catch (Exception ex)
{
    // Display the default error message.
    MessageBox.Show(ex.Message);
}
```

3.42 They make programs more self-explanatory and allow for widespread changes to easily be made to the program.

3.43 `const double DISCOUNT = 0.9;`

3.44 At the top of a class declaration, before any methods

3.45 It's a good programming practice to make fields `private` because `private` fields are hidden from code outside the class. That prevents code outside the class from changing the values of a class's fields and helps prevent bugs from creeping into your program.

3.46 A local variable exists only while the method in which it is declared is executing, but a field exists as long as the form exists.

3.47 `private const decimal INTEREST_RATE = 0.059m;`

3.48 `double product = Math.Pow(12.0, 2.0);`

3.49 `Math.Max`

3.50 `Math.Min`

3.51 The button's Click event handler will execute.

3.52 Click *View* on the Visual Studio menu bar and then click *Tab Order*. Exit tab order selection mode by pressing the Esc key.

3.53 `numberTextBox.Focus();`

3.54 In the Text property, place an ampersand (&) before the character you wish to use as the access key.

3.55 In the Text property, place two ampersand (&&) characters where you would like the single ampersand to appear on the Button control.

3.56
```
resultLabel.BackColor = Color.White;
resultLabel.ForeColor = Color.Red;
```

3.57 None, Tile, Center, Stretch, and Zoom

3.58 The controls inside the GroupBox control are also deleted.

3.59 The TabIndex values of controls inside a group box are organized relative to the GroupBox control's TabIndex property.

3.60 A Panel cannot display a title and does not have a Text property. A Panel's border can be specified by its BorderStyle property.

Chapter 4

4.1 A logical design that controls the order in which a set of statements execute

4.2 It is a program structure that can execute a set of statements only under certain circumstances.

4.3 A decision structure that provides a single alternative path of execution. If the condition that is being tested is true, the program takes the alternative path.

4.4 An expression that can be evaluated as either true or false

4.5 You can determine whether one value is greater than, less than, greater than or equal to, less than or equal to, equal to, or not equal to another value.

4.6
```
if (y == 20)
{
   x = 0;
}
```

4.7
```
if (sales >= 10000)
{
   commissionRate = 0.2;
}
```

4.8 A dual alternative decision structure has two possible paths of execution—one path is taken if a condition is true and the other path is taken if the condition is false.

4.9 When the Boolean expression is false

4.10
```
if (sales >= 50000)
{
   commissionRate = 0.2;
}
else
{
   commissionRate = 0.1;
}
```

4.11
```
if (number == 1)
{
   MessageBox.Show("One");
}
else if (number == 2)
{
   MessageBox.Show("Two");
}
else if (number == 3)
{
   MessageBox.Show("Three");
}
else
{
   MessageBox.Show("Unknown");
}
```

4.12 It is an expression that is created by using a logical operator to combine two Boolean subexpressions.

4.13 F; T; F; F; T; T; T; F; F; T

4.14 T; F; T; T; T

4.15 The `&&` operator: If the expression on the left side of the `&&` operator is false, the expression on the right side will not be checked.
The `||` operator: If the expression on the left side of the `||` operator is true, the expression on the right side will not be checked.

4.16
```
if (speed >= 0 && speed <= 200)
{
    MessageBox.Show("The number is valid");
}
```

4.17
```
if (speed < 0 || speed > 200)
{
    MessageBox.Show("The number is not valid");
}
```

4.18 `true` or `false`

4.19 A variable that signals when some condition exists in the program

4.20 z is not less than a.

4.21 Boston; New York

4.22 `true`

4.23 The variable specified by the second argument

4.24 0

4.25 It specifies an output variable. An output variable is a variable that is passed as an argument to a method, and when the method is finished, a value is stored in the variable.

4.26 Only one

4.27 All of them can potentially be selected.

4.28 When a CheckBox or RadioButton control is selected, or checked, its Checked property is set to True.

4.29
```
switch (choice)
{
    case 1:
        MessageBox.Show("You chose 1.");
        break;
    case 2:
        MessageBox.Show("You chose 2.");
        break;
    case 3:
        MessageBox.Show("You chose 3.");
        break;
    default:
        MessageBox.Show("Make another choice.");
        break;
}
```

4.30 You select the Items property and then use the *String Collection Editor* to enter items.

4.31 When the user selects an item in a ListBox, the item is stored in the ListBox's SelectedItem property.

4.32 When the user selects an item in a ListBox, the item's index is stored in the ListBox's SelectedIndex property. If no item is selected in the ListBox, the SelectedIndex property is set to –1.

4.33 With the `ListBox.Items.Add` method.

4.34 With the `ListBox.Items.Count` property.

4.35
```
namesListBox.Items.Remove("Jamal");
```

4.36
```
namesListBox.Items.RemoveAt(3);
```

4.37
```
namesListBox.Items.Clear();
```

Chapter 5

5.1 An execution of the statements in the body of the loop

5.2 A variable that is used to store the number of iterations that a loop has performed

5.3 A loop that tests its Boolean expression before performing an iteration

5.4 Before

5.5 A loop that has no way of stopping and repeats until the program is interrupted

5.6 6; 5

5.7 Four times

5.8 Initialization, test, and update

5.9 Here is one possible set of answers:

a.
```
count = 0;
```
b.
```
count < 50;
```
c.
```
count++
```
d.
```
for (count = 0; count < 50; count++)
{
   MessageBox.Show("I love to program");
}
```

Here is another possible set of answers:

a.
```
count = 1;
```
b.
```
count <= 50;
```
c.
```
count++
```
d.
```
for (count = 1; count <= 50; count++)
{
   MessageBox.Show("I love to program");
}
```

5.10 It would display the numbers 1, 2, 3, 4, and 5 in message boxes.

5.11 It would display the numbers 0, 100, 200, 300, 400, and 500 in message boxes.

5.12 A loop that tests its Boolean expression after it performs an iteration

5.13 The `while` loop is a pretest loop, and the `do-while` loop is a posttest loop. The `do-while` loop will always iterate at least one time.

5.14 One time

5.15 A file that data is written to. It is called an output file because the program stores output in it.

5.16 A file from which data is read; it is called an input file because the program gets input from the file.

5.17 Open the file, process the file, and close the file.

5.18 A text file contains data that has been encoded as text, using a scheme such as Unicode. A binary file contains data that has not been converted to text.

5.19 Sequential and direct (or random) access. When you work with a sequential access file, you access data from the beginning of the file to the end of the file. When you work with a direct access file, you can jump directly to any piece of data in the file without reading the data that comes before it.

5.20 `StreamWriter`

5.21 `StreamReader`

5.22 Its contents are erased.

5.23 Its contents are not erased.

5.24 The `WriteLine` method writes an item of data to a file and then writes a newline character. The `Write` method writes an item of data to a text file without writing a newline character.

5.25 `File.OpenText`

5.26 A file's read position marks the location of the next item that will be read from the file. When an input file is opened, its read position is initially set to the first item in the file.

5.27 With the `StreamReader` class's `ReadLine` method

5.28 With the `StreamReader` or `StreamWriter` class's `Close` method

5.29 Loop B is written in the correct form.

5.30 Most users of Windows are accustomed to using a dialog box to browse their disk for a file to open or for a location to save a file. An *Open* and/or *Save As* dialog box allows the user of your application to easily specify a file's name and location.

5.31 *InitialDirectory*: Specifies the directory to be initially displayed

Title: Specifies the text displayed in the title bar

Filename: When the user selects a file with the *Open* dialog box, the file's path and filename are stored in the control's Filename property. When the user specifies a location and filename with the *Save As* dialog box, the file's path and filename are stored in the control's Filename property.

5.32 If you call the `Next` method with no arguments, the method returns an integer that is somewhere between 0 and 2,147,483,647. Alternatively, you can pass an argument that specifies an upper limit to the generated number's range.

5.33 A random floating-point number between 0.0 and 1.0 (not including 1.0)

5.34
```
var rand = new Random();
randomNumber = rand.Next(100) + 1;
```

5.35
```
var rand = new Random();
randomNumber = rand.Next(300) + 100;
```

5.36 The system time, retrieved from the computer's internal clock

5.37 It will always generate the same series of random numbers.

5.38 When a form is loaded into memory, before it is displayed

5.39 Double-click any area of the form in the *Designer* window, where there is no other control.

Chapter 6

6.1 A `void` method simply executes a group of statements and then terminates. A value-returning method returns a value to the statement that called it.

6.2 A header and a body

6.3 You are causing the method to execute.

6.4 Control of the program goes back to the part of the program that called the method.

6.5
- The overall task that the program is to perform is broken down into a series of subtasks.
- Each subtask is examined to determine whether it can be further broken down into more subtasks. This step is repeated until no more subtasks can be identified.
- Once all the subtasks have been identified, they are written in code.

6.6 A method uses arguments to pass data into the method using parameter variables.

6.7 Assignment compatibility applies to argument passing in the following ways:

- You can pass only string arguments into `string` parameters.
- You can pass `int` arguments into `int` parameters, but you cannot pass `double` or `decimal` arguments into `int` parameters.
- You can pass either `double` or `int` arguments into `double` parameters, but you cannot pass `decimal` values into `double` parameters.
- You can pass either `decimal` or `int` arguments to `decimal` parameters, but you cannot pass `double` arguments into `decimal` parameters.

6.8 The scope of a parameter variable is limited to the method in which it is declared.

6.9 A named argument is a parameter variable with a default value.

6.10 A copy is passed into the parameter variable, and it does not modify the original argument.

6.11 A reference parameter is a reference to the argument it was passed. Any changes done to the parameter are also done to the argument.

6.12 The calling method can communicate with the called method by passing an argument.

The called method can communicate with the calling method by modifying the value of the argument via the reference parameter.

6.13 The `ref` keyword

6.14 An output parameter works like a reference parameter, but the value passed into it can be uninitialized and it must be set by the method before the method terminates.

6.15 A value-returning method returns a value to the caller. A value-returning method can be used to perform a calculation and then return the result.

6.16 The Boolean values `true` or `false`

6.17 Yes. A method can be written to return any type of data.

Chapter 7

7.1 Value types and reference types

7.2 When you are working with a value type, you are using a variable that holds a piece of data, but when you are working with a reference type, you are using two things:
- An object that is created in memory
- A variable that references the object

7.3 The object is like the kite, and the variable that references the object is like the spool of string.

7.4 Reference type

7.5
```
decimal[] monthlyPay;
```

7.6
```
monthlyPay = new decimal[12];
```

7.7
```
const int SIZE = 12;
double[] monthlyPay = new decimal[SIZE];
```

7.8
```
string [] fullName = { "John", "Quincy", "Adams" };
```

7.9 In any of the following circumstances, you should use a `for` loop to process the array:
- To change the contents of an array element
- To work through the array elements in reverse order
- To access some of but not all the array elements
- To simultaneously work with two or more arrays within the loop

7.10 When an object is no longer referenced, it becomes eligible for garbage collection.

7.11 A reference to the array

7.12 Because arrays are always passed by reference, a method that receives an array as an argument has access to the actual array.

7.13 The receiving method not only has access to the array, but it also has access to the reference variable that was used to pass the array.

7.14 A method of locating a specific item in a larger collection of data

7.15 A technique for scanning through an array and rearranging its contents in some specific order

7.16 The values in the array must be sorted in ascending order.

7.17 200 rows and 100 columns

7.18
```
values[199][99] = 50;
```

7.19
```
int[,] values = { { 12, 24, 32, 21, 42 },
                  { 99,  8, 68, 32, 92 },
                  { 95, 34, 21, 11,  7 } };
```

7.20 In a traditional two-dimensional array, each row has the same number of columns.

7.21
```
int[][] values = new int [3][];
values[0] = new int[3] { 2, 4, 6 };
values[1] = new int[4] { 3, 5, 7, 9 };
values[2] = new int[5] { 5, 9, 11, 17, 21 };
```

7.22 `List<double> valueList = new List<double>() { 1.0, 2.0, 3.0 };`

7.23 `valueList.Add(4.0);`

7.24 `valueList.Clear();`

7.25 Yes. You can use the subscript notation of a `List` to adapt any of the array-processing algorithms so they work with a `List`.

7.26 By getting the value of the control's `Images.Count` property.

7.27 You will find the ImageList control in the Components section of the Toolbox.

7.28
- All the images stored in an ImageList control should be of the same size.
- The images stored in an ImageList control can be no more than 256 by 256 pixels in size.
- All the images stored in an ImageList control should be in the same format (.bmp, .jpg, etc.).

7.29 In the Properties window, click the ellipses button that appears next to the Images property. Click the Add button, and an Open dialog box will appear. Use the dialog box to locate and select the image file that you want to add to the ImageList control.

7.30 If you know the index value for a particular image, you can retrieve that image from the ImageList control and display it in a PictureBox.

Chapter 8

8.1 `$"{number:N2}"`

8.2 `$"{number,10:F1}"`

8.3 `$"{number,10:N1}"`

8.4 `$"{number,-20:N2}"`

8.5 `char letter = 'A';`

8.6 `MessageBox.Show(letterGrade.ToString());`

8.7 `char lastLetter = alphabet[alphabet.Length-1];`

8.8
```
foreach (char character in serialNumber)
{
    MessageBox.Show(character.ToString());
}
```

8.9 `if (char.IsPunctuation(sentence[sentence.Length-1]))`

8.10 `if (char.IsUpper(sentence, 0))`

8.11 `uppercase = char.ToUpper(lowercase);`

8.12 If the argument is already lowercase, the `ToLower` method returns it unchanged.

8.13 `found = dessert.StartsWith("Strawberry");`

8.14 If the substring is not found, the methods will return the integer value `-1`.

8.15 `str2 = str1.Trim();`

8.16 `veggie = vegetable.ToLower();`

8.17 `str2 = str1.Insert(5, "www");`

8.18 `str2 = str1.Remove(0, 6);`

8.19 `str2 = str1.Replace("dollars", "euros");`

8.20 The contents of a `StringBuilder` object can be changed.

8.21 It would be more efficient to use `StringBuilder` objects because it reduces the number of `string` objects that must be created and then removed when they are no longer being used.

8.22
```
var city = new StringBuilder("Regensburg");
```

8.23 `Append`

8.24 `Insert`

8.25 `Remove`

8.26 `Replace`

8.27 The tokens are *apples*, *pears*, and *bananas*. The space character is the delimiter.

8.28 `x` will be assigned 4. The `first` variable will be assigned `"one"`.

8.29
```
string str = "/home/rjones/mydata.txt";
char[] delim = {'/', '.'};
string[] tokens = str.Split(delim);
```

8.30
```
string str = "dog$cat@bird%squirrel";
char[] delim = { '$', '@', '%'};
string[] tokens = str.Split(delim);
```

8.31
```
char[] delimiters = { ',', ';' };
```

8.32 A comma is used as the delimiter for files with the .csv extension.

8.33
```
string result = str.Insert(0, ">").ToUpper().Replace('*', '#');
```

8.34 "ABC-DEF"

8.35
```
string result = String.Format("{0:N}", number);
```

8.36
```
string result = String.Format("{0,10:N1}", number);
```

8.37
```
string result = String.Format("{0,10:F}", number);
```

8.38
```
string result = String.Format("{0,-20:N2}", number);
```

Chapter 9

9.1 Nothing

9.2 The keyword `public` is an access modifier that specifies that the field can be directly accessed by statements outside the structure.

9.3 When you create an instance of a structure, you are creating an object.

9.4 The `new` operator creates an instance of the structure and initializes the object's fields with the default value of 0 or `null` if any of the fields are reference variables.

9.5 Once you have created an instance of a structure, you can access its fields using the dot operator.

9.6
```
motor2 = motor1;
```

9.7
```
Engine[] motors = new Engine[100];
```

9.8 When the items in two data structures are related by their indexes

9.9 January 1, 0001, 12:00:00 AM, 0 milliseconds

9.10
```
DateTime dt = new DateTime(1965, 3, 22);
```

9.11 `MessageBox.Show(dt.ToString("d"));`

9.12 `DateTime dt = DateTime.Now;`

9.13 `DateTime dt = DateTime.Today;`

9.14 False

9.15 An interval of time

9.16 March 25, 1977

9.17 a. `Flower`
b. 1, 2, 3
c. `Flower flora = Flower.Petunia;`

9.18 The `ToString` method

9.19 You can convert an enumerator to its underlying integer type by using a cast operator.

9.20 A key and a value

9.21 `Dictionary <int, string> myDictionary = new Dictionary <int, string>();`

9.22 It stores an element in the `Dictionary`. The element's key is 543 and its value is "Joanne Manchester".

9.23 1. You can use the `[]` operator. For example, the expression `myDictionary[50]` will retrieve the element with the key 50.
2. You can use the `TryGetValue` method.

9.24 Key and Value

9.25 b

9.26 c

9.27 d

9.28 a

9.29 c

9.30 d

9.31 b

Chapter 10

10.1 A class is code that describes a particular type of object. You can think of a class as a code blueprint that can be used to create a particular type of object.

10.2 The first line of a class declaration is known as the class header. It starts with the word `class`, followed by the name of the class. Following the class header is an opening curly brace. Next, you write the class's member declarations. These are the statements that define the class's fields, properties, and methods. A closing curly brace appears at the end of the class declaration.

10.3 A constructor is a method that is automatically executed when an object is created. In most cases, a constructor is used to initialize an object's fields with starting values.

10.4
1. With the project open in Visual Studio, click *Project* on the menu bar and then select *Add Class*….
2. Select *Class* as the type of item in the *Add New Item* window and then change the name that appears in the *Name* text box to match the name of the class that you are creating.
3. Click the *Add* button.

10.5 When you pass an object that is an instance of a class as an argument, the thing that is passed into the parameter variable is a reference to the object.

10.6 A property is a special type of class member that allows an object to store and retrieve a piece of data.

10.7 Properties can be passed only by value.

10.8 If you need to make a property read only, you simply do not write a `set` accessor for the property.

10.9 An error will occur if you try to assign a value to a read-only property.

10.10 When field's value is tightly dependent on other data and that field is not updated when the other data is changed, it is said that the field has become stale.

10.11
```
private int ID { get; set; }
```

10.12 Leave out the `set` keyword in the auto-property declaration.

10.13 A parameterized constructor is a constructor that accepts arguments.

10.14 When a method is overloaded, it means that multiple methods in the same class have the same name, but use different types of parameters.

10.15 Binding

10.16 If you write a class with no constructor whatsoever, the compiler will provide a default constructor.

10.17 The default constructor is a parameterless constructor (it accepts no arguments), and it initializes the object's fields with the value 0. (If any fields are reference variables, they are initialized with the special value `null`.)

10.18 When you create an array of a class type, each element of the array is a reference variable. By default, each element is initialized with the value `null`.

10.19 You can initialize the array element one element at a time, in a loop, or in the declaration statement of the array.

10.20 By writing the class type inside the angled brackets, `<>`, immediately after the word `List`

10.21 A written description of the real-world objects, parties, and major events related to the problem.

10.22 If you adequately understand the nature of the problem you are trying to solve, you can write a description of the problem domain yourself. If you do not thoroughly understand the nature of the problem, you should have an expert write the description for you.

10.23 First, identify the nouns, pronouns, and pronoun phrases in the problem domain description. Then, refine the list to eliminate duplicates, items that you do not need to be concerned with in the problem, items that represent objects instead of classes, and items that represent simple values that can be stored in variables.

10.24 The things that the class is responsible for knowing and the actions that the class is responsible for doing

10.25 In the context of this problem, what must the class know? What must the class do?

10.26 No

10.27 Follow these steps to rename a form:

1. Right-click the form's name in the *Solution Explorer*.
2. Select *Rename* from the pop-up menu.
3. In the *Solution Explorer*, the form's filename should become highlighted. Type the new name, and press Enter.
4. When the dialog box appears, click *Yes* to rename the form.

10.28 Follow these steps to add a new form to a project:

1. Click *Project* on the Visual Studio menu bar and then select *Add Windows Form…* from the *Project* menu. The *Add New Item* window should appear.
2. Near the bottom of the *Add New Item* window, a *Name* text box appears where you can specify the new form's file name. Initially, a default name will appear here. Change the default name that is displayed in the *Name* text box to a more descriptive name.
3. Click the *Add* button.

10.29 If you wish to remove a form from a project and delete its file from the disk, follow these steps:

1. Right-click the form's entry in the *Solution Explorer* window.
2. On the pop-up menu, click *Delete*.

10.30 In your application's code, the first step in displaying a form is to create an instance of the form's class.

10.31 When the user closes the form, control of the application returns to the point where the `ShowDialog` method was called, and execution resumes.

10.32 Only one

10.33 `MyClass.value = 10;`

10.34 `MyClass.Go();`

10.35 No

Chapter 11

11.1 A reference to a `Student` object.

11.2 `Two_D_Space.Point`

11.3 199, 249

11.4 "199, 249"

11.5 No

11.6 An exception is thrown.

11.7 `null`

11.8 It removes unreferenced objects from memory.

11.9 It will cause an error.

11.10 It will not cause an error.

11.11 b

11.12 The will reference the same object.

11.13 b

11.14
```
var point = new Point() { X = 5, Y = 10 };
```

11.15 b

11.16 Because the Name and Email properties are not settable. To use object initialization syntax, the properties must either have a `set` accessor or an `init` accessor.

11.17
```
class Person
{
    public string Name { get; init; }
    public string Email { get; init; }
}
```

11.18 No

11.19 `readonly`

11.20 init-only

11.21 No

11.22 No

11.23 No

11.24 Composition is a strong form of aggregation in which the lifetime of the inner class is dependent on the lifetime of the outer class. This means the outer class is responsible for creating its instances of the inner class. Also, when an instance of the outer class is destroyed, its instances of the inner class are also destroyed.

11.25 An object that is not referenced by a variable.

11.26
- To create an object just for the purpose of passing that object to a method.
- To create an object just for the purpose of returning that object from a method.

Chapter 12

12.1 The base class is the general class, and the specialized class is the derived class.

12.2 One object is a specialized version of the other object.

12.3 It inherits all the base class's members.

12.4 `Bird` is the base class and `Canary` is the derived class.

12.5 Summary of constructor issues in inheritance.

- When you create an instance of a derived class, the base class constructor is executed first, and then the derived class constructor is executed.
- When you create an instance of a derived class, by default the base class's parameterless constructor is automatically executed.
- If you want a parameterized constructor in the base class to execute, you must explicitly call it from the derived class's constructor. You do this by writing the notation `: base(`*`parameterList`*`)` in the derived class's constructor header.
- If the base class does not have a parameterless constructor, the derived class constructor must use the notation `: base(`*`parameterList`*`)` to call one of the base class's parameterized constructors.

12.6 Code in the same class as the member, or any class that is derived from the class.

12.7 Private members of a class cannot be directly accessed by code in classes that are derived from the class. Protected members of a class can be directly accessed by code in classes that are derived from the class.

12.8 I'm a potato.
I'm a potato.

12.9 No. An object of a base class is not a specialized version of a derived class. Although the statement "a dog is an animal" is true, the statement "an animal is a dog" is not true. This is because all dogs are animals, but not all animals are dogs.

12.10 An abstract class is meant to be a base class and never instantiated. It serves as a starting point, providing some members for its derived classes.

12.11 It cannot be instantiated.

12.12 It must override the abstract method.

12.13 It must be overridden in a derived class.

12.14 Simply leave out the `set` accessor.

12.15 If the class is sealed, it is not possible to extend the class through inheritance.

12.16 No

12.17 Static

12.18 The method is static and the first parameter begins with the keyword `this`.

12.19 The object that the method is operating on

12.20 It specifies behavior that a class must have.

12.21 An interface contains only abstract methods and/or abstract properties. An interface cannot be instantiated.

12.22 `class Customer: IRelatable`

12.23 `class Employee: IPayable, IListable`

Chapter 13

13.1 A database management system is software that is specifically designed to store, retrieve, and manipulate large amounts of data in an organized and efficient manner.

13.2 Most businesses use a DBMS instead of creating their own text files because a DBMS is specifically designed to store, retrieve, and manipulate large amounts of data in an organized and efficient manner. When traditional files contain large amounts of data, simple operations such as searching, inserting, and deleting become cumbersome and inefficient.

13.3 The programmer does not need to know specific details about the physical structure of the data because the DBMS handles the actual reading of, writing of, and searching for data. The programmer needs to know only how to interact with the DBMS.

13.4 a. A database is the data that is stored in a database management system and organized into one or more tables.
b. A table holds a collection of related data that is organized into rows and columns.

c. A row is a complete set of information about a single item in a table of a database.
d. A column holds an individual piece of information about the item in a row of a table in a database.

13.5
1. c
2. d
3. f
4. b
5. a
6. e

13.6 A primary key serves as a unique value that can be used to identify a specific row.

13.7 A column that contains unique values that are generated by the DBMS

13.8 Leaving the column empty will result in a error.

13.9 Application
Binding source
Dataset
Table adapter
Data source

13.10 A table adapter connects to a data source and can retrieve data from a table in a data source. It can also update the table in the data source.

13.11 A binding source is a component that can connect user interface controls directly to a dataset.

13.12 A user interface control that is connected to a data source is called a data-bound control.

13.13 Table adapter
Adapter manager
Data set
Binding source

13.14 The `Fill` method causes the table adapter to load data from the database into the dataset.

13.15 Visual Studio provides a *Data Sources* window that lets you see all the data sources in the current project. Click *Project* on the Visual Studio menu bar to display the *Data Sources* window.

13.16 Create a Details view to display on a single row of a database.

13.17 Dragging a table from the *Data Sources* window onto the form creates a DataGridView control (by default), complete with a navigation bar, on the form.

13.18 The project's output folder is the *bin\Debug* folder.

13.19 When you click the smart tag, a tasks panel will pop up, giving you a number of options that you can perform with the control.

13.20 The type of control to which a column is automatically bound is determined by the column's data type.

13.21 To bind a ListBox control to a column, you must set two of the control's properties: DataSource and DisplayMember.

13.22 The table adapter's `Fill` method

13.23 You use the `Select` statement to retrieve the rows in a table.

13.24 Queries

13.25 In SQL, the equal to operator is one equal sign, not two equal signs, and the not equal to operator is `<>`.

13.26 If you need to include a single quote as part of a string, simply write two single quotes in its place.

13.27 The `Like` operator can be used to search for a substring in a column.

13.28 Unlike the `%` character, the underscore represents a single character.

13.29 The `Order By` clause

13.30 The `@` symbol

Chapter 14

14.1 A delegate is an object that works like a reference variable that is referring to a method.

14.2 1. Declare a delegate type.
2. Create an instance of the delegate type and make it reference a method.
3. Use the delegate to call the method.

14.3 It describes the method that the delegate can refer to.

14.4 Yes

14.5 A delegate that refers to multiple methods

14.6 A method with no name

14.7 b

14.8 A lambda expression is a concise expression that creates an anonymous method.

14.9 a. `=>`
b. `x`
c. `x * 2`

14.10 According to the delegate type, the delegate can only take one argument. In this code, the `Sum` delegate is taking two arguments.

14.11 A lambda expression can access the local variables that are declared in the method that contains the lambda expression, as long as those variables are in scope and have been assigned a value.

14.12 A predefined delegate type that invokes a `void` method that accepts 0–16 arguments.

14.13 A predefined delegate type that invokes a method that accepts 0–16 arguments and returns a value.

14.14 A predefined delegate type that invokes a method that accepts one argument and returns a `boolean` value.

Chapter 15

15.1 It gets all the elements with a value greater than 10.

15.2 `IEnumerable<int>`

15.3 The `from` clause iterates over a collection of data elements, assigning each element to an iteration variable.

15.4 The `select` clause determines what will be included in the results of the query, and it returns the results as an `IEnumerable<T>` collection. The `select` clause also determines what data type *`T`* will be.

15.5 The `orderby` clause sorts the results in ascending order. Append the `descending` keyword to the `orderby` clause to sort the results in descending order.

15.6
```
var results = retailItems.Where(item => item.Price < 100.0m)
                                 .OrderBy(item => item.Price)
                                 .Select(item => item.Description);
```

15.7 If the index is out of bounds, the `ElementAt` method throws an exception, whereas the `ElementAtOrDefault` method returns a default value.

15.8 The method throws an exception.

15.9 `Enumerable.Range`

15.10 `Distinct`

15.11 `Take`

15.12 `SequenceEqual`

15.13 An object that represents a database. It has the ability to connect to a database, translate LINQ queries into SQL queries, and execute those SQL queries on the database.

15.14 Table objects represent the tables in a database. A table object is a property of a data context object, and is a collection of entity objects. An entity object represents a row in a table.

15.15 They are properties of an entity object.

15.16 1) Add a connection to the database, 2) add a DBML file to the project, and 3) use the Object Relational Designer to select the tables you want to work with.

15.17 `InsertOnSubmit` and `SubmitChanges`

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APPENDIX E

Installing LINQ to SQL Classes in Visual Studio

LINQ to SQL is not automatically installed with Visual Studio. Before you can use it, you must follow these steps to use the Visual Studio Installer to download and install it.

Step 1: Launch the Visual Studio Installer

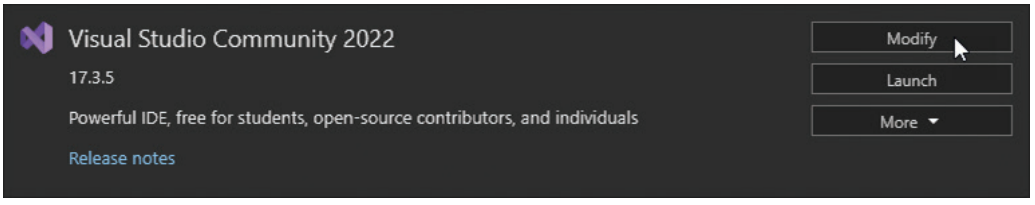
If Visual Studio is running, close it. Then, open the Windows Start menu and scroll down until you see *Visual Studio Installer*. Alternatively, you can start typing *Visual Studio Installer* in the Windows search box. You should see the program listed in the search results.

When you launch the Visual Studio Installer, you will most likely see a screen asking *Do you want to allow this app to make changes to your device?* Click yes.

Step 2: Select Modify in the Visual Studio Installer

When the installer launches, you will see a list of installed Visual Studio products. Find the version of Visual Studio that you are using and select *Modify*. This is shown in Figure E-1.

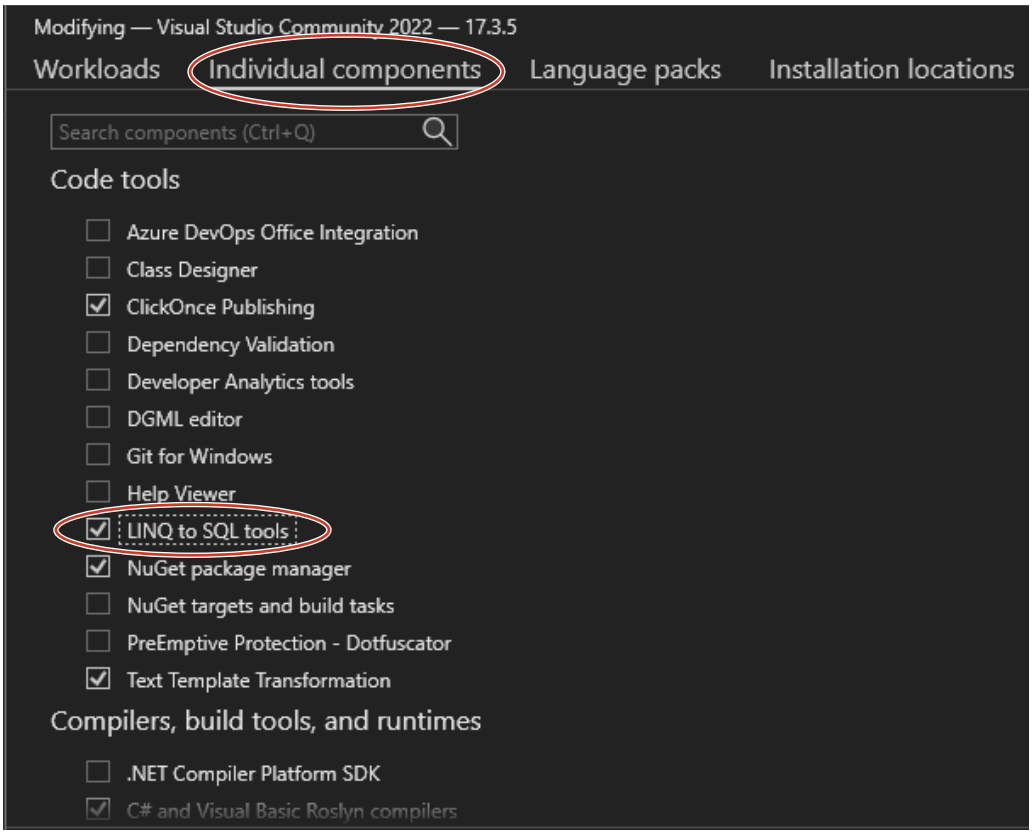
Figure E-1 Select Modify in the Visual Studio Installer



Step 3: Select LINQ to SQL Tools

In the window that appears, select *Individual Components* at the top of the window, and then scroll down until you see *Code Tools*. Under *Code Tools*, select *LINQ to SQL Tools*, as shown in Figure E-2. Click the Modify button in the lower right corner of the window. The installer will download and install the component.

Figure E-2 Select LINQ to SQL Tools



Step 4: Start Visual Studio

After the component has been installed, close the Visual Studio Installer and start Visual Studio.

Glossary

! operator The logical NOT operator.
&& operator The logical AND operator.
.NET A software platform that provides an extensive collection of classes and tools that can be used by a program.
|| operator The logical OR operator.
"is a" relationship A relationship between classes that specifies that one class is a specialized version of the other class.
abstract class A class that is not intended to be instantiated, but to be used only as a base class.
abstract method A method that appears in the base class but expects to be overridden in a derived class.
abstract property A property that appears in the base class but expects to be overridden in a derived class.
accept button A button on a form that is automatically clicked when the user presses the Enter key.
access key A key that is pressed in combination with the Alt key to access a control.
access modifier Specifies how a class member may be accessed by code outside the class.
accessors A special set of methods that provide access to a private field.
accumulator A variable that is used to accumulate the total of a series of numbers.
Action delegate A delegate used to invoke a void method that takes 0–16 arguments.
aggregation methods A method that calculates a single value from a collection of values.
algorithm A set of well-defined, logical steps that must be taken to perform a task.
Alphabetical button Causes the properties in the Properties window to be displayed in alphabetical order.
American Standard Code for Information Interchange (ASCII) A set of 128 numeric codes that represent the English letters, various punctuation marks, and other characters.
anonymous method A method that has no name.
anonymous object An object that is not referenced by a named variable.
application software Programs that are used for everyday tasks, such as word processing, email, text messaging, etc.
argument A piece of data that is passed into a method.
array An object that can hold a group of values of all the same data type.
aspect ratio The ratio of an image's width to its height.
assembler A program that translates an assembly language program into machine language.
assembly language A language that, as an alternative to machine language, uses mnemonics to represent instructions.
assignment operator The = operator assigns the value on its right side to the item on its left side.
assignment statement A statement that assigns a value to an item in memory, such as a variable or a control's property.
Auto Hide A Visual Studio feature that allows a window to collapse into a small tab when it is not in use.
auto-property A simplified syntax for writing a property that only gets and sets its value.
backing field A private field that holds the data that is assigned to a property.
base class In an inheritance relationship, the base class is the general class that other classes are derived from.
binary A numbering system that uses only the digits 0 and 1.
binary digit The smallest unit of storage in a computer's memory, large enough to hold a 0 or a 1.
binary file A file that contains data that has not been converted to text.
binary numbering system A numbering system that uses only the digits 0 and 1.
binary search A search algorithm that repeatedly divides the sorted array in half, eliminating the half that cannot contain the element being searched for.
binding The compiler's process of matching a method call with the correct method.
binding source A component that can connect user interface controls directly to a dataset.
bit A binary digit.
block comment A comment that can span multiple consecutive lines in a program.
body The block of statements that are executed by a loop.
`bool` data type A data type that allows the values true and false.
Boolean expression An expression that can be evaluated as either true or false.
Boolean property A property that can either true or false.
bounding box A thin dotted line that appears around the selected control in the *Designer* window.
braces { and } characters that are used to enclose and group lines of code.
break mode A debugging mode in which the execution of a program is paused.
breakpoint A point in a program's source code that causes the running application to enter break mode.
Button A GUI control that the user can click to cause an action to take place.
byte A unit of storage in a computer's memory large enough to hold a letter of the alphabet or a small number. A byte is made of eight bits.
camelCase A convention by which identifier names are written.
cancel button A button on a form that is automatically clicked when the user presses the Esc key.
capacity The number of characters of characters that a `StringBuilder` can store without reallocating memory.
cast operator An operator that converts a value to a specific data type.
catch block A group of statements that are executed by a `catch` clause when an exception is thrown.
catch clause A clause in a `try-catch` statement that the program jumps to when an exception is thrown.
Categorized button Causes the properties in the Properties window to be displayed in categorized groups.
central processing unit The component of a computer that executes programs. Also known as the CPU.
character literals A character value that is written into a program's code.
check box A control that appears with a small box that can be checked or unchecked.

CheckBox control A control that appears with a small box that can be checked or unchecked.
CheckedChanged event An event that occurs when the user selects or deselects a RadioButton or CheckBox control.
class Code that describes a particular type of object.
class declaration The code that defines a class and its members.
class header The first line of a class declaration.
Click event An event that occurs when the user clicks on a control.
code The statements that a programmer writes in a programming language.
code reuse Writing the code to perform a task once, and then reusing that code each time you need it.
column An individual piece of information about an item, stored in a row.
combined assignment operators Operators that combine a math operation with an assignment statement.
comma separated value (CSV) A file format in which values are separated by commas.
command line interface A computer interface in which the user types commands.
compiler A program that translates source code into a machine language program.
component tray The area at the bottom of the *Designer* that holds nonvisible components.
composition A strong form of aggregation in which the lifetime of the inner class objects is dependent on the lifetime of the outer class objects.
concatenation The operation of appending one string to the end of another string.
concrete class A class that can be instantiated.
conditionally executed To execute a statement or group of statements only if a condition exists.
console interface A computer interface in which the user types commands.
constant field A field that, once given a value, cannot be changed.
constructor A class method that is automatically executed when an object is created.
control structure A logical design of code that controls the order in which a set of statements execute.
controls The individual objects in a graphical user interface.
counter variable A variable that is regularly incremented during each iteration of a loop.
CPU The component of a computer that executes programs.
data context object An object that has the ability to connect to a database, translate LINQ queries into SQL queries, and execute those SQL queries on the database.
data source A source of data.
data type The type of data that a variable can hold.
database An organized collection of data.
database management system Software that is designed to store, retrieve, and manipulate large amounts of data in an organized and efficient manner.
Database Markup Language (DBML) A markup language designed to define database structure.
data-bound control A user interface control that is connected to a data source.
DataGridView control A control that can display an entire database table in a scrollable grid on an application's form.
dataset A component that gets a copy of a table from the table adapter and keeps the copy of the table in memory.
`DateTime` structure A .NET structure that lets you store data about an instance in time.
DateTimePicker control A control that allows the user to select a date from a calendar.
DBML file A file containing database markup.
debug The process of finding and correcting errors in a program.
decimal literal A `decimal` value written into a program's code. A decimal literal is written with the `M` or `m` suffix.
decision structure A programming structure that performs one action if a condition exists, or possibly another action if the condition does not exists.
deconstruction Extracting the fields of a tuple and assigning their values to variables.
decrement To decrease a variable's value.
default argument An argument that is passed to a parameter variable by default, when no explicit argument is provided.
default constructor The constructor that is automatically provided for a class by the compiler when the class has no programmer-defined constructor. The default constructor is parameterless, and performs no actions.
delegate An object that works like a reference variable that is referring to a method.
delegate type A declaration that describes the method that a delegate can refer to.
delimiter A character that separates pieces of data in a string.
derived class In an inheritance relationship, the derived class is the specialized class that is derived from a base class.
design time The time during which the programmer builds an application's GUI and writes the application's code.
Designer window A Visual Studio window in which you design an application's graphical user interface.
Details view A set of individual controls that are bound to the columns in a single row.
dialog box A small window that displays a message to the user.
dictionary A data structure that contains key-value pairs.
digital Describes anything that is stored in binary, using the digits 0 and 1.
digital data Data expressed in binary, which consists of the digits 0 and 1.
digital device Any device that is designed to work with digital (binary) data.
direct access file When working with a direct access file, you can jump directly to any piece of data in the file without reading the data that comes before it.
disk drive A secondary storage device that magnetically encodes data onto a spinning disc.
divide and conquer The process of breaking down a large problem into small manageable pieces.
docked (window) When a window in Visual Studio is docked, it is attached to one of the edges of Visual Studio's main window.
documentation comments Comments that can be used to produce external documentation.
double literal A `double` value written into a program's code.
dual-alternative decision structure A decision structure that provides two possible paths of execution.
element methods A method that returns a specific element from a collection.
elements The storage locations in an array.
End terminal A terminal symbol that marks the end of a flowchart.
ENIAC The first programmable electronic computer.
entity object An object that represents a row in a table.
enumerated data type A programmer-defined data type that consists of a set of named values.
enumerators The named values in an enumerated data type.
event driven A type of program that responds to the actions of the user.
event handler A method that executes when a specific event takes place while an application is running.

exception An unexpected error that occurs while a program is running.
exception handler Code that can handle exceptions when they are thrown.
exception object An object that is created in memory when an exception is thrown. The object contains information about the exception.
executing The process of executing the statements in a program.
execution point While a programming is running in debug mode, the execution point is the next line of code that will execute.
extension method A method that is added onto a class without modifying the class or inheriting from it.
fetch-decode-execute cycle The CPU's process of executing an instruction.
field A piece of data stored in an object, structure, or tuple; a variable declared inside a class, but not inside of any method.
file object An object in memory that is associated with a specific file and provides a way for the program to work with the file.
filename A name that identifies a file on a disk.
filename extensions Short sequences of characters that appear at the end of a filename and are preceded by a period
flag A Boolean variable that signals when some condition exists.
floating (window) When a window in Visual Studio is floating, it is not attached to one of the edges of Visual Studio's main window. It can be freely moved about on the screen.
floating-point notation A binary storage technique used to represent real numbers in memory.
flowchart A diagram that graphically depicts the steps of an algorithm.
focus When an application is running, the control that has the focus is the one that receives the user's keyboard input.
`foreach` loop A loop that is designed to iterate once for each element in a collection of items such as an array.
Form A window that is displayed on the screen.
format item A placeholder for a value that is to be inserted into a format string.
format specifier A special code that causes a value to be formatted in a specific way.
format string A string that will be formatted and possibly have values inserted into it.
formatting string A string that specifies the way a number should be formatted.
fully qualified name The full name of an class or other entity, including any namespaces it might be contained in.
Func delegate A delegate used to invoke a value-returning method that takes 0-16 arguments.
garbage collection A process that periodically runs to remove unreferenced objects from memory.
generation methods A method that generates a new collection of values.
graphical user interface A computer interface that uses graphical elements displayed on the screen. Also known as a GUI.
GroupBox Control A container that can hold other controls.
GUI A computer interface that uses graphical elements displayed on the screen.
hardware The physical components that make a computer system.
has-a relationship A relationship that exists between classes when one class contains an instance of another class.
high-level language A programming language that is closer in nature to human language than machine language.
identifiers An identifier is a name that identifies a variable, control, or other programmer-defined item.
identity column A column that contains unique values that are generated by the DBMS.
`IEnumerable<T>` interface An interface implemented by objects that can be iterated over.
`if-else` statement A statement that tests a Boolean expression and follows one of two possible paths of execution, depending upon whether the Boolean expression is true or false.
`if-else-if` statement An if-else statement in which the else clause causes another if statement to be executed. It works as though the second if statement is nested inside the else clause of the first if statement. A series of if-else statements may be chained together in this way.
ImageList control A control that holds multiple images in memory.
Immediate window A window in the Visual Studio debugger that allows you to type debugging commands.
immutable An object or value that cannot be changed.
immutable class A class that does not allow its properties and/or fields to be changed.
immutable object An instance of an immutable class. An immutable object's fields and/or properties cannot be changed.
implements When a class implements an interface, it must provide all the methods and properties that are specified by the interface.
increment To increase a variable's value.
index A number that identifies an item's position in a collection such as ListBox.
`init` accessor A property accessor that allows the property to be assigned a value only during the object's construction phase.
initialization expression An expression in a for loop that typically initializes counter variable.
initialization list A list of values used to initialize an array.
initialize To assign a value to a variable when the variable is declared.
init-only property A property that may only be assigned a value during the object's construction phase.
input device A device that collects input and sends it to the computer.
input Data that the computer collects from people and other devices.
input file A file from which data is read.
input symbols Symbols in a flowchart that represent input operations.
input validation The process of inspecting input for validity, and rejecting invalid input.
instance A specific object of a structure or a class.
instruction set The entire set of instructions that a CPU can execute.
integer literal An `int` value written into a program's code.
integrated development environment (IDE) An application that provides all the tools necessary to write, test, and debug programs.
IntelliSense A feature of Visual Studio that provides automatic code completion to help you write code faster and more efficiently.
interface A declaration that specifies only abstract methods and/or properties.
interpolated string A special type of formatted string that allows placeholders to be replaced by values at runtime.
interpreter A program that both translates and executes the instructions in a program.
iteration A single execution of a loop.
iteration variable A variable that receives the elements of an array or other collection, one at a time, during the iterations of a foreach loop.
jagged array A two-dimensional array in which the rows can have different numbers of columns.

key-value pair A data structure that consists of two pieces of data a key and a value.
keywords A word that has a specific purpose in a programming language.
Label An object that displays text on a form.
Label control A control that displays text.
lambda expression A shortcut technique for writing an anonymous method.
lambda operator The => symbol.
Language-Integrated Query (LINQ) A set of tools, integrated within .NET, that allow you to query many types of data.
leading spaces Spaces that appear at the beginning of a string.
lifetime The time during which a variable exists in memory while the program is running.
line comment A comment that appears on a single line in the program.
LINQ to Objects LINQ tools that allow you to query data stored in collections such as arrays and lists.
LINQ to SQL LINQ tools that allow you to query data stored in databases.
List A container that can hold a collection of elements.
ListBox control A control that displays a list of items, allowing the user to select one or more of the items.
Load event An event that occurs when a form is loaded into memory at runtime.
local variable A variable that is declared in a method and can be accessed only by statements in that method.
Locals window A window in the Visual Studio debugger that displays all the variables within the scope of the current method.
logic error A programming mistake that does not prevent the program from running, but causes the program to produce incorrect results.
logical AND operator An operator that connects two Boolean subexpressions, and returns true if both of the subexpressions are true. Otherwise, it returns false.
logical NOT operator An operator that reverses the value of a Boolean expression.
logical operator An operator that connects two Boolean expressions to create a compound Boolean expression.
logical OR operator An operator that connects two Boolean subexpressions, and returns true if either of the subexpressions are true. Otherwise, it returns false.
low-level language A programming language that is close in nature to machine language.
machine language The binary instructions that a CPU understands.
main memory Random-access memory, where the computer stores a program and its data while the program is running.
math expression An expression that performs a calculation and gives the value of the calculation.
math operators Operators that perform mathematical calculations.
member declarations Statements that declare a class's fields, properties, and methods.
message box A small window that displays a message to the user.
method body A collection of statements that are performed when a method is executed.
method call A statement that causes a method to execute.
method chaining A coding technique of combining multiple method calls into a single statement.
method header The first line of a method definition.
method syntax A syntax for writing LINQ queries that takes the form of method calls.
methods The operations that an object can perform.
microprocessors A computer CPU stored in a chip.
minimum field width The minimum number of spaces in which a value is displayed.
mnemonics Short words that represent machine language instructions; a key that is pressed in combination with the Alt key.
modal form When a modal form is displayed, it receives the focus and no other form can receive the focus until the modal form is closed.
modeless form When a modeless form is displayed, the user can switch the focus to other forms on the screen without closing the modeless form.
Modifiers property A property that defines a control's access.
modularized program A program that is broken into small units of code, such as methods.
multicast delegate A delegate that references multiple methods.
multiple-alternative decision structure A decision structure that allows your program to follow one of several paths of execution, depending on the value of a variable or expression.
mutable An object or value that can be changed.
mutable class A class that allows its properties and/or fields to be changed.
mutable object An instance of a mutable class. A mutable object's fields and/or properties can be changed.
mutually exclusive selection When only one item in a group of items can be selected at a time.
named argument An argument that is passed to a parameter that is specified by name in the method call.
named constant A name that represents a value that cannot be changed during a program's execution.
named field A tuple field that has been given a programmer-defined name.
namespace A container that holds and organizes code.
nested decision structure A decision structure that is inside another decision structure.
`new` expression An expression that uses the new keyword to create an object in memory and return a reference to the object.
`new` operator Creates an object in memory and returns a reference to that object.
newline character An invisible character that specifies the end of a line of text.
nullable value type A value type that can be assigned null.
numeric literal A number that is written into a program's code.
object initializers Values that are assigned to an object's properties when using object initializer syntax to create the object.
object oriented A programming approach in which a program is a collection of objects that work together.
Object Relational Designer A visual design tool that you use to create the classes needed for LINQ to SQL to work with a particular database.
Object Relational Mapping (ORM) Software that converts a database into a set of objects.
off-by-one error An error that occurs when a loop iterates one time too many or one time too few.
one-dimensional array An array that holds one set of data.
Open dialog box A standard dialog box that allows the user to browse the system and select a file to open.
OpenFileDialog control A control that displays an Open dialog box.
operands The values that an operator performs an operation on.
operating system The programs that control and manage the most fundamental operations of the computer, including managing the system's hardware devices and other programs.

operators A symbol in a programming language that performs an operation on pieces of data.
order of operations Rules that specify the order in which the math operators execute.
out keyword Specifies that an argument is an output variable.
output Data that the computer produces for people and other devices.
output device A device that formats and presents output.
output file A file to which data is written.
output parameter A reference parameter that must be set to a value by the method in which it is declared.
output symbols Symbols in a flowchart that represent output operations.
output variable A variable that is passed as an argument to a method, and when the method is finished, a value is stored in the variable.
overloaded An overloaded method is a method that has the same name as another method, but a different signature.
Panel control A rectangular container for other controls.
parallel relationship When the items in two data structures are related by their indexes.
parameter variable A variable that receives an argument when a method is called.
parameterized constructor A constructor that has parameters.
parameterless constructor A constructor that has no parameters.
parse To analyze a string of characters for some purpose.
`Parse` methods .NET methods that convert a string to a numeric data type.
partitioning methods A method that divides a collection into two sections and returns one of those sections.
Pascal case A naming convention in which the first character of the name is capitalized, and then the first character of each subsequent word in the name is capitalized.
passed by reference When an argument is passed into a reference parameter, the argument is passed by reference.
passed by value When an argument is passed by value, only a copy of the argument's value is passed into the parameter variable.
picture element Also known as a pixel, a tiny dot of color that is part of an image.
PictureBox control A control that displays an image.
pixel A tiny dot of color that is part of an image.
polymorphism An object's ability to take different forms.
postfix mode When a unary operator is written after its operand.
posttest loop A loop that performs an iteration before testing its Boolean condition.
Predicate delegate A delegate used to invoke a method that accepts one argument and returns a Boolean value.
prefix mode When a unary operator is before after its operand.
pretest loop A loop that tests its Boolean condition before it performs an iteration.
primary key A column that can be used to identify a specific row.
primitive data types The most basic data types available in C#.
problem domain The set of real-world objects, parties, and major events related to a specific problem.
processing symbols Symbols in a flowchart that represent steps in which the program performs some process on data.
program A set of instructions that a computer follows to perform a task.
program development cycle The process of creating, testing, and debugging a program.
programmatically To use code to perform an operation.
programmer A person with the training and skills necessary to create computer software.
project A collection of files and resources that make up an application.
project folder A folder that contains the contents of a project.
Properties window A window in Visual Studio that allows you to examine and change the properties of controls.
properties The data stored in an object.
property A special type of class member that allows an object to store and retrieve a piece of data.
protected member A class member that may be directly accessed by methods of the same class or methods of a derived class.
pseudocode Informal statements that describe the steps taken in an algorithm.
pseudorandom numbers Random numbers that are not truly random, but calculated from a formula.
quantifier methods A method that returns true or false to indicate whether one or more elements in a collection match a condition.
queries A statement that retrieves data from a database.
query syntax A syntax for writing LINQ queries that resembles SQL.
radio buttons A GUI control that allows the user to select one choice from several possible choices.
RadioButton control A control that allows the user to select one choice from several possible choices.
RAM The computer's main memory. The CPU is able to quickly access data stored at any random location in RAM.
random access file When working with a random access file, you can jump directly to any piece of data in the file without reading the data that comes before it.
random access memory Also known as RAM, it is the computer's main memory. The CPU is able to quickly access data stored at any random location in RAM.
read position The location of the next item that will be read from a file.
read-only property A property that can be read, but cannot be modified.
rectangular arrays A two-dimensional array in which all the rows have the same number of columns.
reference copy When only a reference to an object is copied, not the object itself.
reference parameter A parameter that receives a reference to the argument.
reference types A reference type variable stores a reference (or pointer) to its data, which is located at a separate memory location.
reference variable A variable that holds a reference to an object in memory.
relational operator An operator that compares two values and determines whether a specific comparative relationship exists between them.
reserved words A word that has a specific purpose in a programming language.
return point The point that control of the program returns to after a method has finished executing.
return type The data type of the value that is returned by a method.
row A complete set of information about a single item, stored in a table.
run time The time during which the application is executing.
running The process of executing the statements in a program.
running total A total of a series of numbers that is kept while a loop iterates through the numbers.

samples A small chunk of a digital audio file.
Save As dialog box A standard dialog box that allows the user to browse the system and select a location to save a file.
SaveFileDialog control A control that displays a Save dialog box.
schema definition file A file that describes the contents of a data set.
scope The part of a program in which a variable may be accessed.
search algorithm A search algorithm is a specific technique for locating an item in a larger collection of data.
secondary storage A type of storage that holds data for long periods time. The computer can fetch data from secondary storage as it is needed.
secure digital (SD) memory card A small external storage device.
seed value A number that initializes the formula for calculating pseudorandom numbers.
Select statement An SQL statement that retrieves data from a database.
selection sort A sorting algorithm that moves the smallest element from the unsorted part of the array to its proper position near the beginning of the array. This is repeated until the entire array is sorted.
selection structure A programming structure that performs one action if a condition exists, or possibly another action if the condition does not exists.
semicolon Marks the end of a programming statement.
sequence structure A set of statements that execute in the order that they appear.
sequential access file A file containing data that is accessed from the beginning of the file to the end of the file. To reach a piece of data in the file, you must read all the data that appears before it.
sequential search algorithm A search algorithm that sequentially steps through the elements of an array until it finds the item it is searching for.
set methods A method that performs a mathematical set operation on two collections of values.
shadows When a local variable or parameter has the same name as a field in the same class, the field is hidden from access within the method.
short-circuit evaluation When a compound Boolean expression is short-circuit evaluated, if the value of the first subexpression is sufficient to give the value of the compound expression, the second subexpression is not evaluated.
signature A method's signature is the method's name and the data types and argument kind (by value, ref, or out) of the method's parameter, from left to right.
single-alternative decision structure A decision structure that provides one alternative path of execution.
single-step To execute an application's code one statement at a time for debugging purposes.
size declarator A number that declares the size of an array.
sizing handles Small handles appearing on a control's bounding box that can be used to resize the control.
software Computer programs are known as software.
software developer A person with the training and skills necessary to create computer software.
software development tools Software that programmers use to create, modify, and test their programs.
solid-state drive A secondary storage device that stores data in solid state memory.
solution A container that holds one or more Visual Studio projects.
Solution Explorer window A window in Visual Studio that allows you to navigate among the files in a solution.
solution file The main file for a Visual Studio solution.
solution folder A folder that contains the contents of a solution.
source code The statements that a programmer writes in a programming language.
source code file A file that contains code.
stale When a field or property's value is dependent on other data and the field or property is not updated when the other data is changed, the field or property has become stale.
standard toolbar A strip displayed in Visual Studio that contains buttons for frequently executed commands.
start terminal A terminal symbol that marks the beginning of a flowchart.
statement An individual instruction in a program.
static class A class that cannot be instantiated, but rather serves as a container for methods and fields.
static field A field that is accessed by the class name, not an object name.
`static` keyword A keyword used to declare a class or class member static.
static method A method that is accessed by its class name, not an object name.
static property A property that is accessed by its class name, not an object name.
Step Into command A debugging command that allows you to single-step through the code in a method.
Step Out command When single-stepping through a method, the Step Out command causes the remainder of the instructions in the method to complete execution without single-stepping.
Step Over command A debugging command that executes a method without single-stepping through its code.
stepwise refinement A software design process in which the overall task is broken down into subtasks, and then each subtask is further broken down into more subtasks.
`StreamReader` class A .NET class that is used to read data from a file.
`StreamWriter` class A .NET class that is used to write data to a file.
string A sequence of characters.
structure A programmer-defined data type that contains one or more items of data.
structured query language (SQL) A standard language for working with database management systems.
subclasses Another term for derived class.
subproperty A property within a property.
subscript A unique number that identifies an element in an array.
substring A string within a string.
superclasses Another term for base class.
`switch` statement A multiple-alternative decision structure that allows your program to follow one of several paths of execution, depending on the value of a variable or expression.
syntax The rules of a programming language.
syntax error A programming mistake that violates the syntax rules of the language.
system software Programs that control and manage the basic operations of the computer.
tab order The order in which the controls on a form receive the focus.
tab order selection mode A mode in the Designer in which a form's tab order may be modified.
table adapter A component that connects to data source and can retrieve data from a table in the data source.
table A collection of related data.
table adapter query An SQL statement that is stored in a table adapter and can be executed simply by calling a method.
table object An object that represents a table in a database.
terminal symbol Oval symbols that appear at the beginning and end of a flowchart.

test expression An expression in a for loop that controls the execution of the loop by testing a condition.
text file A file that contains data that has been encoded as text.
TextBox A GUI control that accepts keyboard input from the user.
TextBox control A rectangular control that accepts user input.
`TimeSpan` structure A .NET structure that lets you store data about an amount of time.
tokenizing The process of breaking down a string into tokens.
tokens An individual piece of data extracted from a string.
Toolbox A window in Visual Studio that allows you to select controls to be placed on a form.
ToolTip A small tip that pops up when you hover the mouse pointer over an item in Visual Studio.
top–down design A software design process in which the overall task is broken down into subtasks, and then each subtask is further broken down into more subtasks.
trailing spaces Spaces that appear at the end of a string.
truncation Dropping the fractional part of a number.
try block A group of statements, inside a `try-catch` statement, that can potentially throw an exception.
`try-catch` statement A statement that can handle exceptions when they are thrown.
`TryParse` methods Methods for converting a string's value to other data types. The TryParse methods determine whether the string can be converted before the conversion is performed.
tuple A data structure that can hold multiple pieces of data of different data types.
two's complement A binary storage technique used to represent signed integers in memory.
two-dimensional array An array that holds multiple sets of data in rows and columns.
Unicode An extensive character encoding scheme that can represent the characters of most languages in the world.
universal serial bus (USB) drives A small external storage device.
update expression An expression that is executed at the end of each iteration of a `for` loop,
user interface The part of the computer with which the user interacts.
user The person that is using a computer program.
utility program A program that performs a specialized task to enhance the computer's operation.
value parameter An implicit parameter available to a property's set accessor. When a value is assigned to a property, the value being assigned is passed into the value parameter.
value types A value type variable has an allocated storage location in memory in which it stores its data. The size of the allocated location depends on the type of data to be stored there.
value-returning method A method that executes the statements it contains, and then returns a value.
`var` keyword A keyword that lets you declare a local variable without specifying the variable's data type. An initialization value is required.
variable A storage location in memory that is represented by a name.
variable declaration A statement that specifies a variable's data type and name.
variable name A name that identifies a variable.
Visual Studio An integrated development environment for creating applications in C#, as well as other languages.
`void` method A method that executes the statements it contains, and then terminates without returning a value.
volatile A type of memory that can be used only for temporary storage because its contents are erased when it is no longer being used.
Watch window A window in the Visual Studio debugger that allows you to add the names of variables you want to watch.
`while` clause The first line of a while loop.
width specifier A value that specifies the width of the field in which a value is displayed.

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